AccDSL

Company Histories, Accounting Interpretations, and Scenario Simulation

Product and method proposal

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This document specifies a system that translates company disclosures into traceable evidence, represents several compatible economic and accounting histories, and simulates their future journals and financial statements under explicit economic scenarios.

The proposed data contracts, methods and acceptance cases specify the intended implementation. Current source and domain limitations remain explicit; this candidate is not a completed product or a claim of full proposal readiness. Proposed libraries can be replaced while preserving those contracts. Sector rule packages require their own accounting review before they can be described as compliant with a reporting standard.

Research materials, source qualifications and review records accompany this document in `docs/design/research` and `docs/design/review`.

Reading this proposal

The starting question is what a company can do after a shock when its published accounts leave part of its history unknown. A fixed-rate borrower and a floating-rate borrower can report the same interest expense today and need different amounts of cash tomorrow. More journal detail helps only when it preserves the contracts, dates, counterparties and decisions that determine that response. Chapter 1 develops this example and the joint family of histories behind it.

The first part follows the simulation problem from evidence to decisions. Chapters 2–4 explain reconstruction, learning and permitted information. Chapters 5 and 6 distinguish financing cash from contractual promises and ownership. Chapters 8 and 9 address how to retain relevant uncertainty without storing a separate company at every branch. The complete company example in Chapter 13, the dated refinancing case in Chapter 14 and the solved risky-debt model in Chapter 26 make the consequences concrete. Their supplied assumptions are part of each answer.

The second part turns those requirements into a proposed accounting system. An analyst can start with the linked statements and actual disclosure in Chapter 28, then follow the accounting chapters appropriate to the company and reporting basis. An implementer should read the same case before the language in Chapter 33, the interfaces in Chapter 39 and the operation reference in Appendix A. The reference gives details that the narrative motivates; it does not replace the economic and accounting premises of an operation.

Basic double-entry accounting, financial statements and probability are assumed. The text introduces the additional notation where it is used. Numerical cases are teaching calculations unless an actual source is identified. A model can be internally consistent while its chosen company description remains incomplete; a successful calculation can also depend on an unverified contract term or behavioral assumption. Chapter 40 separates the checks needed for the specification, an implementation and predictive use. The detailed contents that follow support targeted reference as well as a sequential reading.

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Part I

Company histories and scenario simulation

Chapter 1

The same statements can describe different companies

A financial statement describes selected consequences of a company's activities. A simulation must explain how those consequences can change. These are different tasks. A balance of borrowing tells us how much debt is reported under a particular measurement basis and date. It does not, by itself, tell us when debt resets, who may demand repayment, which assets secure it, or whether management can replace it. Two companies can report the same debt and interest expense and respond very differently to the same increase in market rates. Recovering one plausible journal does not solve this problem. Even a complete journal may omit the contractual fact that determines the response.

This part develops a representation of companies for which evidence is incomplete. Its purpose is useful conditional simulation without concealing what the evidence leaves unresolved. The numerical examples are synthetic teaching cases, not estimates of actual company exposures. They make a distinction visible: a calculation can be internally exact while its interpretation remains conditional on an unverified assumption.

1.1 A rate shock exposes the missing information

Consider a company with cash of 95 and debt of 100 at the forecast origin. Its recent annual interest expense was 5. Suppose debt was outstanding throughout that year, interest was paid when accrued, there were no fees or derivatives, and carrying amount equalled principal. A five per cent fixed-rate loan fits these observations. So does a floating-rate loan whose reference rate plus spread happened to equal five per cent. A mixture fits as well.

Let α be the floating fraction of principal. Assume a two-percentage-point increase applies to that portion for the entire next year, with no cap, floor, maturity or hedge. Next year's interest payment is

$$I(\alpha) = 100(0.05)(1-\alpha) + 100(0.07)\alpha \tag{1.1}$$

$$= 5 + 2\alpha, \qquad 0 \le \alpha \le 1. \tag{1.2}$$

The two terms are payments on fixed and floating portions. Historical observations admit every value of α in the interval. They imply interest between 5 and 7 under the shock, but supply no frequency distribution within that range. Uniform weights would be an additional judgment.

If the interest payment is the only cash movement, closing cash is

$$C(\alpha) = 95 - I(\alpha) = 90 - 2\alpha. \tag{1.3}$$

Every result satisfies $C + I = 95$. The marginal intervals $I \in [5, 7]$ and $C \in [88, 90]$ do not retain this relationship by themselves. The pair $(I, C) = (7, 90)$ lies in the rectangle formed by these intervals but is impossible in the company described. An interface may display intervals; the calculation must retain their common dependence.

Suppose a later note establishes $\alpha \in [0.2, 0.4]$. Then

$$I \in [5.4, 5.8], \qquad C \in [89.2, 89.6], \qquad C + I = 95. \tag{1.4}$$

The family has narrowed because evidence changed. This is not an economic event changing the contract. A reproducible historical forecast needs the earlier evidence family, while a current analysis may use the narrower one.

1.2 Journals and economic histories

For a simple interest payment of 5, a journal debits interest expense and credits cash by 5. Both fixed and floating histories can contain that posting. Storing alternative journals while omitting contracts can therefore collapse economically relevant uncertainty. The journal is essential for accounting consequences, but is not a complete description of rights, obligations and exposure.

Depreciation of 10 may arise from one asset or several vintages with different remaining lives. Equal inventory totals may conceal different expiry dates and replacement costs. Receivables of 50 may consist of five customers owing 10 each or one owing 50. A customer-specific disruption distinguishes these histories even when their published totals agree.

Write an economic history as h, its journal under accounting interpretation b as $J_b(h)$, and its report as $R_b(J_b(h), h)$. The second argument permits notes to reveal facts absent from the journal. Compatibility with report y means

$$R_b(J_b(h), h) \in \mathcal{O}(y), \tag{1.5}$$

where $\mathcal{O}(y)$ includes precision and context. A representation should preserve a joint family of histories, contracts, journals and interpretations. Independently choosing from lists of possible cash, debt and policies can create a company no coherent history supports. Shared evidence need not be copied for each alternative.

1.3 Agreement depends on the question

First restrict attention to histories compatible with the same retained reports. Within that family, two histories may agree under one specified simulation or under every permitted future decision. Agreement for the full set of decisions is stronger than agreement for one selected query. Equal next-year cash under one rate path does not establish equal cash after refinancing or early redemption.

For query q, scenario s and policy π, write the requested result as $F_{q,s,\pi}(h)$. Equality

$$F_{q,s,\pi}(h_1) = F_{q,s,\pi}(h_2) \tag{1.6}$$

permits shared calculation for that question if provenance remains distinct. It does not authorize reuse for a different horizon, shock or policy. When learning changes decisions, agreement for fixed action sequences may itself be insufficient. Later chapters include observation laws in the equivalence condition.

The inverse-problem treatment in Stuart (2010) explains why inference requires an observation relation and, for Bayesian conclusions, prior assumptions. We do not transfer its function-space theory wholesale to accounting. The relevant lesson is that a forward relation need not have a unique inverse. Possible-worlds approaches to incomplete data in Libkin (2014, 2016a) provide language for answers holding across compatible interpretations. Neither identifies a company's actual transactions from totals alone.

1.4 Model completeness is a separate question

A model containing only fixed and floating unsecured loans may enumerate every feasible assignment while omitting convertible debt, leases and guarantees. Completeness within its variables does not establish completeness of the company description. An omitted instrument may dominate the stress response.

A family therefore needs a declared domain: entities, dates, instruments, bases, documents and event types. Its computational status must distinguish an exact characterization, a sampled subset and an unfinished search. Ten witnesses establish ten compatible cases, not that only ten cases exist.

Finite report compatibility also differs from statistical identification. Giacomini and Kitagawa (2021, Sections 2.1–2.3) studies observational equivalence of probability laws and set-identified parameters. Two loan histories fitting supplied reports need not induce the same law of every possible disclosure. Calling them globally observationally equivalent would overstate the evidence. Here they are report-compatible.

1.5 Selecting a convenient history

A fit criterion may choose the smoothest transaction path or the fewest contracts among equally accurate reconstructions. Such a selection can be useful for illustration. It does not turn smoothness or sparsity into an observed fact. It can erase the maturity concentration that makes a stress consequential.

If $\alpha = 0$ and $\alpha = 1$ both fit interest expense 5, a preference for fewer reset clauses may choose zero. It then predicts interest 5 under the shock; the omitted compatible case predicts 7. No historical residual selects the correct response. Repeating the fit does not resolve the missing information.

Bayesian model averaging gives a probabilistic alternative after models, priors and likelihoods are declared (Hoeting et al., 1999). Its weights are not counts of reconstruction rows. Duplicating a fixed-rate record ten times must not increase its probability. This invariance under representational duplication is a useful implementation test.

1.6 Mechanics and management responses

The rate illustration holds financing decisions fixed. A company may repay, refinance, hedge, sell assets or issue equity. Every response needs resources, counterparties and information. Automatically increasing debt to pay all obligations assumes financing capacity. Refusing all adjustment is also a policy choice, even when it is useful for stress analysis.

Shahnazarian (2004) separates financial identities from behavioral relationships. Caiani et al. (2016) specifies action sequences and financial counterparts in an agent-based economy. These ap-

proaches motivate separate accounting and behavior, but do not supply a universally transferable issuer policy. Responses must be estimated, supplied, bounded or left unresolved.

An output can distinguish historical explanations from future policies. One result may assume a floating share between 20 and 40 per cent and existing facilities only; another admits a stated refinancing offer. Averaging these without justified weights erases the question being answered. Conditional results remain informative even when an unconditional forecast is unavailable.

1.7 Trees are explanations, not compulsory storage

A tree displays mutually exclusive events, information arrival and subsequent decisions. It is useful for a small example. The continuum $\alpha \in [0, 1]$ already contains infinitely many compatible companies, so a finite selection of branches is generally an approximation. An interval and symbolic formula can represent the same uncertainty more faithfully and compactly.

Branches must not imply that a signed contract is redrawn each year. Fixed versus floating may be a persistent unknown fact. Future demand may instead generate fresh realizations under a declared stochastic process. A model must preserve the distinction.

Sometimes a formula suffices; sometimes a split at a covenant threshold is needed; sometimes filtering a probability distribution is justified. In difficult cases the result must retain an unresolved region or bounded approximation. None of these choices requires a separate stored company for every imaginable world.

1.8 A useful simulation explains its dependence

For each result a reader should be able to identify its evidence, remaining histories, contractual and behavioral assumptions, and computational status. Balancing a projected balance sheet is weaker than this requirement. The model must explain why cash is available, why a payment occurs and what changes the answer.

The remainder of this part develops reconstruction, learning, scenarios, financing, accounting transitions and tractable computation. Failure cases accompany the mechanisms they test. Their purpose is to expose a specific mistaken inference and its repair, not to imply that a finite catalogue anticipates every future error.

Chapter 2

Reconstructing joint admissible histories

The useful starting point for simulation is a description of what the evidence permits. That description must connect opening balances, transactions, closing balances, notes and contractual facts across time. Without these links, individually plausible entries can be jointly impossible. With unjustifiably strong links, the model can exclude real alternatives because an analyst supplied a convenient default. Reconstruction is consequently both an accounting problem and an inference problem.

2.1 An observation has a context

A reported amount comes with a concept, entity or consolidated group, period, units, measurement basis and precision. Cash at a date differs from cash generated during a quarter. Continuing operations differ from the whole group. Millions differ from currency units. These distinctions determine which equations may use the number; they are not annotations that can be restored after arithmetic.

Let observation o_i contain value y_i, context c_i, publication time p_i and source location ℓ_i. A reconstruction predicts $g_i(h, b; c_i)$ from history h and accounting interpretation b. If the source rounds to increment d_i using a declared rule, compatibility may take the form

$$y_i - \frac{d_i}{2} \le g_i(h, b; c_i) < y_i + \frac{d_i}{2}. \tag{2.1}$$

The half-open endpoint is justified only by the assumed rounding convention. Other tie-breaking rules, signs and reporting practices require a different observation set. The printed absence of decimal places does not establish an exact underlying value.

A debt balance reported as 100 million might therefore permit nearby amounts. A difference of 0.4 million may be immaterial for a broad exposure chart and decisive for a covenant lying inside the reporting interval. Precision should survive extraction so that the query can determine whether it matters. An arbitrary numerical tolerance chosen for solver convenience is different from reporting precision. The first controls computation; the second describes evidence.

2.2 Stocks constrain net movement

For opening balance B_0, signed movements m_k and closing balance B_1, the stock-flow identity is

$$B_1 = B_0 + \sum_{k=1}^{n} m_k. \tag{2.2}$$

It constrains the sum, not the individual events. Receivables increasing from 40 to 50 can reflect credit sales 60 and collections 50, or sales 110 and collections 100. Revenue disclosures narrow the alternatives, but returns, acquisition, currency, taxes and write-offs may add terms.

For an illustrative single-entity bridge, let A denote gross receivables, S_c credit sales, T_c tax amounts included in receivables, K collections, W write-offs, X currency movements and G acquired or transferred balances. Let L be the allowance and N net receivables. Then

$$A_1 = A_0 + S_c + T_c - K - W + X + G, \tag{2.3}$$

$$N_1 = A_1 - L_1. \tag{2.4}$$

Each movement has a defined period and scope. This is a decomposition for the example, not a universal reporting-standard prescription. If only net balances and revenue are observed, interpreting the change in net receivables as uncollected sales silently sets several terms to zero. A later credit stress then begins from a distorted exposure.

The repair is not to invent missing movements. Retain their domains, constrain them with relevant notes, and identify the quantities the evidence does not determine. A zero can be a verified amount, an explicit scenario assumption or an unsupported shortcut. The record must distinguish these meanings.

2.3 A family of compatible histories

Let z collect unknown historical amounts and categorical choices: transaction dates, principals, customers, asset vintages and accounting interpretations. Given evidence D and additional assumptions A, define

$$\mathcal{H}(D, A) = \{z : C_{\text{obs}}(z; D) \wedge C_{\text{acct}}(z) \wedge C_{\text{contract}}(z) \wedge C_{\text{time}}(z) \wedge A(z)\}. \tag{2.5}$$

Observation constraints reproduce the sources within precision. Accounting constraints preserve valid postings under a specified basis. Contract constraints preserve promises and exercise rights. Time constraints preserve sequence and persistence. Assumptions close parts of the model that the evidence leaves open.

This separation makes contradictions interpretable. An empty family may arise from extraction error, inconsistent reports, an overstrong assumption, an unsupported mapping or an impossible contract interpretation. It does not establish misconduct by the company. The reconstruction can be wrong.

Each constraint should retain its origin. A small conflicting subset may identify an annual movement table, an overlapping quarterly flow and a mistaken sign mapping. That is a focused repair task. Removing arbitrary constraints until a solver succeeds would make the result untrustworthy. Relaxation needs a reason, a changed observation model and an explanation of how answers depend on the relaxation.

2.4 Overlapping periods

If half-year revenue is 120 and first-quarter revenue is 50 under the same scope and basis, second-quarter revenue is 70. Adding both reports as independent events counts the first quarter twice. Reports constrain an underlying flow; they are not themselves transactions.

Let f_t denote flow on elementary intervals. A report for period P_i gives

$$y_i = \sum_{t \in P_i} f_t + \varepsilon_i. \tag{2.6}$$

Here ε_i denotes admitted rounding or discrepancy and is not automatically stochastic noise. Multiple reports can constrain the same f_t. Their source identities remain distinct even when their numerical information is redundant. The duplicate can independently confirm a value or reveal a later revision.

A reconstruction must also distinguish year-to-date arithmetic from seasonal interpolation. Deriving a quarter by subtraction is valid under aligned scopes. Dividing an annual flow by four assumes an intra-year pattern. A cash stress during a seasonal trough is sensitive to precisely that assumption. Even when annual totals are exact, the within-year timing may remain unidentified.

2.5 Joint margins conceal concentrated exposure

Suppose total debt is 100. Foreign-currency and domestic-currency debt are each 50. Debt maturing within a year and later debt are each 50. These two disclosures are different partitions of one debt population. They do not give four independent amounts.

Let x denote foreign-currency debt maturing within a year. Compatibility requires

	within a year	later	total
foreign	x	$50 - x$	50
domestic	$50 - x$	x	50
total	50	50	100

$$0 \le x \le 50. \tag{2.7}$$

A 20 per cent rise in the domestic-currency price of the foreign currency raises the domestic-currency value of foreign principal by 10. The additional repayment due within a year is $0.2x$, ranging from zero to 10. Matching both reported margins does not identify near-term cash exposure.

Choosing $x = 25$ selects a midpoint. Assuming independence also yields 25 in this symmetric example, but independence is an extra assumption about the joint exposure. A concentrated currency maturity wall is a case that this assumption can conceal. One scalar with bounds represents all compatible tables more faithfully than a list of arbitrary examples.

Additional dimensions increase the difficulty. Currency, maturity, collateral, lender and rate type may each have published margins. Their joint table can be large, and feasible completions may be numerous. Maintaining constraints on that table preserves the issue even when enumerating it is impractical. Later query-driven elimination can remove dimensions irrelevant to a specified result, provided it retains the relationships they induce.

2.6 Accounting and economic alternatives

Some uncertainty concerns events: whether a sale occurred, when collection happened or which asset was pledged. Other uncertainty concerns recognition and measurement: applicable basis,

estimates and policy interpretation. Their alternatives cannot be combined freely when premises conflict.

A debt cannot be both repaid and continuing to accrue its old contractual interest without a further event explaining the continuation. A historical carrying amount computed under one interpretation cannot be forecast under another for the unchanged instrument without an explicit transition. A presentation difference need not imply a different cash obligation.

Attach interpretation b to its premises and consequences. Evolve economic state under the scenario, then apply the selected accounting rules. Where the applicable interpretation remains unresolved, report conditional statements for each admitted basis. An average of incompatible book measurements is not automatically a valid financial statement under either basis.

2.7 Historical record and execution state

Future calculation rarely needs every historical event in active memory. It may need principal, accrued interest, reset dates, tax basis, asset age and outstanding rights. The evidence history remains important for provenance and later corrections. These are separate storage needs.

The evidentiary record should remain available and versioned. Execution state should contain quantities sufficient for future transitions. Compressing the latter need not delete the former. A derived state can point to supporting constraints and records, and many alternatives can share common events.

An event log alone is not an inference representation. If an event's occurrence is uncertain, a definitive log entry invents certainty. Conversely, constraints without explainable event witnesses make results difficult to inspect. The proposed representation retains symbolic conditions and constructs concrete witnesses for explanations, regression fixtures and counterexamples.

2.8 Feasible, infeasible and undecided

A solver may find a witness, prove no witness exists, or stop without deciding. An adapter may not support the instrument. These outcomes are economically different. A timeout cannot become infeasibility, and unsupported debt cannot become a zero liability.

For the question whether cash can fall below 20, a compatible witness with cash 18 establishes possibility. Failure to find one establishes safety only if the search is complete and successfully proves the relevant bound. A lower bound above 20 applies to the declared family, horizon and policies. Changing any of those premises can invalidate it.

Floating-point feasibility adds another distinction. A small residual is evidence about a numerical calculation, not necessarily a logical certificate. Exact rational examples help detect semantic errors. Larger models may require validated bounds, solver-specific certificates or an explicit undecided status. The method used to establish a conclusion belongs with the conclusion.

2.9 Reconstruction as a source-research guide

The family can identify useful missing evidence. In the currency-maturity example, another total-debt figure contributes little. A cross-classification can determine x and resolve near-term depreciation exposure. The question is which fact changes the decision, not which source is easiest to extract.

Without probabilities, candidate disclosures partition the family and give possible reductions in the answer set. With an admitted observation model, one may compute expected value of information. Assigning equal weights merely to obtain an expectation would confuse these two exercises.

A practical sequence is to construct the important joint constraints, locate uncertainty crossing a decision threshold, and inspect sources capable of resolving it. The source may not exist or may remain inaccessible. The process still identifies the economic consequence of the gap and prevents extraction volume from being mistaken for identification.

2.10 Provenance and multiplicity

Two records can support the same assertion, and one record can support several constraints. Provenance needs to preserve these distinctions without treating duplicate records as additional worlds. The semiring treatment in Green et al. (2007) concerns algebraic provenance for database queries; its value here is a disciplined account of how source contributions combine. It is not a probability assignment to economic histories.

If a source is withdrawn, a derived claim supported by another independent source may survive. If the withdrawn source supplied the only premise for a maturity date, the maturity should become unresolved. Recomputing only displayed totals can miss the effect on future liquidity. Dependencies should extend through contract state and policy eligibility to scenario results.

A stable identity for an economic fact is also different from the identity of a particular source assertion. Two filings can refer to the same loan with different labels. Merging them solely because the labels resemble each other risks combining different facilities; failing to merge verified duplicates can double debt. Entity and instrument reconciliation therefore needs evidence and reversible decisions.

2.11 A method proposal for AccDSL

The proposed reconstruction service accepts an evidence view, a declared domain and an assumption set. It returns a persistent constraint family, its known contradictions or unresolved mappings, and optional witnesses. Queries request possible outcomes, certain conclusions, bounds or distributions under separately admitted measures. A request for one witness should not trigger exhaustive enumeration.

Constraints should be grouped by their source and economic role. Observation intervals and unit conversions can often be checked before expensive solving. Linear bridges and discrete interpretation choices can then be combined, with nonlinear valuation isolated when possible. If a nonlinear method is needed, its completion status remains distinct from the accounting identities it evaluates.

The service should preserve both a stable source snapshot and a reusable symbolic family. A later disclosure can produce a refined version without overwriting the earlier one. A scenario uses a particular version and records which assumptions it adds. This is a concrete way to carry multiple possible journals and economic histories without creating a full independent database for each possibility.

Chapter 3

Learning, persistent uncertainty, and who knows what

A future disclosure may resolve an uncertainty about today's company. It may also reveal nothing relevant. Simulation must allow both outcomes. It must distinguish a change in the company from a change in the analyst's knowledge, and distinguish the analyst's knowledge from information available to managers, lenders and investors. These distinctions determine which actions are possible and which apparent forecasting successes are merely uses of future information.

3.1 Five dates attached to one fact

A loan can be signed on 15 January, take economic effect on 1 February, appear in a report covering the quarter ending 31 March, and become public on 10 May. A system may ingest the report on 12 May. The signature, economic, reporting, publication and ingestion dates answer different questions. A forecast made on 30 April cannot use the 10 May disclosure simply because it describes a March balance.

For a report-based forecast origin t, let D_t contain assertions available by t under the declared availability rule. If the analysis uses public information, publication time is relevant. If it models a lender with private contractual access, the information rule differs. Ingestion time controls reproducibility of the system's own operation; it need not equal the earliest public availability.

A historical replay should therefore specify both the economic date and evidence cutoff. A corrected extraction performed today may accurately recover a number that was publicly available before the old cutoff. A new restatement published today was not available then. The first can repair a data-processing mistake in an old evidence view; the second changes the information set and must not be silently backfilled into the old forecast.

3.2 Persistent unknown facts

Let θ denote a contract characteristic fixed at inception, such as whether a loan is fixed or floating. Let X_t denote changing economic state, including cash, accrued interest and remaining principal. A simple transition is

$$X_{t+1} = T(X_t, \theta, a_t, \varepsilon_{t+1}), \qquad \theta_{t+1} = \theta_t. \tag{3.1}$$

Here a_t is a decision and ε_{t+1} a future innovation under the specified scenario or stochastic process. The second equation is essential. Uncertainty about a fixed contract does not license redrawing it each period. Refinancing or amendment can change the terms, but those are dated events with their own permissions and cash consequences.

Suppose a loan is either fixed at five per cent or floating at the market rate, and the market rate is seven per cent for each of two future years. The possible interest sequences are $(5, 5)$ and $(7, 7)$. Independent yearly resampling adds $(5, 7)$ and $(7, 5)$. Those additional paths describe contract changes that the model did not authorize. Their inclusion can alter liquidity minima, covenant duration and refinancing behavior.

The same persistence issue affects customer identity, asset vintage, tax basis, priority and collateral. A compact state must retain persistent variables even if current totals do not depend on them. Eliminating a variable because it is irrelevant today is unsafe if it changes later observations or events.

3.3 Set-valued evidence updates

When no probability measure is justified, retain a feasible family $\mathcal{H}_t$. If a new compatible hard observation y supplies condition C_y, updating the same historical scope gives

$$\mathcal{H}_t^+ = \{h \in \mathcal{H}_t : C_y(h)\}. \tag{3.2}$$

This operation narrows the old family. It does not advance the economic clock. Forward propagation then maps each remaining history into future possibilities under the transition relation. A smaller historical family can still produce a wider future range when the horizon lengthens or new shocks are admitted.

An empty intersection calls for investigation. A source may conflict with another source, the observation mapping may be wrong, or a model assumption may be too narrow. Selecting one convenient history to restore nonemptiness would hide the contradiction. The result should identify the conflicting premises and preserve the previous view.

A restatement is not always an additional constraint on the old asserted amount. It may replace that assertion in a current evidence view. The system should preserve the original source and link the correction to it. Historical replay can then reproduce the original view, while current reconstruction uses the revised one. Conjoining both as exact truths would manufacture infeasibility.

3.4 Probabilistic filtering requires a model

If a prior measure μ_t and likelihood $L(y \mid h)$ are admitted, Bayesian updating has the form

$$\mu_t^+(dh) = \frac{L(y \mid h)\mu_t(dh)}{\int L(y \mid u)\mu_t(du)}. \tag{3.3}$$

The denominator must be finite and positive. Its role is normalization, not an opportunity to substitute an arbitrary fallback when the observation has zero modeled probability. A zero denominator can reveal a misspecified likelihood, an impossible observation under the model or inappropriate handling of a continuous conditioning event.

For continuous exact observations, ordinary density multiplication must be understood relative to the declared dominating measure and conditional distribution. Adding a tiny Gaussian error

can make a computation run, but it changes the observation model. Its scale should reflect a justified measurement or reporting process, or be explicitly treated as a sensitivity assumption. Numerical convenience does not establish the company's measurement error distribution.

The POMDP formulation in Kaelbling et al. (1998, Sections 3.1–3.4) provides a formal treatment when transition and observation probabilities are known. Its belief state summarizes the history needed for control under those assumptions. An unweighted feasible family is not a probabilistic belief state merely because it is stored in the same data structure. Supporting both requires distinct semantics.

3.5 A finite filter that preserves contract identity

Let θ be an unknown but fixed contract type, x_t a changing hidden operating state, and y_t the analyst's observation. A declared probabilistic model supplies transition probabilities $P_\theta(x' \mid x)$ and observation likelihoods $\ell_t(y \mid \theta, x')$. With joint posterior weights $w_{t-1}(\theta, x)$, the prediction and update are

$$w_t^-(\theta, x') = \sum_x P_\theta(x' \mid x) w_{t-1}(\theta, x), \tag{3.4}$$

$$Z_t = \sum_{\theta, x'} \ell_t(y_t \mid \theta, x') w_t^-(\theta, x'), \tag{3.5}$$

$$w_t(\theta, x') = \ell_t(y_t \mid \theta, x') w_t^-(\theta, x') / Z_t. \tag{3.6}$$

There is no summation that changes θ into another contract type. A contract modification needs an explicit event and a different transition model. If $Z_t = 0$, the result is an evidence conflict under the declared model; normalizing arbitrary fallback weights would answer another question.

An uninformative observation has constant likelihood across states, so its update does not change the predicted relative weights. A missing disclosure is uninformative only when the observation model makes it so. If no prior is justified, propagate a feasible set of joint (θ, x) pairs through the transition relation and evidence restrictions. That procedure produces support, not posterior probabilities. A declared set of priors can instead be updated member by member; zero-likelihood members need an explicit conditioning rule, and one must not replace the result by independently chosen period marginals.

A source correction can remove an earlier restriction. Retaining only the last posterior may then be insufficient: a zeroed type cannot be recovered by multiplying it by another likelihood. Keep the original prior or feasible family, the evidence versions and the transition model, and replay all currently active evidence. Checkpoints can accelerate replay only when their evidence dependencies remain valid. The reference filter compares this recursion with direct path enumeration for four priors and all 27 length-three observation sequences. It also checks rare positive types, uninformative observations, conflict, and a correction that reopens a discarded type. This is an exact finite method; estimating a defensible issuer likelihood and controlling particle approximation remain separate problems.

3.6 An informative and an uninformative disclosure

Return to the fixed-versus-floating loan. Assume, only for this example, prior probabilities of one half each. A report can say "floating" or "fixed." Suppose the report identifies the correct type with probability 0.9 and the wrong type with probability 0.1, independently of other

modeled quantities conditional on the type. After a "floating" report, Bayes' rule gives

$$\Pr(F \mid y = F) = \frac{0.9(0.5)}{0.9(0.5) + 0.1(0.5)} = 0.9. \tag{3.7}$$

These probabilities are supplied assumptions, not estimates from the loan balance. The example isolates the mechanics of updating. If the report were equally likely under both types, its likelihood would cancel and the prior would remain unchanged. More pages of disclosure need not imply more information about the exposure of interest.

In the set-valued version, an exact contract note may eliminate one type. An ambiguous phrase compatible with both eliminates neither. A probabilistic false-positive model is a different treatment from a set of possible interpretations. The analyst should choose the treatment supported by the evidence process and preserve its assumptions.

3.7 Why some ambiguity does not disappear

A long sequence of reports can remain uninformative about a parameter. In the rate example, if fixed and floating rates coincide throughout the observed period, additional interest totals may never distinguish the contract types. Sample size alone does not resolve an unidentified direction.

The set-identification analysis of Giacomini and Kitagawa (2021, Sections 2.1–2.3) separates parameters identified by the observable law from aspects of the prior that data cannot revise. The accounting application is conditional: one must first establish which observations depend on which hidden quantities. It is not enough to invoke a general consistency slogan and expect all uncertainty to vanish.

Persistent ambiguity can also become irrelevant to one query without being resolved. After final repayment, the old rate clause may no longer affect future cash. It may still matter for explaining historical interest, estimating management behavior or evaluating a counterfactual refinancing decision. Removing it from an active cash-flow state is different from asserting that the evidence identified it.

3.8 Managers need not share the analyst's ignorance

Public disclosures may leave θ unknown to an analyst even though management signed the contract. A management policy can legitimately depend on that private fact. If managers refinance only floating-rate debt after a rate shock, their actions differ across histories that the analyst cannot distinguish. That is not foresight; it is a specified information advantage.

Let I_t^a denote actor a's information history and let π_t^a map it to a decision. Then

$$a_t^a = \pi_t^a(I_t^a). \tag{3.8}$$

Two simulated worlds with the same I_t^a must receive the same action under a deterministic policy. For a randomized policy, they must receive the same conditional action distribution. The simulator's access to hidden state does not entitle the actor to use it.

Different actors can have different information. A lender may observe covenant certificates before public investors. A subsidiary may know its operating cash before a parent receives a report. A regulator may receive nonpublic prudential data. The model should specify these information channels where they affect behavior; assuming universal knowledge can make financing appear available too early.

3.9 A simple foresight counterexample

A manager must choose at date zero between two production plans. Plan A yields 10 if next year's demand is high and zero if low. Plan B yields 6 in either state. Assume high and low demand are equally likely and the manager learns demand only after choosing. The expected values are 5 and 6, so the expected-value policy selects B.

A simulator that first draws demand and then chooses the best plan in that realized world obtains 10 in the high state and 6 in the low state, averaging 8. It has valued perfect advance information. The arithmetic is correct for a different decision problem. The target problem permits an average of 6.

The discrepancy is not repaired by drawing more scenarios. It is caused by an information violation. A nonanticipativity check must require the same date-zero action across worlds sharing date-zero information. In a tree formulation this often appears as equality constraints on decisions at indistinguishable nodes. In a symbolic representation it can be enforced by restricting the policy's inputs.

3.10 Signals can have economic value

Suppose the manager can pay 1 at date zero for a perfectly informative demand signal before selecting A or B. With the signal the gross expected payoff is 8 and the net payoff is 7. Without it the payoff is 6. The signal is worth 2 gross and 1 net in this example. These calculations use a declared distribution and feasible action set.

The same logic applies to obtaining a debt schedule before committing liquidity, but real disclosures may be noisy, delayed or nonpublic. A signal's value depends on whether the actor can act after receiving it and whether the relevant action remains feasible. A maturity note arriving after repayment cannot improve that repayment decision, even if it improves a retrospective explanation.

Without probabilities, one can compare worst-case payoffs or regret before and after a partition of the family. The result is not an expected value of information. An interface should name the criterion because different criteria can recommend different research priorities.

3.11 Simulated learning does not revise actual evidence

A scenario may include a hypothetical disclosure at a future date. Inside that run, an actor updates its information and changes its action. The hypothetical report must not enter the analyst's actual evidence store as a fact. Otherwise running a scenario can contaminate later reconstructions with imagined information.

Separate identifiers should distinguish actual source assertions, simulated observations and analyst assumptions. A scenario observation can refer to the source model that generated it and to the hidden state of that run. It cannot acquire the authority of a filed report merely because its text looks like one.

The distinction also matters for conversational systems. An explanation generated from a selected witness should say that it is one compatible history. If the explanation is later ingested as evidence without its conditional status, a possible event becomes a purported fact. Provenance and type checks should prevent that feedback.

3.12 Learning can change the appropriate compression

Two histories may have equal current cash and equal next-period payments, yet differ in the chance of a revealing report. If the report affects a later refinancing choice, their future decision values can differ. A merge based only on current accounting totals discards this difference.

Suppose two contract states both pay 5 next year. In one state, a scheduled certificate will reveal a restrictive covenant before the next refinancing date; in the other, no such information arrives. If management or lenders condition their offers on that certificate, equal immediate cash does not imply equal future financing. The observation process belongs in the transition model used for compression.

This is why exact aggregation later requires equality of the joint law of successor state, observations and requested outputs for each allowed action. Preserving a marginal cash distribution is too weak. A graph may share current arithmetic while retaining separate information states for future decisions.

3.13 Recursive ambiguity and persistent contracts

Bayesian conditioning and a fresh worst-case choice at every date answer different questions. Suppose a persistent binary type θ is either zero or one, and tomorrow a fair binary signal X unrelated to that type is observed. The admitted laws are Q_0 and Q_1: under Q_i, $\theta = i$ with certainty and X remains fair. For the loss $Z = \mathbf{1}\{X = \theta\}$, either law gives expected loss $1/2$, so

$$\max_{Q\in\{Q_0,Q_1\}} \mathbb{E}_Q[Z] = \tfrac{1}{2}, \qquad \mathbb{E}_X\left[\max_{i\in\{0,1\}} \mathbb{E}_{Q_i}[Z \mid X]\right] = 1. \tag{3.9}$$

The second expression selects type zero after signal zero and type one after signal one. It therefore evaluates a new law with $\theta = X$. That law is absent even from the convex hull of the original two laws: every mixture retains independence between the signal and type. No type has physically changed along a realized path, yet dependence has changed across possible paths. Preserving path support alone would miss the error.

Shapiro defines rectangularity through equality between static worst-case evaluation and its nested conditional counterpart (Shapiro, 2016, Definition 2.1, Theorem 2.1 and proof, Section 5). His finite construction pastes conditional distributions across histories; it can enlarge the family. A dynamic program for that enlarged family solves a changed robustness question. Its convenience cannot justify claiming that it solved the original persistent-type problem.

Terminology also requires care. Pichler and Shapiro call a product family a rectangular setting, but explicitly show that its original worst-case functional and its composite conditional functional can differ (Pichler and Shapiro, 2021, Section 4.2, Example 4.7). Computing a conditional law-based risk measure can also differ from conditioning an unconditional ambiguity set. Take six equiprobable states and an observed three-state group with losses $0, 1, 2$. Conditional expected shortfall at level $1/2$ within that group is $5/3$. The unconditional six-state risk envelope permits probabilities at most $1/3$ per state. It can assign $1/6$ to the loss-2 state, zero to the other two group members, and $5/18$ to each state outside the group. This is an admissible probability law; after conditioning on the group it assigns probability one to loss 2. The conditional worst case over that envelope is therefore 2. The difference is between two precise conditional risk definitions, not an arithmetic error (Pichler and Shapiro, 2021, Example 3.2 and equations (3.1)–(3.5)).

Policy consistency and equality of risk values are distinct as well. Xin and Goldberg give moment-constrained inventory examples in which an initially optimal policy ceases to be

optimal after reoptimization, and another in which the optimal policy sets agree although the two optimal values differ (Xin and Goldberg, 2021, Definitions 3.1–3.3, Observations 5–7, proofs in Sections 6.3–6.5). The company model must specify whether management can commit to a policy, what information each later decision uses, and which family of joint laws is retained. For a finite reference problem, enumerate complete admissible laws and observation-measurable policies before choosing an approximation. For larger problems, a set of beliefs over persistent types or an explicit precommitment formulation can retain those restrictions, but its recursive sufficiency and computational cost need a separate derivation. Repeated worst-case optimization remains useful as a stress test when the changed family is stated explicitly.

3.14 Persistent parameters and particle approximation

A particle approximation can reduce computation after a probability model is chosen. It does not create that model. A type with probability 0.01 has probability $0.99^{100} > 1/3$ of disappearing from 100 independent multinomial resampling draws. No new evidence is involved in this loss. Rare but material types therefore need explicit support checks, stratification or separate conditional computations, with the resulting approximation stated.

For state-space models with persistent unknown parameters, SMC^2 combines parameter particles and state filters, with particle-MCMC rejuvenation (Chopin et al., 2013, Section 3.2, equation (7), Proposition 1, Appendix A). Its extended target has the desired parameter posterior as a marginal under the source assumptions. Finite particles still approximate that posterior; exactness of a target marginal does not mean that one finite run represents every possible type. The unbiased full particle likelihood estimate used in this construction must be distinguished from an arbitrary estimated likelihood ratio.

The inspected author-linked modern implementation creates a state filter for each parameter particle, runs that filter through the current observation date when evaluating a proposed parameter, and incorporates a likelihood correction when increasing the state-particle count. These operations have real time and memory costs. Retaining only filtering arrays differs from retaining complete genealogies. This is a candidate for explicitly probabilistic filtering, not a replacement for symbolic unweighted historical ambiguity.

3.15 Changing particles without changing the target

Increasing the number of particles can change an inference result even before new evidence arrives. The reason is that a likelihood estimate and its random construction belong to the state used by particle MCMC. Replacing the estimate with another unbiased estimate does not automatically preserve their joint distribution. A test that examines only current company-type probabilities can miss the error.

Fix an evidence revision v and a persistent type θ. Write its prior as $p(\theta)$ and likelihood as $L_v(\theta)$. A particle representation of size N uses random variables u with law $m_N(u \mid \theta)$ and produces a nonnegative unbiased estimate $Z_{N,v}(\theta, u)$ of that likelihood. Assume the normalizing constant C_v is finite and positive. The joint target weights the random construction by the estimate it produces:

$$\widetilde{\pi}_{N,v}(\theta, u) = \frac{p(\theta) m_N(u \mid \theta) Z_{N,v}(\theta, u)}{C_v}. \tag{3.10}$$

Integrating over u replaces $Z_{N,v}$ by its expectation L_v, giving the desired type posterior. This is the extended-target argument in the retained SMC^2 source, arXiv 1101.1528v3, 27 January 2012 (Chopin et al., 2013, equation (7), Proposition 1 and Appendix A). Unbiasedness establishes

this marginal relation; it does not make an arbitrary finite particle population equal to the posterior.

Now change representation from N to M and draw fresh randomness u' from $m_M(\cdot \mid \theta)$. The augmented proposal law is $\widetilde{\pi}_{N,v}(\theta, u)m_M(u' \mid \theta)$. With backward law $m_N(u \mid \theta)$, the desired augmented target is $\widetilde{\pi}_{M,v}(\theta, u')m_N(u \mid \theta)$. Their ratio is

$$w(\theta, u, u') = \frac{\widetilde{\pi}_{M,v}(\theta, u')m_N(u \mid \theta)}{\widetilde{\pi}_{N,v}(\theta, u)m_M(u' \mid \theta)} \tag{3.11}$$

$$= \frac{Z_{M,v}(\theta, u')}{Z_{N,v}(\theta, u)}. \tag{3.12}$$

The prior, normalizer and auxiliary proposal laws cancel because the type and evidence revision are unchanged. This recovers the exchange correction in Section 3.6.1 of the inspected source. The ratio requires the relevant support conditions. The following example uses strictly positive estimates, so it does not hide a zero-denominator problem inside that cancellation.

Take two types with prior probability 1/2 each and true likelihood one. For type zero the old estimate is 1/2 or 3/2 with equal probabilities; the new estimate is 1/4 or 7/4, again equally likely. Type one's estimate is always one. Each estimator is unbiased. Order the new joint states as $(0, 1/4), (0, 7/4), (1, 1)$. The likelihood-weighted target in equation (3.10) and the distribution produced by unweighted replacement are respectively

$$\pi = \left(\frac{1}{16}, \frac{7}{16}, \frac{1}{2}\right), \qquad q = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right). \tag{3.13}$$

Both assign probability 1/2 to type one. Immediate marginal agreement therefore passes a test that the joint target fails.

To see the consequence, make one valid particle marginal Metropolis–Hastings move. Propose the other type, draw its new estimate, and accept with $\min(1, Z_{\text{proposed}}/Z_{\text{current}})$. From $(0, 1/4)$ the proposal is accepted certainly. From $(0, 7/4)$ it is accepted with probability 4/7. From type one, the low-estimate proposal has total accepted probability $(1/2)(1/4) = 1/8$, and the high proposal has probability 1/2. The transition matrix, with the same state order, is thus

$$K = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 3/7 & 4/7 \\ 1/8 & 1/2 & 3/8 \end{pmatrix}. \tag{3.14}$$

The correct joint law satisfies $\pi K = \pi$ and detailed balance. Starting instead from q, the probability of type one after the move is

$$(qK)_3 = \frac{1}{4} + \frac{1}{4}\frac{4}{7} + \frac{1}{2}\frac{3}{8} = \frac{65}{112} = \frac{1}{2} + \frac{9}{112}. \tag{3.15}$$

The transition is valid for its stated target; the wrongly initialized joint distribution is the problem. This is an exact one-step counterexample, not a claim that a valid PMMH kernel has the wrong stationary distribution. Applying the exchange weight restores the desired joint law in the finite enumeration.

The inspected author-linked implementation includes the exchange correction (Chopin and contributors, 2026, `SMC2.exchange_step`, pinned source). Its new filter is run through the preceding observation before the next incremental weight is added; processing that next observation in both places would count it twice. Rejected Metropolis proposals retain their existing filter state, and repeated resampled selections receive distinct mutable filter copies. These are scoped source-code correspondences, not a run of the full modern package or a performance comparison. The modern waste-free extension requires its own dependence and finite-run analysis.

3.16 A correction can restore an excluded history

Changing the particle representation preserves the likelihood being computed. Correcting a historical disclosure can change that likelihood itself. Suppose two histories have prior probabilities $(1/2, 1/2)$. An old hard observation has likelihoods $(1, 0)$, producing posterior $(1, 0)$. Its replacement has likelihoods $(1/4, 3/4)$. Recomputing from the prior gives

$$\pi_{\text{replacement}} = (1/4, 3/4). \tag{3.16}$$

Multiplying the restricted old posterior by the replacement likelihood and normalizing instead gives $(1, 0)$. The operation conjoins the old observation with the new one. It cannot recover the history the old assertion excluded. The new-to-old likelihood ratio is also undefined on that restored history, where the denominator was zero.

A replacement therefore requires replay from a sufficiently informative preserved checkpoint, or a separately justified bridge with the required support. The inference request must identify the evidence revision and which assertion is replaced. Merely attaching a corrected document to an existing particle population does not establish the corrected posterior. This is the probabilistic counterpart of retaining alternative histories that become admissible after a source correction.

The source's quantitative guarantees require additional premises. Its Appendix B uses uniform transition and observation-density ratio bounds and posterior regularity conditions, and its refreshed-particle idealization is stronger than a finite PMCMC move (Chopin et al., 2013, Proposition 2 and Appendix B). A permanent deterministic type, a hard zero-likelihood accounting observation or an unidentified parameter direction can violate the intended transfer assumptions. Exact finite target calculations establish the distinctions above; they do not establish company-model mixing, scaling, likelihood identification or forecast accuracy.

3.17 Replaying a declared evidence vintage

The preceding correction becomes an executable calculation once the inference request says exactly which information remains active. A document identifier alone is insufficient. Two documents may repeat one observation, and a new document may replace an earlier assertion rather than supply a second independent measurement. The model must retain these relations before it multiplies likelihoods.

Consider a finite set of persistent company types $h \in H$, with declared prior masses $p(h)$. Each active observation occupies a named slot and supplies a likelihood $L_j(h)$. We assume that these observation blocks are conditionally independent given h in this example. Under that assumption, for a selected evidence vintage v with active slots A_v,

$$Z_v = \sum_{h \in H} p(h) \prod_{j \in A_v} L_j(h), \qquad \pi_v(h) = \frac{p(h) \prod_{j \in A_v} L_j(h)}{Z_v}. \tag{3.17}$$

The normalizer must be positive. A zero value means that this finite model assigns zero likelihood to the selected evidence; it does not justify returning an arbitrary distribution. Missing prior probabilities or likelihoods instead leave the calculation unresolved. Neither circumstance supplies probabilities for a family that was specified only as a feasible set.

The selected reference, `FiniteVintageInference/1`, retains the named types, exact rational prior, observation-model identity and factorization assumption. A request also names the actor and information cutoff, the active observation versions and their replacement ancestry. Each observation retains its source content, information identity, slot, likelihood vector, supplied

actor-availability time and any superseded version. Its content identifier changes when any of those fields changes. The semantic information identity answers a different question: whether two records purport to describe the same informational contribution. A replacement keeps that identity even when its likelihood or source changes.

The supplied availability time must be no later than the request's cutoff. An unknown time remains unresolved; a later time is ineligible for that request. The reference uses the existing exact fixed-offset timestamp conversion, including fractional seconds, rather than rounding time to a calendar day. Its convention excludes leap seconds and unspecified offsets. These checks establish consistency with declared availability. Establishing when an actual investor, manager or system could obtain the information still requires a source-selection procedure.

Replacement selects a new head for an existing slot and preserves the chain to the earlier record. Retraction removes that slot's likelihood contribution, equivalently replacing it by one for every type. The request cannot select two heads for one slot, use a missing predecessor, or treat a different information identity as a revision of the same observation. Several unresolved candidate revisions therefore require several conditional requests. Selecting a head does not establish that it is the authoritative version.

The numerical consequences are immediate in the two-type example. After the correction, the posterior is $(1/4, 3/4)$. Suppose a separately declared, conditionally independent signal has likelihoods $(1/2, 1/4)$. Multiplying gives unnormalized masses $(1/8, 3/16)$, whose sum is $5/16$, and hence

$$\pi_{\text{corrected+signal}} = (2/5, 3/5). \tag{3.18}$$

Retracting the corrected observation leaves only the new signal; with the original equal prior the resulting posterior is $(2/3, 1/3)$. Retracting the only observation restores the prior. A type assigned zero prior mass remains at zero unless the prior itself is changed explicitly. Restoring support lost through an erroneous observation is distinct from inventing support absent from the declared model.

Repackaging the corrected observation under another slot would produce a different answer. With the equal prior, counting its likelihood twice gives masses proportional to $(1/16, 9/16)$ and posterior $(1/10, 9/10)$. The reference rejects two active observations that declare the same semantic information identity. An earlier implementation checked distinct slots but omitted this condition: its arithmetic tests passed while the declared information contract remained incomplete. The repaired example makes the resulting overconfidence visible. False distinct identities can still conceal repeated information, just as a false independence assumption can make a well-formed calculation substantively wrong.

A checkpoint retains its complete request and computed result. Reuse requires the same model and prior, actor and exact cutoff, with the old active records forming an unchanged prefix of the new sequence. New independent records can then update that prefix. A replacement that changes the prefix instead causes replay from the preserved prior. This avoids the failed operation that starts from $(1, 0)$ and attempts to recover the excluded type by reweighting it. Corrupt checkpoint contents or incompatible model and information definitions are rejected rather than silently relabelled.

In this finite reference the old checkpoint calculation is itself recomputed before reuse, so the test establishes the posterior relation without claiming a computational saving. Safe reuse from a larger store would require a proved sufficient dependency set. Particle checkpoints would additionally retain the likelihood estimate, proposal and ancestor state required by the extended target. A vector of posterior weights alone is insufficient.

The reference compares complete masses and normalizers over a finite rational test grid and explicit contrary cases, including a correction, retraction, duplicate information, unknown times

and a later fractional-second timestamp. An uninformative observation leaves both types alive. This illustrates a representation in which uncertainty can persist without enumerating a new copy of every historical node. It establishes the exact finite calculation under its declared premises; continuous inference, source completeness and the suitability of the likelihood for a real company remain further questions.

3.18 A correction reaches different decision makers at different times

Replaying the corrected history answers an inference question. It does not yet say which decision maker could use the correction. Suppose the analyst learns on 1 February 2026 that the earlier observation was wrong, while the manager receives that correction on 1 March. On 15 February they possess different information even though they are studying the same company. A simulation of the manager's choice must use the manager's information.

Continue the hypothetical fixed-versus-floating example. Under the original observation the posterior is $(1, 0)$; under the correction it is $(1/4, 3/4)$. Suppose an unhedged exposure costs zero for the fixed type and 20 for the floating type. A perfect hedge costs 6 in either case. These are supplied losses in the same units, and both actions are assumed feasible. They are enough to define this comparison without inventing an observed hedging price or an accounting transaction.

Let p denote the probability of the floating type under the particular actor's available information. The expected losses are therefore

$$L(\text{unhedged} \mid p) = 20p, \qquad L(\text{hedged} \mid p) = 6. \tag{3.19}$$

The original posterior gives an unhedged loss of zero. The corrected posterior gives 15, so the hedge has the smaller declared expected loss. At $p = 3/10$ the two losses both equal 6 and both actions remain minimizers. An implementation must retain that tie unless a separate choice rule is supplied. The manager on 15 February still uses the old posterior, while the analyst can already select the hedge under the corrected one.

The selected information procedure takes an actor and an exact cutoff, selects the available unreplaced observation in each information slot, and passes that request to the finite evaluator. Adding the future correction does not change the manager's stored 15 January request or its decision. Later knowledge can motivate a new decision, but it must not silently replace the information used to reproduce an earlier one.

Unknown availability cannot be treated as known absence. Two incomparable available revisions also leave the request unresolved; choosing the newer file arbitrarily would add an authority rule. The checked examples retain these distinctions, refuse corrupt observation content and fabricated posterior results, and leave unknown action losses unresolved. They select actions from the declared feasible set; they do not execute a hedge or infer its causal effect. Actual actor access, registry completeness, instrument terms and feasible financing still need evidence.

3.19 A proposed information service

AccDSL should represent evidence views and actor information views separately. An evidence view selects actual assertions by availability, revision and scope. An actor view specifies observable state variables, private contractual facts and signal channels at each decision date. Policies receive only that view and permitted memory.

Forward execution first applies events available at the current boundary, constructs observations, updates the relevant information state, and then permits decisions according to the stated timing. Alternative orderings are possible, but they must be explicit and tested. Moving a decision across an observation boundary changes the economic problem.

The service should refuse ambiguous timing where it affects the answer. It can still return a conditional comparison of both orderings. Such a comparison is preferable to silently selecting whichever ordering makes the simulation easier. Exact timestamps, time zones and publication lags matter when events occur on the same calendar date.

3.20 Tests that discriminate information errors

A useful test duplicates a world, changes an unobserved future shock and confirms that the current action remains unchanged. Another changes a privately known contract term and confirms that an informed manager's policy may change while a public analyst's evidence view does not. A third inserts a later restatement and checks that an earlier frozen forecast remains reproducible.

These are semantic tests. They do not require a fitted behavioral model. They establish that information restrictions are implemented as specified. Whether a particular manager actually follows the chosen policy is a separate empirical question. Keeping that distinction prevents a passing information test from being presented as behavioral validation.

A final test removes all informative observations and confirms that persistent uncertainty remains. A system that gradually collapses to a single history merely because a particle filter resampled or an optimizer selected one witness has confused computational concentration with evidence. The retained family and the approximation used for calculation must remain distinguishable.

Chapter 4

Scenarios, interventions, and feasible decisions

A scenario is a specified change to a model, not merely a new column of forecast numbers. It must say what changes, what remains structurally linked, what actors observe and how they may respond. A rate increase imposed on a contract schedule differs from conditioning an empirical model on periods when rates were high. A revenue decline with unchanged prices differs from one caused by price reductions. These distinctions determine the cash flows and the meaning of the result.

4.1 Describe the experiment before calculating it

A scenario request should identify the forecast origin, horizon, evidence view, shock variables, shock timing, transition rules, management policy and requested outputs. It should also specify whether the shock is a deterministic path, a random process or an ambiguity set. A phrase such as "rates up 200 basis points" leaves reset dates, currency curves, spread behavior and duration unresolved.

For a deterministic scenario s, let $u_t(s)$ be the imposed external path. The company evolves as

$$X_{t+1} = T_t(X_t, \theta, a_t, u_{t+1}(s)), \qquad a_t = \pi_t(I_t). \tag{4.1}$$

The transition includes contractual events and operational behavior. The policy uses information available at its decision time. A statement projection is then a reporting map of the resulting events and states. Directly shocking a reported total can be a useful reduced-form approximation, but it requires an explanation of how the corresponding events and balances are reconstructed.

A scenario can also specify a set $\mathcal{S}$ of paths and a collection Π of candidate policies. Let $\mathcal{A} \subseteq \mathcal{H} \times \mathcal{S} \times \Pi$ contain the jointly admitted triples: a compatible history, an allowed shock path and a policy whose information and execution rules are specified for that history and path. The outcome family is

$$\mathcal{Y} = \{F(h, s, \pi) : (h, s, \pi) \in \mathcal{A}\}. \tag{4.2}$$

The joint restriction matters. A lender's offer can depend on the company's contract state; a commodity shock can affect both sales and collateral; a policy may be admissible only for certain information states. Choosing each component independently from its marginal set can violate those dependencies. An admitted triple can still produce a refused funding request, distress or an unresolved result when its declared execution rules allow those outcomes. Admissibility does not mean that every attempted payment succeeds.

4.2 Conditioning and intervention

Suppose sales S and market rates R both respond to an economic factor U. Observing high rates can reveal information about U and therefore about sales. Imposing a rate change while holding the structural equation for sales fixed asks a different question. The distinction is formalized in causal models by separating conditioning from intervention.

In a simple illustration, let $R = U$ and $S = 100 - 10U$, where U is either zero or one. Observing $R = 1$ implies $U = 1$ and sales 90. An intervention setting $R = 1$ replaces the rate equation. It does not, by itself, change U or the sales equation. If $U = 0$ in the starting world, sales remain 100 under that intervention.

This is a local derivation illustrating the distinction in Pearl (2009, Sections 3.2–3.4). It is not an empirical claim that rates have no effect on sales. A richer structural model may include a direct rate effect, feedback and expectations. The point is that the intervention must name which mechanisms change. An observed correlation cannot decide that intervention on its own.

For financial stress work, a reduced-form scenario may intentionally impose several joint paths without claiming a causal intervention. That is a valid conditional exercise if described accurately. It should not be presented as the causal effect of one variable when the exercise changes several variables simultaneously.

4.3 One shock can enter several contracts

A market-rate shock may affect floating debt, deposit income, derivative settlements, pension valuation and refinancing prices. Applying it independently to each reported line can miss offsets or count them twice. The shock should enter the relevant primitive variables, and the affected contracts should generate their consequences.

Suppose floating debt of 100 and a receive-floating/pay-fixed swap of notional 60 share the same reference rate and reset date. Ignoring basis, timing and credit effects, a two-point increase raises debt interest by 2 and swap receipts by 1.2, leaving a net increase in cash interest cost of 0.8. Applying the shock to net finance cost and again to the swap would double count the hedge.

The simplification is material. A different reset date, reference index, currency, collateral agreement or counterparty default changes the result. The correct representation preserves separate contracts and links them through the common rate process. An accounting hedge designation is another relationship; it does not itself create the economic offset.

4.4 Timing changes exposure

For principal N and rate increment Δr applying over year fraction δ, the simple incremental payment is $N\Delta r\delta$. A full-year shock on 100 at two points gives 2. A half-year exposure gives 1. If the debt resets only after maturity, the current contract's payment may not change at all.

The year fraction must use the contract's convention and relevant dates. Replacing actual reset schedules with annual averages can preserve annual interest in a calm period while changing a short-horizon liquidity stress. A payment on the last day of a quarter and one on the first day of the next quarter have different reporting and liquidity consequences even if their annual totals agree.

A simulation should state whether events at a boundary occur before or after a decision.

Refinancing proceeds received after an old debt payment cannot fund that payment unless a bridge or intraday facility is included. Netting same-day totals can conceal this ordering problem.

4.5 Behavioral closure

Accounting identities do not determine production, hiring, investment, dividends or borrowing. A simulation needs rules for these choices. They may be supplied policies, estimated relationships, optimization problems or bounded alternatives. Each has a different evidentiary status.

A fixed payout ratio is easy to calculate but can produce dividends when cash is unavailable or legal restrictions bind. A cash-target policy can call for borrowing beyond committed facilities. An optimal investment policy can require knowledge of demand the manager does not possess. The policy must be evaluated together with feasibility and information, not appended after the financial statements have been balanced.

Let $A_t(X_t, I_t)$ be the feasible action set. A policy is admissible only if

$$\pi_t(I_t) \in A_t(X_t, I_t) \tag{4.3}$$

for the worlds it claims to govern. If the actor cannot observe X_t fully, feasibility may itself be uncertain from that actor's perspective. The model then needs a rule for attempted actions, acceptance by counterparties and consequences of rejection. Assuming successful execution whenever management requests funding is a substantive optimism assumption.

4.6 Attempts, commitments and settlements

An equity issue may be proposed, approved, subscribed, priced and settled at different dates. A loan may be committed but subject to conditions before drawing. A sale of assets may require a buyer and transfer completion. Treating intention as cash receipt collapses these stages.

The event model should distinguish an action request from acceptance, a binding commitment from a contingent one, and settlement from recognition. A financing policy can request 20 while the lender supplies 12. The shortfall remains an economic state requiring a further response. It must not disappear because the request was entered into a forecast cell.

These distinctions also prevent duplicate events. If a commitment is recognized in a schedule and the eventual settlement is recognized again as new borrowing without linking them, principal or cash can be counted twice. Stable contract and event identities support reconciliation from request through settlement.

4.7 Feedback and simultaneous determination

A company's borrowing cost can depend on leverage, while leverage depends on borrowing proceeds and operating decisions. This creates a fixed-point or equilibrium problem. Sequentially evaluating one formula and then the other once is generally not a solution to the simultaneous system.

For illustration, suppose required proceeds are $P = 20$ and face value satisfies $F = P(1 + y)$,

while the yield rule is $y = 0.05 + 0.001F$. Substitution gives

$$F = 20(1.05 + 0.001F), \tag{4.4}$$

$$0.98F = 21, \qquad F = 21/0.98. \tag{4.5}$$

Using the initial guess $F = 20$ in the yield rule and stopping gives a different face value. The discrepancy is small here because the feedback coefficient is small. Near a financing threshold it may change feasibility.

This rule is a teaching assumption, not a market-pricing model. Its purpose is to distinguish simultaneous equations from an event sequence. A proper model must also consider whether a solution exists, whether it is unique, whether the implied offer is feasible, and which equilibrium is selected when several exist. Numerical convergence to one point does not answer all those questions.

4.8 Joint shocks and dependence

A recession scenario can combine lower sales, slower collections, weaker collateral values and tighter funding. Sampling these shocks independently may understate their joint effect. Conversely, forcing every adverse marginal outcome to occur together can describe an intentionally severe stress that is not a probabilistic percentile.

If a joint probability model is used, its dependence structure needs evidence or an explicit sensitivity range. Correlations alone may not determine tail dependence. If a set of scenarios is used, the feasible combinations should be declared. A collection of marginal intervals does not establish that every combination is economically possible.

Historical and future uncertainty can also be dependent. A company with short-term debt may follow a different liquidity policy from one with long-term debt. Treating debt maturity and cash management as independent can generate histories inconsistent with the assumed behavior. The model should either retain the dependence or state that the policies are deliberately varied independently for a conditional stress comparison.

4.9 Counterfactual histories

A counterfactual asks what would have happened under a different action or shock while preserving specified features of an actual or hypothesized world. It requires a rule linking the factual and counterfactual worlds. Independently drawing a new company for each scenario does not preserve that link.

If two policies are compared using the same initial history and exogenous innovations, differences can be attributed to the policies within the model. If histories or shocks are redrawn independently, the comparison includes sampling variation. Common random numbers can reduce variance, but only when the coupling preserves the correct marginal model for each policy.

For unidentified histories, a useful result is the range of policy differences $F(h, s, \pi_1) - F(h, s, \pi_0)$ over the same h. Subtracting an independently minimized baseline from a maximized alternative can combine different companies and overstate the achievable difference. The paired relation should be preserved.

4.10 A threshold example

Suppose cash after interest is $90 - 2\alpha$ and a covenant requires cash at least 89.4. Solving the inequality gives

$$90 - 2\alpha \geq 89.4 \quad \Longleftrightarrow \quad \alpha \leq 0.3. \tag{4.6}$$

After evidence narrows α to $[0.2, 0.4]$, both compliance and breach remain possible. The equality point belongs to compliance under the inclusive covenant in this example. A strict covenant would require a different boundary.

The model can retain two guarded regions without enumerating individual values. The breach region then triggers whatever cure, waiver or acceleration rules the contract specifies. It does not automatically trigger liquidation. An unresolved guard is a reason to preserve conditional outcomes or refine the family, not to choose the midpoint and declare compliance.

4.11 Scenario comparison as a decision aid

A useful comparison reports which conclusions hold across histories, which depend on a particular contract assumption and which depend on a policy. For instance, a rate increase may raise interest for every floating exposure but reduce the value of a fixed-rate liability under a separate valuation basis. These are different quantities and can move in different directions.

The result should separate cash liquidity, book income, book equity, market value and contractual compliance. A positive accounting profit does not guarantee payment capacity; a market-value loss does not necessarily cause an immediate cash outflow. Combining them into an unnamed risk score makes the mechanism difficult to audit.

A proposed dashboard can show a conditional envelope with witness histories at important boundaries and a list of assumptions that change the conclusion. The envelope should retain joint outcomes and timing. A worst cash value at one date and a worst debt value at another may come from different paths; presenting them as one path would mislead.

4.12 Scenario failures and repairs

A scenario can fail because its shock is underspecified, because the model omits an exposure, because behavior uses future information, because financing is infeasible, because the numerical calculation fails, or because the economic assumptions are outside their domain. These failures require different remedies. More Monte Carlo paths cannot repair an omitted contract or an information violation. A more detailed journal cannot repair an unidentified behavioral response.

AccDSL should therefore return a typed result: completed conditional simulation, unresolved history region, unsupported mechanism, infeasible action, economic distress event, or numerical failure. A run may contain several statuses across different regions. Dropping failed regions and renormalizing successful ones changes the target distribution and can systematically remove the worst outcomes.

The next chapters make the financing mechanisms explicit. Debt and equity are especially consequential because they can appear to solve every cash shortage if issuance is represented only as a balancing entry. Their economic terms, counterparties, timing and rights must determine whether the entry may occur.

Chapter 5

Debt is a dated contract, not a balancing amount

Debt issuance joins a cash transaction today to promises about future payments and creditor rights. A journal can record proceeds and a liability without describing those promises. A structural model can describe the promises without supplying the accounting measurement. Scenario simulation needs both, together with a financing policy and a model of whether counterparties accept the proposed issue. Confusing these layers can make a company appear liquid while its funding equation is economically inconsistent.

5.1 Five amounts that should not share one symbol

For a debt instrument distinguish face or promised payment F, gross proceeds P, net cash received $P - f$ after issuance costs f, accounting carrying amount B, and market value V_D. They may coincide in a simple par loan at inception. They need not coincide for discounted debt, transaction costs, accrued interest, modifications or subsequent market changes.

The gross contractual promise is relevant to redemption. Net cash is relevant to liquidity. Carrying amount is relevant to the balance sheet under the selected basis. Market value is relevant to valuation or repurchase. A debt-to-equity covenant may use yet another defined measure. Reusing one variable called debt for all of these quantities creates errors that can remain hidden in a par, no-fee example.

A debt record also needs currency, lender, borrower, issue and settlement dates, maturity, coupon or reset rule, day count, amortization, seniority, collateral, covenants, options and modification rights. Unknown terms remain unknown. A model can support a bounded profile without implementing every clause, but its unsupported terms must not silently default to benign values.

5.2 The one-period pricing equation

Consider a loan issued at date zero and settled at date one. Let D be the date-one default indicator. If there is no default, the lender receives promised amount F. If default occurs, the lender receives actual recovery R, measured in date-one currency units. Let r be a risk-free one-period rate. Under the illustrative assumption that discounting the expected payoff at r is

appropriate, lender break-even is

$$P(1+r) = \mathbb{E}[(1-D)F + DR]. \tag{5.1}$$

This equation is a pricing assumption for the example. Under physical probabilities, discounting at a risk-free rate requires an appropriate risk-neutrality or pricing justification. More generally, with stochastic discount factor M,

$$P = \mathbb{E}[M((1-D)F + DR)]. \tag{5.2}$$

An explicitly justified pricing measure can provide an equivalent formulation. A real-world default probability is not automatically a risk-neutral probability.

Define the promised yield y by $F = P(1+y)$ for positive proceeds and a single final payment. Then

$$P = \frac{F}{1+y}. \tag{5.3}$$

The yield is a ratio of the promised repayment to the price, not the realized return in default. With coupons or multiple dates, a yield-to-maturity equation replaces this simple relation and may have additional conventions. The cash schedule remains the primitive description.

5.3 The convention problem in the supplied note

The user-supplied lecture note by Chak Wong adapts the risky-debt model of Strebulaev and Whited (2012, Section 3.6) to residual-based deep learning (Wong, 2026). It correctly recognizes that equity value, default and debt pricing interact. Its inherited debt convention requires repair before implementation.

The retained December 2012 source's equation (3.26) raises $b'/(1+\widetilde{r})$ at issuance and carries b' into the next period's inherited debt. That use makes b' a promised repayment. Equation (3.27), however, equates $b'(1+r)$ to an expected payoff whose nondefault leg is $b'(1+\widetilde{r})$. That use makes b' cash advanced. Both printed equations were inspected in the retained primary source; the mismatch is not merely a transcription error in the supplied note. The publisher's current correction status has not been cleared.

The consistent face-value convention uses F for the next promised payment and P for issuance cash, as in equations (5.1)–(5.3). Substituting gives

$$\frac{F(1+r)}{1+y} = \mathbb{E}[(1-D)F + DR]. \tag{5.4}$$

Multiplying by $1+y$ yields

$$F(1+r) = (1+y)\mathbb{E}[(1-D)F + DR]. \tag{5.5}$$

The multiplier applies to recovery as well as to the nondefault payoff. Omitting it on recovery changes the equation when recovery is positive. Alternatively one can formulate the entire model in proceeds units, but then future liability state, repayment and recovery arguments must change consistently. Renaming one symbol in the pricing equation is insufficient.

5.4 An exact positive-recovery diagnostic

Hold face at 100, default probability at 0.1, recovery at 40 and the risk-free rate at 0.05. These are synthetic inputs chosen to expose the convention. The expected payoff is

$$0.9(100) + 0.1(40) = 94. \tag{5.6}$$

Thus

$$P = \frac{94}{1.05} = \frac{1880}{21} \approx 89.52381, \tag{5.7}$$

$$y = \frac{100}{P} - 1 = \frac{11}{94} \approx 0.1170213. \tag{5.8}$$

The promised yield exceeds the risk-free rate because the lender does not always receive the promise. The expected payoff discounted at five per cent exactly equals the proceeds by construction.

Under the conflicting printed convention, solving $105 = 0.9(100)(1+y)+0.1(40)$ gives $y = 11/90$. Its implied proceeds are $9000/101$. Applying the actual payoff 94 to those proceeds leaves

$$94 - 1.05\left(\frac{9000}{101}\right) = \frac{44}{101} \approx 0.43564. \tag{5.9}$$

This is a lender break-even residual measured in date-one cash. It is not a rounding error. Zero recovery or no default can conceal the mismatch, so a no-default smoke test is inadequate. Positive recovery is the discriminating case.

5.5 Recovery needs its own definition

Recovery can mean a fraction of face, a fraction of market value before default, a share of realized collateral, or a cash distribution after costs and delays. These meanings are not interchangeable. If recovery is $R = \rho F$, the one-period expected payoff is $F[1 - p + p\rho]$ when default probability is p. If R is an absolute asset distribution, scaling face does not generally scale recovery.

For delayed recovery at date two, discounting it as a date-one payment is wrong. A stochastic discount-factor formulation must use the payment's actual date. Recovery uncertainty can be correlated with default and aggregate conditions; replacing it by an unconditional mean can misprice the claim.

Multiple creditors require an allocation rule. Suppose distributable assets after costs are 70, senior claims are 60 and junior claims are 50. Under an assumed strict priority waterfall without other rights, senior recovery is 60 and junior recovery 10. Paying both creditors 70 per cent would distribute 77 and violate the available asset constraint. The example illustrates conservation and priority, not a legal conclusion for a particular jurisdiction. Actual enforceability, security and restructuring rules need admitted authority.

5.6 Default is a transition, not deletion

An inability or decision not to pay can lead to arrears, cure periods, waiver, acceleration, restructuring, enforcement or liquidation. The contract and legal process determine which transitions are available. A simulator that sets debt and equity to zero at a default flag loses unpaid claims, assets, recovery timing and operating consequences.

A useful state includes scheduled obligation, amount paid, arrears, default status, acceleration status and recovery process. A missed payment can create an arrears balance even if operations continue. A covenant breach can restrict dividends without immediate acceleration. A waiver can restore eligibility subject to fees and revised terms.

Limited liability concerns shareholders' obligations and the value of their claim under the model. It does not require book equity to remain nonnegative. Negative book equity, negative market equity before applying limited liability, cash shortage and legal default are different concepts. Their relationships must be modeled rather than inferred from a single sign test.

5.7 Refinancing creates a new problem

A maturing debt of 100 cannot be repaid by writing a new debt balance of 100 unless a financing transaction supplies the cash in time. Suppose new face 100 sells for 90 and issuance fees are 2. Net proceeds are 88, leaving a shortfall of 12 before considering cash reserves. Treating face as cash hides that shortfall.

If the company requires net proceeds of 100 and the price is 0.9 per unit face with a fee of 2, required face is $(100 + 2)/0.9 = 113.333\ldots$. Increasing face may itself worsen price or violate capacity. The funding calculation is then coupled to the pricing and eligibility problem.

A committed facility may provide capacity but still have draw conditions, currency limits, borrowing-base constraints or expiry dates. An uncommitted facility is not guaranteed funding. A refinancing policy should distinguish request, offer, acceptance and settlement. Failure at any stage leaves a residual funding need and possible distress transitions.

5.8 Cash, debt and net debt

Some structural models use a signed debt variable, with negative debt interpreted as cash. This is economical in a stylized model, but cannot automatically represent a company with cash and debt simultaneously. Restricted cash may not repay a particular obligation. Cash in one subsidiary or currency may not be available to another. Gross debt can affect covenants and refinancing even when net debt is small.

Consider cash 100 and debt 100. Net debt is zero. If cash is unrestricted in the borrowing entity and repayment is allowed without cost, netting may be adequate for a particular valuation question. If cash is pledged to another creditor or held in a subsidiary subject to transfer restrictions, netting destroys liquidity information. The same net balance describes different payment capacity.

AccDSL should therefore preserve gross instruments and cash accounts in the general representation. A net-debt reduction is a query-specific abstraction requiring explicit conditions. It should not become the universal state merely because a numerical paper uses it successfully in a simpler economic model.

5.9 Coupons, accrual and effective interest

Contractual coupon determines a payment schedule. Accounting interest expense can differ because of discount, premium, fees or measurement rules. For a simple illustrative amortized-cost schedule with effective rate i, carrying amount evolves as

$$B_{t+1} = B_t + iB_t - \text{cash coupon}_{t+1} - \text{principal paid}_{t+1}, \tag{5.10}$$

subject to the selected accounting treatment and any additional events. The effective-interest term is not automatically a cash payment.

Suppose a zero-coupon promise of 100 is issued for 90 for one year. With no fees, an effective rate $100/90 - 1$ accretes carrying amount from 90 to 100. Cash is received at issuance and paid at maturity. Charging 10 as both accretion and a coupon payment would double count the financing cost.

Conversely, a floating coupon can reset while an accounting measurement follows its own applicable rules. A simulation should generate contractual cash first from terms, then calculate

accounting consequences under an admitted basis. A generic rate shock on interest expense cannot substitute for these two mechanisms in every instrument.

5.10 Issuance costs and tax timing

A fee can be paid to an intermediary at issuance, deducted from proceeds, financed separately or incurred conditionally. Its cash timing should be explicit. The accounting allocation between expense and carrying amount depends on the applicable standard and facts. A model cannot infer that allocation merely from the fact that cash was paid.

The retained structural source and supplied lecture include a simplifying treatment that credits the present value of a future interest tax benefit at issuance. That can be part of an economic valuation convention. It must not be posted as an actual contemporaneous tax receipt without a separate tax mechanism. A present value is not a cash settlement.

Tax losses, deduction limits, jurisdiction, timing and the availability of taxable income can all affect realization. A scenario that assumes every interest expense produces immediate cash tax savings may overstate liquidity in distress. The proposed design keeps tax cash payments, tax expense, deferred tax and economic tax-benefit valuation separate, then links them under explicit rules.

5.11 Options and amendments

A callable bond gives the issuer an exercise right; a puttable bond gives a holder a right. Their exercise can depend on rates, liquidity, transaction costs and information. Replacing the schedule with the contractual final maturity ignores these options. Exercising automatically whenever it appears favorable assumes a policy and ignores possible constraints.

An amendment can change coupon, maturity, collateral or covenants. It needs an effective date, consent requirements, fees and accounting treatment. A revised schedule should supersede future obligations consistently while retaining the original agreement and past payments. Applying both old and new coupons after amendment creates duplicate obligations; erasing the old contract destroys provenance.

Where a clause is unknown, the family can contain alternative exercise rights or amendment paths. Those alternatives should be activated only by their premises. A model should not give the borrower an option that belongs to the lender, nor assume the lender consents to a concession because it improves borrower liquidity.

5.12 Pricing and policy interact

A debt price can depend on future investment, payouts and refinancing because these affect repayment. The issuer's policy can depend on the price offered. Solving the two separately while holding inconsistent policies fixed can produce a price that is correct for a different company behavior.

The structural-finance survey Strebulaev and Whited (2012, Sections 3.4–3.6) is useful for this interaction. The newer working paper Duarte and Fonseca (2026, Sections 2.1.3–2.1.5 and 3) also solves bond pricing and equity value jointly under its own timing, recovery and tax assumptions. Those assumptions differ from the lecture's model; it is not a drop-in correction.

A general AccDSL financing service can accept a declared price or offer as an input. That supports conditional simulation without claiming equilibrium. An optional equilibrium service may solve pricing and policy together, with separate numerical evidence. The interface should identify which service supplied the offer and what assumptions support it.

5.13 Credit supply and rationing

A fair price does not imply an unlimited quantity is available. Lenders can face capital, liquidity, concentration or mandate constraints. A borrower may be unable to issue enough securities at any admissible terms. A model that solves only lender break-even can miss rationing.

Bank stress and balance-sheet models such as Halaj (2016); Behn et al. (2019) provide examples where financing and adjustment depend on constraints. Their institutional settings should not be transferred mechanically to a corporate issuer. The general lesson is that a financing transaction needs a counterparty and feasible quantity, not only an interest rate.

An offer curve can therefore specify price, maximum quantity, conditions and expiry. A policy chooses among feasible offers or records no acceptable offer. If the curve is uncertain, results remain conditional on alternative credit-supply assumptions. A scenario can deliberately freeze refinancing, but that restriction should be visible as a stress premise.

5.14 What dynamic financing models establish

For longer-lived debt, lender receipts depend on future issuance, investment, priority changes and default choices. A candidate policy π produces a price q^{π}. If management optimizes while holding an incompatible price fixed, the resulting policy may not be the one lenders priced. A simultaneous solution must make the chosen policy optimal under its price and make the price value that policy's promised and default-state payments. A constrained contract-price input is a useful alternative for mechanical stress testing, but its interpretation is conditional on that price and availability.

Benzoni and coauthors make this distinction explicit in a dynamic model with fixed issuance costs (Benzoni et al., 2023, Section 2, Propositions 1–2, Appendix A). Their debt quantity is face value, its price is per unit of face, and shareholders choose default, issuance or inaction. Value matching and smooth pasting generate a candidate barrier policy; off-policy deviations and a supermartingale condition are required for equilibrium verification. The retained model assumes zero default recovery and risk-neutral agents. It therefore illustrates the policy-price consistency problem, but is not the positive-recovery solution required by the preceding example.

The alternative managerial model of Lambrecht and Myers places investment, borrowing, payout and rents in a dated sources-and-uses constraint and imposes a no-Ponzi restriction (Lambrecht and Myers, 2017, Sections 2–4). Its safe-debt solutions depend strongly on managerial preferences and governance. In its exponential-utility example, the liquidation covenant protects lenders under the stated diffusion dynamics. A jump across the boundary, delayed liquidation or costly asset disposal would require a different recovery analysis. A closed-form policy derived under those restrictions is a competitor to consider, not an identified default policy for every issuer.

Dewatripont, Legros and Matthews provide a primary contracting comparison (Dewatripont et al., 2003, Sections 2–6 and Appendix B). Their environment has observable but noncontractible effort, limited liability, monotone investor receipts and efficient renegotiation. Under entrepreneur-offer renegotiation, a debt contract can dominate a more general contract in the stated pure-strategy class. Their appendix also supplies cases in which changed bargaining

power, third parties or mixed messages change that conclusion. Even their investor conversion option need not be an ordinary equity security, as footnote 16 explains. Contract labels cannot replace the exact contingent payments and exercise rights in the company model.

5.15 Equal balances after unequal payment histories

A borrower can recover from a cash shortage before the reporting date. Its closing statements then need not reveal that it failed to pay on time. To isolate the timing, consider a synthetic company with cash 60, inventory 100, debt carrying amount 80 and book equity 80. The contractual promise is 100, payable on 30 June 2027. Assume a zero coupon, no fees or taxes, and one remaining measurement interval that accretes the discount of 20 at maturity. This deliberately simple interval makes the cash and contract comparisons exact; a general effective-interest schedule needs its own measurement rules.

The company issues 40 new shares for one unit of cash each. In the first history, the subscription settles on 29 June. Cash rises from 60 to 100, the discount accretes on 30 June, and the borrower pays the complete promise that day. In the second history, the shares settle on 2 July. The borrower can pay only 60 at maturity. On 1 July an explicit observation records the unpaid 40; the new cash arrives on 2 July and pays the remaining claim on 3 July. No overdraft, emergency loan or retroactive receipt funds the earlier payment.

The following comparison follows the money and claim in the second history. The liability column is positive for presentation, although journal credits use the implementation's signed convention.

After the dated event	Cash	Claim unpaid	Historical miss
30 June: accretion	60	100	0
30 June: partial payment	0	40	0
1 July: missed-payment observation	0	40	40
2 July: share proceeds	40	40	40
3 July: final payment	0	0	40

The zero in the first two historical-miss cells records that the observation has not yet been made. It is not a finding that a payment was timely. The observation changes contract memory without changing cash or posting another expense. The final payment clears the current claim; it leaves the history of the earlier shortfall intact.

Both histories end with the same financial statements under the supplied treatment. Inventory of 100 is the only remaining asset, the debt is zero, and equity is opening equity plus the subscription less the discount expense:

$$A = 100, \qquad L = 0, \qquad E = 80 + 40 - 20 = 100. \tag{5.11}$$

Profit for the illustrated movements is -20. Net financing cash flow is $40 - 100 = -60$ in either history. The late path divides the repayment into $60 + 40$, which changes its timing but not its total. Even agreement across the closing balance sheet, income and cash-flow statements does not answer whether the maturity was paid on time. A simulation query about missed payments must retain the distinguishing history.

The difference also explains why balancing the journal is insufficient. Share proceeds can debit cash and credit capital correctly while a faulty method fails to increase the recorded shares. A missed-payment observation can omit its history update without unbalancing any account. In the checked reference, both omissions pass generic accounting execution and fail the independent

complete-state comparison. Explicit financing conditions subsequently reject these proposed transitions before accepting their successors.

Those conditions have two roles. Snapshot conditions constrain current quantities: claims and arrears are nonnegative, arrears cannot exceed the remaining claim, and under this matured zero-coupon convention the carrying liability equals that claim. Transition conditions constrain how an event changes them. Payment of p gives remaining claim $q' = q - p$, reduces cash by p, and leaves the historical missed amount unchanged. A share subscription must increase both the monetary capital account and the separate share quantity under the supplied issue terms. An internally plausible rewrite of historical miss from 40 to 50 can satisfy snapshot inequalities but violates the transition condition. This is why both kinds of condition are needed.

Opening history creates another distinction. A supplied fiscal opening must satisfy the declared period-start, temporary-balance and flow restrictions; an intermediate state containing this year's payments is not such an opening. Importing a consistent snapshot does not establish that it includes every earlier missed payment. The selected opening method therefore compares it with a separately supplied history premise and preserves unknown completeness. Actual source evidence is still needed to establish that premise.

5.16 A missed payment and the available remedies

The amount unpaid does not by itself identify the creditor's remedy. Apple's 2024 annual report provides a concrete historical description of why the distinction matters. Exhibit 4.1 expressly calls its description incomplete and qualifies it by the 2013 and 2018 indentures; its issuer references concern Apple Inc., rather than automatically every subsidiary (Apple Inc., 2024a, Exhibit 4.1, printed p. 3). We use the summary to identify conditions a simulation must represent. It does not establish an actual issuer default, complete operative terms, or the applicability of those terms to the synthetic company.

The described interest-payment event requires default continuing for 30 days after the installment becomes due. Principal and premium use a different condition: nonpayment when due and payable, including maturity, redemption or declaration. Importing the interest grace period into principal repayment would therefore apply the wrong described condition. The other-covenant route has its own exclusions, a continuing breach and 90 days after qualifying written notice. Holder notice uses at least 25% of outstanding principal under the 2013 indenture and at least 33% under the 2018 indenture; the trustee has a separately described notice route (Apple Inc., 2024a, Exhibit 4.1, printed p. 9).

The thresholds concern the relevant outstanding series, not a fraction of consolidated debt or a count of creditors. Series mapping, denominator, delivery to the required recipients and validity of notice are substantive inputs. No notice and no evidence about notice lead to different answers. Likewise, day-level data at exactly the 30- or 90-day boundary may leave event order unresolved. The conditional reference retains that uncertainty; it does not assert that the complete contract has no answer at the boundary.

The summary next distinguishes declaration from automatic acceleration. Certain continuing nonbankruptcy events permit a declaration by the specified trustee or holder route. Certain bankruptcy events make the obligations due without that declaration. A negative equity balance establishes neither the specified bankruptcy event nor the required category. Moreover, an event for one series does not necessarily constitute an event for another. Neither automatic contagion nor universal independence follows (Apple Inc., 2024a, Exhibit 4.1, printed pp. 9–10).

Cure and rescission must also remain separate. The described rescission of a declaration requires holder action and specified deposits and cure or waiver conditions, with an exception concerning

amounts due solely because of acceleration. Payment can remove current arrears without by itself proving that an earlier acceleration has been rescinded. In the selected synthetic continuation, payment changes the economic claim, reassessment changes the current clause finding, and a separate supported rescission assessment changes the recorded acceleration state. The historical miss remains 40 throughout.

Consent cannot be supplied by a numerical convenience either. The summary requires each affected holder's consent for specified protected changes, including changes to stated maturity and reductions of principal or interest. Its general majority-waiver provision excludes payment defaults and certain individually protected provisions. A waiver concerns the specified past default; it does not license a future one (Apple Inc., 2024a, Exhibit 4.1, printed pp. 11–12). A current contractual default may cease under a valid waiver while the economic record still shows that payment arrived late. Keeping those two statements is consistent.

The surrounding qualifications matter economically. Individual payment enforcement differs from collective proceedings; legal discharge differs from covenant defeasance. The description warns that assets sufficient to fund scheduled maturity may be insufficient after acceleration (Apple Inc., 2024a, Exhibit 4.1, printed pp. 10–13). A simulator must therefore retain the relevant rights, legal owner, timing and usable funding, even when a consolidated closing statement appears fully funded.

The typed conditional implementation translates the retained examples and records selected findings through the common monetary and memory engine. An unresolved finding produces no successor state. Recording a known finding changes no cash, carrying amount, shares or historical missed amount. This establishes a limited representation and execution result. Multiple concurrent clauses, the full instruments, actual notices and amendments, and the economic consequences of acceleration still require their own admitted methods.

5.17 Implementation proposal

The debt module should maintain immutable contract versions, a dated obligation schedule, current economic state and separate accounting measurements. Events include issue, draw, reset, accrue, coupon, amortize, redeem, amend, breach, cure, accelerate, restructure and recover. Each event has preconditions, amounts, dates and successor state.

Existing repository methods already cover bounded issue, accrual, coupon, redemption and conversion calculations, and newer declared methods preserve timing and debt memory. They are useful foundations. They do not by themselves establish a general pricing model, market capacity, legal default process or complete instrument admission. The extension should compose with their checked boundaries rather than describe those capabilities as absent.

The smallest discriminating financing fixture should include proceeds different from face, positive recovery, a fee paid at issuance, a maturity shortfall and an unavailable refinancing offer. A second fixture should add two creditors with priority and delayed recovery. A third should test unknown rate terms and a covenant boundary. Passing these checks establishes the specified mechanics; empirical pricing validity and legal applicability require separate evidence.

Chapter 6

Equity funding, ownership, and dilution

An equity issue can provide cash while changing ownership and future claims. A model that records only an increase in cash and book equity captures one accounting consequence but misses the allocation of control and value. A structural model that charges an issuance cost captures a financing friction but does not automatically create a share register. Scenario simulation needs quantities, prices, rights, holders and dates, together with an explanation of who is willing to subscribe.

6.1 Book equity, market value and share capital

Book equity is the residual of recognized assets and liabilities under an accounting basis. Market equity value is the value assigned to owners' economic claims. Legal share capital is a legally defined component of capital that may be related to nominal share values. These quantities have different meanings. Dividing book equity by shares produces book value per share, not automatically the market subscription price.

A company can have positive book equity and insufficient cash to pay a maturity. It can have negative book equity and valuable future opportunities. Limited liability can bound the value of a shareholder's claim below by zero in a model without forcing the accounting residual to zero. The simulator should retain these quantities separately and define any relation between them.

Let N be the number of existing ordinary shares, V_E the total economic value assigned to their claim, and $p = V_E/N$ a per-share value when shares are homogeneous and $N > 0$. This division presumes that V_E refers to exactly that class. If preferred shares, options or noncontrolling interests are included in total equity value, allocating all of it to ordinary shares is wrong.

6.2 A simple cash issue

Suppose the company issues n new homogeneous ordinary shares at subscription price $p_s > 0$. Gross proceeds are $G = np_s$. If cash issuance costs are f, net cash is $G - f$. With no simultaneous repurchase or conversion, the post-issue count is

$$N^+ = N + n = N + \frac{G}{p_s}. \tag{6.1}$$

An existing holder with n_i shares has ownership fraction n_i/N before and $n_i/(N+n)$ after the issue, assuming that holder does not subscribe. This is percentage dilution. Its economic consequence depends on the price received and the use of the proceeds.

For example, let $N = 100$, pre-issue equity value be 1,000, and issue 20 shares at 10. If the cash adds 200 of value and there are no costs or other effects, post-issue value is 1,200 and each share remains worth 10. The old holders' percentage falls from 100 per cent to 100/120, but their combined value remains 1,000. Percentage dilution is not necessarily a wealth loss.

If the same 20 shares are issued at 8 while pre-issue assets remain worth 1,000, post-issue value is 1,160 and each share is worth $1,160/120$. Old holders then own $100(1,160/120) = 966.666\ldots$. The transfer arises from the discounted subscription price under these assumptions. A model that equates all share-count dilution with value destruction misses the first case; one that ignores issue price misses the second.

6.3 Raising a target amount after a price shock

Suppose net funding need is K and a fixed cash fee is f. At a supplied price p_s, required shares are $(K+f)/p_s$. A price decline therefore increases the share count required to raise the same cash. If authorized issuance or investor demand limits n, the funding target may be infeasible.

Take $K = 90$, $f = 10$ and an initial subscription price 10. The company needs ten shares. After a price shock to 5 it needs twenty. If only fifteen shares may be issued under the assumed authorization, maximum net proceeds become $15(5) - 10 = 65$, leaving a shortfall of 25. Increasing book equity by 90 would conceal both the share constraint and the cash shortfall.

The fee may instead be proportional, for example $f = \lambda G$. Net proceeds are $G(1-\lambda)$ and required gross proceeds are $K/(1-\lambda)$ for $0 \leq \lambda < 1$. A fixed plus proportional fee gives $G = (K + f_0)/(1-\lambda)$. These equations specify different cost structures. The choice should come from a declared offer or empirical assumption, not a silent generic parameter.

6.4 Pricing is not supplied by an accounting identity

The equation $G = np_s$ is an identity once shares and price are specified. It does not determine investor willingness to pay. A subscription price can be an external offer, an assumed discount to an observed market price, or the result of a valuation and demand model. Each supports a different claim.

A public market quote can be stale, illiquid or refer to a different class. A large issue can affect price. A distressed company may face a discontinuous demand schedule or no feasible offer. A model that uses yesterday's price for unlimited new shares assumes unlimited demand at that price. That may be useful in a deliberately simplified scenario, but should not be presented as a financing solution established by the statements.

The same caution applies to an endogenous equity value function. It may value existing shareholders' future payouts under a particular policy and financing friction. Converting that value into a new issue price requires assumptions about how new investors share those payouts and how issuance changes the state. A scalar cost function $\eta(e)$ for external equity is not a complete security issuance model.

6.5 The rights attached to a share

Shares can differ in voting rights, dividends, liquidation preference, conversion rights and redemption terms. An aggregate share count is sufficient only when the distinctions are irrelevant to the question or explicitly normalized. A dollar of preferred funding is not generally equivalent to a dollar of ordinary funding for future payouts.

A simple preference waterfall illustrates the point. Suppose new investors contribute 30 for a nonparticipating preference claim entitled to 30 before ordinary holders, with the option to convert into 25 per cent of ordinary equity. If distributable value is 80, the preference payment 30 exceeds conversion value 20. If value is 200, conversion value 50 exceeds the preference. The exercise threshold is 120 under these assumptions.

This example requires a declared contract: no accrued preference dividend, no senior claims, no participation after the preference and no other options. Adding any of these changes the waterfall. A flat share-percentage allocation would misallocate value below the conversion threshold. The model should retain class-specific rights and exercise policy.

6.6 Options, warrants and contingent issuance

An option can create future shares when exercised. Exercise can bring cash to the company, transfer existing shares or settle in cash, depending on terms. Counting all potential shares as already issued can distort current ownership; ignoring them can distort future dilution.

Let an option cover m shares at exercise price k. In a simplified physical exercise, the company receives mk and issues m shares. The holder's decision depends on share value, expiry, constraints and possibly tax or employment conditions. A purely intrinsic-value rule is a modeling policy, not a universal description of exercise behavior.

Contingent performance awards add dependence on future operating outcomes. A scenario changing revenue can affect both the company's value and the number of shares delivered. Treating share count as independent of the operating shock loses that dependence. Historical statement totals may not identify all award terms, so the family must preserve unresolved conditions.

6.7 Conversion joins debt and equity

A convertible instrument links a liability schedule to ownership changes. At conversion, principal or another contractual amount may be exchanged for shares according to a fixed ratio, a variable price or a contingent formula. Cash settlement alternatives can change liquidity materially.

Suppose debt face 100 converts into ten ordinary shares. Physical conversion removes the specified debt obligation and increases shares by ten under the assumed terms. It does not also pay cash principal 100 unless the contract requires a separate payment. A simulator that schedules both redemption and conversion double settles the same obligation.

Accounting derecognition and allocation can involve carrying amounts different from face and separate components. Those postings need the applicable basis and facts. The economic event should identify exactly which obligations and rights are extinguished or created before the accounting module measures the consequences. Existing bounded conversion methods in the repository can support this event discipline without claiming a universal convertible classification.

6.8 Repurchases and treasury shares

A repurchase uses cash and changes outstanding shares or treasury holdings according to the legal and accounting treatment. A net increase in equity during a period can conceal both gross issuance and gross repurchase. Their timing and prices matter for liquidity and ownership.

Suppose 20 shares are issued at 10 and ten are repurchased at 12. Net cash inflow before costs is 80, while outstanding shares increase by ten if repurchased shares cease to be outstanding. Reconstructing the period as a simple issue of ten shares for 80 uses an implied price of 8 and loses the actual transactions. A future policy inferred from that synthetic price can be misleading.

Repurchase authorization is not settlement. A board may authorize up to an amount while management executes only part. A scenario should not consume cash for the entire authorization unless the policy calls for it and execution is feasible. Conversely, an already binding purchase commitment may create obligations before the cash settlement date.

6.9 Dividends require both claims and capacity

A dividend policy can specify a target payout, but payment depends on declaration, entitlement dates, cash, legal restrictions, covenants and class rights. Retained earnings are not a cash account. A positive accumulated balance does not establish that funds are available in the paying entity.

A declared dividend can create a payable before payment. The event sequence must avoid expensing it as an operating cost in an illustrative ordinary distribution model, while still recognizing the liability and equity movement under the selected basis. Payment reduces cash and the payable. The exact legal and accounting treatment requires its admitted rules.

A cash-constrained policy may reduce or suspend a prospective dividend. It cannot necessarily cancel an already enforceable obligation. Distinguishing intended, declared and paid distributions is therefore important in stress scenarios. A single annual dividend line cannot capture all of these states.

6.10 Ownership and control are related but distinct

Economic ownership percentages do not always determine control. Voting rights, contractual arrangements, dispersed holdings and substantive options can matter. A scenario that changes ownership may alter the consolidation perimeter, but the threshold and assessment are not supplied by a generic percentage rule.

The security register should record holders and rights, while a separate control assessment uses the applicable facts and accounting basis. When control changes, group reporting can change even without a large contemporaneous cash flow. Conversely, a minority issue can raise cash without changing control. The two calculations should communicate without being collapsed.

Historical disclosures may report total shares but not every beneficial holder. A model can represent ownership categories and unknown allocations where sufficient for the query. It must not invent holder identities or treat an unknown allocation as evenly spread. Concentration can matter for approval, takeover and financing decisions.

6.11 Earnings per share is not the financing price

Basic and diluted earnings-per-share calculations use accounting earnings, weighted share counts and rules for potential ordinary shares. They are reporting calculations. They do not determine market value per share or subscription price. A price-to-earnings heuristic is a separate valuation assumption.

Timing matters even in a simple weighted count. If 100 shares are outstanding for half a year and 120 for the other half, the simple time-weighted average is 110. For positive earnings, using closing count 120 understates basic earnings per share relative to that assumed timing. For a loss, it makes loss per share less negative; a zero numerator remains zero. Actual calculations require the applicable conventions, participation rights and events.

Diluted measures may exclude antidilutive instruments under the reporting rules. Exclusion from diluted earnings per share does not mean the contract cannot create future cash or ownership consequences under another scenario. The scenario engine must retain the instrument even when one reporting calculation excludes it.

6.12 An issuance changes future incentives

Financing can change not only cash but the distribution of gains and losses. Existing shareholders, new investors and creditors may prefer different policies after an issue. A model holding behavior fixed is a conditional mechanical simulation. A model claiming an optimal response must specify whose objective is optimized and what constraints apply.

For example, a deeply subordinated new equity contribution can protect creditors while diluting existing owners. Existing owners may refuse it even when total enterprise value rises. A policy maximizing total firm value can therefore differ from one maximizing existing shareholders' value. Treating these objectives as interchangeable can change default or investment decisions.

The structural literature surveyed by Strebulaev and Whited (2012) makes financing frictions and shareholder objectives explicit in stylized models. Transfer to AccDSL requires matching the actor, security rights and timing. A convenient objective function should remain a hypothesis until its application is justified.

6.13 A complete small financing comparison

Consider an immediate cash need of 40, current cash 10 and no available debt facility. Net new funding need is 30. Existing shares are 100 and an admissible offer supplies ordinary shares at price 5 with a fixed cash cost of 5. Raising 35 gross requires seven shares, gives 30 net cash and permits the payment. Closing cash is zero and shares are 107.

If the offer instead caps issuance at six shares, gross proceeds are 30 and net proceeds 25. Available cash is 35, leaving a payment shortfall of 5. The difference is an economic feasibility result. The journal cannot repair it by crediting equity for another 5 without a transaction.

If management can defer 10 of the payment at a cost, the capped issue may become feasible. That is a different policy and contract path, with future obligations. If a shareholder loan supplies the missing 5, it creates debt and related rights. The model should show these alternatives explicitly rather than use a generic balancing source.

6.14 Proposed register and execution methods

A dated security register should identify issuer, class, units, holder or holder category, rights, issue price where relevant, subscription and settlement state, and links to conversion or cancellation events. Outstanding, authorized, treasury and potentially issuable units should be distinct. Changes should reconcile through events.

An equity-financing method should accept a funding request, eligible offers, authorization limits and actor policy. It returns an accepted transaction or a shortfall. The transaction then produces cash, ownership and accounting events. Price formation may be supplied or solved by a separate model; the result records which.

The repository's simple shares-times-price postings and bounded capital events are useful components. The proposed extension adds offer feasibility, class rights and ownership evolution. It should reuse existing atomic posting and timing checks while adding negative cases: zero or negative subscription price, unsupported class, duplicate settlement, insufficient authorization, failed subscription and inconsistent conversion.

6.15 What the equity checks establish

Conservation checks can establish that cash equals settled subscriptions less costs and that share movements reconcile. Waterfall checks can establish that distributions do not exceed available value under the assumed priority rules. Timing checks can establish that ownership changes occur at the declared event date.

These checks do not establish that an issue price is fair, that investors will subscribe, that a legal approval is valid or that management will choose the modeled action. Those claims need their own evidence. The simulator should be able to give useful conditional calculations while keeping these unresolved questions visible.

A useful adversarial fixture issues shares below an assumed pre-issue value, then compares percentage dilution with wealth transfer. Another changes the price while holding funding need fixed and tests an authorization cap. A third combines conversion and redemption to ensure that the same obligation is settled once. These examples expose mistakes hidden by a simple par issuance with unlimited demand.

Chapter 7

From economic events to coherent statements

A scenario produces a sequence of events. Financial statements summarize selected consequences of those events under a reporting basis. Keeping this direction explicit makes it possible to ask whether an output follows from a plausible company history. Starting with independently forecast statement lines and inserting a balancing amount reverses the logic: it can make the accounts add up while leaving the economic transaction unexplained.

7.1 An event carries more than an amount

An event identifies actors, instrument or resource, effective time, settlement time, currency, quantity, conditions and provenance. A sale can transfer goods today, be recognized under a specified rule today, and be collected next month. A loan draw can be approved today and settle tomorrow. A derivative can change value without an immediate cash transfer.

The event layer should therefore represent economic occurrence separately from recognition, measurement and payment. The accounting layer maps admitted events and estimates to postings. The reporting layer aggregates postings into statements. The mapping may create noncash movements, and some economically relevant commitments may be disclosed without a current journal amount.

Pierson (2020, Sections 3.3–3.5) explicitly separates accrual, payment and reporting accumulation in a system-dynamics setting. Its simple debt-delay mechanism is acknowledged as limited. The lesson for AccDSL is the separation of clocks, not the adoption of that simplified debt mechanism as a general contract engine.

7.2 A credit sale and collection

Consider a synthetic sale of goods for 30, with inventory cost 18, recognized at delivery and collected later. Ignore tax, returns and credit impairment for this illustration. At delivery, the postings are a debit to receivables and credit to revenue of 30, and a debit to cost of sales and credit to inventory of 18. Cash has not increased.

At collection, cash increases by 30 and receivables decrease by 30. Revenue is not recognized again. The two events explain why profit and operating cash can differ during the interval. A scenario that delays collection changes receivables and liquidity even if recognized revenue is

unchanged.

If the sale is instead prepaid, receipt creates a different sequence under the selected recognition rule. The same annual revenue and cash collection can conceal different within-year funding needs. An annual-only model can be adequate for some questions, but it cannot claim to resolve a monthly liquidity trough that its time aggregation removes.

7.3 Balanced entries are necessary, not sufficient

Double-entry balance verifies a conservation relation in the accounting representation. It does not establish that the event occurred, that the account mapping is correct or that the measurement basis is appropriate. Debiting cash and crediting revenue balances even when the money was borrowed.

Let j_e be the vector of signed account movements for event e, under a declared debit-credit convention. A balancing check can be written $\mathbf{1}^\mathsf{T} j_e = 0$. This is one invariant. Additional invariants depend on the event: a loan settlement should reconcile cash to proceeds and fees; a physical inventory sale should not consume more units than available unless backorders or short sales are explicitly allowed; a share issue should reconcile units and ownership.

The event's economic and accounting checks should remain separate. A balanced but economically impossible issue must fail transaction admission. An economically valid event with an unresolved accounting treatment can remain in the economic state while its statement consequences are conditional or unsupported. Forcing one layer to masquerade as the other hides useful information.

7.4 Quantity and valuation

Inventory value can change because quantities change, unit costs change, measurement rules change or currency changes. A model using only a monetary stock can confuse these mechanisms. Production capacity and physical shortages depend on quantities, while reported carrying values depend on measurement.

Suppose ten units cost 5 each. Inventory carrying value is 50 under the illustrative cost basis. A replacement-price increase to 8 does not automatically increase that carrying value to 80. If five units are sold and replaced, cash purchases, cost allocation and ending stock depend on the chosen inventory-flow assumptions. A future demand shock can expose differences among vintages with the same initial total value.

The economic state should retain the granularity needed for production and sale decisions: units, location, quality, expiry and ownership where relevant. Accounting state can aggregate or measure those units under its basis. An aggregate sufficient for valuation may be insufficient for a supply interruption.

7.5 Asset vintages and depreciation

Two firms can report the same property carrying amount and depreciation while holding assets of different ages. The remaining useful lives, maintenance needs and replacement dates can differ. A capital-expenditure scenario therefore requires more than a depreciation ratio.

For a simple straight-line asset with cost K, residual value v and life L, annual depreciation is $(K - v)/L$ under the assumed full-year convention. Several combinations of K, v and L can

yield the same charge. The historical charge does not identify replacement timing.

An asset-vintage representation retains acquisition date, cost, accumulated depreciation, remaining life, capacity and relevant impairment or disposal events. If only aggregates are observed, the reconstruction should preserve a family of vintage allocations. A query about next year's depreciation may admit a compact aggregate; a query about replacement cash over five years may require finer structure.

7.6 Working capital is a set of contracts

Receivables and payables are often forecast using days sales outstanding and days payable outstanding. These ratios summarize historical relationships but do not identify individual due dates or counterparties. A ratio-based projection can be a useful approximation when its scope is stated. It cannot automatically describe payment priority or concentration under stress.

A customer delay can raise receivables while the firm still owes suppliers on fixed dates. If the model shifts payables proportionally with sales without checking contracts, it may manufacture supplier financing. Conversely, a model that fixes all payment dates may omit renegotiation that management could pursue. Both are policy choices.

The proposed design permits ratio methods as explicitly reduced models alongside contract schedules. Their outputs should carry the aggregation assumptions, horizon and validation domain. A breach-sensitive liquidity query can require refinement to invoices or cohorts whose due dates cross the critical boundary.

7.7 Currency conversion and cash availability

A foreign-currency account has both native units and a reporting-currency value. Exchange movements can change reported amounts without changing native cash. A debt payment in foreign currency requires the appropriate currency at settlement, not merely sufficient consolidated reporting-currency cash.

Suppose the company holds 100 domestic units and owes 10 foreign units. At exchange rate 8 domestic per foreign, repayment costs 80; at 12 it costs 120. A consolidated cash total of 100 does not establish payment capacity after the depreciation. A hedge, foreign cash balance or exchange facility may alter the result, but each must be represented.

Translation for group reporting differs from converting cash for settlement. The accounting basis determines exchange differences in profit, other comprehensive income or other components. The economic layer tracks actual currencies and conversions. A rate applied for presentation cannot automatically be used as an executable market price for a large transaction.

7.8 Group boundaries and internal funding

A consolidated group can report cash in one entity and debt in another. Consolidation removes some internal balances but does not necessarily make all cash freely transferable. Legal, tax, contractual, regulatory or operational constraints may restrict movement. A group-level cash plug can conceal an entity-level default.

Represent cash accounts and obligations by legal entity, then model permitted transfers as events with dates, costs and approvals. Intercompany loans create claims in both entities. Dividends,

capital contributions and service payments have different legal and accounting consequences. They cannot be substituted solely because each moves cash.

For consolidated reporting, internal claims may be eliminated under the applicable perimeter and basis. The underlying entity records remain necessary for liquidity and counterparty analysis. Eliminating the records themselves would prevent the simulator from identifying which entity can pay which obligation.

7.9 A two-entity liquidity example

Parent P has cash 5 and external debt due 20. Subsidiary S has cash 30 and no current external debt. Consolidated cash is 35, apparently enough for the maturity. Suppose only 10 can legally and operationally transfer from S before the due date under the declared facts. P can then assemble 15 and has a shortfall of 5.

If a permitted intercompany loan of 15 becomes available in time, P can pay. If a dividend of 15 is declared but settles after maturity, it cannot fund that payment without a bridge. If the subsidiary cash is restricted, even the first 10 may be unavailable. The scenario result depends on transfer rights and timing, not just the consolidated total.

A correct consolidated statement after payment can coexist with an incorrect event sequence if the model silently borrowed the cash before it was available. Tests must therefore examine entity cash at event boundaries, not only the final group balance sheet.

7.10 Taxes and losses

Tax expense, cash tax and deferred tax are different quantities. A profit forecast cannot generally multiply pretax income by one rate and use the result simultaneously for expense and immediate payment. Loss carryforwards, payment lags, jurisdiction, interest limitations and temporary differences can separate them.

A simple teaching model may assume immediate tax payment at rate τ on positive taxable income, giving cash tax $\tau \max(T, 0)$. That is a declared simplification. If losses can offset future income, the state must retain the remaining loss balance and expiry or utilization conditions. Forgetting that state can overstate future tax; treating all losses as immediately refundable can overstate current cash.

Accounting recognition of deferred tax assets requires its own admitted rules and evidence. Their carrying amount is not spendable cash. A stress scenario that turns a deferred tax asset into a funding source without an actual refund or financing transaction is economically wrong.

7.11 Estimates, errors and changes of policy

An estimate revision changes the model's measurement of an uncertain quantity. An error correction revises a prior assertion or application. An accounting-policy change can alter recognition or measurement according to the applicable rules. Their timing and comparative effects differ.

The scenario engine should distinguish an economic shock that changes expected credit losses from a correction showing that the old receivables data were wrong. Both can change reported equity, but only one necessarily describes new economic information at the forecast date. The evidence and accounting version relations should make the distinction explicit.

A correction should not silently rewrite every frozen forecast. A current restated view and an as-published historical view can coexist. Their differences are analytically useful: they separate errors in data available at the time from errors in the forecasting model.

7.12 Consolidation and scenario-dependent perimeter

An acquisition, disposal or change of control can alter the reporting perimeter. The group total before and after the event may not be comparable without a bridge. A simulation that treats acquired revenue as organic growth can distort future behavior and validation.

Perimeter state should identify included entities and the relevant control assessment over time. Acquisition events introduce assets, liabilities and consideration under the selected accounting treatment. Disposal events remove or reclassify them and may create gains, losses or continuing interests. These are not ordinary sales receipts applied to an unchanged company.

Unknown deal terms can produce alternative histories. Earn-outs, guarantees and retained interests may affect future cash after apparent disposal. A compact representation that keeps only closing consolidated balances can lose these rights. The source-to-contract-to-event chain should survive even when the initial reporting effect is aggregated.

7.13 Atomic posting and partial failure

A multi-leg event should either apply all admitted accounting movements consistently or remain unapplied with a recorded failure. Posting cash but failing to create debt because a later validation step failed corrupts the state. Similar risks arise when conversion removes debt but does not issue shares.

Atomicity is an engineering property that supports accounting consistency. It does not certify the event's economic validity. Preconditions should be checked before mutation where possible, and the event should retain an immutable input snapshot. Replay should reproduce the same movements under the same rule version.

A failed accounting mapping should not erase a successfully observed economic event from the evidence store. Instead, the system can retain the event and mark its accounting consequences unresolved. This separation permits later repair without inventing or deleting transactions.

7.14 A finding can become stale within one day

Consider the late-funded borrower in Section 5.15. An observation records an unpaid principal clause as satisfied on 2 July. The company then pays the remaining 40 on that same date. A later command asks whether a continuing event permits a new acceleration. Matching the observation's date to the command's date would accept the old unpaid finding, even though the intervening payment changed its premise. Calendar equality does not establish that the financial state is unchanged.

This is a concrete failure in the earlier selected adapter. Its original trace, with payment on the following day, passed. Moving payment to the same day exposed the defect: the stale satisfied finding remained usable. The successor retains the old observation as history but requires reassessment before using it as a current condition. A fresh paid assessment then supports the corresponding absence of a new trigger. It does not retroactively revoke an acceleration already recorded under earlier valid facts.

To express the dependency, let B be the complete typed book and let $P(B)$ retain its economic contents: balances, cash-flow history, nonremedy memory, context, account and type definitions, reporting period and schema. The projection omits the remedy-status record, the list of seen event identifiers and the current observation date. The date is checked separately. The first omission is essential: recording a finding changes remedy status and must not, by that act alone, invalidate the finding it records. Event identifiers alone do not establish an economic change. Other dependencies are retained conservatively, even when a particular clause might use fewer of them.

An observation stores its clause kind k_o, date d_o, evidence revision v_o, and the projected economic state $P(B_o)$, represented by a digest in the implementation. A proposed use at book B and date d, under revision v, requires

$$d = d_o, \qquad P(B) = P(B_o), \qquad v = v_o, \tag{7.1}$$

together with a compatible clause kind and the separately supplied continuing event finding. The same-day payment violates the middle equality. A status recording with no economic effect does not. A later evidence correction can violate the last equality while leaving every balance unchanged.

The evidence revision identifies a supplied immutable view; it is not a certificate that the view contains all relevant facts. A caller that omits a new notice or supplies a false old view cannot be corrected by comparing digests. Source acquisition, availability and revision selection remain separate requirements. Similarly, checking a conservative whole-economic projection can demand reassessment after an unrelated change. Minimal clause-specific dependency analysis would be a further optimization needing its own correctness argument.

Clause kind prevents a different misuse. A satisfied principal-payment clause cannot supply the missing premise for automatic bankruptcy acceleration. The selected implementation rejects that mismatch; a separately supplied qualifying bankruptcy observation can support its own automatic route. Its truth remains conditional on the applicable terms and supplied event finding. The system preserves unresolved applicability rather than turning an unsupported factual inference into a remedy.

This repair illustrates a broader simulation requirement. Assertions about current state need the dependencies under which they were observed, not only an observation timestamp. Reusing a cached covenant finding after a cash transfer, or a borrowing-capacity calculation after a collateral release, has the same risk. Keeping the previous assertion permits historical replay; withholding its current use prevents a past result from deciding a different present problem. Neither operation requires enumerating a separate tree node for every stored copy. Shared data can retain the dependence explicitly.

The checked continuation preserves complete states and the prior missed amount through payment, reassessment and rescission. Unknown or stale findings return without an accepted successor. The reference covers one selected synthetic series and supplied evidence views. Automated source revision, several simultaneous clauses and general rights-changing financial effects remain additional work, rather than consequences of this freshness check.

7.15 Closing a year without erasing a missed payment

A reporting boundary creates another opportunity to confuse present balances with financial history. Return to the late-funded borrower. In the paid case, the final settlement leaves no principal outstanding. In the unpaid case, the company has received the new equity but retains cash 40, outstanding principal 40 and arrears 40 at year-end. Both cases record new shares of

40 and a historical missed payment of 40. The second case must be able to enter the next year with the unpaid obligation intact; requiring every claim to be settled before carrying the book would exclude the scenario being studied.

Closing transfers interest expense of 20 into retained earnings under the selected report mapping. Current-period flows are archived before the new period begins. Carry then initializes the next reporting period without turning that accounting boundary into a payment, waiver or new issue. The selected policy preserves outstanding principal, arrears, share quantities and the old missed amount. It also preserves the prior clause category, observation date, observed economic-state identifier, supplied evidence revision and acceleration record.

This policy is explicit because a generic typed carry operation may permit different treatment of different memory fields. A type-correct reset of the missed amount to zero would destroy financially relevant history while leaving the opening balances coherent. The selected financing wrapper therefore requires preservation for these fields before admitting the next opening. Other memory, including period-specific counters, can require a different policy; preservation of this selected contract record is not a universal fiscal-closing rule.

Preserving the observation does not make it current. The new period and its flow state change the economic dependency in equation (7.1), and the old observation date remains old. A new proposed acceleration consequently requires a new assessment. Even after that assessment, a changed supplied evidence revision requires reassessment again. Carry must not refresh these fields merely to make a later command pass.

The unpaid example continues into 2028 with an explicit fresh observation and a payment of 40. Payment reduces cash and the outstanding claim together; the historical miss remains 40. The separately recorded acceleration also survives unless an applicable rescission or other supported rights-changing event changes it. A principal-payment observation still cannot authorize the automatic bankruptcy route. Those are different contractual premises even when a reporting boundary lies between their observations.

The exact continuation compares the complete closing entries, archived flows, next opening, subsequent payment and next-period statements on two account charts. The unchanged carry wrapper has separate compatibility evidence for the stronger remedy profile that includes supplied evidence revision and clause category. This is explicit compatibility for the selected records, not an implicit migration between earlier and later schemas. The examples demonstrate why financial histories and observation dependencies must travel together across reporting periods. Actual terms, several simultaneous clauses, guarantees and the economic consequences of acceleration still require their own sources and methods.

7.16 Conversion changes a claim and the people who control it

Preserving rights across a reporting boundary is necessary until an event changes them. A conversion provides a contrasting case: cash may stay fixed while debt disappears, new shares are issued and another holder gains control. A simulator that checks cash and balanced journals alone can miss the decision consequence.

Consider a separately supplied fiscal opening on 1 January 2028. Project assets are 100 and cash is zero. Senior debt has face and carrying amount 40. A convertible claim has face 80 and a supplied carrying amount C. There are 40 old shares and no shares held by the convertible creditor. Opening capital is $60 - C$, so the signed balances reconcile. The examples use $C = 80$ and $C = 17000/441$ as two distinct accounting premises. The latter is a diagnostic amount; selecting it does not establish a measurement rule.

The supplied contract says that the conversion signal extinguishes the claim and issues 120 shares to its holder on that date. Under the selected carrying-value transfer, the journal debits the convertible liability by C and credits issued equity by C. The face becomes zero, new shares become 120 and the observed signal and conversion are recorded. There is no cash flow or profit from this assumed transfer. The financial statements then show assets 100, liabilities 40 and equity 60 in either carrying case. Under the other signal, the debt remains outstanding and no shares issue.

Cash-flow and voting fractions follow from the resulting shares under the assumed equal rights per share. After conversion the new holder owns $120/(40 + 120) = 3/4$ and the old holder owns $1/4$. Under this supplied economic governance rule, the new holder therefore has strict majority control. Before conversion the old holder controls all shares. The separate accounting-control assessment, including potential rights before legal conversion, is explained in Section 26.16. These rights follow from the completed transition, rather than from a controller label retained from the opening. Different voting classes would require a different supplied rights rule.

The complete comparisons retain every account, the claim and share record, unrelated memory, event history, journal and report on two account charts. They also compare the resulting rights with a separately authored payoff table. Omitting claim extinction or share issue fails even when a proposed journal balances. Repeating conversion with a fresh event identifier fails because the event has already occurred. Wrong dates and signals fail before effects are accepted. Reporting the equity increase without explaining its noncash movement also fails, while preserving the already accepted conversion.

This example starts from an explicit opening and assumes the accounting transfer. It does not derive that opening from issuance, select a reporting standard, or establish that a real contract has these terms. Its contribution is narrower and useful: the accounting balances and economic rights change together, and the rights can supply the next decision without counting both converted debt and equity as surviving claims.

7.17 Reporting frequency and event frequency

A company may report quarterly while obligations settle daily. A quarterly engine that nets all flows can miss a temporary liquidity failure. A daily engine can still report quarterly by aggregation. The appropriate event resolution depends on the question and the contracts.

Adaptive resolution is possible. A long period with no state-dependent decisions can be aggregated analytically. A period containing maturity, reset, dividend and covenant dates may require explicit boundaries. The aggregation must preserve ordering and any intermediate constraint that matters.

For example, cash 10, a receipt of 20 and payment of 25 produce closing cash 5 regardless of order. If payment comes first and no overdraft is permitted, the path is infeasible despite the positive closing balance. A minimum-cash query requires the event sequence. A closing-balance query alone would not detect the problem.

7.18 Stock-flow consistency across counterparties

When several entities are modeled, a payment by one is a receipt by another, subject to fees, taxes and timing. Keeping counterpart links helps detect money creation or disappearance. In an external-counterparty boundary, the system can record the flow leaving the modeled domain rather than pretending the counterparty does not exist.

The stock-flow consistent approach in Caiani et al. (2016) gives a useful example of explicit sectors and transaction sequence. Its named cash-flow measures must be interpreted in its own definitions: a quantity called operating cash flow in that model subtracts principal repayments and should not automatically be mapped to an accounting statement category. Reusing labels without inspecting definitions is a semantic error.

AccDSL should therefore give each reported or derived quantity a definition and units. A label match is a candidate mapping, not proof of equivalence. This is especially important when connecting economic simulators, accounting standards and regulatory templates.

7.19 A coherent event-to-report method

The proposed method first admits an economic event under its contractual and behavioral premises. It updates quantities, rights and obligations. It then invokes the accounting rule for the declared basis and context, validates the posting bundle, and updates report accumulators. Source and scenario provenance accompany each stage.

At reporting boundaries, the system reconciles opening balances, movements and closing balances, checks units and perimeter, and preserves unresolved amounts or conditional interpretations. It does not force an unsupported movement into a generic other line merely to close the statement. A residual can be displayed as a diagnostic requiring explanation.

This sequence supports both concrete witnesses and symbolic families. If every member of a region has the same event rule, amounts can remain symbolic. If a guard changes the rule, the region must split or retain a piecewise expression. Accounting coherence then applies within every admissible branch, not just to an average statement that no individual history produces.

Chapter 8

Keeping uncertainty without enumerating every world

Twenty binary unknowns have more than a million assignments. A company can have thousands of contracts, continuous unknown amounts and decades of possible events. Explicit enumeration is therefore an unsuitable default. The alternative is to represent the family and compute the answer demanded by a question. This can save enormous work, but cannot guarantee that every problem has a small exact solution.

8.1 Representation size and answer size

A formula containing twenty Boolean variables can describe all 2^{20} assignments without listing them. A solver may answer whether any assignment violates a cash constraint without printing every world. If the user asks for all distinct journals, the output itself can be enormous. No representation can print an exponentially large answer in fixed small space.

Continuous uncertainty sharpens the distinction. The family $0 \leq \alpha \leq 1$ has infinitely many members, but interest $5 + 2\alpha$ is a finite exact answer. Useful computations include symbolic propagation, optimization over a region and classification of regions by behavior. Enumeration is one method, not the definition of correctness.

Darwiche and Marquis (2002) compares knowledge-compilation languages by succinctness and supported queries and transformations. Its definitions of directed acyclic representations, decomposability and determinism explain why sharing and structural restrictions help. Its tradeoffs also warn against assuming that a compact formula makes every query cheap. Compilation can itself be expensive.

8.2 Three different graphs

An evidence graph links assertions to sources and interpretations. A constraint graph links variables through relations. An execution graph links states through dated events. Their edges have different meanings. Source support is not causation; a constraint edge is not a probabilistic independence statement.

An expression DAG can share a repeated discount factor or subtotal. An immutable history graph can share common event prefixes. A transition graph can merge states only when relevant future behavior is equivalent. Confusing these forms of sharing can turn an engineering

optimization into an unjustified economic assumption.

Economic feedback may remain cyclic while expression storage is acyclic. Debt price and investment policy can be simultaneous unknowns. An acyclic residual evaluator does not eliminate the fixed-point problem. The representation and solution method answer different questions.

8.3 Share common structure without merging worlds

If a thousand histories differ only in one loan's reset clause, their source reports, asset register and most events can be shared. Each alternative stores its clause condition and local differences. This reduces storage without identifying the alternatives economically.

A common event prefix can be executed once when every history has the same relevant state and rule up to a boundary. Execution separates at the first differing reset or decision. Persistent data structures retain ancestors and changed fields for explanation and replay.

Sharing must not allow mutation of one branch to alter another. Refinancing in one history cannot rewrite the common original loan. Immutable contract versions and branch-local successor records prevent that failure. Cache identities must include all relevant inputs and rule versions, not merely a displayed company name.

8.4 Elimination must preserve consequences

If a query uses variables x and the full family is $C(x, z)$, exact elimination of z means

$$C_Q(x) \iff \exists z C(x, z). \tag{8.1}$$

Dropping every constraint containing z is generally not this projection. Constraints $x = z + 1$ and $0 \le z \le 2$ imply $1 \le x \le 3$. Removing both constraints leaves x unrestricted. Correct elimination retains the implied interval.

Projection can create dense relationships among remaining variables. A small original representation does not guarantee a small projected one. A dependency slice must include outputs, event guards, feasible actions and observation rules over the horizon. A variable absent from today's report can affect tomorrow's signal or covenant.

An asset identifier irrelevant to interest expense may matter to collateral eligibility. A customer identifier irrelevant to total receivables may matter to concentration. Query-driven elimination therefore follows economic dependencies rather than simply selecting columns present in the requested output.

8.5 Split where behavior changes

If every member of a region follows the same affine transition, propagate it symbolically. When a guard changes the event rule, partition the region or retain a piecewise expression. For cash $90 - 2\alpha$ and requirement 89.4, the split is $\alpha \le 0.3$ versus $\alpha > 0.3$. The equality belongs to compliance under the inclusive rule.

If breach has no effect on the requested output, its consequences may be deferred while the condition is retained. If it changes funding eligibility, it matters immediately. Repeated guards can still create exponentially many distinct behaviors. Guard-based execution avoids unnecessary expansion; it does not prove universal tractability.

A partition should carry its logical coverage: regions are disjoint where intended and their union covers the admitted parent. A missing equality boundary can silently lose a possible company. An overlapping boundary can double probability mass or duplicate an event. Exact rational threshold fixtures are especially useful here.

8.6 Equal totals are insufficient

Two states with cash 50 and debt 100 can differ in tomorrow's maturity. Two states with equal tomorrow payments can differ in a disclosure that changes the next decision. Neither pair can automatically be merged for the corresponding query.

Givan et al. (2003, Section 3.1) shows why equality under fixed action sequences is too weak for general policy equivalence in an MDP: policies react to intermediate states. Section 3.3 develops stochastic bisimulation through equal rewards and transition probabilities to equivalent blocks for each action. The accounting application below adds observation and requested-output preservation; this extension is a local sufficiency argument.

8.7 A sufficient probabilistic condition

Consider a finite horizon. Let x be full state, a an action and s a scenario control. Let $K_t(dx', do, dy \mid x, a, s)$ be the joint law of next state, observation and requested output. Let $\phi_t(x)$ be the compressed state. States sharing a compressed state must have equal feasible actions and current outputs. For every allowed a, s and measurable set B, also require

$$\begin{aligned} &K_t(\{(x', o, y) : (\phi_{t+1}(x'), o, y) \in B\} \mid x_1, a, s) \\ &\quad = K_t(\{(x', o, y) : (\phi_{t+1}(x'), o, y) \in B\} \mid x_2, a, s). \end{aligned} \tag{8.2}$$

Initial distribution and information must be mapped consistently. Policies use only the retained information history and permitted memory. Under these conditions, the compressed model preserves the requested joint output and observation law. Equal means, or equal marginal laws separately, are weaker. Dependence between observation and successor state matters for learning.

8.8 Finite-horizon proof

At the final date, equality of current outputs establishes the result. Suppose the compressed model reproduces continuation laws from date $t + 1$. At date t, an admissible policy selects the same action distribution for identical retained information histories. Equal feasible actions make this choice meaningful in either state.

Equation (8.2) gives the same joint law of next compressed state, observation and immediate output. At each successor, the induction hypothesis supplies the same continuation law. Integrating equal continuation laws against the equal successor law gives equality of the remaining path law. Backward induction reaches the initial date.

This condition is sufficient, not necessary. The finite helper described next checks explicitly supplied finite models; it is not a general checker for arbitrary company-state representations. If managers privately observe a hidden variable removed by ϕ, their policy may not be expressible in the compressed model. The information condition must be strengthened or the state refined. Incorrect kernels cannot be repaired by an otherwise correct aggregation proof.

8.9 Checking a supplied finite aggregation

A lending decision makes the joint-law requirement concrete. Suppose a company starts with cash 10 and observes a green or red signal before deciding whether to lend. A good outcome adds 2 to cash if it lends; a bad outcome subtracts 4. Waiting leaves cash unchanged. In this hypothetical model, good and bad states each have probability one half, and the signal identifies the state correctly with probability 9/10. The policy lends after green and waits after red. The resulting cash movements $+2$, -4 and 0 have probabilities 9/20, 1/20 and 1/2. For example, the profitable lending path needs both a good state and its green signal, so its probability is $(1/2)(9/10) = 9/20$.

Two copies of the same initial state can be merged without changing those paths. But suppose a second state reverses the signal's association with the good outcome. Green and red still each occur half the time, as do good and bad outcomes. The separate distributions therefore agree, while the policy lends in different circumstances. Comparing the joint successor state and signal detects the difference. A separate checked example keeps the successor state fixed and changes dependence between observation and immediate cash movement; separate marginal checks miss that change too.

The finite conformance helper checks this distinction for an explicitly supplied model and proposed partition. At every date it compares permitted actions and current requested outputs within each block. For every action and scenario it groups successor records by the complete triple of successor block, observation and immediate output. In the probability case it sums explicit rational masses and compares the exact totals; each supplied kernel must have total mass one. In the unweighted case it compares sets of triples. Repeated representations of a possibility never supply a probability.

Initial information needs the same care. The input specifies state and initial observation jointly. Aggregation maps the state while retaining that observation and the current requested output. Each later decision uses the retained history of observations, actions and outputs. A checked first-decision example lends after one initial observation and waits after another, even though their two states belong to the same block. Removing the observation would change the permitted policy. A policy asking directly for the hidden state is outside this information interface.

The finite argument follows the preceding induction. At each successor block, the requested current output is fixed by the block. Equality of the joint triples thus gives equal probabilities for each extended observable history. Applying the same admissible policy and continuing the induction preserves the whole path law. For an unweighted relation, compatible path concatenation and union replace probability products and sums. Every quotient transition must be possible from each member of its source block; otherwise an aggregated path might not lift to a path in the original model.

An independently coded path enumerator checks twelve accepted finite cases, including every deterministic lend-or-wait choice after the two signals, duplicate successor states, initial-information decisions and an unweighted model. Eight proposed merges fail with a specific unequal action, output or joint-kernel component. Two deliberately incorrect representative merges also produce different complete path laws. These checks support the supplied finite examples; they do not establish that a company's chosen state contains every economically relevant variable.

The helper refuses missing transitions, invalid probability mass, unrecognized information and incomplete partitions. A comparison budget exhausted before all obligations are checked returns an unknown result with the unfinished comparisons and no approved quotient. Saved results bind the full model, partition, information, outputs and horizon; reuse rechecks the condition. This is a sufficient condition over all declared states, including unreachable ones. Failure may therefore require a finer partition without proving that every aggregation for a particular initial

policy is impossible. The next example shows why persistent contract type is one dependence worth retaining.

8.10 A larger state can preserve the information structure

The distinction between a process and its information becomes especially clear in the work of Beiglböck, Pflügl and Schrott. Their information process recursively augments today's value by the conditional law of tomorrow's information process. Equal laws of suitable Markov lifts characterize their notion of equivalence of filtered processes (Beiglböck et al., 2024, Definitions 2.6–2.8, Proposition 3.6 and Theorem 3.9). A lift may be large or measure-valued. Its existence does not imply that a small sufficient state can be recovered from financial statements. Moreover, transferring strategies requires attention to independent randomization and the available stochastic basis (Beiglböck et al., 2024, Theorem 5.4 and Remark 5.7). An implementer must decide whether an actor may randomize and which observations are available, rather than deducing these permissions from equal outcome laws.

8.11 Unweighted families require relational memory

For an unweighted family, replace probabilities with a relation of feasible successors, observations and outputs. Exact aggregation must preserve joint possibilities and persistent cross-time constraints. One-step marginal sets alone are insufficient.

A fixed-or-floating loan with two-year interest paths $(5, 5)$ and $(7, 7)$ has one-step marginal set $\{5, 7\}$ at each date. Recombining those sets creates $(5, 7)$ and $(7, 5)$, which require unmodeled contract changes. Retaining the latent type or equivalent memory prevents these paths.

An enlarged family can sometimes provide an outer bound for a worst-case query. It is not an exact family of possible companies. A harmful witness from the enlarged set must be validated against original constraints before being described as economically possible. Conservatism does not excuse mislabeling the target.

8.12 Approximation needs a query-specific error

A cash error of 0.1 may be harmless for a chart and decisive at a covenant. For a smooth output, state error can imply an output bound under regularity. For a default indicator, an arbitrarily small change near a threshold can reverse the classification.

If computed cash lies in $[19.95, 20.05]$ and payment required is 20, a midpoint cannot establish feasibility for the whole region. The correct classification is unresolved unless a stronger method determines the sign. Refinement should focus on this boundary rather than indiscriminately increasing precision.

If f is Lipschitz with known constant L in a stated norm and $\|x-\hat{x}\| \le \epsilon$, then $|f(x)-f(\hat{x})| \le L\epsilon$. This follows from the definition. It requires a justified constant and domain. An unknown or very large L can make the bound unhelpful. No such smoothness claim is made for a discontinuous breach indicator.

8.13 Inner and outer approximations

Let $\overline{\mathcal{H}}$ contain the true modeled family and let $\underline{\mathcal{H}}$ be contained in it. Then

$$\inf_{h \in \overline{\mathcal{H}}} C(h) \leq \inf_{h \in \mathcal{H}} C(h) \leq \inf_{h \in \underline{\mathcal{H}}} C(h). \tag{8.3}$$

The outer minimum is a lower bound on the true minimum; the inner minimum is an upper bound. A low cash value from relaxation can be spurious. A high minimum over sampled witnesses does not prove safety.

If the outer lower bound exceeds the payment requirement, the original family is safe for this query under its assumptions. If an inner feasible witness falls below it, failure is possible. Otherwise the status remains unresolved. This three-way answer is more informative than presenting every approximate extremum as exact.

8.14 Counterexample-guided refinement

Begin with a coarse outer model. If it proves the desired bound, the query may finish. If it returns a harmful candidate, validate that candidate against full constraints. A feasible candidate is a genuine witness; an infeasible one identifies a lost dependence and motivates refinement.

The marginal interest-cash rectangle admits $(7, 90)$, but $C + I = 95$ rejects it. Adding that relation removes the spurious combination without enumeration. Other refinements can restore currency-maturity dependence, collateral ownership or persistent contract identity.

This is a proposed reasoning method, not a universal termination result. Refinement may exhaust resources. The returned bounds and unresolved region should remain available, together with the dependencies still preventing a definitive answer. A resource limit is not an economic default or a proof that the region is empty.

8.15 Caching has an economic scope

A cached answer depends on evidence, assumptions, contracts, policy, observation model, horizon, requested outputs and numerical method. A change in any relevant component invalidates it. Company name and scenario label are insufficient cache keys.

An aggregation checked for one-year interest may fail for three-year refinancing. Adding a new observed covenant can invalidate an information summary. The system should record the query class covered by an abstraction and require revalidation before broader reuse.

Provenance remains separate even when computations are shared. Two sources supporting one fact need not duplicate arithmetic, but their independent support matters if one is withdrawn. Execution identity must not replace source identity.

8.16 Trees as selected explanations

A small tree is useful for showing information arrival, choices and counterexamples. It can be generated from the symbolic family for a question. Nodes should identify whether they are exact worlds, samples, regions or hypothetical observations.

A representative tree is not exhaustive unless the finite model and coverage establish that claim. Branch probabilities require a measure. Pruning a display branch must not silently delete it from the calculation or renormalize remaining probabilities. A region can instead remain visible as an unresolved aggregate.

A hybrid interface can show important branches, formulas and bounds together. Users expand regions affecting their decisions. This preserves explanation while avoiding compulsory exhaustive materialization. The underlying family remains authoritative when a new question requires a different partition.

8.17 Budgeted computation

A query should state resources and acceptable result types. Exact values, certified bounds, feasible witnesses and statistical estimates are distinct deliverables. If exact work exceeds budget, return partial bounds and unresolved regions. Do not substitute an unsupported point merely because the interface expects a number.

Priority can go to liquidity, covenant, default and decision boundaries. This is a query-specific strategy and must not remove rare regions from a tail-risk calculation. The next chapter develops sampling and other approximations while preserving the distinction between retained possibilities and sampled evidence.

The layered proposal is symbolic constraints, shared immutable records, dependency-aware elimination, guard-based splitting, justified aggregation and controlled approximation. Each layer has a correctness condition. Choosing a graph instead of a tree is useful engineering, but the mathematical and economic obligations remain.

Chapter 9

Approximate inference and simulation without false certainty

Exact symbolic calculation will not solve every useful company model within available resources. Approximation is therefore necessary. Its legitimacy depends on naming the quantity being approximated and preserving the distinction between numerical evidence and economic possibility. A finite sample can estimate an expectation under a declared distribution. It cannot, by itself, characterize every history compatible with disclosures. A fitted surrogate can interpolate a function well while giving the wrong answer at a default boundary.

9.1 Choose the target first

Different questions require different methods. An expected cash balance, a worst-case cash balance, a breach probability, a feasible breach witness and a policy value are different targets. A method appropriate for one can be misleading for another. Sample averaging targets an expectation; minimizing over samples supplies a witness-based bound on a minimum, not an exact worst case.

Let $Y = F(H, S, \pi)$ under a declared joint measure for history H and scenario S. Monte Carlo draws $Y_1, \ldots, Y_n$ give the sample mean

$$\widehat{m}_n = \frac{1}{n} \sum_{i=1}^{n} Y_i. \tag{9.1}$$

Under independent identical sampling and finite variance, its variance is $\mathrm{Var}(Y)/n$. Those conditions are part of the claim. Correlated paths, importance weights or adaptive sampling require a different variance treatment. The formula does not apply merely because the output file has n rows.

Without a probability model, a finite collection of histories can still be useful for explanation and testing. It should be called a set of witnesses or selected cases. Equal frequencies in that collection do not establish probabilities in the underlying company family.

9.2 Sampling a persistent parameter

If a contract type is fixed but unknown, sample it once for a simulated world and retain it through time. Draw future innovations according to their conditional process. This produces the

intended joint law of parameter and trajectory. Redrawing the type each period approximates a different model.

The distinction remains important after filtering. Resampling particles may duplicate a parameter value as a computational operation; it does not mean the company changed contract. Particle identity and world identity must not be confused. A lineage record can preserve which parameter and past events each particle represents.

If the uncertainty is set-valued rather than probabilistic, selecting parameter values for numerical exploration should retain the original feasible region. Sampled values can reveal failure cases or guide refinement. They should not replace the family with a smaller set and thereby turn unexplored histories into impossible ones.

9.3 Particle filtering under known model assumptions

A particle filter approximates sequential distributions in a specified state-space model. Let x_t be the latent state at date t and y_t the observation then available. The transition density is $f(x_t \mid x_{t-1})$, the observation likelihood is $g(y_t \mid x_t)$, and a proposal q generates the next particle. First consider a sequential importance-sampling step that extends particle i without resampling it. Its updated weight satisfies

$$w_t^{(i)} \propto w_{t-1}^{(i)} \frac{g(y_t \mid x_t^{(i)}) f(x_t^{(i)} \mid x_{t-1}^{(i)})}{q(x_t^{(i)} \mid x_{t-1}^{(i)}, y_t)}. \tag{9.2}$$

The fraction is the incremental weight; multiplying it by the previous weight gives the new weight. It follows by dividing the joint posterior update by the proposal update, as in Doucet and Johansen (2008, Section 3.3, equations (29)–(30)). The proposal must cover the relevant target support. A proposal that never visits a feasible high-risk region cannot recover it by weighting.

Resampling changes this calculation because selecting a parent already uses its weight. Let $W_{t-1}^{(j)}$ be the normalized weights before selection. Suppose child i selects ancestor $A_t^{(i)} = j$ with probability $r_{t-1}^{(j)}$, then draws its next state from $q(\cdot \mid x_{t-1}^{(j)}, y_t)$. Relative to this finite weighted approximation, the joint target mass for ancestor j and new state x is proportional to $W_{t-1}^{(j)} f(x \mid x_{t-1}^{(j)}) g(y_t \mid x)$; the proposal mass is $r_{t-1}^{(j)} q(x \mid x_{t-1}^{(j)}, y_t)$. Their ratio therefore gives

$$\widetilde{w}_t^{(i)} \propto \frac{W_{t-1}^{(A_t^{(i)})}}{r_{t-1}^{(A_t^{(i)})}} \frac{g(y_t \mid x_t^{(i)}) f(x_t^{(i)} \mid x_{t-1}^{(A_t^{(i)})})}{q(x_t^{(i)} \mid x_{t-1}^{(A_t^{(i)})}, y_t)}. \tag{9.3}$$

Both ancestor and state proposals must have positive mass wherever this target has positive mass. For ordinary multinomial resampling, $r_{t-1}^{(j)} = W_{t-1}^{(j)}$, so the first ratio cancels. Equivalently, the resampled children start with equal weights $1/N$ before propagation; the old parent's weight is not multiplied again. The always-resampled and adaptive algorithms in Doucet and Johansen (2008, Sections 3.4–3.5) make that reset explicit. If resampling is skipped, the previous unequal weights are retained. These conventions concern numerical particles, not changes to the company's contract.

For example, suppose two persistent contract types have current weights 3/4 and 1/4, and a new disclosure has likelihoods 1/3 and 1. Bayes' rule gives unnormalized masses 1/4 and 1/4, hence equal posterior probabilities. A four-child resampling allocation with three copies of the first type and one of the second gives aggregate incremental weights $3(1/3) = 1$ and 1, also equal. Multiplying by the old parent weights again would instead give 3/4 and 1/4. This exact

allocation exposes the double counting; a random finite resampling draw need not reproduce the posterior exactly.

Doucet and Johansen (2008, Sections 2–3) explains sequential importance sampling, resampling and degeneracy. Its filtering setup conditions on a specified model. Static unknown parameters are not automatically handled well by adding them to the state and repeatedly resampling. That extension can suffer severe impoverishment.

For AccDSL, particle filtering is appropriate only after prior, transition and observation laws are admitted. It can be an optional numerical view of a history family. The family and source constraints remain available to detect support loss and distinguish probabilistic approximation from exact admissibility.

9.4 Weight degeneracy and genealogical collapse

If nearly all weight concentrates on one particle, a nominal population of ten thousand does not provide ten thousand effective alternatives. A common weight diagnostic is

$$N_{\text{eff}} = \frac{1}{\sum_i (\bar{w}_i)^2}, \tag{9.4}$$

where normalized weights sum to one. It summarizes weight concentration, not every form of approximation error. Uniform weights after resampling can coexist with many duplicate ancestors.

Genealogical collapse matters for historical reconstruction and smoothing. Particles can share the same early history even while their current states differ. If an old unknown contract type is lost, later data cannot revive it without a rejuvenation method or retained symbolic support. A high current effective sample size does not establish historical diversity.

Diagnostics should therefore include unique persistent parameter values or clusters, ancestor diversity, support checks against constraints and sensitivity to proposal design. These are explanatory or veto diagnostics under a stated protocol. No single scalar certifies that the posterior or scenario tail is accurate.

9.5 Rejuvenation and conditional calculation

Move steps can restore diversity if they preserve the intended target distribution. Their acceptance probabilities, conditional variables and constraints must be correct. Arbitrarily perturbing a loan rate or maturity can create a history inconsistent with evidence. Projection back into a feasible set can also alter the target distribution unless its effect is accounted for.

Analytic integration of conditionally tractable components can reduce sampling variance. Doucet and Johansen (2008, Section 4.6) discusses Rao–Blackwellization. In a company model, some accounting accumulations may be deterministic conditional on contract type and shock path, so they need not be sampled separately.

This benefit depends on a valid conditional decomposition. Integrating a variable out while forgetting the dependence it induces on remaining quantities gives the wrong target. The same projection discipline used for symbolic families applies probabilistically: marginalization preserves dependence, whereas independent replacement usually does not.

9.6 Rare events and empty samples

If a breach has probability p and paths are independent, the probability of observing no breach in n draws is $(1-p)^n$. For $p = 0.001$ and $n = 100$, this is about 0.905. Zero observed breaches are therefore unsurprising and do not establish safety.

For an ideal Bernoulli experiment with zero events, a one-sided 95 per cent upper bound solves $(1-p)^n = 0.05$, giving $p = 1 - 0.05^{1/n}$. This is a statistical bound under the sampling assumptions, not a statement about model misspecification. If the simulator omits a default mechanism, more draws cannot detect it.

Tail estimates need enough events and a suitable uncertainty analysis. Extreme quantiles and maxima have high sampling variability. A lower observed maximum in one run is descriptive evidence, not a defensible ranking of methods. The target, sampling design and uncertainty calculation should be specified before interpreting a comparison.

9.7 Importance sampling

Importance sampling deliberately visits rare regions more often. If target density is $p(x)$ and proposal is $q(x)$, an expectation can be estimated using weights $p(x)/q(x)$ where the proposal covers target support. Oversampling distress is useful only if weights and support are handled correctly.

For event indicator $B(x)$, the ordinary importance estimator is $n^{-1} \sum_i B(X_i)p(X_i)/q(X_i)$ with $X_i \sim q$. Self-normalized weights define a different finite-sample estimator. The method should record which is used and how uncertainty is assessed.

A proposal chosen after observing a preliminary sample can be effective, but adaptive procedures require care in interpretation. Selecting only paths that breach and averaging them estimates a conditional severity, not the unconditional breach probability. A report should separate frequency from loss given breach.

9.8 Stress testing without probabilities

A deterministic stress can be useful even when its likelihood is unknown. It asks what happens under the specified path. Reverse stress can search for conditions that produce a shortfall. Neither exercise needs invented probabilities.

For example, solve for the smallest rate increase causing cash below a threshold under each compatible debt history. The answer is a family of thresholds. A minimum across histories identifies a vulnerable compatible case; a maximum identifies how much stress is needed to affect every admitted history, subject to monotonicity and feasibility.

The search should include joint constraints on shocks. Allowing sales, exchange rates and funding spreads to vary independently over arbitrary ranges can produce an economically incoherent reverse stress. A broad stress set may be intentionally conservative, but its interpretation is then a designed adversarial exercise rather than an estimated likelihood.

9.9 Scenario reduction

A large finite scenario set can be reduced by selecting representative paths or aggregating similar ones. Similarity in Euclidean shock distance does not automatically imply similarity in financial

consequences. Two rate paths close in norm can lie on opposite sides of a covenant or option boundary.

The proposed AccDSL method should define reduction error in terms of requested outputs, feasible policies and information timing. If a reduction is intended to preserve an optimization value, it needs evidence for that target. Preserving a few unconditional moments is insufficient for a path-dependent refinancing decision.

Probability-metric reduction provides a precise method when a measure and suitable regularity are available. Its usefulness depends on the financial quantity and decision being approximated. The following derivations distinguish fixed-policy expectations, threshold probabilities and policy optimization.

9.10 What probability-metric reduction preserves

A finite probability model permits a more precise reduction than deleting paths and renormalizing. Let scenarios s_i have admitted probabilities p_i, with $\sum_i p_i = 1$. Choose a smaller retained index set K and map each original scenario to a retained representative $s_{k(i)}$. The reduced probability of representative j is

$$q_j = \sum_{i:k(i)=j} p_i, \qquad j \in K. \tag{9.5}$$

All probability mass remains accounted for, although the distribution has changed. The method approximates a specified measure; it supplies no probabilities for an unweighted family of unknown company histories.

Dupačová et al. (2000, Section 3, Theorem 3.1 and its proof) derives optimal redistribution for a fixed retained subset under the paper's probability-metric assumptions: deleted mass moves to a nearest retained scenario under the admitted cost. The choice of which scenarios to retain remains a separate combinatorial problem. Algorithms 3.6 and 3.7 are backward-reduction and forward-selection heuristics. Their sequential choices do not generally establish a globally optimal subset.

A short local derivation shows what a distance can buy for a financial query. Suppose a fixed-policy cash outcome $f(s)$ satisfies $|f(s) - f(u)| \le Ld(s,u)$ for all admitted pairs, where d is a stated scenario distance and L is a justified constant. Then

$$\begin{aligned} |\mathbb{E}_P f - \mathbb{E}_Q f| &= \left| \sum_i p_i [f(s_i) - f(s_{k(i)})] \right| \\ &\le L \sum_i p_i d(s_i, s_{k(i)}). \end{aligned} \tag{9.6}$$

The first equality substitutes the redistributed masses; the inequality applies the triangle inequality and the assumed regularity. Thus the weighted movement of scenarios bounds this expected-output error. A distance small in interest rates is useful only when the required output responds regularly in that distance.

For example, let the original states be 0, 1 and 2 with probabilities $1/4, 1/2, 1/4$. Keep 0 and 2, and resolve the middle state's distance tie by assigning it to 0. Reduced masses are $3/4, 1/4$. Under absolute distance, weighted movement is $1/2$. For $f(s) = s$, the original mean is 1 and the reduced mean is $1/2$, exactly attaining the bound with $L = 1$. Mass conservation alone does not preserve the mean.

A default indicator is discontinuous. Put equal mass at $1 - \eta$ and $1 + \eta$, where $\eta > 0$, and retain only the first state. Weighted movement is η, tending to zero, but the probability of

$s > 1$ changes from one half to zero. Therefore a small distance does not by itself give a useful default-probability guarantee. One needs additional control of probability near the boundary, a distance tailored to the event, or explicit preservation of the relevant region.

9.11 What sequential transport theorems require

Nested and adapted transport distances address the temporal issue by restricting couplings so that matching paths cannot depend on information that is unavailable at the relevant date. Pflug and Pichler's stability argument uses convex decision sets, convex loss in decisions and uniform regularity in uncertainty (Pflug and Pichler, 2012, Definition 5.1, Theorem 6.1 and proof). A projected action is formed by conditional averaging. If the only permitted issuance sizes are 0 and 1, an average of 1/2 is infeasible. The theorem therefore does not automatically justify merging discrete issuance choices.

Backhoff-Veraguas and coauthors obtain related stability results for financial optimization with bounded predictable strategies and regular payoffs (Backhoff-Veraguas et al., 2020, Theorem 1.5; Lemma 3.7; Theorem 3.10). Their weak transfer permits a different strategy in the other model. Reusing the same strategy needs additional regularity. Their semimartingale distance also has a different cost functional from a simple sum of discrete path deviations; the paper explains why time refinement matters. Its examples of discontinuous strategies and exact superhedging delimit the guarantee. These are useful reasons to formulate the company decision problem precisely, rather than reasons to attach one distance to every accounting query.

9.12 Reduction before choosing a policy

An approximation adequate for one fixed policy may fail after optimization selects a different one. Suppose, more strongly, that every policy in the same feasible, nonanticipative class satisfies

$$|V_P(\pi) - V_Q(\pi)| \le \delta. \tag{9.7}$$

If maximizers π_P and π_Q exist, adding and subtracting their reduced-model values gives

$$\begin{aligned} V_P(\pi_P) - V_P(\pi_Q) &= [V_P(\pi_P) - V_Q(\pi_P)] \\ &\quad + [V_Q(\pi_P) - V_Q(\pi_Q)] + [V_Q(\pi_Q) - V_P(\pi_Q)] \\ &\le 2\delta. \end{aligned} \tag{9.8}$$

The middle term is nonpositive because π_Q optimizes the reduced problem. This elementary bound needs uniform control over the policy class, identical feasibility and consistent information. A bound established only at the final selected policy cannot justify the argument.

Dupačová et al. (2000, Section 2) give stability results for convex stochastic programs under explicit solution-set, integrability and regularity assumptions. Remark 2.2 adds dependence conditions for its multistage setting. Those conditions do not automatically cover discrete default, share conversion, nonconvex financing or private information in a company model. Nor does a close distribution over complete paths establish that intermediate observations and admissible decisions are preserved.

An AccDSL reduction should therefore state the admitted measure, retained scenarios, redistribution rule, distance and query guarantee. It should separately check liquidity and default boundaries and the actor's observation structure. The original joint history family remains available for refinement. Reducing a numerical probability approximation must not erase a

structurally possible contract or turn an unobserved future path into information available today.

9.13 Surrogates and boundary errors

A surrogate can approximate expensive valuations or transitions. Its input domain, training distribution, output definition and constraints must be explicit. A small average prediction error can coexist with large errors in rare distressed states. If those states determine issuance feasibility, the surrogate may be a poor component despite good average accuracy.

Suppose true funding capacity is $c(x)$ and issuance is allowed only when $c(x) \geq K$. An approximation error bound $|\hat{c} - c| \leq \epsilon$ permits safe classification only outside $[K - \epsilon, K + \epsilon]$. Inside that band, refine or return uncertainty. A point prediction alone gives no such protection.

A surrogate should also respect units and economic constraints. Negative principal, duplicated shares or cash creation may be detectable regardless of predictive error. Enforcing constraints can improve mechanical validity, but does not prove that the underlying economic mapping is correct. Both checks remain necessary.

9.14 A quantitative boundary argument

Default, covenant breach and a margin call are threshold events. Let X be exact cash at a decision date, Y its approximation under a specified joint coupling, and c the contractual threshold. All three are in the same currency and date basis. A cash approximation of $-\epsilon$ by $+\epsilon$ has error 2ϵ, yet changes $\mathbf{1}\{X < 0\}$ from one to zero. Small cash error alone does not imply small default-probability error.

Proposition 9.1. *If $\mathbb{E}|X - Y| < \infty$, then for every $\delta > 0$,*

$$|\mathbb{P}(X < c) - \mathbb{P}(Y < c)| \leq \mathbb{P}(|X - c| \leq \delta) + \frac{\mathbb{E}|X - Y|}{\delta}. \tag{9.9}$$

Proof. The absolute difference in event probabilities is at most the probability that their indicators disagree. If $|X - c| > \delta$ and $|X - Y| < \delta$, X and Y lie on the same side of c. Consequently disagreement is contained in $\{|X - c| \leq \delta\} \cup \{|X - Y| \geq \delta\}$. The union bound and Markov's inequality give (9.9). □

The first term is probability mass close to the covenant; the second is the probability allowed for a sufficiently large approximation error. This local derivation is independent of any claim that a particular algorithm supplies the required coupling error. Write $e = \mathbb{E}|X - Y|$. If $e = 0$, then $X = Y$ almost surely and the threshold probabilities agree. For $e > 0$, if X has a density bounded by $M > 0$ near c, the first term is at most $2M\delta$. Choosing $\delta = \sqrt{e/(2M)}$, when that choice lies within the neighborhood where the density bound holds, gives $2\sqrt{2Me}$. Atoms at the threshold and uncontrolled conditional distributions prevent this simplification.

For several dates, a union bound can bound the probability that any breach indicator differs by the sum of the per-date bounds, provided the relevant coupling and cash errors are supplied. If a different breach decision changes subsequent actions and cash flows, those cash errors must already account for that feedback; an open-loop cash comparison does not establish them. When there are no weights over histories, the appropriate output is instead an uncertain boundary region or certified feasible-set containment. A probability bound cannot be manufactured from the proportion of enumerated witnesses.

9.15 A common-policy regret argument

Let $J(\pi)$ and $\widetilde{J}(\pi)$ be expected costs in two models, with the same admissible policy class Π, finite expected costs and finite optimal values. This comparison presupposes that a policy can be implemented with the information and feasible actions in both models. Suppose a uniform bound $\sup_{\pi\in\Pi} |J(\pi) - \widetilde{J}(\pi)| \le \varepsilon$ holds. If $\widehat{\pi}$ is η-optimal for the approximate problem, then

$$\begin{aligned} J(\widehat{\pi}) &\le \widetilde{J}(\widehat{\pi}) + \varepsilon \\ &\le \inf_{\pi\in\Pi} \widetilde{J}(\pi) + \eta + \varepsilon \\ &\le \inf_{\pi\in\Pi} J(\pi) + \eta + 2\varepsilon. \end{aligned} \tag{9.10}$$

The last step follows by applying the uniform bound to an arbitrarily near-optimal policy of the exact problem, so attainment is unnecessary. The result is a regret bound for a common feasible policy. If one only knows that the two infima are close, the first inequality is missing. If a policy lift changes the policy class, a bound for that lift and the relevant comparison class is needed. This is why reporting a small optimized residual or close portfolio value alone does not establish that a compressed financing policy will work in the original company.

9.16 When sampled paths disclose their own futures

With continuous first-period observations, independent sampled paths typically have distinct first values. If each stored first value has only its single sampled continuation, that value becomes an identifier for the sampled future. An optimizer can exploit this unintended information. More paths do not by themselves repair the information structure of the empirical measure. Backhoff and coauthors give an explicit optimal-stopping counterexample (Backhoff et al., 2021, Example 7.1). Their adapted empirical measure quantizes observations so that paths share histories. Strong consistency applies to iid complete paths on a fixed finite horizon and bounded state space; quantitative rates additionally require Lipschitz conditional kernels (Backhoff et al., 2021, Definitions 1.1–1.2, Theorems 1.3, 1.5, 1.7).

A Markov modification improves the horizon exponent only when the process is in fact Markov (Backhoff et al., 2021, Theorem 6.1); constants still depend on the horizon. A company's current balance sheet generally is not a sufficient Markov state when hidden debt terms or asset vintages matter. Nor are reports from one issuer an iid sample of complete company paths. The method suggests how a deliberately generated simulation sample can retain information. Its rates cannot simply be assigned to an observed issuer panel.

9.17 A finite coupling method for discrete decisions

A stronger but directly checkable condition can avoid averaging actions. Consider a finite horizon of T transitions. Keep a persistent contract type θ fixed and let a history record the actor's observations and past actions. In both the full model and a reduced model, require the same feasible actions at paired histories. Specify a relation between their states and couple the initial states so that they belong to this relation, their initial observations agree, and every requested current output agrees almost surely. An equal opening signal alone does not establish these other conditions.

At each paired history and common action, suppose a joint coupling of the next state, observation and reported output leaves the states in the relation and the observation/output records equal

with conditional probability at least $1 - \delta_t$, where $0 \le \delta_t \le 1$. This condition includes every history and action reachable before a mismatch; fitting one historical policy is insufficient. A common deterministic policy chooses the same action while its retained information histories agree. Randomized policies can use the same independent random draws if both models permit that randomization. Neither policy may depend on a discarded private signal.

Let E_t mean that the initial pairing and all comparisons through transition t have succeeded. Under the initial condition, $\mathbb{P}(E_0) = 1$. Integrating the conditional success bound over the surviving histories gives $\mathbb{P}(E_t) \ge (1 - \delta_t)\mathbb{P}(E_{t-1})$. This unconditional recurrence also applies when no histories survive; it does not condition on an event of zero probability. Iterating it gives

$$\mathbb{P}(E_T^c) \le 1 - \prod_{t=1}^{T}(1 - \delta_t) \le \sum_{t=1}^{T} \delta_t. \tag{9.11}$$

The requested path loss must also be preserved by a successful comparison. Assume that the two models use the same measurable loss functional of the retained observations, actions and requested outputs, with values in $[0, B]$. Its two path evaluations therefore agree on E_T and can differ by at most B otherwise. Applying the triangle inequality to the expectation under the chosen coupling yields

$$|\mathbb{E}L - \widetilde{\mathbb{E}}L| \le B\mathbb{P}(E_T^c) \le B\left\{1 - \prod_{t=1}^{T}(1 - \delta_t)\right\}. \tag{9.12}$$

The two expectations refer to the same loss functional under the respective model laws. No convexity or fractional issuance is required. A default indicator is covered when default is one of the compared outputs, or is determined by the matched retained records. A loss depending on a hidden recovery allocation that was discarded is not covered merely because its values are bounded.

Initial and within-pair approximation errors can be included explicitly. If $\mathbb{P}(E_0) \ge 1 - \delta_0$ and the two losses differ by at most ρ on E_T, with $0 \le \rho \le B$, define

$$\Delta_T = 1 - (1 - \delta_0)\prod_{t=1}^{T}(1 - \delta_t), \tag{9.13}$$

$$|\mathbb{E}L - \widetilde{\mathbb{E}}L| \le \rho + (B - \rho)\Delta_T. \tag{9.14}$$

Indeed, with $p = \mathbb{P}(E_T^c) \le \Delta_T$, the coupled expected absolute loss difference is at most $\rho(1 - p) + Bp$. The earlier bound is the special case $\delta_0 = \rho = 0$. These terms have distinct meanings: δ_0 measures failure of the initial pairing; ρ is a loss discrepancy even when the paired path remains valid. With no transitions, identical signals but different opening cash already show why transition errors cannot cover an omitted initial report. Likewise, matching cash paths does not bound an unreported creditor-allocation loss. The initial relation and requested loss must therefore be chosen before the comparison.

For finite kernels with common outcome labels, the smallest one-step mismatch probability is their total-variation distance $\frac{1}{2}\sum_z |K(z) - \widetilde{K}(z)|$. Pairing full states with reduced states requires a coupling whose successful pairs remain in the declared relation; marginal observation agreement is insufficient. An implementable finite procedure first partitions states by persistent type, available information, feasible actions and required outputs. It then compares joint successor kernels into the current blocks for every action, splitting a block whenever these differ. Repeat until no block changes. Each split reduces the number of states per block, so a finite partition stabilizes after finitely many splits. This yields an exact quotient for the declared finite model, not a claim that the model contains every economically possible company.

Approximate sharing uses the same comparison with an explicit error allowance and a checked pairing. Threshold regimes and different feasible actions are kept separate unless their mismatch has been bounded. An unresolved comparison is retained as unresolved, not assigned zero error. For a fixed common policy class, the uniform cost bound in (9.12) also gives the common-policy $2\varepsilon + \eta$ regret bound derived in this chapter. If the persistent types are unweighted, apply the bound within each type. A uniform type-wise bound also bounds the worst-type value, since $|\sup_\theta J_\theta - \sup_\theta \widetilde{J}_\theta| \le \sup_\theta |J_\theta - \widetilde{J}_\theta|$. The adversary does not redraw the contract type at each date.

The exact reference uses three periods, cash observed by management, a hidden fixed type, binary borrowing, and a default penalty. It compares complete paths for all 64 policies in the explicitly restricted class of date-and- zero-cash rules. Removing an irrelevant tag preserves every path. Merging types is rejected. A synthetic change of 1/20 in each period's bad-expense probability gives the conservative cost bound $12\{1 - (19/20)^3\} = 3423/2000$. The largest observed value discrepancies are 4433/8000 and 1/2 in the two types, within the bound. These finite checks support the stated method and expose its conservatism. They do not supply a capacity estimate or a bound for an unenumerated continuous company model.

9.18 Error propagation through time

Small one-step errors can accumulate or amplify through nonlinear feedback. For an illustrative recursion with Lipschitz transition constant L, one-step approximation error at most ϵ and initial error e_0, a bound satisfies

$$e_{t+1} \le L e_t + \epsilon. \tag{9.15}$$

Iterating gives $e_T \le L^T e_0 + \epsilon \sum_{j=0}^{T-1} L^j$. When $L < 1$, the sum is bounded by $1/(1 - L)$; when $L > 1$, it can grow rapidly. This is a local mathematical bound under its assumptions, not a claim that a company model is globally Lipschitz.

Default and option transitions can be discontinuous. Near them, a small state error can select a different regime and invalidate a smooth bound. Validation should therefore include boundary perturbations and multi-period rollouts, not only one-step residuals on typical states.

9.19 Combining exact and sampled components

A hybrid method can keep discrete contract alternatives exact, solve linear accounting relations symbolically and sample future demand conditional on each alternative. It can also bound unresolved histories while estimating conditional means for selected regions. This is often more informative than forcing every uncertainty into one Monte Carlo distribution.

The result must retain which component is exact, bounded or estimated. An exact accounting identity inside a sampled simulation does not make a sample quantile exact. A certified bound on one contract region does not cover omitted regions. Mixed results should be combined using rules appropriate to their evidence type.

A useful report can state that breach is impossible in one region, possible with a verified witness in another, and probabilistically estimated in a third under an admitted measure. These are compatible forms of information. Reducing them to one unlabeled scalar would lose their meaning.

9.20 Stopping rules

A numerical budget should specify what makes an approximation adequate for the decision. A tolerance on expected cash may be appropriate for planning; a confidence interval for breach probability may be needed for a probabilistic statement; a certified lower bound may be needed for a robust safety claim.

Stopping because the sample mean appears stable can miss a rare region. Repeatedly checking a confidence interval without accounting for the stopping design can alter its coverage. A deterministic residual tolerance does not establish statistical accuracy. The stopping criterion should match the target and method.

If the required evidence cannot be obtained within budget, the answer remains provisional or unresolved. This is not a reason to discard the model's useful conditional results. It is a reason to state what is known, what remains uncertain and which computation would discriminate the remaining cases.

9.21 Proposed approximation records

Each approximation should record its target, retained family, measure if any, sampling or discretization rule, random seeds, support assumptions, error estimates, failed regions and method version. A run manifest supports reproduction; the reader-facing result should explain the consequences rather than expose every internal field.

Tests should include a small exact reference, a rare-event fixture, a persistent-parameter fixture, a boundary-sensitive fixture and an intentionally unsupported region. Comparison should use the same model and query. A fast method solving a simplified target is not a faster solution to the original target unless the difference is quantified.

The objective is controlled approximation, not universal exactness. The system can remain useful while refusing an unjustified probability or exact bound. Retaining the original family ensures that future evidence and better methods can revisit uncertainty that today's computation could not resolve.

Chapter 10

Decisions when probabilities and histories remain ambiguous

A simulator can describe possible consequences without selecting an action. A decision method needs an objective, a feasible policy class and a treatment of uncertainty. These are substantive choices. Worst-case liquidity, expected shareholder value and minimax regret answer different questions. No accounting identity selects among them. When uncertainty persists over time, the choice also determines how plans respond to new information.

10.1 Evaluation before optimization

For a fixed policy π, let $V(\pi, h, s)$ be a defined payoff or cost. It may be terminal cash, discounted owner distributions, service continuity or another quantity. Its units and horizon should be stated. Combining several objectives requires weights or constraints that should remain visible.

Evaluating a supplied policy is often the safest first task. It separates economic mechanics from policy selection. Once those mechanics are checked, optimization can search over an admissible class. A numerical optimizer should not be allowed to choose actions outside contractual rights or use information unavailable to the actor.

The distinction also helps interpret historical behavior. A policy that would maximize a modeled objective need not describe what management actually did. Simulation under that policy is a normative or counterfactual exercise unless behavioral evidence supports its use as a forecast.

10.2 Expected value under a declared measure

If a joint probability measure P is admitted, a policy can be evaluated by $\mathbb{E}_P[V(\pi, H, S)]$. The distribution must preserve dependence between history, shocks and signals. A posterior over report-compatible histories is one possible input, but its prior and likelihood assumptions remain part of the result.

If two histories have equal likelihood under all retained evidence, their posterior odds depend on prior odds. More observations of the same uninformative kind do not remove that dependence. The robust Bayesian analysis of Giacomini and Kitagawa (2021) makes this issue explicit for set-identified models. Its application here motivates sensitivity to unresolved prior choices, not an automatic probability for each journal.

A report can show policy rankings under several admitted priors. If every prior gives the same

ranking, that conclusion is more robust to the specified ambiguity. If rankings differ, a single average conceals a material decision sensitivity. The next useful evidence may be the fact that distinguishes those priors' consequences.

10.3 Worst-case evaluation

For an unweighted feasible family $\mathcal{U}$, a robust payoff is

$$W(\pi) = \inf_{u \in \mathcal{U}} V(\pi, u). \tag{10.1}$$

A maximin policy maximizes W. This criterion protects against the worst admitted case. Its conservatism depends on the family and objective. If the family includes implausible combinations introduced by a loose relaxation, the policy may respond to a company that cannot exist.

Robust feasibility is often more interpretable than robust optimization of a single scalar. For example, require every admitted history to maintain cash above zero, then compare expected or conditional costs among feasible policies. If no policy satisfies the requirement, the result should identify the conflicting cases or needed assumptions. It should not silently shrink the family until feasibility appears.

A worst-case result does not need probabilities, but it does need a defensible scope. An unlimited shock set can make every policy fail. That may reveal a badly posed decision problem rather than an economic conclusion. Bounds on shocks and actions require provenance and sensitivity analysis.

10.4 Regret compares actions within the same world

Regret measures the loss relative to the best feasible action for the same realized world. For a one-time action a, define

$$R(a, u) = \sup_{b \in A} V(b, u) - V(a, u). \tag{10.2}$$

Minimax regret chooses the action with the smallest worst regret. The comparator uses the same u, so it should not combine an optimistic world for one action with a pessimistic world for another.

Consider two actions and two states with payoffs $A = (10, 0)$ and $B = (6, 6)$. Maximin selects B because its minimum is 6 rather than zero. Statewise best payoffs are $(10, 6)$, so regrets are $A = (0, 6)$ and $B = (4, 0)$; minimax regret also selects B here. Other payoff tables can separate the criteria. Agreement in one example does not establish equivalence.

In sequential settings, the hindsight comparator may use information the actual actor did not have. That can be intentional in regret analysis, but it must be stated. A claim about implementable policy improvement should compare policies under the same information restrictions.

10.5 Multiple priors

A set $\mathcal{P}$ of probability measures can represent ambiguity about a stochastic model. A lower expected payoff is

$$\underline{V}(\pi) = \inf_{P \in \mathcal{P}} \mathbb{E}_P[V(\pi)]. \tag{10.3}$$

This differs from an unweighted family of deterministic worlds. A prior set can retain restrictions on event probabilities, dependence and learning. Replacing it by all distributions over its support generally enlarges ambiguity and can change the decision.

A singleton prior set reduces to ordinary expected value. A set containing every point mass over a finite family reduces lower expectation to the minimum payoff over that family. Intermediate sets express intermediate restrictions. The representation should preserve which restrictions came from evidence and which express preferences or modeling caution.

Updating each prior by Bayes' rule requires positive conditioning probabilities or a declared extension for null events. Removing priors solely because they yield inconvenient decisions changes the ambiguity model. A likelihood-based restriction can be justified under a specified method, but should not be confused with an arbitrary pruning heuristic.

10.6 Why dynamic consistency matters

A policy selected today may cease to be preferred tomorrow after information arrives. Some changes are rational responses to information. Others arise because the decision criterion evaluates future uncertainty differently from the original plan. To model behavior, one must specify whether the actor commits to the original policy or reoptimizes.

Epstein and Schneider (2002, Definition 3.1 and Theorem 3.2) connects recursive multiple-priors preferences, dynamic consistency and rectangular prior sets under its stated axioms and information filtration. Rectangularity permits specified combinations of one-step marginals and conditional distributions. It is a substantive restriction on the prior family relative to information, not a harmless implementation trick.

The source's theorem is not a statement that every economically natural uncertainty set is rectangular. Nor does it authorize changing a fixed unknown contract at each date. Applying a recursive worst-case operator without examining the allowed cross-time combinations can solve a different decision problem.

10.7 A concrete pasting counterexample

Consider two dates of losses and a hidden type θ that remains fixed. If $\theta = A$, losses are $(0, 10)$. If $\theta = B$, they are $(10, 0)$. The total loss is 10 in either type. Thus the worst total loss over the original two-world family is 10.

If a stagewise procedure independently selects the larger loss at each date, it constructs $(10, 10)$ and reports total loss 20. That path is absent from the original family. The procedure has allowed the adversary to switch type across dates. It computes a worst case over an enlarged family.

Calling the enlarged result conservative does not make it equal to the original target. It may be useful as an outer bound, but the target must be labeled. Retaining θ in the state prevents the unintended switch. This simple example is the deterministic analogue of the dependence issues that rectangularity addresses for prior sets.

10.8 Rectangular enlargement changes the model

A rectangular hull can facilitate recursive evaluation by admitting combinations of conditionals that were not jointly present in the original prior set. The source explicitly discusses enlargement

and its relation to the information structure. In an economic application, added combinations may represent an adversary whose choices adapt over time.

If the original ambiguity concerns one fixed but unknown loan contract, allowing adaptive adversarial changes to that contract is usually a different physical model. One can retain the contract type as part of the state and formulate conditional uncertainty appropriately. If a robust policy intentionally protects against changing terms, those changes should be modeled as amendments, refinancing or legal events.

AccDSL should therefore record whether a robust method uses the original joint ambiguity set, a rectangular enlargement or a commitment formulation. The result should compare the enlargement's effect on important outputs where feasible. A convenient Bellman recursion does not justify hiding that change.

10.9 Commitment and reoptimization

A commitment policy is selected at the initial date as a full rule for future information histories. A reoptimizing actor selects a new continuation policy at each date. These can differ under nonrectangular ambiguity or changing preferences. The simulator should not alternate between them silently.

For a treasury policy approved by a board, some actions may be committed while others remain discretionary. A hedge may be binding after execution; a future dividend target may be revisable. The feasible policy class should represent these commitments. Reoptimizing an already executed hedge as though it never occurred gives the actor an unavailable option.

A useful analysis can compare commitment and reoptimization explicitly. Differences then reveal the value or cost of flexibility under the chosen uncertainty model. They should not be attributed solely to forecasting skill.

10.10 Distributional ambiguity and historical ambiguity

Uncertainty about which debt contract exists differs from uncertainty about next year's demand distribution. Both can enter a decision, but they should retain separate origins. A contract family may be constrained by notes; a demand ambiguity set may be estimated from data or chosen for stress robustness.

Combining them independently can be unjustified if financing structure and demand exposure are related. The joint ambiguity set should preserve known dependence. Conversely, imposing a relation merely because it makes optimization easier can exclude valid histories. The default should be explicit uncertainty, not an invented joint law.

A layered result can report policy performance conditional on contract region and across demand models. This makes it possible to ask whether better contract evidence or better demand estimation would change the decision. A single blended uncertainty score obscures that distinction.

10.11 Risk constraints and probability thresholds

A chance constraint such as $P(C_t < 0) \leq \epsilon$ requires a probability model and a specified time event. A bound on failure at each date differs from a bound on failure at any date. Summing marginal probabilities gives a union bound, not generally the exact path probability.

If failure events are strongly dependent, treating them as independent can distort the probability of ever breaching. Persistent contract uncertainty is one source of this dependence. A company with a floating loan may remain vulnerable in every high-rate period; redrawing the type can alter duration and clustering.

A robust chance constraint requires the probability bound under every measure in an admitted set. It is stronger than a constraint under an averaged measure. Numerical estimation must account for sampling error near the threshold. A point estimate just below the limit is not sufficient evidence of compliance.

10.12 Value of information under ambiguity

A proposed disclosure partitions the history family. Under a probability model, expected information value compares optimized expected payoffs before and after the signal, accounting for its cost and timing. Under maximin or regret, the criterion changes. Information can matter because it enables different feasible actions in different regions.

A signal that arrives after a financing commitment may have little operational value even if it greatly narrows a posterior. A signal about an irrelevant parameter may reduce statistical uncertainty without changing decisions. The useful target is the decision consequence, not entropy reduction alone.

Without defensible probabilities, one can report the range of improvements across possible signal outcomes and criteria. A guaranteed improvement may be zero even when some outcomes are valuable. This is a meaningful conclusion and avoids assigning probabilities merely to produce an expected number.

10.13 Selecting a method for a user question

For disclosure explanation, preserve possible and certain answers. For a supplied stress, evaluate conditional mechanics. For planning under an admitted distribution, estimate expected values and relevant risks. For a safety requirement across compatible histories, use robust feasibility and bounds. For competing actions without probabilities, consider dominance and explicitly chosen robust or regret criteria.

Dominance is particularly useful: if one feasible policy performs at least as well in every admitted world and strictly better in some, it can be preferred under many monotone criteria. If policies cross, the choice depends on preferences or probability assumptions. The simulator should show that dependence rather than invent a universal ranking.

A method-selection interface should ask for a criterion only when the user has requested a decision that the evidence alone cannot determine. It can still complete mechanical comparisons and identify dominant or infeasible options. This preserves progress while keeping value judgments explicit.

10.14 Implementation and validation

A decision record should specify actor, objective, information, feasible actions, uncertainty set or measure, commitment convention, horizon and terminal treatment. Terminal assumptions matter: liquidating assets at book value at the horizon can create an artificial reward and change earlier decisions.

Small exact finite examples should compare direct policy enumeration with recursive methods. Include the fixed-type pasting counterexample, an information-revelation example, a capacity-constrained financing example and a case where expected value and robust choice disagree. These fixtures discriminate semantic errors before expensive numerical optimization.

Passing them establishes the specified criterion and timing mechanics. It does not establish that the criterion describes management or that the uncertainty set is empirically adequate. Those remain separate assumptions or research questions. The system can support several methods without declaring one a universal default.

Chapter 11

Solving the economic equations that the model actually states

A correct financial representation can still be evaluated by the wrong numerical method. Conversely, a numerical method can converge accurately to equations that describe the wrong financing problem. The first obligation is to state the target equations, their timing and their feasible domain. Only then does it make sense to choose a solver or train a neural approximation. The risky-debt convention discussed earlier illustrates why a small residual is insufficient: an inconsistent cash convention can be solved very accurately.

11.1 Separate identities, conditions and approximations

Accounting identities reconcile stocks and flows. Contract equations determine obligations from terms. Behavioral conditions describe choices. Equilibrium conditions connect counterparties and prices. Numerical approximations evaluate these relations with finite resources. They should not be treated as interchangeable equations in an undifferentiated loss.

An accounting identity can often be enforced exactly by construction. A lender pricing equation may involve an expectation and an endogenous default rule. A management optimum may require inequalities and boundary conditions. A reporting standard may require a categorical treatment selected by facts. Each type needs a suitable validation method.

A residual vector can combine them for computation, but the result should still report each component in meaningful units. A small average residual can hide one economically decisive violation, especially when large accounting amounts dominate the scale. A pricing residual of one currency unit and a share-count residual of one share are not directly comparable without a declared normalization.

11.2 The Bellman problem

For a finite-horizon shareholder problem, let x be state, a a feasible action, $d_t(x, a)$ the current net distribution to the modeled owners, and β a discount factor under the chosen valuation formulation. A Bellman equation is

$$V_t(x) = \sup_{a \in A_t(x)} \{d_t(x, a) + \beta \mathbb{E}[V_{t+1}(X_{t+1}) \mid x, a]\} . \tag{11.1}$$

The expectation uses the declared transition law and information. If an action depends on private information or a belief state, the state must be defined accordingly. The terminal value V_T is an assumption with economic consequences, not merely a numerical boundary condition.

A liquidation-at-book-value terminal rule can make asset accumulation artificially attractive. A zero terminal value can induce premature payouts or abandonment. A continuation approximation can be useful, but its accuracy and domain should be checked. Changing terminal treatment while comparing solvers changes the target.

In a stylized limited-liability problem, continuation value may be compared with default or exit value. The choice and its consequences must be specified. A maximum operator alone does not provide a legal restructuring model or creditor recovery schedule.

11.3 Pricing and value must agree on policy

Suppose debt price $P(x, F)$ depends on repayment under future policy π. The policy depends on funding proceeds and therefore on P. A joint solution must satisfy both the policy-value problem and lender pricing for that same policy.

One can write residuals $R_V(V, \pi, P) = 0$ and $R_P(V, \pi, P) = 0$, supplemented by feasibility and optimality conditions. Solving R_V with a fixed old price and R_P with a different policy leaves a mismatch. Alternating iteration can be a candidate algorithm, but convergence and target consistency need checking.

The retained Duarte and Fonseca (2026) working paper uses a lagged value network in part of its pricing training and a subsequent local grid refinement. These are specific numerical choices in its model. They do not supply a general convergence or global-optimality guarantee for AccDSL. Transferring them requires a target-specific protocol and matched timing, recovery, taxes and state definitions.

11.4 A residual can target the wrong expectation

Let $r(x, \varepsilon; \theta)$ be a random equation contribution with parameterized functions θ. Suppose the desired residual is the conditional expectation $m(x; \theta) = \mathbb{E}[r \mid x]$. Squared expected residual is $m(x; \theta)^2$. Squaring a single random contribution and averaging gives

$$\mathbb{E}[r^2 \mid x] = m(x; \theta)^2 + \mathrm{Var}(r \mid x). \tag{11.2}$$

The second objective adds a variance term. If that variance depends on θ, minimizing the two objectives can select different functions. This is a mathematical target mismatch, not merely noisy estimation of the same loss.

With conditionally independent draws $\varepsilon_1, \varepsilon_2$ under the same x,

$$\mathbb{E}[r(x, \varepsilon_1; \theta) r(x, \varepsilon_2; \theta) \mid x] = m(x; \theta)^2. \tag{11.3}$$

Independence is conditional on the common state. Reusing the same draw restores the unwanted variance term. Maliar et al. (2021, Definitions 2.5–2.9 and Algorithm 1) discusses objective formulations for dynamic economic equations, including this distinction. The local derivation explains why the sampling convention must match the intended residual.

11.5 A small objective counterexample

Let $r(\theta, \varepsilon) = \theta - 1 + \theta\varepsilon$, with $\mathbb{E}[\varepsilon] = 0$ and variance one. Then $m(\theta) = \theta - 1$, so the desired squared residual is $(\theta - 1)^2$, minimized at one. The single-draw squared objective is

$$(\theta - 1)^2 + \theta^2 = 2\theta^2 - 2\theta + 1, \tag{11.4}$$

minimized at one half. At that point the expected equation residual is negative one half. The optimizer has correctly minimized the wrong objective.

This exact example is a useful unit test for residual construction. It also shows why training loss alone cannot certify economic equations. Validation should directly evaluate the target conditional residual using independent quadrature or sufficiently controlled sampling.

11.6 Inequalities and complementary slackness

A borrowing or cash constraint can bind. Interior first-order conditions do not apply unchanged at a boundary. Consider minimizing a differentiable cost $f(a)$ subject to $a \geq 0$. With multiplier $\lambda \geq 0$, the Karush–Kuhn–Tucker conditions under appropriate regularity include

$$f'(a) - \lambda = 0, \qquad a \geq 0, \qquad \lambda \geq 0, \qquad a\lambda = 0. \tag{11.5}$$

If $a > 0$, the multiplier is zero and the interior derivative vanishes. If $a = 0$, the derivative need not vanish. A residual enforcing only $f'(a) = 0$ can demand an infeasible negative action.

For $f(a) = (a + 1)^2$, the constrained minimum is $a = 0$, with derivative 2 and multiplier 2. An interior solver seeks $a = -1$, which violates the constraint. Clipping its output to zero may recover the action in this example, but does not establish a general constrained solution or correct derivatives through the clipping operation.

Complementarity formulations, active-set methods and direct constrained optimization are possible approaches. Their conditions are necessary or sufficient only under their stated assumptions. Nonconvex financing problems may have local optima and discrete choices; a small KKT residual does not prove global optimality.

11.7 Discrete choices and nondifferentiability

Default, conversion, refinancing and investment shutdown can be discrete. Their value functions may have kinks or discontinuities. Differentiating a smooth surrogate through these decisions can optimize a smoothed problem rather than the original one.

A smooth approximation may be useful as a warm start. Its smoothing parameter changes the target and should be varied, with final policies evaluated under the original event rules. A soft default indicator that pays a fractional blend of full repayment and recovery does not describe an actual realized default event unless explicitly interpreted as an expectation under a probabilistic model.

Near a threshold, classification error can dominate value error. A network can fit values closely while moving the default boundary enough to affect debt pricing. Validation should include state pairs on both sides of the boundary and pricing consequences under the resulting policy.

11.8 Partial and total derivatives

A financing choice can affect a price directly and through future policy. If $P(a) = P(a, \pi(a))$, the total derivative includes both paths:

$$\frac{dP}{da} = \frac{\partial P}{\partial a} + \frac{\partial P}{\partial \pi}\frac{d\pi}{da}. \tag{11.6}$$

Holding a learned policy fixed computes a partial derivative. It may be the intended derivative in an alternating algorithm, but it is not automatically the total derivative of the equilibrium price.

Automatic differentiation differentiates the implemented computation. It does not decide which economic dependencies should be present. Stop-gradient operations, target networks and detached iterations can deliberately omit paths. Their use should be justified relative to the algorithm and validated against the final target.

An implicit derivative can be derived when a smooth equilibrium equation has a nonsingular Jacobian, but default boundaries and multiple equilibria can violate those conditions. Numerical finite differences can diagnose implementation mistakes away from such boundaries; they do not resolve the economic choice among equilibria.

11.9 Scaling and units

Residuals measured in millions of currency, fractions, shares and utility can differ by orders of magnitude. A loss that sums their squares without scaling effectively assigns arbitrary importance based on units. Changing currency units from dollars to millions can change the trained solution if the formulation is not invariant.

Normalization should be documented and checked. One can scale a cash residual by a meaningful transaction or balance scale, a share residual by outstanding units, and a probability residual by its natural range. Zero or near-zero denominators need explicit treatment. An adaptive scale can itself alter the objective over training.

A unit-rescaling test is useful: transform all monetary inputs consistently and confirm that the implied economic policy and normalized diagnostics transform appropriately. Failure reveals a numerical or modeling dependence on representation rather than economics.

11.10 Training distribution and the queried domain

A learned solver is trained on selected states. If training emphasizes typical solvent states, it may be unreliable near liquidity crises, maturity walls or default. Uniform sampling in raw coordinates is not automatically balanced in economic importance. A high-dimensional box may contain mostly infeasible states.

The training protocol should identify feasible-state generation, boundary coverage, tails, heldout regions and distribution shift. Data from simulated policies can omit states reached by a different policy or scenario. Updating the policy can therefore change the relevant state distribution.

No architecture, learning rate or loss weighting should become a default merely because it worked on another model. Transfer can provide a baseline or warm start. A substantially different target needs its own capacity, optimization, budget and validation protocol. This manuscript proposes no promoted neural default without that evidence.

11.11 Causal generators and feasible policy transfer

Acciaio, Eckstein and Hou construct a time-causal variational autoencoder whose encoder and decoder at date t use only input coordinates available by t. Their retained preprint bounds a *directional* causal transport discrepancy by reconstruction error and a term involving latent relative entropy (Acciaio et al., 2024, Lemma 2.4, Theorem 2.5 and Corollary 2.6). The relative-entropy step requires the reconstructed and generated distributions to have the stated bounded support. Normalizing the observed sample does not establish that a learned decoder satisfies that support restriction for every generated input. A finite training estimate is also different from a population bound.

There is a further issue in transferring the control result. Theorem 3.1 of the retained version states its comparison for a closed convex subset $\mathcal{H}$ of adapted strategies. Its proof takes a policy G on the second process and constructs $\widetilde{G}(x) = \mathbb{E}[G(Y) \mid X = x]$. Causality establishes adaptedness, but adaptedness alone does not establish $\widetilde{G} \in \mathcal{H}$. Here is a direct counterexample to that stated generality. There is one date, $\mu = \delta_1$, $\nu = \delta_0$, and

$$\mathcal{H} = \{h\}, \qquad h(x) = \min(1, \max(0, x)), \qquad Q(x, a) = a. \tag{11.7}$$

The singleton class is closed and convex, and Q is convex in a and uniformly zero-Lipschitz in x. Nevertheless, $V(\mu) - V(\nu) = 1$, whereas the proposed right-hand side with $L = 0$ is zero. The transferred action at $x = 1$ is zero, outside the singleton policy class. This disproves the unrestricted policy-class statement in that retained version. It does not refute the argument when the feasible class is explicitly closed under the conditional transfer. The published edition has not been matched equation by equation, so this is not asserted to be a journal erratum.

For AccDSL, state-dependent borrowing limits, integer issuance sizes and contractual exercise restrictions make closure substantive. A repaired transfer argument must verify feasibility of the mapped policy, or use a method that retains the original feasible actions. The one-sided value comparison is not itself a bound on the performance of the same learned policy after deployment. The inspected paper-linked code has forward LSTM encoder/decoder paths, but also external conditioning variables; those variables require their own availability dates. Its selected conditional decoder uses ReLU and a final linear layer, so the architecture does not enforce the theorem's unit-ball support. These observations identify additional obligations before adoption; no training result is inferred from source-code inspection.

11.12 Controlled transition laws and off-policy support

Saldı and Yüksel distinguish weak and strong kernel-embedding topologies and give a square-wave counterexample to their equivalence (Saldı and Yüksel, 2025, Theorem 4.1 and proof). Let Y be uniform on $[0, 1]$. On alternating half-cells of a fine partition, make a binary successor U deterministically 1 or 0. As the partition becomes finer, smooth integrated tests can approach those of a fair binary kernel. At every particular input, however, the conditional law remains a point mass. For the Kronecker kernel on $\{0, 1\}$, its squared embedding discrepancy from the fair law is $1/2$. The local exact calculation checks this discrepancy and the vanishing error for a linear test, rather than claiming that a few tests prove weak convergence.

Their discounted-control application uses bounded continuous costs, finite actions, continuous transition kernels and a common dominating measure, including for the initial distribution. One route requires a strong essential-supremum condition; the weaker route adds bounded equicontinuous densities (Saldı and Yüksel, 2025, Assumptions 2–4, equation (11), Theorems 5.1–5.2). A default discontinuity, hidden creditor information or an action-dependent financing constraint therefore needs a separate argument. An equally important problem is support: models that

agree under the historically chosen action may disagree under a new action. With action losses $(1, 100)$ in the real model and $(1, 0)$ in an approximate model, historical use of the first action gives identical observations. Optimizing the approximation chooses the second action and incurs regret 99. Agreement under the historical policy alone cannot establish a policy-uniform approximation claim.

The original robustness result used in that application makes another condition explicit. Kara and Yüksel distinguish continuity of optimal values from robustness of the policies attaining them. Their discounted-control argument uses continuous kernel convergence, bounded continuous costs and compact actions. Policy transfer additionally requires each approximate policy to be optimal from every initial state, as when it satisfies the discounted Bellman equation throughout the state space (Kara and Yüksel, 2020, Assumption 1.1, Theorems 4.2–4.4 and Appendix A.5). An optimizer trained only from a historical starting state can leave its actions elsewhere arbitrary; the true transition can reach those neglected states. The paper's counterexample makes precisely this distinction. Its finite-state learning result also assumes continuing coverage of every state–action pair (Kara and Yüksel, 2020, Corollary 5.1). Financial statements generated by one management policy do not by themselves provide that coverage.

11.13 Classical reference methods

Small discretized problems can be solved by direct enumeration, grid search, backward induction or carefully controlled quadrature. These methods can be slow, but they provide transparent references for testing learned or accelerated methods. The reference should be strong enough to answer the same target question.

A weak coarse grid can make a surrogate appear superior merely because the comparator is underresolved. A fair comparison includes convergence or refinement evidence for the reference and matched terminal conditions, constraints and state domains. Comparing different models is not a solver benchmark.

Exact rational examples are especially useful for cash identities and simple pricing. They are not substitutes for numerical validation of high-dimensional expectations, but they can expose sign, timing and unit errors before expensive training. The positive-recovery debt example belongs in this first layer.

11.14 Multiple solutions and selection

A nonlinear system can have no solution, one solution or several. A solver initialized differently may converge to different roots. Reporting the first root as the economic equilibrium requires a selection argument or an explicit conditional interpretation.

A financing market can have feedback in which perceived distress raises spreads and higher spreads increase distress. A stylized fixed point may admit low- and high-spread solutions. Stability under a chosen iteration is a numerical property, not necessarily an economic equilibrium-selection criterion.

The method should search or bound multiplicity where material, report dependence on initialization and preserve unresolved alternatives. If only one local root is found, describe it as such. A failed solver does not establish that no financing equilibrium exists.

11.15 Downstream validation

A learned value function is useful only if its errors are acceptable for the decisions and reports it drives. Validation should evaluate feasible actions, debt price consistency, default classification, cash paths, ownership changes and accounting outputs under heldout scenarios. Low prediction error alone is insufficient.

A pricing error can alter proceeds, which changes investment, which changes default and feeds back into price. A one-step residual screen may miss that loop. Multi-period policy evaluation under an independent method is needed for a claim about the coupled computation.

Stochastic comparisons require uncertainty evidence. One seed or a few short runs can nominate a candidate but cannot establish superiority from descriptive averages alone. Hard failures, descriptive differences and statistically supported rankings should remain separate. This is particularly important when rare defaults dominate economic conclusions.

11.16 A proposed solver interface

The solver receives a typed economic problem: state domain, timing, objectives, constraints, transition and observation laws, price equations and requested accuracy. It returns functions or policies together with target residuals, feasibility diagnostics, unresolved regions and reproducibility information.

A solver should not mutate the economic model to obtain convergence without recording the change. Smoothing, truncation, penalty relaxation and altered terminal values define modified problems. Their relation to the original target must be evaluated. A successful run on a modified problem is not an implicit solution of the original.

AccDSL can support several solvers behind the same problem definition. The interface should preserve method-specific limitations rather than force every solver to claim identical guarantees. A certified finite result, a local numerical solution and a learned approximation can all be useful when labeled accurately.

11.17 The smallest justified numerical program

Begin with a one-period debt model whose face, proceeds and recovery are distinct. Verify cash-leg pricing and limited-liability conventions exactly. Add a finite-state dynamic model with explicit feasible policies and compare enumeration with backward induction. Then add borrowing limits, equity issuance and default transitions one at a time.

Only after these targets are stable should a learned solver be compared on the same equations. The comparison needs a predeclared primary criterion, veto diagnostics, budget ladder and heldout downstream checks. Training settings remain hypotheses until supported on this target.

This staged approach does not postpone the difficult economic questions. It exposes them early in forms that can be checked. The purpose of deep learning is to accelerate a well-defined computation, not to make an ambiguous financing model appear resolved through a low loss value.

Chapter 12

Validation, adversarial cases, and the limits of a simulation

A credible simulator should fail intelligibly when its premises are insufficient. It should also be tested against cases where a plausible implementation gives a confident but wrong answer. The purpose of validation is not to accumulate a large count of passing assertions. It is to distinguish errors in evidence, representation, economics, accounting, information, numerical calculation and interpretation. Each class requires a different test and a different repair.

12.1 What is being validated

There are at least three separate questions. Does the implementation compute the specified model? Is the specified model mathematically and economically coherent? Does the model usefully describe the company or decision under study? A passing unit test can answer part of the first question. It cannot answer the third by itself.

For example, a simulator can correctly implement unlimited borrowing at a fixed rate. That implementation may pass every accounting reconciliation. It remains unsuitable for a claim about refinancing risk unless unlimited borrowing is an appropriate premise. Conversely, a realistic financing model can be implemented with the wrong proceeds convention. The two failures should not be blurred into a generic low accuracy score.

A validation record should state the claimed target, quantity actually computed, comparison method and interpretation. If the target is a worst-case cash bound but the method reports a sampled minimum, the mismatch is explicit. The sampled result may still be useful as a witness; it should not inherit the stronger claim.

12.2 An extensible failure taxonomy

No finite catalogue can prove that every possible simulation failure has been anticipated. New contracts, standards, behaviors and software interactions can create new cases. A systematic taxonomy is nevertheless valuable because it exposes the dimensions along which failures arise and guides new tests.

The taxonomy used here follows the calculation: evidence admission, historical reconstruction, economic state, actor information, scenario definition, feasible behavior, financing, accounting, aggregation, approximation and reporting. Cross-cutting dimensions include time, units, entity,

dependence, precision and provenance. A failure can occupy several categories. A maturity error, for example, affects evidence, timing, financing and liquidity.

Coverage should be expressed by mechanisms and boundaries, not only by scenario count. One thousand random rate paths do not test a missing conversion clause. A single carefully designed positive-recovery case can expose a proceeds-versus-face error that many no-default paths miss.

12.3 Evidence admission failures

A value can be extracted from the wrong row, assigned the wrong sign, scaled by the wrong unit or associated with the wrong period. A note can be omitted because it is a scanned image. A cross-reference can point to an incorporated document outside the downloaded filing. An amended filing can be mistaken for the original.

Tests should include duplicate labels with different contexts, negative values in parentheses, mixed currencies, overlapping periods and rounded totals. The expected result should preserve uncertainty where the source is ambiguous. A parser that always returns a number may look complete while inventing precision.

A source coverage report should distinguish retrieved, parsed, interpreted and modeled content. These are different achievements. Retrieving every page does not establish semantic completeness. A passage that cannot be formalized should remain linked to the model as unresolved material rather than disappearing from the scope claim.

12.4 Reconstruction failures

Historical balances can reconcile while hidden gross flows are wrong. A net borrowing change can conceal large issuance and repayment. A net share change can conceal issuance and repurchase at different prices. An aggregate receivable can conceal concentration. Tests should use pairs of histories with the same observed totals and different future responses.

The family should include both histories when evidence permits them. A selected witness should be labeled as selected. If a regularizer removes one, the result should identify the selection assumption and its consequences. A duplicate encoding of one history must not alter probabilities or set membership.

Infeasibility tests should distinguish source contradiction from solver failure. Deliberately conflicting exact observations should produce an interpretable conflict. Deliberately difficult but feasible constraints should not be reported impossible merely because the budget expires. Unsupported instruments should remain outside the solved profile with their economic significance visible.

12.5 State sufficiency failures

A state can omit a quantity that matters later: accrued interest, remaining loss carryforward, asset age, covenant cure clock, collateral ownership, unpaid dividends or option exercise status. Current statements may still match. A multi-period replay reveals the omission.

A useful test starts from two histories with equal proposed compressed state but different omitted quantities. Apply the same future policy and shocks. If outcomes differ, the compression is insufficient for that query. If observations differ and later decisions change, the information component is insufficient even when immediate cash agrees.

The repair is to retain the missing state or restrict the compression's claimed domain. Adding a balancing adjustment after the divergence hides the cause. A state variable should be justified by future dependence, not by its convenience in a data schema.

12.6 Information failures

Look-ahead can enter through restated data, future prices, a sampled shock passed to a policy, or a compressed state containing private information the actor lacks. It can also enter through training labels if a purported real-time decision uses features calculated from the full future path.

A nonanticipativity test changes future unobserved data while holding current information fixed and checks current actions. A publication test shifts a disclosure after the decision cutoff and checks that it no longer influences that decision. A private-information test permits an informed manager to respond to a known contract while preventing a public analyst from treating it as observed.

These tests must inspect policy inputs and execution, not merely documentation. An interface can claim to hide future information while a cached feature leaks it. Provenance should connect each decision input to an information rule and availability time.

12.7 Scenario-definition failures

A shock can be applied twice, applied to the wrong rate, or applied for the wrong fraction of a year. A currency depreciation can be inverted. A rate shock can affect a fixed coupon that should not reset. A scenario can change both demand and price while being described as a pure volume shock.

Metamorphic tests are useful when exact full outcomes are hard to specify. A zero shock should reproduce the baseline under unchanged policies and information. Splitting one identical contract into two proportional pieces should preserve aggregate cash when no nonlinear clause is introduced. Changing currency units consistently should scale monetary outputs without changing dimensionless decisions.

These invariances have premises. Contract splitting can change fees, concentration limits or legal rights. A test should enforce invariance only where the model says it holds. Otherwise the test itself becomes an incorrect economic requirement.

12.8 Financing failures

Face can be mistaken for proceeds, net cash for gross cash, coupon for effective interest, or book debt for market value. Refinancing can be assumed available without an offer. A fee can be charged twice. Recovery can be distributed beyond available assets. A conversion can coincide with an unlinked redemption.

The positive-recovery diagnostic in Chapter 5 is a primary fixture because it distinguishes two conventions that agree in simpler cases. Additional tests should use discounted issuance, delayed settlement, binding capacity, failed subscription, priority and recovery delay.

A no-funding result is economically meaningful only after the method has correctly evaluated the admitted offers and actions. A numerical pricing failure should be reported separately. If

the model omits a financing channel, the conclusion is conditional on that omission; it is not proof that the real company has no financing option.

12.9 Accounting failures

A journal can balance while mapping borrowing to revenue, treating a valuation change as cash, or recognizing a transaction twice. A reporting aggregation can combine different perimeters or bases. A cash-flow label from an economic model can be mapped incorrectly to a standard's category.

Tests need event-specific expected postings under explicitly supplied rules. They should reconcile physical quantities, cash, obligations and accounting balances. Where authority or facts are unresolved, the expected result may be a refusal or conditional treatment, not a forced entry.

A standard-specific test establishes behavior under the retained authority version and admitted facts. It does not establish universal compliance across jurisdictions and dates. The rule package should preserve applicability conditions, source passages and unresolved interpretations.

12.10 Aggregation and dependence failures

Independent marginal ranges can create impossible combinations. Stagewise worst cases can switch persistent types. A mean state can cross a nonlinear threshold differently from every original state. A scenario reduction can remove the path causing a maturity shortfall.

Tests should retain a full small reference family and compare exact joint paths with the compressed representation. Check outputs, observations, feasible actions and policy values, not just means. Include a case where fixed-action sequences agree but adaptive decisions differ.

A failed exact aggregation test can still motivate an approximate method with bounds. It should not be repaired by weakening the test until the original claim passes. State whether the target changed and quantify the consequence for the decision.

12.11 Approximation failures

Particles can lose a persistent contract type. Importance sampling can omit support. A surrogate can extrapolate beyond training. A numerical tolerance can classify a covenant incorrectly. A Monte Carlo sample can contain no rare defaults while the true modeled probability is positive.

A small exact reference provides a baseline. Coverage checks verify that rare or boundary regions are visited or bounded. Replications and uncertainty intervals assess stochastic estimates. Distinct seeds are not independent evidence if every run shares the same misspecified transition or omitted mechanism.

A computational diagnostic should be assigned a role before interpretation. It may veto validity, explain behavior, trigger repair or support promotion under a declared criterion. A low loss can nominate a candidate; it does not automatically promote it to a default solver.

12.12 Terminal and horizon failures

A model can postpone default beyond the horizon, refinance indefinitely without repayment, or liquidate assets at an unrealistic terminal value. Extending the horizon can reveal that an

apparently safe policy merely shifts the problem.

Tests should compare adjacent horizons under consistent continuation assumptions and inspect obligations remaining at the end. A terminal balance sheet should retain outstanding debt, contingent rights and future commitments. A clean closing cash number is not a complete account of solvency.

The terminal rule should be part of every policy comparison. Using a favorable terminal value for one method and zero for another is an unfair comparison. A finite-horizon model can still be useful, but its conclusion should state what remains outside the horizon.

12.13 Interaction failures

Individually correct modules can compose incorrectly. A debt amendment can change future interest while a cached tax calculation uses the old schedule. A share conversion can update ownership but leave diluted earnings per share on the old count. A group transfer can be recorded in both entities but omitted from elimination.

Composition tests should follow complete event sequences across modules. They should check shared identities, dates, units and versions. A module's standalone test suite cannot certify every interaction. The most valuable sequences are those crossing boundaries: refinancing plus fees plus tax, conversion plus ownership plus reporting, or default plus collateral plus recovery.

A failed interaction should be localized to its violated relation. Re-running a broad test suite without understanding the failure can obscure whether the model, adapter or shared state is wrong. The repair should preserve unaffected evidence and explicitly update dependent results.

12.14 A hierarchy of reference calculations

Start with hand-reconstructible exact cases using rational arithmetic. Then use finite-state enumeration for learning and policy tests. Add independent numerical quadrature or refined grids for continuous small models. Only then compare scalable approximate methods on the same target.

A reference calculation must itself be audited. Coarse grids, truncation and arbitrary terminal values can bias it. Agreement between two implementations sharing the same mistake is not strong validation. Independent derivations and different computational routes are valuable where the economic conclusion is important.

The hierarchy is not a requirement to solve every real company exactly. It establishes local semantics and numerical behavior before applying methods to larger unresolved settings. The large application retains its own evidence and domain limitations.

12.15 Empirical validation

Forecast validation requires a defined prediction target, information cutoff, evaluation period and benchmark. A statement reconstruction evaluated against the same statements it was constrained to match is not an out-of-sample forecast test. A restated target can differ from the information that was observable at prediction time.

Probabilistic forecasts need scoring methods appropriate to their target. The proper-scoring literature represented in the inherited bibliography by Gneiting and Raftery (2007) motivates

evaluating distributions rather than only point means. Applicability still requires a correctly defined observed outcome and a forecast distribution produced without leakage.

Set-valued predictions need measures of coverage and informativeness under a declared evaluation design. A very broad set can cover every outcome while offering little decision value. Narrowness without coverage is also inadequate. The tradeoff should be stated before comparing methods.

12.16 No empirical result is claimed here

The examples in this part are exact or analytic diagnostics of proposed semantics. They are not fitted company forecasts, evidence of profitability or rankings of simulation methods. No neural training campaign or stochastic benchmark is performed as part of this writing task.

This distinction keeps the next research step clear. A future empirical study must choose companies, evidence cutoffs, scenarios, baselines, outcomes and uncertainty analysis. It should preserve the question separately from the implementation checklist and state what result would invalidate the harness rather than merely reject a candidate method.

A negative result can reveal a bad implementation, insufficient tuning, an inadequate observation model or a weak economic mechanism. These explanations require different follow-up. Treating every failure as rejection of the entire direction would waste the diagnostic information.

12.17 Failure behavior is part of the specification

The system should return different statuses for unsupported instrument, contradictory evidence, undecided feasibility, infeasible action, economic default, numerical nonconvergence and completed approximation. A result can contain several statuses across regions.

Unsuccessful regions must not disappear from aggregate results. Renormalizing probabilities after dropping numerical failures changes the distribution, often favorably if difficult cases are distressed. A report should state unresolved probability mass when a measure exists, or unresolved feasible regions otherwise.

A completed run also needs a trace to its assumptions and evidence. Reproducibility requires exact versions of source views, rules, methods and numerical settings. A content hash identifies a file; it does not prove the file's correctness. Both provenance and substantive validation are needed.

12.18 From catalogue to implementation work

The proposed failure catalogue should be machine-readable enough to link each mechanism to a fixture, expected outcome and relevant chapter, while the manuscript explains why the failure matters. New cases can be added without claiming that the catalogue is exhaustive.

Implementation can proceed by vertical slices. A slice reconstructs a bounded history family, applies a scenario and policy, generates events and journals, produces statements and reports unresolved regions. This is more discriminating than implementing isolated formula modules without a source-to-result case.

The first slice should preserve ambiguous debt type, issue proceeds distinct from face, a binding financing limit and actor information. Later slices add ownership rights, group transfer constraints, persistent tax state and approximate inference. Each slice should retain the wider

open scope rather than redefine the mission to match what is easiest to execute.

12.19 What successful completion would mean

For this document, success means an explicit and traceable argument, checked illustrative calculations, preserved prior work, inspected source support and a clear account of proposed methods and their failure cases. It does not mean that all methods are implemented or that the product is ready for unrestricted use.

For the simulator, success is query-specific. It means the implementation computes the declared target within its stated evidence and numerical guarantees, preserves unsupported regions, and explains the economic dependence of the result. Whole-company applicability requires coverage of the actual company's material contracts, events and reporting basis.

The following worked chapters apply these principles to complete small problems. Their purpose is to make the design reconstructible: the reader can follow observations through competing histories, decisions, financing, journals and outcomes, and see exactly where a different assumption changes the answer.

Chapter 13

A complete small company: from reports to divergent stress outcomes

The earlier examples isolate individual mechanisms. This chapter joins them in one synthetic company so that every projected balance can be traced to an event. The company is deliberately simple: one currency, one legal entity, no tax, no dividends, no acquisitions and no other comprehensive income. These exclusions make the arithmetic inspectable. They do not describe a complete real-company model. The point that carries over is the need to preserve compatible economic histories and their joint statement consequences.

13.1 The opening statement

At the beginning of the historical year, the company has cash 40, receivables 30, inventory 50 and property carrying amount 80. Its liabilities are debt 100 and trade payables 40. Equity is 60. All amounts use one currency unit.

Assets	Amount	Liabilities and equity	Amount
Cash	40	Debt	100
Receivables	30	Trade payables	40
Inventory	50	Equity	60
Property	80		
Total	200	Total	200

The equality is an accounting starting point. It does not identify customers, inventory units, asset vintages or debt terms. We initially admit the balances as exact to focus on hidden composition. A later section reintroduces rounding and source uncertainty.

The company operates during the year and reports sales 120, cost of sales 70, cash operating expenses 25, interest expense 5 and depreciation 10. With no tax in this teaching model, profit is 10. The cash-flow information reports customer receipts 130, supplier payments 50, operating-expense payments 25, interest payments 5 and capital expenditure 20. No debt is issued or repaid.

13.2 One compatible event history

One history divides sales into cash sales 80 and credit sales 40. Collections on receivables total 50: 20 from opening receivables and 30 from current-year credit sales. Customer receipts therefore total $80 + 50 = 130$. Purchases on credit total 60. Supplier payments are 50.

The receivables and inventory bridges are

$$A_1 = 30 + 40 - 50 = 20, \tag{13.1}$$

$$I_1 = 50 + 60 - 70 = 40. \tag{13.2}$$

Trade payables become $40 + 60 - 50 = 50$. Property becomes $80 + 20 - 10 = 90$. Cash becomes

$$C_1 = 40 + 130 - 50 - 25 - 5 - 20 = 70. \tag{13.3}$$

Profit of 10 increases equity from 60 to 70. Thus closing assets are $70 + 20 + 40 + 90 = 220$, equal to debt 100, payables 50 and equity 70. Every amount follows from a stated movement.

This is one witness. The reports may not disclose the split of collections between opening and new receivables, and they do not disclose which customers remain unpaid. Other event histories can generate the same statements. The model should not turn the chosen witness into an assertion about what actually happened.

13.3 The journal bundle

Under the illustrative recognition rules, cash sales debit cash and credit revenue by 80. Credit sales debit receivables and credit revenue by 40. Collections debit cash and credit receivables by 50. Purchases debit inventory and credit payables by 60. Consumption of inventory debits cost of sales and credits inventory by 70.

Supplier payments debit payables and credit cash by 50. Operating expenses debit expense and credit cash by 25. Interest payment debits interest expense and credits cash by 5. Capital expenditure debits property and credits cash by 20. Depreciation debits expense and credits accumulated depreciation or the summarized carrying-value account by 10.

Each posting bundle balances. More importantly, the bundles reconcile the economic movements. The inventory purchase is not expensed a second time when cash is paid. Credit sales are not recognized again on collection. Capital expenditure is not simultaneously charged as the full current expense and depreciated. These familiar errors are easy to introduce when separate forecasting formulas are joined without event identities.

The summarized property account hides gross cost and accumulated depreciation. That is acceptable for this bounded calculation, but a later disposal or impairment query may require those components. The representation should retain the option to refine rather than declare the summary universally sufficient.

13.4 What the reports leave unknown

At the forecast origin, receivables of 20 can be concentrated in one customer or spread among ten customers owing 2 each. The debt of 100 can be fixed-rate at five per cent or floating-rate with a historical rate of five per cent. Both alternatives reproduce the historical interest payment and closing debt.

Let c indicate concentration: $c = 1$ for one customer and $c = 0$ for ten equal customers. Let f indicate floating debt: $f = 1$ for floating and $f = 0$ for fixed. For this exercise we admit all four combinations because the specified evidence provides no linking restriction. This is an explicitly declared feasible family, not a claim that real financing structure and customer concentration are statistically independent. No probabilities are assigned.

Other hidden quantities remain fixed in this first stress calculation. Inventory and property do not change during the short stress horizon, and there are no new sales or purchases. This isolates collection and financing exposure. An operating forecast would add production, demand and replenishment policies rather than assume these balances remain static indefinitely.

13.5 Define the shock precisely

At the start of the stress horizon, payables of 50 must be paid. Next, one largest customer (any one of the tied customers in the diversified case) defaults and pays nothing on its outstanding receivable; every other customer pays in full. At the end of the horizon, one full year's interest is paid after market rates rise by two percentage points. There is no hedge, cap, floor or principal maturity during the horizon.

The customer loss is 20 in the concentrated history and 2 in the diversified history. Write

$$L(c) = 2 + 18c. \tag{13.4}$$

Collections are $20 - L(c)$. Interest is $5 + 2f$. After paying suppliers, collecting surviving receivables and paying interest, cash is

$$C_2(c, f) = 70 - 50 + [20 - L(c)] - (5 + 2f) \tag{13.5}$$

$$= 33 - 18c - 2f. \tag{13.6}$$

The formula preserves the joint dependence on concentration and rate type. It is an exact result for the declared event sequence.

13.6 The four outcomes

The resulting cash and equity are

Receivables	Debt	Closing cash	Closing equity
Diversified	Fixed	33	63
Diversified	Floating	31	61
Concentrated	Fixed	15	45
Concentrated	Floating	13	43

Equity falls by customer loss and interest, so $E_2 = 70 - L(c) - (5 + 2f) = 63 - 18c - 2f$. Receivables and payables both become zero. Inventory remains 40 and property 90. Each history satisfies

$$C_2 + 40 + 90 = 100 + E_2. \tag{13.7}$$

The accounting relation is shared, but the economic outcomes differ by 20 across the family. A single historical witness could conceal that range.

The worst cash outcome is concentrated floating debt. The best is diversified fixed debt. These are feasible joint witnesses, not independent marginal extrema assembled from different worlds. If later evidence rules out one combination, the family and its extrema should update accordingly.

13.7 A liquidity threshold changes the question

Suppose a policy requires closing cash at least 14. Three histories meet it; the concentrated floating history ends at 13. Without probabilities, the correct statement is that the policy is not robustly feasible across the admitted family. It is wrong to say the probability of failure is one quarter merely because there are four rows.

The row count depends on representation. The diversified case could be split into ten different customer-name assignments without changing its economic exposure. Equal weighting would then change the apparent failure probability. A measure must be attached to economic states or events with justified weights, not to a convenient enumeration.

If the threshold is a contractual covenant, its consequences require the actual covenant definition and remedy process. If it is an internal treasury target, a shortfall is a policy deviation rather than legal default. The same numerical comparison can support different meanings depending on the source of the threshold.

13.8 A funding response

Suppose a committed facility can supply up to 2 at the final payment date, with no fee or interest within this short example, and management draws only enough to reach cash 14. The concentrated floating case draws 1. Its closing cash becomes 14 and debt 101; equity remains 43. The balance sheet still reconciles: $14 + 40 + 90 = 101 + 43$.

The draw improves current liquidity but creates a future obligation. It does not reverse the credit loss or interest expense. A model that raises equity by the draw amount would misclassify financing. A model that treats the draw as free permanent cash would overstate future capacity.

If the facility is unavailable after the customer default, the response is infeasible. If it settles after interest is due, it cannot fund that payment without a bridge. If its covenant depends on equity falling below 44, the worst history may lose eligibility precisely when it needs funding. These are distinct contract variants that should be modeled as conditional cases.

13.9 A dependence hidden by aggregate data

Suppose a private lender report establishes that the floating loan was granted only when the receivables were diversified. Then the concentrated floating combination is excluded. The remaining minimum cash is 15, so the threshold 14 is satisfied by every retained history.

This new conclusion comes from a joint restriction, not from narrower marginal ranges. Concentration still can be either zero or one, and debt still can be fixed or floating. Marginal sets are unchanged. A representation keeping only those marginals cannot express why the worst combination disappeared.

The example demonstrates the value of constraints over named variables. The restriction can be written $f = 1 \Rightarrow c = 0$. A new source should be attached to it, with availability and scope. If the source is later rejected, the excluded combination returns in the revised family.

13.10 Learning before the payment

Suppose management knows the loan type but the public analyst does not. Management can plan a precautionary draw in the floating history without using future information. The

analyst's forecast must integrate or bound those conditional management actions. Forcing management to use the analyst's ignorance would understate its information.

Now suppose a customer deterioration signal arrives before supplier payment. A policy may delay discretionary capital spending or seek financing in response. If the signal arrives only after payment, those actions may no longer be available. The date and actor access matter as much as the signal's accuracy.

The current stress example has no discretionary spending. Adding one should identify its benefit, cancellation cost and commitment status. An apparently simple cash-saving policy can be infeasible if the expenditure is already contractually committed. The event sequence prevents that distinction from being lost in an annual forecast.

13.11 Historical journal ambiguity beyond contracts

The historical reports also permit different sales and collection timing within the year. One company may collect most cash early and pay suppliers later; another may pay first and collect near year-end. Annual cash flow and closing cash can agree while minimum cash differs.

For example, take opening cash 40 and aggregate receipts 130 with payments 100 in total. If payments of 100 occur before receipts, a company without credit cannot execute the sequence. If receipts arrive first, it can. The annual closing cash of 70 does not distinguish them. A feasible historical reconstruction must respect intermediate cash and financing events where those are within scope.

This does not mean every annual disclosure allows precise daily reconstruction. It means the model should retain timing uncertainty or state that it cannot answer an intra-year liquidity question. Inventing a uniform monthly schedule gives a point path whose apparent smoothness comes from the interpolation assumption.

13.12 Inventory alternatives

Closing inventory value 40 can represent eight units costing 5 or four units costing 10. If a future order requests six physically interchangeable units, the first history has sufficient stock and the second does not. If products are not interchangeable, even total unit count is inadequate.

A monetary demand shock should therefore not be converted automatically into units using an average historical cost. Selling price, cost, product mix and capacity are different quantities. A future price decline can reduce revenue per unit without changing available physical stock. A supply shock can reduce units while increasing unit cost.

The present company case holds inventory fixed to isolate credit and rates. Extending it requires product or cohort state, procurement lead times and sales rules. The aggregate carrying amount remains a reporting consequence of those states. A richer query justifies refinement of the family rather than invalidating the earlier bounded calculation.

13.13 Property and replacement cash

Property of 90 after depreciation can comprise young assets with long remaining lives or old assets near replacement. Equal historical depreciation does not identify next year's capital needs. A policy that sets capital expenditure equal to depreciation assumes a replacement relation.

Suppose one history has a machine requiring replacement next year for 30 and another can continue for five years with maintenance 2. Their current carrying amounts may agree under different acquisition costs and accumulated depreciation. A shock to financing availability has different operating consequences.

A complete extension should retain capacity and maintenance as well as book values. Replacement can change productivity and costs, so it is not only a cash outflow. The model needs a production or service relation and a management policy. The accounting bridge alone cannot supply them.

13.14 Rounding widens the family

The exact amounts used above make arithmetic transparent. If reported receivables 20 are rounded to the nearest unit, the underlying amount may lie in a reporting interval. Customer loss and collections then become functions over that interval. A cash threshold at 14 can be crossed by values that a point reconstruction misses.

Suppose the concentrated floating case's opening cash is itself rounded from an amount between 69.5 and 70.5 under the declared convention. Its stressed cash lies correspondingly between 12.5 and 13.5 if other terms are exact. A facility of exactly 1 may no longer guarantee reaching 14. The required draw can range from 0.5 to 1.5.

Rounding intervals can also be jointly constrained by a total. Treating each line's rounding error independently may admit totals outside the reported balance-sheet interval. The feasible family should retain the total relation rather than merely widen every number separately.

13.15 An explanation the system should produce

For the worst witness, an explanation should identify the concentrated receivable history, floating debt, customer default, rate increase, supplier payment and lack of an eligible funding response. It should show cash 13 and equity 43 with their bridges. It should identify which facts are observed and which are alternative assumptions.

The explanation should not say that the real company will fail. It should say that a compatible history produces the stated shortfall under the specified shock and policy. If the threshold is internal, it should call it a target shortfall. If a covenant consequence is modeled, it should name that consequence and its premises.

A second explanation can identify the source that would exclude the witness, such as a verified customer concentration note or fixed-rate contract. This turns the simulation into a guide for focused investigation. It also makes disagreement productive: a reviewer can challenge a premise rather than merely distrust the final number.

13.16 What this complete case establishes

The case connects reports, compatible histories, events, journals, cash and equity. It shows why identical statements do not identify shock response, why joint restrictions matter, why financing needs a transaction and why timing matters. All numerical relations are small enough for exact checking.

It does not establish a calibrated corporate model, a universal customer-default process or a reporting-standard treatment for every event. Those would require additional facts and authority.

The bounded example is valuable because it exposes the mechanisms without pretending those wider questions are resolved.

An implementation fixture should store the input observations, the four initial history cases, the optional dependence restriction, the event sequence, expected journals and final statements. It should also test the rejected variants: equal branch probabilities without a measure, independent marginal output selection, a late facility draw and a cash-balancing equity plug. These are concrete failures of the target semantics.

Chapter 14

A refinancing crisis: dates, offers, covenants, and recovery

A year-end debt ratio can look manageable while a particular morning's payment is impossible. This chapter follows a synthetic treasury problem through dated cash events. The company has several potential responses, but each has conditions. The purpose is to show how financing feasibility differs from choosing a debt amount in an optimization formula. All offers and legal consequences below are assumptions of the example; they are not estimates or statements of general law.

14.1 The payment calendar

At the opening of day 1, unrestricted cash is 12. A customer receipt of 18 is due on day 5, wages of 10 on day 3, an old loan principal of 30 on day 4 and its coupon of 2 on day 4. Inventory worth 20 in the accounting records can be sold, but only through a specified offer. There is no automatic overdraft.

If all receipts and payments are netted over five days, closing cash is

$$12 + 18 - 10 - 30 - 2 = -12. \tag{14.1}$$

The total shortfall is 12. But the timing problem is larger: after wages, cash is 2 before the day-4 debt payment of 32. The company needs 30 by day 4, even though 18 is expected the next day. A facility that supplies only the final net shortfall cannot bridge the maturity.

The receipt is also uncertain. In history H1 it is contractually due and collected on day 5. In H2 it is subject to customer acceptance and may be delayed to day 9. Both histories fit an opening receivable of 18. The financing decision must distinguish the customer's contractual obligation, actual collection policy and the analyst's uncertainty.

14.2 Three offers

Offer A is a bridge facility settling on day 4 before the old payment, with maximum principal 30 and a fee of 1 withheld at draw. Its repayment of principal plus 1 is due on day 6. Offer B is a term debt issue settling on day 5, priced at 0.9 per unit face, with a fixed fee of 2 and maximum face 40. Offer C is an inventory sale settling on day 4, paying 15 for inventory carried at 20.

These offers have different cash, accounting and future consequences. A supplies short-term cash but creates another near maturity. B can supply longer funding but arrives too late to

pay the old loan without a bridge or deferral. C supplies timely cash and incurs a disposal loss under the illustrative carrying-value treatment, while removing inventory needed for operations.

The offers are not interchangeable columns in a financing plug. Their settlement dates and quantity constraints determine feasible sequences. The company can choose combinations only if the contracts permit them and management accepts their economic costs.

14.3 Why the bridge alone fails

Drawing maximum bridge principal 30 supplies net cash 29 after the withheld fee. Cash before old debt payment becomes $2 + 29 = 31$, one short of 32. A model that treats principal as proceeds would declare the payment feasible and miss the fee-induced shortfall.

Required bridge principal to supply net cash 30 would be 31 under this fixed fee, but the cap is 30. The transaction is infeasible at the needed amount. This conclusion does not require a default-probability model; it follows from offer terms and dates.

If the company can defer wages by 1 under a separately permitted agreement, the bridge becomes sufficient for day 4. But that deferral creates another obligation and may have costs. It must be represented as a negotiated event, not as a spreadsheet reduction of operating expenses.

14.4 The inventory sale plus bridge

Suppose management accepts the inventory sale. After wages, cash 2 plus sale proceeds 15 gives 17. The remaining need for old debt payment is 15. A bridge principal of 16 supplies 15 net after its fee, so day-4 cash reaches 32 and the old obligation is paid exactly.

At day 5, customer receipt 18 arrives in H1. The bridge repayment due day 6 is 17. Paying it leaves cash 1. The old debt is extinguished, bridge debt is extinguished, inventory is lower by 20, and the company has recognized the illustrative sale loss 5 and bridge cost 2 in total under the assumed expense convention.

This path solves the payment problem at an economic cost. The inventory sale can reduce future production or require replacement. A short-horizon model ending on day 6 would miss that operational consequence unless the terminal state retains the missing inventory and relevant commitments.

14.5 Delayed collection changes the result

In H2, the customer does not pay until day 9. After the old debt payment, cash is zero. Bridge repayment 17 is due day 6. The inventory-plus-bridge sequence therefore creates a later shortfall unless another offer or deferral is available.

The term debt issue B can settle on day 5. To raise net cash 17 after a fixed fee 2 at price 0.9, required face is

$$F_B = \frac{17 + 2}{0.9} = \frac{190}{9} \approx 21.1111. \tag{14.2}$$

This is within the face cap 40. If B remains eligible, its proceeds repay the bridge and leave a new longer-term promise. Customer collection on day 9 then raises cash 18. The company has transformed a timing mismatch into longer funding at a cost.

The financing path depends on the continued availability of B. A policy that assumes B simply

because it is numerically sufficient confuses an offer with an executed transaction. Eligibility, acceptance and settlement must be checked at the relevant dates.

14.6 A covenant can remove the rescue

Assume offer B requires inventory carrying amount at least 10 at settlement. The company initially has exactly 20 and the sale C disposes of all of it. After C, the condition fails. B is unavailable in the very path that needed it to refinance the bridge.

A modular simulator can miss this interaction if the asset-sale module reports cash but does not update the collateral or eligibility state used by financing. The cash schedule appears feasible until day 6, then fails. The root error is a missing cross-module dependency, not arithmetic.

If C can sell only half the inventory for 7.5, inventory remains 10 and B stays eligible. After wages and partial sale, cash is 9.5. To reach 32 requires net bridge 22.5, so principal 23.5 is within cap. Bridge repayment is 24.5. In H1, receipt 18 leaves a day-6 need of 6.5; in H2 it leaves a need of 24.5. B can supply either need subject to its own cap.

14.7 Face caps and net funding

Under B, maximum face 40 supplies gross proceeds 36 and net cash 34 after the fee. In H2 the need 24.5 is feasible. Required face is $(24.5 + 2)/0.9 = 265/9$, about 29.4444. In H1 the need 6.5 requires face $85/9$, about 9.4444.

If B's price falls to 0.6 after a market shock, maximum net proceeds become $40(0.6) - 2 = 22$. The H2 need 24.5 is no longer feasible. H1 remains feasible. Thus identical opening statements and the same price shock produce different financing outcomes because collection timing differs.

A sensitivity table should vary offer price and collection history jointly. Reporting only the average required face can hide the infeasible H2 path. A midpoint collection date may produce a third path that neither contract history permits.

14.8 A symbolic feasibility boundary

Let q be B's price per unit face and K the net funding need on day 6. With fixed fee 2 and cap 40, feasibility requires

$$40q - 2 \geq K \quad \Longleftrightarrow \quad q \geq \frac{K + 2}{40}. \tag{14.3}$$

For H1 after partial inventory sale, $K = 6.5$, so the threshold is $8.5/40 = 0.2125$. For H2, $K = 24.5$, so it is $26.5/40 = 0.6625$. These thresholds are exact for the offer structure.

The price may itself depend on face or the asset sale. If so, the linear threshold no longer applies without modification. The symbolic result should carry its premises. A new price model is a new financing query, not merely a different number substituted into an unchanged guarantee.

14.9 Who knows the collection history

Suppose management knows whether customer acceptance has occurred, while the public analyst sees only the receivable total. Management can condition its financing plan on H1 versus H2. The analyst should evaluate that informed policy across the two histories.

If management does not know the eventual delay, it must choose a plan before observing collection. A robust plan must remain feasible in both histories, or specify a contingent action after the day-5 observation. A term issue requested only after nonreceipt may settle too late if the offer has a notice period. The observation date and action lead time determine the policy's feasibility.

A simulator that waits to see H1 or H2 and then retroactively chooses the best day-1 plan gives management foresight. A correct tree branches at the observation time and shares decisions before that branch. A symbolic policy representation enforces the same restriction through its allowed inputs.

14.10 Default after a failed refinancing

Suppose H2 occurs, B is unavailable and bridge repayment 17 cannot be made on day 6. Under a stipulated contract, a two-day cure period begins. The day-9 customer receipt arrives after the cure deadline. Whether the lender can accelerate, waive or restructure is specified by additional terms.

The economic state on day 6 contains unpaid principal and fee or interest according to the contract. It does not set debt to zero. The company may continue operations during cure. If a waiver is granted on day 7 for fee 0.5 and repayment on day 9, the receipt can resolve the arrears. If not, enforcement may proceed.

These branches are not numerical solver failures. They are economic transitions. Conversely, inability to solve a pricing equation for B is not itself the bridge default event. The simulator should preserve the distinction even if both ultimately leave the funding need unresolved.

14.11 A recovery waterfall

Assume enforcement yields distributable assets of 25 after costs. The bridge has senior claim 17 and another unsecured creditor has claim 20. Under the example's strict priority rule, the bridge receives 17 and the unsecured creditor 8. Total recovery equals available assets 25.

If the bridge is secured only on an asset that realizes 12 and has no further recourse under the assumed contract, its recovery may instead be 12. The remaining distribution depends on ownership and priority of the other assets. A single aggregate recovery percentage cannot represent both cases without additional assumptions.

If recovery occurs one year later, creditor valuation discounts it from that date. The accounting treatment of the debtor's unresolved obligation and eventual extinguishment requires its own rules. Cash recovery to creditors and book debt removal at the debtor need not occur at the same instant.

14.12 Why aggregate net debt fails here

At opening, net debt may be calculated as debt less cash. That statistic does not contain the day-3 wage date, day-4 maturity, day-5 receipt or day-6 bridge repayment. It cannot decide feasibility of the sequence.

After the partial sale and term refinancing, net debt may improve when the customer finally pays. Yet the company may have lost productive inventory and paid substantial financing costs.

A net-debt chart alone omits these consequences. A useful result reports cash timing, remaining obligations, inventory capacity and owner value separately.

A compressed state can still be possible. For this bounded problem it must retain the ordered obligation calendar, collection state, inventory eligibility amount and offer terms. The full historical journal need not be replayed at every step, but its economically relevant consequences must survive.

14.13 Discrete sale choices and continuous financing

The inventory offers give three sale choices: none, half or all. They do not make the financing policy set finite. A principal cap permits an interval of bridge draws unless the contract also specifies denominations. To compare policies, retain those amounts as variables and first separate the cases in which each offer is used. For each feasible policy, record minimum cash, final cash, fees, remaining debt and inventory. A policy minimizing fees may differ from one preserving inventory or maximizing the lowest cash balance.

The action domain can be derived directly for this example. Let z be the fraction sold, with $z \in \{0, 1/2, 1\}$, and let a be bridge principal. Every sale choice still needs the bridge before day 4. Write b for term face and u for whether B is issued: $u = 0$ requires $b = 0$, while $u = 1$ requires $0 < b \leq 40$. Hold the supplied price $q > 0$ fixed. For this calculation, both sale and bridge proceeds arrive before the old debt payment; any day-5 customer receipt arrives before B settles, and B's fee is withheld from its proceeds. These are explicit settlement-order assumptions. They do not give the manager permission to decide an issue after its contractual notice deadline.

Let R be the cash collected by day 6: $R = 18$ in H1 and $R = 0$ in H2. Cash just after the old debt payment is

$$C_4 = 12 - 10 + 15z + (a - 1) - 32 = a + 15z - 31. \tag{14.4}$$

Between that payment and bridge repayment, the company receives R and the term issue supplies net cash $qb - 2u$. Repaying principal plus one therefore gives

$$C_6 = C_4 + R + (qb - 2u) - (a + 1) = 15z + R + qb - 32 - 2u. \tag{14.5}$$

The bridge principal cancels from C_6 because the company repays the extra amount it borrowed. It still changes C_4. Under the stated event order, nonnegative cash throughout days 1–6 is equivalent to

$$31 - 15z \leq a \leq 30, \qquad 15z + R + qb - 32 - 2u \geq 0, \tag{14.6}$$

together with the issue/no-issue domain and offer eligibility. To see why no other cash constraint is missing, cash after wages is 2 and the sale and bridge then add nonnegative proceeds. After day 4, the receipt adds cash; cash after B settles must be $C_6 + a + 1$, which is positive whenever $C_6 \geq 0$. The bridge repayment is the final outflow in this comparison. H2's day-9 receipt increases cash later; it cannot repair a negative prefix on day 6. Any term debt remains an obligation beyond this cash horizon.

With the inventory covenant, $u = 1$ also requires $20(1 - z) \geq 10$. For half sale, the bridge interval is $[23.5, 30]$. If B is used, its face must satisfy $qb \geq 8.5$ in H1 or $qb \geq 26.5$ in H2, recovering the earlier price thresholds at the face cap. No sale would require $a \geq 31$, so that entire branch is infeasible. Full sale permits $a \in [16, 30]$; with the covenant it must use $u = 0$, giving $C_6 = 1$ in H1 and $C_6 = -17$ in H2. Without the covenant, the term issue remains a separate possible response.

These intervals describe the continuous amounts; a few selected amounts are only witnesses. For a fixed sale choice and term issue, taking the least bridge principal $a = 31 - 15z$ preserves

feasibility under these assumptions, but it can change the best liquidity policy. In H1 with full sale and no term issue, $a = 16$ leaves cash 0 after the old debt payment and cash 1 after bridge repayment. Taking $a = 17$ instead leaves cash 1 after both payments. Both paths pay the same two units of bridge cost and end with the same cash and no bridge debt, yet their minimum cash balances are 0 and 1. The latter is maximal within this full-sale/no-term comparison because every path ends with cash 1. Restricting the comparison to minimum funding would lose that conclusion. An interest charge depending on the amount borrowed would require a different calculation.

Settlement order also matters for the domain itself. Without the inventory covenant, take full sale, H1, $a = 16$, $q = 0.9$ and $b = 10/9$. Issuing B yields net cash -1 after its fixed fee. Receipt before issue gives a feasible path ending at cash 0. Issue before receipt instead asks the company to pay that one unit while cash is still 0. That order requires the additional prefix condition $C_4 + qb - 2 \geq 0$. The small issue is not proposed as an attractive policy; it exposes an action that a daily net total would mistakenly admit under the opposite order.

A finite enumeration is therefore complete only for an explicitly finite action menu or after a justified reduction for the stated objective. Here one can enumerate sale choices and offer activation, then retain or solve the continuous inequalities inside each branch. With fixed q these are linear constraints. If price depends on face, inventory or the later policy, that continuous problem changes. Across H1 and H2, decisions must still share the same values until the manager's observations permit them to differ. Comparison with another optimizer requires the same amounts, information, settlement order and objective; a better result under different choices answers a different financing question.

14.14 Accounting consequences of the sequence

The old loan repayment reduces cash and the old obligation. Bridge issuance increases cash by net proceeds and creates the specified liability and fee treatment. Inventory disposal removes the asset and records proceeds and loss under the illustrative basis. Term issuance creates a new obligation with proceeds distinct from face.

The exact treatment of debt discounts and transaction costs requires the accounting basis. A cash schedule can be checked without prematurely choosing that treatment. The accounting layer should then reconcile carrying amounts separately from contractual face.

A journal test should ensure that the old debt is repaid once, the bridge is extinguished only when settled or legally discharged, and the term debt remains after bridge repayment. Netting all financing flows into one closing debt amount would make these checks impossible.

14.15 Sensitivity that preserves the same company

Vary B's price while holding the collection history, asset sale and contractual caps fixed. This produces a meaningful price sensitivity for each policy. Vary the collection date while retaining the same receivable identity and amount. This produces a timing sensitivity.

Changing both at once can be a joint stress, but it should be named as such. Comparing a high-price H1 case with a low-price H2 case and attributing the entire difference to price would confound two mechanisms. Paired scenario comparisons should preserve the dimensions not intended to change.

The same discipline applies to recovery. A lower recovery value can affect price at issuance if anticipated, but a surprise enforcement loss after issuance is a different information path. The

model should specify whether creditors knew the recovery change when making the offer.

14.16 A practical implementation slice

This case suggests a bounded event scheduler with exact rational amounts and ordered timestamps. It needs offer admission, net-proceeds calculation, inventory eligibility, collection observation, policy action and arrears transitions. It does not need a neural network to expose the important errors.

The fixture should include both successful and refused paths, with reasons. The refused bridge-only path fails by one due to the fee. The full-sale H2 path loses term eligibility. The half-sale H2 path fails when price falls below 0.6625. The cure-period variant distinguishes late collection from timely cure.

These cases provide a concrete basis for extending the repository's financing methods. They also show what remains outside a small implementation: actual market offers, legal enforceability, operational cost of inventory sale and empirically supported management preferences. The model can explain those dependencies without pretending to have resolved them.

14.17 The price must anticipate who chooses after conversion

The refinancing calendar tells us when an offer can settle. A second issue arises when the security changes the person who chooses later company policy. A price based on the old owner's future choice can be inconsistent with a conversion that transfers control. The following finite example isolates that dependency.

There are three dates: financing on 1 January 2027, a signal and policy choice on 1 January 2028, and terminal distributions on 1 January 2029. Investment costs 60 at the first date. Existing senior face is 40 and new convertible face is 80. The gross discount factor is 21/20 in each of the two intervals. All cash recoveries occur at the terminal date. The equal signal probabilities and the conditional probabilities below are a supplied hypothetical pricing law, also used to evaluate expected cash utility.

Under the debt signal, the new claim remains debt and the old owner chooses policy. Safe policy produces terminal estate 100. Risky policy produces 20 or 220 with equal probabilities. Senior debt is paid first, then the new claim up to its face, and old equity receives the residual. The old owner consequently prefers risky expected cash 50 to safe cash zero. The new creditor's expected cash under that choice is 40.

Under the conversion signal, the new claim is extinguished and its holder receives 120 shares alongside 40 old shares, as in Section 7.16. The new holder now controls policy. Safe estate is again 100; risky estate is 20 or 140 with equal probabilities. After senior priority, safe policy gives the new holder $3(100 - 40)/4 = 45$. Risky policy gives 75/2, so the controlling holder chooses safe.

To make the control distinction consequential, assume the old owner receives a noncash private benefit worth 4 terminal utility units when risky policy is chosen in this conversion branch. Old-owner risky cash has expectation 25/2, giving utility $25/2 + 4 = 33/2$, compared with safe utility 15. Thus a stale old-controller assumption would select risky policy. The benefit is deliberately hypothetical and never enters estate, creditor recovery or accounting equity. At zero private benefit the old owner also prefers safe policy, which checks the mechanism's limiting case.

The consistent continuation is therefore risky under the debt signal and safe after conversion. Let P denote date-zero proceeds paid by the new holder. In this selected competitive-quote comparison, proceeds equal the discounted expected cash the holder will receive:

$$P = \frac{\frac{1}{2}(40) + \frac{1}{2}(45)}{(21/20)^2} = \frac{17000}{441}, \qquad 60 - P = \frac{9460}{441}. \tag{14.7}$$

The two discount intervals matter: the signal date is not a payment date. The second term is the cash the old owner must contribute at financing. Available cash 20 is insufficient, while cash 22 is sufficient.

Affordability and willingness to invest are separate conditions. Under this continuation, the old owner's expected terminal cash is $(50 + 15)/2$. Its date-zero value is $13000/441$. Subtracting the contribution gives $3540/441 > 0$, so participation exceeds the assumed outside utility zero. That positive value does not supply the missing liquid funds in the cash-20 case.

If old control were incorrectly retained after conversion, the new holder's conversion-branch cash would instead be $75/2$. The corresponding quote would be $15500/441$, requiring owner cash $10960/441$, more than 22. Charging the correct-control quote while executing that stale-control policy leaves the lender short by $1500/441$. The discrepancy is a failure to keep rights, continuation and price consistent. It does not establish that conversion is economically superior.

Backward induction agrees with a separate enumeration of all four pure strategies mapping the observed signal to a policy, including every local deviation. A tie variant retains both optimal policies. Equal votes without a supplied governance rule remain unresolved. Policies cannot depend on the terminal realization, which is not known when the choice is made.

The owner's financing contribution is sunk by the policy date, so the continuation ranking in this example is independent of the posted price. This is a finite consistency calculation, not a general fixed-point or market-clearing result. The typed conversion example starts at a separately supplied fiscal opening. The following sections connect the dated funding exchange and identify the source, measurement and control-history conditions needed before issuance, conversion and final liquidation can form one accounting history.

14.18 Funding the project does not yet determine its accounts

The price calculation leaves two different questions. Can the parties deliver the promised cash in the required order? If they can, which amounts should their financial statements recognise between issue and conversion? The first question has a complete answer under the supplied synthetic cash balances. The second still depends on the contract, reporting basis and measurement evidence. Keeping the questions separate prevents a feasible transaction from silently becoming an unsupported accounting history.

At the financing date the new holder pays $17000/441$ to the company and the old owner contributes $9460/441$. Their receipts sum exactly to the project price 60. The company then pays 60 to the seller. The transfer records identify both parties, the date and the same amount in the two legal books. In this example the company starts with no available cash for the purchase, so both receipts must precede payment to the seller. There are six possible orders of these three transfers. The two orders with the seller paid last succeed; the other four fail at the first payment that exceeds available company cash. Positive final net cash is not a substitute for an affordable cash prefix.

The four-party reference retains the company, old owner, new holder and seller books separately. It checks two account charts and both owner cash 22 and cash exactly equal to the required contribution. A refused later transfer preserves the accepted prefix. Counterparty cost-basis

entries are explicit synthetic premises. They demonstrate reciprocal transfers and funding feasibility, not the appropriate measurement of a real convertible instrument or the seller's tax and profit consequences.

The limitation becomes visible if the 5% pricing discount is reused as an accounting yield. Suppose one incorrectly treats the face 80 claim as a plain two-year zero-coupon liability and discounts its entire face at that rate. The resulting liability and residual allocation would be

$$L_0^{\text{forced}} = \frac{80}{(21/20)^2} = \frac{32000}{441}, \qquad P - L_0^{\text{forced}} = -\frac{5000}{147}. \tag{14.8}$$

This negative residual is a diagnostic of that attempted mapping. The plain unconditional cash claim has different payoffs from the contingent instrument used to obtain P. Conversely, simply accreting P at 5% over both intervals gives 85/2, not face 80. Neither calculation supplies a replacement accounting policy. The rate and cash flows appropriate to a liability component require their own evidence.

For the conventional issuer compound-instrument case, the inspected AASB 132 text measures the comparable liability first and assigns the residual to equity. Its application guidance refers to instruments of comparable credit status, substantially the same cash flows and terms, without the conversion option (Australian Accounting Standards Board, 2022d, paragraphs 31–32 and AG31). Those conditions are not established by the economic quote. The typed conversion demonstration instead starts with a supplied carrying amount C. Its balanced transfer from liability to equity remains conditional until classification, intervening measurement and the conversion treatment are supported.

14.19 Issuer and holder classify the same contract differently

The same contract is a claim issued by the company and an investment held by the creditor. Those perspectives need not have matching classification components. The source research therefore begins with the complete settlement terms: who can affect the conversion signal, whether either party can elect cash settlement, the currency and share terms, any redemption provisions, and agreements involving other members of a group. The example supplies none of these facts merely by stating that 120 shares will be issued.

Under the inspected AASB 132 provisions, fixed-for-fixed analysis is subject to the definitions and exceptions for qualifying own-equity instruments. A genuine contingent liability settlement beyond both parties' control raises the separate conditions in paragraph 25. A derivative with settlement choices has its own rule in paragraph 26. The non-derivative compound-instrument assessment in paragraphs 28–32 cannot be inferred from a fixed share count alone (Australian Accounting Standards Board, 2022d, paragraphs 15–16, 22–26 and 28–32). These passages identify the questions to answer; they do not classify the underspecified fixture.

The holder's analysis is different. For an in-scope financial-asset host, AASB 9 assesses the hybrid as a whole. Its fixed-number convertible-bond example links the return to issuer equity and fails the solely-principal-and- interest condition (Australian Accounting Standards Board, 2026i, paragraphs 4.3.2 and B4.1.14, Instrument F). Issuer liability/equity components therefore cannot simply be copied into the holder's asset book. Initial fair value, transaction costs, subsequent measurement and relevant exceptions remain separate dependencies. Likewise, AASB 132 AG32 expressly addresses conversion at maturity; it alone does not justify the fixture's 2028 conversion before its 2029 terminal date.

The resulting holder interest also raises a reporting-perimeter question. Section 26.16 follows the separate-statement, consolidation and business-acquisition alternatives, including the possibility

of control before legal conversion. Its repaired dependency analysis preserves uncertain control histories rather than choosing whichever journal balances. The retained source compilations do not yet establish the complete accounting authority for the fixture's 2027 issue and 2028 conversion. Classification, measurement, control history and applicable edition therefore remain linked questions with distinct evidence requirements.

14.20 Evidence over an interval is stronger than two snapshots

To support uninterrupted control, an analyst needs evidence covering the intervening period. An empty event list supplies no such evidence. The selected interval reference requires an explicit assertion that a control status holds throughout a specified half-open interval, together with its subject, reporting basis, source identity and availability to each actor. This is a representation of an asserted coverage premise. It does not certify that the assertion is true or that the source collection is complete.

For example, let one source assert no control from 30 December to 31 December, and another assert control from 31 December to 1 January. A half-open interval includes its starting instant and excludes its ending instant, so the shared boundary is assigned once. If both assertions are eligible for the selected actor view, they conditionally support a gain at that boundary. If the second source is missing, the second interval remains unresolved. If it conflicts with another eligible assertion, the conflict remains visible rather than being resolved by whichever record was loaded last.

Effective time and information time do different jobs. A correction can describe control during December yet become available to an analyst in January and to a manager only in February. The analyst's January view can use it; the manager's January view cannot. A later available revision replaces the corresponding earlier assertion only within its declared subject and interval. Unknown actor availability remains unresolved. The selected profile refuses partial-interval replacement, whose meaning would require an additional rule.

The implementation reuses the exact fixed-offset timestamp parser and partitions the query at every eligible interval boundary. It retains every covered, uncovered, unknown or conflicting piece. Equivalent timestamp offsets denote the same instant, and adding a future correction leaves an earlier view's control conclusions unchanged. The checked cases include all two-cell combinations of control, no control and unknown control, as well as revision conflicts, coverage gaps, missing ancestry and corrupted source identities.

This gives the simulator a precise way to carry unresolved control history alongside otherwise balanced company books. It does not yet connect an admitted real history to acquisition measurement, changing group membership or complete financial statements. The next financial composition must supply those methods and their dated authority while preserving the source and actor-view distinctions. Neither a successful funding transfer nor a complete conditional interval partition can supply the missing accounting evidence.

Chapter 15

A recapitalization: cash raised, rights transferred, and incentives changed

A financing plan can save a company while reducing the value held by its existing owners. It can also dilute their percentage without reducing their value. Distinguishing these outcomes requires a security-level calculation. This chapter follows a synthetic recapitalization with ordinary shares, a preference offer and a convertible claim. The terms are stipulated so the arithmetic can be reconstructed. They illustrate mechanisms rather than legal or valuation conclusions for a real issuer.

15.1 The company before financing

There are 100 ordinary shares. Existing operating assets have a stipulated economic value of 600, cash is 20 and a senior debt payment of 80 is immediately due. For the first comparison, assume asset value is unchanged by financing and there are no other claims. Before paying debt, residual economic equity value is $600 + 20 - 80 = 540$, or 5.4 per ordinary share.

The company nevertheless cannot pay 80 from cash 20. It needs 60 of additional cash. Positive economic equity value does not eliminate this liquidity problem. Selling assets, borrowing, issuing equity or negotiating deferral are different possible responses. We first study an ordinary-share offer.

Book assets may have a different carrying amount. Nothing in the calculation requires book equity to equal 540. If the model uses book equity to set a subscription price of 5.4, it would be making a separate valuation assumption. Here 540 is a supplied economic valuation used only for the illustrative allocation.

15.2 A fairly priced ordinary issue

Suppose investors subscribe at 5.4 with no fees, and the company raises exactly 60. New shares are $60/5.4 = 100/9$, about 11.1111. Cash becomes 80 and is paid to the senior creditor. Remaining assets are 600, there is no debt, and shares total 1000/9.

Post-transaction value per share is

$$\frac{600}{1000/9} = 5.4. \tag{15.1}$$

Existing holders retain value 540, and new holders own value 60. Their percentages change, but

no value is transferred under these assumptions. The creditor receives its contractual 80. The cash injection solves liquidity without creating extra value by itself.

This result relies on the supplied fair price, homogeneous rights, no costs and unchanged operating value. It is a useful benchmark because deviations can be attributed to specific mechanisms. It is not a prediction that a distressed company can always issue at fair value.

15.3 A discounted issue

Now suppose the offer price is 3, with the same 60 cash target and no fees. Twenty shares are issued. After paying debt, value remains 600 but shares total 120. Each share is worth 5 under the stipulated valuation. Existing holders own 500, a loss of 40 relative to the benchmark; new investors paid 60 for claims worth 100.

The transfer is not an additional operating loss. It is an allocation effect of selling claims below the assumed post-transaction value. A journal recording cash and equity does not reveal it. The ownership register and valuation assumptions are needed.

If investors' required price of 3 reflects information that operating assets are worth less than 600, the benchmark valuation is wrong. The calculation should not accuse the offer of unfairness without validating the valuation premise. Scenario analysis can show transfers conditional on alternative asset values and prices.

15.4 Fees change both quantity and value

Add a fixed cash issuance fee of 6. To net 60, the company must raise 66. At price 3, it issues 22 shares. After fees and debt payment, operating assets remain 600, cash is zero and shares total 122. Existing owners' value is $600(100/122)$, about 491.8033.

The fee destroys 6 of total value relative to an otherwise identical financing if it buys no additional benefit in this model. The discounted price transfers value between old and new owners. These effects should be separated. Calling both dilution obscures which mechanism could be changed by renegotiation or a different financing channel.

If the fee is paid later, current funding need differs and a payable remains. If it is financed by issuing additional shares to an intermediary, share count changes without the same cash payment. The offer must state the form and timing of compensation.

15.5 An authorization constraint

Suppose only fifteen new ordinary shares may be issued under the admitted authorization. At price 3 and fee 6, maximum net cash is 39. Existing cash 20 then supplies only 59 toward debt 80, leaving a shortfall of 21. The ordinary issue cannot solve the maturity under these terms.

At price p, feasibility requires $15p - 6 \geq 60$, so $p \geq 4.4$. This is an exact offer threshold. It does not establish that investors will pay 4.4. A market model or a new offer is needed to answer that question.

An optimizer that increases shares beyond fifteen has changed the feasible set. An accounting routine that credits capital for the missing 21 without issuing a security has invented financing. A correct result records the shortfall and considers other admitted actions.

15.6 A rights offer

Suppose existing holders have a right to subscribe proportionally at the discounted price. If every holder subscribes in proportion, their relative ownership remains unchanged, but they contribute new cash. The issue can still have fees and execution risk.

A holder who does not subscribe may sell the right if trading is permitted and a buyer exists. The value of the right depends on issue terms and market value. A model that treats all existing holders as equally able and willing to contribute assumes away liquidity and participation differences.

For AccDSL, a rights offer needs entitlement, subscription, transfer and expiry events. Unexercised rights may lapse or be allocated under a backstop arrangement. The company receives cash only from settled subscriptions or the backstop, not from the mere existence of rights. Historical statements may not reveal the eventual participation pattern.

15.7 A preference alternative

Consider instead an investor providing 60 for a nonparticipating preference claim. At a future exit, the investor may take 90 before ordinary shareholders or convert into 20 per cent of ordinary equity. Assume no senior debt remains, no interim dividends, no other rights and distributable value V. The investor receives

$$P(V) = \max\{\min(90, V), 0.2V\}. \tag{15.2}$$

Ordinary holders receive $V - P(V)$. The first term respects limited available value: a preference of 90 cannot pay 90 if only 50 exists. Conversion dominates the full preference when $0.2V > 90$, or $V > 450$.

For $V = 50$, the investor takes 50 and ordinary holders receive zero. For $V = 300$, the investor takes 90 and ordinary holders receive 210. For $V = 600$, conversion pays 120 and ordinary holders receive 480. The same 60 of initial funding creates very different allocation across outcomes from a simple ordinary issue.

15.8 Participation changes the waterfall

A participating preference can take a preference amount and then share in remaining value. Suppose the investor receives up to 90 first and then 20 per cent of the residual, with no cap. Its payoff is

$$P_{\text{part}}(V) = \min(90, V) + 0.2\max(V - 90, 0). \tag{15.3}$$

At $V = 300$, this gives $90 + 42 = 132$, compared with 90 for the nonparticipating offer. At $V = 600$, it gives 192, compared with 120 under conversion in the earlier contract.

These are different securities, not two calculation methods for one security. A parser that extracts only the word preference and the amount 90 will miss the distinction. The contract admission model must preserve participation, conversion and caps where they affect the query.

If a participation cap applies, another region boundary appears. The engine can represent the payoff as a piecewise function and split only where the active right changes. A sample of a few exit values is useful for illustration but not a complete characterization of the payoff.

15.9 Value conservation in each state

For every exit value, distributions to all claimants must not exceed distributable assets under the assumed waterfall. In the nonparticipating example, $P(V) + [V - P(V)] = V$ exactly. This is a conservation check independent of whether the offer is fairly priced.

A common implementation error pays the preference and also allocates 20 per cent of total value as though conversion were simultaneous. That double counts mutually exclusive rights. Another pays 90 when $V < 90$, creating cash. Both can be detected with low-value boundary cases.

Fair pricing requires discounting and a distribution or valuation operator over future payoffs. The waterfall itself supplies state-contingent claims, not their current price. An investor's willingness to pay 60 can be taken as a declared offer without claiming that a model has proved its fairness.

15.10 A convertible debt alternative

Suppose an existing note has face 80 due at exit and can convert into sixteen ordinary shares, with no other debt. If conversion is optional for the holder and the company has 100 ordinary shares before conversion, conversion gives a fraction 16/116 of total equity after conversion. The holder compares that claim with the debt payoff under the specified default and priority rules.

If distributable asset value is high enough that debt is paid in full, conversion value is $16V/116$. It exceeds 80 when $V > 580$. Below that threshold, the holder may prefer debt. At low asset values, debt recovery is limited by available assets and seniority. The conversion threshold can change if other securities exist.

A mistaken calculation using $16V/100$ ignores the new shares in the denominator and gives a threshold of 500. That is a dilution error. A second mistake pays debt 80 and also issues sixteen shares. The conversion event must extinguish the specified debt claim if that is what the contract says.

15.11 Cash settlement and issuer options

Some instruments permit cash settlement of conversion value or allow the issuer to choose a settlement form. That changes liquidity and ownership. A model that always assumes physical shares may understate cash need; one that always assumes cash may overstate dilution avoidance.

The choice belongs to the contract and the authorized actor. If the issuer can choose, its policy must consider cash capacity and economic cost. If the holder chooses, management cannot select the favorable form merely to balance the forecast.

An unknown settlement clause should produce conditional outcomes. The system can show a physical-settlement branch and a cash-settlement branch with their premises. It should seek the clause when the distinction changes a liquidity conclusion. This is a high-value source-research target.

15.12 Control and approvals

A recapitalization can shift voting control even when economic ownership changes modestly. A preference investor may receive veto rights or board appointment rights. Those rights can affect future financing, dividends and asset sales. Treating them as irrelevant because they have no immediate journal amount can distort later policies.

The model need not simulate every governance detail for every query. It should retain material rights and declare when behavior assumes their exercise or nonexercise. If a future equity issue requires investor consent, the feasible action set must include that condition.

A public-data analyst may not know whether consent will be given. This is future behavioral uncertainty, distinct from uncertainty about whether the consent right exists. Combining both into one unspecified branch makes it harder to identify useful evidence or policy assumptions.

15.13 Incentives after financing

Suppose ordinary holders choose between a safe project with exit value 300 and a risky project yielding 100 or 600 with equal probability. Under the nonparticipating preference, ordinary payoff from the safe project is 210. Under the risky project, ordinary payoffs are 10 and 480, with mean 245.

Total expected exit value of the risky project is 350, above 300, so both total value and ordinary value favor it in this particular example. Change the low outcome to zero and high outcome to 500. Total expected value is 250, below 300, while ordinary expected payoff is $[0 + 400]/2 = 200$, also below 210. The rankings still agree. These examples should not be used to claim a general conflict.

To exhibit a conflict, consider a senior fixed claim of 250 and ordinary residual value. A safe project with value 300 gives ordinary holders 50. A risky project with values zero and 500 equally likely has total expected value 250 but ordinary expected payoff 125. Ordinary holders prefer the risky project while total expected value is lower. The contract allocation creates the divergence under the stated objective and probabilities.

15.14 Why the actor objective matters

A policy maximizing total enterprise value selects the safe project in the final example. A policy maximizing ordinary equity selects the risky one. A lender pricing debt must anticipate the policy actually modeled. Pricing under the safe policy and then letting shareholders choose the risky one is inconsistent.

Covenants or control rights may constrain the choice. Their presence changes the feasible set and possibly the price offered initially. This joins the security register to behavioral modeling. The register is not merely a reporting table; it determines who receives gains and losses and who can authorize actions.

The example is a local derivation of incentive conflict, not a calibrated prediction. It gives a discriminating fixture for a structural solver: the actor objective, priority and feasible actions should reproduce the stated ranking. A solver returning the same policy for both objectives without a binding constraint is suspect.

15.15 Book entries and economic allocation

The cash issue creates cash and accounting equity under the applicable treatment. The preference or convertible may require classification and component allocation that cannot be inferred from the simplified payoff alone. Those accounting decisions need the standard, jurisdiction and complete terms.

Economic valuation can proceed conditionally on a specified payoff while accounting classification remains unresolved. Conversely, a classified instrument can still have uncertain market price. The system should allow these statuses to coexist. Forcing both layers into one known or unknown flag loses useful work.

A book-equity bridge should reconcile issue proceeds, fees under the chosen treatment, retained earnings, distributions and other movements. A separate ownership bridge reconciles units and rights. A valuation bridge explains changes in economic claims. The three bridges communicate but should not be substituted for one another.

15.16 Historical ambiguity in a capital movement

Suppose a filing reports net cash from equity transactions of 60 and an increase of twenty ordinary shares. This could be a simple issue at 3 with no costs. It could also be an issue of thirty shares at 4 and repurchase of ten at 6, again netting to 60 and twenty shares. The closing share count and net cash agree.

The histories differ in gross funding, repurchase behavior and prices. If future policy extrapolates issuance discounts or repurchase commitments, the distinction matters. A note disclosing gross proceeds and repurchases can narrow the family. Without it, a single implied price of 3 is not established as an actual transaction price.

This example parallels the receivables and debt bridges. Net movements constrain sums but do not identify gross events. The security register should therefore permit unresolved transaction families while preserving observed opening and closing counts.

15.17 An end-to-end fixture

A useful fixture starts with the 100-share company, the debt maturity and the cash need. It evaluates ordinary issuance at fair and discounted prices, fixed fees, authorization limits, preference rights and conversion. Each transaction updates cash, liabilities, shares and rights before producing statements.

Expected failures include using book value as an unstated subscription price, ignoring the authorization cap, counting conversion shares against the old denominator, paying mutually exclusive preference and conversion rights together, and treating an unsigned offer as settled cash. Each failure has a specific input that exposes it.

The fixture should also retain unresolved accounting classification as a legitimate result where full terms are absent. A financing calculation can be mechanically complete while a statement presentation remains conditional. That distinction is preferable to inventing a universal accounting rule to make the example look finished.

15.18 What the case adds to simulation design

Equity funding is a transaction among actors with claims, rights and constraints. Its effect cannot be reduced to an external-equity cost scalar when ownership and future decisions matter. Nor can share count alone establish value transfer. Prices, fees, rights and valuation premises jointly determine the result.

The proposed AccDSL design therefore keeps a dated security register, offer and settlement states, class-specific payoff rules and actor objectives. It can use a simple homogeneous-share profile when appropriate, but retains the boundary that tells a user when richer terms are required. This makes dilution and financing consequences explicit without pretending that every security must always be simulated at maximum detail.

Chapter 16

A decision before and after disclosure: an exact learning example

Learning matters only through a specified relation between hidden facts, observations and decisions. This chapter constructs that relation explicitly. The example is finite, so every policy and probability can be checked by enumeration. It demonstrates why two companies with equal current statements can require different information states, and why preserving only unconditional future cash distributions is insufficient for adaptive decisions.

16.1 Two hidden exposures

A company must choose a hedge before a future settlement. Its exposure type is either A or B. Under type A, hedge A avoids a loss of 10 while hedge B avoids none. Under type B, hedge B avoids a loss of 10 while hedge A avoids none. Each hedge costs 2. Doing nothing costs nothing and leaves the loss 10.

For the first calculation, assume type A and B each have probability one half. This probability is a supplied teaching assumption. The company has enough cash to execute either hedge and can choose at most one. Current financial statements do not distinguish the types.

Net loss after the correct hedge is 2, after the wrong hedge is 12, and after no hedge is 10. Without further information, either hedge has expected loss 7, so a risk-neutral expected-loss policy chooses either hedge rather than doing nothing. Symmetry gives two equally optimal actions. The model should not invent a preference between them.

16.2 A noisy signal

Before choosing, the actor can receive a signal labeled A or B. Let the signal be correct with probability q, symmetrically across types, with $q \geq 1/2$. Bayes' rule with equal priors gives posterior probability q for the signaled type. Following the signal yields expected loss

$$L_{\text{signal}}(q) = 2 + 10(1 - q) = 12 - 10q. \tag{16.1}$$

The first term is hedge cost. The second is the loss when the signal is wrong. At $q = 0.9$, expected loss is 3; at $q = 0.5$, it is 7, matching no informative signal.

The gross value of the signal relative to the best uninformed hedge is $7 - (12 - 10q) = 10q - 5$. A signal costing k is worthwhile under this criterion when $k < 10q - 5$. At $q = 0.9$, the maximum

willingness to pay is 4. This calculation requires the signal to arrive before the action and the actor to be able to respond.

16.3 A complete joint probability table

For $q = 0.9$, the four type-signal combinations have probabilities

Type	Signal	Probability	Loss following signal
A	A	0.45	2
A	B	0.05	12
B	A	0.05	12
B	B	0.45	2

The expected loss is $0.9(2) + 0.1(12) = 3$. The signal is marginally A or B with equal probability. Type is also marginally A or B with equal probability. The information lies in their dependence.

If a compression retains these two marginals but makes them independent, it changes each joint cell to 0.25. Following the signal then has expected loss 7. The compressed model has destroyed the signal's information while preserving both marginal distributions. This is why the joint successor-observation law appears in the exact aggregation condition.

16.4 Two observation technologies with equal current reports

Consider company state U, which will receive the informative signal with $q = 0.9$, and state V, which will receive an uninformative signal with $q = 0.5$. Their current cash, debt and income are identical. Their hidden type priors are identical. Before signal arrival, a fixed hedge has the same expected loss 7 in both.

An adaptive policy following the signal has expected loss 3 in U and 7 in V. Merging U and V because current reports and fixed-action values agree loses a decision-relevant difference. The future observation technology is part of state for this question.

This finite example gives an accounting-oriented counterpart to the policy-versus-action-sequence distinction in Givan et al. (2003, Section 3.1). The numerical construction is local. The source supplies the general warning that policies react to intermediate information and therefore require a stronger equivalence than fixed sequences.

16.5 What happens when the signal arrives late

Suppose the hedge must be committed at date 1 and the signal arrives at date 2. The informative technology no longer improves that hedge decision. A policy that follows the date-2 signal but executes the hedge at date 1 is infeasible.

The expected-loss value returns to 7 for the best uninformed hedge, even though the analyst eventually learns much more about the type. Information has retrospective explanatory value but no value for this already committed action. A simulation must represent action deadlines and settlement lead times, not only observation dates.

If the contract permits cancellation or adjustment at date 2 for cost c, some signal value can return. The calculation then compares adjustment benefits with c and available liquidity. The ability to revise is a contract term or policy permission, not a default property of every action.

16.6 A cancellation calculation

Suppose hedge A is chosen initially. After receiving the signal, the actor may switch to B at extra cost c. With signal B, posterior probability of type B is q. Keeping A has conditional expected loss $2 + 10q$. Switching has conditional expected loss $2 + c + 10(1 - q)$.

Switching is preferred when

$$2 + c + 10(1 - q) < 2 + 10q \quad \iff \quad c < 20q - 10. \tag{16.2}$$

At $q = 0.9$, the threshold is 8. The switching opportunity is exercised only after signal B under the symmetric setup. Its unconditional value must also account for the probability of that signal.

If the initial hedge cannot be unwound and the second hedge creates an offsetting or additional exposure rather than replacing it, the formula changes. A model should not assume that a new action erases an old contract. This is the same event-persistence issue seen in debt amendments.

16.7 Unknown signal quality

Now suppose evidence supports only $q \in [0.6, 0.9]$. The expected loss from following the signal lies in $[3, 6]$ under the same prior and symmetry assumptions. The gross information value lies in $[1, 4]$. A signal priced at 0.5 improves expected loss for every admitted q; one priced at 3 is worthwhile only for some qualities.

These are bounds over a parameterized probabilistic family. They are not probabilities over signal quality. Assigning a uniform distribution to q would add another assumption. A robust decision can use the worst admitted quality, while a Bayesian decision can use a prior over it if justified.

If the quality depends on exposure type, one scalar q is insufficient. Let sensitivity and specificity differ. Posterior probabilities then depend on the prior and both error rates. A report that is usually correct overall can still be misleading for a rare type. The observation model should preserve the relevant asymmetry.

16.8 Unequal prior probabilities

Let prior probability of A be p and signal accuracy remain symmetric at q. After signal A, the posterior is

$$p_A^+ = \frac{pq}{pq + (1 - p)(1 - q)}. \tag{16.3}$$

After signal B, it is

$$p_B^+ = \frac{p(1 - q)}{p(1 - q) + (1 - p)q}. \tag{16.4}$$

The actor chooses hedge A when the posterior probability of A exceeds one half, hedge B when below, and is indifferent at equality under the symmetric loss structure.

If p is very high, even signal B may not overturn the prior. Following the signal mechanically is then suboptimal. For example, $p = 0.95$ and $q = 0.9$ gives posterior A after signal B equal to $0.095/(0.095 + 0.045)$, about 0.6786. Hedge A remains preferred.

This illustrates why a signal label is not itself a decision. The policy combines the observation with prior information and payoffs. In an unweighted reconstruction, there may be no justified

prior at all, so a posterior calculation would be inappropriate until that modeling choice is admitted.

16.9 An unweighted version

Suppose both types remain possible and no probabilities are assigned. An exact signal identifying the type allows the correct hedge in every world, producing loss 2. Without a signal, either hedge has worst-case loss 12, while doing nothing has worst-case loss 10. A minimax-loss policy therefore chooses no hedge before information and the correct hedge after exact information.

The criterion differs from the earlier expected-loss decision, which chose a hedge without a signal. This is not a contradiction. The uncertainty representation and decision rule changed. A system should report the dependence rather than label one answer universally optimal.

If a noisy signal can be wrong in either type and only support is known, both types may remain possible after either signal. The worst-case benefit can be zero. A probability of correctness above one half is a probabilistic restriction; it cannot be inferred from support alone.

16.10 Management and public investors

Suppose management already knows the type from internal contracts, while public investors see only aggregate statements. Management can choose the correct hedge with loss 2 even before public disclosure. The analyst should simulate that informed behavior if it is the declared policy.

A public announcement can then affect investor beliefs and financing offers without changing management's hedge choice. The same information event has different economic consequences for different actors. A single global belief state cannot represent this distinction.

If management's action itself is observable, investors may learn from it. A hedge choice can signal the hidden type under an assumed policy. That inference depends on the policy model. If management sometimes hedges for other reasons, the action is not a perfect signal. Endogenous disclosure and signaling require a richer model than an exogenous likelihood.

16.11 Endogenous observations

A company can choose when and what to disclose within its obligations. A lender can request information before making an offer. An auditor can discover discrepancies with probability depending on effort. These observations depend on actions and hidden state.

The observation kernel should then be written conditional on action as well as state. Treating disclosure as independent can distort the value of monitoring or the selection of companies that reveal information. A missing report may itself be informative only under an explicit model of why reports are missing.

AccDSL need not assume a strategic disclosure game in every application. It should allow exogenous observation models as a bounded profile and state when they are inadequate. If the query concerns financing responses to disclosure, the endogenous channel may be material and should remain an open modeling requirement.

16.12 Persistent type and repeated signals

Suppose two conditionally independent signals of accuracy 0.9 arrive about the same fixed type with equal prior. Two A signals give posterior A

$$\frac{0.9^2}{0.9^2 + 0.1^2} = \frac{81}{82}. \tag{16.5}$$

An A followed by B returns posterior one half under the symmetry and independence assumptions. These updates concern one persistent type, not a newly drawn type for each signal.

If the signals are copies of the same underlying source, conditional independence is false. Counting a duplicated report as independent evidence would overstate confidence. Two identical copies should not square the likelihood unless the observation process actually supplies independent information.

Source provenance therefore matters statistically. A news summary quoting a filing is not necessarily another independent measurement of the loan type. The system should preserve derivation links so that a likelihood model can avoid double counting.

16.13 Correlated errors

Suppose both signals share a common extraction error. With probability 0.1 the type label is reversed in the source, and both downstream reports repeat it. Two matching reports then have accuracy 0.9, not 81/82. The agreement supplies no additional independent confirmation.

A model with separate independent error terms would produce unjustified confidence. The remedy is a shared latent error or a source-dependence relation. If the dependence is unknown, sensitivity over plausible observation models is more defensible than assuming independence because it is convenient.

The same issue arises when several financial databases reproduce one filing or when multiple ratios use the same extracted amount. More records do not necessarily mean more evidence. The model should count informational content through its observation relation, not file count.

16.14 A belief-state calculation

In the finite probabilistic example, the posterior probability p_t of type A summarizes the evidence relevant to the hedge decision, provided the signal model and loss structure remain fixed. Expected losses are $2 + 10(1 - p_t)$ for hedge A, $2 + 10p_t$ for hedge B and 10 for no hedge.

The policy chooses the smallest of these expressions. Hedge A is preferred to B when $p_t \geq 1/2$. It is preferred to no hedge when $p_t \geq 0.2$. Hedge B is preferred to no hedge when $p_t \leq 0.8$. Thus at least one hedge is preferred throughout $[0, 1]$ in this particular cost structure, with the choice switching at one half.

This compact belief is sufficient because the future depends on the past only through the fixed type posterior under the declared model. If signal quality is itself unknown or past actions create commitments, p_t alone may cease to be sufficient. The state must then include those additional quantities or their joint belief.

16.15 Compression tests from the example

A proposed compression should preserve the joint type-signal distribution, action availability, costs and timing. It should reproduce expected loss 3 for informative signals and 7 for uninformative signals. It should refuse to merge those technologies for an adaptive policy query.

A separate test can merge states differing only in an irrelevant customer label when neither payoffs nor observations depend on that label. This establishes a valid reduction within the small model. Positive and negative examples are both needed: a system that never merges is safe from one error but does not solve the computational problem.

Changing the query to one fixed hedge can make a coarser compression sufficient. The approval should be scoped to that query. Reusing it for information acquisition or adaptive hedging without rechecking would repeat the original mistake.

16.16 What the case teaches about trees

The full tree has hidden types, signals and actions. Hidden type branches need not be visible to the actor. Signal branches define information partitions, and actions before a signal must be shared across indistinguishable worlds. A drawing that places separate decisions under each hidden type can accidentally imply forbidden knowledge.

A belief-state or symbolic implementation can encode the same problem without storing every path. It must still preserve the information partition. The tree is a useful explanatory projection of that structure, not its only possible storage form.

The central requirement is therefore not one point per possibility. It is preservation of the joint relation among persistent facts, observations and feasible decisions. This example gives exact numbers against which a proposed representation can be tested before it is applied to a much larger company.

Chapter 17

Continuous history families: projection, optimization, and threshold refinement

A finite list of histories is often an inadequate description even of a small company. Unknown rate exposure, customer allocation, asset age and settlement timing can vary continuously. The right question is frequently whether a conclusion holds over a constrained region. This chapter develops several exact small calculations and shows where they cease to be adequate. The methods are proposals for bounded profiles, not a claim that every company reconstruction is a linear program.

17.1 An affine exposure family

Let x be foreign-currency debt maturing soon and a the floating share of total debt. Suppose disclosed margins imply $0 \le x \le 50$ and $0 \le a \le 1$. A simplified next-period cash requirement under currency depreciation d and rate increase r is

$$Q(x, a) = Q_0 + dx + 100ra. \tag{17.1}$$

If d and r are nonnegative and the two unknowns are jointly unconstrained beyond their bounds, the maximum is $Q_0 + 50d + 100r$ and minimum is Q_0. These extrema occur at feasible corners. No grid is needed.

Now add a disclosure that the floating portion is entirely within the foreign-currency debt maturing soon, so $100a \le x$. The feasible family is no longer a rectangle. The minimum remains Q_0, but the maximum floating share is one half, giving maximum $Q_0 + 50d + 50r$. The relation changes the worst case even though debt total remains 100.

The condition is a stipulated example, not an inference from the margins alone. It illustrates why exact bounds must optimize over the joint constraints. Independently maximizing every marginal amount can overstate or understate the relevant result depending on signs and relationships. A constraint-aware bound carries the assumptions that make its corner feasible.

If the query includes a hedge whose receipt increases with x, the coefficient on x can change sign. Monotonicity should be derived from the full expression, not assumed from a variable's label. A larger foreign-currency exposure can have a smaller net cash effect when a linked hedge changes at the same time.

17.2 Linear programs as a bounded exact method

When observations, bridges and query are linear in continuous unknowns, a bound can be formulated as a linear program. Let z collect amounts and let $Az \le b$, $Ez = e$ define the family. The minimum cash query is

$$m = \inf\{c^{\mathsf{T}} z + c_0 : Az \le b, Ez = e\}. \tag{17.2}$$

The formulation separates a mathematical target from a solver. A solver can return an optimum, infeasibility, unboundedness or an unresolved numerical status. Each requires interpretation. Unboundedness may reveal a missing economic constraint rather than a truly unlimited company exposure.

A dual certificate can support a bound when its assumptions and numerical accuracy are checked. For a simple example, constraints $x \ge 0$, $y \ge 0$ and $x + y \le 10$ imply $2x + y \le 20$ because $2x + y \le 2(x + y) \le 20$. Equality is attained at $(10, 0)$, so the maximum is exactly 20. The inequality proof and feasible witness establish both sides.

A reported optimum without a feasible witness or trustworthy bound is weaker. Floating-point tolerances can matter near a covenant. Rational reconstruction or validated interval checks may be appropriate for a small certificate. The economic model should not change its threshold merely to accommodate solver tolerance.

Linear programming does not resolve categorical interpretations, nonlinear prices or integer security units automatically. Those additions can require mixed-integer, nonlinear or hybrid methods. The profile should identify which relations remain outside the exact solver and how their unresolved regions are retained.

17.3 Projection can preserve a useful small relation

Suppose receivables satisfy $A_1 = A_0 + S - K - W$, with nonnegative write-offs W, collections K and known credit sales S. If only A_0, A_1, S matter to a future query, eliminating K and W leaves $K + W = A_0 + S - A_1$. That sum is determined while its allocation is not.

A credit-loss stress depends on W or surviving customer exposure; a pure cash-history query depends on K. Eliminating both into an unnamed residual can make neither query answerable. A better compressed record retains the identified sum and the feasible allocation interval.

If $A_0 = 30$, $S = 40$ and $A_1 = 20$, then $K + W = 50$. With a note limiting write-offs to at most 5, collections lie in $[45, 50]$. A cash-flow disclosure of collections 48 then identifies write-offs 2 within this simplified bridge. Each additional source narrows the joint relation.

If currency movements or acquisitions are later admitted, the relation changes. A cached inference $W = 2$ must be invalidated unless those terms are established zero. This is an example of model expansion changing an earlier identified quantity. The system should preserve the old inference with its premises rather than silently rewriting its meaning.

17.4 Intervals and dependence

Interval arithmetic can bound expressions quickly, but repeated variables can cause overestimation. If $x \in [0, 1]$, the expression $x - x$ is exactly zero. Treating its two occurrences independently gives interval $[-1, 1]$. The dependency loss is entirely computational.

For the financial expression $C = 95 - I$ with $I \in [5, 7]$, interval evaluation correctly gives $C \in [88, 90]$. But evaluating $C + I$ from separate intervals gives $[93, 97]$ instead of exact 95. A relational representation or symbolic substitution preserves the identity.

Overestimation can be acceptable for an outer bound if it is labeled and used with the correct direction. It can become useless near a threshold or create spurious defaults. Refinement can retain shared symbols, affine forms or polyhedral constraints rather than independently widening every intermediate amount.

Nonlinear functions add further issues. A naive interval for $x(1 - x)$ on $[0, 1]$ is $[0, 1]$, while the exact range is $[0, 1/4]$. Completing the square, $x(1 - x) = 1/4 - (x - 1/2)^2$, identifies the maximum. A small analytic reformulation can be more valuable than a large random sample.

17.5 Piecewise contract payoffs

A floating coupon with floor l and cap u can be written $N\delta \min(u, \max(l, r + s))$ under a stated schedule. Unknown reference rate r partitions into floor, interior and cap regions. The function is continuous but has slope changes.

Suppose principal is 100, year fraction one, spread zero, floor 3 per cent and cap 6 per cent. For $r \in [0.02, 0.08]$, interest ranges from 3 to 6. A linear approximation using the midpoint rate five per cent and an unconstrained slope predicts 2 to 8, missing both contractual limits.

If the query is only the interest range, monotonicity gives exact endpoints. If the query includes a refinancing choice triggered by interest above 5.5, a split at $r = 0.055$ is needed in the interior region. If the cap changes after a covenant event, contract state must retain that dependence.

A piecewise formula can therefore represent many paths compactly. The implementation needs complete and disjoint guards, correct boundary inclusion and a record of the clause version. A missing cap is an economic-model error; an incorrect inequality at the cap boundary is an implementation error. Their repairs differ.

17.6 Nonmonotone outcomes

Many financial outputs are not monotone in a primitive shock. A rate increase raises floating interest but can increase income on cash, reduce a fixed-rate debt’s market value or trigger a favorable option exercise. A customer demand increase can improve revenue while creating a working-capital funding need.

Consider cash after a profitable sale with collection delay. Producing q units costs $4q$ immediately, while revenue $6q$ is collected later. Current cash falls as q rises even though eventual profit rises. A solver maximizing profit may select a quantity that violates interim liquidity.

If financing capacity is B and opening cash C_0, feasible production satisfies $4q \le C_0 + B$ before collection. The eventual profit objective $2q$ does not remove this constraint. Aggregating the period so that revenue and cost net immediately changes the feasible set.

A query-driven method must therefore include intermediate constraints and not rely on a generic sign assumption. Monotonicity can be proved locally within a regime and fail when a guard changes. The method should partition or return a qualified bound rather than extrapolate the local result globally.

17.7 Integer quantities and minimum lots

Shares, bond lots and physical units may be discrete. If a company needs net cash 10 and shares sell for 3 with no fees, a continuous calculation requires 10/3 shares. If only whole shares can be issued, four shares raise 12. The excess cash and dilution are real consequences of the lot constraint.

A minimum issuance size can make a small funding need expensive or infeasible. A fixed fee can make splitting one issue into two change net proceeds. These effects break the proportional invariance that holds for a simple linear loan.

The model should specify whether quantities are continuous approximations or exact units. For large public share counts, a continuous approximation may be immaterial for some queries. For a closely held company or indivisible asset, it may dominate the decision. The approximation needs a bound tied to the question.

17.8 Branch-and-bound interpretation

A mixed discrete-continuous family can be explored through bounds on regions. If a region's best possible result cannot improve the incumbent for an optimization query, it can be pruned under the method's correctness conditions. If a region is infeasible, it can be removed with evidence.

Pruning for computational convenience is different. A low-probability region cannot be removed from a worst-case query merely because it is unlikely. A region far from current data may still be admissible if the observation model permits it. The pruning reason belongs in the result record.

For a safety query, an outer lower cash bound above the requirement can close a region as safe. A feasible witness below it establishes possible failure. Regions whose bounds straddle the threshold remain unresolved and receive refinement priority. This approach can concentrate work where it changes the answer.

The procedure may still face exponential growth. Resource exhaustion should preserve the incumbent witnesses, certified bounds and unresolved regions. Reporting only the best found point hides the remaining uncertainty and can reverse the meaning of a worst-case result.

17.9 Sensitivity of identified sets

A small source revision can change a feasible family discontinuously when constraints meet at a boundary. Suppose exact observations require $x = 1$ and an interpretation requires $x \leq 1$. The family is nonempty. Revising the observation to $x = 1.001$ makes it empty if no tolerance or other interpretation is admitted.

This does not mean the company changed abruptly. It may reveal that exact equality was too strong for a rounded report. Observation precision and numerical tolerance should be audited before interpreting a discontinuity economically.

Conversely, a small genuine change can cross a contractual threshold and have a large effect. Smoothing it away would hide a real mechanism. The analyst must distinguish measurement uncertainty from contractual discontinuity. Both can occur near the same number but require different treatment.

A useful sensitivity report varies source precision, interpretation choices and economic shocks

separately. It identifies whether a conclusion is driven by evidence, modeling assumptions or a threshold in the contract. This is more informative than a single generic tolerance parameter.

17.10 A bounded implementation profile

An initial exact profile can support rational linear constraints, finite interpretation choices and piecewise affine event rules. It can return feasible witnesses, exact or certified bounds, and guarded outcome formulas. Nonlinear valuation and unsupported clauses remain explicit extensions.

The profile should reject incompatible units, unbound variables and missing event premises before solving. It should preserve unknown amounts as variables with justified domains. Arbitrary large bounds inserted to make a solver finite should be labeled as computational assumptions and tested for sensitivity.

A later nonlinear profile can use interval methods, constrained optimization or sampling, but must retain the original query target and distinguish local from global results. A local minimum of cash is not a global worst case. A successful solver exit code is not a proof of completeness.

The practical advantage of a bounded profile is that its claims can be checked. It does not redefine the whole mission to exclude difficult contracts. Instead, it provides a reliable core and a precise account of which additional methods are needed for broader company simulation.

17.11 Sharing a rate family without sharing away a borrowing right

Return to the capped floating coupon, but now ask whether the company can pay it. The company starts with cash of 5.5, other assets of 94.5 and debt of 100. Its equity is initially zero. One admissible history has a fixed coupon of 5; another has the floating coupon

$$I(r) = 100 \min\{0.06, \max\{0.03, r\}\}, \qquad r \in [0.02, 0.08]. \tag{17.3}$$

At the earlier observed rate of five per cent, both histories report interest of 5. A later rate of six per cent separates them: fixed interest stays at 5 while floating interest reaches 6. A journal history that retains only the earlier interest amount cannot answer the new payment question.

Each history also has two possible contractual interpretations. A committed facility may be available, or no facility may be available. In this stipulated example the facility lends 1, charges an immediate fee of 0.25 and can be drawn only when interest exceeds 5.5. These terms define a small test case; they are not inferred from the earlier statements. The four combinations of coupon type and facility availability remain separate possibilities. No probability is attached to them.

There is one decision date. Interest accrues before the decision, and the manager submits either a payment order or a combined draw-and-pay order. These are indivisible instructions. An instruction that fails a prerequisite is refused. Contingent instructions, such as *draw only if the bank permits it*, would enlarge the action set and require a fresh analysis.

17.11.1 The amount and the right determine different conditions

A payment order requires enough existing cash. A draw-and-pay order requires the contractual right, the strict trigger and enough cash after the fee. Write $b = 1$ when the facility is committed

and $b = 0$ otherwise. Feasibility is therefore

$$\text{pay:} \quad 5.5 - I \geq 0, \tag{17.4}$$

$$\text{draw and pay:} \quad b = 1, \quad I > 5.5, \quad 5.5 + 1 - 0.25 - I \geq 0. \tag{17.5}$$

Because interest never exceeds 6, an eligible draw leaves at least 0.25 cash. At interest exactly 5.5, ordinary payment remains possible and the draw is forbidden. Replacing the strict trigger with a weak inequality would change the contract at that point. At interest 6, the two facility interpretations have identical interest amounts but different feasible actions: the committed company can draw, while the company without the facility can execute neither order.

Accrual is independent of that execution result. The company first debits interest expense and credits interest payable by I. If payment refuses, cash remains 5.5 and interest payable remains I; equity has fallen to $-I$. A failed payment does not erase the expense or the creditor's claim. If payment succeeds, it removes the payable and cash becomes $5.5 - I$. If the draw succeeds, borrowing adds 1 of cash and debt, the fee reduces cash and equity by 0.25, and payment removes the payable. The resulting balance sheet satisfies

$$\underbrace{(6.25 - I) + 94.5}_{\text{assets}} = \underbrace{101}_{\text{debt}} + \underbrace{(-I - 0.25)}_{\text{equity}}. \tag{17.6}$$

These journal rules are supplied premises for the example. They demonstrate how refusal and contractual memory enter the comparison; they do not determine a reporting treatment for an actual instrument.

17.11.2 A continuous comparison can be exhaustive

For floating debt, the coupon changes formula at rates 0.03 and 0.06, and the payment trigger changes at 0.055. Suppose also that a coarse signal tells the manager whether the rate is at most 0.04. The six points

$$0.02, \quad 0.03, \quad 0.04, \quad 0.055, \quad 0.06, \quad 0.08 \tag{17.7}$$

then delimit every relevant change. Keep each point as a singleton and each interval between successive points as an open interval. The eleven resulting regions are disjoint and cover the entire closed rate domain. The fixed-coupon alternative can use fewer regions because it has no rate-dependent coupon or payment trigger.

Within each open region, every financial amount is affine and every signal and eligibility condition is constant. Two implementations therefore agree throughout that region when their affine coefficients and discrete results agree. At singleton regions, compare their values directly. If one implementation uses a coarser partition, compare both on a common refinement and establish that each refined region lies entirely inside one region of each implementation. Sampling a midpoint can identify a known affine piece, but it cannot establish that the list of pieces is complete.

The checked reference makes this comparison for the full journal, cash flows, balance sheet, payable, facility right, observations and action outcomes. Its independent comparator explicitly enumerates all coupon types, facility interpretations and the eleven rate regions. The smaller representation retains 32 mode-specific regions and shares 14 identical calculations. Every original mode and region still points to its calculation. These counts describe this example: sharing reduces duplicated results here, but supplies no general bound on memory use or runtime.

Even this small construction exposed a consequential implementation mistake. Solving the draw-liquidity equation using the interior formula $I = 100r$ gives a root at $r = 0.0625$. That formula applies only below the cap at 0.06. The root is irrelevant to the capped region and must not be inserted as though the interior formula still governed there. A root finder must check the domain of the formula that generated its candidate boundary.

Open regions also require careful treatment of extrema. On $0.055 < r < 0.06$, cash after drawing and paying is $6.25 - 100r$, so

$$\inf C = 0.25, \qquad \sup C = 0.75. \tag{17.8}$$

Neither endpoint is attained within that open region. The point $r = 0.06$ attains cash 0.25, while $r = 0.055$ belongs to the ordinary-payment region. Reporting an interval without its endpoint status loses a distinction needed for strict contractual conditions.

17.11.3 Information can prevent an otherwise affordable action

For the moment retain only histories with a committed facility. Every detailed history admits one of the two orders. This does not imply that the manager can select a feasible order using the information available at the decision date.

Under the coarse signal, *high* means only that $r > 0.04$. It includes floating rates with interest below 5.5, where payment is possible but drawing is forbidden, and rates above 0.055, where drawing is possible but ordinary payment fails. The manager must issue the same order at indistinguishable histories. Neither order works throughout this signal class.

Now suppose a second, explicit pre-decision signal reveals whether interest exceeds 5.5. The manager can pay when the trigger is false and draw and pay when it is true. This rule is feasible throughout the committed-facility family. Its feasibility comes from the additional information, not from a better numerical approximation. If unavailable facilities are restored to the family, some high-rate floating histories admit neither order. Even complete information cannot create borrowing capacity.

The exact condition is elementary in this finite action setting. Let $H(o)$ contain all retained histories with observation o, and let $A(h)$ be the actions feasible at history h. A deterministic policy feasible throughout the family exists exactly when

$$\bigcap_{h \in H(o)} A(h) \neq \varnothing \quad \text{for every observed class } H(o). \tag{17.9}$$

Necessity follows because one observation selects one action, which must work at every compatible history. For sufficiency, choose an action from each nonempty intersection. The finite collection of signals then defines a total policy. This argument depends on the supplied observation map, two atomic orders and robust feasibility across the family. It is not a conclusion about contingent instructions or a probability-weighted objective.

17.11.4 What remains safe when calculation is unfinished

A computation budget can stop evaluation of individual regions without removing them from the family. In the reference, the partition is constructed first and transitions are evaluated when requested. If only one region has been evaluated, the others remain explicitly unresolved. The coupon still has the valid outer bound $3 \leq I \leq 6$, but that bound alone does not identify a feasible payment policy. Eager construction of the partition itself can become expensive when more continuous variables or contractual alternatives are added.

Approximation needs a separate justification. Suppose an independently established bound gives $|I - \widehat{I}| \leq \varepsilon$ throughout a region. A strict trigger $I > K$ is certainly true when $\widehat{I} - \varepsilon > K$, and certainly false when $\widehat{I} + \varepsilon \leq K$. In the remaining band the bound permits both answers, so the region must be refined or left unresolved. The conclusion follows by comparing the entire allowed interval with K; sampled errors do not supply its uniform premise.

Observation thresholds need their own error bounds. Preserving the interest trigger while changing which rate signal the manager sees can still change the feasible-policy set. A reused calculation must therefore retain its rate domain, contract interpretation, clause version, horizon, query, observation map, action class and precision. Changing any of these requires reconsidering the grouping. The example provides an exact one-rate, one-date reference for that discipline. Persistent hidden terms across several dates, higher-dimensional regions, nonlinear guards and defensible approximation bounds remain further method problems.[1]

17.12 Repayment changes tomorrow's feasible payments

The previous payment calculation ends as soon as interest has been settled. A facility used to make that payment may have to be repaid before the next coupon. To see why the horizon matters, carry the same company forward for one more date. The fixed or floating coupon interpretation persists, as does the existence or absence of the facility. The two reference rates may vary independently within the earlier interval. This rectangular rate family is an assumption for the example, not a probability model.

The facility permits one draw of 1 during the two-date horizon and charges the same immediate fee of 0.25. A first-date draw must be repaid on the second date, after an operating cash receipt and before the second coupon. Repayment extinguishes the facility debt but does not restore the right to draw. The original principal of 100 matures beyond the horizon. We omit tax and interest on the facility to isolate the timing of cash and the persistence of the drawing restriction.

Let I_1 and I_2 denote the two coupons and R the operating cash receipt on the second date. In the fixed interpretation both coupons are 5. In the floating interpretation each is calculated from its own reference rate using the earlier floor and cap. Receiving one coupon of 5 does not by itself reveal the debt type: floating debt can produce the same amount. Nor does it reveal whether the facility exists.

17.12.1 The first order leaves a different second opening

Ordinary payment on the first date leaves cash $5.5 - I_1$. Drawing and paying leaves $6.25 - I_1$ and facility debt of 1. On the second date the operating receipt increases cash and income by R, after which any outstanding facility debt is repaid. Cash available before paying the second coupon is therefore

$$C_2 = \begin{cases} 5.5 + R - I_1, & \text{after ordinary first-date payment,} \\ 5.25 + R - I_1, & \text{after a first-date draw.} \end{cases} \tag{17.10}$$

The second line follows by adding R to $6.25 - I_1$ and subtracting the repayment of 1. Although the facility improved first-date liquidity, its fee leaves less cash available over the complete two-date sequence.

[1]The derivation and independently authored comparison are retained in `data/accounting-master/continuous-guards-20261026/`. The accepted result is in `attempt-02/`; the earlier failed partition and its repair remain available separately.

Compare two supplied receipt amounts, $R = 6$ and $R = 7$. When both coupons are 6, the first payment requires a draw and leaves cash 0.25. The consequences at the second date are:

Operating receipt	Cash after repayment	Coupon due	Payment result
6	5.25	6	Refused; payable 6 remains
7	6.25	6	Paid; cash 0.25 remains

The receipt of 6 cannot fund the second coupon. Omitting repayment would incorrectly show 6.25 cash. Allowing another draw would also reverse the answer: the extra 1 less its 0.25 fee exactly fills the 0.75 shortfall. Both mistakes erase a consequence of the first decision.

Before either payment order, the second coupon accrues as interest expense and interest payable. A refused order leaves that payable and the earlier receipt, repayment and fee entries intact. If a first-date order already refuses, this particular reference returns that incomplete prefix without inventing second-date states. Modeling later default, recovery or renegotiation requires additional events.

17.12.2 The relevant boundary now joins both rates

Following a first-date draw, ordinary second payment requires $C_2 - I_2 \geq 0$. Substituting the cash expression gives

$$I_1 + I_2 \leq 5.25 + R. \tag{17.11}$$

For a receipt of 6, the boundary is $I_1 + I_2 = 11.25$. Within the interior floating-coupon pieces it becomes a diagonal boundary in the two rates, $100r_1 + 100r_2 = 11.25$. Separate bounds on the two marginal rates do not describe which joint histories cross it. Equality permits payment; larger coupon totals cause refusal.

For a receipt of 7, the corresponding bound is 12.25. Each coupon is at most 6, so their sum is at most 12 and ordinary second payment is feasible throughout the first-draw region. After ordinary first payment, $I_1 \leq 5.5$; even the receipt of 6 then supplies at least 6 before the next coupon. The failure with the smaller receipt is thus concentrated in histories that needed an earlier draw and then face a sufficiently large combined coupon.

The exact comparison first takes the products of the earlier point and open rate intervals. It then splits those regions at the later linear cash conditions. Every resulting balance and journal amount is affine in (r_1, r_2), while each action condition has a fixed truth value. The independent financial calculation uses the closed-form cash expressions above. It checks that accepted regions cover exactly the feasible domains, that refusals cover the remainder, and that both calculations produce the same complete journal and financial state.

The geometric calculation also retains strict boundaries. On a bounded region described by rational linear inequalities, its closed relaxation is generated by finitely many vertices. The average of all those vertices satisfies a required strict inequality whenever at least one vertex satisfies it strictly: every other vertex has nonpositive slack, so their average remains strictly negative. If all vertices lie on its boundary, no point of the closed region can satisfy that inequality strictly. Applying this reasoning to every strict constraint decides whether the original region contains a point. Extremum attainment is checked by adding the optimizing equality and testing the resulting face in the same way. These arguments apply to the supplied bounded linear regions; a nonlinear or unbounded problem needs another method. The financial comparator shares these geometric primitives, so its agreement does not constitute an independent solver validation.

17.12.3 More cash does not reveal a hidden contract

With a committed facility and a receipt of 7, the refined first-date signal supports the same initial rule as before: pay when the trigger is false and draw when it is true. On the second date the manager observes whether the coupon exceeds 5.5 and whether available cash covers it, together with the earlier signal and chosen order. Ordinary payment is feasible throughout this family.

Some histories also permit a second-date draw: the first coupon was paid without drawing, the facility remains unused and the second coupon exceeds 5.5. There are two such reachable observation classes in this example, and either ordinary payment or drawing is feasible in each. Consequently four policies satisfy the stipulated feasibility conditions. This count does not rank their value. Optional borrowing incurs the fee and leaves debt beyond the current horizon.

Under the coarse first-date signal, the manager still cannot distinguish histories requiring opposite first orders. Raising the later receipt does not repair that information problem. Retaining histories without a facility also leaves some first payments unfunded. A correct continuation therefore preserves contractual memory, cash timing and the information available to the actor together. Renewing the facility, redrawing the debt type or revealing an unobserved trigger would change the problem being solved.[2]

17.13 Connecting the history to accounting execution

A financial path can be correct in a small mathematical model and still be translated incorrectly into a company's accounting system. The preceding example treats equity as a single amount after income and expenses. An accounting book usually distinguishes capital, operating income, interest expense and fees. Those distinctions matter when the simulator must produce an income statement, cash-flow statement and retained claim history as well as a closing balance sheet.

The checked connection uses the existing common accounting executor with explicit definitions of the selected events. It does not put the earlier Python posting calculation inside that executor. The comparison instead asks whether independently specified methods produce the same full accounts, journals, cash-flow origins and contractual memory. Two different account charts test whether a method depends on accounting roles rather than convenient account names.

17.13.1 Equity agrees only after the accounts are mapped

Let B_k be an account's signed balance, positive for a debit and negative for a credit. Assets and expenses then normally have positive balances, while liabilities, capital and income normally have negative balances. In this example the reduced equity E used in the earlier calculation corresponds to

$$E = -B_{\text{capital}} - B_{\text{income}} - B_{\text{interest expense}} - B_{\text{fee expense}}. \tag{17.12}$$

For example, an operating receipt of 7 gives the income account a credit balance of -7. Two coupons of 6 give interest expense a debit balance of 12, and a facility fee adds expense of 0.25. With no opening capital, the reported equity after profit is $7 - 12 - 0.25 = -5.25$. This equals the reduced model's equity without hiding the income statement inside the capital account.

[2]Complete symbolic states, accepted-domain checks, geometric diagnostics and the retained journal-identity repairs are in `data/accounting-master/two-date-continuous-memory-20261026/`. The accepted run is `attempt-04/`. This is an exact small reference; demand-driven two-date compression, checkpoint/resume guarantees, source revision invalidation and integration with the common typed executor remain separate work.

An interest accrual I adds I to expense and $-I$ to interest payable. Under the mapping, equity falls by I and the creditor's claim rises by I. Paying the coupon reduces cash and removes the payable, leaving profit unchanged. A combined draw-and-pay instruction has the following signed account changes:

$$\Delta B_{\text{cash}} = 0.75 - I, \qquad \Delta B_{\text{facility debt}} = -1, \tag{17.13}$$

$$\Delta B_{\text{fee expense}} = 0.25, \qquad \Delta B_{\text{interest payable}} = I. \tag{17.14}$$

Their sum is zero. The cash-flow record separately retains borrowing of 1, the fee payment of 0.25 and the interest payment of I. Netting those amounts in the journal's cash entry must not erase their origins in the cash-flow statement.

Repayment removes the facility liability and reduces cash by the same amount. Its accounting balance becomes zero, but the separate memory that the facility has already been used remains true. Each method also checks that interest payable agrees with accrued coupons less paid coupons, that the event occurs at the declared phase and date, and that its prerequisites are satisfied. A refused payment leaves the accepted prefix unchanged, including the coupon already accrued.

For the concrete execution comparison, the two decisions are assigned to 30 June and 31 December 2027 within one supplied reporting period. The coupon amounts remain the one-period charges of the earlier example. These dates test ordering; they do not supply an actual instrument's day-count convention. Each comparison checks the complete account balances, income statement, balance sheet, gross cash flows, journals and memory. Applying the equity mapping also recovers the earlier independent financial calculation.

17.13.2 A checked path is not a checked continuous region

The common executor accepts concrete rational amounts. The selected comparison therefore uses exact points at the coupon cap, at the first trigger, and at and immediately to either side of the later joint cash boundary. Those executions exercise important failure mechanisms. They do not by themselves establish that every rate in a continuous region produces the correct accounting result.

Within a region where the guards are fixed, the amount relations above are affine. If the opening accounts map correctly, each event has the stated affine effect, and the same guard, timing and memory conditions govern both descriptions, induction over the accepted events gives the same mapped financial quantities. The conditions carry the argument: a wrong guard or lost drawing right can change which event occurs even when its monetary formula is correct. A general checker must verify those conditions and every supported method, rather than replace that obligation with a larger sample of rates.

The concrete evidence at this stage has two parts. The exact continuous reference establishes its financial formulas and feasible domains. The concrete common-engine comparison checks complete accounting execution at declared points. The explicit mapping and local event derivation motivate the selected symbolic certificates in Section 17.14; extending them to general methods remains a separate problem. Source-backed terms and actual accounting applicability also remain distinct from agreement between two synthetic representations.

17.13.3 Stopping computation must preserve what remains unknown

Suppose the simulator has executed the first coupon and then reaches its computation budget. A useful saved result contains the complete accepted book and journal, the next command, and

the commands still pending. It must also retain the alternative histories and regions that have not been evaluated. A successful payment on one path supplies no reason to delete a different path that may fail tomorrow.

The selected checkpoint reference validates a saved concrete prefix by replaying it from its declared opening and method registry. It compares the complete transition records and current state before continuing. The tested interruptions recover the same result as uninterrupted execution, including the first-date and second-date refusal prefixes. Altering cash while keeping the book balanced, changing paid-coupon memory or removing the used-facility flag fails that comparison. Merely recomputing a checksum on the altered record would not establish that the altered history followed from its opening.

A zero budget attempts no new financial event. It can still incur the cost of checking the saved prefix. More generally, the budget in this reference counts new event attempts, while replay validation costs additional work. A practical implementation would need a separately justified way to reduce that cost. The present result establishes a small exact recovery comparison, not an efficient or crash-durable storage system.

The queue also distinguishes a completed concrete program from a resolved continuous region. For the two-date example, the declared roster contains four persistent contract combinations, 11×11 rate-atom pairs, four order pairs and two receipt hypotheses:

$$4 \times 121 \times 4 \times 2 = 3{,}872 \quad \text{region and instruction obligations.} \tag{17.15}$$

A finite witness cannot mark any of those whole regions complete. Their identities remain unresolved under this concrete checkpoint even when individual concrete jobs succeed or refuse; Section 17.14 supplies separate certificates for the selected whole regions. Within the finite job queue, pending, completed and refused jobs form an exact partition. A refused job retains both its payable and its unexecuted commands; it is a completed assessment of that particular instruction sequence, not proof that every alternative has failed.

Finally, reuse depends on what was asked and what information was available. The saved record names the source revision, clause, actor's observation map, available actions, horizon, query, precision and method. A changed scope refuses reuse under the old record. An earlier query can still use its earlier explicitly supplied scope, preserving the distinction between later knowledge and information available at the time. Those bindings do not authenticate the source or establish when a real manager received it. Source admission, extension beyond the selected symbolic domains, efficient refinement and durable recovery require their own evidence.[3]

17.14 Proving a result for an entire rate region

The second coupon gives a useful test of what a simulation has actually established. Suppose the company drew its once-only facility at the first date. After the operating receipt and repayment, cash available for the second coupon is $5.25 + R - I_1$. A payment succeeds exactly when

$$I_1 + I_2 \le 5.25 + R. \tag{17.16}$$

At $I_1 = I_2 = 6$, a receipt of 6 leaves only 5.25 before the payment, so the coupon remains unpaid. A receipt of 7 leaves 6.25 and permits payment, with 0.25 left over. A simulator that happens to test only the latter case has learned nothing about the former. Even a large set of

[3]The complete common-engine and checkpoint comparisons are retained in `data/accounting-master/guarded-memory-runtime-20261026/`, respectively under `attempt-01/` and `checkpoint-attempt-01/`. The former checks concrete accounting execution; the latter checks replay and queue preservation. Neither promotes a concrete witness to a whole-region certificate.

successful sample executions does not exclude an omitted boundary or an incorrect condition between the samples.

The selected example now has a stronger, exact connection to the declared accounting methods. It proceeds in three stages: establish each method's equations, substitute those equations through the dated sequence, and divide the rate domain wherever a condition changes. This division concerns computational work. It does not assert that a company, a manager or an observer learns which historical explanation is true.

17.14.1 Compare the method with an equation, not another sample

An affine monetary expression is a constant plus a weighted sum of quantities. For this example, those quantities include account balances, accrued and paid coupons, event amounts and the supplied parameters I_1, I_2, R. Write an expression as

$$f(x) = c + \sum_j a_j x_j, \tag{17.17}$$

where each coefficient and the constant are exact rationals. The accounting language expresses these calculations through constants, references, addition and subtraction. A constant contributes to c; a reference contributes one coefficient; addition and subtraction combine the coefficient lists. Induction on these four constructions shows that the resulting list represents precisely the original expression. Two identical lists therefore give the same amount for every known typed valuation, rather than merely for the values tested in a run.

For instance, the combined borrowing and payment changes cash by $3/4 - I_t$. The corresponding coefficient of the due coupon is -1 and the constant is $3/4$. Doubling both interest expense and the credit to interest payable would still balance the journal. It would change these independently specified account relations and fail the comparison. Balance is a necessary accounting constraint, but it cannot establish that the right transaction was recorded.

The comparison also preserves the conditions on the transaction. A numerical comparison is represented by its exact left-minus-right affine expression and its comparison sign. The facility trigger $I_t > 11/2$ retains a strict sign; replacing it by $I_t \geq 11/2$ fails even if a chosen set of sample rates misses equality. Boolean conditions retain their operands, and memory assignments retain their source and destination. The same check covers cash-flow origins: a borrowing receipt, fee and interest payment cannot be replaced by one net cash movement.

Parameters enter through a separate selected template. Its coupon and receipt amounts, contract mode and facility right become ordinary typed literals before execution by the common runtime. Exact placement checks bind the parameters to the opening state, event inputs and selected-amount conditions. The surrounding method records retain their account maps, units, dates, schemas, declared dependencies and refusal policies. This is a certificate for the inspected selected profile. It does not extend the runtime to arbitrary symbolic programs or prove an unrestricted compiler correct.

17.14.2 A correct event still needs a correct predecessor

Local agreement leaves a temporal question. When a method reads the payable or the facility balance, does it receive the state produced by the preceding accepted event? Substitution answers that question for the selected sequence. Every posting and memory assignment reads the same pre-event state; the resulting changes become visible together. The next event receives the complete new state, including accrued and paid coupon history.

After a first-date draw and payment, cash is $25/4 - I_1$. The receipt adds R, and repayment subtracts the borrowed unit. Substituting successively gives

$$C_{1+} = \frac{25}{4} - I_1, \tag{17.18}$$

$$C_{\text{receipt}} = \frac{25}{4} - I_1 + R, \tag{17.19}$$

$$C_{2-} = \frac{21}{4} - I_1 + R. \tag{17.20}$$

The second accrual changes expense and the payable, leaving this cash unchanged. Substituting C_{2-} into the payment condition $I_2 \le C_{2-}$ yields equation (17.16). Repayment removes the facility liability while leaving the used-facility flag true. Omitting that memory would admit a second draw which the contract in this example forbids.

The checked substitution compares all signed accounts, memory fields, gross flows and conditions with independent complete formulas. It includes both refusal positions. A first refusal leaves the first coupon accrued; a second refusal retains the accepted first-date events, receipt, repayment and second accrual. It does not replace the final payable by zero merely because the proposed payment appeared in the command sequence.

At this stage a branch can be written down even when no rate makes it feasible. For example, the formal second-draw branch can be constructed after a first draw, but its unused-facility condition is false. Distinguishing an algebraically described branch from a possible history is the task of the next stage.

17.14.3 Why an empty branch needs a proof

Within each rate atom the capped coupon is constant or linear in the corresponding rate. A point atom uses its exact constant coupon. Substituting those pieces into the conditions gives inequalities in two variables, r_1 and r_2. Each comparison divides the current region into the points satisfying it and the points satisfying its exact negation. For $f \le 0$, the other child is $-f < 0$; for $f < 0$, it is $-f \le 0$. Equality belongs to one child, never to both and never to neither.

A discarded child carries a short arithmetic reason why its inequalities cannot hold together. Suppose the child requires

$$a_i^{\mathsf{T}} r + c_i \le 0, \qquad i = 1, \dots, m, \tag{17.21}$$

with some signs possibly strict. Choose rational weights $\lambda_i \ge 0$ that cancel both variable coefficients:

$$\sum_i \lambda_i a_i = 0. \tag{17.22}$$

Adding the weighted inequalities then requires $\sum_i \lambda_i c_i \le 0$. If that constant is positive, the requirement is impossible. A zero constant also proves impossibility when at least one strict inequality receives positive weight: the sum would require $0 < 0$. These implications follow directly by multiplying and adding the original inequalities. The verifier checks the weights against those original rows rather than trusting a solver's declaration that a branch is empty.

The distinction matters even in one dimension. The pair $x < 0$ and $-x \le 0$ has no solution: equal unit weights produce $0 < 0$. Weakening the first inequality to $x \le 0$ admits $x = 0$. A certificate that discards the strictness would incorrectly remove that boundary point.

The reference producer eliminates the two variables with exact rational arithmetic and retains the weighted combinations it uses. A separate checker verifies each proposed contradiction. A retained leaf instead carries a point satisfying every inequality, including every strict sign. That

point proves the leaf is nonempty. The affine method and sequence argument, together with the predicates along the branch, establishes the financial relation throughout the leaf; the witness alone does not.

17.14.4 What can now leave the unfinished-work list

For the declared two-date model, all 3,872 rate-region and instruction obligations have a checked partition. The partition contains 918 nonempty accepted leaves and 2,960 nonempty refusal leaves; 4,772 additional children are discarded with checked contradiction certificates. These are computational counts, not probabilities. They depend on the selected atom refinement and cannot be interpreted as the likelihood of repayment or default.

The saved proof archive is re-read before its jobs enter a new certified queue. Each entry binds the source revision, clause, method and sequence certificates, domain, query and horizon. A pending job leaves the queue only with the corresponding verified proof. Zero-budget and interrupted JSON continuations retain the same final partition as uninterrupted processing. Checking the saved proof registry costs work in addition to the budget for newly completed jobs.

The earlier concrete checkpoint correctly kept these regions unresolved under its own evidence. The new result supplies separate whole-region certificates; it does not reinterpret the earlier witness as a proof. The distinction is useful beyond this example: completing a computation should mean satisfying the evidence obligation attached to that job, not merely reaching its last instruction.

This result concerns known exact values, the supplied coupon and once-only facility terms, two bounded rates and the selected event sequence. It leaves a further decision problem. An actor can choose differently only when the actor's information distinguishes the relevant histories. Consequently, a list of financially feasible region-specific actions is not yet an admissible policy. Section 17.15 binds these leaves to the supplied observation map and checks decisions that must agree across indistinguishable histories. Section 17.16 supplies a conditional source-revision and actor-access binding for the selected public-document profile; actual company evidence and broader information structures remain separate requirements. Higher-dimensional and nonlinear relations, unknown-value semantics, actual accounting authority, numerical quantisation and crash-durable recovery require additional evidence.[4]

17.15 A feasible action for each world is not yet a policy

Return to the first coupon, when cash is 5.5 and the once-only facility is available. Two companies can produce the same coarse observation that the market rate is high. In one, a fixed coupon is 5; in the other, the floating coupon at the cap is 6. Paying is feasible for the first company, while drawing is barred by the strict trigger. The second company cannot pay from existing cash and needs the facility. If the manager sees only the coarse high signal, the two individually feasible instructions do not form a decision rule the manager can follow. The rule asks for information that the manager does not have.

This is the difference between a financially possible action and an information-feasible policy.

[4]The local derivations and immutable comparisons are retained under `data/accounting-master/symbolic-method-certificate-20261026/`, `symbolic-path-composition-20261026/` and `exact-region-certificates-20261026/`. The last directory includes failed diagnostics, the exact partition archive, an independent weighted-sum checker and separate saved-archive/queue validation. No new literature proposition is asserted by these local calculations.

Let w denote a possible history and $O_t(w)$ the manager's observation history when the date-t instruction is chosen. An admissible policy must satisfy

$$O_t(w) = O_t(w') \quad \Longrightarrow \quad \pi_t\big(O_t(w)\big) = \pi_t\big(O_t(w')\big). \tag{17.23}$$

Here O_t includes remembered earlier observations and the manager's own earlier instructions. The condition gives one decision to each information cell. It does not require that the underlying histories become identical or that the manager eventually learns the contract type.

17.15.1 Refining an observation without inventing knowledge

The selected first coarse signal reports whether $r_1 \le 0.04$. The refined signal adds whether the first coupon exceeds 5.5. In the pair above, refinement distinguishes the two required first instructions. It still does not reveal the facility right or generally identify a persistent coupon type. At the second date the manager remembers the first signal and instruction, then observes whether the second coupon crosses the trigger and whether the cash then available covers it. The receipt hypothesis is held fixed for each comparison.

A financial region need not have a single observation label. For example, the condition that drawing plus cash covers a coupon is weaker than the condition that existing cash alone covers it. A region on which a draw is financially possible can therefore straddle the manager's ordinary cash-coverage observation. The checked construction splits that region on the cash-coverage inequality and its exact complement. Empty children carry the contradiction proofs of the preceding section; nonempty children retain their symbolic financial outcomes. Whole-atom proofs likewise establish that the first low/high signal and each coupon-trigger signal are constant where they are used. The observation is thus valid throughout the cell, rather than assigned from one representative rate.

This timing is consequential. The first decision does not receive the second rate or a later cash balance. The second decision uses cash before its proposed payment. Using terminal cash would let the result of an action determine what the manager supposedly observed before choosing it. Including the hidden facility right in the signal would change the information problem, even if every resulting journal still balanced.

17.15.2 Which instructions survive every indistinguishable history?

For each possible first decision rule, retain every history consistent with that rule. A first-date refusal rejects the rule. At each reachable second-date information cell, intersect the allowed instructions across the retained histories. An empty intersection means there is no action that works in every history the manager still considers possible. Otherwise choose one instruction from each cell's intersection. Cells that cannot be reached under the first rule need no arbitrary decision in this comparison.

An independent formulation expresses the same condition as Boolean restrictions. Let X_o say whether the first instruction at signal o is draw, and Y_h the second instruction at remembered history h. A certified first-date refusal under instruction a forbids $X_o = a$. A certified second-date refusal under first instruction a and second instruction b forbids their joint selection:

$$(X_o \neq a) \ \vee \ (Y_h \neq b). \tag{17.24}$$

Enumerating the satisfying assignments and retaining only reachable second-date choices gives the complete policy set. This independently authored constraint calculation agrees with the action-intersection calculation and with the earlier finite-policy reference. Agreement concerns whole policies, not only the number found.

Across the eight supplied problems—receipts 6 or 7, coarse or refined observations, and committed or mixed facility rights—only the receipt-7, refined-observation, committed-right problem has robustly feasible policies. It has four. The first instruction is fixed by the refined trigger. After a first draw, the exhausted facility cannot be used again and the second instruction must be pay. After a first pay, two reachable second-date cells allow either pay or draw, giving $2 \times 2 = 4$ reachable policies. The model supplies no preference or objective that ranks them.

The failures have different causes. Coarse information combines incompatible first instructions. A family containing an unavailable facility includes a high-coupon world with no feasible first instruction. Receipt 6 leaves the double-high path short after repayment, despite knowing the trigger and having the original right. Removing such a world or ignoring its refused branch changes the robust question; it does not solve it.

These results concern the explicit family of possible histories and the supplied observation maps. They assign no probabilities to that family, and a feasible robust policy is not a forecast or an optimal policy. Applying the calculation to an actual manager next requires evidence about source revisions, access times and observations. That evidence must preserve what was available when a decision was made rather than replacing it with a later corrected account.[5]

17.16 Which source could this actor use at this date?

The four feasible policies in the preceding example depend on two supplied interpretations: which receipt and facility-right family is possible, and which signals the decision maker observes. A revised document can change either interpretation. It can also leave both interpretations unchanged while correcting the evidence supporting them. The simulator needs to distinguish those cases before it decides which calculation to repeat.

Consider a correction released on 3 January to a document used for a decision on 2 January. The correction may improve a retrospective account of the company, but it cannot be inserted into what the actor knew on 2 January merely because the simulator downloads it later. Conversely, a document may have been available to the actor on 1 January but entered the simulator on 3 January and passed internal review on 4 January. On 4 January the system can reconstruct the earlier information set if the retained evidence supports that access. It has not established that the reconstruction itself was run on 2 January.

These are distinct dates with distinct subjects. Publication concerns the document's release. Access concerns a particular actor. Ingestion and review concern the information system. Effective time concerns the economic period or event to which the interpretation applies. A publication timestamp alone does not establish personal access; a download timestamp does not establish either the original publication or the document's economic effect.

17.16.1 Select revisions before selecting the policy problem

The selected construction uses an immutable record for each supplied interpretation. The record identifies the source bytes, the interpretation's subject, the previous revision it replaces, its effective time, and separate observations of publication, access, ingestion and review. The caller supplies the evidence accepted for these timing observations. Hashes preserve the identity of those bytes and observations; they do not establish their truth.

[5]The complete observation refinements, whole-atom proofs, two policy formulations, source locks and eight complete comparisons are retained in `data/accounting-master/certified-observation-policy-20261026/attempt-01/`. The reference checks all 3,872 saved financial jobs, with 1,766 complementary observation refinements. These are exact calculations for the selected synthetic information structure.

For a request, retain revisions that are effective for the chosen economic event, available to the actor by the decision time, and ingested and reviewed by the system's reconstruction cutoff. Known later revisions are excluded from that earlier view. A missing relevant time leaves the selection unresolved. Within a subject, select a revision only if it is the unique remaining head of its replacement chain. Two incomparable available corrections preserve a conflict. Retraction of a required interpretation leaves a missing input rather than silently restoring its earlier assertion.

This procedure is deliberately restricted to a declared source scope. An unrelated management note cannot change the coupon interpretation merely because it appears in the same registry. A relevant correction must be inspected again on each request, however. Comparing a cached view identifier without checking for newly available revisions could miss exactly the change that should invalidate the result.

In the reference example, one required subject supplies receipt 6 or 7 and the mixed or committed facility-right family; the other supplies the coarse or refined observation scheme. These choices map to the eight previously certified problems. They preserve the same possible persistent contract types, complete financial rows and rejected actions within the selected family. No convenient prior or likelihood is introduced to make a source selector callable. The question remains whether one policy works in every history the actor cannot distinguish.

The timing checks reuse the existing source and policy-information mechanisms. The selected adapter accepts explicit whole-second UTC instants and public-document evidence with a separate supplied actor-access observation. It refuses finer or incomplete clock representations instead of rounding them into eligibility. Section 17.17 extends this selected reference to exact clock ranges and an explicit private-delivery route; broader information practices remain separate. An unresolved access record is therefore a substantive result about this request, not an instruction to substitute the publication date.

17.16.2 Reuse a calculation without reusing an obsolete source binding

Suppose a replacement document supports exactly the same receipt, facility-right family and observation scheme. The set of robust policies can remain unchanged. Nevertheless, a report that still cites the replaced source would give the wrong explanation of which evidence supported the current request. Numerical equality and evidential identity answer different questions.

Let F denote the retained financial relations and certified rows, s the selected receipt and facility family, and o the observation scheme. Write the complete reachable policy calculation as

$$\mathcal{P}(F, s, o). \tag{17.25}$$

Its numerical inputs do not include a document's filename or the system's ingestion date. Those records determine whether the interpretation (s, o) is available for the actor and request. The reported result therefore needs both $\mathcal{P}(F, s, o)$ and the source, revision, actor and cutoff records that justify using these inputs.

The checked dependency relation makes this separation explicit. Certified financial evidence supports the chosen financial scenario; that scenario and the observation scheme support the policy set. Separate source bindings attach the selected financial and observation interpretations to their revision histories. The final actor binding joins the numerical policy set to both source bindings and to the decision, reconstruction and effective cutoffs.

A same-meaning correction changes a source binding and the final actor binding while preserving the numerical policy set. A change from coarse to refined information preserves the financial relations but requires a new observation and policy calculation. A changed receipt or facility

family changes the financial scenario and its dependent policy problem. Changing only the actor or cutoff requires a new actor binding even when the selected interpretations and numerical answer happen to be identical. A correction known to arrive after the old cutoff changes none of that earlier view's selected dependencies.

Reuse is justified by those dependencies, not by observing that the new and old policy counts coincide. The reference compares the entire reused policy set and its complete world-row population with a fresh calculation. It also requires the saved inputs and method bytes to match expected identities supplied outside the transported bundle. Removed or duplicated rows, changed hidden rights or signals, altered policy references and rewritten prior results are rejected. Recomputing an inner checksum after changing a saved result does not make it the previously accepted result.

The scope remains conditional. The caller must establish which source and timing evidence is accepted, and a complete history family for an actual company may require many more subjects than these two. The result shows how revision-sensitive selection can coexist with numerical reuse while preserving the robust question. It establishes neither authenticated storage nor crash recovery, and the small exact comparison supplies no performance ranking.[6]

17.17 A date range can leave several information views possible

A filing marked as published on 2 January does not establish whether a manager had read it at the beginning of that day. Nor does a correction received within a narrow interval around a decision establish whether the original or corrected interpretation was available for that decision. These are questions about the order of events. Assigning midnight or the interval's midpoint would settle them by assumption.

The selected successor retains that uncertainty explicitly. An exact timestamp is a point; a calendar day in UTC is a half-open interval from its midnight to the next midnight. A supplied time range has the same half-open convention. An unknown access time remains unknown. Exact rational seconds preserve differences smaller than a microsecond and equivalent fixed-offset spellings without passing through a floating-point or microsecond clock.

If the access time lies in $[a, b)$, a decision at b is certainly after access. A decision at a can coincide with access or precede it: the lower endpoint belongs to the interval. The distinction disappears if an implementation replaces the interval with a single representative time. It also disappears if a strict upper bound is inadvertently changed to a closed one.

17.17.1 Availability depends on the route and on joint chronology

A public-document retrieval and a private delivery impose different conditions. In the first route the source must have been released publicly before the claimed access. In the second, an actor-addressed delivery can establish access before any public release. The private route neither invents a public timestamp nor makes the document available to other actors. Both routes still require evidence about that actor's access; publication alone cannot supply it.

The earlier whole-second reference did not distinguish these routes. One of its retained cases places actor access before a later public release. Under the new, explicit public-retrieval premise, that chronology is inconsistent. It can be meaningful under a private-delivery premise.

[6]The immutable source-view cases are retained in `data/accounting-master/certified-source-views-20261026/`; the separate dependency and complete cold-versus-reused comparisons are in `data/accounting-master/source-policy-invalidation-20261027/`. Both preserve their supplied source scope and withhold actual-source, accounting, real-actor and forecast admission.

Preserving the difference is necessary: changing the access mechanism changes the question rather than improving timestamp precision alone.

Clocks are also dependent. Authorship precedes ingestion, ingestion precedes review, and authorship precedes access. Public retrieval additionally follows publication. The selected revision profile requires nondecreasing authorship, ingestion, review and same-actor access along a replacement chain, while permitting retrospective effective dates. These are declared premises of this bounded source model. A document practice that violates one of them requires another profile; a corrected economic effective date is not forced forward merely to preserve a convenient time ordering.

An unknown clock need not make every conclusion unknown. If a private access time is supplied before the decision and the document was necessarily authored before that access, the chronology can establish the relevant authorship inequality without identifying its timestamp. The result must retain the unknown-time marker. Logical implication from other premises does not turn an unobserved time into an observed fact.

17.17.2 Partition clock possibilities without enumerating timestamps

Let x_e be the time of event e, measured in exact UTC seconds on the retained calendar axis. Point and interval observations, chronology and cutoff conditions become difference inequalities. A non-strict edge represents

$$x_v - x_u \le c, \tag{17.26}$$

with a strict variant when the right endpoint is excluded. A fixed origin converts an absolute bound into the same form. The source identities, actors, access routes and original precision remain attached to these mathematical restrictions; the inequalities alone do not establish the source facts.

For example, publication before access contributes $x_{\text{publication}} \le x_{\text{access}}$, while access by the decision contributes $x_{\text{access}} \le t_{\text{decision}}$. Keeping both inequalities in one system prevents the calculation from accepting a combination that separate per-clock classifications would mistakenly permit.

For each revision, eligibility is a conjunction of its required actor, system and effective-time guards, $g_1, \ldots, g_m$. The complementary cases can be written as a disjoint first-failure partition:

$$\neg(g_1 \wedge \cdots \wedge g_m) = \neg g_1 \ \vee \ (g_1 \wedge \neg g_2) \ \vee \ \cdots \ \vee \ (g_1 \wedge \cdots \wedge g_{m-1} \wedge \neg g_m). \tag{17.27}$$

Each branch preserves the earlier satisfied guards. Repeating the construction for the relevant revisions covers every clock assignment admitted by the original ranges and chronology. Eligibility and the selected revision heads are constant within each surviving branch. The procedure therefore represents families of timestamps symbolically; it does not need an explicit tree point for every possible instant.

An empty branch requires a checked contradiction. Summing original difference inequalities around a cycle cancels all clock variables. A negative right-hand sum is impossible; a zero sum is also impossible if at least one inequality in the sum is strict. A nonempty branch instead carries exact rational clock values satisfying every original inequality. The checker verifies those values directly and reconstructs every complementary branch before accepting the partition.

The witness establishes nonemptiness, while the guards and complete partition establish the scope of the conclusion. A few feasible timestamp examples would establish neither exhaustive coverage nor the absence of an omitted information view. Conversely, reaching the computation budget leaves an unfinished temporal family. The selected reference then reports incompleteness and does not present the already examined views as the full set.

17.17.3 Different timing views can support different policies

Return to receipt 7 and the committed facility. Suppose a correction changes the available observation scheme from coarse to refined, but the actor's access interval straddles the first decision. There are two remaining timing views. In the view where the correction has not yet arrived, the coarse signal leaves incompatible first instructions in the same information cell and no robust policy is feasible. In the view where it has arrived, the refined signal supports the four previously derived policies. The complete financial row population is retained in both calculations.

The timing evidence supplies no probability for either view. Selecting the four-policy answer would discard the other possibility. Reporting the two answers separately is therefore a substantive conclusion. Constructing one decision rule that is admissible when even the observation mechanism is uncertain requires a common specification of what the actor sees and how actions are represented across those mechanisms; a naive intersection of differently labelled policy tables is insufficient.

The exact comparison verifies interval boundaries, equivalent offsets, unknown and untrusted clocks, public and private access, correlated correction times, conflicting revisions and retractions. The saved partitions retain unresolved source outcomes as well as selected ones. Their scope is the supplied chronology and access model, not an empirical statement about how a real company releases information. General revision practices, authenticated evidence and durable recovery remain separate questions; Section 17.18 develops complete-family reuse and common decisions across possible observation mechanisms.[7]

17.18 Preserving possible views does not yet choose a common policy

The correction interval leaves two questions. A calculation must first preserve every information view consistent with the supplied clocks. The decision maker must then choose instructions that can actually be followed when the relevant view may itself be unknown. Keeping both policy tables answers the first question. It does not, by itself, answer the second.

17.18.1 Share calculations while retaining their different supports

A complete temporal family carries the original clock restrictions, its checked partition, every nonempty branch and the source outcome on that branch. Several branches can select the same revision; several different revisions can also imply the same financial and observation problem. The numerical calculation can then be stored once, provided that all the distinct source and timing supports remain attached to it. Removing a repeated numerical answer must not remove a possible history.

The selected binding reference therefore compares complete policy sets and their full financial-row populations. A same-meaning correction can reuse the numerical result while changing the source binding. A changed financial assumption or observation scheme requires its corresponding numerical problem. Even two empty policy sets retain their different scenario meanings.

[7]The derivation, original-edge checker, complete saved temporal partitions, independent finite order comparisons and full policy sets are retained in `data/accounting-master/source-time-intervals-20261027/`. The reference preserves its failed namespace diagnostic, the explicit public-route difference from the whole-second profile, and the six-revision and computation-budget limits. No clock probabilities or actual actor-access admission are introduced.

Their common cardinality, zero, is not sufficient evidence that the underlying questions are interchangeable.

The request and saved family are checked against expected identities supplied outside the transported record. The request includes the actor, original sources, clock observations, caller's admitted evidence and computation budget. The family retains the full temporal proof and each unresolved outcome. These checks detect alteration and missing support relative to that supplied record. They do not authenticate a document or establish that the actor actually received it.

A budget stop needs special care. The earlier temporal procedure does not save a partial partition that can be resumed. The binding reference therefore preserves the entire original clock domain as unresolved and supplies no completed-view policy answer. This conservative remainder keeps all possibilities, at the cost of repeating work later. Describing it as an incremental frontier would claim a computational capability that has not been implemented.

17.18.2 Decisions must agree on the histories the actor can distinguish

Let z denote a possible observation mechanism and w a possible financial history. At decision date t, the actor receives a message $m_t(z, w)$ from that mechanism. The message may differ across histories, and the meaning of a given message may differ across mechanisms. Let $\eta_t(z)$ record a mechanism identifier only when the model explicitly makes it observable by that date. Without such an observation, η_t has the same null value for every mechanism.

For the two-date example, the information used by a deterministic policy is

$$h_1(z, w) = \big(m_1(z, w), \eta_1(z)\big), \tag{17.28}$$

$$h_2(z, w, a_1) = \big(h_1(z, w), a_1, m_2(z, w, a_1), \eta_2(z)\big). \tag{17.29}$$

The second history retains the first message and the actor's own instruction. It contains no future message, hidden financial state or internal proof identifier. The action at each date is a function of the corresponding history, $a_t = \pi_t(h_t)$. Consequently,

$$h_t(z, w) = h_t(z', w') \quad \Longrightarrow \quad \pi_t\big(h_t(z, w)\big) = \pi_t\big(h_t(z', w')\big). \tag{17.30}$$

This is the common-policy restriction: indistinguishable available histories require the same instruction. It applies across possible mechanisms as well as across financial histories within one mechanism. Feasibility must then hold for every retained world under the selected stress question. No probability distribution is needed for that requirement.

Internal source hashes can identify z for the analyst without identifying it for the actor. Putting such a hash into h_1 would add information and could change the answer. Conversely, if a real message identifies its source or interpretation, omitting that information would solve a different problem. The observation declaration must say which case is intended and why.

17.18.3 Four policies in each view can leave no common instruction

Consider again receipt 7, the committed facility and the refined financial problem. Its complete row population supports four policies. For a concrete test, call the first messages L, H and X. Mechanism A sends these messages for, respectively, low conditions without a trigger, high conditions without a trigger, and high conditions with a trigger. Mechanism B exchanges the meanings of H and X. These are deliberately supplied synthetic encodings; they are not an empirical claim about a company's communications.

Each known encoding preserves the original four policies by relabeling their observations. The difficulty appears when the actor sees H but does not know which encoding applies. Under A, the instruction must be pay; under B, it must be draw. A common policy cannot satisfy both requirements. The two individually feasible problems therefore have no jointly feasible policy when the mechanism is hidden.

If the encoding is identified before the first decision, the two message histories are distinguishable. One of four policies can be selected for each encoding, giving $4 \times 4 = 16$ feasible joint policies. If the encoding is identified only at the second date, the earlier contradiction remains. Later learning cannot repair a first instruction that had to satisfy incompatible requirements at the time it was chosen.

The conclusion depends on when the incompatibility arises. In a separate elementary liquidity example, the only first instruction is wait. At the second date, a company with cash 1, an amount due of 1 and no facility must pay. Another possible company has cash 0, the same amount due and a facility of 1, so it must draw. Two mechanisms exchange the meanings of the messages describing those circumstances. Hidden message meaning leaves no common second instruction; revealing the mechanism at the second date restores one feasible policy because waiting had created no earlier conflict. If the second message itself has an unambiguous meaning, learning can resolve the uncertainty without revealing a mechanism's name. This example concerns stipulated cash feasibility, not the accounting recognition of a facility or a complete company report.

A different case shows why late revelation can expand a policy set even when the mechanisms have the same financial meaning. Two possible correction heads can both select the same refined problem. All four of its policies share the same first instructions; only their later choices differ. Learning the source identity at the second date then permits four continuations for each source, giving 16 joint policies. When that identity is never observed, the continuations must agree and only four policies remain. A first attempt incorrectly expected four in both cases; the complete-set comparison exposed the omitted late distinction, and the corrected independent relabeling preserves the failed attempt as evidence.

17.18.4 What a complete conditional answer must retain

The original coarse/refined correction example remains different from the swapped-encoding example. The coarse member already has no feasible policy. Identifying that member does not supply the refined signal it lacks. The joint policy problem therefore stays infeasible even if the actor is told which mechanism applies. An unresolved source outcome for which no observation map is specified instead leaves the policy question unresolved; it is not another zero-policy result.

The checked finite method retains the full financial population for every mechanism and solves the shared-history constraints in two ways: enumeration of feasible instructions and an independent formulation of forbidden action combinations. It also compares the complete policy sets with explicit relabelings of the original policies. The checks distinguish a missing action row from an available safe action, and preserve the original temporal supports when equal numerical answers share storage.

These results explain how unresolved uncertainty can persist without requiring a separate explicit node for every time or transaction possibility. Symbolic temporal restrictions represent many possible times; shared numerical results avoid repeated calculation; common histories impose the information limits on decisions. Each operation preserves a different relation and needs its own justification. General observation languages, longer horizons, empirical actor access and durable recovery require further work. The saved partial-frontier extension for the temporal profile is developed in Section 17.20.[8]

17.19 A visible label need not identify the observation mechanism

In the preceding example, a first message H requires pay under one interpretation and draw under another. An additional label can resolve the conflict if it tells the actor which interpretation applies. Suppose, however, that both interpretations attach the same label, *reviewed*, to that message. The actor sees a label but still cannot distinguish the two cases. Declaring a label visible and declaring a mechanism identified are different assumptions.

The distinction changes the feasible policy set. With distinct labels supplied before the first decision, the two four-policy problems admit 16 joint policies. With the same label supplied in both, the conflicting instructions remain and no common policy is feasible. A simulator that automatically inserts its internal mechanism identifier into the actor's message would report the first answer while pretending to solve the second problem.

The selected observation declaration therefore records the actual token presented to the actor. It permits distinct mechanisms to emit the same token. Source identities and temporal proof labels remain attached to the analyst's explanation, but do not enter the decision history unless the declaration supplies a separate observation with that meaning. The numerical comparison keeps these cases apart even though both could informally be described as a labeled source.

[8] Complete-family binding, 26 family-change cases, 26 adversarial controls and saved independent replay are retained in `data/accounting-master/temporal-family-binding-20261027/`. The 21 cross-mechanism policy cases, independent full-set comparisons and corrected late-revelation expectation are retained in `data/accounting-master/cross-mechanism-policy-20261027/`. All message mappings and visibility regimes are explicit synthetic assumptions; neither reference introduces probabilities, actual actor knowledge or accounting permission.

17.19.1 A message map must use information available at that date

Hiding internal identifiers is insufficient if the message itself encodes an unavailable fact. A map that takes the entire financial history as its input could send one message when a hidden borrowing right exists and another when it does not. The resulting policy would exploit that right even if the source observations never revealed it. Giving the message an innocuous name would not remove the additional information.

The finite reference avoids that ambiguity by restricting what each map can read. At the first date, the coarse view supplies the market signal; the refined view supplies that signal and the current coupon-trigger observation. At the second date, the map receives the already specified trigger and cash-coverage observations. The actor retains the first message and its own earlier instruction. Hidden contractual rights, complete world identifiers and later observations cannot be introduced as new first-date map inputs.

For each selected source outcome, the declaration must map every supported observation key to a message. A missing key is an unfinished declaration, not permission to remove the corresponding financial history. Two entries for the same key are also invalid. These restrictions establish a closed finite interpretation of the existing observations. They do not establish that those observations describe a particular company's actual information practices, nor do they define a general language for arbitrary messages.

This ordering is useful when designing richer models. First specify which quantities can be observed and when; then specify how the observation is represented to the actor; finally constrain decisions to agree on indistinguishable histories. Reversing the order can make a desirable policy feasible by designing messages around the hidden state that policy would need to know.

17.19.2 Source, actor and time remain separate premises

A claimed visible tag carries an actor, the selected source revision, its content identity, a token and an observation time. The first decision must agree exactly with the temporal family's cutoff; the second occurs strictly later. A tag claimed at the first decision must have been available by that instant. Equivalent fixed-offset spellings describe the same instant, whereas a tag received one ten-millionth of a second afterward is late under the selected exact-time convention.

Unknown tag availability leaves the declaration unresolved. So does a supplied observation whose chronology cannot be established from the retained authorship bounds. The current reference treats this conservatively: it does not discard inconvenient clock possibilities or silently refine the source family with a new tag observation. A later method may condition or refine that family under explicitly admitted additional evidence, but it must preserve the distinction between the original question and the revised one.

These unresolved cases differ from a complete observation model with no feasible policy. In the latter, the available histories are specified and every possible instruction fails somewhere in the retained financial family. In the former, the model has not established which histories the actor may distinguish. Returning the same empty policy list for both would hide a missing information premise behind an apparently decisive financial conclusion.

A content identity detects alteration relative to expected bytes; it does not show that the actor received them. Likewise, the synthetic tag records used in the comparison test the consequences of supplied assumptions. They are not observations of a real company's managers or investors. That distinction remains necessary even when every mapping and timestamp check passes.

17.19.3 Reuse the calculation only when the decision problem is unchanged

A corrected source can preserve the same financial meaning and messages. The numerical policy calculation can then be reused while its source explanation changes. A different actor can also have the same supplied observation map; the actor binding must change even if the resulting mathematical problem is identical. Neither case licenses removal of the old and new source supports from the retained temporal family.

Changing a message label is different. A policy is a rule for responding to the labels the actor receives. Relabeling H as another token changes that rule's interface even if four policies remain feasible and a mathematician can construct a one-to-one relabeling between the sets. The selected reference recomputes this case. It does not claim to have proved and implemented a general translation between equivalent policy representations.

The reuse comparison consequently includes the complete financial and action population, the actual first and second messages, the meanings of the actions, and the numerical methods. Identical repeated constraints can share computation while keeping every source outcome and temporal proof support attached. Distinct problems with zero feasible policies retain distinct identities. A future correction that is unavailable at the earlier cutoff remains in the source record without replacing the earlier view.

The independent checks reconstruct the messages from the supplied declarations and original observations, compare full policy sets from two formulations, and reject missing or duplicated keys, hidden-state inputs, wrong actors or times, altered sources and forged prior results. They also check changed second-date labels without changing first-date messages. These examples make the dependency rule concrete: identical counts are not enough, but a changed source explanation need not force an unchanged numerical problem to be solved again.

For this observation-binding profile, the remaining computational difficulty is how to continue an unfinished temporal search while retaining its unexamined possibilities. Section 17.20 develops that extension for the selected temporal domain. The conservative full-domain remainder already prevents a partial result from masquerading as complete; a checked, resumable partition would additionally preserve useful completed work. Actual message provenance, richer observation languages and refinement by new tag evidence remain distinct extensions.[9]

17.20 Stopping a search without losing the possibilities it has not examined

A correction may have reached the manager before a payment decision, or afterward. An interval for the delivery time can preserve both possibilities without assigning either a probability. If the source record contains several corrections, public releases and private deliveries, the exact temporal calculation may exhaust its work allowance before checking every combination. The unfinished calculation still has a useful result, provided it keeps the completed conclusions together with an exact description of what remains unexamined.

The conservative earlier method returned the whole original clock domain whenever its search stopped. That answer preserved uncertainty but discarded useful completed proof work. A resumable calculation can retain more. It must save the partition it has already established, including every unfinished part. Saving only the branches that produced interesting financial

[9]The finite declarations, complete mapping comparisons, 22 main cases and 30 adversarial controls are retained in `data/accounting-master/observation-map-binding-20261027/`. A separate saved audit rechecks all main cases and four additional source, exact-time and token cases. The reference retains full source supports under numerical sharing and makes no claim of actual actor knowledge or authenticated evidence.

views would change the question on the next run.

17.20.1 Each split must cover the entire timing domain

Consider one cell D of admissible event times under the supplied chronology and access assumptions. For the next source revision, let $g_1, \ldots, g_m$ be its eligibility conditions. A condition may require that publication or actor access occurred no later than the decision, or that ingestion occurred no later than the system's reconstruction date. The revision is eligible only if all these conditions hold.

A complete split keeps one eligible cell and one cell for each possible first failed condition:

$$D_{\text{yes}} = D \cap \bigcap_{j=1}^{m} g_j, \tag{17.31}$$

$$D_{\text{no},k} = D \cap \bigcap_{j<k} g_j \cap g_k^{\text{c}}, \qquad k = 1, \ldots, m. \tag{17.32}$$

Here a condition also denotes the set of time assignments satisfying it, and g_k^{c} denotes its exact complement. Every assignment in D either satisfies every condition or has a unique first failure. These alternatives therefore cover D and are pairwise disjoint. Their union is the original cell, not a sample of it. Some children may be empty, and a contradiction proof can establish that fact later.

The word *exact* matters at a decision boundary. If g_k says that access occurred at or before time d, its complement says strictly after d. Replacing that complement by *at or after* would put the boundary in two children. Replacing the original condition by a strict inequality would remove the boundary from the eligible child. The selected reference retains rational event times and strict inequalities, so the preceding submicrosecond and fixed-offset examples keep their original meanings.

The search resolves one pending cell at a time. It first checks whether the cell is empty. A nonempty cell with no remaining revision to classify yields a selected or unresolved source view. Otherwise it installs the complete family of children above. Only after all those children exist may the calculation pause. This logical rule prevents a work limit from turning a complete split into a partial list of favorable branches.

17.20.2 Pending does not mean feasible, and known does not mean complete

A saved node has one of four meanings: its domain remains unchecked; a contradiction proves it empty; a witness establishes a terminal source view; or a witness supports an expanded cell whose children form the complete partition. An unchecked child need not contain a possible time assignment. It may already be inconsistent with an earlier chronology restriction, but the calculation has not yet established that. Retaining it is conservative; deleting it without checking is unsupported.

In the selected example, resolving the root installs eight children. Seven are later proved empty. With a one-node work allowance, all eight remain explicitly pending. The result therefore neither asserts eight feasible worlds nor pretends that the single eventual view is already established. The eight cells are a representation of unresolved work under known restrictions.

Once some terminal views are known, they can be displayed together with their source supports. Agreement among them still does not establish that the whole family has that view. An unexamined cell could contain a conflicting correction, a retraction or a different access history. Completion requires that no pending cell remains, including cells that eventually prove empty.

The same distinction controls policy interpretation. A policy that fails in an already established view fails for any larger family containing that view. A policy that succeeds in every established view may still fail in an unexamined one. Thus an incomplete temporal search can sometimes establish a counterexample before it can establish universal feasibility. Connecting that asymmetry to the numerical policy calculation requires an explicit treatment of the pending remainder; a saved search alone does not supply it.

17.20.3 Resume the original question and preserve its proofs

The checkpoint binds the complete source request, actor, trust declarations, timing assumptions, method identities and execution allowance. On reading it, a separate checker reconstructs each branch's constraints from the original domain and its path. It does not accept a saved mutable constraint list as a substitute. It verifies all completed witnesses or contradictions, the exact complement structure, source outcomes, support paths and work counts.

The reference uses a fixed left-first depth-first order. This makes the resumed tree directly comparable with the existing uninterrupted calculation. It also allows the checker to reject a purported completed branch placed after an earlier unfinished one. The order is a comparison convention, not a claim that this schedule is the fastest way to find useful financial conclusions.

A step allowance specifies how much new work one call may perform. A separately bound total execution cap determines when further work must stop. Reaching that cap retains the pending tree and reports a capped calculation; it does not turn unresolved cells into completed results. The test with a cap of two preserves the same original question and performs no repeated solver work when resumed at that cap. Changing the cap or the source assumptions requires an explicitly different, checked request to the continuation interface.

Across the 47 selected temporal requests, step allowances of one, two, seven and 4,096 produce exactly the same final proof trees, outcomes and source supports as uninterrupted search. Earlier certificates remain unchanged at every pause. A separate reread checks 1,060 saved checkpoints and rejects altered partition cells, proofs, source premises and completion declarations. These comparisons establish the selected finite continuation property.[10]

This representation avoids a separate explicit node for every real-valued event-time assignment. It does not remove exponential growth in the number of distinguishable revision combinations. Nor does installing all children before a logical pause establish crash-safe file storage, authenticated evidence or actual actor knowledge. Larger revision families, better schedules, justified approximate reduction and durable storage require their own arguments and comparisons. The useful advance here is narrower: unfinished computation retains both what has been proved and every domain that has yet to be checked.

17.21 Fixing the policy question before the search finishes

A manager who may pay an obligation or draw on a facility needs a rule for acting on the information available at each decision. The preceding search can establish some possible information histories while leaving others unexplored. To compare policies during that search, their meaning must remain fixed. If each newly discovered history changes what a policy is allowed to observe, an apparent reduction in the feasible set may instead reflect a change in the

[10]The implementation, independent structural checker, 188 complete comparisons, 43 main adversarial controls and five additional saved-record controls are retained in `data/accounting-master/temporal-frontier-20261027/`. The main comparison makes 10,146 assertions; the saved-record audit makes 3,191. The source locks and manifests preserve the exact inputs, methods and comparison scope.

question.

The first requirement is therefore a declaration of possible messages before temporal search. In the finite example, start with the original registry of financial assertions and observation schemes. For each assertion-valued observation scheme, declare the first- and second-decision messages for every raw observation obtainable from the original financial assertions. Cross those financial and observation assertions conservatively. Some combinations may later prove temporally impossible; including them at this stage avoids selecting a favourable source history in advance.

The declaration binds the manager, decision dates, source records and meanings of pay and draw. A second-decision key also retains the manager's own earlier action, as in the preceding policy example. Internal source-head identifiers remain evidence of support and do not become messages to the manager. A retraction supplies no replacement assertion payload, but it does not justify removing every message associated with the original assertion before its timing has been examined. This is a declared finite vocabulary, not a finding that a real manager received every listed message.

17.21.1 A fixed domain makes partial restrictions comparable

Let U denote the declared finite set of decision keys, including the decision stage and the required own-action memory. A total rule assigns either pay or draw to every key in U. Let Π_U be the set of these rules. Assignments to keys that are never reached still belong to a total rule; they have no effect on that rule's realised path. Consequently, several total rules can represent the same policy on the reachable histories.

At search stage t, let H_t contain only the selected financial and information histories supported by completed temporal branches. For a history h, write $V_h(\pi)$ when following rule π selects no refused action in that history. This condition uses the complete finite action rows for that selected history, including first-decision refusal and second-decision refusal after the first action. The rules surviving the known restrictions are

$$C_t = \{\pi \in \Pi_U : V_h(\pi) \text{ for every } h \in H_t\}. \tag{17.33}$$

These are sets of decision rules, not monetary balances or probability distributions. The relation is a local definition under a fixed message declaration and the supplied action semantics. Unresolved source outcomes are retained separately; they cannot supply a selected history or establish its financial constraints.

Resuming the same checked search retains the previously completed histories. Thus $H_t \subseteq H_{t+1}$. A rule satisfying every condition for the larger set also satisfies every condition for its subset, so

$$C_{t+1} \subseteq C_t. \tag{17.34}$$

This inclusion explains the asymmetric conclusions. Once known constraints eliminate every rule, adding more histories cannot restore a rule under the same declaration. If some rules survive, a pending branch may still eliminate them. In the coarse/refined correction example, the known constraints already eliminate all rules after four of the 25 temporal nodes have been resolved. The baseline instead retains four total rules and establishes conditional feasibility only when the search completes with selected source outcomes throughout.

17.21.2 Three reasons that a surviving rule may still be unresolved

Unfinished search, unresolved source meaning and an inadequate domain allowance are different limitations. A two-node search cap retains 15 pending branches in the baseline. Those branches

have not all been shown to contain possible histories, but they must remain represented until checked. The surviving rules therefore remain candidates.

A search can also finish while a required assertion remains unavailable, retracted or in conflict with another head. Unknown access leaves four rules surviving the known constraints, yet one source outcome remains unresolved. Retraction and conflicting-head cases likewise do not become feasible merely because all temporal branches have been processed. An inconsistent clock system is different again: it admits no temporal assignment. The 512 unconstrained rule assignments in that example do not establish a feasible policy for a possible company history.

Finally, the conservative domain may exceed the stated computational allowance. The finite profile permits at most 12 binary decision keys, hence at most 4,096 total rules. Renaming only the low message in one correction produces 12 keys and eight surviving total rules. Renaming all correction messages requires 18 keys, so the reference retains the original request and returns an unresolved domain budget without truncating the keys or enumerating a preferred subset. The extra rule count in the smaller example is not evidence of better information or better economic performance.

The checked partial calculation binds each result to the original request, declaration, domain, methods and temporal frontier. It retains the supporting branches, unresolved source outcomes and pending temporal domains. Changing a message, source, actor or domain cannot reuse the old result as the answer to the new question. A future extension rule may justify some reuse, but that relation must itself be stated and checked.[11]

This construction joins a fixed decision problem to an unfinished search without treating uncertainty as missing arithmetic. Longer horizons, additional actions, changing declarations and broader observation languages still require their own comparisons.

17.22 Changing a message without confusing the decision problem

A manager may receive a differently worded notice carrying the same information, or a notice that distinguishes two situations previously described alike. These changes have different consequences for a policy. The preceding requirement to fix message meaning does not prohibit either change. It requires the simulator to state which relation connects the old decision problem to the new one, and to identify the new result accordingly.

A consistent one-to-one renaming preserves the original distinction between every pair of messages. The finite comparison checks that renaming across every raw observation in every original source head, including heads not selected at the current frontier. It also preserves actor, decision dates, action meanings, source identities and second-stage memory of the manager's first action. Each total rule then has exactly one renamed counterpart. The first- and second-action refusal conditions have the same truth values after the key positions are permuted, so the full surviving rule sets correspond exactly. This is the same finite decision problem under different spellings; it is not an expansion of what the manager knows.

[11] The original-registry domain comparison is retained in `data/accounting-master/prospective-policy-domain-20261027/`; its 10 cases include explicit overflow and conflicting-head cases. The partial comparison in `data/accounting-master/prospective-partial-policy-20261027/` checks 12 cases at 110 frontiers, with independent full-rule enumeration and 31 refusal controls. The candidate takes no complete-family comparison as an input. Exact sets and endpoint projections, rather than the counts alone, support these finite conclusions.

17.22.1 A refined notice allows the old distinction to be recovered

Now suppose a correction distinguishes a low message that was previously shared with another source history. The new message identifies which of those situations occurred. A manager can ignore that distinction and follow the old rule, or potentially act differently in the two situations. To make this precise, each refined message must map to one old message, consistently across all original raw observations. Combining this message map with the unchanged decision stage and own-action memory gives a map from new policy keys to old keys.

Write U_o and U_n for the old and new finite key domains, and $e : U_n \to U_o$ for this erasure map. Require e to reach every old key. An old rule π_o assigns pay or draw to each old key. Its lifted rule $L_e\pi_o$ chooses $\pi_o(e(u))$ at a new key u: the manager acts as if the additional distinction had not been received. New keys that erase to the same old key form a *fiber*. A rule constant on each fiber makes the same choice at all of those keys.

For the same checked temporal frontier, let C_o and C_n be the rules satisfying the old and new known constraints. The comparison preserves each underlying action row and its refusal meaning, and checks that erasing the new forbidden assignments gives exactly the old forbidden assignments. Under those conditions,

$$\{L_e\pi_o : \pi_o \in C_o\} = \{\pi_n \in C_n : \pi_n(u) = \pi_n(v) \text{ whenever } e(u) = e(v)\}. \tag{17.35}$$

To see the first direction, a lifted rule makes the old choice at each corresponding raw observation. A refused first action or refused pair of successive actions would therefore also have been refused under the old rule. Conversely, a new rule constant on fibers defines a unique old rule because every old key has a preimage. The same correspondence of refused assignments then shows that this old rule satisfies the old constraints. The relation concerns exact finite rule sets under supplied message semantics; it assigns neither probabilities nor monetary values to information.

17.22.2 Additional rule assignments need a reached-action comparison

In the correction-specific low-message example, the old declaration permits four total rules and the refined declaration permits eight. Lifting preserves all four old rules. The other four also produce distinct sets of reached action rows across the complete selected family. Here the new distinction therefore permits additional feasible behavior under the stipulated information design. In another example, collapsing the first messages leaves no feasible rule, while restoring the refined messages permits four. Neither comparison establishes that those messages are actually available to a real manager or that any policy maximises value.

The source-unknown examples show why counting total rules is insufficient. With conflicting source heads, a refinement increases the unconstrained assignments from 32 to 2,048, but there is no selected history on which to demonstrate a new action. Every rule has the same empty reached-row signature. Under inconsistent clocks, the corresponding counts are 32 and 512, yet no admissible time assignment exists. An empty signature is a set used for comparison, not a realised or feasible company behavior. With unknown access, additional choices can be exhibited on the selected completed histories while the overall source family remains unresolved.

The comparison therefore retains both the full rule sets and their reached-row signatures. Pending branches, unknown source outcomes and inconsistent clocks keep their original meanings. The refined result has its own declaration and domain identity, even when its feasible rules correspond exactly to an older result.[12]

[12]Exact renaming is checked in `data/accounting-master/policy-domain-transport-20261027/`: 22 frontier comparisons and 12 refusal controls. Erasure and lifting are checked in `data/accounting-master/`

This gives two precise ways to compare changed declarations. Broader changes to sources, decision times, actions or the observation language need additional arguments before earlier policy work can be reused.

17.23 A correction arriving between two decisions

A manager can make the first payment before receiving a correction, then consult the corrected information before the next financing decision. The first action belongs to the earlier information history. Replacing both observations with the latest source view would rewrite that history; treating the two dates independently could invent a sequence that no delivery time permits. The simulator needs a joint sequence of source views before it can interpret the two-stage policy.

Consider the supplied correction from a refined observation scheme to a coarse one. Let the first decision be on 2 January and the second on 3 January. If the correction is received before both dates, both select the coarse scheme. If it arrives between them, the first selects refined and the second coarse. Arrival after both leaves refined at both dates. When its access-time interval spans both decisions, the joint family therefore contains

$$\{(\text{coarse}, \text{coarse}),\ (\text{refined}, \text{coarse}),\ (\text{refined}, \text{refined})\}. \tag{17.36}$$

Each pair lists the first and second selected source schemes. These are possible sequences under the supplied chronology, not probabilities. Each marginal date admits both schemes, so their Cartesian product would also include coarse followed by refined. That fourth pair is impossible here: the same correction cannot have arrived before the first decision yet remain unavailable at the later one. A later retraction or a different replacement would be another source event requiring its own representation.

17.23.1 Keep the source clocks shared while changing the decision cutoff

The joint calculation uses the same source records, actor, access clocks and chronology at both dates. Its system reconstruction cutoff is fixed and may lie after both decisions; that cutoff describes the evidence inventory used by the analyst, not information available to the manager at the time. Each stage separately tests authorship, publication and actor access against its own decision date and applies its effective cutoff. The first increment keeps one financial assertion fixed while observation schemes can change.

The exact temporal partition first classifies eligibility at the first date and then at the second, carrying the accumulated clock constraints throughout. A completed branch records both selected or unresolved source views. An independent checker reconstructs the constraints and both head selections. The resulting joint pairs agree with the authored before, between and after cases, including exact decision boundaries and a submicrosecond arrival immediately after the first decision. Their projections agree with independently checked marginals, while the joint counterexample above tests the dependence that marginal agreement alone cannot establish.

Late retraction and conflicting heads leave the first view selected and the second unresolved. Unknown access retains every timing sequence admitted by the supplied chronology. It does not assert when the manager actually received the correction. Inconsistent clocks admit no joint

`policy-observation-refinement-20261027/`: 17 comparisons and 10 controls, with whole-set equality and explicit reached-action comparisons. Split or merged labels are outside the renaming relation; a false refinement direction, altered source or dropped pending branch is refused. Both profiles remain finite conditional design evidence.

sequence. If the joint search exhausts its allowance, this first profile retains the entire original clock question as unresolved; it does not yet save a resumable joint frontier.[13]

17.23.2 Use the selected map at each stage

A prospective policy domain must now cover more than same-head combinations. Before inspecting completed joint outcomes, the declaration crosses the original financial head with every assertion-valued first-stage observation head and every assertion-valued second-stage head. It constructs the first message from the first selected map and the second message from the second selected map. The second decision key retains the actual first message and the manager's own first action. Internal head identities remain support records, not messages visible to the manager.

In the inherited raw observation schema, coarse versus refined changes only the first observation. The second raw observation remains the trigger and cash-coverage pair. Selecting a different source head at the second date therefore does not, by itself, add second-stage information. An explicit change to a second-message token tests whether the correct stage map is used, while remaining a supplied interface assumption.

That changed-token case needs 15 keys in the conservative original domain, exceeding the 12-key allowance. It remains unresolved without dropping mixed-stage combinations. A separate diagnostic collapses the first messages and uses 7 keys. Its encoded second messages differ when the wrong first-stage head is substituted for the second-stage head, so the test detects that wiring error. The correctly wired diagnostic nevertheless has no feasible policy. Correct source-to-message binding and policy feasibility are separate conclusions.

With the original mappings, arrival between the decisions or after both leaves 16 total rules on the conservative 11-key domain. Some assignments concern keys not reached in the selected family; these are not 16 demonstrated reached behaviors. Arrival before both dates supplies the coarse first view and leaves no feasible rule. The interval containing all three sequences also has no common rule because it includes that coarse-first history. Unknown access preserves the same conditional family and its infeasibility, without turning unknown access into an observed fact.

A required retracted or conflicting source view leaves the overall policy question unresolved, even when candidate assignments satisfy every known numeric constraint. Inconsistent clocks and either temporal or policy-domain budget exhaustion supply no feasible-policy answer.[14]

These results bind a complete joint source family to two-stage decisions. Continuing an unfinished joint search, changing financial rules between dates and admitting actual actor information remain separate tasks.

17.24 What a paused joint search can establish

Suppose the search stops after checking some of the correction's possible arrival times. A financing rule may survive those cases and fail in an unexamined case. To interpret the partial

[13] The exact shared-clock comparison in `data/accounting-master/two-decision-source-views-20261027/` checks 11 cases and 12 refusal controls, preserving full joint partitions, source support and separately checked marginal results. Its 208 assertions include the impossible Cartesian-product sequence and the conservative budget remainder.

[14] `data/accounting-master/joint-stage-policy-20261027/` retains 16 cases,204 assertions and 7 refusal controls. A separate raw-observation mapper checks complete encoded rows; an independent policy oracle checks full rule sets. Same-head cases reproduce the earlier mapping, and the stage-specific token diagnostic tests preserved first-action memory. The 15-key case and the separately collapsed 7-key case are both retained.

answer, the modeller needs both the source sequences already checked and the clock assignments still awaiting examination. The shared-clock construction in the preceding section supplies the latter without choosing a separate history at each date.

Each unfinished branch retains the constraints accumulated on the original clock variables. At a branch concerning a particular source and decision date, the alternatives are that every eligibility condition holds or that a named condition is the first to fail. These alternatives cover the parent branch without overlap. The search creates all of them before pausing. It can therefore resume from the first pending branch while preserving every other pending branch, the first decision's eligibility choices and the second decision's remaining tests. A short allowance changes how much has been examined, not the question being asked.

The saved state distinguishes a per-call allowance from a total node cap. Reaching the total cap leaves a capped, incomplete result with its pending domains. A checker reconstructs each branch's constraints from its path, verifies the rational feasibility or contradiction evidence and recomputes both selected source views at completed leaves. Checking only the number of leaves would miss a lost or duplicated support region. In the bounded examples, resuming one node at a time, seven at a time or without intermediate pauses produces the same complete tree, joint outcomes and support paths.

17.24.1 Surviving rules can shrink before the search is complete

Keep the prospective message domain fixed before examining these outcomes. Let Π be its finite set of total two-stage decision rules: each rule assigns an action at every declared first- or second-stage message key, including keys that a particular source sequence might never reach. After k units of search work, let K_k contain the distinct fully selected source pairs found so far. For a pair h, let $F(h) \subseteq \Pi$ contain the rules satisfying every financial row supplied by that pair. The candidates consistent with the known pairs are

$$C_k = \Pi \cap \bigcap_{h \in K_k} F(h), \qquad K_k \subseteq K_{k+1} \implies C_{k+1} \subseteq C_k. \tag{17.37}$$

This is a local set-theoretic consequence of adding restrictions in the same rule universe. Initially no source pair may have been selected, in which case the empty intersection imposes no restrictions and $C_0 = \Pi$. That initial survival says nothing about financing feasibility. Each newly selected pair can remove a rule; none can restore a rule already rejected by a known pair. The index k measures search work, not elapsed financial time, so this relation does not describe how a manager's information evolves.

If the known restrictions leave no rule, no completion that merely adds further source pairs can supply a rule satisfying all of them. Conversely, a nonempty C_k remains a partial answer while any source region is pending. In the correction example, a completed coarse-first pair can already remove all candidates, whereas observing only refined-first pairs cannot establish that no coarse-first pair remains. A changed source declaration or message domain would alter the comparison and requires a separate justification for reusing old results.

The returned conclusion also preserves the reason a family is unresolved. A retracted or conflicting required source remains an unresolved source view; the overall result keeps that status even if the known restrictions separately rule out every candidate. A complete timing family with unknown actual access is instead evaluated conditionally on all its supplied possible arrival sequences. Inconsistent clocks supply no admissible joint sequence and are not treated as a feasible empty family. Exceeding the prospective key allowance leaves the policy population unassessed. Only a completed search with selected source content and surviving rules supports conditional feasibility under these declarations.

17.24.2 Check the reason for each rejected rule

A saved list of surviving rules is insufficient evidence by itself. For every excluded total assignment, the result names a known financial row that refuses it. A first-stage refusal must match the rule's first action at that row's message. A second-stage refusal must also match the second action, using the second key that retains the actual first message and action. The checker validates these witnesses and requires the surviving and rejected assignments together to cover the entire fixed rule universe. It also reconstructs the conclusion, unknown source pairs, pending domains and support paths from the checked source search.

The finite comparison checks every saved endpoint and selected partial results against an independently evaluated policy population. Altered refusal witnesses, premature feasible conclusions, missing pending evidence and changed identities are rejected. The result checker still uses the established source-to-row mapping and policy oracle; it is not an independent redevelopment of all accounting and observation semantics.[15]

This establishes logical continuation for the supplied two-date question. It does not test crash-safe file storage or establish actual information delivery. Transporting a result to changed observation declarations, extending the horizon and admitting company evidence require further work.

17.25 Changing the observation declaration

The fixed-domain result raises a practical question when a modeller revises the observation declaration. A message may acquire a new label, or a formerly pooled observation may be separated into more informative messages. Both changes alter the encoded decision keys, but they have different consequences. A new spelling can preserve the same choices exactly. A new distinction can let a manager choose different actions where the old declaration required the same action. The comparison must say which change has occurred before reusing a computed policy set.

17.25.1 A different label must preserve the entire decision key

Consider replacing every first-message label with a distinct new label and doing the same for the second messages. The substitution must be global across source heads: two equal old messages must remain equal, and distinct old messages must remain distinct. A substitution chosen separately for each source could reveal the source's identity and therefore change the information available to the manager.

The second-stage key contains the first message and the manager's first action as well as the current message. Consequently, renaming the first message also changes its occurrence inside second-stage memory. The joint comparison generates this full key correspondence from all original financial and observation-head triples, before inspecting completed source outcomes. If the correspondence is a bijection, the old total action assignment can be permuted to the new key order. Inverting the permutation recovers every assignment.

The refusal evidence travels with the assignment. A rejected rule keeps the same financial-row identity, but the checker must confirm that the renamed rule still takes the actions that row

[15]The complete evidence is retained in `data/accounting-master/joint-temporal-frontier-20261027/`. Attempt 02 checks 11 requests, 33 completed search sequences, 662 saved checkpoints and 848 policy prefixes in 16 cases. Attempt 03 checks the 16 saved policy endpoints and 47 selected prefixes with 33 alteration controls, including wrong refusal witnesses and false feasible conclusions. The earlier failed import is preserved separately. These counts describe coverage of the supplied finite cases, not a general error rate.

refuses. The transported surviving population agrees with fresh evaluation under the new declaration, and the source supports, pending clock regions and unresolved source outcomes remain unchanged. An overflowing policy domain stays unassessed; label equivalence does not remove its computational limit.[16]

17.25.2 A new distinction permits an old rule to ignore it

Now suppose a refined declaration distinguishes messages that a coarse declaration pools. The financial rows, action meanings, source chronology and decision dates are unchanged. Let D_c and D_f be the coarse and refined sets of full decision keys. A surjective map $e : D_f \to D_c$ erases the new distinctions. It must also erase the first-message distinction wherever that message occurs in second-stage memory. Surjectivity ensures that every coarse decision key still has a refined counterpart.

A coarse rule π assigns an action to each key in D_c. Its lift $\pi^{\uparrow}$ on D_f ignores the added distinction: at a refined key z, it takes the coarse action at $e(z)$. Let C_c and C_f be the candidate populations for the same selected source pairs under the two declarations. With unchanged financial refusal conditions, the checked relation is

$$\pi^{\uparrow}(z) = \pi(e(z)), \qquad \{\pi^{\uparrow} : \pi \in C_c\} = \{\rho \in C_f : e(z) = e(z') \Rightarrow \rho(z) = \rho(z')\}. \tag{17.38}$$

The right-hand side selects refined rules that take the same action whenever the new messages collapse to the same coarse key. To see why the equality holds, substitute the coarse action at $e(z)$ for each refined action in a refused financial row. Erasing that row's decision keys produces exactly the coarse refusal condition. A lifted coarse candidate therefore passes every refined row. Conversely, a refined candidate constant on each erasure class defines a unique coarse rule, and the same substitution shows that it passes the coarse rows. The equality concerns the same finite decision problem under two declared information restrictions.

In the between-date correction case, restoring the original first-message distinctions from an artificially collapsed first message changes the candidate population from zero to 16 total rules, with four additional reached-row signatures. Restoring the second-message distinctions changes two total candidates to 16, with three additional signatures. A signature records the financial rows reached by a rule's actions; it is not a probability distribution or an observed company path. The comparison verifies the reached rows of each individual lifted rule, so equality of aggregate signature sets cannot conceal a wrong pairing.

The distinction matters particularly when required sources are unresolved. In the late-conflict example, refinement expands eight total assignments to 2,048, yet no financial row is selected and no additional reached-row signature appears. Those assignments do not establish feasible company behavior. Inconsistent clocks produce no assessed policy population at all. If a refined domain exceeds the key allowance, its policy comparison remains unassessed even when the coarse domain has been evaluated.[17]

These are conditional comparisons of supplied observation declarations. The extra distinctions must still be justified as information available to the actor. Their finite effects on permitted actions do not measure utility or predictive accuracy. Both relations also retain one financial source view across the two decisions; allowing a financial correction to change that view requires an account of what was believed earlier and which economic history continues afterward.

[16] The bounded joint comparison in `data/accounting-master/joint-domain-transport-20261027/` checks 34 comparisons across nine declarations and 16 alteration controls. Eight comparisons retain unassessed domain-budget results. The assessed comparisons check full populations, preserved refusal witnesses and inverse roundtrips, including the first message inside second-stage memory.

[17] `data/accounting-master/joint-observation-refinement-20261027/` retains 46 comparisons across 12 declaration pairs, with 14 alteration controls. Nine comparisons remain unassessed. Assessed cases verify full lifted populations, constant-class equality, erased refusal conditions and per-rule reached rows. First-stage, second-stage and simultaneous refinements are separate cases; exact renaming is an equality control.

17.26 A revised report and the economic history it describes

A financial report can change between the two decisions without changing the economic event it describes. Suppose an earlier assertion reports an operating receipt of 7 and a later assertion corrects it to 6. Selecting the available assertion at each date answers an information question. It does not yet determine whether the receipt was always 6, whether an agreement actually changed, or what information the manager received. Those questions must be separated before the selected assertions can support a simulation.

The joint source calculation can now retain a different financial head at each decision. A correction available between the dates selects the old assertion first and the corrected assertion second. With an interval-valued availability time, the supported sequences can be old/old, old/new and new/new. The reverse new/old sequence is impossible under this one-way correction history, even though it would appear in an unconstrained Cartesian product of the two marginal selections. Conflict and retraction retain unresolved content. None of these source selections, by itself, specifies a financial policy.[18]

17.26.1 Continuity concerns the complete carried state

The financing example gives a precise way to distinguish an assertion revision from a change in economic history. Opening cash is $11/2$. The first coupon I_1 is accrued and settled; the operating receipt R arrives next; any first-period facility borrowing is repaid; and the second coupon I_2 is then accrued and settled. A draw supplies 1 and immediately costs a fee of $1/4$. Let d_1 and d_2 indicate a draw at the respective decisions. The once-only right and the other payment guards still determine whether an action sequence is permitted.

After the first payment, cash is $11/2 - I_1 + 3d_1/4$. Receipt and repayment change it to $11/2 - I_1 + R - d_1/4$: repaying the principal removes the first draw's unit of cash, leaving its fee. The second settlement therefore gives

$$C_{\text{final}} = \frac{11}{2} + R - I_1 - I_2 - \frac{d_1}{4} + \frac{3d_2}{4}. \tag{17.39}$$

The last term includes the second draw's principal because the sequence ends at the second settlement. It does not erase the associated liability. Under receipt 6, fixed coupons 5 and 5 with pay/pay give final cash $3/2$ and no facility fee. Floating coupons 6 and 5 with draw/pay give final cash $1/4$ and a fee of $1/4$. Both examples satisfy the inherited pre- and postconditions. The equation also describes other action sequences algebraically; their guards must be checked separately. All these monetary amounts are supplied synthetic units.

Before the receipt arrives, the complete states under $R = 6$ and $R = 7$ coincide in this example. That includes the book, contractual memory and recorded flows. Attaching the receipt-6 continuation to the identical pre-receipt prefix therefore reconstructs exactly the complete receipt-6 path. Rejecting every change of a receipt parameter would miss this valid correspondence. The correspondence can support an explicitly assumed correction of the same event; it cannot establish that the revised report is true.

After receipt settlement, the two states differ. Replacing 7 by 6 then removes one unit of cash, changes the income credit and rewrites the recorded operating receipt. Both books balance, but the complete states no longer join. A different contractual right is visible even earlier: the

[18]The checked financial-source selection companion is retained in `data/accounting-master/joint-financial-source-20261027/`. It reproduces 11 protected prior cases and checks 13 new cases, with 341 assertions and 16 alteration controls. Its declared financial-policy binding remains unassessed.

right is carried in opening memory, whether or not it changes the numeric accounts. An actual amendment consequently needs an explicit event and its accounting treatment.[19]

17.26.2 An explicit interpretation selects a complete world

For a supplied same-event correction, the analyst can declare the corrected head, a fixed receipt and a contractual-right family. This declaration supplies a hypothesis about the whole economic history. The selected heads at both dates must lie in the corrected head's receipt-only ancestry, and the right family must remain unchanged. Retaining the superseded assertion as evidence does not impose both 7 and 6 as simultaneous exact facts about one receipt; imposing those incompatible facts would leave no world.

The binding then selects the complete financial rows for the declared world while retaining the dated source pairs, their supports and any unresolved temporal remainder. Receipt 6 with the committed right binds 980 action-region rows; receipt 7 binds 968. These are complete action-region row families, not counts of feasible policies. Their populations agree with the protected reference problems. Unknown interpretation, actual amendment, changed rights, uncovered ancestry, conflict or retraction, inconsistent clocks and incomplete search receive no financial row set.[20]

The corrected head may be available to the analyst who specifies the hypothesis without having been available to the earlier manager. Neither that head's identity nor the assumed world's identity is automatically an observable message. An adaptive policy still needs an explicit account of the information available at each decision and must choose the same action whenever that information is the same. The next comparison must therefore preserve the distinction between the analyst's economic hypothesis and the manager's dated observations.

17.27 Dated observations and a common decision rule

The analyst's receipt assumption does not tell us what the manager can observe. Two explicit alternatives make the distinction testable. The first retains the original numeric observations: the first coupon signal and the later trigger and cash-coverage signal. The second pairs each numeric message with the receipt reported by the financial assertion selected at that date. These are supplied information assumptions. A reported future receipt is neither cash already received nor proof of the eventual amount.

For a correction arriving between decisions, the supplemented first message reports 7 and the second reports 6, even when the analyst evaluates a complete receipt-6 world. Putting the later correction into the first message would give the manager information too early. At the second decision, the key still includes the first message and the manager's first action. That memory is needed even if the current numeric signal coincides with one reached after a different earlier choice.

Several supplied source/world alternatives can then be evaluated together. The alternatives share an actor, dates and action meanings. Their identities remain internal: equal visible decision keys require the same action in every alternative. Finding a policy separately in each

[19] `data/accounting-master/financial-assertion-continuity-20261027/` checks 16 symbolic cases against an independent full-state recurrence: 112 boundary comparisons, 594 assertions and 48 rejected post-receipt joins. It also checks the two viable receipt-6 examples.

[20] The `SuppliedReceiptCorrectionWorld/1` declaration and its checks are retained in `data/accounting-master/financial-source-world-binding-20261027/`: 20 cases, 289 assertions and 25 alteration controls. Twelve cases bind rows and eight remain unresolved or inconsistent. The declaration records its provenance as a supplied synthetic assumption.

world is insufficient. For example, if one world requires pay and another requires draw at the same first message, each has a policy in isolation but there is no common rule. The retained economic-world declaration still requires a correction ancestry compatible with its supplied world; this comparison pools independently specified alternatives rather than inferring several truths from one corrected report.

The prospective domain is formed before selecting favorable source outcomes, using the original source heads and the declared complete worlds. Adding the reported receipt can expand this domain beyond the twelve-key enumeration allowance. The larger domain is retained intact. A size-limit result says that the complete policy population has not been enumerated; it says nothing about whether a policy exists. The original same-world observation-correction example retains exactly its 16 policies, and pairing its messages with a constant reported receipt preserves that number.[21]

17.27.1 An action assignment or a contradiction can settle existence

The two available actions give a smaller logical question than enumerating every rule. Associate a Boolean variable x_i with each complete decision key: 0 denotes pay and 1 denotes draw. A financial row refused at the first decision forbids one assignment $x_i = a$. A row refused at the second decision forbids the joint choice of its first and second actions, $x_i = a$ and $x_j = b$. A rule avoids those refusals exactly when it satisfies, respectively,

$$x_i \neq a, \qquad (x_i \neq a) \ \vee \ (x_j \neq b). \tag{17.40}$$

The second clause says that at least one of the actions leading to the refused row must change. It is equivalent to the implications

$$(x_i = a) \Rightarrow (x_j = 1 - b), \qquad (x_j = b) \Rightarrow (x_i = 1 - a). \tag{17.41}$$

A unary refusal similarly implies $(x_i = a) \Rightarrow (x_i = 1 - a)$. These clauses preserve the original financial rows and observation keys. They change how existence is decided, without changing which decisions are allowed.

A proposed total assignment is checked directly against every refused row. If none is reached under that assignment, it supplies one common policy for the resolved financial family. It does not enumerate other policies or choose an optimal one. Conversely, suppose implications justified by refused rows lead from $x_i = 0$ to $x_i = 1$ and also from $x_i = 1$ to $x_i = 0$. Either value contradicts a consequence of choosing it. The two paths therefore certify that no assignment can avoid all those refusals. A graph calculation can discover the paths, but verification checks each edge against its cited financial row.

In the between-date receipt-6 example, a three-edge contradiction has a direct decision interpretation. At the first numeric message X paired with reported receipt 7, paying is refused, so the rule must draw. After that action, a later message reports no trigger, inadequate coverage and receipt 6. Paying is refused there, and drawing is also refused. The two second-stage refusals make the required first draw impossible. This identifies incompatible requirements in the supplied decision problem. It does not yet tell us whether a particular economic path is impossible even with full information, or whether indistinguishable paths demand different actions.

[21] `data/accounting-master/revised-source-actor-messages-20261027/attempt-02/` preserves 47 comparisons, 394 assertions and 20 alteration controls. Independent reconstruction checks the dated reports, complete first-stage memory, protected policy population and shared-action restrictions. Unresolved sources and worlds retain their separate statuses.

The certificate comparison decides thirteen domains above twelve keys, including a satisfying rule for a 30-key reverse-receipt declaration and contradictions for receipt-6 declarations with 38 keys and a concurrent-correction declaration with 54 keys. The previous enumeration results remain unchanged. Missing world rows receive no certificate. A contradiction found within a resolved subset refutes that subset, while a satisfying assignment there cannot establish feasibility for unresolved alternatives.[22]

These results make the next question more specific. A rejected common rule calls for examining the financial histories responsible for its contradiction. That diagnosis must distinguish economic infeasibility from an observation that hides a consequential difference before proposing a repair to the model or its information assumptions.

17.28 When the financial history itself cannot be settled

The receipt-6 contradiction contains an economic impossibility that additional information cannot remove. Its two second-stage refusal rows share the exact rate point 19/320 and 43/800. Under the existing floating-coupon formula, these rates give $I_1 = 95/16$ and $I_2 = 43/8$. Checking all four action sequences at this one point avoids confusing two different financial histories that happen to produce the same message.

Opening cash $11/2 = 88/16$ cannot pay the first coupon 95/16, so either sequence beginning with pay is refused immediately. Drawing supplies one unit and costs 1/4, leaving first-settlement cash 5/16. Receipt 6 followed by repayment of the borrowed principal leaves 85/16, while the second coupon is 86/16. Draw/pay therefore fails by 1/16. Draw/draw also fails: the once-only facility has already been used, and the second coupon is below its strict draw trigger. All four sequences fail even for a manager who knows both coupons in advance.

At the same rate point with receipt 7, draw/pay succeeds and ends with cash 15/16. This is an explicit counterfactual, not a replacement for the assumed receipt-6 world. The earlier viable receipt-6 examples also remain valid. The diagnostic establishes one impossible point in the original family, rather than impossibility at every point. Later algebraic cash from a sequence whose first payment was refused is not an executed outcome.[23]

17.28.1 An exact boundary on the committed-right coupon domain

Keep the committed right, opening cash, once-only borrowing rule, fee and repayment timing unchanged. Let both coupons range over $[3, 6]$, and take R to be either 6 or 7. If $I_1 \le 11/2$, the first payment can be made without borrowing. Cash after the receipt is then $11/2 - I_1 + R \ge R \ge 6$, enough to pay every permitted second coupon. Thus pay/pay is viable throughout this first region.

If $I_1 > 11/2$, paying first is impossible, while the trigger permits a first draw. Since $I_1 \le 6$, the net draw of 3/4 leaves nonnegative cash after that payment. Receipt and repayment then leave

$$C_{\text{before second}} = \frac{11}{2} - I_1 + R - \frac{1}{4} = R + \frac{21}{4} - I_1. \tag{17.42}$$

[22] `data/accounting-master/policy-existence-certificates-20261027/` checks all 47 retained source/world comparisons and 260 small Boolean formulas against exhaustive assignments, with 1,083 assertions and 22 alteration controls. The 31 certificates comprise six assignments and 25 contradictions; two contradictions apply only to resolved subsets. Each small-domain result agrees with the protected full policy population.

[23] `data/accounting-master/policy-failure-diagnosis-20261027/` retains ten exact path comparisons, 337 assertions and seven alteration controls. Complete states and all guards agree with the independent recurrence; the two second-stage refusal rows use the same rational witness.

The first borrowing has used the facility, so the second coupon must be paid from this cash. Draw/pay is therefore viable exactly when $I_1 + I_2 \le R + 21/4$. Combining the two regions, the economically infeasible set is

$$\mathcal{N}_R = \left\{ (I_1, I_2) \in [3,6]^2 : I_1 > \frac{11}{2}, \quad I_1 + I_2 > R + \frac{21}{4} \right\}. \tag{17.43}$$

For receipt 6, the sum boundary is 45/4. Equality is viable: the second payment leaves exactly zero cash. Equality at the first trigger is also viable, because $I_1 = 11/2$ can be paid without drawing. For receipt 7, the sum boundary is $49/4 > 12$, above the largest possible coupon sum, so $\mathcal{N}_7$ is empty on this domain.

The result concerns the continuous coupon square. Exact affine certificates check every inherited guard of pay/pay in the first region and draw/pay in the second, and exclude every action sequence in $\mathcal{N}_R$. The three regions are exhaustive and disjoint because they split first on $I_1 \le 11/2$ and then on the sum inequality. Rational boundary examples illustrate the result, while the region certificates establish it.[24]

On the viable subset, the original refined first trigger suffices for a common rule: draw first exactly when that trigger is true, then pay second. The manager does not need the future coupon to choose the first action. This rule cannot settle the excluded points, which remain part of the original model. The result does not cover absent or mixed borrowing rights, arbitrary receipts, or a changed facility contract.

A financially impossible settlement is still a possible company history to represent. The unpaid coupon, the earlier draw and repayment, and the dates of failed payments must remain available to subsequent accounting and recovery events. Refusing a payment instruction must therefore preserve the outstanding obligation and the failed history rather than remove it from later statements.

17.29 Keeping an unpaid coupon through close and later payment

The refused second payment leaves a complete financial state to report. In the receipt-6 draw/pay history, the company has cash 85/16 and owes the full second coupon $43/8 = 86/16$. The difference 1/16 measures the additional cash needed to pay in full; it is not the outstanding liability. No part of the refused payment was executed. The accepted history ends after the second coupon accrual, with the facility principal repaid and the once-only borrowing right still used.

The retained report contains income 6, expenses 185/16 and loss 89/16. Assets are 1597/16, liabilities 843/8, and equity including the loss is $-89/16$. These reconcile because $843/8 - 89/16 = 1597/16$. Operating cash movement is $-3/16$. Financing cash movement is zero, but its gross record contains both a draw of 1 and repayment of 1. Netting those entries would conceal the use and repayment of the facility. The refused event does not enter the accepted-event history; its diagnostic and all unexecuted commands remain separately available.[25]

[24] `data/accounting-master/financing-viability-boundary-20261027/` checks two receipt domains, 366 guard implications, eight action-region exclusions and eight boundary examples, with 315 assertions and ten alteration controls. Emptiness certificates are verified independently of their producer.

[25] `data/accounting-master/refused-payment-reporting-20261027/` preserves the failed first verifier attempt and the corrected successful second attempt: ten cases, 396 assertions and twelve adverse controls. The method wrapper retains a refusal diagnostic, but the refused transition contains no after-state, journal or cash flow. Complete states and reports agree with independent arithmetic and the symbolic recurrence.

17.29.1 Closing the period preserves the debt and the missed decision

The stopped state is dated 31 December 2027, the last day of the declared period. A selected closing operation transfers the temporary income and expense balances into equity and opens the next period on 1 January 2028. Under the signed-book convention, the loss produces a debit capital balance 89/16, representing negative equity −89/16. Cash remains 85/16, the signed interest-payable balance remains −43/8, and the second coupon's amount paid remains zero. The facility balance stays zero and its used flag stays true.

The opening period has no cash flows yet. Earlier flows remain in the closed book, and an associated archive retains the original refusal, the unexecuted commands and the identity of the new opening. Carrying balances alone would lose the distinction between a missed payment and an ordinary unpaid accrual. Conversely, automatically resetting the used flag would invent a fresh borrowing right. The generic typed carry engine can express such a reset, but this selected coupon-preservation rule rejects it. States stopped on 30 June are left unassessed by this year-end operation; they are not silently advanced to December.[26]

17.29.2 A new instruction can reduce the claim without rewriting history

Suppose an explicitly supplied instruction pays 85/16 on 2 January 2028. This amount uses the carried cash; it is not supplied by a new loan or inferred managerial policy. The selected late-payment method reduces cash and the unpaid coupon by the same amount, and adds that amount to cumulative payment toward the second coupon. Let C be current cash, U the positive unpaid coupon and P its cumulative amount paid. For an instructed partial payment q, the accounting and memory changes are

$$C' = C - q, \qquad U' = U - q, \qquad P' = P + q, \qquad 0 < q < U, \quad q \le C. \tag{17.44}$$

Here $U = I_2 - P$, with the originally accrued coupon I_2 unchanged. Thus $U' + P' = I_2$: payment discharges part of the existing claim and creates no second expense. The strict inequality $q < U$ gives this selected method a genuinely partial settlement domain. Full settlement would require a distinct supported instruction and sufficient cash.

For the supplied instruction, $C' = 0$, $U' = 1/16$ and $P' = 85/16$. The earlier action remains `none`, because the original second payment was refused. A new `partial2` phase records the later settlement, while the used right and original refusal remain intact. Reusing the old policy solver with this new phase would require a separate compatibility argument; cumulative payment does not turn the original failed policy into a successful one.

Two supplied payments of 85/32 on 2 and 3 January produce the same final balances and cumulative payment. They produce two dated cash flows and a different last-event date, so equality of financial balances is not equality of histories. The report classifies these payments as operating under the inherited synthetic convention. Complete comparisons on both account charts retain the distinction between the two instruction schedules.[27]

The three operations answer different questions: a refusal states what was not executed, a close carries the accepted balances into the next period, and a later instruction supplies a new

[26] `data/accounting-master/coupon-carry-20261027/` compares six final-day carries and four nonfinal refusals, with 406 assertions and sixteen adverse controls. The checks cover the entire closing journal, closed and opening books, report and refusal archive. The close is provisional recorded-balance carry, not evidence of complete fiscal reporting.

[27] `data/accounting-master/coupon-partial-settlement-20261027/` contains eight complete-state, journal, flow and report comparisons, 446 assertions and twenty adverse controls, with native financial fallback disabled. Source identities, selected-method identities and the full original refusal archive are checked. These finite cases support the stated selected implementation; they are not a universal conformance proof for every partial-payment schedule.

event. None supplies missing contractual terms. Penalties, waivers, legal default, additional funding and actual accounting applicability require their own evidence and selected rules. Event completeness and external source authenticity also remain unresolved. The simulator can therefore preserve the distressed company history without inventing its recovery.

17.30 Which payment histories have the same future consequences?

The residual claim is a property of the cash-only payment rule, not an accident of dividing the payment into one or two instalments. Let $S_i = \sum_{j=1}^{i} q_j$ be cumulative payment after i accepted instructions. Each positive amount q_j reduces cash and the unpaid coupon equally and increases the paid amount by the same quantity. Starting from $C_0 = 85/16$, $U_0 = 86/16$ and $P_0 = 0$, repeated substitution gives

$$C_i = \frac{85}{16} - S_i, \qquad U_i = \frac{86}{16} - S_i, \qquad P_i = S_i, \qquad U_i - C_i = \frac{1}{16}. \tag{17.45}$$

The zero-payment prefix satisfies these identities. If they hold after i instructions, subtracting q_{i+1} from cash and claim, and adding it to payment, replaces S_i with $S_i + q_{i+1}$ in all three expressions. This proves the recurrence by induction. It also preserves $U_i + P_i = 86/16$, the originally accrued coupon.

Because the unpaid claim always exceeds cash by $1/16$, any positive affordable payment $q \le C_i$ is strictly below U_i. For a finite schedule of positive payments, affordability is therefore equivalent to its total being at most $85/16$: every earlier prefix sum is smaller. This statement concerns the economic amounts. Dates, identities and runtime support impose separate conditions. At cash exhaustion the company still owes $1/16$; further cash-only payment cannot remove that claim. A separate cash inflow or a supported change to the claim would be needed.

The selected method's actual expressions, account bindings, cumulative memory write and preserved fields match this recurrence on both charts. Twenty complete runtime comparisons include nonuniform and same-day payments, unequal rational fractions, a tiny first payment, and a schedule leaving exactly $1/2^{64}$ in cash. The last residual is retained rather than rounded to zero. These finite checks support the selected implementation while the induction establishes the algebraic result.[28]

17.30.1 A date or an event identity can distinguish equal balances

Now compare a payment of 1 on 2 January with payments of $1/2$ on 2 and 3 January. Both leave cash $69/16$, unpaid coupon $35/8$ and cumulative payment 1. Their contract memory agrees. Their current dates, accepted event identities and dated gross flows do not.

A fresh instruction dated 2 January can follow the first history. It cannot follow the second, whose last accepted event is already dated 3 January. Similarly, an instruction using the second instalment's event identity can be fresh in the one-payment history and a replay in the split-payment history. The runtime rejects the replay without changing balances. Thus equal financial totals do not preserve the full set of permissible instructions.

[28] `data/accounting-master/coupon-schedule-invariant-20261027/` retains four method inspections, 517 assertions and eight controls. The runtime permits at most 64 commands per program, bounds its accumulated trace at 8 MiB, and imposes record-size and exact-rational limits. A 65-command schedule whose total equals available cash is refused by the command limit. Such a resource refusal is not economic infeasibility; satisfying one limit alone does not guarantee execution.

A common fresh payment of 1 on 4 January succeeds in both histories and leaves cash 53/16, claim 27/8 and cumulative payment 2. Numeric statements and activity totals agree, but complete reports retain different dated gross flows. This gives a useful restricted result. For the same original obligations and rules, equal cumulative payment, identical future amounts, dates no earlier than either current date, and identities unused in either history, the financial recurrence remains equal whenever bounded execution accepts. It does not make the complete histories equal or establish compatibility with every future method, observation or policy.[29]

17.30.2 Resume from the current book and report from the period opening

Execution and reporting need different dates here. The original period opening supplies the starting cash and empty current-period flow history for a cumulative report. The current book supplies the balances, last event date, used identities and contract memory for the next instruction. Replacing one with the other would either forget accepted events or misstate the reporting interval.

The present declared-program runner validates a fiscal opening at the period start with no flows. It therefore refuses a midperiod current book as its opening. The selected continuation uses the same typed event engine directly for one instruction, retains the actual prefix unchanged, and reports the resulting book against the original period opening. This is a checked single-step interface. A general program-resumption interface still needs explicit bindings for both books, the verified prefix and the remaining commands; it cannot be implemented by resetting a date or discarding prior flows.

17.31 Resuming selected payments without losing their history

A selected same-period runner now extends the checked coupon prefix through several supplied instructions. Its plan retains the original period opening for reports and the current book for execution. It also identifies the original source, selected method registry, verified prefix, complete future instruction list and resource limits. A checkpoint's cursor counts only accepted new instructions. The earlier coupon refusal remains in its archive; it is not replaced by the outcome of a later payment.

For a successful three-instruction suffix, processing one event per call, two per call or the whole suffix produces the same terminal checkpoint. The complete accepted journals, dated flows, current book and cumulative report agree with independent arithmetic and uninterrupted execution. This comparison also covers a later unaffordable payment, duplicate identity and backwards date. In each case, execution retains the accepted prefix and leaves the refused instruction first in the remaining list.[30]

An execution pause has a different meaning from a refusal. A call may consume its allowed number of instructions while another valid instruction remains; this returns a budget pause and can continue from the checked cursor. An economic or chronology refusal stops before the

[29] `data/accounting-master/coupon-continuation-scope-20261027/` retains 24 complete continuation comparisons, 514 assertions and ten controls on the four chart/action origins. Original refusal archives and complete reports are checked against the external identified prefix. Altering a prefix and assigning it a new identity does not bypass the independent comparison.

[30] `data/accounting-master/coupon-program-resume-20261027/` retains sixteen cases and 48 chunk comparisons, with 824 assertions and eighteen adverse controls. A separate audit adds 493 assertions, twelve controls and two exact byte-boundary comparisons. The first attempt is preserved: its comparator incorrectly required identical duplicate-refusal wrappers. The generic runner detects a duplicate command before calling the event engine; the selected runner retains the engine's duplicate-event diagnostic. Both reject the same instruction and preserve the same financial state.

offending event, which cannot be silently skipped on a subsequent call. Changing that event requires a newly identified plan. A complete, refused or resource-limited checkpoint remains unchanged when submitted again under its existing plan.

Resource limits apply to the entire selected suffix, not afresh to each chunk. The inherited command ceiling is 64 and the default trace allowance is 8 MiB. A deliberately smaller allowance exposes the boundary exactly: 82,883 bytes fit the first accepted trace row, but not the second. An allowance of 82,882 bytes accepts no row. The runner retains the next command and reports the unchanged accepted state in both cases. Splitting execution into additional calls does not enlarge the allowance. These limits govern this diagnostic implementation; they neither determine the financial shortfall nor prove that a larger economically feasible schedule is impossible.

The checkpoint's hash is necessary for identifying the supplied record but is insufficient for validating it. The selected interface recomputes the bounded accepted prefix from the identified source, registry and instruction list, then compares the complete checkpoint. This detects altered request metadata even when a narrower financial comparison would still find the same balances. It also detects a changed opening, skipped cursor, missing trace, erased refusal, modified report or enlarged plan budget. Recomputed validation does not append a second accepted event or duplicate a cash flow.

This evidence establishes a selected same-period partial-coupon interface. It does not migrate the general program runner, support every method or reporting-period transition, or establish durable exactly-once commits under concurrent writers. Repeated prefix computation is also a cost, not a scalability result. The distinction between original opening, accepted current book and remaining instructions nevertheless supplies a concrete starting point for broader resumption contracts without erasing the company's earlier history.

Chapter 18

Reference computations that distinguish a useful approximation from a wrong answer

A method can be tested most clearly when a small version of its target is known exactly. This chapter constructs such targets for cash bounds, breach probabilities, policy comparison and residual minimization. The calculations are analytic reference cases. They are not a benchmark ranking and do not supply empirical evidence about a real company. Their role is to expose errors that a large simulation can conceal behind apparently stable averages.

18.1 An exact distribution from a declared prior

Return to cash $C = 90 - 2\alpha$, and now deliberately assume α is uniform on $[0, 1]$. This assumption is not inferred from the financial statements. It creates a transparent probabilistic reference. Cash is uniform on $[88, 90]$, with mean 89 and variance $4/12 = 1/3$.

For a covenant requiring cash at least 89.4, breach occurs when $\alpha > 0.3$. Its probability is 0.7. The expected cash conditional on breach is the mean of the interval $[88, 89.4)$, namely 88.7. These quantities answer different questions: unconditional mean, breach frequency and severity conditional on breach.

A simulation with the right mean but the wrong breach probability is not accurate for covenant analysis. For example, a two-point approximation placing equal mass at 88 and 90 has mean 89 but breach probability one half. Matching the first moment does not preserve threshold events.

The exact distribution also gives quantiles. Under the usual left-continuous generalized-inverse convention, the p-quantile is $88 + 2p$ for $0 < p < 1$, with endpoint values defined by continuous extension. The convention should be explicit when discrete mass points are introduced, because quantile definitions can differ at jumps.

18.2 A discrete grid with known bias

Approximate α by equally weighted midpoints $(i - 1/2)/n$, for $i = 1, \ldots, n$. The mean of the grid remains one half, so mean cash is exactly 89 for every n. Breach frequency is the fraction of midpoints above 0.3. For $n = 4$, midpoints are 0.125, 0.375, 0.625 and 0.875, giving breach probability 0.75.

The mean check passes perfectly while the breach estimate differs from 0.7. More grid points need not reduce that error at every refinement. With $n = 5$, the strict breach event includes only 0.5, 0.7 and 0.9, so its estimate is 0.6. The absolute error rises from 0.05 at $n = 4$ to 0.1 at $n = 5$. At $n = 10$ the estimate is exactly 0.7, whereas $n = 11$ gives $8/11$. Alignment with the threshold matters even as the grid becomes finer.

A bound explains what refinement does guarantee. For a threshold $t \in [0, 1]$, let $p_n(t)$ be the fraction of the n midpoints strictly above t. A midpoint is at or below t precisely when $i \le nt + 1/2$. There are therefore $\lfloor nt + 1/2 \rfloor$ such points, and

$$p_n(t) = 1 - \frac{\lfloor nt + 1/2 \rfloor}{n}, \qquad \bigl|p_n(t) - (1 - t)\bigr| \le \frac{1}{2n}. \tag{18.1}$$

To obtain the bound, write $k = \lfloor nt + 1/2 \rfloor$. The defining inequalities $k \le nt + 1/2 < k + 1$ give $-1/2 \le nt - k < 1/2$. Dividing by n bounds $p_n(t) - (1 - t)$. Thus the maximum permitted error shrinks to zero, although the actual error can rise between successive grids. The result holds for this uniform midpoint rule and one threshold event; a convergence study must still examine the event actually being queried.

If the threshold lies exactly on a grid point, strict versus inclusive inequality changes the finite approximation. The continuous reference assigns zero probability to one point, but a discrete approximation assigns positive mass. This is a concrete reason to preserve boundary semantics during discretization.

A deterministic grid is not an independent random sample. Standard Monte Carlo standard errors do not describe its quadrature error. Its uncertainty should be assessed through analytic bounds or refinement, not by treating grid points as independent observations.

18.3 A nonlinear guard with a narrow region

Let α remain uniform on $[0, 1]$, but suppose a special covenant fails only when $|\alpha - 0.5| < 0.01$. The exact probability is 0.02. A coarse endpoint or midpoint grid can miss this region entirely.

An independent random sample of size 100 has probability 0.98^{100}, about 0.133, of observing no failures. A deterministic grid can have either zero or several hits depending on alignment. The two methods fail for different reasons: sampling variation versus systematic discretization geometry.

A query-aware method should detect the guard and refine around its roots. If the guard is known analytically, solving its boundary is more efficient than hoping random paths discover it. Sampling remains useful when the region is complex, but the model should exploit available structure.

The example also warns against dropping a region because its volume is small. A two per cent event can dominate expected loss if severity is large. Probability and consequence must be evaluated together.

18.4 A rare-loss mean

Suppose loss is zero with probability 0.999 and 1,000 with probability 0.001. Expected loss is 1, while the most common observed loss is zero. In 100 independent draws, the chance of no loss is 0.999^{100}, about 0.905. Most such short runs report a sample mean of zero.

The variance is $0.001(1000^2) - 1^2 = 999$. The standard error of the sample mean at $n = 100$ is $\sqrt{999/100}$, about 3.16. A single observed mean cannot support a precise estimate. A normal approximation is also questionable in a sample with very few tail events.

This target can test whether a simulator reports uncertainty honestly. A zero-event run should not declare zero risk. A rare-event method should be checked against the known probability and mean, including its weighting and interval procedure.

The example does not establish how rare real defaults are. Its parameters are chosen to make sampling failure visible. A real application needs an admitted event law and severity model, with model uncertainty separate from Monte Carlo error.

18.5 Importance sampling on the rare event

For the same two-state target, choose a proposal that samples the loss state with probability 0.1 instead of 0.001. The target-to-proposal weight is 0.01 in the loss state and $0.999/0.9 = 1.11$ in the zero state. The ordinary weighted loss contribution is therefore 10 in the loss state and zero otherwise.

Its expectation under the proposal is $0.1(10) = 1$, the correct target mean. Its variance is $0.1(10^2) - 1 = 9$, much smaller than 999 in this constructed example. The reduction follows because the proposal deliberately visits the only important rare state.

This is a mathematical comparison for a known finite distribution, not a general performance claim for importance sampling. In a high-dimensional model, a poor proposal can create extreme weights or miss support. A proposal tuned to one failure mode may neglect another. The finite case provides a correctness test for the weighting formula.

If the implementation averages the unweighted sampled loss, it obtains expectation 100 under the proposal, not target expectation 1. If it reports the proportion of sampled defaults as the target probability, it reports 0.1 instead of 0.001. These large errors can look stable as sample size grows because the wrong quantity is being estimated accurately.

18.6 Paired policy differences

Suppose two policies have costs $Y_A = Z + 2$ and $Y_B = Z + 1$, where Z is a common random operating cost. The true difference $Y_A - Y_B$ is exactly 1 in every world. Using the same draw of Z gives zero sampling variance for the difference. Independent draws add unnecessary noise.

This is an ideal illustration of common random numbers. In a real policy comparison, the policies can change transition probabilities or event timing. The coupling must preserve each policy's correct marginal distribution. Reusing a random seed alone does not guarantee that corresponding random draws represent the same economic shock after control flow diverges.

A stable shock identity can help: demand innovation for a named period and customer should remain associated with that economic event rather than the next call to a global random-number generator. Otherwise adding a harmless diagnostic draw can change all subsequent scenarios.

The paired comparison should also hold the initial history fixed. Subtracting policy A on one reconstructed company from policy B on another confounds policy and history. If the goal is a range of policy differences, optimize the paired difference over the same family.

18.7 A sampled minimum and a true minimum

Let $C(\alpha) = 90 - 2\alpha$ on $[0, 1]$. The true minimum is 88 at $\alpha = 1$. A sample whose largest value is 0.9 has minimum cash 88.2. This is an upper bound on the true minimum because the sample is a subset of the family.

If a payment requirement is 88.1, the sampled minimum suggests no sampled failure while the true family contains failure. An optimization or analytic monotonicity argument identifies it immediately. More random sampling may eventually approach the endpoint, but no finite sample alone proves the exact bound.

Conversely, a relaxed outer model allowing $\alpha \in [0, 1.1]$ gives minimum cash 87.8. That is a lower bound on the original minimum, but its witness is not a feasible original company. Inner and outer methods can bracket the answer. The bracket's width, not a single attractive number, describes unresolved computation.

18.8 A local optimizer misses another basin

Consider an illustrative cash function $C(x) = (x^2 - 1)^2 + 0.1x$ on $[-2, 2]$. It has regions near both negative and positive one where cash is relatively low. A local optimizer initialized near one basin may not find the global minimum.

The function is a numerical diagnostic, not a proposed cash-flow mechanism. Its purpose is to test whether a method labels a local result honestly. A global claim requires exhaustive interval bounds, verified comparison of all stationary and boundary candidates, or another suitable certificate.

In a financial model, analogous nonconvexity can arise from fixed issuance fees, default options, tax thresholds or discrete investment. Repeated random starts can find more candidates but do not automatically prove completeness. The result should distinguish found local solutions from a verified global bound.

A benchmark should include such a case only if the target profile admits nonconvexity. Requiring a linear solver to solve it would be an unfair capability test. Conversely, describing a general solver while testing only convex examples leaves a material gap.

18.9 Residual scaling can alter a compromise

Suppose an approximation cannot satisfy two residuals simultaneously: $r_1(\theta) = \theta$ and $r_2(\theta) = \theta - 1$. Minimizing $r_1^2 + r_2^2$ gives $\theta = 1/2$. If the first residual is multiplied by 100 because currency units changed, minimizing $10000\theta^2 + (\theta - 1)^2$ gives $\theta = 1/10001$.

The different result is caused by weighting, not new economic information. In a well-specified exact system with sufficient capacity, both residuals may be zero and scaling would not change the exact solution. Under approximation, inconsistency or finite optimization, scaling affects the compromise.

This diagnostic motivates reporting component residuals in original units and testing unit invariance. A normalized aggregate loss can be useful for training, but it should not hide a cash pricing error or constraint violation. A single loss threshold needs an economic interpretation.

18.10 Expected residual versus mean squared contribution

The earlier example $r = \theta - 1 + \theta\varepsilon$ with mean-zero unit-variance innovation gives desired squared expected residual $(\theta - 1)^2$ and single-draw squared objective $(\theta - 1)^2 + \theta^2$. Their minimizers are one and one half.

A reference implementation can evaluate both objectives analytically and by Monte Carlo. It should verify that independent paired draws estimate the former expectation and identical draws estimate the latter. The test should compare the target functions, not merely whether training loss decreases.

A gradient check can add another layer. The desired derivative is $2(\theta - 1)$; the other derivative is $4\theta - 2$. At $\theta = 1/2$, the second vanishes while the first is negative one. Automatic differentiation can return the correct derivative of the wrong implemented objective.

This case is particularly relevant to learned equilibrium conditions because it separates stochastic estimation from target definition. Increasing batch size makes the wrong objective estimate more precise; it does not repair the missing expectation structure.

18.11 A boundary optimum

Minimize $(a + 1)^2$ subject to $a \geq 0$. The answer is $a = 0$, objective 1. An interior first-order condition gives $a = -1$, objective zero, which is infeasible. A constrained solver should return the boundary with the appropriate multiplier.

A penalty objective $(a + 1)^2 + \rho \min(a, 0)^2$ has a negative optimum for every finite positive ρ, namely $a = -1/(1 + \rho)$. The violation shrinks as the penalty grows but is not exactly zero. Reporting the penalized optimum as a feasible original solution without checking the constraint is wrong.

Penalty methods can be useful algorithms. Their residual feasibility must be assessed against the original constraint and the required tolerance. A small violation may be immaterial for some economic quantities and decisive when it represents issuing more shares than authorized or paying cash that does not exist.

18.12 Designing a fair method comparison

A comparison should use the same evidence family, objective, policy class, timing, terminal condition and accuracy target. A faster method that omits a recovery term is not a faster solution of the original pricing model. A method trained on a smaller domain should not be compared as though it covers the full stress region.

The baseline ladder can include exact finite enumeration, a refined classical numerical method, the plain proposed approximation and any enhanced version. Tuning budgets and stopping criteria should be declared. A deliberately weak comparator can establish only a narrow improvement over that comparator.

For stochastic methods, use replications and uncertainty analysis appropriate to the claim. If several candidates pass hard validity screens but differences are not statistically resolved, describe them as viable under current evidence. Runtime, tail error and objective values from a few seeds remain descriptive unless the predeclared analysis supports ranking.

18.13 What to retain from a run

A run should preserve the model and data version, command, environment, seeds, hardware mode, wall time, outputs and the plan defining the question. It should record which diagnostics are validity vetoes, which trigger repair and which merely explain behavior.

The result note should identify whether a failure invalidated the harness, implementation, target or candidate. A candidate failing a planned stress can motivate a repair phase. It does not automatically reject the broader research direction. A broken harness, by contrast, prevents interpreting the candidate comparison.

For the present manuscript, these are proposed experiments and exact reference calculations. No stochastic ranking is claimed. The retained arithmetic script checks selected deterministic examples and records its scope. The package companion `companions/reference-computations` reproduces its complete checked values and adds the midpoint-boundary examples above. The all-threshold conclusion rests on the floor argument in equation (18.1), not on the number of rational examples tested. Future numerical runs should use these cases to establish semantics before making claims about scalable performance.

Chapter 19

What the literatures contribute, and where their assumptions stop

The simulation problem crosses several literatures because it combines inference from incomplete reports, accounting representation, dynamic corporate finance, sequential information and computational approximation. No inspected source supplies the entire system. The useful synthesis identifies the problem each source actually solves and the additional work needed for AccDSL. A source's notation or software can be valuable without becoming a universal model of a disclosed company.

19.1 Economic events and accounting views

McCarthy's REA model provides a useful architectural antecedent. Resources, events and agents, linked through stock-flow, duality and control relations, organise economic activity before particular accounting views are produced (McCarthy, 1982, pp. 560–570). Double entry can be one external view. Claims can be derived from imbalances or retained as base records when their uses warrant it. The paper also recognises the practical need for aggregation and the fact that measurement and periodicity are not settled by a database schema (McCarthy, 1982, pp. 568–576). These distinctions support an event-and-contract representation with journal views. They do not imply that disclosures identify a unique event history, or supply the likelihood of observing a disclosure.

19.2 Inverse problems and incomplete information

Stuart (2010) develops Bayesian inverse problems with explicit forward models, priors and analytical conditions. It is relevant because statements are observations generated by hidden economic histories under a reporting map. The source does not reconstruct company journals or justify a particular prior over contracts. Its contribution is a disciplined distinction between the observation relation and the assumptions needed for inference.

Libkin (2014, 2016a) studies incomplete databases and answers across possible worlds. This perspective fits the requirement to preserve alternatives without assigning probabilities. A certain answer holds across admitted worlds; a possible answer has a witness. Translating that logic to finance requires a domain of economic histories and valid accounting relations, not merely null values in a table.

The two approaches are complementary. A feasible family expresses what has not been ruled

out. A probability measure expresses relative weight under a model. A Bayesian posterior can be computed on a family when priors and likelihoods are justified, but the family itself should not be discarded when those assumptions are uncertain.

19.3 Set identification and prior sensitivity

Giacomini and Kitagawa (2021, Sections 2.1–2.3) separates parameters identified by the observable distribution from remaining directions whose conditional priors are not revised by data. The retained technical reading uses the April 2020 working-paper version associated with the later publication. That version identity matters when citing equations and claims.

For company simulation, the immediate implication is conditional: first determine which hidden features the observation model identifies. A long history of interest expense may leave fixed-versus-floating terms unresolved when rates have not varied. Additional data of the same uninformative kind need not eliminate the ambiguity.

The local history family is not automatically the identified set of the source's statistical model. Finite compatibility with reports is weaker than equality of full observable laws. The manuscript therefore uses report-compatible histories unless the stronger identification argument has been established. This avoids borrowing a technical term to imply more evidence than exists.

19.4 Model averaging

Hoeting et al. (1999) surveys Bayesian model averaging, in which uncertainty across models is represented by posterior weights. This can be useful when alternative economic or observation models are explicitly specified. It also makes clear that averaging requires model priors and likelihoods, not branch counts.

The practical difficulty for AccDSL is defining model identity. Ten syntactic encodings of one contract should not receive ten times the weight of a single encoding of another. A prior over economic structures must be invariant to irrelevant representation choices or at least expose its dependence.

Model averaging can also conceal a decision-relevant minority case if the report shows only a mean. Conditional outcomes, tail measures and sensitivity to prior weights remain important. When weights are not defensible, the unweighted family is the appropriate retained representation.

19.5 Causal scenarios

Pearl (2009, Sections 3.2–3.4) distinguishes observational conditioning, interventions and counterfactual reasoning. This distinction matters whenever a financial scenario is described as the effect of a shock. A model fitted to periods with high rates need not identify the effect of changing rates while other mechanisms remain fixed.

AccDSL can support deterministic stress paths without claiming causal identification. Such a run answers a conditional mechanical question. A causal claim additionally needs structural equations, intervention semantics and assumptions about confounding and invariance.

Counterfactual comparisons require a coupling between factual and alternative worlds. Preserving the same hidden history and exogenous innovations is one modeled coupling. Independently drawing a new company for each policy changes the comparison. The causal literature supplies the conceptual distinction; the company-specific structure remains to be justified.

19.6 Corporate microsimulation

Shahnazarian (2004) links financial stocks through accounting bridges and separately specifies behavioral relationships for incorporated businesses. The inspected sections include stock movements, investment, disposal, depreciation and tax-related distinctions. The source's Swedish data and tax setting are not modern universal issuer defaults.

Its relevance is methodological: identities constrain movement while behavioral equations determine choices. A simulator can preserve that separation while using different data, accounting rules and policies. Importing estimated coefficients without target-specific evidence would be an unsupported transfer.

The source also illustrates why state includes more than current statement totals. Tax reserves, capital and other accumulated quantities affect future responses. AccDSL's state design should identify such memory by mechanism rather than simply copying a fixed list of variables from one model.

19.7 Accounting in system dynamics

Pierson (2020) develops accounting reporting within a system-dynamics model. Accrual, payment and reporting accumulation are separate, and the retained author model exposes equations for cash, receivables, payables and report timing. These are useful implementation concepts.

The source's payables payment rule uses a delay approximation, and its financing and inventory mechanisms have stated limits. A first-order payment delay is not an invoice-level due-date schedule. Reporting values expressed as annualized rates must be converted before comparison with period amounts.

The source therefore contributes a bridge between continuous simulation stocks and discrete reports. It does not establish a complete instrument-level financing engine or empirical validity for a particular company. AccDSL should reuse the separation of clocks while admitting more detailed schedules when the query requires them.

19.8 Agent-based stock-flow consistency

Caiani et al. (2016) models heterogeneous agents, loans and capital vintages with an explicit within-period sequence. The retained paper and selected official code show production, credit, payments, taxes and distributions connected through counterpart balance changes. This supports the importance of event ordering and financial counterparts.

The source begins from an artificial economy with its own initialization and calibration. That is not a reconstruction of an observed issuer from disclosures. Its reported simulation properties are author results, not experiments reproduced for AccDSL.

A particularly useful source-to-target warning concerns its operating-cash-flow label: the inspected definition subtracts scheduled principal repayments. That quantity can be appropriate for its credit-demand mechanism, but is not automatically the same as a financial-statement operating cash-flow category. Definitions must travel with values across adapters.

Selected code inspection is also bounded. The retained implementation revision exposes event switches, debt payment calculations and bankruptcy paths. It does not certify every formula, the entire runtime or correspondence with the paper-era executable. Before reuse, zero-rate schedules, timing, numerical domains and accounting adapters need targeted checks.

19.9 Financial planning and simultaneous equations

The retained Gentry working paper studies strategic financial planning from management inputs such as earnings-per-share growth, investment growth, leverage and financing costs (Gentry, 1974). It asks an inverse planning question about returns needed to achieve objectives. Its managerial distributions and independence assumptions are part of that model, not facts about every company.

The retained scan lacks a referenced derivation appendix in the inspected trailing pages, and correspondence with the later journal version has not been established. The manuscript therefore does not rely on an unchecked derivation as a validated algorithm. This is a source-specific limit, not a reason to dismiss the planning approach.

Gentry and Hoque (1990) organizes cash-flow components and estimates simultaneous behavioral equations. Its accounting reconciliation, definitions and behavioral relationships perform different jobs. The retained reading identifies a normalization inconsistency in one displayed cash term and records the local algebraic normalization explicitly. It does not silently claim a corrected edition.

The reported sample fits do not establish out-of-sample intervention validity. Firm-level data, a complete instrument matrix and code were not available for local replication. The useful comparison is architectural: a simultaneous system can propose related annual amounts, while a scenario engine still needs dated events, feasibility, information and accounting treatment.

19.10 Dynamic corporate finance

Strebulaev and Whited (2012) surveys dynamic corporate-finance models with investment, financing frictions and default. The risky-debt section makes debt pricing and equity value interdependent. This is essential for any model claiming endogenous financing terms.

The retained equations also illustrate why source inspection cannot stop at a citation. The face/proceeds convention mismatch in equations (3.26)–(3.27) must be reconciled before implementation. A positive-recovery calculation exposes it; a no-default example can conceal it. The present manuscript uses distinct cash and promise symbols and retains the source qualification.

Other simplifications require equally explicit transfer decisions. Signed net debt can collapse cash and borrowing. Present-valued tax benefits are not actual tax receipts. Aggregate issuance costs do not create legal share rights. A structural model can be internally useful under these assumptions while remaining insufficient for a full accounting simulator.

19.11 Bank funding and adjustment

Halaj (2016) models portfolio choices under uncertain funding and constraints. Behn et al. (2019) links bank behavior to capital and liquidity restrictions using its own supervisory-data setting and structural assumptions. These sources show that funding is constrained behavior, not an automatic balancing liability.

A corporate issuer does not inherit their bank-specific objective or regulatory measurements. Their estimated parameters are not identified from public company statements merely because the balance-sheet labels resemble those of banks. AccDSL must distinguish accounting, prudential and economic quantities.

The transferable design principle is a policy interface that receives state, information and

constraints jointly. Funding availability can change under stress and affect asset decisions. Applying that principle to another sector requires sector-specific rules and evidence, not a relabeled bank equation.

19.12 Contract algorithms

The retained ACTUS technical specification separates schedules, state transitions and payoffs (ACTUS Financial Research Foundation, 2026). This is a promising component boundary for contractual cash flows. It does not supply operating behavior, accounting recognition, consolidation or a justified future risk process.

The inspected specification and service code leave important adapter conventions to be resolved: timezone mapping, midnight interpretation, same-date event order, day-count implementation, currency scaling and risk observations. A service fallback to current wall-clock time plus a horizon is incompatible with reproducible historical simulation unless overridden explicitly.

The selected service requests a particular core dependency but does not establish a fully resolved, matched implementation and dictionary version. No runtime adoption follows from the inspection alone. A conditional algebraic correspondence for a simple borrower loan is useful evidence, but not complete executable conformance.

19.13 Sequential decisions under partial observation

Kaelbling et al. (1998) provides the belief-state formulation of planning under partial observation with specified transition and observation probabilities. It explains how information history can be summarized for control. The relevant sections were inspected in the retained author preprint, including rendered pages because the copy is scanned.

AccDSL adds two complications. Some uncertainty is unweighted and cannot be represented as a single probability belief without assumptions. Different actors can have different private information. A public analyst's belief is not necessarily the state on which management acts.

The POMDP formulation is therefore a method for an admitted probabilistic subproblem, not the universal semantics of all company uncertainty. It is especially useful for testing nonanticipativity, signal value and belief updates in finite reference cases.

19.14 Exact aggregation and knowledge compilation

Givan et al. (2003) gives equivalence and minimization results for MDPs, including policy-preserving stochastic bisimulation. Its fixed-action counterexample explains why equal outputs under preselected action sequences can be too weak. AccDSL's local extension preserves observations and requested outputs as part of a joint successor law.

Darwiche and Marquis (2002) maps representational tradeoffs among compiled logical languages. It supports sharing and query-aware structure while warning that succinctness and tractable operations trade off. It does not prove that arbitrary financial histories can be compiled cheaply.

Together these sources support a careful answer to state explosion: share common structure, preserve joint constraints and aggregate only under a scoped sufficiency condition. They do not support the claim that replacing a tree with a graph eliminates exponential worst cases.

19.15 Filtering and approximation

Doucet and Johansen (2008) explains particle filtering, smoothing, degeneracy and conditional analytic integration in the retained tutorial version. Its technical treatment motivates weighted particles and diagnostics when a state-space probability model is specified.

Unknown static contract parameters require additional care. Repeated resampling can remove types without any evidence ruling them out. A high effective sample size after resampling does not guarantee historical diversity. The retained feasible family should therefore remain available alongside the particle approximation.

Dupačová et al. (2000, Sections 2–4) studies probability-metric scenario reduction in the retained 2000 preprint. Theorem 3.1 solves redistribution for a fixed subset; the subsequent subset-selection algorithms are heuristics. The stability results depend on the stochastic program and its regularity, while the reported electrical-load experiment measures distribution distances in its own setting. It does not establish company default accuracy or preserve every adaptive policy. The local derivations in Chapter 9 explain the resulting expected-output and policy bounds, and the discontinuous-threshold counterexample shows their limits.

19.16 Dynamic ambiguity

Epstein and Schneider (2002) derives recursive multiple-priors preferences under stated axioms and rectangularity relative to an information filtration. The definition, theorem and relevant proof structure were inspected in the retained September 2002 preprint. The result concerns a particular preference representation.

Its relevance is the danger of using stagewise worst-case recursion with an uncertainty set that restricts cross-time dependence. Rectangular enlargement can change the admitted joint laws. A fixed unknown contract cannot silently become an adversary choosing a new type each period.

AccDSL should offer commitment, original-set and recursive robust formulations only with their semantics stated. The choice can alter policy and should remain visible. Dynamic consistency is not a label that makes every robust calculation equivalent.

19.17 Learned economic solvers

Maliar et al. (2021) formulates dynamic economic equations for deep learning, with careful distinctions among residual objectives and sampling constructions. Its definitions and algorithm are relevant to the supplied lecture's proposed solver. The double-sampling derivation in this part shows why the target objective matters before training begins.

Duarte and Fonseca (2026) is a newer structural-estimation working paper with jointly solved financing and value components, lagged-network training and local refinement. Its assumptions, state and pricing conventions differ from the lecture model. Its numerical organization is a candidate, not a verified AccDSL implementation or general global-optimality result.

Official code for these learned methods has not been comprehensively audited for this local integration. Consequently, the manuscript makes no code-equivalence, convergence or performance promotion claim. Any implementation decision must add that inspection and a target-specific numerical protocol.

19.18 Disclosure selection and the observation likelihood

Imagine equal prior probabilities of a good and bad type. If good types disclose with probability 0.9 and bad types with probability 0.1, then disclosure raises the probability of a good type to 0.9, while silence reduces it to 0.1. Treating a missing disclosure as no information would retain 0.5 in both cases. The discrepancy comes from omitting the selection mechanism, not from numerical approximation. These probabilities are an authored example, not estimates of actual issuer disclosure behaviour.

Let Y denote potentially observable information, M the pattern of observed components, $f_\theta(Y)$ its model and $g_\phi(M \mid Y)$ the publication or missingness mechanism. The likelihood of the realised data is

$$L(\theta, \phi) = \int f_\theta(y_{\text{obs}}, y_{\text{mis}}) g_\phi(m \mid y_{\text{obs}}, y_{\text{mis}})\, dy_{\text{mis}}. \tag{19.1}$$

Here the realised pattern m fixes which coordinates are observed; the integration is over the missing coordinates only. Discrete missing components use a sum instead. If, for the realised observation, $g_\phi(m \mid y)$ is constant over all compatible missing components, it factors out of this integral. Together with distinct parameter spaces this yields the relevant likelihood proportionality; a Bayesian factorisation also needs the appropriate prior independence. Seaman and coauthors distinguish these realised-MAR conditions from the everywhere-MAR conditions relevant to repeated-sampling comparisons (Seaman et al., 2013, Theorems 1–3, Section 5.4 and Appendix). These are sufficient conditions for specified inferential equivalences. They do not demonstrate that corporate missingness satisfies them.

19.19 When silence has an equilibrium interpretation

A separate local illustration makes non-disclosure endogenous. Let value V be uniform on $[0, 1]$. A manager knows V with probability $1/2$, independently of value; otherwise the manager cannot disclose it. Informed disclosure is truthful and costless, the manager maximizes the contemporaneous price, and risk-neutral investors price at conditional expected value. Suppose an informed manager discloses above cutoff a. Silence occurs with probability $(1 + a)/2$. Its unconditional value contribution is $1/4 + a^2/4$, giving silence price

$$P_{\text{silence}}(a) = \frac{1 + a^2}{2(1 + a)}. \tag{19.2}$$

An informed manager at the cutoff is indifferent, so $a = P_{\text{silence}}(a)$. Multiplying through and collecting terms gives $a^2 + 2a - 1 = 0$, hence $a = \sqrt{2} - 1 \simeq 0.4142$. Types above that value disclose and those below it prefer the silence price. The uninformed manager explains why silence does not identify the worst type. This derivation is a local example; the blocked Dye original is not claimed to have been reproduced.

Nor is favorable selection universal. Suijs models a firm that knows its return but is uncertain how investors compare it with an outside opportunity (Suijs, 2007, Sections 3–4, Proposition 2 and Appendices A–B). Under the stated conditions a profitable firm can disclose lower returns and withhold higher returns. Multiple partial-disclosure equilibria are possible. The implication for AccDSL is to represent alternative disclosure mechanisms and test their consequences, rather than hard-code one direction of selection.

19.20 What changes under an intervention

A mechanical rate stress replaces an input while holding specified other relations fixed. A conditional forecast computes outcomes among states where an observed rate has a chosen value. A causal intervention changes the mechanism generating the rate and then propagates the remaining structural relations. These calculations can disagree.

For a finite counterexample, let U be a fair binary variable and observe $X = U$, $Y = U$. An alternative structural model has $X = U$, $Y = X$. Both give exactly the same observed joint distribution. Under an intervention setting $X = 1$, the first model retains $\mathbb{E}[Y] = 1/2$ while the second gives $\mathbb{E}[Y] = 1$. Observational agreement does not identify the intervention response. A forecast labelled as causal therefore needs the structural assumptions that distinguish these possibilities, or must retain the set of responses. Source corrections and simulated observations also have different time meanings: a hypothetical future disclosure updates an actor inside that run, not the analyst's actual present evidence.

19.21 What remains to be researched

The survey is broad but not exhaustive. Important remaining questions include scenario-reduction guarantees for adaptive company policies, robust filtering with static unidentified parameters, strategic disclosure, multi-creditor restructuring, real-company policy estimation and empirical forecast evaluation. Some require additional primary sources; others require derivations and data specific to the target company.

The earlier Francis–Rowell simultaneous-equation lead remains technically uninspected because attempted routes did not provide the paper. It cannot be declared inferior or irrelevant on that basis. The missing Gentry derivation and source-version questions also remain explicit.

A source gap blocks the unsupported claim, not all independent work. The methods already derived here can be specified and checked while the literature queue continues. This keeps research honest without allowing an inaccessible paper to become a reason to stop every implementation task.

19.22 The resulting method proposal

The synthesis favors a joint history family, dated contract and ownership state, actor-specific information, explicit scenario semantics, feasible policy generation, event-to-accounting translation and query-driven computation. Each component has a distinct literature basis and a distinct local obligation.

No single source validates their composition. The worked cases and proposed tests supply initial evidence for the interfaces, while broader implementation and empirical studies remain necessary. The design should therefore preserve the origin and scope of each assumption instead of presenting the synthesis as an established off-the-shelf theory.

The next chapter turns this method proposal into concrete records and algorithms. Its engineering choices remain replaceable. What must survive replacement is the meaning of evidence, uncertainty, information, financing and result status developed throughout this part.

Chapter 20

Implementing the proposed methods without losing their meaning

The preceding chapters define what a result means. Implementation should preserve that meaning across extraction, reconstruction, simulation and reporting. The engineering choices in this chapter are proposals and can be replaced. The essential requirements are stable: evidence is not an assumption, a history family is not a sample, actor information is not global state, a financing request is not cash, and numerical failure is not default.

20.1 Evidence records

An evidence assertion needs a source identifier and exact location, value or text, concept, entity, dimensions, units, basis, period and precision. It also needs publication and ingestion information, extraction method and review status. A correction links to the assertion it revises instead of deleting it.

The record should preserve original text alongside normalized values. A normalized number can be checked against the source, and an unresolved phrase remains visible. A mapping from source label to model concept is a separate interpretation with its own support. This allows the mapping to change without pretending the source changed.

An evidence view selects assertions for a particular cutoff and purpose. It can represent as-published, currently restated or analyst-reviewed information. The view should be immutable once used for a frozen result. A new correction creates a new view and an explicit dependency change.

20.2 Assumptions as first-class inputs

An assumption record identifies the quantity or relation supplied, its scope, provenance and role. Examples include a fixed management payout policy, an offer price, a prior over debt types or a terminal liquidation rule. A convenient default should be marked as a hypothesis or baseline rather than disguised as observed evidence.

Assumptions can conflict. A policy allowing unlimited borrowing conflicts with an offer cap if both claim to govern the same action. The system should detect or expose such inconsistencies. Precedence rules must be explicit; silently favoring whichever input was loaded last is unsafe.

Sensitivity should vary assumptions by identity. If one assumption supplies both customer

delay and credit loss, changing only one derived field can produce an inconsistent variant. A structured assumption relation makes the intended intervention and its dependencies visible.

20.3 History families

A history family contains shared economic records, persistent unknowns, domains, constraints, interpretation alternatives and source support. It records the domain for which completeness is claimed and the method used to construct it. Optional witnesses are members, not substitutes for the family.

Continuous variables should retain units and bounds. Discrete choices should retain mutually exclusive premises where appropriate. Cross-time identity should connect a loan, customer or asset across periods. A new contract version is a dated event, not a reassignment of an old unknown.

The family can expose queries for feasibility, witnesses, possible answers, certain answers and bounds. A probability operation requires an attached measure record. The API should not offer an unconditional mean for an unweighted family without requesting or selecting an explicitly declared measure.

20.4 Contract and security state

A contract record identifies parties, currency, dates, obligations, rights and clauses. Its execution state records remaining principal, accrued amounts, past exercise and breach status. Accounting carrying values are separate measurements linked to the contract.

A security register identifies classes, units, holders or categories, rights and settlement status. It distinguishes authorized, subscribed, issued, outstanding, treasury and potentially issuable units where the profile needs them. Conversion links the extinguished claim to created shares.

Unknown clauses should remain typed unknowns or alternatives. An unsupported clause should block only claims depending on it, with an explicit reason. A simple loan profile can still calculate unaffected payments while refusing a general default or option valuation. This is more useful than either inventing a default clause or refusing every calculation.

20.5 Actor information

Each actor has an information specification: observable variables, private facts, report channels, lags and memory. Policies receive a view constructed from that specification. They should not receive the full simulated world and rely on programmers to remember which fields are forbidden.

A policy invocation records its decision time and input view. Tests can then check that changing unobserved future data leaves the action unchanged. A manager's private contract knowledge can be included without exposing it as actual public evidence.

Randomized policies need a declared randomization source independent of unavailable future information. Reusing a shock draw as a policy coin can create unintended dependence. Economic random variables should have stable identities and explicit coupling, not depend on incidental call order.

20.6 Scenario definitions

A scenario identifies external paths or distributions, interventions, horizon, event timing and policy treatment. It distinguishes a deterministic stress from observational conditioning and a causal intervention. A scenario can also define alternative future observation processes.

Shock units should be typed: a two-percentage-point rate increase differs from a two per cent proportional increase. Currency rates need orientation. Price levels differ from returns. The scenario layer should reject ambiguous conversions rather than rely on labels.

A scenario version should include the underlying risk-process and observation versions. Changing a future disclosure rule can change policy values even when the rate path is unchanged. The cache and provenance system must recognize that dependency.

20.7 Event proposals and admission

A policy proposes an action. The event service checks contractual rights, resource availability, counterparty acceptance and timing. An accepted action creates dated events; a rejected action returns a reason and leaves the state unchanged except for any explicitly modeled attempt cost.

Financing proposals should include requested cash or quantity, eligible offers and settlement conditions. The service calculates proceeds, fees, promises and ownership consequences. It does not infer unlimited supply from a balance identity.

Admission can be conditional over a history region. If an unknown covenant makes an offer feasible in some histories and infeasible in others, the service returns guarded outcomes. It should not choose one witness merely to avoid branching.

20.8 Ordering events

The scheduler orders economic events by effective time and declared precedence. Same-time events can still have dependencies: proceeds must arrive before they fund payment, and a reset can occur before or after accrual according to contract terms. A total ordering chosen arbitrarily by event identifier can change results.

A partial-order representation can express independent events and necessary precedences. If simultaneous equations are genuinely required, they should be passed to a solver rather than forced into a sequential order. The result should distinguish an unresolved ordering from numerical nonconvergence.

Time conversion needs explicit precision and timezone rules. A local midnight, 23:59:59 and next-day midnight are not interchangeable in a boundary-sensitive query. The existing exact timing work in the repository is a foundation; external contract adapters must preserve its distinctions or state their loss of precision.

20.9 Accounting rule application

After an admitted economic event, the accounting service selects a rule under the declared basis, jurisdiction, entity facts and effective period. It produces a complete posting bundle and supporting measurements. Missing authority or facts can yield conditional treatments or a typed refusal.

Posting bundles are validated atomically. They reconcile debits and credits, currencies, account contexts and event-specific quantities. An event identifier prevents duplicate application. Reversal and correction events refer to the original bundle rather than editing it invisibly.

The reporting service aggregates within a specified perimeter and period. It preserves source-to-event-to-journal-to-line provenance. A consolidated report can eliminate internal balances while retaining the underlying entity records for liquidity and legal-entity analysis.

20.10 Symbolic execution

A symbolic state contains expressions and constraints rather than one numeric company. An event whose rule is common across the region can update expressions directly. A guard partitions the region only when its truth changes the relevant event or output.

The executor should preserve coverage of the parent region and distinguish disjoint alternatives from overlapping descriptions. Overlap can duplicate probability mass if weights are attached. A symbolic union should therefore use semantic region identity rather than merely concatenating records.

Simplification can share subexpressions and eliminate variables under checked projection rules. Any approximation records an outer or inner relation, error target and scope. A witness extracted from an outer approximation must be validated before being presented as a possible history.

20.11 Optional probability measures

A measure record identifies support, weights or densities, parameterization, conditioning variables and source assumptions. It is attached to economic variables and histories, not to the number of stored nodes. Duplicate encodings should not change the measure.

Updates record likelihood and evidence identity. Reusing the same evidence twice should be detected where the model treats it as one observation. Correlated sources require a joint observation model or an explicit approximation.

Simulation can carry both the retained feasible family and an approximate measure. A particle filter's support loss does not delete the original family. A user can request probability estimates under one measure and robust bounds over the family without conflating them.

20.12 Query planning

A query names outputs, dates, actors, scenarios, policies and acceptable guarantees. The planner traces dependencies through contracts, guards, observations and reports. It selects exact symbolic calculation, optimization, enumeration or approximation according to the admitted profile.

The planner should prefer the smallest method that answers the question. An affine interest range does not require a million paths. A policy-sensitive signal problem cannot be answered by a fixed-path cash table. A worst-case query cannot be answered by an unlabeled Monte Carlo minimum.

Resource budgets are explicit. If the selected exact method cannot finish, the planner may return bounds or ask for an allowed approximation according to the request. It should not

silently downgrade a guarantee. Useful partial results include verified witnesses and certified safe regions with unresolved remainder.

20.13 Result records

A result identifies its target and scope, evidence and assumptions, conditional regions, values or formulas, measure if any, numerical status and provenance. It can contain exact values, intervals, distributions, witnesses and unresolved statuses together. Their semantics should be explicit.

Possible failure, certain failure and unresolved failure are different answers. A breach is different from default. An infeasible proposed action differs from a company with no feasible action in the modeled class. An unsupported mechanism differs from numerical failure. These distinctions should survive both machine output and reader-facing explanation.

Aggregates must retain unresolved regions or mass. If some paths fail numerically, reporting only successful paths can bias results. The system should state what portion of the admitted family or probability remains unevaluated and avoid renormalization unless that conditional target is explicitly requested.

20.14 A small request sequence

A user asks whether cash can fall below 14 after a rate and customer shock. The system selects an evidence view and reconstructs the four-history family of Chapter 13. It applies the stated events, obtains cash formula $33 - 18c - 2f$, and solves the threshold condition.

The result returns the concentrated floating witness with cash 13 and the other three outcomes. It states that no probabilities are attached. If the user adds a committed facility, the event service checks its timing and eligibility, creates a draw in the required region and recomputes debt and cash.

If a new source excludes the concentrated floating combination, the family version changes and the query's dependency triggers recomputation. The old result remains reproducible. This sequence demonstrates the intended workflow without requiring every possible company to be instantiated as a separate database.

20.15 DSL and typed-library authoring

A textual language can make scenario and contract definitions readable and reviewable. A typed library can offer direct programmatic construction. Both should lower to the same intermediate representation and be compared on the same tasks. The DSL is not justified merely by its existence.

Equivalent authored inputs should produce equivalent IR and results. Syntax convenience should not alter uncertainty or timing. A textual default that silently assumes a rate or probability would be especially dangerous because it can look like a declarative fact.

The existing repository includes language, expression, method and event work that should be reused where it matches these semantics. New records should be added only when an included operation cannot be expressed adequately. The design should avoid rebuilding a parallel incompatible representation for the new chapters.

20.16 Adapters to external components

An external contract engine can propose schedules and payoffs. An optimization package can solve a declared mathematical problem. A probabilistic library can approximate a posterior. Each adapter needs input and output semantics, version identity, numerical conventions and failure mapping.

The ACTUS inspection illustrates the boundary: contract events do not supply accounting journals, and local-time schedules do not automatically preserve exact timestamps. A double-valued payoff is not exact rational evidence merely because its printed decimal matches a fixture.

Adapters should be checked against small independent cases and preserve unsupported outcomes. A service exception must not become zero cash flow. A missing risk series must not silently carry forward an unrelated value. Dependency and license review remain necessary before selecting a runtime, but no external adoption is claimed by this manuscript.

20.17 Incremental implementation

The first vertical slice should support a bounded history family, exact event timing, simple debt and equity offers, actor information and coherent statements. It should include both successful and refused paths. The worked company and refinancing cases provide expected outcomes.

The next slice can add class rights, conversion, group transfers and persistent tax or asset state. Later work can add nonlinear pricing, partially observed policy optimization and controlled approximation. Each stage should preserve the full mission's open requirements rather than imply that a small slice closes whole-company representation.

A slice is complete only when its source-to-result behavior is checked and its unsupported scope is explicit. A collection of isolated module tests is useful but insufficient for composition. The critical interactions are where financing, information and accounting meet.

20.18 Review and release evidence

A release should preserve model and source versions, exact fixture results, numerical validation where applicable and rendered documentation. Engineering conformance, accounting authority, scientific validity and human comprehension remain separate evidence categories.

A clean PDF or passing arithmetic script does not establish product readiness. A reader-facing explanation must still make assumptions and derivations understandable. A human review can identify comprehension problems that mechanical checks miss; its absence does not prevent provisional drafting but does prevent claiming that acceptance occurred.

The stable documentation links should point to a built, identified candidate with an honest status. Historical manuscripts and hashes remain available. This allows the design to advance without rewriting evidence from earlier stages or confusing the newest document with a completed system.

Chapter 21

Groups, counterparties, and contagion across otherwise coherent statements

A company group is not a single bank account. Consolidation can eliminate internal balances for reporting while cash remains held by particular entities, currencies and custodians. Stress can also propagate through guarantees, collateral and common counterparties. These mechanisms make group simulation a natural test of whether the model preserves economic rights beyond accounting totals. The examples here stipulate transfer and priority rules; actual legal applicability requires source-specific authority.

21.1 Consolidated cash and entity cash

Consider a parent P and subsidiary S. P holds cash 5 and owes external debt 20 tomorrow. S holds cash 30 and owes external suppliers 8 the following day. The consolidated cash balance is 35. It exceeds the parent's debt maturity, but that fact does not establish that P can pay.

Suppose S may transfer at most 10 before tomorrow because the remainder is restricted or needed under a binding local rule. P can assemble 15 and has a shortfall of 5. A consolidated cash forecast that deducts 20 from 35 reports 15 remaining and misses the entity failure.

If a transfer of 15 is permitted instead, P can pay but S's remaining cash must still cover its own obligations. The transfer is not free group liquidity: it changes where resources are held and can create an intercompany claim or distribution. The event should be represented in both entities before consolidation.

21.2 Intercompany loan mechanics

Suppose S lends 15 to P. S records an intercompany receivable and cash outflow; P records cash and an intercompany payable. P then repays external debt 20 using its original 5 plus the loan. Consolidated cash falls from 35 to 15 and external debt falls by 20. Internal claims eliminate in the consolidated statement under the stipulated perimeter.

The internal loan nevertheless matters for future entity liquidity. P owes S according to its terms. If S later needs cash, it cannot spend a consolidated elimination entry. A model that deletes internal contracts after consolidation loses repayment dates and possible stress transmission.

Interest on the internal loan can affect entity tax and distributable earnings even when consolidated interest eliminates. Currency differences can create further effects. The general engine should retain legal-entity records and produce consolidation as a reporting view rather than using the consolidated view as its only state.

21.3 A guarantee creates contingent support

Suppose P guarantees S's external supplier obligation of 8. The guarantee produces no cash payment while S pays normally. If S fails, P may be required to pay under the stipulated terms. The group's reported balance sheet may not reveal the same exposure as a direct loan.

A scenario that stresses S can therefore affect P even when there is no planned intercompany transfer. The guarantee needs beneficiary, covered obligation, trigger, cap, expiry and settlement timing. A vague statement of support is not automatically an enforceable guarantee. Conversely, omitting a binding guarantee because it has no current principal balance understates contingent funding need.

If P's payment gives it a recovery claim against S, that successor claim should be recorded. Paying the guarantee and also leaving the supplier fully paid by S would double settle the obligation unless the model includes reimbursement or another event. Counterparty identities make this check possible.

21.4 A common collateral pool

Two loans may claim the same collateral. A model that assigns the full collateral value independently to each overstates recovery. Let collateral realize 50, with secured claims 40 and 30. Under an assumed first-ranking claim of 40 and second-ranking claim of 30, recoveries are 40 and 10.

If the claims rank equally and share proportionally, recoveries are 50(40/70) and 50(30/70). Both allocations conserve 50 but differ by creditor. Priority is an economic and legal fact, not an accounting balancing choice.

The collateral record should link each asset to encumbrances and ranking. Unknown ranking creates alternative recovery histories. An aggregate secured-debt amount does not identify whether collateral is shared, separate or overpledged. A stress result should preserve that ambiguity when it affects pricing or liquidity.

21.5 Collateral value and borrowing capacity

A borrowing-base facility can limit drawn debt to a fraction of eligible collateral. Suppose eligible receivables are 100 and the advance rate is 0.8, giving capacity 80. Drawn debt is 70. A customer default removes 20 from eligibility, reducing capacity to 64 and creating a borrowing-base deficiency of 6.

The credit loss has two effects: it reduces asset value and can trigger a cash repayment or collateral substitution requirement. A model that records only the impairment misses the liquidity consequence. A model that records a repayment without the contractual deficiency rule invents one.

Eligibility can depend on age, concentration, jurisdiction and disputes. Two receivable pools with equal net carrying value can have different borrowing capacity. The reconstruction may

need invoice cohorts or symbolic eligibility amounts even when the financial statement reports one total.

21.6 A margin call before cash income

A derivative can economically hedge a future exposure while requiring collateral today. Suppose a company has a fixed-rate asset whose economic value rises when rates fall and a derivative position that offsets that value movement but requires cash margin after an adverse mark. The ultimate hedge can reduce value risk while increasing short-term liquidity need.

The cash call depends on collateral agreement, thresholds, minimum transfer amounts, eligible assets and settlement timing. It cannot be inferred from the derivative's year-end fair value alone. A scenario that nets the derivative and underlying value change may correctly describe net economic exposure while missing gross cash settlement.

The simulator should track valuation, collateral posting and final payoff as distinct events. Posted collateral may remain an asset or claim under the applicable treatment; it is not necessarily an expense. Reusing the fair-value change as a cash flow can create double counting.

21.7 Currency trapped in the wrong account

Suppose P owes 10 foreign units and holds 100 domestic units. At an executable exchange rate of 8 domestic per foreign, conversion needs 80. Under a depreciation to 12, it needs 120. If conversion capacity is limited or markets close before payment, even a favorable reporting-currency total may not be usable.

Now suppose S holds 10 foreign units but cannot transfer them to P in time. Consolidated foreign cash equals the payment, yet P still faces a shortfall. A model should retain both currency and entity location. Netting by currency across the group can be as misleading as netting all currencies into one reporting amount.

A permitted currency swap or intercompany transfer can solve the problem, but it introduces counterparties, settlement risk and possible costs. The model should generate those transactions rather than assume frictionless conversion as an unnamed default.

21.8 Settlement risk

Two offsetting payments on the same date need not settle simultaneously. A company may pay one currency before receiving another. A counterparty failure between the legs can leave an exposure. A net daily cash amount does not describe the gross settlement sequence.

For a bounded example, P pays 80 domestic units at noon and expects 10 foreign units at 15:00. Its foreign debt is due at 14:00. Even if both legs settle as promised, the receipt is too late. A settlement service with payment-versus-payment or a bridge can change the sequence, but must be represented.

Exact intraday simulation is unnecessary for every query. It becomes necessary when ordering affects payment feasibility. The method should allow event resolution to increase around material boundaries without pretending that an annual model resolved them.

21.9 Cross-default and acceleration

A default on one obligation can trigger rights under another. The triggering event may be nonpayment, acceleration, insolvency or a threshold amount. A generic flag called default cannot substitute for the actual cross-default clause.

Suppose S misses 8 and P has a loan of 50 whose clause triggers only for subsidiary defaults exceeding 10. The event does not meet that threshold. If the clause instead aggregates several defaults totaling 12, it may trigger. The scope, aggregation period and cure rights matter.

Acceleration changes the payment calendar and can create a liquidity spiral. The model should record the trigger, exercise or automatic effect, new due date and notice requirements under the stipulated contract. It should not accelerate every group obligation whenever any branch is distressed.

21.10 A contagion sequence

Combine the borrowing-base example with a parent guarantee. A customer default reduces S's eligible receivables from 100 to 80. The facility demands repayment 6. S has cash 4 and cannot pay the full amount. Under the example's guarantee, P pays the remaining 2. P's cash falls from 5 to 3.

If P has an independent payment of 4 later that day, the support creates a parent shortfall of 1. Consolidated asset loss alone does not show the sequence. The guarantee transfers liquidity pressure across entities.

If the guarantee is capped at 1, S remains in arrears by 1 and P retains cash 4. The parent can meet its payment, but subsidiary distress persists. Different guarantee terms produce different locations and timing of default even when consolidated net assets are similar.

21.11 Behavioral contagion

Counterparties may react to distress before a contractual trigger. Suppliers can shorten terms, customers can delay orders, lenders can decline new credit and employees can leave. These responses are behavioral assumptions or estimated mechanisms. They should not be inferred automatically from a balance-sheet loss.

A scenario can include a supplier-term tightening rule conditional on a public rating event. The rule needs the event's timing and actor information. If suppliers have not observed the distress, immediate tightening may give them unavailable knowledge. If they have private signals, those should be specified.

Behavioral contagion can amplify losses, but an arbitrary amplification coefficient is not evidence. The model should expose sensitivity and distinguish contractual consequences from discretionary reactions. This separation makes the result more useful for identifying which mechanisms require empirical research.

21.12 Endogenous prices and fire sales

Selling assets to meet cash needs can depress their prices when the market is illiquid or several firms sell simultaneously. A fixed-price sale model may then overstate proceeds. An endogenous price rule creates feedback between liquidity need, quantity sold and realized value.

For illustration, let sale price per unit be $p(q) = 10 - 0.1q$ for $0 \leq q \leq 50$. Proceeds are $10q - 0.1q^2$, which increase on this interval but at a decreasing rate. Raising 200 requires solving the quadratic rather than dividing by the initial price 10.

The formula is a teaching assumption. It should not be treated as an estimated market impact curve. Its role is to expose how quantity and price can interact. Beyond the stated domain, the rule may be nonsensical. A solver should not extrapolate it to negative prices to force a solution.

21.13 Feedback across several sellers

If price depends on total sales by several entities, each policy affects the others' proceeds. Sequentially solving each entity at an unchanged price can be inconsistent. A simultaneous market-clearing or declared sequential mechanism is needed.

The agent-based stock-flow literature provides examples of interacting counterparties and explicit event order (Caiani et al., 2016). It does not imply that a specific market-impact function is valid for the company under study. The local mechanism must be stated and tested.

Multiple equilibria or no feasible clearing outcome can arise. A numerical failure to find one should remain distinct from a proved absence. If the model selects one equilibrium by a behavioral adjustment rule, that rule is part of the scenario and should be retained in provenance.

21.14 Consolidation changes after distress

A restructuring can change control, ownership or the reporting perimeter. A subsidiary may be sold, placed into a process that changes control, or receive new capital. The group statements then require a perimeter transition under the applicable basis.

The economic event should identify assets, liabilities, consideration and retained rights. The accounting layer determines derecognition, gains, losses and continuing interests. A generic default event that removes the entire subsidiary from consolidation without an authority-based control assessment is unsupported.

The scenario can remain useful with conditional perimeter outcomes if facts are unresolved. The result should show which cash and obligations remain in each legal entity and how each accounting interpretation reports them. A single consolidated number can conceal materially different legal exposures.

21.15 A group-state proposal

The state should include legal entities, ownership and control assessments, currency accounts, external and internal contracts, collateral pools, guarantees and transfer permissions. It should also retain actor-specific information and event calendars. Consolidated reporting is a derived view.

This does not require every query to load every field. Dependency-aware execution can select the relevant subgraph. A parent liquidity query follows transfers and guarantees; a consolidated revenue query follows perimeter and eliminations. The underlying representation preserves the links needed when the question changes.

Unknown links can be represented as alternatives. A guarantee's existence, cap or trigger may be unresolved. The query can report outcomes conditional on those terms and identify the source that would resolve them. Inventing no guarantee as a default would systematically favor benign outcomes.

21.16 Tests for group coherence

A transfer test should show equal and opposite cash movements before fees and consistent internal claims. A consolidation test should eliminate internal balances without deleting entity state. A collateral test should prevent recovery exceeding the shared pool. A timing test should prevent late cash from funding an earlier obligation.

A guarantee test should create the correct successor claim or loss under the stipulated rules and avoid double payment. A currency test should preserve native units through conversion and reporting. A cross-default test should distinguish threshold, cure and acceleration.

These tests are stronger than a final consolidated balance-sheet equality. They examine the mechanisms that can make a group fail despite apparently sufficient aggregate resources. Passing them establishes the specified small cases, while actual legal and regulatory applicability remains tied to source evidence.

21.17 What cannot be compressed away

For group liquidity, entity location, currency, transfer rights and dates are often essential. For recovery, collateral and priority are essential. For behavior, information and contractual discretion are essential. A compression that removes them may still reproduce current totals while changing stress outcomes.

The practical method is to retain a rich underlying state and derive query-specific summaries. A summary approved for one purpose can be refined for another. This avoids both extremes: duplicating the entire group for every possibility and collapsing the group into totals that cannot explain who can pay whom.

Chapter 22

When the model itself is incomplete: missingness, misspecification, and research priorities

Preserving all histories inside a model does not protect against a missing mechanism outside it. A feasible family can be exact and still omit a guarantee, a customer concentration or a financing restriction. This chapter examines how to represent that limitation without abandoning useful analysis. The aim is to make conditional conclusions precise and to direct further work toward omissions that can change decisions.

22.1 Unknown value versus absent mechanism

An unknown interest rate can be represented by a variable if the contract form is known. An unknown existence of a guarantee requires an alternative contract structure. A mechanism the language cannot express is a different limitation. These cases should not share one generic null.

A missing value within a supported form can often be bounded or inferred. A missing structural alternative can change the dimension and event logic of the family. An unsupported mechanism may require a new model component before any bound is meaningful. The result should identify which kind of gap it faces.

For example, allowing a loan rate between zero and twenty per cent does not cover a missed conversion right. A wider rate interval cannot reproduce the ownership change caused by conversion. Numerical sensitivity over existing parameters is not a substitute for structural coverage.

22.2 The disclosure package defines an evidence boundary

A company filing can incorporate other documents by reference or discuss risks without quantifying them. A reconstruction based only on primary statements may omit material terms in notes, exhibits or subsequent events. The evidence package should state which documents and passages were considered.

Coverage has stages: retrieval, parsing, interpretation and model admission. A page can be retrieved but not parsed; parsed but not understood; understood but outside the current executable profile. These distinctions matter for the completeness claim.

An unresolved passage should remain linked to relevant model areas. A guarantee discussion may block a claim that all contingent obligations are represented. It need not block a simple historical revenue reconciliation. Scope should be query-sensitive rather than all-or-nothing.

22.3 Missingness can be informative

A missing disclosure may result from immateriality, reporting exemptions, extraction failure or deliberate omission within a strategic disclosure model. These causes have different implications. Treating every missing value as zero assumes the most benign explanation. Treating every missing value as arbitrarily large can make analysis useless.

The first task is to identify the source of missingness where possible. If a table was not extracted from a scanned page, the remedy is data processing. If the filing does not disclose a contract term, the uncertainty is substantive. If a disclosure is legally required under certain facts, inference from absence requires actual authority and evidence about compliance; it cannot be assumed.

A probabilistic model of missingness can be added when justified. It should state whether missingness depends on hidden exposure or actor behavior. An assumption of missing at random is not a universal default for corporate disclosure. Without support, retain alternative explanations rather than fabricate a likelihood.

22.4 Conflicting sources

An annual report, interim filing and debt exhibit can disagree. The disagreement can reflect different dates, scopes, currencies, amendments or errors. A model that takes the latest number without reconciling context can discard valid historical information.

Source precedence should depend on the claim. A later amendment may supersede future contract terms while leaving past payments unchanged. A restated financial statement may revise a comparative amount. A management presentation may use a nonstandard metric rather than contradict a filing.

The evidence view should preserve both assertions and the interpretation linking them. If conflict remains unresolved, the family may contain alternative source-admission views or return an explicit contradiction. It should not average incompatible facts merely because both are numeric.

22.5 Measurement error and reporting discretion

A discrepancy between predicted and reported amounts can arise from rounding, estimation, extraction error or an incorrect economic history. A single Gaussian residual model collapses these mechanisms. It may be convenient statistically but can imply implausible support and independence.

Rounding often gives bounded intervals. Extraction errors can be discrete shifts in sign or scale. Accounting estimates can have structured dependence on assumptions and information. Reporting discretion may be strategic. The observation model should reflect the mechanism relevant to the claim.

A robust likelihood or contamination model can be considered, but its parameters need provenance and sensitivity analysis. It should not become a way to make every contradiction

disappear. A large residual may be evidence that the model omitted a transaction rather than merely noisy reporting.

22.6 Structural alternatives

Suppose revenue is modeled as price times quantity. A different model includes subscription renewals, usage charges and cancellation options. Both may fit historical totals. A demand shock can produce different future responses because the contract structures differ.

The family can include alternative structural models with shared observations and explicit scope. Comparing their conditional outcomes identifies model risk. Assigning posterior weights requires model priors and likelihoods; otherwise the alternatives remain unweighted.

A structural alternative should be concrete enough to simulate or bound. A vague statement that anything could happen is not useful. The modeler should identify plausible omitted mechanisms, their source support and the smallest discriminating evidence. This makes uncertainty actionable without claiming completeness.

22.7 Behavioral parameter uncertainty

A collection-delay parameter estimated from historical data can be uncertain and unstable. A recession can change customer behavior, credit terms or the company's enforcement policy. Treating the estimate as a fixed fact understates uncertainty.

Separate estimation uncertainty within a behavioral form from uncertainty about the form itself. A confidence interval around a constant collection rate does not cover a regime switch in which major customers stop paying. Conversely, a broad regime set may be unnecessarily pessimistic for a short, well-understood contract schedule.

The scenario should state whether behavior is held fixed, estimated conditionally, changed by a stress rule or optimized. Each treatment answers a different question. An empirical policy estimated under one financing regime may not transfer after covenants bind.

22.8 Extrapolation

A model calibrated in low-rate periods can be applied to high-rate stress. Contractual cash formulas may remain valid if terms are known, while behavioral relationships may not. The extrapolation status should be component-specific.

A floating coupon formula with a known cap can be evaluated outside historical rates because the contract supplies the rule. A regression of sales on rates may have no support in that region. A neural surrogate trained only below the cap may also fail even though the underlying contract formula is simple.

The model should distinguish extrapolation of an economic law, a behavioral estimate and a numerical approximation. Each requires different evidence. A global warning that the scenario is extreme is less useful than identifying which components are unsupported and how that affects the result.

22.9 Unknown unknowns and explicit coverage

A model cannot enumerate mechanisms it has not conceived. It can, however, maintain a coverage map organized by assets, obligations, rights, entities, timing and information. Review can ask whether material disclosures have a place in that map.

Stress testing the representation itself is useful. Introduce an unexpected guarantee, a delayed tax payment or a conversion clause and see whether the system can preserve it as unresolved rather than dropping it. A robust data model should fail visibly when a new mechanism is outside its profile.

This is different from claiming that a checklist proves completeness. The coverage map supports systematic discovery. It remains extensible as new cases arise. The manuscript's failure catalogue should be read in that spirit.

22.10 Sensitivity to the family boundary

Suppose a result is safe for every admitted history. The conclusion can still depend on the family's bounds. If floating share is limited to 0.4 by a weak assumption rather than a source, expanding it to one can change safety.

A useful analysis identifies active assumptions: those whose relaxation changes the conclusion or its bound. This can be done through alternative runs, dual information in suitable optimization problems or counterexample search. The method should not infer economic importance solely from a numerical coefficient.

An assumption may be inactive for one query and decisive for another. A maturity bound irrelevant to next month's interest can dominate a one-year refinancing stress. Sensitivity records should therefore be tied to query and horizon.

22.11 Partial identification of a decision

The evidence may not identify every parameter but still identify the preferred action. If one policy dominates another across all compatible histories, further historical detail is unnecessary for that comparison. Conversely, a narrow parameter interval can still straddle a decision threshold.

This distinction directs research effort. The goal is not always to recover the true journal uniquely. It may be enough to establish a robust decision or a bounded exposure. A history family is useful precisely because it can show which unresolved details matter.

The system should therefore report both parameter uncertainty and decision sensitivity. A large uncertainty set need not imply indecision, and a small set need not imply safety. The geometry relative to policies and thresholds is what matters.

22.12 Research priorities from counterexamples

A verified adverse witness can reveal the missing fact that drives risk. If the witness requires a near-term foreign-currency maturity, the next source target is a currency-maturity cross-classification. If it requires a guarantee, inspect the guarantee terms. If it requires refinancing refusal, investigate committed capacity and draw conditions.

The witness should be accompanied by competing interpretations and checked consequences. This is useful internal preparation before asking a professional a specific question. An unresolved label alone is not a review-ready brief.

The repository's internal-preparation requirement therefore fits the method: finish feasible source retrieval, calculations and competing interpretations first. No professional engagement or expenditure is authorized by the existence of an open accounting or legal question. Ordinary reader feedback is a separate need.

22.13 Model revision without losing history

When a missing mechanism is added, old results may change. Preserve the old model and result with their scope, then create a revised version. This allows users to distinguish a forecast error under the old model from a correction to the target definition.

A new guarantee can expand the family or add obligations. A new disclosure can narrow it. A corrected accounting mapping can change reported outcomes without changing economic cash. These revisions have different meanings and should not all be called data refreshes.

Dependency tracking should identify which results need recomputation. A source affecting one loan should not require rebuilding unrelated evidence, but its effects may propagate through liquidity, default and group guarantees. The dependency graph must follow economic consequences as well as direct formulas.

22.14 Communicating model risk

A useful explanation states the result, the assumptions that drive it and the specific unresolved mechanisms that could change it. It does not bury the conclusion under generic warnings. For example, "cash remains above 14 for all retained histories provided the facility remains drawable after customer default" identifies a meaningful condition.

If that condition is not established, say so directly and show the alternative result. Avoid presenting a conditional safe outcome as unconditional safety. Equally, do not describe every unresolved mechanism as inevitable disaster. The family should distinguish possible, necessary and unassessed consequences.

The reader should be able to see whether more evidence, a different policy or a better computation would resolve the issue. These are different next steps. A numerical approximation gap cannot be closed by another disclosure if the model already knows the relevant fact; a missing contract clause cannot be closed by more solver iterations.

22.15 A defensible stopping point

A query can be complete within a declared scope even when the entire company model is incomplete. Completion means that the requested target has been computed with the stated guarantees and that material excluded mechanisms are disclosed. It does not mean all uncertainty has disappeared.

A broader product-readiness claim requires broader coverage and evidence. The 150–300-page treatment is a documentation deliverable, not proof of that readiness. The manuscript remains a candidate whose assumptions, sources and proposed methods require continued implementation and human review.

This distinction permits useful progress. It avoids both false certainty and paralysis. The system can answer bounded questions accurately, retain uncertainty honestly and identify the concrete work needed to support stronger conclusions.

Chapter 23

Memory across reporting periods: taxes, vintages, arrears, and future rights

Two companies with the same current statements can differ because their past creates different future rights and obligations. A simulation that keeps only current totals forgets this memory. This chapter follows several small examples over more than one period. The examples use stipulated accounting and tax rules solely to expose state requirements. They are not claims about a particular jurisdiction's current law.

23.1 A loss carryforward

Assume a teaching tax system with rate 25 per cent, no refund for current losses, unlimited use of carried losses against future positive taxable income and no expiry. Let L_t be unused losses entering period t and T_t taxable income before using them. For positive T_t, used losses are $U_t = \min(L_t, T_t)$. Cash tax and next loss balance are

$$\text{Tax}_t = 0.25 \max(T_t - L_t, 0), \tag{23.1}$$

$$L_{t+1} = \max(L_t - \max(T_t, 0), 0) + \max(-T_t, 0). \tag{23.2}$$

The second equation covers both profit and loss periods under these simplified rules. A positive income reduces available losses; a negative income adds its magnitude.

Company A enters with losses 40 and company B with none. Both have the same current cash, assets and book equity because the example assumes no recognized deferred tax asset. Next period taxable income is 30. A pays no cash tax and retains losses 10. B pays 7.5. Current statements alone do not identify that difference.

In the following period taxable income is again 30. A uses its remaining 10 and pays 5; B pays 7.5. Over two periods, tax is 5 versus 15. Omitting loss memory overstates A's tax. Treating the original loss as an immediate refund would instead overstate its initial cash. Both errors can occur in simple effective-tax-rate forecasts.

23.2 Limits and expiry change the recurrence

Now impose a rule that losses can offset at most 50 per cent of positive taxable income in a period. With $L_0 = 40$ and $T_0 = 30$, usable losses are 15, tax is 3.75 and remaining losses are 25.

The earlier recurrence is no longer correct.

An expiry date requires vintage-specific loss balances. A single total 40 cannot determine which amount expires next year. Two companies with the same total can have different future tax capacity. The state may need a vector by expiry cohort, jurisdiction and restriction.

A change of ownership can affect utilization under some actual systems, but no such rule is assumed here without authority. The point is structural: tax rights depend on facts beyond a total balance. The model should admit the applicable rule and preserve the state it requires.

23.3 Deferred tax is a measurement, not the loss pool

A deferred tax asset may be recognized only under specified accounting criteria. Its carrying amount can differ from the mechanical tax value of losses. A valuation allowance or recognition restriction can change book equity without changing the legal loss pool.

Suppose the legal losses remain 40 while a recognized deferred tax asset is reduced from 10 to 4 under the chosen assessment. The reduction is not a cash tax payment. The future tax computation still needs the legal loss balance and applicable utilization rules.

Conversely, recognizing an asset of 10 does not create 10 of spendable cash. A liquidity scenario should use actual tax payment or refund dates. The accounting measurement and economic tax rights are linked but distinct states.

23.4 Asset vintages with the same carrying amount

Consider two machines under a simple straight-line model with zero residual value. Machine A originally cost 100 and has two of ten years remaining, so its carrying amount is 20 and annual depreciation 10. Machine B originally cost 40 and has one of two years remaining, also carrying 20 but annual depreciation 20. Equal current carrying amount does not imply equal next-year expense.

If the reported current depreciation is also constrained to match, combinations of several machines can still create ambiguity. For example, a portfolio can mix assets with different remaining lives whose total charge is the same. The next-period charge changes when a cohort reaches the end of life.

The state should retain acquisition or remaining-life cohorts where depreciation, maintenance or replacement depends on them. A single average life may be adequate for a smooth aggregate forecast but can miss a replacement wall. The approximation should be checked against the horizon and decision.

23.5 Replacement and operating capacity

An asset can be fully depreciated while still productive, or retain book value while becoming unusable. Depreciation is not a direct physical failure process. Replacing capital expenditure with depreciation assumes a relationship that must be justified separately.

Suppose A's machine can operate for two more years without replacement and B's fails after one year unless 30 is spent. Both enter with carrying amount 20. A financing shock in year two affects B's production but not A's in the same way. A model using only carrying amount cannot distinguish them.

A richer state includes capacity, maintenance, failure or retirement rule and replacement lead time. These can be uncertain. The history family may retain alternative vintage and condition profiles compatible with the same accounting data. Their scenario outcomes remain conditional rather than being collapsed into one average machine.

23.6 Arrears are not current-period expense again

Suppose a loan coupon of 5 accrues in period one but is not paid. Under the illustrative accrual treatment, interest expense and a payable increase by 5. In period two, paying the arrears reduces cash and the payable. It does not recognize the same interest expense again.

If penalty interest accrues on arrears, that is an additional contractual event with its own base, rate and dates. The state must retain unpaid amounts and payment allocation rules. A payment may be applied first to fees, interest or principal according to the admitted contract.

A model that stores only total debt can lose this allocation. Paying 5 might reduce principal when it should settle interest, altering future coupon calculations. The resulting statements may still balance, but the contract state becomes wrong. Multi-period replay exposes the error.

23.7 A payment allocation example

At a date, principal is 100, accrued interest 5 and fee arrears 2. A payment of 6 is made. Under the stipulated order fees first, then interest, then principal, fees become zero, interest becomes 1 and principal remains 100.

Under principal-first allocation, principal becomes 94 while interest and fees remain 7. Total obligation falls by 6 in both cases, but future interest and default status can differ. A total-debt bridge cannot distinguish the two. The allocation rule is part of the contract.

If the rule is unknown, both allocation histories may remain possible. A scenario should not choose the order that minimizes future cost unless management actually has that right. Creditor-controlled allocation cannot be optimized by the borrower without permission in the model.

23.8 Cure periods and clocks

A breach can start a cure period measured in business or calendar days. The state needs the trigger date, deadline, required remedy and whether cure has occurred. Merely retaining a Boolean breach flag loses the deadline.

Suppose a payment is missed on day 1 and can be cured within three calendar days, with the deadline defined at the end of day 4 under the stipulated convention. A receipt on day 4 can cure if settlement precedes the cutoff; one on day 5 cannot. An annual or monthly timestamp cannot resolve this distinction.

The actual contract may count days differently, include notices or exclude holidays. The simulator should not infer a convention from a generic date library. Exact boundary fixtures should accompany the admitted timing rule.

23.9 Options survive quiet periods

An option can remain out of the money for many reporting periods and then become decisive under a shock. A call right, put right, conversion feature or renewal option should not disappear because it had zero exercise activity historically.

Two loans can have identical coupon and principal histories while one permits early redemption at par and the other imposes a premium. A rate decline creates different refinancing choices. Current book debt and past interest do not identify the option.

A state summary can omit the option only for a query where its exercise cannot affect outputs over the horizon. That claim may rely on bounds showing the option never becomes relevant. Once the scenario or horizon changes, the summary must be reconsidered.

23.10 A refinancing option value example

Suppose a company owes a fixed payment of 110 next year. It can refinance today at a cost of 102 if it has a call right, and current cash is sufficient. Ignoring discounting for this one-period comparison, the right can save 8. Without the right, the old obligation remains.

If exercising requires a fee of 9, the saving disappears and the policy should not exercise under a cost-minimizing objective. If the fee is uncertain between 0 and 12, the exercise decision is conditional. A model that always redeems at par gives every borrower an option it may not possess.

The example does not price the option probabilistically. It shows a state-contingent decision consequence. Pricing would require a distribution or valuation operator over future rates and policies, along with the contractual exercise schedule.

23.11 Estimates carry information history

An impairment estimate can depend on forecasts and discount rates available at the reporting date. A later recovery in value does not mean the earlier estimate was necessarily an error. A restatement that corrects a mistaken fact is different from an estimate revised because new information arrived.

A historical simulation should preserve the information used for each estimate. If it recomputes every past estimate using today’s knowledge, it changes the as-reported history and can introduce look-ahead into validation. A current restated view can be useful, but should be named separately.

The economic asset state and accounting estimate can therefore have different version histories. A new estimate changes measurement without necessarily changing physical assets or cash. A model should track the relation rather than force every valuation change into an economic transaction.

23.12 Inventory age and obsolescence

Inventory with the same quantity and carrying value can have different remaining shelf life. A demand slowdown can make one cohort obsolete while another remains saleable. A pooled average age can conceal the threshold.

Suppose ten units expire next month and ten otherwise identical units expire next year. Both pools are carried at 50. If no sales occur for two months, the first may lose all saleable value under the stipulated physical rule while the second remains available. Historical cost equality does not preserve the response.

A cohort model can retain quantities by expiry and location without individual serial numbers when units within a cohort are interchangeable for the query. This is a justified aggregation opportunity. If a recall affects a particular batch, the cohort key must include batch identity or be refined.

23.13 Customer behavior has memory

A collection process can depend on invoice age, previous delinquency, dispute status and payment promises. A single receivable total or average collection ratio may omit these predictors. Two companies with equal receivables can have different next-month receipts.

A Markov model can be appropriate if the chosen state captures the relevant dependence under its assumptions. Calling the process Markov does not make a deficient state sufficient. If repayment depends on the duration of arrears, include age or a suitable duration state.

An empirical model should test whether the proposed summary predicts the relevant transitions and whether stress changes that relation. A purely contractual due-date model answers a different question from a behavioral collection model. Both can be retained with separate roles.

23.14 Horizon truncation

A policy can look safe over one year by issuing debt due just after the horizon. The closing cash is positive, but the future obligation remains. A terminal report should show that maturity and avoid treating its proceeds as permanent wealth.

Suppose a company borrows 20 at year-end and repays 22 one month later. A one-year cash objective rewards the 20 inflow unless terminal debt is valued or constrained. A policy optimizer can exploit this omission even if the event mechanics are correct.

Possible remedies include extending the horizon, adding a justified continuation value, requiring terminal funding coverage or reporting residual obligations separately. Each changes the decision problem in a stated way. No terminal rule is universally correct for all company questions.

23.15 A sufficient state is query-relative

For a one-day payment query, asset age may be irrelevant while settlement dates are essential. For a five-year capital plan, vintages and replacement needs matter. For tax cash, legal loss pools matter; for book equity, deferred tax measurement also matters.

The underlying representation should preserve these quantities where supported, while the active execution state can be reduced for the query. A sufficiency argument names the allowed scenarios, policies, outputs and horizon. It is not a permanent declaration that discarded details never matter.

This is how compact simulation can coexist with rich historical evidence. The system need not replay every old event at every step, but it must retain the memory that determines future events and the provenance needed to reconstruct it after revisions.

23.16 Implementation tests across periods

A loss-pool test should distinguish cash tax from deferred tax measurement and preserve expiry cohorts when required. A vintage test should produce different replacement paths from equal carrying amounts. An arrears test should prevent double expense and apply payments in the stipulated order.

A cure-clock test should distinguish receipts before and after the deadline. An option test should preserve unexercised rights through quiet periods. A horizon test should retain obligations beyond the final reporting date. Each case targets a specific missing state variable.

The tests should also verify legitimate aggregation. Identical loans with the same terms and dates can sometimes be combined additively. Identical inventory units can be pooled by cohort. The goal is sufficient memory, not maximal detail for its own sake.

Chapter 24

Evaluating a real-company simulator: what evidence would support its use

The exact cases in this part establish useful semantics, but a real-company simulator needs a broader evaluation. It must represent disclosures, preserve important uncertainty, generate coherent events and provide decision-relevant results. These objectives cannot be reduced to one extraction score or one forecast error. This chapter proposes an evaluation design and explains what each result would and would not establish. No empirical evaluation is claimed to have been performed here.

24.1 Define the unit of evaluation

A unit can be a source assertion, a contract, a reporting period, a company history family, a scenario query or a policy decision. Accuracy at one unit does not automatically transfer to another. Correctly extracting a debt total does not establish that maturity and option terms are represented.

A company-period case should identify its document package, entity perimeter, accounting basis, evidence cutoff and materiality scope. A scenario case adds shocks, policies, information and requested outputs. These definitions prevent evaluation from quietly changing the target after seeing results.

The unit should also preserve unresolved content. If a system refuses an unsupported instrument, that can be correct behavior within a bounded profile. The evaluation should distinguish honest refusal from silent omission, while still recording the coverage limitation for the full mission.

24.2 Source-to-model traceability

For each material modeled assertion, a reviewer should be able to reach the source passage and interpretation. Conversely, material source passages should have a disposition: represented, irrelevant to the declared query, unresolved or outside the executable profile.

This bidirectional trace catches different failures. A model assertion without a source may be an assumption or hallucination. A source passage without a disposition may be an omission. A correct numeric total can coexist with a missed covenant note that changes the stress result.

The evaluation should sample across difficult formats and concepts, including scanned notes, tables with overlapping dimensions, incorporated documents and revisions. A convenient set

of clean tables is insufficient evidence for whole-filing coverage. The selection procedure and omissions should be recorded.

24.3 Annotation and disagreement

Human annotations of financial meaning can disagree. A term may be ambiguous, a source may conflict, or the accounting treatment may depend on facts not disclosed. The reference set should permit unresolved or alternative annotations rather than force artificial unanimity.

Adjudication should record the competing interpretations and the evidence that resolves them. A majority vote is not necessarily an authority on a technical accounting or contractual question. For some cases the correct reference is a conditional family.

Ordinary comprehension review and expert technical review serve different purposes. A reader can identify unclear exposition without certifying accounting treatment. An accountant can assess a specific treatment without validating a probabilistic forecasting model. Evaluation should not collapse these roles into a generic human-approved label.

24.4 Historical reconstruction checks

A reconstruction should reproduce observed statements within admitted precision and context. It should also satisfy cross-statement bridges and contract constraints. These are necessary checks, but a model explicitly fitted to the reports can pass them while selecting the wrong hidden history.

Use withheld disclosures where possible. For example, reconstruct from primary statements and selected notes, then test whether the retained family includes a contract detail revealed in a withheld exhibit. The procedure must avoid using that detail indirectly through another source.

A family that excludes the revealed truth under correct source interpretation has failed coverage. A family that includes everything can pass inclusion trivially but be uninformative. Evaluation therefore needs both coverage and useful restriction, interpreted relative to the available evidence.

24.5 Synthetic truth and real uncertainty

Synthetic companies provide known complete histories and controlled disclosure maps. They allow tests of whether the system preserves all intended alternatives and rejects impossible ones. They are valuable for semantic coverage and adversarial construction.

Their limitation is that the generator defines the world. A system matching that generator may fail on mechanisms the generator omits. Synthetic results should not be presented as real-company empirical validity. The generator's assumptions and diversity need review.

Real disclosures supply realistic ambiguity but often lack a fully observed ground truth. Evaluation can then assess traceability, consistency, withheld facts, conditional predictions and decision consequences. The two evidence sources are complementary rather than interchangeable.

24.6 Forecast origins and leakage

A forecast should be reconstructed using information available at its origin. Later restatements, realized prices and revised macro data can leak into features or labels. The evidence store needs

dated versions to prevent this.

A rolling-origin evaluation can produce forecasts at several historical dates and compare them with subsequent observations. The model's parameter estimation and tuning must also respect the cutoff. Training on the full sample and then calling earlier predictions out of sample is incorrect.

Company events such as acquisitions or changes of reporting basis can alter comparability. The evaluation should bridge those changes or define a target that remains meaningful. A forecast error caused by a perimeter mismatch is not the same as an economic prediction error.

24.7 Conditional scenario evaluation

A deterministic stress scenario may never occur exactly. Its mechanical correctness can still be tested through contract formulas, accounting invariants and controlled cases. Empirical validation of conditional responses requires suitable observed variation or another identification strategy.

A model predicting cash under a stipulated 200-basis-point shock cannot be validated by comparing it with actual cash after a period containing many other shocks without accounting for those differences. Such a comparison may be descriptive, but it does not isolate the rate response.

The evaluation should distinguish unconditional forecasting, conditional stress mechanics and causal intervention claims. Each has a different comparator and evidence requirement. Calling all three simulation accuracy obscures the target.

24.8 Probabilistic forecasts

If the system supplies a predictive distribution, evaluate both calibration and informativeness with methods appropriate to the outcome. A forecast assigning broad mass everywhere may be calibrated in a weak sense while offering little value. A sharp forecast can be badly miscalibrated.

Proper scoring rules, discussed in Gneiting and Raftery (2007), provide a principled way to reward honest predictive distributions under their conditions. The chosen score should match the outcome type and decision context. A score on annual earnings does not automatically validate default probability or liquidity tails.

Probabilities conditional on one reconstructed history should be distinguished from probabilities averaged across histories. If history weights are subjective or ambiguous, report sensitivity. A good score under one prior does not establish that the prior is uniquely justified.

24.9 Set-valued forecasts

An unweighted family produces a set of possible outcomes. Evaluation can ask whether later observed outcomes fall within it and how wide or structured the set is. A rectangular interval display may obscure joint restrictions, so multivariate coverage should respect the actual feasible set.

If observed cash and interest each lie within their marginal intervals but violate the modeled relation $C + I = 95$, the joint model has failed for that observation. Marginal coverage alone would miss it. Conversely, a wide marginal interval may coexist with a precise relation that is

useful for a decision.

A set can be evaluated against a declared decision task: does it correctly identify policies that are robustly feasible or expose the history driving a failure? This should supplement, not replace, source and mathematical checks. A useful decision result can still rest on an incomplete family.

24.10 Policy evaluation

A policy should be evaluated under the information and action constraints it would actually face. A perfect-foresight benchmark can provide an upper bound in some settings, but it is not an implementable comparator. A no-action baseline may be useful but can be too weak to establish practical value.

A fair comparison includes a credible existing policy or classical method, with equivalent access to data and tuning. If the proposed method receives a richer contract representation, separate the benefit of representation from the benefit of its optimizer. Otherwise a solver improvement claim may actually be a data improvement.

Policy value estimates need uncertainty analysis and attention to rare failures. A higher mean payout can come with unacceptable liquidity risk. The primary criterion and vetoes should be declared before selecting a winner. Descriptive advantages from a few simulations should not be promoted to a default policy.

24.11 Ablations that answer real questions

An ablation can remove contract detail, actor information, persistent memory or joint constraints to test their contribution. The comparison should preserve other conditions. Removing several features at once can show that the bundle matters but not which mechanism explains the change.

For example, compare a full debt schedule with an aggregate debt balance while holding pricing and policy fixed. Then test the effect of adding endogenous refinancing. This sequence separates contract-state sufficiency from behavioral response.

Expected failures can be informative. If the aggregate model misses a maturity wall, that supports the need for timing in that query. It does not prove that every query requires instrument-level detail. The conclusion should match the demonstrated mechanism and scope.

24.12 Stress coverage

A validation set should include ordinary operations, near-boundary cases and severe but admitted stresses. Important dimensions include rates, currency, customer concentration, maturity, collateral, offer capacity, settlement delay and information timing.

A Cartesian product of every dimension may be infeasible or economically incoherent. Use the joint model to identify feasible combinations and targeted adversarial cases. Cover interactions known to matter, such as collateral loss reducing funding exactly when cash falls.

Coverage should record mechanisms tested and regions untested. A large path count is not a substitute. Ten thousand smooth scenarios can leave a single conversion or cure-period boundary unchecked.

24.13 Numerical reliability

For approximate methods, validation should compare against exact or highly controlled small references, assess convergence or refinement, and inspect downstream decisions. Solver tolerances should be translated into economic quantities where possible.

A residual below a threshold can still be unacceptable if it changes default classification. A slightly larger residual in a large nonbinding balance may be immaterial. The evidence contract should state which diagnostics veto use and which explain remaining error.

Repeat stochastic comparisons with appropriate seeds and uncertainty estimates. If candidates remain statistically indistinguishable after validity screens, report that. A method can be viable without being proven superior. Default promotion requires stronger evidence than an optional experimental path.

24.14 Failure accounting

Evaluation must count unsupported, refused and numerically failed cases. Excluding them from average error can make a fragile method look accurate on the easy subset. Report coverage and conditional performance together.

A refusal can be correct if the request exceeds the admitted profile, but excessive refusal limits usefulness. A numerical failure is different from a correctly identified economic shortfall. A result table should preserve these categories rather than assign all failures the same placeholder number.

If an approximation leaves probability mass unresolved, include that mass in risk interpretation. If the model is unweighted, report unresolved regions and the bounds they permit. Renormalizing completed cases is a different conditional target and should be labeled as such.

24.15 Reproducibility and independent checks

A reproducible run records source snapshots, model versions, assumptions, methods, seeds and environment. It also preserves exact commands and outputs. Reproduction verifies that the same computation can be obtained; it does not independently validate the model.

Independent checks should use different routes where feasible: algebra versus code, event-level versus aggregate reconciliation, or one solver versus a certified reference. Shared data and code can share errors. A second model reading the same incorrect assumption is not independent economic evidence.

Rendered document review is also distinct. It can reveal missing symbols, broken references and unreadable derivations. A clean build cannot establish that a reader understands the argument. The manuscript should remain provisional until actual human comprehension evidence is available.

24.16 A proposed staged evaluation

First establish exact semantics with the synthetic cases in this part. Then evaluate source-to-model coverage on a bounded real disclosure package, preserving unresolved content. Next test historical reconstruction against withheld facts and carefully defined reporting bridges.

Only after those stages should a forecast or policy study claim economic performance. Its plan must identify target companies, time cutoffs, baselines, tuning protocol, primary criterion and uncertainty analysis. A failure at an earlier stage can invalidate interpretation of the later results.

The stages are dependencies, not excuses to stop independent work. Source research, documentation and bounded implementation can continue while empirical data or human review remain pending. The full mission remains open until its specified evidence exists.

24.17 What a defensible conclusion sounds like

A strong narrow conclusion might state that a method reproduces all exact cash and ownership transitions in a declared profile and preserves the tested information restrictions. Another might state that a predictive distribution outperforms a named baseline under a predeclared score with uncertainty evidence on a stated sample.

Neither conclusion should be expanded into universal accounting compliance or company forecasting ability. A failed candidate can be rejected without rejecting the entire uncertainty-preserving approach. A successful small model can justify the next experiment without becoming a production default.

The evaluation program should make those distinctions easy to maintain. That is the practical counterpart of the manuscript's central idea: preserve what is known, preserve what remains possible, and make every stronger conclusion earn its evidence.

Chapter 25

Failures at the joins: how individually plausible methods produce a wrong simulation

Many serious errors occur where two reasonable methods meet. A contract engine can calculate coupons correctly while an accounting adapter treats them as expense under an incompatible basis. A particle filter can implement its weights correctly while a policy sees hidden future state. A consolidation routine can balance while a liquidity engine spends cash in the wrong entity. This chapter examines these joins and proposes checks that target the interaction rather than one isolated module.

25.1 Reconstruction followed by valuation

Suppose historical debt carrying amount is 100 and an analyst reconstructs face as 100. A valuation engine then prices the promise correctly. If the carrying amount included an unamortized discount or accrued interest, the promise was misidentified before pricing began. The valuation solver can return a precise answer to the wrong contract.

The interface should require a distinction among observed carrying amount, inferred face and supplied valuation terms. If face is unknown, pricing should operate conditionally over its feasible values or request a supported mapping. A test should include face 110, carrying amount 100 and proceeds different from both.

The repair is at the reconstruction boundary, not by adjusting the discount rate until price matches book value. Such calibration would hide the missing contractual amount and could distort every later scenario. A book-to-contract bridge should explain the difference with admitted movements and measurement rules.

25.2 Valuation followed by cash forecasting

A bond’s market value falls after rates rise. A cash forecast that treats the decline as a cash inflow because the liability became cheaper creates money unless the company actually repurchases or settles the debt. The valuation change is not itself a financing event.

If repurchase is allowed, it requires a market offer, available cash, quantity and settlement date. The company may realize an economic benefit by paying less than face, but it still spends cash

to do so. Accounting gain or loss depends on carrying amount and the applicable treatment.

A discriminating test changes market rates while holding contractual payments and policy fixed. Cash should not change merely from revaluation unless a contract or collateral clause creates a payment. A second test enables repurchase and checks the explicit transaction.

25.3 A hedge followed by accounting classification

An economic hedge can offset cash exposure while failing a particular hedge-accounting designation. Conversely, a qualifying designation does not guarantee perfect economic offset. A simulator that equates the two can misstate both cash and reported income.

The contract layer should calculate the underlying and derivative cash flows separately, including basis and timing. The accounting layer applies designation, effectiveness and measurement rules under the admitted basis. Unknown designation affects reporting, while unknown notional or reset terms affect economic exposure.

A test can use equal notionals but different reset dates. Annual net interest may appear small while interim collateral or coupon cash differs. Another can hold economic cash fixed and vary the accounting designation to show that reporting changes without inventing a new cash flow.

25.4 A credit loss followed by a funding rule

An impairment model may reduce receivable value after customer deterioration. A funding model may continue using the old gross receivable as collateral. Each module can pass its own local checks while the joined model overstates borrowing capacity.

The interface should distinguish accounting allowance, contractual eligibility and realized collection. A borrowing base may exclude overdue invoices regardless of allowance, or apply a separate haircut. These relations need explicit terms. The correct funding reduction is not automatically equal to the accounting impairment.

Use a case where an invoice becomes ineligible before an accounting loss is recognized, and another where an allowance changes but eligibility does not. The tests prevent accidental identification of two different quantities. The repair is a shared event and state relation, not a single reused field called net receivables.

25.5 A source update followed by a cached policy

A new note reveals a near-term maturity. The reconstruction updates debt timing, but a cached financing policy was optimized for a later maturity. Reusing it can produce an infeasible payment path while all displayed balances use the new source.

Cache invalidation must follow semantic dependencies. A policy depends on contract schedule, offers, information, horizon and objective. A hash of the displayed statement totals is insufficient because the totals may be unchanged.

A regression test changes only the maturity clause while preserving current balances. The policy and relevant scenario result should be recomputed or marked stale. Unaffected historical arithmetic may remain cached. This tests precise invalidation rather than indiscriminate recomputation.

25.6 A restatement followed by backtesting

A current data pipeline supplies restated historical earnings to a forecast evaluation that claims to use information available at the time. The forecasting code can be correct while the data view leaks later knowledge.

The evaluation needs source versions and publication cutoffs, including parameter-estimation inputs. A restatement can be used as a later target if that is the defined outcome, but it cannot silently become an earlier predictor.

A test should run an old forecast before and after adding a later restatement. The frozen as-published prediction remains identical, while a separately named restated analysis can change. This is a direct check of evidence semantics at the validation boundary.

25.7 A particle filter followed by robust optimization

A particle approximation may retain only a small subset of plausible contract types. A robust optimizer then treats those particles as the entire feasible family and reports a worst case. The filter approximated a distribution; it did not prove that discarded histories were impossible.

The optimizer should receive the retained family for robust queries, or explicitly report a sample-restricted worst case. Particle witnesses can supply an incumbent adverse case, while symbolic constraints or other methods bound the unexplored region.

A test deliberately gives a low-weight but feasible history a severe cash outcome. Resampling may remove it. The robust result should remain sensitive to it unless evidence excludes it. A probabilistic expected-value result can weight it according to the admitted measure, but that is a different target.

25.8 Scenario reduction followed by policy optimization

A reduced scenario set can preserve mean revenue while removing a signal pattern that changes the optimal financing decision. Optimizing on the reduced set can then produce a policy that fails on omitted paths.

Reduction should preserve or bound decision-relevant outcomes and information structure. A representative scenario is not merely a nearby vector of shocks. Its observation timing, feasible actions and threshold events matter.

A useful test has two close rate paths on opposite sides of a covenant and a refinancing action available only before the breach is observed. A reduction that merges them must either retain conditional behavior or report an approximation bound that covers the decision loss. Mean matching alone is insufficient.

25.9 A policy optimizer followed by event execution

An optimizer can choose an issue amount based on continuous shares and immediate settlement. The event engine enforces integer lots and a two-day settlement lag. The optimized plan is no longer feasible. Rounding and delaying it after optimization can alter the objective materially.

The optimization problem should include the same action domain and timing as execution, or explicitly account for a projection and its effect. A post hoc repair that borrows the residual

cash is another policy and needs its own feasibility.

A test can use a funding need just above a lot boundary and a maturity before settlement. It should reject the original plan or reoptimize under the correct constraints. Reporting the optimizer's objective before execution repair would overstate achieved value.

25.10 An accounting rule followed by an economic policy

A dividend policy based on book profit may distribute cash that is unavailable because profit includes noncash gains or uncollected sales. The accounting calculation can be correct; the policy is infeasible without a cash constraint.

Conversely, a cash-rich company can have accounting restrictions on distributions under the applicable rules. A cash-only policy can violate those restrictions. The feasible action set should combine cash, legal, contractual and class-right constraints where admitted.

A test should include positive profit with negative operating cash and another with positive cash but a binding distribution restriction. Neither a profit ratio nor a cash balance alone defines feasible dividends. The policy must receive the relevant state and rules jointly.

25.11 Consolidation followed by credit analysis

A consolidated group can eliminate an intercompany loan while a creditor's claim remains against one legal entity. A credit model using only consolidated net assets can assign support that the creditor cannot access.

The credit analysis needs borrower, guarantor, security and transfer relationships. Consolidated reporting remains useful but is not a legal recovery waterfall. The same assets can support different claims under different entity structures.

A fixture should compare two groups with identical consolidated statements but different guarantees or asset ownership. Recovery and payment capacity should differ where the stipulated rights differ. If the model returns the same result, it has probably compressed away a material legal-entity relation.

25.12 A default model followed by accounting close

A default indicator can trigger an accounting routine that zeros every liability and equity balance. This makes the balance sheet close neatly while erasing continuing obligations, recovery claims and operating assets.

The economic default event should create a process state. Accounting consequences follow actual or stipulated modification, impairment, settlement and extinguishment events under the basis. Default alone does not supply every measurement.

A test should leave unpaid debt outstanding during a cure period, then compare repayment, waiver and restructuring paths. Another should include delayed recovery. The system should preserve obligations until the admitted event changes them, even if a value function assigns zero to shareholders.

25.13 A pricing measure followed by a risk report

A debt price may be computed under a pricing measure, while a risk report asks for real-world default frequency. Reusing pricing probabilities as forecasts confuses valuation with statistical prediction.

The measure identity should travel with every expectation. A risk-neutral probability can be useful for price consistency, but does not automatically describe actual event frequency. A physical probability model can estimate frequency but requires a risk adjustment to price claims unless special assumptions apply.

A test with different physical and pricing probabilities should produce different expected losses while preserving the declared price. The report should refuse an unlabeled probability when the measure is ambiguous. This is a semantic type check with direct financial consequences.

25.14 A numerical tolerance followed by a legal threshold

A solver treats a constraint violation of 10^{-6} as acceptable. A downstream routine interprets the approximate value as exact compliance with a strict contractual threshold. The numerical convention has silently become a legal conclusion.

The result should carry an error interval or tolerance. If the interval crosses the threshold, classification is unresolved unless another calculation determines the sign. The contract's inclusive or strict comparison remains unchanged.

A test at equality and on both sides by less than the solver tolerance reveals this interaction. Exact rational cases can establish intended boundary semantics. Floating-point production methods then need an explicit policy for translating their error into a classification.

25.15 A language model followed by evidence admission

A language model can propose a plausible missing journal or summarize a contract. Those outputs are interpretations or hypotheses until supported. If they are inserted into the evidence store as facts, subsequent inference can become circular: the model later cites its own generated explanation.

The admission layer should retain source anchors and distinguish extracted text, normalized interpretation and generated suggestion. Unsupported suggestions can be useful search leads or candidate templates. They should not narrow the family merely because the text is fluent.

A test should submit a plausible but unanchored maturity date. The system should preserve it as a proposed assumption or reject factual admission. If a genuine source later supports it, the new assertion can narrow the family with traceable provenance.

25.16 A user interface followed by decision interpretation

An interface may display one attractive central path while hiding unresolved regions. Users can reasonably interpret it as the forecast even when the calculation produced a family. The presentation then overstates certainty without any numerical bug.

The default display should identify conditional paths, ranges, measures and unresolved cases in proportion to their importance. It should not label an arbitrary witness most likely. A central

estimate needs a stated selection rule or probability model.

A useful review asks whether the reader can identify the history driving a shortfall, the assumption that could remove it and the status of the computation. If those answers require searching internal logs, the explanation is insufficient even if the underlying data are preserved.

25.17 Composition contracts

Each boundary should specify input meaning, units, timing, information, allowed uncertainty and failure behavior. A contract engine returns obligations and payoffs; an accounting engine returns postings under a basis; a policy engine returns admissible actions; an inference engine returns families or measures.

These interfaces should be tested end to end with the worked cases. A local adapter can be mechanically correct while mapping the wrong concept. The test must compare the resulting economic and accounting consequences, not only schema validity.

Version compatibility also matters. A changed rule can alter event order or measurement. Results should identify the exact component versions and invalidate dependent caches. A stable interface name does not establish semantic compatibility across versions.

25.18 A complete rate, covenant and ownership path

An exact combined example shows why the financing steps must be dated. Two possible companies each start with cash 16 and loan face 100. One pays a fixed coupon of 5; the other pays 7 after the specified floating-rate reset. Each then makes an operating payment of 6. A covenant tests cash of at least 5 immediately after these payments and before financing. Breach forbids new borrowing but permits an equity cure. Management requires closing cash 8. These are assumed contract terms, not a claim about a standard covenant.

The fixed-rate company has cash $16 - 5 - 6 = 5$, meets the covenant and needs 3. A candidate debt issue with maximum face 5, conditional solvency probability 4/5, recovery 2 and gross risk-free return 21/20 has proceeds 88/21. Holding that quoted per-face price fixed for this conditional stress, issuing face 315/88 raises exactly 3. The floating-rate company has cash $16 - 7 - 6 = 3$, breaches the covenant and needs 5. Debt is unavailable; even an improper borrowing override at the maximum face would raise only $88/21 < 5$.

Suppose the permitted equity cure has fee 1, pre-issue economic equity value 20 and 100 incumbent shares. A gross subscription of 6 provides net cash 5. The post-issue equity value is 25, new investors own 6/25, and the common share price is 19/100. Their 600/19 new shares have value 6, while incumbents retain value 19. Both companies finish with cash 8, but their new debt and ownership differ. Merging them on closing cash would conceal those differences. Testing the covenant only after the cure would erase the earlier breach and wrongly admit the borrowing alternative.

This example holds quoted financing prices and the pre-issue equity valuation fixed to expose event ordering and allocation. It does not solve the feedback from financing to default and prices. A source correction introduces another interaction: withdrawing an unsupported fixed-rate classification can reopen the floating-rate history, invalidating an earlier single-history result. If management later receives a rate notice, its admissible policies can change even when the analyst's reported balances do not. Shared computations must therefore depend on evidence version, actor information, contracts and the requested outcomes, as well as current cash.

25.19 Executing promises, carrying amounts and ownership

The combined example can be executed through the existing typed company runtime once its state distinguishes contractual face from carrying amount. For the fixed-rate history, the authored initial-measurement convention posts cash of 3 and a liability carrying amount of 3, while contractual face increases by 315/88. The closing liability carrying amount is 103 and outstanding face is 9115/88. Subsequent effective-interest measurement and settlement require further methods; substituting carrying amount for the promise would lose the very exposure the example is intended to preserve.

For the floating-rate history, subscription proceeds of 6 increase cash and contributed capital; the direct issue cost of 1 reduces both. The typed state retains 100 incumbent shares and 600/19 new shares. Under this explicitly synthetic posting convention, both histories report assets of 112. The fixed history has liabilities of 103 and book equity of 9; the floating history has liabilities of 100 and book equity of 12. These book amounts differ from the economic equity values used to price the subscription. No reporting-standard admission follows from the balanced journals.

The reference adapter executes coupon payment, operations, covenant observation and financing in that order, using the unchanged common company executor. It rejects borrowing after the floating-rate breach, proceeds inconsistent with the supplied per-face quote, a zero issue price, duplicate issuance and payment without sufficient cash. A mutation that omits the share-count update still produces balanced accounts and completes the accounting path; the independent ownership comparison detects the error. Accounting conservation alone therefore cannot validate a financing method.

The same runtime also executes the earlier information example. Starting with cash 6, a fixed coupon 5 leaves 1, while a floating coupon 7 cannot be paid. A hedge costing 1 caps the coupon at 5. An informed manager hedges only the floating history and retains cash 1 or 0; a manager who must act identically in both histories can hedge both and retain 0. Withdrawing the earlier fixed-only evidence reopens the payment-refusal branch. The negative cash in the algebraic shortfall calculation denotes unmet funding; the runtime refuses the payment atomically and keeps the opening cash, rather than recording an impossible settlement. A missed-payment event and any ensuing default process would need separate declared methods.

25.20 Prioritizing interaction tests

The highest-value interactions are those that can create cash, remove obligations, leak information or drop adverse histories. They should be tested early because they can make every downstream chart look plausible. Positive-recovery pricing, settlement timing, ownership conversion and joint-history preservation are examples.

Other interactions can be staged according to the active company and query. A bank capital rule may be irrelevant to a simple industrial-company fixture but essential to a bank funding scenario. The full mission retains these domains without forcing every first implementation to solve all of them at once.

The catalogue remains extensible. Each newly discovered failure should identify the target, actual behavior, mechanism, discriminating case and repair. That practice turns failures into durable knowledge and prevents the same convenient shortcut from reappearing under another module name.

25.21 Credit schedules after financial write-off

Financial closure leaves a separate question about the history of scheduled receipts. The retained credit examples already record uncollected amounts and portions removed by write-off. A dated reconciliation now checks how those amounts relate to the original schedule, actual collections and any remaining modeled receipts. These quantities are nominal currency amounts. They are neither discounted carrying values nor an assessment of enforceable legal rights.

Let d denote a scheduled payment date and k the position in the recorded event sequence. Write O_d for the original scheduled amount, $\Delta_{k,d}$ for cumulative signed changes to the modeled receipt schedule, $C_{k,d}$ for collections, $U_{k,d}$ for uncollected amounts recorded when a due date is resolved, $W_{k,d}$ for portions removed by financial write-off, and $R_{k,d}$ for the remaining schedule. Each dated record satisfies

$$O_d + \Delta_{k,d} = C_{k,d} + U_{k,d} + W_{k,d} + R_{k,d}.$$

A collection transfers the scheduled amount at that date from R to C and U. A proportional write-off transfers the specified portion from R to W. A replacement schedule changes R and Δ by the same dated amount. Accrual changes the financial measurements but leaves this nominal relation unchanged. These observations derive the identity from the admitted events; they do not establish which events should be admitted for a real loan.

In the partial-write-off example, the original schedule is 66 on 1 January 2026 and 363/5 on 1 January 2027. The identified one-fifth portion removed on 1 July 2025 contains 66/5 and 363/25. Subsequent collections of 44 and 55 leave uncollected amounts 44/5 and 77/25. At the two dates separately,

$$66 = 44 + \frac{44}{5} + \frac{66}{5}, \qquad \frac{363}{5} = 55 + \frac{77}{25} + \frac{363}{25}.$$

Their sum is $693/5 = 99 + 297/25 + 693/25$. The full-write-off example instead retains the whole original schedule among the removed portions and keeps its zero-valued remaining entries. The record distinguishes those entries from missing dates.

The performing prepayment illustrates why the signed revision belongs in the identity. Under its supplied unchanged-terms finding, the modeled schedule changes from 55 and 121/2 at the two annual dates to 105 at the first date. The changes are therefore $+50$ and $-121/2$. The later receipt of 105 reconciles to that explicit replacement. Calling the replacement a legal amendment, or silently comparing the receipt with the old total, would change the meaning of the example.

The read-only companion checks all 57 stages of the nine retained histories against their original schedules and archives, with independently authored endpoints. It retains event order even when two events share a date, zero uncollected amounts, write-off portion identities, and the exact source and financial-result identities. Twenty-five controls reject missing or duplicated portions, changed dates, hidden replacements and unsupported assertions of forgiveness or accounting admission. The existing financial journals remain unchanged. Reproducible calculations and their limits are recorded in `data/accounting-master/credit-claim-continuity-20261027/result.md`.

All nine examples finish with a financially closed book. Adding a further event through the existing native interface produces `event_after_close`. That refusal describes the implemented interface; it does not determine whether a recovery can occur. A later recovery needs its own dated receipt evidence, allocation and supported accounting treatment. The present record preserves the earlier history while leaving forgiveness and later recovery unassessed.

25.22 What a later recovery would require

A receipt after financial write-off raises a new accounting question. The original EU source used by the retained examples treats write-off as a derecognition event (IFRS 9, paragraph 5.4.4), while its disclosure requirements explicitly contemplate written-off assets still subject to enforcement (IFRS 7, paragraphs 35F(e) and 35L). These provisions explain why the history of a removed portion should remain available. They do not determine the legal position of a particular borrower or supply evidence of a later receipt.

The distinction also matters when choosing a recognition rule. The IFRS Interpretations Committee's March 2019 decision on curing a credit-impaired financial asset addresses the adjustment of a recognised loss allowance, including discount unwinding. It requires the difference considered in that request to be presented as a reversal of impairment losses rather than interest revenue. Applying that answer to cash received after full write-off would require a further argument: the checked decision does not explicitly address that fact pattern. The retained purchased or originated credit-impaired cases also need their own applicability analysis. Exact inspected passages, source versions and unresolved questions are retained in `data/accounting-master/credit-recovery-scope-20261027/source-map.md`.

Before a later event can affect the book, the model needs the receipt's amount, currency, transaction and information dates, source availability and allocation. An allocation to a previously written-off portion must identify that portion; an allocation to a different supported claim must preserve that distinction. The applicable recognition and presentation treatment remains a separate proposal. Cash-flow classification also depends on the entity and the nature of the receipt, so a synthetic operating or investing label cannot answer it.

The first intake therefore records missing evidence explicitly. For each of the nine closed histories, it retains the complete dated nominal record and leaves the new receipt and its allocation unknown. Zero-valued historical components remain visible; an unknown new receipt is not replaced by zero. Actual prior collection identities submitted as new receipts are rejected because those collections are already represented. The checked intake has no financial effects and accepts no positive recovery in this version. Its 50 adverse controls test preservation and refusal boundaries, not recognition of an actual recovery. The reproducible result is in `data/accounting-master/credit-recovery-intake-20261027/result.md`.

This limited interface makes the remaining work concrete. A positive recovery method needs authentic receipt evidence, a supported allocation and a justified post-derecognition treatment. Those additions must preserve the original write-off finding and financial history, rather than retrospectively rewriting them to match a later outcome.

25.23 Issued and held results before dated reporting

An insurance calculation can preserve the right total while losing the relation between issued contracts and reinsurance held. The retained covered-loss example already calculates both sides under a supplied one-half quota and zero rates. Its estimate change reduces issued claims by 4 and expected held recoveries by 2. The issued loss reversal therefore contributes 4 to profit, while the held remeasurement contributes an expense of 2. The held loss-recovery component falls from 4 to 2 as a tracked component; that movement is not a second expense at the estimate-change stage.

A complete stage ledger now exposes the consequence through settlement. It starts with cash 247/2, an issued liability of 120, held fulfilment cash flows of 49/2 and held cost CSM of 6. Thus opening equity is

$$\frac{247}{2} + \frac{49}{2} + 6 - 120 = 34.$$

The estimate change raises equity to 36. Issued service subsequently releases revenue of 100 and the remaining loss component of 16 against claims expense of 114, adding service profit of 2. Held service records recovery income of 22, premium allocation of 53/2 and loss-recovery allocation of 2, for a loss of 13/2. Final equity is therefore

$$34 + (4 - 2) + 2 - \frac{13}{2} = \frac{63}{2}.$$

Cash reaches the same amount through the separate payment of 114 and receipt of 22. The held component's later service allocation remains present even though there is no additional component expense at estimate revision. Combining those two stages would conceal the distinction.

The second retained case reduces covered claims by 8, exhausts the held loss-recovery component and ends with cash and equity of 67/2. Its zero component and zero later release remain explicit. Both examples reconcile every account, journal, cash movement and component origin at opening, estimate change, issued service and held service. Their complete native results and independently authored intermediate balances agree; 22 adverse controls test double expenses, lost origins, net-only substitutions and unsupported admission.

These are ordered accounting stages, not dated fiscal reports. The source calculation supplies neither event dates nor separate identifiers for the issued and held groups. The existing annual reporting adapter accepts only the two retained issued-loss schedules; direct calls confirm that it does not admit these recovery-change examples. The stage ledger therefore leaves dates and separate group identities unknown and keeps that report-connection gap open. Its complete results and the consumer-call audit are retained in `data/accounting-master/insurance-recovery-stage-book-20261027/result.md` and `data/accounting-master/insurance-shared-group-audit-20261027/result.md`. A dated extension must supply chronology, group identity and a compatible fiscal opening before these stages can become a company-book execution.

Chapter 26

A fully solved risky-debt model before any neural approximation

A small structural model can make the financing equations completely transparent. This chapter solves one with two future asset states. It is deliberately restrictive: one period, one creditor, no tax, no fees initially, no bankruptcy costs, no interim distributions and risk-neutral pricing with zero risk-free rate. These assumptions isolate the relationship among promised repayment, proceeds, default, recovery and equity value. They are not a general corporate-finance specification.

26.1 Assets and claims

At the final date, assets available to creditors and shareholders are 60 in the low state and 140 in the high state, each with probability one half. A debt contract promises face $F \geq 0$. The creditor receives the smaller of face and available assets. Shareholders receive the positive residual.

$$D(A, F) = \min(F, A), \tag{26.1}$$

$$E(A, F) = \max(A - F, 0). \tag{26.2}$$

For every state, $D(A, F) + E(A, F) = A$. This conservation relation is exact under the stipulated priority and absence of costs. Default occurs when assets are insufficient for the promise, with equality treated as full payment.

Debt proceeds at issuance are the expected creditor payoff because the pricing rate is zero and risk neutrality is assumed. Economic equity value is the expected shareholder payoff. Therefore

$$P(F) = \tfrac{1}{2}\min(F, 60) + \tfrac{1}{2}\min(F, 140), \qquad V_E(F) = 100 - P(F). \tag{26.3}$$

The expected asset value 100 is allocated between the claims. Face does not equal proceeds once default becomes possible.

26.2 Solve the price schedule by regions

For $0 \leq F \leq 60$, debt is paid in both states, so $P(F) = F$. For $60 < F \leq 140$, the creditor receives 60 in the low state and F in the high state, so $P(F) = 30 + F/2$. For $F > 140$, the

creditor receives all assets in both states, so proceeds remain 100.

$$P(F) = \begin{cases} F, & 0 \le F \le 60, \\ 30 + F/2, & 60 < F \le 140, \\ 100, & F > 140. \end{cases} \tag{26.4}$$

The function is continuous but its slope changes. Promising another unit raises proceeds by one in the safe region, by one half in the intermediate region and by zero beyond the highest asset payoff. An unlimited face amount does not create unlimited funding.

Equity value is $100 - F$ in the first region, $70 - F/2$ in the second and zero in the third. These are economic values under the model. They are not a balance-sheet measurement rule. The model has not specified accounting carrying amounts.

26.3 A funding need of 80

Suppose an admitted transaction requires gross proceeds of 80. This need lies above the safe-debt capacity 60. Solve $30 + F/2 = 80$, giving face 100. In the low state the lender receives 60 and shareholders zero. In the high state the lender receives 100 and shareholders 40.

Expected lender payoff is 80, matching proceeds. Expected shareholder payoff is 20. The promised yield is $100/80 - 1 = 25$ per cent, although the risk-free rate is zero. That promised yield compensates for the modeled default loss; it is not the realized return in both states.

The low-state recovery is 60, or 60 per cent of face. If the model instead specifies recovery as a fixed fraction of face independent of available assets, it is a different recovery model and may violate conservation at some face values. Recovery assumptions cannot be swapped silently.

26.4 A funding need of 110

No face value raises 110 in this model because expected distributable assets are only 100 and the creditor cannot receive more than assets. A root finder searching for $P(F) = 110$ should report no solution under the model, or a validated infeasibility conclusion if its method establishes it.

A numerical optimizer that pushes face to a large upper bound has not solved the funding problem. Reporting that bound as issued debt would invent proceeds. The flat price region is a useful test of solver behavior when increasing the promise no longer raises cash.

The company could still have an alternative outside this model: new equity, collateral from another entity, a guarantee or a changed investment plan. The no-solution conclusion is conditional on the admitted financing channel and asset process. It is not proof that the real company has no option.

26.5 Fees and net cash

Add a fixed issuance fee of 5 paid at inception. To net 80, gross proceeds must be 85. Solving $30 + F/2 = 85$ gives face 110. Net cash is 80, expected creditor payoff 85, and expected equity payoff 15 under the unchanged asset distribution.

If the fee instead equals five per cent of gross proceeds, required gross proceeds are $80/0.95 = 1600/19$. Face is $2(1600/19 - 30) = 2060/19$, about 108.4211. The two cost structures give different promises.

If the fee is itself paid out of final assets rather than at inception, the asset payoff to claims changes and pricing must be recomputed. Merely subtracting the fee from current proceeds while also reducing final assets would charge it twice. The event dates determine which equation it enters.

26.6 Recovery costs

Suppose default destroys 10 of assets before creditor recovery. For $60 < F \leq 140$, low-state recovery becomes 50 and high-state payment remains F, so $P(F) = 25 + F/2$. Raising 80 now requires face 110.

At face 110, expected shareholder payoff is 15 and expected creditor payoff 80. Their sum is 95, below expected assets 100 by expected default cost 5. The conservation relation now includes the cost recipient or destroyed value: creditor payoff plus shareholder payoff plus default cost equals assets in each state.

A test that insists creditor plus shareholder value always equals 100 would be wrong after adding bankruptcy costs. Invariants must follow the model's resource accounting. Conversely, a model with no costs that loses value unexplained has a defect.

26.7 Endogenous investment changes the price

Suppose management can choose between project S with assets 100 in both states and project R with assets zero or 180 equally likely. Expected assets are 100 for S and 90 for R. Let face be 80. Under S, creditor payoff is 80 and equity payoff 20. Under R, creditor expected payoff is 40 and equity expected payoff 50.

Shareholders prefer R if they choose after debt proceeds are fixed and maximize their residual value. Total expected asset value is lower under R. A lender anticipating R prices face 80 at 40, not 80. Pricing as though S will be chosen while allowing shareholders to choose R is inconsistent.

A covenant restricting project choice, a different security structure or a commitment can change the feasible policy. The model must say whether project choice is committed before issuance, observed by lenders or chosen afterward. Timing determines the financing equilibrium.

26.8 A committed project comparison

If management commits credibly to S before issuance, face 80 raises 80. If it commits to R, face 80 raises 40. To raise 80 under R, face must satisfy $F/2 = 80$ while $F \leq 180$, giving face 160. Equity expected payoff is then 10.

The comparison is now between different promises financing the same proceeds under different committed projects. S leaves equity value 20; R leaves 10. Under the stated objective and no other benefits, S is preferred ex ante. This differs from the post-issuance asset-substitution choice at fixed face.

The example demonstrates why a structural model must distinguish issuance decisions from later policy. A single optimization over all variables can accidentally grant commitment that the actor does not possess. A sequential model must encode which choices are binding when the lender prices the debt.

26.9 A simple equity alternative

Suppose the company can instead sell a fraction u of the final asset claim for fair value under the original 60/140 distribution, with no fees. The price is $100u$. Raising 80 requires selling 80 per cent, leaving existing owners expected value 20.

This matches the expected residual value after issuing face 100 debt in the frictionless original example, but the state payoffs differ. With equity financing, old owners receive 12 in the low state and 28 in the high state. With debt financing, they receive zero and 40. Equal expected value does not imply equal risk or incentives.

Adding taxes, costs, control rights or different investor preferences can change the comparison. Those mechanisms must be specified rather than inferred from the equality in this simplified benchmark. The benchmark is valuable because it identifies exactly which additions drive a difference.

26.10 Unknown asset distribution

Suppose reports do not identify the low-state asset payoff and it may be 40 or 60, while the high payoff remains 140 and probabilities remain one half by assumption. For a promised face 100, proceeds are 70 or 80. A funding plan requiring 80 succeeds only in the second history.

If lenders know the asset type privately, they offer different prices across histories. If lenders share the analyst's uncertainty and use a prior, pricing requires that prior and the information structure. Averaging proceeds without stating who knows the type can misrepresent both the offer and management's choice.

An unweighted robust funding calculation asks whether a face can raise the required cash under every admitted pricing history. At face 120, proceeds are 80 in the low-asset history and 90 in the other. But the higher promise changes shareholders' payoffs. A robustly feasible financing plan need not maximize expected owner value under a selected prior.

26.11 Learning before issuance

A verified asset report arriving before issuance can identify the low-state payoff and narrow the price schedule. The same report arriving afterward cannot retroactively change settled proceeds, although it may affect secondary-market value or future refinancing.

The simulator should retain issuance-time information for the original price and current information for later valuation. Repricing the original cash receipt after new information would rewrite history. A market-value update is a separate measurement, not a new issuance event.

This distinction is particularly important in backtests of debt pricing. Using later realized recovery information to set the earlier issue price creates look-ahead. The model can use that information to evaluate the forecast, but not as an original input.

26.12 Two creditors, policy consistency and financing refusal

In a finite one-period model, policy consistency can be checked by enumeration. Specify a candidate policy π, the distribution of the terminal estate under that policy, the priority of each claim, and a pricing measure and discount factor. Allocate the estate once, calculate each

lender's expected discounted receipt, and use those receipts as candidate issue proceeds. At those fixed quoted proceeds, compare the issuer's payoffs from every feasible policy. Retain the candidate only when its conjectured policy is a best response and all funding and participation constraints hold. Ties must be retained. If no pure-policy pair passes, report that result for the declared class; neither a balancing loan nor an assumed mixed equilibrium repairs it automatically.

For a concrete local example, existing senior debt has face 40 and newly issued junior debt has face 80. A project costs 60. The safe policy produces estate 100; the risky policy produces estate 20 or 220 with equal probabilities. Both pricing and shareholder valuation use the stated risk-neutral probabilities and gross discount factor 21/20. Absolute priority in this illustration gives receipts

$$(s, j, e) = \big(\min(A, 40), \min((A - 40)^+, 80), (A - 120)^+\big). \tag{26.5}$$

Thus the safe estate pays $(40, 60, 0)$, and the risky estates pay $(20, 0, 0)$ and $(40, 80, 100)$. Each allocation sums to its estate. The candidate junior proceeds are 400/7 under the safe policy and 800/21 under the risky policy.

At a posted quote P, existing shareholders must subscribe $60 - P$ to cover the investment. Their net value from policy π is $\mathbb{E}[e_\pi]/(21/20) - (60 - P)$. They prefer risk at either quoted price. Consequently, the safe-policy quote is inconsistent with their best response: if risk is chosen, junior lenders receive present value 800/21 after advancing 400/7, a shortfall of 400/21. The risky quote is consistent within the pure-policy class, but requires shareholder cash 460/21. If shareholders have only 20, the project remains unfunded. The safe quote would need less cash, but its incentive problem remains. Even an enforceable safe-policy commitment would still need shareholder participation: here its zero terminal equity receipt does not cover their positive subscription. Liquidity, incentives, lender valuation and voluntary participation are distinct restrictions.

The same separation applies to restructuring. Given estate 70 and creditor outside values 50 and 20, an offer 45/25 conserves value but the senior creditor refuses. An offer 50/20 meets both constraints. A mandatory fee 1 reduces the available estate to69 and makes simultaneous participation impossible under these stated outside values. This is a feasibility result. Selecting a unique offer requires bargaining rights or another decision rule; it does not follow from total value conservation.

26.13 Why positive surplus can still end in liquidation

Hart and Moore's retained 1991 working paper makes the distinction between productive value and credible repayment particularly clear. The entrepreneur can withdraw human capital, the creditor can seize specified assets, and renegotiation follows an explicit offer game. Returns and liquidation values are deterministic, private savings have a particular seizure treatment, and one creditor controls the assets (Hart and Moore, 1991, Sections 2–3 and Appendix A1–A4). Under those assumptions, Lemma 1 derives the creditor's continuation reservation value and requires both future nonrepudiation and cash feasibility. The bargaining rule determines a result that cannot be obtained merely by balancing accounts.

In their Example 1, take two remaining operating receipts of 2 and 26 and liquidation rents of 8 in each period. The creditor's reservation value is 16, while the final credible payment is at most 13. The entrepreneur therefore needs to provide at least 3 before the final date. The first operating receipt provides only 2, so private wealth $w \geq 1$ is necessary. At $w = 1$, an immediate payment of 1 followed by payments 2 and 13 satisfies the cash and repayment conditions. At $w < 1$, it cannot. Liquidation occurs despite operating value 28 exceeding liquidation value 16

(Hart and Moore, 1991, Example 1, printed p. 16). The reference calculation reproduces this boundary with exact rational amounts. It is a useful counterexample to assuming that positive total surplus guarantees refinancing or restructuring.

Their repayment-path analysis also reinforces the distinction between an observed payment and its contractual explanation. A long-term contract and rolling short contracts can support the same path under the paper's conditions; the latter may additionally require access to outside refinancing (Hart and Moore, 1991, Proposition 2 and printed pp. 27–28). A future market closure can therefore distinguish histories that looked equivalent before the shock. For a multi-creditor company, the analyst must additionally specify priority, control and consent rights, collateral and intercreditor restrictions. The single-creditor result supplies neither an equity-issuance model nor a general legal restructuring procedure.

26.14 Reference-solver requirements

The finite model has a closed-form piecewise price schedule, so a numerical method can be checked at safe, risky, boundary and flat-price regions. Required points include face zero, 60, 100, 140 and above 140. Funding roots should be checked for needs below 60, between 60 and 100, exactly 100 and above 100.

A neural approximation can fit the schedule, but average error is not the only criterion. It should preserve monotonicity, upper funding capacity and boundary classification within stated error. A small positive price slope beyond 140 can falsely imply that an arbitrarily large face eventually raises 110. Downstream feasibility is the decisive check for that use.

A smooth approximation around the kinks can be useful for training, but final results should be evaluated against the original payoff. The source literature on learned economic solvers provides candidate algorithms; this exact model provides a local target-specific test. No training result is claimed here.

26.15 How the example answers the financing concern

Debt issuance is not a choice of a balance-sheet number. It is a contract whose cash price depends on future payoff and policy. Equity issuance is not merely negative dividends; it allocates future claims and potentially control. The finite model makes those relations explicit before numerical approximation.

For AccDSL, an external offer can be accepted as a conditional input, while an endogenous financing module can solve a stated pricing-policy problem. Both routes must keep face, proceeds, recovery, owner claims and information dates separate. This provides a defensible starting point for extending the supplied lecture rather than importing its conflicting convention into a larger simulator.

26.16 Conversion also changes the accounting question

The economic financing model specifies who receives cash and who chooses the project. Accounting for those events requires additional distinctions. In the supplied contingent-conversion extension, a creditor receives 120 shares alongside 40 existing shares. The resulting 75 per cent voting fraction explains the extension's assumed economic control rule. It does not by itself establish the date control begins for accounting, the holder's reporting perimeter, or the amount

to recognise in its accounts.

The following analysis uses Australian standards as a conditional design case. It does not select a reporting basis for an actual company. The retained editions and their application periods matter: the inspected AASB 10 compilation covers annual periods beginning in 2026 before 2027, and the inspected AASB 3 compilation covers starts from 1 July 2026 before 2027. The synthetic issue and conversion occur in 2027 and 2028, so a complete applicable edition and entity-scope chain remains necessary. The source paragraphs identify the questions that must be answered, while the missing facts remain unresolved.

26.16.1 Control can begin before the shares arrive

Under AASB 10, control requires power over the investee, exposure or rights to variable returns, and the ability to use that power to affect the returns. Power concerns existing rights giving the current ability to direct the activities that significantly affect those returns. Majority votes support that conclusion in the circumstances specified by the standard, but the votes must be substantive and contrary rights or arrangements can matter (Australian Accounting Standards Board, 2026c, paras. 6–18, B22–B28, B35–B37).

This is a question about rights and relevant decisions, not just a percentage. A lender's protective covenant can restrict a borrower without giving the lender power over its activities. Conversely, an apparently contingent decision right is not automatically protective. If the activities that significantly affect returns arise only after a particular event, the right to direct those activities can matter before the event occurs (Australian Accounting Standards Board, 2026c, paras. B26–B28, B53).

Potential voting rights introduce the same timing issue into a convertible instrument. AASB 10 requires consideration of substantive potential rights, including those arising from convertibles, options and forwards. Whether rights are substantive depends partly on whether they can be exercised when decisions about the relevant activities need to be made (Australian Accounting Standards Board, 2026c, paras. B24, B47–B50). It follows that the simulator needs a control assessment before conversion as well as after it. Setting the first control date equal to the share-issue timestamp would conceal this question.

The acquisition date for an in-scope business combination is the date control is obtained. It can precede or follow the legal closing date when the relevant facts support that timing. Consolidation similarly begins when control is obtained (Australian Accounting Standards Board, 2026c, para. 20) (Australian Accounting Standards Board, 2026h, paras. 8–9). Thus the economic conversion, the rights assessment and the accounting acquisition date require distinct records linked by evidence. A date cannot be supplied merely to make the annual schedule continuous.

26.16.2 Two control assessments do not determine the intervening history

Suppose the investor controls the company at both observation dates. Control may have continued throughout, or it may have been lost and later regained. Likewise, two observations of no control permit an intervening acquisition followed by a disposal. An unknown earlier assessment must preserve both acquisition and continued-control possibilities. The reporting period therefore needs a history of changes and evidence about the intervals between them; an empty event list does not establish that no change occurred.

The accounting consequences differ. Changes in ownership while control continues are equity transactions under the inspected AASB 10 paragraphs. Loss of control instead requires

the specified derecognition, retained-interest measurement, treatment of accumulated other comprehensive income and resulting gain or loss (Australian Accounting Standards Board, 2026c, paras. 20, 23, 25, B96, B98–B99). A later return to control raises a new acquisition assessment. The same closing voting fraction cannot select among these histories.

A boundary comparison exposed this distinction in the first conditional reference: unknown prior control incorrectly selected an acquisition route, and a known loss omitted the loss-of-control dependencies. The successor preserves all nine combinations of controlled, not controlled and unknown endpoints, together with possible intervening changes. Each interval remains unresolved until additional evidence narrows it. Its history classes preserve the alternatives relevant to these dependency questions; they do not enumerate every possible event sequence. A supplied measurement-reference string likewise identifies a dependency without establishing the measurement itself.

26.16.3 The holder's investment and the group's assets are different reports

In separate financial statements, AASB 127 permits investments in subsidiaries, joint ventures and associates to be accounted for at cost, under AASB 9, or using the equity method, subject to its requirements and consistent treatment within each investment category. Certain required fair-value treatments carry through to the separate statements (Australian Accounting Standards Board, 2022c, paras. 3–10, 11A). Choosing a cost policy does not provide the initial cost amount. The earlier price paid for a convertible claim cannot silently become the post-conversion measurement without the relevant transaction and measurement analysis.

For consolidated reporting, a parent normally consolidates controlled subsidiaries, with specified exemptions and exceptions. An investment entity generally measures qualifying subsidiary investments at fair value through profit or loss, while a non-investment-entity subsidiary providing services related to its parent's investment activities can fall within the consolidation exception to that exception (Australian Accounting Standards Board, 2026c, paras. 4–4B, 27–33). The investor's entity status, the subsidiary's activities and the complete exemption conditions therefore belong among the required facts. None is determined by the 75 per cent fraction.

Consolidation combines the relevant assets, liabilities, income, expenses and cash flows, eliminates the parent's investment against the relevant subsidiary equity, and removes intragroup transactions and balances. It also requires consistent policies and appropriate reporting dates (Australian Accounting Standards Board, 2026c, paras. B86–B93). Potential rights used in a control assessment do not automatically set the proportion allocated to non-controlling interests: the standard separately addresses existing ownership and instruments that already provide access to ownership returns (Australian Accounting Standards Board, 2026c, paras. B89–B91).

This creates a concrete implementation consequence. The existing selected group-period reference assumes full ownership and the same supplied control edges and membership at opening and closing; it supplies no non-controlling-interest method. A company acquired during that period violates the assumption. A correct continuation must distinguish the period before control, the acquisition measurements and the period consolidated afterward, rather than force the new subsidiary into the opening group or omit its acquisition effects.

26.16.4 A controlled project is not automatically a business combination

Obtaining control does not settle whether the acquired set is a business. AASB 3 distinguishes an in-scope business combination from acquiring an asset or group of assets, a common-control combination, and specified investment-entity acquisitions. An asset-group acquisition under

its scope paragraph does not create goodwill (Australian Accounting Standards Board, 2026h, paras. 2–3).

The optional concentration test can establish that an acquired set is not a business when substantially all of the gross-asset fair value is concentrated as specified. Failing that test does not establish that the set is a business; the further assessment remains necessary. The underlying analysis requires inputs and a substantive process, with different criteria depending on whether outputs are present (Australian Accounting Standards Board, 2026h, paras. B7A–B12D). A project described by a terminal cash payoff supplies neither the workforce and process evidence nor the fair-value composition needed to decide the question.

For an in-scope business combination, the acquisition method requires identifying the acquirer and control date, recognising and measuring the identifiable assets, liabilities and non-controlling interest, and determining goodwill or a bargain-purchase gain. The goodwill calculation uses the specified acquisition-date components, including any relevant previously held interest (Australian Accounting Standards Board, 2026h, paras. 4–10, 32–33). The synthetic funding price, a 5 per cent pricing discount and a balanced four-party journal do not supply these missing measurements. They remain separate economic evidence.

26.16.5 Keep dated authority and missing facts visible

The inspected AASB 18 consequential amendments change specified AASB 3 presentation and disclosure references, add title-related footnotes in AASB 10 and AASB 127, and alter accompanying AASB 10 guidance. They do not in that instrument rewrite the core control, business-definition or separate-policy paragraphs used here (Australian Accounting Standards Board, 2024a, Appendix D, pp. 57, 70, 101–102). That bounded observation cannot establish that no other amendment applies.

AASB 2024-4 defers specified sale-or-contribution amendments involving an investor and its associate or joint venture until annual periods beginning in 2028, with earlier application permitted. It is not a general acquisition-of-control rule (Australian Accounting Standards Board, 2024b, paras. 6–11). AASB 2026-3 changes a downstream associate/joint-venture fair-value election, while AASB 2026-4 contains entity-specific presentation and application changes and references AASB 2026-2 (Australian Accounting Standards Board, 2026d, p. 5) (Australian Accounting Standards Board, 2026f, paras. 1–3). Completing that entity and edition chain is still required before actual 2027 or 2028 admission.

The repaired conditional dependency reference consequently records supplied, dated assessments of rights, control, reporting scope and acquisition type. Unknown endpoints retain alternative control histories; missing measurements stay named requirements. It distinguishes a separate cost-policy choice from evidence of cost, and a positive supplied control assessment from actual source-backed admission. Its current results authorize no posting. This is the useful conclusion for scenario design: unresolved accounting alternatives must remain linked to the rights, dates, entities and measurements that could distinguish them, instead of being collapsed into whichever journal first balances.[1]

[1] Retained official sources, paragraph and physical-page anchors, the conditional analysis and refusal cases are in `data/accounting-master/conversion-control-scope-20261026/`. The original 21 scenarios are retained alongside its adverse boundary cases. The successor in `data/accounting-master/control-boundary-repair-20261026/` checks 30 cases and 23 invalid-input controls. These establish bounded dependency behavior, not actual accounting correctness or complete standards coverage.

26.17 A control history must fit the reporting method

A supplied history of control can answer the continuity question without yet supplying a method for consolidated reporting. The existing selected group-period calculation supports a stable perimeter and full ownership. A conditional history must therefore be checked against that scope before the simulator uses the calculation. Matching opening and closing control edges cannot replace the intervening history.

The checked connection carries the original history request, actor, exact interval and source identities into the group-report request. It identifies the same parent and subsidiary in both. It also requires an explicit relation between the history's reporting basis and the book-composition measurement basis. Similar labels do not establish that those two premises mean the same thing.

Time precision matters here too. The book period uses calendar dates, whereas the control history can use exact instants. The reference requires an explicit fixed-offset calendar interpretation and compares the resulting midnight boundaries exactly. A partial-day acquisition cannot be rounded into whole-period control merely because the annual report's date fields contain no time of day.

The outcome depends on the whole interval. Missing coverage, conflicting evidence or unresolved actor availability leave the control-history requirement unresolved. Continuous absence of control stays outside this parent–subsidiary reporting route, even if an ownership record says 100 per cent. A gain or loss requires changing-perimeter methods; this includes a loss followed by regain when the opening and closing assessments are identical. Continuous control with partial ownership requires non-controlling-interest methods beyond the present full-ownership reference.

Only continuous supplied control together with a separate full-ownership premise fits the existing method's control and ownership scope. Even that result is a candidate for further report checks. It supplies no missing legal-book report, acquisition measurement or accounting authority, and the scope check itself produces no consolidated statement. The original report can still refuse a request whose control scope fits but whose other inputs are absent or incompatible.

This separates two sources of failure. If a valid unchanged-period fixture is refused, the existing report or its inputs need diagnosis. If a history contains a mid-period acquisition and the stable-perimeter report refuses it, the refusal exposes a missing financial method. The next work must specify acquisition or loss accounting, dated member and flow allocation, ownership and non-controlling interests, and the required measurements. A source-history check cannot manufacture those amounts or authorize a journal merely because its identities and dates match.[2]

26.18 Acquiring control changes the measurement and the period

Suppose the financing arrangement eventually gives the creditor control of the debtor. The debtor's legal accounts do not thereby become the creditor's accounts, and the new shares do not supply an acquisition-date balance sheet. Consolidated reporting now needs to explain which assets and liabilities enter the group, at what amounts, and from which instant the acquired company's income belongs in the group result. Multiplying its annual statement by a

[2]The conditional history-to-group connection, complete expected routes and unchanged/changed-perimeter baseline checks are retained in `data/accounting-master/control-group-boundary-20261027/`. The selected comparison covers 23 cases and 15 refusal controls. Every scope result withholds source, accounting, posting and report authorization; it is preparation for financial-method composition.

75 per cent shareholding would answer none of those questions. It would both omit part of the controlled company's activity and bring in activity from before control.

The smallest continuation considered here is one gain of control in an ordinary business acquisition, with a single reporting currency, aligned reporting dates and no subsequent loss of control or change in ownership rights during the selected interval. These are limits of the first method. A purchase of assets that is not a business, a combination under common control, an investment-entity exception and separate financial statements require their own treatments. The business assessment and acquirer identification therefore precede the acquisition calculation. The acquisition date follows the assessed date of obtaining control, rather than the share-issue date by definition (Australian Accounting Standards Board, 2026h, paras. 2–9).

26.18.1 The acquisition balance needs its own measurements

Consider the consolidated acquisition worksheet at that date. Let A be the aggregate recognised acquired assets and L the aggregate assumed liabilities, both expressed as positive magnitudes in the same currency unit. Their amounts follow the applicable recognition and measurement rules, including item-specific exceptions; an unchanged legal carrying amount is not automatically the required acquisition amount. Let T denote measured consideration transferred, Q measured non-controlling interests, and P the acquisition-date fair value of a qualifying previously held equity interest. Under the ordinary business-acquisition branch, the excess described in AASB 3 paragraph 32 is

$$E = T + Q + P - (A - L). \tag{26.6}$$

The calculation compares the consideration and interests in the acquired business with its recognised net assets. A positive excess can determine goodwill only when the route and measurements support that recognition. A negative excess requires the reassessment specified in paragraphs 34–36 before a bargain-purchase gain is recognised. Computing $\max(E, 0)$ and putting the remainder into profit would skip that requirement (Australian Accounting Standards Board, 2026h, paras. 10–21, 24–36).

The distinction between T and P matters particularly in the conversion example. A creditor may hold debt immediately before obtaining shares. That debt is not, merely because it converts, the previously held equity interest addressed by paragraph 42. Nor does the issuer's transfer of a carrying amount from debt to equity determine the holder's acquisition consideration. The contract and its financial-instrument treatment must establish what is transferred and which effects belong to the business combination. Pre-existing relationships and separate transactions require their own assessment (Australian Accounting Standards Board, 2026h, paras. 42, 51–53 and B50–B53). The face amount, an economic price and a carrying amount cannot silently fill the same measurement slot.

Non-controlling interests introduce a further measurement choice. For qualifying present ownership interests, a proportional-share election gives $Q = q(A - L)$, where q is the non-controlling fraction. A fair-value election instead requires a distinct measurement of Q (Australian Accounting Standards Board, 2026h, para. 19). Substituting the proportional expression into equation (26.6) gives

$$E = T + P - (1 - q)(A - L). \tag{26.7}$$

For a conditional 75 per cent ordinary interest, $q = 1/4$ and the acquired net-asset contribution is $3(A - L)/4$. This algebra does not establish that a particular preference share or other right qualifies for the election. With full ownership, $q = 0$ and no non-controlling interest remains. An unspecified prior interest does not become $P = 0$; that zero requires an explicit no-prior-equity premise under the selected route.

26.18.2 Only the controlled interval contributes subsidiary results

After the acquisition worksheet is established, consolidation incorporates the controlled company's income and expenses from the date control is obtained. Those subsequent amounts follow the acquisition-date basis: for example, depreciation uses the acquired asset amount recognised in the consolidated statements (Australian Accounting Standards Board, 2026c, para. 20 and B86–B88). A legal-book expense may therefore need a group adjustment even when the legal journal itself is correct. The simulator must preserve the legal books and put the adjustment in the separate consolidation worksheet.

Write R for the subsidiary's legal profit over the exact post-control interval, and D for the signed additional expense required to express that interval on the group basis. Positive D reduces profit; negative D increases it. Let O be the legal other comprehensive income over the same interval and U its signed group adjustment. All four quantities have the same currency unit. The adjusted subsidiary contribution is

$$\pi = R - D, \qquad \omega = O + U. \tag{26.8}$$

This is a period and measurement reconciliation, not a forecast or a rule for dividing an annual total. A whole-year R cannot be prorated by elapsed months unless an independently supported method justifies that approximation. A journal recorded only by calendar date also needs an ordering premise when control changes within that date; rounding the control instant would conceal the ambiguity.

Under the deliberately limited assumption of constant ordinary present ownership rights, the owners' and non-controlling shares of π are $(1 - q)\pi$ and $q\pi$, and similarly for ω. Actual allocation must consider the relevant rights and the potential-voting-rights exception; the fraction of issued shares is not a universal allocation rule (Australian Accounting Standards Board, 2026c, B89–B95). Let V be distributions to non-controlling interests, expressed as a positive reduction, and M the other signed, supported changes in those interests within this limited branch. Their selected reconciliation is then

$$N_{\text{close}} = Q + q(R - D) + q(O + U) - V + M. \tag{26.9}$$

Starting with measured acquisition-date NCI, the equation adds its share of adjusted profit and OCI and then accounts for direct changes. A sufficiently large loss may make the closing amount negative; clipping that deficit at zero would contradict the attribution requirement (Australian Accounting Standards Board, 2026c, B94). The equation cannot establish whether the supplied M or adjustment population is complete. Ownership transactions with continued control, and a later loss of control, need their separate methods rather than an unexplained entry in M.

26.18.3 Cash and revisions retain their own dates

Acquisition also changes the cash included in the group. The conversion of debt to equity is a noncash transaction, but the acquired company may already hold cash. Its cash enters the consolidated perimeter even when the conversion itself creates no receipt or payment. The method must retain cash consideration, acquired cash and noncash consideration separately, with the applicable acquisition cash presentation net of acquired cash. Acquisitions and disposals are not netted against one another merely because both change control (Australian Accounting Standards Board, 2026b, paras. 39–44B). Cash transfers before control are not retrospectively intragroup flows; transfers afterwards require supported counterpart matching before elimination.

The acquisition worksheet must also distinguish new information about conditions at the acquisition date from later events. Provisional amounts and measurement-period adjustments follow their own rules, including the one-year maximum, rather than a general permission to rewrite the past (Australian Accounting Standards Board, 2026h, paras. 45–50). A revised measurement can therefore have an acquisition-date effect and a later information-availability date. The earlier actor view must remain reproducible, using the temporal distinctions developed in Chapter 17.

26.18.4 A transaction date does not select an edition

The conditional AASB analysis preserves several dates because they answer different questions. For example, the retained AASB 2026-2 compilation was prepared on 21 October 2026, has a compilation date of 31 December 2027, and generally applies to annual reporting periods beginning on or after 1 July 2029, with earlier application permitted under its terms. These dates cannot be substituted for one another. Its amendments to AASB 3 and AASB 10 concern application of the Conceptual Framework; entity classification and elections remain necessary inputs (Australian Accounting Standards Board, 2026e, paras. 3, 18–19, 25; compilation details). The inspected interaction with AASB 2026-4 also imposes a specific AASB 18 condition for early use of that amendment (Australian Accounting Standards Board, 2026f, paras. 3, 25).

Other branches remain visible. AASB 20's Appendix D adds a business-combination treatment for regulatory assets and liabilities, with its own application timing and an additional dependency when applied before AASB 18. The retained appendix alone does not establish the entity's eligibility or the complete transition treatment (Australian Accounting Standards Board, 2026g, Appendix D, amendments to AASB 3). AASB 1061's Appendix C changes tier wording relevant to AASB 10; it does not supply the full Tier 3 scope or measurement rules (Australian Accounting Standards Board, 2026a, Appendix C, amendment to AG1). Similarly, the inspected sale/contribution amendment chain affects a particular loss-of-control branch, rather than replacing every acquisition method (Australian Accounting Standards Board, 2024b, paras. 6–8). The dated authority matrix retains these alternatives and exact paragraph anchors. A future transaction's policy must be checked against its entity, annual period, elections and then-current applicable requirements.

26.18.5 What the selected reference establishes

The conditional expression reference binds the original control-history request and the exact gain-of-control instant before constructing these relations. Each amount remains an explicitly named unknown, with its role, currency, unit, timing, measurement basis and source identity. Future or unknown source availability remains unresolved for the selected actor. Missing assets, consideration, tax effects or adjustment methods cannot be supplied by the fact that an equation balances.

Independent coefficient maps check both NCI elections, qualifying prior equity, the full-ownership limit and the post-control reconciliations. Adverse requests test prior debt labelled as equity, pre-control profit leakage, sub-day rounding, inconsistent currency or ownership, source alteration and attempted insertion of an unmeasured face value. All source, accounting, posting and report authorization remains withheld.[3]

[3]The dated source review is retained in `data/accounting-master/acquisition-period-authority-20261026/`; its 13 primary PDFs have retained identities and inspected anchors. The symbolic design and independent expected expressions are in `data/accounting-master/acquisition-symbolic-20261027/`; the selected check covers 25 cases and 21 refusal controls. These deterministic checks establish the stated local behaviour, not source truth or complete acquisition accounting.

The next financial-method comparison must expand the aggregate unknowns into an identified item population, preserve every legal account, and reconcile a complete dated acquisition worksheet and period report against independently authored expectations. That comparison must include consideration and separate transactions, acquisition-basis expenses, NCI, cash, eliminations, taxes and revisions. The expressions developed here identify the measurements that comparison needs; they do not replace it.

26.19 A complete small acquisition makes the adjustments visible

The unknown amounts in the preceding section protect the real financing question from invented measurements. A separate synthetic example can now test whether the proposed accounting machinery combines those amounts correctly. Every amount in this example is stipulated in one currency unit. It is not a valuation of the creditor or debtor. The example assumes an ordinary business acquisition, a 75 per cent present ownership interest, aligned reporting periods, no prior interest or relationship, and a gain of control on 1 July 2025. Tax recognition and the measured tax amounts are supplied premises; the calculation tests their propagation.

At the start of the year the parent has cash 300 and capital 300. Immediately before acquisition, the subsidiary has cash 50, equipment 70, payables 20, contributed capital 80 and pre-control accumulated profit 20. Its net assets are 100. The parent pays 120 in cash and records an investment 120, leaving its own cash 180. Those legal accounts remain separate and balanced. The subsidiary's accumulated profit 20 belongs to the acquired equity; it does not become current post-control group profit.

The stipulated acquisition basis changes equipment from 70 to 90 and introduces a deferred-tax liability 5. Acquired recognised assets are therefore 140 and liabilities 25, giving net assets 115. With proportional non-controlling interests, $Q = 115/4$ and the positive excess in equation (26.6) is $120 + 115/4 - 115 = 135/4$. Conditional recognition of that excess is an explicit premise of this positive-excess example. The consolidated acquisition worksheet removes the parent's investment 120 and the subsidiary's acquired capital 80 and accumulated profit 20. It adds the equipment adjustment 20, the tax liability 5, goodwill 135/4 and NCI 115/4. Each entry retains its own origin; none is an unexplained balancing amount.

The resulting acquisition balance can be reconstructed from cash 230, equipment 90 and goodwill 135/4, against payables 20, deferred tax 5, parent capital 300 and NCI 115/4. The legal books have not changed. Only the separate consolidation worksheet contains the acquisition adjustments.

26.19.1 The closing report follows the controlled part of the year

After control the subsidiary receives external revenue 40 in cash and records legal depreciation 7. Its legal closing cash is 90 and equipment 63; its post-control legal profit is 33. The acquisition basis requires an additional depreciation 2 and a stipulated release of tax liability 1/2. Thus the group profit contribution is $33 - 2 + 1/2 = 63/2$. The original pre-control profit 20 is still excluded. This is a comparison of complete dated accounts, not a proportional allocation of a whole-year income number.

The closing group holds cash 270, equipment 81 and goodwill 135/4. Its payables are 20 and deferred-tax liability 9/2. Assets total 1539/4, liabilities 49/2 and equity 1441/4. Owners receive three quarters of profit, 189/8, while NCI receives 63/8. Closing owners' equity is 2589/8 and closing NCI 293/8; together they equal 1441/4. The expense and tax adjustments therefore

affect both the asset/liability measurements and the allocation of the controlled-period result.

Choosing fair-value NCI changes this example's acquisition measurement. If the stipulated Q is 40, goodwill is 45 and closing NCI 383/8; the same post-control profit still receives the stipulated 75/25 attribution. Full ownership instead removes NCI, gives goodwill 5 under the same total consideration, and assigns all profit to the owners. A further loss case adds a subsidiary operating expense 200 with a matching external payable. It leaves cash unchanged but produces closing NCI $-107/8$. The reference retains that deficit rather than truncating it, consistently with the attribution boundary already described (Australian Accounting Standards Board, 2026c, B94).

26.19.2 An equity-funded purchase need not consume cash

Hold total stipulated consideration at 120 and vary only its composition. A cash-only purchase uses cash 120 and no new parent equity. An equity-only purchase issues parent equity 120 and uses no cash. A mixed purchase pays cash 50 and issues equity 70. This controlled comparison isolates the accounting consequences of the supplied components; it does not establish that the securities have equivalent market values.

The subsidiary brings cash 50 into the consolidated perimeter in each case. Net acquisition cash is therefore -70, 50 or 0 for the cash-only, equity-only and mixed purchases. With the same later operating receipt 40, closing group cash is 270, 390 or 340. In the equity-only case, acquired cash 50 appears despite the absence of a purchase cash payment. Issuing shares does not create a fictitious cash receipt followed by a payment. The issued equity is retained as a separate noncash consideration component and increases parent capital, while the cash report retains cash paid and cash acquired separately (Australian Accounting Standards Board, 2026b, paras. 39–44B).

This newly issued parent equity must not be confused with the financing example's prior creditor claim. The cash/equity comparison assumes no prior relationship. Extinguishing or converting an existing debt requires the separate consideration, financial-instrument and pre-existing-relationship analysis discussed earlier. Renaming the debt as an equity payment would bypass that unresolved analysis.

26.19.3 A later measurement changes the worksheet and the information view

Suppose the synthetic acquisition equipment amount was provisional. A later proposed measurement changes it from 90 to 100, together with explicitly reviewed dependent amounts: the acquisition tax liability becomes 15/2, additional controlled-period depreciation becomes 3, and the tax release becomes 3/4. These values are supplied together; a revised equipment amount cannot silently inherit unrelated tax or expense assumptions. With proportional NCI, net assets become 245/2, NCI 245/8 and goodwill 225/8. Post-control profit becomes 123/4, owners' profit 369/16 and NCI profit 123/16. In the cash-only case, closing assets are 3105/8, liabilities 107/4 and equity 2891/8. Owners' equity 5169/16 plus NCI 613/16 gives the same total. All original legal accounts remain unchanged.

Those financial effects are conditional on the proposed information being eligible to revise acquisition-date measurements. A later event, a nonprovisional item, a different effective date or information outside the applicable measurement period cannot be treated as the same adjustment merely because the proposed worksheet balances (Australian Accounting Standards Board, 2026h, paras. 45–50). The selected reference receives those dispositions as explicit premises; it does not decide their truth.

Availability introduces a different question. In the example the new information is discovered on 15 January 2026 and reaches the selected actor on 1 February 2026. A 10 January query retains the original even if the exact delivery time is unknown, because delivery cannot precede discovery. A 20 January query likewise retains the original when the February delivery is known. At delivery, eligible supplied information can select the revised conditional report. If delivery is unknown after discovery, both information views may remain possible. If the supplied disposition is a later event, unknown delivery does not make retrospective acquisition adjustment an eligible financial alternative; only the original worksheet remains, with another-method work conditional on the actor receiving that evidence.

The original report must also survive an unsupported proposed method. An undiscovered revision might produce a negative excess that the current positive-excess reference cannot handle. It cannot make the earlier report invalid. Once the revision could be available and eligible, the system retains the original and the complete proposed request with its refusal and missing-method requirement. It does not select the original as if the new information had been resolved, and it does not fabricate a revised result. This distinction preserves history while making the next accounting-method obligation visible.

26.19.4 The comparison remains narrower than company accounting

The selected comparisons inspect every original account, acquisition item, adjustment origin, adjusted account and report field, including cash and parent/NCI equity movements. The repaired revision reference also rebinds the embedded source and control queries to the requested actor and cutoff. Earlier passing candidates were rejected when discriminating cases exposed a pre-discovery alternative, an ineligible adjustment under unknown access, and a future failed candidate that erased the original. Those failures remain in the evidence history.[4]

This remains a four-item synthetic acquisition with stipulated tax treatment, one currency, ordinary rights, one actor and one proposed revision. Prior interests, actual debt conversion, loss or later ownership changes, intragroup trade, general item mappings, foreign exchange, revision chains and complete statutory disclosures need further methods and evidence. The next comparison must add post-control intragroup transactions without leaking pre-control activity or losing the acquired measurement basis.

26.20 An internal sale becomes group profit only through an outside transaction

The acquisition worksheet now has a second task: keep two valid legal books from counting a transaction within the group as income from outside it. Continue the synthetic 75 per cent cash acquisition, but add inventory costing 20 and matching capital of 20 to the parent's opening balances. After obtaining control, the parent sells the whole lot to the subsidiary for 30 on credit. The parent records revenue 30, cost of sales 20 and a receivable 30. The subsidiary records inventory 30 and a payable 30. Each legal book balances, yet their sum includes a profit of 10 before any customer outside the group has bought the goods.

Consolidation removes the reciprocal receivable and payable in full. It also removes the internal revenue 30 against cost of sales 30, then charges the unrealised margin 10 to cost of sales and

[4]The complete cash-only and cash/new-parent-equity comparisons are retained in `data/accounting-master/acquisition-worksheet-20261027/` and `data/accounting-master/acquisition-consideration-20261027/`. The active revision evidence is `data/accounting-master/acquisition-revision-prefix-repair-20261027/`: 26 full cases and 15 refusal controls. Earlier revision profiles are preserved as superseded candidates after their counterexamples. All references withhold actual source, accounting, posting and report authorization.

reduces inventory by 10. After these two profit adjustments, the parent's original cost of sales 20 contributes zero, and the group carries inventory at its original cost of 20. Multiplying the adjustments by the parent's 75 per cent ownership would leave internal profit in group income. Full elimination and attribution of the subsidiary's result are separate operations (Australian Accounting Standards Board, 2026c, paras. B86–B89).

The original acquisition adjustments still apply. Equipment retains the acquired basis and the additional depreciation; the subsidiary's pre-control profit remains outside current group profit. The internal sale takes place after control, so it needs no retrospective reassignment of a pre-control transaction. If the subsidiary subsequently pays the 30, its cash payment and the parent's cash receipt cancel as separate intragroup flows. Settlement removes the reciprocal balances from the legal books, but does not realise the inventory profit for the group.

26.20.1 Removing profit does not remove an external tax obligation

Tax depends on more than the consolidated carrying amount. AASB 112 compares that amount with the appropriate tax base, which may depend on consolidated or separate tax returns. Its illustrative examples include intragroup profit eliminated from inventory without an equivalent adjustment for tax purposes (Australian Accounting Standards Board, 2023b, para. 11 and illustrative examples, B11). The accounting rule does not tell us which tax law applies to this company.

To expose this dependency, stipulate that the parent's margin of 10 is taxed now at one quarter, that the subsidiary keeps a tax base of 30, and that the same rate applies when it uses the deduction. The parent therefore has a current tax expense and external payable of 5/2. The group inventory has carrying amount 20 and tax base 30, leaving a deductible temporary difference of 10. If the supplied recoverability conditions permit recognition, the deferred tax asset is also 5/2. Its income offsets the current expense in this example; the external payable remains. If recognition is explicitly withheld, the asset is zero and group profit is lower by 5/2. The reference retains these alternatives separately. If the tax treatment is unknown, it retains the pretax worksheet and requires the missing tax method before selecting a final report.

These are supplied tax and attribution premises. Recognition requires the appropriate taxable-profit assessment, and measurement depends on enacted or substantively enacted rates and expected recovery (Australian Accounting Standards Board, 2023b, paras. 24–29, 46–51 and 58). In this downstream example the parent's current tax and the consolidation tax adjustment are attributed to parent owners. That premise does not resolve upstream transactions or every group tax arrangement. The retained source editions support the conditional design; they do not select an actual entity's reporting package or tax law.

With the deferred asset recognised, the unsold-inventory case closes with assets 1629/4, liabilities 27 and equity 1521/4. Parent equity is 2749/8 and NCI is 293/8. Group cash remains 270, whether the internal payable is settled or outstanding. Inventory and the extra opening parent capital account for 20 of the change from the acquisition-only case; the separate tax asset and payable account for 5/2 on each side. No balancing amount is needed.

26.20.2 An outside sale releases only the corresponding part of the margin

Now let the subsidiary sell half the lot to an outside customer for 20 cash before the year closes. The example stipulates a qualifying customer contract, transfer of control and a fixed price, with no returns or additional obligations. A cash receipt alone would not establish those facts (Australian Accounting Standards Board, 2022f, paras. 9, 31–38 and 47). The subsidiary recognises revenue 20 and inventory cost 15. Inventory cost is expensed when the related revenue

is recognised (Australian Accounting Standards Board, 2021a, para. 34).

The group still eliminates the full internal sale of 30. Only half of its margin remains in unsold inventory, so the closing unrealised-profit adjustment is 5. The remaining group inventory is 10, while the sold half costs the group 10. The outside sale thus adds group pretax profit of 10: revenue 20 less the original group cost 10. Of this increment, the subsidiary's own sale contributes 5 and the release of the parent's margin contributes 5.

Continue the supplied tax premises. The parent's current tax remains 5/2; the subsidiary's profit of 5 adds current tax 5/4. Only the unsold margin supports the remaining deferred asset, now 5/4. Relative to the acquisition case without inventory trading, the group earns an additional after-tax profit of 15/2. Half arises in the subsidiary after its current tax, and half belongs to parent owners after the downstream consolidation adjustments. Ordinary NCI therefore participates in the subsidiary's 15/4 increment, not in the parent's released margin.

The full closing comparison gives group profit 39: owners receive 483/16 and NCI receives 141/16. Assets are 416, liabilities are 113/4, and equity is 1551/4. Parent equity of 5603/16 plus NCI of 601/16 gives the same total. Cash is 290, including the outside receipt of 20. That operating receipt remains in the group cash report even if the separate internal payment of 30 is also made.

26.20.3 The dated journals make the closing report explainable

The comparison retains every legal entry, including the supplied opening trial balances, the explicitly stipulated pre-control receipt and interim close, acquisition consideration, controlled-period receipts and depreciation, internal sale and settlement, outside sale and current taxes. Replaying these dated journals recovers the opening, acquisition and closing accounts. Their origins distinguish an outside receipt from a reciprocal internal payment; the consolidation entries remain in a separate worksheet. A set of closing balances could not establish that history on its own.

No resale, half resale and full resale are compared under each selected NCI election and consideration mix. Unknown tax preserves the missing dependency instead of completing the report with zero. These results remain a one-lot, same-period downstream example with supplied revenue, tax and ordinary-rights premises.[5] Carrying unsold inventory into another year creates a new question: the next report must release profit from opening group equity and carry its tax basis, rather than repeat the original acquisition or eliminate a sale that occurred in the prior year. That requires a separate two-year comparison.

26.21 The next year starts from closed group equity

Suppose the first year ends with the entire lot still inside the group. The next report cannot eliminate that year's internal sale as if it had happened again. The parent's legal margin has been closed into retained profit, and the group's opening inventory already excludes it. The first task is therefore to carry the two legal books and the consolidation adjustments into the new year without changing the preceding report.

The legal close transfers each entity's income and expense balances into retained profit. The group close also transfers the acquired-basis depreciation and related tax effect into the subsidiary's

[5] The complete downstream comparison is retained in `data/accounting-master/acquisition-intragroup-20261027/`; the dated journal and resale comparison is in `data/accounting-master/acquisition-intragroup-realisation-20261027/`. Both preserve full independent account, journal and report expectations and keep source, posting and report authorisation false.

retained result. NCI's first-year profit is then moved out of owners' retained profit and added to the NCI already recognised at acquisition. In the unsold-inventory example, group opening equity remains 1521/4: parent owners hold 2749/8 and NCI holds 293/8. Carrying the original acquisition NCI alone would omit its first-year profit; adding the whole closing NCI again would count part of it twice.

A separate opening worksheet records the account-by-account difference between the closed legal books and those closed group accounts. It retains the investment/equity elimination, acquired equipment basis, goodwill, inventory margin and tax adjustments, with their original journal sources. This is a carried balance, not another acquisition. Its complete accounts must reproduce the first year's closing assets, liabilities, cash and both equity interests before any second-year transaction is evaluated.

26.21.1 Releasing a carried margin changes current cost of sales

In the second year, let the subsidiary sell the full lot outside the group for 40 cash, under the same supplied revenue-recognition premises. Its legal cost is 30, but the group opening inventory is 20. The legal sale removes 30 from the subsidiary's inventory; the opening group adjustment still contains a reduction of 10. Releasing that carried adjustment debits inventory 10 and credits current cost of sales 10. The group therefore reports cost 20 and zero closing inventory. Debiting opening retained profit a second time would duplicate an adjustment already present in the opening worksheet.

Under the supplied tax assumptions, the subsidiary's current tax on its legal profit is 5/2, and the carried inventory deferred tax asset of 5/2 reverses. The inventory transaction consequently contributes after-tax group profit 15. The subsidiary contributes 15/2 after its current tax; the remaining 15/2 is the parent-attributed release of the downstream margin, net of the deferred tax reversal. These attribution premises remain explicit. An upstream seller would require a different comparison.

26.21.2 Other acquired assets continue to affect the year's result

The equipment did not disappear at the inventory sale. A second-year comparison that freezes it would omit a known accounting dependency. Continue the synthetic example with separately supplied full-year legal depreciation 14, additional acquired-basis depreciation 4 and a tax-liability release of 1. The amounts extend the first year's half-year pattern, but are declared schedules, not inferred useful lives or tax deductions. AASB 10 requires the subsidiary's consolidated expenses to retain the acquisition basis; it does not determine these particular amounts (Australian Accounting Standards Board, 2026c, para. B88).

Equipment contributes a net expense of 17: legal depreciation 14 plus the additional 4, less tax income 1. Combined with inventory profit 15, second-year group profit is therefore -2. The subsidiary-attributed result is $15/2 - 17 = -19/2$, so ordinary NCI receives $-19/8$. Parent owners receive three quarters of that result plus the parent-attributed margin release, giving 3/8. Their sum is -2. The negative NCI result is retained rather than clipped (Australian Accounting Standards Board, 2026c, para. B94).

The closing group holds cash 310, equipment 63 and goodwill 135/4, with no remaining inventory or inventory deferred tax asset. Assets total 1627/4. Liabilities are 57/2: the original external payable 20, acquired deferred tax liability 7/2, and unpaid current taxes of 5. Equity is consequently 1513/4, equal to opening equity 1521/4 less the current loss of 2. The earlier tax payable has not vanished, and no payment has been invented to make the roll-forward balance.

The reference compares complete closed openings, the carried adjustment worksheet, second-year legal and group journals, closing accounts and cash origins. No-sale and partial-sale cases retain the appropriate inventory and tax amounts while still recognising the supplied equipment expenses. Unknown tax continuation retains the prior report and proposed request but selects no second-year final report.[6] This remains a one-lot downstream comparison with supplied continuing control, ordinary rights and financial schedules. The next discriminating case must place the seller inside the partially owned subsidiary and check which profit adjustments then belong to NCI.

26.22 When the subsidiary earns the internal margin

Now let the subsidiary sell the inventory to the parent. The group must still eliminate the full internal sale, but the profit being removed was earned in the company partly owned by NCI. Applying the minority percentage to that company's unadjusted legal profit would allocate profit that the group has not yet earned from an outside customer.

Keep the acquisition on 1 July 2025 unchanged. On 1 September, after control has begun, the subsidiary buys inventory for 20 cash from an outside vendor. On 1 October it sells the lot to the parent for 30 on credit. Buying the lot after control matters: inventory is not inserted into the acquired net assets, and the original goodwill and acquisition NCI remain unchanged. The subsidiary's cash falls by 20. This external payment survives consolidation even if the parent later settles the internal payable of 30. Settlement creates a payment by the parent and a receipt by the subsidiary; both internal cash flows are eliminated in full.

Before any outside resale, the parent's legal inventory is 30 and the subsidiary has legal trading profit 10. Consolidation removes the reciprocal receivable and payable, eliminates internal revenue 30, and reduces inventory by the unrealised margin 10. Group inventory is therefore 20. Neither the inventory adjustment nor the reciprocal-balance elimination is multiplied by the parent's 75% ownership. Full elimination precedes the allocation of the subsidiary's adjusted earnings between parent owners and NCI (Australian Accounting Standards Board, 2026c, paras. B86, B89–B94).

26.22.1 Tax recognition and attribution answer different questions

Continue with explicitly supplied separate tax bases and 25% rates. The subsidiary owes current tax 5/2 on its legal margin. If the parent's inventory tax base remains 30, the difference from the group carrying amount 20 can support a deferred tax asset of 5/2 when the recognition conditions are met. The comparison assumes those conditions in one branch and explicitly withholds recognition in another. The tax base and recognition conditions remain distinct from the inventory elimination (Australian Accounting Standards Board, 2023b, paras. 11, 24 and illustrative example B11).

For this example, attribute the inventory elimination and its deferred tax effect to the subsidiary seller's earnings. This is a supplied attribution premise; the cited paragraphs do not themselves determine the example's tax sharing. With recognised deferred tax, removing profit 10 and adding tax income 5/2 offsets the subsidiary's after-tax legal margin 15/2. Its consolidated profit returns to 63/2, the acquired-basis operating result already derived. NCI receives one quarter, or 63/8, and parent owners receive 189/8.

[6] The full comparison and protected inputs are retained in `data/accounting-master/acquisition-intragroup-carry-20261027/`. The pre-execution audit repaired the omitted equipment schedule and avoided a duplicate debit to opening retained profit before running the candidate.

The group now holds cash 250, inventory 20, equipment 81, goodwill 135/4 and the inventory deferred tax asset 5/2. Total assets are 1549/4; liabilities are 27, including the unpaid current tax. Equity is 1441/4. If no inventory deferred tax asset is recognised, the current tax remains an expense. Group profit becomes 29, of which NCI receives 29/4 and parent owners 87/4. Assigning that entire tax expense to parent owners would be wrong under the stated subsidiary-attribution premise. If tax is unresolved, the reference retains the pretax worksheet and the outstanding tax questions but produces no final financial report.

26.22.2 Outside resale releases the subsidiary's margin

Let the parent sell a fraction r of the lot outside the group on 1 December for $40r$ cash, where this finite comparison permits $r = 0$, $1/2$ or 1. Its legal cost is $30r$ and its current tax is $5r/2$, so its own after-tax trading contribution is $15r/2$. The subsidiary's original margin 10 is now realised only in proportion r. Write d for the remaining recognised inventory deferred tax asset: $d = 5(1 - r)/2$ in the recognised branch and zero in the unrecognised branch. Under the supplied attribution, the two trading contributions and resulting NCI profit are

$$\pi_S^{\text{trade}} = 10r - \frac{5}{2} + d, \qquad \pi_P^{\text{trade}} = \frac{15r}{2}, \qquad \pi_{\text{NCI}} = \frac{1}{4}\left(\frac{63}{2} + \pi_S^{\text{trade}}\right).$$

The subsidiary contribution combines realised margin, its unchanged current tax, and the deferred benefit still recognised for unsold inventory. With recognised deferred tax it simplifies to $15r/2$. Half resale therefore adds 15/4 of profit to each company: group profit is 39 and NCI profit is 141/16. The group holds cash 270 and inventory 10; total assets are 396, liabilities 113/4 and equity 1471/4. The external vendor payment and outside customer receipt remain visible separately in the cash record.

Equal recognised-tax trading contributions happen to make several profit totals match the downstream example. That numerical coincidence is not a reason to forget which company earned the margin. The unrecognised-tax branch, the separate entity accounts and the carried opening equity distinguish the two directions. Complete legal journals, account worksheets, cash origins and attribution results are compared across the finite cases.[7] The remaining question is how the subsidiary-attributed margin and NCI move through retained profit when the outside sale occurs in a later year.

26.23 Carrying the subsidiary's margin into the next year

At the first upstream year end, suppose the parent still holds the whole lot. The subsidiary's legal profit is 81/2: operating profit 33 plus internal margin 10 less current tax 5/2. Closing its legal income accounts therefore puts 81/2 into post-control retained profit. The parent has earned no trading profit. These legal retained amounts cannot simply be copied into group equity.

With the inventory deferred tax asset recognised, the subsidiary's consolidated result is 63/2. The group close leaves 189/8 of that result in owners' retained profit and adds 63/8 to acquisition NCI of 115/4. Opening NCI is consequently 293/8. The account-by-account opening bridge reconciles these closed group accounts with the closed legal books, preserving the investment elimination, acquired basis, inventory adjustment and tax. It does not acquire the subsidiary again. Without the inventory deferred tax asset, the corresponding subsidiary result is 29:

[7] The exact upstream comparisons, malformed-request controls and protected sources are in `data/accounting-master/acquisition-intragroup-upstream-20261027/`. The dates, lot, customer-control facts, rights and tax premises are synthetic; no actual source or accounting admission is inferred.

owners retain 87/4 and closing NCI is 36. The tax recognition decision has already affected the equity that must be carried forward.

26.23.1 The parent's sale releases profit attributed to the subsidiary

In year two the parent sells the full lot outside the group for 40 cash. Its legal books recognise cost 30 and current tax 5/2, leaving its own after-tax trading profit 15/2. The group opening inventory adjustment still reduces the parent's inventory by 10. Releasing it requires a debit to the parent's inventory and a credit to the subsidiary's current cost of sales. The debit removes the carried adjustment after the legal sale; the credit locates the realised margin in the subsidiary seller's earnings. A second charge to opening retained profit would duplicate the opening bridge.

Where the inventory deferred tax asset was recognised, its 5/2 reversal is also attributed to the subsidiary under the supplied premise. The subsidiary's trading contribution is therefore 15/2. Continue the explicit equipment schedules: legal depreciation 14, additional acquired-basis depreciation 4, and acquired tax-liability release 1. They contribute a net subsidiary expense of 17, so its total second-year result is $-19/2$. NCI receives $-19/8$; parent owners receive 3/8 after adding the parent's own trading result. Group profit is -2. The negative NCI result remains part of the allocation (Australian Accounting Standards Board, 2026c, paras. B88, B94).

Without a recognised inventory deferred tax asset, there is no asset to reverse. The subsidiary instead contributes margin 10 less equipment expense 17, a loss of 7. NCI receives $-7/4$ and parent owners receive 9/4; group profit is 1/2. This branch earned 5/2 less in the first year and 5/2 more in the second. After full realisation, the two branches therefore reach the same closing equity under the stipulated tax assumptions. Their yearly earnings are different.

Both branches end with cash 290, equipment 63, goodwill 135/4, and no inventory or inventory deferred tax asset. Assets are 1547/4. Liabilities of 57/2 include the original payable 20, remaining acquired deferred tax liability 7/2, and unpaid current taxes 5. Equity is 1433/4, split between owners 324 and NCI 137/4. The prior tax payable is carried; no settlement is invented to obtain these balances.

Complete opening and closing accounts distinguish the attribution even where aggregate profits coincide. Partial resale retains the appropriate margin and tax; an unresolved opening or continuing tax method retains the prior result and proposed continuation without a second-year final report.[8] These acquisition examples all started with no prior interest in the subsidiary. When an investor already holds an interest before obtaining control, the opening investment, acquisition consideration and any remeasurement must be distinguished before the same consolidation logic can be applied.

26.24 An investment already held before control

An investor may already own shares when it obtains control of a business. Suppose the old holding is carried at 35, its acquisition-date fair value is 40, and the investor pays 80 for additional shares. These are three different amounts. The old carrying value describes the opening books; 40 measures the previously held equity at the acquisition date; 80 is the new consideration. Treating 35 as new consideration would misstate the acquisition calculation. Treating the increase from 35 to 40 as cash would invent a receipt.

[8]The upstream carry comparisons, complete expected accounts and rejection controls are retained in `data/accounting-master/acquisition-intragroup-upstream-carry-20261027/`. Continuing control, ordinary rights, depreciation and tax schedules are supplied conditions. Actual source and accounting admission remain unresolved.

For an acquisition achieved in stages, the previously held equity interest is remeasured at acquisition-date fair value. The resulting gain or loss enters profit or loss or other comprehensive income as appropriate; prior OCI follows the basis that would apply to a direct disposal of that interest (Australian Accounting Standards Board, 2026h, paras. 41–42). That rule makes the prior accounting classification consequential. It cannot be selected merely by observing that the investor did not previously control the investee.

26.24.1 No control does not establish the prior accounting method

In the example, the old holding represents 20% of voting power. AASB 128 presumes significant influence at 20% or more unless its absence can be clearly demonstrated. Board representation, participation in policy decisions and other relationships can matter (Australian Accounting Standards Board, 2025, paras. 5–8). Joint control is a separate question. The simulator therefore needs explicit relationship assertions before it can use the restricted AASB 9 route below. A percentage and a no-control assertion are insufficient.

For this synthetic case, stipulate both no significant influence and no joint control immediately before acquisition, together with an AASB 9 fair-value- through-profit-or-loss classification. These are supplied premises, including the rebuttal of the influence presumption, rather than findings about an actual company. An unknown or positive influence or joint-control assertion leaves the method unresolved in this finite example. The particular FVTPL election available to specified venture-capital and similar investors is not presumed (Australian Accounting Standards Board, 2025, para. 18).

Separate financial statements require their own accounting-method choice: AASB 127 identifies cost, AASB 9 and the equity method as alternatives within its conditions (Australian Accounting Standards Board, 2022c, para. 10). Here, explicitly supply an AASB 9/FVTPL separate-book schedule and a fair value of 120 for the combined investment both at acquisition and at closing. The equality $120 = 40 + 80$ in this example is not a valuation rule for a controlling holding. The combined measurement is independently supplied. Also stipulate zero current and deferred tax on the prior-interest gain. This premise leaves the acquired equipment's existing tax schedule intact.

26.24.2 The prior gain belongs to the parent and uses no cash

The parent opens with cash 265, the old investment 35 and capital 300. Immediately before the additional purchase it debits investment 5 and credits its own remeasurement profit 5. It then pays 80 for a further 55% interest: investment is debited 80 and cash credited 80. The resulting 75% holding and control are separately stipulated. The parent's acquisition-date books now contain cash 185 and investment 120. The first entry is noncash; only the second entry pays for new shares.

Retain the earlier subsidiary's acquisition-date recognised assets 140 and liabilities 25, together with the proportional-NCI election and ordinary ownership rights. Identifiable net assets are $140 - 25 = 115$. NCI is therefore $115/4$, while the acquisition calculation combines new consideration 80, previously held equity at fair value 40, and that NCI. Goodwill is

$$80 + 40 + \frac{115}{4} - 115 = \frac{135}{4}.$$

This is the acquisition-method calculation under the supplied measurements (Australian Accounting Standards Board, 2026h, para. 32). The old carrying value 35 does not enter it. The parent investment of 120 is eliminated against the acquired equity and the acquisition adjustments; the parent's gain of 5 remains its own profit.

Continue the earlier subsidiary operating and equipment schedules, without an intragroup inventory sale. Its acquired-basis post-control profit is 63/2. The parent's prior gain is added separately when group profit is calculated:

$$\Pi_{\text{group}} = 5 + \frac{63}{2} = \frac{73}{2}, \qquad \Pi_{\text{owners}} = 5 + \frac{3}{4}\frac{63}{2} = \frac{229}{8}, \qquad \Pi_{\text{NCI}} = \frac{1}{4}\frac{63}{2} = \frac{63}{8}.$$

All three quantities are profits for the illustrated period, in the same monetary units as the accounts. Owners receive the whole parent gain and their share of subsidiary earnings. NCI participates only in the subsidiary result under the supplied ordinary rights. Allocating a quarter of the parent's remeasurement gain to NCI would change the stated economic interest.

The complete closing worksheet contains cash 275, equipment 81 and goodwill 135/4, giving assets 1559/4. Liabilities are 49/2 and equity 1461/4. Owners' closing equity is 2629/8; acquisition NCI 115/4 plus its earnings 63/8 gives closing NCI 293/8. These sum to total equity, which also equals assets less liabilities. The complete legal journals and consolidation worksheet retain the origins of the cash movements; the remeasurement never appears as a cash flow.

Changing the old carrying value to 40 or 45 produces zero gain or a loss of 5 at the same fair value 40. In these comparisons opening cash changes to 260 or 255 so that opening capital remains 300; this is an explicit change in the synthetic starting position. The new cash payment remains 80, and acquisition NCI and goodwill remain unchanged. Group profit becomes 63/2 or 53/2, respectively. This comparison isolates the signed prior-interest result without attributing it to the subsidiary's outside shareholders.

26.24.3 An equity-FVOCI election changes where the gain appears

The same difference between prior fair value and carrying value need not enter profit. For an eligible equity investment with the irrevocable OCI election, AASB 9 sends the applicable fair-value change to OCI. Amounts presented there are not subsequently recycled to profit or loss; a transfer within equity is permitted (Australian Accounting Standards Board, 2026i, para. 5.7.5 and B5.7.1). Applying the direct-disposal basis in the acquisition rule must preserve that distinction.

Write C for the old carrying value, P for its acquisition-date fair value, and O for accumulated prior OCI. The conditional current update is $P - C$. Under the eligible equity-FVOCI branch it enters OCI, leaving the reserve $O + P - C$. An explicit transfer choice either retains that reserve or transfers it within equity, with no recycling gain in profit. The FVTPL branch instead puts $P - C$ in parent pretax profit and requires zero prior OCI in this finite specification. These symbolic presentation branches do not supply a complete FVOCI tax or legal-book worksheet. Missing classification, relationship, measurement, tax or timely source premises remain unresolved.[9]

A debt claim held before a conversion is not previously held equity simply because the conversion produces shares. Its extinguishment or other settlement and any pre-existing relationship require their own transaction analysis; AASB 3 separates qualifying acquisition consideration from separate transactions (Australian Accounting Standards Board, 2026h, paras. 51–52 and B50–B53). That distinction determines the next extension of the financing example.

[9]The preserved expressions, repaired relationship scope and complete FVTPL worksheet are recorded in `data/accounting-master/acquisition-prior-interest-20261027/`, `data/accounting-master/acquisition-prior-interest-scope-20261027/`, and `data/accounting-master/acquisition-prior-worksheet-20261027/`. The earlier expression algebra alone did not establish classification scope. The repaired scope records supplied assertions, not actual source admission.

26.25 One debt exchange, two sets of accounts

Suppose a company extinguishes debt of 100 by issuing its own shares, valued at 70, to the lender. The borrower removes a liability and issues equity; the lender removes a financial asset and receives a different asset. Neither side receives cash. A gain for one party and a loss for the other can arise, but the two parties need their own classifications, measurements and journals. The shares received also raise a separate question about the lender's subsequent relationship with the borrower.

26.25.1 The debtor must first establish the scope of the exchange

AASB Interpretation 19 addresses a debtor's renegotiated financial liability settled by issuing equity. It does not address creditor accounting. It also excludes an existing shareholder acting as an owner, common control before and after the exchange together with equity-distribution or contribution substance, and share settlement in accordance with the liability's original terms (Australian Accounting Standards Board, 2020, paras. 2–3). An original contractual conversion therefore cannot enter this debtor example merely because it produces shares.

The compound conditions matter. An existing shareholding alone does not say that the creditor acts as an owner in this transaction. Common control alone does not establish the specified equity substance. The conditional scope calculation keeps yes, no and unknown assertions separate. A known exclusion remains effective even when another condition is unknown; otherwise missing scope facts leave the method unresolved. The same discipline applies to source availability at the stated information cutoff.

For the finite example, supply a renegotiated, whole-liability extinguishment within the Interpretation's scope, a financial liability and an issue of own equity. The borrower opens with cash 30, equipment 100, debt 100 and contributed capital 30. There is no other activity, interest, transaction cost, current or deferred tax, OCI or cash leg during the illustrated interval. Each absence is an explicit synthetic premise.

When the issued equity's fair value can be reliably measured, the debtor uses that fair value. Only when it cannot be reliably measured does the Interpretation direct measurement by reference to the extinguished liability's fair value (Australian Accounting Standards Board, 2020, paras. 6–7). Carrying amount is not a substitute for missing fair value. At the extinguishment date, the difference between the liability carrying amount and consideration paid enters profit or loss (Australian Accounting Standards Board, 2020, paras. 5, 9).

At the supplied equity value 70, the debtor debits debt 100, credits newly issued equity 70 and credits gain 30. Closing cash remains 30 and equipment 100; liabilities are zero. Equity of 130 comprises opening capital 30, new equity 70 and current profit 30. Operating, investing and financing cash flows are all zero, while the noncash debt-for-equity exchange is identified separately. The Interpretation's conceptual analysis using an issue for cash followed by repayment does not establish actual cash movements in this example (Australian Accounting Standards Board, 2020, BC13, BC25).

An equity value of 100 gives no gain; 120 gives a loss of 20. If equity fair value explicitly cannot be measured reliably and liability fair value is independently supplied as 60, the measured equity is 60 and the gain is 40. All four cases leave assets and total equity at 130 but change the division between contributed equity and earnings. A partial exchange needs an allocation between the extinguished portion and any modification of the remaining debt; it is outside this complete whole-debt worksheet (Australian Accounting Standards Board, 2020, paras. 8, 10–11).

26.25.2 The creditor must separate impairment from disposal

Now consider the lender's books. Supply a loan measured at amortised cost, with the relevant business-model and contractual-cash-flow conditions, followed by expiry of the entire contractual claim. These are distinct accounting premises (Australian Accounting Standards Board, 2026i, paras. 3.2.3, 4.1.2). For the shares received, stipulate no control, significant influence or joint control, an AASB 9 FVTPL classification and no OCI election. No-control alone would again be insufficient to establish that investment method.

The creditor opens with cash 30, gross loan 100, no loss allowance and capital 130. Before the exchange, an independently supplied expected-loss assessment requires an allowance of 30. The creditor debits impairment expense 30 and credits the allowance 30, leaving a net loan balance of 70. An allowance update belongs in impairment profit or loss; it must not disappear into a later net disposal calculation (Australian Accounting Standards Board, 2026i, paras. 5.2.2, 5.5.8).

At exchange, shares with supplied fair value 70 replace that net claim. The creditor debits shares 70 and the allowance 30, and credits gross loan 100. There is no disposal gain or loss. Total profit for the interval is a loss of 30, entirely from impairment. Closing cash 30 plus shares 70 gives assets 100; opening capital 130 less the loss gives equity 100. Cash flows remain zero. The complete record shows both journals and the intermediate assessed loan, so a correct closing total cannot conceal an omitted impairment step.

To see why this matters, supply a different assessment: allowance zero before receiving the same shares worth 70. The disposal then gives a loss of 30 and the total loss is again 30. This is a different conditional accounting history. The illustration does not infer expected losses from the share value or certify either assessment for an actual borrower. With allowance zero, shares worth 100 or 120 instead produce zero disposal result or a gain of 20. With allowance 30 and shares worth 60, impairment loss 30 is followed by disposal loss 10. Whole-asset derecognition and amortised-cost gains and losses have their own requirements (Australian Accounting Standards Board, 2026i, paras. 3.2.12, 5.7.2).

Initial share recognition also needs a measurement distinct from the debtor's fallback. The example supplies share fair value and initial transaction price as equal, with unchanged fair value at closing. A difference between initial fair value and transaction price invokes the additional conditions in AASB 9; the finite worksheet leaves it unresolved rather than automatically recognising a day-one gain (Australian Accounting Standards Board, 2026i, paras. 5.1.1–5.1.1A, B5.1.2A). Missing share value cannot be replaced by the debtor's liability value. Zero tax, fees, interest, other activity and cash legs remain supplied premises.

The matched 70, 100 and 120 examples give opposite total debtor and creditor results under their specified carrying amounts, share values and absence of other effects. That agreement does not make the accounting methods identical. It is the complete component accounts, measurement premises and dated source bindings that permit the comparison.[10]

If the new shares confer control, the creditor's noncontrolling worksheet no longer supplies the acquisition accounting. The debt settlement, consideration for the business and any pre-existing relationship must be distinguished before goodwill or consolidated statements can be measured. The next question is how to preserve that separation through the control-changing exchange.

[10]Scope truth tables and complete conditional statements are retained in `data/accounting-master/acquisition-debt-relationship-20261027/`, `data/accounting-master/debtor-conversion-worksheet-20261027/`, and `data/accounting-master/creditor-conversion-worksheet-20261027/`. The creditor plan was repaired before execution to make impairment and initial measurement explicit. Unknown premises retain the request without a report; actual source and accounting admission remain open.

26.26 Separating consideration from the other parts of an exchange

Obtaining control through a debt exchange raises a question that the separate debtor and creditor worksheets cannot answer alone: which amounts belong to the acquisition of the business? A transaction may also settle an existing relationship or pay for services. Before measuring goodwill, the company must distinguish consideration for the acquired business from effects accounted for separately. The acquisition guidance requires that separation and discusses the reasons for a transaction, who initiated it and its timing; no one indicator by itself settles the classification (Australian Accounting Standards Board, 2026h, paras. 51–53 and B50–B53).

Consider an explicitly supplied example in one currency at one measurement date. Four identified components are assigned values of 80 for acquisition consideration, 20 for separate debt settlement, 5 for acquisition costs and 3 for another separately accounted transaction. Their inventory sums to 108, while the declared consideration bucket contains only 80. The difference is not a residual to be forced into goodwill. Nor does this example infer the 20 settlement amount from a loan's face or carrying value: both the amount and its assigned purpose are supplied premises.

The finite calculation gives each component a distinct identity and retains its source, availability date, amount and assigned accounting purpose. It requires an explicit assertion that the supplied inventory is complete for the question. Reordering the four components leaves the same records and sums. An unknown settlement amount is not zero; missing allocation, coverage or source-timing premises leave the requested totals unresolved. Relabelling debt as a previously held equity interest or as acquisition consideration cannot establish the missing accounting premise.

These checks establish arithmetic conditional on the supplied partition. They do not determine the appropriate measured allocation for the intended conversion-to-control transaction. The inspected acquisition paragraphs explain separation and settlement of pre-existing relationships, but do not by themselves establish that loan-specific allocation. The remaining work is to support the relationship treatment, transaction assignments and measurement basis before constructing goodwill and acquired-basis statements.[11]

[11] `data/accounting-master/conversion-control-transaction-20261027/` retains the source inquiry, 33 finite cases including all 24 orderings of the four components, 21 refusal controls and 351 assertions. The calculation produces no goodwill, journals or consolidated report. The saved source inquiry distinguishes an irrelevant supplemental document from unavailable further support; neither is evidence that applicable guidance does not exist.

Part II

AccDSL product and implementation design

Chapter 27

The product decision

A company's accounts describe the financial consequences of its activities, but they leave much of the activity itself unobserved. Revenue combines many sales, a borrowing balance combines contracts with different terms, and a provision combines estimates about uncertain future payments. Aggregation is necessary for reporting. It also makes the inverse task—recovering economic history from published reports—fundamentally different from entering known transactions into a ledger.

AccDSL will preserve this distinction. Its first product is an evidence-backed representation of a company's historical record. Its second is a set of compatible histories and current economic states. Its third is a collection of conditional simulations, each explaining how a particular history, accounting treatment and business response produces future journals and reports. These products share identifiers and provenance, so an analyst can move from a forecast balance to the assumption or disclosure responsible for it.

The recommended design is a small, typed accounting language over a shared intermediate representation, combined with an evidence database, contract calculators and replaceable inference engines. A language earns its place by making recurring accounting decisions explicit and checkable: which entity and period an amount belongs to, whether it is known or assumed, which rule applies, and how an event changes a balanced ledger. The language does not supply missing facts or determine a company's future behaviour.

27.1 Why journals are necessary but insufficient

Consider a company with a borrowing of 100 currency units and annual interest expense of 5. Assume that the entire principal was outstanding throughout the year, interest was paid at year-end, and there were no fees, defaults, hedges or foreign-exchange effects. Both a fixed-rate loan at 5% and a floating-rate loan whose rate happened to be 5% generate the same entries:

Account	Debit	Credit
Cash, on initial borrowing	100	
Borrowing liability		100
Interest expense, at year-end	5	
Cash		5

Even the full historical journal would not distinguish these two contracts. Now impose a scenario in which the floating rate resets immediately to 7% for the whole next year. The fixed

loan produces interest of 5; the floating loan produces interest of 7. If the disclosure is also compatible with any fixed-rate fraction $\alpha \in [0, 1]$, future interest is

$$I(\alpha) = 100[0.05\alpha + 0.07(1 - \alpha)] = 7 - 2\alpha. \tag{27.1}$$

Here α denotes the fixed share. In Chapter 1 the same letter denoted the floating share. These are complementary coordinates: the fixed share here equals one minus the floating share there. Substituting that relation in $7 - 2\alpha$ gives the earlier $5 + 2\alpha$ when its α is read as the floating share. Both conventions describe interest of 5 for wholly fixed debt and 7 for wholly floating debt; a numerical share must be complemented when transferring it between these examples.

The historical accounts are compatible with the whole interval $[5, 7]$ of future interest. This is a range of feasible amounts. It becomes a probability distribution only when a distribution for α is supplied. Neither the number of generated interpretations nor a language model's confidence provides that distribution.

Journals remain essential because simulated economic events must have coherent accounting consequences. However, the simulated state must also contain contract terms, maturities, customer or borrower cohorts, collateral, unrecognised commitments, and management decisions. These determine what the journals will be under a changed economy. Reconstructing every historical transaction is unnecessary when a smaller state preserves the responses needed for the intended scenarios. Establishing that smaller state is an explicit modelling task, examined in Chapter 32.

The analogy with competing legal interpretations is useful at the level of representation. Several readings may survive the evidence and require separate reasoning. In accounting, some alternatives concern the meaning of a disclosure; others concern undisclosed facts or future behaviour. The system must identify which kind of uncertainty is involved. A known contractual term must not become an arbitrary interpretive choice merely because another choice is convenient for a simulation.

27.2 Users, questions and deliverables

An accounting analyst needs to inspect a number in its original table, verify its entity, currency and period, and see how it reconciles with the notes. A financial modeller needs to distinguish a reported balance from assumptions about its composition. A language engineer needs deterministic semantics and tests. A researcher needs to compare explanations under the same scenario, including explanations that fit history equally well.

The first software release must answer the following complete workflow. An analyst loads a set of reports and specifies an information cutoff. AccDSL preserves the original documents and extracts source-linked observations. It identifies reconciliation failures and constructs an explicitly bounded family of compatible histories. The analyst declares additional assumptions where simulation requires them. The system executes each requested scenario and returns statements, journal traces, explanation-specific distributions or bounds, and unresolved limitations. When a later filing arrives, the analyst can compare the revised interpretation with the earlier information set.

Acceptance of this design document is a separate milestone. It requires an implementable specification, adequate source and conformance evidence, and independent reader, implementation and domain review. It does not establish that the software workflow has been delivered or that forecasts are validated. The implementation sequence is described in Chapter 40.

Requirement	Required behaviour
R1 Evidence	Every observed amount or narrative assertion retains its document version, location, extraction method and original representation.
R2 History	Queries distinguish economic time, reporting period, publication time and database ingestion time; restatements remain recoverable.
R3 Meaning	Entity, perimeter, dimensions, currency, measurement basis, standard version and accounting concept accompany values.
R4 Ambiguity	Discrete alternatives and continuous unknowns retain their joint constraints across accounts and periods. Missing values never mean zero.
R5 Accounting	All committed journals balance in the book's functional currency, and balances reconcile to explicitly classified movements.
R6 Economics	Contracts, exposures and behavioural assumptions drive events. Accounting identities alone never determine scenario responses.
R7 Results	Conditional results survive by interpretation and scenario. Weighted results require declared, normalised probability measures.
R8 Explanation	A report cell can be traced to journals, events, rules, assumptions and evidence, including consolidation adjustments.
R9 Extensibility	Standard, industry and company packages extend shared types through versioned interfaces, with explicit unsupported cases.
R10 Reproduction	A saved run records source versions, model versions, parameters, calendars, seeds and numerical conventions.

These are acceptance requirements. Python, SQL storage, a particular parser, and a particular constraint solver are implementation choices. Initial support will cover a non-financial company with cash, receivables, payables, inventory, fixed assets, ordinary debt and ordinary equity. Banks and insurers must be representable in the core data model from the outset, but their executable measurement and behavioural packages follow dedicated milestones. Until a package supports an instrument, the result is explicitly incomplete.

27.3 What constitutes a useful first implementation

The first demonstrator should reconstruct three reporting dates and the intervening annual and interim flows for one company, including at least one restatement or revised comparative. It must preserve a borrowing-composition ambiguity and execute the fixed/floating example through both journals and all affected statement lines. A synthetic company supplies known hidden history; a public company supplies realistic disclosure and extraction problems. Their roles differ: the synthetic case tests preservation and rejection of known alternatives, while the public case tests traceability and practical coverage.

Success is a reproducible chain of evidence and computation, with identified gaps. A convincing-looking complete ledger reconstructed from incomplete disclosures would fail this product contract if its assumptions were hidden. Forecast accuracy requires a later evaluation with historical information cutoffs and suitable baselines; it is not established by the first demonstrator.

Chapter 28

A complete company and a real disclosure

28.1 What a balance must remember

Northstar Components is a synthetic seller of one homogeneous good. Its disclosures, transactions and contract terms are specified together, so we can check whether reconstruction preserves a known possible history. All amounts in this example are currency units, not thousands. The company has no foreign currency, subsidiaries, leases, returns or other comprehensive income. These are declared features of the example, not assumptions to impose on an imported company. Its source package and expected results are supplied in `fixtures/northstar.json`.

At the beginning of 2024, Northstar holds cash of 60, receivables of 20, inventory of 30 and plant costing 100 with accumulated depreciation of 20. It owes suppliers 20 and lenders 100. Share capital is 50 and retained earnings are 20. Thus assets of 190 equal liabilities of 120 plus equity of 70. Inventory consists of six units costing 5 each. The existing plant depreciates by 10 a year, with no residual value. The loan remains outstanding throughout 2024; both possible contract types charge 5% during that year.

The six months ending June and December provide two successive observation periods. Here the chosen accrual convention assigns one half-year to each. Cash-flow dates within a half-year are not recovered from a period aggregate; the synthetic history supplies a feasible ordering in which inventory is purchased before delivery and customer receipts precede the payments they fund. More detailed liquidity questions require more detailed dates.

Movement	First half	Second half
Delivered sales on credit	110	150
Customer collections	110	135
Inventory purchases on credit	70	85
Supplier payments	65	80
Cost of goods delivered	75	90
Payroll paid	20	22
Plant purchased at period end	10	12
Depreciation	5	6
Interest accrued and paid	2.5	2.5
Tax expense	1.5	5.9
Tax paid	3.5	3.9
Dividends paid	4	5

Each row has an economic meaning before it has a posting. Delivery of the good transfers control and creates an unconditional receivable; invoicing is assumed simultaneous in this example. Collection settles that receivable. Purchases increase inventory and trade payables; paying the supplier settles the payable. Delivery separately removes inventory at cost. Plant purchases are capital expenditure, while depreciation consumes its service potential. The addition of 10 enters service after June and has a five-year life, producing one unit of second-half depreciation. The addition of 12 enters service after December and has a twelve-year life.

Tax is deliberately simple: 20% of positive profit before tax, with all these expenses deductible and no deferred tax. The first-half payment exceeds its tax expense by 2, leaving prepaid tax; that prepayment is used in the second half. A genuine corporate tax package would need jurisdiction, tax basis, losses, timing differences and payment schedules. The example's tax rule governs its own numbers only.

28.2 Recovering movements without inventing invoices

Suppose the annual report discloses sales of 260, cost of sales of 165 and the opening and closing working-capital balances, while the collection and purchase totals are hidden. Receivables rise from 20 to 35, inventory falls from 30 to 20, and trade payables rise from 20 to 30. Let C denote collections, P purchases and S supplier payments, all in currency units over the same year. Under Northstar's explicit no-write-off, no-acquisition and no-currency-effect scope, the three bridges are

$$35 = 20 + 260 - C, \tag{28.1}$$

$$20 = 30 + P - 165, \tag{28.2}$$

$$30 = 20 + P - S. \tag{28.3}$$

Substitution gives $C = 245$, $P = 155$ and $S = 145$. These are identifiable aggregate movements within this model. They do not identify which invoice was paid on which day. If acquisitions or write-offs are admitted, their movement variables enter the same bridges and these totals may cease to be unique.

The loan remains different. Write F and V for fixed and floating principal. The source imposes $F + V = 100$, $F, V \geq 0$ and $0.05F + 0.05V = 5$. The last equality adds no information about the split. Both endpoints and every intermediate allocation survive. Northstar therefore has identified working-capital totals and an unidentified financing exposure in the same joint historical family. A solver must not turn its convenient witness $F = 0$ into the company's recovered contract.

The model's historical slots are one delivery, collection, purchase, supplier settlement, payroll, plant addition, depreciation, interest, tax payment and dividend per half-year. Each slot represents a classified aggregate event. The interim filing supplies the information for allocating the annual movements between the two half-years; the annual bridges alone identify only their totals. The template for collection contributes $+C$ to cash and $-C$ to receivables; its state effect reduces the unpaid receivable pool by C. The purchase template adds inventory quantity $P/5$, inventory cost P and a payable P. A cash-balance identity by itself cannot substitute for either template.

28.3 The complete historical statements

The corrected statements follow from those movements. Temporary income and expense accounts close to retained earnings at each half-year boundary; annual income is the sum of the two

periods' classified movements, excluding closing entries.

Balance	Opening	June	December
Cash	60	65	74.6
Receivables	20	20	35
Inventory	30	25	20
Plant, net	80	85	91
Prepaid tax	0	2	0
Total assets	190	197	220.6
Trade payables	20	25	30
Debt	100	100	100
Share capital	50	50	50
Retained earnings	20	22	40.6
Liabilities and equity	190	197	220.6

Annual revenue of 260 less cost of sales 165, payroll 42, depreciation 11 and interest 5 gives profit before tax of 37. Tax expense is 7.4 and profit is 29.6. Other comprehensive income is zero under the declared scope, so total comprehensive income is also 29.6. Equity moves from 70 to $70 + 29.6 - 9 = 90.6$ after dividends.

The cash-flow statement supplies an independent reconciliation. Collections 245 less supplier payments 145, payroll 42, interest 5 and tax payments 7.4 produce operating cash of 45.6. Investing cash is -22 and financing cash is -9, giving a cash increase of 14.6. The indirect route gives the same operating cash: profit 29.6 plus depreciation 11, minus the receivables increase 15, plus the inventory decrease 10 and the payables increase 10. The prepaid-tax balance is zero at both annual endpoints. The two cash routes use different information and therefore provide a useful check on classification.

28.4 A correction changes what was knowable

The original interim filing reported revenue of 120 and receivables of 30. Ten units of revenue had been recorded before delivery. Its tax expense was 3.5, profit 14 and equity 80. The annual filing subsequently corrected the interim revenue and receivable by 10 and recognised the excess tax payment as prepaid tax of 2. Corrected interim profit is 6 and equity is 72, as above. The original statement remains an accurate record of what was published, even though its revenue recognition was wrong under the stated delivery rule.

For the temporal fixture, the original report becomes available on 1 August 2024 and is ingested on 3 August. The correction and annual report become available on 2 March 2025 and are ingested on 5 March. An investor-information query at 1 September 2024 00:00 UTC returns 120. For this synthetic example, take UTC as the reporting timezone and use an instant cutoff of 3 March 2025 at 00:00 UTC, after the correction's entire publication-date interval has ended. That query returns 110 when evaluated using the subsequently acquired archive. A query that additionally asks what the database contained on 4 March still returns 120. Publication and ingestion answer different questions.

The annual total 260 and corrected interim total 110 share revision group `v2`, so second-half revenue is $260 - 110 = 150$. Subtracting the original 120 would yield 140, but that result mixes incompatible revisions. The query must reject the subtraction unless a reviewed compatibility relationship supports it. A later date alone does not establish such a relationship.

28.5 One economic scenario, several complete futures

For 2025, assume demand for 30 units at price 10, unit purchase cost 5 and a target closing inventory of eight units. Opening inventory is four units, so the purchase decision is $30+8-4=34$ units, costing 170. Deliveries cost 150, leaving inventory of 40. Collect 90% of current sales and 10 of old receivables: $270+10=280$, leaving receivables of 55. Pay old suppliers 30 and 130 of the new purchases, leaving payables of 40. Payroll is 50, capital expenditure of 20 occurs at year-end and dividends are 8.

These decisions are inputs to a conditional forecast. They are not estimates of demand or optimal management responses. No new borrowing is available. Under the existing asset schedules, depreciation is $10+10/5+12/12=13$. The new year-end asset begins depreciation in the next period. Let $\alpha = F/100$. With the floating rate resetting to 7%, interest is $I = 7-2\alpha$. Profit before tax is $300-150-50-13-I = 80+2\alpha$, giving tax $16+0.4\alpha$ and profit $64+1.6\alpha$.

2025 statement amount	All fixed	All floating
Revenue	300	300
Interest expense	5	7
Tax expense	16.4	16
Profit and comprehensive income	65.6	64
Operating cash flow	48.6	47
Investing cash flow	−20	−20
Financing cash flow	−8	−8
Closing cash	95.2	93.6
Receivables / inventory / net plant	55 / 40 / 98	55 / 40 / 98
Assets	288.2	286.6
Payables / debt	40 / 100	40 / 100
Equity	148.2	146.6

The joint symbolic result is cash $93.6+1.6\alpha$, profit $64+1.6\alpha$ and equity $146.6+1.6\alpha$, with the same $\alpha \in [0,1]$ throughout. Selecting the smallest cash and largest equity independently would describe no feasible company. A requested distribution requires a measure over that same historical family; the accounting equations do not supply one.

The later method schedules use the floating share x rather than the fixed share α used here: $x = 1-\alpha$. Both parameterisations describe the same joint family. Its 2025 opening cash is 74.6 and equity is 90.6; the forecast spends 20 on period-end capital expenditure, distributes 8 and settles its stipulated tax in the same year. These teaching assumptions are retained when the later lease example introduces a different tax-payment schedule.

28.6 Keeping the same company through every calculation

The first inventory purchase shows what a reproducible company calculation must retain. Before the purchase, Northstar has six units costing 30 and owes suppliers 20. It buys fourteen more units for 70 on credit. The journal debits inventory and credits trade payables by 70; the resulting inventory is twenty units costing 100, payables are 90, and cash remains 60. A balanced journal accompanied by twenty-one units of inventory is wrong under the stipulated unit cost of 5. The quantity change and the monetary posting must therefore be checked against the same purchase and opening state.

Delivery follows purchase. Its four journal lines recognise the receivable and revenue together with cost of sales and the reduction in inventory. For the first half-year, delivery of fifteen units leaves five units costing 25. Separating the revenue and inventory effects into independently committable events could expose a state in which the customer has the goods but the seller still reports them. Treating delivery as one economic event prevents that intermediate inconsistency. Collections then precede the payments they fund. This order establishes a feasible aggregate history; it does not recover the dates of individual invoices.

The portable company example retains twenty-two successive states and journal steps: ten economic events and a closing entry in each half-year. Each step identifies its prior state, economic inputs, accounting template and resulting state. The corrected June statements and December statements refer to this same chain. The December closing state, with cash 74.6 and equity 90.6, is also the explicit opening of the 2025 projection. Consequently, changing a historical purchase or tax payment cannot leave the future opening silently unchanged. These are proposed histories; publishing them as candidates does not post them to an accounting book.

The rate comparison checks a different connection. Recall that x is the floating share of the unchanged principal of 100. For an imposed floating rate r, the fixed part earns interest at 5%, so annual interest is

$$I(x; r) = 100(1 - x)0.05 + 100xr = 5 + (100r - 5)x. \tag{28.4}$$

At 7%, increasing x from zero to one adds 2 to interest; at 9%, it adds 4. Under the stipulated 20% tax rule, profit and cash therefore fall by 1.6 or 3.2 respectively. The second rate supplies a useful control: two authoring frontends might agree because they share an erroneous constant. Independent arithmetic must recover the changed slope from the changed rate. The reference exchanges retain the exact expression for every $x \in [0, 1]$, including its dependence across cash, profit, assets and equity. They attach no probabilities to the endpoints or to intermediate debt allocations.

The opening also determines what kind of forecast this is. The corrected December and interim information becomes public in March 2025, after the January origin of the scenario. Using it is an archival conditional exercise: given that corrected opening, what follows from these imposed decisions and rates? It cannot establish what an investor could have forecast in January. An investor-information request at that earlier cutoff rejects the later disclosure dependency. Separately, a management policy cannot read a future realised collection merely because the simulation scheduler already stores it. An imposed scenario assumption and a later observed outcome have different information meanings.

Partial information must survive the same chain. If the fraction of current sales collected is missing, delivery can still determine revenue and cost of sales. Cash, receivables and the ability to fund the proposed settlements remain unresolved. The partial report identifies its affected lines and retains the full debt family. If the tax method is unavailable, profit after tax, retained earnings and the final closing entry are unresolved as well. A zero in a temporary account would incorrectly suggest that closing had been completed. The portable values therefore carry an explicit reason and its outstanding obligation wherever the calculation is unavailable.

There is a similar distinction in reading a note. The separate authored phrase “substantially all customers pay promptly” does not imply a 90% threshold. An analyst may impose an all-payers assumption for a conditional calculation, but its decision must preserve the unquantified source reading. Likewise, interpreting a displayed 100 as nearest-unit rounding requires an explicit rounding assumption. With ties rounded upward, that assumption permits values in $[99.5, 100.5)$; absent the assumption, the display alone supplies no such interval. Neither decision grants permission to post a journal.

Finally, a correct purchase must survive an uncertain acknowledgement. The request identifies Northstar, its book and branch, and the particular purchase; it also fixes the opening state, expected revision and all input versions. Suppose the client sends the request but receives no response. If the store committed nothing, an exact retry may commit the purchase. If the store already committed it, the same retry returns the original receipt without another purchase. The client cannot distinguish these possibilities from a timeout.

Receipt lookup therefore precedes comparison with today's head, after the original request has passed its input and authorization checks. A later, separately identified purchase may have advanced the book, but it does not erase the first receipt. Replaying the first request still returns that receipt. Changing its request identifier or budget changes its full fingerprint and conflicts with the already recorded business event. A new event presented against an obsolete opening is instead rejected as stale. Every rejection leaves the committed states, journals, quantity changes and receipt index unchanged.

Review decisions, saved search progress and imports have their own owners and receipts. Correcting the reported revenue invalidates dependent current uses, including a selection that depended on the old set of eligible readings; it does not rewrite committed purchases. Saving one searched region of $[0, 1]$ preserves the other region and the next position to visit. Resuming with a larger budget updates that saved work, while a changed model requires a different calculation. These distinctions make the complete company example reconstructible from its inputs and record references. The operation-specific fields and exact documentary exchanges are described in Section A.1; they still require implementation and actual fault testing before any claim of database durability.

28.7 One company close across several accounting methods

Separate examples can each balance while disagreeing about the cash they share. A return settlement may draw on the same bank account that must redeem a bond; an internal sale may increase a parent's earnings without increasing the group's earnings. To see these interactions, consider a second synthetic company: a parent, P, and its subsidiary, S. P owns 80% of S at the beginning of 2025 and controls it throughout the year. Both use USD, the same calendar and the same accounting policies. All amounts are currency units. The opening balances are supplied facts; the example begins after acquisition and debt issuance.

P opens with cash 500, inventory 60, plant 100, a return-recovery asset 16, an investment in S of 80 and a deferred tax asset of 10. It owes a refund liability of 40, matured debt of 100 and a deferred tax liability of 4. Its capital 600, original compound-instrument equity 10 and retained earnings 12 give equity 622. S opens with cash and capital of 100. Eliminating P's investment against its share of S's opening capital leaves non-controlling interests (NCI) of 20. The group's opening assets are 786, liabilities 144 and equity 642.

This case has a deliberately narrow tax law: a supplied enacted rate of 25%, deductible wages, matching book and tax depreciation, and the return timing specified below. Recovery of the stated deferred tax assets is supported by a supplied finding. P controls the timing of reversal of taxable outside-basis investment differences, with no reversal expected in the foreseeable future, so the IAS 12.39 exception applies conditionally. These are teaching premises, not a statement of any jurisdiction's current law. The recognition, consolidation and cash-flow rules use the retained historical standard text (European Commission, 2023a, IAS 12.24,39,46–51; IFRS 10.B86–B94).

28.7.1 Cash paid and expense incurred answer different questions

The bond's original liability proceeds were 90 and its original equity component was 10. Accrued interest of 10 has already increased the opening liability to 100. Redemption on 2 January therefore debits debt 100 and credits cash 100. It creates no new 2025 interest expense. Under this company's historical IAS 7 operating election, the payment is financing principal 90 and operating interest 10. Interest paid must still be disclosed separately even when the associated expense belongs to another period (European Commission, 2023a, IAS 7.31–33).

A component adapter must consequently preserve the original debt cash basis and the accumulated liability. Carrying amount alone cannot reconstruct the cash-flow split. Nor can an event labelled "accrue" simply be moved across a year boundary: the opening states whether its expense has already been recorded. Here P's cash falls to 400, the debt becomes zero and original equity remains 10.

The return liability comes from a previous ten-unit sale. Each unit originally cost 12 and can be refunded for 20. Expected returns were two units; estimated handling cost 3 and deterioration 1 per unit gave a recovery asset $2(12 - 3 - 1) = 16$ and refund liability $2(20) = 40$. Two units actually return. Handling is 4 per unit, deterioration is 2 and the returned stock's supplied net realisable value is 20. All return rights are now closed.

The stock recovered is $\min\{2(12 - 2), 20\} = 20$. P pays refunds 40 and handling 8, releases the refund liability 40 and recovery asset 16, and records inventory 20. Because the actual handling is explicitly a selling expense, the remaining entries are a cost-of-sales credit of 4 and selling expense 8. Their net expense is 4; there is no revenue adjustment when actual refunds equal the liability. This separates the recovery estimate from the cash actually spent and avoids charging handling twice (European Commission, 2023a, IFRS 15.B21–B27; IAS 2.16).

Tax timing supplies the next link. The original sale was already taxable, and its original inventory cost was deducted. On return, tax deducts the refund 40, restores recovered inventory 20 to tax stock and deducts handling 8. Thus the return reduces taxable income by 28, compared with book expense 4. The difference is 24. Opening deferred tax asset 10 and liability 4 reverse, giving deferred tax expense 6. A zero-tax return example would miss this interaction.

28.7.2 The subsidiary's inventory contains the parent's margin

During the first half P also pays wages 10 and records depreciation 10. It then sells its other inventory, which cost 60, to S for 140 on credit. P recognises revenue 140 and cost of sales 60; S records inventory and a payable of 140. The legal entries are necessary: P and S have different owners' claims and separate obligations. But the group has not yet sold that inventory to anyone outside the group.

S sells two-fifths of the lot externally for 84, recording legal cost of sales 56, and pays P the full internal invoice of 140. The diagram distinguishes the two sales. Only the second crosses the group boundary.

After the external sale, S carries three-fifths of its purchase price, or 84. The same physical stock cost the group only $3(60)/5 = 36$. Its unrealised internal profit is therefore $84 - 36 = 48$. IFRS 10.B86(c) requires full elimination of intragroup amounts and of profit still contained in assets; it also directs the resulting temporary differences to IAS 12.

For clarity, let T be the internal price, C the group's original cost and s the fraction sold externally. The remaining internal profit is

$$U = (T - C)(1 - s) = (140 - 60)(1 - 2/5) = 48. \tag{28.5}$$

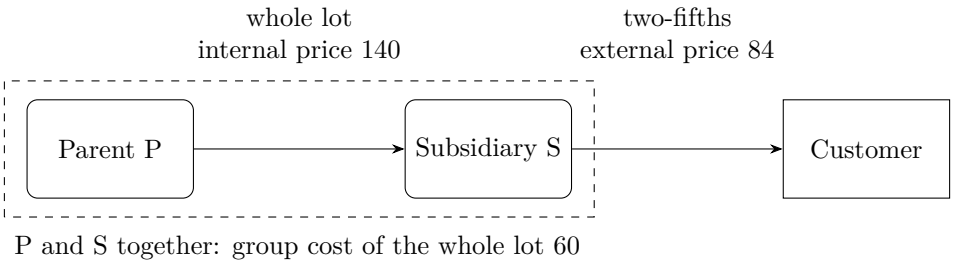


Figure 28.1: The internal price changes the legal books; the external sale determines group revenue.

The group removes internal revenue 140 and reduces the combined legal cost of sales by $T - U = 92$. S's tax base remains 84, while group carrying amount is 36. The deductible difference 48 gives a deferred tax asset 12 at the supplied 25% rate. The after-tax elimination is consequently 36. If all the stock were sold outside the group, $s = 1$, both the remaining profit and this deferred tax asset would be zero.

P's first-half pretax profit is $80 - 4 - 10 - 10 = 56$. The return timing difference reduces taxable income to 32, giving current tax 8; deferred tax expense 6 leaves P profit 42. S earns 28 before tax and 21 after current tax 7. Both current-tax amounts are paid. Group profit is therefore

$$\Pi_{H1} = 42 + 21 - 48 + 12 = 27. \tag{28.6}$$

The downstream margin originated in P's sale. Its after-tax elimination belongs to P's shareholders; NCI receives 20% of S's legal profit 21, or 4.2. At June the group has cash 511, inventory 56, plant 90 and deferred tax asset 12, totalling 669. It has no remaining external liability in this case. Equity is parent 644.8 plus NCI 24.2. These totals reconcile the same shared legal books used by the return and debt methods.

28.7.3 Buying more shares changes who owns the equity

On 1 July P pays 20 to the outside shareholders for another 10% of S. P now owns 90%, and control is unchanged. In P's separate books, its cost-policy investment increases by 20 and cash decreases by 20. S records nothing: the payment was between its shareholders. The group's accounting must instead describe a change in the relative claims of P's owners and NCI.

Immediately before purchase, S's equity is 121 and NCI owns 20%, carrying 24.2. P buys half of that interest. The NCI reduction is 12.1, not the cash price 20. IFRS 10.23 and B96 put the difference directly in parent equity:

$$\Delta NCI = -(0.90 - 0.80)121 = -12.1, \tag{28.7}$$

$$\Delta E_P = -(20 - 12.1) = -7.9. \tag{28.8}$$

No gain, goodwill or remeasurement is introduced. IAS 7.42A–42B classifies the group cash payment as financing, while the parent's separate purchase of an investment is investing under its own cash-flow presentation (European Commission, 2023a, IFRS 10.23,B96; IAS 7.16,42A–42B; IAS 27.10).

The purchase date matters for later earnings. First-half NCI profit remains 4.2; it is not recomputed at 10%. The new fraction applies to subsequent profit. Recording only an ending ownership percentage would lose that history and allocate the year's earnings incorrectly.

28.7.4 The external sale releases both margin and tax

In the second half S sells the remaining lot for 126. Its legal cost is 84, so legal pretax profit is 42 and tax is 10.5. The group's cost is only 36. The earlier elimination of 48 now reverses, and the associated deferred tax asset 12 is consumed. P separately sells its recovered inventory for 40 at cost 20 and records a further depreciation charge 10. Its second-half pretax profit is 10 and tax is 2.5. All current taxes are paid.

The resulting group profit is

$$\Pi_{H2} = (10 - 2.5) + (42 - 10.5) + 48 - 12 = 75. \tag{28.9}$$

NCI receives 10% of S's after-tax profit 31.5, or 3.15. The group release again belongs to P's shareholders because it reverses P's downstream margin. Annual NCI profit is $4.2 + 3.15 = 7.35$, and annual parent profit is $102 - 7.35 = 94.65$. Closing equity can now be reconstructed without using an asset total as a balancing plug:

$$E_P^{close} = 622 + 94.65 - 7.9 = 708.75, \tag{28.10}$$

$$NCI^{close} = 20 + 7.35 - 12.1 = 15.25. \tag{28.11}$$

Their sum is 724, equal to closing cash 644 and plant 80. The inventory has been sold, the return and debt obligations settled, and the temporary differences in this case reversed.

The comparison below follows the derivation rather than replacing it. The balance-sheet columns are closing amounts; income and cash columns are flows over the stated interval.

	First half	Second half	Full year
External revenue	84	166	250
Cost of sales	20	56	76
Profit before tax	36	100	136
Current tax expense	15	13	28
Deferred tax expense/(benefit)	-6	12	6
Profit after tax	27	75	102
NCI share of profit	4.2	3.15	7.35
Operating cash flow	1	153	154
Financing cash flow	-90	-20	-110
Closing cash	511	644	644
Closing group equity	669	724	724

Cash provides an independent check on the joins. External receipts are $84 + 126 + 40 = 250$. Operating payments are refunds 40, handling 8, wages 10, interest 10 and tax 28, totalling 96. Financing payments are debt principal 90 and the NCI purchase 20. Thus

$$C^{close} = 600 + (250 - 96) - (90 + 20) = 644. \tag{28.12}$$

The internal receipt 140 and payment 140 are removed from their respective gross totals. Merely observing that they net to zero would overstate both group receipts and payments. Group adjustments never alter either entity's legal cash ledger.

28.7.5 The implementation consequence of a joined close

Each method receives the current shared state, not its own duplicate cash balance. The internal sale and settlement each update both legal books as one transaction; an unmatched invoice or

one-sided publication is refused. A report must link an uninterrupted receipt sequence from its opening state to its closing state. Consolidation recomputes the required worksheet adjustment from legal balances and lot history at each date. It does not append the June elimination again in December.

The companion specification, `CompanyComposition/1`, fixes these inputs, events and outputs in closed records. An independent calculation reconstructs the legal and group reports, including the adverse cases of balanced but wrong journals, missing tax findings, changed ownership identities and partial publication. Its two narrow migration examples preserve the original source owners and dates; they do not rename a previous synthetic company into P. Section A.27 gives the exchange details.

This joined case establishes the stated calculation and record design for the supplied opening. It does not reconstruct the prior acquisition or issuance, nor does a corporate group with no bank or insurance business specify those sectors' combined statements. Foreign currency, wider ownership rights, actual tax authority and the remaining accounting methods retain their own required evidence. Those limits identify what the next case must add without obscuring what the shared company close already demonstrates.

28.7.6 Carrying the company into its next fiscal periods

The next period begins with the company's existing assets, obligations and owners. The parent closes its 2025 profit of 49.5 into retained earnings, raising that balance from 12 to 61.5; the subsidiary closes its profit of 52.5 into retained earnings of 52.5. The parent still holds cash 491.5, plant 80 and investment 100, financed by equity 671.5. The subsidiary has cash and equity of 152.5. Eliminating the investment gives group equity 724, of which 708.75 belongs to the parent and 15.25 to non-controlling interests. Closing the income accounts does not earn the 2025 profit a second time.

Consider a hypothetical continuation under the same historical accounting principles and a supplied tax law. The companies change their reporting year-end to June to align with their operating cycle. A six-month period from 1 January to 30 June 2026 is followed by two July–June years. The source requires the reason for the short period and a warning that its amounts are not entirely comparable with a full year (European Commission, 2023a, IAS 1.36–38). These scenario dates do not establish the standards or tax law that will actually govern a future reporting period. Control, the parent's 90% interest, common policies and the outside-basis tax exception are expressly renewed for each period. The earlier control finding ended at the start of this continuation.

In the short period the parent receives revenue of 20, pays operating costs of 110 and depreciates plant by 10. Book and tax depreciation are equal, so both its accounting loss and its tax loss are 100. Cash is 401.5 and plant is 70. The supplied law permits the parent to carry this loss forward until 30 June 2028, but provides no carryback or transfer to the subsidiary. A deduction that may be used later is not automatically an asset today. IAS 12 requires probable taxable profit against which the loss can be used; recent losses require convincing other evidence or suitable taxable temporary differences (European Commission, 2023a, IAS 12.34–37).

Suppose convincing evidence supports 40 of taxable profit in the next period, and the enacted rate expected on recovery is 25%. Recognise a deferred tax asset of $40 \times 25\% = 10$, leaving 60 of the loss without a recognised asset. The parent's after-tax loss is 90. The subsidiary separately earns 20 in cash and incurs tax of 5, which remains unpaid. Its equity rises to 167.5. Group cash is 574, plant 70 and deferred tax assets 10, against a current tax liability of 5:

$$574 + 70 + 10 - 5 = 649 = 724 - 75.$$

The group's loss is 75, comprising the parent's loss of 90 and the subsidiary's after-tax profit of 15. Non-controlling interests receive 1.5 of that profit. The loss attributable to the parent's shareholders is therefore 76.5. With ten unchanged ordinary shares and no potential shares, basic and diluted earnings per share are both $-76.5/10 = -7.65$. This is the result for the six months; doubling it would invent a full-year result.

At the start of the next period, a newly enacted law changes the current and recovery rates to 30%. The previously recognised 40 of recoverable loss is now worth 12, so remeasurement adds an asset and tax benefit of 2. This is a rate effect on the existing supported quantity. It does not establish that more of the original loss is recoverable. Both rate and quantity must be traced separately (European Commission, 2023a, IAS 12.47,56,60,79–82).

The parent then receives revenue of 100, pays costs of 20 and depreciates plant by 20, earning pretax profit of 60. All 60 is covered by the available tax loss, leaving no current tax and an unused loss of 40. Of the 60 used, 40 had supported the opening asset; consuming it reverses the remeasured asset of 12. The other 20 had no recognised asset to reverse, although its use now saves current tax of $20 \times 30\% = 6$. At the new close, evidence supports 30 of the remaining 40 for next-period use, producing a new asset of 9. The period's deferred tax expense is

$$\underbrace{-2}_{\text{rate benefit}} + \underbrace{12}_{\text{previously recognised loss used}} - \underbrace{9}_{\text{remaining loss now supported}} = 1.$$

Equivalently, the asset falls from the opening 10 to the closing 9. The decomposition explains that change instead of concealing the rate benefit inside a net number. The parent's after-tax profit is 59.

The subsidiary again earns 20, incurs tax of 6 and pays the previous liability of 5. Its closing cash is 187.5 and its current liability is 6. Paying the old liability creates no new tax expense. The parent closes with cash 481.5, plant 50 and the deferred asset of 9, as well as its unchanged investment. After consolidation, group equity is

$$669 + 50 + 9 - 6 = 722 = 649 + 73.$$

The parent's shareholders receive profit $59 + 0.9(14) = 71.6$, while non-controlling interests receive 1.4. Earnings per share are 7.16. The first period's asset of 10 and loss of 90 remain in its comparative figures; later information changes the current estimate, not the old report (European Commission, 2023a, IAS 8.34–38).

In the last year the parent receives 60, pays costs of 20 and depreciates plant by 20. Pretax profit of 20 uses another 20 of the loss. The remaining 20 expires at year-end. Of the asset of 9 brought forward, 6 is consumed by the deduction actually used and 3 is lost when the remaining supported deduction expires. Deferred tax expense is 9, leaving parent profit of 11. The subsidiary earns 20, incurs tax of 6 and pays both that amount and its opening liability of 6. Group profit is 25; 23.6 belongs to the parent's shareholders, giving earnings per share of 2.36. Cash 717 and plant 30 are now the group's only assets, with equity 747 and no tax liability or loss asset. The unused loss of 20 remains in the history as an expired right, not as an unreported zero.

The complete cash bridge provides a separate check on the three closes. Receipts of 240 less operating cash costs of 150 and tax payments of 17 increase cash by 73, from 644 to 717. Depreciation of 50 explains why cumulative profit is only 23. That profit increases equity from 724 to 747. The parent's retained earnings finish at 41.5 and the subsidiary's at 95.5. At group level, parent retained earnings are 125.35 and the earlier ownership transaction remains a separate equity reduction of 7.9. Adding these to parent capital and original compound equity of 610 gives parent equity 727.45; non-controlling interests hold the remaining 19.55.

If the earlier evidence had instead supported no probable recovery, the deferred asset would have been zero. The same actual deductions would still be available under the supplied law. Group equity would have been 639, then 713, then 747; period profit would have been −85, 74 and 34. Recognition changes when the benefit enters profit, while the realised tax payments and terminal equity coincide in this example. Missing evidence is a different situation: it leaves the recovery assessment unresolved.

Tax presentation preserves the same legal distinctions. The parent's deferred asset cannot be netted against the subsidiary's current liability. Current tax offset requires an enforceable right and an intention to settle net or simultaneously. Deferred tax offset also requires the same authority and either the same taxable entity or the specified settlement intention across the relevant future periods (European Commission, 2023a, IAS 12.71–76). Consolidation alone supplies none of these conditions. Each close therefore carries the loss's owner, authority, expiry, supported recovery and rate evidence alongside the financial balances. Multiple loss vintages, different recovery-rate schedules and tax arising in other comprehensive income or equity require those additional origins before this successor can represent them.

28.8 A bank and insurer share a group close

A bank's loan receipts, an insurer's claims and a parent's guarantee can all affect the same group's cash without belonging to the same legal taxpayer or the same shareholders. Consider a parent P that owns 80% of bank B and insurer I. Control and equal ordinary ownership rights are supplied facts. P reports in USD; both subsidiaries keep their books in a foreign currency, FCU. They apply the same retained historical EU accounting profile. Their opening is 1 January 2025; the two intervals close on 1 January 2026 and 1 January 2027. Events described as first- or second-period events occur at those respective closing dates. This explicit calendar also fixes which ownership fraction receives each period's profit.

P begins with cash 500, investments 80 in each subsidiary and an external financial-guarantee liability 6, so its equity is 654. B has cash 300, customer deposits 200 and equity 100. I has cash 200, insurance liability 100 and equity 100. Of I's cash, 20 is a demand deposit at B; the same deposit is included in B's liability 200. The deposit is identified by its creditor I, debtor B, currency, amount and matching closing dates. No interest or movement occurs in it during these two periods. Opening FX is 1 USD per FCU.

The group therefore starts with cash $500+300+200-20 = 980$, liabilities $200+100+6-20 = 286$, and equity 694. Eliminating the parent's investments leaves NCI $20 + 20 = 40$, alongside the parent's equity 654. The two component calculators' notional funding amounts do not become new legal capital. Only their identified loan and insurance movements enter B's and I's books. Adding the isolated component balance sheets would answer a different question.

28.8.1 Three taxpayers, three different profit origins

At opening B advances 102 and receives an integral loan fee 2. The loan's initial carrying amount is 100. Its receipts are 55 at the first close and 60.5 at the second. At the supported effective interest rate 10%,

$$L_1 = 100(1.10) - 55 = 55, \tag{28.13}$$

$$L_2 = 55(1.10) - 60.5 = 0. \tag{28.14}$$

Interest income is 10 and 5.5. The contractual cash record separately identifies principal receipts 50 and 52, and coupons 5 and 8.5. Thus principal receipts return the 102 advanced, while

coupons 13.5 plus the integral fee 2 equal lifetime effective-interest income 15.5. The first coupon is 5, although that period's effective-interest income is 10. Cash classification cannot be recovered by renaming the interest-income line. The admitted bank policy classifies these lending flows as operating and preserves their separate gross origins (European Commission, 2023a, IAS 7.14–15,31–33; IFRS 9.B 5.4.1–B 5.4.4).

I uses a supplied loss-component service schedule. Its opening liability 100 incurs finance expense 10 and releases claims/risk adjustment 55, closing at 55; the second finance expense is -5, followed by releases 50, closing at zero. Actual claims paid are 50 and 45. Insurance profit before tax is -5 and 10. Neither the loss component nor the risk-adjustment release is another cash payment.

For P, the guarantee beneficiary is an unrelated external creditor. A supported scope finding places this contract under IFRS 9. Its opening recognized amount is 6. Supported service income is 2 in each period, while independently supplied expected-credit-loss measurements are 7 and 3. Let U_t be the unamortized initial amount and A_t that allowance. Subsequent liability is

$$G_t = \max\{U_t, A_t\}, \qquad U_1 = 4, \quad U_2 = 2. \tag{28.15}$$

This is the higher-of rule in IFRS 9.4.2.1(c), conditional on the stated classification and measurements (European Commission, 2023a). It gives liabilities 7 and 3. The excess above U_t is 3 and 1, respectively. Its changes, 3 and -2, are impairment expense and reversal. Subtracting them from service income 2 gives guarantee profit -1 and 4. No guarantee settlement occurs during these periods, and consolidation does not remove an external liability.

Tax now connects these movements to legal equity. The supplied law taxes book profit at 20%, including guarantee changes, with usable loss carryforwards. There is separate convincing evidence supporting the losses' recovery; a future book profit alone is not that evidence (European Commission, 2023a, IAS 12.34–36,46–47). There is no loss pooling between P, B and I. Outside-basis exceptions and absence of tax on translation are also explicitly supported premises, not conclusions about an actual jurisdiction.

For each taxpayer let ℓ denote the opening unused tax loss and p the period's pretax profit. Under this finite supplied law, loss used is $u = \min\{\ell, \max(p, 0)\}$, current tax is $0.2[\max(p, 0) - u]$, and closing loss is $\ell - u + \max(-p, 0)$. The recognized DTA is 20% of that loss. Deferred tax expense equals opening DTA minus closing DTA; hence the tax benefit on a new supported loss is negative expense, without a current cash refund.

B pays tax 2 and 1.1 and earns after-tax profit 8 and 4.4. I's first loss 5 creates DTA 1 and after-tax loss 4. Its next profit 10 uses that loss, pays tax 1 and reverses DTA 1, leaving after-tax profit 8. P's first loss 1 creates DTA 0.2 and after-tax loss 0.8. Its next profit 4 uses the loss, pays 0.6 and reverses 0.2, leaving after-tax profit 3.2. Before any ownership purchase, the foreign legal books therefore close as follows; all amounts in this table are FCU.

	B, first	B, second	I, first	I, second
Cash	253	312.4	150	104
Loan	55	0	0	0
Deferred tax asset	0	0	1	0
Deposits or insurance liability	200	200	55	0
Equity	108	112.4	96	104

For example, B's first cash is $300 - 102 + 2 + 55 - 2 = 253$, and $253 + 55 - 200 = 108$. I's first cash is 150, while its equity is $150 + 1 - 55 = 96$. The tax benefit belongs in equity through profit even though it has not produced cash.

28.8.2 Currency and ownership follow different dates

The exact first-period transaction rate is 1.1 and the closing rate 1.2 USD/FCU; the second rates are 1.25 and 1.3. Opening loan origination uses the opening rate 1. The foreign recognition and cash events use these supplied transaction rates; closing translation follows them at the stated closing rates. No annual average is inferred. Under IAS 21.39–41, closing assets and liabilities use closing rates, income uses transaction-date rates, and the resulting translation difference enters OCI (European Commission, 2023a).

With no foreign capital contributions or dividends in the period, a subsidiary whose opening equity is E, after-tax profit is p, and opening, event and closing rates are r_0, r_e, r_c, has translation difference

$$\text{OCI} = E(r_c - r_0) + p(r_c - r_e). \tag{28.16}$$

To derive it, subtract translated opening equity Er_0 and translated profit pr_e from translated closing equity $(E + p)r_c$. The first term revalues the opening net assets; the second reconciles the two rates applied to this period's retained profit. Thus B's first translated profit is 8.8 and OCI is $100(0.2) + 8(0.1) = 20.8$. I's translated profit is -4.4, with OCI $100(0.2) - 4(0.1) = 19.6$. Together with P's loss 0.8, group profit is 3.6 and OCI 40.4.

Only after those first-period results does P buy another 10% of B for USD 15. Control continues. B's translated net assets are $108(1.2) = 129.6$, so the NCI carrying amount removed is 12.96. Cash consideration exceeds that amount by 2.04, charged directly to parent equity. No goodwill or profit is created by this retained-control transaction (European Commission, 2023a, IFRS 10.23,B 96). P's separate investment rises from 160 to 175 and cash falls from 500 to 485; B makes no legal entry. First-period profit and OCI still use 80% parent ownership. Bank attribution changes to 90% only for the second interval.

First-period NCI profit is $0.2(8.8 - 4.4) = 0.88$, and its OCI is $0.2(20.8 + 19.6) = 8.08$. Consequently,

$$NCI_1 = 40 + 0.88 + 8.08 - 12.96 = 36, \tag{28.17}$$

$$E_{P,1} = 654 + (3.6 - 0.88) + (40.4 - 8.08) - 2.04 = 687. \tag{28.18}$$

The sum 723 also follows from $694 + 3.6 + 40.4 - 15$. This second bridge tests the equity transaction independently of the closing asset total.

In the second interval, B's translated profit is 5.5 and OCI $108(0.1) + 4.4(0.05) = 11.02$. I's translated profit is 10 and OCI $96(0.1) + 8(0.05) = 10$. Group profit is 18.7 after adding P's 3.2, and OCI is 21.02. NCI receives $0.1(5.5)+0.2(10) = 2.55$ of profit and $0.1(11.02)+0.2(10) = 3.102$ of OCI. Closing NCI is 41.652, with parent equity 721.068 and total equity 762.72.

28.8.3 Cash translation does not equal translation OCI

At the first close, I's deposit 20 appears as 24 in translated cash and as 24 in B's translated deposits. The group eliminates both; neither legal book changes. Group cash is $485 + (253 + 150 - 20)(1.2) = 944.6$. Adding loan 66 and DTAs 1.4 gives assets 1012; liabilities are $(200 + 55 - 20)(1.2) + 7 = 289$. At the second close the deposit elimination is 26. Group cash becomes $484.4 + (312.4 + 104 - 20)(1.3) = 999.72$, and liabilities are $(200 - 20)(1.3) + 3 = 237$. There is then no loan or DTA.

Cash must also be reconstructed from its own dated origins. First-period operating cash is opening net advance -100, plus first-period loan receipt, bank tax and insurer claims translated at 1.1:

$$CFO_1 = -100 + (55 - 2 - 50)(1.1) = -96.7. \tag{28.19}$$

The NCI purchase is financing cash −15, under IAS 7.42 A–42 B. The currency effect on B's cash is $300(0.2) - 100(0.2) + 53(0.1) = 45.3$; on I's cash it is $200(0.2) - 50(0.1) = 35$. The internal deposit's currency effect 4 must also be eliminated. Thus cash FX is 76.3 and

$$980 - 96.7 - 15 + 76.3 = 944.6. \tag{28.20}$$

In the second period, operating cash is $(60.5 - 1.1 - 45 - 1)(1.25) - 0.6 = 16.15$. B's cash currency effect is 28.27, I's 12.7, and the internal deposit's 2, giving net 38.97. The bridge is $944.6 + 16.15 + 38.97 = 999.72$. IAS 7.28 requires the exchange effect needed to reconcile opening and closing cash separately from cash flows (European Commission, 2023a). It is not the OCI 40.4 or 21.02, which also reflects noncash net assets.

Keeping all rates at 1 removes both cash FX and translation OCI, while preserving the legal tax and operating calculations. Omitting the NCI purchase leaves both subsidiary books unchanged, raises each subsequent group cash balance by 15 and retains 20% NCI participation in B's second-period results. These comparisons reveal errors that a single balanced closing statement could hide. The companion `SectorCompany/1` report preserves separate component contexts, matched deposit identities, legal books, tax origins, dated ownership and gross cash movements; it refuses missing support or mismatched inputs without publishing a partial result. This finite composition supplies the calculation just derived. Prudential capital, actual tax law, general guarantee settlement and nonlinear insurance valuation still require their own methods.

28.9 Apple: source labels do not identify exposure

The real reference package is Apple's 2024 Form 10-K and first-quarter 2025 Form 10-Q, for fiscal dates 28 September and 28 December 2024 respectively (Apple Inc., 2024a, 2025c). The annual balance sheet also supplies 30 September 2023. The annual filing is archived as the company's PDF from AnnualReports; the interim filing is archived from SEC EDGAR with its inline XBRL. Apple's Note 1 describes a 52/53-week year ending on the last Saturday of September. Calendar-year assumptions would therefore misstate the periods.

The package inventory preserves every annual PDF page and every interim top-level section, all thirteen annual notes and all ten interim notes. It records internal cross-references and externally incorporated material, including the proxy statement, as distinct obligations. Statement and debt facts below are structured; policies, other notes and exhibits retain their source locations and specific modelling obligations. This is complete inventory coverage of the acquired files, with partial semantic formalisation. It does not claim to reconstruct every Apple contract or to have acquired every incorporated external document.

Apple's annual balance sheet reports assets 364,980, liabilities 308,030 and equity 56,950, in millions of US dollars. Note 9 reports term-debt principal 97,341, unamortised discount/premium and issuance costs of −321, and hedge accounting adjustments of −358. Thus its carrying amount is $97{,}341 - 321 - 358 = 96{,}662$, split into current 10,912 and non-current 85,750. Fair value is separately disclosed as 88.4 billion. None of these three measurements may replace another merely because a table calls it debt.

Note 9 identifies the notes as fixed-rate, then states that interest-rate swaps effectively convert a portion to floating rates. Note 4 reports interest-rate contract notional of 14,575 at the annual date. A notional is neither a liability carrying amount nor sufficient information for an exact interest sensitivity: reset timing, remaining periods, curve conventions and hedge coverage still matter. The cross-reference connects the derivative to the borrowing without erasing either contract.

At the interim date, total term-debt carrying amount is $10{,}848 + 83{,}956 = 94{,}804$. Cash is

30,299, total assets 344,085, liabilities 277,327 and equity 66,758. The interim debt note's rounded "94.8 billion" and the statement's 94,804 million can coexist; their precision differs. Treating them as conflicting exact numbers would create an artificial infeasibility.

These balances leave two practical financing questions open: how much cash can fund a payment, and when the payment falls due. The annual cash and cash equivalents balance of 29,943 includes 2.6 billion held in escrow and restricted from general use for the State Aid tax obligation. Note 4 states the inclusion explicitly, and Note 7 connects the escrow to that obligation (Apple Inc., 2024a, Notes 4 and 7, printed pp. 37 and 39). Subtracting the reported restricted component gives $29{,}943 - 2{,}600 = 27{,}343$ million as an indicative remainder at the displayed amounts. The restricted component is rounded to tenths of a billion, so this subtraction does not establish an exact freely available balance. Interpreting the reported cash balance as excluding the escrow is wrong relative to this footnote. Correcting that interpretation changes funding calculations without changing the issuer's reported number. It also requires revisiting any liquidity result that relied on the exclusion.

The debt movement supplies a second reason to preserve a calculation's missing terms. The first-quarter cash flow statement reports repayments of term debt of 1,009 million, while carrying amount falls by $96{,}662 - 94{,}804 = 1{,}858$ million. At the displayed amounts, the reconciliation is

$$94{,}804 = 96{,}662 - 1{,}009 - 849.$$

The 849 million is the residual after the reported cash repayment; it does not identify an exchange loss, an income-statement charge or a particular journal. Explaining it requires the missing movement information, including applicable currency, discount/premium and hedge-accounting effects. The quarterly cash flow statement still reconciles independently: operating inflow 29,935 plus investing inflow 9,792 less financing outflow 39,371 gives the reported cash increase of 356, from 29,943 to 30,299 million (Apple Inc., 2025c, statement of cash flows and Note 6). A balanced cash bridge therefore does not identify the debt history.

The annual filing's securities exhibit adds terms that a debt-total query cannot supply. Exhibit 4.1 describes two euro-denominated series bearing the year 2025 in their names. Its reported outstanding principal is dated 18 October 2024, after the annual balance-sheet date. The description concerns Apple Inc. as issuer, rather than every subsidiary in the consolidated group (Apple Inc., 2024a, Exhibit 4.1, pp. 3–4).

Series	Principal (EUR m)	Fiscal year	Scheduled maturity
0.875% 2025 Notes	1,250	2025	24 May 2025
0.000% 2025 Notes	1,000	2026	15 November 2025

Apple's disclosed 52-week fiscal 2025 ends on 27 September 2025 under its last-Saturday convention. The November maturity therefore belongs to fiscal 2026 even though the bond's name says 2025. The annual Note 9 principal-payment schedule uses fiscal years, whereas the instrument's stated maturity is a calendar date. Joining those fields on their year number would put a billion euros of scheduled principal into the wrong fiscal bucket. Neither the later holding date nor the euro principal may be silently substituted for a September US-dollar consolidated balance.

Suppose the disclosed faces remain outstanding until scheduled maturity, with no early redemption, default or additional withholding-related payment, and euro remains the payment currency. The annual coupon on the 0.875% series is then the reported face multiplied by the stated rate:

$$1{,}250 \times 0.00875 = 10.9375 \quad \text{million EUR}.$$

Its final scheduled principal plus annual coupon is consequently 1,260.9375 million EUR; the zero-coupon series has 1,000 million EUR of scheduled principal and zero stated coupon. Both scheduled maturity dates fall on a Saturday. Exact settlement requires the applicable payment-adjustment provision and business calendar. Exhibit 4.1 is expressly a summary qualified by the 2013 and 2018 indentures; its payment-currency exceptions and redemption rights also matter. The retained 2021 indenture cannot replace those earlier contracts. The calculation establishes conditional scheduled amounts, while their exact settlement dates and subsequent holdings remain unresolved.

Those missing facts have different consequences for the financial statements. Principal and coupon payments reduce cash, but settling principal differs from recognising interest expense. The latter also depends on carrying amount, effective-interest accrual and the treatment of discounts and hedges; Note 9 expressly includes applicable amortisation and hedging adjustments in effective rates. Exchange rates, hedge designations and tax treatment can affect profit, other comprehensive income, retained earnings and closing debt differently. The cash flow statement additionally needs the payment classification and any associated hedge settlements. A usable answer thus carries the conditional euro schedule together with these unresolved dependencies. A dollar company forecast must supply consistent holdings, exchange, hedge and tax assumptions across all affected statements.

To demonstrate a source-to-forecast calculation, freeze the interim balance sheet and apply a *marginal* financing stress for a following 13-week period: a 200-basis-point rate increase, no principal change, no offsetting hedge beyond the explicitly selected exposure, immediate cash settlement and no tax effect. An assumed effective floating exposure $E = 14{,}575$ million produces $0.02 \times (13/52)E = 72.875$ million additional interest. This exposure uses the annual notional as a conditional input; it is not asserted to be the interim net sensitivity. Under that condition, debit interest expense and credit cash by 72.875. The resulting marginal statements have cash 30,226.125, assets 344,012.125, liabilities 277,327, equity 66,685.125, profit and comprehensive income of -72.875, and operating cash flow of -72.875 relative to the frozen baseline. The balance-sheet difference equals the equity difference exactly.

If exposure is instead only restricted to $0 \leq E \leq 14{,}575$, the conditional added interest lies in $[0, 72.875]$ before monetary rounding. That restriction is a stress assumption and is not claimed to bound every real Apple exposure. This real case verifies evidence, context, mappings and a coherent marginal propagation. Northstar supplies the complete-company forecast; a complete Apple forecast remains dependent on its unmodelled operations and undisclosed terms.

28.10 Apple: five statements constrain one year

The debt example leaves a larger question: can the reported profit, changes in equity and cash flows describe the same financial year? Apple's five consolidated statements supply a useful starting point. Their fiscal 2024 amounts refer to the year ending 28 September 2024; the opening balance sheet is dated 30 September 2023. All monetary amounts below are millions of US dollars. The operations statement's per-share amounts and weighted share counts keep their separate units (Apple Inc., 2024a, printed pp. 29–33).

Products revenue of 294,866 and services revenue of 96,169 sum to 391,035. Their costs, 185,233 and 25,119, leave gross margin of 180,683. Research and development of 31,370 and selling, general and administrative expense of 26,097 reduce that amount to operating income of 123,216. Other income of 269 and an income-tax provision of 29,749 give net income of 93,736 (Apple Inc., 2024a, printed p. 29). This same net income enters both the equity statement and the operating-cash reconciliation. It is one period result appearing in three statements, so it must

not be added three times.

Other comprehensive income supplies the next connection. Foreign-currency translation contributes 395 after tax. Derivative fair-value changes of −832 and reclassification adjustments of −1,337 contribute −2,169; marketable-debt fair-value changes of 5,850 and reclassifications of 204 contribute 6,054. Thus OCI is $395 - 2{,}169 + 6{,}054 = 4{,}280$, and total comprehensive income is $93{,}736 + 4{,}280 = 98{,}016$ (Apple Inc., 2024a, printed p. 30). Adding the period OCI to opening accumulated other comprehensive loss of −11,452 gives the closing −7,172. That accumulated balance, rather than the period flow, enters the balance sheet.

Despite positive comprehensive income, total equity falls. The equity statement explains the difference: common-stock issuance adds 1,423 and share-based compensation adds 12,034; withholding reduces the capital component by 3,993 and retained earnings by 1,612; declarations reduce retained earnings by 15,218 and repurchases by 95,846 (Apple Inc., 2024a, printed p. 32). These movements, apart from comprehensive income, total −103,212. The complete equity reconciliation is therefore

$$62{,}146 + 98{,}016 + 1{,}423 + 12{,}034 - 3{,}993 - 1{,}612 - 15{,}218 - 95{,}846 = 56{,}950.$$

The components reconcile separately: capital rises from 73,812 to 83,276, the accumulated deficit moves from −214 to −19,154, and accumulated other comprehensive loss moves from −11,452 to −7,172. These are the three closing equity components reported on the balance sheet. Their agreement prevents a model from explaining the correct total with the wrong allocation among capital, retained earnings and accumulated OCI.

The same change appears in net assets. Table 28.1 compares the reported endpoints; its changes are derived subtractions (Apple Inc., 2024a, printed p. 31). Assets rise by 12,397 while liabilities rise by 17,593, leaving a 5,196 decrease in equity, exactly the decrease from the equity reconciliation.

	30 September 2023	28 September 2024	Change
Assets	352,583	364,980	12,397
Liabilities	290,437	308,030	17,593
Equity	62,146	56,950	−5,196

Table 28.1: The balance-sheet change agrees with the complete equity movement. The agreement constrains a history without identifying its transactions.

Cash follows a different reconciliation. Starting from the same net income, the cash-flow statement adds depreciation and amortisation of 11,445, share-based compensation expense of 11,688 and other adjustments of −2,266. These three adjustments sum to 20,867. Its six operating-asset and liability adjustments are net receivables −3,788, vendor non-trade receivables −1,356, inventory −1,046, other assets −11,731, payables +6,020 and other liabilities +15,552. They sum to 3,651, giving operating cash of $93{,}736 + 20{,}867 + 3{,}651 = 118{,}254$ (Apple Inc., 2024a, printed p. 33). The full indirect bridge is thus present; none of its broad adjustment rows has been replaced by a presumed change in a similarly named balance-sheet account.

Investing cash is also positive: security purchases of −48,656, maturities of 51,211, sales of 11,135, plant-acquisition payments of −9,447 and other investing cash of −1,308 sum to 2,935. Financing uses 121,983: employee-award tax payments of 5,441, dividend payments of 15,234, share repurchases of 94,949, debt repayments of 9,958 and other outflows of 361 outweigh net commercial-paper proceeds of 3,960. The source shows a dash for net term-debt issuance; the displayed total treats that row as making no contribution, while the source reading retains the

dash. The three activity totals give

$$118{,}254 + 2{,}935 - 121{,}983 = -794, \qquad 30{,}737 - 794 = 29{,}943.$$

The opening and closing amounts here are cash, cash equivalents and restricted cash and cash equivalents, as defined by the cash-flow statement. Opening balance-sheet cash is only 29,965, so the cash-flow population exceeds it by 772; both closing amounts are 29,943. Consequently, the balance-sheet line falls by 22 while the cash-flow total falls by 794. The earlier 772 remains unallocated in the retained evidence. It cannot be relabelled as the closing 2.6 billion escrow, which Note 4 explicitly includes in reported cash and equivalents (Apple Inc., 2024a, printed p. 33 and Note 4, printed p. 37).

These joins expose the missing transaction links more precisely. Annual receivables rise by 3,902, leaving 114 after adding their −3,788 indirect adjustment. Inventory rises by 955, leaving −91 after its −1,046 adjustment. Payables rise by 6,349, exceeding their 6,020 adjustment by 329. Even the exact opposite vendor-receivable movements identify neither gross purchases nor collections. The notes examined below must supply the relevant recognition, valuation, currency and settlement information; the residuals themselves supply no such explanation.

Nor can the equity rows simply replace financing cash flows. The reported repurchase reduction of 95,846 exceeds cash repurchases of 94,949 by 897; withholding reductions total 5,605 against cash tax payments of 5,441; and equity compensation of 12,034 differs from the 11,688 expense adjustment. Cash dividends of 15,234 exceed declarations of 15,218 by 16. Those differences remain unassigned to particular settlements, fees, taxes or recognition periods (Apple Inc., 2024a, printed pp. 32–33). The later note examples retain their own evidence and conditional histories.

A reconstruction must now satisfy the statement relationships together. Correct total cash cannot excuse wrong equity components, and correct net assets cannot excuse omitted cash-flow adjustments. The five statements constrain a common history, but their agreement does not reveal all gross events, dates, counterparties or accounting treatments. Those unresolved links determine which historical alternatives can be supported and what additional facts a conditional future path still needs.

28.10.1 The following quarter must start from the same balances

The next filing makes the time connection testable. Its balance-sheet comparison is 28 September 2024, and all 27 monetary rows reproduce the annual closing column. Its income, comprehensive-income, equity and cash-flow comparisons instead cover the three months ending 30 December 2023. Those earlier-quarter flows cannot become the opening amounts for the quarter ending 28 December 2024. The interim statements are explicitly unaudited (Apple Inc., 2025c, printed pp. 1–5).

Quarterly products revenue of 97,960 and services revenue of 26,340 total 124,300. Cost of sales of 66,025 and operating expenses of 15,443 leave operating income of 42,832; other expense of 248 and tax expense of 6,254 give net income of 36,330. Translation OCI of −625, derivative OCI of 2,435 and marketable-debt OCI of −1,427 add 383, giving comprehensive income of 36,713 (Apple Inc., 2025c, printed pp. 1–2). The resulting equity movement begins at the same 56,950 that closed the annual record. Compensation adds 3,383, withholding removes 1,891 from capital and 1,102 from retained earnings, declarations remove 3,819 and repurchases remove 23,476. Together these other movements are −26,905, so $56{,}950 + 36{,}713 - 26{,}905 = 66{,}758$. Capital, accumulated deficit and accumulated OCI also join separately at both endpoints (Apple Inc., 2025c, printed pp. 3–4).

Assets decline by 20,895 to 344,085 and liabilities decline by 30,703 to 277,327. Their difference gives the same 9,808 equity increase. This check connects the period result to the balance sheet without confusing declining assets with a loss. The retained earnings deficit improves by 7,933, capital rises by 1,492 and accumulated OCI rises by 383; these three changes also sum to 9,808.

The cash reconciliation supplies a different account of the quarter. Depreciation and amortisation of 3,080, compensation expense of 3,286 and other adjustments of −2,009 add 4,357 to net income. The six operating asset/liability adjustments are +3,597, +3,166, +215, +939, −6,671 and −11,998, in the same reported categories used for the annual reconciliation. They total −10,752, so operating cash is $36{,}330 + 4{,}357 - 10{,}752 = 29{,}935$. Investing cash adds 9,792 and financing uses 39,371, leaving a 356 increase from 29,943 to 30,299 (Apple Inc., 2025c, printed p. 5). Both cash-flow endpoints now match the corresponding balance-sheet amounts. That numeric match still does not make every dollar unrestricted.

Adjacent flows may now be combined for a deliberately longer question. Over fiscal 2024 and the following quarter together, comprehensive income is 134,729 and the other equity movements total −130,117. Over that same interval, operating, investing and financing cash total 148,189, 12,727 and −161,354. The two cumulative reconciliations are

$$62{,}146 + 134{,}729 - 130{,}117 = 66{,}758, \qquad 30{,}737 + 148{,}189 + 12{,}727 - 161{,}354 = 30{,}299.$$

These are derived amounts for the interval from 1 October 2023 through 28 December 2024, not an issuer annual statement or a trailing-twelve-month result. Endpoint balances are carried forward, not added. EPS and weighted share averages also cannot be summed as if they were dollar flows.

The quarter preserves several distinctions that a common schema might otherwise erase. The annual equity table contains a common-stock issuance row and the annual cash-flow table contains a term-debt issuance row; the interim tables omit both. The interim reported totals reconcile over their own disclosed rows. A comparison record may describe those omissions, but must not manufacture two issuer-reported zeros. The prior-year quarter's term-debt repayment dash is likewise retained in its own dated column (Apple Inc., 2025c, printed pp. 4–5).

Transaction uncertainty remains across the joined interval. In the current quarter, the receivable change plus its indirect adjustment leaves −174; inventory leaves −160. The payable change is −7,050, differing from the −6,671 adjustment by −379. Cash dividends exceed declarations by 37, and cash repurchases exceed the equity reduction by 130. None identifies the missing settlement, valuation or timing components. The subsequent note inquiries must retain these relationships together when testing historical alternatives. Combining individually compatible note examples does not by itself establish that they share a compatible cash, income and equity history.

28.11 Apple: revenue recognition and cash collection

An iPhone sale can create revenue today and an obligation to provide services later. Apple's annual revenue note separates delivered hardware and bundled software from the right to receive product-related services such as iCloud, Siri and Maps. It allocates the price and related discounts by relative stand-alone selling prices. For the bundled services those prices are estimates, because observable separate prices are unavailable. The hardware portion is recognised when control transfers, generally on shipment; the service portion is deferred and recognised straight-line over the estimated period of provision (Apple Inc., 2024a, Note 2, printed p. 35). A historical model consequently needs the allocation and service cohort as well as the sale date and amount. Neither the product label nor the customer payment supplies those missing terms.

The note describes collection shortly after control transfer or the start of service delivery. It does not specify one collection lag for every contract. Products revenue also reflects expected returns, price protection and customer incentives. Revenue excludes taxes collected for remittance to governments (Apple Inc., 2024a, Notes 1–2, printed pp. 34–35). Together these qualifications prevent a simple conversion of reported revenue into gross customer receipts.

Presentation creates a second distinction. Where Apple obtains control of a third-party product before sale, it reports the gross amount billed. For third-party applications sold through the App Store, its stated treatment records only the retained commission in Services revenue. Services also already includes recognition of the deferred value of services bundled with products (Apple Inc., 2024a, Note 2, printed p. 35). Adding that recognition to reported Services a second time would double count it; treating Services as gross App Store customer spending would change the measured quantity.

The product and geographic tables describe different divisions of the same sales. The five product/service rows sum to 391,035 million dollars in fiscal 2024 and 124,300 million in the quarter ended 28 December 2024. The geographic segment sales sum to those same totals. They must not be added together. Apple says the product mix is generally similar across segments, except that iPhone represents a moderately higher share in Greater China. That qualitative statement does not identify a product-by-region matrix (Apple Inc., 2024a, Notes 2 and 13, printed pp. 35–36 and 46–47) (Apple Inc., 2025c, Notes 2 and 10, printed pp. 6 and 11).

28.11.1 The movement that the note leaves unidentified

At 28 September 2024 Apple reports total deferred revenue of 12.8 billion dollars. At 28 December it reports 13.2 billion, and says that quarter's net sales include 3.7 billion recognised from the deferred balance at the September date (Apple Inc., 2025c, Note 2, printed p. 6). The balance sheet's December current deferred-revenue line is 8,461 million, with 8,249 million at September (Apple Inc., 2025c, printed p. 3). Those are current liabilities; the note's larger amounts are totals. The precision differs too: the total is displayed to tenths of a billion, whereas the current line is displayed in millions.

Let D_0 and D_1 denote opening and closing total deferred revenue. During the quarter, let A be additions to deferred revenue, R_o revenue released from the opening balance, and R_n revenue released from amounts added during the quarter. Let M collect other signed movements, positive when they increase the liability. These symbols describe a balance reconciliation; M is not an assertion that a particular adjustment occurred. In billions of dollars,

$$D_1 = D_0 + A - R_o - R_n + M, \qquad A - R_n + M = D_1 - D_0 + R_o \simeq 13.2 - 12.8 + 3.7 = 4.1. \tag{28.21}$$

The source identifies R_o, not the whole release $R_o + R_n$. The 4.1 billion residual therefore measures net additions after new-origin releases and other movements at the displayed amounts. Calling it gross new deferrals, billings or cash collected would be wrong relative to what has been calculated.

Rounding also matters. If each of the three displayed inputs is rounded to the nearest 0.1 billion, a closed outer envelope for that residual is $[3.95, 4.25]$ billion: the lower endpoint uses $13.15 - 12.85 + 3.65$, and the upper uses $13.25 - 12.75 + 3.75$. This is a conditional precision calculation, not a verified issuer rounding rule or a bound on gross collections. Its endpoints leave the treatment of exact rounding ties unspecified.

Two constructed histories make the missing movement visible. For this illustration alone, use the displayed amounts as exact inputs and set $M = 0$. Keep the reported quarter's total revenue at 124.3 billion. The table supplies two different choices for A and R_n; other recognised revenue is the remainder needed to match that total.

Quarterly amount (USD billion)	History 1	History 2
New deferrals, A	4.1	5.1
Release from those new deferrals, R_n	0	1.0
Release from the opening balance, R_o	3.7	3.7
Other recognised revenue	120.6	119.6
Closing deferred revenue	13.2	13.2
Total recognised revenue	124.3	124.3

Both histories satisfy the selected note totals, yet their new deferrals differ by one billion. They are witnesses within this aggregate model, not reconstructed Apple contracts or a fit to every financial-statement constraint. Additional contract, cohort, return and movement information can rule out or refine them. No probability attaches to either merely because two examples were written down.

The corresponding recognition entry debits deferred revenue and credits revenue when service is provided. Receipt or billing of consideration that creates the deferral instead credits deferred revenue and debits cash or, where an unconditional right to payment exists, a receivable. Collection of that receivable changes cash and receivables without creating revenue a second time. For example, hold History 1's other revenue cash-settled and omit costs, taxes, refunds and other movements solely to isolate this subsystem. Receiving its 4.1 billion of new consideration gives revenue-side cash inflow 124.7 billion. If that new consideration is receivable at quarter end, cash inflow is 120.6 billion and receivables increase 4.1 billion. In both cases the asset increase is 124.7, matched by a 0.4 increase in deferred revenue and 124.3 of revenue before omitted expenses. These conditional entries explain the identification problem; neither inflow is an estimate of Apple's actual cash collections.

28.11.2 A dated recognition expectation is not a cash forecast

At both September and December, Apple expects 64% of the then-reported total deferred revenue to be realised in less than a year, 25% within one to two years, 9% within two to three years and 2% after three years (Apple Inc., 2024a, Note 2, printed p. 36) (Apple Inc., 2025c, Note 2, printed p. 6). These are two dated expectations about two balances. Multiplication at the displayed December values gives $13.2 \times 0.64 = 8.448$ billion in the first bucket. The corresponding September calculation gives 8.192 billion. Their difference does not measure revenue recognised during the intervening quarter: the dates, remaining horizons and contract populations have changed.

The percentages divide an expected amount by recognition horizon. They do not assign probabilities to alternative histories, fix collection dates or forecast all future sales. The 8.448 billion calculation also cannot replace the reported 8.461 billion current liability. Rounded amount and percentage disclosures do not justify an exact equality between those figures, and the query must retain the source's distinct description of each quantity. A future calculation that requires monthly service or receipts needs finer timing and payment evidence.

The note leaves another part of future service unquantified. For certain long-term arrangements, Apple has not yet delivered the service, has no right to bill for it and attributes the unbilled consideration entirely to that undelivered service. It reports neither revenue nor an amount for those obligations in this passage (Apple Inc., 2024a, Note 2, printed p. 35). A query must retain an unknown amount; setting it to zero or equating it with recorded deferred revenue would erase the distinction the company makes.

These disclosures therefore yield several different results. The total and current deferred balances are reported observations. The 4.1 residual is a derived net movement with unresolved

components. The recognition buckets are management's dated expectations, and the excluded consideration is unquantified. Contract-level allocations, service lives, new-origin releases and collection facts remain necessary for a particular future journal path. The annual and interim periods also stay separate: the December 2024 quarter follows the September 2024 fiscal year, so subtracting it from that annual revenue cannot produce the year's final nine months. The complete revenue-note reading makes these consequences explicit while the wider Apple disclosure package remains only partly formalised.

28.12 Apple: commitments and the cash needed to fulfil them

The same distinction between recognition and cash arises when Apple buys capacity and components. A lease liability measures one part of an existing obligation; a purchase commitment describes goods or services still to be acquired. A funding projection needs the payments arising from both, together with new orders and operating decisions. Adding every disclosed amount would mix measurements and could count the same requirement twice.

28.12.1 From a lease liability to future payments

At 28 September 2024, Apple's lease note reports the following amounts in millions of dollars (Apple Inc., 2024a, Note 8, printed p. 42). The maturity table first totals undiscounted payments and then deducts imputed interest to reach the recognised liability. The right-of-use assets are reported separately.

Amount (USD million)	Operating	Finance	Total
Undiscounted lease payments	13,518	1,272	14,790
Less: imputed interest	1,984	376	2,360
Recognised lease liabilities	11,534	896	12,430
Right-of-use assets	10,234	1,069	11,303

Thus $14{,}790 - 2{,}360 = 12{,}430$ reconciles the maturity and liability measurements. The recognised liability is already included in the balance sheet; it is not an extra cash requirement to add to 14,790. Its current components total $1{,}488 + 144 = 1{,}632$ million, whereas the fiscal-2025 maturity bucket is 1,991 million. A current carrying amount and a gross payment bucket answer different questions. The right-of-use asset likewise need not equal the remaining liability; these reported balances do not disclose the complete amortisation, prepayment and modification history.

Apple also reports 849 million of fixed payments under additional leases that had not commenced. It expects them to commence in 2025–2026, with terms from less than one year to twenty-one years. Adding this separately disclosed amount to the maturity total gives $14{,}790 + 849 = 15{,}639$ million. That sum is consistent, under a nearest-0.1-billion rounding assumption, with the 15.6 billion of fixed lease obligations in management's cash-requirements discussion (Apple Inc., 2024a, printed pp. 25 and 42). Numerical compatibility does not establish a contract-by-contract reconciliation. It neither adds 849 to the recognised liability nor identifies its commencement and payment dates.

The remaining lease information cannot supply those dates mechanically. The note describes original terms typically no longer than ten years, with renewal options, some reasonably certain to be exercised. Its weighted-average remaining term of 10.3 years describes a different quantity. The reported 3.1% discount rate, generally based on estimated incremental borrowing rates, does

not supply every contract's rate or the timing hidden in "thereafter". Applying that one rate to invented year-end payments would introduce new assumptions. The policy combining lease and nonlease components also has a stated scope: corporate, data-centre and retail facilities (Apple Inc., 2024a, Notes 1 and 8, printed pp. 34 and 42).

Fixed and variable costs add another distinction. For 2024, Apple reports 2.0 billion of fixed operating-lease cost, 1.9 billion of fixed operating-lease cash payments and 1.0 billion of noncash right-of-use assets obtained for lease liabilities. Variable costs across its leases were 13.8 billion; payments of that kind primarily depend on purchases of output from the leased assets (Apple Inc., 2024a, Note 8, printed p. 42). Historical expense is therefore not a fixed future cash schedule. Forecasting it requires output, price and settlement assumptions. Adding it to manufacturing commitments without identifying their relationship could double count procurement requirements.

28.12.2 Why a fall in commitments does not identify cash spent

The purchase note and management discussion disclose different views. Note 12 lists 11,226 million of future payments under unconditional purchase obligations whose remaining terms exceed one year. Its fiscal-2025 row is 3,206 million (Apple Inc., 2024a, Note 12, printed p. 45). The term restriction selects contracts; it does not place every payment beyond the coming year. The note describes these as off-balance-sheet commitments, primarily involving supplier arrangements, intellectual property, content and distribution rights.

Management's discussion provides additional near-term evidence. At the same September date it reports manufacturing purchase obligations of 53.0 billion, of which 52.9 billion is payable within twelve months. Other purchase obligations total 12.0 billion, with 4.1 billion within twelve months; that description includes capital assets as well as supplier, content and rights arrangements (Apple Inc., 2024a, printed p. 25). A model confined to Note 12 would miss those reported aggregates. Conversely, subtracting 11.226 from 12.0 and labelling the difference short-term contracts is unsupported: the category and term populations have not been reconciled, and the precision differs.

At 28 December 2024 manufacturing obligations are 39.8 billion, all payable within twelve months. The interim report says contractual cash requirements have not changed materially since the annual filing except for commercial paper, manufacturing obligations and the State Aid tax payable (Apple Inc., 2025c, printed p. 16). That qualitative statement is not an exact December lease measurement or evidence that every lease remained unchanged. The two manufacturing totals are dated observations over changing orders and payment horizons. Their 13.2 billion difference does not identify purchases or cash paid during the quarter.

To see what is missing, consider an explicitly simplified model in billions of dollars. Let O_0 and O_1 denote a comparable population of undelivered fixed-price orders at the two dates. Let N be new orders and F the value removed when goods are delivered. Assume no cancellation, price or currency change. Treat the displayed 53.0 and 39.8 as exact inputs for this illustration. These assumptions define the example; the disclosure does not establish this roll-forward convention for every reported obligation. The balance relation is

$$O_1 = O_0 + N - F, \qquad F - N = 53.0 - 39.8 = 13.2. \tag{28.22}$$

With no new orders, delivery is 13.2. With 5.0 of new orders, delivery is 18.2. Both leave the same closing undelivered amount of 39.8. The source balances alone cannot choose between these conditional movements; actual contract changes and delivery evidence could do so.

Payment is a further choice. Suppose the first history's 13.2 of delivered goods all remain unsold inventory, with no opening payables, prepayments or tax. Paying on delivery debits inventory

and credits cash by 13.2: total assets and profit do not change within this subsystem. Delivery on credit instead debits inventory and credits trade payables by 13.2: assets and liabilities rise together, with no immediate cash or profit effect. Settling that payable later debits the payable and credits cash; consumption, sale or impairment would require further entries. These examples isolate recognition and settlement. They are not estimates of Apple's actual purchases or payments, or a fit to all its financial statements. No probability follows from writing two histories.

Delivery assumptions also need the narrative that surrounds the purchase table. Apple describes single or limited sources for some components, industry-wide shortages and price fluctuations, and capacity or yield constraints for new technologies. Suppliers may favour common components over custom ones. Its named manufacturing locations describe outsourced production, not a product-by-revenue-region allocation or ownership map (Apple Inc., 2024a, Note 12, printed p. 46). A scenario in which input availability changes may alter output, variable lease payments, inventory and receipts together. The passage motivates those dependencies without identifying their size or probabilities.

The same note, and the interim contingency note, preserve management's dated assessment of unresolved legal claims. At each date, management states that there was not at least a reasonable possibility that Apple had incurred a material loss, or a material loss greater than a recorded accrual (Apple Inc., 2024a, Note 12, printed p. 46) (Apple Inc., 2025c, Note 9, printed p. 11). It does not make litigation exposure zero. Contract terms, supply constraints, commencement, delivery and settlement must therefore remain in the historical record alongside reported balances. They determine which future journal paths can be supported when a funding question is asked.

28.13 Apple: resources available for payment

The procurement and settlement choices just described determine when Apple needs money. Funding those payments requires a second account: what resources can be used, how securities become cash, and when customers pay. Annual and interim Note 4 distinguish these questions even though a single total might appear to answer them. The following table combines the notes' measurement and balance-sheet classifications. Amounts are in millions of US dollars at the displayed precision; the December date is the end of the first fiscal quarter, not another annual endpoint (Apple Inc., 2024a, Note 4, printed pp. 36–37) (Apple Inc., 2025c, Note 4, printed pp. 7–8).

Resource / measurement	30 Sep 2023	28 Sep 2024	28 Dec 2024
Cash and cash equivalents	29,965	29,943	30,299
Current marketable securities	31,590	35,228	23,476
Non-current marketable securities	100,544	91,479	87,593
Total fair value	162,099	156,650	141,368
Total adjusted cost	173,752	160,600	147,088
Net unrealised gains / losses	−11,653	−3,950	−5,720

Cash plus the two securities rows equals total fair value. Adjusted cost plus net unrealised gains and losses provides a second reconciliation of that same total; it is not an additional resource to add to it. The source calls the column *adjusted cost*. Relabelling every investment as amortised-cost debt would erase the different instruments represented in the table.

28.13.1 Restriction changes what can be spent

At September 2024, cash includes 2.6 billion dollars held in escrow, while marketable securities include a further 13.2 billion. Both are restricted from general use and designated for the State Aid tax obligation discussed earlier. Subtracting their displayed centres from the total gives $156{,}650 - 2{,}600 - 13{,}200 = 140{,}850$ million. MD&A separately reports unrestricted cash, equivalents and securities of 140.8 billion (Apple Inc., 2024a, printed pp. 25–26 and Note 4). The apparent difference does not establish conflicting disclosures: the restrictions and the MD&A total are rounded to tenths of a billion, whereas the table is in millions. If each amount is rounded to its nearest displayed quantum, the subtraction has possible bounds of 140,749.5 and 140,950.5 million. Those bounds overlap the 140,750 to 140,850 million rounding range for the MD&A total. This is a conditional compatibility check, not an issuer-stated rounding convention or a confidence interval.

The December footnote makes the distinction between maturity and restriction especially clear: *current* marketable securities include 3.3 billion dollars of restricted escrow funds, and cash includes 126 million. At the displayed centres, total resources less this disclosed escrow are $141{,}368 - 3{,}300 - 126 = 137{,}942$ million. Cash plus current securities is 53,775 million; removing the same escrow gives 50,349 million. These are indicative subtractions, not exact freely spendable balances. Under the same nearest-rounding hypothesis the first subtraction ranges from 137,891 to 137,993 million. Interim MD&A supplies no numerical unrestricted total (Apple Inc., 2025c, Note 4 and printed p. 16). A current classification therefore cannot serve as permission to spend an asset, and excluding escrow does not establish immediate liquidation at the reported price.

Maturity supplies a different kind of timing information. Note 1 tests the three-month cash-equivalent maturity at *purchase*, while annual Note 4 classifies marketable debt as available-for-sale and assigns current or non-current status using each instrument's underlying maturity (Apple Inc., 2024a, printed pp. 34 and 37). Among non-current debt excluding mortgage- and asset-backed securities, the September maturity shares are 86%, 10% and 4% for 1–5, 5–10 and more than 10 years. The separate mortgage- and asset-backed population has shares of 14%, 9% and 77%. At December these become 85/11/4% and 13/10/77%, respectively (Apple Inc., 2025c, Note 4, printed p. 8). Each triple divides its own population; the two triples cannot be added. Nor do contractual maturity bands establish a sale date, sale price or cash receipt schedule. Management's stated belief in sufficient resources also relies on ongoing operating cash and continued access to debt markets. It is a dated funding assessment, rather than a solvency guarantee derived from the table.

28.13.2 Security turnover differs from valuation

Securities can fund payments through sales or maturities, but a fall in their reported balance need not equal cash released. Fiscal 2024 cash proceeds from maturities and sales are 51,211 and 11,135 million, while purchases use 48,656. Net cash released is therefore $51{,}211 + 11{,}135 - 48{,}656 = 13{,}690$ million. The marketable-security balance falls by only $126{,}707 - 132{,}134 = -5{,}427$ million (Apple Inc., 2024a, cash-flow statement, printed p. 33, and Note 4). To expose the missing reconciliation, let M_0 and M_1 be total current and non-current marketable securities at the period's opening and closing. Let C be maturity and sale proceeds less purchases, positive when cash is released. Define R as the unallocated difference between the stock movement and

that cash measure, all in millions over the same period:

$$\begin{aligned} M_1 - M_0 &= -C + R, \\ \text{Fiscal 2024:} \quad -5{,}427 &= -13{,}690 + 8{,}263, \\ \text{First quarter 2025:} \quad -15{,}638 &= -13{,}335 - 2{,}303. \end{aligned}$$

For the quarter, the cash measure is $15{,}967 + 3{,}492 - 6{,}124 = 13{,}335$ million (Apple Inc., 2025c, cash-flow statement and Note 4). The relation defines a residual; it does not explain it. Cash sale proceeds differ from the carrying amount removed on disposal, and changes in measurement or classification may also enter the stock. Apple's specific-identification policy for securities sold requires the disposed lots to recover their cost (Apple Inc., 2024a, Note 1, printed p. 34). Neither the positive 8,263 nor the negative 2,303 identifies a fair-value, currency or reclassification journal.

The valuation columns help separate another tempting explanation. Across the annual dates, total adjusted cost falls by 13,152 million and net unrealised gains and losses rise by 7,703, producing a total fair-value decline of 5,449. Across the quarter, the corresponding changes are $-13{,}512$ and $-1{,}770$, producing $-15{,}282$. These totals include cash and equivalents, whereas M_0 and M_1 include only marketable securities. They also differ from the comprehensive-income statement's net-of-tax marketable-debt amounts: 6,054 million for the year and $-1{,}427$ for the quarter, including reclassification adjustments (Apple Inc., 2024a, comprehensive-income statement, printed p. 30) (Apple Inc., 2025c, comprehensive-income statement, printed p. 2). Endpoint unrealised balances and period OCI differ in population, tax and timing; their numerical differences cannot be assigned to a tax charge or realised gain without the missing movements.

Even reported cash needs matching populations. Annual cash-flow cash, cash equivalents and restricted cash opens at 30,737 million, exceeding opening balance-sheet cash and equivalents of 29,965 by 772. Both close at 29,943. Thus the cash-flow change is -794, while the balance-sheet line changes by -22; the cause and location of the earlier 772 remain unallocated. The quarter's endpoint populations agree, giving the same 356 million increase (Apple Inc., 2024a, 2025c, balance and cash-flow statements).

Valuation risk is likewise distinct from a payment. The notes describe Level 2 prices or models with significant observable inputs and counterparties that *generally* have high credit ratings. Annual Item 7A reports that a hypothetical 100-basis-point rate rise at all tenors would reduce investment fair value, including associated derivatives, by 2,755 million at September 2024 and 3,089 million at September 2023 (Apple Inc., 2024a, printed p. 27). Those are dated issuer scenarios, not cash outflows or loss probabilities; the issuer's valuation model has not been replicated. Interim Item 3's no-material-change referral does not issue a new December sensitivity.

28.13.3 Future receipts need customers and collection dates

Receivables may turn into cash later, but their movement does not identify collections. Annual net trade receivables rise by 3,902 million, while the indirect cash-flow adjustment is $-3{,}788$. Their sum leaves an unallocated 114 million difference. In the quarter, the balance change of $-3{,}771$ and adjustment of $+3{,}597$ leave -174. Vendor non-trade changes match the opposite indirect adjustments exactly: $+1{,}356/-1{,}356$ annually and $-3{,}166/+3{,}166$ quarterly. Even that match supplies neither gross sales nor total collections (Apple Inc., 2024a, 2025c, balance and cash-flow statements). Sales, allowances, currency and noncash movements still need their own evidence before receipts can be dated.

Concentration describes who might pay, but overlapping descriptions cannot be summed as separate exposures. Carriers account for 41%, 38% and 33% of total trade receivables at the three successive dates. At December the note also identifies two customers individually accounting for at least 10%, with shares of 18% and 11%. It does not say whether they are carriers (Apple Inc., 2024a, Note 4, printed p. 38) (Apple Inc., 2025c, Note 4, printed p. 9). Let A be those two customers, B the carrier group, and $s(S)$ the share of total trade receivables owed by customers in a set S. Each customer belongs wholly to a set or does not. Treat the displayed percentages as exact centres for this illustration. Inclusion-exclusion gives

$$s(A \cup B) = s(A) + s(B) - s(A \cap B) = 29\% + 33\% - s(A \cap B).$$

If both large customers are carriers, their 29% is already inside the 33% carrier share, so the union is 33%. If neither is a carrier, the union is 62%. For a concrete construction, allocate receivables across eleven hypothetical customers with shares $18, 11, 9, 9, 8, 7, 4, 9, 9, 8, 8$ percent. These sum to 100 and only two reach 10%. Carriers can be the 18, 11 and 4 customers, totalling 33, or the first 9, 9, 8 and 7 customers, also totalling 33. The same marginal disclosures therefore permit both endpoint unions. If only one large customer is a carrier, the union is 44% or 51%; exact-centre membership does not permit an arbitrary continuously divisible overlap. These are conditional percentage examples, not recovered customer identities or probabilities of loss. The note's trade denominator is not automatically the balance sheet's *net* receivable line, so multiplying it by that line would add an unsupported measurement assumption. Required credit support or collateral applies to *certain* customers; it does not establish universal recovery.

Vendor non-trade receivables have a separate economic origin and denominator. Apple buys components from suppliers and sells them to manufacturing vendors that make subassemblies or final products for Apple. Those component sales are excluded from product net sales; any gain reduces product cost of sales when Apple sells the related final products. The two largest reported vendor shares are 44% and 23% at September 2024, and 43% and 24% at December, each pair totalling 67% on its own dated vendor population (Apple Inc., 2024a, 2025c, annual and interim Note 4). Equal combined shares do not establish unchanged counterparties, component cohorts or collection risk. A funding forecast consequently needs settlements and restrictions on assets alongside payment dates on obligations. Keeping reported assertions distinct from those missing facts is the starting point for representing the company's historical record.

28.14 Sales geography is not a map of costs or assets

The liquidity question leaves a related operating question. If demand weakens in one place, which part of Apple's sales and profit is exposed? Apple reports five geographic segments, but a segment is a management reporting division, not a list of incorporated companies, factories or product lines. Reading its two segment notes as though those were the same partitions would make a regional scenario appear more precise than the filing permits.

28.14.1 What the regional labels measure

Apple's fiscal 2024 Note 13 names Americas, Europe, Greater China, Japan and Rest of Asia Pacific. Americas covers North and South America. Europe also includes India, the Middle East and Africa; Greater China includes China mainland, Hong Kong and Taiwan. Apple says regional net sales are *generally* based on customer location and sales through retail stores in those geographic locations. Its segments sell similar products and services, but are managed separately around customers, distribution partners and market conditions (Apple Inc., 2024a, Note 13, printed pp. 46–47). An India-customer sales shock therefore belongs with Apple's

Europe segment under this disclosure's general attribution rule, even though India is in Asia. Neither this mapping nor the label "Greater China" locates the legal seller, the manufacturing partner or the cash-receiving entity.

The prior revenue reading established that product and regional net sales are two classifications of the same 391,035 million dollars in fiscal 2024 and 124,300 million in the first quarter of fiscal 2025; the totals cannot be added. The full Note 13 and interim Note 10 tables confirm five reported regional sales and operating-income pairs in each dated column. For example, Greater China has fiscal 2024 sales of 66,952 million and segment operating income of 27,082 million; in the quarter ended 28 December 2024 the amounts are 18,513 and 8,160 million (Apple Inc., 2024a, Note 13, printed p. 47) (Apple Inc., 2025c, Note 10, printed p. 11). The quarter is a distinct three-month period, not a fourth of the fiscal-year observation. Apple's fiscal calendar can contain 53 weeks, as it did in fiscal 2023, and its interim MD&A describes historically higher first-quarter sales associated in part with the holidays (Apple Inc., 2024a, 2025c, Note 1 and MD&A).

Management attributes the annual Greater China sales decline mainly to lower iPhone and iPad sales and reports an unfavourable renminbi effect. For the first quarter of fiscal 2025 it attributes the decline mainly to iPhone, while describing a favourable currency effect (Apple Inc., 2024a, 2025c, annual and interim segment operating performance). These are dated issuer explanations, not measured contributions of each driver or an estimated response equation. A future region-specific demand or exchange-rate path needs product, currency and contract information that the regional totals alone do not supply.

28.14.2 Why segment income cannot be treated as fully allocated profit

Apple defines each segment's operating income as third-party sales less the related cost of sales and directly attributable operating expenses. The annual total of segment operating income is 162,044 million, but the consolidated operating statement reports 123,216 million. Note 13 supplies the bridge: 31,370 million of research and development and 7,458 million of other corporate expenses, net, are subtracted. The latter category includes general and administrative compensation, various nonrecurring charges and other separately managed costs. The reported relation is

$$162{,}044 - 31{,}370 - 7{,}458 = 123{,}216 \quad \text{million dollars, fiscal 2024.} \tag{28.23}$$

The first-quarter fiscal 2025 bridge has the same form: $53{,}174 - 8{,}268 - 2{,}074 = 42{,}832$ million. Both endpoints agree with the consolidated statements of operations (Apple Inc., 2024a, statements of operations and Note 13, printed pp. 29 and 47) (Apple Inc., 2025c, statement of operations and Note 10, printed pp. 1 and 11). The annual "other corporate" amount is not identical to the 26,097 million selling, general and administrative expense line in the consolidated statement. Equating the two would change the reported classification.

Equation (28.23) identifies the company-wide cost pool, not how it would be attributed to regions. To see the difference, take a hypothetical allocation of the 38,828 million total corporate bridge across Americas, Europe, Greater China, Japan and Rest of Asia Pacific of respectively 14,000, 10,000, 5,000, 4,000 and 5,828 million. Moving 1,000 from Americas to Greater China produces a second allocation of 13,000, 10,000, 6,000, 4,000 and 5,828 million. Both are nonnegative, sum to the same corporate total and reproduce the consolidated 123,216 million. Yet Greater China's hypothetical profit after these costs is either $27{,}082 - 5{,}000 = 22{,}082$ or $27{,}082 - 6{,}000 = 21{,}082$ million. Apple does not report either fully allocated figure, and the filing supplies no rule preferring one allocation. These constructions establish numerical nonidentification within the stated nonnegative illustration; they do not assert that all real cost allocations are nonnegative or that either construction describes Apple's internal books.

Forecasting regional profit also requires the behaviour of directly attributable and corporate costs under the scenario.

28.14.3 Country sales and long-lived assets answer different questions

The annual note separately presents country sales for fiscal 2024, 2023 and 2022 and country long-lived-asset stocks at the ends of fiscal 2024 and 2023. It separately reports countries individually accounting for at least 10% of the respective totals and groups the rest; that selection rule does not make "Other countries" a single location. The 2024 displayed amounts illustrate the different populations:

Country class	2024 net sales	Long-lived assets
	USD millions	
U.S.	142,196	35,664
China, including Hong Kong and Taiwan	66,952	4,797
Other countries	181,887	5,219
Total	391,035	45,680

The China country-sales row equals Greater China segment sales in all three annual columns. The note's China footnote includes Hong Kong and Taiwan, so the equality is a useful check on these displayed populations. It does not turn that row into a legal-entity or production-location ledger. The U.S. country-sales row is below Americas segment sales by 24,849 million in 2024; assigning that difference to particular countries requires the precise country-to-region attribution and customer facts. Europe's segment label cannot be substituted for a European-country total because it includes India, the Middle East and Africa (Apple Inc., 2024a, Note 13, printed pp. 46–47).

The 45,680 million long-lived-asset total equals net property, plant and equipment at the 2024 balance date, and the 2023 totals both equal 43,715 million at their displayed precision (Apple Inc., 2024a, Notes 5 and 13, printed pp. 39 and 47). The equality is a checked comparison, not a definition of every asset's nature, owner or productive capacity. Country asset stocks are measured at an instant, while sales cover a year. Further, Apple says the information supplied to its chief operating decision maker for segment assessment excludes asset information (Apple Inc., 2024a, Note 13, printed p. 46). Thus neither a segment return on assets nor an allocation of production capacity to the five sales regions is identified by the disclosed tables. The statement that hardware is made substantially by outsourcing partners located primarily in China mainland, India, Japan, South Korea, Taiwan and Vietnam concerns another population, and does not repair that missing asset/segment mapping (Apple Inc., 2024a, Note 12, printed p. 46).

28.14.4 The same totals permit different product exposure

Note 2 reports 201,183 million of fiscal 2024 iPhone sales. Its four other product and service rows sum to 189,852 million. Along with Greater China's 66,952 million regional total, these are *margins* of an unreported product-by-region table. Let z be iPhone sales assigned to Greater China, with all amounts in millions of dollars at the displayed centres. The four cells for Greater China versus the other four regions must be

	iPhone	Other products and services
Greater China	z	$66{,}952 - z$
Other four regions	$201{,}183 - z$	$122{,}900 + z$

(28.24)

The row and column sums in (28.24) hold for any nonnegative cells; they do not select z. For example, $z = 36{,}000$ and $z = 38{,}000$ both meet the displayed numeric margins. For an explicit extension, let S_r be each other region's reported sales; those four regions sum to $391{,}035 - 66{,}952 = 324{,}083$. Assign each $(201{,}183 - z)S_r/324{,}083$ of iPhone sales; these amounts sum to the remaining iPhone total and are below S_r. Then split every region's remaining sales across the four non-iPhone categories in their disclosed company-wide proportions. This produces all five product/service and five region margins exactly. These proportional splits are mathematical witness constructions, not issuer allocations. The source adds that the product mix was *generally* consistent across segments, with a *moderately* higher iPhone share in Greater China. Both numerical examples put Greater China's iPhone share above the combined other-region share. The qualitative words have no disclosed tolerance, so neither matrix is certified as a complete interpretation of that statement (Apple Inc., 2024a, Note 2, printed pp. 35–36). The interim note repeats the qualitative comparison for its own dated quarter (Apple Inc., 2025c, Note 2, printed p. 6); it does not supply the missing cross-classified cells.

If a scenario *assumes* a 10% decline in Greater China iPhone sales alone, the two illustrative fiscal 2024 matrices imply 3,600 and 3,800 million less sales, respectively. The scenario percentage is a teaching assumption, not an estimated shock; neither sales change determines profit, cash flow, currency translation or future behaviour. The gap between those two conditional answers comes from the unobserved joint allocation, even before cost responses are considered. A reproducible model should retain the reported marginals, the qualitative mix statement, any admitted cross-classification and its source, and the unresolved allocation when the evidence does not determine it.

Finally, the annual and first-quarter MD&A state Apple's then-planned fourth-quarter fiscal 2025 adoption of ASU 2023-07 with retrospective transition (Apple Inc., 2024a, 2025c, Recent Accounting Pronouncements). Those were dated plans, not the later disclosure itself; they do not establish the standard's operative provisions without inspection of that authority. The retained fiscal 2025 annual filing now supplies comparative 2024 segment expenses, examined in Section 28.17. A historical query keeps this later presentation separate from the older filing's content. The regional calculations here are checked source-bound comparisons and conditional witnesses, not a reconstructed Apple segment ledger or a forecast-ready regional model.

28.15 From geographic sales to legal entities

A regional sales forecast still leaves open which company sells the goods, owns the inventory or owes the supplier. Those identities matter when a scenario changes a particular entity's tax, funding or supply arrangements. Apple's geographic segments cannot answer that question: their sales are generally attributed by customer location and retail-store geography. The consolidation policy and subsidiary exhibit supply a different kind of information, about the companies represented in the group statements.

Annual Note 1 says that the consolidated financial statements include the accounts of Apple Inc. and its wholly owned subsidiaries. The filing also defines "Company" and "Apple" collectively in that way, unless otherwise stated. Interim Note 1 repeats the wholly-owned scope for its condensed statements and requires them to be read with the fiscal 2024 annual statements and notes (Apple Inc., 2024a, definition of Company and Note 1, printed pp. 1 and 34) (Apple Inc., 2025c, Note 1, printed p. 5). This is positive evidence about the reported consolidation perimeter. It is not a set of separate parent-company accounts, nor does it identify every legal entity or ownership link within that perimeter.

Exhibit 21.1 names the following subsidiaries and their jurisdictions of incorporation. The labels

below are transcribed from the exhibit, including the separate U.S. state jurisdictions (Apple Inc., 2024a, Exhibit 21.1, retained PDF p. 114).

Subsidiary	Jurisdiction of incorporation
Apple Asia Limited	Hong Kong
Apple Asia LLC	Delaware, U.S.
Apple Canada Inc.	Canada
Apple Computer Trading (Shanghai) Co., Ltd.	China
Apple Distribution International Limited	Ireland
Apple India Private Limited	India
Apple Insurance Company, Inc.	Arizona, U.S.
Apple Japan, Inc.	Japan
Apple Korea Limited	South Korea
Apple Operations International Limited	Ireland
Apple Operations Limited	Ireland
Apple Operations Mexico, S.A. de C.V.	Mexico
Apple Pty Limited	Australia
Apple Sales International Limited	Ireland
Apple South Asia (Thailand) Limited	Thailand
Apple Vietnam Limited Liability Company	Vietnam
Braeburn Capital, Inc.	Nevada, U.S.
iTunes K.K.	Japan

The exhibit expressly limits the list. Its footnote states: "Pursuant to Item 601(b)(21)(ii) of Regulation S-K, the names of other subsidiaries of Apple Inc. are omitted because, considered in the aggregate, they would not constitute a significant subsidiary as of the end of the year covered by this report." This preserves Apple's stated reason and date for the omission; it does not establish a numerical significance test for this model. The eighteen rows cannot be treated as an exhaustive group roster or as proof that an omitted subsidiary is irrelevant to every entity-specific question.

The same filing makes that incompleteness concrete. Its tax-risk discussion mentions Ireland and Singapore, "where a number of the Company's subsidiaries are organized," yet no Exhibit 21.1 row has Singapore as its jurisdiction (Apple Inc., 2024a, Risk Factors, printed p. 16). The evidence supports subsidiaries organized there while leaving their names and ownership paths unresolved. Creating a company named "Apple Singapore" would turn a geographic reference into an invented identity.

28.15.1 Membership does not determine the ownership path

Wholly-owned membership and immediate ownership answer different questions. For a simple hypothetical example, a subsidiary could be owned directly by the parent, or through another subsidiary that the parent wholly owns. If every link is wholly owned, both arrangements preserve ultimate wholly-owned membership while producing different immediate parents. This logical example does not select either arrangement for any company in the table. Neither the similar names of Apple Operations Limited and Apple Operations International Limited nor their common incorporation jurisdiction establishes an ownership link between them.

A model can therefore retain each named subsidiary, its reported jurisdiction and the general consolidation assertion without drawing unsupported edges. It should record an immediate parent, share class, ownership interest and effective interval only when the evidence identifies them. The September 2024 exhibit is not a control history for earlier years or an updated list at 28 December 2024. Repetition of the general policy in the interim note does not show that every subsidiary and ownership path remained unchanged. Determining the dated group

requires incorporation, acquisition, disposal, reorganisation and control evidence as relevant; it cannot be recovered from the fiscal calendar alone.

28.15.2 Selling, manufacturing and owning are separate relationships

Apple India Private Limited illustrates the distinction between incorporation and sales attribution. The exhibit places its incorporation in India, while Apple's segment definition places India in Europe. Under the general customer-location rule, an India-customer sale belongs in that sales region; neither statement identifies its legal seller or assigns all India sales to that subsidiary. Even a complete subsidiary list would not supply the missing seller-by-customer-region relationship.

The Business discussion also separates distribution channels: direct sales through Apple's stores and sales force accounted for 38% of fiscal 2024 net sales, and indirect channels for 62%. The latter include third-party cellular carriers, wholesalers, retailers and resellers (Apple Inc., 2024a, Markets and Distribution, printed p. 2). These are sales channel shares, not ownership shares or counts of companies. A trading relationship does not by itself establish that a counterparty belongs in the consolidated group.

Production adds a further distinction. Apple says substantially all its manufacturing is performed in whole or in part by outsourcing partners, primarily in China mainland, India, Japan, South Korea, Taiwan and Vietnam. It has also invested in manufacturing process equipment, much of which is held at certain outsourcing partners, and made prepayments under long-term supply agreements (Apple Inc., 2024a, Risk Factors, printed p. 8). Its MD&A describes partners manufacturing subassemblies and performing final assembly and testing, alongside purchases from component suppliers (Apple Inc., 2024a, Manufacturing Purchase Obligations, printed p. 25). Equipment custody, a supply contract and corporate control are thus separate relationships. A disruption at a partner may affect product supply, equipment recoverability and prepayments without supplying evidence that the partner should be consolidated. The affected assets, contracts and counterparties must be identified before amounts can be assigned to that scenario.

28.15.3 Tax and currency queries need facts beyond incorporation

The jurisdiction column also leaves entity tax residence and functional currency unresolved. Functional currency is the currency in which an entity measures its accounts; it is a different question from the dollars in which Apple presents its group statements or the currency of a particular invoice. The group foreign-exchange discussion describes Apple as a net receiver of currencies other than the U.S. dollar, with pricing and hedging responses to exchange-rate movements. It supplies no entity-by-entity functional-currency assignment (Apple Inc., 2024a, Foreign Exchange Rate Risk, printed p. 27). An Ireland incorporation row therefore does not justify entering euros as the entity's functional currency, just as the India row does not establish rupees.

Tax evidence has its own dates and units of analysis. Annual Note 7 describes the historic tax allocation of profits of Irish branches of two subsidiaries without naming those subsidiaries in that passage. The MD&A describes the resulting group payment obligation and restricted escrow (Apple Inc., 2024a, Note 7, printed pp. 39–40; State Aid Decision Tax Payable, printed p. 26). The four Ireland rows in Exhibit 21.1 do not identify which entities had those branches, their historic tax residence or the jurisdictional tax bases used in current accounts. A name match or country match would bypass the legal and temporal facts needed for the tax calculation.

These distinctions determine what a supported query can return. The retained sources answer

questions about the stated consolidated scope and the named subsidiaries' incorporation jurisdictions. Entity-specific tax, currency and intercompany questions require additional records: legal counterparties, dated ownership and control, compatible entity books, functional currencies, tax bases and matched transactions. Group eliminations then require the applicable accounting methods as well as those facts. The later IFRS group examples explain their own assumptions; they do not establish the operative US GAAP treatment for Apple. Keeping those dependencies explicit preserves useful source knowledge while leaving the unanswered entity questions open.

28.16 Legal proceedings beyond the booked liability

Knowing the companies within a group does not identify all the events that could change their earnings. A legal proceeding can require a payment, but it can also change how a company sells its products or earns licensing revenue. Apple's reported App Store injunction illustrates the second channel: it concerns developers' external purchase links on the U.S. storefront, rather than a disclosed amount of damages. A forecast that represents legal exposure only as a liability balance would leave that operating consequence outside the model.

The annual Item 3 and interim Part II Item 1 each discuss Digital Markets Act investigations, the Department of Justice lawsuit, Epic Games and other legal proceedings. These are dated issuer reports, with different kinds of statements inside each matter: allegations, reported decisions, pending motions, management's position and possible consequences. Keeping those distinctions allows a model to use what is reported without deciding the case itself (Apple Inc., 2024a, Item 3, printed p. 18) (Apple Inc., 2025c, Part II, Item 1, printed p. 18).

28.16.1 A loss assessment answers a narrower question than exposure

Annual Note 12 and interim Note 9 give the same qualified assessment: "In the opinion of management, there was not at least a reasonable possibility the Company may have incurred a material loss, or a material loss greater than a recorded accrual, concerning loss contingencies for asserted legal and other claims" (Apple Inc., 2024a, Note 12, printed p. 46) (Apple Inc., 2025c, Note 9, printed p. 11). The sentence identifies whose assessment it is, concerns an incurred material loss, and allows for a recorded accrual. It does not disclose a total legal accrual, allocate an accrual to one of the named matters, or assert that all exposure is zero. Nor does it supply a numerical probability or materiality threshold.

The annual MD&A separately reports Apple's recognition policy: the company records a liability when it is probable that a loss has been incurred and the amount is reasonably estimable, a determination requiring significant judgment (Apple Inc., 2024a, Legal and Other Contingencies, printed p. 26). That policy explains why a pending proceeding and a recognised liability are different facts. It is issuer-policy evidence; applying the operative US GAAP requirements to a particular case still needs the relevant authority and case facts. An allegation cannot supply those facts, and a possible change in future sales is not itself the incurred loss described by the policy.

Settlement adds a further distinction. Annual Item 3 says certain matters settled during the fourth quarter of fiscal 2024 without an individually or aggregate material impact on financial condition or operating results. The interim passage makes that statement about the first quarter of fiscal 2025. These are two period-specific assertions, not the same settlement observation copied forward. Neither passage identifies the matters, settlement amounts, payment dates or earlier accruals. Both retain the warning that adverse resolutions above management's

expectations could materially affect the reporting period. A payment schedule or an expense roll-forward therefore remains unresolved even though the settlement statements themselves are known.

28.16.2 The remedy, jurisdiction and procedural stage matter

The Digital Markets Act discussion reports two investigations opened by the European Commission on 25 March 2024. Article 5(4) concerns developers' communication and promotion of offers and conclusion of contracts with end users; Article 6(3) concerns default settings, app uninstallation and a browser choice screen on iOS. The Commission's 24 June announcement is described as preliminary findings alleging an Article 5(4) breach and the opening of a third investigation into new contractual requirements for third-party developers and app marketplaces. Apple states that it believes it complies and continues to change its compliance plan in response to Commission engagement. The preliminary allegation and Apple's position remain separate assertions.

Both filings say that, if the Commission makes a final violation determination, it can issue a cease and desist order and may impose fines up to 10% of Apple's annual worldwide net sales. They also say an order would apply immediately during an appeal unless a stay is granted. This preserves a conditional consequence, its stated sales base and the stay qualification. The percentage is not a fine already imposed, an expected loss or a booked liability. Even a conditional fine calculation would need the applicable operative law, relevant sales period and base, and assumptions about the determination and remedy. None is supplied by multiplying a convenient annual sales figure by the reported cap.

The Department of Justice discussion reports the civil antitrust lawsuit filed on 21 March 2024 in the District of New Jersey by the DOJ and state and district attorneys general. It alleges monopolization or attempted monopolization in the markets for "performance smartphones" and "smartphones." The DOJ seeks equitable relief; related state and federal civil litigation seeks monetary damages and other nonmonetary relief. Apple states that it has substantial defenses and intends to defend itself. Treating all these proceedings as one quantified damages claim would erase both the different requested relief and the absence of disclosed amounts.

Epic's case contains a reported court finding as well as unresolved procedure. Apple says the Northern District of California court found certain App Store Review Guidelines provisions to violate California's unfair competition law and issued an injunction against prohibiting developers' external purchase links. The stated scope is the U.S. storefront of the iOS and iPadOS App Store. Apple reports implementing a compliance plan and filing its statement on 16 January 2024. Epic's motion disputing that plan and seeking enforcement is reported as pending, opposed by Apple; Apple also reports filing a motion to narrow or vacate the injunction on 30 September 2024. These passages do not establish that the disputed plan was judicially accepted or quantify its sales effect. An EU-wide scenario cannot inherit the U.S. injunction merely because both matters concern the App Store.

The interim bodies repeat the substantive DMA, DOJ and Epic account, with acronym or trademark presentation differences. Repetition is evidence of what Apple reported in each filing, not proof that every intervening legal fact was unchanged. The September 30 motion also falls after the annual September 28 period-end. Event date, reporting period and the date on which the report became available must therefore remain distinct in a historical information query.

28.16.3 A partner's remedy can change Apple's revenue

The risk discussion connects these matters to operations. Both filings report changes to U.S. developer communications and EU changes to distribution, payment processing, fee structures and developer tools. They also describe licensing revenue from Google and other search providers, whose arrangements are subject to investigations and proceedings (Apple Inc., 2024a, Risk Factors, printed pp. 14–15) (Apple Inc., 2025c, Part II, Item 1A, printed pp. 19–20). The interim adds that Google was found to have violated U.S. antitrust laws on 5 August 2024 and that the DOJ proposed remedies ranging from restrictions on licensing agreements to prohibiting commercial search-distribution terms with Apple for up to ten years. Apple says that, if implemented, those remedies could materially affect its ability to earn that licensing revenue. The finding is an earlier event newly described in this retained comparison; the remedies are proposed, not reported here as imposed.

That channel differs from an Apple fine. The affected economic state would include the search agreement, its revenue terms, duration, geography and substitution possibilities. Likewise, an App Store remedy requires the affected storefront, developer terms, fees, customer purchases and behavioural response. The filings identify these channels without supplying the contract economics needed to calculate them. They do not justify applying one loss percentage to all Services revenue or treating every customer as affected.

A conditional calculation should consequently specify which matter and remedy it assumes, when that remedy takes effect, which contracts it changes and how the parties respond. A monetary-loss calculation additionally needs recognition facts, any opening accrual and the settlement and payment schedule; expense recognition and cash payment can occur in different periods. The State Aid tax charge and escrow settlement, developed in Section 35.7, retain their own tax records and must not be counted again as a DMA loss. Until the missing facts and applicable treatments are supplied, the model can retain the reported proceedings and affected queries as unresolved, without assigning probabilities or producing numerical journals. This preserves useful knowledge about future exposure while keeping the distinction between a known assertion and a supported financial consequence.

28.17 When later disclosures change the historical query

A later filing can reveal more about an earlier period without identifying a different economic history. For example, a new segment-expense disclosure may make a previously hidden cost visible, while the transactions that generated that cost remain the same. To use that information correctly, a model must retain the original report, the later presentation and the date at which the additional detail became available. Apple's announcement and its later segment tables show the difference between a planned disclosure and the comparative information eventually reported.

Annual and interim MD&A describe planned changes to tax and segment disclosures; the interim adds expense disaggregation and a subsequent effective-date clarification (Apple Inc., 2024a, Recent Accounting Pronouncements, printed p. 26) (Apple Inc., 2025c, Recent Accounting Pronouncements, printed p. 17). The following table preserves Apple's stated adoption plans. The quarters are fiscal, and the descriptions are issuer reports of the updates, rather than an independent determination of their operative requirements.

Update	Reported disclosure change	Apple's stated plan
ASU 2023-07	Segment expenses and information about the chief operating decision maker	Fiscal 2025 Q4, retrospective
ASU 2023-09	Income-tax reconciliation and taxes-paid detail	Fiscal 2026 Q4, prospective
ASU 2024-03, clarified by 2025-01	Expense components and selling-expense amount and definition	Fiscal 2028 Q4, prospective

The tax discussion says the rate reconciliation will include additional information and detail for items meeting a quantitative threshold. Taxes paid will be disaggregated into federal, state and foreign taxes, with further detail for significant individual jurisdictions. It gives neither the threshold value nor the future jurisdictional amounts. Those cash-tax populations also remain distinct from tax expense, payable and deferred balances; their reconciliation is developed in Section 35.7. An announced disclosure change cannot be used to allocate an old tax total among jurisdictions by assumption.

The earlier segment discussion concerns expenses that are significant and regularly provided to the chief operating decision maker, together with that person's title, position and use of segment profit or loss in assessing performance and allocating resources. That anticipated disclosure could answer questions about the information used by management. It did not yet identify the regional allocation of corporate expenses or supply segment assets. The adoption sentence was a dated plan at that information cutoff. The retained fiscal 2025 annual report now lets us check what followed.

Apple reports that it adopted ASU 2023-07 for its 2025 annual period using a retrospective transition method (Apple Inc., 2025a, Note 1). Its Note 13 presents 2024 and 2023 comparative segment expenses. Take the Americas in fiscal 2024. The earlier Note 13 reports net sales of 167,045 million and operating income of 67,656 million, but no separate regional cost of sales or selling-and-marketing amount (Apple Inc., 2024a, Note 13, printed p. 47). The later comparative gives 89,587 million of cost of sales and 9,802 million of selling and marketing for that same region and fiscal year (Apple Inc., 2025a, Note 13, printed p. 47). The displayed relationship is

$$167{,}045 - 89{,}587 - 9{,}802 = 67{,}656 \quad \text{(USD millions)}.$$

The 2025 table gives corresponding expense rows for the other four regions. Together, their disclosed selling-and-marketing expenses total 18,639 million. The five operating incomes still sum to 162,044 million; after the displayed corporate research and development of 31,370 million and general and administrative amount of 7,458 million, the company reports the same 123,216 million of operating income. The 2024 report called its 7,458 million line "other corporate expenses, net" and qualified its contents. The identical displayed amount is not, by itself, proof that the two labels have identical scope. Both original and later assertions retain their source identities.

This later presentation answers a prior-period disclosure question without identifying new fiscal-2024 transactions or creating a correction journal. Reading the named 2024 source leaves the regional expense detail unresolved; reading the later source supplies its 2025 comparative assertions. An investor-information view additionally needs evidence that the exact chosen source version was publicly available by its requested cutoff. The accepted corpus and filing-event metadata do not prove that for the retained 2024 PDF or the later issuer-hosted HTML. Nor do the new regional expense rows reveal the cross-classified product-by-region sales, entity cash allocation, or behavioural response needed by the earlier regional stress example.

The interim expense passage reports ASU 2024-03's issuance in November 2024 and ASU 2025-01's clarification in January 2025. It describes purchases of inventory, employee compensation, depreciation and intangible-asset amortization, as applicable, within certain expense captions,

a qualitative description of the remaining amounts, and both the amount and the company's definition of selling expenses. This describes disclosure populations; it does not provide their values or make selling expenses identical to the entire selling, general and administrative line. Purchases included in an expense disclosure likewise cannot be identified with cash paid to suppliers without the relevant timing and population facts.

The observed segment comparatives show why the two source-content views must remain separate: the original cannot be overwritten even when a later filing reports the same amounts with more detail. This comparison checks the affected periods, dimensions, displayed units and issuer labels; it does not use the phrase "retrospective transition" as an amount. The prospective tax and expense plans still supply no backward allocation of missing components. Their exact transition requirements need the operative updates and any applicable amendments.

The January clarification illustrates the timing problem directly. It occurs after the interim period ended on 28 December 2024 but is described in the interim report. An event can therefore belong to the information in a report without occurring inside its reporting period. A historical forecast cutoff needs evidence of availability and the exact retained version, not merely the period-end or the update's issuance month. Likewise, the earlier announced segment adoption did not become an observed adoption until the later filing was checked. The tax and expense plans remain future claims until their own later evidence is inspected.

28.17.1 Unchanged critical estimates do not mean constant amounts

Annual Critical Accounting Estimates says management makes judgments, assumptions and estimates affecting reported amounts, using historical experience and other assumptions it considers reasonable to judge carrying values. Its uncertain-tax discussion says reserves can be adjusted as facts and circumstances change, and that final outcomes may differ from those reserves. Its legal-contingency policy is the incurred-loss and estimability condition examined above (Apple Inc., 2024a, Critical Accounting Estimates, printed p. 26). These are descriptions of judgment and policy, not a disclosed numerical estimation model or an assurance that the estimates will prove accurate.

The interim references both annual and interim Note 1 and reports no material changes to critical accounting estimates since the annual filing (Apple Inc., 2025c, Critical Accounting Estimates, printed p. 17). That statement preserves the annual dependency and its materiality qualifier. It cannot make every carrying amount, reserve or underlying assumption exactly constant, and it supplies no numerical tolerance for a comparison. A model should retain the reported policy and qualified continuity assertion alongside the actual dated balances, then request the missing estimate or transition facts when the query needs them. These estimate passages supply policy and continuity assertions; they do not establish numerical stability, current compliance or the tax and expense details still absent from these sources.

28.18 Claims against another party can still affect Apple

The estimate policy leaves a further question: whose claim can give rise to a cost for Apple? A company can face exposure through an agreement even when the claim is against someone else. Apple's annual risk discussion says its agreements sometimes include indemnification provisions that can subject it to costs and damages in the event of a claim against an indemnified third party (Apple Inc., 2024a, Risk Factors, printed p. 13). The direction matters: the passage describes a possible obligation of Apple arising from another party's claim, rather than compensation owed to Apple. A model limited to proceedings naming Apple as defendant would miss that

contractual channel.

The statement supplies neither an indemnified party nor a particular agreement. It also leaves the covered conduct, trigger, duration and limit unknown. To connect a third-party claim to a dated obligation, the model would need those terms and the facts that activate them. The existence of indemnification language somewhere in Apple's agreements cannot establish that a named case is covered, that a payment is due, or that a liability has been recognised. Those facts remain distinct from the management loss assessment examined in Section 28.16.

The full annual passage also describes unresolved ordinary-course claims and investigations, an increase over time in their number and alleged magnitude, and patent claims concerning cellular-enabled products. Apple reports defending infringement actions in U.S. and international courts, whose plaintiffs frequently seek broad injunctions and substantial damages. These are issuer statements about claims and requested relief. They do not identify a complete matter population, decide the infringement allegations or quantify the payments and product restrictions that will result.

Defense is a separate source of cost. The filing states, "Regardless of the merit of particular claims, defending against litigation or responding to government investigations can be expensive, time-consuming and disruptive to the Company's operations." Thus a scenario in which Apple successfully resists a claim need not be a scenario with no defense effort or disruption. The model therefore needs the response services, their amounts and dates, and related payment obligations. The damages outcome cannot identify those expenses or payments.

Settlement can alter recurring operations as well as resolve a dispute. Apple says it may enter agreements or other arrangements to settle litigation, without assurance of acceptable terms or that litigation will not occur. Those agreements can increase cost of sales and operating expenses, change business practices and limit products and services. A conditional settlement therefore needs its payment terms and its continuing effects on royalties, costs or offerings. An adverse resolution can additionally involve monetary damages, disgorgement, corporate measures or an injunction. The annual passage explicitly includes a resolution against an indemnified third party above management's expectations. Neither that possibility nor the separately reported quarterly settlements supplies a contract, amount or probability for a forecast.

28.18.1 Insurance describes a possible recovery, not a net loss

Apple says it maintains insurance for certain types of claims, while warning that coverage may be insufficient for all losses or all claim types. The qualification leaves both the covered population and the amount recoverable unresolved. A model cannot deduct an assumed insurance payment from damages or an accrued liability merely because insurance exists. It needs the policy period, insured party and beneficiary, covered event, exclusions, deductible, limit, insurer and recovery terms, together with facts about the claim. The applicable recognition and presentation treatment must also be established before a recovery amount becomes an accounting entry. The passages identify none of those amounts or terms.

Other annual insurance statements concern different exposures. The business-interruption discussion warns of recovery expenditures and lost sales, especially when critical components have single or limited sources, while saying insurance for certain types of losses may be insufficient (Apple Inc., 2024a, Risk Factors, printed p. 6). The data-security discussion says coverage is intended for certain aspects of those risks and may not cover all losses or claim types (Apple Inc., 2024a, Risk Factors, printed p. 12). The trade receivable discussion instead says a substantial majority of outstanding trade receivables lack collateral, third-party bank support or financing arrangements, or credit insurance (Apple Inc., 2024a, Risk Factors, printed p. 16). These

statements cannot be combined into one insured share of Apple's exposures. The receivable qualifier is qualitative; it identifies neither every uncovered customer nor a percentage. Insurance bought by Apple also does not make these exposures an insurer's issued-contract measurement problem.

The interim risk preamble incorporates the annual risks and states that there have been no material changes except those set out below (Apple Inc., 2025c, Part II, Item 1A, printed p. 19). It preserves the annual dependency and an express exception; it does not assert identical policies, claims, coverage or loss amounts at the later date. Keeping the original risk, the later qualified statement and the separately reported updates allows a historical query to retain what was actually disclosed. The resulting account connects direct proceedings, contract indemnities, defense, settlement, operating restrictions and possible recovery without collapsing them into a single loss balance. Calculating their consequences still requires the dated contracts, claim facts and accounting treatment for each channel.

28.19 A regulation can change the feasible operating plan

A legal exposure need not begin with a claim or end with a payment. An import restriction can prevent a product from reaching a customer, leaving the company to change its supply route, offering or production plan. Apple's annual and interim general-regulation risk both say that its products could be banned, delayed or prohibited from importation despite procedures intended to satisfy regulatory requirements. They also warn that measures may arrive with little or no advance notice and that Apple may be unable to mitigate all adverse impacts (Apple Inc., 2024a, Risk Factors, printed pp. 13–14) (Apple Inc., 2025c, Part II, Item 1A, printed p. 19). This describes a possible constraint on operations, separate from the damages, defense and insurance channels examined above.

The two reports preserve a broad list of regulated subjects: among them antitrust, privacy, consumer protection, intellectual property, financial services, trade, foreign exchange controls, cash repatriation, tax and employment. The interim adds national security to that list. It also expands the discussion of new and changing laws and regulations to name executive orders, directives and enforcement priorities. These additions appear in the cost-and-business-change, interpretation and specialized-application passages. They identify the scope of Apple's later risk statement without identifying a particular new instrument, effective date, jurisdiction or violation. A model cannot create a dated legal event merely because a subject is newly named in this comparison.

Both reports describe how changing requirements can increase costs, restrict a product or feature, alter its design, affect customer demand and change the business or supply chain. They describe further risks as products enter specialized applications such as health and financial services or expand the use of machine learning and artificial intelligence. The annual combines these consequences and the compliance-policy discussion in one paragraph; the interim splits and expands it. Paragraph count therefore cannot measure additional economic exposure. Conversely, the notice-and-import paragraph repeats the same substantive wording. That repetition establishes what Apple disclosed in each report, rather than identical underlying laws, products or supply routes.

The policy qualification is essential to the mechanism. Apple says it has implemented policies and procedures designed to ensure compliance, while giving no assurance that employees, contractors or agents will not violate laws, regulations or company procedures. Its separate import passage gives no assurance that programs will prevent a violation or a claim of one. Policy existence, compliance in a particular case, a claim of violation and effective mitigation

are thus different facts. Treating the reported procedures as proof of compliance would remove precisely the uncertainty the source preserves.

A conditional operating scenario should start with the actual or assumed measure and its effective date, legal scope, affected entity, product and jurisdiction. It then needs the available supply routes, inventory, lead times, customer demand and management response. For an assumed import delay, those facts determine whether delivery is postponed, rerouted or lost, and which storage, production or sourcing costs change. Recognition, payment and any later claim must follow their own facts and treatments. The filings identify these possible channels but supply neither a tariff rate nor a universal sales reduction. Their reporting dates also do not establish when a particular rule arose or when these exact report versions became available. Keeping disclosed risk, operative authority, assumed restriction and response separate lets the model retain a useful operating question without inventing its answer.

28.20 Demand, disruption and recovery follow different paths

An open supply route is useful only if a customer still wants the product. Conversely, a customer may be ready to buy while a supplier cannot deliver a critical component. The operating scenario introduced above must therefore describe both demand and the ability to serve it. Apple's annual risk discussion connects global and regional economic conditions to consumer spending, the financial condition of its business partners and its own access to finance. Its separate interruption discussion describes obstacles to manufacture, delivery and service provision (Apple Inc., 2024a, Risk Factors, printed pp. 5–6). These passages give reasons to retain several possible operating histories behind the same reported financial totals.

Apple says that sales outside the United States account for a majority of net sales and that a majority of its supplier facilities are outside the United States. The two statements concern different populations. A sales region does not locate a factory, and a factory's location does not identify the customers whose orders depend on it. The segment definitions examined earlier generally attribute sales by customer and retail-store location, while manufacturing purchase obligations describe purchases from outsourcing partners and component suppliers (Apple Inc., 2024a, Note 13, printed p. 46; Liquidity and Capital Resources, printed p. 25). Linking the two requires the actual product, component, supplier and route relationships. Geographic labels alone cannot create them.

The macroeconomic passage describes several possible effects of the same economic conditions. Recession, unemployment, inflation, tighter credit, higher interest rates and currency fluctuations can weaken consumer confidence and spending. Fiscal and monetary policy, market volatility and changes in income or asset values can also affect demand. Suppliers, manufacturers, logistics providers, distributors, carriers, other channel partners and developers may instead face financial instability, difficulty obtaining credit or insolvency. Apple further identifies trade-receivable collectibility, derivative-counterparty and financial-institution failure, limited access to new debt, reduced liquidity and declining financial-instrument fair values. The disclosure permits a scenario to connect these channels through common conditions; it supplies neither a joint probability law nor a requirement that every channel deteriorate together.

An interruption raises a different operating question: which activity becomes unavailable, and for how long? Apple's discussion includes political and trade disputes, geopolitical conflict, terrorism, natural disasters, public-health issues and industrial accidents. It says substantially all manufacturing is performed wholly or partly by outsourcing partners located primarily in China mainland, India, Japan, South Korea, Taiwan and Vietnam. Trade restrictions can raise input costs, require different suppliers or business relationships, or lead the company to cease

offering affected products, services or third-party applications. Changing the supply chain may itself be costly and disruptive, and restrictions may arrive with little notice. Apple reports that past U.S.–China tensions led to tariffs and other restrictions affecting its business. This historical assertion is distinct from its conditional warning about further escalation; a future scenario still needs a specified measure, effective date and affected trade.

The same passage describes facilities in disaster-prone locations and asserts that climate change is increasing the frequency or intensity of certain natural disasters and extreme weather. It also identifies fire, power shortages, nuclear and other industrial accidents, hostile acts, ransomware, labor disputes and health emergencies. These are Apple's stated exposures, without facility-specific hazard estimates. Its requirement that suppliers maintain safe workplaces is separate from whether an accident occurs. The company reports that major public-health issues, including COVID-19, have adversely affected demand, operations and distribution, and warns of possible future effects. Travel restrictions, freight limitations and interruptions to production ramps supply concrete mechanisms; the paragraph does not quantify an historical loss or a future duration.

Consider two explicitly assumed histories for a particular product. In one, the supply route remains available but customers reduce orders. In another, orders remain but a critical component cannot arrive. The first requires a demand response; the second requires an inventory and delivery response. They can also occur together. To compare the histories, the model needs product-level orders, available components and finished goods, lead times, contractual commitments and feasible management actions. An assumed alternate supplier must have capacity and suitable terms before it becomes an available response. A sales delay or lost order follows from those facts and assumptions; it cannot be obtained by applying one percentage to every reported balance.

Recovery adds another date. Apple warns that an interruption can require substantial time and expenditure to resume operations, with lost sales and greater consequences where critical components have single or limited sources. Note 12 separately explains that even components available from several suppliers can face industry-wide shortages and commodity-price fluctuations (Apple Inc., 2024a, Note 12, printed p. 46). Ending an assumed interruption therefore need not restore production or sales immediately. Recovery spending can precede or follow resumed deliveries. The model must retain the dates of the interruption, restored capacity, deliveries, expenditure and settlement separately; the applicable accounting treatment then determines the journals. The insurance qualification already examined in the litigation discussion does not collapse those dates into a net loss or guarantee reimbursement.

The reported receivables, investments, debt and liquidity resources constrain particular dated histories, but they do not determine these operating responses. The investment sensitivity discussed above measures a specified interest-rate effect. Chapter 36 examines the distinct debt and currency measures; none identifies the broader disruption response. The interim's incorporation of annual risk factors, except for the changes it sets out, likewise does not establish unchanged facilities, orders or funding conditions (Apple Inc., 2025c, Part II, Item 1A, printed p. 19). For the historical record, the useful distinction is now precise: retain Apple's assertions, the relationships needed to interpret them, and the analyst's scenario assumptions as separate information. Alternative demand, supply and recovery histories can then remain visible until evidence narrows them.

28.21 Product transitions change the relevant history

Keeping a supply route open does not ensure that it supplies the product customers now want. A new offering can replace demand for an earlier one while the company still holds its components, finished goods or purchase commitments. Apple's product-transition risk explicitly says that new products, services and technologies may replace existing offerings and produce lower revenues and lower profit margins (Apple Inc., 2024a, Risk Factors, printed p. 7). A model that equates a successful release with an improvement in financial performance would remove a possibility the source asks it to preserve.

The preceding competitive-market discussion explains why a transition is not merely a production date. Apple describes price competition, short product life cycles, changing standards, rapid adoption of technology and price-sensitive customers. It says that developing almost the whole product solution requires substantial research and development, without assurance of the expected returns or successful marketing. Its reported intellectual-property portfolio also does not assure protection in every jurisdiction. The discussion of competitors imitating products or infringing rights remains an issuer assertion; it does not decide a particular infringement dispute. Apple's description of minority share in three global device markets and of competitors able to sell at little or no profit supplies competitive context, rather than a numerical market share or a forecast of Apple's margins. Services face a further set of alternatives, including free content. The company's expectation of more intense competition is a dated expectation, not an observed future result.

The transition passage identifies several separate conditions for a new offering: development must succeed on time, customers must accept it, production must reach suitable quantities and costs, and application software or other third-party support must be available. New technology can create ramp-up or quality problems. Purchase commitments and inventory must also be managed in line with anticipated demand. Note 12 makes the supply condition concrete: new products often use custom components available from only one source; capacity can be constrained while yields mature or manufacturing expands, and a supplier may instead concentrate on common components (Apple Inc., 2024a, Note 12, printed p. 46). A forecast of customer acceptance cannot establish that those components will be available.

Suppose, as an illustration, that customers switch from an earlier product to its replacement. If existing components can be used in the replacement and the supplier can expand output, the transition presents one set of feasible production choices. If the components are incompatible or the new component has constrained capacity, the same assumed customer response presents another. Neither history is identified by aggregate revenue or inventory. The comparison requires the old and new product definitions, component compatibility, inventory cohorts, committed purchases, cancellation terms, lead times and available capacity. Prices, quantities and costs would then be separate scenario inputs or model consequences. The risk language gives no substitution fraction, write-down amount or probability with which to combine the histories.

Even a delivered replacement has more than one financial date. Apple's revenue policy generally connects product revenue to transfer of control, while allocated revenue for bundled services is deferred over the estimated service period; collection follows its own terms (Apple Inc., 2024a, Note 2, printed p. 35). Development, announcement, production, shipment, service provision and cash receipt therefore cannot share a single "launch" timestamp. The reported policy must be applied to the relevant contract and performance obligations. Any inventory disposition, commitment change or development-cost treatment needs its own facts and applicable accounting method. Preserving these distinctions lets the model ask whether an assumed transition is feasible and how its financial consequences would arise, while leaving actual product histories unresolved.

28.22 A supply agreement does not establish available output

Once a transition identifies the components a product needs, the next question is whether they can arrive on the required date. An agreement to purchase them does not answer that operating question. Apple's component-availability risk says that supply agreements cover many components but gives no assurance that they can be extended or renewed on similar terms, or at all. It also identifies supplier financial difficulties, business failure and industry consolidation as possible limits on supply (Apple Inc., 2024a, Risk Factors, printed p. 8). The historical record must therefore distinguish the agreement, its effective term and the supplier's ability to perform. Renewal and delivery need their own evidence before a scenario can rely on them.

More suppliers do not eliminate every shortage. Apple describes significant supply and pricing risks for single or limited sources, while warning that even components available from several sources can face industry-wide shortages and commodity-price fluctuations. Its example of past semiconductor demand and supply shortages reports an adverse effect on obtaining components and products on commercially reasonable terms. The warning that such disruptions could recur preserves a future possibility; the passage supplies no duration, quantity shortfall or loss with which to calibrate it. A scenario may specify a common industry constraint, but it must state that assumption rather than derive independence or a diversification benefit from the number of suppliers.

Custom components present a different constraint. A sole source may initially have insufficient capacity while yields mature or manufacturing expands. Alternatively, the supplier may give priority to common components. A yield problem and a production-allocation decision can both delay the same product, yet the possible responses differ. Apple separately reports past adverse effects from delayed or constrained component supply and delayed shipments of completed products by outsourcing partners. Component production, final assembly and delivery remain separate activities; an improvement in one need not establish completion of the others. The Note 12 supply discussion and the outsourcing passage retain the same distinction between sources, capacity, partner performance and control (Apple Inc., 2024a, Note 12, printed p. 46; Risk Factors, printed p. 8).

For an explicitly assumed delay, compare restoring the current source with switching to another. The disclosure says that consequences can depend on the time needed to obtain sufficient quantities from either route. A named alternative is therefore only a proposed response until component compatibility, available quantities, lead times, price and contractual terms are established. Existing purchase obligations do not identify those facts or the release of the old commitment. The model should retain both responses with their separate prerequisites and delivery dates, leaving an infeasible or unresolved response visible. Only after the relevant events and accounting facts are specified can the resulting purchases, settlements or inventory movements generate journals. The filing does not select a route or assign probabilities to the alternatives.

28.23 Available output can still create losses and obligations

Obtaining components and completing delivery do not establish that a product can safely serve its intended use. Apple's defects discussion covers its hardware and software, third-party components and products, and service outages, slowdowns and errors. It reports that services have sometimes failed to perform as anticipated. It also warns that defects can create personal-injury, property or environmental risks, that specialized uses such as health can increase those risks, and that complex technologies such as artificial intelligence can expose users to harmful or inaccurate content. Detection and repair of every defect are not assured (Apple Inc., 2024a,

Risk Factors, printed p. 9). For the model, a delivered unit, an acceptable unit and a repaired unit are different states. The source identifies possible transitions between them, not a defect rate or a particular incident.

The financial consequences can extend well beyond inventory. Apple names product liability, recalls, replacement or modification, write-offs of inventory, property, plant and equipment or intangible assets, warranty and other expenses, litigation costs and regulatory fines. Quality can also affect reputation, acceptance, demand and the timing of new offerings. These effects must not all become one inventory expense. A customer obligation and a possible supplier recovery are also separate. The outsourcing passage says partner arrangements may include defect-expense reimbursement while Apple generally remains responsible to the consumer for warranty and out-of-warranty service (Apple Inc., 2024a, Risk Factors, printed p. 8). Identifying a supplier does not establish a reimbursement right, amount or receipt date.

Excess stock raises a different question even when its quality is adequate. The inventory-risk passage says Apple records write-downs for product and component inventories that are obsolete, exceed anticipated demand, or have cost above net realizable value. It separately describes necessary cancellation fee reserves for orders of excess products and components. Inventory already held and an order not yet received are different exposures; cancellation depends on the actual order terms. Note 1's first-in, first-out policy identifies the reported cost-flow method, rather than a lot's realizable value or a cancellation charge (Apple Inc., 2024a, Risk Factors, printed p. 9; Note 1, printed p. 34).

Apple also describes impairment reviews when circumstances indicate that long-lived assets, including capital assets held at suppliers and inventory prepayments, may not be recoverable. It reports a write-down for excess carrying value over fair value if impairment is determined. This is the issuer's description of its assessment; an executable calculation still needs the actual asset classification, assessment basis and applicable accounting method. The company's belief that the named assets and commitments are currently recoverable is accompanied by no assurance against future charges. It neither proves a future impairment nor permits the model to erase that possibility. An inventory cost-flow calculation, a cancellation obligation and an asset impairment assessment cannot be substituted for one another.

The link to product transitions is the forecast used before delivery. Apple orders components and builds inventory ahead of announcements and shipments; manufacturing purchase obligations cover forecast requirements, typically for periods up to 150 days. It warns of excess or insufficient output, or failure to use firm commitments fully, when forecasts are wrong. The qualifier "typically" prevents that period from becoming a universal contract limit. Nor does the description make the separately reported Note 12 obligations with remaining terms above one year the same population as manufacturing obligations (Apple Inc., 2024a, Risk Factors, printed p. 9; Note 12, printed p. 45).

Consider an explicitly assumed reduction in expected demand. Two companies could hold the same reported inventory amount while one holds saleable components for the new offering and the other holds incompatible earlier components, defective finished goods or commitments it cannot cancel cheaply. Their feasible responses and possible losses differ. Lot identity and cost, quality, compatibility, forecast dates, anticipated demand, realizable value, order terms and customer or supplier obligations distinguish the histories. Recognition and measurement then require their own evidence. The disclosure does not supply those facts or probabilities, so the model preserves the conditional histories without inventing a reserve or treating the forecast as observed customer demand.

28.23.1 Service provision requires continuing access

Adequate components and saleable inventory still leave a service question: can the company continue offering the software and content its customers want? Apple says its products and services require third-party intellectual property, while giving no assurance that the necessary licences will be available on commercially reasonable terms or at all. Its digital-content arrangements can be short and need not renew on similar terms. Content providers can also require new rights-management or security technology before access can continue (Apple Inc., 2024a, Risk Factors, printed pp. 9–10). The relevant resource is therefore partly a dated right to offer particular content or functionality. Physical capacity alone cannot establish that right.

Developer support has a separate economic cause. Apple believes customers' hardware purchases depend partly on available applications, whose continued development and maintenance are not assured. It describes developers comparing platforms by market position, anticipated revenue, expected growth and the cost of developing and updating software (Apple Inc., 2024a, Risk Factors, printed p. 10). An application can consequently become unavailable or unsuitable even while the hardware remains usable. A scenario needs the affected applications and developers' response; an estimate of component capacity cannot supply either.

Access also depends on how a service uses data and takes payment. Apple reports restrictions on collecting, using and transferring personal information, including transfers within its international group. It separately warns that failure to follow payment-card security standards could increase transaction costs or remove the ability to process cards (Apple Inc., 2024a, Risk Factors, printed p. 15). These issuer descriptions identify possible constraints. Which constraint applies depends on the data, entity, jurisdiction, agreement and effective rule.

Suppose a content agreement permits a specified offering only until its expiry date. Under one explicitly assumed history it renews; under another it expires without replacement. Identical physical resources then permit different offerings. The comparison needs the service, customer population, territory, contract term and allowed substitutions, together with any new cost and demand response. A financial calculation must apply the relevant revenue, payment and remedy terms to the service actually provided. Apple's risk statements supply neither those contract facts nor probabilities for the alternatives. The historical record must preserve the continuing relationships as well as the reported balances if it is to support these different futures.

28.24 A useful answer before the company's history is complete

The Apple examples now give us enough information to pose a more demanding question: what can one connected answer say about the reported company and its possible cash requirements? The answer must carry both the reported amounts and the limits of the histories used to explain them. Otherwise a complete-looking statement could conceal a missing liability, a hypothetical retirement or an unavailable settlement date.

For the retained annual source, the selected statements report net income of 93,736 and other comprehensive income of 4,280, giving comprehensive income of 98,016 million dollars. Assets of 364,980 less liabilities of 308,030 give closing equity of 56,950. The following quarter reports net income of 36,330, other comprehensive income of 383 and closing equity of 66,758 (Apple Inc., 2024a, 2025c, consolidated financial statements). These are source-located readings of displayed amounts. Their units, periods, group perimeter and statement populations accompany them. A share count or a per-share amount keeps its own unit; it cannot enter a monetary total merely because it sits next to one.

The connected result retains the 102 selected annual rows, 100 quarterly rows and their separate

opening balance-sheet comparisons. It also retains all 23 note dispositions, 70 known references, seven package obligations, 32 proposed source assertions and seven additional frontier entries from the frozen inventory. A note's remaining facts stay attached to it. These records preserve the declared package; they neither discover every possible reference nor establish that every retained sentence has a formal meaning.

A query for the selected displayed statement tokens can therefore finish even though the question about actual whole-company histories remains unresolved. The family question specifies the issuer, sources and target histories, but has no admitted member list or probability measure. The distinction matters: an unresolved feasible remainder does not mean that no history exists, and two illustrated histories do not mean that the real company has exactly two possible histories. The combined answer is partial because the financial explanation has obligations that the reading task does not.

Preserving a source token also requires restraint in arithmetic. The annual cash-flow row for debt issuance displays a dash. Its observation remains a reported dash. The retained financing reconciliation separately states the narrow convention under which that dash contributes zero to its displayed sum. This convention does not turn all dashes into zeros. Similarly, the annual cash-flow opening amount of 30,737 differs from the opening balance-sheet cash amount of 29,965 by 772. That population difference is an explanatory residual, not permission to create a receipt or payment for 772.

28.24.1 Matching annual rows leaves different liabilities to settle

Two conditional asset histories illustrate why the connected result must retain journals as well as totals. In H1, gross additions of 10,529 are paid with 9,447 of cash and a capital payable of 1,082. Retirements remove gross cost of 6,000 and accumulated depreciation of 5,636, leaving a loss of 364. H2 adds and retires another 1,000 of gross cost: additions become 11,529, retirements 7,000, loss 1,364 and capital payable 2,082. Depreciation is 8,200 in both branches. The extra acquisition and retirement cancel in net property, plant and equipment; cash is unchanged. The asset journals alone leave H2 with 1,000 more liability and 1,000 less profit and equity.

Now suppose a distinct non-tax expense liability exists: its amount is q in H1 and $q - 1{,}000$ in H2, with $q \geq 1{,}000$. Each branch debits expense and credits that liability. This is an independent economic premise, not an event inferred from a statement residual. Suppose also that the retirement loss and this expense share the product-cost row, both liabilities share the same other-current-liability bucket, the extra asset cost has the same gross class, and current and deferred tax, all other activities, shares and distributions are unchanged. Then the expense difference of $-1{,}000$ offsets the extra retirement loss, and the liability difference offsets the extra capital payable.

The indirect cash-flow presentation still needs a specified allocation. Let $\alpha \in [0, 1]$ allocate the expense-liability adjustment once between the noncash adjustment and the working-capital adjustment. If Δ means H2 minus H1, the two displayed-row differences become

$$\Delta\text{other adjustments} = 1{,}000 + \alpha(-1{,}000) = 1{,}000(1 - \alpha), \tag{28.25}$$

$$\Delta\text{other liabilities} = (1 - \alpha)(-1{,}000) = -1{,}000(1 - \alpha). \tag{28.26}$$

The two differences sum to zero for every allowed α. At $\alpha = 1$, each is zero and all 102 selected annual row differences vanish under these assumptions. At $\alpha = 0$ or $1/2$, the same total operating cash has a different presentation in those two rows. Here α is a reporting assumption, not a probability over histories. Matching differences also does not construct an absolute feasible remainder for the company. The illustrative journals remain conditional subsystem histories.

Their future consequences remain joint. With $q = 2{,}000$, carrying-value settlement of the two liabilities gives the following changes, in millions, if both payments occur in the first future period:

Branch	Operating cash	Investing cash	Total cash
H1	−2,000	−1,082	−3,082
H2	−1,000	−2,082	−3,082

Paying the capital liability is an investing movement; paying the separately accrued expense liability is an operating movement. Each payment debits its liability and credits cash. Under the explicit assumption of no incremental tax, currency or interest effects, it creates no new profit, other comprehensive income or equity movement. Sufficient unrestricted funding is assumed; no opening cash balance is manufactured.

Taking the lowest operating amount from H1 and the lowest investing amount from H2 would produce −4,082. Neither history produces that total. The result must return the two pairs, not independently selectable marginal endpoints. Moving capital payment to the second period changes first-period cash to −2,000 in H1 and −1,000 in H2; moving the expense payment instead changes it to −1,082 and −2,082. Thus equal eventual cash does not identify its timing. The six specified paths retain both branch identity and payment period.

When dates are missing, the response keeps the conditional eventual amounts and marks period timing unresolved. It distinguishes these calculated amounts from reported tokens. A request for future depreciation remains unsupported without the asset cohorts, tax state and other required facts. A probability request remains unsupported without an admitted law. Neither refusal erases the amounts that a narrower question can answer.

28.24.2 The information used belongs to a particular view

The later annual source provides a different kind of progress. Its FY2024 Americas comparative shows sales of 167,045, cost of sales of $-89,587$ and selling and marketing expense of $-9,802$, giving operating income of 67,656 (Apple Inc., 2025a, Note 13, FY2024 Americas comparative). The selected original source lacks those two separate expense rows. An original-source query therefore returns unresolved components, while a query explicitly naming the later source can return them. The result links the later view separately; it does not silently enlarge the original view or assert that measurement scope was unchanged. Later detail is not, by itself, an accounting-error correction.

Named-source access answers what retained bytes contain. Investor-information and system-as-of queries additionally require accepted clocks for those exact bytes. Where publication, ingestion or review time is unestablished, the historical query remains partial with a named timing obligation. A filing event alone cannot date a separately converted PDF byte for byte. This prevents the later comparative detail from becoming evidence that an earlier investor could already have used.

JPMorgan's reference conflict illustrates the same discipline in another company. In the retained 2023 report, Item 15 faithfully says page 156; Item 8 and the auditor heading support printed page 163, physical page 165, for navigation (JPMorgan Chase & Co., 2024, Items 8 and 15; printed pp. 156 and 163). The wrong target is printed page 156, physical page 158. Both use the same two-page offset, so changing the offset cannot reconcile the targets. A proposed navigation correction preserves the literal statement. If a counterfactual earlier view had accepted the wrong target, its selected page set, bounded search and coverage summary would

need reconsideration, while the literal and an independently sourced statement amount could remain usable. A one-page negative search could never establish absence from the whole filing.

That review proposal has its own issuer and source dependencies. It supplies no Apple financial constraint, no posting permission and no observed human decision. The connected financial result and the separate review proposal can therefore be inspected together without joining their companies' histories. Section A.28 gives the exact exchange locations and rejection boundaries that preserve these distinctions.

Chapter 29

Representing a company's historical record

Northstar's corrected receivable, Apple's debt measurements and its deferred revenue show why the source record must retain more than a label and a number. Section 28.11 distinguishes a reported balance, a derived movement, a dated expectation and an unquantified obligation. Section 28.12 adds the contract and delivery facts needed to turn a commitment into a future payment. The procedures below determine which assertions enter a model and which remain unresolved.

29.1 The record is a collection of assertions

The original record includes financial statements, notes, accounting policies, auditor opinions, management discussion, segment tables, security terms, regulatory returns and subsequent announcements. A complete ingestion process accounts for every page and every extracted block. A complete financial model is a stronger claim: it would require economically relevant information that public reports may never disclose. AccDSL therefore reports document coverage, semantic coverage and simulation coverage separately.

Each source span receives a disposition: structured observation, narrative assertion, relationship, policy evidence, contextual material, duplicate, or unresolved extraction. "Unresolved" remains visible. A commitment buried in a note is not discarded because it has no current balance-sheet line. An auditor's qualification is retained as evidence about the record; it does not become an automatic numerical haircut. Management's forecast is a dated assertion with an author and modality, not an observed future outcome.

An observation is an immutable record with the fields below. Values inferred from observations are separate records, linked by a derivation. This avoids overwriting a company's statement with an analyst's preferred reconstruction.

Field group	Minimum contents
Identity and source	Observation ID; document hash; filing identifier; PDF page and bounding box or HTML/XBRL locator; literal text; extraction version.

Field group	Minimum contents
Reporting context	Legal entity; reported group perimeter; statement or note; instant or interval; standard and taxonomy versions; dimensions such as segment, product, geography, seniority or measurement category.
Amount	Original string; parsed number; sign interpretation; unit; scale; currency; precision or rounding interval; missing-value category.
Meaning	Source label; taxonomy QName when present; one or more proposed concept mappings; whether the assertion is reported, estimated or conditional.
Timing	Economic validity when known; reporting period; publication and availability timestamps; ingestion timestamp; correction or supersession links.
Reliability and review	Extraction confidence, reviewer decision, source status and known contradictions, all distinct from a probability over histories.

A narrative assertion adds actor, predicate, object, scope, quantifier and modality. For example, "substantially all debt is fixed rate" has a source span and a qualitative quantifier. It does not assert $\alpha = 1$. A reviewed interpretation can introduce a numerical restriction such as $\alpha \geq a$, but the threshold a remains an analyst assumption unless the source defines it. Boilerplate policy language also does not prove that a company holds a particular instrument.

29.2 Different names, different measurements

An accounting concept is more than a row label. "Debt" can refer to legal principal, amortised carrying amount, undiscounted contractual cash flows or a fair-value disclosure. These amounts can all be correct at the same date. Likewise, "revenue" may be reported gross or net, a segment measure may differ from the consolidated measure, and an alternative performance measure may exclude selected expenses. A mapping must state its relationship: equivalent meaning, component of, broader than, presentation reclassification, or measurement transformation.

The common vocabulary will preserve source concepts as first-class identifiers. Company extensions may inherit a broad economic category without being declared identical to a standardised concept. For instance, an extension can be a component of borrowings while retaining an unresolved split between lease liabilities and other financing. Many-to-many mappings are allowed; each mapping has applicable dates, dimensions and evidence. An aggregation mapping does not silently become a measurement conversion.

Standards differ in recognition, measurement, presentation and disclosure. Consequently, conversion between IFRS, US GAAP and Chinese Accounting Standards for Business Enterprises (ASBE) is not generally a dictionary lookup. The original basis remains preserved, and any comparison basis is a separate derived book. The same principle applies to tax and prudential capital. A tax loss carryforward, accounting deferred tax asset and regulatory capital deduction can concern the same economic loss while belonging to different measurement systems.

29.3 Time, revisions and information availability

At least four times matter. A machine may enter service in November; its cost appears in the accounts for the year ending December; the annual report is published in March; and AccDSL ingests the report in April. A later report may revise the December comparative. These times answer different questions and must remain distinct.

The storage model uses validity intervals for economic assertions and transaction intervals for database versions, following the distinction in Snodgrass and Ahn (1986). Publication and availability timestamps are additional domain fields. Database transaction time records what AccDSL contained; it cannot substitute for what investors could have known. An "as known on" query applies an explicit availability rule and does not include a later restatement simply because its economic period is earlier.

AccDSL supports three named views: the original filing view; the latest restated view, with a stated information cutoff; and a harmonised research view, with explicit transformations. No view changes the stored source facts. Corrections have links to the replaced assertion and a reason. Where two sources conflict without a clear supersession relationship, both survive with a conflict record.

Instant values and interval flows have different types. A balance at 30 June is not a six-month flow. Calendar intervals use half-open boundaries $[t_0, t_1)$ internally; a filing period ending 30 June maps to the next boundary at 1 July in its reporting calendar. Quarter-only earnings may be derived from year-to-date values only if the periods, basis, perimeter, units and revision vintage are compatible. The derivation retains both source observations and their correlated rounding uncertainty. Fifty-two/fifty-three-week calendars, stub periods and reporting-date changes require explicit calendar objects.

Accounting-policy changes, estimate changes and error corrections also have different temporal effects. The applicable rule package decides whether a change is prospective, requires restatement, or adjusts opening balances. The engine never infers that effect from a generic "revision" flag.

29.4 The company boundary changes

Legal entity, reporting group and economic exposure are separate dimensions. Acquisitions, disposals, changes in control, joint arrangements, special-purpose vehicles and non-controlling interests can change the reporting perimeter. A group's increase in property need not be capital expenditure; it may reflect an acquisition, a currency translation or a revaluation. The bridge between opening and closing property must identify these movements before a scenario model can extrapolate investment.

The company graph therefore stores dated ownership, control and counterparty relationships. Control is an accounting conclusion with supporting evidence, not a rule that blindly consolidates every majority-owned entity. Group reporting combines entity books with a separate adjustment and elimination book. Intragroup receivables and payables require matching counterparties, currencies and dates. Unmatched eliminations remain unresolved; a balancing "plug" is not a recovered transaction.

Acquired opening balances, goodwill, purchase-price allocation, divested operations and non-controlling interests each have their own event types. Analyses can preserve reported scope or define a constant perimeter, but the latter is a derived modelling assumption, not the original record.

29.5 Further problems that the representation must handle

Several difficulties extend beyond differences in standards, industries and company labels. Recognition and settlement occur on different dates: a sale, invoice, cash receipt and impairment can belong to four periods. Net balances hide gross flows: repeated borrowing and repayment can leave the same closing debt. Netting can conceal offsetting exposures and liquidity needs. Different contractual maturities and repricing dates can share the same current carrying amount. Guarantees, undrawn credit facilities, litigation and covenants can be economically important before an asset or liability is recognised.

Foreign currency introduces transaction currency, functional currency and presentation currency, together with different translation conventions for stocks and flows. Currency translation effects must not be mistaken for cash receipts. Inventory and depreciation depend on vintages or lots; an aggregate stock and average age may be insufficient under a shock. Tax bases can evolve differently from carrying amounts. Deferred tax recognition requires assumptions about recoverability, rather than multiplication by a tax rate alone.

Public information is selectively disclosed and rounded. Duplicate XBRL facts may agree, overlap within rounding bounds or contradict one another. A dash can mean zero, immaterial, unavailable or inapplicable. Tables can mix units, span pages and embed signs in parentheses. Text and table references can be wrong in the source itself. AccDSL retains these distinctions and gives an analyst a local repair path without pretending that extraction confidence settles economic interpretation.

Finally, strategic behaviour matters. A firm may refinance, cut investment, raise equity, change prices, breach a covenant or fail. A bank may preserve a capital ratio by shrinking assets; an insurer may change reinsurance or asset allocation. Such responses cannot be deduced from the balance-sheet identity. They require explicitly calibrated or assumed policies, and alternative policies must remain separate when the available record does not distinguish them.

29.6 Account for a disclosure before translating it

The importer first freezes a filing package: documents, byte hashes, reporting dates, availability evidence and incorporated references. It compares the document's contents page and statement index with detected section and note headings. A missing heading, table continuation or referenced exhibit creates an obligation to resolve; successful text extraction cannot discharge it. Each PDF page and each HTML content unit then has a stable locator and a disposition. Child units such as a table or note belong to a parent page/section; the importer records overlap instead of counting the same content twice.

Three denominators remain separate. Acquired-document coverage counts source units with a disposition, including unresolved units. Semantic coverage counts identified assertions whose meaning and context have been reviewed. Simulation coverage counts the operations and exposures required by a specified query that have supported state and rules. The first can be complete while the other two are partial. An assertion inventory can itself be incomplete; until the page/section reconciliation is reviewed, semantic coverage is a lower-bound inventory rather than a percentage of all financial meaning.

Cross-reference traversal is a work-list algorithm. Start with the filing's statement and note references, resolve each target against the acquired manifest, add newly discovered targets, and stop when no unvisited target remains or a declared resource limit is reached. Document hash plus target locator prevents cycles from expanding forever. A target has one of four results: resolved, duplicate of a resolved source, unavailable, or ambiguous. The last two

remain obligations. Apple's proxy and prior-filed agreements show why closure over acquired files is weaker than closure over incorporated disclosures. A manifest may declare a bounded acquisition scope, but must show these outstanding references beside its completeness status.

A narrative translator records the actor, predicate, object, quantifier, modality, applicable entities/dates and qualifications before proposing any constraint. "All notes are fixed-rate" constrains contractual coupon type; it does not constrain the combined note-and-swap exposure. "May repurchase" creates an authorised action with limits, not an obligatory payment. "Substantially all" remains qualitative unless a separately reviewed assumption gives it a threshold. A conditional guarantee becomes an obligation with a trigger and exposure, not a present liability merely because its maximum amount is known. Unsupported predicates retain their literal text and links to affected balances. The reviewer accepts the translation separately from accepting that the text was extracted correctly.

29.7 Mappings must establish a financial relationship

A mapping record identifies source concepts, target concept, relationship, applicable entity/basis/period/dimensions, evidence, reviewer and any required transformation. Equivalence permits substitution only when all these contexts match. A component relationship permits inclusion in a sum only through a declared disjoint partition. A broader concept supplies containment, not an equality. Reclassification changes presentation while preserving the stated population and total; measurement transformation must supply an actual formula and inputs. These relationships cannot compose indiscriminately: component followed by equivalence remains component, while two measurement transforms compose only if their intermediate bases and information dates agree.

For example, Northstar's product revenue might split into 180 domestic and 80 overseas, while its customer channel split is 200 wholesale and 60 retail. Both classify the same 260. Adding all four numbers gives 520 and is invalid. The partition records therefore share a population ID but have different axes. To remove an axis, `sumOver` requires evidence that members are disjoint and exhaustive for that context. Where the source supplies only named segments, an explicit unallocated member carries the residual; it is not assumed zero. Geography-by-product cells cannot be recovered from the two marginals without further constraints or assumptions.

Apple supplies another counterexample: its vendor non-trade receivables arise from component sales to manufacturing vendors, whose gains reduce product cost of sales when the final products are sold (annual Note 4, p. 38). They cannot be relabelled as ordinary revenue-generating trade receivables without changing meaning. Likewise gross derivative assets and liabilities cannot be silently netted. Company extensions can define these distinctions and dated presentation changes; they cannot override mandatory recognition rules in a selected standards package.

29.8 Selecting original and revised assertions

The query procedure first filters assertions by the declared availability cutoff, then by an optional database-time cutoff. A day-only filing timestamp is conservatively available at the next local midnight in its named timezone; any more precise rule needs evidence. Within the eligible set, explicit supersession removes only the targeted assertions in matching contexts. The superseding assertion must itself be eligible. Follow the acyclic graph transitively; retain all maximal assertions if two eligible corrections conflict. File order and ingestion recency never decide which is true.

The original view selects the initial assertion chain in a named filing; the latest-available view selects eligible maximal versions; the harmonised view adds reviewed transformations to

those selections. A revision to revenue does not automatically revise every note or change the company's accounting basis. An estimate change takes effect in future measurements from its specified date. An error correction may change a comparative and an opening balance bridge, each as a new assertion with its own lineage. Northstar's explicit query fixture in Chapter 28 checks all these clocks without assuming that the most recently stored row is always the appropriate history.

29.9 What an accepted reading permits

Apple's inventory policy illustrates the distance between reading a sentence and reconstructing a business. Its reported use of FIFO supports a proposed cost-formula reading for the inventory and reporting contexts covered by that policy. It does not reveal the quantities and ages of the actual lots, establish that no write-down occurred, or supply the authority for every US accounting treatment. Those are separate inputs to a cost and impairment calculation. Similarly, a straight-line depreciation policy leaves useful lives, residual values, components and available-for-use dates to be established. Section 35.5 tests those limits against the reported asset balances, depreciation and cash payments, then constructs two different asset books compatible with the selected amounts.

The initial source preparation retains six located Apple policy passages: consolidation of Apple and wholly owned subsidiaries; comparative reclassification; FIFO; straight-line depreciation; the interim statement's estimates and annual-report reference; and its 52/53-week fiscal calendar. Each proposed reading carries the original document hash, extraction hash and exact character slice. These are company assertions. The consolidation sentence, for example, cannot exclude every partly owned entity without a control analysis. The calendar sentence does not make every future fiscal year 52 weeks or make a contractual interest year 365 days. The exact quotations and proposed predicates are retained in `../methods/source-translations-v1.json`; the meaning explained here is not an invented human acceptance of them.

A reading progresses through separate decisions. First, extraction review asks whether the located words and numbers were transcribed faithfully. Translation review asks whether the proposed predicate preserves their scope, quantifier, exceptions and reporting context. A model then admits that predicate, or an explicitly labelled assumption, for a particular purpose and information view. Accounting eligibility subsequently asks whether a method applies to the actual event. A faithful transcription of an erroneous company assertion is still faithful transcription. The four decisions therefore cannot share a single "approved" flag.

An explicit presumption in a standard also differs from filling an unknown with a convenient answer. Article 6 of the retained official ASBE 21 text requires both practical ability throughout use and economic benefit for a substantive substitution right. It specifies a presumption when that right is difficult to determine. Applying that conditional rule requires its stated circumstances; it does not transform missing lessor information into a reported fact. Nor may a translator assign arbitrary numerical thresholds to "substantive", "probable" or "highly probable".

29.10 When a faithful reading preserves a wrong reference

An analyst looking for the auditor's report in JPMorgan Chase's 2023 Form 10-K receives two incompatible directions. Item 15 says that the consolidated financial statements, notes and auditor's report listed in Item 8 are "set forth commencing on page 156". Item 8 identifies that same set as appearing on pages 163–309. Both statements are printed in the retained

issuer-hosted PDF (JPMorgan Chase & Co., 2024, Items 8 and 15, printed pp. 36 and 40). Correctly copying each sentence therefore preserves the disagreement.

The page numbering does not explain it. Printed page 156 is physical PDF page 158 and contains management's discussion and analysis. Printed page 163 is physical page 165 and begins the report of the independent registered public accounting firm. The two-page offset applies to both locations. Item 15's explicit reference to the material listed in Item 8 also rules out treating it as a direction to a broader discussion beginning in MD&A (JPMorgan Chase & Co., 2024, printed pp. 156 and 163). For this document and target, the page-156 direction is wrong. The supported navigation correction is page 163; it is a separate interpretation of the retained document, not a replacement for what Item 15 actually says.

The review must retain those two meanings of correctness. The source record keeps the PDF, both located sentences, their unchanged extractions and images of the referenced pages. A proposed extraction decision can accept 156 as the faithfully transcribed number. A translation can likewise record that Item 15 directs the reader there. Neither decision accepts the proposition that the auditor's report actually starts there. That target proposition is refuted by the inspected page. A separate navigation reading points to 163, supported by Item 8 and the report heading. This reading permits locating the report; accounting admission still requires the relevant financial assertions, contexts and methods.

Consider the resulting review sequence as a specification example. Propose both literal readings and retain their common target and document identity. Accept their transcription and attribution at the stated layers. Reject 156 as the verified start location, keeping the rejected proposal and its reason, and propose 163 for navigation admission. The source disagreement remains visible after that repair. It does not require keeping 156 and 163 as equally admissible start locations: an `AcceptAlternative` action for 156 at the target-location layer would fail the evidence check. Uncertainty is preserved when evidence leaves alternatives open; a refuted reading is preserved as review history.

Different queries now have different exact answers. Asking what Item 15 says still returns 156, with its locator and the conflict annotation. Asking for the supported start of the auditor's report returns printed 163, physical 165, with both supporting passages. Asking whether a particular accounting treatment is valid remains outside this navigation decision. Nor can a search confined to page 156 conclude that the entire filing contains no auditor's report. Its proper negative result is only that the report was not found on the selected page.

Suppose an earlier analysis had nevertheless accepted 156 as the correct start. This is a counterfactual mistaken decision, not an observed human review in this project or a permitted new admission after the target check. Repair creates a new reading and an explicit `Supersede` decision in a new evidence view. The old accepted decision, its reason and its results remain available under the old view. The corrected view cannot reuse a cached result whose justification still depends on that old navigation choice.

That dependency includes the pages selected for a search. A cached "not found on the selected page" answer must be recomputed when the admitted location changes, even if its output contains no extracted assertion from the new page. A coverage conclusion derived from that answer changes with it. By contrast, the literal Item 15 quotation remains unchanged. A separately located statement value whose provenance does not depend on the repaired reference may also be reused after its other dependencies are checked. An annotated display of the unchanged quotation still needs to refresh when its conflict or review status changes; preserving the text does not preserve every display derived from it. The correction therefore invalidates dependent navigation, search and coverage results; it does not mechanically invalidate every financial amount in the filing.

This case supplies an actual source-reference conflict and a proposed account of its review consequences. The source inspection is author work; the review actions above are expected behaviour, with human adjudication and software execution still pending. The finding establishes no financial misstatement, publisher erratum or historical availability of the retained bytes. Those questions require their own evidence.

29.11 Three meanings of disclosure coverage

The initial Apple annual/interim preparation inventories 125 quantities, 13 annual notes, 10 interim notes, 70 exhibit references and seven further package obligations. In that starting assessment, all 23 notes retain unresolved semantic content; none is recorded as fully formalised. The later worked readings now trace representative revenue, commitment, asset, tax and funding consequences, while retaining missing facts and wider package obligations. Downloading a note, reviewing its material passages and using it in a complete company model remain different achievements.

Three populations make that distinction inspectable. Retained-unit coverage partitions acquired documents into nonoverlapping pages or smaller leaf units, including duplicates and unreadable material. Assertion-review coverage counts the separately enumerated material assertions and their review decisions. Query-exposure coverage identifies the sources, legal facts, methods and authority that a particular result requires. A fully downloaded annual report can have incomplete assertion review, and a reviewed debt total can leave a refinancing query unresolved because an incorporated agreement is missing. No weighted average of these populations establishes completeness.

Preparation begins with declared annual and interim roots and a policy for following incorporated references. Each edge is recorded as satisfied, duplicate or cyclic, outside the declared scope with a reason, inaccessible, or still unexpanded. A bounded traversal saves its unexplored frontier and budget so it can resume. An unknown external extent remains unknown. Acquired proxy and registration documents improve the Apple record, but do not prove that every executed amendment, issuance term and exhibit dependency has been closed. The present real package remains incomplete for that reason.

Container choice can itself create a false missing-source diagnosis. Apple's retained annual PDF contains securities and award exhibits absent from the retained HTML container; other incorporated award forms are present in earlier reports in the retained corpus. Locating those bodies resolves a source-access question. It still leaves their qualifications and outgoing references to be interpreted. Conversely, the cash restriction in Section 28.9 shows why a located note can change the meaning of an already extracted figure: correct arithmetic alone would not detect an erroneous "unrestricted" label.

29.12 Displayed precision does not determine a rounding rule

Suppose a statement displays 100 in millions. The scale establishes how the literal is read; it does not establish how the issuer rounded its underlying amount. Under an explicitly admitted nearest-rounding assumption, an integer display m at scale s permits the conservative compatibility enclosure

$$s(m - \tfrac{1}{2}) \leq y \leq s(m + \tfrac{1}{2}).$$

The closed endpoints accommodate an unknown tie rule within that nearest-rounding model. A specified tie rule can exclude an endpoint. Truncation of a positive amount instead permits

$sm \le y < s(m+1)$. Thus the illustrative display 100 corresponds to 99.5–100.5 million under the first assumption and 100–101 million, excluding 101, under the second. These are different restrictions on a history.

The retained Apple arithmetic uses the first enclosure as a conditional compatibility assumption; the issuer's rounding convention has not been established. Without that assumption or source support, the record states known display precision and unknown rounding convention and supplies no hard half-unit constraint. A one-unit difference between a displayed total and the sum of displayed components is consequently not enough to establish a contradiction. The source value is scaled once, while its literal and precision metadata remain available for inspection.

29.13 Public availability and system possession are different views

Northstar's correction explains why a result needs a reporting vintage. A historical forecast also needs to establish when the exact document bytes were available. A report downloaded today may carry an old date on its current web page; that does not prove that the same bytes were public on that date. Public availability evidence and an ingestion receipt answer different questions.

An investor-information query supplies a cutoff K. Each admitted source must have supported public availability no later than K and evidence linking the selected bytes to that availability. Later ingestion is allowed. Later extraction review can read eligible old bytes, but an interpretation that uses later information fails the cutoff on its own dependencies. This view establishes public availability, not actual investor awareness or the ability to trade.

A system-as-of query instead supplies an instant S and requires an immutable receipt showing possession of those bytes by S. Publication cannot replace that receipt. An accepted-layer query also requires review decisions visible by S; a raw-layer query may return a then-unreviewed document. A query asking both what was public and what the system possessed must pass both filters. Unknown publication alone need not block a pure possession question.

Archival exploration permits inspection of retained sources with incomplete timing. With a cutoff, unknown publication yields an unresolved candidate, not a cutoff-qualified selection. Without a cutoff, the archive may display the retained lineage, explicitly without a historical-knowledge claim. A known later version is excluded from an earlier view and cannot suppress an eligible old assertion. An unresolved potentially relevant version keeps the affected answer unresolved even when an older assertion is displayed.

Date-only evidence requires particular care. For publication date D in a named timezone, let $[a_D, b_D)$ be the interval from local midnight to the next local midnight. A cutoff before a_D excludes the source; a cutoff inside the interval leaves availability unresolved; a cutoff at or after b_D supports admission on this date evidence. No arbitrary noon or fixed 24-hour duration is inserted. Missing timezone leaves the instant comparison unresolved.

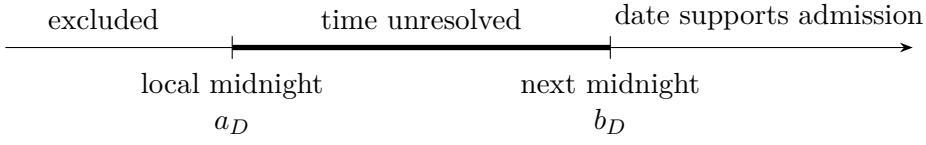


Figure 29.1: A dated publication supports historical admission only after its possible publication interval has ended. The line marks uncertainty about an instant, not a delay imposed on an already known timestamp.

Revision selection follows admission. Original selection returns eligible roots; latest-available selection returns eligible assertions without an eligible superseding descendant for the same context. Unrelated roots and multiple live leaves remain competing assertions. A harmonised view adds explicit transformations with their own admitted sources, method and opening bridges. File order never selects a winner.

The synthetic interim example has original first-half revenue 120, revised first-half revenue 110 and revised annual revenue 260. Compatible subtraction gives second-half revenue $260 - 110 = 150$. The arithmetic $260 - 120 = 140$ is easy but mixes incompatible vintages. An original-view display may show both source amounts while refusing that derived flow. A change in calendar or perimeter likewise requires a specific bridge.

The financial-amount comparisons investigated so far have not supplied a qualifying unresolved same-context measurement contradiction. The reference conflict in Section 29.10 establishes the narrower source-navigation finding. Known revisions, different reporting bases and synthetic conflicts remain useful cases of their own kinds. A financial conflict case must first match entity, perimeter, concept, dimensions, unit, basis, period, versions and precision, and investigate extraction and supersession. The absence of a qualifying measurement conflict in a bounded search is not evidence that the corpus contains no conflicts.

29.14 A revised fact changes a conclusion without rewriting its history

A correction can change the amount that a model now explains while leaving both its earlier explanation and the company's committed journals intact. The bank and insurance examples later in this proposal make the distinction concrete. The bank needs the credit-risk baseline of its irrevocable commitment even after a loan is drawn. The insurer needs to know whether its reinsurance treaty preceded the recognition of an onerous group. Neither fact can be replaced by the time a document happens to arrive.

Consider a constructed bank record that supplies no original commitment baseline. The analyst can retain its cash flows and conditional loss scenarios, but cannot establish the required original-to-current credit comparison. A later record identifies the original commitment risk and supports the separately supplied finding of significantly increased credit risk. With the other assumptions in Section 37.6, the funded allowance is 4 and equity is 16. The additional record resolves the missing history; it does not reset the baseline to the draw date. The relevant AASB 9 provisions retain the irrevocable commitment as the initial comparison and treat the resulting asset as its continuation (Australian Accounting Standards Board, 2022i, paragraph 5.5.6 and B5.5.47). The probability tree remains an authored assumption, and the credit finding is not inferred merely because a probability doubled.

Suppose, for this constructed record, exact bytes were publicly available on 2 January, ingested into the system on 5 January, and accepted for the required reading on 6 January. An investor-information reconstruction with a 3 January cutoff may use those bytes if their public availability is verified, even though this particular system obtained them later. A system-as-of query for 3 January cannot use them. On 5 January a raw system query may retrieve the document, but a query for an accepted reading still lacks the 6 January decision. Later extraction is compatible with the earlier investor cutoff only if its interpretation introduces no later information. These dates are authored specification fixtures; they establish no actual historical availability.

A treaty correction separates knowledge from measurement

In Section 38.6, treaty R entered before group B's recognition produces an initial reinsurance asset of 85, initial equity of 85 and year-one equity of 83. If R instead enters immediately afterward, with all other stated facts unchanged, the initial asset is 70, initial equity is 70 and year-one equity is 74.75. The loss-recovery recognition condition causes the difference (Australian Accounting Standards Board, 2022g, paragraph 66A and B119C). Both histories eventually reach cash of 88.3 on the same supplied claim and recovery path. Agreement on that terminal amount therefore cannot establish correct recognition timing.

Now give the two order assertions different information dates. A constructed January disclosure says the treaty preceded recognition; a February correction says it followed recognition and explicitly replaces that order assertion. A latest-available query cut off in January retains the January assertion. The February record is known to be too late and cannot suppress it. A February query may select the correction and produce the later-entry measurement. The correction concerns the order assertion for B and R; it does not replace every other assertion in the company's report.

A competing record with unknown public availability has a different effect from a record known to be published after the cutoff. The known late record is excluded. The unknown record leaves an unresolved dependency if it might affect the requested conclusion, even while an older eligible assertion is displayed. Similarly, an asserted publication date without evidence linking the retained bytes to that date does not establish historical admission. If only the common calendar date of treaty entry and group recognition is known, their order remains unresolved; the model cannot choose 85 merely because that treatment is more favorable.

An extraction correction is different again. A source may consistently say that R followed recognition while its first accepted extraction mistakenly records the opposite. Correcting the reading creates a new reading and review decision linked to the same source bytes. The old result remains retrievable as a historical error; reproducing it does not make it correct. An issuer's correction instead changes the source assertion and usually its bytes. The system records which change occurred before deciding which derived results require reassessment.

Invalidate the selection, preserve the committed event

A result depends on the candidates considered by its selector as well as the candidate ultimately chosen. Adding a possible competing treaty record changes that candidate population. Reusing a cached selection solely because its chosen assertion identifier is unchanged would miss the new obligation. The proposed dependency graph therefore includes a version for candidate-set membership. The change propagates from membership through selection and interpretation to measurement and the resulting statement. A fresh selection may exclude a known late competitor and calculate the same amount, but it records that decision under a new view.

Newly created descendants must check their complete ancestry too. Otherwise a report created after an invalidation scan could inherit a stale measurement without appearing on the scan's earlier list. In contrast, a bank result whose complete dependency set contains no changed insurance inputs can remain reusable. Neither a shared company label nor an unchanged displayed value establishes independence. Missing ancestors leave reuse unestablished; a cycle or missing required reference rejects the affected dependency chain.

The retained explanation of a result uses its original evidence view. It can show a current warning that a dependency has changed, but it cannot replace the old source with today's correction and still purport to explain the old calculation. Requesting a different view requires a new calculation. This preserves the distinction between the company's historical record, what

the system then concluded, and what a later reconstruction supports.

The same rule protects committed accounting events. The connected insurance example commits financing, service, incurrence and settlement as four separate events. After settlement, retrying the exact original financing request returns its original receipt and reports no additional commit; today's evidence revision does not turn that retry into a new assessment. Changing the request while retaining the business-event identity is a conflict. The request's original bindings, historical admission and required bytes must still validate before receipt lookup.

A new conclusion that the treaty came later does not itself authorize an entry reducing the committed reinsurance asset. A correction proposal must identify its accounting treatment, effective period, affected accounts and opening-balance consequences under the applicable authority. Without that support, the proposed assessment returns `TREATMENT_UNSUPPORTED`, retains the revised conditional calculation and creates no journal. Versioning supplies the record of a correction; it cannot supply the accounting rule that makes the correction appropriate.

Chapter 30

What the literature contributes

The research problem crosses accounting, databases, programming languages, econometrics and document understanding. Each field resolves part of it. The review below distinguishes results that can be adopted directly, engineering ideas that need adaptation, and evidence that does not establish the intended product capability. Source versions and technical anchors are recorded in the accompanying research ledgers. The search includes work outside the original collection, with public-index searches and backward and forward citation searches through 2 October 2026. It is a scoped design survey, rather than a claim to enumerate every paper in these fields.

30.1 Accounting structure and company dynamics

Nehmer and Robinson (1997), especially Sections 2–5, supply a useful algebraic starting point: balances occupy a structured state space, transactions change that state, and aggregation relates detailed accounts to reports. This supports a distinction between conservation constraints and permitted transactions. An entry can conserve the accounting total and still be an inadmissible transaction. AccDSL adopts that distinction in its posting validator. Its particular event vocabulary, evidence model and inverse-history semantics are new design decisions, not results established by the algebraic paper.

Doshi (2024) develops an accounting DSL with time, transaction logs and correction mechanisms. The relevant material is in Chapters 4 and 7–9 of the 2024 thesis. It demonstrates why corrections and dates belong in the language rather than in informal comments. The thesis does not establish recovery of an entire company history from public reports or inference over economically distinct histories. Babcinețchi et al. (2022), pp. 342–345, offer a much smaller accounting language and grammar proposal. It is useful as a syntax precedent, but it provides insufficient semantics or evaluation to govern AccDSL by itself.

Shahnazarian (2004) connects accounting dynamics with behavioural and econometric equations. Sections 2.3–2.8 describe the accounting relationships; the optimisation and econometric discussion in Chapter 3 and the data and evaluation material in Chapters 4 and 8 show how the model is used with standardised Swedish corporate data. This is an important precedent for simulating companies under changed conditions. Its standardised tax-data starting point differs from heterogeneous public-report extraction, and its working-capital funds formulation should not be imported as a general modern cash-flow statement implementation. The transferable idea is the explicit separation of accounting evolution and behavioural assumptions.

Pierson (2020), Sections 3–4 and the reporting structures, makes that separation operational in system dynamics. Receivables, payables and wages accumulate differences between accrual

and payment flows; reporting-period accumulators turn instantaneous flows into statements. AccDSL can use these structures as examples of coherent movement schedules. The model's payment delays, inventory approximations and simplified equity treatment remain modelling choices. They are not universal defaults for companies or evidence that an aggregate state is sufficient for all shocks.

Bank models demonstrate why journal generation needs economic state. Halaj (2016), Section 3, combines funding shocks, fire sales, loans, securities and risk constraints. The homogeneous portfolios and specified funding process make the model tractable, but limit instrument-level interpretation. Behn et al. (2019), Section 2, make managers' choices over asset growth, debt, risk and payouts depend on capital and liquidity constraints. These models motivate interfaces for management policy and regulatory calculations. Their parameterisations must be calibrated and evaluated for the target application; passing their accounting identities would not validate an AccDSL forecast.

A stock-flow model makes the role of counterparties especially clear. The recovered journal paper by Caiani et al. (2016) describes heterogeneous loans and capital vintages, then orders production, credit and payments within each period (Sections 2–3). Its accounting checks precede comparisons with empirical regularities. These distinctions support keeping economic state, event order and accounting validation separate in AccDSL. The authors' steady-state opening in Section 4 and Appendix A serves an artificial economy; it cannot replace a company's observed and unresolved opening positions. Nor can a shared label establish a shared measurement: their operating cash flow in Section 3.1.3 subtracts loan principal repayments. The inspected author-linked code makes that subtraction too. AccDSL must retain this debt-service measure under a separate concept and reconcile its components before comparing it with statement operating cash flow.

Financial planning asks a related but different question. In the retained 1974 working paper, Gentry (1974) asks what new investments must earn to attain management's chosen earnings-per-share growth. Dividend and borrowing decisions interact with that objective over successive years. This suggests an explicit interface for managerial objectives and financing responses, but does not determine customer demand or collection dates. Its independently sampled managerial inputs are assumptions, not default probability laws. The 1979 journal version has not been checked for equivalence, and the appendix derivation referenced in the working paper was not located in the retained extraction. The present comparison therefore uses the planning structure, not a reconstructed return equation.

The recovered October 1990 working paper by Gentry and Hoque (1990) provides a distinct simultaneous-equation comparison. Its twelve signed cash-flow components come from income statements and balance-sheet changes. Each relative component divides by total cash flow. The nine-component free cash flow after working capital includes investment, dividends and fixed coverage expenditures as well as operating and working-capital flows (printed pp. 4–6). Those definitions differ from simply selecting a modern statement's operating cash-flow line.

Let S be the sum of those nine normalized components, N the normalized financing flow, O the normalized other-asset/liability flow, and C the normalized change in cash and marketable securities. The paper's equation (1) then gives

$$S + N + O - C = 0, \qquad S = C - N - O.$$

The second equality moves financing and other flows to the right; it neither chooses a financing instrument nor supplies a payment date. Different financing and cash changes can reconcile the same S while leaving different future obligations. Equation (7) of the retained paper prints its cash term without the normalization star used in equation (1). Every term in the displayed local derivation is explicitly normalized. The source notation ambiguity remains recorded, and

a zero total-cash-flow denominator has no supplied evaluation rule.

The paper separately proposes five behavioral equations linking operating flows, receivables, inventory, payables and investment (printed pp. 10–11). It compares OLS and three-stage least squares using 117 December-year-end companies. The exhibits report selected significant coefficients; the discussion describes specification changes based on system fit (printed pp. 13 and 19). Reported weighted R-squared therefore describes the fitted sample comparisons. The retained paper supplies neither a held-out forecast test nor the complete instrument matrix, company data and estimation code needed for local reproduction. Its interpretation of earlier models remains the authors' characterization where those models have not been inspected.

For AccDSL, this comparison separates a calculation, a reconciliation and a decision rule. A definition computes an amount from its inputs; an identity checks consistency among amounts; a behavioral rule proposes an economic decision under stated information and assumptions. A simultaneous behavioral system would also need a solution method, selection among solutions and explicit failure semantics. The current expression evaluator solves no such system. The paper instead motivates keeping these requirements distinct from the dated events and accounting operations that produce financial statements.

These methods suggest a common division of work. Financial-planning equations, stock-flow models and behavioural policies can propose economic actions. Contract terms and dated obligations determine events; recognition and measurement rules determine their accounting effects; reporting combines the resulting movements. This is a provisional architectural judgment for the initial company profile, not a finding that one simulation method dominates the others. The Francis–Rowell simultaneous-equation alternative remains uninspected, and a general library selection still needs exact interface and runtime evidence.

McCarthy's 1982 REA model remains a foundational source gap for the relation between resources, events and agents. Its uninspected technical results are not used here. AccDSL's event and counterparty records are specified and justified in Chapters 33 and 34. A later multi-company equilibrium engine would require a broader stock-flow literature and a separately justified economic model.

30.2 Incomplete information and provenance

Financial disclosures are incomplete databases in a substantive sense. Multiple underlying records can yield the same published observations. Libkin (2014), Sections 2–6, explains the semantics of incomplete databases, labelled nulls, conditional tables and certain answers. Repeated occurrences of the same unknown must share a value. Conditional tables can represent dependence between alternatives that independent missing cells cannot express. This is directly relevant to a debt note and a cash-flow table that refer to the same unknown repayment.

AccDSL adopts shared symbolic unknowns and conditions. It extends the database idea with accounting constraints, continuous amounts, temporal transitions and economic event models. A "certain" answer means a statement that holds for every history in a nonempty declared feasible family. This is Boolean certainty; a table of individually certain rows can lose existential information and must not imply that omitted rows are absent. The payable example in Section 32.2 develops this distinction. The family itself may omit unknown mechanisms; certainty is therefore always relative to a model scope. Imieliński and Lipski's original 1984 paper is a foundational backward lead whose publisher copy was inaccessible in this campaign. The inspected Libkin paper supplies the technical discussion actually used here.

Green et al. (2007), Definitions 3.1–3.2 and Section 4, treat provenance as annotations that compose through relational operations. Combining sources and alternative derivations can

therefore have formal semantics, rather than being a list of unrelated citations. This is useful for report aggregation and explanation queries. Positive relational operations covered by provenance semirings do not by themselves specify every operation AccDSL needs: negation, nonlinear valuation, optimisation and probabilistic conditioning require their own derivation records. Accordingly, the implementation starts with a typed dependency graph and may use semiring annotations for supported relational subqueries. A dependency edge is not a posterior probability.

In *Temporal Databases*, Snodgrass and Ahn (1986) distinguish historical validity from the time facts are stored. Their snapshot, rollback, historical and temporal examples support the need to recover both old business facts and old database beliefs. AccDSL adds public availability time because a database insertion date cannot establish whether a historical investor could have seen a filing.

30.3 Inverse problems, identification and probability

Stuart (2010), Section 2.2, starts from an observation equation $y = G(u)$ and distinguishes inversion, regularisation and Bayesian inference. The discussion makes explicit that a regularising norm or prior is an additional assumption. AccDSL has a similar observation problem, although its unknowns include discrete accounting interpretations and event structures as well as continuous parameters. The infinite-dimensional well-posedness theorems in that paper are not automatically applicable to this mixed, potentially discontinuous model.

The relevant issue is often identification rather than computation alone. Giacomini and Kitagawa (2021), Sections 2.1–2.6 and Theorem 1 of the archived 2020 working version, show how a class of priors can preserve uncertainty within an identified set. A likelihood that is flat over the unresolved part does not learn the conditional prior over that part. Posterior lower and upper probabilities summarise a class of posterior beliefs. The paper's regularity conditions and application to structural vector autoregressions do not prove that an arbitrary accounting program admits the same algorithms or guarantees. The design borrows the distinction between learnable quantities and assumption-dependent distributions; the simple accounting bounds in Chapter 32 are derived directly.

Hoeting et al. (1999), equations (1)–(3), give Bayesian model averaging as a weighted sum of model-conditional distributions, with weights determined by model priors and marginal likelihoods. Sections 3 and 5 discuss computation and prior specification. This supports an optional weighted mode, provided AccDSL specifies the likelihood, priors and candidate model space. It does not justify giving equal mass to every syntactically different journal reconstruction. Two encodings of the same economic history must not gain extra weight by duplication. Model averaging also remains conditional on the considered model family, rather than repairing missing mechanisms.

Scenario semantics need similar care. Pearl (2009), Section 3.2.1 and equations (5)–(6), distinguishes an intervention that replaces a structural equation from ordinary conditioning. AccDSL will distinguish a supplied exogenous path, an observational conditional forecast and an intervention. A rise in rates imposed by a stress test is not automatically distributed as the collection of historical periods in which rates happened to be high. Identifying a causal effect requires assumptions beyond reproducing accounts.

30.4 Languages for rules and contracts

van Deursen et al. (2000) survey DSL benefits, risks and implementation options. Mernik et al. (2005), Sections 1.1 and 2.2–2.5 of the archived CWI report, treat application libraries as serious competitors and distinguish deciding, analysing, designing and implementing a DSL. Its implementation decision tree explicitly includes embedding in a host language. The literature therefore supports a conditional design decision: a DSL is attractive when domain constraints and repeated notation justify it, but an ordinary typed library may be cheaper and equally expressive. AccDSL will implement one shared representation accessible from both a Python API and a textual language, so syntax does not become a prerequisite for testing the accounting model.

Merigoux et al. (2021), Sections 3–4 and Figures 8–9, provide Catala's scopes, exceptions, default calculus and translation into a conventional functional target. Its explicit empty and conflict outcomes are valuable precedents for accounting rules with exceptions. They are different from probabilistic uncertainty. AccDSL will preserve this distinction: conflicting applicable rules produce a diagnostic, while an evidence-supported alternative reading produces a conditional model branch. The paper's compiler correctness results do not establish correctness of our accounting rules or of an automatic translation from disclosures.

The inspected LegalMath code offers reusable ideas in its shared typed interpretation, partial observations, source references and validity/knowledge times. Its ordinary frontend execution selects a reading. AccDSL needs an additional execution mode that retains a family of readings, continuous unknowns and economic state. A bounded candidate generator is useful for search, but its candidate limit does not prove completeness.

Peyton Jones et al. (2000), Sections 3–5, build complex financial contracts from combinators and separate their compositional meaning from a chosen valuation model. The ACTUS technical specification similarly separates contract states, scheduled events, state transitions and payoffs, with risk factors supplied through an interface (ACTUS Financial Research Foundation, 2026). These are strong reasons to represent contractual obligations independently of accounting classifications. Neither contract cash flows nor a derivative price determine the recognition, measurement and presentation rules of a complete IFRS or US GAAP book.

Reusing such a component also requires agreement about dates and editions. The inspected ACTUS service requests core 1.1.0, but the accessible Java mirror identifies itself as 1.0-beta. The separately retained dictionary identifies version 1.4. Their correspondence is unproved. The mirror rounds timestamps to hours and counts complete elapsed days for Actual/365-Fixed. Northstar's authored draw at 11:00 on 4 January and interest settlement at 09:00 on 6 January are 46 hours apart: this formula counts one day, whereas the example's calendar-date convention counts two. The resulting interest formula values differ by a factor of two. Naming the day-count convention without its time rule would conceal a material modelling choice.

The inspected sources also disagree on fee/interest ordering and on maturity payoff when accrued interest or fees remain. These are findings about the retained editions, not an evaluation of the uninspected core 1.1.0. The exact source identities, function readings and counterexamples are recorded in `docs/design/research/actus-core-correspondence-20261023.md`. ACTUS remains a candidate contract-event component. Adoption requires a matched specification, dictionary and implementation plus explicit accrual, settlement and ordering conventions; the company accounting design can proceed while that dependency remains unresolved.

30.5 Extraction and financial reasoning

The document literature supports a staged ingestion process. PubTables-1M (Smock et al., 2022), in its canonicalisation procedure and evaluation section, distinguishes table detection, structure recognition and functional analysis. Its GriTS evaluation compares reconstructed table structure. TableBank (Li et al., 2020), Sections 3–4, uses document structure to obtain weak supervision for table detection and recognition. GTE and FinTabNet (Zheng et al., 2021), Sections 3–5, provide a financial-document setting and company-level dataset splits. These sources help select extraction components and evaluation units. Their table scores do not establish that a debt note has been interpreted correctly across years or that every off-balance-sheet exposure has been found.

The Docling report (Auer et al., 2024), Section 3, describes PDF parsing, page layout analysis and table structure recognition within a document-conversion pipeline. The locally archived version lists Christoph Auer first; this version difference is recorded in the catalogue. Docling is an ingestion candidate whose output must retain source locations and undergo accounting validation. Its code and model licences must be considered separately when packaging a deployment.

TableFormer makes the division between structure and text concrete. Nassar et al. (2022a), Sections 4–5, describe a model that receives a cropped table image and predicts its HTML structure together with a bounding box for each cell. The cell's words and numbers come from separately supplied PDF text; they are not generated by the structure decoder. Matching the predicted cells to that text is itself consequential. The supplementary algorithm can remove duplicate columns, attach previously unmatched text to a cell, or create a new cell (Nassar et al., 2022b, Section 2). A plausible grid therefore still needs its text assignments and structural corrections checked against the source.

The paper's evaluation preserves this distinction. Table 4 reports TEDS, a similarity measure based on tree edit distance, for structure *with content* on PubTabNet PDF documents containing cropped tables: 95.4 for simple tables, 90.1 for complex tables and 93.6 overall. The content comes from PDF cells; Table 2 separately reports structure-only results (Nassar et al., 2022a, Tables 2 and 4, p. 4620). These are results for that input population and text supply. They do not establish extraction accuracy for complete financial filings, or whether a cell has been assigned the correct entity, period, unit or accounting meaning. Those questions require the source-linked accounting checks developed here.

FinQA (Chen et al., 2021), Sections 3–6, provides questions with executable reasoning programs and supporting evidence. TAT-QA (Zhu et al., 2021), Sections 2–4, combines table cells and text spans with aggregation operations, and explicitly treats numerical signs in evaluation. DocFinQA (Reddy et al., 2024) extends financial questions to long documents and evaluates retrieval/context choices. FinBen (Xie et al., 2024) broadens the collection of financial tasks and model evaluations. These benchmarks can test retrieval and local numerical reasoning. Their question-answer format does not require a complete multi-period company representation, joint alternative histories or scenario-dependent journal generation.

A directly relevant recent competitor is the auditable RAG architecture of Cosentino et al. (2026). Sections 3.1–3.2 describe persistent extraction, enrichment, retrieval and provenance records; Section 4.6 checks citation resolvability and failure recovery. Section 5 limits the interpretation of those results to the evaluated document-intelligence tasks. AccDSL should reuse the idea of persistent evidence and recoverable processing, while adding executable accounting and economic semantics. No performance comparison with that architecture has been run here.

30.6 Solvers and probabilistic execution

de Moura and Bjørner (2008), Sections 2–3, describe Z3's interfaces and theory-solving architecture. It supports choosing an SMT solver for logical choices and arithmetic constraints, but does not make arbitrary nonlinear simulation tractable. The first solver fragment is deliberately bounded: finite choices, exact linear arithmetic and selected calendar relations. Unsupported formulas remain in the representation with a solver diagnostic.

van de Meent et al. (2018) develop probabilistic programs through sampling, observations and inference; the inspected material includes the first-order language and weighted sampling discussion. Stan (Carpenter et al., 2017), its model language and sampler-output discussion, and Pyro (Bingham et al., 2019), Sections 2–3, show different ways to separate a probabilistic model from inference machinery. They are candidates for optional inference over an already-defined economic model. Exact accounting identities and discrete rule selection should not be turned into nearly satisfied likelihood penalties merely to fit an inference library. Nor should successful sampler execution be described as evidence that all interpretations were explored.

30.7 The remaining design problem

The combined literature provides algebra for journals, semantics for missing information, temporal storage, provenance, contract languages, rule languages, inference methods and extraction tools. The unresolved integration problem is to retain their distinctions while connecting them: a disclosure constrains economic histories; a history supplies contract and behavioural state; a scenario drives events; accounting rules produce journals; reporting rules render statements; and every step remains explainable.

The design is therefore an integration proposal with several local mathematical constructions. Its originality and empirical advantage have not been established by this survey. Its practical merit will depend on the acceptance cases, accounting review and comparative implementation work in Chapter 40.

Chapter 31

Architecture and reuse

31.1 One representation, several authoring methods

The architectural centre is a typed intermediate representation (IR), not a text parser. A document extractor, a human editor, a Python library and the AccDSL compiler all create the same records. This allows a modeller to begin with a reviewed structured file and later add convenient notation. It also allows source extraction to improve without changing accounting semantics.

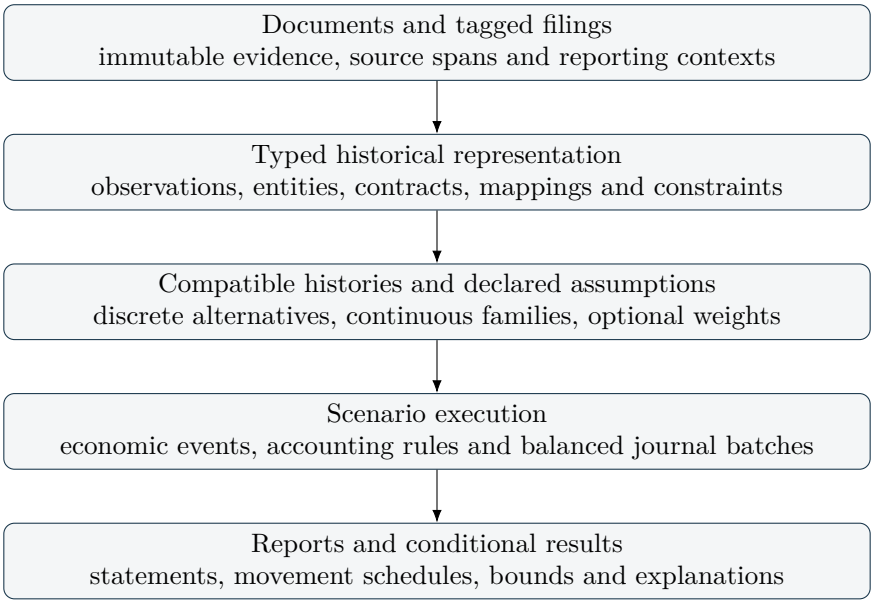


Figure 31.1: The simulation path. Every transformation records its inputs and assumptions; the source record remains available throughout.

Evidence storage and economic modelling must remain separate. If a company reports 100 of borrowing, a model can assume two loans of 60 and 40 without creating a new reported fact that the company disclosed those loans. The contract library consumes the assumed terms, the accounting module consumes the resulting events, and the explanation graph records the assumption at each affected result.

31.2 Why a DSL, and when another approach is better

The strongest baseline is a typed schema, SQL storage and ordinary Python code. That combination can implement the whole product. Its disadvantage is that repeated accounting requirements can become conventions scattered across functions: implicit currency conversion, missing period checks, hidden default values and duplicated rule selection. A bounded language makes these requirements part of typing and compilation. Its costs are language maintenance, diagnostics, migration and training.

Approach	Useful role	Limitation for this product
Typed schema and Python	Reference executor, flexible economic models, initial authoring API	Domain restrictions require disciplined APIs or a validator; arbitrary code complicates static analysis and provenance.
XBRL and OIM	Tagged evidence, contexts, dimensions, taxonomies and exchange	Reported facts and reporting relationships do not recover hidden economic histories or specify scenario behaviour.
Ledger languages	Recording known transactions, balance checks, export and differential testing	Alternatives, source interpretation and contract exposure need additional semantics.
Knowledge graph / SQL	Relationships, lineage, temporal queries and evidence navigation	A graph or schema alone does not define event execution or the probability meaning of alternatives.
Rules / Catala	Dated policy logic, defaults, exceptions, traceable rule evaluation	Ledger state, event scheduling and ensembles require an enclosing runtime.
Probabilistic program	Priors, likelihoods and inference over model parameters	Needs declared probability measures; cannot turn unspecified ambiguity into probability or supply accounting rules.
Spreadsheets	Familiar review and small transparent financial models	Versioned evidence, shared unknowns and branching histories are difficult to govern as the model grows.
AccDSL over shared IR	Common domain checks, reusable templates, explicit alternatives and deterministic posting semantics	Requires implementation and evaluation; broad notation without stable semantics would add cost.

The decision is to implement the IR, validators and Python reference API first, then the textual frontend. The same conformance cases must pass through both frontends. If the DSL fails to reduce ambiguity in representative authoring tasks, the IR and library remain useful. This is a reversible product decision, not a claim that a new language is inherently better than a library.

31.3 Open-source components

The following survey uses official repository descriptions, licence files and selected source interfaces archived on 2 October 2026. It is a source inspection, not an executed integration or performance evaluation. The machine-readable tool ledger records retrieval paths and available version metadata. Dependencies must be pinned when implementation begins; a branch name observed during this survey is not a release requirement.

Component	Proposed use and checked boundary	Licence / decision
Arelle	Parse and validate XBRL/iXBRL. Inspected `ModelFact`, context and unit accessors. Preserve taxonomy extensions and full contexts when importing.	Apache-2.0 core; first-choice adapter.
Docling	PDF layout, text and table conversion. Inspected converter interface; retain page coordinates and extraction settings.	MIT code; assess model licences separately. Candidate.
Table Transformer	Table extraction baseline associated with PubTables-1M; check finance-domain performance before selection.	MIT; repository marked archived in inspected metadata. Baseline, not a maintenance default.
Beancount	Posting and transaction data types; export selected concrete branches and compare balances.	GPL-2.0 project metadata; integration and redistribution choice needed. Optional adapter.
Ledger	Known-journal reference and export. Inspected transaction balancing code and manual date/assertion behaviour.	BSD-style licence; optional reference tool.
hledger	Alternative ledger import/export and accounting checks. Repository documentation inspected.	GPL-3.0 project metadata; optional process adapter.
Catala	Potential backend for deterministic policy modules; defaults and exceptions have useful semantics.	Apache-2.0; prototype only after core execution semantics stabilise.
ACTUS	Contract schedules, payoffs and transitions; external risk-factor interface. Verify event schedules against ACTUS tests.	Service Apache-2.0; technical specification CC-BY-SA-4.0. Candidate contract adapter.
QuantLib	Calendars, cash flows and valuation. Floating coupon source separates fixing dates and pricing. Accounting treatment remains external.	Modified BSD-style licence; candidate valuation adapter.
OpenFisca	Versioned tax-benefit formulas as a rule-system reference; inspected variable/formula metadata. Corporate-tax coverage must be supplied.	AGPL-3.0; optional, not the core runtime.

Component	Proposed use and checked boundary	Licence / decision
Z3	Finite alternatives, logical compatibility, linear constraints and conflict explanations.	MIT licence text; first solver candidate.
DuckDB	Local analytical queries over extracted facts and results.	MIT; candidate alongside SQLite transactional storage.
Stan	Continuous probabilistic models and diagnostics with an explicit target. Inspected model interface and paper.	BSD-3-Clause; optional inference backend.
Pyro / PyMC	Alternative probabilistic backends for model-specific inference. Pyro design paper and project descriptions inspected.	Apache-2.0 / Apache-2.0 licence texts; optional, no default selection.

The licence entries identify the inspected projects, not a legal conclusion about every combination of code, models and datasets. The default product architecture can communicate with optional tools through explicit adapters. This does not, by itself, settle redistribution obligations. That question is resolved when a concrete dependency and distribution method are selected.

XBRL's Open Information Model supplies standardised facts with concepts, entities, periods, units and dimensions (XBRL International, 2021). Those facts are an excellent ingestion source. AccDSL retains their original contexts and decimals rather than flattening them immediately into a company-by-year table. Calculation relationships can help validate reporting totals, but are not a general description of the transactions that generated those totals.

31.4 Responsibilities that AccDSL must implement

AccDSL owns the evidence and temporal model, typed accounting concepts, shared unknowns and alternatives, rule-package selection, journal validation, consolidation provenance and report explanations. It also owns the distinction between feasible sets and distributions. These responsibilities cannot be delegated implicitly to whichever parser or solver is convenient.

External components receive narrow requests and return typed responses. A contract adapter returns scheduled events, cash-flow quantities, state changes and source/model references; it does not directly mutate the ledger. A valuation adapter returns a measure, currency, date, method and numerical status; the accounting package decides whether and where to recognise the change. A solver returns feasibility or bounds with scope and completeness status. A language model proposes source-linked mappings or interpretations; it cannot commit an unsupported assertion as fact.

The architecture consequently permits deterministic reference execution, alternative solvers and later acceleration without changing the meaning of a historical observation or a journal entry. Performance work begins only after the same semantic cases pass through these interfaces.

Chapter 32

Historical compatibility and future simulation

32.1 Define what is observed before inverting it

Let D_k be the disclosures available under information cutoff k. Each numeric observation d_i has a context: entity or group perimeter, concept, measurement basis, dimensions, currency, period and precision. These contexts are part of the observation operator, not additional columns that can be discarded after extraction.

A candidate history h contains an opening economic and accounting state, events through the last historical date T, contract terms, accounting policies, consolidation relationships and latent quantities. Let M denote the declared model scope: its event templates, admissible rule versions and allowed unknowns. It excludes arbitrary events that the chosen model has no way to represent. The reporting operator $O_i(h)$ produces the quantity to which observation i refers, including its basis and perimeter.

Reported numbers are often rounded. If a number v_i is rounded to the nearest unit q_i, an illustrative admissible interval is $[v_i - q_i/2, v_i + q_i/2)$ under the corresponding tie convention. This half-open endpoint choice is used only when that tie rule is declared; Section 29.12 gives the conservative closed enclosure when nearest rounding is assumed but the tie rule is unknown. The actual rounding convention, scale and XBRL accuracy metadata govern the imported constraint. A disclosure with unspecified precision does not acquire a convenient tolerance silently. Write $\mathcal{I}_i$ for the justified observation interval, and $C_M(h)$ for the joint accounting, contractual and semantic constraints. Then the historical compatibility set is

$$\mathcal{H}_M(D_k) = \{h \in \mathcal{H}_M : C_M(h) \ \wedge \ O_i(h) \in \mathcal{I}_i \text{ for every applicable } i\}. \tag{32.1}$$

Qualitative assertions enter as reviewed logical conditions, rather than numeric intervals. Incompatible source contexts are never forced into the same equation.

Equation (32.1) distinguishes three outcomes. An empty set means that the observations and declared model are inconsistent. A singleton means that this model identifies a history at its chosen resolution. A larger set preserves unresolved alternatives. An empty set does not prove that the company's accounts are false: a wrong extraction, missing event type or incorrect perimeter can cause the inconsistency. A singleton does not prove that the model contains the real history.

32.2 A certain liability need not identify its creditor

Suppose the admitted records establish exactly one payable of 100 currency units, but leave its creditor unresolved between A and B. For this constructed example the declared family has two histories: in h_A the company owes A 100 and owes B nothing; in h_B it owes B 100 and owes A nothing. These are complete alternatives within the example, not two sampled points from a wider family. The question *does a payable exist?* has the answer yes in both histories. The question *is A owed 100?* does not.

Now ask for the rows of the payable schedule. If $Q(h)$ returns those rows, then $Q(h_A) = \{(A, 100)\}$ and $Q(h_B) = \{(B, 100)\}$. Their intersection is empty: no creditor–amount row occurs in both. Reading that empty table as the company's complete schedule would wrongly assert that it owes nothing. Reading the union as its complete schedule would instead assert that it owes 200. Evaluating the total within each history gives 100 in both. The uncertainty concerns the creditor, not the existence or total of the payable.

This distinction is central to Libkin's critique of intersection-based answers. His example retains a tuple with an unknown second component; discarding it loses knowledge that a tuple with the stated first component exists. An intersection read under a closed-world interpretation can also add an unsupported absence claim (Libkin, 2014, Section 3 and the opening of Section 6). A table of individually certain rows is therefore a restricted view of knowledge, not generally a replacement for the possible answer schedules.

For a nonempty feasible history family $H = \mathcal{H}_M(D_k)$ and a total, deterministic query Q on its histories, retain the family of answers

$$\mathcal{A}_Q(H) = \{Q(h) : h \in H\}.$$

Each answer remains linked to the histories that produce it. Equality of payable schedules alone does not license merging histories for another query. For a row-valued query, a separately requested certain-tuples view is

$$T_Q(H) = \bigcap_{h \in H} Q(h).$$

Membership in $T_Q(H)$ asserts that a row appears in every answer. Omission asserts only that the row is not established in every answer; it does not assert absence from every answer. A total computed after taking this intersection answers a different question from totals computed within histories. In the payable example, $T_Q(H)$ is empty while $\{\text{total}(Q(h)) : h \in H\} = \{100\}$.

For a proposition P about a concrete answer, Boolean certainty has the separate meaning

$$\text{Certain}_H(P \circ Q) \quad \Longleftrightarrow \quad \forall h \in H,\ P(Q(h)).$$

Taking P to mean that the schedule contains at least one payable proves existence without identifying A or B as the creditor. Allowing a third history with no payable would change that answer. The two-history result consequently depends on the admitted evidence and scope that require the liability to exist, not merely on the two rows chosen for display.

Libkin distinguishes a logical description of knowledge shared by all answers from an object representing their greatest lower bound in an information ordering. These representations need not exist in the chosen language; ordinary tuple intersection is not their general definition (Libkin, 2014, Section 5.3, equations (6)–(8)). AccDSL's answer family and Boolean propositions above are local definitions. They do not assert that every family has a finite relational representation. A symbolic answer must carry its conditions and preserve the stated family, or declare the restriction or approximation it makes.

Input completeness and output meaning must also be explicit. A model that fixes the available event templates and contract slots imposes a scope restriction; it does not establish that a public filing lists every economic event. Nor does that restriction entitle a display to interpret every omitted row as false.

Libkin's primary proof compares information by reverse inclusion of possible completions: an object is more informative when fewer complete objects remain compatible with it. For a query to be evaluated directly on an incomplete object, the chosen output interpretation must preserve that order as input information increases. The query must also respect the relevant renamings, and its output domain must have a representation system whose formulas describe each object's complete interpretations (Libkin, 2014, Sections 6.1–6.2, equations (9)–(10)). In Theorem 2, the proof first shows that evaluating the query preserves exactly the statements true in all completed answers; it then uses those statements to establish the greatest-lower-bound answer as an object (Libkin, 2016a, Definitions 4–7 and Theorem 2, pp. 11–18). Monotonicity alone is insufficient, and SQL null comparisons need not meet the genericity condition (Libkin, 2016a, Section 7.1, pp. 18–19).

The relational examples make the difference between input and output assumptions concrete. Positive existential queries have the stated result with open-world inputs and answers. A guarded positive fragment also has it with *closed-world inputs and open-world answers*; the latter is an interpretation of omitted result rows, not a claim that the input filing lists every economic event (Libkin, 2016a, Proposition 9 and Corollary 3, p. 20). These results do not authorize evaluating arbitrary financial negation, aggregates or nonlinear constraints by treating unknown amounts as ordinary constants. AccDSL has not established the theorem's abstract domain, information order and representation-system conditions for every financial output. The constructed payable example above instead follows directly from its two declared histories.

A related relational method illustrates what a sound but incomplete answer means in practice. For marked-null tables under closed-world set semantics, Libkin's modified three-valued evaluator returns a certainly positive answer when it yields 1 and a certainly negative answer when it yields 0; its value 1/2 carries no guarantee (Libkin, 2016b, Theorems 4.1 and 5.3, pp. 8–12). Let $T = \{(1, \bot)\}$ and ask whether there is a y for which $T(1, y)$ and either $y = 1$ or $y \neq 1$. Every completed table answers yes, yet the modified evaluation leaves the answer at 1/2: its null comparison and its negation are both unknown. The paper gives this case as a missed certain answer for ordinary SQL; the modified evaluator's outcome follows from its atomic rules. An inconclusive evaluation therefore does not show that two possible histories give opposite answers. AccDSL requires witnesses on both sides to establish that stronger claim; a timeout or unsupported operation remains undecided. The relational guarantee itself does not transfer to financial totals, nonlinear valuations, temporal histories or scenario conditioning.

The query contract records whether it returns a proposition, individually certain rows, or a family of answers, together with the family scope and completeness and the interpretation of omitted rows. Unsupported evaluation and unfinished search keep their existing failure or unresolved status. In particular, a sampled answer family cannot be relabelled as the full family merely because its displayed rows agree.

32.3 Why published statements cannot identify all transactions

The simplest derivation uses fixed account mappings and a fixed basis. Let $z \in \mathbb{R}^n$ contain unknown historical movement amounts. For m disclosed aggregates, let $A \in \mathbb{R}^{m \times n}$ be the aggregation matrix and $y = Az \in \mathbb{R}^m$. If there is a nonzero vector v with $Av = 0$, then

$$A(z + v) = Az + Av = y.$$

Both movement vectors give the same reports whenever both satisfy the remaining constraints. Thus the feasible family is the affine set $z + \ker A$ intersected with those constraints. Accounting requirements can shrink this family, but do not generally eliminate it. For example, cash collections in a receivables bridge are constrained by

$$AR_T = AR_0 + \text{credit sales} - \text{collections} - \text{write-offs} + \text{other classified movements}.$$

If the note does not distinguish acquisitions, currency effects and write-offs, several collections paths may fit the same closing balance.

There is a second ambiguity beyond aggregation. In Equation (27.1), identical historical movements arise from different contractual reset terms. Even recovering z exactly would leave these terms unidentified. AccDSL therefore infers economically relevant states and contract terms alongside journal movements. A linear inverse solver over statement lines alone cannot solve this second problem.

32.4 The accounting state and its invariants

For one entity book with n_a accounts in one functional currency, let $b_t \in \mathbb{Q}^{n_a}$ be signed account balances, with debit amounts positive and credit amounts negative. Exact decimal amounts can be represented as rationals with a power of ten denominator. A journal batch j_t contains the signed functional-currency amounts posted by one atomic event or explicitly grouped event batch. Its balance condition is

$$\mathbf{1}^{\mathsf{T}} j_t = 0, \qquad b_t = b_{t^-} + j_t. \tag{32.2}$$

If $\mathbf{1}^{\mathsf{T}} b_0 = 0$, substitution gives $\mathbf{1}^{\mathsf{T}} b_t = \mathbf{1}^{\mathsf{T}} b_{t^-} + \mathbf{1}^{\mathsf{T}} j_t = 0$. Induction establishes conservation through every committed batch. Opening balances must therefore be supplied as a balanced, explained entry or a balanced imported state with unresolved composition explicitly identified.

This proof establishes a bookkeeping invariant. It does not establish that an amount belongs in the chosen account, that a contract exists, or that an accounting standard permits a treatment. Posting validation additionally checks account roles, ownership, currency conversions, applicability of rules and event preconditions.

The balance vector includes temporary revenue and expense accounts during the period. Period closing transfers their net balance to equity through balanced entries. A reporting function can include current profit in equity before closing, or use the closed book afterwards, but must not include it twice. Income and cash-flow reports use classified movements and exclude mechanical closing transfers from operating flows. Consequently statements are functions of balances, journals, contract state and reporting rules:

$$Y_{[u,v)} = R(b_{v^-}, J_{[u,v)}, X_{v^-}, \rho, [u, v)). \tag{32.3}$$

Here ρ specifies the reporting basis and perimeter, and X contains economic state beyond ledger balances. Cash-flow classification is part of R, not recoverable from the closing cash balance alone.

Each book has its own functional currency. Quantities such as shares, tonnes and foreign-currency units are typed separately and never added to this balance equation. A foreign-currency posting stores its original units and the functional-currency valuation with its rate source. Translation into a group's presentation currency and consolidation adjustments occur in a separate book under explicit rules.

32.5 Simulate events from economic state

For a chosen history h, the state at T is $X_T(h)$. It contains balances, open contracts, lots or cohorts, unpaid accruals, commitments and other quantities needed for future evolution. Let s identify a scenario definition, U_t^s its supplied economic drivers, a_t management's actions, and ε_t future innovations. Let θ denote the declared behavioural and event-generation parameters, with values specified by the chosen history, scenario and model assumptions. A behavioural policy π chooses a_t using the permitted information $\mathcal{F}_t$:

$$a_t = \pi(X_t, U_{\le t}^s, \mathcal{F}_t; \theta), \tag{32.4}$$

$$e_{t+1} = \mathcal{G}(X_t, a_t, U_{t:t+1}^s, \varepsilon_{t+1}; \theta), \tag{32.5}$$

$$(j_{t+1}, m_{t+1}) = \mathcal{A}(e_{t+1}, X_t, r_{t+1}), \tag{32.6}$$

$$X_{t+1} = \mathcal{T}(X_t, e_{t+1}, j_{t+1}, m_{t+1}). \tag{32.7}$$

The generator $\mathcal{G}$ emits events, such as a sale, rate reset, payment or default. The accounting function $\mathcal{A}$ applies rule package r and returns journal postings plus measurement annotations m. The transition $\mathcal{T}$ updates economic and ledger state atomically. A default can therefore change the loan's legal status, expected recoveries, recognised allowance and cash collections through related, but distinct, records.

This ordering is a reference abstraction. Within a date, event dependencies and phases specify the actual order. Expectations used to measure an allowance or provision must be calculated from the information permitted at that date. They cannot use the realised future path of the simulation. An inner expectation model may use quadrature, analytic formulas or nested simulation; its approximation and seed belong in the measurement annotation.

Some scenarios specify a deterministic macroeconomic path while leaving idiosyncratic defaults random. Others supply a joint stochastic driver model. A risk-neutral measure used to value a derivative is distinct from the real-world measure used to simulate business outcomes. Every stochastic adapter declares its measure and conditioning information. The accounting function receives the resulting valuation with that declaration.

32.6 Preserve sets and distributions separately

Let $K_{h,s}$ be the probability law of future reports produced by the forward model conditional on history h, scenario s and its declared behavioural assumptions. A deterministic simulation is the special case of a point mass. When no probabilities over historical explanations are justified, the natural answer is a family of laws,

$$\mathcal{K}_{M,s}(D_k) = \{K_{h,s} : h \in \mathcal{H}_M(D_k)\}. \tag{32.8}$$

The system can report each explanation, bounds on a deterministic outcome, or lower and upper event probabilities across this family. Those are different result types. A range of conditional means is not a confidence interval, and an envelope of conditional quantiles is not itself a single predictive distribution.

If a defensible probability measure $\mu(dh \mid D_k, M)$ supported on $\mathcal{H}_M(D_k)$ is supplied, histories and report outcomes must have specified measurable spaces. Assume that $K_{h,s}$ is a probability kernel: it is a probability measure on report outcomes for each h, and $h \mapsto K_{h,s}(B)$ is measurable for each measurable event B. The predictive law is then

$$\mathbb{P}(Y^s \in B \mid D_k, M) = \int K_{h,s}(B)\, \mu(dh \mid D_k, M). \tag{32.9}$$

For discrete explanations this is the usual weighted sum. For continuous unknowns, integration uses the declared measure over those unknowns, not a count of solver solutions. Bayesian weights require a prior and a likelihood; expert weights remain labelled as elicited. A likelihood must not treat a statement total and all its components as independent repeated evidence. Deterministic equality constraints require a reduced parameterisation or proper conditional-measure treatment, rather than an undocumented Gaussian penalty with near-zero variance.

Suppose the fixed-loan explanation receives weight w and the floating-loan explanation receives $1 - w$, solely as an illustration. Under the rate shock,

$$\mathbb{P}(I = 5) = w, \qquad \mathbb{P}(I = 7) = 1 - w, \qquad \mathbb{E}[I] = 7 - 2w.$$

The expected expense is useful but hides the two different outcomes. For a general scalar output with finite second moment, put $m(h) = \mathbb{E}[Y \mid h]$. Write $Y - \mathbb{E}[Y] = (Y - m(h)) + (m(h) - \mathbb{E}[Y])$. The cross term vanishes because $\mathbb{E}[Y - m(h) \mid h] = 0$. Taking expectations of the squared expression gives

$$\mathrm{Var}(Y) = \mathbb{E}_h[\mathrm{Var}(Y \mid h)] + \mathrm{Var}_h(\mathbb{E}[Y \mid h]). \tag{32.10}$$

The first term is uncertainty within a chosen explanation; the second captures differences between explanations. This decomposition requires the measure μ. Without it, the system reports the family in Equation (32.8), rather than manufacturing a variance.

Weights themselves can be uncertain. For an admissible class $\mathcal{W}$ of measures, AccDSL can compute

$$\underline{p}(B) = \inf_{\mu \in \mathcal{W}} \int K_{h,s}(B)\, d\mu(h), \qquad \overline{p}(B) = \sup_{\mu \in \mathcal{W}} \int K_{h,s}(B)\, d\mu(h).$$

If every measure supported on the feasible histories is allowed and point masses are included, these bounds reduce to the infimum and supremum of $K_{h,s}(B)$ over histories: averages cannot lie outside the range of their integrands, and point masses attain, or approximate, the extremes. Narrower weight restrictions require an explicit justification and a separate optimisation problem.

32.7 Resolution and computational limits

For a requested horizon H and scenario family $\mathcal{S}$, histories may be grouped only when they have the same conditional law for all requested outputs:

$$h_1 \sim_{H,\mathcal{S}} h_2 \quad \Longleftrightarrow \quad K_{h_1,s} = K_{h_2,s} \text{ on those outputs for every } s \in \mathcal{S}.$$

This definition explains why individual sales can sometimes be replaced by cohorts, while fixed and floating loans usually cannot be merged for an interest-rate stress. Exact equivalence can be difficult to prove. A grouping justified only by approximate numerical agreement carries an error criterion and is not labelled exact. Source provenance remains available even when execution shares a common state.

Bounds require comparable honesty. A linear program can provide certified extrema for a supported linear model. A set of feasible sampled histories provides observed extrema: its minimum is no smaller than the true infimum, and its maximum is no larger than the true supremum. It therefore supplies an inner exploration of possibilities, not a certified outer bound. Interval methods may provide outer bounds that no single history attains. Every result records which of these meanings applies.

The initial implementation offers exact bounded choices and linear constraints, symbolic continuous families such as $\alpha \in [0, 1]$, and explicit scenario execution at selected feasible states. General nonlinear partial identification, automatic model discovery and exhaustive enumeration of all economic histories are research extensions, not promises made by the core language.

32.8 A solver formulation for ordinary rounded disclosures

The debt example used an exact principal. Real disclosures often do not. Suppose debt is shown as 100 and interest as 5, each rounded to the nearest unit with the declared half-open intervals $D \in [99.5, 100.5)$ and $I_0 \in [4.5, 5.5)$. Using an unknown fraction α and unknown total D would introduce the product αD. That expression is bilinear and is outside the promised linear fragment.

Instead use principal amounts F and V. The compatible family satisfies

$$F, V \geq 0, \quad 99.5 \leq F + V < 100.5, \quad 4.5 \leq 0.05(F + V) < 5.5. \tag{32.11}$$

Future interest is $I_1 = 0.05F + 0.07V$. Its infimum is $0.05 \times 99.5 = 4.975$, attained with $F = 99.5, V = 0$. Its supremum is $0.07 \times 100.5 = 7.035$, approached as $F = 0$ and V tends to 100.5 from below, but not attained. A closed interval is a valid enclosure, but its upper endpoint must not be reported as attainable. The exact solver check separately tests that no better value exists and whether each endpoint is attainable.

For $D > 0$, the amount representation is equivalent to the fraction representation through $D = F + V$ and $\alpha = F/D$. At zero principal the fraction is undefined and must be omitted or treated as irrelevant, not recovered. A known fraction restriction $\alpha \geq a$ becomes $F \geq aD$ when $D > 0$ and a is fixed. If both a and D are unknown, that product remains nonlinear. Reformulation has a domain of validity; it is not permission to send arbitrary formulas to a linear solver.

The supported exact fragment contains rational linear equalities, weak and strict inequalities, and finite Boolean or enumeration choices. Guard partitions become finite disjunctions. Posting in integer cents introduces a different integer problem if performed before bounding. Consequently the request specifies whether bounds apply before quantisation, after a monotone declared rounding map, or to a bounded integer model. Approximate numerical valuations carry error metadata outside the exact rational kernel. Solver timeout, unsupported theory and unbounded objective have distinct responses; none is relabelled as infeasibility.

32.9 Building the family with explicit templates

The reconstruction input is a model-scope record, not an invitation to invent unlimited journals. It lists event templates, slots per entity and period, allowed contract classes, source mappings, opening-state domains, calendars, resolution and excluded operations. A template gives an incidence vector for postings and a transition for economic state. Northstar's annual bridges arise by substituting its collection, purchase and payment templates into opening-plus-movement identities.

For each allowed slot, create an activity flag and nonnegative amount; an inactive slot has zero amount. Instantiate the template's posting and state constraints, unify variables referring to the same observation or economic contract, and match report expressions to source domains through validated context mappings. Add cross-note restrictions and choice-group dependencies to this same formula. No separate account-by-account reconstruction is allowed. Mark every constraint with its evidence, rule or assumption origin.

Enumerate the finite structural choices in a deterministic order. For each cell, ask for feasibility and retain the continuous formula, rather than enumerating real-valued solutions. An infeasible cell preserves a sufficient conflicting subset; an undecided cell preserves its formula and reason. Search is complete within the declared scope only when every finite cell has been decided, every applicable observation has been matched or explicitly qualified, and no resource truncation

remains. A witness establishes existence in one cell. It does not establish complete search, and complete search does not establish that the scope includes the company's real history.

In Northstar the declared templates identify annual collections, purchases and supplier payments, but retain every $F \in [0, 100]$. Adding a verified note that $F = 60$ narrows this family by conjunction. Adding a proposed "all floating" reading to that revised view produces a rejected cell with the conflicting note attached. Adding a rule that expenses a capital contribution would instead fail treatment admission: inconsistent evidence and unsupported accounting have different diagnoses.

32.10 Which details may safely be combined

A resolution record names the outputs, horizon, scenario family, grouping key, required state, transformation and exact or approximate status. Safe exact transformations are deliberately few. Cash balances in the same book and currency can be added. Homogeneous inventory lots may be combined under a weighted-average policy if ownership, cost basis, impairment state and future costing treatment coincide. Debt principal can be added only within a group sharing currency, coupon/reset function, fixing and payment calendars, maturity, options, guarantees and measurement method. The transition and reporting maps must be additive on that group.

The last condition supplies a short proof obligation. If a transition obeys $T(x_1 + x_2, u) = T(x_1, u) + T(x_2, u)$ and reporting is additive for every admitted driver u, induction preserves equality after grouping. This does not apply to covenant thresholds, counterparty netting, nonlinear guarantees or lot-specific impairment. Two fixed loans with the same dates and rate can share an execution state under this test. The fixed/floating pair cannot; neither can two policies with different guarantees merely because their current reserves match.

A changed horizon, scenario, rule version or requested output invalidates the resolution approval and reruns the sufficiency check. Extending beyond a debt maturity requires schedule detail that an earlier short-horizon view could omit. Approximate cohorts retain the detailed source references and a stated error criterion tied to outputs. Unless that error is bounded or evaluated, the grouping remains an approximation awaiting evidence, not an exact quotient of the history family.

32.11 A finite probability model that can actually be evaluated

Optional probability weighting begins with disjoint economic hypotheses and a declared joint measure. As an illustration, assign fixed and floating hypotheses prior probabilities 0.6 and 0.4. The stress itself leaves these weights unchanged and gives mean interest $0.6(5) + 0.4(7) = 5.8$. Suppose genuinely new observed evidence has likelihood 0.2 under fixed and 0.8 under floating. The unnormalised posterior masses are 0.12 and 0.32; normalisation by 0.44 gives 3/11 and 8/11. These likelihoods are illustrative inputs, not extraction-confidence scores.

The finite reference algorithm enumerates joint hypothesis and innovation states, evaluates their declared joint mass and any observation likelihood once, checks the nonzero normalising constant, and propagates the entire statement vector on each state. It sums masses only across disjoint events. Two overlapping labels such as "fixed share at least one half" and "all fixed" are query views and cannot become two independent mixture components. Totals and their components share an observation block so their evidence is not counted twice. Continuous inference needs an additional parameterisation, density or conditional measure and an assessed numerical method; the finite algorithm supplies no guarantee for those extensions.

Under the prior, before the new observation is admitted, if the floating branch has an unsupported valuation operation, its mass 0.4 remains unresolved. The run may report the fixed result with successful mass 0.6, but cannot report 5 as the unconditional mean by renormalising away the failure. A separately requested mean conditional on successful execution must show that conditioning event and its mass. For unweighted families, an unresolved branch blocks a complete envelope over the family. Economic default is retained as a model outcome with remaining obligations, rather than being treated as a computational error to remove.

32.12 A query keeps its domain, endpoints and memory

The same displayed amount can arise from different input domains, and the same ending cash can arise from different histories. A query therefore carries its domain, operation order, endpoint status and requested state. The examples below make those fields concrete before the implementation is asked to reuse a summary.

Real amounts, rounded outputs and a cent grid are different models

Return to the debt family in which nonnegative fixed-rate principal F and floating-rate principal V satisfy

$$99.5 \le F + V < 100.5, \qquad 4.5 \le .05(F + V) < 5.5.$$

The future interest query is $I = .05F + .07V$. The lower endpoint is attained at $F = 99.5, V = 0$, giving $I = 4.975$. The upper endpoint is the supremum $.07(100.5) = 7.035$, but it is excluded because the reported total is strictly below 100.5. The exact real image is $[4.975, 7.035)$, with an attained lower endpoint and an unattained upper endpoint. A solver result must carry both facts; writing a closed interval would change the set. To establish that there are no gaps, hold the total at 99.5 and vary V from zero to 99.5. Then $I = 4.975 + .02V$ covers $[4.975, 6.965]$. The all-floating histories with totals from 99.5 up to, but excluding, 100.5 cover $[6.965, 7.035)$. Their union is the stated interval. Conversely, $.05(F + V) \le I \le .07(F + V)$ excludes values outside those bounds.

Now ask for a display rounded to cents, using nearest rounding with exact halfway values away from zero. Every displayed value from 4.98 through 7.03 has a witness in the real family. The lower raw endpoint displays as 4.98. The excluded raw supremum would display as 7.04, but no admissible amount reaches it. Thus the rounded image has 206 values, 4.98–7.03. This is an image of the real family, not evidence that the underlying amounts are cents. For any cent amount x in the image with $x \le 6.965$, a witness is $V = (x - 4.975)/.02$ and $F = 99.5 - V$. Above 6.965 choose $F = 0, V = x/.07$. These formulas also show why enumerating cent displays has not discretised the unknown principal.

A different model restricts F and V to integer cents before computing interest. Its maximum is 7.0343 at $V = 100.49$, and its rounded display still runs from 4.98 to 7.03. Equal displays do not prove equal latent families: the real model admits values between 7.0343 and 7.035 that the cent-grid model excludes. A stored result must therefore retain the model identity and input quantum.

Signed and tie cases expose the same boundary. Negating the real image gives $(-7.035, -4.975]$; its rounded outputs are the cents from -7.03 through -4.98. The lower endpoint remains excluded. The interval $[0, 1)$ can round to 1.00 because 0.995 is an admitted value, even though 1 itself is not. A certified enclosure $[1.0049, 1.0051]$ crosses the 1.005 tie and cannot select one posting without more information. Unknown tie rule, zero denominator, and an unproved

nonzero denominator are separate failures. Rounding each of two 0.005 lines before adding gives 0.02, while adding first and rounding gives 0.01. Operation order belongs to the query method.

A projection is not a reusable transition state

For a concrete counterexample, take two separately retained histories in which the timing of a reinsurance treaty changes when income is recognised. Both end with cash 88.3. Their initial and first-year equities are 85 and 83 for the timely treaty, but 70 and 74.75 for the late treaty. These are the supplied comparison values here; Section 38.6 derives them from the separate issued and held contracts. The terminal-cash projection is therefore equal while the transition states and service results are different. A query requesting only terminal cash may project both histories to 88.3; a later query requesting first-year equity, loss recovery or treaty timing must revalidate the resolution and retain the full group states.

The bank commitment supplies the same warning. Funded gross loan 100 does not contain the original commitment date, credit baseline or observed draw. The funded allowance 4 cannot be recomputed from that scalar alone. If the original records were preserved, refinement can recover the state. If they were never known, the result is `DETAIL_UNAVAILABLE` or measurement unresolved, not a fabricated staging history. A source revision or changed query creates a new manifest; it does not silently resume an old result.

Conditional mass remains attached to unresolved outcomes

For a small authored law, assign prior masses $1/3, 2/3$ to immutable timely and late histories and likelihoods $3/4, 1$ to a newly admitted observation. The evidence normaliser is $11/12$, so posterior masses are $3/11$ and $8/11$. The invalid pooled treatment is not a third law atom: it was rejected as unsupported accounting treatment. If the late history's execution is unresolved, its mass remains $8/11$; reporting the timely result with probability one silently conditions on successful execution. If both first-year equities are available under that law, their mean is $(3/11)83 + (8/11)74.75 = 77$. With only the timely branch evaluated, the known weighted contribution is $(3/11)83$, not the complete mean. The same event IDs, measure provenance and unresolved mass must travel with the output.

These examples do not add a general solver or a cent-based accounting policy. They specify the distinctions that a compatible implementation must preserve: input domain, output operation, endpoint attainment, event order, retained economic memory, conditional mass and failure status. A changed output, horizon, grouping rule or scenario therefore returns `RESOLUTION_REVALIDATION_REQUIRED` until sufficiency is checked again.

32.13 Numerical precision is part of a financial method

The affine family is useful for Northstar because the scenario fixes prices, rates and operating coefficients while the debt share remains unknown. Other contracts require roots or discount factors. Their numerical treatment must state what is exact, what is enclosed and when a posted amount is determined. An accounting name on a library call does not make its result exact.

Consider positive payments C_i at positive times t_i and a positive price P. The effective annual rate solves

$$f(r) = \sum_i \frac{C_i}{(1+r)^{t_i}} - P = 0, \qquad r > -1.$$

Each term discounts a payment to the valuation date. In this positive-payment case,

$$f'(r) = -\sum_i \frac{t_i C_i}{(1+r)^{t_i+1}} < 0.$$

Consequently a certified sign-changing bracket contains exactly one root. The method uses stated bracket bounds and an expansion cap, evaluates signs with outward error bounds, and bisects until the requested rate or value enclosure is reached. An uncertified sign, unavailable bracket or exhausted accuracy budget yields an unresolved numerical result. Mixed-sign cash flows can have several internal rates of return; the preceding monotonicity argument does not apply and cannot justify returning the first Newton iterate that converges.

Net proceeds 100 and a payment 121 in two years give exactly 10%. An immediate fee of 2 changes proceeds to 98 and changes that root. Annual-effective 21% for half a year has the exact accumulation factor $\sqrt{1.21} = 1.1$, so interest on 100 is 10. Simple proration would give 10.5 and computes a different contract. At annual-effective 10%, the half-year factor $\sqrt{1.1}$ is irrational. For rational time p/q and positive $1 + r$, exact rational power comparisons can enclose the positive qth root of $(1 + r)^p$. A negative p first requires the reciprocal. General exponential discounting needs a separately specified error-controlled method.

A posting quantum of 0.01 does not eliminate this uncertainty. The whole certified monetary enclosure must lie in one rounding cell under the admitted tie rule. An enclosure from 1.0049 to 1.0051 crosses a half-cent boundary and does not establish one posted amount. Refine within a declared budget or retain rounding unresolved. This rule separates evidence for a numerical answer from the accounting authority to use that answer.

32.14 Updating a law without counting evidence twice

Two financial histories can be compatible with the same reports without being equally probable. Once an explicitly justified law is supplied, genuinely new evidence may change it. For finite histories h, prior mass $p(h)$ and an admitted observation likelihood $L(D \mid h)$, Bayes' rule gives

$$p(h \mid D) = \frac{p(h)L(D \mid h)}{\sum_g p(g)L(D \mid g)}.$$

The denominator is the probability of that evidence under the declared model. If it is zero, the condition is impossible in that model; dropping incompatible histories and renormalising an empty collection cannot repair it.

For two histories with assumed prior masses 1/3 and 2/3 and likelihoods 3/4 and 1, the unnormalised masses are 1/4 and 2/3. Their sum is 11/12, so the posterior masses are 3/11 and 8/11. If the second history's cash calculation fails, the first still has mass 3/11 and unresolved cash retains mass 8/11. A report giving the successful history probability one would answer a different, success-conditioned question.

An observation has a semantic information identity as well as a file locator. Reimporting the same observation through another file does not create a second measurement. Repeating an identical admitted condition is idempotent: it keeps the law and information lineage unchanged. Submitting an already consumed observation as a fresh likelihood block is rejected as duplicate information. Equal selected member sets alone do not prove that two conditions have the same meaning. An admitted conditional kernel may divide a region's existing mass when it is refined; the number of new representatives never determines that division.

Continuous inference adds a measure choice. One must specify coordinates z, domain, base measure, prior, observation mechanism and any transformation Jacobian. Stuart (2010, Section

2.2, equations (2.5)–(2.8)) supplies the inverse-problem formulation, but does not establish identification or easy sampling for a mixed accounting model. In particular, an exact accounting equality can confine a model to a lower-dimensional set. Multiplying an ordinary Lebesgue density by its zero-volume indicator is not a posterior construction. One must parameterise the constrained model with its declared measure or supply an actual measurement-error model. Adding convenient small noise changes the target.

32.15 A continuous example whose posterior can be checked

A linear Gaussian model supplies a tractable comparison for later numerical inference. Assume the unknown economic quantity has a Gaussian prior $z \sim \mathcal{N}(m_0, V_0)$ with positive definite covariance V_0. Let the observation be $y = Hz + \epsilon$, where ϵ is independent Gaussian noise with positive definite covariance R. The measurement-error assumption is substantive: it describes this example, not rounding in every financial statement.

Multiplying prior and likelihood adds their quadratic forms. Terms that depend on z are

$$\begin{aligned}&(z - m_0)^\top V_0^{-1}(z - m_0) + (y - Hz)^\top R^{-1}(y - Hz)\\ &\quad = z^\top (V_0^{-1} + H^\top R^{-1} H)z - 2z^\top (V_0^{-1} m_0 + H^\top R^{-1} y) + \text{constant}.\end{aligned}$$

Completing the square therefore gives posterior covariance V and mean m through

$$V^{-1} = V_0^{-1} + H^\top R^{-1} H, \qquad V^{-1} m = V_0^{-1} m_0 + H^\top R^{-1} y.$$

The prior and observation contribute precision and precision-weighted information. An implementation should solve factored linear systems rather than explicitly form these inverse matrices, and reject an invalid covariance or unsupported rank condition.

For $m_0 = 0$, $V_0 = 4$, $H = 1$, $R = 1$ and $y = 2$, posterior precision is $1/4 + 1 = 5/4$, variance is $4/5$ and mean is $(4/5)2 = 8/5$. A conditional cash quantity $100 + 10z$ has mean 116 and variance 80. A second *independent* measurement of 2 would give precision $9/4$, variance $4/9$ and mean $16/9$. Reusing the first measurement would produce those same numbers but would be wrong for the stated information. The analytic example can distinguish numerical error from information double counting.

For a bounded non-Gaussian scalar model, quadrature requires a stated domain or tail truncation and sufficient integrand error bounds to enclose the normalizer and requested moments. Without a positive lower normalizer bound, the ratio remains unresolved. Mixed models can enumerate finite choices and integrate each conditional continuous model, but combining branches requires their evidence factors, not draw counts. Optional MCMC needs target-specific parameterisation, tuning, independent chains, tractable correctness comparisons, convergence and Monte Carlo error checks. This proposal supplies no generally validated sampler or universal tolerance.

Nor can a more elaborate inference engine identify a quantity that the observation model never sees. If reports constrain only total debt 100, the likelihood may be constant in its floating share x. Its conditional prior then remains unchanged. A uniform prior is a chosen assumption, not a discovery from those reports. The inspected working version of Giacomini and Kitagawa (2021, Sections 2.1–2.3, equations (2.2)–(2.4)) motivates retaining posterior classes where a conditional prior cannot be revised. For Northstar there is a simple direct deduction: under all probability measures on $[0, 1]$, the expectation of $5 + 2x$ lies in $[5, 7]$, with endpoints attained by point masses. This deduction imports no asymptotic theorem from that paper's structural-vector-autoregression setting.

32.16 A stress is not an observed condition or an intervention

The interest-rate stress in Northstar fixes a future driver path. It supplies no new evidence about the historical debt share. An observational forecast instead conditions a joint law on information. A causal intervention needs structural equations and an exogenous disturbance law, because it replaces the mechanism determining a chosen variable while leaving other mechanisms in place (Pearl, 2009, Section 3.2, equations (5)–(9) and (15)–(17)).

An elementary example makes the difference numerical. Let U equal zero or one with probability $1/2$ each, and assume

$$X = U, \qquad Y = 2X + U.$$

Observing $X = 1$ establishes $U = 1$ and hence $Y = 3$. Intervening to set $X = 1$ replaces the first equation. It leaves the original law of U, so Y is 2 or 3 with equal mass and mean 2.5. Selection of a high-investment company and an imposed investment policy can therefore answer different questions even under one fully specified model. Financial filings alone do not establish these causal equations or identify their effects.

An intervention request must name its target, replacement equation, structural parents and disturbance law and satisfy acyclicity or a separately supported equilibrium rule. Correlated disturbances remain joint. The initial company profile supports imposed exogenous stress; the continuous and causal methods described here require separately admitted extension schemas and signatures. They do not silently expand the initial finite-law interface.

At every decision date, behavior and measurement receive the admitted historical view and permitted simulated past. The scheduler's knowledge of a complete future stress path is not permission for a manager to foresee it. A perfect- foresight policy needs an explicit additional assumption. Common random numbers can improve a paired numerical comparison while establishing no causal effect in actual company data.

Chapter 33

The accounting language

33.1 What the current language can express

A company model needs to keep disclosures, possible histories and chosen methods distinct as its accounts evolve. A reported balance constrains a history; it does not reveal every contract or transaction that generated it. A collection policy supplies a behavioural assumption; it does not establish what a customer will pay. An accounting method turns an eligible event into state changes and postings; a balanced result alone does not establish that the method applies. The language makes these distinctions explicit so that reports retain the meaning of the facts and assumptions behind them.

The current design has several checked, bounded language profiles. Together they cover source-constrained histories, conditional company paths, selected shared-book operations and complete method authoring. Their interfaces are explicit, but they are not yet one complete whole-company grammar. The earlier source compositions in Sections A.54.17 and A.54.18 make complete methods, registries, state invariants and full event intent inspectable. Section A.54.19 extends that authoring to the selected opening checks, ordered eligibility and parameterized construction and assessment recipes. One company source contains those algorithms together with explicit events, reports and carry commands; its imports hold the opening facts, schemas and statement mapping. The selected path reuses typed evaluators without calling the native lifecycle or operating-method builder. This finite correspondence does not complete the wider financial language or establish accounting applicability. Section A.54.20 extends the opening contract to live carried claims under an explicit supplied-population declaration and the independently selected full invariant. The source states that premise without inferring completeness from a successful continuation. Nested term histories require a further distinction: an earlier overdue assessment uses the due date applicable then, not the current due date. Section A.54.21 supplies owner-qualified queries, typed per-claim and per-assessment rules, and their combined semantic checks. Section A.54.22 binds them to an explicit selected state contract and direct source for prospective amendments, assessments, collections and carry. Its selected ordered-program envelope now binds the complete source and request sequence, retains each successful prefix, and preserves the first refused request and all unexecuted commands. Two relocated programs reproduce the complete retained results. This bounded synthetic continuation preserves the separate requirements for actual-source and accounting admission. Financing requires the same distinction between a consistent snapshot and a faithful history. Section A.54.23 preserves missed-payment and share memory through settlement, separates conditional remedies from economic effects, and rejects clause findings made stale by a same-day payment.

To represent the historical record, company method selection first binds to a source-constrained

partial state (Section A.10). Statement observations have distinct relation roles rather than one undifferentiated set of totals (Section A.11). Selected balances are tested jointly under explicit rounding and identity hypotheses (Section A.12), with opening, movement and closing dates kept separate across statements (Section A.13). These declarations can expose inconsistency or missing information without inventing the underlying books.

Several histories can remain consistent with those observations. Section A.14 represents finite conditional families with symbolic openings and explicit measurement attachment. Typed economic quantities and constraints at each event give those histories economic content (Section A.15); cash and quantity observations attach to the same history (Section A.15.2). Unknown monetary amounts remain linked to selected method prerequisites without choosing a single witness (Section A.16). The retained return example follows every branch through settlement and sale, including funding and intermediate observations (Section A.17). Outcome queries preserve which results are possible or necessary and their exact finite bounds (Section A.18). Stored definitions and results can be recovered across restart, while the archive distinguishes stored evidence from verified execution (Section A.19).

Explicit future assumptions then determine how a conditional company path evolves. The earliest matched text frontend handles the declared debt fragment. The funded-year extension adds identified claim allocations, credit terms and dated events; its imported annual projection still supplies the other operating and accounting rules (Section A.3). The operating-input example instead derives purchases, sales, claims, expenses and closing statements from a historical opening, authored quantities, prices, asset cohorts and contract terms (Section A.4). It imports no forecast journals: a price or service-date change has consequences in the calculation under the example's stated method restrictions.

Identified operations can repeat and carry unpaid claims and asset cohorts into later years. Northstar's three-period path includes revised assumptions and an interrupted year (Section A.5). Its text frontend preserves the same typed records, named settlement allocations and incomplete paths (Section A.6). Named, unweighted scenarios keep the alternative assumptions separate. Their collection policies have pure typed expressions and bounded reusable functions with matched text declarations (Section A.7). Function calls preserve units, company context and the distinction between declared and actually used inputs. They reuse a calculation without supplying missing facts or creating endogenous events.

Combining methods requires compatible economic state as well as account names. Portable module identities expose the additional state required by a shared-book return method (Section A.8); its matched authoring and finite execution appear in Section A.9. Heterogeneous lease rights require explicit account allocations and funding conditions (Section A.20). Aggregate balances may leave those allocations unidentified, producing different complete or interrupted outcomes (Section A.21). Actual rounded lease disclosures constrain some possibilities before individual contract state is known (Section A.22).

A common caller does not establish a common book. The return-and-lease example first establishes compatible state, then follows both kinds of event through one declared cash balance (Section A.23). Its recorded statements reconcile while retaining unresolved tax assessment and incomplete fiscal coverage (Section A.24). The complete opening and mixed event sequence have matched text and typed forms (Section A.25). Section A.26 explains why the original tax method refuses that mixed state, then develops explicitly conditional successors for assessment, payments, period carry and annual programs. Those successors preserve unknown declarations and stopped prefixes. They do not establish the company's actual tax policy or complete fiscal history.

Methods state their prerequisites before applying financial effects. Reusable selected requirements have matched typed and text declarations (Section A.53). Exact input binding, dimensional

products and guarded applicability require explicit version selection (Section A.54). Collection coverage, identity order, signed totals and state-domain findings add constraints that scalar types cannot express (Section A.55). Typed calculations and atomic state updates then support exact source-bound replay of the retained return example (Section A.56). Complete method text includes these prerequisites, monetary calculations, postings and memory changes (Section A.54.9). The common program preserves that source and its explicit owners; it does not silently replace a newly written method with a named demonstration.

The historical AccDSL Core 0 proposal remains the teaching baseline. It describes entities and books, accounts, observations, shared unknowns, constraints, alternative interpretations, contract instances, accounting rules and reporting mappings, with pure expressions and a restricted posting effect. Appendix B preserves its grammar and examples. The proposed portable records and first company profile refine that baseline. Numerical and forecasting algorithms can live in typed plugins; unrestricted numerical code need not be reimplemented in the DSL. General recursive or simultaneous algorithms, endogenous decisions, source admission and broader accounting profiles remain separate language and method work.

The historical Core 0 compiler has six ordered stages: parsing with source locations; name and module resolution; type and temporal-context checking; lowering to the IR; constraint and rule-applicability analysis; and execution-plan construction. It never silently supplies an economic assumption to make a program compile. A valid historical representation can still require additional assumptions before a particular simulation can run. The proposed portable compilation sequence is specified separately in Section 33.4. Structural compilation of a current program likewise does not execute its financial methods or resolve unknown eligibility findings.

The IR and evaluator define meaning separately from surface spelling. Formatting that preserves the lowered method preserves its semantic digest. The complete authored program also retains the original method source, so its program digest changes when those bytes change. In the checked common-program example, adding one space leaves the lowered method and materialized financial request identical while changing the source and program digests. Source spans and reviewer decisions can change evidence provenance without changing the numerical model. Evidence, semantic-model and execution identities therefore remain distinct; a field named `hash` must identify what it actually binds.

For the preserved Core 0 examples, the complete grammar is `spec/core-0.lark`; the original design directory retains `spec/core-ir.schema.json` and `spec/core-contract.md`. They define reference resolution, expression arity, type and context checks, operations, failures and identity. A structural schema check alone does not establish those semantic properties. This revision uses ASCII identifiers and UTF-8 strings, with explicit lexical rules for currency suffixes, dates and decimal literals. Parsing the complete demonstration checks the grammar; it does not establish an implemented compiler or solver. The following type distinctions apply before any of those operations can be interpreted.

33.2 Types that protect meaning

The following table uses the preserved Core 0 type vocabulary. Its distinctions motivate the more precise portable types introduced below.

Type	Meaning and checks
`Money<C>`	Exact amount in currency C. Addition requires identical currency; conversion requires a dated rate and policy.

Type	Meaning and checks
`Quantity<U>`	Units of shares, goods or other commodities. Quantities cannot be added to money or used as a functional-currency posting.
`Rate<B>` and `YearFraction`	Rate with basis B, plus a separately computed accrual fraction. Simple and compound accrual functions are distinct.
`Instant`, `Period`, `Calendar`	Instant observations and flows over intervals; interval operations preserve calendar conventions.
`Account<K,Book>`	Account role K and owning book. Cross-book effects require separate balanced batches linked by an economic event.
`Observation<T>`	Source assertion with contextual and precision metadata; accessing its value retains its constraint and provenance.
`Symbol<T,D>`	Shared unknown with domain D; repeated references denote the same value in a history.
`Choice<E>`	A finite structural choice over enumeration E. No probability is implied.
`Distribution<T>`	Explicit probability measure with parameter, conditioning and measure metadata. It is not a Symbol or a Choice.
`SourceRef`, `RuleRef`, `ModelRef`	Versioned evidence or implementation references; unresolved references prevent concrete execution.
`Measurement<T>`	Value plus valuation date, basis, method, assumptions and numerical status. Recognition remains a rule decision.

The portable profile spells the rate type `Rate<convention>` and makes both calendar and accrual convention explicit in `YearFraction<calendar, convention>`. `InstantValue<T>` and `Flow<T, [start,end)>` distinguish dated stocks from interval flows; their context carries the calendar. An approximate numerical component of a historical `Measurement<T>` uses `Approximation` with its method, error description and environment references. An exact measurement remains exact; valuation context and recognition authority are separate in either case.

Core expressions use exact integers and rationals, Boolean logic, enumerations, dates, records, lists, comparisons and registered pure functions. A decimal literal is parsed exactly, not first converted through binary floating point. Numerical plugins can return approximate measurements, but must declare their error or numerical status and the rounding used when posting money.

Missingness is a tagged type with distinct values: `notDisclosed`, `notApplicable`, `unreadable`, `withheld`, and `unresolvedMeaning`. Zero is an ordinary known number. A missing datum can be replaced by a declared Symbol or assumption in a model branch; the source observation stays missing. An unknown is not necessarily random, and an estimate is not necessarily missing. On portable import, `notDisclosed`, `unreadable`, `withheld` and `unresolvedMeaning` become `Unknown` with the corresponding reason and obligation references. `notApplicable` becomes the explicit `NotApplicable` variant, never an unknown or zero.

Addition of monetary observations also requires compatible entity, perimeter, time, dimensions and basis. These checks operate on contextual expressions in addition to the currency type. An analyst can deliberately aggregate dimensions or transform a basis using an explicit function. For example, `sumOver(segment, ...)` records the dimension being removed and requires the segment partition to be declared; it does not assume every disclosed segment total is additive.

33.3 A portable record has an exact meaning

A constructor and a textual frontend can describe the same financial model only if they exchange values whose identity is independent of a Python object or an incidental JSON writer. The proposed portable record therefore has six fields: `kind`, `schema`, `key`, `revision`, `body` and `dependencies`. Kind and schema identify a registered version; key is its stable, schema-scoped logical name; revision is a nonnegative integer. The body obeys that schema's required, optional and nullable fields. Unknown fields reject.

A typed reference has exactly kind, schema and digest. The digest covers the complete target record. Every body reference is also listed in dependencies; additional dependencies need a schema-declared role. The submitted closure is finite and acyclic. A receipt can refer to its request and old state, but the old state cannot refer forward to that receipt. Stable business names can relate counterparties without creating a cycle of content references.

Financial quantities use a reduced rational with signed numerator text n and positive denominator text d, with $\gcd(|n|, d) = 1$ and zero represented by $0/1$. The decimal token 2.50 therefore denotes $5/2$ exactly. Revisions, ordinals and integer counts remain integers; a Boolean is never an integer. Approximate measurements instead retain decimal text, method, error description and environment. A rounded decimal cannot silently acquire exact status.

An absent required field is invalid. Unknown financial content is an explicit value with a reason and outstanding evidence references. It can be retained, but cannot satisfy a method that requires the missing fact. Null is permitted only where the particular schema gives it a meaning; not applicable, unknown and zero are distinct. A candidate also differs from a value admitted in a specific immutable view. Admission checks the actual decision and its ancestors; wrapping a candidate in an "admitted" type is insufficient.

Portable bytes use UTF-8 JSON with ASCII object keys sorted by their byte order, no insignificant whitespace, no floats, duplicate keys, nonfinite numbers or invalid Unicode scalar values. Decoded strings retain their code points. An escaped Unicode character and the same literal character encode equally; composed and decomposed spellings remain different unless an explicit named transformation relates them. Global NFC normalisation would silently change some source text and is therefore not part of this encoding.

The writer emits literal non-ASCII scalars, escapes quotation marks and reverse solidus, and escapes control characters U+0000–U+001F. It uses the five named JSON escapes where defined and lowercase `\u00xx` for the remaining controls; solidus is not escaped. Integers use minimal base-ten spelling without plus signs, leading zeroes or negative zero. Dependencies, module declarations by key, dimensions by axis and explicit all-of/any-of operand sets are sorted after rejecting duplicates. Other arrays, including journal lines, event schedules and diagnostic priority, retain order.

With these rules, the record digest is

$$\mathrm{SHA256}\big(\mathrm{UTF8}(\text{“accdsl.portable.v1/”} + \mathrm{schema}) \parallel \texttt{0x00} \parallel \mathrm{canonical}(\mathrm{record})\big).$$

This recognises representational identity, not economic equivalence or source truth. Original documents retain a separate hash of their raw bytes. A blob reference names that byte hash, algorithm and length; it is not a typed record reference. A submission record links those bytes to media type, transport and decoder, preserving a malformed request without pretending it was valid.

Historical Core 0 and replay examples keep their original encodings and IDs. A declared legacy import retains original bytes, decoder, old identity, portable output and field-by-field comparison. If one historical record maps to several portable records, an import mapping

names every output and conversion. Import does not grant financial admission in a new view. This separation permits old examples to remain reproducible while the proposed portable representation becomes precise enough for two implementations to exchange.

33.4 The first company profile and its authoring boundary

The proposed `CorporateJointHistory/1` profile connects one legal entity and book per branch to annual/interim sources, revisions, working capital, payroll, tangible assets, simple financing, an explicitly declared tax model and complete statements. Its types preserve entity, perimeter and currency even where an operation is outside this initial scope. A real company cannot become eligible for the profile by dropping its subsidiaries, leases or contingencies.

Typed constructors must express every included declaration in the portable IR without opaque Python functions. For this original portable profile, the first matched text frontend covers the bounded debt and constraint fragment; its whole-company typed model has a wider domain than that frontend. Later bounded profiles add text authoring for repeated multi-year operations (Section A.6), a mixed return-and-lease book (Section A.25) and method requirements (Section A.53). Each retains its own grammar, typed records and execution limits. Together they provide several checked authoring routes, while a common language covering all company methods remains unfinished. Domain scope and authoring syntax are therefore separate decisions. Section 39.7 adds a finite description of dependencies between retained native exchanges, preserving their distinct state and publication effects without supplying a general method dispatcher. The following Core 0 listing teaches the shared debt family under its preserved historical grammar. It is not a promise that the first new parser supports every historical declaration. The matched portable debt example uses floating share x, while the following listing uses fixed share α. The substitution $x = 1 - \alpha$ preserves the entire family, including its endpoint results.

Backend admission examines expanded expressions. Multiplication of an unknown quantity by an unknown price is bilinear even when hidden inside a function called revenue. The initial exact reconstruction fragment admits rational affine constraints, finite choices and bounded integers under explicit templates. Unsupported functions return a capability diagnostic. A type-correct historical representation may therefore be retained even when a requested forecast still requires an unavailable method or an additional admitted assumption.

The portable `CompileRequest` supplies source kind, source bytes or a typed module, schema and profile references, the full package lock, evidence/time view and requested capability target. Its stages are strict decoding and source locations; name/import resolution; dependency and cycle checks; static type and context checks; admission-layer checks; expanded-operation capability analysis; and canonical record production. The earlier six-stage workflow describes the historical Core 0 frontend; it does not add execution planning to this portable compiler or permit it to call a solver.

The result contains candidate IR, source map, unresolved inputs, capability analysis and a validation receipt bound to every input. Invalid structure or types yield no executable IR. Valid structure with an unresolved financial input can yield a retained candidate whose affected target has execution admission false. Source maps retain decoded Unicode code-point half-open ranges for text or constructor paths for typed input. Formatting can change source-map identity without changing financial-model identity.

For a matched example, both authoring routes must describe debt 100, floating share $x \in [0, 1]$ and conditional interest $100((1 - x)/20 + 7x/100) = 5 + 2x$. Endpoints are 5 and 7; $x = 1/2$ gives 6. Changing the floating assumption from 7% to 9% changes the upper endpoint to 9.

Independent arithmetic is needed because two frontends could share the same wrong constant. Agreement between them alone does not establish correctness.

33.5 A small model with a continuous interpretation family

The following source uses a synthetic observation whose full context is stored in the evidence registry. The example imports the demonstration debt package, which uses one-year simple accrual, no fees and immediate rate resets. It is not an IFRS rule package.

All dependencies are instantiated in `fixtures/demo-debt-package.json`: the source reading, balanced opening state, currency precision, contract state and dates, reset and interest operations, closing and report mappings, request and expected outputs. `fixtures/borrowing.acc` is the complete source, including the retained earnings account and an `opening` declaration. The shorter listings below explain its declarations in stages. They do not replace that full source.

```
module BorrowingExample version "1";
use DemoDebt version "1";
source note7 = external("synthetic-note-7");
entity Acme functional USD;
book Local entity Acme basis DemoDebt version "1";
account Cash : Asset book Local;
account Debt : Liability book Local;
account InterestExpense : Expense book Local;

observation debt : Money<USD> instant @2025-12-31
  = 100 USD from note7;
symbol alpha : Real in [0, 1]
  because "The fixed-rate proportion is not disclosed";
```

The remaining declarations use this same unknown fraction in both loan components. A change to the fraction therefore changes the whole debt mix consistently.

```
let fixedPrincipal = alpha * value(debt);
let floatingPrincipal = (1 - alpha) * value(debt);
constraint split:
  fixedPrincipal + floatingPrincipal == value(debt);

contract loans : DebtMix book Local {
  principal = value(debt);
  fixed_fraction = alpha;
  fixed_rate = 0.05 per year;
  initial_floating_rate = 0.05 per year;
}
alternative allFixed { assume alpha == 1; }
alternative allFloating { assume alpha == 0; }
scenario rateShock {
  set market_rate = constant(0.07 per year);
}
```

The parent model retains every $\alpha \in [0, 1]$. The two named alternatives are useful conditional views of that family, not a declaration that the endpoints exhaust it. Requesting the parent family's exact bounds produces $[5, 7]$ for the next year's expense under `rateShock`. Requesting a conditional run of `allFixed` supplies $\alpha = 1$. Requesting a weighted mixture requires a separate weight model; the two labels do not create equal weights.

An observation declaration is a reference to evidence plus a proposed typed reading. The

compiler checks that the registry context agrees with the declared period and type. If the evidence is rounded, `value(debt)` denotes a shared constrained value rather than a silently exact 100. This synthetic registry explicitly declares an exact value, which keeps the illustration deterministic.

33.6 Posting rules and explicit exceptions

An event type declares the information needed by a rule. The example below assumes an event representing interest accrued and paid on the same date. Separate accrual and settlement events are needed when those dates differ.

```
event InterestPaid(principal: Money<USD>,
                   rate: Rate<annual_simple>,
                   fraction: YearFraction) book Local;

rule recogniseInterest for InterestPaid as e
when e.principal >= 0 USD {
  let charge = roundMoney(simpleInterest(e.principal, e.rate, e.fraction),
                          currencyScale(USD), "half_even");
  require charge >= 0 USD;
  post {
    debit InterestExpense charge;
    credit Cash charge;
  }
}
```

The rule cannot perform file I/O, mutate another entity or sample a random number. It returns a proposed batch. The runtime verifies balance and all preconditions before committing state. Negative interest requires an explicit supported rule; it is not mishandled by this intentionally restricted example. Rounding is a declared demonstration convention, not a universal accounting default.

Rules can declare `overrides ruleName`. The resulting override graph must be acyclic. For a concrete event, the evaluator determines all applicable rules and removes any that are overridden by another applicable rule. Exactly one maximal rule must remain for a required recognition operation. An empty set of applicable rules produces `UnsupportedTreatment`; more than one maximal rule produces `RuleConflict`. File order and a hidden numeric priority never break ties.

If a guard depends on a symbolic value, the solver partitions the current feasible family into the guard and its negation, subject to the existing constraints. This is a logical partition, not a random draw. An incomplete solver returns `UnknownApplicability`, not a guessed rule. Statically provable disjointness or total coverage can be cached; otherwise the runtime performs the same checks on every concrete event.

33.7 Alternative readings, assumptions and updates

An alternative adds constraints and possibly a supported semantic mapping to its parent model. Its record contains a rationale, supporting spans, opposing evidence, author, creation time and scope. Child alternatives inherit all parent constraints; they cannot quietly discard an inconvenient disclosure. A deliberate source rejection is a separate review decision with an explicit reason and a new evidence view.

Two independent alternatives may coexist, while two mutually exclusive readings of the same text require an explicit choice group. The representation stores these relationships in a constraint graph. It does not take a Cartesian product blindly. For example, interpreting an instrument

as a liability affects both its balance-sheet account and the treatment of its periodic payments; those choices must be linked through one shared treatment variable.

New evidence narrows an existing family by conjunction when it concerns the same definitions and reporting basis. If it changes the basis or corrects an earlier assertion, the system creates a revised evidence view and records the relationship. Posterior reweighting, when enabled, is a separate operation with likelihood provenance. A branch that becomes inconsistent is retained as historically considered and rejected, with the evidence that rejected it.

33.8 Modules and stable intermediate representation

A package identifies its namespace, semantic version, dependencies, effective dates, jurisdiction, supported entity classes, recognition operations and measurement policies. Standard packages define allowable treatments; industry packages add economic objects and specialised rules; company packages map source concepts and declare evidenced elections. A company extension cannot override a mandatory standard rule merely because its package is loaded last. An override is valid only at a declared extension point and within the selected standard's policy domain.

The IR has separate node classes for evidence, contextual observations, expressions, domains, constraints, alternatives, contracts, economic events, measurement results, rules, journal batches, report mappings and runs. Each node has a stable identifier and explicit input references. No expression stores executable host-language text. Plugins are referenced by versioned identifiers with typed signatures. This permits a second implementation to execute the same supported program without evaluating arbitrary embedded code.

The IR serialisation version is independent of the DSL version and of any accounting-standard edition. Migrations preserve the original payload and record changed meanings. A syntax migration can be automatic when semantics are unchanged. A changed recognition policy requires a new model version, even if the source file still parses.

33.9 Errors are part of the interface

The compiler and runtime return typed diagnostics with a source span, affected entity and period, dependency chain and suggested local action. The principal diagnostic categories have distinct recovery paths:

Diagnostic	Meaning and next action
`TypeMismatch`, `ContextMismatch`	Resolve incompatible units, periods, entities or measurement bases; do not coerce implicitly.
`MissingEvidence`	Supply a source or label a model input as an assumption.
`InfeasibleHistory`	Inspect the conflicting evidence and model constraints; preserve the rejected candidate.
`RuleConflict`, `UnknownApplicability`	Resolve rule scope or preserve a justified conditional partition.
`UnsupportedTreatment`	Add a reviewed capability or leave the affected output incomplete.
`UnbalancedBatch`	Reject the atomic update and diagnose the posting rule.
`SolverIncomplete`, `NumericalFailure`	Preserve the target and inputs; repair or change the computational method.
`InsufficientSimulationState`	Add declared conditional assumptions or request a family of results.

Warnings can describe low extraction confidence or an approximate economic model, but they do not replace these errors. A required unknown input prevents a concrete unconditional result. The analyst can add a declared conditional assumption, request bounds, or leave that output incomplete. Every path has a different, explicit result type; there is no generic fallback to zero.

Chapter 34

Reconstruction and execution procedures

34.1 Ingest, reconcile and construct histories

The historical procedure starts with a filing manifest. It records documents, reporting dates, publication dates, revisions and expected statements and notes. Tagged filings are parsed through an XBRL adapter; PDF and HTML sources are segmented into text, tables, captions and references. The original bytes and extraction output remain immutable. Re-running an improved extractor creates a new extraction version with links to the same source.

For each table, the importer first resolves headers, units, signs, periods and dimensions. It then proposes concept mappings. A deterministic validator checks subtotal arithmetic, duplicate facts, comparative consistency and cross-note references. Every proposed repair creates a new reading. The system may suggest that an OCR value is 10 rather than 70, but only a reviewed correction or an explicit alternative allows that reading into a validated model.

Reconciliation proceeds from local totals to movement schedules. A debt bridge may contain opening debt, issuance, repayment, acquired debt, interest accretion, currency effects, fair-value changes and closing debt. Every component receives a movement type. An unexplained residual is stored as a residual with a source and period; it is not automatically classified as repayment or issuance. A model can introduce an unknown decomposition of that residual, subject to evidence and economic constraints.

The reconstruction engine then performs the following finite procedure within a declared model scope:

1. Build shared variables for observations, opening states, contract terms and supported movement amounts. Attach original rounding domains.
2. Add identities, roll-forwards, account restrictions, contractual relationships and cross-period links. Add each semantic reading as a condition on the same joint model.
3. Propagate types, units, intervals and simple equalities. Reject context mismatches before invoking a numerical solver.
4. Ask the solver for feasibility. On infeasibility, return a sufficient conflicting subset of constraints, its source spans and model assumptions. A solver's conflict core need not be the smallest possible explanation.
5. Retain finite alternatives and continuous solution families symbolically. Materialise

example histories only when needed for review or execution.

6. Check whether each requested simulation has sufficient economic state. Missing maturities, reset dates or cohort composition become declared assumptions or explicit unresolved inputs.

Unrestricted journal enumeration would be unbounded: arbitrary offsetting transactions can be inserted without changing reports. The model therefore declares allowed event types, time resolution, relevant accounts, lot/cohort resolution and limits on latent contract instances. These limits describe the represented family; they are not evidence that all real possibilities were searched. The compiler canonicalises syntactic duplicates and equivalences established by its supported rewrite rules. It does not decide arbitrary economic equivalence. Probability measures attach to declared economic hypotheses or parameters, never to a count of encodings; adding a duplicate representation therefore cannot create extra probability mass or coverage.

The first implementation uses symbolic linear bridges and finite structural choices. Nonlinear economic models can be evaluated on feasible states through plugins. A solver timeout preserves the unresolved family and returns `SolverIncomplete`; it does not prune the family as impossible. Search budgets belong to run configuration and are recorded with the results.

34.2 Condition a run without rewriting history

A run selects an evidence view and cutoff, a historical family, a forecast starting date, scenario definitions, accounting packages, management policies, output queries and numerical settings. A run may use a concrete history, an exact symbolic family, a finite set of conditional histories, or weighted draws from a declared distribution. The output records which mode was used.

Suppose no debt maturity is disclosed. An analyst may run a conditional case that assumes all debt matures after the forecast horizon. The run is valid under that assumption even though the historical representation remains incomplete. The interface must display the assumption next to the result and provide sensitivity cases when the assumption is material. It must not label the assumed maturity as a discovered historical fact.

Management behaviour normally uses only information available at the decision date. A perfect-foresight stress case is a separate, explicitly labelled policy. Scenarios should share the same underlying innovations when estimating paired differences where appropriate, with stable stream identifiers so branch order does not change the draws. Common random numbers improve the comparability of paired simulations; they do not imply that every branch has the same economic event counts or shock distribution.

34.3 Schedule and commit events

An annual cash reconciliation leaves a practical question unanswered: can the company pay when each obligation falls due? Northstar opens its projection with cash 74.6, plans collections of 280 and supplier payments of 160. If collections arrive first, that supplier payment can be made from cash. If the supplier must be paid first, the company is short by 85.4. Adding the same annual receipts and payments gives the same unconstrained year-end cash in both orders. Only one order establishes this payment's feasibility when no eligible financing is available.

This is a conditional timing example, not a newly inferred invoice history. The original Northstar projection places collections before payments. The alternative schedule shows why aggregate annual drivers need dated obligations before they can support a liquidity conclusion. Unknown

collection dates or unknown facility access leave feasibility unresolved; neither implies default. The proposed schedule annotation in Section A.2 makes these distinctions explicit.

An explicit borrowing decision can repair this timing shortage, but its costs must follow through the accounts. Suppose the scenario supplies an eligible facility and chooses a draw of 85.4 on 4 January, before paying the supplier. Collections arrive on 5 January and the draw is repaid on 6 January. Under an authored 5% simple Actual/365 contract, two calendar days of interest cost $85.4(0.05)(2/365) = 427/18250$. With the demonstration's immediate 20% tax method and deductible interest, annual profit, equity and closing cash each fall by $854/45625$. The principal inflow and repayment remain separate financing movements even though they net to zero. The full-year comparison in Section A.3 accounts for those costs explicitly.

The commitment, collection dates and tax treatment are scenario assumptions. A proposed draw is not cash already received. If it settles and a later payment fails, the borrowing remains; refusing the payment cannot erase the earlier event. In the conditional reference example, moving payroll before collections produces a cash shortage of 50. Payroll remains unapplied while the completed draw and supplier settlement remain in the state.

A report of that failed path must distinguish the available snapshot from the requested annual statements. Cash immediately after the supplier payment is known to be zero. Year-end cash and profit remain unresolved because the later operations have not completed. The report retains the last successful event, remaining obligations and known opening balances; it does not rename the interrupted state a year-end result. The operation-driven example in Section A.4 checks this distinction alongside the complete cash and equity bridges.

The same distinction governs a sequence of years. An incomplete year supplies no closing state from which the next year can start. Earlier completed years remain available, while a revision of their assumptions requires the affected later years to be recomputed. Section A.5 checks these dependencies with repeated claims and asset cohorts, rather than carrying only annual account totals. Section A.6 then distinguishes an exchanged partial outcome from a closing state, and a fixed collection instruction from an explicit rule that regenerates it after a price change. Section A.7 retains those outcomes together, then distinguishes a partial financial year from suspended computation and checks member-boundary recovery with exact retries. Section A.8 separates this preservation from trusted accounting replay and explains why a module must use the company book's existing cash, tax history and inventory memory.

Each event carries an economic occurrence time, accrual interval when relevant, settlement time, entity, contract or cohort identifier, currency, inputs and provenance. Its ordering key consists of date, declared phase, dependency edges and a deterministic tie identifier. The reference phases are: opening and scheduled contractual changes; market observations and rate resets; business events and decisions; accrual and measurement; settlement; period closing and reporting. A contract can require a more specific dependency. These phases are an engineering starting point, not a universal description of market timing.

Two events may share a tie order only if their order is either specified or irrelevant to the result. If payment before remeasurement and payment after remeasurement differ, an unresolved ordering is a model ambiguity or an error. The engine does not conceal it behind alphabetical event IDs.

```
execute_event(operation_request):
    validate exact input closure, bindings and historical admission
    validate separate posting authorization, journal and delta
    lookup stable entity/book/branch/business-event key
    if exact committed fingerprint: return original receipt
    if different committed fingerprint: return event conflict
    compare expected opening state and head generation
```

```
    atomically recheck key/head and commit journal, delta, state, receipt
    return receipt, emitted dependent events and diagnostics
```

The atomic unit includes the economic state and its accounting effects. If posting fails, neither state changes. A repeated event ID with identical inputs returns the existing result; the same ID with different inputs is a conflict. Here the inputs include the request identifier, expected head and execution settings, not only the financial payload. Section 34.5 defines the stable key and full fingerprint. Consolidation runs after entity books reach the required date and stores its adjustments separately. A failure in one scenario branch cannot mutate another branch's state.

34.4 From an economic event to permission to post

A delivery can change revenue, inventory and receivables at once. Before any book changes, the method needs to know which contract was performed, which opening state it consumes, which evidence supports that reading and which accounting profile applies. Balance alone cannot supply those facts. AccDSL therefore separates the economic basis, permission to use a treatment, proposed entries and the transaction that makes them durable.

The event basis binds an immutable event to opening state, context, branch, time view, chart, authority, method, evidence, admissions, assumptions and view. Its business identity is the four-tuple of entity, book, branch and business event. The branch declares whether it is synthetic or real. A synthetic teaching profile can justify a teaching posting; it cannot authorize a real book merely because the resulting arithmetic is identical.

The method first evaluates reporting framework, jurisdiction and dates; then entity and transaction scope; contract and recognition facts; elections; numerical capability; and account/context bindings. Every material finding survives. A false required fact is ineligible, an unknown required fact is unresolved, and an event outside the method is outside scope. A known false finding must not disappear behind a later unknown one. Missing operative authority never becomes source-supported permission.

The native operation `accounting.assess@1` remains read-only assessment. Its `assess_only` request and false posting permission cannot be promoted by editing a flag. The separate `accounting.authorize@1` operation produces a `PostingAuthorization` from the complete basis and a supported posting profile. It permits posting only when the treatment is eligible, the method is fully specified for the event and the branch mode matches the profile's evidence scope. An unresolved or unsupported assessment is still a useful completed assessment, but it contains no permission to post.

An authorized proposal contains both a journal and an independent economic delta, each bound to the same basis and authorization. Journal lines name account role, debit or credit, positive exact quantity, currency/context, entity, book, event and method. Zero lines are omitted; a negative effect reverses sides. The journal balances within each currency and book. The delta separately checks quantities and contractual memory: inventory movements consume available lots, cash follows settlement, share counts follow ownership events and debt face amount remains distinct from carrying value.

The commit arguments have exactly four references:

```
AccountingCommitArguments/v1 {
  basis_ref, authorization_ref, journal_ref, delta_ref
}
```

The expected opening state and branch generation appear once, in the outer operation request.

A reference to a native assessment in the authorization field fails its reference type. False posting permission fails authorization. An unknown posting schema returns unsupported operation with no journal commit. All bindings are checked before looking up a prior receipt.

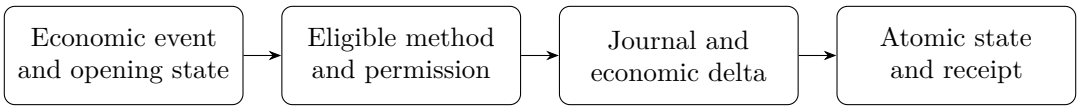


Figure 34.1: Each step answers a different question. Accounting assessment may inform the permission decision, but only the final transaction changes the book.

For a delivery, the basis identifies the exact performance obligation, quantity, allocation and control finding. The opening state contains cumulative revenue, billing, collection and inventory. The selected method derives the recognition and lot movements; the report later consumes that same committed path. Combining a journal from one method version with a delta or opening state from another is invalid even if the totals happen to balance.

34.5 A lost acknowledgement must not create a second sale

Suppose a sale commits but the response is lost. The client knows the request, but cannot tell whether the server committed it. Retrying under a new event ID could duplicate revenue. Retrying the original request must instead return the original immutable receipt.

The lookup key is

$$(\text{entity},\ \text{book},\ \text{branch},\ \text{business event}).$$

Payload, event version, opening state and request settings belong to the complete fingerprint, not this key. A renamed run or bumped event version cannot evade a conflict. A separately commissioned hypothetical projection uses a new branch namespace; reopening the same projection uses its existing one.

A correction of an already committed event is a new business event linked to its predecessor and an admitted correction method, such as reversal and replacement or a dated adjustment. It preserves the original receipt. The accounting treatment must apply to the actual correction; absent treatment authority leaves it unresolved rather than licensing a generic reversal.

Commit first validates input shape, closure, bindings and authorization in the requested historical view. It then looks up the stable event key. An equal fingerprint returns the old receipt with no additional commit; a different fingerprint returns an event-identity conflict. Only an absent key proceeds to the expected-state and generation comparison. This order is essential: after a successful sale the current head has changed, so checking freshness before replay would reject the very retry needed to resolve the lost response.

A new candidate is assessed and constructed without mutation. One transaction then rechecks key absence and the expected head, inserts journal, delta, closing state, dependencies and receipt, and advances the head. It commits all or none. If another client has committed the identical event, return that receipt. A different event that advanced the head makes this request stale. A changed payload under the same key conflicts. Diagnostic or scratch records never count as economic transitions.

A timeout consequently has three possible observations: committed, not committed or unknown. Unknown retains the original key, fingerprint and expected head for receipt lookup or exact retry. Refreshing the head changes the request and cannot be presented as the same retry. Missing committed dependencies are an integrity failure, not permission to reconstruct an approximately similar receipt.

34.6 Funding, tax and other simultaneous relations

Some financial models have circular dependencies. Interest affects taxable profit; tax and investment affect funding needs; funding affects interest. The reference executor first tries to resolve these through explicit timing. For example, beginning-of-period debt can determine current interest, with new borrowing affecting the next accrual period. That is acceptable only when the model's timing assumption is stated and appropriate.

Where the economic model genuinely requires simultaneous decisions, the dependency graph identifies the strongly connected component and delegates it to a declared equation or optimisation solver. The solver must return a solution, residuals, admissibility checks and uniqueness or multiplicity status. A contraction iteration requires a justified convergence condition; an arbitrary fixed iteration count is not a mathematical solution. Multiple admissible equilibria become alternatives when material. No solution produces a model outcome such as financing failure, or an explicit numerical failure, depending on the cause.

Funding policy must include available facilities, limits, covenants, collateral and allowed capital actions. The program cannot issue unlimited debt to keep cash positive. An inability to meet payments may be a legitimate default outcome. Default ends or changes contractual and accounting processes under the relevant event rules; it is not quietly discarded as a bad simulation draw.

34.7 Worked example: identical history, different future journals

Return to the company in Chapter 27. Immediately after the historical interest payment and closing, cash is 95, debt is 100 and equity is −5. There are no other assets, liabilities or transactions. Next year's expense is 5 in the all-fixed branch and 7 in the all-floating branch under the rate shock. The generated entries and closing statements are:

Output	All fixed	All floating
Debit interest expense / credit cash	5	7
Closing cash	90	88
Closing debt	100	100
Current-year profit	−5	−7
Closing equity after profit transfer	−10	−12
Cash change	−5	−7

The balance-sheet identity holds in each branch: $90 = 100 - 10$ and $88 = 100 - 12$. The demonstration reporting profile classifies interest paid as operating cash flow. A different permitted reporting profile changes that classification, while preserving the cash movement and source event. This example proves the intended distinction and its arithmetic; it does not estimate the probability that either debt structure describes a real company.

34.8 Worked example: a bond has several relevant amounts

A two-period zero-coupon bond raises cash of 100 and promises 110.25 at maturity. Assume a 5% effective rate per period, no transaction costs, no default, and amortised-cost treatment under a demonstration rule. The effective-interest schedule follows directly from $100(1.05)^2 = 110.25$:

Date	Opening liability	Interest accretion	Closing liability
Issue	0	0	100
End of period 1	100	5	105
End of period 2, before payment	105	5.25	110.25
Redemption	110.25	0	0

Issuance debits cash and credits the liability by 100. Each accretion debits interest expense and credits the liability. Redemption debits the liability and credits cash by 110.25. Legal face amount, carrying amount and market fair value remain separate fields. A market-rate shock may change fair value without changing this contractual payment schedule or this demonstration amortised-cost path.

Cash-flow reporting uses the actual payment event and its economic components; the annual accretion is non-cash. The applicable reporting package decides how to classify the maturity payment, rather than inferring it solely from the liability debit. Fees, modifications, early redemption, embedded options, hedges and foreign currency require additional event and measurement rules.

Ordinary equity issuance similarly separates cash, share quantity and legal capital. Ten new shares at 12, with par value 1, produce cash of 120, share capital of 10 and premium of 110 under a simple par-value profile. Shares outstanding increase by ten units. Direct issuance costs of 3 reduce the specified equity account and cash under the chosen illustrative treatment. The accounting rule and the cost's eligibility must be explicit; this example does not prescribe every jurisdiction's treatment.

34.9 Reporting and explanations

The reporting engine produces the balance sheet, income statement, comprehensive income, cash-flow statement, changes in equity and note schedules when the necessary modules are present. Each output carries its basis, perimeter, period, scenario and interpretation. A comparative table cannot combine an original prior-year figure and a restated current-year comparative without displaying that choice.

An explanation is a dependency graph with typed edges: extracted from, interpreted as, assumed by, constrained by, generated by, measured by, posted by, aggregated into and supersedes. A cash-flow line must lead back to payment events; a fair-value movement must lead back to valuation inputs; a consolidated amount must show eliminations. A provenance path establishes how a result was obtained. It does not by itself establish that the underlying economic model is correct.

34.10 The corporate reference package as operations

The Northstar example defines a closed, reproducible corporate package rather than a universal standard. Its events separate performance, invoicing and settlement even where the example makes the first two simultaneous. Every payload includes entity/book, currency, economic date, period, source or assumption IDs and an information-view ID. Amounts are exact before the declared posting quantisation. Quantities and money are checked separately.

Event	Preconditions and state change	Posting
Deliver/invoice	Goods owned and available; control transferred; quantity removed at selected cost; unconditional amount due added	Debit receivable, credit revenue; debit cost of sales, credit inventory
Collect	Receipt no greater than selected unpaid receivable; reduce that balance; excess must be separately classified	Debit cash, credit receivable
Receive goods	Accepted owned quantity at supported unit cost; create supplier obligation and inventory lot	Debit inventory, credit payable
Pay supplier	Identified obligation and sufficient available cash; reduce both	Debit payable, credit cash
Payroll	Service period and cash obligation evidenced or assumed	Debit payroll expense, credit cash; accrual variant uses a payable
Acquire plant	Control obtained, cost known, in-service date/life/residual recorded; add asset cohort	Debit plant cost, credit cash or payable
Depreciate	In-service asset with remaining depreciable amount; apply dated schedule, capped by that amount	Debit depreciation, credit accumulated depreciation
Interest	Contract, reset state and accrual fraction known; create interest obligation, then settle if cash available	Debit interest, credit interest payable; debit payable, credit cash
Current tax	Reference tax base positive; 20% teaching rule; apply prepaid tax before further payment	Debit tax expense; credit prepaid tax or payable; settlement separately affects cash
Dividend	Explicit authorised distribution, sufficient funds and permitted capital action	Debit retained earnings, credit dividend payable; then settle
Close	Required period operations completed; preserve original movements	Transfer temporary income/expense balances to retained earnings

The current example combines accrual and settlement when they share a date, but the separate obligation remains the general operation. Customer advances are debit cash/credit contract liability and become revenue only on the supported performance event. Inventory costing requires an admissible policy; negative inventory and collection beyond an identified claim are rejected. Non-cash acquisitions, write-offs, impairment and currency movements require their own classified templates. An unknown residual never enters one of these operations merely to force a reconciliation.

Reporting maps asset and contra-asset balances into the balance sheet and liability/equity balances with their opposite sign. Before closing, current profit is included once in equity; after closing it is already in retained earnings. Income uses period movements in income/expense accounts, and OCI uses its own designated accounts. Cash flow sums cash legs by economic operation: Northstar's collections, suppliers, payroll, interest and taxes are operating, plant

payments investing, and debt/equity proceeds or repayments and dividends financing. Non-cash depreciation is excluded from direct cash flow and added back only in the indirect reconciliation. Note schedules use the same contracts, lots and classified movements as the statements.

34.11 Scenario meaning and information access

An `ExogenousStress` supplies a path or joint driver law while holding the historical evidence model fixed. `ObservationalConditioning` additionally supplies actual observations and a joint probability model; the finite method in Chapter 32 evaluates the conditional distribution when its normalising mass is positive. `StructuralIntervention` requires a named structural model and replaces designated equations, retaining its exogenous noise. An absent structural model gives `UnsupportedScenario`, not an intervention inferred from an accounting identity (Pearl, 2009, Section 3.2.1). The textual Core 0 `scenario` declaration is an exogenous stress only; the other modes require the explicit typed API records.

At each decision date the runtime constructs an information view containing eligible disclosures, committed state through that date, currently observed drivers and the declared forecasting model. Future realised innovations and future outcomes are inaccessible. A stress path may be announced in advance, but this is an explicit visibility policy. Otherwise management sees only its current and past values. A bank's allowance at the opening date may use a forecast distribution of defaults; it must not inspect the outer simulation's future realised default. Inner valuation randomness has a separate stream domain from outer realised outcomes.

Funding follows a declared waterfall: available cash, committed eligible facilities up to their limits, and separately authorised capital actions. Each action checks collateral, covenants and settlement timing. If an amount remains due, record an economic funding failure and the unpaid obligation. If required valuation or accounting is unsupported, stop the affected dependencies as incomplete. The branch manifest preserves the last committed state, completed outputs, unavailable outputs and reason. These distinctions let an analyst see a meaningful bad outcome without confusing it with broken computation.

Chapter 35

Accounting methods for an operating company

35.1 A common core with distinct accounting books

The common core governs evidence, time, money, events, journals and provenance. Reporting standards govern recognition, measurement and presentation within that core. The design preserves three levels of meaning: economic terms such as a promise to pay cash; an accounting treatment under a particular standard; and a company's reported label or extension. These levels are connected by explicit rules and mappings, rather than collapsed into one universal chart of accounts.

A rule package includes a standards register with jurisdiction, issuer class, effective dates, transition provisions, permitted elections, paragraph references, implementation version and tests. The package selector uses the entity, reporting basis, period and adopted elections. It must distinguish a standard's mandatory effective date from permitted early adoption. For example, the IFRS Foundation's IFRS 18 overview states an effective date of 1 January 2027 with earlier application permitted (IFRS Foundation, 2026c). A 2026 historical report cannot be assigned that presentation regime solely because the latest installed package includes it.

IFRS, US GAAP and ASBE packages share economic contracts and events where appropriate, but produce separate accounting books. A reconciliation between books consists of explained adjustments with origins in differing treatments, not a renaming of accounts. The first implementation can preserve all three source bases even while only a selected subset has executable rules. Such a source is represented as observed information with unsupported simulation treatments, rather than being silently converted to the supported basis.

The official IFRS overviews for IFRS 9, IAS 32 and IFRS 17 establish the relevant scope of financial-instrument measurement, presentation and insurance contracts (IFRS Foundation, 2026d,a,b). They are orientation sources, not substitutes for reviewed paragraph-level implementations. The retained official Ministry of Finance ASBE 21 page now supports selected lease provisions. The official 2017 ASBE 22 text also supports the selected credit-loss, interest and write-off distinctions worked through in Chapter 37 (Ministry of Finance of the People's Republic of China, 2017). Entity-specific issuance and amendment chains and broader ASBE authority remain incomplete. Chapter 38 compares the dated ASBE 25 premium-allocation and incurred-claim provisions with the retained AASB 17 text (Ministry of Finance of the People's Republic of China, 2020). The Ministry's 2025 explanation supports the stated entity-class timetable, while Ping An's reports supply an actual adoption and comparative-restatement example (Section 38.24). Direct verification of the operative issuance and implementation

notices remains open, including the detailed conditions of the later simplifications. Current ASC access is also unresolved. Historical amendments and issuer policies inform proposed comparisons without substituting for operative authority. These source limits are addressed explicitly in the following dated-method discussion.

35.2 Non-financial companies

The initial industry package covers sales, collections, purchases, payments, inventory movements, payroll, capital expenditure, depreciation and basic financing. Revenue recognition is a separate rule from invoicing and cash collection. A service contract may generate a performance event before an invoice, creating a different asset from an unconditional receivable. Customer advances create obligations before revenue. Returns, rebates and variable consideration need separate assumptions and subsequent adjustments.

Inventory state includes quantity, cost, location, ownership and lot or cohort. A costing method is selected within the reporting package's permissible domain. An aggregate stock can support a simple model only if its approximation is declared. A commodity-price shock can expose differences between old and new inventory that an average value suppresses. Fixed-asset state likewise requires acquisition dates, components, useful lives, residual values, depreciation method, tax basis and impairment history when those affect the requested forecast.

Industry extensions add economic mechanisms without bypassing the accounting core. Construction requires performance and completion states; extractive industries require reserves, depletion and restoration obligations; utilities may require rate-regulation treatment; software businesses may require contract liabilities and development-cost decisions. The module manifest states what is represented, what is executable, and what remains aggregated.

35.3 A matched inventory case across reporting profiles

Standards packages need operation-level decisions. For interchangeable goods, the inspected AASB 102 text incorporating IAS 2 permits FIFO or weighted average in paragraph 25 and explains both in paragraph 27 (Australian Accounting Standards Board, 2015). The reproduced ASBE 1 text specifies FIFO, weighted average or specific identification in Article 14; its official Ministry version could not be retrieved in this review, so that reproduction has provisional source status (Ministry of Finance of the People's Republic of China, 2006a). Caterpillar's company-authored 2024 report states that it principally uses LIFO and supplies a FIFO comparison (Caterpillar Inc., 2024, Note 1.C). That is evidence of a disclosed US-GAAP company policy; full ASC 330 paragraph authority remains an acquisition task. It is not a rule that every US company must use LIFO.

Take the same purchases in three separate books: ten units at 5, followed by ten at 7, then delivery of twelve units. FIFO charges $10(5) + 2(7) = 64$ to cost of sales and leaves inventory 56. LIFO charges $10(7) + 2(5) = 80$ and leaves inventory 40. Periodic weighted average uses cost $120/20 = 6$, charges 72 and leaves 48. The physical deliveries and supplier cash flows agree; profit and carrying inventory differ by the selected costing rule. A concept mapping cannot remove that difference. The FIFO/average selector in an IAS 2 profile rejects LIFO as an unsupported election under the inspected paragraph. The US company profile admits its evidenced LIFO policy while retaining its narrower authority. An ASBE package remains source-provisional until the attributed text and applicable amendments are officially verified.

Inventory impairment illustrates another required memory. Under inspected AASB 102 para-

graphs 28–34, an item costing 56 written down to net realisable value 50 records expense 6. If the conditions reverse and net realisable value becomes 54, reversal is 4, capped by original cost; the previous write-down must therefore remain in state. The reproduced ASBE 1 Article 19 also addresses reversal within the earlier provision. This paragraph is not used to assert the US treatment, whose primary authority was unavailable. Source limitations constrain the treatment claim, while the separate-book method and exact costing calculations remain fully specified.

35.4 Operating events

The operating methods use the selected revenue, inventory and asset provisions of Australian Accounting Standards Board (2022f, 2015, 2022b). The lease and gross-tax case uses the dated lease, income-tax and cash-flow compilations (Australian Accounting Standards Board, 2023c,b,a). Each rule still needs the entity, period, transaction facts and elections described below.

Every event uses the event-basis, authorization and commit contracts described in Section 34.4. The event inputs described here are required in addition to identity and date. Each object identifier resolves to its contract, context and admitted source or assumption. A purported cash collection does not establish a sale; a purchase order does not establish receipt.

Revenue, billing and collection are separate transitions. A delivery event names the contract, performance obligation, quantity, allocated price and control evidence. Recognition first releases a related contract liability; earned consideration not yet unconditional creates a contract asset. Both credit revenue. A missing control finding or unresolved price constraint leaves the recognition unresolved.

An invoice or right-to-payment event names the contract, bill and unconditional amount. It debits the receivable and credits an existing contract asset, with any excess recorded as a contract liability. A collection names that receivable, settlement and amount and debits cash while crediting the receivable. An excess payment needs a separate customer-advance method; it cannot be credited to revenue merely because cash arrived. These transitions retain cumulative performance, billing and collection independently.

Purchasing starts when the company receives owned goods, not when it places an order. The event records lot, quantity, unit cost and ownership evidence, debits inventory and credits the payable. Consigned or unreceived goods cannot pass that ownership condition. Consumption or sale identifies an available lot allocation, quantity and admitted cost formula; it debits cost of goods sold or an eligible production asset and credits inventory. Overselling does not create stock. Supplier payment separately debits the payable and credits actual cash. Insufficient cash blocks that settlement and can trigger a separately modelled default; it never creates a hidden loan.

A tangible-asset event records component cost, residual value, useful life, depreciation method and the date it becomes available for use. Initial debit to PPE has a corresponding cash or payable credit; depreciation later debits expense or an eligible asset and credits accumulated depreciation. Employee service records earned pay, deductions and employer obligations for the service period, separately from settlement. Unknown labour or tax law cannot become a zero payable.

Closing derives revenue, expenses, OCI and owner movements from those events using admitted statement mappings. It checks cash, inventory, financing and tax roll-forwards rather than introducing a balancing entry. These event fields form the method's required inputs; each identifier resolves to the same contract, context and evidence used by its authorization.

AASB 15 paragraphs 9, 22, 31, 35–43, 47–59, 73–86 and 105–108 provide the selected revenue conditions; the retained compilation dated 31 December 2022 applies to annual periods beginning on or after 1 January 2023 and before 1 July 2026, covering the stated for-profit FY 2024 example. Variable consideration is estimated under a selected method and admitted only subject to the significant-reversal constraint, not recognized solely because its unconstrained expectation is positive. Principal versus agent, returns, warranties, contract modifications and distinct bundles require their own scoped decision records. If their facts are unresolved, retain alternative eligible methods; do not borrow the simple delivery rule by label.

The following schedules make the distinction between performance and billing explicit. For one performance obligation with price 100, cumulative revenue 60, unconditional bills 50 and collection 30, AR=20, CA=10, CL=0 and cash=30. Revenue 60 is represented by those three assets. Billing before performance changes intermediate balances but not the final result. For bills 80 instead, AR=50, CA=0, CL=20; net assets remain 60. Cumulative performed-service cost 40 then 70 posts 40 then 30, not 110. A downward revision follows the correction method and keeps the old result tied to its original view.

For this simple contract, let R be cumulative recognised revenue, B unconditional billings and C collections, with $0 \le C \le B$. The unpaid bills determine accounts receivable (AR); performance in excess of billing is a contract asset (CA), while billing in excess of performance is a contract liability (CL). After each event,

$$\text{AR} = B - C, \quad \text{CA} = \max(R - B, 0), \quad \text{CL} = \max(B - R, 0), \quad \text{cash received} = C.$$

This recurrence applies per contract, with no impairment, variable consideration or unbilled prepayment; it is not permission to offset unrelated contracts. For billing 80 before delivery, debit AR 80/credit CL 80. Recognizing service 60 later debits CL 60 and credits revenue 60. Collecting 30 debits cash 30/credits AR 30, leaving AR 50 and CL 20, without a second receivable or an artificial contract asset. If service comes first, recognize CA 60/revenue 60, then bill AR 80 against CA 60 and CL 20. Both orders end at the same state. Cash received before an unconditional bill needs an explicit advance event that debits cash and credits CL and records the advance applied to later billing; it cannot pass through collect(AR).

35.4.1 A payment before the bill must be applied only once

The preceding recurrence deliberately excludes an advance before billing. Suppose a customer pays 30 towards a service priced at 100. No service has yet been performed and no payment was already due. The company now has cash 30 and an obligation to provide the service, so it debits cash and credits a contract liability. Crediting revenue would balance the entry but recognise performance that has not occurred. Paragraphs 105–108 of the retained AASB 15 distinguish this payment from an unconditional receivable (Australian Accounting Standards Board, 2022f).

Now let the company earn 60 before issuing a bill. Recognition releases the liability of 30 and creates a contract asset of 30, with a credit to revenue of 60. The later invoice shows a gross amount of 80 and applies the customer's advance of 30, leaving a new unconditional right to 50. Its entry debits the receivable by 50, credits the contract asset by 30 and credits a contract liability by 20. Creating a receivable of 80 would count the prepaid 30 again. When the customer pays another 20, cash is 50, the receivable is 30 and the contract liability is 20. Their net contribution to assets is the earned 60.

The method therefore keeps the invoice's gross amount separate from the right still owed. Let G be cumulative gross billings, A cash advances, K the amount of those advances applied to bills, and C later collections of receivables. Each application names both its advance and its

invoice; the remaining unapplied credit is $A - K$. As before, R is cumulative earned revenue. The consideration already paid or made unconditionally due, counted once, is $T = G + A - K$. For this one fixed-price contract,

$$\text{AR} = G-K-C, \qquad \text{CA} = \max(R-T, 0), \qquad \text{CL} = \max(T-R, 0), \qquad \text{cash received} = A+C.$$

The minus K prevents the applied advance from appearing both in the cash payment and in the bill. Since $\text{CA} - \text{CL} = R - T$, the control

$$(A + C) + (G - K - C) + \text{CA} - \text{CL} = G + A - K + R - T = R$$

recovers the earned amount. This derivation uses a single contract, with $0 \le K \le \min(A, G)$, $0 \le C \le G - K$ and $0 \le R, T \le 100$. It provides no authority to offset different contracts or to ignore impairment, financing, tax, cancellation or refund terms.

Each event must carry its contract, identity, date, currency and admitted facts. In this restricted extension, advances precede all performance and billing; later payments settle identified receivables. A revenue increment first releases an existing contract liability and then creates a contract asset. For a new gross bill g that applies unused advance credit k, only $g - k$ creates a receivable. That amount releases the existing contract asset first; any remainder increases the contract liability. The invoice consumes its identified credit even if the advance has already been earned. A duplicate credit application, an overcollection or missing unconditional-right evidence cannot pass by adjusting a balancing account.

Billing before performance gives the same final balances in the example but a different intermediate history. The invoice creates a receivable of 50 and adds 50 to the existing contract liability of 30; earning 60 then releases 60 of that liability. After the later collection of 20, both paths have the balances above. Completing the service adds revenue 40, releasing the remaining liability of 20 and creating a contract asset of 20. A final bill for 20 and collection of the remaining 50 leave cash 100 and no contract or receivable balance. The actual sequence is retained even where endpoint amounts agree.

A wholly prepaid invoice exposes another implementation requirement. If its gross amount and applied advance are both 30, it creates no new monetary right and needs no accounting lines. It still consumes an identified credit. The proposed authored method therefore declares a state-only journal proposal: empty lines, zero account changes and explicit changes to the invoice and advance allocations. Authorization checks that the net new right is zero; the same atomic commit and replay rules protect the allocation from reuse. An empty journal cannot carry a nonzero monetary effect. The native acquisition schema requires nonempty lines and does not implement this case; an unavailable authored schema returns an unsupported-operation result without changing state.

Take the wholly prepaid invoice in a synthetic book containing only cash 30 and a contract liability of 30. No performance has occurred. Advance A1 has 30 unused; there is no invoice yet. The new invoice I1 records gross billings of 30 and applies all of A1. Cumulative G and K both rise from zero to 30, so $T = G + A - K$ remains 30. Cash, revenue, receivables and contract balances do not change. But A1 is now fully used and I1 is paid. Omitting this transition would let a later invoice claim the same credit again.

The authored advance-allocation profile therefore pins a closed journal variant and a separate economic delta. In addition to their common context, basis and authorization references, their distinguishing fields are:

```
JournalProposal {
  effect: "state_only", lines: []
}
EconomicDelta {
  account_changes: zero for every chart account,
  before_contract: advance A1 unused 30; no invoice,
  after_contract: A1 used 30; invoice I1 gross 30,
                  applied 30, unpaid 0; allocation A1->I1
}
```

This display abbreviates the required common references; the pinned schema bundle supplies the closed field lists and complete portable records. The independent calculation checks both levels. It derives zero account changes from the contract equations and verifies that the advance, invoice and allocation agree with the new cumulative values. Empty lines with a proposed receivable increase of 1 fail that reconciliation. Zero account changes with the allocation omitted also fail: the required economic effect is missing.

Authorization uses the identified unused advance, a fresh invoice identity, the zero unpaid amount and the admitted contract bounds. A caller cannot replace those findings with a true permission flag. The profile supports a synthetic book only; its historical source support and authored facts provide no authorization for a real company's posting. An assurance that the empty journal balances would establish none of these conditions.

For a valid first event E1, the atomic transaction publishes the state-only proposal, independent delta, new contract state, receipt and event-index entry, and advances the branch generation once. The receipt is evidence that the allocation was committed, even though every account balance stayed the same. The stable identity remains the entity, book, branch and business event. Its fingerprint covers the whole portable request, including the original expected head, generation, request identifier and execution settings.

Retrying the identical request after the commit returns the original receipt. The lookup must precede comparison with today's head; otherwise a successful retry would wrongly fail because its own earlier commit advanced the branch. Changing I1 to I2 under E1 instead conflicts, as does changing the request identifier while keeping the amounts. A genuinely new event E2 submitted against the old opening fails freshness. Rebuilding E2 against the current opening cannot rescue the reused credit: A1 has no unused amount, so authorization fails before lookup. These are different reasons for preserving the current book unchanged.

On replay, the response identifies the original committed event but reports no additional commit by this invocation. A lost acknowledgement remains unknown until lookup under the original identity resolves it; inventing a new event identity could double the allocation. Exports and recovery must also retain the economic transition. A format limited to monetary journal lines must preserve the state-only event separately or report that a lossless export is unsupported. Dropping it because revenue and cash did not change would make the recovered advance available twice. The complete documentary exchanges exercise these distinctions; transaction durability and concurrent execution remain implementation checks.

These are revenue and cash contributions before service costs and tax. Their missing measurements remain inputs to the complete company statements. The next example retains a different obligation: cash already received may have to be returned even after the goods have been delivered.

35.4.2 A return right survives the sale

Consider 100 goods sold for 10 each, with a former inventory cost of 6 each. The buyer obtains control, pays cash and has a contractual right to a refund for returned non-defective goods. Assume an eligible Australian for-profit contract in 2024, within the retained AASB 15 compilation's application period. The contract, control, price-estimation and significant-reversal conditions are separately satisfied; immediate cash receipt alone would not establish them. The numbers are synthetic measurements, not estimates of Apple's returns.

If the company initially expects 10 goods to be returned, it expects to retain 900 of the 1,000 received. AASB 15 B20–B25 requires revenue for the expected entitlement, a refund liability and a separate asset for the right to recover the goods (Australian Accounting Standards Board, 2022f). With no expected recovery costs or deterioration in this example, the refund liability is 100 and the recovery asset is 60. The company debits cash 1,000, credits revenue 900 and credits the refund liability 100. It removes inventory 600 by debiting cost of sales 540 and the recovery asset 60. Pretax profit is therefore 360. Recognising the full 1,000 as revenue would omit a known refund obligation, although the entry balances.

For a homogeneous cohort of N units, unit price p, former unit cost c and an admitted estimate q of returns, the initial no-recovery-cost case gives

$$\begin{aligned} \text{revenue} &= (N-q)p, & \text{refund liability} &= qp, \\ \text{cost of sales} &= (N-q)c, & \text{recovery asset} &= qc. \end{aligned}$$

The estimate and its measurement date are retained with the cohort. Paragraphs 53–59 require an appropriate estimation method and the constraint on a significant reversal; q is not a probability supplied by the accounting standard. Missing evidence for that measurement leaves the amount unresolved. After actual returns begin, the state separately records refunded units, remaining expected returns, remaining liability and recovery rights. A later outcome cannot be inserted into the original information view.

Suppose the sale occurs in 2024 and new information at year end raises expected returns to 12. The liability increases by 20, reducing revenue by 20. The recovery asset increases by 12, reducing cost of sales by 12. Closing revenue is 880, cost of sales 528 and pretax profit 352. The change in expectations does not itself establish an error in the original estimate. If the opening inventory of 600 was financed entirely by opening equity of 600, the resulting pretax balance sheet has cash 1,000 and recovery asset 72, against refund liability 120 and equity 952.

In 2025, assume the return period ends with eight accepted returns and no remaining legal right or unresolved claim. Before final settlement, remeasure the liability from 120 to 80 and the recovery asset from 72 to 48. The first change credits revenue by 40; the second debits cost of sales by 24. Refunding 80 then debits the liability and credits cash. Receiving the eight goods debits inventory by 48 and credits the recovery asset by 48. Settlement does not recognise the sale or its cost a second time. The 2025 pretax profit contribution is 16, closing cash is 920 and inventory is 48. With no other movements, equity is 968: opening 600 plus cumulative profit 368.

The cash and accrual statements can now be checked independently. Operating cash is 1,000 in 2024 and −80 in 2025. In the indirect method, the first year adds the inventory reduction and liability increase to profit while subtracting the recovery-asset increase. The second year reverses the relevant balance movements:

$$\begin{aligned} \mathrm{CFO}_{2024} &= 352 + 600 - 72 + 120 = 1{,}000, \\ \mathrm{CFO}_{2025} &= 16 - 48 + 72 - 120 = -80. \end{aligned}$$

There is no investing or financing cash flow, OCI or subsequent owner transaction in this slice. Tax has not been computed: these pretax contributions must enter the separate current/deferred-tax method before they can become complete after-tax company statements.

Recovery costs change the asset without necessarily changing the refund. For the 12-unit estimate, expected recovery costs of 1 per unit and deterioration of 0.5 give a recovery asset of $12(6 - 1 - 0.5) = 54$. Cost of sales becomes 546 and pretax profit 334, while the refund liability remains 120. The cost inputs are measurements, not automatic zeros; estimating them does not pay cash early. Paragraph B25 also requires separate presentation of the asset and liability. Showing only their net 48 in the original no-cost case would conceal both the gross right to recover goods and the gross obligation to refund cash.

Scope conditions matter as much as the arithmetic. Paragraph B26 treats an exchange for the same type, quality, condition and price differently from a return; B27 sends a defective-product replacement to warranty guidance. The refund method cannot accept those events simply because the goods move back to the seller. It also needs the contractual return window, accepted-unit identities, actual refund and recovery measurements, and evidence that any remaining rights have expired before clearing the balances. An unknown estimate yields unresolved, absent control makes recognition ineligible, and an event of a different kind remains outside this method. Inventory costing and the condition of recovered goods continue to have their own requirements.

35.4.3 A warranty can promise another year of service

A returned product and a warranty claim need different histories. The return example unwinds a sale when a customer exercises a refund right. A warranty may instead promise to keep a product working, or provide a service after the product has been delivered. The company must identify which promise it made before deciding how much of a customer's payment it has earned. The following cases use the same retained AASB 15 compilation and historical Australian for-profit scope as the return example.

Suppose a customer buys a product and a separately purchasable service warranty for a combined 108. Comparable standalone selling prices at contract inception are 100 for the product and 20 for the warranty. Both are distinct promises; the product works independently of the additional service. The contract has neither a discount attributable to just one promise nor variable consideration. Under paragraphs 73–86, the 12 bundle discount is therefore allocated in proportion to the standalone prices. The product receives $108 \times 100/120 = 90$ and the service receives $108 \times 20/120 = 18$. An invoice's labels would not by themselves establish those standalone prices or justify a different discount allocation (Australian Accounting Standards Board, 2022f).

On 1 July 2024 the company receives 108 and transfers control of the product, whose inventory cost is 60. First, receipt debits cash and credits the contract liability by 108. Delivery then debits that liability and credits product revenue by 90, while debiting cost of sales and crediting inventory by 60. Profit is 30 and a liability of 18 remains. Recognising all 108 on delivery would credit revenue for service the company still owes.

The service warranty runs for twelve monthly service periods to 30 June 2025. The company stands ready uniformly; the customer receives and consumes the benefits as service is supplied. The admitted measure gives each completed service month one twelfth of the allocation and faithfully depicts performance. These facts support over-time recognition under 35(a), 39–45 and B14–B19; the existence of a twelve-month date range alone would not suffice. At 31 December 2024, half the service has been provided, so half of 18, or 9, is earned. Debit the contract liability and credit service revenue by 9. Actual service costs of 3 are expensed and

paid. Opening inventory and equity were both 60, with no other opening balances. Year-one revenue is now 99, expenses 63 and pretax profit 36. Cash 105 equals the remaining liability 9 plus equity 96.

The second half-year releases the remaining liability of 9 and incurs another 3 of paid service costs. Its profit is 6 and closing cash and equity are both 102. Over the complete contract, revenue 108 less costs 66 gives profit 42. The cash statement follows the same history: operating cash is 105 in the first period and -3 in the second. The independent accrual bridges are $36 + 60 + 9 = 105$ and $6 - 9 = -3$, using the inventory reduction and changes in the contract liability. There is no OCI or subsequent owner transaction in this pretax slice; the complete company calculation still needs tax and any other obligations.

The contract record must therefore retain the two promises, the original allocation and its price evidence, delivery identity, coverage, progress measure, cumulative service revenue and actual costs. Each service update posts the difference between newly supported cumulative revenue and revenue already recognised. Replaying an accepted delivery cannot release the product allocation twice. Unknown progress is not zero performance: paragraphs 44–45 require a separate assessment of reasonable measurement and, where applicable, cost recovery. This percentage method cannot invent those findings.

The classification also has a substantive boundary. A separately purchasable warranty is a distinct service under B29. Without that option, B30 sends an assurance-only warranty to AASB 137 unless it provides additional service. Law, duration and promised tasks help distinguish the cases under B31; there is no universal number of months that makes a warranty a service. If assurance and additional service cannot reasonably be accounted for separately, B32 treats the two warranties together as one performance obligation. Harm or infringement compensation under B33 is not such a service. The historical AASB 137 population-measurement case in Section 35.4.13 supplies the separate assurance method. Actual coverage, claims, timing and reimbursement evidence remain required; missing facts cannot become a zero provision or convert assurance into this service example.

35.4.4 Changed promises can affect future units or past progress

A contract can change after some performance has already been recognised. The accounting consequence depends on what remains to be supplied. Consider an original prepaid order for ten distinct units at 10 each. Four have been delivered, earning 40. The parties approve an enforceable amendment adding six units for 48. Comparable standalone prices remain 10 each, and no appropriate contract-specific adjustment makes the 48 addition reflect those prices. The addition therefore fails the separate-contract condition in paragraph 20. Because the remaining units are distinct from those already delivered, paragraph 21(a) reallocates the consideration for the remaining performance (Australian Accounting Standards Board, 2022f).

The undelivered original units carry 60 of consideration. Adding the new 48 gives 108 for twelve remaining units, whose standalone prices are equal. Thus their new unit allocation is

$$\frac{\text{unrecognised original consideration} + \text{additional consideration}}{\text{remaining equal-price units}} = \frac{60 + 48}{6 + 6} = 9.$$

The first four units keep their revenue of 40. Spreading the entire revised price over all sixteen units would instead give $148/16 = 9.25$ and rewrite their revenue as 37. That calculation is wrong for the stated facts: this modification reallocates remaining performance, not completed deliveries.

Inventory and cash make the consequences concrete. Start with sixteen units costing 6 each, financed by opening equity of 96; six units were initially uncommitted. The first prepayment

debits cash and credits a contract liability by 100. The first four deliveries release 40 of revenue and remove inventory cost 24. Immediately before amendment, cash 100 plus inventory 72 equals the liability 60 plus equity 112. Receipt of the additional 48 increases cash and the liability. Approving the changed order itself earns no new revenue and creates no new stock.

Deliver six more units by the first year end. They release liability and revenue of 54, with inventory cost 36. Total year-one revenue is 94, cost of sales 60 and profit 34. Cash 148 and inventory 36 equal liability 54 and equity 130. Delivering the final six in the next year earns another 54 at cost 36, leaving cash and equity 148 with no inventory or contract liability. That period's profit is 18 but no cash arrives: the customer already paid. The operating cash bridges are $34 + 60 + 54 = 148$ and $18 + 36 - 54 = 0$. Lifetime revenue 148 less inventory cost 96 gives profit 52.

If the additional six units instead cost 60 and meet paragraph 20's standalone-price condition, the addition is a separate contract. The first four still earn 40; the six remaining original units earn 60 and the new six earn 60. Both contracts keep their own allocation records. The difference between these cases is the admitted price and scope assessment, not permission to choose whichever result produces the preferred margin.

An integrated promise can have the opposite effect on past cumulative revenue. Suppose a customer controls work in progress as the company creates it, satisfying paragraph 35(b). The company has one integrated performance obligation, and its remaining work is not distinct from work already supplied. Original price 100 and supported output progress of 40% yielded revenue 40. An approved enforceable amendment changes that same promise to price 130; the revised faithful output measure is 30% of the expanded work. Under 21(b), both the price and progress change the cumulative amount earned:

$$\text{revised cumulative revenue} = 130 \times 0.30 = 39, \qquad \text{catch-up} = 39 - 40 = -1.$$

The new work makes the completed work a smaller share of the promise. Revenue decreases by 1 at amendment even though the total contract price increases. The old 40 remains attached to its original information view; the modification does not establish that the earlier estimate was an error.

For a complete prepaid example, start with cash and equity 50. Receive 100 and pay service costs 30 before amendment. Recognition of 40 leaves cash 120, liability 60 and equity 60. At amendment, receive the additional 30 and debit revenue 1 while crediting the liability 1. Cash is now 150, the liability is 91 and equity is 59: cumulative revenue 39 less costs 30 has earned profit 9. Operating cash since opening is 100, also $9 + 91$. Completing the promise next period releases revenue 91 and incurs further paid service costs of 60. Period profit is 31, operating cash is $31 - 91 = -60$, and closing cash and equity are 90. Lifetime profit is $130 - 90 = 40$. The output progress percentages were independently supported; they were not inferred from these cash costs.

Both modifications need the approved rights, effective date, previous and revised promise, consideration, performance measurements and source versions. The distinct-goods method freezes the fulfilled deliveries and replaces the remaining allocation; the integrated method proposes only the difference between revised cumulative revenue and the amount already recognised. Economic state and journals change atomically. Different billing schedules can require a contract-asset adjustment instead of the prepaid liability entries used here. A request that has not been approved leaves the existing contract operative under paragraph 18. Unknown enforceability stays unresolved; approved scope with undetermined price requires paragraph 19's variable-consideration estimate and constraint, not a price of zero.

Paragraph 21(c) also admits mixed remaining obligations, requiring an explicit allocation consistent with the two principles above. The examples do not force such a contract into one

category. Price-estimate changes without a contract modification follow 87–89. A later price change after modification needs paragraph 90: variable consideration promised before a 21(a) modification can still be allocated to the earlier obligations under 90(a). It cannot automatically be spread over the twelve remaining units merely because their current allocation is 9 each.

35.4.5 The same cash can accompany different revenue

Suppose an entity receives 120 from a customer and sends 100 to a supplier. It keeps 20. Those cash movements cannot tell us whether it sold the supplier's good or arranged a sale by that supplier. Under B34–B38, the decision starts with the specified good or service and whether the entity controls it before transfer to the customer. Responsibility for delivery, inventory risk and price discretion inform that decision; none is a substitute for control, and momentary legal title does not by itself establish it (Australian Accounting Standards Board, 2022f).

In the principal case, the entity controls the identified good before customer transfer, can direct its use and obtain its benefits, and is responsible for delivering it. Acquisition on credit debits inventory 100 and credits the supplier payable 100. Customer transfer and cash receipt debit cash 120 and credit revenue 120; the inventory leaves through cost of sales 100. Settling the supplier debits the payable and credits cash 100. From zero opening balances, the result is cash and equity 20, revenue 120 and profit 20.

In the agent case, the supplier controls the good until transfer; the entity only arranges the sale and earns a fixed commission of 20 when that service is completed. Assume the entity controls the collected cash and owes the supplier 100 under its remittance agreement. Funds held outside its balance sheet in a fiduciary capacity are outside this example. Receipt therefore debits cash 120, credits the supplier payable 100 and credits commission revenue 20. Settlement clears the payable against cash 100. The agent has no inventory or supplier cost of sales. Its final cash, equity and profit are again 20, but revenue is 20.

Even the intermediate balance sheets agree: before remittance, both have cash 120, a payable 100 and equity 20. The different control facts explain the different revenue. The method must retain those facts and classify each specified promise separately, since the same entity can act as principal for one promise and agent for another. An unknown control finding leaves classification unresolved; the receipt cannot decide it. Both cash legs remain in the ledger. Whether the cash-flow statement presents receipts and remittances gross or net requires its own AASB 107 conditions, which do not follow merely from the revenue classification. These examples establish the cash change and pretax profit contribution while preserving that separate presentation question.

The customer cases separate earned revenue from billing and cash. The cost side requires a second history: which goods and asset components remain, what they cost, and which losses or revaluations they have already absorbed. Ordinary cost and replacement cases establish that history before production and changed obligations make it more complex.

35.4.6 Inventory and assets retain their cost histories

Inventory uses identified cost for non-interchangeable items or the admitted cost formula for interchangeable items. AASB 102.9, 23, 25 and 28–34 support lower of cost/net realizable value, relevant cost formulas and write-down/reversal conditions. A method never chooses LIFO merely to fit a reported balance. In a second lot example, purchases are 10 units at 2 and 10 at 3, followed by sale of 12 units. FIFO gives cost 26 and stock 8 units/cost 24. Weighted average gives cost 30 and stock 20. Under the same given selling price 48, margin differs 22 versus 18. Both alternatives require their own permitted policy, consistent application and evidence;

matching total cash does not remove the difference. NRV 16 for the remaining FIFO lot records expense 8, inventory allowance 8; recovery to NRV 22 reverses 6, leaving allowance 2. Recovery to 30 is capped at original cost 24. US LIFO remains a separately sourced question.

The retained AASB 116.7/15, 29–30 and 48–62 support cost and depreciation. Asset 100, residual 10, three-year even use gives 30 per full year. A year-two estimate change after first-year depreciation to residual 10 and three remaining years gives $(70-10)/3=20$ prospectively; an error in original cost is a correction, not an estimate change. Component replacements derecognize the replaced component's remaining amount before recognizing its replacement; no perpetual double asset. Borrowing-cost capitalization, impairment, provisions, employee benefits and share-based payment are specified as additional scoped methods below, not silently assigned zero in real-company cases.

35.4.7 Preparing equipment and selling its first samples

Preparing a machine can produce two things of value: equipment capable of operating as intended, and goods that can already be sold. Their costs and receipts answer different accounting questions. Necessary delivery, installation and functional testing contribute to the machine's cost under AASB 116.16–17. Paragraph 20A requires the proceeds from selling items produced during preparation, and those items' costs, to enter profit under the applicable standards. It measures the goods' costs using AASB 102 (Australian Accounting Standards Board, 2022b, 2021a). The samples' margin therefore does not reduce the cost of the machine.

Consider a synthetic Australian manufacturer in calendar 2024, starting with cash and contributed capital of 300. It pays 100 for equipment, 5 for its initial delivery, 12 for necessary installation and 3 for an engineering service that tests and calibrates its physical performance. Probable benefits and reliable cost measurement are admitted. These four amounts give PPE cost of 120. The engineering service uses different resources from those recorded as sample production below. Each receipt or resource has an identity and an assigned purpose; sharing a resource would require an evidenced allocation, not two recognitions of its full cost.

The sole sample batch uses normal materials costing 4 and directly attributable contracted conversion costing 2. The example has no other production overheads. Its goods cost is thus 6 under AASB 102.10–15. In a real factory, paragraph 13 requires fixed production overheads to be allocated using normal capacity; low output does not justify loading every unused resource onto a few units. Unknown overheads would leave the goods' cost unresolved here. Separate abnormal production waste of 1 is expensed under paragraph 16(a). Staff training of 3 and advertising of 2 are also period expenses, excluded from PPE by AASB 116.19. All these amounts are paid in cash.

On 30 June, the manufacturer sells the entire batch for cash of 8. These are ordinary manufactured goods, outside the inventory standard's agricultural, financial-instrument and broker-trader exceptions. Net realisable value is 8, with no further completion or selling costs, so cost 6 needs no write-down. The approved, enforceable customer contract identifies the goods and price, has commercial substance and probable collection. Delivery, acceptance, title and control transfer together on that date. There are no remaining promises, returns or price concessions in this case. These facts admit the revenue treatment under AASB 15.9 and 31–38; cash receipt alone would not establish it (Australian Accounting Standards Board, 2022f).

The entries preserve both assets. Acquisition and preparation debit PPE 120 and credit cash 120. Sample production debits inventory 6 and credits cash 6; the excluded costs debit period expense 6 and credit cash 6. At sale, debit cash 8 and credit revenue 8, then debit cost of sales 6 and credit inventory 6. AASB 102.34 links the goods' expense to recognition of the related revenue. The batch's cost history remains available even though its closing stock is zero. Before

depreciation, cash is 176, PPE is 120 and the loss is 4.

The machine becomes capable of its intended operation on 1 July but remains idle through 31 December. No further production occurs in this example. Management's admitted estimate is a five-year useful life, zero residual value and uniform consumption of benefits, including obsolescence while idle. The straight-line method uses equal monthly charges. Paragraph 55 starts depreciation when the asset is available for use and does not suspend this charge during idleness. It also allows a usage-based method to have zero depreciation when production is zero; that is a different admitted consumption pattern. Here the six available months give

$$\text{depreciation} = \frac{120 - 0}{60} \times 6 = 12, \qquad \text{closing PPE} = 120 - 12 = 108.$$

Debit depreciation expense 12 and credit accumulated depreciation 12. Profit is $8 - 6 - 6 - 12 = -16$, so equity closes at $300 - 16 = 284$. Cash 176 plus net PPE 108 gives the same total. The initial capital is already in the opening balance; it is not a financing receipt in this year.

Cash-flow presentation follows the purchases and sale. PPE payments of 120 are investing outflows under AASB 107.16(a). The sample receipt of 8, goods payments of 6 and period-cost payments of 6 give operating cash flow of $8 - 6 - 6 = -4$; profit of -16 plus noncash depreciation of 12 gives the same result under the indirect method. Paragraphs 14 and 18–20 support this classification and reconciliation (Australian Accounting Standards Board, 2023a). Closing cash is $300 - 120 - 4 = 176$.

Several wrong treatments would still balance. Netting the samples' cost and proceeds into PPE gives $120 + 6 - 8 = 118$, depreciation 11.8, net PPE 106.2 and a loss of 17.8; cash remains 176. It also shifts the sample margin from operating cash flow into investing cash flow. Waiting for productive use would omit depreciation 12 and overstate both PPE and equity by 12. Capitalising the excluded costs would instead produce cost 126 and closing PPE 113.4. A balanced trial balance cannot distinguish these errors from the required treatment. The method needs the cost purpose, goods population, sale conditions, consumption pattern and availability date.

The four retained compilations have an overlapping application period that includes this calendar 2024 example: AASB 102 compilation 3, AASB 116 compilation 6, AASB 15 compilation 8 and AASB 107 compilation 5. The example assumes no early adoption and isolates the pretax cost and sale mechanism; taxes, financing, grants, restoration obligations and later impairment require their own facts. With initial cost established, replacing part of the machine must next remove the cost history of the part that has gone.

35.4.8 Replacing a component removes an old carrying amount

An asset can remain in use while one of its components is replaced. The company then owns a new component and has given up the remaining benefits of the old one. Recording only the new purchase would preserve an asset that no longer exists. The register therefore needs component identity, original cost, accumulated depreciation, remaining life and the dates of replacement and use. Paragraphs 13 and 70 of the retained AASB 116 require the replaced part's carrying amount to be removed when an eligible replacement is recognised (Australian Accounting Standards Board, 2022b).

Consider a synthetic company with cash and contributed capital of 200 on 1 January 2024. It pays 100 for an administrative asset: a shell costing 80 with an eight-year useful life, and a significant component costing 20 with a two-year life. Both are available immediately, have zero residual values and consume benefits evenly. The company admits the cost model. These are authored facts, including recognition eligibility and the absence of any additional impairment;

real assets require evidence for those assessments. Administrative use makes depreciation an expense. A production asset might instead contribute to inventory cost under paragraph 49.

Acquisition debits the two asset costs by 80 and 20 and credits cash by 100. Paragraphs 43–44 require separate depreciation of significant components; the full year's charge is $80/8+20/2 = 20$. Debit depreciation expense 20 and credit each component's accumulated depreciation by 10. At 31 December 2024, cash is 100, net PPE is 80 and equity is 180 after the loss of 20. No revenue or other operating activity is included in this asset-only example.

On 1 January 2025 the company replaces the old component. Its remaining carrying amount is $20 - 10 = 10$. Debit accumulated depreciation 10 and disposal loss 10, and credit the old component's cost 20. The replacement costs 30, is paid in cash and is available immediately for three years of even use, again with zero residual. Its entry debits new PPE 30 and credits cash 30. Cash is now 70 and PPE is $70 + 30 = 100$; equity is 170. Retaining the old component would instead report PPE of 110 and miss the loss of 10, although both acquisition entries would balance.

The requirement to remove the old part also applies when it was not depreciated separately. Paragraph 70 permits replacement cost as an indication of the old part's cost at acquisition if its carrying amount cannot practicably be determined. That does not make today's invoice the old part's remaining value. Acquisition-date cost and consumption would still need a supported estimate. Ordinary servicing has a different consequence: paragraph 12 expenses it as incurred. A separately assumed maintenance payment of 4 would debit maintenance expense and credit cash by 4. Capitalising it would balance but overstate both PPE and profit by 4 under those facts.

During 2025 the shell and new component each depreciate by 10. Immediately before sale on 31 December, gross active cost is $80 + 30 = 110$, accumulated depreciation is $20 + 10 = 30$, and net PPE is 80. Cash remains 70 and equity is 150. The buyer then obtains control and pays 90, with no selling costs or sale-and-leaseback arrangement. Remove gross cost 110 and accumulated depreciation 30, debit cash 90, and credit disposal gain 10. Paragraphs 67–72 locate the disposal at control transfer and measure the gain as net proceeds less carrying amount. The gain is not ordinary revenue; routine sales of formerly rented assets have a separate rule at 68A.

Pretax amount	2024	2025 including sale
Depreciation expense	20	20
Replacement loss	0	10
Disposal gain	0	10
Profit	−20	−20
Closing cash / net PPE	100 / 80	160 / 0
Closing equity	180	160
Operating cash	0	0
Investing cash	−100	60

The cash-flow statement separates the 2025 purchase of 30 from the sale proceeds of 90. Under the retained AASB 107.16(a)–(b), both are investing flows. The indirect operating reconciliation removes depreciation and the non-operating loss and gain from profit:

$$\mathrm{CFO}_{2024} = -20 + 20 = 0, \qquad \mathrm{CFO}_{2025} = -20 + 20 + 10 - 10 = 0.$$

Lifetime investing cash is $-100 - 30 + 90 = -40$, equal to the equity reduction once no assets remain. An ordinary servicing payment would instead be operating cash; paragraph 16 does not permit investing classification merely because an expense relates to equipment (Australian Accounting Standards Board, 2023a).

These cash-flow comparisons concern calendar 2024 and 2025. The retained AASB 107 compilation is dated 31 December 2023 and its official application record covers annual starts from 1 January 2024 to before 1 January 2026. The retained AASB 116 compilation is dated 31 December 2022 and covers annual starts from 1 January 2023 to before 1 January 2027. Their different intervals must remain attached to the respective rules. Tax has not been computed in these pretax statements; no tax-law or complete reporting claim follows.

35.4.9 Revaluation needs each asset's past gains and losses

Revaluation changes the amounts to be depreciated and can change profit and other comprehensive income (OCI) differently. A class total alone cannot determine the split. For the Australian for-profit case here, paragraphs 39–40 refer to the earlier decreases and surplus of the same asset. The Australian not-for-profit exceptions permit different class-level treatment; they cannot be imported into this case (Australian Accounting Standards Board, 2022b).

A second synthetic company owns two administrative machines, A and B, which form the entire class for which it elects the revaluation model. They became available on 1 January 2023 at costs 100 and 40, with useful lives of five and four years, zero residuals and even use. Cash after their acquisition was 60, funded by historical capital of 200. Every fair value used below is an explicitly assumed reliable measurement. Paragraphs 29, 31 and 36–38 require the class policy, sufficiently regular measurements and coverage of the whole class; they do not infer missing fair values from the ledger. No additional impairment is assumed after the separate assessment required by paragraph 63.

The 2023 history establishes the opening for the 2024 statement comparison. After depreciation of 20 and 10, A and B carry 80 and 30. At 31 December 2023 their assumed fair values are 120 and 24. A increases by 40, recorded in OCI and accumulated in its revaluation surplus. B decreases by 6; with no surplus of its own, that decrease is expense in profit or loss. The net class increase of 34 therefore cannot all be OCI. Such an entry would omit B's expense of 6 and understate OCI by the same amount while leaving total equity unchanged.

For clarity, this company chooses paragraph 35(b)'s treatment of accumulated depreciation. At each revaluation it first debits accumulated depreciation and credits gross PPE by the accumulated amount, leaving the net amount as the adjusted gross balance. It then posts the revaluation difference. The prior gross cost, accumulated charge and original-cost history remain in the asset register. This choice changes neither the net revaluation nor its allocation between profit and OCI.

The opening 2024 balances are consequently cash 60, PPE $120 + 24 = 144$, capital 200, retained earnings -36 from depreciation and B's loss, and surplus 40. Assets and equity both equal 204. The register also remembers B's unreversed profit-or-loss decrease of 6. This prior-year derivation uses the retained AASB 116 scope; it does not assert a 2023 cash-flow presentation under the later AASB 107 compilation.

Let L_i denote an asset's outstanding earlier revaluation decrease in profit or loss, and S_i its remaining surplus credit. For an increase $u_i > 0$, the part reversing that earlier loss goes to profit; the rest goes to OCI. For a decrease $d_i > 0$, existing surplus absorbs the reduction through OCI before any excess reaches profit. The operative amounts are

$$\begin{aligned} \text{profit gain}_i &= \min(u_i, L_i), & \text{OCI increase}_i &= u_i - \min(u_i, L_i), \\ \text{OCI decrease}_i &= \min(d_i, S_i), & \text{profit loss}_i &= d_i - \min(d_i, S_i). \end{aligned}$$

An increase reduces L_i by its profit gain and adds its OCI increase to S_i. A decrease reduces S_i by its OCI decrease and adds its profit loss to L_i. These are revaluation-history rules; the

separate impairment reversal ceiling is considered in the impairment method. No surplus is transferred during use in this example. If that permitted transfer were elected, the later surplus available to absorb a decrease would be smaller.

For 2024, the revalued amounts are depreciated over the remaining four and three years. The charges are $120/4 = 30$ and $24/3 = 8$, leaving carrying amounts 90 and 16. The company then revalues the whole class to 45 and 22. A falls by 45: its surplus absorbs 40 through OCI and the remaining 5 is a profit-or-loss expense. B rises by 6, reversing its own earlier loss through profit. After eliminating accumulated depreciation, gross PPE is 45 and 22, with surplus zero. Profit is $-38 - 5 + 6 = -37$ and OCI is -40. Cash 60 plus PPE 67 equals closing equity 127. A now retains a prior revaluation loss of 5; B retains none.

In 2025, the remaining three-year life of A and two-year life of B give depreciation $45/3 = 15$ and $22/2 = 11$. Carrying amounts become 30 and 11. At 31 December their assumed fair values are 50 and 11. A's increase of 20 first reverses its loss of 5 through profit, then adds 15 to OCI and surplus. B's valuation is included even though its monetary adjustment is zero. Before sale, cash 60 and PPE 61 equal equity 121.

Immediately after that measurement, the buyer obtains both machines and pays their same-date values, 50 for A and 11 for B. Debit cash 61 and remove PPE 61; the disposal gain is zero. The company elects the transfer permitted by paragraph 41: debit A's surplus 15 and credit retained earnings 15 directly. This transfer creates no income, new OCI or cash. Routing it through profit would falsely raise 2025 profit from -21 to -6.

Pretax amount	2024	2025 including sale
Depreciation	38	26
Revaluation loss / gain in profit	5 / 6	0 / 5
Disposal gain	0	0
Profit / OCI	-37 / -40	-21 / 15
Closing cash / net PPE	60 / 67	121 / 0
Closing retained earnings / surplus	-73 / 0	-79 / 0
Closing equity	127	121
Operating cash	0	0
Investing cash	0	61

The 2024 equity movement is $204 - 37 - 40 = 127$. In 2025, profit of -21 and OCI of 15 leave total equity at 121. Retained earnings move from -73 to -79: the year's loss reduces them by 21 and the direct transfer increases them by 15. That internal transfer does not change total equity. Operating cash follows independently from the year's profit and noncash effects:

$$\mathrm{CFO}_{2024} = -37 + 38 + 5 - 6 = 0, \qquad \mathrm{CFO}_{2025} = -21 + 26 - 5 = 0.$$

The entire 61 sale proceeds appear in investing cash. Tax consequences still need legal tax bases, applicable rates and the profit/OCI origin of each difference under AASB 112, as AASB 116.42 requires.

AccDSL must therefore preserve the class and its members, each valuation's date and evidence, original cost and accumulated charges, individual surplus and loss history, component exits and the chosen surplus-transfer policy. An absent valuation leaves that class remeasurement unresolved; it is not permission to select only appreciating assets. A missing disposal date or old component value likewise prevents the affected calculation. The journals and these economic changes belong to the same authorised transition. The zero adjustment for B illustrates why a valuation record may change even when no monetary line is needed. Together with the

state-only advance-allocation rule above, this preserves the information needed by the next depreciation, revaluation or disposal.

35.4.10 Employee benefits

The employee-benefit, impairment, provision, award and valuation methods use the selected IAS 19, IAS 36, IAS 37, IFRS 2, IFRS 13 and IAS 23 provisions in the original 2023 EU incorporating regulation (European Commission, 2023a). This is historical mechanism support; later amendments and actual entity facts remain separate applicability inputs.

DefinedContributionState retains service periods, promised contribution formula, contributions due/paid and applicable legal terms. For earned service requiring contribution 10, debit employee expense 10/credit payable 10; payment settles it. Calling a plan defined contribution needs the limited-obligation facts; a guaranteed benefit or additional funding obligation cannot disappear by name.

The defined-benefit obligation (DBO) measures accrued employee benefits under the stated actuarial assumptions. DefinedBenefitState retains employee/cohort service, benefit formula, projected salary, retirement/turnover/mortality assumptions, discount-curve currency/term, obligation, plan-asset fair value, asset ceiling, contributions, benefits and remeasurements. IAS 19.67–74 requires projected unit credit and the stated service attribution; .83 supplies discount-rate requirements. Assumptions are separately admitted measurements, not constants provided by the accounting rule. A one-employee toy plan promises 100 for each completed service year, paid after year two, with certain completion, zero salary growth and supported 10% discount. At year-one end accrued benefit 100 due in one year has PV 1000/11. In year two that opening obligation accretes 100/11 and the second earned benefit adds 100; obligation 200 then settles 200. Merely using current salary times years without timing would give the wrong year-one obligation.

A separate funded-plan roll-forward supplies an independent statement example: opening DBO 100, plan assets 80 and net liability 20; current service 12; interest 5 on DBO, interest 4 on assets at 5%; contributions 10 at year end; benefits 8 paid from plan at year end; actuarial obligation loss 7; actual asset return 6. DBO closes 100+12+5−8+7=116, assets 80+6+10−8=88, net liability 28. Expense is 12+1=13, OCI loss is 7−(6−4)=5. Debit service expense 12/credit net pension liability 12; debit interest expense 1/credit liability 1; debit OCI 5/credit liability 5; debit liability 10/credit cash 10. End liability 20+13+5−10=28. Plan-paid benefits reduce both DBO and plan assets and do not cause a second employer cash payment. IAS 19.113, .120–127 supports these components; remeasurement OCI is not recycled to profit under 122. A surplus requires the asset ceiling and applicable minimum funding analysis, not recognition of any calculated negative liability.

Unknown actuarial inputs give conditional obligations or measurement_unresolved. Curtailment, settlement, amendment and multi-employer/insured arrangements need their own fact conditions and measurement-date remeasurement. A buyout is not settlement if obligations remain. Sensitivities vary one assumption only when that conditional meaning is explicit; correlated risks require joint scenarios.

35.4.11 Impairment and provisions

ImpairmentState retains asset/CGU boundaries, carrying amounts, allocated goodwill, corporate assets, recoverable-amount inputs and previous impairments/depreciation. IAS 36.9–10 determines indicators and mandatory annual tests. Recoverable amount is the higher of value in use and fair value less disposal costs (.18); it is not the lower of two pessimistic scenarios. VIU

needs the supported cash flows, forecast horizon/growth, discount and exclusion of inappropriate financing/tax or uncommitted restructuring effects under 33–57. Compute conditional values if those estimates are unknown; do not infer them from historical profitability.

For a cash-generating unit (CGU), assets excluding goodwill 100 plus goodwill 20, recoverable amount 90: loss 30 first removes goodwill 20 and then reduces other assets 10 subject to their floors under 104–105. Debit impairment 30/credit goodwill 20/credit asset allowance 10. If the asset floor is 95 the requested reduction to 90 is impermissible; apply the remaining allocation/recognition conditions, not a forced plug. For a standalone cost-model asset with carrying 80, recoverable 60, recognize 20 loss. At a later measurement before further depreciation, supported recovery to 90 permits reversal only 20 to the no-impairment 80 ceiling (.117–119). Goodwill impairment is not reversed (.124). Different dates require the hypothetical unimpaired depreciation path, not the old nominal cost as cap.

An asset floor changes the allocation of a unit's loss

A cash-generating unit can lose value while one of its assets retains a substantial disposal value. Allocating the unit's loss in proportion to asset balances just once may then reduce that asset too far. IAS 36.104–105 first allocates loss to goodwill, then to the other assets pro rata, while protecting each asset from a reduction below the highest of zero, its fair value less costs of disposal if measurable, and its value in use if determinable. A share that cannot be assigned to one asset is redistributed to the others (European Commission, 2023a).

Consider a synthetic cost-model unit containing equipment A with carrying amount 60, equipment B with carrying amount 40, and allocated acquired goodwill 20. Cash 80 lies outside the tested unit. With no liabilities, opening equity is 200. The unit boundary and goodwill allocation are assumed to have been admitted under paragraphs 66–80; there are no shared corporate assets, noncontrolling interests or liabilities requiring an adjustment to this comparison. In a real case those judgments precede the calculation. In particular, paragraph 78 requires consistent treatment of a liability that is included in the unit's recoverable value; subtracting it from only the carrying amount would compare different populations.

At measurement date T_0, admitted value in use is 70 and fair value less disposal costs is 65, so the unit's recoverable amount is 70. Individual values in use are supported as not determinable, while the individual disposal values are 55 for A and 5 for B. These are supplied valuation assumptions, not values inferred by an allocation routine. They establish floors of 55 and 5. A missing valuation would be a different state: the program may omit a term because its non-determinability is admitted, but may not replace an unexamined input with zero.

The unit's loss is $60 + 40 + 20 - 70 = 50$. Goodwill takes the first 20. An unconstrained allocation of the remaining 30 would assign $30 \times 60/100 = 18$ to A and $30 \times 40/100 = 12$ to B. A, however, can absorb only $60 - 55 = 5$. Its excess share of 13 passes to B, whose total loss becomes $12 + 13 = 25$, within its capacity $40 - 5 = 35$. The closing balances are therefore

$$A = 60 - 5 = 55, \qquad B = 40 - 25 = 15, \qquad \text{goodwill} = 20 - 20 = 0.$$

Together they equal the unit's recoverable amount of 70. Debit impairment expense 50, credit goodwill 20, and credit the separate asset impairment allowances by 5 and 25. Cash 80 plus equipment 70 equals equity 150. The unconstrained 18/12 entry would also balance, but would leave A at 42, below its admitted floor of 55. Clipping A's loss to 5 without redistributing the excess would instead recognize only 17 of the required remaining 30.

For an implementer, each round has a remaining amount and an active set of assets. Compute shares from their pre-allocation carrying amounts. Fix each share that reaches an asset's available capacity, remove that asset from the active set, and redistribute the remaining amount.

A round that needs further redistribution removes at least one asset, so the procedure terminates after finitely many such removals. For positive carrying weights b_i and capacities c_i, its allocation has the form

$$x_i = \min(c_i, \lambda b_i), \qquad \sum_i x_i = \min\left(q, \sum_i c_i\right),$$

where q is the requested allocation and $\lambda \geq 0$ gives the common proportional rate among assets that have not reached their limits. In the example A is fixed at 5 and B receives the remaining 25. A source floor above an asset's existing cost gives zero reduction capacity; it does not create permission to increase that asset during an impairment calculation.

The capacity limit also explains how to handle an unsuccessful allocation. A diagnostic input requesting 50 against capacities 5 and 35 can allocate only 40, leaving 10. That arithmetic does not establish that the supplied unit and individual valuations are jointly coherent. It calls for checking measurement assumptions, the unit boundary and any separate obligations. Paragraph 108 permits a liability for a remaining unit loss only if another IFRS requires one. AccDSL must retain the residual and the unresolved decision; it cannot invent a payable to make the unit reach a requested total. This diagnostic is not a second admitted company example.

A recovery is limited by the depreciation that would have occurred

Suppose both pieces of equipment have five remaining years of even use and zero residual values at T_0. Under paragraph 63, their revised carrying amounts give next-year depreciation $55/5 = 11$ and $15/5 = 3$. At T_1, one full year later, they carry 44 and 12. The company must also retain the path that would have occurred without impairment. On that path depreciation would have been $60/5 = 12$ and $40/5 = 8$, leaving 48 and 32. Restoring a past loss therefore cannot mean restoring the old nominal balances.

Assume a supported change in expected future cash flows now gives the unit a recoverable amount of 90. Individual recoverable amounts remain admitted as not determinable. Paragraphs 114–116 require the change in estimates; mere passage of time and discount unwinding would not support this reversal. Paragraphs 122–123 allocate a unit reversal over assets other than goodwill pro rata, subject to each asset's recoverable amount if determinable and its depreciated no-impairment amount. Goodwill itself cannot be restored under paragraph 124 (European Commission, 2023a).

The requested increase is $90 - (44 + 12) = 34$. A has room for only $48 - 44 = 4$; B has room for $32 - 12 = 20$. The same redistribution procedure therefore recognises $4 + 20 = 24$, leaving equipment at 48 and 32 and goodwill at zero. The recognized total of 80 is below the unit's value of 90. That difference does not identify an omitted asset, authorize goodwill recognition or imply cash has been received.

Debit the impairment allowances by 4 and 20 and credit reversal gain 24. The original allowances of 5 and 25 consequently retain balances of 1 and 5. Accumulated depreciation is 11 and 3. The register now reconciles as

$$\text{PPE} = 100 - \underbrace{(11 + 3)}_{\text{depreciation}} - \underbrace{(1 + 5)}_{\text{impairment allowances}} = 80.$$

The remaining allowances record the different past depreciation paths. Erasing all 30 of the original allowances would produce PPE of 86, exceeding the no-impairment ceiling by 6. Future annual depreciation is $48/4 = 12$ and $32/4 = 8$, consistent with the remaining lives.

Pretax amount	T_0 impairment	Following year to T_1
Impairment expense	50	0
Depreciation / reversal gain	0 / 0	14 / 24
Profit / OCI	−50 / 0	10 / 0
Closing cash / PPE	80 / 70	80 / 80
Closing goodwill / liabilities	0 / 0	0 / 0
Closing equity	150	160
Operating cash	0	0

The first column isolates the impairment measurement; the second follows the next full year. With no cash or owner movements in this slice, the operating cash reconciliations are $-50+50 = 0$ and $10 + 14 - 24 = 0$, removing the noncash loss, depreciation and reversal under IAS 7.20. Equity moves from 200 to 150, then to 160. Tax remains a separate calculation under IAS 36.64 and IAS 12. These technical rules use the retained original 2023 EU text, with its date corrigendum; the relative dates do not assert current reporting admission or a complete later amendment review.

The method thus needs actual and no-impairment depreciation histories, measurement dates, individual valuation-status judgments, goodwill identity and evidence of the changed estimate. There is also an explicit numerical boundary. If every eligible asset has zero carrying amount but a positive reversal is requested, the positive-weight procedure above has no denominator. It returns an unresolved allocation basis and preserves the history; it does not choose equal weights without an additional justified interpretation. That extension, along with shared corporate assets and other scope interactions, remains distinct from the fully specified positive-weight case.

Recognising a separate obligation

ProvisionState retains obligating past event, present legal/constructive duty, outflow assessment, estimate distribution, settlement dates, risk/discount basis, reimbursements and evidence. IAS 37.14 requires present obligation, probable outflow and reliable estimate; .36–47 governs best estimate/risk/discount. An intended future operating expenditure without that duty is not a provision. Assume an established obligation to pay 121 in two years, deterministic amount, appropriate 10% rate and no double-counted risk: recognize expense/provision 100, accrete interest 10 then 11, settle 121. The passage-of-time increase is borrowing cost under 60. If the obligation is an eligible asset restoration cost, initial debit follows that asset's method rather than always expense; later changes need the applicable restoration interpretation. Contingent liabilities/disclosures remain visible when recognition conditions are not met; no generic 0.5 threshold may substitute for the relevant standard's required analysis.

These cost, benefit and obligation measurements supply the starting states for the following changes. Production first asks which costs the output can carry; later examples change estimates, rights and obligations while retaining their earlier measurement histories.

35.4.12 An idle factory does not make its remaining stock more valuable

Producing fewer goods leaves the factory's fixed costs spread over fewer units, but that arithmetic does not establish their inventory cost. Consider a factory that normally makes 100 good units each year after allowing for planned maintenance. Its annual fixed production overhead is 180. If it makes only 60 units, dividing all 180 by 60 would assign 3 to each unit, rather than the

normal 1.8. The unsold goods would then carry part of the cost of idle capacity into the next year.

AASB 102.13 prevents that result. Fixed production overhead is allocated using normal capacity; low output does not increase the amount per unit, and unallocated overhead is expensed. Abnormally high output requires a lower allocation per unit so that inventory is not measured above cost (Australian Accounting Standards Board, 2021a). Normal capacity is an estimate of average production under normal conditions, including planned maintenance. It requires evidence separate from the actual output or desired margin.

Use a synthetic manufacturer in calendar 2024 and the retained AASB 102 compilation dated 31 December 2021. Its documented application interval includes that year; subsequent amendment and actual-entity admission remain separate. The company makes one ordinary manufactured product, with no opening stock, work in progress, joint products or normal production loss. One material kit produces one good unit. A kit costs 2; direct labor costs 1 and variable production overhead costs 1 per good unit, all paid in cash. These goods are outside the agricultural, financial-instrument and broker-trader exceptions. Periodic weighted average is used for this homogeneous batch under paragraphs 25–27.

The fixed 180 consists of cash factory management and maintenance of 120 and depreciation of 60. Opening factory PPE is 600, with ten remaining years, zero residual value and admitted uniform time-based consumption. There is one component and recoverability has been separately established. The depreciation method therefore remains unchanged across the output scenarios. Paragraph 12 includes factory depreciation and production administration among production overheads; AASB 116.48–49 allows depreciation to enter another asset's carrying amount when appropriate (Australian Accounting Standards Board, 2022b). Recording it in inventory does not make it a second cash payment.

Keep excluded costs visible. A separately documented accident destroys six unused kits costing 12 before labor or conversion. They are additional to the kits used for good output and have no salvage value. Unrelated head-office administration of 10 and period selling services of 8 are also paid in cash. The abnormal loss and these unrelated expenses do not bring the good units to their present location and condition, so paragraphs 10–16 exclude them from inventory. Factory management and head-office administration cannot be classified solely from the word *administration*.

All completed goods have supported net realisable value of 10 per unit, with no further completion or sale costs. The example therefore still applies the lower-of-cost-and-NRV requirement rather than suspending it. Delivered units sell for cash of 10 each under an approved, enforceable single-product contract: rights and payment terms are identified, collection is probable, the contract has commercial substance, and delivery, acceptance, title and control transfer coincide. There are no remaining promises, returns or concessions. These facts admit the existing AASB 15.9 and 31–38 delivery method (Australian Accounting Standards Board, 2022f); cash receipt by itself would not do so.

Allocate the cost that production can carry

Let Q denote good completed units and S the units sold, with $0 \le S \le Q$. For this single cost pool, normal capacity is 100 and fixed overhead is 180. Each good unit first receives its direct and variable cost of $2 + 1 + 1 = 4$. When output is low, the fixed allocation is 1.8 per unit; when output is abnormally high, the total allocation is capped at the 180 actually incurred. For $Q > 0$, this gives

$$f(Q) = \frac{180}{\max(100, Q)}, \qquad u(Q) = 4 + f(Q), \qquad U(Q) = 180 - Qf(Q),$$

where f is fixed overhead per unit, u is admitted unit cost and U is unallocated overhead expensed now. This compact formula implements the stated homogeneous, constant-cost example. Multi-product pools, work in progress or changed production resources require their own allocation. With no output, there is no unit cost to divide by: stock is zero and all 180 is current expense.

At low output $Q = 60$, the good units carry fixed overhead of $60 \times 1.8 = 108$, leaving 72 as current expense. Unit cost is 5.8. Selling 40 units removes inventory cost 232; the other 20 remain at 116. Abnormal waste 12 and the unrelated administration/selling cost 18 are also current expenses. Revenue 400 therefore leaves pretax profit $400 - 232 - 72 - 12 - 18 = 66$. Normal output of 100 absorbs all 180 at the same unit cost of 5.8. Abnormally high output of 150 instead allocates $180/150 = 1.2$ per unit, giving unit cost 5.2.

The comparison below changes output and sales explicitly. All arms retain the same fixed costs and the separately stated accident. They do not have equal total production spending: direct and variable costs change with good output. Sold-stock cost means the carrying amount removed on sale; AASB 102.38 also includes unallocated overhead and abnormal production costs in the broader inventory expense.

	Low	Normal	High	Idle
Good units / sold units	60 / 40	100 / 80	150 / 120	0 / 0
Unit cost	5.8	5.8	5.2	n/a
Unallocated fixed expense	72	0	0	180
Sold-stock cost	232	464	624	0
Closing inventory	116	116	156	0
Broader inventory expense	316	476	636	192
Pretax profit	66	306	546	−210
Closing cash	1010	1250	1450	850
Closing PPE	540	540	540	540
Pretax net assets	1666	1906	2146	1390

Opening cash is 1000 and PPE is 600, so opening net assets are 1600. There are no payables, borrowing or owner movements in this slice. Tax effects are uncomputed: the last row is a pretax net-assets reconciliation, not after-tax equity, distributable profit or a complete statutory set of statements. This boundary avoids manufacturing tax law to finish the example. Each arm has separately reconciled cash, stock and PPE; an after-tax extension must supply the relevant tax bases, rates and timing.

Preserve both the cost origin and the cash movement

The low-output journal path starts with 66 kits costing 132. Debit raw materials and credit cash 132. The accident debits abnormal-loss expense 12 and credits raw materials 12. Transferring the 60 good kits to production moves raw-material cost 120 into finished goods; direct labor 60 and variable overhead 60 enter the same goods account against cash. No raw materials remain, and the goods initially carry 240.

The fixed costs first enter a production clearing account: debit it 120 against cash and 60 against accumulated depreciation. Then debit finished goods 108 and unallocated-overhead expense 72, crediting the clearing account 180. Its balance is now zero, and finished goods cost 348. The separate administration and selling entry debits expense 18 against cash. On delivery, debit cash and credit revenue 400, then debit sold-stock expense and credit finished goods 232. The final 116 is the cost of 20 unsold goods, not a residual chosen to make the books balance.

Cash receipts of 400 less payments of $132 + 60 + 60 + 120 + 18 = 390$ increase cash by 10. The whole factory's depreciation is 60 even though part remains in inventory. To reconcile pretax profit with operating cash, add that noncash consumption and subtract the full inventory increase:

$$\Delta\text{cash} = 66 + 60 - 116 = 10.$$

The stock includes depreciation of $20 \times 60/100 = 12$. An equivalent reconciliation adds only the 48 of depreciation reaching current profit, but then subtracts only the cash-cost part of stock, $116 - 12 = 104$: $66 + 48 - 104 = 10$. Adding back 48 while subtracting all 116 would wrongly report a cash decrease of 2. The two valid methods differ in presentation, not in the underlying cost population.

Selling all 60 good units in the low-output arm provides a second check. Production and its payments are unchanged. Revenue becomes 600, sold-stock expense 348, profit 150 and cash 1210; inventory becomes zero. The additional 200 cash receipt releases inventory cost 116 and increases profit by 84. Total pretax net assets become 1750. This is why a forecast needs both production and sell-through, rather than an inventory margin inferred from total factory spending.

Balanced books can still defer the wrong costs

If low output absorbed all fixed overhead using $180/60 = 3$ per unit, unit cost would be 7. Unsold stock would be 140 and profit 90, each 24 above the admitted result; cash would remain 1010. The books would balance, but they would postpone idle-capacity expense in the remaining goods. At high output, applying 1.8 without a cap would allocate 270 of fixed cost when only 180 was incurred. Crediting a variance of 90 to balance that over-allocation would leave stock 174 and profit 564, both overstated by 18. A balanced variance cannot create an incurred cost.

Capitalizing abnormal loss 12 and unrelated period costs 18 into the 60-unit batch would defer $30 \times 20/60 = 10$ of excluded expense in unsold stock. This is a separate classification error from the capacity error. Each resource therefore retains a purpose and identity, its membership in the cost pool, and its cash or noncash origin. The state also retains capacity evidence, good and lost units, batch cost and quantity, and the specific sale removals. A duplicate resource or negative stock is invalid; a revised capacity estimate creates a new supported allocation decision instead of overwriting an earlier result.

Unknown normal capacity leaves the fixed allocation unresolved. Unknown NRV prevents a complete closing measurement, and missing control-transfer facts prevent revenue recognition. Work in progress, joint products and heterogeneous cost populations require a different specified model rather than silent use of this unit formula. The proposed procedure and checked case define those boundaries; they do not establish production-engine conformance or an actual manufacturer's accounting treatment.

35.4.13 Changed obligations and unfinished production

A company can change its restoration estimate before a machine wears out, promise to repair sold products, or end a period with unfinished goods. Each condition changes a different part of its accounts. The following cases extend the preceding operating methods through those transitions. They retain exact illustrative amounts and separate economic evidence from bookkeeping.

Restoration, production and uncertain-tax methods use the retained historical EU text; assurance and mixed contracts use the inspected AASB compilations for synthetic 2024/2025 Australian

for-profit reporters. The AASB 137 PDF covers annual periods beginning from 1 January 2023 to before 1 July 2026; its website gives a different end-period summary. Both admit these particular historical calendar years. A real application still needs its own source and period admission. Amounts are pretax unless an explicit tax calculation follows.

Restoration estimates change the asset and the obligation differently

The restoration record retains the related component and liability, cost/revaluation policy, gross cost/accumulated depreciation, remaining life/residual, current cash-flow estimate and rate, prior estimate identity, same-asset revaluation surplus/prior profit deficit, hypothetical cost-model carrying amount, impairment evidence and whole-class valuation review. IFRIC 1.2 (European Commission, 2023a, IFRIC 1.1–8) requires the original liability to have been included in the asset's cost. A standalone litigation obligation is not admitted to this asset-adjustment method.

Separate passage of time from a new estimate. Under the cost model, a liability increase adds to PPE; a decrease reduces PPE only to zero, with excess in profit. An increase prompts assessment of impairment indications and an IAS 36 test when required. Depreciate the adjusted carrying amount less admitted residual over the remaining useful life. Discount unwind is finance expense, never capitalised under IFRIC 1.8. Once useful life is exhausted, later estimate changes go to profit under 7 regardless of asset model. Actual settlement reduces the obligation and cash; a difference from the last carrying amount requires a dated remeasurement.

With cash 100, PPE 80 and provision 20, opening equity is 160. An admitted new provision 50 adds 30 to PPE. Recoverable amount 90 requires impairment 20: PPE 90, cash 100, liability 50 and equity 140. Four years remaining and zero residual gives annual depreciation 22.5. The subsequent depreciation-only illustration assumes a separately supplied zero discount rate and no further liability change; PPE 67.5 and equity 117.5 follow. In another branch, PPE 30/provision 80 and a new provision 30 require Dr provision 50 / Cr PPE 30 / Cr profit 20. Negative PPE 20 would be wrong. Cash 100 then equals liability 30 plus equity 70.

Under the revaluation model, a liability increase first consumes a same-asset surplus through OCI, then affects profit; a decrease reverses a same-asset profit deficit and otherwise increases OCI. A decrease exceeding the hypothetical cost-model carrying amount has an immediate profit component. Determine any necessary class revaluation first so that these splits use the correct history. With PPE 150/provision 50/surplus 20, an increase 35 gives OCI loss 20 and profit loss 15; PPE stays 150 only because a supplied class valuation review supports it. Cash 100 then gives equity 165 and liability 85. Surplus on another asset cannot be borrowed. In a separate decrease case, PPE 25, hypothetical cost 40, earlier profit deficit 15 and provision 80 → 30 yield excess 10 plus deficit reversal 15 in profit and 25 in OCI. Cash 100+PPE 25=provision 30+equity 95. Using 25 as the cost floor would compute the wrong split. The prior deficit explains why 25 and 40 differ; no fresh impairment is inferred from that difference.

A full cash history opens cash 100/PPE 120/provision 20/equity 200. Two-year straight-line life and zero residual give depreciation 60 in year 1. At 10%, unwind 2 raises the provision 22; new supported estimate 33 adds 11 to PPE. Close PPE 71/provision 33/equity 138, with pretax loss 62 and no cash change. Year 2 depreciation 71 and unwind 3.3 give loss 74.3; actual settlement 36.3 leaves cash/equity 63.7 and zero PPE/provision. Across both years depreciation 131 plus finance 5.3 equals cumulative loss 136.3. The total cash bridge is exact; IAS 7 classification of restoration settlement needs a separate fact-specific policy. The method therefore returns a cash-classification obligation rather than inventing a complete categorized cash-flow statement. A fully exhausted PPE 0 with provision 20 → 25 gives expense 5 even if a historic revaluation reserve exists.

Assurance is a measured obligation alongside the product sale

The assurance record stores covered sales/population, legal or constructive promise, coverage/task findings, claim history, remaining-outcome law, timing/discount, reimbursement contract and virtual-certainty evidence. AASB 15.B 28–B 33 determines whether assurance, service or an inseparable combined service is present. AASB 137.14/24 (Australian Accounting Standards Board, 2022e, paras. 14,24,36–62) admission precedes measurement. For a large population 39 uses weighted outcomes; individual claim probability does not replace the population outflow judgment. Correlation matters for distributions even when marginal means are sufficient to add expected cost. A single lawsuit uses 40's different analysis.

Consider 100 products with admitted per-product costs 0,10,40 and probabilities 3/4,1/5,1/20. The population estimate weights each possible repair cost and then sums across the covered products:

$$100\left(\frac{3}{4}0+\frac{1}{5}10+\frac{1}{20}40\right)=400.$$

A low claim probability for one product therefore does not eliminate the population obligation. The entire population has a present assurance duty, probable outflow and reliable estimate; discount is immaterial over the short horizon. Opening cash 1000 and inventory 600 are funded by equity 1600. Cash sales 2000 and inventory expense 600 accompany warranty expense/provision 400. An independently enforceable quarter-cost reimbursement is virtually certain on settlement, so recognise a separate asset 100/benefit 100. Do not net it against the provision. Paragraph 54 permits net expense presentation.

Pay claims 100 and receive reimbursement 25. A new remaining-claims estimate 250 replaces the old 300, releasing 50; reduce the remaining reimbursement 75 to 62.5, expense 12.5. Year 1 cash 2925+reimbursement 62.5=provision 250+pretax equity 2737.5. Profit 1137.5 follows from 2000 − 600 − 400+100+50 − 12.5. Direct operating cash 1925 also equals profit 1137.5+inventory release 600+provision increase 250 − receivable increase 62.5. Year 2 settlement 250 and receipt 62.5 leave cash/equity 2737.5 with no further profit under the realised amounts. Without virtual certainty, no reimbursement asset is recognised; retain the separate uncertainty. Missing claim evidence does not permit zero provision or the printed 400 by default.

A mixed amendment needs two revenue calculations

The modification record retains the original allocation/performance, enforceable change, remaining distinctness findings, amended allocations, faithful measures of progress and new receipts. AASB 15.21(c) (Australian Accounting Standards Board, 2022f, paras. 18–21,39–45,73–90) requires consistent application of 21(a)/(b) to the affected promises; it does not supply a universal price split. The following exact case deliberately removes discount-allocation ambiguity.

A contract includes construction 100 and four distinct inspection visits 5 each, at their evidenced standalone prices. Inspections concern another operating line and can occur before the new construction finishes. The original expected construction cost 80; actual eligible cost 40 and faithful cost-to-cost progress give 50% and revenue 50. One visit adds revenue 5 and cash expense 2. The 120 price was prepaid. The approved amendment adds integrated construction 40 and two visits 10; supported standalone allocations therefore become 140 and 30. New total construction cost 100 makes cumulative progress 40%. The amended construction allocation gives

$$\text{cumulative revenue}=140\frac{40}{100}=56,\qquad \text{catch-up}=56-50=6.$$

This corrects only the construction promise. The first visit's 5 remains; five remaining distinct visits share 25 prospectively. The new 50 receipt itself earns no revenue.

With opening cash/equity 100, two additional visits by year-end yield revenue 71, expense 46, profit 25, cash 224 and contract liability 99. The indirect operating bridge 25+99=124 matches 170 received − 46 paid. Complete construction for cash 60 and three visits for 6 next year: revenue 84+15=99, profit 33 and cash/equity 158. Lifetime profit 58=price 170 − construction 100 − visit costs 12. Keeping old progress would recognise 70 for construction; applying 40% to the entire 170 would wrongly recognise distinct future visits. Missing allocation/progress or enforceability blocks the numeric transition. The one-year-or-less financing expedient under 63 is explicitly elected for this example, not assumed for long prepaid contracts.

The invoice and the work need different clocks

The builder can earn the same construction and inspection revenue while being paid on a different schedule. Retain the preceding allocations, costs and approved amendment, but place all work and payments in calendar 2024. At inception and after the amendment, the expected interval between each payment and its related service is at most one year, and the builder elects the financing expedient. Receivables and contract assets have a separately assessed zero impairment allowance in this example. These findings matter: AASB 15 requires the financing assessment and AASB 9 treatment of the rights to consideration; eventual collection alone establishes neither (Australian Accounting Standards Board, 2022f, paras. 63,105–109).

Suppose the customer initially advances only 30. Opening cash and capital of 100 become cash 130 and a contract liability of 30. In April the builder pays construction costs of 40 and visit costs of 2, earning the same revenue of 55 as before. Cash falls to 88. The advance covers the first 30 of that work; the remaining 25 is a contract asset because the right to payment still depends on a billing milestone. Profit is 13 and equity is 113.

Now the milestone makes a gross invoice of 80 unconditional. The invoice credits the advance of 30 already received, leaving a receivable of 50. The contract asset disappears. But the builder has earned only 55 of the 80 now paid or due, so a contract liability of 25 also appears. Figure 35.1 isolates this change. The receivable and contract liability coexist: one is an unconditional payment right and the other is work still owed under the same contract.

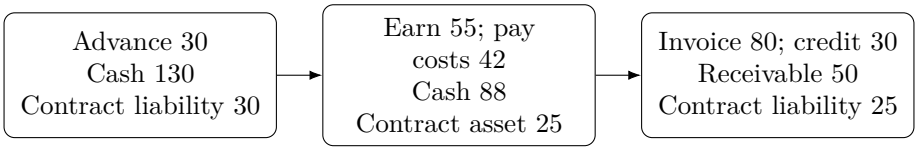


Figure 35.1: The invoice changes the payment right without new work or new cash. Cash stays at 88 and cumulative revenue at 55 when the final box is reached.

The advance must therefore retain its identity after receipt. Crediting it reduces the new amount collectible; it does not refund cash or recognise revenue again. Let G be cumulative gross invoices, K the advances credited against those invoices, and C later collections of the resulting receivables. The amount still collectible is

$$\mathrm{AR} = G - K - C.$$

For the invoice just issued, this is $80 - 30 - 0 = 50$. Collecting 20 changes cash to 108 and the receivable to 30, with no change in revenue. Each credit names both its advance and invoice; each collection names its invoice. A second credit cannot reuse an advance that has already been consumed.

To determine the contract balance, count consideration already paid or unconditionally due once. If A is cumulative advance cash, adding it to gross invoices would count credited advances twice. Subtract K to obtain $T = G + A - K$. Compare this with cumulative earned revenue R:

$$T = G + A - K, \qquad \mathrm{CA} = \max(R - T, 0), \qquad \mathrm{CL} = \max(T - R, 0).$$

At invoicing, $T = 80 + 30 - 30 = 80$, giving the liability $80 - 55 = 25$. Collection leaves T unchanged: cash replaces a right that was already unconditional. Together these relations imply, for cumulative paid costs E and opening capital 100,

$$\mathrm{cash} + \mathrm{AR} + \mathrm{CA} - \mathrm{CL} = 100 + R - E.$$

Substitution shows why: cash is $100 + A + C - E$, the receivable is $G - K - C$, and the net contract asset is $R - G - A + K$. The advance, invoice, credit and collection terms cancel, leaving earned profit in equity. The cancellation checks the arithmetic; the contractual findings determine which amounts belong in it.

The May amendment now adds construction revenue of 6 without a receipt, reducing the liability from 25 to 19. A new, separately identified advance of 50 raises cash to 158 and the liability to 69. Two further visits earn 10 and cost 4, leaving cash 154, receivables 30, contract liability 59 and profit 25. The remaining gross invoice of 90 credits the new advance of 50. Its net right of 40 increases total receivables to 70; paid-or-due consideration reaches 170, so the contract liability becomes $170 - 71 = 99$.

Collect the first invoice's remaining 30, then complete construction and the last three visits for cash costs of 66. Completion earns the remaining 99 and removes the contract liability. Cash is 118, the second invoice still carries 40, and equity is 158. Collecting that 40 leaves cash and equity of 158, as in the prepaid example. Customer cash is advances $30 + 50$ plus collections $20 + 30 + 40$, totalling 170. Paid costs are 112 and profit is 58. The gross invoices of 170 describe the rights created, not an additional receipt.

Full prepayment provides a useful limiting case. If the final invoice credits all 170 already received, it creates no receivable and no monetary journal; it still records which advances the invoice consumes. Conversely, invoicing before any work can create a receivable and contract liability together. Both schedules preserve the same earned amounts under these terms. The finite method retains these billing identities alongside the two performance calculations. Nonzero impairment, refundable excess payments, contract losses and tax require their own measurements before this example can be joined to a more general company book.

Unfinished production carries component-specific completion

The production record adds started/completed/scrapped quantities, material insertion point, conversion stages, opening WIP by component, observed progress, actual eligible costs, normal capacity, fixed/variable overhead classification, NRV, and the costing-policy version. Equivalent units are a derived policy justified by this homogeneous process; IAS 2 (European Commission, 2023a, IAS 2.12–18) prescribes rational cost allocation and the normal-capacity constraint, not this exact spreadsheet convention.

Start 100 units with no opening WIP or abnormal waste. Finish 80;20 are half converted, with all material already inserted. Material 1000 is spread over 100 material units; variable conversion 450 over 90 equivalent conversion units. Fixed cash overhead 600 has admitted normal capacity 120 equivalent units, so its rate is 5, assigned 450 and expensed idle 150. The material denominator is 100 units, while conversion uses only 90 equivalent units:

$$80 + 20\left(\frac{1}{2}\right) = 90, \qquad \text{completed unit cost} = 10 + 5 + 5 = 20.$$

Separate denominators keep missing conversion work out of inventory cost. Completed cost 1600 and WIP 300 exhaust assigned cost 1900: WIP has material 200 and conversion 100. Sell 50 finished units at 30: revenue 1500, cost 1000 and idle expense 150 give profit 350; finished 600 plus WIP 300 remain. Opening cash/equity 3000 leaves cash 2450 and equity 3350.

In the next production month, the remaining 10 equivalent conversion units require variable 50 and fixed 50; independently supplied normal capacity 10 admits both. Add 100 to WIP 300 and transfer completed 400. The remaining 50 finished units then sell 1500 at total cost 1000. Cumulative cash/equity 3850 and profit 850 follow; no WIP remains. A different opening WIP, spoilage policy or completion pattern needs its own allocation. Reusing 100 as conversion units overstates completion; capitalising all 600 fixed cost conceals 150 idle expense. NRV is separately assessed for both unfinished and completed stock, including costs to complete.

Joint products and delayed supplier payment

The joint-production record retains inseparable joint costs, split-off quantities, separate subsequent costs, allocation-basis evidence, selling values/dates and NRV. For joint cost 300, A 60 at split-off price 4 and B 40 at 9 give values 240/360. The selected rational and consistent relative-sales method assigns 120/180, or unit costs 2/4.5. Selling 30 A and 10 B gives revenue 210, cost 105 and profit 105. Opening cash 500 leaves cash 410, unsold A 60/B 135 and equity 605. Weighting by mass instead changes the admitted policy; equal weights are not a missing-value fix.

Separately, production 300 with an independently established immaterial byproduct of NRV 20 allocates main inventory 280 and byproduct 20. Sale of that byproduct for 20 removes its 20 carrying amount; it earns no margin here. Opening cash 500 ends cash 220+main 280=equity 500. Retaining main 300 as well as byproduct 20 invents cost; deducting 20 without recording the unsold byproduct loses an asset.

The supplier record names the original creditor, normal-credit purchase price, actual terms, financing assessment, initial liability, effective rate, payment schedule and actual payments. Normal cash price 100 and a fixed payment 121 in two years imply 10% effective interest under the supplied terms. Record inventory and payable 100; interest 10 then 11 raises the liability 121. Ordinary inventory is not a qualifying asset in this case. Sell for cash 150 in year 1: opening cash/equity 100 becomes cash 250, payable 110 and equity 140. Year 2 interest 11 and payment 121 leave cash/equity 129; lifetime profit 29. Under the explicit historical IAS 7 election, both original supplier payment and interest are operating cash. Year 1 indirect 40+payable increase 110=150; year 2 $-$ 11 $-$ 110= $-$ 121. A substituted finance provider or altered arrangement requires reclassification analysis.

Uncertain tax has its own measurement decision

Uncertainty about whether a tax authority will accept a treatment affects both current tax and the tax bases used for deferred tax. The calculation must retain gross DTA/DTL, recoverability, rate changes, lease interactions and the original profit, OCI or equity allocation. It also needs a measurement decision about the uncertain treatment. Under the historical EU IFRIC 23.6–14 (European Commission, 2023a, IFRIC 23.6–14,A1–A5) method, the uncertain-tax record adds treatment grouping, filing position, authority/examination scope, full-information acceptance finding, candidate outcomes, predictive-measure choice, consistent current/deferred bases, reassessment event and disclosure obligations. Assume examination with full knowledge; do not multiply exposure by an estimated chance of avoiding examination.

Synthetic law ultimately deducts equipment 100, with the timing uncertain. Book depreciation

20 leaves PPE 80; cash operating revenue 100 makes pretax profit 80. Opening cash/equity 500, purchase 100 and cash revenue 100 leave cash 500. The enacted rate 25% is supplied, all tax amounts remain payable, and all origins are profit. If full deduction 100 is probably accepted, current tax 0 and DTL 20 follow. If outcomes 100/20 are binary with full acceptance probability 1/5, and most-likely amount better predicts resolution, use deduction 20: current 20 and DTL 0. In a third branch, deductions 100/40/20 at 1/5,3/10,1/2 and a supported expected-value choice give an expected deduction of

$$\frac{1}{5}(100) + \frac{3}{10}(40) + \frac{1}{2}(20) = 42.$$

Taxable profit and the remaining tax base are both 58. Current tax is $58(25\%) = 14.5$, while the taxable temporary difference is $80 - 58 = 22$, giving a deferred tax liability of $22(25\%) = 5.5$. All branches have total tax 20, profit 60 and equity 560; they differ in the timing liability. These equal totals arise from this particular full-deduction law.

If new evidence immediately resolves the expected-value branch at deduction 40, increase current tax $14.5 \to 15$ and reduce DTL $5.5 \to 5$. The two 0.5 expense/benefit entries net to zero. Pay 15: cash 485+PPE 80=DTL 5+equity 560. A later resolution must also include intervening depreciation and other facts; silence alone is not new evidence of acceptance. Real law, probability and predictive suitability remain required. The printed hypothetical rate or distribution cannot fill them.

Combining the new operating contributions

Combine only the five explicitly compatible EU year 1 contributions: WIP, joint products, byproduct, supplier financing and lifetime restoration. Give each independent lots/contracts/assets but one reporting entity, currency and period; sum their openings, without intercompany elimination or transferring AU cases. Opening cash 4200, PPE 120, provision 20 and equity 4300 become cash 3430, inventory 1375, PPE 71, liabilities 143 and pretax equity 4733. Pretax profit 433 reconciles equity $4300 \to 4733$; total assets 4876 equal liabilities 143+equity 4733. The same cash movement follows from the indirect reconciliation:

$$433 + 60 - 1375 + 110 + 2 = -770.$$

Starting from pretax profit 433, add depreciation 60, deduct inventory growth 1375, add the supplier liability increase 110 and add noncash restoration interest 2. The 11 restoration estimate adds both PPE and provision and is excluded from cash. This is a pretax operating slice; new tax effects are unresolved, so a complete after-tax report cannot be issued from it. The separate uncertain-tax example is not silently grafted onto this different law/fact pattern.

35.4.14 Inspections, surrendered rights and changed employee benefits

A replacement inspection, a smaller leased area and a pension curtailment all remove something from an existing arrangement, but they remove different economic rights or obligations. The accounting must identify what survives before measuring the new arrangement. The following methods use the retained historical EU text and synthetic pretax amounts. Actual authority, contract and actuarial facts remain necessary for application.

Replacing the inspection component

IAS 16.14 (European Commission, 2023a, IAS 16.12–14,43–55) distinguishes a major inspection required for continued operation from the physical parts inspected. Capitalize its cost as a

replacement only if the recognition criteria hold; derecognize the remaining previous inspection cost. That obligation persists even if the original purchase did not separately identify an inspection component. The estimated cost of a comparable future inspection may indicate the original component cost; it is not automatically today's replacement expenditure or an additional historical purchase.

The component record carries asset identity, inspection cohort, original cost allocation and evidence, availability date, residual value, consumption pattern, useful life until the next inspection, accumulated depreciation/impairment and the replacement event. Allocate the initial asset's existing total cost among its components; do not increase that total while recognizing the old inspection. Advance depreciation to the replacement date, remove the old carrying amount, add eligible new cost and begin the new component's depreciation when available. IAS 16.43–55 requires separate significant-component depreciation and retains depreciation during idle periods unless the applicable consumption method gives zero. Ordinary servicing is expense under IAS 16.12, not a new inspection asset. Other production-cost rules may place depreciation in inventory; these examples assume a non-production use, so charges enter profit.

Suppose original equipment cost 120 includes an inspection component 20. The remaining physical part costs 100, has ten-year straight-line life and zero residual; the inspection has four-year life and zero residual. After two full years the physical part carries 80 and the old inspection 10. A qualifying inspection performed early costs 24 and is available immediately for four years. Derecognize the old 10 as loss, capitalize 24, and retain physical carrying amount 80. The equipment now carries 104. One subsequent full year charges 10 physical and 6 inspection depreciation, leaving 88. Expensing all 24, retaining both inspection cohorts, or depreciating 104 uniformly over the physical life would compute different targets.

With opening cash 100 and equipment 90, equity 190, replacement leaves cash 76, equipment 104 and equity 180. Subsequent depreciation 16 leaves equity 164 and equipment 88. An otherwise separate routine service costing 3 reduces cash and profit by 3 and leaves equipment unchanged. An unavailable old-cost allocation does not justify writing off zero: reconstruct a supported component estimate or retain the unresolved derecognition amount. Revaluation/impairment histories must first update the component carrying amounts under their own methods.

Reducing a lease and paying usage-dependent rent

IFRS 16.44 (European Commission, 2023a, IFRS 16.38–46) first tests whether an added right and commensurate consideration make a separate lease. A scope reduction cannot meet that increase-in-scope test. For a modification that is not separate, paragraph 45 requires revised contract allocation, term and discount rate at the effective date. Paragraph 46(a) requires right-of-use derecognition and the gain/loss for a partial or full termination; the remaining liability remeasurement adjusts the surviving right-of-use asset. A mere decrease in price with unchanged scope belongs to paragraph 46(b), not automatic derecognition.

Represent the terminated rights explicitly, including the physical/use/time units and their carrying-cost allocation. In a homogeneous space lease with an unchanged remaining term and proportional rights, let fraction q of the rights be surrendered. The illustrative proportional allocation removes qA of the right-of-use carrying amount A and qL of the pre-modification liability L; the termination gain is $q(L - A)$. This allocation requires the homogeneous-rights facts. Do not use the payment reduction as q or apply floor-area proportions to heterogeneous equipment or a pure term change without a supported allocation.

For $A = 80$ and $L = 100$, surrender 25% of identical space with no termination fee. Remove right-of-use 20 and liability 25; recognize gain 5. The remaining $A = 60$ and $L = 75$ are then remeasured. Suppose the remaining contract requires one payment 70.4 after one year and the

newly supported discount rate is 10%. Its present value is 64. Reduce the remaining liability by 11 and the asset by 11, giving asset 49 and liability 64. The complete journal debits liability 36, credits the asset 31 and credits gain 5. It is wrong to send the entire liability reduction 36 to income or to omit the termination gain by adjusting the asset 36.

With opening cash 120/ROU 80/liability 100/equity 100, the event leaves cash 120, ROU 49/liability 64/equity 105. Over the one remaining year, zero residual and straight-line use give depreciation 49, interest 6.4 and payment 70.4. Closing cash and equity are 49.6, with no lease balances. Present-value calculation and the right-of-use consumption pattern are independent assumptions.

A separate fee of 2 per actual unit of usage, not index/rate dependent, not in-substance fixed and not already in the liability, is recognized under IFRS 16.38(b) when the condition occurs. Three units create expense 6/payable 6; payment reduces cash/payable 6. It is not a lease-liability remeasurement. Qualifying production use can instead include that cost in another asset under the applicable standard. Missing usage evidence, uncertain payment character, heterogeneous surrendered rights or a termination penalty require their own measurement before the conditional example can be applied.

Curtailment, settlement and the asset ceiling

The benefits record separates obligation D, fair-valued plan assets P, asset- ceiling adjustment, recognized net liability/asset, actuarial assumptions, remaining-year service cost, discount rate, legal discharge and actual payments. For a liability $D - P$ is positive. If there is a surplus, the recognized asset is limited to the admissible economic benefit C; the added ceiling adjustment is $\max(P-D-C, 0)$. This simple representation assumes no further minimum-funding adjustment; any applicable IFRIC 14 obligation remains a separate admission dependency. The example supplies C, not a method for inferring legal recoverability.

IAS 19.99 (European Commission, 2023a, IAS 19.99–112,120–126) requires current assumptions and asset fair values immediately before and after amendment, curtailment or settlement. First recognize pre-event remeasurements, then the change in benefits/covered employees, then any later settlement. Past service cost arises from an amendment or curtailment and is recognized at the earlier date specified by paragraph 103; a related restructuring or termination-benefit recognition can therefore precede the formal change. The examples have no such earlier trigger. A settlement discharges obligations; ordinary scheduled benefits included in assumptions are not a settlement gain event. Buying insurance while retaining an obligation is also not settlement under paragraphs 111–112.

Measure past service and settlement gain/loss without the asset ceiling, then remeasure the ceiling under paragraph 101A. Service cost and settlement enter profit unless eligible capitalization applies; actuarial and ceiling remeasurements enter OCI under paragraph 120 and are not later recycled under paragraph 122. For the rest of the year, use updated service assumptions and the post-event net position/discount rate, allowing for subsequent contributions and benefits, under paragraphs 122A–126. An unchanged beginning-of-year interest base would miss this required reset.

Consider opening $D = 1{,}000$, $P = 900$, net liability 100 and company cash 500/equity 400. At the event date, current measurement gives $D = 1{,}100$, $P = 920$: actuarial loss 100 less asset gain 20 creates OCI loss 80 and liability 180. A curtailment reduces the freshly measured obligation by 150 to 950. Recognize past-service gain 150, leaving net liability 30. A later transaction fully discharges a measured obligation 200 for plan assets 170 plus company cash 20. Settlement price 190 gives gain 10; D falls to 750, P to 750, liability to zero and cash to 480. Profit 160 and OCI−80 reconcile equity from 400 to 480.

For the remaining half-year, assume current service cost 30, annual discount rate 5%, simple half-year interest accrual and no contributions or benefit payments. Gross obligation and asset interest are both 18.75, so net interest is zero. Closing $D = 798.75$, $P = 768.75$, net liability 30, cash 480 and equity 450. Full period profit 130 and OCI−80 reconcile the change 50. Using the original net liability 100 for half-year interest would incorrectly add 2.5 expense. The simple accrual convention is an illustrative assumption, not a universal actuarial compounding rule.

To see the ceiling ordering, take D 100/P 130 and recoverable economic benefit 10. The surplus 30 yields recognized asset 10 and ceiling adjustment 20. An amendment reduces D by 20 to 80, while current evidence changes C to 15. Recognize past-service gain 20 without the ceiling. The new surplus 50 has ceiling adjustment 35, so its increase 15 is an OCI loss. The net recognized asset rises only 5 to 15. Recording just gain 5 would misclassify profit and OCI even though closing equity agrees.

Multi-employer information is a classification decision

IAS 19.32–36 (European Commission, 2023a, IAS 19.32–41) requires classification from actual plan terms, including constructive obligations. With adequate reliable allocation, a participant in a defined-benefit multi-employer plan recognizes its share of obligation, assets and cost. If the specific paragraph 34 information condition holds, account as if defined contribution and retain the paragraph 148 disclosures. Insufficient information is not proof that the plan economically has no defined-benefit exposure.

Paragraph 37 separately requires any contractual surplus/deficit-funding asset or liability and resulting profit effect. A synthetic agreed deficit payment 121 due in two years, with a supplied appropriate discount rate 10%, has liability 100 and expense 100 initially, followed by interest 10 and 11 and settlement 121. Ordinary current-service contributions are distinct. Do not omit the deficit contract because ongoing contributions use defined-contribution accounting or recognize the nominal 121 immediately while also accruing interest.

Withdrawal or wind-up liabilities follow IAS 37 under paragraph 39. A plan among entities under common control is not a multi-employer plan under paragraph 40; paragraph 41's plan-wide information and separate-entity charge/allocation rules must be applied instead. The record must therefore retain employer relationships, allocation reliability, contributions, deficit agreements, withdrawal terms and required disclosures. Unknown relationship or contractual terms block the classification, not the preservation of the underlying evidence.

35.4.15 Opening work carries both cost and completed effort

The earlier production example began with an empty factory account. An opening work-in-progress balance changes both the cost available and the work still needed. Materials may already be present even though conversion remains unfinished. Treating the opening units as entirely new would charge that effort twice; ignoring their costs would lose part of the inventory.

Consider homogeneous units with all materials added at the start and conversion effort incurred evenly. Ten opening units contain material cost 100 and conversion cost 50 and are halfway through conversion. The factory starts another 90 units, incurring material cost 900 and conversion cost 510. It completes 80 units and leaves 20 units halfway through conversion. Assume the opening units are completed first, no spoilage, no idle-capacity cost, and net realizable values above cost. Physical units reconcile as $10 + 90 = 80 + 20$; available cost is $100 + 50 + 900 + 510 = 1560$. These are supplied production facts, not estimates derived from a reported inventory total.

Under FIFO process costing, retain the opening 150 and allocate current cost only to current work. Completing the ten opening units requires five equivalent conversion units. Seventy units are started and completed, and the closing twenty require ten equivalent conversion units. Current conversion effort is therefore $5 + 70 + 10 = 85$, at $510/85 = 6$ per equivalent unit. Current material cost is $900/90 = 10$ per new unit. Completing opening work costs $150 + 5 \times 6 = 180$; the seventy new completed units cost $70 \times (10 + 6) = 1120$. Transfer 1300 from WIP to finished goods and retain $20 \times 10 + 10 \times 6 = 260$ in WIP. Completion is not a sale, so this transfer does not itself create cost of sales or a cash payment.

Weighted-average process costing pools the opening and current costs. Material cost remains 10 per equivalent unit, but conversion cost becomes $(50 + 510)/(80 + 10) = 56/9$. Completed goods then cost $11680/9$ and closing WIP $2360/9$, still totaling 1560. Posting at cents gives 1297.78 and 262.22 with a preserved rounding allocation. The methods distribute the same total cost differently because the opening conversion cost differs from the current-period rate. Neither method is selected merely to obtain a preferred closing inventory. IAS 2.23–27 requires specific identification where applicable and consistent FIFO or weighted-average treatment for the relevant interchangeable inventories; the equivalent-unit calculations are the local implementation of those cost policies (European Commission, 2023a, IAS 2.10–16, 23–27, 34).

For several stages, transfer the preceding stage's reconciled cost into the next stage as a separate cost component. Record when that component enters the process; it need not follow the completion percentage of later materials or conversion. Each stage retains opening units and costs, transfers in, completed transfers out, closing units, progress by component, inspection points, losses and recoveries. An impossible completion fraction, unexplained physical loss, missing opening cost or inconsistent transfer amount blocks the cost allocation rather than becoming an unexplained unit-cost adjustment.

Abnormal loss cannot become a higher cost of the good units

Normal process loss and abnormal waste need different treatment. IAS 2.16(a) excludes abnormal wasted materials, labour and other production costs from inventory. Its normal-capacity rule also prevents idle fixed overhead from being hidden in a higher unit cost (European Commission, 2023a, IAS 2.12–16, 21). The normal-loss estimate must therefore be supported before observing the actual loss, with its inspection stage and recoveries specified.

In a separate fully processed batch, 100 input units cost 1000. Assume normal yield is 95 units, all lost units have consumed the same full effort at inspection, there is no salvage, and this cost pool contains no fixed overhead. The normal unit cost is $1000/95 = 200/19$. If actual good output is 90, five units of loss are normal and five abnormal. Good inventory receives $90 \times 200/19 = 18000/19$, while abnormal loss of $1000/19$ is expensed. At cents these are 947.37 and 52.63, which reconcile to 1000. Allocating all 1000 to the ninety good units would conceal the abnormal loss. Earlier inspection, partial conversion, recoverable scrap or a different supported normal yield changes the calculation; those facts cannot be supplied by the final good-unit count.

35.4.16 A pension surplus is useful only to the extent it is available

The preceding pension examples supplied a recoverable economic benefit. IFRIC 14 specifies how the plan's terms, statutory requirements and funding basis constrain that benefit. Separate an unconditional right to a refund from the ability to reduce future contributions, and separate funding for past service from funding for future service. An IAS 19 surplus can coexist with a shortfall measured on the funding basis (European Commission, 2023a, IAS 19.64;

IFRIC 14.7–24).

Start with plan assets 130 and defined-benefit obligation 100, giving surplus 30. Suppose the employer has an unconditional right to the surplus but obtaining it costs the plan 5. The available refund is 25 and the recognized asset is 25, with a ceiling adjustment of 5. Costs depend on the permitted refund route, including settlement costs if winding up the plan is necessary. A right conditional on an uncertain event outside the employer's complete control does not establish this refund benefit. For a refund defined as the whole surplus or a proportion of it, IFRIC 14.15 makes no additional time-value adjustment to that refund (European Commission, 2023a, IFRIC 14.11–15).

Contribution reductions require a different calculation. Use employer service costs, excluding employee-funded amounts, over the shorter of expected plan life and entity life. Assumptions must be consistent with the benefit obligation and the situation at the reporting date. In a two-period example with zero discounting solely for illustration, future employer service costs are 20 and 20. Without a prepayment, required future-service funding would be 12 and 25. Their net difference is $8 - 5 = 3$. Suppose a prior prepayment separately reduces future required contributions by 7. The available reduction benefit is then

$$7 + \max\{0, (20 - 12) + (20 - 25)\} = 10.$$

The floor applies to the combined paragraph 20(b) amount, not separately to each period: ignoring the second period's excess would incorrectly produce 15. With no minimum funding requirement, use the available future employer service costs under paragraph 16 instead (European Commission, 2023a, IFRIC 14.16–22).

The refund and contribution routes may be mutually exclusive. In that case the available benefit is the greater of 25 and 10, not their sum 35; the recognized asset is also capped by the actual surplus. If the refund right fails but the contribution route remains supported, the asset is 10 and the ceiling adjustment is 20. Compatible combinations require their actual feasible terms and cannot consume the same surplus twice.

Past-service funding can require a liability before any payment. Suppose plan assets are 120 and the IAS 19 obligation is 100, but none of the surplus can ever be refunded or reduce contributions. The recognized pension asset is zero. A separate funding-basis assessment now requires a past-service contribution of 10 that will also be wholly unavailable after payment. IFRIC 14.23–24 requires the resulting liability when that obligation arises. With no intervening interest or other pension changes, recognize an OCI ceiling loss of 10 and a net pension liability of 10. On payment, debit that liability and credit cash 10. Plan assets become 130, but the recognized asset remains zero and payment causes no second loss or gain. Starting company cash and equity of 100 become cash 90 and equity 90. The obligation's recognition and the payment are distinct events (European Commission, 2023a, IFRIC 14.23–24; IAS 19.120(c), 126–127).

35.4.17 Restoration cash follows the activity being paid for

Return to the exhausted asset whose restoration liability is settled for 36.3. To complete its cash-flow example, specify the payment facts: the company pays wages, materials and contractors to remove the exhausted asset and restore the site; it acquires no new long-term asset and repays no borrowing. The proposed classification for this conditional case is an operating outflow of 36.3. This follows the activity definitions and supplier/employee examples in IAS 7.6 and 14. The earlier noncash creation of an asset and provision is not a cash outflow, and unwinding the provision does not by itself establish that part of the eventual contractor payment is interest paid (European Commission, 2023a, IAS 7.6, 12–16, 31–33, 43–44; IFRIC 1.8).

The competing interpretation classifies the payment as investing because the provision's initial measurement entered the asset's cost. The July 2012 IFRS Interpretations Committee staff paper examines comparable rehabilitation facts, presents both approaches and favors operating classification. That is support for the proposed reasoning, not a binding Interpretation or proof of universal treatment (IFRS Foundation staff, 2012, Appendix A, A35–A43; status notice on p. 1). The design records the chosen interpretation and its facts so that a different admitted rule can change the category without rewriting the legal journal or total cash.

Under the proposed branch, opening cash 100 plus operating cash flow minus 36.3, with no investing or financing flow, gives closing cash 63.7. In year 2 the indirect reconciliation is loss 74.3, plus noncash depreciation 71 and unwinding 3.3, less provision settlement 36.3, giving operating outflow 36.3. Under the competing investing branch, operating cash flow is zero and investing outflow 36.3; closing cash is identical. The classification difference is economically visible even though both cash bridges balance.

A new productive asset, actual borrowing repayment, separately evidenced interest payment, or restoration caused by current inventory production requires its own assessment. The proposed operating branch completes the specified exhausted-asset example; it is not a mandatory default for every restoration liability or evidence of an actual issuer's compliance. All three additions require explicit successor records and exchanges. They preserve the closed `CorporateJointHistory/1` profile rather than silently enlarging its accounts or event vocabulary.

35.4.18 Impairment across a group and the origin of tax

The following methods use the retained historical EU 2023/1803 text and the separately retained November 2023 IAS 12 amendment. Each example is conditional on its declared authority, perimeter, allocation and tax assumptions. These texts do not establish current issuer compliance or cross-GAAP equivalence.

Testing assets that serve more than one business

IAS 36.100–102 (European Commission, 2023a, IAS 36.100–108) identifies corporate assets by their contribution to multiple cash-generating units without independent cash inflows. Do not invent a separate recoverable amount for a headquarters just to make every row independently testable. When a reasonable consistent allocation to a unit exists, include that share with the unit. When it does not, first test the individual unit excluding the corporate asset, recognize that loss, then test the smallest larger group to which an allocation is reasonable and consistent. The second test uses the already adjusted carrying amounts.

The impairment record retains the hierarchy of units, legal/group/perimeter identities, corporate asset links, allocation basis, carrying and recoverable amount dates, goodwill allocation, individual floors and prior loss origins. The calculation is postorder: individual tests precede wider group tests. IAS 36.104 applies loss first to goodwill, then pro rata to other asset carrying amounts. Under paragraph 105, no asset falls below its measurable fair value less costs of disposal, determinable value in use, or zero, whichever is highest. Allocate a capped asset's unabsorbed loss again over the eligible remainder. Do not create a balancing liability for an unallocated remainder unless another standard actually requires it. Revised depreciation follows the adjusted amounts.

For a conditional example, unit A has two inseparable assets of 40 each, unit B two of 60 each, and headquarters 40. Headquarters cannot be reasonably allocated to either unit alone, but it serves their combined group. Unit A's recoverable amount 70 first causes loss 10, allocated 5 to each A asset. B's recoverable amount 130 causes no loss. The group's adjusted carrying

amount is then $70 + 120 + 40 = 230$. Given group recoverable amount 207, additional loss 23 allocates 3.5 to each A asset, 6 to each B asset and 4 to headquarters. Closing amounts 31.5, 31.5, 54, 54, 36 sum to 207. No individual asset's value in use or fair value less disposal costs is determinable in this example, so only the zero floor binds. The unit-level recoverable amount 70 is not an individual A-asset floor.

Using the original group carrying amount 240 in the second test would repeat the first loss 10. Treating headquarters as costless would omit its exposure. These are different errors even if both versions produce balanced journals. The calculation supplies an allocation method, not a valuation method for the assumed recoverable amounts.

Partial goodwill and the non-controlling interest

IAS 36 Appendix C (European Commission, 2023a, IAS 36.C1–C9) distinguishes the consolidated carrying amount from the amount used for an impairment test. When acquisition-date NCI was measured as its proportionate share of identifiable net assets, the reported goodwill omits NCI goodwill, while the unit's recoverable amount includes its contribution. C4 therefore requires a notional gross-up before the test. C8 prevents recognizing an impairment of goodwill that was never recognized. Neither the notional goodwill nor its notional impairment is posted as an asset or expense.

Retain the acquisition's NCI measurement election, ownership and profit-allocation history, unit allocation and evidence for the notional amount. Division of parent goodwill by ownership is a conditional illustration when proportional goodwill is justified; it must not silently assume away a control premium or differing synergies. For a subsidiary that is itself a unit, C6 allocates loss on the same basis as profit. C7 separately allocates a larger unit's goodwill losses by the parts' relative goodwill and identifiable-asset losses by their relative net identifiable carrying amounts, then allocates within each part. Preserve the NCI-bearing and parent-only parts rather than applying one ownership fraction to the entire group.

Assume the parent owns 80%, acquired proportionate-share NCI, has recognized goodwill 40 and supports proportional notional goodwill 50. Identifiable unit assets carry 200. If recoverable amount 220, test 250 against 220 and allocate loss 30 entirely to goodwill. The recognized goodwill loss is 24; the NCI notional loss 6 is not recorded. Reported goodwill falls from 40 to 16 and reported assets total 216; adding the remaining notional NCI goodwill 4 gives test carrying amount 220. Demanding the reported amount equal 220 would improperly capitalize that 4.

If recoverable amount instead falls to 180 in a separate branch, test loss 70 removes notional goodwill 50 then identifiable assets 20. Recognized loss is 60: parent goodwill 40 plus asset loss 20. Of the recognized loss, 56 belongs to the parent and 4 to NCI. The unrecognized NCI-goodwill loss 10 never enters either profit or NCI equity. Closing identifiable assets 180 and goodwill 0 reconcile the reported decrease from 240 to 180. Goodwill impairment cannot subsequently reverse under IAS 36.124; a rising valuation is not a new recognized goodwill asset.

A larger-unit branch adds a wholly owned part with goodwill 20 and identifiable assets 100. Total test carrying amount 370 includes subsidiary notional goodwill 50 and parent-only goodwill 20. At recoverable amount 270, loss 100 first removes all goodwill 70, then allocates 30 to identifiable assets in their $200 : 100$ ratio. Recognized loss 90 comprises goodwill 60 and assets 30; attribution is parent 86 and NCI 4. The notional 10 remains unrecognized. Asset floors and allocation of remaining goodwill must be respected when a branch does not exhaust goodwill.

Revalued impairment and its related deferred tax

IAS 36.60–64 (European Commission, 2023a, IAS 36.60–64,117–124; IAS 12.61A–65) treats a revalued asset's impairment as a revaluation decrease: charge the same asset's existing surplus through OCI before charging profit. Afterward, determine deferred tax from the revised carrying amount and tax base. IAS 12.61A attaches current and deferred tax to the related item even when the item was recognized in a different period. Preserve the origin of each temporary difference so a later tax-rate change does not move all tax to profit by default. Where exact attribution is exceptionally impracticable, paragraph 63 permits a reasonable allocation or a more appropriate method; document that circumstance.

Take an asset carrying 140, tax base 100, gross same-asset revaluation surplus 40 and related deferred tax liability 10 at a supplied 25% rate. There is no passage of time or tax-base movement. Recoverable amount 90 causes loss 50: OCI 40 and profit 10. The old DTL 10 reverses in OCI. A deductible difference 10 gives DTA 2.5 in profit only if recovery is probable; that condition is supplied here. Net OCI loss 30 plus net profit loss 7.5 equals the net-asset decrease 37.5, from $140 - 10 = 130$ to $90 + 2.5 = 92.5$. Offset presentation is not assumed: DTA and DTL eligibility must follow the applicable jurisdictional and legal-offset rules.

In the same no-time-passage illustration, recoverable amount later recovers to 130. IAS 36.117–120 limits reversal to the no-impairment amount and reverses the prior profit loss 10 first, with the remaining 30 through OCI. Reversing DTA 2.5 charges profit; establishing DTL 7.5 charges OCI. Net profit gain 7.5 and OCI gain 22.5 sum to 30, the increase from 92.5 to 122.5. A revaluation increase beyond the no-impairment amount is a separate IAS 16 event, not unlimited reversal; future depreciation must update both the reversal cap and carrying amount.

Separately, if the original carrying 140/tax base 100 asset's admitted future tax rate changes from 25% to 30% before any impairment, DTL rises from 10 to 12. The additional tax 2 relates to the earlier revaluation and is charged to OCI, not profit. The example assumes enacted/substantively enacted rates and the same manner of recovery; it does not infer tax law or settlement method from the accounting carrying value.

Acquisition and investment tax are separate questions

IAS 12.66 (European Commission, 2023a, IAS 12.39–51,66–68) requires acquisition-date temporary differences on identifiable assets and liabilities to enter acquisition accounting, subject to recognition rules. For a simple full acquisition paying 150, identifiable fair-valued assets 150 with tax base 100 and an admitted 25% rate create DTL 12.5. Net identifiable assets 137.5 give goodwill 12.5. There is no recursive DTL on the initial recognition of that goodwill under paragraph 15(a). This exception is not permission to omit the DTL on the identifiable asset. The treatment of an acquirer's own previously unrecognized DTA, and acquiree benefits recognized later, follows paragraphs 67–68; the former does not automatically change acquisition goodwill.

Outside-basis differences on investments are tracked separately from temporary differences inside the subsidiary's asset book. Under IAS 12.39, a taxable difference avoids DTL only to the extent the investor controls reversal timing and reversal is probably absent in the foreseeable future. Both conditions are required. For a taxable difference 60 and supplied 20% tax on the expected reversal, DTL is 12 when the exception does not hold; it is not recognized where both conditions demonstrably hold. Unknown control or distribution intent is unresolved, not evidence for the exception. An associate investor usually cannot infer control from a minority shareholding. Under paragraph 44, a deductible difference requires probable foreseeable reversal and available taxable profits to recognize a DTA. The source gives asymmetric tests; copying

the liability exception to assets is wrong. Rates and recovery paths follow paragraphs 46–51.

Pillar Two is not another deferred-tax rate

The original September 2023 compilation predates the specific amendment adopted by Commission Regulation (EU) 2023/2468 (European Commission, 2023b, arts. 2–3; annex 4A,88A–88D,98M) of 8 November 2023, published 9 November 2023. Its Article 3 enters into force the day after publication. Article 2 and the annex state the application and retrospective rules. The retained amendment adds IAS 12.4A, 88A–88D and 98M. For legislation within paragraph 4A, an entity neither recognizes nor discloses information about related deferred-tax assets/liabilities, and discloses that it applied this exception. It separately discloses current Pillar Two tax expense/income. Enacted or substantively enacted legislation not yet effective requires known or reasonably estimable exposure information; unavailable information requires the specified explanation and assessment progress.

The record therefore separates ordinary current/deferred tax from Pillar Two current tax, exception status, applicability authority and exposure disclosures. A supplied in-scope current top-up tax 7 creates current expense/payable 7 and later payment 7. A hypothetical Pillar Two temporary difference 40 does not create a DTL 6 by multiplying by 15%. That is the wrong target under the exception. Ordinary IAS 12 temporary differences continue under their own rules. The current top-up amount 7 is an illustrative input, not a calculation of an actual GloBE liability. Jurisdictional enactment, group thresholds, covered taxes, allocation, safe harbours and current-law measurement remain separate required facts/methods before actual application. The inspected amendment supplies the accounting boundary; it does not itself calculate tax due for a company.

35.4.19 A shorter useful life can change two different histories

A carrying amount does not reveal whether the estimate behind it was sound. Suppose an administrative asset costs 120, becomes available for use on 1 January 2023 and has no residual value. The company records straight-line depreciation of 10 in each of 2023 and 2024, leaving PPE of 100. On 1 April 2025, management concludes that the asset will be retired on 31 December 2028. That conclusion alone cannot tell us whether to change future expense or correct the earlier accounts. We need the information that supported the old estimate, including what was available when those accounts were authorized.

Consider two explicitly synthetic histories. In the first, twelve years was a reasonable useful life at acquisition and at both year-end reviews. A newly enacted operating restriction on 1 April 2025 now limits use to the end of 2028. That restriction neither existed nor was anticipated at the earlier dates. In the second history, the reliable, approved schedule at acquisition already specified six years, but a transcription error entered twelve. The original schedule was available and could reasonably have been obtained and considered at each authorization. Six years remains appropriate; the new discovery exposes the old mistake. These histories differ in facts, not in an accounting election available on the same known evidence.

AASB 108.5 makes that availability condition part of the definition of a prior-period error. Paragraphs 32–34A distinguish new information and developments from correction of an error; paragraphs 36–38 apply estimate changes to the current and relevant future periods (Australian Accounting Standards Board, 2022a). AASB 116.50–51 and 60–62 require systematic allocation, review of useful life and a method reflecting consumption (Australian Accounting Standards Board, 2022b). Here there is one component, even consumption, no capitalization into another asset, and an independently satisfied recoverability assessment. The shorter life does not replace

that assessment.

Dates determine which accounts are being corrected. The 2024 statements were authorized on 31 March 2025; the discovery occurs the following day. The 2023 statements were authorized on 31 March 2024. No 2025 interim statements have been issued. Thus the error affects two authorized annual reports and an unissued current quarter. Moving discovery before the 2024 authorization would change the treatment of that report under paragraph 41. The example concerns newly arising April conditions in the estimate history; it makes no separate determination about events revealing conditions that existed at the earlier reporting date.

The historical method comparison uses the retained December 2022 AASB 108 and 116 compilations and September 2023 AASB 112 compilation. It does not establish a later amendment chain or actual-company applicability. Assume a private, for-profit Tier 1 entity with no publicly traded ordinary or potential ordinary shares, no filing for public issuance and no voluntary EPS presentation. Those facts place this illustration outside the inspected AASB 133.2–3 scope (Australian Accounting Standards Board, 2021b). A public or voluntarily presenting entity needs the EPS restatement as well; the financing example's share schedule cannot simply be borrowed for this company.

New information changes the remaining allocation

Use equal monthly depreciation. In the estimate history the first three months of 2025 still consume $10 \times 3/12 = 2.5$, leaving 97.5 on 1 April. There are 45 months from April 2025 through December 2028. The remaining carrying amount, rather than original cost or a reconstructed earlier estimate, is allocated across those months:

$$d_{\text{Apr–Dec 2025}} = \frac{97.5}{45} \times 9 = 19.5, \qquad d_{2025} = 2.5 + 19.5 = 22.$$

Year-end PPE is $100 - 22 = 78$. Each full year from 2026 through 2028 has depreciation $97.5 \times 12/45 = 26$; the future allocations $19.5 + 26 + 26 + 26 = 97.5$ exhaust the April balance. The reported 2023 and 2024 depreciation remains 10 in each year. Revising those amounts to 20 would turn later information into a fact supposedly known earlier.

The estimate disclosure states the new restriction, the revised remaining life and the affected amounts under AASB 108.39–40. Relative to continuing the old schedule, 2025 depreciation increases by 12. Depreciation in each of 2026–2028 increases from 10 to 26; after retirement it is zero instead of the old schedule's 10 in each of 2029–2034. These changes redistribute the same total depreciable cost. They are consequences of the supplied retirement date, not forecasts inferred from the single reported balance.

An old error changes comparative profit and opening equity

In the error history, correct annual depreciation has always been $120/6 = 20$. Assume the error is material and all period-specific and cumulative effects are practicably determinable. The 2025 report presents 2024 as its comparative year. AASB 108.42 requires correction of that comparative and of the opening balances for errors before its start; paragraph 46 excludes the prior-period correction from 2025 profit. We therefore need two linked amounts: the error accumulated before 1 January 2024 and the additional error during 2024.

Tax makes the equity effect smaller than the PPE correction. Supply a conditional tax law that permits deductions of 20 annually in 2023–2028, with an enacted rate of 25% throughout the relevant recovery periods. The asset is recovered through use, and initial book and tax bases are both 120. There are no other differences or uncertain positions. Each year the company earns

cash operating income of 50 before depreciation; current tax is paid on the following 31 March. These are authored tax assumptions, not statements of Australian tax law. AASB 112.7, 15–17, 46–47 and 51 support the tax-base and rate method; paragraph 53 prohibits discounting the deferred tax (Australian Accounting Standards Board, 2023b).

The originally reported 2023 PPE of 110 exceeded tax base 100 by 10, giving a deferred tax liability (DTL) of 2.5. In 2024, reported PPE 100 exceeded tax base 80 by 20, giving DTL 5. Correct PPE is instead 100 and 80 respectively, equal to the tax bases, so those DTL balances disappear. Original pretax profit of 40 carried current tax 7.5 and deferred expense 2.5, leaving profit 30. Correct pretax profit of 30 carries current tax 7.5 and no deferred expense, leaving profit 22.5 in each year. Cash tax does not change: the tax deductions were correct throughout.

At the earliest comparative opening, 1 January 2024, reduce PPE by 10, DTL by 2.5 and retained earnings by 7.5. During comparative 2024, the extra depreciation of 10 and removal of deferred expense 2.5 reduce that year's profit by another 7.5. Thus the cumulative 1 January 2025 adjustment is PPE minus 20, DTL minus 5 and retained earnings minus 15. The opening and comparative movements sum to the same cumulative correction; they are not two separate postings of it.

To see the balance-sheet consequence, start with capital and cash of 200, then pay 120 for the asset at acquisition. The following figures retain the original publication alongside its corrected comparative. Initial capital remains 200; all later equity changes here are retained profit.

	2024 original	2024 restated	2025 error	2025 estimate
Depreciation	10	20	20	22
Profit after tax	30	22.5	22.5	21
Cash	172.5	172.5	215	215
Net PPE	100	80	60	78
Current tax payable	7.5	7.5	7.5	7.5
DTL	5	0	0	4.5
Total equity	260	245	267.5	281

The estimate column starts from the valid original equity of 260. Its 2025 pretax profit is $50 - 22 = 28$. Tax base is 60, so PPE 78 gives DTL 4.5, a decrease of 0.5 from the opening DTL. Current tax 7.5 less that deferred benefit gives tax expense 7 and profit 21. Compared with the old 2025 schedule, the estimate reduces pretax profit by 12 and after-tax profit by 9. The error column instead starts from corrected equity of 245 and adds current profit 22.5. Equal cash of 215 therefore does not establish equal earnings or equal equity.

The discovery entry joins the books to the restatement

The system discovers the mistake in April; it does not rewrite the time at which it learned the fact. Its correction records 1 April as the posting and knowledge date while attributing the prior-period amount to opening equity. Debit retained earnings 15, debit DTL 5 and credit accumulated depreciation 20. AASB 112.61A(b) and 62A(a) place the associated tax correction directly in equity with the prior error, rather than crediting current tax expense (Australian Accounting Standards Board, 2023b).

There is also a current-quarter error. Before discovery the books have already charged depreciation 2.5 and deferred tax expense 0.625 for January–March. Correct depreciation for those three months is 5. Debit current depreciation expense 2.5 and credit accumulated depreciation 2.5; debit DTL 0.625 and credit current deferred tax expense 0.625. That second tax entry reverses the current quarter's erroneous DTL increase. It has a different origin from the prior-period DTL

correction of 5. After both corrections, April PPE is 75 and DTL is zero. Depreciation 15 over the remaining nine months leaves PPE 60; full-year current depreciation is $2.5 + 2.5 + 15 = 20$, not 40.

The comparative is a presentation of the corrected earlier periods, not an instruction to post the correction a second time. A correction identity binds the entity, affected periods, original close, accepted treatment and target restatement. An exact retry returns the original receipt before testing later staleness. A changed payload under that identity is a conflict; a new request identifier cannot post the same target correction again. Original publications remain retrievable and explicitly identified as containing the error. These are proposed semantic compositions of the existing time, dependency and commit rules, not new portable request encodings or evidence of a running accounting engine.

The cash bridge provides a useful independent check. Cash after acquisition is 80; 2023 income increases it to 130. Subsequent annual income of 50, less payment of the previous year's tax 7.5, gives 172.5 and 215. Under the case's operating-cash convention, corrected 2025 profit 22.5 plus noncash depreciation 20 gives 42.5. The estimate history gives the same cash increase from profit 21 plus depreciation 22 less the deferred-tax benefit 0.5. Neither the opening-equity correction nor the comparative restatement creates cash.

When the classification cannot yet be made

Missing original-life documentation is not proof that management made an error, and finding a later shorter-life estimate is not proof that the old estimate was reasonable. The unresolved case retains both questions until evidence distinguishes them. A change of measurement basis requires the policy-change procedure. Paragraph 35's instruction for difficulty distinguishing a policy change from an estimate change does not resolve an error-versus-estimate ambiguity.

Nor does a missing local file establish impracticability. Paragraphs 43–45 distinguish inability to determine period-specific effects from inability to determine the cumulative effect; paragraphs 50–53 require the relevant effort and prohibit hindsight. Such a case needs its own supported starting date and correction route. The fully practicable example above cannot supply them. Unknown materiality also leaves the material-restatement route unestablished; it does not authorize ignoring a known error.

The error disclosure identifies the transcription mistake, comparative depreciation and tax/profit effects, the affected closing balance-sheet items and the earliest opening adjustment under paragraph 49. Cash and current tax are unchanged. If EPS is required, its effect must be added using that entity's actual numerator and share history. Missing tax law, original availability or accepted correction treatment blocks the related net-equity or posting claim. Balanced entries alone cannot supply those facts.

The year is the same; the report is different

An analyst asking for 2024 profit can legitimately encounter both 30 and 22.5. The first is what the company originally published; the second is the corrected comparative in its later report. The useful-life error explains the difference, but does not decide which number belongs in an analysis made at a particular date. Each answer needs the identity of the report and the time at which that report was available.

Keep the same synthetic asset, income and tax assumptions. To make the information dates explicit, suppose the original 2024 report was authorized, published and entered into the system on 31 March 2025. Management discovers the error on 1 April and accepts its correction on

2 April. The 2025 annual report is authorized on 31 March 2026, published on 2 April and entered into the system on 3 April. These publication and ingestion dates are authored for the example. They are not observations about a real issuer.

A latest-available query imposes two cutoffs. The public cutoff limits the reports an outside analyst could obtain; the system cutoff limits the reports actually recorded in the system. With a public cutoff of 1 April 2026, the answer for 2024 still comes from the original report, even if the system is queried much later. With a public cutoff of 2 April and a system cutoff of 2 April, the new report is public but has not yet entered the system, so the same original report remains the answer. Once both cutoffs reach its respective dates, the 2025 edition supplies corrected 2024 profit of 22.5. An explicit request for the original edition can still retrieve 30 afterward. Discovery of the error does not backdate publication of its correction.

Now reconstruct the positions shown in that later report. For this separate historical source comparison, IAS 8.42–49 and IAS 1.40A–D in the retained EU 2023/1803 text support the particular retrospective steps described here (European Commission, 2023a). The corresponding IAS 12, IAS 16 and IAS 33 provisions were checked for this example's tax, depreciation and private-entity EPS assumptions. That comparison does not establish equivalence of entire standards or the law applicable to a later actual filing.

The current reporting date is 31 December 2025 and the comparative date is 31 December 2024. Because the error also has a material effect on the opening position of the comparative period, IAS 1.40A–B requires a third statement at 1 January 2024. Materiality at that opening is a separate finding; the mere existence of a restatement does not establish it. Starting with the corrected 2023 close gives the following three positions, in currency units:

	1 Jan. 2024	31 Dec. 2024	31 Dec. 2025
Cash	130	172.5	215
Net PPE	100	80	60
Current tax payable	7.5	7.5	7.5
Deferred tax liability	0	0	0
Equity	222.5	245	267.5

Each column balances. From the earliest opening, annual profit of 22.5 raises equity first to 245 and then to 267.5. Cash rises by 42.5 each year because receipts of 50 exceed payment of the preceding year's tax by that amount. Depreciation of 20 explains why this cash increase exceeds profit. Correcting earlier depreciation and deferred tax changes neither those receipts nor the fiscal deductions.

The reconciliation to originally reported equity must preserve where each part of the error arose. Original equity at the start of 2025 was 260. The correction before 2024 reduces it by 7.5; the correction to 2024 profit reduces it by another 7.5. Only then is current profit added:

$$\underbrace{260}_{\text{original opening 2025}} - \underbrace{7.5}_{\text{error before 2024}} - \underbrace{7.5}_{\text{error during 2024}} + \underbrace{22.5}_{\text{2025 profit}} = 267.5.$$

The two corrections together are the opening-equity adjustment of 15 already explained above. Subtracting that 15 again would count the same error twice. The report presents the corrected history; requesting it does not create another journal or charge the prior error to 2025 profit.

The earliest opening also has a precise informational limit. Although it equals the corrected 2023 closing position, it is not a full restated 2023 income statement in the 2025 edition. A query for the latest published full-year 2023 column therefore still selects the 2024 edition, which contains that earlier comparative. The return value must identify it as original, even though later evidence establishes an error. A request for a reconstructed 2023 income statement

would ask a different question. Keeping the period, edition and information cutoffs together prevents the system from silently substituting the answer to that other question.

35.5 Apple: asset balances do not identify asset movements

The preceding methods need receipts, lots, useful lives and settlement dates. Apple's disclosures supply some of the balances those events must explain, but they do not supply the events themselves. Its FIFO and straight-line policies narrow the permitted cost calculations without identifying the actual inventory lots or asset cohorts (Apple Inc., 2024a, Note 1, printed p. 34). Comparing the notes with the cash-flow statement shows why filling those missing inputs by subtracting balances can produce a wrong history.

35.5.1 An inventory adjustment is not purchases or payments

Between 28 September and 28 December 2024, components fall from 3,627 to 2,792 million dollars while finished goods rise from 3,659 to 4,119 million. Total inventory falls from 7,286 to 6,911: the component changes, −835 and +460, sum to −375 million (Apple Inc., 2025c, Note 5, printed p. 9). Those two classes do not reveal quantities, unit costs or the transfers and sales that produced the balances.

The quarter's cash-flow statement reports +215 million for inventories within its reconciliation of net income to operating cash (Apple Inc., 2025c, printed p. 5). That is an indirect reconciliation adjustment, not total cash paid to suppliers. Even replacing it with minus the stock change would give +375, a different quantity. The difference $215 - 375 = -160$ million remains unexplained by these selected disclosures. In fiscal 2024, inventory rose from 6,331 to 7,286, while the cash-flow adjustment was −1,046; relative to the −955 stock-change shortcut the difference is −91 million (Apple Inc., 2024a, printed pp. 31 and 33). Neither residual is an identified currency effect, write-down or other journal entry. Production, consumption, ownership, valuation and payment histories must be established before they can explain the differences.

35.5.2 Testing a proposed fixed-asset history

The fixed-asset notes distinguish three measurements. The table shows gross property, plant and equipment (PP&E), its accumulated depreciation and the remaining carrying amount, all in millions of dollars (Apple Inc., 2024a, Note 5, printed p. 39) (Apple Inc., 2025c, Note 5, printed p. 9).

Amount	30 Sep 2023	28 Sep 2024	28 Dec 2024
Gross PP&E	114,599	119,128	120,615
Accumulated depreciation	70,884	73,448	74,546
Net PP&E	43,715	45,680	46,069

Accumulated depreciation is shown here as a positive deduction: gross less that deduction equals net. During fiscal 2024, gross cost increases 4,529, accumulated depreciation increases 2,564 and net PP&E increases 1,965. Yet the note reports PP&E depreciation expense of 8.2 billion, and the cash-flow statement reports capital payments of 9,447 million. Its broader depreciation-and-amortisation adjustment is 11,445 million (Apple Inc., 2024a, printed pp. 33 and 39). These amounts differ in meaning; the 8.2 billion also has coarser displayed precision.

The change in accumulated depreciation cannot be substituted for the year's depreciation charge.

Consider a deliberately limited asset book to test a tempting reconstruction. Let G_0, G_1 be opening and closing gross cost and B_0, B_1 accumulated depreciation, recorded as positive amounts. Let A be gross additions, R the original cost of retired assets, d the depreciation charge and S the accumulated depreciation removed with those assets. Assume there are no other movements and all depreciation is expensed. For this illustration, use the displayed amounts as exact, including $d = 8{,}200$ million. The two balances must then satisfy

$$G_1 = G_0 + A - R, \qquad B_1 = B_0 + d - S, \qquad 0 \le S \le R. \tag{35.1}$$

The inequality requires a nonnegative carrying amount for the retired assets in this cost model. If additions are equated with the 9,447 of cash payments, the gross balance requires $R = 9{,}447 - 4{,}529 = 4{,}918$. The depreciation balance requires $S = 8{,}200 - 2{,}564 = 5{,}636$. The proposed book removes more accumulated depreciation than original cost, by 718 million. It is infeasible under these combined assumptions. That finding rejects the proposed history; it does not establish an error in Apple's filing or identify which assumption must change.

The result is not sensitive to small rounding differences under a specified rounding model. If the million-dollar balances and payment are rounded to the nearest million, and 8.2 billion to the nearest 0.1 billion, the excess $d - 9{,}447 + (45{,}680 - 43{,}715)$ lies in the closed outer envelope $[666.5, 769.5]$ million. Its lower endpoint is $8{,}150 - 9{,}447.5 + 45{,}679.5 - 43{,}715.5 = 666.5$. This is a conditional precision check, not a verified issuer rounding rule; the exact tie convention remains unknown.

Allowing additions to be partly unpaid provides two simple repairs. Retain the same selected opening and closing balances, 8,200 depreciation and 9,447 cash payments. Assume retirement at the start of the year with zero proceeds, depreciation of surviving assets, and new acquisitions at year end. There are no opening capital payables, taxes or other asset movements. The figures below are constructed histories, not estimates of actual Apple transactions.

Annual movement (USD million)	History 1	History 2
Gross additions	10,529	11,529
Gross cost retired	6,000	7,000
Accumulated depreciation removed	5,636	5,636
Retired carrying amount, expensed	364	1,364
Cash paid for additions	9,447	9,447
Additions remaining payable	1,082	2,082

Both histories satisfy equation (35.1); their gross additions can also be allocated to match the three reported asset classes. For History 1, acquisition debits PP&E cost 10,529 and credits cash 9,447 and a capital payable 1,082. Depreciation debits expense and credits accumulated depreciation by 8,200. Retirement debits accumulated depreciation 5,636 and loss 364, while crediting gross PP&E cost 6,000. Net PP&E rises 1,965 and cash falls 9,447, so total assets fall 7,482. Liabilities rise 1,082 and equity falls 8,564 through the two expenses: $1{,}082 - 8{,}564 = -7{,}482$. History 2 has the same asset change but liabilities rise 2,082 and equity falls 9,564. These are coherent subsystem consequences, not a fit to all of Apple's statements or an exhaustive family of real asset histories. Neither history receives a probability or preference from this arithmetic.

The following quarter supplies a new snapshot, not the missing annual movements. Its gross, accumulated and net changes are 1,487, 1,098 and 389 million, whereas capital payments are

2,940 and broad depreciation/amortisation is 3,080 (Apple Inc., 2025c, printed pp. 5 and 9). The inspected interim note does not separately identify PP&E depreciation expense. The lease note adds a related distinction: finance-lease right-of-use assets are included in PP&E, while operating-lease right-of-use assets are in other noncurrent assets. The 1.0 billion of noncash right-of-use acquisitions covers the disclosed lease population (Apple Inc., 2024a, Note 8, printed p. 42); adding it all to PP&E or to cash capital expenditure would erase that distinction.

The method therefore preserves reported balances, cash payments, expense and unresolved movement components separately. A future cost or funding query still needs lot and asset ages, useful lives, residual values, impairment and settlement facts. The real filing has supplied constraints and ruled out one simple history, while leaving several different explanations open.

35.5.3 Local alternatives need different company histories

The two asset books expose a constraint on combining explanations. Both fit the selected PP&E movements and capital payments. Can they therefore be inserted interchangeably into one otherwise unchanged history of Apple? The journals above answer no. Relative to History 1, History 2 retires an additional 1,000 of gross cost without removing additional accumulated depreciation, so it recognises an additional 1,000 loss. It acquires an additional 1,000 of gross cost on credit. Those changes cancel in net PP&E and leave cash unchanged, but they add 1,000 to liabilities and subtract 1,000 from profit and equity. Amounts remain millions of dollars, under the same conditional asset-book assumptions, including exact illustrative depreciation of 8,200 and no tax effect within that subsystem.

Let P_i denote the vector of statement effects generated by asset History i, and let R_i denote the effects of the rest of the company's history. For this comparison, order each vector as asset change, liability change, profit, equity change and cash change. Both full histories must explain the same reported vector Y. If an admissible remainder exists for each branch, their necessary relation is

$$P_1 + R_1 = Y = P_2 + R_2, \qquad R_2 - R_1 = -(P_2 - P_1) = (0, -1{,}000, +1{,}000, +1{,}000, 0).$$

Thus the rest of History 2 must contribute 1,000 less liability growth and 1,000 more profit and equity growth, with no additional asset or cash change. Holding the remainder fixed is wrong relative to the common-statement target. The balance-sheet identity alone fails to detect the error: adding 1,000 to liabilities and subtracting 1,000 from equity still leaves their sum unchanged.

The reported annual totals make the required differences concrete. Apple's asset growth is 12,397, liability growth is 17,593, profit is 93,736 and equity change is −5,196 (Apple Inc., 2024a, printed pp. 29–32). Since History 1 contributes liability growth of 1,082, profit of −8,564 and equity change of −8,564, its hypothetical remainder must contribute 16,511, 102,300 and 3,368 respectively. History 2 instead requires 15,511, 103,300 and 4,368. In both cases the remaining asset change must be 19,879. These are necessary subtraction results. They do not supply the transactions, expense classifications, taxes or liability assignments that would make a remainder admissible.

The distinction matters for cash as well. In the isolated asset book, depreciation and the retirement loss are noncash expenses. Adding both back to its negative profit gives zero operating cash; acquisitions use 9,447 of investing cash. Switching branches therefore adds 1,000 to the noncash loss and 1,000 to the corresponding addback, leaving subsystem cash unchanged. If that loss were mapped into the annual cash-flow statement's broad Other adjustment of −2,266, the remaining contributions to that row would have to be −2,630 in History 1 and −3,630 in History 2 (Apple Inc., 2024a, printed p. 33). That mapping is itself conditional:

the source does not identify either retirement loss or the composition needed to offset it. A matching operating-cash total cannot substitute for the reported adjustment rows.

Nor may a remainder erase the cash-population difference from Section 28.10. After the common asset payments, the required remaining balance-sheet cash change is 9,425, whereas the required remaining cash-flow change is 8,653. Their 772 difference is inherited from the disclosed opening populations. Calling it another cash receipt would manufacture a transaction rather than explain the record.

A proposed completion must now identify the liability class containing the capital payable, the expense class containing the loss, the tax consequences and the noncash cash-flow mapping, and satisfy the detailed statement rows under the same assignments. It must also retain the asset cohorts and payment terms needed at the next date. The December snapshot does not establish that either hypothetical annual payable remained outstanding or identify when it was settled. Failure of the unchanged-remainder construction therefore calls for a coupled historical model. It rejects that construction without deciding whether either local asset book has an admissible full-company completion.

35.5.4 An offset must explain the individual rows

The required offset can be tested without pretending that its transaction has been observed. Suppose a separate economic obligation would justify an expense and an unpaid liability of q in History 1 and $q - 1{,}000$ in History 2, with $q \geq 1{,}000$. These are alternative amounts in two histories; they are not a release booked within either history. For a numerical example, take $q = 2{,}000$: each branch debits expense and credits its own liability, for 2,000 or 1,000 respectively. The difference supplies the 1,000 lower liability and 1,000 higher profit required of History 2's remainder. The economic obligation is an additional premise. Subtracting reported totals has not established that it exists.

Even conditional on that premise, a matched total is insufficient. Assume both the retirement loss and the new expense enter product cost, and both the capital payable and the new liability enter the same non-tax current liability category. History 2's extra loss then offsets its smaller expense within that expense row, and its larger capital payable offsets its smaller expense liability within that liability row. Other operations, distributions, share counts and OCI remain fixed in this comparison. Equal tax expense and tax balances are a further condition: equal pretax profit does not imply equal deductions, tax bases or deferred tax.

The indirect cash-flow statement exposes another restriction. If the expense liability is treated as an ordinary operating accrual, its 1,000 smaller increase reduces the working-liability adjustment by 1,000. The extra retirement loss increases the noncash addback by 1,000. Total operating cash flow is unchanged, but the separately reported Other adjustment and Other current and non-current liabilities adjustment each change. That construction fails to reproduce Apple's detailed cash-flow statement, even though it balances and matches total operating cash (Apple Inc., 2024a, printed p. 33).

A different conditional reporting map identifies the expense liability's recognition as a separate noncash adjustment. Its amount must then be excluded from the working-liability adjustment to avoid adding it back twice. Let α be the fraction of this hypothetical recognition allocated to the separate noncash adjustment; the remaining $1-\alpha$ enters the operating-liability adjustment. This parameter describes an assumed allocation, not a probability or a freely selectable accounting policy. With the retirement loss also assigned to Other, the differences between History 2 and History 1 are

$$\Delta\text{Other} = 1{,}000 - \alpha\,1{,}000, \qquad \Delta\text{WorkingLiabilities} = -(1-\alpha)\,1{,}000, \qquad \Delta\text{CFO} = 0.$$

For $\alpha = 0$, the two rows differ by +1,000 and −1,000. For $\alpha = \frac{1}{2}$, they still differ by +500 and −500. Both row differences vanish only at $\alpha = 1$ within this construction. Including the full liability movement as well as the separate addback would instead understate the History 2 minus History 1 operating-cash difference by 1,000.

Under the common expense and liability mappings, equal taxes and the disjoint noncash mapping with $\alpha = 1$, all five annual statements are unchanged by the combined transformation. Fixed shares also preserve EPS; fixed OCI and distributions carry the profit equality into comprehensive income and the equity statement. This is a conditional invariance result. It assumes a feasible full-company History 1 exists and that the transformed events remain admissible. It neither constructs that base nor proves that either complete history is compatible with all notes. Assigning the two expenses to different rows, the liabilities to different categories, or different tax consequences immediately breaks the relevant equality.

The notes make these remaining conditions specific. Note 5 separates gross PP&E by asset class: the extra acquisition and retirement must cancel within each affected class to leave that table unchanged. Note 6 separates 26,601 of current income taxes payable from 51,703 of other current liabilities, and 9,254 of non-current income taxes payable from 36,634 of other non-current liabilities. A total-liability offset cannot choose its own location inside those amounts. Note 8 identifies finance-lease right-of-use assets of 1,069 within PP&E and distinguishes lease liabilities and noncash lease acquisitions. Those disclosures do not identify the hypothetical capital purchases or expense liability. Nor do Note 12's purchase commitments and contingency assessment establish the assumed obligation or its amount (Apple Inc., 2024a, printed pp. 39, 42, 45–46).

The distinction therefore survives the repair: an algebraic offset is available; a row-preserving transformation is available under stated economic, tax and reporting assumptions; an actual Apple completion is unestablished. To admit one, the historical model needs the underlying obligation, gross asset classes, expense and liability assignments, tax bases and noncash treatment, together with a feasible rest of the company. The subsequent quarter additionally needs settlement dates and surviving asset cohorts. Identical annual statements do not make those future responses identical.

35.5.5 Equal liabilities can require different cash paths

The conditional repair leaves different obligations beneath the same annual liability total. In the $q = 2{,}000$ illustration, History 1 owes 1,082 for capital purchases and 2,000 for the separate expense; History 2 owes 2,082 and 1,000 respectively. Each owes 3,082. Consider what paying those balances would do, conditional on the economic and reporting assumptions needed for the preceding construction. Assume payment at carrying value, sufficient unrestricted cash, and no additional interest, foreign exchange, remeasurement or tax effect. Capital settlement is classified as investing cash and expense settlement as operating cash in this example. Neither the obligation nor its settlement terms have been identified in Apple's disclosures.

If both obligations are paid at the first subsequent date, each payment debits its liability and credits cash. No new expense arises: the expense or acquisition was recognised in the prior period. Both histories reduce cash and liabilities by 3,082, leave assets other than cash unchanged, and leave profit, OCI and equity unchanged. The indirect operating-cash calculation includes the expense liability's decrease once; it excludes capital settlement from the operating-liability adjustment. The cash-flow statement nevertheless distinguishes the histories. Writing each ordered pair as operating cash followed by investing cash, the joint alternatives are

$$(\mathrm{CFO}, \mathrm{CFI})_{H_1} = (-2{,}000, -1{,}082), \qquad (\mathrm{CFO}, \mathrm{CFI})_{H_2} = (-1{,}000, -2{,}082).$$

Both pairs sum to −3,082. Selecting the smaller outflow from each category would instead produce (−1,000, −1,082), whose total is only −2,082. No branch produces that pair. Selecting both larger outflows would invent an extra 1,000 payment. The category intervals can describe marginal possibilities, but their Cartesian product contains paths excluded by the shared history.

Timing makes the distinction relevant even to total cash. Suppose capital obligations are paid at the first date and expense obligations at the second. History 1 then pays 1,082 followed by 2,000; History 2 pays 2,082 followed by 1,000. Relative to History 1, History 2 has 1,000 less cash after the first date and the same cumulative cash after the second. Paying the expense obligations first reverses the first-date difference. An aggregate liability of 3,082 suffices for the narrow query of total undiscounted cash after both payments, under these fixed assumptions. It does not suffice for cash by date or by statement category. Retaining liability kind, amount, settlement terms and joint branch identity is therefore necessary for those queries.

The actual following quarter constrains a proposed continuation without identifying it. Apple reports PP&E payments of 2,940, operating cash of 29,935, investing cash of 9,792 and total cash growth of 356 in the quarter ended 28 December 2024 (Apple Inc., 2025c, printed p. 5). If both hypothetical obligations settled in that quarter, the remaining PP&E payments would have to be 1,858 under History 1 or 858 under History 2. The rest of the company would have to contribute operating cash of 31,935 or 30,935, and investing cash of 10,874 or 11,874, respectively. After unchanged financing cash of −39,371, either required remainder has cash growth of 3,438; subtracting the hypothetical 3,082 settlement recovers the reported 356. These are necessary conditions on unconstructed remainders. Positive remaining payments and a matching cash identity do not establish the underlying transactions or prove that either continuation fits the complete quarterly record.

This calculation extends the conditional history into explicit journals and statement effects while keeping actual-history admission unresolved. If cash is insufficient, the model must retain the resulting funding or default question; it cannot invent a loan to preserve the path. A later depreciation or production query requires different information again: asset cohorts, remaining lives, residual values, capacity and tax bases. The settlement example supplies none of those quantities. A change in the query must therefore reopen the state-sufficiency check, even when the annual statements and the final cash total coincide.

35.6 Lease and tax integration

A lease connects asset use, a financing obligation and income tax. The following schedule extends the operating-company example by separating these three measurements. The retained lease and tax dossiers supply the dated provisions, modification case and assumptions. The official lease compilation is dated December 2023. The right-of-use (ROU) asset represents the admitted use of the leased asset. Its cost, the lease liability, accumulated depreciation, remaining payments, rate version and option judgments are separate state fields. Tax retains carrying amounts, legal tax bases, reversal dates, enacted/substantively-enacted rate evidence, recovery support, and original profit/OCI/equity allocation. The deferred tax asset (DTA) and deferred tax liability (DTL) are retained gross before any permitted offset presentation. A liability schedule is not a tax-base schedule.

For the two annual 121 payments at 10%, initial liability and ROU are 210. Year-one interest is 21, liability after payment 110, depreciation 105 and ROU 105. Year two interest is 11, and both balances finish at zero. A same-asset modification just after year one creates two future payments 144 at revised 20%, PV 220, so liability increases 110 and ROU increases 110 to 215. A payment index reset has different rate rules; it cannot reuse that modification blindly.

To demonstrate connection to company statements, assume a synthetic entity with opening cash/equity 500, no other opening balances, and annual cash service revenue 400 less cash operating costs 100, all recognized in their years. There are no dividends, other assets, debt, losses or permanent differences. Add the base lease above. Assumed tax law deducts cash lease payments against taxable income, attributes those deductions to the lease liability, gives zero tax bases to ROU/liability, and taxes profits at enacted 25%. Sufficient usable profits and all recognition conditions for DTA are assumed. Tax is paid the following year. No current-period tax payment is invented from tax expense.

At commencement debit DTA 52.5 and credit DTL 52.5 in addition to the lease recognition. At year-one close DTA is 27.5 and DTL 26.25. The DTA reduction 25 is deferred expense; DTL reduction 26.25 is deferred benefit, net benefit 1.25. Taxable profit is 300−121=179; current tax/payable 44.75. Pretax book profit is 300−105−21=174, total tax 43.5 and profit 130.5. At year two, deferred expense 1.25 combines with current tax 44.75 to give total tax 46 on pretax profit 184.

Closing balance	2024	2025
Cash	679	813.25
ROU asset	105	0
Deferred tax asset	27.5	0
Lease liability	110	0
Deferred tax liability	26.25	0
Current tax payable	44.75	44.75
Equity	630.5	768.5

2024 assets 811.5 equal liabilities 181 plus equity 630.5; 2025 assets 813.25 equal liabilities 44.75 plus equity 768.5. For this case, elect financing classification for lease interest under the retained AASB 107 regime. The operating and financing cash totals below depend on that presentation election. Direct operating cash is 300 then 255.25 after the first tax payment; financing lease cash is −121 each year; no investing cash arises. Indirect operating cash in year one is profit 130.5 plus depreciation 105 plus interest 21 minus deferred benefit 1.25 plus unpaid current tax 44.75 = 300. Year two gives profit 138 +105 +11 +1.25 =255.25. The second tax payable settles next year. Zero OCI and no owner transactions mean the equity change equals cumulative profit. This case tests gross tax, cash timing and financial-statement coherence, not a real entity's tax law.

35.7 Apple: recognising tax and settling the obligation

The lease example separates a tax charge from the payment that follows it. Apple's fiscal 2024 disclosures show why that separation matters when several tax jurisdictions, deferred balances and a large disputed obligation enter the same accounts. Its annual income-tax expense was 29,749 million dollars; net cash paid for income taxes was 26,102 million (Apple Inc., 2024a, printed pp. 29 and 33). The 3,647 difference is not, by itself, an increase in tax payable. Deferred tax and other balance movements must first be distinguished from current tax and settlement.

35.7.1 The expense and balance reconciliations answer different questions

Note 7 divides the 2024 expense between current and deferred amounts for federal, state and foreign taxes. In millions of dollars, the reported components are as follows (Apple Inc., 2024a, Note 7, printed p. 40). Parentheses in the source represent negative expense, or a tax benefit.

Tax category	Current	Deferred	Total
Federal	5,571	−3,080	2,491
State	1,726	−298	1,428
Foreign	25,483	347	25,830
Sum	32,780	−3,031	29,749

The current charge is therefore 32,780 and the deferred benefit 3,031. The tax payable has its own scope. Note 6 reports current income taxes payable of 26,601 and noncurrent income taxes payable of 9,254 at 28 September 2024, against 8,819 and 15,457 a year earlier. Combining both maturity classes gives 35,855 against 24,276, an increase of 11,579 (Apple Inc., 2024a, Note 6, printed p. 39). Current expense less net cash taxes is only $32{,}780 - 26{,}102 = 6{,}678$. Reclassifying an amount between current and noncurrent cannot explain the difference in their combined total.

To expose the missing movements, let T denote that combined tax payable and N the net deferred tax asset reported in Note 7. Let C be current tax expense, D deferred tax expense (negative for a benefit), and P net cash taxes paid, all for the same annual period and in millions of dollars. Define U and V as the residuals needed to connect the reported amounts:

$$\Delta T = C - P + U, \qquad \Delta N = -D + V. \tag{35.2}$$

These are diagnostic definitions, not assertions that every tax balance has the same recognition or settlement scope. The first relation gives $U = 11{,}579 - 6{,}678 = 4{,}901$. Setting that residual to zero would produce the wrong closing payable. Assigning it to currency translation, uncertain tax positions or another journal would require the missing transaction and classification evidence.

The deferred balance illustrates the second relation. At September 2024, gross deferred tax assets of 34,873 less a valuation allowance of 8,866 leave 26,007. Deducting deferred tax liabilities of 6,805 gives net deferred tax assets of 19,202, compared with 17,251 a year earlier (Apple Inc., 2024a, Note 7, printed p. 41). The increase is 1,951, so $V = 1{,}951 - 3{,}031 = -1{,}080$. The deferred benefit in profit does not explain the whole net-balance movement. Its allocation among currency, equity, other comprehensive income and other causes remains unresolved here.

Note 6's 19,499 deferred-tax-asset presentation is also different from Note 7's 19,202 net position. Their difference of 297 is an arithmetic consequence, not a separately disclosed liability line or authority to offset jurisdictions. The reported temporary differences—including research expenditure, deferred revenue, leases and depreciation—need their own tax bases and recovery conditions. The note reports indefinitely available Irish foreign-tax and California R&D credit carryforwards, but also a valuation allowance for those credits and some other differences. An indefinite term does not establish that an asset is recoverable (Apple Inc., 2024a, Notes 6–7, printed pp. 39–41).

35.7.2 A recognised charge, restricted assets and a later payment

Apple attributes its large 2024 State Aid charge to the September 2024 court decision confirming the European Commission's recovery decision. This is the issuer's account of the dispute and its accounting consequence. Note 7 describes a 15.8 billion dollar payable to Ireland, partly offset by a 4.8 billion US foreign tax credit and an 823 million decrease in unrecognised tax benefits (Apple Inc., 2024a, Note 7, printed pp. 39–40).

The narrative calls the net charge 10.2 billion; the statutory-rate reconciliation gives the more precise amount 10,246 million. Subtracting the displayed narrative centres gives $15{,}800 - 4{,}800 - 823 = 10{,}177$, which is 69 million lower. That is compatible with the coarser inputs: if the two

billion-dollar amounts are rounded to the nearest 0.1 billion and 823 to the nearest million, the component calculation lies in the closed outer interval [10,076.5, 10,277.5] million, containing 10,246. The 10.2 billion charge is compatible with it as well. This is a conditional rounding check, with the issuer's precise rule and tie convention unverified; the centre difference does not establish a conflicting source assertion.

The tax charge does not make the settlement funds disappear at recognition. At 28 September 2024 the escrow comprised 2.6 billion dollars of cash and cash equivalents and 13.2 billion of current marketable securities, restricted from general use (Apple Inc., 2024a, Notes 4 and 7, printed pp. 37 and 39). The annual liquidity discussion separately identifies the obligation and the funds held to settle it. The following quarter reports both a payment and the remaining obligation (Apple Inc., 2024a, printed p. 26) (Apple Inc., 2025c, printed p. 16).

State Aid obligation and settlement	EUR billion	USD billion
Obligation at 28 Sep 2024	14.2	15.8
Payment during the following quarter	11.0	11.9
Obligation at 28 Dec 2024	3.3	3.4

The three rows are different dated quantities. In dollars, a movement equation with opening obligation 15.8, payment 11.9 and closing obligation 3.4 requires an additional net movement of $3.4 + 11.9 - 15.8 = -0.5$ billion. Under nearest rounding to 0.1 billion, its outer interval is $[-650, -350]$ million, excluding zero. The same calculation in euros gives $3.3 + 11.0 - 14.2 = 0.1$ billion, with outer interval $[-50, 250]$ million, which includes zero. Neither calculation supplies a common exchange rate or an exact exchange-gain entry. The interim tax discussion mentions foreign-currency revaluation effects related to the decision, but does not allocate this 500 million dollar residual (Apple Inc., 2025c, printed p. 16).

The remaining funds also retain their asset classifications. At December 2024, restricted cash and cash equivalents were 126 million dollars and restricted current marketable securities 3.3 billion (Apple Inc., 2025c, Note 4, printed p. 8). Their displayed sum, 3,426 million, is compatible with the rounded 3.4 billion obligation. It is not all cash available for ordinary operations. The opening 2.6 billion of restricted cash alone could not have funded the reported 11.9 billion payment; securities sales, maturities, transfers and valuations need their own histories.

For an isolated settlement illustration, suppose 11,900 million of restricted cash is available and the amount paid equals the liability's carrying value at settlement. Debit tax payable and credit restricted cash by 11,900. Assets and liabilities each fall by 11,900; equity and current-period tax expense do not change in this posting. This assumes the required cash conversion has already occurred and omits any associated gain, loss or fee. It explains settlement of an existing obligation without pretending to reconstruct Apple's funding and currency entries. The quarter separately reports tax expense of 6,254 and net cash taxes of 18,651 million (Apple Inc., 2025c, printed pp. 1 and 5); neither should be replaced with the Ireland payment or treated as another copy of the preceding year's charge.

35.7.3 Uncertain tax positions do not supply a payment distribution

The last part of Note 7 preserves another population: gross unrecognised tax benefits. Its 2024 movement is $19{,}454 + 1{,}727 - 386 + 2{,}542 - 1{,}070 - 229 = 22{,}038$ million. The increases concern prior- and current-year positions; the decreases concern changed prior-year positions, settlements and expired limitation periods. Interest and penalties are excluded from this rollforward (Apple Inc., 2024a, Note 7, printed p. 41). The 1,070 settlement-related decrease is a movement in that gross balance, not an independently identified total of cash paid to tax authorities.

Of the rounded 22.0 billion gross amount, the issuer says that 10.8 billion, if recognised, would affect its effective tax rate. It also describes a reasonably possible decrease of approximately 5–13 billion over the next twelve months, primarily concerning transfer pricing and deemed repatriation tax, while noting uncertainty about examination timing. These are different conditional statements. The range is neither a probability law nor a promised cash payment or expense release. A forecast still needs jurisdictional outcomes, recognition, interest, penalties and settlement dates. Retaining those unknowns keeps the tax note connected to the company's funding needs without turning management's dated assessment into an invented future transaction.

35.8 Returning to the company forecast

The Northstar case remains the broader ordinary-company teaching baseline: collections, purchases/payments, inventory, payroll, PPE, financing, tax, statements, annual/interim and revision views are already specified. Its 2024 collection/purchase/payment solution is 245/155/145. The 2025 forecast in the Northstar reference fixture opens with cash 74.6 and equity 90.6, spends 20 on capital expenditure at period end, pays dividend 8 and uses a base floating rate 7%. Its simplified tax is settled in the year earned. The running case uses fixed-rate share α; here floating-rate share $x = 1 - \alpha$ describes the same uncertainty. For 2025 the same unknown floating fraction $x \in [0, 1]$ determines every statement amount:

$$\begin{aligned} I(x) &= 5 + 2x, & \text{pretax profit}(x) &= 82 - 2x, \\ \text{tax}(x) &= 16.4 - 0.4x, & \text{profit}(x) &= 65.6 - 1.6x, \\ \text{operating cash}(x) &= 48.6 - 1.6x, & \text{closing cash}(x) &= 95.2 - 1.6x, \\ \text{equity}(x) &= 148.2 - 1.6x, & \text{assets}(x) &= 288.2 - 1.6x. \end{aligned}$$

Receivables 55, inventory 40, net PPE 98, payables 40 and debt 100 complete the balance sheet. Liabilities are 140 for every x, so the common cash/equity slope preserves the accounting identity. There is no default distribution on x. At a 9% floating rate the all-floating history has interest 9, profit 62.4, cash 92, equity 145 and assets 285. The annual cash bounds still cannot establish daily payment solvency without event-date cash flows.

The lease/tax example extends the interaction evidence without changing the original Northstar fixture or pretending that its teaching 20% tax formula is an IAS 12 method. In a real-company package, each required method is admitted separately and unsupported operations remain visible. The corporate methods in this chapter cover pensions, impairment, provisions, valuation and share-based payment; omission from a finite company case never proves the actual company has none.

35.9 Share-based compensation

Ordinary employee benefits and impairment have established how service and past measurements remain in state. Share awards add settlement choices and vesting conditions; the valuation and capitalization methods that follow address the inputs and costs those cases require. These extensions retain the historical EU source scope stated with the earlier foundations.

AwardState distinguishes equity versus cash settlement, grant/vesting/service dates, granted units, grant-date fair value, service/nonmarket/market conditions, expected units vesting, cumulative expense and changes/cancellations. IFRS 2.7–11, 15–23 supports the employee equity-settled method. With 100 units, grant value 3, two-year service condition and all

expected to vest, year-one cumulative expense 150 and equity 150. At vesting only 80 satisfy the service condition: cumulative 240, second-year expense 90. Debit employee expense and credit equity. Failure of a market condition is different under 21; if the employee satisfies other vesting conditions, it does not automatically reverse the grant-date measured service expense. Unknown conditions cannot be represented by a single forfeiture percentage without their type and evidence.

Cash-settled awards remeasure the liability under 30–33. Same 100 expected units, half service and current unit fair value 4 gives liability/expense 200 at year one; 80 actual units and value 5 at settlement gives total 400, second expense 200, then debit liability 400/credit cash 400. Reusing the equity grant value 3 would be wrong. Actual award valuation requires the specified model/market inputs, option rights, volatility, dividends and calibration; the fair values 3/4/5 are case inputs. Modifications and settlement-choice arrangements require their own conditions, with the original minimum expense and increment analyzed under the applicable provisions rather than overwriting the original grant.

35.10 Fair value and capitalization boundaries

A ValuationRequest names the unit of account, market/date, asset/liability characteristics, principal/accessible market, technique, input hierarchy, counterparty/own non-performance risk where relevant, calibration and sensitivity. IFRS 13.24, .42–43 and .67–80 guide the selected measurement. Quoted identical asset price 7 for 10 units gives 70 in a qualifying Level 1 case; a blockage discount based only on holding size cannot reduce it to 63 under 69/80. A present-value illustration 111 due next year at an independently supported 11% rate gives 100; that equality does not establish the rate, market comparability or accounting classification. Historical EIR, an impairment discount, risk-neutral pricing and a physical expected return answer different questions.

Borrowing-cost capitalization requires a qualifying asset, eligible costs, commencement/suspension/cessation dates and the specific/general-borrowing calculation, before moving interest into cost. IAS 23.8–25 supplies these rules in the retained historical EU text: for specific borrowing, eligible actual costs less temporary investment income; for general borrowing, the applicable weighted rate and cost ceiling. Commencement requires expenditures, borrowing costs and necessary preparation activities; extended inactive periods can suspend it, and substantial completion ends it. Assume specific borrowing 100 outstanding during a full eligible construction year at 5%, with an admitted expenditure schedule and temporary investment income 1 on temporarily unused funds during that period. Net capitalized cost is 4. Real admission still needs the dated standard, actual schedule and eligibility evidence. It is not a default 5% add-on to all assets. Internally developed intangibles likewise need research/development separation and all applicable recognition criteria; a forecast of future sales does not establish an asset.

The extension interface returns the stated measured components, proposed journal, economic delta, disclosures and unresolved inputs under the event-basis, authorization and commit contracts. Calculation success does not permit another operation to fill a missing authority/fact with a default. Real-company acquisition/tax/employee/valuation evidence and the full exceptions remain explicit internal preparation, with proposed answers and competing treatments retained for any later authorized domain review.

35.11 When a component reaches its next accounting event

A recognised balance also promises a subsequent method. A refund liability must eventually be settled; an impaired asset may recover; a modified lease must continue through payments and depreciation. The following methods specify those transitions under stated facts. They use the retained historical IFRS text, with current authority and actual-company measurement kept as separate admission questions (European Commission, 2023a).

35.11.1 Actual returns can differ from the estimate

Section 28.7 follows two expected and actual returns. More generally, let L be the opening refund liability, A the recovery asset, P the actual refund payment, K the actual selling-handling cost and V the returned inventory admitted after deterioration and its NRV ceiling. When all rights have closed, the debit-positive entries are

$$\Delta\text{cash} = -P - K, \qquad \Delta\text{inventory} = V, \tag{35.3}$$

$$\Delta\text{refundliability} = L, \qquad \Delta\text{recoveryasset} = -A, \tag{35.4}$$

$$\Delta\text{revenue} = P - L, \qquad \Delta\text{costofsales} = A - V, \tag{35.5}$$

$$\Delta\text{sellingexpense} = K. \tag{35.6}$$

The debit-positive revenue amount is negative when releasing an excessive refund estimate increases revenue. These entries sum to zero. The separation of selling cost follows the explicit handling classification and IAS 2.16; IFRS 15.B25 governs the recovery asset. If handling instead creates eligible inventory cost, that needs its own admitted classification.

With $L = 40$ and $A = 16$, one actual return pays refund 20 and handling 4, and recovers inventory 10. Revenue increases 20, cost of sales is 6 and selling expense 4. Three actual returns instead pay refund 60 and handling 12, recover inventory 30 and reduce revenue by 20; cost of sales is credited 14. If two returns recover inventory worth only 14, the inventory ceiling raises cost of sales to 2. Unknown handling, inventory NRV or an open return window prevents final settlement rather than supplying a zero.

35.11.2 A reversal ceiling does not supply an allocation weight

IAS 36.122–123 allocates a cash-generating unit's reversal to assets other than goodwill, proportionately to their carrying amounts, subject to the individual ceilings. Suppose assets A and B carry 40 and 60 and goodwill is 10. The unit's recoverable amount rises to 170, allowing a total increase of 60. A's individual ceiling is 45 and B's is 150. The first proportionate allocation would assign 24 to A and 36 to B. A can accept only 5; reallocating the remaining 19 to B gives increases of 5 and 55. Goodwill remains unchanged.

Writing w_i for the original carrying amounts and c_i for the remaining headroom, the method distributes a remaining reversal R among eligible assets by $Rw_i/\sum_j w_j$, caps each amount at c_i, then redistributes any remainder among assets still below their ceilings. This terminates when the reversal is exhausted or no headroom remains. It requires a positive sum of eligible weights whenever a positive amount remains.

If all eligible carrying amounts are zero, the denominator is zero. Neither equal weighting nor weighting by headroom follows from the inspected source. The specification therefore retains an unresolved allocation with unchanged books. A different rule applies to an individual asset: a zero carrying amount, no-impairment carrying value of 80 and recoverable amount of 120 permits an individual reversal to 80 under IAS 36.117–119. That individual calculation does not resolve the allocation among several zero-carrying assets.

35.11.3 A lease reduction must identify the rights surrendered

A single physical percentage is insufficient when leased rights have different values. Consider right A with right-of-use asset 90 and liability 100, and right B with asset 30 and liability 50. The supplied modification terminates B, releasing asset 30 and liability 50 for a gain of 20. It leaves A with one payment of 121 in a year at a revised 10% rate. A's revised liability is $121/1.1 = 110$; the increase of 10 adjusts its asset from 90 to 100. These are the scope-reduction and remeasurement steps in IFRS 16.45–46.

A further same-day reduction surrenders an admitted portion of A carrying asset 10 and liability 11. The remaining liability before remeasurement is 99. Revised consideration of 99 due in one year has present value 90, so the remeasurement reduces both liability and asset by 9. The final asset is $100 - 10 - 9 = 81$, liability 90 and additional gain 1. A year later, payment 99 contains interest 9 and principal 90; the supplied systematic depreciation amount of 81 exhausts the asset.

The event must therefore carry the actual surrendered asset and liability for each right, the revised payment schedule and the rate. The finite schedule uses annual-end payments and a checked one-year interval. Missing attribution, an unadmitted depreciation basis, a different calendar or a negative remaining asset requires a distinct finding or method. It cannot be repaired by applying the same physical fraction to every balance.

35.11.4 Contract costs survive more than one service period

IFRS 15.91–104 first distinguishes eligible contract costs from amounts governed by another standard. Acquisition costs must be incremental and recoverable. Fulfilment costs outside other standards must relate directly to the contract, create or enhance resources for future performance and be recoverable. Waste, past service and the specified administrative costs are expensed. The event records these findings before calculating the posting.

For example, expense waste 5 and capitalise eligible fulfilment cost 100 for five equal service units. Delivery of one unit amortises 20, leaving 80. After other applicable asset impairments have been recognised, credit-adjusted remaining consideration 70 less direct future cost 20 supports a contract-cost asset of 50: recognise impairment 30. Delivery of another unit amortises $50/4 = 12.5$, leaving 37.5. Without the impairment, the unamortised balance would now be 60. If circumstances improve enough to support 60, reversal is limited to $60 - 37.5 = 22.5$; the final three units amortise 60.

That ceiling needs a separate no-impairment balance, not merely the present asset. Subsequent CGU assessment under IFRS 15.103 remains required after the component test. A successful component calculation does not establish that the wider CGU has no impairment.

The lost asset and the obligation still to be met

Writing down a contract-cost asset does not necessarily recognize the whole loss on the contract. The asset represents expenditure already incurred. The company may still have to spend more to perform than the customer will pay. IAS 37.66–69 requires a provision for that remaining onerous obligation, after recognizing impairment of assets used in fulfilling the contract (European Commission, 2023a).

Suppose a company has cash 200, an eligible unamortized contract-cost asset 20 and capital 220. One service remains to be delivered at completion. The customer will pay 100; incremental future cash costs are 100, and an allocation of other directly related contract cash costs adds 30. The existing asset's amortization is separate from those future payments. There are no advances,

taxes, foreign exchange or reimbursements, and the time value of money is immaterial. Assume the required other-asset impairment tests have been completed and the later CGU assessment finds no additional loss. These are necessary supplied facts, not consequences of the calculation.

Future direct costs 130 exceed the remaining consideration 100, so none of the existing cost asset is recoverable. Recognize impairment 20 first. If the company fulfills the contract, a further cash loss of $130 - 100 = 30$ remains. Assume it can instead cancel by paying enforceable compensation 50, with no remaining benefit or obligation. The least unavoidable loss is then 30, giving a provision 30. The total immediate loss is $20 + 30 = 50$, and equity is $220 - 50 = 170$. Cash 200 less provision 30 confirms that amount.

The distinction also gives a compact calculation. Let A_0 be original unamortized contract cost, B the remaining collectible consideration, D the complete direct future cash cost, and K the available cancellation payment. Before any service is delivered, the asset after its test, the remaining fulfilment loss and the provision are

$$A' = \min\{A_0, \max(B - D, 0)\}, \tag{35.7}$$

$$F = \max\{D + A' - B, 0\}, \qquad P = \min\{F, K\}. \tag{35.8}$$

Here $A' = 0$, $F = 30$ and $P = 30$. The asset loss $A_0 - A'$ has already been recognized and does not enter the provision again. A provision of 50 obtained by adding the old asset 20 to the remaining cash loss 30 would charge the asset loss twice. Comparing the gross cost 130 with the cancellation payment 50 before subtracting revenue 100 would also be wrong: cancellation does not earn the performance receipt.

The cost population matters as much as the order. IAS 37.68A includes both incremental costs and allocations of costs directly related to fulfilling contracts. Ignoring the allocated 30 would incorrectly produce no provision. That paragraph includes depreciation among possible allocations; this example uses cash costs to keep the cash reconciliation visible. Unrelated general administration is excluded from these direct costs. If the company later pays 15 of such administration, it records a separate period expense and cannot use this contract's provision to absorb it.

Now complete the service, receive 100 and pay 130. Use provision 30 against the expenditure for which it was recognized: debit the provision 30 and current service expense 100, and credit cash 130. Recognize revenue and the cash receipt 100. There is no additional completion loss. Cash is $200 + 100 - 130 = 170$, the cost asset and provision are zero, and equity remains 170. This use of the provision follows IAS 37.61–62; it is not permission to release provisions against unrelated expenses. With the separate administration payment, final cash and equity would instead be 155.

A cheaper exit changes the amount recognized before management acts. If cancellation costs 20, the provision is 20 and total immediate loss is 40. Cancelling then pays 20 and leaves cash and equity 180. Management could still choose to fulfill for commercial reasons. In that history, performance creates an additional loss 10 and final cash is again 170. Measuring the minimum unavoidable loss does not choose the company's action. A freely cancellable contract has no onerous provision, although a worthless cost asset still requires its own write-down.

Estimates must be kept current. Suppose the original future cost 130 later falls to 90 before any service occurs. Consideration 100 now supports asset 10: reverse 10 of its impairment, within the original unamortized ceiling, and release the provision 30. The cumulative loss becomes 10. Completion then amortizes asset 10, pays 90 and earns 100, leaving cash and equity 210. IFRS 15.104 limits the asset reversal, while IAS 37.59 requires the separate provision review. Partial service, material discounting and further CGU losses would change this calculation and require their own dated measurements.

The same history discipline applies to repeated revenue modifications. Under IFRS 15.21(a), eight remaining distinct units with consideration 80, followed by four additional units for 60 that do not form a separate contract, leave 12 units and consideration 140. Delivery of three recognises 35. Adding three more units for 30 then leaves 12 units and consideration 135, recognised on their delivery. Earlier revenue is not reallocated.

For a single integrated obligation, IFRS 15.21(b) instead requires the cumulative effect. A contract priced at 200 and 25% complete has revenue 50. Changing price to 260 and admitted progress to 20% changes cumulative revenue to 52, a catch-up of 2. Later 50% progress yields 130. A further modification to price 300 and progress 40% reduces cumulative revenue to 120, a catch-up of negative 10; completion ultimately brings it to 300. Arbitrary mixtures of distinct and integrated promises require their own allocation, not a selection of whichever result is convenient.

35.11.5 A tax benefit retains the origin of the underlying item

Acquisition-related recognition depends on information as well as dates. Under IAS 12.68 and IFRS 3.45–46, a newly recognised acquired deferred tax benefit from acquisition-date information within the open measurement period reduces goodwill, with any excess recognised in profit. A benefit of 20 against goodwill 5 therefore reduces goodwill by 5 and gives tax income 15. The measurement period cannot exceed one year; elapsed time alone does not establish that the new information concerns acquisition-date facts.

Other benefits retain their independently identified origin. In particular, IAS 12.67 excludes an acquirer's own deferred tax benefit from acquisition accounting; it does not place every such benefit in profit. IAS 12.61A–63 still routes tax associated with OCI or direct-equity items to that origin. An acquirer-owned benefit of 20 with an OCI origin therefore adds DTA 20 and OCI 20, leaving goodwill and profit unchanged. Missing origin remains unresolved.

Mixed tax schedules retain one lot per entity, authority, temporary difference, recovery-period rate, recognition finding and origin. For a signed difference D_i, positive for deductible and negative for taxable, admitted rate r_i and supported deductible recognition fraction f_i, the signed deferred amount is $D_i r_i f_i$; taxable lots use $f_i = 1$. Changes go to the traced profit, OCI or equity account. A supplied schedule can thus move gross DTA from 30 to 19 and DTL from 20 to 30 while allocating expense 5 to profit, 10 to OCI and 6 to direct equity.

Gross asset and liability amounts stay available even when presentation is net. The finite offset route requires the same taxable entity and authority and an enforceable right to set off current tax under IAS 12.74. It does not implement the separate cross-entity simultaneous-settlement route. Neither a blended rate nor an assumed right of setoff can substitute for those findings.

35.12 Rule selection

The proposed selection procedure distinguishes operative authority from historical source support. The retained sources include official AASB compilations, the original EU incorporating regulation and official ASBE lease and income-tax texts (Australian Accounting Standards Board, 2022i,f, 2023c,b; European Commission, 2023a; Ministry of Finance of the People's Republic of China, 2019, 2018, 2006b, 2022). Historical US lease additions are separately identified in the retained amendment source (Financial Accounting Standards Board, 2016b).

AuthorityRequirement contains issuer/framework, jurisdiction, entity class, annual-period/application interval, paragraph and local source Ref, incorporating instrument, amendment chain, transition, elections and unresolved dependencies. RuleSelection binds those requirements to

the accounting event and evidence view. The algorithm evaluates chronological applicability first, then scope and facts, then available elections/transition, then measurement/presentation. If two supported rules can apply, retain the explicit alternative and required election; if one is contradicted by facts, mark ineligible. Missing current authority gives unresolved, never eligibility through a familiar rule name.

An unchanged historical report remains an assertion under its own basis and vintage. A harmonized comparison creates an adjustment book and reconciliation, with changed recognition/measurement explained. Label mapping alone cannot convert US GAAP to IFRS or ASBE. The four clocks and source version remain part of each rule dependency; later knowledge cannot leak into an original as-of view. Transition entries reconcile opening assets, liabilities, tax and equity, rather than changing old source bytes or plugging an unexplained amount.

35.13 IFRS and incorporating jurisdictions

The retained official AASB texts provide explicit IAS/IFRS incorporation for their stated entity/version scope. AASB 9/15 compilation dates are 31 December 2022 with the source covers' application window; AASB 16 compilation 8 is 31 December 2023, AASB 112 compilation 7 is 29 September 2023. AASB 102's older retained edition is selected historical support for inventory, not current 2024 applicability certification. The retained EU 2023/1803 original provides official EU-adopted IFRS 3/10/IAS 19/21/36/37/IFRS 2/13 passages inspected in this method investigation; its historical adoption scope and later amendments remain separate. The retained amendment investigation supplies additional source evidence, not a blanket acceptance of all rules in these standards.

Initial real-method admission is therefore claim-specific: the selected revenue, lease/tax, financing, credit, group/FX and insurance cases must resolve each named applicability prerequisite. For synthetic demonstrations, declare the chosen dated provisions and assumptions and use the synthetic book mode. No assertion of whole-IFRS compliance follows from those entries.

The base lease in the saved dossiers (two annual 121 payments, 10% rate, no qualifying exemptions or extra costs) has liability 210 initially, 110 after year one, 0 after year two. IFRS-equivalent cost-model ROU is 210/105/0; expenses 126 then 116. Income tax needs its own legal deductions and gross temporary differences, as the connected corporate case demonstrates. Revenue timing, inventory cost formula, financial instrument classification, control and contract boundaries are required rule predicates, not accounting options chosen after seeing which outcome fits best.

35.14 US GAAP proposed methods and authority limitation

The saved official FASB ASU 2016-02 Section A and later 2020-05/2021-09 amendments contain inspected historical ASC 842 additions. Their covers explicitly say an ASU is not authoritative. Current Codification access returned 403 again in this source investigation; the proposed methods cannot be marked current-authority-admitted. ASU amendments, historical FAS 109, issuer accounting policy and a model's memory are all insufficient substitutes for the operative ASC text and full applicable amendment chain.

The same lease facts include ten-year remaining equipment life, fair value 1000, alternative use to lessor, no ownership transfer, no purchase option, two-year term and PV 210. Under the inspected historical 842-10-25-2/25-3 provisions, the stated facts support the proposed operating classification. Historical 842-20-25-6 and 35-3 yield single lease expense 121 per year,

liability 210/110/0 and ROU 210/ 110/0. First year debit lease expense 121/credit liability 21/credit ROU 100; payment debits liability 121/credits cash 121. Next expense 121 decomposes into liability interest 11 and ROU reduction 110. The finance comparison uses the separate depreciation 105 plus interest 21/11 path. Changing classification facts is not a free election. The 75%/90% heuristics in the historical guidance cannot become universal requirements. This complete numerical contrast is useful despite the remaining current-authority limitation.

Proposed ASC 740 temporary-difference treatment retains tax base, reversal dates, enacted rates, valuation allowance and source-specific allocation/presentation. The saved historical FAS 109 and ASU 2019-12 provide inspected enactment support, not a completed current ASC 740 implementation. Under the matched PPE case, book 800/tax 600 at assumed 25% yields DTL 50 and current tax 75 on taxable 300. A future 30% rate that is merely substantively enacted but not legally enacted cannot be imported from the AASB 112 rule into this proposed US treatment; the declared 2025 closing temporary difference is 300, giving DTL 75 at the enacted 25% rate versus DTL 90 at 30%. This deferred-tax comparison is separate from current tax 75 on taxable profit 300. Actual legal enactment remains a separate jurisdictional fact.

The proposed CECL method for in-scope amortised-cost assets estimates lifetime expected cash shortfalls from origination using past/current/forecast evidence, with a documented reversion method beyond a reasonable/supportable forecast horizon and applicable instrument exceptions. It retains allowance roll-forwards, write-offs, recoveries, purchased-credit-deteriorated handling and off-balance commitments separately. It must not copy IFRS 9's 12-month/SICR staging to the same record. A candidate contrast: lifetime discounted loss 6 with 2 attributable to default events in the next year gives IFRS general Stage 1 candidate 2 versus CECL lifetime candidate 6, conditional on scope and method support. No current ASC 326 text was retrieved, and the attempted official amendment path returned 403; this contrast remains an explicitly unadmitted proposed reading, not a source-verified US requirement or posting example. Full current ASC 326, insurance 944, consolidation 810, FX 830 and relevant later amendments are specific internal source tasks before corresponding real admission.

35.15 ASBE lease adoption and deferred tax

For an ASBE lease, selecting the lease standard and selecting the related income-tax treatment require different application dates. Article 68 of the revised ASBE 21 states a general date of 1 January 2019, but the Ministry of Finance's 2018 issuance notice specifies the enterprise groups. Paragraph I requires application from that date for enterprises listed both domestically and overseas, and for overseas-listed enterprises preparing financial statements under IFRS or ASBE. Other enterprises applying ASBE must apply the revision from 1 January 2021 (Ministry of Finance of the People's Republic of China, 2018, issuance notice, para. I).

Paragraph II permits earlier application for enterprises whose parent or subsidiary is listed overseas and prepares its overseas financial statements under IFRS or ASBE. Their early date cannot precede simultaneous application of the specified 2017 financial-instruments and revenue standards. Paragraph III discontinues the 2006 lease standard and its application guidance when the enterprise applies the revision. Thus a 2020 reporting date alone does not establish use of the revised lease standard for an enterprise in the second group: its qualifying early adoption would need evidence. The retained official Chinese texts include the 2018 notice and attachment and the Ministry's later ASBE 21 page, published on 28 October 2019 (Ministry of Finance of the People's Republic of China, 2018, 2019).

Articles 4–8 require control of identified use, economic benefits and direction; article 6 includes both conditions for substantive substitution and the express presumption when difficult to

determine. Articles 14–18 specify commencement, ROU cost and PV, using implicit rate or incremental borrowing rate if the former cannot be determined. Articles 20–23 separate cost-model depreciation and periodic interest; articles 25–29 distinguish reassessment/index changes/modifications and their discount-rate rules. Articles 30–33 define the exemptions; a two-year term is not short-term. Articles 60–67 specify transition choices and disclosures; the chosen transition cannot be inferred from a present closing liability (Ministry of Finance of the People's Republic of China, 2018, attached standard, arts. 4–8, 14–33, 60–67).

For the declared base lease, these inspected recognition/measurement provisions support proposed ROU 210/105/0 and liability 210/110/0 with expense 126/116, under the same stated facts, depreciation assumption and fixed rule version. A same-asset modification is tested against 28–29, rather than labeled a separate lease without increased scope and commensurate consideration. A 2019 first-adopter using article 61(2) records cumulative opening changes without adjusting comparatives, while 61(1) is retrospective; comparison datasets must retain that distinction and the entity's actual election.

The tax question is why equal opening lease balances can still require separate deferred-tax assets and liabilities. The Ministry's original ASBE 18 defines tax bases and temporary differences in articles 4–9 and contains initial-recognition exceptions in articles 11 and 13. Article 13 also limits a deferred-tax asset to the probable taxable profit available to use the deduction. Articles 16–23 govern measurement, recovery or settlement assumptions, subsequent review, allocation and noncurrent classification (Ministry of Finance of the People's Republic of China, 2006b). The original text must be read with later interpretations.

Interpretation 16 removes the initial-recognition exemption for a single transaction that is not a business combination, initially affects neither accounting profit nor taxable income or loss, and creates equal taxable and deductible temporary differences. Section I(a) expressly includes qualifying initial lease liabilities and right-of-use assets; the corresponding deferred-tax liabilities and assets are recognized separately under ASBE 18. Equal carrying amounts alone do not establish equal temporary differences, and the asset's recoverability condition still applies. Section IV makes this treatment effective from 1 January 2023; Section I(b) permits early application from the year of issuance, 2022, with disclosure (Ministry of Finance of the People's Republic of China, 2022, secs. I(a)–(b), IV). This tax date is distinct from the enterprise's lease-standard adoption date. The operative text now supports the treatment previously illustrated by the retained official 2023 tax application case.

For the synthetic two-year lease, retain the assumed payments of 121, 10% discount rate and straight-line depreciation. All amounts use the same illustrative monetary unit. Assume that tax deductions are allocated to the liability, both tax bases are zero, the applicable tax rate is 25%, and sufficient taxable profit satisfies the deferred-tax asset recognition condition. Also assume that the transaction meets Section I(a)'s conditions and falls within its application period. At commencement the ROU asset and liability are each 210. The taxable ROU difference gives a deferred-tax liability (DTL) of 52.5; the deductible liability difference gives a deferred-tax asset (DTA) of 52.5. The entry debits DTA and credits DTL by those amounts. At the first annual close, liability 110 and ROU 105 give DTA 27.5 and DTL 26.25. Reducing the DTA by 25 creates deferred-tax expense; reducing the DTL by 26.25 creates a benefit, leaving net benefit 1.25. Both balances reverse by expiry. Applying the old exemption to omit both opening deferred balances is wrong for these qualifying facts and dates: it conceals the required gross balances and the later tax effect.

Transition also needs more than an opening journal entry. Section I(b) requires adjustments for qualifying transactions between the earliest comparative opening and application. For qualifying existing balances at that opening, cumulative effects adjust opening retained earnings and the relevant financial-statement items. Consider a separate transition example with cash 500, ROU

105, lease liability 110 and equity 495, with no deferred tax yet recognized. Under the same assumed tax bases, rate and recoverability, recognition debits DTA 27.5, credits DTL 26.25 and credits opening retained earnings 1.25. The resulting balance sheet is

$$\underbrace{500 + 105 + 27.5}_{\text{assets } 632.5} = \underbrace{110 + 26.25}_{\text{liabilities } 136.25} + \underbrace{495 + 1.25}_{\text{equity } 496.25}.$$

The equity adjustment records the cumulative effect at the comparative opening. If the remaining lease year falls within the comparative year, the DTA of 27.5 and DTL of 26.25 reverse during that year, giving comparative deferred-tax expense 1.25. The opening adjustment alone therefore does not complete the transition. Neither ASBE 21's article 61(2) lease-adoption election nor Interpretation 16 Section III's no-comparative-adjustment rule for its separate share-payment topic may be substituted for Section I(b)'s tax transition.

These are gross recognition and measurement amounts. Separate recognition and noncurrent classification do not determine whether an actual balance sheet may offset DTA and DTL. That decision needs the applicable presentation provisions and entity/tax-authority facts. Actual tax law, tax bases and rates, recoverable profit, the enterprise's adoption and elections, and later amendments remain to be established before applying the example to an issuer. Full ASBE 22 credit, consolidated statements, group combinations and insurance revisions also retain specific source dependencies; similar amounts do not establish full IFRS equivalence.

35.16 Reading an actual lease transition

Ping An's 2019 interim report shows what happens when the dated lease rules meet an actual disclosure. The group reports adoption of revised ASBE 21 on 1 January 2019 and leaves its 2018 comparative financial statements unadjusted. Its stated listings in Shanghai and Hong Kong place it within the enterprise class covered by the 2018 notice's 2019 application date (Ping An, 2019, pp. 112–113); (Ministry of Finance of the People's Republic of China, 2018, notice, para. I). This supplies an actual adoption fact for the representation. The later deferred-tax treatment described above still requires its own applicable date and tax facts.

The immediate reconstruction problem is to keep three observations apart: the original December presentation, the January adoption balances and the June closing balances. A right-of-use asset records the group's recognised right to use leased property; the lease liability measures the associated recognised payment obligation. Table 35.1 retains the amounts and their dates. All amounts in this example are displayed millions of renminbi; arithmetic on them does not establish exact underlying amounts or the issuer's rounding rule.

Table 35.1: Adoption balances and later carrying balances answer different questions.

Observation	Right-of-use assets	Lease liabilities
31 December 2018, unadjusted comparative	—	—
1 January 2019, reported at adoption	18,500	17,667
30 June 2019, carrying balance	17,390	17,212

The comparative dashes reproduce the presentation under the previous basis; they do not say that the group had no lease contracts in December. The January amounts come from the policy-change note, whereas the June amounts come from the consolidated balance sheet (Ping An, 2019, pp. 99–100, 112). Subtracting a December dash treated as zero from the June liability would mix recognition at transition with six months of subsequent activity.

The opening asset exceeds the liability by $18{,}500 - 17{,}667 = 833$. That difference alone does not identify an increase in retained earnings. Article 61(2) permits specified alternative measurements of the right-of-use asset for former operating leases, including a liability-based amount adjusted for prepaid rent; its cumulative approach adjusts retained earnings and other relevant financial-statement items (Ministry of Finance of the People's Republic of China, 2018, attachment, art. 61(2)). To see why the other items matter, suppose solely for illustration that pre-existing prepaid rent of 833 is reclassified. Debiting the new right-of-use asset by 18,500 and crediting the new liability by 17,667 and prepaid rent by 833 balances the entry with no equity movement. Assigning the same residual entirely to retained earnings would instead require a different assumption about all other transition adjustments. The report does not identify the illustrative prepayment or supply a complete opening journal. Reconstruction therefore retains the unexplained components rather than creating an equity plug.

The policy note nevertheless narrows the permitted interpretation. It says the opening liability discounts remaining payments at the lessee's incremental borrowing rate on 1 January 2019. It names five simplifications: treating operating leases with a remaining term shorter than twelve months as short-term leases; excluding initial direct costs at transition; using one discount rate for leases with similar characteristics; using actual option exercise and other recent information to determine the lease term; and assessing pre-existing onerous contracts instead of a separate right-of-use impairment test. These correspond to the specified provisions of Article 63, with the rate and direct-cost choices appearing in the same numbered provision (Ping An, 2019, p. 113); (Ministry of Finance of the People's Republic of China, 2018, attachment, art. 63). The record should preserve the reported elections without assigning an unreported rate or deciding

which asset-measurement alternative was used for each lease. The note also calls the difference between discounted prior operating-lease commitments and the recognised opening liability "not material". That qualification supplies neither an equality nor a numeric tolerance.

The June maturity disclosure adds a different measurement of the obligation. Under its heading for undiscounted contractual and expected cash flows, the lease-liability row gives 1,515 within three months, 5,885 in three to twelve months, 11,048 in one to five years and 485 beyond five years. Those numeric cells sum to 18,933; the immediate and no-term columns contain dashes (Ping An, 2019, pp. 164–165). The first two dated buckets sum to 7,400, but disclose no payment dates within those buckets. The 18,933 total and the 17,212 carrying liability can coexist because they measure different things. Their difference, 1,721, is not the interest expense for the half-year and does not by itself determine a discount rate.

Even on the post-adoption basis, a change in a balance is not a complete account of the transactions that produced it. The right-of-use asset falls by $17{,}390 - 18{,}500 = -1{,}110$ and the liability by $17{,}212 - 17{,}667 = -455$. Additions, remeasurements, disposals and other movements can accompany depreciation or payments. The first difference therefore does not identify depreciation, and the second does not identify cash paid. The consolidated cash-flow statement's finance-leasing business borrowing and receivable lines concern the group's financing activities as a business; their labels do not make them payments on its own lessee obligations (Ping An, 2019, pp. 106–107).

The same care prevents a false narrative conflict. In its major-contract discussion, the company says that it has no reportable custody, contracting, lease or other major-contract matters for the period (Ping An, 2019, p. 86). Preserving the company reference and the reportable-major-contract qualification gives a different assertion from saying that the consolidated group has no leases. Removing those qualifiers would create the apparent contradiction with the balance sheet.

For the proposed representation, the useful result is a set of dated, source-supported balances, policy elections and maturity restrictions. A query for the reported June carrying liability can return 17,212 with its basis and precision. A query for future scheduled payments must keep the disclosed maturity buckets and leave their internal dates unresolved. A query for the complete transition journal or the half-year's lessee cash payments must retain the missing inputs. These distinct answers let actual disclosures constrain histories without pretending that every balanced completion is the company's history.

35.17 Matched-case outcomes and decision

Input changed or withheld	Required consequence
Framework changes with same lease facts	Preserve different expense/ROU paths and source strength, not just rename accounts
Method source present but target entity's adoption unknown	Unresolved applicability, no real posting authorization
ROU/liability initially equal	Still evaluate gross tax differences and later divergent balances
Historical rule supports method but a later reporting year is requested	Recheck operative amendments/transition; no automatic extension
Issuer policy discloses LIFO, regulator text unavailable	Retain issuer assertion; do not upgrade it into universal US authority
Foreign basis label replaced by local label	Reject as a standards conversion without recognition/measurement bridge

These decisions establish a common procedure and specific proposed treatments. The recovered ASBE lease notice and tax interpretation support the stated application schedules and conditional example. Broader authority coverage remains incomplete: operative US sources, ASBE sector methods, actual tax law and presentation conditions, and the applicable amendment and entity-election evidence still require internal investigation.

Chapter 36

Financing, ownership and the group boundary

A company can preserve the same operating assets while changing who supplies its funding and who shares its results. The following cases move from debt cash flows to ownership, then to the reporting boundary of a group. Each numerical example states its own contract; the coupon-bearing convertible below and the later zero-coupon compound example are different instruments.

36.1 Debt and hybrid financing

The contract layer for a debt instrument stores issuer and holder perspective, currency, issue and settlement dates, issue proceeds and legal face amount. Its payment terms include coupon/index/spread, reset and fixing dates, day-count convention, payment calendar, amortisation schedule and maturity. Its rights and protections include seniority, collateral, covenants, options, conversion terms, guarantees and linked hedges. Missing terms are explicit unknowns. A single carrying amount cannot stand in for this record.

The accounting layer stores the classification decision and its evidence, initial measurement, carrying amount, accrued interest, effective-rate method, unamortised fees or discounts, fair-value disclosures, impairment or credit adjustments where applicable, and presentation mappings. Contractual interest, effective-interest expense, realised cash payments and fair-value changes are separate quantities. The zero-coupon example in Chapter 34 provides a minimal conformance case for this separation.

The lifecycle must support issuance, recognition, accrual, reset, coupon, principal repayment, modification, derecognition, default, recovery and conversion. A modification event first determines which accounting operation the selected standard requires; it does not always mean issuing a new loan. An embedded option can affect valuation and classification without being exercised. A derivative linked as a hedge requires both the economic contract and any applicable accounting designation; economic risk reduction alone does not establish hedge-accounting eligibility.

Hybrid instruments make interpretation particularly consequential. IAS 32's overview discusses liability/equity classification and the separation of a compound instrument (IFRS Foundation, 2026a). The executable classification must consider the full terms and the selected standard version. US GAAP or ASBE treatment must be implemented under its own rules rather than inferred from an IFRS classification. Where the terms are not disclosed sufficiently, alternative

classifications can coexist only if each is actually compatible with the known terms and applicable rules.

The contingent-conversion case in Section 14.18 connects a four-party funding sequence to these classification questions. Its supplied economic control rule is separate from accounting control and the date at which control is obtained. Sections 14.19 and 14.20 explain the missing source, measurement and interval-history evidence before a changing group perimeter can be reported.

36.1.1 Apple: debt balances and cash repayment

Apple's borrowing shows why the distinction between a contract and its carrying amount matters for a funding forecast. A fall in reported debt might suggest how much cash the company repaid. Its annual disclosure lets us test that interpretation by separating principal from the adjustments included in the carrying amount. The following amounts are in millions of US dollars at the displayed precision; changes run from 30 September 2023 to 28 September 2024 (Apple Inc., 2024a, Note 9, printed pp. 43–44).

Term-debt component	2023	2024	Change
Principal	106,572	97,341	$-9,231$
Net premium, discount and issuance costs	-356	-321	$+35$
Hedge accounting fair-value adjustment	$-1,113$	-358	$+755$
Carrying amount	105,103	96,662	$-8,441$

The cash-flow statement reports no net term-debt issuance proceeds and cash repayments of 9,958 million for fiscal 2024 (Apple Inc., 2024a, printed p. 33). To connect that cash movement to the table, first calculate the difference between the change in reported principal and net term-debt cash flow. It is $-9{,}231 - (-9{,}958) = 727$ million. Adding the reported measurement-component changes then gives the carrying-amount movement. With Δ denoting closing less opening balance, the two displayed-amount identities are

$$\Delta\text{principal} = -9{,}958 + 727 = -9{,}231,$$
$$\Delta\text{carrying amount} = -9{,}958 + 727 + 35 + 755 = -8{,}441.$$

The 727 is an unallocated balance-to-cash difference. It does not identify an exchange movement or a journal. Likewise, the 35 is the change in a net unamortised balance, rather than an identified issuance fee or interest charge; the 755 is a change in the debt's hedge adjustment, rather than a cash receipt. Together they explain 790 of the 1,517 million difference between the carrying-amount movement and net cash repayment. A forecast that used the 8,441 carrying decline as repayment would therefore compute the wrong cash amount for this observed year. The cash statement's proceeds measure is net; it must not automatically become legal face issued in another period.

The maturity schedule answers a different funding question. Its six future principal buckets sum to 97,341 million, with 10,930 in fiscal 2025. The current carrying amount is 10,912. Their 18 million difference does not allocate a discount or hedge adjustment to the current portion. The measurements differ, and their time boundaries must also be retained: Note 1 supplies the fiscal-year default, the issuance table explicitly labels its 2024–2062 range as calendar years, and MD&A separately reports amounts payable within twelve months. MD&A also reports future term-debt interest payments of 38.5 billion dollars, including 2.6 billion within twelve months (Apple Inc., 2024a, printed pp. 25, 34 and 43). Those cash requirements cannot replace period interest expense or a present value. The dated Level 2 fair-value disclosures likewise measure

something different: the annual note reports 88.4 billion dollars at September 2024, while the interim note reports 82.7 billion at December 2024 alongside a rounded carrying amount of 94.8 billion (Apple Inc., 2025c, Note 6, printed p. 10).

36.1.2 Net borrowing can conceal refinancing

Commercial paper exposes a second limitation of aggregate financing data. Apple describes unsecured short-term promissory notes used for general corporate purposes, including dividends and repurchases. The annual note says their maturities are generally less than nine months; that description does not identify each note or trace individual borrowed dollars to a use. At the annual statement amounts, commercial paper increases from 5,985 to 9,967 million, a rise of 3,982, while net cash proceeds are 3,960. The resulting 22 million difference remains unallocated. During the following quarter the balance falls from 9,967 to 1,995, while net cash repayments are 7,944; the corresponding unallocated difference is −28 million (Apple Inc., 2024a, Note 9 and balance and cash-flow statements) (Apple Inc., 2025c, Note 6 and balance and cash-flow statements).

Even a complete explanation of those differences would leave refinancing activity unresolved. Annual Note 9 reports net cash for paper with maturities of at most 90 days, but separate proceeds and repayments for longer maturities. In fiscal 2022, for example, the longer-maturity proceeds of 5,948 and repayments of 7,257 give net repayments of 1,309 million. Adding short-maturity net proceeds of 5,264 gives total net proceeds of 3,955. For fiscal 2024 the longer-maturity cash rows are zero and the short-maturity net inflow is 3,960. Neither that net amount nor a closing balance reveals how many issues and redemptions occurred between reporting dates.

Consider a hypothetical incremental borrowing of 1,000 million US dollars, issued at par and repaid after 60 days. Stipulate simple interest of 5% using actual days over a 360-day year, with no fees, taxes, hedges or early settlement. The rate is an assumption of this example; Apple's disclosed 5.00% weighted rate at September 2024 does not establish the rate on such a transaction. Issuing and repaying the principal adds $1{,}000 - 1{,}000 = 0$ to net principal cash flow and leaves zero principal outstanding, yet supplies 1,000 million between those dates. The cost over the 60 days is

$$\text{interest} = 1{,}000 \times 0.05 \times \frac{60}{360} = \frac{25}{3} \approx 8.3333 \quad \text{million US dollars.}$$

Issue debits cash and credits principal 1,000. Accrual debits interest expense and credits interest payable 25/3. Settlement debits principal 1,000 and interest payable 25/3, and credits cash 3,025/3. Each entry balances; after settlement the incremental liability is zero, while cash and equity are each lower by 25/3 relative to the otherwise funded baseline.

This example isolates what a net principal-flow observation fails to reveal. It is not a reconstructed Apple history: interest expense, interest cash and other disclosures could distinguish or exclude the added borrowing. Within that scope, it shows why a financing model needs dated issuance, accrual and redemption events. Matching a net financing row does not establish either intraperiod liquidity or funding cost.

36.1.3 A rate scenario needs an exposure definition

The contract's coupon is also insufficient to describe exposure after hedging. Apple's annual debt note states that interest-rate swaps effectively convert some fixed-rate US-dollar notes to floating rates, while cross-currency swaps effectively convert some foreign-currency notes to

US-dollar notes. The effective-rate ranges include interest, discount or premium amortisation and applicable hedge adjustments. These are relationships between separate contracts and measurements, not a single rate that can be applied to total principal (Apple Inc., 2024a, Note 9, printed p. 43).

The filing supplies a useful, more specific constraint. Annual Item 7A reports that a hypothetical increase of 100 basis points at all tenors, including associated derivatives, would increase annual term-debt interest expense by 139 million dollars at September 2024 and 194 million at September 2023 (Apple Inc., 2024a, printed p. 27). Multiplying Note 4's 2024 interest-rate contract notional of 14,575 million by 1% instead gives 145.75 million. It does not reproduce the reported sensitivity, even allowing for nearest rounding of the displayed amounts. The difference cannot be assigned to a particular reset date or swap feature from these aggregates. For 2023 the same shortcut gives $19{,}375 \times 0.01 = 193.75$, consistent with rounded 194; that numerical match does not validate the shortcut or recover the issuer's model.

The scenario's measure matters as much as its shock. In the same table, the 2024 investment-portfolio response is a 2,755 million decline in fair value. It cannot be added to an annual interest-expense increase as though both were the same loss measure. At December 2024 the reported interest-rate notional is still 14,575 million, while the carrying amount of term debt subject to fair-value hedges is a rounded 13.3 billion, compared with 13.5 billion at September. These aggregates describe different populations or measurements. The interim statement that market risk has not changed materially refers the reader to annual Item 7A; it does not publish an updated 139 million sensitivity (Apple Inc., 2025c, Note 4 and Part I, Item 3).

Hedge accounting and cash presentation also remain distinct. Annual Note 1 presents derivative assets and liabilities at gross fair values. Note 4 classifies derivative cash flows with the items hedged, which are *generally* operating activities; that qualifier does not classify every derivative settlement as operating. The maximum disclosed horizon for hedging debt-related foreign-currency cash-flow variability is 18 years, and Apple states that some exposures may be left unhedged and that offsets may be partial (Apple Inc., 2024a, Notes 1 and 4, printed pp. 34 and 37–38).

Finally, Item 7A's currency-risk statistic has its own population. Apple reports a Monte Carlo estimate, at 95% confidence over one day, of 538 million dollars at September 2024 and 669 million at September 2023 for adverse fair-value changes in its foreign-currency derivatives. The assessment excludes underlying currency exposures, whose changes generally offset derivative changes. This derivative VaR is neither a hard maximum loss nor a measure of the company's net currency risk; the issuer's simulation has not been replicated here. A financing simulation can retain these dated scenarios as disclosed constraints, but still needs actual holdings, currencies, payment calendars, reset terms and hedge relationships to calculate future cash and accounting consequences. The next question is how financing choices change the quantities and claims of the company's owners.

36.2 Equity and ownership

Equity has both monetary accounts and security quantities. The security register records share class, issued and outstanding quantities, treasury holdings, par or stated value where applicable, voting rights, dividend rights, redemption features, options and conversion relationships. Monetary equity accounts include contributed capital, reserves, retained earnings and other components required by the selected basis. These records evolve together through issuance, repurchase, cancellation, split, dividend and conversion events.

A stock split changes quantities and per-share measures without necessarily changing aggregate

monetary equity. A cash dividend affects distributions and cash rather than current operating expense under the corresponding ordinary equity treatment. Repurchases and employee share awards require their own rules. Earnings per share uses a dated quantity schedule and rules for potential ordinary shares; simply dividing profit by the closing share count is insufficient when shares change during the period.

Funding simulations must connect the quantity register to market assumptions. An equity issue at a scenario-dependent price changes cash raised, shares and dilution. The model must state whether it targets a fixed cash amount, a fixed share count or a capital ratio. These are different management decisions even when the historical statement would report the same aggregate proceeds.

36.2.1 A share history changes the denominator

A company can earn the same profit with different numbers of shares outstanding during the year. Dividing by the closing count would attribute the entire year's earnings to shares that may have existed for only a few months. Basic earnings per share instead relates earnings attributable to ordinary shareholders to the time those shares were outstanding. Diluted earnings per share asks a further, conditional question: how would that measure change if eligible potential shares had become ordinary shares? Neither calculation exercises an option or converts a bond in the company's actual accounts.

The following synthetic case uses IAS 33 in the retained original EU Regulation 2023/1803, including its integral Appendix A. It develops a method under that historical text; applying it to a real issuer requires the issuer's reporting basis, operative version and instrument terms. It is separate from the Apple disclosures that follow (European Commission, 2023a, IAS 33, paragraphs 2–4, 9–12, 30–36). Assume one fully paid ordinary-share class, one consolidated reporting population, and a 365-day year from 1 January to 31 December 2025. Monetary amounts are currency units; share counts are numbers of shares, so EPS is currency units per share.

The year opens with 1,000 ordinary shares and no treasury holdings. On 1 April, consideration becomes receivable for 200 new shares at 12 each and is paid that day: cash and contributed equity rise by 2,400. On 1 July, a two-for-one split doubles every outstanding ordinary share. On 1 October, the company buys 300 shares for 7 each and holds them in treasury. Cash and equity fall by 2,100; issued shares remain 2,400, treasury shares become 300, and outstanding shares become 2,100. The split contributes no additional resources. IAS 33 therefore restates earlier share counts onto the new share basis, whereas the resource-changing issue and repurchase enter the average from their respective dates (European Commission, 2023a, IAS 33, paragraphs 19–21, 26–29).

Interval, including first date	Days	Actual outstanding	Split-adjusted
1 January–1 April	90	1,000	2,000
1 April–1 July	91	1,200	2,400
1 July–1 October	92	2,400	2,400
1 October–1 January 2026	92	2,100	2,100

Each interval excludes its end date, so no day is counted twice. The basic denominator is consequently

$$D = \frac{90(2000) + 91(2400) + 92(2400) + 92(2100)}{365} = \frac{162480}{73} \approx 2225.7534.$$

The closing count of 2,100 answers a different question. Nor should the split-adjusted count overwrite the historical register: there really were 1,000 outstanding shares in January. If the comparative year had constant shares of 1,000 and ordinary earnings of 6,000, its previously calculated EPS of 6 would be presented as 3 after this split. The same retrospective principle applies to a split occurring after year-end but before authorization of the statements. A later cash issue does not receive that treatment (European Commission, 2023a, IAS 33, paragraphs 64–65, 70(d)–71).

36.2.2 Attribute earnings before testing potential shares

Suppose consolidated profit is 15,000 from continuing operations and 18,000 in total. Non-controlling interests receive 1,000, all from continuing operations. A cumulative, nonparticipating preference class, classified here as equity, has a current-year entitlement of 2,000. Deduct that entitlement even though it has not been declared. A payment of 500 of prior-year arrears is not another deduction from this year's earnings. These are supplied attribution facts; the EPS calculation does not infer them from cash dividends (European Commission, 2023a, IAS 33, paragraphs 12–14 and A1).

Earnings attributable to the parent's ordinary shareholders are thus $15000 - 1000 - 2000 = 12000$ from continuing operations and $18000 - 1000 - 2000 = 15000$ in total. Discontinued operations contribute 3,000. Basic continuing EPS is $12000/D \approx 5.3914$; basic total EPS is $15000/D \approx 6.7393$. Their difference, approximately 1.3479, is basic discontinued EPS. Profit already includes the interest and tax effects described below; adding interest again at this stage would be wrong.

There are three potential-share series. First, cash options have been outstanding all year: 200 shares at an exercise price of 4 before the split, and 400 at 2 afterward. Use a supplied average market price of 5 on the post-split basis. Earlier prices must be converted to that same basis. The options require no future employee services and contain no performance condition, debt tender, redemption requirement or cash-settlement alternative.

Assumed exercise would provide $400(2) = 800$. At the average market price, 800 corresponds to 160 fairly priced shares. Only the remaining $400 - 160 = 240$ shares enter the diluted denominator; there is no earnings adjustment. Adding all 400 shares would ignore the assumed proceeds. IAS 33 describes this as the difference between assumed issue and the shares represented by proceeds at the average price (European Commission, 2023a, IAS 33, paragraphs 45–47 and A4–A5). The calculation records neither cash of 800 nor an actual share issue.

Second, a convertible debt series is issued on 1 July. Its original terms give the holder a voluntary right to convert at any time before its 1 January 2026 maturity into 600 post-split ordinary shares. Assume the debt is denominated in the issuer's functional currency and the conversion right is equity-classified, with no additional derivative features. Conversion settles solely in those shares; the unconverted debt retains the cash payments specified below. No terms are changed and no additional consideration is offered to induce conversion. For this case, supplied issue proceeds of 10,000 divide into a liability of 9,200 and conversion equity of 800. The recognized effective-interest expense for the remaining year is 800, with a related tax benefit of 200 and no other earnings consequences. The original terms pay a coupon of 400 on 1 October and principal of 9,200 plus a final coupon of 400 on 1 January 2026. The supplied effective rate is 1/23 per 92-day period: discounting those payments gives $400/(24/23) + 9600/(24/23)^2 = 9200$. Interest of 400 first raises the liability to 9,600; the October coupon returns it to 9,200, and another 400 of interest gives a year-end liability of 9,600 before January settlement. The intervening payment explains why the two interest amounts are equal. These supplied amounts include the relevant cost and discount effects; substituting a coupon alone need not do so. Instrument classification

and component valuation are separate accounting inputs, illustrated by the compound-bond example below.

Assumed conversion from issue eliminates the net expense of $800 - 200 = 600$ and adds shares only for the 184 days that the instrument exists. Its contributions are

$$\Delta N_b = 600, \qquad \Delta D_b = 600\frac{184}{365} = \frac{22080}{73}, \qquad \frac{\Delta N_b}{\Delta D_b} = \frac{365}{184} \approx 1.9837.$$

IAS 33 requires the after-tax interest effect and any consequential earnings changes, together with this issue-date weighting (European Commission, 2023a, IAS 33, paragraphs 33–39, 49–50). For example, a profit-sharing expense caused by removing interest would also change the numerator; it cannot disappear merely because a coupon and a tax rate are available.

Third, the preference class can convert into 400 post-split ordinary shares and has existed throughout the year. Its assumed conversion adds back the current entitlement of 2,000. Earnings per incremental share are therefore $2000/400 = 5$. The class has no participation in residual earnings; participating rights would require a different attribution calculation.

36.2.3 The order of dilution changes the answer

Continuing-operations earnings attributable to ordinary shareholders, 12,000 here, control admission. Each series is assessed separately in ascending order of earnings per incremental share: options at zero, debt at about 1.9837, then preference shares at 5. Recalculate after each admitted series; comparing every instrument only with basic EPS loses this dependency (European Commission, 2023a, IAS 33, paragraphs 41–44).

The options reduce continuing EPS from approximately 5.3914 to 4.8667. The debt then reduces it further:

$$D_{\text{dil}} = D + 240 + \frac{22080}{73} = \frac{202080}{73}, \qquad E_{\text{dil,cont}} = \frac{12000 + 600}{D_{\text{dil}}} = \frac{7665}{1684} \approx 4.5517.$$

The preference class would increase this result because its incremental ratio, 5, exceeds 4.5517. Exclude it. Yet $5 < 5.3914$, so it would have appeared dilutive if tested against basic EPS alone. That is why the order matters.

The admitted population also determines the other EPS amounts. Total earnings increase from 15,000 to 15,600 through the same debt addback; discontinued earnings remain 3,000. Thus

$$E_{\text{dil,total}} = \frac{15600}{202080/73} = \frac{4745}{842} \approx 5.6354, \qquad E_{\text{dil,disc}} = \frac{3000}{202080/73} = \frac{1825}{1684} \approx 1.0837.$$

Continuing and discontinued EPS sum to total EPS before presentation rounding. The report retains that reconciliation, each admitted numerator and denominator increment, and the excluded preference series. Basic and diluted EPS receive equal prominence; negative EPS is still presented (European Commission, 2023a, IAS 33, paragraphs 66–70).

Two sign changes test the control number. If continuing ordinary earnings are instead a loss of 5,000 while total ordinary earnings remain a profit of 15,000, none of these three series is admitted: each would reduce the loss per share from continuing operations. Total profit cannot reverse that decision. Conversely, retain continuing earnings of 12,000 but change total ordinary earnings to a loss of 9,000. Options and debt still enter, so diluted total EPS uses $-9000 + 600 = -8400$ over the same diluted denominator. It is less negative than basic total EPS. That is the consequence of the continuing-operations control, illustrated explicitly in the standard's Appendix A (European Commission, 2023a, IAS 33, paragraphs 41–43 and A3).

36.2.4 One comparison rule for a finite set of series

The preference shares failed because the options and debt had already reduced continuing EPS. An implementation needs to reproduce that changing comparison for any finite number of series. Assume that attribution and instrument-specific calculations have supplied each series' effects on the same reporting population, ordinary class, currency scale and retrospective share basis. Assume also that these effects are fixed and additive: admitting one series does not change another's conversion terms or earnings consequences. These assumptions fit the three series above; participating rights or mutually exclusive conversion choices require further analysis before this procedure.

Write the current continuing ordinary earnings as N and weighted shares as $D > 0$. For one candidate, let a be its signed continuing-earnings increment and $b > 0$ its already time-weighted share increment. Both the current and enlarged denominators are positive. A strict decrease in EPS is therefore equivalent to

$$\begin{aligned} \frac{N+a}{D+b} < \frac{N}{D} \quad &\iff \quad D(N+a) < N(D+b) \\ &\iff \quad aD < Nb \quad \iff \quad \frac{a}{b} < \frac{N}{D}. \end{aligned}$$

The cancellation uses $DN = ND$ and does not divide by earnings, which may be zero or negative. It consequently handles a loss without a separate sign shortcut. For example, if $N = -20$, $D = 10$, $a = -15$ and $b = 5$, the candidate changes EPS from -2 to $-35/15 = -7/3$: it increases the loss per share and is dilutive. A zero earnings increment with the same positive share increment would reduce that loss and be excluded. Equality in the comparison is neutral and is also excluded under the strict dilution rule (European Commission, 2023a, IAS 33, paragraphs 41–44).

Sort candidates by their exact ratio a/b in ascending order. Break equal ratios by a stable series identifier, then test each against the current N, D. On admission, replace N, D with $N+a, D+b$ before proceeding. Use rational arithmetic for this decision; rounding an incremental ratio to the displayed precision can change an equality into an apparent decrease. The debt in the example has the exact ratio $365/184$, and the preference shares are compared with $7665/1684$, not with the earlier basic ratio. The share increment already contains its time weight; the selector must not apply the issue-date fraction a second time.

Why does this order suffice under the additive assumptions? Admission makes the new ratio a weighted average of the old ratio and a/b, strictly between them. The new ratio remains above the admitted ratio. Every earlier admitted ratio is therefore below the final ratio, say q. At the first exclusion, the next ratio is at least q; every later ratio is at least as large and is excluded too. If S is the selected set and N_0, D_0 the basic amounts,

$$N_0 - qD_0 + \sum_{i \in S} (a_i - qb_i) = 0.$$

Each selected summand is negative; every unselected summand is nonnegative. Replacing S by any other subset can only increase the left side. Dividing by that subset's positive denominator shows that its continuing EPS is at least q. This is a local derivation for fixed independent effects, not a claim that arbitrary instruments satisfy those assumptions. Neutral series can create other subsets with the same minimum; strict admission excludes them.

Total earnings have their own supplied increment for each series. Carry those increments through the same selected population, then subtract continuing earnings to obtain discontinued earnings. Selecting again on total profit would contradict the continuing-operations control just illustrated (European Commission, 2023a, IAS 33, paragraph A3).

36.2.5 Keep unresolved instruments in the report

The proposed `OrderedEPS/1` report accepts attributed basic earnings, weighted ordinary shares and a separately supported complete instrument inventory. Each item remains visible as a candidate, a supported no-effect item, an unresolved item, or an item requiring an unsupported method. A no-effect item has zero earnings and share increments; it is reported without dividing by zero. A zero-share item with a nonzero earnings effect requires another method. An empty inventory also requires its own completeness finding. These findings are inputs to the calculation: the report cannot discover omitted instruments from a list supplied as complete.

The request fixes the entity, book and dated head, evidence view, reporting population and period, ordinary class, share basis, currency and scale, and historical rule version. Basic and instrument records repeat a hash of that context. Every premise also points to exact source bytes and a source anchor. Matching hashes establish which evidence was used; interpretation, completeness and operative authority still require their own findings. Potential-share windows must lie within the period and must not overlap. A converted series also identifies its subsequent actual-share window and confirms that those actual shares already enter basic EPS. Only the earlier potential-share window is added by the diluted calculation. The date checks prevent overlap; instrument methods must still establish the amount and weighting of each increment.

A completed conditional report returns basic and diluted continuing, total and discontinued EPS, their numerators and common denominators, and the ordered comparison for every series. Each comparison retains its increments, the amounts before the test, the amounts if admitted, and the admission decision. Missing attribution can prevent both basic and diluted EPS. Missing instrument consequences or an incomplete inventory preserve an independently ready basic result but leave diluted EPS unresolved. Known dependent effects or participation terms produce an unsupported result; both kinds of obstacle are reported together when present. Invalid bindings, malformed fractions, duplicate or omitted inventoried series, and overlapping windows reject the request. None of these outcomes returns a partial selection as complete.

The exact fields and result states are specified in Section A.57. This is a read-only proposed report: it creates no journal or share and changes no book. Its finite reference checks cover the supplied arithmetic, ordering and refusal behavior. Applying it to a company still requires instrument-specific preprocessing, complete facts and applicable authority; joining it to a company's dated earnings, tax and share records is a further implementation obligation.

36.2.6 Actual conversion, expiry and incomplete terms

Actual events divide the year differently. Suppose the debt pays its coupon and then converts under its original terms on 1 October; interest ceases that day. Before conversion, recognized interest is 400 with a tax benefit of 100. The holder exercises the equity right recognized at issue; this is not a newly negotiated exchange of debt for shares.

IAS 32.AG32 expressly addresses conversion at maturity. For an original holder option exercisable at any time before stated maturity, BDO describes a widely used practice interpretation that treats the permitted exercise date as maturity for this purpose. Under that interpretation, the liability's carrying amount at conversion transfers to equity, without a conversion gain or loss or acceleration of the remaining interest (BDO, 2026, section 4.5, p. 19). This synthetic case selects that interpretation. Paying the October coupon reduces the liability to 9,200: debit the liability and credit ordinary equity by 9,200. The original conversion-equity component of 800 remains in equity; the remaining 400 of interest is not recognized on conversion.

The interpretation is technical practice literature, rather than express early-exercise wording in AG32. IFRIC 19.3(c) excludes conversion under original terms from its renegotiated debt-

for-equity treatment, but does not itself specify the early-conversion amount. A transaction outside exercise of that unchanged equity right needs separate analysis of IFRS 9's general extinguishment and consideration comparison. Actual-company use requires the complete terms, classification, applicable authority and a consistent accounting-policy decision under the IAS 8 hierarchy (European Commission, 2023a, IAS 32, paragraphs 28–32 and AG32–35; IFRS 9, paragraphs 3.3.1–3.3.3; IFRIC 19, paragraphs 2–3; IAS 8, paragraphs 9–13). The maturity-conversion company case in Section 36.2.7 continues to use AG32's explicit maturity rule. The tax benefit of 100 and absence of an additional conversion tax effect are supplied earnings facts for this example; the practice interpretation does not establish tax deductibility, tax bases or current versus deferred tax.

After conversion, 600 actual shares enter basic EPS for 92 days. The diluted calculation adds only the preceding 92 days of potential shares and the preconversion net expense of 300. There are therefore $600(92+92)/365$ additional share-years in the diluted comparison, with no overlap. Under the otherwise identical supplied earnings assumptions, basic ordinary earnings rise by the 300 of avoided later expense, and the remaining addback of 300 restores the same fully diluted numerator as before. Treating the potential shares as outstanding for all 184 days *in addition* to the actual shares would double count (European Commission, 2023a, IAS 33, paragraphs 21(c), 36–38).

Expiry has no subsequent actual-share leg. If the options instead lapse on 1 October, their contribution is $240(273/365)$, using a supplied average price of 5 for this variant's relevant period. They do not vanish from the earlier part of the year. An exercise price equal to or above the average price gives zero incremental shares in this simple option profile. More generally, a conversion that leaves continuing EPS exactly unchanged is not dilutive: paragraph 41 requires a decrease in profit per share or an increase in loss per share.

Employee awards require another input even when their other terms fit the option calculation. Under paragraph 47A, assumed proceeds include the IFRS 2 value of future goods or services. If that supplied value were 400 across these 400 options, the assumed price would be $2+400/400 = 3$, and the increment at market price 5 would be 160 shares. This conditional arithmetic neither values the services nor calculates award expense. Performance-based awards follow the contingent-share provisions instead; share-or-cash alternatives, written puts, rights issues and participating or partly paid shares also need their own methods (European Commission, 2023a, IAS 33, paragraphs 47A–63, A2 and A13–A16).

An EPS assessment therefore references a dated book, reporting population, share class, attributed earnings, share history, prices and complete instrument terms. It produces report amounts and their reconciliation, without changing the book. Missing average prices, future-service values, conversion consequences or settlement terms leave diluted EPS unresolved even when basic EPS is available. Missing attribution or ordinary-share timing can prevent both. Known complex terms select the relevant further method; they are not grounds for silently dropping the instrument. These distinctions let the following issuer disclosures supply facts without being mistaken for a complete method or a complete history.

36.2.7 Carry the conversion through the whole company

The EPS calculation must use earnings and shares from the same company history. Return to the parent and subsidiary in Section 28.7. Their unchanged annual operations produce group profit 102, of which 7.35 belongs to non-controlling interests and 94.65 to the parent's ordinary shareholders. Group cash is 644 and equity 724. The opening matured convertible was already redeemed in January; the new instrument below is a separate July issue. Supply a register of ten parent ordinary shares outstanding from the start of the year, with no treasury shares,

splits, preference claims or discontinued operations. Capital 600 is a monetary balance and does not establish that share count.

On 1 July the parent receives 100 for a new instrument that matures on 1 October. Its original terms require a coupon of 4, followed by either principal repayment of 92 or conversion of that principal into six ordinary shares. The fixed conversion feature qualifies as equity in the parent's functional currency; no fees, participation or other consequential earnings effects are assumed. At the supplied comparable 92-day debt rate of 1/23,

$$L_0 = \frac{4 + 92}{1 + 1/23} = 92, \qquad E_0 = 100 - 92 = 8.$$

Issue debits cash 100 and credits liability 92 and conversion equity 8. Effective interest of 4 raises the liability to 96. Paying the coupon returns it to 92. At maturity, original-terms conversion transfers that 92 to ordinary equity and issues six shares without cash; redemption pays 92 instead. The original equity 8 remains equity in either world (European Commission, 2023a, IAS 32, paragraphs 28–32 and AG32). This October-maturity contract differs from the preceding January-maturity early-conversion illustration.

Tax requires its own facts. Assume a synthetic fiscal allocation of 92 to the debt and 8 to equity, effective-interest deductibility at accrual, a 25% rate, and no additional tax on coupon payment, redemption or conversion. The liability's tax base then follows its carrying amount. No new temporary difference arises under these supplied premises. A rate alone would not justify that conclusion: IAS 12.23 describes jurisdictions whose initial tax base equals the combined proceeds, producing a deferred tax liability charged to equity. Missing or different tax-base findings prevent this particular company extension; they require the corresponding tax method (European Commission, 2023a, IAS 12, paragraphs 8, 23, 58 and 61A).

The parent's annual taxable income falls from 42 to 38. Current tax falls by 1; after the 8 already paid in the first half, its second-half payment is 1.5. The subsidiary's tax and both companies' existing deferred-tax consequences are unchanged. Group pretax profit falls from 136 to 132, current tax from 28 to 27, and deferred-tax expense remains 6. Group profit is therefore 99. Non-controlling interests still receive 7.35 because the new interest belongs to the parent. Ordinary earnings are $99 - 7.35 = 91.65$, not group profit 99 or the parent's separate-company profit 46.5.

The two settlements have identical earnings but different cash and ownership consequences. With conversion, group cash becomes $644 + 100 - 4 + 1 = 741$ and equity becomes $724 + 100 - 3 = 821$: issue and conversion add 100 of equity, while after-tax interest reduces earnings by 3. With redemption, cash becomes $644 + 100 - 4 - 92 + 1 = 649$ and equity becomes $724 + 8 - 3 = 729$. In both worlds, external operating receipts remain 250 and payments become 99; financing receipts are 100. Financing payments remain 110 with conversion and rise to 202 with redemption. Net cash movements of 141 and 49 reconcile to opening cash 600. Noncash conversion is excluded from those cash flows.

Each potential-share window runs from 1 July to 1 October, or 92 days. Conversion alone creates six actual shares for the subsequent 92 days. The basic denominators are therefore

$$D_{\text{basic,conv}} = 10 + 6\frac{92}{365} = \frac{4202}{365}, \qquad D_{\text{basic,redeem}} = 10.$$

The same potential-share increment 552/365 and after-tax earnings addback 3 are tested in both worlds. Their incremental ratio is $3/(552/365) = 365/184$, below each world's basic EPS, so the series is admitted. The complete results are

$$E_{\text{dil,conv}} = \frac{94.65}{4754/365} = \frac{138189}{19016}, \qquad E_{\text{dil,redeem}} = \frac{94.65}{4202/365} = \frac{138189}{16808}.$$

The denominators differ because only actual conversion leaves fourth-quarter shares outstanding. Adding potential shares after the October redemption would invent a period in which the instrument no longer existed. Adding potential shares after conversion as well as the actual shares would count the same days twice (European Commission, 2023a, IAS 33, paragraphs 21(c), 33–38).

The documentary `CompanyEPSJoin/1` bridge checks this connection by replaying the complete legal journals, recalculating the parent's tax from the combined books, and reconstructing ordinary and potential shares day by day. A proposed EPS request must match that company's parent-attributed earnings, dated shares, book head and evidence view before the ordered selector runs. Even a self-consistent hash cannot rescue a request for the wrong population or period. Missing tax or share-history findings leave the join unresolved; different contract or tax-base methods remain unsupported. The precise exchange appears in Section A.58.

36.2.8 A different tax base changes the equity at issue

The company example's matching tax bases were supplied assumptions. Keep proceeds 100, debt value 92 and residual conversion equity 8, but now suppose the applicable fiscal treatment assigns the liability a tax base of 100. IAS 12.23 expressly describes this situation: the separately recognized equity component creates a taxable temporary difference, and its deferred tax is charged directly to equity on initial recognition (European Commission, 2023a, IAS 12, paragraphs 23 and 61A).

At the supplied rate of 25%, the initial deferred tax liability is

$$T_0 = \frac{1}{4}(100 - 92) = 2, \qquad E_{\text{net}} = 8 - 2 = 6.$$

After the ordinary issue entry, debit conversion equity 2 and credit deferred tax liability 2. Cash 100 then equals debt 92 plus deferred tax 2 plus net equity 6. Posting that 2 to current tax expense would balance the journal but place the issue-date tax effect in the wrong component.

More generally, let issue proceeds be P, supplied debt value L, and residual equity $E = P - L$. Under this specific fiscal treatment the liability's tax base is P, so at rate t,

$$T_0 = t(P - L) = tE, \qquad E_{\text{net}} = (1 - t)E, \qquad P = L + T_0 + E_{\text{net}}.$$

With the alternative matching tax base L, this initial difference is zero. A zero tax rate also produces zero deferred tax; neither the 25% example nor the debt value 92 is a universal validity condition. Other fiscal bases need their own supported analysis instead of interpolation between these two examples.

This issue-date calculation leaves later earnings to be derived. IAS 12.23 states that subsequent changes in this deferred tax liability are recognized in profit or loss under paragraph 58. The tax bases, deduction timing and settlement consequences determine those changes. In particular, the initial tax charge to equity cannot simply be substituted for the earnings tax effect used in diluted EPS. The finite `CompoundTaxBasis/1` reference calculation checks the two initial-base profiles, their equity origin and refusal of missing or unsupported inputs; it does not supply that later tax schedule or alter the preceding company's reported results.

36.2.9 Follow the tax difference into earnings

The initial charge to equity does not tell us the later cost of financing. That cost depends on how the debt grows and what the tax authority allows the issuer to deduct. To expose the

distinction, consider a separate July issue whose principal at October maturity is 100, with a coupon of 4. Its proceeds are again 100 and its initial debt value is 92. The supplied comparable rate for the 92-day period is now 3/23, so

$$L_0 = \frac{100 + 4}{1 + 3/23} = 92, \qquad I = 92\frac{3}{23} = 12, \qquad L_{\text{after coupon}} = 92 + 12 - 4 = 100.$$

The accounting interest of 12 includes both the coupon of 4 and the debt discount of 8. These are different terms from the preceding principal-92 instrument. At maturity the holder may instead convert the principal into six ordinary shares after receiving the coupon; the original conversion equity remains equity under IAS 32.AG32.

Now supply the fiscal treatment. The tax authority treats the initial 100 entirely as debt and permits a deduction for the coupon when it accrues. It allows no separate deduction for the accounting discount. The fiscal liability therefore rises to 104 before payment and falls to 100 when the coupon is paid. Assume sufficient taxable income to use the deduction and no additional tax on either maturity redemption or original-terms conversion. These are explicit illustrative fiscal facts, not consequences of a known tax rate. At the supplied rate of 25%, they give the following sequence.

Stage	Debt carrying amount	Tax base	Deferred tax liability
Issue	92	100	2
Interest accrued	104	104	0
Coupon paid	100	100	0
Principal settled	0	0	0

The first row reduces conversion equity from 8 to 6. The next row has a different earnings consequence: debit deferred tax liability 2 and credit deferred tax income 2. IAS 12.23 explicitly places these subsequent changes in profit or loss. The coupon deduction also reduces current tax expense by $4/4 = 1$. Thus the financing cost after tax is

$$I + \Delta T_{\text{current}} + \Delta T_{\text{deferred}} = 12 - 1 - 2 = 9.$$

There is no second expense of 2 for the initial equity charge. Conversely, using only the coupon would miss the discount accretion and its tax reversal (European Commission, 2023a, IAS 12, paragraphs 23, 58 and 61A; IAS 33, paragraphs 33–34).

The relation extends to the specified one-period fiscal treatment. Let P denote both proceeds and principal at maturity, L the initial debt, C the coupon and t the unchanged tax rate. Gross conversion equity is $E = P - L$; effective interest is $I = P + C - L = E + C$. The initial deferred liability tE reverses as book and tax amounts meet, while the coupon supplies a current deduction of C. Therefore

$$I - tC - tE = (1 - t)I.$$

A zero coupon still leaves discount accretion and deferred tax income. If $E = 0$, the deferred difference vanishes and only the coupon deduction remains. Different fiscal bases, changing rates or restricted deductions require their own schedule before this identity can be used.

Return to the same parent and subsidiary, preserving all their original operations and ownership changes. The new finance expense reduces group pretax profit from 136 to 124. Current tax falls from 28 to 27 and deferred tax expense from 6 to 4, leaving group profit $124 - 27 - 4 = 93$. Non-controlling interests still receive 7.35 because this financing belongs to the parent, so parent ordinary earnings are $93 - 7.35 = 85.65$. Conversion leaves cash $644 + 100 - 4 + 1 = 741$ and equity $724 + 6 + 100 - 9 = 821$. Cash redemption instead leaves cash $644 + 100 - 4 - 100 + 1 = 641$

and equity $724 + 6 - 9 = 721$. The issue-date tax charge is visible in the net equity contribution of 6; the later reversal is already included in the earnings cost of 9.

Diluted EPS uses the consequential earnings change as well as the dated shares. Under the explicit assumption of no other earnings or tax effects, assumed conversion removes interest 12, current tax relief 1 and deferred tax income 2. Its earnings addback is consequently 9, restoring parent earnings to $85.65 + 9 = 94.65$. The initial equity charge is outside that numerator. Six potential shares still run for 92 days; actual shares after October appear only in the conversion world. Using the same day counts as before gives

$$E_{\text{dil,conv}} = \frac{94.65}{4754/365} = \frac{138189}{19016}, \qquad E_{\text{dil,redeem}} = \frac{94.65}{4202/365} = \frac{138189}{16808}.$$

The incremental ratio $9/(552/365) = 1095/184$ is below both worlds' basic EPS, so the series is dilutive under continuing-operation control. The diluted fractions happen to equal those of the earlier instrument because both addbacks restore the same earnings and use the same share dates. The contracts have different actual earnings and principal obligations; equal diluted EPS does not make them economically interchangeable (European Commission, 2023a, IAS 33, paragraphs 32–38 and 41).

The documentary `CompoundTaxRollforward/1` calculation reports separate carrying amounts, fiscal bases, tax origins and journal movements. `CompanyTaxEPS/1` resolves the fixed company inputs and reconstructs the legal books before accepting an EPS projection. It refuses a coupon-only addback, a missing deferred-tax consequence, the wrong parent earnings or overlapping share periods. Missing fiscal findings remain unresolved, and different rights or tax schedules require another supported method. These calculations specify the supplied one-period case; early-conversion measurement, changing tax rates, loss limitations and operative jurisdictional authority remain separate obligations.

36.2.10 Apple: shares through time

Apple's disclosures show why the ownership record needs dates as well as quantities. In fiscal 2024, its opening 15,550,061 thousand common shares, less 499,372 thousand repurchased and plus 66,097 thousand issued net of employee-tax withholding, give 15,116,786 thousand closing shares. That bridge accounts for the change in the reported stock. It does not say how long each share participated in the year's earnings. The reported basic weighted average is 15,343,783 thousand shares (Apple Inc., 2024a, Notes 3 and 10, printed pp. 36 and 44).

The unit conversion matters before any timing inference. Net income is 93,736 million dollars, whereas the denominator is in thousands of shares. Thus $93{,}736 \times 1{,}000/15{,}343{,}783 \approx 6.109$ dollars per share, consistent with reported basic EPS of \$6.11. Using closing shares instead gives about \$6.201: it computes a different ratio. Adding the reported 64,312 thousand incremental shares from dilutive awards gives the diluted denominator of 15,408,095 thousand and EPS of about \$6.084, reported as \$6.08. These are checks using displayed amounts. Conditional nearest-rounding intervals also overlap the reported two-decimal EPS; the issuer's tie convention and undisclosed exact amounts remain unverified.

The first quarter of fiscal 2025 makes the distinction again. Income of 36,330 million divided by 15,081,724 thousand basic shares, with the same factor of 1,000, gives about \$2.409. The diluted denominator adds 69,141 thousand shares and gives about \$2.398, consistent with the reported \$2.41 and \$2.40 (Apple Inc., 2025c, Note 3, printed p. 7). Those denominators describe three months; adding them to other quarterly averages would not produce an annual share count. Period lengths and consistent measurement rules are needed to combine averages.

Even a weighted average leaves dates unresolved. To see exactly what it constrains, consider a

deliberately simplified common-share model for the 364 days of fiscal 2024. Apple reports a 52-week year ending 28 September 2024 (Apple Inc., 2024a, Note 1, printed p. 34). Let S_0 be the opening number of ordinary shares, and let ΔS_j be a signed change after d_j whole days have elapsed since 1 October 2023. Positive changes issue shares; negative changes repurchase them. An event then affects the remaining $364 - d_j$ days. Assume no splits or other quantity changes, and aggregate all net issuance into one event. The average is the opening holding plus each change weighted by its remaining time:

$$\overline{S} = S_0 + \sum_j \Delta S_j \frac{364 - d_j}{364}, \qquad 0 \le d_j < 364. \tag{36.1}$$

All quantities in this equation are ordinary shares and event days are integers. An immediate change receives full weight; a change after 363 days receives only one day's weight. Closing shares depend on the sum of changes, while equation (36.1) also constrains their timing. Neither relation separately identifies every change and date.

The construction must also respect information outside the annual EPS note. Apple reports fourth-quarter purchases of 35,697, 42,910 and 33,653 thousand shares in the intervals 30 June–3 August, 4–31 August and 1–28 September, respectively (Apple Inc., 2024a, Item 5, printed p. 19). A history putting all repurchases in the middle of the year fails that disclosure even if its annual average is correct. The later interim EPS note also reports a first-quarter fiscal 2024 basic average of 15,509,763 thousand shares. Both constraints are included in the following two authored quantity histories.

Event	History 1	History 2
First-quarter purchase	−40,298 (45)	−40,298 (45)
First-quarter purchase	−40,298 (46)	−40,298 (46)
Net issuance	+66,097 (182)	+66,097 (91)
Further purchase	−131,136 (179)	−15,643 (159)
Further purchase	−175,380 (180)	−290,873 (160)
June–August interval purchase	−35,697 (290)	−35,697 (290)
August interval purchase	−42,910 (322)	−42,910 (322)
September interval purchase	−33,653 (350)	−33,653 (350)

The table expresses changes in thousands of shares; parentheses give elapsed days, not negative signs. Each quantity is multiplied by 1,000 when used in equation (36.1). Treat the displayed source centres as exact solely for this illustration. Both histories start at 15,550,061,000 shares, repurchase 499,372,000, issue a net 66,097,000 and end at 15,116,786,000. Their first 91 days average 15,509,763,000 shares: the two equal early purchases reduce the average by $40{,}298{,}000(46 + 45)/91 = 40{,}298{,}000$. For the full year, the signed time-weighted changes sum to −75,085,192,000 share-days in each history. Dividing by 364 reduces the opening count by 206,278,000, giving the reported 15,343,783,000 average. An independent sum of daily balances gives the same result.

After 100 elapsed days, however, the first history has 15,469,465,000 shares and the second 15,535,562,000. The earlier issuance makes the difference. The fourth-quarter purchase dates lie inside their reported intervals, but none of these illustrative dates is an estimate of Apple's actual transactions. This establishes non-uniqueness for the stated quantity constraints. Prices, award-level settlement and monetary journals impose further conditions; the two examples are not a complete reconstruction of the filing or an exhaustive family, and no probability attaches to either. A query for shares on an intermediate date must therefore retain the timing alternatives or obtain additional evidence.

36.2.11 Awards, repurchases and cash measure different movements

The next quarter shows how another disclosure can narrow an unknown. Note 7 describes about 100 million repurchased shares, while Item 2's interval table gives 41,627, 32,784 and 25,379 thousand, totalling 99,790 thousand. The balance sheet reports 15,040,731 thousand closing shares against 15,116,786 thousand at the preceding year end. At the displayed centres, the quantity bridge requires a net other increase of $15{,}040{,}731 - 15{,}116{,}786 + 99{,}790 = 23{,}735$ thousand shares (Apple Inc., 2025c, printed pp. 3, 10 and 20). This narrows the total without identifying the events composing it. Repurchase authority is different again: both the note and programme footnote explicitly say that the programme does not oblige Apple to acquire a minimum amount of shares.

Restricted stock units have their own movement record. In thousands of RSUs, the fiscal 2024 bridge is $180{,}247 + 80{,}456 - 87{,}633 - 9{,}744 = 163{,}326$: opening awards plus grants, less vesting and cancellations. The next quarter gives $163{,}326 + 61{,}391 - 35{,}683 - 2{,}278 = 186{,}756$ (Apple Inc., 2024a, Note 11, printed pp. 44–45) (Apple Inc., 2025c, Note 8, printed p. 10). These gross award movements are different from net common shares issued or incremental shares in the diluted EPS calculation. Apple reports that most vested RSUs were net share settled and that approximately 31 million shares were withheld for employee taxes in 2024. Subtracting that rounded amount from 87.633 million vested RSUs does not explain the 66.097 million net issuance line in full.

The annual report describes several award types under the 2022 employee plan and the usual terms of its RSUs. In that reported practice, RSUs generally vest over four years subject to continued employment, settle one-for-one in common shares and carry dividend-equivalent rights. "Generally" cannot determine the vesting dates of every grant. The prior plan ceased authorising new grants in the third quarter of 2022; that does not say its outstanding awards vanished. Likewise, the approximately 24 million RSUs excluded as antidilutive in fiscal 2023 are a period-specific disclosure, not a universal exclusion rule (Apple Inc., 2024a, Notes 3 and 11, printed pp. 36 and 44–45). A future dilution calculation needs the applicable award terms and EPS rules as well as quantities; the arithmetic above does not validate that methodology.

Monetary statements impose another set of distinctions. The following amounts are millions of dollars, with reductions and payments shown as positive magnitudes (Apple Inc., 2024a, printed pp. 32–33 and 45) (Apple Inc., 2025c, printed pp. 4–5 and 10).

Reported measure	Fiscal 2024	Fiscal 2025 Q1
Repurchase reduction in equity	95,846	23,476
Cash-flow repurchases	94,949	23,606
Employee-tax withholding reduction in equity	5,605	2,993
Cash-flow employee-tax payments	5,441	2,921
Share-based compensation in equity	12,034	3,383
Share-based compensation expense	11,688	3,286

The withholding reductions combine the contributed-capital and retained-earnings rows: $3{,}993 + 1{,}612 = 5{,}605$ annually and $1{,}891 + 1{,}102 = 2{,}993$ in the quarter. These differences remain unallocated here; assigning them to excise tax, transaction fees or unsettled trades would require further evidence. The narrative's \$95.0 billion and \$23.3 billion purchase amounts also retain their own precision and transaction scope. The separate employee-tax settlement narrative cannot silently replace the cash-flow line.

Finally, grant-date value, vesting-date value, compensation expense and future unrecognised cost answer different questions. Apple's stated policy recognises expense on a straight-line basis for its estimate of awards that ultimately vest (Apple Inc., 2024a, Note 1, printed p. 34). Its

unrecognised RSU cost increases from $19.4 billion over a weighted-average 2.4 years to $29.2 billion over 2.9 years at the quarter end. Dividing either amount by its weighted period does not disclose a constant annual expense schedule or a cash liability: different cohorts have different remaining recognition periods, and future vesting estimates can matter even under the stated straight-line policy. The ownership model therefore needs linked award cohorts, recognition periods, share events and cash settlements. Keeping those dates and measurements distinct allows an actual later grant or settlement disclosure to narrow the histories without overwriting the reported EPS or inventing a missing event.

36.3 A plan's capacity is different from an employee's award

The share register has already separated awards, vesting and common shares issued. The documents governing an award explain why those quantities can move differently. Apple's 4 March 2022 Form 8-K supplies the 2022 Employee Stock Plan, a time-based restricted-stock-unit form and a performance-award form as Exhibits 10.1–10.3. These are the counterparts of three references in the annual report. The filing reports shareholder approval and effectiveness on 4 March 2022. The plan also records Board adoption on 9 November 2021 and uses that earlier date as its defined Effective Date when setting its term. The two dates perform different jobs; replacing them with one undifferentiated date would lose information needed to interpret the plan (Apple Inc., 2022, Items 5.02 and 5.07; Exhibit 10.1, Sections 1 and 15(a)).

Start with the capacity to make awards. Section 5(a) supplies 510 million shares plus specified amounts available or recycled from the 2014 plan, subject to a maximum of 1,274,374,682. That maximum is neither the current unused capacity nor a promise to issue that many common shares. Section 5(b) counts a full-value award at two reserve shares for each underlying share, whereas exercised options and stock appreciation rights use one reserve share for each share to which the exercise relates. Options and stock appreciation rights count their gross underlying amount even when fewer shares are delivered (Apple Inc., 2022, Exhibit 10.1, Section 5(a)–(b)).

A small hypothetical award makes the distinction concrete. Suppose 100 restricted stock units were initially charged against the reserve at two for one, all 100 vest and 20 shares are withheld for employee taxes. Assume no other adjustments. The employee receives 80 shares. Under the plan's full-value withholding rule, the 20 withheld shares return 40 reserve shares, because the original charge used that ratio. The two quantities are therefore

$$\underbrace{100 - 20 = 80}_{\text{common shares delivered}}, \qquad \underbrace{2(100) - 2(20) = 160}_{\text{net reserve shares used}}.$$

The reserve records authority used, rather than shares in the employee's account. An option exercise covering 100 shares can also deliver 80 after withholding, yet uses 100 reserve shares under the gross-count rule; the option withholding does not restore capacity. The same observed delivery therefore cannot identify the reserve movement without the award type and counting rule. Cash settlement creates another distinction: the plan excludes the shares that would otherwise have been delivered from its share count. These are consequences of Section 5(b), not an assumed employee tax rate, cash payment or compensation expense (Apple Inc., 2022, Exhibit 10.1, Section 5(b)).

The 2025 proxy illustrates why the reporting population must travel with each number. At 28 September 2024 its equity-compensation table reports 125,973,077 shares subject to outstanding awards under the 2022 plan, 37,387,521 under the 2014 plan and 10,612 under the director plan, totalling 163,371,210. The 2014 amount includes 39,984 options and 37,347,537 restricted stock units. Removing those options leaves 163,331,226 shares subject to restricted stock units across the three listed populations. The annual note's 163,326 thousand restricted stock units is a

different reported amount; these disclosures alone do not identify a reconciliation. The proxy separately reports 1,022,428,528 shares available under the 2022 plan, 66,924,428 under the employee purchase plan and 4,110,122 under the director plan, totalling 1,093,463,078. That total includes different plans from the outstanding-award total and is governed by their counting conventions (Apple Inc., 2025b, printed p. 97 and table footnotes). A reconstruction must retain these populations rather than subtracting unlike totals to infer grants, dilution or unused authority.

Capacity still does not tell the employee what will be delivered or when. Section 9 permits restricted-stock-unit agreements to differ: the agreement specifies vesting, settlement may be in shares, cash or a combination, and delivery may be deferred under its terms. Dividend-equivalent rights may be granted; the units themselves carry no voting rights. The filed time-based form then makes a more specific promise. Its Section 7 provides for shares equal to vested units less the applicable tax-related withholding, delivered promptly and no later than two and one-half months after vesting, subject to the stated conditions. The plan's wider permission does not replace that form's narrower promise (Apple Inc., 2022, Exhibit 10.1, Section 9; Exhibit 10.2, Sections 7 and 11).

The form's grant notice leaves the recipient, quantity, award date and vesting schedule blank. Those blanks matter: the annual report's general four-year practice cannot fill every individual notice. For example, the 2025 proxy describes named executives' 1 October 2023 time-based awards as vesting in three equal annual instalments beginning 1 April 2026, over approximately four and one-half years. It also describes termination terms under which earned or vested rights and delivery dates must be kept separate: Cook's retirement treatment does not accelerate the originally scheduled delivery of shares or dividend equivalents (Apple Inc., 2022, Exhibit 10.2, grant notice and Section 8) (Apple Inc., 2025b, printed pp. 51 and 53–54). These examples establish that a general vesting convention cannot substitute for the applicable award; they do not identify every employee's terms or a complete amendment history.

For the simulation, the consequence is concrete. Retain the plan version and reserve rule alongside the actual award terms, vesting conditions, units earned, settlement medium, delivery date and withholding choice. A permitted plan alternative becomes a branch for a particular award only when its terms or explicitly stated unresolved facts allow it. Otherwise cash-settlement paths would be invented for a share-only promise, or vesting would create shares before their contractual delivery. Tax amounts, accounting expense and diluted EPS require their own facts and methods. The performance-award form poses the next question: how are units earned when the contractual performance schedule itself has not been filled in?

36.4 Performance conditions determine units before settlement

The performance form turns the distinction between a plan and a grant into a calculation problem. Its notice leaves the target quantity, grant date, performance period and vesting schedule blank. Exhibit A likewise leaves the percentile thresholds and payout percentages blank, although it specifies interpolation between thresholds, a cap when Apple's absolute return is negative, and rounding to whole units. The form therefore supplies part of the method without supplying an individual award's inputs (Apple Inc., 2022, Exhibit 10.3, notice and Exhibit A).

The 2025 proxy provides a concrete schedule for the named executives' 1 October 2023 performance awards. It describes a three-year performance period from the beginning of fiscal 2024 through the end of fiscal 2026, with vesting on 1 October 2026 subject to the agreements. This is a dated description of that grant population, not the result of the performance period.

The proxy separately reports results for awards granted on 27 September 2020 and vested on 1 October 2023. Those older results must not be attached to the newer grants merely because the same proxy reports both (Apple Inc., 2025b, printed pp. 51–53).

For the October 2023 schedule, first suppose that the relative-return percentile has been supplied correctly. Write it as p on a scale from 0 to 100, so that $p = 70$ means the seventieth percentile. Let $m(p)$ be the number of units earned per target unit before applying the negative-return cap and employment conditions. The proxy gives zero below the twenty-fifth percentile, 25% of target at that percentile, 100% at the fifty-fifth and 200% at the eighty-fifth or above. Between those points it requires straight-line interpolation (Apple Inc., 2025b, printed p. 52). In these units,

$$m(p) = \begin{cases} 0, & p < 25, \\ \dfrac{1}{4} + \dfrac{p - 25}{40}, & 25 \le p < 55, \\ 1 + \dfrac{p - 55}{30}, & 55 \le p < 85, \\ 2, & 85 \le p \le 100. \end{cases}$$

The first slope is $(1 - 1/4)/(55 - 25) = 1/40$; the second is $(2 - 1)/(85 - 55) = 1/30$. Thus a supplied percentile of 40 earns 0.625 units per target unit, while 70 earns 1.5. The threshold at 25 is a jump, not a line running from zero at some unreported lower percentile. If two unresolved rank calculations produced only $p = 24.9$ or $p = 25$, a hypothetical target of 1,000 units would yield either zero or 250 units before other conditions. Replacing those two possibilities by every quantity from zero to 250 would create unsupported outcomes.

Relative performance is only one input. Let R be Apple's absolute total shareholder return over the same performance period, expressed as a fraction. The source caps vesting at 100% of target whenever $R < 0$, regardless of relative rank. For this part of the calculation the multiplier is

$$g(p, R) = \begin{cases} \min\{m(p), 1\}, & R < 0, \\ m(p), & R \ge 0. \end{cases}$$

At the supplied percentile 70, a hypothetical target of 1,000 units therefore gives 1,500 units when the return is nonnegative but only 1,000 when it is negative, assuming the other vesting conditions are met. A company can rank well in a falling market: the rank does not identify the sign of its return. The example avoids fractional units; Exhibit A's nearest-whole-unit rule still needs an explicit treatment of exact half-unit ties before implementing all inputs (Apple Inc., 2022, Exhibit 10.3, Exhibit A).

Computing R introduces its own source requirements. The form uses twenty-trading-day beginning and ending average prices, dividend reinvestment, and adjustments for specified capital changes. Consider a deliberately simple path with no dividends, distributions or corporate actions. Suppose each of the twenty beginning prices is 100; nineteen ending-window prices are 80 and the final price is 120. The beginning average is 100 and the ending average is $(19 \times 80 + 120)/20 = 82$, giving $R = (82 - 100)/100 = -18\%$. Substituting only the first and last prices would instead give $+20\%$. With the supplied percentile held at 70, that substitution would wrongly remove the cap and increase the hypothetical award from 1,000 to 1,500 units. Even in this simple case, the observation window changes a contractual consequence (Apple Inc., 2022, Exhibit 10.3, Exhibit A, Beginning Price and Ending Price).

The general return and rank calculation requires more than those two averages. The form defines the comparison group using initial S&P 500 membership and continued listing, with specific bankruptcy, successor and merger provisions. It ranks Apple ahead of a comparison member with equal return. That tie rule does not by itself specify a unique numerical percentile

formula; a library's default rank conversion cannot silently supply one. Dividends enter the return through reinvestment, noncash distributions need values, and corporate actions need the stated adjustments. A calculation must document how those provisions are implemented, including avoiding a second addition of dividends to a price series that already includes them. Unresolved ambiguities belong with the Committee's determination under the form; they are not permission to invent historical inputs (Apple Inc., 2022, Exhibit 10.3, Exhibit A).

Employment conditions then determine which otherwise earned units survive. Under Section 8 of the filed 2022 performance form, ordinary termination generally cancels unvested rights. Death or disability before vesting instead allows a fraction of the performance-qualified award to vest on the specified vesting date. The fraction is the total performance-period days less the days remaining at severance, divided by total days. For an expressly hypothetical eligible fraction of 3/5 and the uncapped 1,500-unit result, 900 units survive to that date; if the negative-return cap first reduces the result to 1,000, 600 survive. These calculations use the filed form's terms and assume no other modification, leave adjustment or recoupment event (Apple Inc., 2022, Exhibit 10.3, Sections 3–4 and 7–9).

The proxy's termination summary must remain alongside that versioned form, rather than overwrite it. It describes pro-rata *accelerated* vesting for named executives other than Cook upon death or disability, and different provisions for Cook. The 2022 form's Section 8 instead points to the vesting date. An applicable executed agreement, amendment or further disclosure is needed to establish which timing governs a particular executive. This is an unresolved applicability and timing distinction between the inspected sources, not an established contradiction for one identified award (Apple Inc., 2025b, printed p. 53).

Dividend-equivalent rights add a cash consequence to the same conditional award. The filed form credits ordinary cash dividends using the outstanding target units and pays the vested proportion in cash, subject to withholding and the other terms. Suppose one dividend of $0.25 occurs while all 1,000 target units are outstanding before severance, and there are no other dividends or adjustments. The initial credit is $250. A 1.5 performance multiplier with full eligibility pays $375 gross; under the assumed 3/5 eligible fraction, the vested proportion is 900/1,000, so $225 is payable. Applying the negative-return cap reduces that conditional payment to $150. This cash right is different from the hypothetical dividend reinvestment used to measure shareholder return; adding it to that return would count a different entitlement (Apple Inc., 2022, Exhibit 10.3, Section 5(b)).

The model consequently needs distinct stages: determine the applicable award and observations; calculate the return and rank under evidenced conventions; apply the schedule and cap; determine surviving rights; and apply settlement and withholding terms. The checked examples cover selected consequences with explicit inputs. They do not recover the October 2026 outcome, reproduce all individual awards, value compensation expense or establish diluted EPS. Those questions require further data and their own accounting methods.

36.5 The applicable agreement changes which rights survive

The timing question can be narrowed by reading the later forms already incorporated into Apple's filings. The annual exhibit list identifies a CEO performance form effective 25 September 2022, filed as Exhibit 10.2 with the quarter ended 31 December 2022. It also includes general and CEO performance forms effective 29 September 2024 as Exhibits 10.20 and 10.22. These are different versions and participant categories, even though their notices use similar headings and leave the individual grant particulars blank (Apple Inc., 2024a, annual exhibit list; Exhibits 10.20 and 10.22) (Apple Inc., 2023, Exhibit 10.2).

The 2024 general form does not remove the earlier timing distinction. Section 8 still provides that qualifying death or disability before vesting allows the performance-qualified award, multiplied by the severance fraction, to vest on the Vesting Date. Section 7 requires share delivery as soon as administratively practical thereafter, within two and one-half months and subject to its conditions. Thus neither inspected general form supplies a rule to deliver the shares immediately on severance. The proxy's description of accelerated vesting remains a summary whose application to a particular award needs further evidence (Apple Inc., 2024a, Exhibit 10.20, Sections 7–8) (Apple Inc., 2025b, printed p. 53). Later filing alone is not evidence that an earlier grant was amended.

The CEO forms change the surviving entitlement. In both the September 2022 and September 2024 versions, Section 8(b) allows unvested units to remain eligible for the Vesting Date following death, disability, or qualifying retirement, subject to the Committee's stated authority. These clauses do not apply the general form's severance fraction. Retirement combines an age of at least 60 and at least ten years of service with the stated service scope; the retirement exception normally also requires termination on or after the first anniversary of the Award Date. Earlier retirement does not receive that exception unless the Committee determines otherwise. Continued eligibility still depends on performance and does not make the eventual number of shares known on departure (Apple Inc., 2023, Exhibit 10.2, Sections 3–4 and 8) (Apple Inc., 2024a, Exhibit 10.22, Sections 3–4 and 8).

To isolate the effect, reuse the hypothetical 1,000-unit target and a performance multiplier of 1.5, with no modification, recoupment or other condition changing the result. Suppose death or disability occurs at a point giving a general-form severance fraction of 3/5. Under the general form, $1{,}000 \times 1.5 \times 3/5 = 900$ units remain eligible to vest. Under the CEO form's stated continuation provision, all $1{,}000 \times 1.5 = 1{,}500$ units remain eligible. With a negative-return cap reducing the multiplier to one, the corresponding quantities are 600 and 1,000. These comparisons hold the performance result fixed to expose the contract difference; they are not facts about any executive's award.

The difference carries into cash rights without creating an earlier payment. With the same hypothetical $250 dividend-equivalent credit established before departure, the uncapped gross payments are $225 under the general-form fraction and $375 under the CEO continuation provision. The units and cash rights remain linked to the same applicable terms and performance conditions. The CEO forms' Section 7 anchors delivery to the Vesting Date, and Section 5(b) ties dividend equivalents to the award's vesting and payment terms. Moving those amounts to the severance date would change the specified cash and share path (Apple Inc., 2023, Exhibit 10.2, Sections 5(b), 7–8) (Apple Inc., 2024a, Exhibits 10.20 and 10.22, Sections 5(b), 7–8).

A second boundary is independent of the performance calculation. An employee aged 60 with ten years of service has not necessarily satisfied the CEO form's first-award-anniversary condition. Conversely, passing that anniversary does not establish the age and service conditions. The record therefore needs the applicable agreement, Award Date, termination date and reason, service history and any relevant Committee decision. Unknown values remain unknown; they cannot be converted into automatic eligibility or forfeiture.

The later forms retain unfilled performance schedules and the same kinds of return, comparison-group and adjustment dependencies. They do not fill the rank convention or establish the actual October 2023 award terms by themselves (Apple Inc., 2024a, Exhibits 10.20 and 10.22, Exhibit A). The appropriate result of this comparison is a narrower conditional model: the general and CEO forms support different survival rules, both preserve later delivery, and the proxy's timing description still needs an award-specific explanation. A source-reference match establishes which form was retained; applicability establishes which form can govern a financial path.

36.6 A compound bond exposes classification dependencies

Consider an issuer receiving 100 for a two-year bond paying annual coupons of 5 and principal 100 at maturity, with a holder option to convert that principal into ten ordinary shares. Assume its currency is the issuer's functional currency, no other options, fees or contingent settlements, and that the conversion feature qualifies as equity under the selected rule. A comparable liability without conversion yields 10%. The inspected IAS 32-equivalent text requires separate liability and equity components and measures equity as the residual (Australian Accounting Standards Board, 2022d, paragraphs 28–32, AG30–AG32).

The liability's present value is

$$L_0 = \frac{5}{1.1} + \frac{105}{1.1^2} = \frac{11050}{121} \approx 91.3223, \qquad E_0 = 100 - L_0 \approx 8.6777.$$

Issue debits cash 100 and credits these two components. Under the stated effective-interest schedule, the first coupon leaves liability $1.1L_0 - 5 = 1050/11 \approx 95.4545$; after the second coupon it is 100. Redemption debits liability and credits cash 100. Alternatively, conversion at maturity debits liability and credits equity 100 while issuing ten shares; the original equity component remains in equity. AG32 supports the absence of a gain or loss on that maturity conversion. Exact rational valuations are kept before the declared monetary rounding.

A variable-number share settlement, cash alternative or other clause can change classification. The contract record must therefore retain these terms; it cannot select liability/equity accounts independently of the coupon, conversion and settlement rules. Early repurchase and induced conversion invoke separate operations: the inspected AG33–AG35 text describes allocation and the induced-conversion consideration difference. They are not implemented by changing the maturity date of the simple example. US-GAAP and ASBE classification require their own dated authority; the IFRS-oriented split is never silently reused under another basis.

36.7 Schedules and initial classification

The following liability and ownership methods use selected dated AASB 9 and AASB 132 provisions (Australian Accounting Standards Board, 2022i,d). Issuer, holder, contractual cash flow and accounting measurement remain separate throughout the schedule.

DebtContract contains parties, currency, principal, issue/settlement dates, payment calendar, fixed/index/spread/reset terms, compounding/day-count, business-day adjustment, principal changes, fees, options, security, priority and legal default/remedy terms. Each term has evidence/admission provenance. The economic schedule emits due cash independently of the recognition method. A market-value adapter may value that schedule but cannot choose amortised cost, FVOCI, FVTPL or liability/equity classification.

An investor's classification evaluates business model and contractual cash-flow features; a debt label alone cannot establish the AASB 9.4.1.2/4.1.2A conditions. At initial recognition the selected 5.1.1 rule measures fair value and qualifying costs as appropriate. An unknown day-one observable-value condition yields unresolved under 5.1.1A/B5.1.2A. The retained acquisition cases distinguish FVTPL expensed fees, amortised-cost fees, original EIR and POCI credit-adjusted EIR. No rate method inferred from market value can overwrite the contract rate.

For an issuer with a plain amortised-cost debt, net proceeds 100 and one legally due payment 121 at the end of two annual periods, the exact EIR solves 100=121/(1+r)^2, hence r=10%. Issue debit cash 100/credit debt 100; year-one debit interest 10/credit debt 10; year-two debit interest

11/credit debt 11; redemption debit debt 121/credit cash 121. Outstanding contractual payment stays 121 until settled even while carrying value is 100 or 110. Total interest 21 equals cash outflow 121 minus proceeds 100. An immediate transaction fee 2 changes net proceeds to 98 and the EIR root; retaining 10% is wrong. The numerical root method is in the numerical-method discussion; an irrational rate is not silently an exact rational.

A fractional period uses the declared convention. Annual-effective 21% over half a year produces factor 1.1 and interest 10 on 100; simple proration gives 10.5 and is wrong for that contract. A floating coupon resets from the admitted fixing available at the reset date, not a later observed rate. Actual day-count and holiday-calendar versions are immutable inputs. Unknown fixings preserve conditional paths, rather than defaulting to the latest value.

36.8 Modification, extinguishment and default

At a modification, retain original and replacement terms, counterparty, fees, effective date, legal-release evidence and original EIR. Determine derecognition before measurement. AASB 9.3.3.1–3.3.3 and B3.3.1–B3.3.6 govern the selected issuer-liability cases. The quantitative ten-percent test compares the PV of new terms, including specified borrower/lender fees, at original EIR against remaining original PV. It is not a universal asset test, nor proof that every change below ten percent is immaterial; changed currency, rights and other qualitative terms need the applicable analysis. Insufficient evidence remains unresolved. Third-party payment without legal release does not extinguish debt.

On the first anniversary above, old remaining 121 has PV 110. Replace with 133.1 due at the original maturity date, one year after the modification, with no fees and an explicit supported substantial-terms conclusion: PV 121 differs by exactly 10%, satisfying the inclusive test. Assume separately supported new fair value 118; debit old debt 110, debit extinguishment loss 8, credit new debt 118. The new EIR is $133.1/118-1$, not automatically the old rate. If the replacement is instead a stated non-substantial same-currency reduction to 115.5, no fees or other changes, PV 105 leads to debit debt 5/credit modification gain 5 under the B5.4.6 recalculation mechanism, then interest 10.5 and settlement 115.5. The actual change below ten percent still needs the qualitative finding. Repurchase of carrying amount 110 for 108 extinguishes the repurchased obligation: debit debt 110/credit cash 108/credit gain 2. Unamortised fees belong in the carrying amount before that comparison, not a second gain adjustment.

If cash available is 80 against debt 121 due, record the due obligation and an unpaid/default economic state; payment 80 leaves contractual unpaid 41 only when that partial payment actually occurs. Do not cancel the remaining liability, recognize a waiver or issue a loan unless a separately admitted event authorizes it. Covenant acceleration, cross-default, cure periods, collateral realization and restructuring require the actual legal terms and priority waterfall. A scenario may explore specified alternatives with explicit assumptions; it may not silently select the cheapest settlement as an accounting fact.

36.9 Equity and compound instruments

EquityInstrument retains class, voting/distribution/liquidation rights, issued, treasury and outstanding counts, par/legal-capital requirements and conversion terms. Share counts are integer quantities separate from currency balances. Issue 100 ordinary shares at 2 with no fees: debit cash 200/credit equity 200, issued and outstanding 100. Repurchase 10 at 3 for treasury: debit treasury-equity 30/ credit cash 30; outstanding 90, issued 100, treasury 10. Reissue 5 at 4: debit cash 20/ credit treasury-equity 15/credit other equity 5; outstanding 95, treasury 5.

There is no trading profit under the selected AASB 132.33 rule; a statutory par split requires that jurisdiction's facts. Dividend declaration creates a liability only at the admitted point at which the obligation exists; proposal of a dividend is not itself payment or automatically an expense.

For a qualifying fixed-for-fixed issuer conversion option in functional currency, AASB 132.28–32 separates liability fair value and residual equity. Assume total issue proceeds 100, standalone debt fair value 90, zero coupon, maturity two years and cash redemption 100 or fixed 10 ordinary shares; no extra derivative, tax or fee. Issue debit cash 100/credit debt 90/credit conversion-equity 10. Effective interest accretes debt 90 to 100. At maturity conversion under original terms debits debt 100 and credits issued-share equity 100; the original equity 10 stays in equity (a permitted internal transfer is not income). Shares increase 10 and no redemption cash leaves. Redemption instead debits debt 100/credit cash 100; unexercised equity is not remeasured into profit. AG 32 supports no conversion gain/loss in that stated case. Variable share count, cash alternatives, foreign currency and changed conversion terms require classification/modification analysis; they cannot be forced into this example.

Financial statement mappings separate interest, modification/extinguishment results, owner transactions and cash settlement. A financing-liability note reconciles opening carrying amount, cash proceeds/repayments, effective interest, fees, conversions, modifications and FX. Total shares and earnings-per-share denominators need dates and dilution conditions; outstanding shares at year end are not a substitute for a weighted period denominator. Full option valuation, hedge accounting, EPS and legal-capital regimes remain explicitly additional methods; the proposed records preserve their required inputs.

36.10 Changed conversion terms change the accounting question

A maturity conversion, an early repurchase and an inducement to convert all remove or alter debt, but they do not allocate the resulting consideration in the same way. The issuer must retain the old conversion rights, the revised rights, the date of amendment and the actual settlement. A record that says only *converted* loses the facts needed to distinguish these events. The following cases use the retained Australian standards for explicit calendar-2024/2025 for-profit Tier 1 illustrations, without optional later amendments. They are pretax: tax consequences remain uncomputed.

For an early repurchase with unchanged conversion privileges, the issuer allocates the consideration between the liability and equity components using the method of the original separation. The liability allocation uses its current fair value; the difference from its carrying amount enters profit or loss, while the equity allocation stays in equity (Australian Accounting Standards Board, 2022d, AG33–34 and IE39–46). The standard's example has a face amount of 1,000, half-yearly coupons of 50 and ten payments remaining. The original half-year discount rate is 5.5%; the contemporaneous comparable nonconvertible debt rate is 4%. For either rate r, the remaining debt value is

$$V(r) = 50\frac{1-(1+r)^{-10}}{r} + 1{,}000(1+r)^{-10}. \tag{36.2}$$

The first term values the ten coupons; the second values principal at the last payment. The source rounds the carrying and current fair values to 962 and 1,081. Repurchase consideration of 1,700 then allocates 619 to equity and produces debt settlement expense of 119. The entries debit debt 962, expense 119 and equity 619, and credit cash 1,700. Expensing the whole $1{,}700 - 962 = 738$ would misclassify the equity purchase. The original equity component of 60 remains within equity; it does not cap the 619 deduction. Exact present-value fractions and the

source's rounded presentation remain separate in the accompanying calculations.

An inducement asks a different valuation question. At the amendment date, compare the fair value of what the holder receives under the revised conversion terms with what the holder would have received under the original terms. Their difference is an inducement loss (Australian Accounting Standards Board, 2022d, AG35 and IE47–50). In the retained example, changing the conversion price from 25 to 20 on face amount 1,000 increases shares from 40 to 50. At the amendment-date share quotation of 40, the additional ten shares are worth 400. A later quotation of 50 would give 500 and would use the wrong measurement date.

For an explicitly immediate, unconditional share-only conversion, suppose the liability's admitted carrying amount is 960 and the original equity component is 60. This carrying amount is a separate illustrative input. The resulting entry debits debt 960 and inducement expense 400, and credits issued equity 1,360; 50 shares are issued. Original conversion equity remains in equity. The 400 expense offsets the corresponding part of the issued equity in total equity, while derecognizing the debt increases net equity by 960. There is no cash movement. Noncash financing disclosure preserves the event without inventing a financing receipt (Australian Accounting Standards Board, 2023a, 43–44).

This immediate case does not determine the recognition timing of a delayed, contingent or cash inducement. Those terms must first establish the obligation. Nor does an ability to settle in shares establish equity classification. A derivative with a net-cash alternative fails the all-equity-alternatives condition, and an obligation to deliver a variable number of shares worth a fixed sum does not represent a residual ownership interest (Australian Accounting Standards Board, 2022d, 21 and 26–27). The system must refuse the fixed-for-fixed method when these facts are missing or contradicted, while retaining the instrument and its unresolved classification.

36.10.1 An offer, an amendment and delivery can occur on different dates

The immediate conversion above combines several events that need not coincide. An issuer can announce an offer before it changes any enforceable right; the holder can accept a binding amendment before shares are delivered. Retain the announcement, approval or acceptance conditions, legally effective amendment, discharge of the old debt, and delivery separately. The following conditional examples use IAS 32 and IFRS 9 in the retained September 2023 EU compilation. Their dates and terms are synthetic. IAS 32.AG35 measures the inducement at the amendment date, while IFRS 9.3.3.1 and B3.3.1 remove the debt when it is discharged, cancelled or expires, including a qualifying legal release by the creditor. Administrative delay alone does not keep a discharged debt outstanding (European Commission, 2023a, IAS 32.13, AG35; IFRS 9.3.3.1–3, B3.3.1–6).

Suppose an offer is announced on 1 January 2024 but, under its supplied legal terms, changes no rights until approval and the holder's binding acceptance on 5 January. Acceptance releases the issuer from the old cash obligation in return for 50 fixed ordinary shares; delivery occurs on 20 January. There is no cash alternative, further condition or variable-share obligation. The ordinary shares satisfy the equity definition. The old terms would have delivered 40 shares. Share quotations are 30, 40 and 50 on the three dates, respectively. The incremental ten shares give an amendment-date loss of $10(40) = 400$. Using the announcement quote would give 300; using the delivery quote would give 500. Both would measure the wrong date.

With debt carrying amount 960 and original conversion equity 60, debit debt 960 and inducement expense 400, and credit equity to be issued 1,360 on 5 January. Original conversion equity remains within equity. On 20 January, transfer 1,360 from equity to be issued to issued equity; the new quotation does not remeasure that equity-classified obligation (European Commission,

2023a, IAS 32.15–16, 21–22, AG32–35). Starting with cash 1,020, debt 960 and equity 60 gives closing cash and total equity 1,020: original equity 60 plus new equity 1,360 less loss 400. Neither the announcement nor administrative delivery creates another expense or cash flow. Recognition within equity does not settle the separate IAS 33 question of when shares enter the weighted denominator; retain the actual participation and issue terms for that assessment.

Change the accepted consideration to the original 40 fixed ordinary shares plus a fixed cash bonus of 100 payable later. Assume zero discount and no intervening interest in this short illustration, with the old debt legally released at acceptance. The incremental consideration and loss are 100. Debit debt 960 and expense 100; credit equity to be issued 960 and bonus payable 100. Equity is then $60 + 960 - 100 = 920$, with cash 1,020 and the payable 100. Later payment debits the payable and credits cash 100, leaving cash and equity 920 without a second loss. The payment buys the debt holder's early conversion and is a financing outflow under the stated facts; the share exchange is disclosed as noncash financing (European Commission, 2023a, IAS 32.AG35; IAS 7.6, 17, 43–44).

These entries require the supplied legal and classification findings. A binding instrument with a contingent cash clause can already be a liability under IAS 32.25; it is not equivalent to the expressly nonbinding offer above. An obligation to deliver shares worth a fixed cash amount uses a variable quantity and is also a liability. A derivative's cash settlement alternative fails equity classification unless all alternatives qualify as equity (European Commission, 2023a, IAS 32.19, 21, 25–27). The procedure first determines whether rights changed, then classifies each new obligation, measures the inducement at the effective amendment, and separately tests debt discharge. If the debt has not been released, retain and measure that debt and assess the actual modification under IFRS 9; do not copy the 960 debit from this case. Unknown approval effects, contingent consideration values or discharge terms remain unresolved inputs to that procedure, with the original history intact.

36.11 A hedge reserve is neither cash nor a second gain

Suppose a company hedges a highly probable inventory purchase. A derivative gain can arrive before the inventory affects earnings. Cash-flow hedge accounting preserves that timing through a separate equity reserve, but only after the relationship qualifies. The designation must identify an eligible instrument and item, the hedged risk, the risk-management objective and the assessment method at inception. An economic relationship, the absence of credit-risk dominance and an appropriate actual hedge ratio are separate requirements (Australian Accounting Standards Board, 2022i, 6.2–6.4 and B6.4.1–19). Favorable price correlation alone does not establish them. Unknown designation or valuation information remains unresolved; it cannot be repaired by assigning zero or placing a gain in other comprehensive income.

Take one documented, offsetting exposure and a derivative worth zero at designation. The illustration includes the whole derivative, no excluded option or forward component and no intermediate reserve release. Let D_t be its cumulative gain, including settled gains, and H_t the cumulative change in the hedged item's present value attributable to the designated risk. Their signs refer to their economic values. With admitted offsetting changes, the cumulative effective gain E_t and ineffectiveness I_t are

$$E_t = \operatorname{sgn}(D_t)\min\{|D_t|, |H_t|\}, \qquad I_t = D_t - E_t. \tag{36.3}$$

The reserve follows the lower absolute cumulative amount; effective changes enter other comprehensive income and the remainder enters profit or loss (Australian Accounting Standards Board, 2022i, 6.5.11). This formula handles the stated offsetting case; a complex exposure outside it needs its own measurement method. In particular, one same-direction period is not itself

proof that no economic relationship exists. Item valuation must include time value and only the item's relevant features; a hypothetical derivative cannot import features that exist solely in the actual hedging instrument (Australian Accounting Standards Board, 2022i, B6.4.5 and B6.5.4–6).

Initially, supplied values $D_1 = 12$ and $H_1 = -10$ produce effective gain 10 and profit 2. At the next measurement, $D_2 = 15$ and $H_2 = -14$ produce effective gain 14 and cumulative profit 1. The second period therefore records a derivative increase of 3, other comprehensive income of 4 and a loss of 1. Applying the minimum only to that period's changes would incorrectly give effective gain 3. The retained cumulative values are essential economic memory.

Now buy the inventory for 114 and settle the derivative for cash of 15. Settlement removes the derivative asset; it does not by itself release the 14 reserve. Because the forecast transaction has produced a nonfinancial asset, debit the reserve 14 and credit inventory 14. Inventory cost becomes 100. This direct basis adjustment is not an other-comprehensive-income reclassification (Australian Accounting Standards Board, 2022i, 6.5.11(d)(i)). Selling the inventory for 130 subsequently produces revenue 130 and cost of sales 100.

With opening cash and equity of 200 and no other movements, the full sequence has cumulative pretax profit 31, other comprehensive income 14 and a separate equity-to-inventory basis transfer of −14. Closing cash and pretax equity are both 231. The equity reconciliation is

$$200 + 31 + 14 - 14 = 231, \tag{36.4}$$

where omitting the direct basis transfer would overstate equity. Gross operating cash is $-114+15+130 = 31$: the identified hedge's cash flows follow the inventory position's classification (Australian Accounting Standards Board, 2023a, 16). A matching terminal cash total does not excuse a wrong interim reserve, profit or basis.

If the hedged payments are interest, the reserve instead enters earnings when the interest affects earnings (Australian Accounting Standards Board, 2022i, 6.5.11(d)(ii)). With an effective gain of 8, derivative settlement of 8 and cash interest of 18, release 8 from the reserve into earnings. Expense is then 10. The release has an OCI reclassification of −8, but no second cash receipt. The illustrative nonfinancial company consistently elects operating classification for interest under the retained rule; net operating cash is −10 (Australian Accounting Standards Board, 2023a, 31–33). A later presentation regime needs its own applicability decision.

Stopping hedge accounting also requires a reason. If the future cash flows remain expected, the accumulated reserve remains until the flows occur, subject to the unrecoverable-loss rule. If they are no longer expected, the reserve is reclassified immediately to profit or loss. No longer highly probable is not the same as no longer expected (Australian Accounting Standards Board, 2022i, 6.5.11(d)(iii) and 6.5.12). The worked alternatives keep a gain reserve of 10 in the first case and release it in the second. A loss reserve of 10 with an unrecoverable portion of 4 instead releases expense 4 and leaves a loss reserve of 6. Each decision changes a named exposure and reserve; it cannot be inferred from the derivative's cash settlement alone.

36.11.1 Fair-value adjustments and benchmark changes

An admitted fair-value hedge of amortised-cost debt uses a different route. Both the derivative change and the risk-attributable debt adjustment enter profit or loss; the latter changes the debt carrying amount (Australian Accounting Standards Board, 2022i, 6.5.8). For debt initially carried at 100, a derivative gain of 8 and debt-value increase of 10 give a derivative asset of 8, debt of 110 and net loss of 2. No cash-flow hedge reserve is created.

Once the debt ceases receiving hedge adjustments, the retained adjustment must be amortized using a recalculated effective interest rate (Australian Accounting Standards Board, 2022i,

6.5.10). Suppose contractual principal of 100 and coupon of 21 are payable together one year later. From carrying value 110, the recalculated rate is $121/110 - 1 = 10\%$, giving effective interest expense 11 before settlement. Cash interest remains 21 and principal remains 100; the difference between cash interest and effective expense amortizes the 10 adjustment. Reusing the old 21% rate would over-accrete the liability. Missing contractual dates or payments leave that calculation unresolved.

Benchmark reform is not a general exemption from this analysis. The retained standard's rate-update expedient applies to necessary reform changes whose new cash-flow basis is economically equivalent to the old basis. Additional changes then receive the ordinary modification or derecognition analysis (Australian Accounting Standards Board, 2022i, 5.4.5–9). Comparing one 4% coupon with 3.8% plus a 0.2% spread verifies one number, not equivalence of all future terms. The amendment record must retain the legal change, rates, spreads, reset terms, dates and the supporting equivalence analysis.

A qualifying change to formal hedge designation is neither automatic discontinuation nor a new relationship. Existing cash-flow reserves follow the alternative benchmark under the specified conditions rather than being reset to zero. Other qualifying requirements continue to apply; additional changes and mixed benchmark groups need their own assessment (Australian Accounting Standards Board, 2022i, 6.9.1–10). The rate-specific 24-month exception for a separately identifiable risk component likewise requires a supported expectation and ends on the stated adverse reassessment (Australian Accounting Standards Board, 2022i, 6.9.11–13).

The exact conditional journals and adverse alternatives are retained in `data/accounting-master/finite-method-decisions-20261015/financing/`; the field and event specification is `docs/design/proposal/methods/financing-changes-20261015.md`. These cases specify the financial consequences of admitted values and eligibility findings. The next section extends the treatment to excluded components and rebalancing. Actual valuation, delayed or contingent inducements and group hedges retain their named method and evidence obligations. Supplying a market value is useful input; it is not a completed valuation method.

36.12 Excluded hedge components and rebalancing

The following conditional methods use the retained AASB 9 compilation for calendar-2024/2025 Australian for-profit Tier 1 illustrations, without optional later amendments. Actual authority admission and component valuation remain separate requirements.

36.12.1 The option premium pays for protection

A purchased option can protect an inventory purchase even if it expires with no intrinsic value. Its time value is the price of that protection. When only intrinsic value is designated, applying the earlier whole-derivative method would mix the price of protection with the payoff that offsets the purchase. AASB 9.6.5.15 (Australian Accounting Standards Board, 2022i, 6.5.15 and B6.5.29–33) separates them. The designated intrinsic-value movement follows the applicable hedge method; the related time-value movement accumulates in a different equity component. The nature of the hedged item determines when that cost affects earnings. A transaction such as an inventory purchase uses a cost attached to that transaction. Protection over a period uses systematic, rational amortisation over the period in which the intrinsic-value hedge adjustment can affect earnings. The life of the option alone does not decide which route applies (B6.5.29–30).

The event record must identify the instrument, designated component, underlying item and

risk, critical terms, actual and aligned time values at designation and every measurement date, intrinsic-value changes, separate reserve balances, the transaction or time-period classification, amortisation policy, and each basis transfer or reclassification. Existing intrinsic-value effectiveness requirements remain necessary. Unknown component valuations cannot be replaced with a total derivative value and a zero time value.

Suppose a qualifying inventory hedge begins with cash and equity of 200. Buy an option for 6, entirely time value. Cash is 194 and the option asset is 6; there is no initial expense. The first supplied valuation is intrinsic value 12 and aligned time value 4. The option is worth 16. Its increase of 10 comprises an effective intrinsic gain of 12 and a time-value loss of 2, both in OCI but in separate reserves. At purchase and expiry, intrinsic value is 20 and time value zero. The option is worth 20, the intrinsic reserve is 20 and the cost reserve is negative 6. The last period's change of 4 comprises intrinsic gain 8 and time-value loss 4.

Buy the inventory for 120 and settle the option for 20. Settlement removes the option asset and leaves cash 94. The intrinsic reserve reduces inventory cost by 20. Removing the negative cost reserve adds 6 to inventory cost:

$$120 - 20 + 6 = 106.$$

Both direct transfers affect the asset and equity; neither is a second cash flow or an OCI reclassification. Cash 94 plus inventory 106 equals equity 200 after the transfers. Selling for 130 gives profit 24 and cash 224. The cash bridge is $200 - 6 - 120 + 20 + 130 = 224$. The equity bridge is opening 200, profit 24, cumulative OCI 14, and a direct basis transfer of negative 14. Omitting the premium would report cost 100 and overstate profit by 6. For a forecast sale instead, the accumulated cost enters earnings when the sale does. An unrecoverable portion enters earnings immediately under 6.5.15(b)(iii), rather than waiting for a transaction that cannot recover it.

36.12.2 Protection over time and imperfectly matching terms

Consider separately an option that protects an existing inventory holding for two equal periods. Admit aligned initial time value 6 and a rational even allocation over those periods. Intrinsic-value and inventory movements are zero in this illustration, isolating the protection cost. After one period, supplied time value 2 gives an OCI loss of 4. Amortisation of 3 is reclassified to expense, leaving the cost reserve at negative 1. After expiry, the final time-value loss of 2 and second amortisation of 3 leave no reserve; cumulative expense is 6. Immediately expensing each observed time-value fall would instead charge 4 and 2. That is a different timing policy. If the intrinsic-value hedging relationship is discontinued after the first period, 6.5.15(c) requires the remaining net reserve loss of 1 to enter earnings immediately. The remaining option asset of 2 is still an asset; discontinuation does not destroy it.

When critical terms differ, B6.5.32 requires an aligned option valuation using terms that perfectly match the hedged item. The hypothetical option is a measurement input for the related cost, not a replacement of the actual contract. Under B6.5.33(a) (Australian Accounting Standards Board, 2022i, B6.5.32–33), if actual time value exceeds aligned time value at inception, base the cost reserve on aligned changes and put the difference between actual and aligned changes in profit or loss. For initial actual/aligned values 6/4 and later values 3/3, actual loss 3 separates into OCI loss 1 and profit loss 2. At expiry, cumulative losses 6/4 leave OCI loss 4 and profit loss 2, before the required transfer or reclassification.

If actual time value is initially lower, B6.5.33(b) limits the reserve by the lower cumulative change of actual and aligned time value; the remainder of actual changes enters earnings. In a

same-direction decline example, initial values 2/3 become 1/2.5. Their cumulative losses are 1 and 0.5, so the cost reserve holds loss 0.5 and profit holds loss 0.5. At expiry, cumulative losses are 2 and 3: the reserve loss becomes 2 and cumulative profit from the mismatch returns to zero. The second period therefore records OCI loss 1.5 and a profit gain of 0.5, while actual value falls 1. Applying a fresh lower-of rule only to each period would lose this reversal. These numbers test declining positive time values; they do not define a signed-value selector for every collar or complex combination. Such cases must retain the source-defined alignment and valuation analysis rather than infer it from this small example.

A zero-cost collar can have zero initial time value and nonzero intermediate time value. B6.5.31 still requires the related changes in OCI. A supplied movement from zero to 1.2 and back to zero produces OCI gain 1.2 and loss 1.2; zero initial premium is not permission to omit both. The terminal transaction cost adjustment and initial-cost amortisation are zero in that example.

36.12.3 The actual option and the hedged price need separate calculations

The time-value comparison also has to fit the risk being hedged. An actual option may have a different exercise price from the price level the company has designated. Its payoff then differs from the one-sided change in the hedged purchase. IFRS 9.B6.5.5 permits a hypothetical derivative to measure that item but prohibits importing features that belong only to the actual instrument. B6.5.32 similarly measures aligned time value using an option whose critical terms match the item (European Commission, 2023a, IFRS 9.B6.5.5, B6.5.32–33). The following complete transaction uses the retained historical EU compilation; it is a distinct conditional example from the preceding AASB-profile illustrations.

Consider a one-unit inventory purchase with a designated one-sided price risk above 110. The company buys a call with strike 100. At designation, spot is 90 and both calls have zero intrinsic value. The actual option costs 6, entirely time value; the hypothetical call with strike 110 has aligned time value 4. These component values and the qualifying designation are supplied. The company pays only the actual premium. The hypothetical option is a calculation of the hedged risk, not another acquired asset.

At a later spot of 115, actual intrinsic value is $115 - 100 = 15$, while the hedged item's aligned intrinsic value is $115 - 110 = 5$. Suppose actual and aligned time values are both 3. The actual option is worth 18 and the hypothetical option is worth 8. The effective intrinsic gain is 5; the remaining actual intrinsic gain of 10 enters profit. Because initial actual time value exceeded aligned time value, B6.5.33(a) gives a cost reserve change of $3-4 = -1$ and time-value mismatch profit of $(3 - 6) - (-1) = -2$. Thus total reserve is 4 and profit is 8, together accounting for the actual option's gain of $18 - 6 = 12$.

The distinction can be expressed directly from the contractual price levels. Let x be the observed price, q the common quantity, K_A the actual strike and K_H the hedged price level. The two intrinsic values are

$$I_A = q\max(x - K_A, 0), \qquad I_H = q\max(x - K_H, 0).$$

With zero initial intrinsic values, these are also the cumulative intrinsic gains. In the supplied qualifying relationship, the intrinsic reserve is $R_I = \min(I_A, I_H)$, and intrinsic mismatch profit is $I_A - R_I$. Writing actual and aligned time values as T_A, T_H, with designation values T_{A0}, T_{H0}, the admitted cumulative-loss cases give

$$R_T = \begin{cases} T_H - T_{H0}, & T_{A0} > T_{H0}, \\ -\min(T_{A0} - T_A, T_{H0} - T_H), & T_{A0} < T_{H0}. \end{cases} \qquad \Pi = (I_A - R_I) + (T_A - T_{A0} - R_T).$$

Here R_T is the separate cost reserve. Both time values remain between zero and their own initial values in this profile; the second branch compares cumulative losses. It is not a signed-value rule for every collar or opposed change. Matching initial values are admitted here only with fully matched terms and matching subsequent component values. The intrinsic payoff identity does not price either time value.

At expiry spot 130, actual intrinsic value is 30 and aligned intrinsic value is 20, while both time values are zero. The intrinsic reserve is 20, the cost reserve is -4, and cumulative profit remains $10 - 2 = 8$. Start with cash and capital 200: pay premium 6, pay the inventory supplier 130, and receive actual option settlement 30. Cash closes at 94. The two basis transfers give

$$\text{Inventory cost} = 130 - 20 - (-4) = 114, \qquad 94 + 114 = 200 + 8.$$

The intrinsic reserve reduces the inventory cost, while the negative cost reserve adds the related protection cost. Both transfers are separate from cash under IFRS 9.6.5.11(d)(i) and 6.5.15(b)(i) (European Commission, 2023a, IFRS 9.6.5.11, 6.5.15). If actual intrinsic value 30 were copied into the hedged-item calculation, inventory would become 104 and profit -2. Cash 94 plus inventory 104 would still equal capital 200 less the loss 2. The book would balance while measuring the wrong hedged risk.

For the lower-actual-time-value branch, reverse the strikes: actual strike 110, hedged price 100, and initial actual/aligned time values 2 and 3. At spot 105, actual intrinsic is zero and aligned intrinsic is 5. With time values 1 and 5/2, the smaller cumulative loss is 1/2; the cost reserve and mismatch profit are each $-1/2$. At spot 115, time values 1/2 and 1 give cumulative losses 3/2 and 2. The cost reserve is now $-3/2$ and cumulative time mismatch profit is zero. The intervening profit gain of 1/2 reverses the earlier mismatch. Applying a new lower-of comparison only to each period would leave a reserve of -1 and miss that reversal.

At expiry spot 120, actual/aligned intrinsic values are 10 and 20, and time values are zero. The intrinsic reserve is 10, the cost reserve is -2, and cumulative profit is zero. From capital 200, cash closes at $200 - 2 - 120 + 10 = 88$ and inventory at $120 - 10 + 2 = 112$. The full premium, supplier payment and actual settlement are again visible; the hypothetical payoff of 20 creates neither an asset nor cash.

The read-only `AlignedOptionComponents/1` report binds each observation to both contract identities and checks its intrinsic/time decomposition. Six paths include matched terms, out-of-the-money expiry and doubled quantities. Unknown component values, copied critical terms, omitted premiums, mixed rule profiles and unsupported cumulative changes produce no partial report. Actual pricing, qualifying findings, general signed components and subsequent company or sector integration retain their separate requirements.

36.12.4 A zero-cost collar can have a signed interim cost reserve

Equal option premiums can conceal a changing net time value. A purchaser may pay for a call that limits price increases and finance it by writing a put that gives up price decreases below a lower strike. Even when those premiums cancel, the two options can have different time values before expiry. IFRS 9.B6.5.31 expressly includes a purchased/written option combination with net nil time value at designation: its related time-value changes enter other comprehensive income, although the cumulative change over the full relationship is zero (European Commission, 2023a, IFRS 9.B6.5.31–32).

Consider a highly probable purchase of ten units. A purchased call has strike 11 and a written put has strike 9, both expiring on the purchase date. Spot is initially 10, so both intrinsic values are zero. The call premium paid and put premium received are each 6. They are two gross cash flows and two actual contract balances, even though their net cash effect is

zero. Assume external contracts, a qualifying combined designation that is not a net written option, documented matching risk components, complete values and continuing qualification. IFRS 9.6.2.5–6 permits joint designation subject to those eligibility restrictions; equal premiums alone do not establish them (European Commission, 2023a, IFRS 9.6.2.1–6, 6.3.7, 6.4.1).

Let q be the common quantity, $L < U$ the lower put and upper call strikes, and S the spot price at a measurement date. Denote the positive call asset by C and positive put liability by P. Their supplied nonnegative time values are T_C and T_P. The contract payoffs give

$$I_C = q\max(S-U,0), \quad C = I_C + T_C,$$
$$I_P = q\max(L-S,0), \quad P = I_P + T_P.$$

The put enters the purchaser's net position with a negative sign. In this finite measurement profile, zero discount and the supplied matching item-risk measure make the hedged-item change exactly opposite to $I_C - I_P$. No carry, credit or basis difference remains. The lower-of test therefore gives the signed intrinsic reserve $R_I = I_C - I_P$, with no ineffectiveness, while the fully aligned cost reserve is $R_T = T_C - T_P$. Consequently

$$C - P = R_I + R_T.$$

The reserves retain different origins. A zero sum cannot justify dropping either one. This use of the matching item measure follows the declared designation; a hypothetical derivative cannot import instrument features absent from the item (European Commission, 2023a, IFRS 9.6.5.11, B6.5.5).

One intermediate valuation has spot 11.7, call asset 15 and put liability 3. Call intrinsic value is 7, leaving call time value 8; the put has no intrinsic value and time value 3. Thus $R_I = 7$ and $R_T = 5$. At a later valuation, spot is 8.3, the call asset is 3 and the put liability is 15. Put intrinsic value is now 7 and its time value is 8, giving $R_I = -7$ and $R_T = -5$. The earlier common-loss restriction for purchased-call time values cannot describe this signed reversal.

These synthetic marks also have a simple consistency check. At zero discount, terminal-price weights on $(6, 9, 10, 11, 14)$ of $(1/10, 1/10, 1/10, 1/5, 1/2)$ give mean price 11.7, mean call payoff 15 and mean put payoff 3 for ten units. Reversing the weights gives mean price 8.3 and option values 3 and 15. Equal weights give the inception price 10 and premiums 6 each. These are separate mathematical witnesses for the supplied marks; they are neither estimated probabilities nor a joint transition or forecast model across dates.

At expiry, both time values vanish. The purchaser pays the supplier qS, receives call payoff I_C and pays put payoff I_P, as distinct cash flows. The final inventory cost, after the intrinsic reserve is included in its basis, is

$$B = qS - I_C + I_P = q\min\big(\max(S,L),U\big).$$

Below 9 the written put raises the ten-unit cost to 90; between 9 and 11 the cost is the supplier price; above 11 the call limits it to 110. Terminal prices 6, 9, 10, 11 and 14 therefore give costs 90, 90, 100, 110 and 110. With initial capital 300, cash closes at 210, 210, 200, 190 and 190. Derivative balances, both reserves and earnings close at zero, so cash plus inventory remains 300. The time-value basis adjustment is zero because $R_T = 0$ at expiry, exactly as B6.5.31(a) requires. Its interim OCI history remains part of the account.

`AlignedZeroCostCollar/1` returns the two actual contracts, both signed reserves, proposed balanced journals and all five gross cash origins. It requires sufficient cash at each supplied payment step: call premium before put premium receipt, then supplier payment before derivative settlements. A future observation, missing qualification, wrong component value or netted supplier amount returns no partial report. The method supplies no option pricing model, tax treatment or cash-flow-activity classification. Unmatched actual/aligned changes under B6.5.33 and reserve attribution to heterogeneous stopped purchase portions require separate support.

36.12.5 Forward elements and currency basis spreads

AASB 9.6.5.16 (Australian Accounting Standards Board, 2022i, 6.5.16 and B6.5.34–39) permits the corresponding treatment when the forward element is excluded from a spot-designated forward, or a foreign-currency basis spread is excluded. The accounting election is explicit; designation of spot alone does not automatically make the excluded component an OCI reserve. Apply the transaction/time-period distinction and aligned-value rules in B6.5.34–39. For the forward element, the inception comparison uses absolute actual and aligned amounts. Retain their signed fair values and the changes used for the reserve; do not compare signed inception amounts as though a negative forward element were necessarily smaller in economic magnitude. B6.5.39 carries the same guidance to currency basis spreads. The exact valuation of unmatched forward elements and collars remains a required method for those applications.

36.12.6 Rebalancing preserves the history of the hedge

A changed hedge ratio can preserve the original risk-management objective. It does not erase the accounting before the change. Under 6.5.5–6.5.6 and B6.5.7–21 (Australian Accounting Standards Board, 2022i, 6.5.5–6.5.6 and B6.5.7–21), first measure and recognise ineffectiveness under the existing designation, then adjust designated quantities prospectively. Distinguish ordinary fluctuations around a valid ratio from a changed relationship that requires a new ratio. A changed objective requires discontinuation of that relationship; a counterparty default is not repaired by changing quantities. Update the documented expected sources of ineffectiveness after rebalancing.

Before a change, cumulative derivative gain 12 and offsetting item loss 10 give reserve 10 and profit 2. Immediately before rebalancing, admitted values 18 and negative 14 instead give reserve 14 and cumulative profit 4. Record the period's derivative gain 6 as reserve gain 4 and profit gain 2 before adjusting the quantities. A reset to equal amounts at the new designation date would hide the accumulated ineffectiveness.

Store each retained, added and removed volume with its designation date and valuation reference. An added item layer begins at the rebalancing date; the earlier layer keeps its old reference. The source's example has 100 tonnes designated at forward price 80 and adds 10 tonnes when the price is 90. In a zero-discount illustrative comparison at price 95, their changes are $100(95 - 80) + 10(95 - 90) = 1550$. Applying the old price to all 110 tonnes gives 1650, inventing a gain of 100 on the added layer. Actual measurement must use the admitted risk, dates, discounting and contractual terms.

Adding an instrument similarly starts its change at the new designation date without resetting the old instrument. Decreasing designated instrument volume leaves any retained undesignated derivative at fair value through profit or loss unless another qualifying designation applies. For example, removing 10% of a homogeneous instrument from the designation leaves its total carrying value unchanged; of a subsequent total fair-value gain 10, the undesignated 1 enters profit and the designated 9 follows the continuing hedge's effectiveness test. This split does not assert that all 9 is effective. Decreasing item volume invokes partial-discontinuation rules for the removed part. Whether its forecast cash flows remain expected determines the reserve's later release; removal from designation alone does not mean cancellation.

36.12.7 A new ratio does not restart cumulative measurement

The origin of an added layer matters again at every later measurement. Consider a continuing hedge of purchases with an unchanged hedging instrument. The original purchase volume is

100 tonnes, designated at a forward price of 80 currency units per tonne. For this illustration, assume zero discounting and one common price risk; actual valuations must support those simplifications. When the price reaches 90, the cumulative loss on the hedged purchase is 1000. A separately supplied cumulative derivative gain of 1200 gives reserve 1000 and cumulative ineffectiveness income 200.

Suppose the documented economic relationship now requires ten more tonnes to be designated, while the risk-management objective remains unchanged. Recognize the old designation's ineffectiveness first, then add those tonnes at their current price of 90. At a later price of 95, the cumulative hedged-item loss is

$$100(95 - 80) + 10(95 - 90) = 1550.$$

If the cumulative derivative gain is then 1400, IFRS 9.6.5.11 requires a reserve of $\min(1400, 1550) = 1400$, leaving cumulative ineffectiveness zero. The derivative has gained only 200 since the last observation, but reserve must increase by 400: debit derivative 200, debit ineffectiveness expense 200 and credit reserve 400. The earlier income of 200 reverses (European Commission, 2023a, IFRS 9, paragraphs 6.5.11(a)–(c), B6.5.8 and B6.5.17).

Restarting the lower-of calculation at the ratio change gives a different, incorrect answer. It adds the smaller of the new derivative gain 200 and new item loss 550 to the old reserve, leaving reserve $1000 + 200 = 1200$ and profit 200. That journal can balance, yet it fails the cumulative rule. Rebalancing continues the relationship; it does not turn successive measurement intervals into separate hedges.

For repeated additions, retain each layer's quantity q_ℓ and designation forward price f_ℓ. At current price p_t, the signed cumulative change in the hedged purchase is

$$H_t = -\sum_\ell q_\ell(p_t - f_\ell).$$

The minus sign records a loss on the purchase when its price rises. Let D_t be the separately measured cumulative derivative gain. In this finite offsetting case, the signed reserve and cumulative profit are

$$R_t = \operatorname{sgn}(D_t)\min(|D_t|, |H_t|), \qquad I_t = D_t - R_t.$$

At each observation post only ΔD_t, $-\Delta R_t$ and $-\Delta I_t$ to derivative, reserve and gains, using debit-positive signs. Their sum is zero because $\Delta D_t = \Delta R_t + \Delta I_t$. Adding a layer at its current price gives that layer an initial change of zero, so it creates no immediate journal and does not reset earlier amounts. The same argument applies at each subsequent addition.

For example, add another 20 tonnes at 95 after the preceding measurement. At price 97 the cumulative item loss is $100(97-80)+10(97-90)+20(97-95) = 1810$. A cumulative derivative gain of 1900 then gives reserve 1810 and profit 90. Later price and derivative reversals are evaluated from these same origins. They can reverse previously recognized ineffectiveness rather than merely add another positive amount. Fewer intermediate observations leave the final cumulative amounts unchanged when the designation history and final valuations are the same.

The documentary `RepeatedItemRebalancing/1` report applies this add-only item method to a finite dated event list. Each addition requires an immediately preceding same-date measurement, a new layer identifier, the current designation price and updated eligibility/documentation findings. Missing findings are unresolved; different risk or valuation methods and false eligibility findings are unsupported. Changes in units, dates or layer origins reject the whole proposed report without posting. The offset arithmetic does not itself establish an admissible hedge ratio. Removed item volume, changed instrument volume and the reserve's treatment when the purchase occurs require the separate rules described above.

36.12.8 An added derivative has its own starting value

Increasing the instrument volume introduces a second measurement origin. IFRS 9.B6.5.19 retains the existing instrument's history and measures the added instrument's change from the date of rebalancing. If the company pays for an instrument with a positive fair value, that payment establishes an asset under IFRS 9.5.1.1. Only changes after designation contribute to its hedge gain (European Commission, 2023a, IFRS 9, paragraphs 5.1.1 and B6.5.19).

Return to the original 100-unit exposure designated at price 80, measured at price 90 with derivative gain 1200, reserve 1000 and cumulative profit 200. Here the item volume stays at 100. Suppose changed circumstances require adding an eligible external derivative A, acquired and designated together for its fair value of 200. The risk-management objective is unchanged, the revised ratio is eligible and the documentation is updated. Debit derivative A 200 and credit cash 200. Total derivative carrying value is now 1400, but the cumulative hedge gain remains 1200: the company acquired the additional 200 of value with cash. Reserve and profit remain 1000 and 200.

At price 95, suppose the original instrument is worth 1400 and A is worth 220. The original instrument has gained 1400 since its zero-value designation; A has gained $220 - 200 = 20$. The combined gain is therefore 1420, while total carrying value is 1620. The item loss is $100(95 - 80) = 1500$, giving reserve 1420 and cumulative profit zero. Since the preceding measurement, derivative value rose by 220, reserve rose by 420 and earnings fell by 200. The entry reverses the earlier ineffectiveness. Substituting carrying value 1620 for gain 1420 would give reserve 1500 and profit 120, both incorrect for this designation history.

The general relation separates the acquired value from later change. Let b_j be instrument j's fair value when it is acquired and designated, and $V_j(t)$ its current value. The original zero-value instrument has $b_0 = 0$. With no settlement or removal, define total acquired value B, cumulative hedge gain D_t and total derivative carrying value V_t by

$$B = \sum_j b_j, \qquad D_t = \sum_j \big(V_j(t) - b_j\big), \qquad V_t = B + D_t.$$

An acquisition increases both B and V_t by the same amount, leaving D_t unchanged. Later changes in value alter D_t. The reserve calculation uses D_t and the cumulative hedged-item change H_t:

$$R_t = \operatorname{sgn}(D_t)\min(|D_t|, |H_t|), \qquad I_t = D_t - R_t,$$

within the same offsetting, supplied-valuation conditions as the preceding example. The item layers keep their original designation prices; an instrument purchase does not reset them.

Acquire a second additional derivative for 50 after that measurement. At the next observation, price 97, let the original instrument and the two additions have values 1900, 260 and 60. Carrying value is 2220; cumulative gain is $1900 + (260 - 200) + (60 - 50) = 1970$. The item loss is 1700, so reserve is 1700 and cumulative profit 270. With opening cash 20000, the two acquisitions leave cash 19750. Cash plus derivatives is $19750 + 2220 = 21970$, equal to capital 20000, reserve 1700 and profit 270. The acquisition payments belong in that reconciliation even though they created no earnings.

Different instruments can later have opposite carrying signs. If, at price 79, the values become -200, 100 and 20, the company has derivative assets 120 and a derivative liability 200. Their signed sum is -80, whereas cumulative hedge loss is $-80 - 250 = -330$. The item gain is 100, leaving a reserve loss 100 and cumulative loss 230. Cash 19750 plus assets 120 equals liability 200 plus equity 19670. Keeping the separate positions preserves their meaning; the signed total alone gives no permission to offset assets and liabilities in the statements.

The documentary `AddedInstrumentLayers/1` report checks each instrument's identity, acquired value, designation date and subsequent marks. Missing marks, unfunded purchases and unsupported designation changes are refused. Marks and eligibility are supplied findings, with all effects of the whole instruments included. This acquisition example assumes equal transaction price and fair value, no fees and no earlier local book position. An already-owned derivative, excluded valuation components, instrument removals and settlement need their own histories and methods. The unchanged objective and updated designation remain substantive conditions; changing quantities merely to suppress observed ineffectiveness is not justified by these calculations (European Commission, 2023a, IFRS 9, paragraphs 6.2.1–6.2.6, B6.5.12–15 and B6.5.21).

36.12.9 Reducing an instrument preserves the purchase and its earlier hedge history

An instrument addition starts a new contribution to hedge gains. An instrument reduction stops part of the future contribution. It does not erase the gains already accumulated or necessarily reduce the purchase being hedged. IFRS 9.B6.5.18 leaves measurement of the hedged item unchanged when the instrument volume is reduced; B6.5.8 requires ineffectiveness to be measured immediately before rebalancing and treats rebalancing as continuation (European Commission, 2023a, IFRS 9.B6.5.8, B6.5.16–18).

Suppose two eligible external derivatives jointly hedge a highly probable purchase of 100 units, originally designated at price 10. Derivative A has nominal quantity 60 and B has quantity 40. Both initially cost zero and have zero value. At the first observation, the purchase price is 11, A is worth 60 and B is worth 80. These complete values and the qualifying relationship are supplied facts. The total derivative gain is 140, the hedged purchase change is -100, and the lower-of calculation gives reserve 100 and cumulative profit 40.

Now suppose a documented change in the relationship requires reducing A's designated quantity by 30 while the risk-management objective remains unchanged. Half of A remains designated, all of B remains designated, and the purchase stays at 100 units. The company still owns both derivatives. The removed half of A is worth 30 at the reduction date. Taking 30% of the aggregate value 140 would give 42, which is wrong: A and B have different values per nominal unit. IFRS 9.6.2.4(c) permits a proportion of an entire instrument, and 6.2.5 permits joint designation of combinations (European Commission, 2023a, IFRS 9.6.2.4–6.2.5). Those permissions do not make every instrument unit financially interchangeable.

The reduction itself changes no value, reserve or profit. A's earlier gain of 60 remains in the history, including the 30 attributable to the portion now removed. Only subsequent changes on that portion are outside the hedge and enter profit under B6.5.16. At a later price of 12, suppose A is worth 120 and B is worth 100. The gains accumulated while designated are

$$G = 140 + \tfrac{1}{2}(120 - 60) + (100 - 80) = 190.$$

The removed half of A contributes a further gain $U = \frac{1}{2}(120 - 60) = 30$ outside the hedge. Together, $G + U = 220$, the full value of the two derivatives. The purchase change is still measured on all 100 units from price 10, giving $H = -200$. The reserve is 190 and cumulative profit is $(190 - 190) + 30 = 30$. Profit has fallen from 40 to 30 since the previous observation: reversal of earlier ineffectiveness exceeds the new undesignated gain by 10.

This history can be represented without treating the remaining nominal ratio as a valuation rule. For each instrument j, let $V_j(t)$ be its whole value, $w_j(t)$ its remaining designated fraction, and a_{jk} the fraction removed at date u_{jk}. All fractions refer to that instrument's original nominal amount. With zero initial value and no intermediate sale or settlement, its designated and

undesignated gains are

$$G_j(t) = w_j(t)V_j(t) + \sum_k a_{jk}V_j(u_{jk}),$$
$$U_j(t) = \sum_k a_{jk}\{V_j(t) - V_j(u_{jk})\}, \qquad w_j(t) + \sum_k a_{jk} = 1.$$

The first line retains each stopped portion's value at the date its contribution ceased. The second assigns only its later change to profit outside the hedge. Adding the lines gives $G_j + U_j = V_j$. An equivalent chronological calculation adds $w_j\Delta V_j$ to designated gains and $(1 - w_j)\Delta V_j$ to undesignated gains between observations. Changing w_j after a completed measurement creates no jump. This is a local deduction of prospective rebalancing and cumulative measurement, rather than an allocation rule for an unidentified reserve.

Sum the component histories to obtain G and U. While the continuing hedge has offsetting cumulative instrument and item changes, IFRS 9.6.5.11 gives

$$R = \operatorname{sgn}(G)\min(|G|, |H|), \qquad \Pi = (G - R) + U.$$

The reserve remains attached to the unchanged purchase. Deleting a removed instrument's prior gain would reset history. Applying the matched purchase-portion method in the following Section 36.12.10 would instead remove purchase units and change the target. Neither describes this instrument-only reduction. Missing marks, uncertain qualification, an altered objective or an unidentified portion prevents this calculation from being admitted.

The identity of the removed instrument matters even when the nominal result looks identical. Remove 30 units of B instead of A. The stopped fraction is $30/40 = 3/4$, worth 60 at the reduction date. The remaining designated nominal is again 70. At the same later marks, however,

$$G = 140 + (120 - 60) + \tfrac{1}{4}(100 - 80) = 205, \qquad U = \tfrac{3}{4}(100 - 80) = 15.$$

Reserve is now 200 and profit is $(205 - 200) + 15 = 20$. The same remaining nominal ratio therefore produces a different reserve and earnings history.

Complete the original A-reduction case by receiving the 100 units at price 12 and settling both derivatives at the final marks. With opening cash and capital 2000, pay the supplier 1200, receive 120 on A and receive 100 on B. Closing cash is 1020. Inventory initially costs 1200; transferring reserve 190 into its basis under IFRS 9.6.5.11(d)(i) leaves inventory 1010 (European Commission, 2023a, IFRS 9.6.5.11). Cash plus inventory is 2030, equal to capital 2000 plus cumulative profit 30. The supplier payment, two settlements and basis transfer remain distinct entries. In the B-reduction case, the same cash 1020 joins inventory 1000 and profit 20 instead.

The `InstrumentReduction/1` report retains component marks, designated fractions, stopped values, prior gains and settled cash. Eight authored paths also check unchanged marks, no reduction, proportional component reductions, a smaller item change, a sign reversal and simultaneous derivative assets and liabilities. All future changes on owned undesignated portions remain visible. The method requires a complete purchase-and-settlement bridge and returns no partial journals on refusal. Supplied valuation, recoverability, qualifying facts and documentation remain substantive prerequisites. Actual and aligned option time values, tax, cash-flow activity classification and allocation among heterogeneous discontinued purchases require their separate methods.

36.12.10 A stopped hedge portion leaves a reserve behind

Return to the original forecast purchase of 100 tonnes at an initial forward price of 80. At the first observation, price is 90 and the derivative has gained 1200: the cash flow hedge reserve is

1000 and cumulative ineffectiveness is 200. Now management finds that 20 tonnes are no longer highly probable, while the remaining 80 tonnes continue to satisfy the hedge conditions. The derivative is still owned. Stopping hedge accounting for a portion therefore changes where its later gains go, even if no contract is sold and no cash changes hands.

IFRS 9.B6.5.27(b) requires partial discontinuation for the forecast volume that is no longer highly probable; paragraphs 6.5.6 and B6.5.22 make the change prospective. Paragraph B6.5.23 prevents voluntary discontinuation of a relationship that continues to qualify. These are the inspected historical EU 2023/1803 provisions (European Commission, 2023a). The example supplies the qualification and documentation findings; a percentage reduction alone cannot establish them.

For this case, the 100 tonnes have identical per-unit exposures, one original measurement date and one price origin. The identified 20 tonnes correspond to 20% of the entire derivative, whose proportional designation is permitted by paragraph 6.2.4(c). This homogeneity matters. Multiplying both sides of the lower-of calculation by a nonnegative fraction a gives

$$\min\{a|D|, a|H|\} = a\min\{|D|, |H|\},$$

where D is cumulative derivative gain and H the cumulative change in the hedged cash flows. With $a = 0.2$, the existing amounts divide as follows, without creating another gain or resetting the original price:

	Continuing 80 tonnes	Stopped 20 tonnes	Total
Derivative value and gain	960	240	1200
Cash flow hedge reserve	800	200	1000
Prior ineffectiveness	160	40	200

The stopped portion now carries two distinct origins: derivative value 240 and accumulated reserve 200. Suppose a later observation gives price 95 and whole-derivative value 1400. The continuing derivative portion has gained $0.8 \times 1400 = 1120$ since original designation; its hedged cash-flow change is $-80(95 - 80) = -1200$. Its reserve is therefore 1120 and its cumulative ineffectiveness is zero. The earlier continuing ineffectiveness of 160 has reversed.

The stopped derivative portion is now worth $0.2 \times 1400 = 280$. Its additional gain of $280 - 240 = 40$ enters profit under paragraph 5.7.1; it does not increase the stopped reserve. Alongside the prior stopped ineffectiveness of 40, it leaves total cumulative profit of 80. If the stopped purchase is still expected, total reserve is $1120 + 200 = 1320$. Applying the lower-of rule to the whole original position would instead give reserve 1400 and profit zero. That calculation would incorrectly keep hedging the portion whose designation has ended.

The distinction between “highly probable” and “expected” decides what happens to the frozen 200. Paragraph 6.5.12 explicitly allows a forecast transaction to remain expected after it ceases to be highly probable. Its reserve then remains until the cash flows occur, subject to the unrecoverable-loss rule. If the stopped purchase is no longer expected, the remaining 200 moves immediately from the reserve to profit. Total reserve and profit become 1120 and 280. A query or later valuation does not itself make that expectation judgment.

Follow both outcomes into inventory, using the supplied later price and settlement values. If all 100 tonnes arrive, supplier payment is 9500 and the remaining reserve of 1320 reduces initial inventory cost to 8180. With opening cash and capital of 20000, receiving derivative settlement of 1400 leaves cash $20000 - 9500 + 1400 = 11900$. Cash plus inventory is 20080, equal to capital plus profit 80. If the stopped 20 tonnes are canceled, only 80 tonnes arrive: payment 7600 less reserve 1120 gives inventory 6480, while cash is 13800. Net assets of 20280 then equal capital

plus profit 280. The released reserve cannot also reduce inventory. Supplier and derivative cash movements remain separately identified.

A loss reserve needs an additional check. Reverse the price path: the first observation is 70 with derivative value −1200, followed by price 65 and value −1400. Before any loss-recovery adjustment, continuing and stopped reserves are −1120 and −200, with total profit −80. Suppose new evidence shows that 60 of the stopped loss will not be recovered. Paragraph 6.5.11(d)(iii) requires that 60 to enter profit immediately, leaving stopped reserve −140 and total profit −140. If the full purchase then occurs, payment 6500 plus the remaining total loss reserve of 1260 gives inventory cost 7760. If the stopped purchase is canceled instead, only its remaining loss of 140 is released. Charging the original 200 again would count the earlier 60 twice.

This allocation follows from identified homogeneous portions. Different price origins, unmatched instrument reductions or a heterogeneous reserve need their own allocation analysis. The example also assumes an eligible external non-option derivative, supplied complete valuations, recoverability before the specified loss finding, no fees and a pretax reporting scope. Those facts make the reserve history reconstructible; they do not provide a pricing model or a general rule for splitting any accumulated reserve.

36.12.11 Different designation origins require their own gain histories

A reduction of five tonnes does not identify a reserve when the tonnes entered at different prices. Consider two forecast purchase origins, A and B, each of ten tonnes. A was designated at a price of 8 and B later at a price of 9. Each has its own documented match to an external linear instrument from the date of designation. The instruments entered at zero value and cost. Under the supplied model there is no discounting, carry, credit or basis difference, and each instrument's gain exactly offsets its named item's risk change. Prices are currency units per tonne; gains and all balances below are currency units. These are synthetic measurement assumptions, rather than a procedure for deriving actual forward values from spot prices.

At a subsequent measurement price of 11, A's instrument has gained $10(11 - 8) = 30$, while B's has gained $10(11 - 9) = 20$. The relationship's reserve is 50. If five tonnes from A cease to qualify, their recorded effective gain is $5(11 - 8) = 15$. If five from B cease to qualify, it is $5(11 - 9) = 10$. In both cases the stopped quantity is one quarter of the total. Multiplying 50 by that nominal fraction gives 12.5, which is wrong for either identified history. Quantity without the designation origin has lost information needed for this calculation.

The accounting basis remains the historical IFRS 9 text in EU 2023/1803 (European Commission, 2023a). Paragraph B6.5.17 preserves an added volume's own designation date and reference price; paragraph B6.5.20 leaves the measurement of continuing quantities unaffected by removal. Paragraphs 6.5.6 and B6.5.22–28 make cessation prospective when qualification fails. The example supplies the required documentation, individual eligibility, management as a group and proportional variability under paragraphs 6.4.1 and 6.6.1–3. It also supplies the finding that the stopped forecast volume is no longer highly probable and that the rest continues to qualify. Choosing a quantity in a report does not establish those facts.

The retained gain histories support a narrower deduction than a general allocation of a heterogeneous reserve. Let origin j have quantity q_j, designation date d_j and original price f_j. At a supplied measurement price S, its cumulative instrument gain and the opposite change in its hedged purchase cash flows are

$$G_j = q_j(S - f_j), \qquad H_j = -G_j.$$

Only origins already designated at that date enter the sums. All active origins' gains have the

same weak sign at each supplied measurement, and there is no historical ineffectiveness. For the relationship as a whole, $D = \sum_j G_j$ and $H = \sum_j H_j = -D$. Consequently the cumulative signed lower-of calculation required by paragraph 6.5.11 gives

$$R = \operatorname{sgn}(D) \min\{|D|, |H|\} = D.$$

At zero, both reserve and ineffectiveness are zero. The independently documented, fully effective origin gains therefore reconcile to the relationship's reserve. This reasoning retains the relationship calculation; it does not substitute a sum of independent lower-of measurements. The match must have existed from designation, rather than be invented later to explain an aggregate reserve. The derivation establishes one traceable attribution for this exact history, without establishing uniqueness for general histories.

Suppose quantity z, strictly between zero and q_k, stops from named origin k at a same-date measured price S_d. Keep the actual instruments. Denote the frozen gross gain on that portion by F_0, and the continuing quantities by $q'_k = q_k - z$ and $q'_j = q_j$ otherwise. Their original dates and prices remain in force. At a later price S,

$$\begin{aligned} F_0 &= z(S_d - f_k), \\ R_c &= \sum_j q'_j (S - f_j), \\ U &= z(S - S_d). \end{aligned}$$

Here R_c is the continuing relationship's reserve, again obtained from its exact offset and cumulative lower-of test. The stopped instrument portion's change U after cessation enters profit under paragraph 5.7.1; it cannot enlarge the stopped reserve. The identity follows by splitting the original k-origin quantity:

$$q_k(S - f_k) = (q_k - z)(S - f_k) + z(S_d - f_k) + z(S - S_d).$$

Adding the other origins gives total actual derivative value $D = R_c + F_0 + U$.

To follow later expectation or loss findings, let L be cumulative signed reserve released to profit. A released gain makes L positive; a released loss makes it negative. Remaining stopped reserve F and cumulative profit P then satisfy

$$F = F_0 - L, \qquad P = U + L, \qquad D = R_c + F + P.$$

The distinction between no longer highly probable and no longer expected remains the one in paragraph 6.5.12. Expected cash flows retain F, subject to the immediate unrecoverable-loss requirement in paragraph 6.5.11(d)(iii). Cancellation releases the entire remaining F once. If a loss was released earlier, only the balance remains available for cancellation or an eventual inventory transfer.

Follow the two origins to a terminal price of 12. A's actual instrument settles for 40 and B's for 30 in every history; neither instrument was sold when hedge accounting stopped. If five tonnes from A were stopped and then canceled, the continuing reserve is $5(12 - 8) + 10(12 - 9) = 50$. The stopped instrument's post-cessation gain is $5(12 - 11) = 5$, and releasing its frozen gain of 15 gives profit 20. If instead five tonnes from B were stopped and canceled, continuing reserve is $10(12 - 8) + 5(12 - 9) = 55$. Releasing its frozen gain of 10 gives profit 15. For example, canceling A debits the stopped reserve by 15 and credits profit by 15; the later gain of 5 is a separate instrument remeasurement.

With opening cash and capital of 300, both canceled histories buy fifteen tonnes for 180. Paragraph 6.5.11(d)(i) transfers the attached reserve into the nonfinancial asset's initial cost. Supplier cash and the two actual instrument settlements remain gross:

	Stop A, then cancel	Stop B, then cancel
Continuing reserve before purchase	50	55
Profit	20	15
Supplier payment	180	180
Actual instrument receipts	70	70
Inventory cost	130	125
Final cash	190	190
Cash plus inventory	320	315

The identical derivative cash does not make the histories interchangeable. Their different original effective gains have different consequences when the associated forecast purchases are canceled.

If the stopped purchases remain expected and occur instead, twenty tonnes cost 240. Stopping A leaves reserve $50 + 15 = 65$ attached; stopping B leaves $55 + 10 = 65$. Both then have inventory cost 175, cash $300 - 240 + 70 = 130$, and profit 5. The equality of these final totals is consistent with different intermediate reserve origins. An implementation must retain those origins because later cancellation or loss findings act on the named stopped amount.

The same signs work for losses. With the original prices still 8 and 9, stop five tonnes from A at price 7. Their frozen reserve is -5. A supplied finding that 2 will not be recovered releases -2 to profit and leaves -3 attached. If the purchase remains expected and the terminal price is 6, continuing reserve is $5(6 - 8) + 10(6 - 9) = -40$, the stopped instrument's later gain is $5(6 - 7) = -5$, and profit is $-5 - 2 = -7$. Purchasing twenty tonnes for 120 and attaching reserve -43 gives inventory 163. Actual instruments settle at $-20 - 30 = -50$, leaving cash 130 and net assets 293, equal to capital less the loss of 7. If cancellation follows the partial loss release, only the remaining loss of 3 moves to profit. A subsequent rise in instrument value does not restore the 2 already released from the stopped reserve.

More generally, let Q be continuing purchase quantity plus z only if the stopped purchase still occurs. Immediately before purchase, the attached reserve is $B = R_c + F$. For opening cash and capital C_0,

$$I = QS - B, \qquad C = C_0 - QS + D, \qquad C + I = C_0 + (D - B) = C_0 + P.$$

Thus the inventory basis I, final cash C and profit reconcile without using net settlement to conceal financing. The finite example pays the supplier before settling each instrument in origin order, so every intermediate cash balance must also be nonnegative. Capital of 240 exactly funds the retained twenty-tonne purchase at 12; 239 does not, even though the later instrument receipts would make final cash positive. This payment order is a supplied convention, rather than a universal legal requirement.

The associated read-only method, `OriginMatchedCessation/1`, retains the item and instrument identities, original dates and prices, each supplied measurement, signed releases and the final basis transfer. It proposes balanced journals without posting them. Missing qualification or matching facts leave the report unresolved; contradictory values are rejected. Historical ineffectiveness, opposed active-origin gains, acquired nonzero origins, additional designations, repeated stops and a failure of continuing loss recoverability lie outside this finite method. A negative resulting inventory basis is also refused: the accounting for that extension has not been established here. General heterogeneous attribution remains open. Paragraph B6.6.14 addresses allocation across income-statement lines and does not supply the missing stopped-reserve measurement rule.

36.12.12 When the hedged purchase becomes inventory

A hedge reserve does not remain in equity indefinitely once the forecast purchase takes place. When that purchase produces inventory, IFRS 9.6.5.11(d)(i) requires the accumulated reserve to be included directly in the inventory's initial carrying cost. This transfer is not a reclassification through profit or loss. The carrying cost reaches earnings later, when the goods are sold, under IAS 2.34 (European Commission, 2023a, IFRS 9, paragraph 6.5.11(d)(i); IAS 2, paragraphs 10–11 and 34).

Return to the three purchase layers: 100 tonnes at designation price 80, ten at 90 and twenty at 95. Suppose all 130 tonnes are received when the supplier price is 97. The supplier is paid $130 \times 97 = 12610$. At that date the accepted cumulative derivative gain is 1900, reserve gain 1810 and ineffectiveness income 90. Assume the derivative is fully settled for its carrying value, with no fees or further valuation movement. The terms permit that settlement; no such right is inferred merely from the numbers.

Record the supplier purchase at 12610, then receive 1900 and remove the derivative asset. To transfer the reserve, debit the cash-flow hedge reserve 1810 and credit inventory 1810. Inventory therefore carries

$$K = 12610 - 1810 = 10800.$$

The 90 of previously recognized ineffectiveness remains in profit. Starting with cash and equity of 20000 before the hedge, cash after purchase and settlement is $20000 - 12610 + 1900 = 9290$. Cash plus inventory, $9290 + 10800 = 20090$, equals opening equity plus that profit of 90. The reserve transfer creates no new income or cash.

The same direction rule handles a reserve loss. Take instead the later observation with supplier price 80, derivative liability 500, reserve loss 400 and cumulative ineffectiveness loss 100. All 130 tonnes cost 10400; settling the derivative requires a further payment of 500. The reserve is a debit balance, so transfer it by debiting inventory 400 and crediting the reserve 400. Inventory again carries $10400 + 400 = 10800$. Cash is $20000 - 10400 - 500 = 9100$, and cash plus inventory equals 19900, consistent with the loss of 100. These are two different terminal observations, not two optional treatments of the same unchanged purchase.

In signed notation, let C be the supplier's cost, R the cumulative reserve gain (negative for a loss), D the signed derivative settlement receipt and $I = D - R$ the cumulative ineffectiveness income. Then

$$K = C - R, \qquad \Delta\text{Cash} = -C + D.$$

The basis transfer is debit reserve R, credit inventory R, reversing both directions if $R < 0$. It changes neither cash nor earnings. Suppose all the goods are subsequently sold for cash S, with recognition conditions satisfied and no further costs. Cost of sales is K, so cumulative pre-tax profit becomes

$$S - K + I = S - (C - R) + (D - R) = S - C + D.$$

For sale proceeds of 14500, the gain case ends with profit 3790 and cash 23790; the loss case ends with profit 3600 and cash 23600. The purchase and sale are operating cash flows. IAS 7.16 classifies the cash flows of an identifiable hedge consistently with the hedged position, so the derivative settlement here is also operating, while retaining its own gross receipt or payment origin (European Commission, 2023a, IAS 7, paragraph 16).

Checking only those final totals would miss an important error. Crediting the positive reserve directly to profit at purchase would leave inventory at 12610 and pre-sale profit at 1900. It would reach the same total profit after the inventory was sold, because the excessive inventory cost would then become excessive cost of sales. The interim inventory and profit are nevertheless

wrong. The negative-reserve case exposes the equivalent sign error. Intermediate balances establish the accounting route, not just agreement between final cash and earnings.

The documentary `HedgeInventoryClose/1` report reconstructs the accepted hedge before checking the purchase population, price, date and settlement amount. It retains purchase, settlement, basis transfer and sale journals separately, with pre-sale books and gross cash origins. All acquisition costs, recognition conditions and net realisable value support are supplied findings. Missing findings leave the result unresolved; partial purchases, different valuations or insufficient cash under the declared payment order require another method. The report is cumulative and pre-tax: these examples do not determine an annual tax charge, non-controlling interests or a complete company's consolidated results.

36.12.13 Selling part of the hedged inventory

The purchase has absorbed the entire hedge reserve, but some goods may remain on the shelf when the company next reports. Their cost stays in inventory until sale, subject to a write-down if net realisable value falls below that cost. The sale does not release another portion of the hedge reserve: that reserve was already removed when all the goods were acquired.

Take the preceding purchase of 130 units with adjusted cost 10800, and suppose the units are ordinarily interchangeable, with the same nature and use. There is no opening stock or additional purchase. We apply weighted-average cost, one of the formulas permitted by IAS 2.25 and explained in IAS 2.27. The original hedge designation layers describe when the exposure entered the hedge; they do not identify separate physical inventory lots. For this one uniform batch, the unit cost is

$$c = \frac{10800}{130} = \frac{1080}{13}.$$

Selling 52 units for 5800 therefore recognizes cost $52c = 4320$ and leaves $78c = 6480$ in inventory. In the positive-hedge case, cumulative pre-tax profit is $90 + 5800 - 4320 = 1570$. The earlier 90 of hedge ineffectiveness remains in profit; the adjustment of purchase cost has already done its separate work.

Now suppose reliable evidence supports net realisable value of 70 per unit for the 78 units still held. This is a supplied estimate after completion and selling costs, with the same selling circumstances for every unit. Inventory falls from 6480 to $78 \times 70 = 5460$, and the company recognizes a write-down of 1020. IAS 2.28–31 requires the estimate to reflect the goods and their purpose; different contracts or selling circumstances would require a more detailed population. Retaining gross adjusted cost 6480 and an allowance 1020 makes the next sale easier to follow.

Sell 26 of these units for 2900. IAS 2.34 expenses their current carrying amount, $26 \times 70 = 1820$. The sale removes 2160 of gross cost and 340 of allowance. The 52 units still held therefore have gross cost 4320, allowance 680 and carrying amount 3640. If later evidence of changed circumstances raises net realisable value to 90 per unit, IAS 2.33 permits a recovery only up to original cost. Inventory returns to 4320, so the reversal is 680. Reversing the original 1020 would restore 340 relating to goods already sold and would overstate both remaining inventory and profit.

For the general calculation within this batch, let q be units still held and A their current allowance. Gross cost is qc, and carrying amount is $qc - A$. A sale of x units removes gross cost xc and allowance xA/q, giving

$$\text{cost of sales} = x\left(c - \frac{A}{q}\right), \qquad A_{\text{remaining}} = \frac{q-x}{q}A.$$

The first expression charges the carrying amount of the sold goods; the second keeps only

the allowance attributable to unsold goods. At the next supported valuation v', the required allowance is

$$A' = (q - x)\max(c - v', 0), \qquad \Delta\text{inventory expense} = A' - A_{\text{remaining}}.$$

A positive difference is a new write-down. A negative difference reduces inventory expense, subject to the recovery conditions in IAS 2.33. Because A' cannot be negative, the reduction cannot exceed the allowance for goods still held. Neither entry affects cash or recreates the cash-flow hedge reserve.

If the final 52 units are sold for 5800 after the recovery, total cost of sales is $4320 + 1820 + 4320 = 10460$. Net write-down expense is $1020 - 680 = 340$, so total inventory expense remains 10800. Total receipts 14500 give cumulative profit $90 + 14500 - 10800 = 3790$ in the positive-hedge case and $-100 + 14500 - 10800 = 3600$ in the negative case. These agree with the earlier complete-sale results, but the intervening inventory and earnings differ. Final agreement alone cannot establish that those earlier amounts were correct.

The documentary `HedgeInventorySubsequent/1` report reconstructs the purchase from the retained hedge history, then carries quantity, gross cost and allowance through dated sales and valuations. Missing recognition or recovery findings remain unresolved; excess sales, inconsistent dates and substituted opening amounts are refused. The calculations are cumulative and pre-tax. They require a homogeneous batch and supplied valuation evidence; partial forecast purchases, additional inventory lots, tax and company-wide presentation retain their own methods (European Commission, 2023a, IAS 2, paragraphs 25, 27–34).

These procedures retain the history needed when a hedge changes. Combining exposures creates a further timing question: sales and expenses can offset economically while affecting different reporting periods. The following net-hedge case retains those gross transactions; the award methods then distinguish service conditions and settlement rights. Actual rights, complex component valuation and complete successor exchanges remain separate financing obligations.

36.12.14 A net hedge still needs its gross transactions

A company expecting dollar sales and dollar expenses may manage only their net currency exposure. Hedge accounting still needs both gross amounts and the periods in which they affect earnings. IFRS 9.6.6.1 and B6.6.1–8 require individually eligible items and actual group risk management. For offsetting cash-flow positions, the designated risk must be foreign currency risk, and the designation must specify nature, volume and the relevant reporting periods. An abstract label such as “net dollars 20” is insufficient (European Commission, 2023a, IFRS 9.6.6.1–3, B6.6.1–8).

Use a euro-functional company with highly probable external cash sales of USD 100 and payments of USD 80 for services consumed immediately. The designation identifies those transactions, their earnings periods and a forward to sell USD 20 for EUR 20. Assume actual net risk management, an eligible external derivative, contemporaneous documentation and all effectiveness conditions. Interest, discount, forward points and credit effects are zero in this illustration; the whole forward is designated. At inception the rate is EUR 1 per dollar and the derivative is worth zero. At execution the rate is EUR 1.10, so sales are EUR 110, service expense EUR 88 and the forward loses EUR 2. No inventory asset is created in this example.

The effectiveness comparison retains the gross changes: the expense’s currency effect of minus 8 and the forward loss of 2 offset the sales effect of plus 10. IFRS 9.B6.6.5 and B6.6.9 require the offsetting item to enter that comparison alongside the instrument, without recognizing a forecast transaction before it occurs. When both transactions affect the same period, recognize the effective forward loss of 2 in OCI and subsequently release it as a separate hedge expense of

2. Revenue remains 110 and service expense 88; profit is $110 - 88 - 2 = 20$. The hedge result is presented separately because the offsetting risks affect different statement lines (European Commission, 2023a, IFRS 9.6.5.11, 6.6.4, B6.6.5, B6.6.9, B6.6.15). Showing revenue 108 would preserve profit but misstate the required presentation. Opening cash and equity 200 become 220 after gross operating receipts 110, service payments 88 and forward settlement 2. The reserve and derivative liability both close at zero. The designated forward follows the operating classification of the hedged flows (European Commission, 2023a, IAS 7.16).

Move only the service expense into the next period, retaining the same EUR 1.10 rate and a forward that settles with that expense. The designation already identifies these different earnings periods. At the first close the company has cash revenue 110 and a forward liability of 2. The revenue line must still use its transaction-date spot rate. To reflect the net hedge's timing, recognize a separate hedge expense of 10 when the sale occurs, with the corresponding credit to the cash-flow hedge reserve. Together with the forward's OCI loss of 2, the reserve is a credit of 8. First-period profit is 100 and OCI is 8; cash 310 less the derivative liability 2 equals equity 308. The 8 reserve carries the offset relating to the later expense; it is neither an additional derivative asset nor recognized service income (European Commission, 2023a, IFRS 9.B6.6.9, B6.6.15).

In period 2, record service expense and payment 88, settle the forward liability for 2, and release the reserve 8 as separate hedge income. Profit is minus 80 and OCI minus 8. Cash falls by 90 to 220, which also equals $308 - 80 - 8$. Across both periods profit remains 20 and OCI nets to zero. Releasing only the derivative loss when the sale occurs would give first-period profit 108 and miss this timing rule. A later rate change would require new measurements of all relevant components; the constant second-period rate isolates the recognition issue without supplying a general valuation model.

With sales and service payments both USD 100, the position is nil and no derivative is needed. Such a designation is allowed only under all four conditions in IFRS 9.6.6.6: an actual rolling net-risk strategy, changes in net size over that strategy's life, normal use of eligible instruments and hedge accounting when the position is nonnil, and otherwise inconsistent recognition of the offsetting risks. Conditional on those facts, the earlier-sale example has revenue 110, separate hedge expense 10 and reserve 10 in period 1. Expense 110 and separate hedge income 10 in period 2 leave lifetime profit and cash flow zero. Equal gross totals alone do not establish eligibility (European Commission, 2023a, IFRS 9.6.6.6, B6.6.3, B6.6.6, B6.6.10).

The group record therefore preserves each gross exposure's identity, reporting entity, currency, risk, amount, recognition period and admitted valuation, plus the strategy, designation, instrument, effectiveness calculation and reserve history. A layer also needs an identified and tracked overall pool and the conditions in 6.6.3. A group without offsetting risks apportions the instrument's net gain or loss systematically to the affected lines without grossing it up; a fair-value hedge adjusts each relevant asset or liability separately (European Commission, 2023a, IFRS 9.6.6.3–5, B6.6.11–16). A group of items is not permission to ignore consolidated-group external-party requirements: retain 6.2.3 and 6.3.5–6's distinct tests. These methods need explicit successor records and complete exchanges; the existing closed financing profiles are preserved until those additions are specified.

36.13 Conditions, settlement choices and the earnings that each share receives

The following conditional methods use IFRS 2 and IAS 33 in the retained September 2023 EU compilation. They are separate from the Apple contractual examples and do not determine their

US GAAP treatment. Employee counts, fair values, service patterns and participation rights below are explicit synthetic inputs. Valuing the awards and admitting an entity's applicable rules remain distinct requirements.

36.13.1 A failed condition does not always reverse an expense

An employee's failure to complete required service differs from a share-price target that is not reached. For employee equity awards, IFRS 2.10–21 (European Commission, 2023a, IFRS 2.10–23) measures the services using grant-date fair value. Service and non-market performance conditions affect the estimated number of instruments expected to vest; market conditions enter fair value. Revise the former estimate as information changes and use the actual eligible number at vesting. If a market condition fails but the other vesting conditions are met, the service expense remains. Freezing the quantity at grant and subsequently marking every equity award to the current share price would get both sides of this distinction wrong.

The award record therefore separates grant and measurement dates, recipient and employing entity, each service/performance/non-vesting condition and who can choose to fulfil it, service already received, expected eligible quantity, grant-date valuation and its incorporated conditions, settlement rights, cumulative expense, equity and any liability. A contract's use of the word performance does not decide whether the condition is a market condition. Missing terms or fair value remain unresolved rather than selecting the most convenient expense pattern. Services are expensed in these examples; qualifying goods or services that create an asset require their own recognition treatment.

Give 100 employees ten equity instruments each, with grant-date fair value 3 per instrument and three years of equal service. At the first year-end, 90 employees are expected to complete service; at the second, 80; at vesting, 75 have completed it. The cumulative expenses are

$$90(10)(3)/3 = 900, \qquad 80(10)(3)(2/3) = 1600, \qquad 75(10)(3) = 2250.$$

The annual charges are therefore 900, 700 and 650, credited to award equity. If an additional non-market performance condition ultimately fails for all awards, cumulative expense becomes zero and previously recognised expense reverses. In a separate award whose supplied grant-date fair value of 3 already incorporates a market hurdle, failure of that hurdle does not reverse the 2250 for the 75 employees who satisfy the other conditions. Equal supplied fair values in these examples do not imply that different contracts have the same economic value. Once an equity award has vested, later non-exercise does not reverse total equity under paragraph 23; a transfer within equity remains possible.

Cancellation also differs from forfeiture. Paragraphs 27–28 (European Commission, 2023a, IFRS 2.27–28A) preserve the minimum original grant-date expense, subject to failure of the specified non-market vesting conditions, and accelerate unrecognised service expense on cancellation during vesting. For one grant with total grant-date value 300, one of three equal service years gives expense 100. An entity cancellation that is not a forfeiture accelerates the remaining 200. If it pays 330 when the repurchased award's fair value is 320, debit equity 320 and expense 10, and credit cash 330. Total service/cancellation expense is 310. Starting from cash and capital 1000 leaves cash and equity 670 after all entries. Calling the event a service forfeiture would wrongly reverse the first 100.

Non-vesting conditions enter fair value under paragraph 21A. However, when the entity or counterparty can choose whether to meet such a condition, failure to do so during vesting is a cancellation under paragraph 28A. Consequently the condition's classification and the party's control of it must be preserved together. Beneficial modifications require their incremental

value analysis under paragraph 27 and Appendix B; this cancellation example does not replace that calculation.

36.13.2 Cash obligations change as the award's value changes

For a cash-settled award, IFRS 2.30–33D (European Commission, 2023a, IFRS 2.30–33D) requires remeasurement at each reporting date and settlement. Service and non-market vesting conditions affect quantity; market and non-vesting conditions enter the current valuation. Apply the service fraction to the current eligible quantity and fair value, then charge the difference from the previously recognised liability. The final cumulative expense equals cash paid, unlike the fixed grant-date measurement of an equity award.

Use the preceding employee quantities and ten rights per employee, but admit cash-settled fair values of 3, 6 and 8 at the three year-ends. Cumulative liabilities are 900, 3200 and 6000. The annual expenses are 900, 2300 and 2800. After vesting, a settlement-date value of 9 raises the liability to 6750 and adds expense 750. Paying 6750 removes the liability. A model that freezes the original value of 3 would leave expense 2250 and an unrecorded cash obligation of 4500. The changing values are supplied, not estimated from this example.

36.13.3 Who chooses settlement matters

If the employee can demand cash or shares, paragraphs 35–40 (European Commission, 2023a, IFRS 2.35–43) treat the award as a compound instrument. Measure the debt component first, then the equity component taking account of the cash right that must be surrendered for shares. Do not add the standalone values of both alternatives. Over the service period, remeasure the debt component under the cash-settled method and recognise the grant-date equity component under the equity-settled method.

For an immediately vested award, admit debt value 100 and incremental equity value 20. Recognise expense 120, liability 100 and equity 20. At settlement, remeasure the liability to 130 and charge expense 30. If the employee takes cash, pay 130 and retain the original 20 within equity. If the employee takes shares, transfer the liability 130 directly to equity, leaving total award equity 150. Both branches have cumulative expense 150; only the cash branch has a cash outflow. The values are an admitted compound valuation, not a claim that standalone alternative values uniquely determine the equity part.

If the entity chooses, paragraphs 41–43 first ask whether it nevertheless has a present obligation to settle in cash. Commercially insubstantial share settlement, a past practice or stated policy of paying cash, or normally paying cash whenever requested can create that obligation. A written share option therefore does not by itself establish an equity award. If no cash obligation exists, use equity-settled accounting. At settlement, cash is an equity repurchase, except that choosing the higher-valued alternative creates expense for the excess. For an immediately vested grant with admitted value 100, settlement cash 130 and alternative shares worth 120 require a debit to equity of 120, additional expense 10 and cash credit 130. A 130 expense would count the repurchase as remuneration again.

These cash choices are distinct from statutory withholding. Under paragraphs 33E–33H, an otherwise equity award can retain equity classification when shares are withheld to satisfy a legally imposed employee-tax withholding obligation. Preserve the legal obligation and required quantity. The exception does not cover an arrangement without that obligation or excess shares paid out in cash. It cannot justify classifying every net-settled award as equity.

36.13.4 A beneficial amendment adds a second service-cost clock

Cancellation accelerates an award's remaining cost. A beneficial amendment instead normally preserves the original grant-date cost and adds value measured at the modification date. IFRS 2.B42–B44 requires the original minimum, subject to failure of the relevant original non-market vesting conditions, and recognizes the beneficial change. The comparison uses the award's value immediately before and after modification at the same date (European Commission, 2023a, IFRS 2.27, B42–B44). A higher current value is not a replacement grant-date value.

Grant 100 instruments with fair value 3 each for three years of equal service. All ultimately vest. After the first year, a beneficial repricing raises their supplied fair value from 2 to 4 per instrument, with the original vesting date unchanged. Original total cost is $100(3) = 300$. Incremental cost is $100(4 - 2) = 200$, spread over the two remaining service years. The first year's expense is 100; each later year records original cost 100 plus incremental cost 100. Total expense and award equity are 500. Replacing the grant with its current value of 400 would omit 100; subtracting the grant-date value 3 from the modification-date value 4 would also use the wrong comparison. Each annual entry debits service expense and credits award equity. With no cash or other movements, the increase in that equity component offsets the reduction in retained earnings, leaving total equity unchanged.

The two service periods need not end together. In a separate modification, leave the original 100 instruments unchanged and add 20 instruments worth 4 each that require only the second year's service. IFRS 2.B43(b) measures the additional 80 at the modification date over their own service period. Annual charges are now 100, 180 and 100, totaling 380. Spreading the additional 80 over both remaining original years would understate year 2 by 40. This example does not shorten the original award's service condition: a beneficial change to that condition must itself be reflected under B43(c). An adverse value-only change generally leaves the original 300 cost, while reducing the number of instruments invokes the cancellation rule for the affected portion (European Commission, 2023a, IFRS 2.B43(b)–(c), B44).

A change from cash settlement to equity settlement follows a distinct rule. Suppose two of three equal service years have passed. The cash award liability, remeasured at the modification date, is 180. Replacement equity has a total modification-date fair value of 300 and retains the third year's service requirement. IFRS 2.B44A recognizes $300(2/3) = 200$ in equity for services already received, removes the liability 180 and charges the difference 20 immediately. Debit liability 180 and expense 20; credit award equity 200. The remaining year charges 100 to expense and equity. Prior cash-award expense 180 plus the modification charge 20 plus later service 100 gives lifetime expense 300, with no cash payment and no remaining liability. If the correctly remeasured liability had instead been 220, the immediate difference would be an expense credit of 20. It is not automatically an extra positive cost (European Commission, 2023a, IFRS 2.B44A–B44C).

Keep the original grant, every modification's before-and-after valuation, eligible quantity, original and modified service conditions, settlement classification, cumulative expense and component balances together. Apply each event once and recognize the change from the previously recorded cumulative amounts. B44B requires the modified vesting period when applying the cash-to-equity rule; B44C extends it to an identified replacement for a cancelled cash award. Supplied fair values in these examples do not provide an option-pricing method. Missing valuation or legal terms prevent application without erasing the original award or assuming that it costs zero.

36.13.5 Participation allocates the numerator before division

IAS 33.A13–A14 (European Commission, 2023a, IAS 33.A13–A16) allocates earnings according to the actual dividend and undistributed-earnings rights of the classes. First deduct the declared and contractually required dividends from the period's allocable result. Allocate the remainder as if it were distributed, using the participation formula and any caps. Add each class's dividends back to its allocation, then divide by the instruments to which that allocation belongs. A class's right to profits does not establish a right or obligation to share losses; preserve the actual terms. Unknown participation rights block the allocation.

Suppose parent-attributable profit is 120, 100 ordinary shares participate with weight one, and ten nonconvertible participating shares have weight two. Assume all are outstanding throughout the period and there are no senior claims, caps or other instruments. Declared dividends are 20 and 4. Remaining profit 96 allocates over total weight $100 + 2(10) = 120$: ordinary shares receive 80 and participating shares receive 16. Adding dividends gives class earnings 100 and 20, so per-share earnings are 1 and 2. Allocating all 120 to the ordinary shares would overstate their EPS as 1.2. Dividends are an allocation of this profit, not a second expense. A capped or senior participation formula needs its own piecewise allocation; the simple weights cannot replace it.

Diluted EPS then applies the relevant potential-share rules and dilution tests; it is not the same calculation as award expense. Under paragraphs 47A–48 (European Commission, 2023a, IAS 33.47A–60), future goods or services form part of the assumed proceeds for employee options. A separate case with profit 100, 100 ordinary shares, ten options, cash exercise price 5, period-average future service value 2 per option and average share price 10 has proceeds 70. Seven shares are assumed repurchased, so incremental shares are three and diluted EPS is 100/103. Omitting future services would instead add five. This comparison assumes the awards are outstanding for the whole period and pass the applicable dilution test; performance-conditioned awards also require paragraphs 52–57's period-end contingency assessment.

Under paragraphs 58–60, an entity cash/share choice presumes share settlement if dilutive, whereas a holder choice uses the more dilutive outcome. Required numerator adjustments preserve the profit effect of assumed equity treatment. For a separate nonparticipating contract, let reported profit be 90 on 100 shares after liability-remeasurement expense 10 that would be absent under wholly equity classification, and let assumed share settlement add 20 shares. With no tax effect or other changes, cash gives EPS 0.9 and share settlement gives $100/120 = 5/6$. The share outcome is more dilutive. Adding shares while leaving profit at 90 would produce 0.75 and compute the wrong assumed world. Continuing-operations control, instrument ordering, time weights and antidilution remain the earlier EPS method's requirements, not assumptions waived by this example.

36.13.6 A senior dividend and a cap change who receives the next unit of profit

The proportional example leaves every additional unit of earnings to the same sharing rule. A senior entitlement and a cap change that allocation. Consider 100 ordinary shares and 20 nonconvertible participating preference instruments, all outstanding throughout the year. The supplied terms give the preference class a cumulative current-year entitlement of 20, then equal participation per instrument in positive residual earnings, up to total class earnings of 40. The ordinary shares receive what remains. The preference class does not share residual losses. Equity classification and these rights are assumed facts of this example; the earnings allocation does not declare a dividend or create a payable. IAS 33.14(b) includes the current period's cumulative entitlement whether declared or not, while A13–A14 require allocation by the actual

participation rights (European Commission, 2023a, IAS 33.14(b), A13–A14).

At allocable profit 140, deduct the senior 20 first. The remaining 120 divides over 120 equal participation units, so the preference class receives another 20 and reaches its cap. Ordinary earnings are 100 and EPS is 1. At profit 80, the residual is only 60: the preference class receives 20 plus 10, leaving 50 for ordinary shareholders and EPS of one half. At profit 200, an uncapped residual allocation would give the preference class another 30. Its cap allows only 20, leaving ordinary earnings of 160. The ordinary shareholders receive every further unit once the cap binds.

To state that rule precisely, let E be earnings available to these two classes after all other required attribution, n the ordinary population, and m the total residual participation weight of the preference class. Let f be its senior period entitlement and c the maximum additional participation. Thus this example has $m = 20$, $f = 20$ and $c = 20$; the total cap is $f + c = 40$. The rights give

$$p = \min\left(c, \frac{m \max(E - f, 0)}{n + m}\right), \qquad E_P = f + p, \qquad E_O = E - f - p, \qquad r = \frac{E_O}{n}.$$

Here p is residual participation, E_P and E_O are the two class allocations, and r is ordinary EPS. The two allocations sum to E. If earnings are zero, the current cumulative entitlement still assigns 20 to the preference class and -20 to the ordinary class; ordinary EPS is $-1/5$. At earnings of -40, ordinary earnings are -60, giving EPS $-3/5$. These negative ordinary amounts follow from the senior right and the absence of residual loss sharing, rather than a rule that every class must receive a nonnegative allocation.

Assumed issuance now changes both sides of the ordinary EPS fraction. Return to profit 140 and add 100 options with exercise price 10 and average market price 20. The IAS 33.45–47 treasury-share calculation gives $100 - 100(10)/20 = 50$ incremental ordinary shares (European Commission, 2023a, IAS 33.45–47). Assume no further service proceeds, no earnings effect and a full-year eligible option population. A14 includes the assumed shares in the ordinary population. With $n = 150$, the preference class's residual participation falls to $120(20)/170 = 240/17$, below its cap. Consequently

$$E_P = 20 + \frac{240}{17} = \frac{580}{17}, \qquad E_O = 140 - \frac{580}{17} = \frac{1800}{17}, \qquad r = \frac{1800/17}{150} = \frac{12}{17}.$$

The preference class receives less because more ordinary shares now share the residual. Keeping ordinary earnings at 100 would instead give 2/3, which is wrong for these participation rights. The numerator cannot be frozen merely because the option has no direct earnings adjustment.

Add a full-year convertible debt series whose complete after-tax earnings effect is 6 and whose incremental ordinary population is 10. With both series assumed, total earnings become 146 and the ordinary population becomes 160. The preference class receives $20 + 126(20)/180 = 34$; ordinary earnings are 112 and EPS is $112/160 = 7/10$. This is below 12/17, so the debt remains dilutive after the option. Debt alone would give ordinary EPS $(146 - 20)/(110 + 20) = 63/65$. The same conversion has different ordinary numerator effects depending on the shares already assumed.

There is nevertheless a simple ordering rule for this restricted family. Assume continuing and total earnings coincide, rights remain fixed, and each series independently adds a complete nonnegative gross after-tax earnings amount a_i and a positive ordinary share amount b_i. The gross amount is measured before the two-class allocation. For $E \geq f$, substituting the participation formula gives

$$r(E, n) = \max\left(\frac{E - f}{n + m}, \frac{E - f - c}{n}\right).$$

The first fraction applies before the cap binds; the second applies once the preference class receives its full cap. An assumed series adds the same a_i to either numerator and b_i to either denominator. If current EPS is $r \geq 0$, both resulting fractions fall below r exactly when $a_i - rb_i < 0$. If that expression is positive, the previously active fraction rises; equality leaves EPS unchanged. Thus order complete gross effects by a_i/b_i, recalculate the allocation, and include a series only when it strictly lowers ordinary EPS. IAS 33.41–44 provides the continuing- operations control and separate-series dilution objective (European Commission, 2023a, IAS 33.33–44); the formula here is a local derivation for the stated rights and additive gross effects.

The ordering also finds the minimum over this family's combinations. For any trial bound r, requiring both fractions to be at most that bound imposes two inequalities with the same varying sum $\sum_i (a_i - rb_i)$. Including every negative term minimizes that sum in both inequalities, so a minimizing combination exists among the prefixes in ascending a_i/b_i order. An included series leaves EPS above its own ratio, and all later ratios are at least as large. Once a series no longer lowers EPS, later ones cannot restore dilution. Below the senior entitlement, nonnegative earnings additions and positive share additions cannot deepen the ordinary loss per share. Equality is excluded by the strict dilution condition.

The read-only `CappedParticipationEPS/1` method uses exact rational amounts and returns the allocation before and after each trial assumption. It requires explicit rights, complete effects, context and period findings; unknown findings leave the answer unresolved, and unsupported or contradictory terms produce no partial EPS result. Its 18 authored cases include both cap regimes, zero and negative earnings, equality, reversed input order and the nonparticipating limit. An independent calculation enumerates all 49 finite exercise combinations. This supports the specified calculation, while the existing fixed ordinary-numerator method retains its separate assumptions.

Complete gross effects are the consequential restriction. IAS 33.35 points out that removing interest expense can increase a nondiscretionary employee profit-sharing expense. If that expense depends on the combined conversions, single-series effects need not add. Conditional exercise or changed rights can alter the possible assumptions as well. Those cases need their own joint earnings and eligibility calculation before any dilution ordering is justified.

36.13.7 Conversion can change another expense and depend on another instrument

Reallocating a fixed amount of earnings is only one way that assumed conversion can change the numerator. Removing interest expense may increase the earnings on which employee remuneration depends. IAS 33.35 expressly requires the resulting change in nondiscretionary profit-sharing expense to enter the EPS calculation, together with the related tax effects under paragraphs 33–34 (European Commission, 2023a, IAS 33.33–35). Calculating each conversion once in isolation can therefore miss a consequence of converting two instruments together.

Suppose profit before a contractual employee bonus is 90. The bonus is half of profit above 100, and complete tax expense is one fifth of profit after the bonus. These are supplied terms for the example. There are 100 ordinary shares, with no participation rights or other attribution; continuing and total earnings coincide. The reported bonus is zero, tax is 18 and earnings are 72. Each of two full-year convertible debt series would remove interest expense of 20 and add 20 ordinary shares. Assume all other conversion consequences have been accounted for.

Converting either series alone raises profit before the bonus to 110. The bonus becomes 5, tax becomes 21 and earnings become 84. The complete after-tax increase is 12, and the hypothetical EPS is $84/120 = 7/10$. Converting both raises profit before the bonus to 130. Now the bonus is

15, tax is 23 and earnings are 92, giving hypothetical EPS $92/140 = 23/35$. Adding the two isolated increases would instead give $72 + 12 + 12 = 96$. That four-unit overstatement comes from failing to recalculate the common bonus and its tax consequence.

For a set S of assumed conversions, let I_i denote interest saved by series i, and b_i its incremental ordinary shares. Write P for profit before the bonus, H for its threshold, q for the bonus rate and t for the complete proportional tax rate. The supplied terms imply

$$\begin{aligned} P(S) &= P + \sum_{i \in S} I_i, & B(S) &= q \max(P(S) - H, 0), \\ T(S) &= t\{P(S) - B(S)\}, & N(S) &= P(S) - B(S) - T(S), \\ D(S) &= n + \sum_{i \in S} b_i, & r(S) &= N(S)/D(S). \end{aligned}$$

Thus the first conversion adds 12 to earnings, while the second adds only $92 - 84 = 8$. With $q = 0$, the bonus disappears and the two after-tax interest effects become additive. The formula describes this remuneration and tax model; it does not decide whether a real obligation meets recognition requirements or establish its tax treatment. The loss illustrations use zero tax rather than assuming that every loss creates a recoverable tax asset.

A complete hypothetical calculation still needs the applicable dilution rule. For example, change only initial profit before the bonus to 80. Basic earnings are 64 and EPS is 16/25. Either conversion alone gives earnings 80 and EPS 2/3, an increase. Both together give earnings 88 and EPS 22/35, below basic EPS. Merely taking the smallest of these values would bypass the individual-series requirements of IAS 33.41–44 (European Commission, 2023a, IAS 33.41–44). The `ConsequentialEPSWorlds/1` method therefore reports the complete four worlds and leaves final dilution selection unevaluated. The earlier additive ordering proof does not apply to these joint earnings effects.

The official illustrative ordering example provides a useful comparison. AASB 133 Example 9 orders options, bonds and preference shares by fixed incremental earnings per share of 0, 1.5 and 4. Options and bonds reduce the continuing-operations ratio to $13{,}000{,}000/4{,}020{,}000 = 650/201$; adding preferences would raise it to $970/281$, so they are excluded (Australian Accounting Standards Board, 2021b, Example 9, pp. 28–30). The earnings increments there do not depend on another assumed conversion. Examples 7 and 12 explain contingencies, actual conversion dates and the continuing-operations control number, but do not add a rule for the shared threshold expense above.

Fixed standalone ranking followed by complete recalculation, and ranking by the next marginal effect, both reject the first conversion in the 80-profit example. With only two series, these calculations do not distinguish those ranking policies in general. The inspected examples therefore do not settle when changing increments should be reordered or a rejected issue reconsidered. The unresolved selection is kept separate from the correct conditional world amounts; lower combined EPS alone does not establish an admissible selection.

There is a separate, explicit dependency for contingently issuable potential shares. IAS 33.57 first tests whether the instruments may be assumed issuable under paragraphs 52–56, then applies their instrument-specific dilution rules. Its final sentence also requires assumed exercise or conversion of similar outstanding potential shares that are not contingently issuable (European Commission, 2023a, IAS 33.52–57). A satisfied contingency alone is insufficient.

Consider profit 100 and 100 ordinary shares. A similar outstanding option series contains 20 options at strike 6; a contingent option series contains 20 at strike 5. Average market price is 10. For this conditional example, similarity has been separately admitted; different exercise prices neither establish nor automatically disprove that finding. Both agreements cover the full

reporting period, with no future-service proceeds, earnings effects, participation or additional dependencies. The contingent option is excluded from basic shares. Its nonfinancial condition requires five stores, assessed at the reporting-period end while the contingency period remains open.

Under paragraphs 52 and 56, use the period-end condition status. Five observed stores qualify; a forecast of a fifth opening next year does not. The outstanding option supplies $20-20(6)/10 = 8$ incremental ordinary shares. Its exercise is dilutive, reducing EPS to $100/108 = 25/27$. With the condition satisfied, the contingent option supplies another $20-20(5)/10 = 10$ shares, giving diluted EPS $100/118 = 50/59$. Both have zero earnings increments, so testing the prerequisite first respects their tied incremental-earnings ratios while enforcing paragraph 57.

Now give the similar outstanding option strike 12, retaining the example's explicit similarity finding. It is out of the money at average price 10 and its exercise is not assumed. The contingent option would reduce EPS to $100/110 = 10/11$ in isolation, but paragraph 57's prerequisite is unsatisfied. Both are excluded and EPS remains 1. If instead only four stores are observed, the original outstanding option remains dilutive but the contingent option is excluded, leaving EPS $25/27$. At zero or negative earnings neither zero-earnings-effect option strictly reduces EPS or increases loss per share.

The `ContingentOptionEPS/1` method records both tests and each exclusion reason. Missing similarity or complete terms leaves the answer unresolved; future observations, incompatible periods and unsupported conditions cannot produce a partial result. Seven consequence cases and eight option cases check these distinct calculations. Arbitrary linked conversions and a general selection method for nonadditive earnings remain separate source and derivation requirements, with the complete conditional worlds preserved for that work.

36.13.8 An unpaid participating dividend can change ordinary earnings

Assumed conversion is not the only hypothetical transaction that changes earnings per share. Allocating undistributed profit between ordinary and participating shares can itself have a tax consequence. Suppose participating dividends are tax deductible and the resulting benefit belongs to ordinary shareholders. An allocation that divides reported profit solely by the base participation ratio misses that additional ordinary entitlement. The June 2017 IFRS Interpretations Committee decision requires the ordinary numerator to include the ordinary holders' share of the tax benefit from the hypothetical distribution required by IAS 33.A14. This applies whether an actual benefit is recognised in profit or directly in equity (IFRS Interpretations Committee, 2017a).

The accompanying example makes the consequence visible before any dividend is paid. Participating and ordinary holders share base distributions in the ratio ten to one. The tax deduction is 30% of the participating distribution, and its entire benefit belongs to ordinary holders. In the first year, profit before that benefit is 330 and no dividend is declared. The ordinary base share is $330/11 = 30$, while participating holders' hypothetical allocation is 300. Distributing that 300 would create a benefit of 90 for ordinary holders. Their basic-EPS numerator is therefore $30 + 90 = 120$, although no cash dividend or actual distribution-related tax benefit arose in that year (IFRS Interpretations Committee, 2017b, pp. 1–5).

To carry this allocation across periods, distinguish the profit being shared from dividends that may distribute earlier earnings. Let P be current profit before the participating-dividend tax benefit, r the participating- to-ordinary base-rights ratio, and t the tax rate. Define the class fractions $a = r/(1 + r)$ and $b = 1/(1 + r)$. If the base dividend declared in the current period is D, participating holders receive aD and ordinary holders receive bD. In the source's payout pattern the actual tax benefit $H = taD$ is also distributed to ordinary holders in that

period. All quantities here are money except the ratios and tax rate; classification, deductibility, recognition and these rights are supplied facts.

First put the actual tax benefit on a consistent basis for the EPS allocation. Reported profit is $P + H$ when that benefit is recognised in profit, or P when it is recognised directly in equity. Adding the latter benefit for this allocation gives adjusted profit $G = P + H$ in either case. Subtract all declared class dividends, including the ordinary tax-benefit dividend:

$$R = G - aD - bD - H = P - D.$$

This signed residual is the base profit still to allocate under the hypothetical distribution. It can be negative when the actual dividend includes earlier years' earnings. The participating residual is aR, the ordinary base residual is bR, and the ordinary holders' further tax adjustment is taR. Adding declared and residual amounts gives

$$\begin{aligned} N_O &= bD + H + bR + taR, \\ N_P &= aD + aR, \qquad \mathrm{EPS}_{\mathrm{basic}} = N_O/n, \end{aligned}$$

where N_O and N_P are the ordinary and participating earnings allocations and $n > 0$ is the supplied weighted-average ordinary denominator. These are the example's three steps: adjust and subtract declared dividends, allocate the remaining profit and its tax consequence, then add the declared allocations back to each class (IFRS Interpretations Committee, 2017b, Steps 1–3, pp. 3–5).

The example's following two years show why the signed residual matters. Base profits are 550 and 110. The second year distributes the first year's base profit of 330; the third distributes the remaining base profit of 660. All actual tax benefits are paid to ordinary holders. The complete allocation is

	Year 1	Year 2	Year 3
Current base profit P	330	550	110
Base dividend D	0	330	660
Actual benefit H	0	90	180
Residual $R = P - D$	330	220	−550
Residual tax adjustment taR	90	60	−150
Ordinary earnings N_O	120	200	40
Participating earnings N_P	300	500	100

In year 3 the ordinary dividend is $60 + 180 = 240$. Its remaining base allocation is -50, and the tax adjustment on that residual is -150, giving $240 - 50 - 150 = 40$. Setting the negative adjustment to zero would give 190 and count part of a benefit already attributed to earlier earnings again. The -150 is an EPS-allocation reversal; it is neither a negative cash dividend nor evidence that every tax loss creates a recognised asset.

The dependence on payout timing cancels under these fixed rights. Substituting $H = taD$ and $R = P - D$ into the class allocations yields the local identities

$$\begin{aligned} N_O &= bD + taD + (b + ta)(P - D) = (b + ta)P, \\ N_P &= aD + a(P - D) = aP, \\ N_O + N_P &= P + taP = G + taR. \end{aligned}$$

Thus current ordinary earnings depend on current profit and its associated hypothetical benefit, even when the declared dividend distributes older profit. At zero tax the formula reduces to the ordinary base share bP; with no participating entitlement it reduces to P. Total class

allocations can exceed reported profit because they include the hypothetical tax consequence. For example, year 1 allocates $120 + 300 = 420$ while reported profit is 330. Those two quantities serve different purposes and must not be forced to match by posting the hypothetical difference.

`ParticipatingTaxEPS/1` specifies this basic-EPS calculation with exact rational amounts and a supplied denominator. It checks actual class dividends, the benefit amount and recognition location, and an opening base-profit balance sufficient to fund any prior-period distribution. It permits a negative post-dividend residual while restricting current base profit to nonnegative amounts. Missing rights or tax findings leave the result unresolved; inconsistent payouts, future information and unsupported findings produce no partial numerator. Its output contains no dividend, tax asset or journal entry.

The interpretation also says that a tax benefit shared by both classes must follow the applicable rights (IFRS Interpretations Committee, 2017b, p. 2, footnote (a)). That is a different allocation from the ordinary-only benefit used here. Retained actual benefits, changing or capped rights, loss-tax recognition and nonadditive diluted selection require their own supported treatment. In particular, a fixed after-tax earnings input to the earlier capped-participation method cannot silently stand in for the hypothetical benefit calculated here.

36.14 Control and an elimination book

The consolidation procedure selects the control graph at each date, aligns supported accounting policies and presentation currencies, combines controlled entity books, and posts acquisition, elimination and non-controlling-interest adjustments in a separate book. A control conclusion needs evidence and a rule version. Ownership percentage alone is insufficient. Unknown purchase allocation, currency translation or unmatched intragroup differences remain named unresolved adjustments rather than unexplained balancing entries.

For a simplified acquisition of a different subsidiary from the goodwill case that follows, assume control is established, all net assets are already at the relevant acquisition measurements, no tax or goodwill arises, and non-controlling interest is measured proportionately. A parent has cash 60, investment 40 and equity 100; an 80%-owned subsidiary has cash 30, plant 20 and equity 50. Consolidation debits subsidiary equity 50, credits investment 40 and credits non-controlling interest 10. Group assets are cash 90 plus plant 20, and group equity is 100 attributable to the parent plus 10 to other owners.

A subsequent intragroup loan of 10 shifts cash from parent to subsidiary, creates a receivable and payable, and is eliminated by debiting payable and crediting receivable 10. Group totals remain 110. An unmatched loan currency or settlement date would block that exact elimination. If the parent instead sells its entire interest for 44 immediately after the initial state and loses control, the group removes subsidiary cash 30, plant 20 and non-controlling interest 10, recognises cash consideration 44 and gain 4; remaining cash/equity is 104. More complex acquisitions and retained interests require additional measurement operations; this example fixes the structure and arithmetic without certifying a complete consolidation standard.

36.15 Perimeter and consolidation state

The acquisition, control and currency methods below use the inspected IFRS 3, IFRS 10 and IAS 21 provisions in the retained 2023 EU text (European Commission, 2023a). The acquisition with goodwill is a new example; it does not change the preceding zero-goodwill case.

ControlAssessment records investor/investee, relevant activities, rights and their substantive/pro-

tective character, variable returns, ability to use power, principal/agent analysis, effective interval and evidence. IFRS 10.5–9 requires power, returns and their connection, with reassessment on changed facts. An 80% share count is evidence to analyze, not an automatic substitute for control. Unknown control yields separate conditional perimeters and unresolved group results; no entity disappears merely to make reconciliation easy.

PerimeterVersion binds control intervals, ownership classes, report cutoffs, functional/presentation currency, policy-alignment adjustments and the exact source entity books. ConsolidationAdjustment is a distinct group book with matched counterparty/event references; legal-entity journals remain intact. The procedure is: select time-compatible entity books; apply supported policy alignment; translate them; add like balances/movements; eliminate investment against acquisition equity; eliminate intragroup balances, income/cost and unrealised gains; allocate profit/OCI to owner/NCI; reconcile all statements and movement notes. IFRS 10.19–23/B86–B96 support these steps. Reporting-date differences need B92–B93's conditions and intervening adjustments; a generic nearest-date join is invalid.

36.16 Acquisition and ownership case

Assume a business combination, no common-control transaction, parent obtaining control and 80% of ordinary residual interests for cash 96. Acquiree identifiable assets fair value 160 and liabilities 60 give net 100. There is no prior interest, contingent consideration, tax difference, separately unrecognized intangible, transaction cost or impairment; those absences are synthetic facts. Select proportionate-share NCI under IFRS 3.19: NCI 20 and goodwill 96+20−100=16 under .32. In parent books debit investment 96/credit cash 96. In the consolidation overlay, debit acquiree acquisition equity 100 and goodwill 16, credit investment 96 and NCI 20. Only post-acquisition income enters group performance. Asset measurement exceptions, deferred tax and a bargain gain reassessment under .36 would change the calculation; goodwill cannot absorb an unexplained missing balance.

If post-acquisition net profit is 20, all accrued in retained earnings with no dividend/OCI, NCI increases 4 to 24 and parent's attributable profit is 16. Parent buys another 10% for 15 while retaining control: transfer 12 from NCI (10% of current net 120), credit cash 15 and debit parent equity 3. Remaining NCI 12; goodwill 16 unchanged; no new profit. IFRS 10.23/B96 makes this an equity transaction. This example has equal ordinary rights; preference rights need their actual allocation and cannot use a simple ownership fraction.

In a separate alternative starting before that additional purchase, sell all 80% for 130 when net assets 120, goodwill 16, NCI 24 and OCI zero. On loss of control, group gain=130+24−(120+16)=18. Derecognize individual assets/liabilities and goodwill, remove NCI, recognize consideration and gain. The separate parent book's investment gain is not added again. If 20% is retained and 60% sold for 100 with retained fair value 35, group gain=100+35+24−136=23; recognize the retained investment 35. Its subsequent associate/financial-asset method needs separately established significant-influence/classification facts. IFRS 10.B98–B99 also requires the appropriate prior-OCI treatment; this case assumes none.

36.17 Intercompany inventory and tax

Assume parent's external inventory cost 60 is sold to its controlled subsidiary for 100 on credit, all still held at report date, tax ignored in the base case. Parent records AR 100/revenue 100 and COGS 60/inventory 60; subsidiary inventory 100/ AP 100. Consolidation debits AP

100/credits AR 100 and debits revenue 100/credits COGS 60 and inventory 40. Group inventory 60, revenue and COGS zero, and no group profit arises. When subsidiary later sells externally for 120, group COGS is 60 and profit 60, not 20; release the previously eliminated 40 in the continuing adjustment roll-forward. Reversing a closing overlay and recreating a new one must not recognize the profit twice.

If the purchaser's independently supported tax base is 100, group carrying amount 60 creates deductible difference 40. At assumed applicable 25% and with all IAS 12 DTA conditions satisfied, recognize group DTA 10/deferred-tax benefit 10; reverse on outside sale. The original tax expense and cash obligations in the seller's tax jurisdiction remain. Different rates, recoverability or tax-law transfer rules change the answer; tax is not simply eliminated with group profit. For an upstream subsidiary sale, attribute elimination and related tax effects to the appropriate subsidiary/NCI performance; downstream parent's profit elimination does not mechanically reduce NCI by the same fraction.

36.18 Functional currency versus group presentation

EntityCurrencyState names the functional currency and its evidenced rationale, foreign-currency nominal balances, monetary/nonmonetary classification, historical or fair-value measurement dates, rates and sources. IAS 21.21–23 and .28 require transaction-date translation and closing-rate remeasurement of monetary items; historical-cost nonmonetary assets retain transaction-date rates. A functional currency change is prospective under .35–37. Hyperinflation, lack of exchangeability and hedge accounting require separate applicable methods; do not import the historical .26 text into a later amended period without checking it.

Example: buy goods for foreign 100 at functional 1.2 per foreign unit; debit inventory 120/credit payable 120. Closing rate 1.3 gives payable 130 and FX expense 10, while inventory remains 120 before any NRV test. Pay at 1.25: debit payable 130/ credit cash 125/credit FX gain 5. Cumulative expense 5 matches 125 cash less 120 asset cost. Translating the inventory again to 130 would be wrong for this declared basis.

Group presentation is a different operation under IAS 21.39–41. A subsidiary has opening net assets 100, equity 100, no debt or distributions, annual profit 20 entirely earned at rate 1.25, and closing assets 120. Opening historical rate 1.2 and closing 1.3 give translated assets 156, contributed/opening equity 120 and profit 25; translation OCI 11 reconciles the difference. It is explained as $100\times(1.3-1.2)+20\times(1.3-1.25)=11$, not a free balancing plug. If ownership is 80%, parent/NCI shares of profit are 20/5 and OCI 8.8/2.2 under the stated equal rights. Use transaction rates; an average is acceptable only as a justified approximation, not when significant fluctuation invalidates it. Goodwill and fair-value adjustments follow IAS 21.47's foreign-operation treatment, rather than freezing them automatically in the parent's currency.

Group reports retain the translation reserve by entity, ownership and event. Loss of control requires the appropriate reserve reclassification under IAS 21.48 and IFRS 10.B99; a no-control-loss change transfers the relevant attribution in equity under applicable conditions. Cash translation effects reconcile cash separately from operating/investing/financing movements. An intercompany monetary balance's exchange result cannot simply be eliminated as if no currency exposure existed; IAS 21.45 and net-investment .32 require the substantive classification.

36.19 A foreign subsidiary from acquisition to disposal

The preceding examples establish acquisition, translation and disposal separately. Their interaction matters because a foreign subsidiary carries more than translated net assets into a sale: goodwill and accumulated exchange differences also have owners and dates. A sale can leave total equity correct while putting the wrong amount in profit. We follow one subsidiary through these steps using the selected historical IFRS 3, IFRS 10, IAS 21 and IAS 7 provisions in the retained EU text (European Commission, 2023a).

A parent starts with USD 300 of cash and equity. USD is its functional and presentation currency; the subsidiary's functional currency is EUR. At acquisition the parent pays EUR 96 for 80% of equal ordinary interests, when each euro buys USD 1.20. It obtains substantive power over the relevant activities, exposure to variable returns and the ability to affect them. The other owners have no substantive rights that defeat that control. Neither investment-entity treatment nor a common-control transaction applies.

The acquired business already produces outputs and includes a skilled organised workforce performing a process critical to continuing them, as required by the assumed IFRS 3.B12C(a) facts. The optional concentration test is not elected. Identifiable fair values are EUR 20 of cash, inventory of 140 and a payable of 60. There is no other separately identifiable asset, tax difference, previous holding, contingent consideration or acquisition cost in this synthetic case. The workforce itself is subsumed into goodwill under B37; calling a residual goodwill would not justify omitting an identifiable customer relationship or other asset in a real acquisition.

Identifiable net assets are $20 + 140 - 60 = 100$ euros. The parent chooses proportionate-share measurement for the non-controlling interest (NCI) under IFRS 3.19. NCI is therefore EUR 20 and goodwill under paragraph 32 is $96 + 20 - 100 = 16$ euros. This recognised goodwill belongs to the acquired parent interest; no NCI goodwill has been recognised. The parent's separate entry debits investment and credits cash by USD 115.2. The consolidation adjustment debits acquisition equity 120 and goodwill 19.2, and credits the investment 115.2 and NCI 24. Immediately after acquisition, group cash is 208.8, inventory 168, goodwill 19.2 and the payable 72. Net assets of 324 equal parent equity 300 plus NCI 24.

The subsidiary then sells all the inventory for EUR 160, receives cash, and pays its EUR 60 payable. Both events occur at USD 1.25 per euro. The specified event interval has no further expense, impairment or distribution. In the subsidiary's books, revenue 160 less cost of sales 140 gives profit 20; cash closes at $20 + 160 - 60 = 120$ euros. Translated revenue is 200, cost of sales 175 and profit 25. Parent and NCI shares of profit are 20 and 5. The translation worksheet translates the opening balances to the event rate before recording these movements; it does not charge cost 175 against unadjusted acquisition inventory 168 and leave a temporary negative stock. These are presentation adjustments, not a remeasurement of historical-cost inventory in the subsidiary's own functional currency.

At reporting, and at an immediate disposal considered below, the rate is USD 1.30 per euro. Subsidiary cash translates to 156. IAS 21.47 also treats the acquisition goodwill as an asset of the foreign operation: its EUR 16 becomes USD 20.8. To explain the currency reserve, translate the opening net assets and the subsequently earned profit from their respective rates:

$$\underbrace{100(1.30 - 1.20)}_{\text{opening net assets}} + \underbrace{20(1.30 - 1.25)}_{\text{later profit}} = 11.$$

Goodwill adds $16(1.30 - 1.20) = 1.6$. The total translation gain in other comprehensive income (OCI) is 12.6, but it is not divided mechanically 80/20. Only the net-asset gain is shared: parent OCI is $0.8(11) + 1.6 = 10.4$, and NCI OCI is $0.2(11) = 2.2$. This follows from the chosen proportionate NCI measurement and the separate attribution required by IAS 21.41 and IFRS

10.B94. NCI closes at $24 + 5 + 2.2 = 31.2$, equal to 20% of the subsidiary's identifiable net assets of 156. Parent equity is $300 + 20 + 10.4 = 330.4$. Group cash 340.8 plus goodwill 20.8 equals their combined equity of 361.6.

Cash has its own translation bridge. At acquisition, payment of 115.2 less cash acquired of 24 gives investing outflow 91.2. The subsequent operating receipt of 200 less payment of 75 gives operating inflow 125. Exchange changes increase the translated value of the acquired cash by $20(1.30 - 1.20) = 2$ and of the later EUR 100 net receipt by $100(1.30 - 1.25) = 5$. IAS 7.25–28 reports that total exchange effect of 7 separately from transaction cash flows. Thus $300 - 91.2 + 125 + 7 = 340.8$. The cash exchange effect is 7; the total translation OCI is 12.6. They measure different populations.

36.19.1 Losing control removes the whole subsidiary

Suppose an unanticipated transaction immediately after reporting transfers control to a buyer. No earlier held-for-sale condition is assumed in this event slice; a real planned disposal needs that assessment. Compare two alternatives: sell the whole 80% interest for USD 169, or sell 60% for 126.75 and retain 20% with independently admitted fair value 42.25. Both transactions remove the seller's substantive power. The equal price per share is an illustrative valuation assumption, not an accounting requirement. It makes proceeds plus retained value equal to 169 in each alternative.

IFRS 10.B98 removes all subsidiary assets, liabilities and NCI when control is lost, and measures any retained investment at fair value. If C is cash consideration and V the retained fair value, the gain before the currency reserve is reclassified is

$$C + V + \underbrace{31.2}_{\text{NCI removed}} - \underbrace{156}_{\text{subsidiary cash removed}} - \underbrace{20.8}_{\text{goodwill removed}} = 23.4.$$

For the full sale, debit cash 169 and NCI 31.2, credit subsidiary cash 156 and goodwill 20.8, and credit gain 23.4. In the retained-interest alternative, replace the debit to cash 169 with cash 126.75 and retained investment 42.25. The subsidiary still holds its own bank account; removal of cash 156 is a change in the consolidated reporting population, not a payment by the parent.

IAS 21.48–48B and IFRS 10.B99 then reclassify the parent's entire accumulated exchange reserve of 10.4 into profit: debit that reserve and credit disposal gain. Total disposal gain is $23.4 + 10.4 = 33.8$. NCI's reserve of 2.2 is already within the NCI balance of 31.2 removed above; it is derecognised without being recycled to profit. Retaining 20% does not justify recycling only the fraction corresponding to the shares sold. Paragraph 48A treats a loss of control as a disposal even when a minority interest remains.

The two alternatives therefore have equal total ending equity but different cash and investment positions:

USD amount	Sell all 80%	Sell 60%, retain 20%
Cash consideration	169	126.75
Retained investment	0	42.25
Disposal gain including recycling	33.8	33.8
Net disposal cash flow	13	−29.25
Closing group cash	353.8	311.55
Closing equity	353.8	353.8

IAS 7.39–42 reports acquisition and loss-of-control cash separately, each net of the whole subsidiary's cash. Net disposal cash is $169 - 156 = 13$ for the full sale, but $126.75 - 156 = -29.25$

when 20% is retained. A positive parent receipt can therefore accompany a negative consolidated disposal cash flow. The retained investment's 42.25 is not cash. The two cash bridges are $300 - 91.2 + 125 + 7 + 13 = 353.8$ and $300 - 91.2 + 125 + 7 - 29.25 = 311.55$.

For the whole interval, operating profit 25 plus disposal gain 33.8 gives profit 58.8, attributed 53.8 to the parent and 5 to NCI. Total OCI is $12.6 - 10.4 = 2.2$, all attributed to NCI; period OCI does not vanish merely because closing NCI is zero. The equity roll-forward is

$$300 + \underbrace{24}_{\text{NCI at acquisition}} + 58.8 + 2.2 - \underbrace{31.2}_{\text{NCI at disposal}} = 353.8.$$

Parent equity also follows directly from $300 + 53.8 = 353.8$. In the full sale, the parent's separate cost-based investment gain is $169 - 115.2 = 53.8$, already equal to its cumulative consolidated profit in this example. Adding that separate gain to the group result would count the investment return twice.

Intermediate statements are essential checks. Freezing goodwill at USD 19.2 understates both goodwill and parent OCI by 1.6 before disposal. Yet its larger pre-recycling disposal gain and smaller recycled reserve cancel, leaving the same final gain of 33.8. Dividing goodwill translation 80/20 would misattribute 0.32 to NCI. Recycling all 12.6 while still removing NCI 31.2 would overstate profit by 2.2 and leave a spurious negative parent reserve; recycling only the sold fraction in the retained-interest case would leave 2.6 after control is lost. Final total equity alone misses these different failures.

The case uses relative event dates and historical technical support. Actual valuation, tax, impairment, acquisition exceptions and applicable amendments remain required inputs. A retained 20% interest also needs a separate significant-influence and classification assessment before subsequent measurement; its percentage does not supply that conclusion. The same separation of legal-entity balances and group adjustments explains why eliminating an intragroup loan can leave a currency exposure.

36.20 Keeping the group calculation attached to its books

The disposal example gives the parent a cash receipt while removing cash from the consolidated balance sheet. A software implementation must know whose cash each entry describes. If it posts the removal of the subsidiary's 156 to the parent's legal book, it records a payment the parent never made. If it leaves the subsidiary's cash in the group after control ends, it overstates the group's assets. The consolidation calculation therefore needs its own book and exact references to the legal books it reads.

Call the legal entities Parent-P and Subsidiary-S, and the reporting entity Group-P. Parent-P's functional currency is USD; Subsidiary-S's is EUR. Their identifiers remain distinct even if two imported records have the same display name. Each legal-book snapshot identifies its entity, book, currency, relative date and signed balances. At acquisition the parent's investment 115.2 and cash payment stay in its own USD book. The subsidiary's opening cash 20, inventory 140 and payable 60 stay in its EUR book. Group-P's consolidation entries refer to both snapshots without altering either. This implements the separation between combining balances and eliminating the parent's investment against subsidiary equity (European Commission, 2023a, IFRS 10.B86–88).

The selected perimeter has equally precise dependencies. Its control assessment records the investor, investee, substantive rights, variable returns and ability to use power, together with the interval for which those findings hold. Ordinary ownership of 80% and proportionate NCI measurement are separate inputs. The currency record distinguishes the subsidiary's functional

currency from the group's presentation currency. A rate of 1.30 is usable here only when its units are USD per EUR and its date is the closing date R. An inverse quote or a later rate is a different input, even when its label looks familiar. The relative A, O, R and D calendar defines this synthetic history; it supplies no evidence of when an actual investor could have known its facts.

The group needs more state than the initial corporate example. Goodwill, the parent's translation reserve and NCI have distinct balances, while profit, OCI and cash movements retain their period histories. The proposed `GroupForeignOperation/1` profile supplies those fields under its own schema lock. It preserves the common request and result fields rather than silently adding accounts to the closed corporate profile. Loading this extension would still require an implementation to declare support for it; there is no automatic conversion of a corporate state into a group state.

Follow the existing history through that representation. At A the group records the acquisition. At O it translates the opening balances before the subsidiary's operating entries. At R it finishes cash and goodwill translation and allocates operating profit. A and O are intermediate worksheet states; they are not presented as fully closed interim reports. The R state retains cash 340.8, goodwill 20.8, parent translation reserve 10.4 and NCI 31.2, together with profit 25, OCI 12.6 and the separate cash exchange effect 7. Closing the income accounts does not erase those movements.

Full disposal and disposal with a retained 20% interest then start from separate copies of that same immutable R state. Their branch identities prevent the second alternative from being posted after the first. Each disposal proposal identifies its opening state, control change, source books, cash consideration, retained fair value, rates and accounting method. Its permission to post is conditional on this synthetic profile and exact proposal. It is not permission to post to an actual company's books.

In the retained-interest alternative, the parent's receipt is 126.75. The group removes subsidiary cash 156, producing investing cash $126.75 - 156 = -29.25$. The retained investment 42.25 is a separate noncash asset. Both disposal alternatives remove the entire NCI balance 31.2, including its translation reserve 2.2. Only the parent's reserve 10.4 is recycled into the disposal gain. These are checks on the meaning of the journal, beyond the fact that its debits equal its credits (European Commission, 2023a, IFRS 10.B98–99; IAS 21.48A–B; IAS 7.39–42).

The report reads one branch and one dependency view. In both alternatives, period profit is 58.8, of which 53.8 belongs to the parent and 5 to NCI. Period OCI is 2.2, attributable to NCI. Ending NCI is nevertheless zero, because the disposal removes its accumulated balance. The equity movement must therefore retain both its introduction and removal:

$$300 + 24 + 58.8 + 2.2 - 31.2 = 353.8.$$

Reading only the closing NCI account would lose the subsidiary's contribution while it was controlled. Reading only final equity would also miss the difference between full-sale cash 353.8 and retained-interest cash 311.55 plus investment 42.25.

Publication follows the common accounting transaction rule. The request identifies the group owner and expected opening revision; its business-event identity includes the branch. Before publishing, an implementation must validate the source and currency bindings, posting authorization, ordered journals and independently specified state changes. It then publishes the closing state, journal, economic delta, receipt and new group head together. The parent and subsidiary legal-book owners remain untouched.

An acknowledgement can be lost after that publication. Retrying the exact original request then finds the existing receipt and has no second effect, even though its original opening revision

is now old. If publication never occurred, the same request can commit once. Changing the request identifier, budget or expected head changes the fingerprint and cannot masquerade as that retry. A newly proposed event with an outdated opening is rejected. The documentary packet preserves both complete acknowledgement histories and an invalid partial-publication example; later runtime tests must exercise those failures in the chosen storage system.

A corrected exchange-rate source creates a new evidence view. It makes the old group's calculation unsuitable for a request using the corrected view, but does not overwrite the old books or remove the old explanation. Unknown control or an unknown rate instead returns the missing-input obligation, with no fabricated zero amount. The companion group-binding specification supplies exact records and exchanges for these cases. Actual control, valuation, tax, policy alignment and subsequent accounting for the retained investment retain their separate evidence requirements.

36.21 Eliminating the loan, retaining the currency exposure

An intragroup loan disappears from the consolidated balance sheet, yet its exchange gain can remain in group profit. The reason becomes visible when we follow both the loan and the money it financed. IAS 21.45 requires the exchange result to survive elimination; paragraph 32 determines when it belongs in other comprehensive income instead (European Commission, 2023a, IAS 21.32–33, .45).

Consider a parent whose functional and presentation currency is USD and a wholly owned subsidiary whose functional currency is EUR. Control and both functional currencies are established assumptions. At an opening rate of USD1.20 per EUR, the parent's balances are cash 56, an investment in the subsidiary carried at historical cost 24, and a EUR100 loan receivable worth 120. Its equity is therefore 200. The subsidiary holds EUR120 in cash, owes EUR100 to the parent and has EUR20 of paid-in equity. At the opening rate, its equity matches the parent's investment. Eliminating that investment and the loan leaves group cash and equity of 200.

We isolate an exchange-rate remeasurement of these established balances, holding every other measurement input fixed. This is neither the initial valuation of a new loan nor a complete reporting period. Interest, tax, impairment, hedges and new transactions are outside this calculation; IAS 21.24 leaves the underlying carrying amount to the relevant measurement standards. Thus EUR100 is a given monetary carrying amount, not an assumption that an interest-free long-term loan must be recognised at its face value (European Commission, 2023a, IAS 21.23–24, .50). All following statement amounts are in USD unless explicitly marked EUR.

Now let the rate rise to USD1.30 per EUR. The parent remeasures its receivable from 120 to 130: debit the receivable 10 and credit exchange gain 10. The subsidiary records no exchange gain on its EUR loan in its own EUR books. For group presentation, its cash becomes $120 \times 1.30 = 156$ and its payable becomes $100 \times 1.30 = 130$. Its historical capital remains 24. The remaining 2 is translation OCI, explained by its EUR20 net assets and the rate change: $20 \times (1.30 - 1.20) = 2$ (European Commission, 2023a, IAS 21.28, .33, .39–41).

Consolidation now debits the translated loan payable 130 and credits the parent's receivable 130. It also debits the subsidiary's historical capital 24 and credits the parent's investment 24. These entries eliminate the internal balances; neither entry removes the parent's exchange gain. Group cash is $56 + 156 = 212$, with no external liability in this example. Opening equity 200, exchange profit 10 and translation OCI 2 reconcile to the same 212. Deleting the profit merely because the receivable was internal would leave equity short by 10. The group still owns the euros whose dollar value has risen (European Commission, 2023a, IFRS 10.B86; IAS 21.45).

The loan's settlement facts can change the presentation of that 10. In the ordinary-loan case, settlement is planned in the foreseeable future, so the gain stays in consolidated profit. In a separate conditional case, assume settlement is neither planned nor likely in the foreseeable future and the loan forms part of the net investment under IAS 21.15. This is a substantive condition, not an election created by calling a loan long term; trade receivables and payables are explicitly excluded. The parent's separate gain is still 10. In the consolidated statements, debit exchange gain 10 and credit OCI 10, giving total OCI of 12 and no exchange profit. The two cases in Table 36.1 have identical group cash and total equity but different profit and OCI (European Commission, 2023a, IAS 21.15, .32–33).

Table 36.1: Settlement facts change where the exchange result appears.

USD amount	Ordinary loan	Qualifying net-investment loan
Parent's separate exchange gain	10	10
Consolidated exchange profit	10	0
Consolidated OCI	2	12
Consolidated closing cash and equity	212	212

Neither case includes a cash receipt. The same EUR120 remains in the subsidiary, so its translated value rises by $120 \times (1.30 - 1.20) = 12$. IAS 7.28 presents this exchange effect separately from operating, investing and financing cash flows. Group cash reconciles as opening 200, transaction cash flows zero and exchange effect 12, giving closing 212 (European Commission, 2023a, IAS 7.25–28). Adding a further gain of 12 would count the same currency movement twice.

The representation must therefore retain both entities' functional currencies, the monetary balance, the settlement assessment and the origin of each OCI component. If the settlement facts are unknown, these conditional presentations remain separate even when they agree on total equity. The group adjustment does not overwrite the parent's separate journal, and the OCI history remains available for the applicable later disposal treatment. An equal consolidated total cannot establish that the histories answer the same profit, cash-flow or disposal question.

36.22 An elimination belongs to an identified transaction

Equal balances need not describe the same debt. Suppose the parent records a receivable of 100 from subsidiary S, while another subsidiary T reports a payable of 100. Cancelling those numbers would balance the worksheet while misstating both relationships. The group must first establish the seller, buyer and economic event that connect the two entries. IFRS 10.B86 requires the elimination of intragroup transactions; it does not make equality of amounts evidence that two entries are counterparts (European Commission, 2023a, IFRS 10.B86–88, B92–93).

For each proposed elimination, retain the reporting group and perimeter version, both legal entities and book revisions, the common economic-event reference, invoice or settlement allocations, transaction currency, reporting cutoff and history branch. The two books' local event identifiers can differ; an evidenced mapping connects them to the common transaction. For balance elimination, use the perimeter at the reporting date; for income and cash-flow elimination, use the intervals during which control exists. A receivable created before control began can be intragroup at the reporting date without making the original sale an intragroup revenue transaction for that period. Acquisition measurement and subsequent movements must remain distinct. Align accounting policies, reporting dates and currency measurements before comparing balances. A reporting-date exception needs the conditions and intervening

adjustments in IFRS 10.B92–93.

One invoice may correspond to several ledger lines. An explicit allocation of 40 and 60 can exhaust a receivable of 100; reusing the 40 creates an invalid allocation of 140. Each source amount is consumed at most once in a given consolidation version. A receivable of 100 against a payable of 90 leaves an unresolved difference of 10 until a settlement, cutoff correction or other supported adjustment explains it. Equal labels, a balancing entry or a stale source revision cannot supply that explanation. Unknown matches remain visible and prevent a complete-consolidation result for that view. These identity rules are implementation choices for preserving the source meaning; they are not a claim that IFRS prescribes a database key.

Now follow one correctly matched inventory sale. The wholly owned subsidiary begins with cash and capital of 50. The parent holds cash 90, inventory 60 and the investment 50, with equity 200. All amounts use one currency; taxes, impairment and other transactions are absent by assumption. The parent sells the inventory to the subsidiary for 100 on credit. The subsidiary then sells 40 percent of those goods to an external customer for cash 60, recording cost 40 and retaining inventory 60. The parent's profit is 40 and the subsidiary's profit 20, but the group has earned only 36: external revenue 60 less the original cost 24 of the goods sold.

Eliminate the investment 50 against subsidiary opening equity 50 and the payable 100 against receivable 100. Of the internal markup 40, the fraction still in inventory is 60 percent. The performance adjustment therefore debits internal revenue 100 and credits cost of sales 76 and inventory 24. Combined cost of sales 100 becomes 24, inventory 60 becomes 36, and profit 60 becomes 36. Closing group cash 200 plus inventory 36 equals opening equity 200 plus profit 36. The same result follows directly from the external history: opening group cash 140 plus customer receipt 60, with inventory cost 60 less 24. The adjustment retains the lot and external-sale references needed to release the remaining markup when those goods leave the group. Actual tax bases, upstream NCI allocation and evidence that an intragroup loss reflects impairment require their own consequences under IFRS 10.B86(c). The earlier foreign-loan case also remains: eliminating a monetary balance does not automatically erase its exchange gain (European Commission, 2023a, IFRS 10.B86(c); IAS 21.45).

36.23 Restating purchasing power before translating currency

An inventory cost and a bank balance of the same nominal amount do not retain the same purchasing power. IAS 29 addresses this problem when the functional currency belongs to a hyperinflationary economy. Its paragraph 3 requires judgment across economic indicators; three-year inflation approaching or exceeding 100 percent is one indicator, not an automatic test. The selected assessment, effective period and general price index must therefore accompany the entity book (European Commission, 2023a, IAS 29.1–4, 8–19, 24–28, 37–39).

For historical-cost nonmonetary amounts, retain each acquisition or measurement date and restate from that date's index to the closing index. Restate income and expense from recognition dates, and contributed equity from the relevant contribution dates. Monetary balances remain at their closing nominal amounts; contractually index-linked amounts first follow their contractual adjustment. Amounts already measured at closing current values are not indexed again. After restatement, apply the relevant recoverability or net-realizable-value limit. Missing dates or indices require the specific permitted estimation procedure and its support; an unexplained common multiplier is insufficient.

Consider a period beginning with cash 100, inventory 60, fixed nominal debt 80 and contributed capital 80. All arose at the opening date, and there are no transactions, interest, tax or impairment during the period. Assume IAS 29 already applies and the general index rises from

100 to 150. Inventory becomes 90 and capital 120; cash 100 and debt 80 remain unchanged. Holding cash loses 50 units of closing purchasing power, while owing the fixed debt gains 40. The net monetary loss is therefore 10. If M denotes the constant excess of monetary assets over monetary liabilities, and I_0, I_1 the opening and closing indices, this special case gives

$$L_{\text{mon}} = M\left(\frac{I_1}{I_0} - 1\right) = (100 - 80)(1.5 - 1) = 10.$$

Positive L_{mon} is a loss. A net monetary liability reverses the sign; an unchanged price index gives zero. The restated balance sheet reconciles as $100 + 90 - 80 = 110 = 120 - 10$. The restatement entry debits inventory 30 and monetary loss 10 and credits restated capital 40. If inventory's supported closing net realizable value is 84, a separate write-down 6 reduces equity to 104. Indexing both monetary balances would incorrectly increase net assets by 10 and conceal the purchasing-power loss.

The constant-exposure formula does not apply to arbitrary turnover. For a piecewise constant nominal exposure, split the history at every monetary change. If M_j is the net monetary amount held between index observations I_{j-1} and I_j, its loss expressed at closing index I_T is $M_j(I_j/I_{j-1} - 1)I_T/I_j$; sum those interval losses and separately handle contractually indexed balances. Recognition times and index observations determine those intervals. IAS 29.27 also permits a weighted-average estimate; its approximation must be identified. Reconcile the result with restated nonmonetary amounts, equity and income rather than inventing a residual profit. For example, external cash income 30 at index 125 changes the preceding net monetary exposure from 20 to 50. The two losses are 6 and 10, totaling 16; restated income 36 and monetary loss 16 reconcile closing net assets 140 to restated opening capital 120. Cash-flow items likewise use closing purchasing-power units. In the no-flow case, opening cash expressed in closing units is 150 and ending cash 100; the purchasing-power reconciliation of minus 50 is not a payment (European Commission, 2023a, IAS 29.26–34).

First application requires earlier acquisition history. IFRIC 7.3–5 requires restatement as though the economy had always been hyperinflationary and a particular ordering for opening deferred tax. In a separate inventory example, cost 40 at index 80 becomes 50 at the opening index 100 and 75 at the closing index 150. Assume a fixed nominal tax base 40, tax rate 20 percent, profit-origin differences and no recognition exception. Opening deferred tax is $(50 - 40) \times 0.2 = 2$, then restated to 3. Closing deferred tax is $(75 - 40) \times 0.2 = 7$; the period expense is 4. Indexing a nominal opening deferred-tax balance of zero would miss this sequence (European Commission, 2023a, IFRIC 7.3–5; IAS 29.32).

Only then translate the restated statements. At a supplied closing rate of USD 0.20 per local unit, the first example gives cash 20, inventory 18, debt 16, capital 24 and monetary loss 2: net assets 22 equal 24 minus 2. Translating unrestated inventory 60 would report 12 and understate the asset by 6. IAS 21.42–43 uses the closing rate for these restated current-period amounts. When the presentation currency is not hyperinflationary, prior comparative amounts remain those presented in the relevant prior-year statements. Thus prior cash 100 translated at the prior rate 0.25 remains USD 25, even though the current local comparative is restated. Keep these two comparative representations distinct (European Commission, 2023a, IAS 21.42–47; IAS 29.34–36).

36.24 A quoted rate may not be available for the required purpose

A subsidiary may obtain foreign currency to import goods while being unable to remit its net investment. Exchangeability therefore belongs to a currency direction, measurement date and

purpose. The inspected EU amendment 2024/2862 requires application no later than financial years beginning on or after 1 January 2025; IAS 21.60L permits earlier application with disclosure. The following method uses that amendment, rather than extending the older paragraph 26 rule about temporary lack of exchangeability to later periods (European Commission, 2024, Article 2; IAS 21.8–8B, 19A, 26, 60L–M).

Record the entity, both currencies, measurement date, purpose, amount needed, amount obtainable, permitted mechanisms and their enforceability, normal administrative delay and evidence for the finding. The assessment concerns ability, including indirect exchange, rather than management's intention. For individual transactions, the purpose is realization or settlement of those transactions; for presentation-currency translation it is realization or settlement of net assets or liabilities; for a foreign operation it is realization or settlement of the net investment. Assess the relevant direction and purpose separately. Obtaining only an insignificant amount fails the exchangeability test. The ratio 5 available to 1000 needed is evidence for a judgment, not a threshold supplied by this design (European Commission, 2024, IAS 21.A2–A10).

When exchangeability fails, IAS 21.19A requires an estimate reflecting an orderly exchange between market participants at the measurement date under prevailing economic conditions. A rate observable for another purpose, or the first subsequent exchange rate, can be used without adjustment only if it meets that objective. Examine incentives or penalties, restricted purposes, interventions, update frequency, elapsed time and inflation. Another estimation technique may adjust observable rates, including those from mechanisms that would not establish exchangeability. It must preserve the inputs, adjustments and reasoning supporting the measurement-date objective. Absence of that support leaves the rate unresolved; the design does not manufacture a universal estimator (European Commission, 2024, IAS 21.A11–A17).

For a conditional arithmetic illustration, assume an admitted estimate of USD 0.08 per local unit for a foreign operation's net investment. Local net assets 1000 translate to 80. A preferential import rate 0.10 would produce 100; a later rate 0.09 following changed conditions would produce 90. Neither alternative is interchangeable with the admitted estimate merely because it is observable. This calculation exposes the consequences of the choice; the numerical differences do not prove that 0.08 meets paragraph 19A.

Initial application has its own journal origin. Suppose an affected monetary payable of 100 foreign units moves from 10 to 12 functional units per foreign unit at initial application. The liability increase 200 adjusts opening retained earnings under paragraph 60M(a). In a separate nonhyperinflationary foreign-operation translation example, changing the rate for net assets 1000 from 0.10 to 0.08 reduces translated equity 20; paragraph 60M(b) assigns the initial effect to accumulated translation differences. Hyperinflationary functional-currency equity items also use the estimated initial rate. Comparatives are not restated for this amendment. Keep these transition adjustments distinct from subsequent-period profit or OCI (European Commission, 2024, IAS 21.60M).

The accompanying disclosures identify affected currencies and restrictions, transactions and carrying amounts, rates, estimation methods and assumptions, and exposure to risk. A foreign operation also needs the additional information in paragraph A20, including relevant support arrangements (European Commission, 2024, IAS 21.57A–B, A18–A20). These methods extend the documentary group design. They do not silently add fields or supported events to the closed `GroupForeignOperation/1` exchange profile; its successor must explicitly bind the additional histories and assessments before an implementation can claim these operations.

Chapter 37

Credit losses and the bank balance sheet

The funding and default distinctions become central in a bank. A loan creates an asset, customer payments affect settlement liquidity, and a forecast of credit loss affects an allowance before cash is lost. The first case connects funding to a simple allowance; the later dated loan schedule separates credit stages, interest recognition and eventual recovery.

37.1 Banks: accounting capital is not regulatory capital

A bank package needs instrument and cohort states for loans, securities, deposits, wholesale funding, repos, derivatives, collateral and commitments. For interest-rate analysis, currency and repricing buckets matter alongside remaining maturity. For credit analysis, exposure, default state, loss severity, recoveries, guarantees and their dependencies matter. For liquidity analysis, deposit behaviour, collateral eligibility, encumbrance and payment timing matter. These states can be represented at an explicitly chosen aggregation level; the public record rarely identifies every contract.

37.1.1 A reported group boundary

JPMorgan Chase's filings make the boundary between a bank, its parent and its securities issuers concrete. In the 2022–2025 reports, Item 1 identifies JPMorgan Chase Bank, N.A. as the principal bank subsidiary and describes an intermediate holding company, JPMorgan Chase Holdings LLC (the IHC). The reports also describe J.P. Morgan Securities LLC as a principal non-bank subsidiary and place J.P. Morgan Securities plc and J.P. Morgan SE beneath the bank. These are reported relationships for the stated dates, not a complete legal-entity graph (JPMorgan Chase & Co., 2023, 2024, 2025, 2026).

The attached Exhibit 21 tables qualify their scope: they list significant legal-entity subsidiaries, some parents of significant subsidiaries and certain entities designated for resolution planning, while excluding subsidiaries that would not be significant in aggregate under the stated SEC definition. Their indentation shows how the issuer presents the rows, but it does not supply ownership percentages, tax residence, functional currency or unrestricted transfer rights. A model therefore keeps a reported display relationship separate from a confirmed control edge.

Exhibit 22.2 adds a different edge. It names JPMorgan Chase Financial Company LLC as issuer of Global Medium-Term Notes, Series A, and JPMorgan Chase & Co. as guarantor. Financial

Company LLC does not appear in those qualified Exhibit 21 lists. The 2022–2023 text describes Series A notes as currently offered and sold under one cited registration; the 2024–2025 text also says that the issuer may issue debt securities and warrants under a later cited registration. The descriptions say that the relevant securities are fully and unconditionally guaranteed, but they give no outstanding principal, guarantee valuation or accounting recognition. They also do not establish that a warrant was issued or that an earlier note was redeemed.

The same reports state that the parent depends on subsidiary cash flows to pay its securities. The bank's dividends are subject to regulatory restrictions, and the IHC cannot pay dividends or extend credit to the parent when stated capital or liquidity thresholds, or management and board limits, apply. This means that consolidated cash, a subsidiary's assets and the parent's distributable funding are different state variables. The example gives the simulator a place to record the legal entity, guarantee and conditional funding path before it assigns a cash movement; it does not infer a balance or a probability of default.

37.1.2 Parent income, cash and conditional transfers

Note 33 makes the funding distinction measurable for 2025. Its parent-only income statement reports net income of US$57,048m. The parent-only cash-flow statement subtracts US$71,163m of subsidiary and affiliate net income, reports a US$14,115m parent company net loss in that reconciliation, then adds US$50,000m of cash dividends from subsidiaries and US$8,583m of other operating adjustments to reach US$44,468m of operating cash. Neither the income figure nor the dividend alone is the period's ending cash.

At 31 December 2025, the parent reports US$39m of cash and due from banks and US$9,751m of deposits with banking subsidiaries, together US$9,790m. The cash-flow footnote says this measure includes US$8.0bn of intercompany time deposits. The same parent-only balance sheet reports US$669,449m of equity-method investments in bank and bank-holding subsidiaries; that asset is not parent cash. Operating cash of US$44,468m and investing cash of US$34m, less US$44,512m used in financing, leave closing cash and deposits US$10m below their opening balance. These dated amounts distinguish accounting income, existing cash and conditional subsidiary transfers. A future debt-service calculation still needs payment dates, permissible transfers, deposit terms and funding actions (JPMorgan Chase & Co., 2026, pp. 19, 310–311).

Note 26 supplies a dated transfer capacity, which differs from both cash held and cash already received. It reports that, at 1 January 2026, JPMorgan Chase Bank, N.A. could pay approximately US$30.1bn in dividends to the parent without prior approval from its relevant banking regulators. That is a capacity of the named bank at that date; the US$50,000m above is cash received from subsidiaries during 2025. The note also describes regulators' power to prohibit unsafe or unsound dividends and the IHC's separate capital, liquidity and management restrictions. It says the bank's 2026 earnings will supplement dividend capacity, without estimating those earnings. Turning the reported capacity into a future cash receipt still requires a dividend decision, payment date and the applicable conditions (JPMorgan Chase & Co., 2026, pp. 4, 292).

The same note reports US$29.0bn of consolidated restricted cash at 31 December 2025: US$19.4bn segregated for securities and cleared-derivative customers, plus US$9.6bn of non-US central-bank reserves and cash held for other general purposes. Its balance-sheet footnote allocates the total to US$27.8bn of deposits with banks and US$1.2bn of cash and due from banks. Those classifications describe the same restricted-cash total. Subtracting it from the parent's US$9,790m would mix consolidated and parent-only perimeters and cannot determine the parent's unrestricted cash. Other pledged cash and securities are disclosed separately; their category overlap must be established before adding them to a restricted-resource total

(JPMorgan Chase & Co., 2026, p. 292).

37.1.3 Dividend declarations and payout restrictions

For a future cash dividend, the model needs the bank's declaration as well as the eventual payment. The OCC's national-bank dividend rules, retrieved for 1 January 2026, use the declaration date to determine compliance. They define retained net income as net income less all dividends declared in the period. Absent OCC approval under the earnings provision, current-year declarations, including the proposal, are compared with current-year net income to date plus the previous two years' retained net income, less specified required transfers. The current year is the calendar year of declaration. A cash-flow statement of dividends paid therefore cannot supply the declaration history by itself (Office of the Comptroller of the Currency, 2026f, 12 CFR 5.61–5.64).

The preceding two years are not always enough. Section 5.64(c)(2) provides conditional offsets for retained losses caused by dividends exceeding a single year's net income. Those offsets can require the third and fourth preceding years and follow a specified order; other earnings deficits do not qualify for that adjustment. An implementer must preserve the cause of the loss and the allocation of offsets before computing the earnings restriction. Capital, undivided-profit and supervisory conditions remain separate: an earnings calculation alone cannot establish a permissible transfer (Office of the Comptroller of the Currency, 2026f, 12 CFR 5.63–5.65).

Selecting the applicable capital rule also requires more than a calendar lookup. The final leverage rule published on 1 December 2025 is effective 1 April 2026 but permits early adoption from 1 January (OCC, Federal Reserve Board, and FDIC, 2025, p. 55248). JPMorgan's 2025 Form 10-K reports that the Firm elected early adoption effective 1 January 2026 (JPMorgan Chase & Co., 2026, p. 91). The rule's publication, general effective date, permitted election date and the issuer's reported election answer different questions. The later filing provides evidence about the election; it does not establish that the election was publicly known on 1 January. A historical information query must retain that distinction even when reconstructing an earlier regulatory state.

The amendment removes the additional 6% supplementary-leverage condition from the affected national banks' well-capitalized classification and adds separate leverage-buffer payout provisions. That classification change does not remove the distinct undercapitalization test or establish unrestricted distributions (OCC, Federal Reserve Board, and FDIC, 2025, pp. 55287–55288). The reported Firm election also does not supply every bank-specific application fact. A permissible dividend calculation still needs the relevant capital definitions, distribution limits, supervisory conditions and bank-level inputs. The US$30.1bn remains the issuer's dated capacity disclosure, rather than a recomputed legal limit.

One of those inputs, eligible retained income, differs from the retained net income used in the national-bank earnings rule. Part 3 takes the greater of two amounts: Call Report net income for the four preceding calendar quarters, net of distributions and associated tax effects not already reflected in that income; and the average quarterly net income over those four quarters. Its maximum payout amount for the current quarter multiplies this eligible income by the applicable payout ratio. The relevant capital buffers use the previous quarter's closing ratios, so the declaration, income window and capital-measurement date must remain distinct (Office of the Comptroller of the Currency, 2026d, 12 CFR 3.11(a)(2)–(4)).

The transactions being limited also extend beyond a cash dividend. Part 3's distribution definition includes dividend declarations or payments on Tier 1 capital instruments and specified capital repurchases or redemptions, with conditional exceptions for qualifying replacements (Office of the Comptroller of the Currency, 2026b, 12 CFR 3.2). The quarterly limit also covers

discretionary bonus payments and the creation of obligations to make the specified payments. A bank's event history must therefore identify the instrument, declaration, obligation and settlement, then reconcile their regulatory treatment; a cash-dividend total cannot substitute for that history (Office of the Comptroller of the Currency, 2026d, 12 CFR 3.11(a)(4)).

A hypothetical bank in the first quarter of a year illustrates the income distinction. Suppose each of the preceding four quarters has Call Report net income of US$10m. Assume its only distributions in that preceding calendar year were US$38m of declared dividends, with no associated tax adjustment. Part 5 retained net income for that year is then US$2m: US$40m less US$38m. Part 3 eligible retained income is US$10m, the greater of that US$2m remainder and the US$10m quarterly average. These are two calculations for the same hypothetical history, not interchangeable names for one balance.

Now suppose the bank falls within the amended rule's covered population and applies that rule. Take its previous-quarter capital conservation buffer as 1.5%, its applicable countercyclical amount as zero, its leverage buffer standard as 1%, and its measured leverage buffer as 0.2%. These are supplied example inputs, rather than estimates from a public balance sheet. The rule's Table 1 gives a 40% payout ratio, which alone would produce US$4m from the US$10m eligible income. Table 2 gives zero because the leverage buffer is at most one quarter of its standard. The required lower ratio therefore gives a zero table-based maximum, absent the specified OCC permission. Conversely, a table entry saying that no payout-ratio limitation applies does not mean a 100% ratio or permission under every other rule (Office of the Comptroller of the Currency, 2026d, 12 CFR 3.11(a)(4), (c), Tables 1–2).

37.1.4 Systemic indicators and the applicable surcharge

Replacing the example's supplied leverage-buffer standard with an observed requirement needs a specific surcharge. For a bank within the amended rule's covered population, the standard is the lesser of 1% and, if applicable, half of the controlling global systemically important holding company's most recent required Method 1 surcharge, subject to the specified effective-date provisions. The holding company's overall surcharge instead takes the greater of its Method 1 and Method 2 surcharges. The overall figure alone does not identify the input this bank rule requires. Annual determination and the date an adjusted surcharge takes effect also remain separate (Office of the Comptroller of the Currency, 2026d, 12 CFR 3.11(a)(1)(v)) (Board of Governors of the Federal Reserve System, 2026c, 12 CFR 217.403(a)–(d)).

Method 1 compares twelve reported systemic indicators with published global amounts, applies their specified weights and caps the substitutability category's contribution at 100 basis points (Board of Governors of the Federal Reserve System, 2026b, 12 CFR 217.404). The denominators have their own measurement and publication dates. The Federal Reserve's notice published on 16 December 2025 supplies the 2025 denominators using year-end 2024 global data, converted from euros into dollars at the stated 31 December 2024 rate. The preceding notice, published on 18 December 2024, uses year-end 2023 data (Board of Governors of the Federal Reserve System, 2025, pp. 58245–58246) (Board of Governors of the Federal Reserve System, 2024, pp. 102908–102909). The release-year label therefore does not identify the underlying observation year or the date a bank's capital requirement changes. The published amounts are in dollars; reported indicator values must be brought to the same unit before division.

Reporting precision adds another constraint. The retained FR Y-15 general instructions require dollar amounts in thousands and permit rounding to the nearest million, with adjusted detail so the components add to the totals. Its payment-activity instructions instead say payment totals are not rounded, specify accuracy according to their magnitude and allow known overestimates when precise totals are unavailable. A reconstruction cannot treat every reported input as an

exact value or give all components independent rounding intervals (Board of Governors of the Federal Reserve System, 2026d, GEN-3–4 and C-1; physical pp. 7–8, 25).

Rounding the resulting score is a separate operation. In the background to its March 2026 proposal, the Federal Reserve describes scores as rounded to the nearest basis point: fractions below one-half round down and fractions at or above one-half round up (Board of Governors of the Federal Reserve System, 2026a, p. 14909, n. 9). The earlier description appears in the 2015 final rule's preamble, within its account of the preceding proposal (Board of Governors of the Federal Reserve System, 2015, p. 49084, n. 22). These passages provide an official explanation of the convention; they do not complete the amendment history needed to establish its applicability on a chosen date. The 2026 proposal also contains a separate ceiling formula for Method 2 surcharge bands and changes to indicator averaging and denominator exchange rates. Those proposed changes cannot be substituted for the rule applicable to the bank and date being reconstructed (Board of Governors of the Federal Reserve System, 2026a, pp. 14917–14922).

The report's quarter-end date differs from its submission deadline and public availability; amendments, later resubmissions and confidentiality can change what a historical reader could know. The retained instruction file itself combines a September 2021 cover with a January 2026 reporting-scope page. Applying it consequently requires the relevant page and reporting vintage, alongside the applicable rule and institution facts. These sources explain the input requirements; they do not yet supply a verified bank-specific score or a complete reporting-rule history (Board of Governors of the Federal Reserve System, 2026d, cover and GEN-1–5; physical pp. 1, 5–9).

37.1.5 Debt measurement, payment rights and succession

The payout rules constrain cash leaving the bank. Debt contracts determine a different set of claims on the group: who owes a payment, when it falls due, and what a holder can require after a breach. Debt amounts therefore need their own measurement and contractual histories. Note 20 puts US$253,039m of long-term-debt carrying value under Parent company for 2025, while Note 33's parent-only balance sheet shows US$254,633m. The US$1,594m difference equals Note 20's separately displayed junior-subordinated-debt subtotal. The same comparison gives US$1,603m in 2024. This exact two-year arithmetic match does not establish how that row relates to the parent-only balance, or which legal entity owes each instrument. Its accounting and contractual relation still needs source support.

Timing requires a different measure. Note 20 places US$14,298m of parent carrying value in its under-one-year bucket; Note 33 reports US$14.3bn of parent long-term debt contractually maturing in 2026. They align at the footnote's displayed precision, but carrying value includes premiums, discounts, costs and valuation adjustments rather than simply principal payable. For the separately reported consolidated zero-coupon population, Note 20 gives US$18.1bn of carrying value against US$57.6bn of aggregate principal at the notes' respective maturities. The firm's US$435,206m consolidated long-term-debt total and US$47.6bn of subsidiary debt and structured notes guaranteed by the parent are also separate from the parent's own debt balance; the latter guarantee amount supplies neither a liability valuation nor a payment forecast. A dated parent debt-service path still needs instrument terms, cash dates and permissible funding transfers (JPMorgan Chase & Co., 2026, pp. 281–282, 310–311).

The complete Form 10-Ks describe Series A as the issuer's direct, unsecured general obligations, ranking equally with its other unsecured and unsubordinated debt. Claims under the parent guarantee generally rank behind those of subsidiary creditors. Their summary of the 2016 Indenture requires the parent, on demand after issuer nonpayment, to make full and punctual payment of principal, interest and other amounts due: a guarantee of payment, not collection (Sections 14.01–14.02). It excludes amounts the issuer is excused from paying or beyond those actually due. Issuer defenses and counterclaims carry over, except those arising solely from insolvency or bankruptcy status. These descriptions refer to the Indenture and applicable prospectus supplements; they specify reported terms but do not supply a current balance, valuation or accounting recognition.

The complete filings also describe one particular Series A population: the Callable Fixed Rate Notes due 10 June 2032. They report US$5,000,000 of aggregate principal outstanding at 31 December 2022 in the 2022 filing, at 31 December 2023 in the 2023 filing, and at 31 December 2024 in both the 2024 and 2025 filings. The last report repeats a 2024 as-of date; this passage supplies no 2025 year-end balance. The notes were issued on 10 June 2021 and bear 2.60% annual interest payable semiannually on 10 June and 10 December. They mature on 10 June 2032 unless an earlier contractual outcome applies. The issuer may redeem all, but not part, on 10 June 2031 at principal plus accrued and unpaid interest. Upon acceleration after a continuing event of default, the stated amount is US$1,000 per US$1,000 principal plus accrued and unpaid interest. The descriptions report NYSE listing under JPM/32, DTC global book-entry issuance, and the possibility that a reopening increases aggregate principal outstanding (JPMorgan Chase & Co., 2023, 2024, 2025, 2026).

These dated amounts are reported principal, not carrying or fair value. Nor do they establish the accounting classification or value of the parent guarantee, the entire Series A population, or an unchanged balance between the stated dates. Each filing's Exhibit 4.4 identifies the 19 February 2016 Indenture and incorporates it by reference to Exhibit 4(a)(7) of Form S-3 file 333-209682, filed 24 February 2016. The complete filings summarize that instrument; its text and applicable supplements or amendments require their own inspection before being used as complete legal terms.

The filings then describe general Series A terms that may apply to securities including the Callable Notes. In their summary, unpaid interest must continue for 30 days to become an event of default; principal and other amounts default when due. An issuer covenant breach has a 90-day period after the specified written notice. For listed continuing payment, issuer-covenant or guarantee events, the trustee or holders of at least 25% of the aggregate principal then outstanding for the relevant series may, by written notice to the issuer and guarantor, declare principal and accrued interest due. Specified continuing issuer insolvency events accelerate the debt automatically without a declaration. Guarantor insolvency alone is excluded and does not authorize acceleration under this description, while a payment default by the guarantor remains a separate possibility (JPMorgan Chase & Co., 2023, 2024, 2025, 2026). A history must therefore keep the failure, notice, elapsed time, actor, declaration and changed due schedule distinct, rather than record one group-wide default flag.

Collective remedy control also differs from an individual's right to payment. The summaries condition an ordinary holder remedy action on prior trustee notice, a request by holders meeting the stated 25% threshold, 60 days of trustee inaction, indemnity and no inconsistent majority direction. They separately protect each holder's right to receive and sue for payment when due against impairment without that holder's consent. The summary permits certain amendments without holder consent under Section 8.01. For consent-based modifications under Section 8.02, it specifies a majority in aggregate principal amount of each affected outstanding series, whereas listed changes to maturity, amounts, currency, payment method, redemption, payment suit or specified holder repurchase rights, voting threshold or adverse guarantee terms require each affected holder's consent. These thresholds attach to different actions and voting populations, not a generic creditor vote. The 2022 default-and-waiver subsection omits the trustee-conflict paragraph that the 2023–2025 versions place there. The 2022 filing discusses a potential default, trustee conflict and possible resignation later under its trustee heading. This change in placement does not establish a contract amendment. The original Indenture and series-specific supplements remain uninspected (JPMorgan Chase & Co., 2023, 2024, 2025, 2026).

Reported succession terms give another reason to keep the obligor separate from the debt claim. Under the general Series A summary, a successor to the issuer in a covered merger, consolidation or transfer of all or substantially all assets must be a U.S. corporation or limited liability company that expressly assumes due payment and the issuer's covenants. A successor to the guarantor must instead be a U.S. corporation that assumes the full guarantee and the guarantor's covenants. The continuing issuer or guarantor is the other stated route. Each party's route also requires that no default or event that would become a default with notice or time has occurred or would be continuing immediately after the transaction. The descriptions deem certain resolution-related transfers of the guarantor's material assets to meet the all-or-substantially-all category; they do not define a numeric percentage (JPMorgan Chase & Co., 2023, 2024, 2025, 2026).

The same summaries say a recapitalization, change of control or highly leveraged transaction alone receives no additional holder protection from these merger provisions; a particular supplement may provide a put or higher interest. The model therefore needs a dated successor, assumed payment and covenant duties, separate issuer/guarantor conditions and any evidenced series-specific protection. Succession itself does not prove repayment, accounting derecognition or unrestricted access to the group's cash. These are reported terms that may apply, pending inspection of the original Indenture and applicable supplements.

37.1.6 Loan creation, commitments and expected credit losses

Loan origination can debit loans and credit a customer deposit in the bank's book. Subsequent transfer to another bank produces settlement flows. Treating all loan growth as an immediate cash outflow would therefore encode a specific funding assumption, not a universal banking identity. A system-wide model must also link counterparties and settlement assets; single-company simulation records external counterparties without claiming to close the whole economy.

An allowance calculation is a dated measurement based on forecast information. For a single horizon, let EAD be a fixed exposure at default, PD the probability of default, and LGD the mean loss fraction conditional on default. Ignoring discounting, the illustrative expected loss in currency units is

$$L = EAD \times PD \times LGD.$$

With exposure 100, default probability 0.02 and loss given default 0.4, it produces 0.8. Changing those inputs to 0.04 and 0.5 produces 2, so the allowance increase is 1.2 if there are no other movements. This calculation is a demonstration model. A standard-specific expected-credit-loss implementation must determine horizon, staging or other classification, discounting, forward-looking information, write-offs and purchased-credit-impaired or other special cases. The simple formula is not an IFRS 9 or US GAAP compliance implementation.

The exposure entering an allowance calculation also needs a defined population. JPMorgan's 2025 report gives four different amounts for lending-related commitments at 31 December 2025:

Reported quantity	US$ millions
Contractual lending-related amount	1,817,307
Commitments in the allowance-methodology table	603,947
Allowance for lending-related commitments	5,071
Lending-related carrying value	6,278

Notes 13 and 28 define these populations and measurements separately (JPMorgan Chase & Co., 2026, pp. 258–262, 295–297). The contractual total contains consumer and wholesale commitments; the allowance table excludes specified populations. The commitment allowance is reported in accounts payable and other liabilities. Carrying value can also contain guarantee liabilities, fair-value measurements and deferred fees or purchase adjustments, as applicable. For standby letters of credit and other financial guarantees, the report explicitly reconciles a US$175m allowance and a US$441m guarantee liability to US$616m of carrying value. Treating either the contractual total or carrying value as the allowance would compute the wrong quantity.

The report also distinguishes contractual capacity from expected funding. Customers may leave facilities unused, refinance them or let them expire; some commitments can be reduced or cancelled under their contracts or law. The reported policy records no allowance for future draws on unconditionally cancellable commitments. That exclusion concerns the loss measurement; a liquidity scenario still needs the conditions under which a customer can draw, the bank can cancel, and a payment requires settlement funds. Those conditions and their dependence on borrower distress cannot be recovered from a commitment total. Comparisons also need the same reporting basis: specified wholesale amounts are net of risk participations here but gross in Federal Reserve filings. Parent guarantees of subsidiary debt are expressly outside the Note 28 table (JPMorgan Chase & Co., 2026, pp. 258, 295–296, 300).

The historical FASB amendment ASU 2016-13 helps explain these distinctions. Its paragraph 326-20-30-11 measures off-balance expected credit losses over the contractual period of a present obligation to extend credit, except for an obligation the issuer may cancel unconditionally. For a covered commitment, the estimate considers both whether funding will occur and the expected loss on commitments that do fund. Example 10 keeps funded credit-card loans within loss measurement while excluding an allowance on unconditionally cancellable undrawn credit, even though the bank has sometimes advanced funds before detecting default. Paragraph 460-10-30-5 separately requires a contingent expected-loss liability and a noncontingent fair-value guarantee liability for covered financial guarantees. These are historical amendment provisions, not verified current Codification; applying them to the 2025 report requires the actual contracts and later rule history (Financial Accounting Standards Board, 2016a, PDF pp. 2, 108, 118, 144, 173–176).

A commitment to acquire loans makes the time distinction especially clear. The report dates the Apple Card forward purchase commitment to 30 December 2025, its announcement to 7 January 2026, and expected closing to approximately 24 months later. The commitment disclosure includes about US$104bn of estimated credit exposure at closing, including about US$23bn of estimated drawn loans, while the reported transaction allowance is US$2.2bn (JPMorgan Chase & Co., 2026, pp. 47, 260, 296). The expected drawn loans at that future closing are not loans already owned at the 2025 reporting date. A reconstruction therefore retains the forward commitment, the future portfolio estimate and the dated allowance as different facts. A forecast must supply closing and subsequent draw assumptions before generating loan and settlement events. These reported policies and amounts support the distinctions; their application to a new calculation still requires the operative accounting rules and actual contract facts.

37.1.7 Reconciling accounting equity to regulatory capital

Regulatory capital is computed in a separate prudential layer with its own perimeter, eligibility, deductions, risk weights and transition provisions. Accounting equity feeds a reconciliation to capital; it is not automatically Common Equity Tier 1. Liquidity and capital constraints influence the management policy, which in turn generates asset sales, funding or capital events. A breached ratio can be an economically meaningful scenario outcome. No hidden capital injection is permitted to repair it.

The United States national-bank rules show why that reconciliation needs records for each instrument. Under the common-stock criteria in the retained January and April 2026 texts, classification as equity under GAAP is only one condition. The instrument must also satisfy conditions concerning payment and direct issuance by the bank, priority in insolvency, loss absorption, permanence, dividend discretion, purchase financing and guarantees. The rules distinguish common equity Tier 1, additional Tier 1 and Tier 2, each with its own eligible elements and deductions. A security's name and its balance-sheet category therefore cannot supply its regulatory treatment (Office of the Comptroller of the Currency, 2026c, 12 CFR 3.20(a)–(d)).

Regulatory recognition can also change while a debt remains outstanding. The general Tier 2 instrument criteria require an original maturity of at least five years. At the beginning of each of its last five years, the amount eligible for Tier 2 falls by 20% of the original amount, net of redemptions; the instrument is excluded when less than one year remains. Calls and early redemptions have further conditions, including OCC approval. This reduction in eligible capital does not itself reduce the accounting debt. The contract, issuance and redemption history must support both amounts separately (Office of the Comptroller of the Currency, 2026c, 12 CFR 3.20(d)(1), including footnote 7).

Deductions require similarly specific records. Section 3.22 deducts goodwill and specified other assets from common equity Tier 1, with separate rules for their tax effects. Its general rule permits an associated deferred tax liability to reduce an asset deduction only if that liability would be extinguished when the asset is impaired or derecognized. The same liability cannot offset two assets; separate conditions govern offsets against deferred tax assets, and the chosen netting treatment must be consistent across reporting periods unless the OCC approves a change. Accumulated other comprehensive income has its own adjustments and election conditions, including limits on which institutions may opt out and when they must elect. Reconstructing capital consequently needs asset-specific tax links, instrument facts, institution classification and the actual election history. The reported equity and aggregate deferred-tax balances do not determine that reconstruction (Office of the Comptroller of the Currency, 2026a, 12 CFR 3.22(a), (b), (e)).

37.2 A bank's cash, allowance and capital in one path

The bank reference state contains settlement cash 30, gross loans 100, allowance 0.8, deposits 110 and equity 19.2. Originating a loan of 20 first debits loans and credits the borrower's deposit. A subsequent transfer of 15 to another bank debits deposits and credits settlement cash. Loan origination and external cash use are consequently distinct events. Receive loan interest 6 in cash and credit deposit interest 2 to customer balances. Remeasure the allowance to 2.4, posting an additional loss of 1.6. The resulting state is cash 21, gross loans 120, allowance 2.4, deposits 117 and equity 21.6. Net assets are $21 + 120 - 2.4 = 138.6 = 117 + 21.6$.

Now 25 of deposits runs off. The declared funding policy uses cash first, then a committed facility with limit 10. It draws 4 and settles 25, leaving cash zero, deposits 92 and wholesale debt 4. A runoff of 40 instead leaves 9 unpaid even after the maximum facility draw. The simulator records default and retains the unpaid deposit obligation. It does not assume unlimited central-bank or shareholder funding.

The prudential calculation is a separate, illustrative policy. Deduct 2 from accounting equity to obtain capital 19.6; apply a 75% risk weight to gross loans for risk-weighted assets 90. The resulting ratio is 19.6/90. Neither the deduction nor the weight is a Basel default. Actual prudential packages require jurisdiction, exposure class, eligibility, buffers and dated rules, with a reconciliation from the accounting book. The structure follows the need to model capital and liquidity jointly in Behn et al. (2019, Section 2), without transferring that paper's estimated preferences or shock parameters to this bank.

37.2.1 An allowance affects capital and exposure in different ways

The illustrative ratio above shows where a prudential calculation belongs. Applying a particular rule requires a further step: reconcile its numerator and denominator to the same bank's accounting records. An allowance already reduces accounting equity through expense, yet a specified portion may enter Tier 2 capital. The regulatory loan exposure can meanwhile remain gross of that allowance. Using the net loan balance for every purpose would erase these distinctions.

Consider a separate synthetic US national bank, with amounts in millions of US dollars. Its conditional reporting window is January–March 2026. For this calculation, freeze the retained 1 January 2026 national-bank rules; this assumption does not establish the rules actually applicable to a 31 March filing. The bank has no subsidiaries, minority interests, AOCI, deferred taxes, other capital instruments or deductions apart from goodwill. It is outside the advanced-approaches, Category III, market-risk and community-bank leverage-ratio regimes, and has no transition or special supervisory adjustment. Its common stock is supplied as satisfying all thirteen conditions in section 3.20(b)(1). These are explicit case facts, not conclusions drawn from a security's name (Office of the Comptroller of the Currency, 2026c, 12 CFR 3.20(b)(1)–(2)).

Settlement cash is a direct Federal Reserve balance. The loans are ordinary, unsecured, performing corporate exposures measured at amortized cost: no purchased credit-deteriorated loans, nonaccrual, ninety-day delinquency, HVCRE, securities, derivatives or credit mitigation. The bank has adopted CECL. Its allowance amounts are supplied accounting measurements, not estimates made here, and all qualify as adjusted allowances for credit losses, or AACL. There is no allocated transfer risk reserve or off-balance credit exposure. These restrictions are consequential: a deposit at another commercial bank and a nonaccrual corporate loan need different exposure analysis (Office of the Comptroller of the Currency, 2026b, 12 CFR 3.2, definitions of AACL and carrying value).

The opening book contains cash 30, gross loans 100, goodwill 2 and allowance 0.8 against deposits

110, common stock 20 and retained earnings 1.2. Assets net of allowance are $30+100+2-0.8 = 131.2$, so equity is 21.2. Originate a loan of 20 by crediting the borrower's deposit, transfer 15 externally, receive earned loan interest 6, and credit deposit interest 2. Remeasure the supplied allowance to 2.4, recording credit-loss expense 1.6. The income statement reports $6 - 2 - 1.6 = 2.4$, which a separate closing entry transfers to retained earnings. Cash becomes 21, loans 120, goodwill 2, allowance 2.4 and deposits 117; net assets 140.6 equal deposits 117 plus equity 23.6. The cash bridge is $30 - 15 + 6 = 21$. The loan origination, deposit-interest credit and allowance change do not move settlement cash.

Construct the exposure base before applying the allowance cap

For these ordinary loans, the regulatory carrying-value definition does not subtract the associated allowance. The exposure amount is therefore the gross loan balance. Section 3.31 multiplies that amount by the applicable weight; section 3.32 assigns 100% to the stated corporate exposures and zero to the direct central-bank balance. Goodwill is deducted from CET1 and excluded from risk-weighted assets. With no associated deferred tax liability, its deduction is the full 2 (Office of the Comptroller of the Currency, 2026h, 12 CFR 3.31(a)–(b)) (Office of the Comptroller of the Currency, 2026g, 12 CFR 3.32(a)(1)(i)(A), (f)(1), (k)) (Office of the Comptroller of the Currency, 2026a, 12 CFR 3.22(a)(1), (e), (g)).

Let G be gross loans, A eligible AACL and B the risk-weighted base used to limit the allowance admitted to Tier 2. For this restricted book, $B = G$. The capital rule caps that admission at 1.25% of the relevant base. The official reporting instructions make the order concrete: Part I item 42 uses Part II item 26 for the cap; Part II subsequently subtracts the excess allowance in arriving at its total. Define admitted allowance T, excess X and final standardized risk-weighted assets D accordingly:

$$T = \min\{A, 0.0125B\}, \qquad X = A - T, \qquad D = B - X.$$

The last equality uses this example's exclusions. In a broader book, the cap base and pre-excess standardized denominator need not coincide. The instructions exclude section 22(a) deductions such as goodwill from the cap base but include specified section 22(c)–(f) deductions, which remain excluded from the final standardized denominator. The case has none of those other deductions or market-risk amounts (Office of the Comptroller of the Currency, 2026c, 12 CFR 3.20(d)(3)) (Federal Financial Institutions Examination Council, 2024a,b, Part I, item 42, pp. RC-R-27–28; Part II, items 26 and 28–31, pp. RC-R-109–110).

These retained instructions are the December 2024 files, with March 2024 revision footers on the cap pages. They corroborate the ordering in the specified historical rule calculation; they do not supply a complete amendment history for an actual later filing. The source date and the conditional reporting date remain separate.

Initially $B = 100$, the cap is 1.25 and all the allowance 0.8 enters Tier 2, leaving $D = 100$. At the final pre-runoff state, $B = 120$, the cap is 1.5, and excess allowance is $2.4 - 1.5 = 0.9$. Thus $D = 120 - 0.9 = 119.1$. CET1 is equity less goodwill, $23.6 - 2 = 21.6$. There is no additional Tier 1 capital, so Tier 1 is also 21.6; total capital is $21.6 + 1.5 = 23.1$. The following reconciliation retains the different measures rather than giving them one label.

Measure	Opening	Before runoff
Accounting equity	21.2	23.6
Goodwill deduction	2	2
CET1 and Tier 1	19.2	21.6
Eligible allowance in Tier 2	0.8	1.5
Total capital	20	23.1
Allowance-cap base	100	120
Excess allowance	0	0.9
Standardized risk-weighted assets	100	119.1

CET1 and Tier 1 ratios are $21.6/119.1 = 72/397$, about 18.14%; the total-capital ratio is $23.1/119.1 = 77/397$, about 19.40%. These are conditional ratios, not a determination of payout permission or all applicable capital requirements (Office of the Comptroller of the Currency, 2026e, 12 CFR 3.10(b)(1)–(3)).

Several plausible substitutions give the wrong quantity. Net loans 117.6 are the accounting amount, not this regulatory exposure. Deducting the full allowance from risk-weighted assets loses the rule's distinction between admitted and excess allowance. Deducting it again from CET1 counts an expense twice. Including goodwill in the exposure base reverses its required exclusion. Finally, using D to calculate its own allowance cap gives the circular equation $T = (117.6 + T)/80$, hence $T = 117.6/79$, rather than the required 1.5. Solving that equation accurately cannot make it the specified calculation.

A capital ratio does not settle the next cash payment

Now suppose 25 of deposits runs off. The supplied funding policy draws 4 within an available facility limit of 10, then pays 25. Cash becomes zero, deposits 92 and borrowing 4. Net assets 119.6 still equal liabilities 96 plus equity 23.6. Capital and risk-weighted assets are unchanged: the disappearing central-bank cash has zero weight, and no additional income or expense is assumed. The facility is borrowing by this bank, not an off-balance extension of credit by it to a customer.

If instead 40 is demanded, the same facility supplies at most 10. The bank pays 31 and leaves 9 unpaid, retained within its deposit liability of 86. Borrowing is 10 and cash zero. The capital ratios remain the same, although the payment obligation has failed. No capital or cash adjustment may be invented to remove that default. The two runoff outcomes are alternative histories of this bank, not balances to be combined.

The ordinary leverage ratio asks for another denominator: average total consolidated assets as reported for the relevant window, less specified Tier 1 deductions. Neither closing assets 140.6 before runoff nor 119.6 after it determines that average. If a separate, properly matched input supplies average assets 145, the denominator is $145 - 2 = 143$ and the ratio is $21.6/143 = 108/715$, about 15.10%. Without that input, the leverage ratio remains unresolved while the supported risk-based ratios remain available (Office of the Comptroller of the Currency, 2026e, 12 CFR 3.10(b)(4)).

The proposed `RegulatoryAssessment` therefore references the exact bank, book, perimeter, accounting state, information view, rule snapshot, capital eligibility, exposure classifications, deductions, allowance admission and reporting window. Average-assets evidence is a separate dependency. Its result names each numerator and denominator and records which outputs remain unresolved. Unknown stock eligibility blocks the dependent capital amounts; an incorrect perimeter rejects the input; a missing average blocks leverage alone. The assessment creates no accounting entry. A changed allowance, instrument, exposure class, average or rule

requires reassessing the affected outputs while retaining the historical result under its original assumptions.

For a richer credit adapter, partition borrower cohorts by currency, vintage, credit state, remaining maturity, collateral and measurement category. At date t, supply scenario probabilities, conditional default-time probabilities, exposure paths, conditional loss severity and discount factors. Its expected loss calculation has the declared form

$$L_t = \sum_s p_s \sum_{k \in H} d_{t,k} \Pr(\tau = k \mid s, \mathcal{F}_t) \mathbb{E}[\mathrm{EAD}_k \mathrm{LGD}_k \mid \tau = k, s, \mathcal{F}_t].$$

The conditional product expectation retains dependence between exposure and severity. The horizon H concerns eligible default times; associated cash shortfalls may continue later. The rule package decides twelve-month versus lifetime default scope, credit-state transfers, interest basis and write-off recognition. Changes in allowance must be attributed to origination, derecognition, transfers, revised expectations, currency and write-offs. Contract and prudential support can be specified now, while full IFRS 9/CECL/ ASBE classification and paragraph-level validation remain later obligations.

37.3 Classification comes before the loan calculation

A loan's contractual principal is not enough to determine its carrying amount. The dated amortised-cost examples that follow require both an eligible contractual cash-flow pattern and a business model for collecting those flows. These are findings about the instrument and the way assets are managed, not labels chosen to obtain a preferred profit figure. The following procedure uses the retained historical IFRS 9 clauses in EU Regulation 2023/1803; the surrounding Australian and Chinese examples retain their own stated jurisdiction and date boundaries.

First establish the financial asset's terms and the level at which the bank actually manages it. Evidence includes how performance is reported to key management, how risks are managed and how managers are compensated. An intention to retain one asset is insufficient. Sales must be explained by their reasons, timing, volume and place in that management objective; the source supplies no universal sale-count threshold (European Commission, 2023a, IFRS 9.4.1.1–3, B4.1.1–4B).

Next ask whether contractual payments are solely principal and interest on the outstanding principal. Principal here begins with fair value at initial recognition. Interest compensates for time value, credit risk, other basic lending risks and costs, and a profit margin. Equity-linked returns and leverage introduce different exposures. For a hybrid whose host is a financial asset, assess the entire contract under IFRS 9.4.3.2; the issuer's separate liability and equity components do not automatically become the holder's measurement units (European Commission, 2023a, IFRS 9.4.1.3, 4.3.2, B4.1.7A–9, B4.1.14).

If that cash-flow test passes, a hold-to-collect business model leads to amortised cost, while a model achieved by both collection and sales leads to fair value through other comprehensive income (FVOCI) for debt. Other financial assets generally use fair value through profit or loss (FVTPL), subject to the separate equity election below. At initial recognition, an otherwise eligible asset may be irrevocably designated FVTPL when the specified accounting-mismatch condition is met. Preserve that designation and its reason. A later change in one asset's intended sale date is not by itself a business-model change authorising reclassification (European Commission, 2023a, IFRS 9.4.1.2–5, 4.4.1, B4.1.2A–3).

37.3.1 The same price decline can reach profit and OCI differently

Consider a debt asset purchased for 100 and now worth 98. Supply cumulative ECL of 3, with no interest accrual, foreign-exchange movement, tax or fees in this measurement comparison. Opening cash and equity are each 200; purchase leaves cash 100. The three cases below have different admitted business models or an eligible FVTPL designation. They compare reporting consequences, rather than three freely selectable policies for one factual case.

At amortised cost, record impairment expense and allowance 3. Gross asset 100 less allowance 3 gives net asset 97, so cash 100 plus net asset 97 equals equity 197. A subsequent derecognising sale for 98 debits cash 98 and allowance 3, credits gross asset 100 and recognises gain 1. Cumulative profit is $-3 + 1 = -2$, and closing cash and equity are 198.

For debt FVOCI, the balance-sheet asset is 98. Record the fair-value decline 2 in OCI. Impairment expense 3 enters profit, with its allowance counterpart credited to OCI rather than deducted again from the asset's fair value. Net OCI is therefore $-2 + 3 = 1$; profit is -3, and total comprehensive income is -2. Cash 100 plus asset 98 equals equity 198. Reporting asset 95 by subtracting the allowance once more would count the credit effect twice. On a derecognising sale for 98, remove the asset and reclassify cumulative OCI gain 1 to profit. Cumulative profit then equals -2, with OCI cleared (European Commission, 2023a, IFRS 9.5.5.1–2, 5.7.10–11).

Under FVTPL, the decline from 100 to 98 is expense 2. There is no separate IFRS 9 loss allowance added to that fair-value loss. The sale at the same price creates no further gain. All three examples ultimately close with cash and equity 198, but their interim carrying amounts and placement of income differ. Reconciliation must preserve the chosen route rather than infer it from the equal final cash.

An equity investment is a distinct instrument. Where the irrevocable FVOCI election is available and made at initial recognition, a fall from 100 to 98 gives OCI loss 2 and no fair-value expense in profit. It excludes an investment held for trading and specified contingent consideration of an acquirer in a business combination. On sale, the cumulative loss is not recycled to profit; a transfer within equity is permitted. Dividends follow their separate recognition conditions, including whether they clearly represent recovery of part of the investment's cost. With no dividends in this example, cash and equity again end at 198 while cumulative profit is zero (European Commission, 2023a, IFRS 9.5.7.5–6, B5.7.1–1A). Copying the debt recycling entry into this case would wrongly create expense 2 on sale.

37.3.2 A familiar interest label can conceal a different exposure

A floating rate need not faithfully represent the passage of time. Suppose principal 100 pays monthly interest but resets each month to a one-year rate. A comparison instrument would reset monthly to a one-month rate, with the other relevant terms and credit quality matched. In one supplied six-month scenario, hold the contractual annualised rate at 12% and the benchmark rate at 6%. Each month's interest is respectively 1 and 0.5; six payments differ by 3. With principal repaid at month six, total undiscounted receipts are 106 and 103.

This calculation identifies the difference to investigate. IFRS 9's modified-time-value test compares undiscounted contractual and benchmark cash flows both by reporting period and cumulatively over the instrument's life, considering reasonably possible scenarios. A current yield curve, discounted present-value match or an invented percentage cutoff cannot settle whether the difference could be significant. Retain the benchmark construction, scenarios, period totals and significance reasoning before making that finding (European Commission, 2023a, IFRS 9.B4.1.9A–9E).

Nonrecourse terms likewise do not automatically fail the cash-flow test. The holder examines

the underlying assets or cash flows to determine whether the contractual restriction introduces payments inconsistent with basic lending. Contractually linked tranches require distinct tests of the tranche's terms, underlying pool and relative credit-risk concentration, including the permitted instruments in the pool. If the holder cannot perform the assessment required by B4.1.26, that clause requires FVTPL. A missing document in this system, however, is initially an unresolved evidence requirement; it is not proof that the holder made, or could not make, that assessment (European Commission, 2023a, IFRS 9.B4.1.15–26).

The method therefore records the governing rule, management evidence, contractual findings, designation or election, and remaining unknowns before selecting an interest, impairment and disposal procedure. Unsupported classification leaves those dependent calculations conditional. A supplied fair value or loss allowance can illustrate the reporting arithmetic while its valuation or credit model still requires separate specification and validation.

37.4 State and event order

The dated credit methods use the selected amortised-cost, interest and impairment provisions of AASB 9 (Australian Accounting Standards Board, 2022i). The following loan has different inputs from the earlier liquidity example, so its allowance and final equity are calculated separately.

BankState contains cash/reserves by settlement venue and currency; deposits by customer, access rights and maturity; assets by contract, gross carrying amount, allowance, original EIR, accrued cash flows, collateral and credit state; undrawn commitments; funding lines and covenants; accounting equity; regulatory mappings; and evidence/information views. LoanState separately retains origination credit risk and current risk, SICR finding, credit-impaired/POCI status, forbearance, modification, write-off and recovery history. Aggregating by today's loan balance alone loses both staging evidence and later response to shocks.

Each time step first admits information known then, forms permitted credit and liquidity forecasts, chooses behavior under that information, and schedules due events. Accrual, actual settlement, impairment measurement and regulatory queries have separate business identities. A deposit received from outside debits cash/reserves and credits deposit liability. A loan credited to the borrower's deposit debits loans and credits deposits without initially transferring reserves; a transfer to another bank then debits deposits and credits reserves. A loan paid directly outside instead debits loan/credits reserves. These two funding paths cannot be treated as identical cash events. Deposit interest debits interest expense/credits deposit or accrued payable; withdrawal requires settlement funds.

37.5 Expected credit loss

For each admitted scenario j and date t, calculate contractual due flows minus the cash expected to be received, including supported collateral/enforcement cash and timing; discount the shortfall under the applicable original EIR or specified exception. With mutually exclusive exhaustive scenario weights w_j, ECL is the sum of w_j times those present-value shortfalls. Default-time hazards must be converted to disjoint default-time probabilities; summing cumulative PDs over dates counts the same default repeatedly. Collateral recoveries cannot be credited both in LGD and as a second recovery cash flow. A shorthand PD×LGD×EAD is admissible only if its definitions and discounting reproduce the specified cash-shortfall calculation; it is not a universal replacement.

Expected credit loss (ECL) measures the discounted expected cash shortfall. A significant

increase in credit risk (SICR) changes the general method's default horizon. Purchased or originated credit-impaired (POCI) assets follow a separate method. For the general approach, twelve-month ECL considers lifetime shortfalls arising from possible defaults during the next twelve months; it is not just unpaid cash in those twelve months. SICR uses the change in lifetime default risk since initial recognition, reasonable forward-looking information and the applicable rebuttable presumptions. Do not install a two-times-PD or 30-day rule as an unconditional universal trigger. Lifetime ECL follows a supported SICR finding; cure needs a supported reversal of that finding. Subsequently credit-impaired assets use original EIR on amortised cost for interest in subsequent periods; POCI has a separate credit-adjusted-rate and cumulative-change method.

The 2017 ASBE 22 text makes the two horizons explicit. Article 47 defines credit loss as the present value of all contractual cash flows less expected receipts; a late receipt can therefore create a loss even when its nominal amount is paid in full. Article 48 limits twelve-month ECL by the date of the possible default event, while retaining the resulting cash shortfalls over the asset's life. Article 52 assesses changes in default risk, rather than changes in the allowance, and Article 58 requires probability weighting, time value and reasonable, supportable information (Ministry of Finance of the People's Republic of China, 2017, Articles 47–48, 52, 58). These provisions support the following dated comparison. Article 84 states 1 January 2018, but applying the standard to a particular company also requires its issuance schedule, subsequent amendments and adoption facts.

The general measurement interface requires scenario_cashflows, law_ref, information_view, horizon_admission, credit_assessment and discount_rule. An unweighted history set can produce conditional losses or loss bounds, but cannot produce an unbiased probability-weighted ECL by silently using equal weights. Return measurement_unresolved for the ECL claim. Trade receivables and certain other assets use the separately applicable simplified approach; the general loan staging rule must not overwrite AASB 9.5.5.15–16.

37.6 A commitment carries its credit history into the loan

A promise to lend can create a credit-loss provision before it creates a loan or consumes cash. Once the customer draws, the uncertainty about whether funding will occur has been resolved, but the customer's earlier credit history has not disappeared. The distinction connects the commitment populations reported above to the funded-loan method that follows. AASB 9.5.5.6 dates the initial impairment comparison from the irrevocable commitment; B5.5.47 treats the resulting financial asset as a continuation of that commitment (Australian Accounting Standards Board, 2022i).

Consider a synthetic bank with cash 200, deposits 180 and equity 20. It grants an irrevocable commitment allowing one immediate draw of 100 or lapse. The contract is fixed rather than revolving, cannot be net settled, is not designated at fair value through profit or loss, and does not promise a below-market loan. No fee or separate noncontingent commitment liability is assumed. The bank will hold the resulting loan to collect principal and interest. Its admitted initial fair value is 100, equal to its cash price, and its payment of 121 after two years gives an effective interest rate of 10%. These are explicit measurement and classification facts. The credit probabilities below do not themselves establish market value; a different initial value or a below-market commitment would require the separate analysis under AASB 9.2.3, 4.2.1(d) and 5.1.1–5.1.1A.

All commitment and draw snapshots are ordered on 1 January 2024, before any material interest accrues. Initially, the probability of a full draw is one half. Conditional on drawing, default

occurs in the first year with probability 0.1, in the second with probability 0.1, or does not occur with probability 0.8. These are mutually exclusive outcomes, not cumulative default probabilities. Either default produces collection of 108.9 at the contractual two-year maturity; no default produces 121. The present value of the cash shortfall from either default is

$$\frac{121 - 108.9}{1.1^2} = 10.$$

The first-year default's loss is paid out at a later date, yet belongs in twelve-month expected credit loss (ECL). The horizon selects possible default events; it does not truncate their later cash shortfalls. Including expected draws as required by B5.5.30–31 gives an initial provision of $0.5 \times 0.1 \times 10 = 0.5$. Debit impairment expense 0.5 and credit the commitment provision 0.5. The nominal limit 100 is neither a funded asset nor an impairment expense.

Before the draw, a new forecast and a separately supported significant increase in credit risk (SICR) change the conditional default probabilities to 0.2 in each year and 0.6 for no default. The borrower is not credit-impaired. Doubling a probability is not a universal SICR rule; the credit finding is a distinct input in this illustration. With draw probability still one half, lifetime ECL is now $0.5(0.2 + 0.2)10 = 2$. Debit a further impairment expense of 1.5 and credit the provision by 1.5.

The customer now actually draws the whole 100. To see what changes, the joint probabilities immediately before funding are 0.5 for no draw, 0.1 for draw and first-year default, 0.1 for draw and second-year default, and 0.3 for draw without default. Observing the draw removes the first outcome and conditions the other three on their total probability of 0.5. The funded loan therefore has lifetime ECL

$$\left(\frac{0.1}{0.5} + \frac{0.1}{0.5}\right) 10 = 4.$$

Keeping the old draw probability would still report 2, which measures the earlier uncertain commitment. Resetting the funded loan to twelve-month ECL would also give 2, now by omitting the second default window. Equal wrong amounts can conceal different errors. The original commitment date, credit baseline, current finding and actual draw must all survive the transition.

For a direct payment outside the bank, debit gross loan 100 and credit cash 100. In the same draw event, debit the existing provision 2, debit additional impairment expense 2 and credit the loan allowance 4. The first 2 moves an existing loss estimate from a liability into a deduction from the funded asset; the other 2 reflects the information that funding has occurred. AASB 7.B8E distinguishes commitment provisions from financial-asset allowances, while 35H–35I and B8D require an explanation of their movements (Australian Accounting Standards Board, 2022h). Funding and its impairment transition are committed together under the existing event procedure, even though the entries are explained separately here.

There is no remaining undrawn capacity. Cash 100 plus gross loan 100 less allowance 4 gives net assets 196, equal to deposits 180 plus equity 16. Cumulative impairment expense is $0.5 + 1.5 + 2 = 4$. Keeping the old provision of 2 as well as the full funded allowance would count the loss twice and leave equity 14. Using allowance 2 would instead leave equity 18. All three sets of accounts can balance; only one measures the admitted funded exposure.

The route through the borrower's bank account has a different intermediate state. Debit gross loan 100 and credit the borrower's deposit 100, with the same provision-to-allowance transition. Cash stays 200, deposits become 280, and equity is 16 after allowance 4. If the borrower then transfers 100 to another bank, debit deposits and credit settlement cash by 100. The final state matches direct disbursement. Creating the deposit did not itself consume settlement funds; the later transfer did.

Alternatively, the customer lets the binding window expire without drawing. Debit the provision 2 and credit impairment expense 2. Cash 200, deposits 180 and equity 20 remain, with no loan. This is an admitted lapse, not an invented cancellation right. A partial or revolving draw needs separate funded and remaining-exposure measurements. B8E allows combined allowance presentation when their losses cannot be separately identified; any excess over the gross funded asset is a provision. Neither that condition nor the longer revolving exposure period in AASB 9.5.5.20 follows merely from calling something a credit facility.

The probability tree is an authored assumption, not a calibrated forecast or an observed default sample. Unknown probabilities can support conditional losses but cannot produce a probability-weighted ECL without a measure. Missing original credit history leaves staging unresolved, and missing initial valuation leaves the funded measurement unresolved. The retained AASB 9 compilation 5 and AASB 7 compilation 7 cover the stated historical 2024 case. Accounting equity of 16 does not determine regulatory capital: the applicable jurisdiction, perimeter, deductions and exposure rules must be supplied independently. The next example follows a funded loan through interest, deterioration, collection and write-off using different stated facts.

37.7 Full loan and allowance roll-forward

Assume a fixed-rate non-POCI amortised-cost loan held to collect its contractual cash flows, originated for 100, with one terminal payment 121 two years later, original EIR 10%, no fees, collateral, modifications, FX or taxes. Since no payment falls due before maturity, an earlier default in this example is a separately established qualitative credit event, such as financial distress under the admitted default definition; it is not a missed instalment. Article 52 of ASBE 22 requires a definition consistent with credit-risk management and consideration of qualitative indicators (Ministry of Finance of the People's Republic of China, 2017). At origination the assumed lifetime scenarios are: default in the first twelve months with probability 0.1, default during the following year with probability 0.1, and no default with probability 0.8. Either default gives terminal collection 60.5. The first twelve-month ECL includes lifetime loss from the first scenario only: $0.1\times(121-60.5)/1.1^2=5$; initial lifetime ECL would be 10. Debit loan 100/ credit cash 100; debit impairment 5/credit allowance 5. Gross 100, allowance 5, net 95.

At year-one accrual the loan has survived the first window and is not yet credit-impaired: debit gross loan 10/credit interest 10. Survival alone would give remaining default probability $0.1/0.9=1/9$. A subsequent, separately admitted credit forecast revises that conditional probability to 0.2; it is an authored forecast update, not a Bayes calculation from survival alone. Its lifetime weights are 0.8/0.2, with a separately established SICR finding; $\text{ECL}=0.2\times60.5/1.1=11$. Debit impairment 6/ credit allowance 6. Gross 110, allowance 11, net 99. The ratio change alone is not the defined SICR test. Only twelve months remain, so twelve-month and lifetime ECL both equal 11 under this revised forecast. This amount cannot distinguish the two classifications.

Immediately afterward, a distinct, ordered credit-impairment event changes expected maturity collections to 60.5. This is a forecast mean, not a guaranteed or maximum recovery. Required allowance is $110-60.5/1.1=55$; debit impairment 44/credit allowance 44. Net 55. ASBE 22 Article 59(5) measures this non-POCI credit-impaired asset by gross balance less discounted expected receipts. Articles 38 and 39(2) then make amortised cost, after the allowance, the basis for subsequent-period interest (Ministry of Finance of the People's Republic of China, 2017).

The bank is assumed still to have reasonable prospects of fuller recovery before maturity. The separate finding that some contractual cash will not reasonably be recovered first arises at settlement on this branch. If that finding arose earlier, ASBE 22 Article 43 would require

the corresponding earlier write-off. Expected collections alone do not justify postponing an already-required write-off.

In the next full year interest revenue is 55×10%=5.5. The gross contractual accretion is 11. One consistent decomposed posting debits gross loan 11, credits interest revenue 5.5 and allowance 5.5: gross 121, allowance 60.5, net 60.5. The allowance increase of 5.5 is attributed to discount unwind in its roll-forward, with no additional impairment expense in this posting. This reconciles gross and net-interest bases without counting the same expected loss again. Collect 60.5: debit cash/credit gross 60.5. When no reasonable expectation of remaining recovery is established, debit allowance 60.5/credit gross 60.5. Both balances reach zero. A mere delinquency is not that write-off finding. Subsequent unexpected cash recovery is recorded with its own recovery-income method and evidence; it does not restore an old live receivable by default.

Moment	Cash, opening 200	Gross loan	Allowance	Deposit liability	Equity, opening 20
Initial loan and allowance	100	100	5	180	15
Year-one accrual and SICR	100	110	11	180	19
Credit impairment	100	110	55	180	−25
Maturity accrual	100	121	60.5	180	−19.5
Collection and write-off	160.5	0	0	180	−19.5

Total interest 15.5 less impairment 55 is loss 39.5, equal to 100 advanced minus 60.5 received. The negative equity state survives; no missing branch or capital plug restores solvency. These five dates jointly test allowance, recognition and bank balance-sheet arithmetic. The same loss cannot also be deducted directly from gross before the stated write-off.

Buying a claim whose credit loss is already in the price

The preceding loan became credit-impaired after origination. A purchased or originated credit-impaired asset, abbreviated POCI, has a different starting point: expected credit losses already enter the cash flows used to determine its credit-adjusted effective interest rate. Charging those same initial losses again would reduce the asset twice.

Suppose the bank buys a fixed claim of 121, due in two annual periods, for 50. Its initial expected collection is 60.5. Assume separately established POCI status, amortised-cost classification and acquisition fair value equal to the price, with no fees, interim payments, modifications, foreign exchange or tax. Assume that reasonable prospects of fuller recovery remain until the terminal assessment, so no earlier full or partial write-off is required. Expected cash below the contractual amount does not establish that some portion has no reasonable prospect of recovery. Paragraphs 5.4.4 and B5.4.9 require that separate assessment. The original credit-adjusted rate r, with positive compounding factor $1 + r$, must discount expected collection to the amount paid:

$$50 = \frac{60.5}{(1+r)^2}, \qquad r = 10\%.$$

The contractual claim remains 121. Its discounted expected shortfall is $(121 - 60.5)/1.1^2 = 50$, but this is already incorporated in the purchase measurement and rate. The additional loss allowance at acquisition is zero. IFRS 9.5.5.13 recognizes only subsequent cumulative changes in lifetime expected credit losses for POCI assets; paragraph 5.4.1(a) applies the credit-adjusted rate to amortised cost from initial recognition. These are the inspected historical EU 2023/1803 provisions, together with the Appendix A rate definition and B5.4.7 (European Commission, 2023a). A large discount or high credit risk alone does not establish POCI status.

With unchanged expectations, first-year interest is $50 \times 10\% = 5$, leaving carrying amount 55. Now suppose a supported credit reassessment raises expected terminal collection to 88. With one year remaining, its present value at the original rate is $88/1.1 = 80$. The increase $80 - 55 = 25$ is an impairment gain. The following year's interest is $80 \times 10\% = 8$, bringing carrying amount to 88 if expectations remain unchanged. Improved credit does not replace the original rate or switch this POCI asset to the ordinary gross-interest method. The cure rule in paragraph 5.4.2 refers specifically to assets that became impaired *after* initial recognition under paragraph 5.4.1(b).

To reconcile the allowance as well as the net asset, distinguish three amounts. Let G_t be carrying amount before the subsequent loss adjustment, L_t the signed allowance, and $N_t = G_t - L_t$ amortised cost after that adjustment. With no receipts before maturity, the original purchase measurement accretes to $G_t = 50(1.1)^t$. If E_t is expected terminal collection at annual date t, then

$$N_t = \frac{E_t}{1.1^{2-t}}, \qquad L_t = G_t - N_t = \frac{(121 - E_t) - (121 - 60.5)}{1.1^{2-t}}.$$

The last expression compares current and initial expected shortfalls at the same reporting date. It explains both signs. At year one, expected collection of 44 gives $N_1 = 40$ and $L_1 = 15$, an impairment loss of 15. Expected collection of 88 instead gives $N_1 = 80$ and $L_1 = -25$, an impairment gain of 25. That negative allowance is a favorable adjustment to the asset, not a liability owed by the bank. Paragraph 5.5.14 expressly requires favorable lifetime-loss changes to enter profit even when expected losses fall below those reflected at acquisition. Imposing a zero floor would leave carrying amount at 55 and suppress that gain.

The passage of time must also respect the interest basis. Over the next unchanged year, gross accretion is rG, allowance unwind is rL, and net interest is their difference:

$$rG - rL = r(G - L) = rN.$$

In the favorable case, these amounts are 5.5, -2.5 and 8. One balanced decomposition debits the gross asset 5.5 and the signed allowance account 2.5, and credits interest 8. The allowance becomes -27.5, so gross 60.5 less allowance -27.5 gives net 88. The 2.5 is discount unwind within this interest reconciliation; it is not a second credit-revision gain. The impaired case with net 40 instead earns interest 4, while gross accretion 5.5 and allowance unwind 1.5 give net 44.

Finally, suppose the favorable case collects only 70 at maturity, with no reasonable expectation of further recovery. Measuring the remaining net asset at the known receipt produces a final impairment loss of $88 - 70 = 18$. Total interest is $5 + 8 = 13$, net impairment gain is $25 - 18 = 7$, and total profit is $13 + 7 = 20 = 70 - 50$. The cash reconciliation agrees: starting with cash and capital of 200, purchase and collection leave $200 - 50 + 70 = 220$, equal to capital plus profit.

Immediately before collection, gross carrying amount is 60.5 and the signed allowance is -9.5, giving net 70. Receiving cash 70 and removing those two balances clears the financial asset without another gain. The contractual amount not collected is nevertheless $121 - 70 = 51$. The no-further-recovery finding supports financial derecognition under paragraph 5.4.4; it does not establish that the debtor's legal obligation has been discharged. The record retains that contractual shortfall and leaves legal release unassessed.

This terminal-claim example supplies credit forecasts to isolate their accounting consequences. It does not estimate their probabilities or accuracy. Intermediate payments, altered contractual terms, fees and earlier write-offs need their own dated transitions. Positive expected recovery also cannot justify postponing a required partial write-off. The original credit-adjusted rate and the distinction between interest and subsequent credit changes remain the starting conditions for those methods.

Receipts and write-offs between annual dates

The terminal claim isolates how an initial credit loss differs from a later change in that loss. A bank also needs to know what happens when cash arrives in instalments and a portion must be written off before the next year end. Consider a separate purchased credit-impaired asset acquired on 1 January 2025. The price is 98 and an integral transaction cost is 2, giving initial carrying amount 100. Contractual receipts are 66 on 1 January 2026 and 72.6 on 1 January 2027. Initial expected receipts are 55 and 60.5. Assume the same separately established POCI and amortised-cost classifications as before, and no modification, foreign exchange or tax. Opening cash and capital are 300.

The credit-adjusted effective rate discounts the expected receipts to the amount invested, including the integral cost. Here it is 10% because

$$\frac{55}{1.1} + \frac{60.5}{1.1^2} = 50 + 50 = 100.$$

The corresponding contractual present value is 120. Its excess of 20 over the expected value is embedded in the acquisition measurement; the subsequent- change allowance again starts at zero. The Appendix A definition and B5.4.1–4 require the fee and cost classification to precede rate calculation (European Commission, 2023a, IFRS 9, Appendix A and B5.4.1–4). Calling the 2 a fee does not, by itself, establish that it is integral.

For intermediate dates, this example supplies ACT/365F with annual compound interest: count actual elapsed days and divide by 365. This is a declared calculation convention, rather than a day-count rule prescribed by IFRS 9. Let $F(s,t)$ be the accumulation factor from date s to date t, and retain G, L and $N = G - L$ for gross carrying, signed allowance and net carrying. During an interval without cash or new credit information, both balances accrete at the original rate:

$$F(s,t) = (1+r)^{\mathrm{days}(s,t)/365}, \qquad G_t = G_s F(s,t), \quad L_t = L_s F(s,t).$$

Their difference therefore earns interest $N_s\{F(s,t) - 1\}$. At a credit reassessment, the bank discounts the complete remaining expected receipts e_d, each at its own date d:

$$N_t = \sum_{d \geq t} \frac{e_d}{F(t,d)}, \qquad L_t = G_t - N_t.$$

The change in N_t due to revised expectations is impairment income or expense; the preceding passage-of-time change is interest. This distinction implements the original-rate requirement and subsequent-loss treatment in 5.4.1(a), 5.5.13–14 and B5.5.45 (European Commission, 2023a).

There are 181 days from acquisition to 1 July 2025. Write $f = (11/10)^{181/365}$. Immediately before new information, gross and net carrying are both $100f$. Suppose expected receipts fall to 44 and 48.4. Each is four-fifths of its initial expectation, so their combined present value is $80f$. The bank recognizes an impairment loss of $20f$, leaving $G = 100f$, $L = 20f$ and $N = 80f$.

Now add a separate finding: an identified, proportional 20% portion of the whole claim has no reasonable prospect of recovery. Its expected cash is zero; the stated allocation assigns it 20% of gross carrying. The financial write-off is consequently $W = 20f$. Reducing both gross and

allowance by W leaves gross $80f$, allowance zero and net $80f$. There is no second impairment charge. Paragraphs 5.4.4 and B5.4.9 require this earlier write-off when that finding applies, even though the surviving claim still has positive expected receipts (European Commission, 2023a). The proportional allocation is an explicit assumption of this example; another identified portion needs its own supported allocation.

The removed contractual receipts are 13.2 and 14.52, leaving 52.8 and 58.08 in the financial claim. The bank retains the removed amounts and leaves legal forgiveness unassessed. This distinction matters when a later recovery or legal-enforcement question arises. A lower expected total alone would not have supplied the no-recovery finding or identified the portion to remove.

At 1 January 2026, the remaining gross balance has grown to 88. Collection of 44 leaves gross and net carrying of 44. Immediately afterward, a supported improvement raises the final expected receipt to 55, within the surviving contractual 58.08. Its present value is $55/1.1 = 50$, so the bank records an impairment gain of 6 and a signed allowance of $44 - 50 = -6$. The following year's net interest is 5. Just before the final collection, gross is 48.4, allowance is -6.6, and net carrying is 55. Receiving 55 and removing gross and allowance together closes the asset without another gain. Final cash is

$$300 - 98 - 2 + 44 + 55 = 299.$$

The lifetime loss is therefore 1. The period origins give the same answer: cumulative interest is $20f - 7$ and net impairment is $-20f + 6$, whose sum is -1. The cancellation does not make either interim amount an exact rational number.

A performing asset provides a useful comparison. Suppose the bank instead advances 102 and receives an integral origination fee of 2, with an independent zero-credit-loss assessment. The same expected contractual receipts 55 and 60.5 imply the same initial carrying amount 100 and effective rate 10%. On 1 July, a revised estimate under an existing prepayment right changes the remaining receipt to 105 on the following 1 January. Assume this is a non-credit estimate change, not a contractual modification. Under B5.4.6, the bank recomputes gross carrying at the original rate:

$$G_{\text{new}} = \frac{105}{1.1/f} = \frac{1050}{11} f, \qquad G_{\text{new}} - 100f = -\frac{50}{11} f.$$

The negative catch-up enters profit separately from credit impairment. Subsequent interest brings the asset to 105 for collection. Total profit is $2 - 102 + 105 = 5$. Omitting the integral fee would fit a different rate to 102 and would misstate this comparison.

Dates also change the numerical problem. Fractional powers such as f are generally irrational, so a rational calculator must preserve a bound rather than label a rounded decimal exact. Reduce the day fraction to p/q and choose rational endpoints a and b on a declared decimal grid. The following integer-power comparisons establish containment:

$$a^q \le (1 + r)^p \le b^q \quad \Longrightarrow \quad a \le F(s, t) \le b.$$

All quantities are positive, which justifies taking the monotone qth root. Interval arithmetic then carries these bounds through discounting and each balance; the requested scale and accumulated widths remain part of the result. A leap-day interval uses its actual number of days, rather than an implicit annual-anniversary convention. Zero and negative rates above -1 are admitted under the same convention.

For positive future receipts, present value decreases continuously as the rate increases. Certified endpoint signs can therefore enclose a unique effective rate. Such a bracket is useful numerical evidence, but its midpoint is not an exact rate. The two financial examples require an exactly established initial PV identity and use bounds for the intervening dates. Propagating an uncertain rate through a general financial schedule remains a separate method.

The dated records retain cash, write-off and catch-up origins so gross carrying can be reconstructed from their independently compounded contributions. Terminal allowance clearing is also an origin; omitting it would leave the gross reconciliation wrong when the allowance is negative. Every event needs its information date, unchanged-terms assessment and any required write-off finding. A missing receipt or unsupported finding returns no partial report. These finite methods supply dated accounting consequences; credit forecasting, arbitrary write-off allocation and the wider bank-company joins remain open.

37.7.1 A rate bracket must travel with the loan

The preceding dated examples have a convenient property: their initial cash flows identify an exactly representable effective rate. Ordinary schedules need not do so. Consider an asset acquired on 1 January 2025 for 98 with integral acquisition costs of 2. Expected receipts are 50 on 1 January 2026 and 60 on 1 January 2027. Its initial carrying amount is 100, including those costs. For a supplied POCI classification, the original credit-adjusted rate discounts these initial expected receipts; subsequent interest applies that rate to amortised cost (European Commission, 2023a, IFRS 9.5.4.1(a),B5.4.1–2,B5.4.7–8).

Retain the supplied ACT/365F convention and annual compounding. Both intervals here are exactly 365 days. Write $x = 1 + r$, where $r > -1$ is the original annual rate. The initial measurement requires

$$100 = \frac{50}{x} + \frac{60}{x^2}, \qquad 100x^2 - 50x - 60 = 0, \qquad x = \frac{5 + \sqrt{265}}{20}.$$

The positive solution gives a rate of about 6.3941%. That decimal is useful for reading the result; substituting it for the original rate would change every later valuation. An implementation must carry a justified numerical bound instead of presenting the rounded rate as exact.

More generally, let $c_i \geq 0$ be initial expected receipts at strictly positive ACT/365F times t_i, with at least one positive receipt. Their present value is

$$P(r) = \sum_i \frac{c_i}{(1+r)^{t_i}}, \qquad P'(r) = -\sum_i \frac{t_i c_i}{(1+r)^{t_i+1}} < 0.$$

Thus present value is continuous and strictly decreasing on $r > -1$. Certified endpoint values $P(r_-) \geq 100$ and $P(r_+) \leq 100$ enclose the unique original rate. Mixed-sign receipts do not have this guarantee and are outside this particular method. The bracket expresses computational uncertainty about one rate, not different economic histories or probabilities assigned to them.

The same rate must govern every later date. For elapsed days $d \geq 0$, define the accumulation factor $F(r, d) = (1 + r)^{d/365}$. It increases with r, so

$$F(r_-, d) \leq F(r, d) \leq F(r_+, d).$$

Each endpoint factor is itself enclosed by the integer-power calculation used in the preceding dated method. Adding, subtracting and multiplying these intervals, and taking reciprocals only of positive factors, encloses the corresponding financial calculation. Starting with exact gross carrying 100 and allowance zero, the argument applies successively to accrual, receipts, credit revisions, non-credit catch-ups and equal gross/allowance write-offs. The original rate is never refitted when expected credit changes.

The POCI example makes the distinction concrete. Supply contractual receipts of 60 and 75, separate from the initial expectations of 50 and 60. After 181 days, let $f = x^{181/365}$. Reducing expected receipts to 40 and 48 reduces net carrying from $100f$ to $80f$, a loss of $20f$. A supported

finding that one fifth of the entire claim has no reasonable recovery prospect writes off gross and allowance of $20f$ together. Remaining gross and net are both $80f$; the contractual terminal receipt remaining after the write-off is 60. That contractual ceiling matters even though the accounting write-off does not itself establish legal forgiveness (European Commission, 2023a, IFRS 9.5.4.4,B5.4.9).

At the first annual date, collect 40. The initial quadratic implies $80x - 40 = 48/x$, so remaining gross is $48/x$. If the supported expected terminal receipt then improves to 60, net carrying is $60/x$ and signed allowance is $-12/x$. Favorable changes in a POCI asset's lifetime loss expectations can produce this negative allowance (European Commission, 2023a, IFRS 9.5.5.13–14). One year later, gross is 48, allowance is -12, and net is 60. Collecting 60 and clearing the remaining gross and allowance closes the asset without another gain. Opening cash 300 becomes $300 - 98 - 2 + 40 + 60 = 300$; cumulative profit is zero.

For this calculation, a certified rate bracket of width 2^{-40} gives monetary intervals narrower than 5.38×10^{-10} currency units. The profit interval contains zero, while cash is exactly 300. These are bounds under the declared numerical budget. A wider rate interval can make the financial result too imprecise; the method then reports that failure instead of silently rounding a balanced result.

The returned bounds share the same original rate and cash history. Their separate endpoints cannot be freely combined into a journal or a set of jointly attainable statements. The documentary report therefore retains the root certificate, its original cash-flow target, all dated origins and the width of each amount. It proposes no postings from interval endpoints. Direct remaining-cash present values and compounded gross origins check the same history through different expressions. This extends the numerical admission of the finite performing and POCI profiles; their supplied classification, recovery, day-count and information-date requirements remain. Floating rates, mixed-sign root selection and general modification or write-off allocation methods require their own treatment.

For a non-derecognized loan modification, AASB 9.5.4.3 requires gross PV at original EIR and a modification result, then allowance and SICR reassessment under 5.5.12. For example gross 110 changed to a single future 110 at original 10% gives revised gross 100 and modification loss 10 before any allowance change. ASBE 22 Articles 42 and 56 similarly separate remeasurement of a non-derecognised asset from its credit-risk reassessment (Ministry of Finance of the People's Republic of China, 2017). This is separate from the financing method's issuer-liability ten-percent test. A derecognition decision requires its own asset conditions and cannot reuse that threshold.

37.8 Changed terms and recovered credit

A borrower can receive easier terms without recovering its ability to pay. A bank can also recover its expectation of payment without recovering all the credit quality present at origination. These are different changes to the loan. The following historical Australian examples use the retained AASB 9 compilation for synthetic 2024–2025 reporting periods. Probabilities, legal findings and credit assessments are supplied assumptions; taxes and fees are excluded.

Start from cash 100, a gross loan 110, allowance 55 and deposits 180. Net assets 155 leave equity -25. At the 2024 close, the bank replaces a payment 121 due one year later with 110 at the same date. Assume the change preserves the existing asset's contractual rights and does not qualify for derecognition. The original annual effective interest rate is 10%. Discounting

the amended contractual payment gives

$$G_{\text{modified}} = \frac{110}{1.1} = 100, \qquad \text{modification loss} = 110 - 100 = 10.$$

Debit modification expense 10 and credit gross loan 10. This calculation changes the contractual carrying amount before considering expected defaults. Paragraph 5.4.3 requires that separation. Fees would require an additional carrying-value adjustment and amortisation method (Australian Accounting Standards Board, 2022i, paragraph 5.4.3).

The bank separately expects full 110 with probability 4/5 and zero with probability 1/5. Expected cash 88 has present value 80, so the required allowance is $100 - 80 = 20$. Debit the old allowance 35 and credit impairment reversal 35. Cash 100 and net loan 80 now equal deposits 180 and equity 0. Neither this reversal nor the new payment terms establish recovery from credit impairment. For a loan that remains recognised, the comparison for a significant increase in credit risk retains the original, unmodified origination terms (Australian Accounting Standards Board, 2022i, paragraph 5.5.12 and B5.5.27).

A distinct, ordered credit review at that same close changes the supplied full-payment probability to 9/10. Expected 99 has present value 90; allowance falls 20 to 10, adding an impairment reversal 10 and leaving equity 10. Compare two histories at this point. In one, reinstatement of the borrower's operating licence and verified repayment capacity support a finding that the asset is no longer credit-impaired. The improvement is objectively related to an event after credit impairment. A significant increase in risk relative to origination nevertheless remains. In the other history, the supplied credit assessment still finds impairment despite the same improved cash expectation. A forecast mean alone does not decide between these findings.

The distinction changes the next year's income presentation. For the cured asset, paragraph 5.4.2 returns subsequent interest to gross 100: debit gross loan 10 and credit interest 10. At maturity, the expected shortfall 11 is no longer discounted, so debit impairment expense 1 and credit allowance 1. For the still-impaired asset, paragraph 5.4.1(b) applies the original rate to net 90: debit gross loan 10, credit interest 9 and credit allowance 1. Both histories reach gross 110, allowance 11 and net 99. Interest is 10 in one history and 9 in the other, with the compensating difference in impairment expense (Australian Accounting Standards Board, 2022i, paragraphs 5.4.1–2).

The borrower then pays 110, an outcome admitted by both supplied laws. Debit cash 110 and allowance 11, credit gross 110 and impairment reversal 11. Cash 210 equals deposits 180 plus equity 30. Profit since the opening is 55, equal to 110 received less the opening net loan 55. Equal closing wealth therefore cannot detect the wrong interest basis. Returning from lifetime to twelve-month expected credit losses would require a separate reversal of the significant-risk finding under paragraphs 5.5.3–12.

Ending contractual rights requires another decision. Suppose a different loan has gross 100 and allowance 20, and an enforceable agreement releases all remaining rights for an actual payment 85. There is no retained interest or replacement instrument. Debit cash 85 and allowance 20, credit gross 100 and gain 5: consideration 85 exceeds net carrying amount 80 by 5. Without the release, the same payment cannot by itself establish full derecognition (Australian Accounting Standards Board, 2022i, paragraphs 3.2.3 and 3.2.12).

For a transfer, the bank first identifies whether the unit is the whole asset or an eligible part, then assesses transfer conditions, risks and rewards, and control. A fixed-price repurchase can retain the risks rather than establish a sale. The issuer-liability ten-percent comparison does not answer these asset questions (Australian Accounting Standards Board, 2022i, paragraphs 3.2.2–7). A replacement asset needs fresh classification, initial measurement and a purchased-or-originated credit-impairment assessment; a distressed replacement is not automatically an ordinary new

loan with twelve-month losses (Australian Accounting Standards Board, 2022i, B5.5.25–26). Partial transfers and continuing involvement use paragraphs 3.2.13–23 (Australian Accounting Standards Board, 2022i, 3.2.13–23); the later revolving-capacity and transfer section develops those conditional methods.

37.9 Collateral received after maturity

Collateral changes expected receipts through its amount, timing and enforcement costs. Consider a separate loan advancing 100 on 1 January 2024 and requiring 121 on 1 January 2025. Its contractual half-year accumulation factor is 1.1, equivalent to 21% effective annual interest. This rate convention differs from the preceding annual 10% loan.

The supplied initial law assigns probability 3/5 to full payment at maturity and 2/5 to a default first observed within twelve months. In the default branch, collateral receipts arrive on 1 July 2025, one half-year after maturity. Possible net receipts 121, 66.55 and 12.10 have conditional probabilities $1/4, 1/2, 1/4$, respectively. Their conditional mean is 66.55. The bank has supported reasonable prospects of fuller recovery before the sale; 66.55 is not a known upper limit. These probabilities are illustrative assumptions.

The expected loss compares the present value of contractual cash with that of all expected receipts. The full-payment value is $121/1.1^2 = 100$; the default-branch mean has value $66.55/1.1^3 = 50$. Thus

$$\mathrm{ECL}_0 = 100 - \left(\frac{3}{5}\,\frac{121}{1.1^2} + \frac{2}{5}\,\frac{66.55}{1.1^3}\right) = 20.$$

The extra half-year belongs in the discounting. Omitting receipts after maturity produces the wrong loss 40. B5.5.55 expressly includes collateral cash beyond maturity and deducts obtaining and selling costs; B5.5.43 limits twelve-month losses by the default date, not by the recovery date. Here all possible defaults occur in the first year, so twelve-month and lifetime losses coincide (Australian Accounting Standards Board, 2022i, B5.5.43–44 and B5.5.55).

Start with cash 200, deposits 180 and equity 20. Advance 100 and recognise allowance 20. On 1 July 2024, first accrue gross interest 10. Rolling the original expectation forward gives net 88 against gross 110, so allowance rises 20 to 22 through impairment expense 2. A separately observed qualitative credit-impairment event then establishes the collateral-only expectation: $66.55/1.1^2 = 55$. Increase allowance 22 to 55 through impairment expense 33. The event is an admitted financial-distress finding, not an invented missed instalment before any instalment is due.

During the next half-year, original-rate interest on net 55 is 5.5. Debit gross loan 11, credit interest 5.5 and allowance 5.5. At contractual maturity gross 121 less allowance 60.5 leaves net 60.5. There is no contractual post-maturity interest, unamortised fee or premium. The initial 100-to 121 difference has been fully amortised. Over the final half-year the expected receipt value rises 60.5 to 66.55, giving net interest 6.05. Under these terms, gross remains 121 and allowance falls 60.5 to 54.45: debit allowance 6.05 and credit interest 6.05. This follows from the Appendix A definitions and B5.5.33's gross-minus-discounted-receipts measurement. It is a local derivation for these terms, not a rule that every overdue contract stops accruing contractual interest (Australian Accounting Standards Board, 2022i, Appendix A, paragraph 5.4.1(b), B5.5.33).

The middle recovery occurs: an enforcement agent obtains 79.86, deducts costs 13.31 already included in the forecast, and remits 66.55. Debit cash 66.55 and credit gross loan 66.55. Gross and allowance now both equal 54.45. At this sale, a separate supported exhaustion finding establishes that no further recovery is reasonably expected. Paragraph 5.4.4 requires write-off:

debit allowance 54.45 and credit gross 54.45. An earlier such finding would require an earlier write-off; settlement is not a general waiting rule (Australian Accounting Standards Board, 2022i, paragraph 5.4.4).

Final cash 166.55 equals deposits 180 plus equity −13.45. Total interest 21.55 less net impairment expense 55 gives loss 33.45, also equal to 100 advanced less 66.55 received. The unpaid legal claim 54.45 remains in the contract history because no release has been supplied. Accounting write-off does not by itself waive that claim. The bank never recognised the agent's collateral as a separate owned asset, and never counted its proceeds a second time.

37.10 Revolving capacity, loan fees and the rights retained after transfer

These conditional methods extend the preceding full-draw example using the retained historical AASB9 and AASB7 compilations for explicit calendar-2024/2025 Australian for-profit Tier 1 cases. Legal rights, initial values and credit loss estimates are supplied assumptions. None is inferred from a facility label or from the fact that a receipt reached the bank's account.

37.10.1 A repayment can restore capacity without erasing credit history

A facility has both a funded balance and a right to request more funding. Record the legal limit, draw requests and actual settlements separately, together with the replenishment rule, expiry or cancellation conditions, currency, original credit-risk comparison date and current credit assessment. If the facility permits repeated drawings, paying down one loan may restore capacity. It does not establish a new low-risk borrower or reset the original credit history. Determine the resulting loan's classification and initial measurement from its business model, cash-flow terms and value, rather than assuming every funded amount qualifies for amortised cost.

For a partial draw, remeasure funded and remaining undrawn losses using the information available after the draw. Do not allocate the old provision by the percentage drawn unless the admitted loss model actually implies that result. AASB 9.5.5.6 and B5.5.47 (Australian Accounting Standards Board, 2022i, 5.5.6 and B5.5.47) preserve the commitment's impairment history. Where the two losses can be identified separately, AASB 7.B8E (Australian Accounting Standards Board, 2022h, B8E) presents the loan allowance and commitment provision separately. Where they cannot, the combined loss is recognised against the funded asset, with any excess over its gross carrying amount presented as a provision. Inseparability is a measurement fact, not an election to conceal a missing exposure model.

Start with cash 200, deposits 180 and equity 20. A revolving limit of 100 has an admitted undrawn provision of 2. No interest accrues between the following ordered events. The customer draws 40 externally; cash becomes 160 and gross loan 40. Conditional on the draw, supplied funded ECL is 3 and remaining undrawn ECL is 1.5. Debit impairment expense 2.5 and the old provision 0.5, and credit allowance 3. The result is allowance 3, provision 1.5 and equity 15.5. Cash 160 plus net loan 37 equals deposits 180, provision 1.5 and equity 15.5. The undrawn capacity is 60; it is not an asset of 60.

The borrower repays 10 externally. The contract restores that capacity, so cash is 170, loan 30 and available capacity 70. New supplied ECL values are funded 2.4 and undrawn 1.8. Reduce allowance by 0.6, increase provision by 0.3 and recognise a net impairment reversal of 0.3. Then a new draw of 20 gives cash 150, loan 50 and capacity 50. Supplied funded and undrawn ECL of 4 and 1.2 require allowance increase 1.6, provision reduction 0.6 and expense 1. Total loss

balances are 5.2, cumulative impairment is 5.2 and equity is 14.8. Cash 150 plus net loan 46 equals deposits 180 plus provision 1.2 plus equity 14.8. Reusing the first funded estimate or treating the second draw as a new pristine commitment would compute a different exposure.

For a separate inseparable-loss example, gross funded loan 4 and combined ECL 6 require allowance 4 and provision 2. The rule does not create a negative net loan of 2, nor does it write off the contractual claim of 4. The reporting record preserves the combined estimate and the reason a separate attribution cannot be made.

A revolving facility does not necessarily use its short cancellation notice as its loss horizon. Under AASB 9.5.5.20 and B5.5.39–40 (Australian Accounting Standards Board, 2022i, 5.5.20 and B5.5.39–40), some jointly managed drawn/undrawn facilities expose the lender beyond that notice period because normal risk-management actions do not prevent those losses. Admit the exposure period from the facility's features, comparable historical exposure/default experience and expected risk actions. A one-day cancellation clause and a three-year assumed horizon are both insufficient alone. Missing evidence leaves the horizon unresolved; the mechanical capacity rollforward does not calibrate it.

37.10.2 Fee purpose determines whether it enters yield

Under AASB 9.B5.4.1–4 (Australian Accounting Standards Board, 2022i, B5.4.1–4), fees integral to origination or a probable specific lending arrangement enter effective interest for instruments measured at amortised cost. A fee's name does not settle its purpose. Servicing fees, commitment fees when a specific loan is unlikely, and qualifying syndication fees instead follow AASB15. Preserve the services promised, probability of a specific loan, instrument classification, timing, recipient and any allocation between distinct purposes. For instruments at fair value through profit or loss, the source's treatment of integral fees differs; do not copy the amortised-cost deferral.

Suppose a probable specific loan attracts commitment fee 2. Receive cash 2 and defer the fee; receipt alone is not service revenue. On origination, advance contractual principal 100 and incorporate the deferred fee in the loan's carrying amount, giving 98. A single contractual receipt of 107.8 one year later gives effective interest

$$r = 107.8/98 - 1 = 10\%, \qquad 98r = 9.8.$$

Accrete 9.8 and collect 107.8. Net cash from the fee and loan is $2 - 100 + 107.8 = 9.8$, equal to interest income. A supplied zero loss allowance isolates fee timing in this example; fee receipt does not imply zero credit risk. If the commitment instead expires without a loan, B5.4.2(b) recognises the fee as revenue on expiry. This branch contains no loan carrying amount.

If a fee is outside effective interest, determine revenue under the separate customer-service method. Calling it a service fee is not permission to recognise it immediately. For a separately admitted two-period stand-ready service with equal service each period and total consideration 2, the same cash receipt releases revenue 1 per period from a contract liability. A different obligation or satisfaction pattern changes that timing. The exact services must be supplied before this branch can be applied.

37.10.3 Selling a part requires identifying what was sold

AASB 9.3.2.2 (Australian Accounting Standards Board, 2022i, 3.2.2–3.2.23) first decides whether the derecognition test applies to a part or the entire asset. Eligible parts include specifically identified cash flows, fully proportionate shares of all cash flows, and fully proportionate shares

of specified cash flows. The first or last collections from a pool do not become a qualifying pro rata part merely because their nominal amount is 40%. Then establish a transfer or qualifying pass-through arrangement and evaluate risks and rewards. If neither substantially all risks and rewards are retained nor transferred, evaluate control through the transferee's practical ability to sell. The retained legal and cash-flow evidence must support each branch.

When an eligible part is derecognised, 3.2.13–14 allocates the old carrying amount by the relative fair values of the transferred and retained parts at the transfer date. Retained servicing rights enter that analysis; inadequate servicing compensation can require a servicing liability. A nominal principal ratio is not a substitute for relative fair value.

Assume an eligible identified cash-flow part, completed transfer and admitted derecognition. The larger asset carries 100; transferred and retained parts have supplied fair values 45 and 55, and cash consideration is 46. Allocate carrying amounts 45 and 55. Debit cash 46, credit the asset 45 and recognise gain 1. The retained asset carries 55; net assets rise from 100 to 101. If the sold principal fraction happened to be 40%, using that fraction would report retained carrying value 60 and gain 6, despite the supplied relative values. This example excludes servicing, fees, tax and prior OCI and assumes no separate allowance at the measurement instant; those features need their own retained allocations, not silent deletion from a real transfer.

If substantially all risks and rewards are retained, 3.2.15 instead keeps the entire asset and recognises a liability for the proceeds. Receiving 46 then gives cash 46, unchanged loan 100 and liability 46, with no sale gain. The same cash receipt can therefore have a different balance sheet. Subsequent asset income and liability expense are recognised separately, not offset.

37.10.4 Continuing involvement preserves the guarantee exposure

Where neither substantially all risks nor rewards are transferred or retained and control remains, 3.2.16–22 requires continued recognition to the extent of involvement and an associated liability. A guarantee case follows B3.2.13(a) (Australian Accounting Standards Board, 2022i, B3.2.13(a)): the retained asset is the lower of its carrying amount and the maximum consideration that could be repaid; the initial associated liability is that guarantee amount plus the fair value of the guarantee. Subsequent loss allowance reduces the retained asset, while the initial guarantee fair value enters profit as the obligation is satisfied under AASB15 principles. Options and partial retained interests require their own measurement branches; the guarantee formula is not a general transfer formula.

Admit a transferred asset carrying 100, cash received 103 including fair value 3 for a guarantee, maximum repayment 20, and the required risk/reward/control findings. Retain an asset of 20 and recognise associated liability 23. The entry debits cash 103 and the retained asset 20, credits the old asset 100 and liability 23, and has no initial gain. Cash 103 plus asset 20 less liability 23 preserves net assets of 100. Derecognising everything would wrongly report gain 3 and omit the exposure.

At a later date, admit satisfaction of one third of the guarantee service and loss allowance 2. Release guarantee income 1 by reducing the associated liability to 22; recognise impairment expense 2 and allowance 2. Net assets are $103 + (20 - 2) - 22 = 99$, matching net expense 1. If the guarantee then expires with no claim and no remaining obligation, reverse that allowance 2 and remove retained asset 20 against liability 22, recognising the remaining guarantee income 2. Cash and net assets are 103 and cumulative net income 3. An actual claim, changed valuation or unsatisfied obligation would require its actual settlement analysis; expiry cannot be presumed from a convenient reporting date. The continuing asset and liability are presented separately.

37.11 A transferred asset can still contain an option exposure

The guarantee example retains a possible credit loss. A call instead retains the right to buy an asset back; a written put obliges the bank to buy it if the counterparty exercises. Neither name determines whether a sale has occurred for accounting purposes. The transfer assessment first identifies the cash flows being tested and whether substantially all risks and rewards were retained or transferred. Only the intermediate case proceeds to control and, where control remains, continuing involvement. A transferee's practical ability to sell the whole asset unilaterally, without additional restrictions, matters. In particular, a call over an asset readily obtainable in the market need not prevent derecognition. Deeply in-the-money and out-of-the-money options can also lead to different risk-transfer findings (European Commission, 2023a, IFRS 9.3.2.2–16, B3.2.6–9, B3.2.16).

The following fair-value cases assume that this assessment requires continuing involvement and that the underlying asset is measured through profit or loss. They use the historical IFRS 9 text retained in EU Regulation 2023/1803. Amounts are in one currency, with no tax, coupons or transaction costs. Fair values and option time values are supplied measurement inputs. Applying the cases to an actual transfer requires its rights, market access, applicable rules and valuation evidence.

37.11.1 The asset less the liability represents the retained call

Suppose an asset worth 80 is transferred for 75 while the bank retains a call with exercise price 95 and value 5. The call is out of the money: exercise would currently cost more than buying the asset. Its entire value of 5 is therefore time value. Under the admitted continuing-involvement branch, the bank keeps an asset of 80 and recognises an associated liability of 75. Their difference is precisely the call's value. Opening cash 200, asset 80 and equity 280 become cash 275, asset 80, liability 75 and unchanged equity. The receipt debits cash and credits the associated liability; it creates no sale gain in this constructed example.

Let S denote the asset's current fair value, K the call exercise price and T_c its supplied time value, all in the same currency. The call's intrinsic value is $\max(S-K,0)$. The source's two measurement branches can consequently be written together as

$$A = S, \qquad L = \min(S,K) - T_c, \qquad A - L = \max(S-K,0) + T_c.$$

Here A and L are the recognised asset and associated liability. When the call moves into the money, the liability uses the exercise price instead of the higher asset value. This preserves the same net call exposure (European Commission, 2023a, IFRS 9.B3.2.13(c)).

At a later measurement, take $S = 100$ and $T_c = 4$. The asset becomes 100, the liability 91 and the net exposure 9. Asset income 20 and liability expense 16 produce net profit 4. At expiry, take $S = 110$ and zero time value. The asset becomes 110 and the liability 95, producing further net profit $10 - 4 = 6$. Physical exercise pays 95 and removes the liability; the already recognised asset remains. Closing cash 180 plus asset 110 equals equity 290, the original 280 plus profit 10. Recognising a separate call of 5 at transfer would have counted the same right twice. IFRS 9.B3.2.14 prohibits that duplication, while 3.2.22 preserves separate asset/liability and income/expense presentation (European Commission, 2023a, IFRS 9.3.2.19, 3.2.22, B3.2.14).

37.11.2 A written put caps the retained asset; a collar combines rights

For a written put, the bank bears the downside below the exercise price but does not retain the upside above it. With exercise price K and put time value T_p, the source therefore limits the

recognised asset and measures the associated liability as

$$A = \min(S, K), \qquad L = K + T_p, \qquad A - L = -\big(\max(K - S, 0) + T_p\big).$$

The negative net amount is the full put obligation (European Commission, 2023a, IFRS 9.B3.2.13(d)). For $S = 120$, $K = 100$ and $T_p = 5$, asset 100 and liability 105 represent a net obligation of 5. Choose transfer proceeds 125, equal to the underlying value plus compensation for that obligation. From cash 200, asset 120 and equity 320, the entry debits cash 125, credits asset 20 and credits liability 105. Equity stays 320. Keeping the full asset value 120 would overstate the retained asset by 20.

If the asset falls to 90 and time value is 4, the recognised asset is 90 and liability 104. Net loss is $10 - 1 = 9$. At expiry with asset value 80, the liability becomes 100 and the further loss is $10 - 4 = 6$. Exercise pays 100 and removes that liability. Cash 225 plus asset 80 equals equity 305, reconciling the total loss 15. This is a physical-settlement case; different cash-settlement rights require their own derecognition analysis.

A collar combines a retained call at an upper exercise price K_c with a written put at a lower price K_p. Let P be the *full* put value, $\max(K_p - S, 0) + T_p$. The asset continues at fair value, and the associated liability is

$$A = S, \qquad L = \min(S, K_c) + P - T_c, \qquad A - L = \big(\max(S - K_c, 0) + T_c\big) - P.$$

Thus the recognised net position is the call less the put (European Commission, 2023a, IFRS 9.B3.2.13(e)). With $S = 100$, call strike 120, put strike 80 and time values 5 and 1, the liability is 96 and net asset 4. From cash 200, asset 100 and equity 300, proceeds 96 produce cash 296, asset 100 and liability 96 without an initial gain.

Two subsequent paths show why the full put value matters. At $S = 130$, call time value 2 and put time value 1 give liability 119 and net asset 11. At $S = 70$, call time value 1 and put time value 2 give full put value 12, liability 81 and net liability 11. Using only the put's time value in the latter calculation would understate the associated liability by 10. If these respective asset values persist to expiry, both time values vanish. The upper path pays 120 to exercise the call, closing with cash 176, asset 130 and equity 306. The lower path pays 80 on the exercised put, closing with cash 216, asset 70 and equity 286. Cumulative income is respectively 6 and -14, measured from the initial net option value 4.

37.11.3 Amortised cost uses two interest schedules

The fair-value formulas cannot be copied into an amortised-cost case. IFRS 9.B3.2.13(b) instead accretes the associated liability from proceeds to the transferred asset's gross carrying amount at option expiry, using effective interest. Any difference between that liability and the exercise price enters profit when the option is exercised (European Commission, 2023a, IFRS 9.B3.2.13(b)).

Take the source's gross asset 98, proceeds 95 and gross asset 100 at expiry, here one year later. Assume a zero loss allowance, no intervening coupons or cash flows, and collection of the asset after option expiry. The asset's effective rate is $100/98 - 1 = 1/49$, producing income 2. The liability's rate is $100/95 - 1 = 1/19$, producing expense 5. They cannot share a single rate merely because they concern the same transfer. If a written put is then exercised at 102, debit liability 100 and loss 2, and credit cash 102. Opening cash 200, asset 98 and equity 298 become cash 193, asset 100 and equity 293. Interest expense net of income is 3 and exercise loss is 2. The zero allowance isolates the financing calculation; it does not infer an absence of credit risk from the transfer.

37.12 Funding, capital and decision policies

Liquidity simulation settles the dated net payment queue by venue/currency, respecting encumbrance, available cash, committed line terms and admissible actions. Haircut/eligibility and transferability come from a versioned policy; market value alone is not available liquidity. With cash 20, due withdrawal 30 and an admitted usable facility 8, even a full draw leaves shortfall 2. Record the default/unresolved settlement according to the event's legal model. Raising exactly 2 of new equity without an investor event is an invented financing source. Interbank contagion requires explicit counterparties and settlement priorities; it cannot be inferred from aggregate deposits.

37.12.1 A draw request and three cash histories

Make the facility example precise. Cash 20, securities 50 and gross loans 80 less allowance 10 give assets 140; deposits 120 leave equity 20. A synthetic agreement limits borrowing to 10 and accepts one unencumbered security lot with market value 10 and a 20% haircut. The other securities 40 are ineligible. With no existing drawing, usable capacity is $\min(10, 10(1-0.2)) = 8$. The agreement is an authored assumption, not evidence of a central-bank entitlement. The collateral remains the bank's asset; pledging it does not create another asset or sell the existing one.

Requesting 8 creates no cash journal. Confirmed receipt 8 debits cash 8 and credits borrowing 8. Confirmed receipt 6 instead posts 6, leaving the unfilled request visible. A third history retains a pending request and independent evidence that no receipt reached the bank before the withdrawal deadline. Its cash stays 20. An acknowledgement timeout alone cannot establish that third history: the receipt may have occurred even though the acknowledgement was lost.

Assume the deposit contract permits partial payment of the due withdrawal 30 and classifies the unpaid portion as a liquidity default without waiving it. Pay the cash actually available. The three outcomes are:

Confirmed borrowing	8	6	0
Cash before withdrawal	28	26	20
Withdrawal paid	28	26	20
Closing deposits	92	94	100
Unpaid due, within deposits	2	4	10

Each payment debits deposits and credits cash. In every column final cash is 0 and securities 50 plus net loans 70 give assets 120. Deposits plus borrowing total 100, leaving equity 20. Profit and OCI are 0. The unpaid amount is part of the deposit liability, so adding it again would overstate liabilities. Under the specified activities, deposit settlement is operating cash and the separate borrowing receipt is financing cash (Australian Accounting Standards Board, 2023a, paragraphs 14–17).

The corresponding records must retain facility version, borrower, currency, settlement venue, eligibility, encumbrance, valuation date, haircut and existing drawings. Capacity, request and settlement have distinct identities. A proposed request exceeding capacity is rejected. An independently evidenced unexpected actual receipt must nevertheless be recorded, together with the control violation; admission rules cannot justify omitting money that arrived. The existing publication and replay rules apply to the cash and borrowing as one transaction. A repeated identifier with identical content has no second effect; a changed payload conflicts. Lost acknowledgement preserves the possible committed and uncommitted histories until reconciled.

Credit and collateral records likewise retain the original loan, current and original terms, dated risk assessments, information cutoff, receipt law, discount convention and source dependencies. Missing legal rights or credit findings produce unresolved or explicitly conditional measurements. Actual credit calibration, fees, continuing involvement and revolving collateral remain separate method obligations. The historical Australian cases do not automatically satisfy the distinct US prudential profile in Section 37.2.1.

A RegulatoryAssessment reads an accounting view and an independent jurisdiction, date, perimeter, eligibility, deductions and exposure-weight policy. It outputs capital components, RWA/exposure measures, liquidity measures and reconciliations with their distinct denominators. Illustratively, equity 20 minus an assumed regulatory deduction 2 yields capital 18; assumed RWA 125 gives 14.4%. Those inputs are synthetic, not a Basel rule, national capital approval or default threshold. Selected dated US national capital rules are inspected in the surrounding bank inquiry. A connected accounting-to-capital reconciliation for the same bank still requires its applicable perimeter, eligibility, deductions and exposure facts; the illustrative inputs above do not supply those facts.

Behavior under capital/liquidity constraints is a separate economic model. Specify feasible actions, objectives, information set and ties before solving; failure to find an action returns policy_failure or default under the stated contract. Halaj/Behn sources can motivate candidate behavior but cannot calibrate it for an individual bank. A forecast experiment must compare simple fixed-book, contractual runoff and explicitly estimated policy baselines with time-safe data, before claiming a predictive improvement. CECL and other bases retain separate authority/method branches in the dated-authority discussion.

Chapter 38

Insurance service, finance and claims

An insurer receives cash before much of its service is delivered. Cash premium, service revenue and profit therefore follow different schedules. A short coverage example first connects gross insurance to reinsurance. The subsequent two-year cases introduce unearned profit, loss allocation and finance one change at a time.

38.1 Insurance: cohorts and future service

An insurer requires policy cohorts or groups with issue dates, coverage, premiums, benefits, expenses, lapses, claims development, guarantees and reinsurance relationships. Asset portfolios need their own contracts and valuations. The liability state must distinguish expected future cash flows, discounting, risk adjustments or margins, unearned profit measures where applicable, incurred claims and movements between opening and closing states. The IFRS 17 overview's distinction between measuring future cash flows and recognising profit as service is provided motivates this separation (IFRS Foundation, 2026b).

The module API accepts policy/cohort state, economic assumptions, non-economic assumptions and information date. It returns a measurement vector, movement attribution and proposed recognition events. Experience differences, changes in assumptions, finance effects and service effects remain separately identifiable. Reinsurance held has its own record and measurement; it is not an automatic percentage offset to every insurance liability.

Scenarios can alter discount curves, mortality, morbidity, lapse rates, inflation, catastrophe losses and policyholder behaviour. Their joint dependence and cohort exposure must be declared. A portfolio total can match history while hiding very different guarantee and duration profiles. This makes alternative economic histories especially valuable, but also limits what public-data reconstruction can identify. A full insurance package needs actuarial and accounting review beyond the core release.

38.2 Insurance service and reinsurance in a complete cohort

The example uses one-year policies and a one-year proportionate treaty starting when those policies are recognised, with uniform service for each. Both groups qualify for premium allocation through the one-year conditions in paragraphs 53(b) and 69(b); paragraphs 62–62A determine treaty recognition. Assume no acquisition cash flows, investment components, significant financing or onerous remaining coverage. Claims and recoveries are expected within a year of the claims being incurred. The inspected AASB 17 text incorporating IFRS 17 specifies the movements in paragraphs 53–59, allocation in B126 and separate reinsurance requirements in 63–70A (Australian Accounting Standards Board, 2022g). Gross and ceded groups retain distinct identities.

Begin with capital and cash of 20. Receive premium 120 and recognise a remaining-coverage liability of 120. Pay reinsurance premium 24 and recognise a coverage asset of 24. After half the uniform service, recognise revenue 60 by releasing gross coverage and reinsurance expense 12 by releasing its coverage asset. Estimated incurred claims are 40, of which 30 is paid; the remaining-claim risk adjustment is assumed to be 2. Gross claims expense is therefore 42 and incurred-claims liability is 12. Assume the treaty reimburses half the gross expected claims, recovery cash of 15 has arrived, the risk transferred is valued at 1, and reinsurer non-performance has zero expected effect in this example. The reinsurance claims asset is $20 + 1 - 15 = 6$, and recovery income is 21.

Cash is $20 + 120 - 24 - 30 + 15 = 101$. Assets are cash 101, remaining reinsurance coverage 12 and claims recovery 6, totalling 119. Liabilities are remaining gross coverage 60 plus incurred claims 12, totalling 72. Profit is $60 - 42 - 12 + 21 = 27$ and equity is $20 + 27 = 47$. These balances reconcile without netting reinsurance against the insurance liability. If counterparty credit risk changes, the recovery measurement changes through its own inputs; it does not mechanically halve every gross liability or risk adjustment.

Put the cohort on an explicit calendar. The balances above specify a half-service observation but do not determine its date or the sequence of transactions. For a reproducible dated version, supply coverage from 1 January 2024 to 1 January 2025, exclusive, and observe the book at the start of 2 July 2024. There are 183 elapsed days in this 366-day coverage year, so uniform service gives exactly one half. These dates are synthetic premises, not a chronology recovered from an insurer's disclosures.

Recognise the issued and held groups separately on 1 January and record their premium receipts and payments then. At the July observation, first release issued coverage and recognise incurred claims, then record the claim payment; release held coverage, recognise the recovery measurement, and record the recovery receipt. Together with the two premium movements, these are eight ordered events. The group identifiers and the held-to-issued link accompany every movement. Their order and opening balances are checked at each step, so repeating a release cannot create another period of revenue.

This dated version gives the same cash 101, assets 119, liabilities 72 and profit 27. Gross receipts are 135 and payments 54; their difference of 81 reconciles opening cash 20 to closing cash. The example supplies an operating cash classification. Changing transferred risk from 1 to 1.5 increases only the held recovery measurement and profit by 0.5; a separate non-performance effect of 2 reduces them by 2. Neither change alters the recorded cash or nets the two groups. The proposed company-book interface and complete interim report are specified in Section A.7. They retain unknown actual accounting admission, complete notes and the unobserved remainder of the year.

Longer-duration business adds a fulfilment-cash-flow vector, discount curve, risk adjustment,

contractual service margin, loss component and coverage-unit schedule. Select portfolio and annual cohort, assess onerous grouping, project cash flows within contract boundaries, discount with the declared financial assumptions, measure non-financial risk separately, and attribute subsequent movements to new business, experience, assumptions, finance and service. Future-service changes adjust the supported service-margin operation subject to the loss rules; current/past-service effects enter their specified result components. Unsupported variable-fee, transition or loss-recovery operations must stop those dependencies explicitly. A PAA example does not implement these longer-duration methods by implication.

An actuarial reserve adapter also needs a concrete estimation contract. For cumulative claims $C_{i,k}$ by origin year i and development year k, the chain-ladder point method estimates each factor from matched observed cells, $\widehat{f}_k = \sum_i C_{i,k+1} / \sum_i C_{i,k}$, then projects the latest cumulative claim along the remaining factors. A triangular example with rows $(100, 150, 180)$, $(120, 180)$ and (140) gives factors 1.5 and 1.2, outstanding amounts 36 and 112 for the unfinished rows, and total 148. Mack (1993, Sections 2–3, equations (1)–(3)) makes explicit the conditional-mean and variance assumptions and independence across origin years; its aggregate-error discussion retains dependence from shared estimated factors. A single final-development observation cannot supply an empirical tail variance without an additional assumption. The adapter therefore returns the point projection and marks uncertainty estimation unavailable in this tiny example. It supplies neither a full predictive distribution nor an IFRS risk adjustment. Calendar effects, inflation, changing claims handling and tail selection require separate sensitivity and validation before real use.

38.3 Group state and general measurement

The following cases apply selected general-measurement and service-allocation provisions from the retained AASB 17 compilation (Australian Accounting Standards Board, 2022g). They extend the premium-allocation example without presuming actuarial calibration or a complete insurer model.

The general method distinguishes fulfilment cash flows (FCF), including an explicit risk adjustment (RA), from the contractual service margin (CSM), which holds unearned profit. The liability for remaining coverage (LRC) concerns future service; the liability for incurred claims (LIC) concerns service and insured events already experienced. Keeping these components separate makes a change in future estimates different from an unpaid claim.

Contract records retain insured risks, premiums/benefits, service and settlement dates, options/guarantees, cancellation/repricing rights and acquisition costs. Group admission evaluates portfolio/similar-risk management, onerous status and required issuance cohorts under AASB 17.14–24, recognition under 25–28 and boundary under 34. Cash outside the boundary belongs to future contracts, not this group's profit. GroupState retains expected dated cash flows, current discount curves, inception rates, explicit risk adjustment, CSM, loss component, LRC/LIC, coverage units, service experience and assumption vintages. Same premium totals are not sufficient state for merging different guarantees or claim tails.

General measurement follows 32–38: fulfilment cash flows are probability-weighted future net outflows, adjusted for timing/financial risk without double counting, plus explicit non-financial risk adjustment. CSM removes initial unearned gain under its conditions; an initial net loss is recognized under 47, not negative CSM. Paragraphs 40–44 separate remaining coverage and incurred claims. For a nonparticipating group, accrete CSM at the specified locked-in rate, apply future service changes at their required rate to CSM up to its available amount, record any excess loss and allocate current service over current plus future coverage units under B119. B96–B97 distinguish future service from incurred-claim changes, financial changes and current experience; not every revision is a CSM adjustment.

A MeasurementInput must resolve cash-flow/law, discount, risk-adjustment method, service

grouping and coverage-unit evidence. Missing probabilities or RA support permit conditional sensitivity cases but block a source-supported measurement. Risk adjustment is the compensation the entity requires under 37; a standard deviation, confidence quantile or arbitrary percentage is only a candidate method until justified for that entity. Disclosures and the confidence-level equivalent, where applicable, are separate obligations. Market-consistent financial inputs and entity risk compensation must not be conflated.

38.4 Complete two-year general-measurement case

Take a two-year nonparticipating group with one premium 120 received at inception, claims expected/actually incurred and paid 40 each year, zero discount rate, RA 10 initially released 5 with each equal service year, no acquisition costs, investment component, lapses, taxes, finance changes or reinsurance. Coverage units are 1 in each year because equal benefits and coverage are assumed. Before premium payment, expected net cash outflow is 80−120=−40, RA 10, FCF −30 and CSM 30, with net initial carrying amount zero. After the premium debit cash 120/ credit LRC 120. Its components are remaining expected claims 80, RA 10 and CSM 30.

Each year releases expected service claims 40, RA 5 and CSM 15: debit LRC 60/ credit insurance revenue 60. Incur claims 40: debit service expense 40/credit LIC 40; payment debits LIC 40/credits cash 40. End of year one: cash 80, LRC 60 (claims 40, RA 5, CSM 15), LIC 0 and equity/profit 20. End year two: cash 40, liabilities 0, cumulative equity/profit 40. Revenue 120 and claims 80 reconcile to premium 120 less claims cash 80. No investment component has been mistaken for revenue.

At year-one close, a separate pre-next-service change increases the remaining claim estimate from 40 to 60, RA unchanged. FCF rises 20; consume remaining CSM 15 and recognize service loss 5. LRC rises from 60 to 65, comprising FCF 65, CSM 0 and loss component 5 as a tracked portion, not an added liability on top. Debit service loss 5/credit LRC 5 with component transfer CSM→FCF 15. A decrease of 8 in the revised future claim estimate, before any additional service/finance movement, first reverses loss 5 and restores CSM 3. FCF 57 plus CSM 3 gives LRC 60. Credit loss reversal 5/debit LRC 5. This tests 48–52 ordering. Later systematic allocation of loss component to releases/finance must reduce it to zero by coverage end; it cannot be amortized as revenue a second time.

In a separate case, an already-incurred unpaid claim estimate rises 40 to 48. Debit service expense 8/credit LIC 8; do not consume future-service CSM. A discount-rate change is financial under B97; its profit/OCI route follows the selected permitted finance policy. The zero-rate base case does not remove the need for separate current and locked-in rate calculations in a real group.

For the unreversed onerous alternative, carry the loss component through the remaining service year. Opening LRC 65 comprises future claims 60 and RA 5, with CSM 0 and loss component 5 within that LRC. A declared proportional allocation assigns 5/65 of each expected claim/RA release to the loss component, consistent with paragraphs 49–52 and B124's revenue exclusions. The expected claim release allocates 60/13 to loss reversal and the RA release 5/13; the total is 5, leaving loss component 0. Debit LRC 65, credit revenue 60 and credit loss reversal 5. Separately incur and pay claims 60 through LIC. Period profit is 60−60+5=5; combined with first-year profit 20 and the earlier onerous loss 5, lifetime profit is 20. Closing cash 20 equals equity 20 and all liabilities are zero. Treating all 65 as revenue while also reversing loss 5 overstates profit by 5.

38.5 Nonzero discount and service allocation

A second complete case keeps premium 120, two equal coverage units and RA 10, but expected and realized claims are 44 after one year and 48.4 after two years. Both the inception and current discount curves remain a flat 10% throughout; there are no new estimates, options, fees, taxes, investments or OCI election. This RA schedule is an authored measurement input, not an actuarial calibration: it declines by 5 per year, with its entire change in the service result under 81. All other insurance finance effects go to profit or loss. The retained source supports the mechanics in 44, 80–88, B119 and B123–B124.

At inception the present value of the two expected claims is

$$\frac{44}{1.1} + \frac{48.4}{1.1^2} = 40 + 40 = 80.$$

FCF before premium is 80+10−120=−30, CSM 30, and after premium LRC 120. In year one the claim PV accretes 8; CSM accretes 3 at the locked-in rate. Allocate the resulting CSM 33 across the two current/future units, releasing 16.5. Revenue is expected claims 44 plus RA 5 plus CSM 16.5=65.5. Service expense 44 and finance expense 11 leave profit 10.5. After paying 44, cash 76 and LRC 65.5 (claim PV 44, RA 5, CSM 16.5) reconcile to equity 10.5.

In year two claim PV accretes 4.4 and CSM accretes 1.65. Release all remaining CSM 18.15, RA 5 and expected claims 48.4, giving revenue 71.55. Service expense 48.4 and finance expense 6.05 leave profit 17.1. Final cash/equity 27.6 equals premium 120 minus claims 92.4; all group liabilities are zero. Total revenue 137.05 less finance expense 17.05 equals premium 120. Setting revenue equal to cash premium each year or releasing 15 of CSM each year would ignore the specified financing. These are separate component calculations followed by debit finance expense/ credit LRC for accretion, debit LRC/credit revenue for service and the stated LIC recognition/settlement entries. Cash is deliberately assumed uninvested.

38.6 Two issued groups and their reinsurance share one balance sheet

An insurer's accounts contain several groups at once. A profitable group cannot lend its unearned margin to an onerous group merely because both belong to the same company. Reinsurance may recover part of the loss, but does so through a separately measured contract. The following extension puts those three measurements on one balance sheet and follows their cash through the two service years. It uses the retained AASB 17 rules for an Australian private-sector insurer's calendar 2024 and 2025 periods.

Start with cash and equity of 100. Group A is the preceding nonzero-rate example: premium 120, claims of 44 and 48.4 at the two year ends, initial claim present value 80, risk adjustment 10 and CSM 30. Group B receives premium 100 and expects claims of 66 and 72.6 at the same dates. At the same 10% rate, B's claim present value is $66/1.1 + 72.6/1.1^2 = 120$. Its separately supplied risk adjustment is 10, so it recognises a loss of 30 and has no CSM. After premium receipt its remaining-coverage liability is 130, including a loss component of 30. Both groups contain contracts issued at the beginning of 2024 and provide two equal annual coverage units. Their different profitability requires separate groups under paragraphs 16 and 47; assume no pricing-law exception under paragraph 20. Estimating cash flows together does not authorise pooling their CSMs (Australian Accounting Standards Board, 2022g, paragraphs 14–28, 32–38 and 47).

The current and inception discount curves remain at the assumed 10%. Each issued group's risk adjustment releases 5 per year, entirely in the service result under paragraph 81. All insurance

finance enters profit or loss under paragraph 88(a). Claims are incurred and settled at each year end, with realised claims equal to the admitted expectations on this path. Cash is uninvested. There are no acquisition costs, investment components, taxes, currency effects, new contracts or estimate changes. The rates, expected claims, service units and risk adjustments are supplied conditional measurements; these calculations do not estimate them from insurer data.

Now add treaty R, entered into before B is recognised. It covers all B claims at 50%, with no exclusions, limits, settlement delay or expected non-performance. Its premium of 70 is paid immediately. Expected recoveries are 33 and 36.3, with present value 60. A separately supplied transferred-risk adjustment of 5 releases 2.5 each year. Its equality to half B's adjustment is an assumption of this example, not a general quota-share valuation rule. Without the initial loss-recovery adjustment, the net cost held in R's CSM would be $70 - 60 - 5 = 5$, and the asset after payment would be 70.

Because R was entered into in time and covers all B's contracts, paragraphs 66A and B119C–B119D recognise recovery income of $30 \times 0.5 = 15$. The adjustment increases R's cost CSM to $5 + 15 = 20$, so its complete remaining-coverage asset after payment is $60 + 5 + 20 = 85$. Its loss-recovery component of 15 is already within that asset. It is not another receivable. Debit B's initial service loss 30 and credit its liability 30; debit R's asset 15 and credit recovery income 15. Receive gross premiums 220 and pay the reinsurance premium 70. Cash is 250, issued liabilities are 120 and 130, and the reinsurance asset is 85, leaving equity 85 (Australian Accounting Standards Board, 2022g, paragraphs 60–68, 78 and B119C–B119F).

Financing moves the loss component as well

B's loss component must be allocated across claim releases, risk release and finance under paragraphs 50–52. For this unchanged group, use the initial proportion $\alpha = 30/130 = 3/13$ for all three. If F is the period's financing of B's future claim value and S is its expected claims plus risk-adjustment release, its loss component K follows

$$K_{\text{close}} = K_{\text{open}} + \alpha F - \alpha S.$$

This is a declared systematic allocation for these facts. It preserves $K = \alpha$ times B's remaining-coverage liability and reaches zero at coverage end. New estimates, coverage or risk patterns would require their own reassessment of the allocation; the example does not prescribe this ratio for every insurer (Australian Accounting Standards Board, 2022g, paragraphs 49–52 and B124).

In year one, B's claim value accrues finance of 12. Its loss component therefore increases by 36/13, before service releases claims 66 and risk adjustment 5. Of that release of 71, 213/13 reverses the loss and 710/13 is insurance revenue. The closing liability is $130+12-71 = 71$, comprising future claim value 66 and risk adjustment 5. The loss component is $30 + 36/13 - 213/13 = 213/13$, within that liability. Debit finance expense 12 and credit B's liability; then debit the liability 71, credit revenue 710/13 and credit loss reversal 213/13. Separately incur and pay claims 66 through the incurred-claims liability.

R's claim value accrues 6 and its CSM accrues 2. The asset gains finance income 8. Its CSM of 22 is then allocated across two equal current and remaining service units, releasing 11. Recovery 33, risk release 2.5 and CSM release 11 reduce remaining coverage by 46.5; the closing asset is $85 + 8 - 46.5 = 46.5$, comprising 33, 2.5 and 11. Its loss-recovery component tracks half B's loss component: finance adds 18/13, service reverses 213/26, and the closing component is 213/26. No component is added on top of the measured asset. In this unchanged, fully covered case, the component reversal is represented by the scheduled cash-flow and risk release; paragraph 66(bb) must not produce a second CSM reversal for it.

The service entry debits reinsurance premium-allocation expense $46.5 - 213/26 = 498/13$ and loss-recovery reversal 213/26, and credits R's remaining-coverage asset 46.5. Separately recognise recovery income 33 and an incurred-recovery asset 33, then collect the cash. Thus R's service result is $33 - 46.5 = -13.5$, with finance income 8 shown separately. The component determines presentation within that service result; it does not change its total. Paragraph 86 also prevents the reinsurance premium allocation from reducing issued insurance revenue (Australian Accounting Standards Board, 2022g, paragraphs 66–67, 80–88, B72, B119 and B119F).

A retains its preceding result: revenue 65.5, claims expense 44 and finance expense 11 give profit 10.5, with remaining coverage 65.5. B has service profit 5 and finance expense 12, giving a further loss of 7. Adding R's loss of 5.5 makes the year-one movement in equity -2, after the initial net loss of 15. Cash is $250 - 44 - 66 + 33 = 173$, and $173 + 46.5 - 65.5 - 71 = 83$ confirms closing equity.

Settlement and timing expose different errors

In year two A produces profit 17.1 as before. B incurs finance expense 6.6, then releases expected claims 72.6 and risk adjustment 5. Its loss component first increases by 99/65, then reverses 1164/65, reaching zero. Of the total release 77.6, revenue is 776/13 and the remainder is loss reversal. Paying claims 72.6 leaves B with no liability and period loss 1.6. R accrues finance income $3.3 + 1.1 = 4.4$, then releases recovery 36.3, risk adjustment 2.5 and all CSM 12.1. Its asset release is 50.9; its loss-recovery component also reaches zero. Collecting 36.3 leaves reinsurance service expense 14.6 and net expense 10.2. Total year-two profit is $17.1 - 1.6 - 10.2 = 5.3$.

The following balances are after each date's premium or claim settlements. All incurred claims and incurred recoveries have been settled; the exact event sequence preserves those intermediate balances before the cash entries.

	Initial receipt/payment	Year one	Year two
Cash	250	173	88.3
A: remaining-coverage liability	120	65.5	0
B: remaining-coverage liability	130	71	0
R: remaining-coverage asset	85	46.5	0
Equity	85	83	88.3

The final cash of 88.3 equals initial capital 100 plus gross premiums 220, less reinsurance premium 70 and gross claims 231, plus recoveries 69.3. Cumulative profit is $-15 - 2 + 5.3 = -11.7$. The two issued groups and the held treaty finish with every measurement component at zero.

Suppose instead that R is entered into immediately after B's recognition, with no intervening service, insured event or financing and otherwise the same prospective coverage. It is a valid different history. It cannot recover the original recognised loss under paragraph 66A because it fails B119C's ordering condition. R then starts with cost CSM 5 and asset 70, with no loss-recovery component. Opening equity is 70. Its year-one closing asset is $33 + 2.5 + 2.75 = 38.25$, giving total equity 74.75. Year-two cash and equity still finish at 88.3. The difference is the timing of recognition, not a difference in cash. Paragraph C20A's transition exception does not apply to this ordinary new-business example.

Incorrectly pooling A and B produces a third history of reported amounts. Their combined premiums 220 equal claim present value 200 plus the two supplied risk adjustments 20. A pooled calculation would therefore show neither CSM nor initial loss, conceal B's loss of 30,

and suppress the timely treaty's recovery adjustment of 15. It reports equity 100 initially and 91.25 after year one, although final cash is again 88.3. Those accounts balance but violate the separate-group recognition requirements. Likewise, adding the recovery component to the full asset overstates opening equity by 15, while omitting financing from loss allocation makes the component run out before coverage ends. Intermediate group identity, component movements and recognition time are essential checks; terminal cash alone cannot distinguish these mistakes.

Actual use also requires the treaty's boundary, collectability, measurement inputs and eligible contract mapping. Partial coverage needs the systematic loss allocation in B119E; a nominal quota cannot supply it. Changes in those facts or in the estimates require new measurement events under paragraphs 66–67, rather than extending this fixed schedule unchanged. The shared event procedure commits journal and component movements for each dated event together, while preserving separate issued and held group identities. (Australian Accounting Standards Board, 2022g, paragraphs 24, 63–67, 78–86 and B119C–B119F)

38.7 Commit each insurance event as a complete book transition

Separate group measurements still have to produce one dated company balance sheet. In the preceding example, R's loss-recovery component depends on B's loss component. Publishing B's financing while retaining R's opening measurement would combine amounts from different stages of the calculation. The proposed close operation therefore declares A, B and R as the required group set for each event in this example. Each group retains its own identity and measurement; the event publishes their consequences together in one entity, book and branch. This is a declared dependency for these facts, not a requirement to update every group in every insurance transaction.

The year end contains four events, in order: finance, service, claim and recovery incurrence, then settlement. Each has its own business-event identifier and immutable opening state. Keeping those boundaries preserves the incurred balances and the distinction between earning income and moving cash. The following intermediate states follow from the preceding journals. Coverage columns show liabilities for A and B and an asset for R.

After event	Cash	A coverage	B coverage	R coverage	Equity
Opening	250	120	130	85	85
Finance	250	131	142	93	70
Service	250	65.5	71	46.5	160
Incurrence	250	65.5	71	46.5	83
Settlement	173	65.5	71	46.5	83

Service releases the coverage amounts before the separate claim expenses and recovery income are recognised. Incurrence creates liabilities of 44 and 66 and a recovery asset of 33. Those balances explain why equity is 83 at that stage even though cash remains 250. Settlement clears them and moves cash by $-44 - 66 + 33 = -77$, leaving equity unchanged. Collapsing the four events into a single final balance would conceal this distinction.

A valid component calculation is part of permission to post

The proposed operation uses the common `OperationRequest` for `accounting.commit@1`. Its four arguments remain references to the accounting basis, posting authorization, journal proposal and economic delta. The expected company head remains in the common request envelope. Each

group measurement identifies the group, issued or held role, covered group where applicable, event phase, opening and proposed closing components, book, branch, time, information view, method and source. A complete measurement has actual component values; an unresolved one has no closing value and gives its reason.

Before looking up a previous receipt, the operation validates the complete group set, common ownership and information view, admitted evidence, measurement calculation and posting authorization. If R is unresolved, the whole declared event lacks permission. A and B's completed proposals remain available for diagnosis, but they do not form a closing company state. A missing group, duplicate group, wrong branch or different method also prevents lookup. The existing receipt index cannot turn malformed historical input into an eligible request. This proposed profile admits only the authored synthetic case; submitting it to a real book fails that permission check.

For an eligible new event, the journal and economic delta must then agree with the measurements. During finance, R's asset rises by 8 because its claim present value gains 6 and its cost CSM gains 2. Its loss-recovery component separately rises by 18/13, as a component within the asset. An extra debit to R's asset and credit to finance income for 18/13 would balance, yet overstate both the asset and income. The operation rejects that proposal. It also rejects a proposal whose journal is correct but whose economic delta leaves the recovery component at its old value. The monetary accounts and component memory must both be correct.

Publication inserts the journal, complete component and account changes, closing company state, their dependency records and edges, and the immutable receipt, then advances the company head in the same transaction. The finite exchange examples list those published records separately from the global registry of proposed records. Merely retaining a rejected measurement for inspection therefore does not make it part of the book. A rejected request publishes none of its proposed transition. This requirement has one legal book as its owner; it supplies no transaction guarantee across different books or entities.

Retry the event without repeating its accounting

The business-event lookup key remains the entity, book, branch and business-event identifier. After input validation, an equal request fingerprint returns the original receipt before testing the freshness of today's head. Suppose all four events have committed, advancing the head from generation zero to generation four. Retrying the original finance request returns its original receipt and closing state at generation one. The response separately reports the observed generation-four head and that this invocation made no additional commit. It does not apply finance again or replace today's state with the old closing state.

Equality covers the complete request, including its request identifier, budget, original expected head and dependency identities. Changing any of those bytes while retaining the business-event key is a conflict, not an exact retry. A previously unseen event with an old head is stale. The transaction repeats both the key and head checks before publication: an identical request that has just won returns its receipt; different request bytes for that event conflict; a different event that advances the head makes this request stale. In each case the competing transaction's changes are distinguished from any changes made by the requester.

If the acknowledgement disappears, the client cannot infer whether the transaction committed. The exchange therefore records a transport observation with unknown commit status, the original submission and its exact request bytes. It does not invent a received operation result. One possible storage state still has the opening head and no receipt; the other has the complete closing state and receipt. Resubmitting the exact request commits once in the first case and returns the old receipt in the second. A timeout supplies no reason to change the event identifier,

opening head or economic facts.

These finite examples check the equations, complete record bindings and required decisions for both storage states. They specify observable behavior; actual transaction isolation, crash recovery and durability need later implementation tests. Their completed coverage is a decision on one request, with no interpretation as probability mass over financial histories. The underlying source applicability and supplied actuarial assumptions of the connected-group example continue to govern its accounting meaning.

38.8 PAA, reinsurance and valuation methods

The preceding premium-allocation case states separate eligibility and recognition timing for gross insurance and reinsurance. PAA under 53–59 requires the one-year coverage or nonmaterial-difference condition; a short payment lag is not the eligibility test. LRC starts with received premium and applicable acquisition adjustments, then releases insurance revenue under the prescribed service pattern. LIC still measures incurred obligations using the applicable fulfilment-cash-flow rules; the allowed short-settlement discount simplification needs its own conditions. Emerging onerous facts require a loss assessment; a premium balance cannot be presumed sufficient.

A one-year policy can leave claims unpaid for several years. The distinction between the period of insurance service and the period until settlement is therefore essential even when PAA is available. The official ASBE 25 text separates three timing tests. Article 50 permits PAA when each contract's coverage is no longer than one year, or when the stated no-material-difference condition holds. Article 54 concerns a different interval: the time between each portion of service and the related premium due date. If that interval does not exceed a year, the remaining-coverage liability need not be adjusted for financing. Article 56 instead measures from claim incurrence to expected payment or receipt. Its option to omit the time-value and financial-risk adjustment requires settlement within a year of incurrence (Ministry of Finance of the People's Republic of China, 2020, Articles 50, 54, 56). The retained AASB 17 compilation makes the corresponding distinctions in paragraphs 53, 56 and 59(b).

Consider a separate group of issued policies with exactly one year of coverage. All amounts are illustrative currency units. Premium 120 is due and received when coverage starts, with no earlier onerous trigger; group recognition is therefore at that date under Article 16. Assume valid insurance grouping, uniform risk release, no acquisition costs, investment component, reinsurance, tax, other expenses or other movements, and no facts indicating onerous remaining coverage. Elect the permitted remaining-coverage financing exception. After half a year, recognise revenue 60 and retain remaining coverage of 60. Articles 53 and 57 support this roll-forward and time allocation; Article 57 requires a different allocation when risk release is not predominantly time-based (Ministry of Finance of the People's Republic of China, 2020). These assumptions isolate the claim measurement from other changes in the group's result.

At that reporting instant a claim is first incurred, with an expected payment of 60.5 exactly two years later. This is a mean estimate, not a certain settlement. Assume an appropriate flat annual effective discount rate of 10% and a separately estimated non-financial risk adjustment of 5 in current measurement units. The inputs are authored for the example; applying them to an insurer would require market and actuarial evidence. Articles 21, 23, 25 and 26 require current probability-weighted cash flows, appropriate discounting and a separate non-financial risk adjustment (Ministry of Finance of the People's Republic of China, 2020). The claim's cash-flow present value is $60.5/1.1^2 = 50$, so its liability is $50 + 5 = 55$. The risk adjustment is not discounted a second time. The claim has only just been incurred, so no intervening finance

accrual enters this measurement.

Record debit cash 120/credit remaining coverage 120 on premium receipt, then debit remaining coverage 60/credit insurance revenue 60 for the service provided. Recognising the claim gives debit insurance service expense 55/credit incurred claims 55, consistent with Article 31. The group then has cash 120, remaining coverage 60, incurred claims 55 and a cumulative result of 5: $120 - 60 - 55 = 60 - 55 = 5$. These are balances attributable to the group at mid-year, not a complete insurer forecast. Using the undiscounted $60.5 + 5 = 65.5$ for this two-year claim would overstate the required liability by 10.5 and understate the result by the same amount. The shortcut is wrong under the stated rules and assumptions. A zero rate would remove that discounting difference while leaving the separate risk adjustment intact. Articles 86(3) and 97 also require disclosure of the distinct claim components and the PAA/financing choices (Ministry of Finance of the People's Republic of China, 2020).

This comparison uses identified editions. The Ministry page reproduces the 2020 ASBE 25 text; Article 120 states 1 January 2023, but that sentence does not settle every entity group's adoption schedule. Section 38.24 distinguishes the Ministry's later explanation of that timetable from a company's reported adoption. The retained December 2022 AASB 17 compilation applies to annual periods beginning on or after 1 January 2023 but before 1 July 2026. The shared result in this example establishes a scoped measurement comparison. Applying it to a particular entity still requires its applicable rules and contract facts; later amendments and broader equivalence require separate evidence.

38.8.1 Acquisition costs and claim finance follow different clocks

The one-year coverage example establishes PAA eligibility without shortening the claim's settlement period. Acquisition costs add another choice. Under the historical IFRS 17 text, costs incorporated into the PAA liability reduce its opening balance and are subsequently amortised as an expense. Immediate expense is permitted under 59(a) when each contract's coverage period at initial recognition is no more than one year (European Commission, 2023a, IFRS 17.53–59).

Consider a separate eligible one-year group with premium 120 and acquisition cash paid 12 at recognition. Opening cash and equity are each 200. Assume uniform service and risk release, no investment component, tax or FX, and the permitted LRC financing exception because the service-to-premium-due interval is at most a year. No claims are incurred before coverage ends, and the base case has no onerous indication. Cash after receipt and costs is 308. Deferring the acquisition cost gives opening LRC $120 - 12 = 108$ and equity 200. Expensing it gives LRC 120 and equity 188.

At mid-year, both cases recognise revenue 60. In the deferred case, debit acquisition expense 6 and credit LRC 6 for amortisation, then debit LRC 60 and credit revenue 60. LRC is $108 + 6 - 60 = 54$, and cumulative profit is 54. In the expensed case, LRC is 60 and cumulative profit is $60 - 12 = 48$. After the second half's service, both LRC balances are zero and both cases have recognised revenue 120 and total acquisition expense 12. Equal final costs do not justify erasing the interim choice. Time allocation also requires the supplied uniform risk pattern; B126–127 instead uses expected incurred-service-expense timing when the risk-release pattern differs significantly (European Commission, 2023a, IFRS 17.55,59(a),B126–127).

An adverse mid-year branch requires a fresh onerous calculation. If current FCF for the remaining coverage is 70 while deferred-cost LRC is 54, recognise loss 16 and increase LRC to 70. Adding the entire FCF 70 to the existing 54 would overstate the liability by 54. The measurement must use the discount treatment required by paragraph 57, including its connection to the LIC simplification (European Commission, 2023a, IFRS 17.57–58).

Return to the base case. At coverage end a claim is incurred, with expected payment 48.4 two years later, current discount rate 10% and separately supplied risk adjustment 2. Initial LIC is $48.4/1.1^2 + 2 = 42$; cash 308 less LIC 42 leaves equity 266. The claim-incurrence rate of 10% differs from the group's assumed inception rate of 5%. If the portfolio elects to split LIC finance between profit and OCI, B133 uses the claim-incurrence rate specified in B72(e)(iii), not the inception rate (European Commission, 2023a, IFRS 17.59(b),B72(e)(iii),B133).

One year after incurrence, suppose the current rate falls to zero and stays there. The claim's current present value becomes 48.4, while a separate schedule using the claim-incurrence rate reaches 44. Finance expense is 8.4 in total: 4 in profit and 4.4 in OCI. Assume the risk adjustment releases 1 in each of the two years, wholly in the service result under paragraph 81. First-year LIC is then 49.4 and equity 258.6. In the second year, current claim value remains 48.4, so total finance is zero; the claim-incurrence schedule reaches 48.4 through profit expense 4.4 and OCI income 4.4. Release the remaining risk adjustment and pay the claim. Final cash and equity are 259.6, equal to opening 200 plus premium 120 less acquisition costs 12 and settlement 48.4. Cumulative OCI is zero. Using the inception rate would give first-year profit finance of only 2 and the wrong allocation, even if the total liability were correct.

38.8.2 Reinsurance and direct participation

Reinsurance has its own group, boundary, recognition time, counterparty risk, cash-flow, CSM/cost and possibly loss-recovery component under AASB 17 paragraphs 60–70A. Do not multiply every gross insurance liability by a quota share: excluded claims, limits, deductibles, timing, reinstatements and counterparty non-performance matter. Claim 100 with an eligible 50% recovery but expected 90% collectability gives conditional expected recovery 45, not 50; altering transferred insurance risk and reinsurer non-performance are different inputs. This simple calculation does not establish treaty recognition or RA by itself. Gross liability 100 remains; reinsurance recovery is a separate asset/measurement with its own authority.

For direct participation, test B101's conditions using the actual underlying items, substantial-share and variation judgments. Retain underlying fair values, entity fee share and guarantee/risk-mitigation choices; paragraphs 45 and B101–B118 specify the variable-fee adjustments to CSM, with onerous and mitigation exceptions. A local component illustration of a qualifying share: underlying-value increase 20 and entity's fixed 10% share gives candidate variable-fee change 2; eligibility, guarantees and other FCF changes must be evaluated before posting. The full option/guarantee valuation and transition calculations are further supported profiles, with source and numerical obligations, not an assumed linear fee model.

Eligibility depends on the contract and the insurer's expectations at inception. The policyholder must participate in a clearly identified pool, with the required substantial share of fair-value returns and substantial variability in payments. The variability assessment covers the contract's duration on a probability-weighted present-value basis. A minimum guarantee can bind in some scenarios without automatically excluding direct participation. Later market movements do not reopen the classification; contract modification follows its own rules. The identified reference pool also need not be held by the insurer, so the contractual link and the actual asset holdings require separate records (Australian Accounting Standards Board, 2022g, paragraphs B101–B108 and 72).

A small settlement example shows why the guarantee matters even when the stated share is fixed. Consider only a maturity-benefit component, in one currency at one date. Assume a held pool allocated to this component has amount U, and the policyholder is owed the larger of 90 and $0.9U$. These are illustrative contractual choices. The amount left from the pool after payment is $U - \max(90, 0.9U)$. Writing the payment as $0.9U + \max(90 - 0.9U, 0)$ shows that the amount left is $0.1U - \max(90 - 0.9U, 0)$: the nominal retained share less the extra payment needed to meet the minimum.

Pool amount	Policyholder payment	Amount left
80	90	−10
100	90	10
120	108	12

Moving from 80 to 100 leaves 20 more after payment: the ordinary share rises by 2 and the extra floor payment falls by 18. Moving from 100 to 120 leaves only 2 more because the minimum no longer binds. The negative amount at 80 is a funding shortfall for this component. These unweighted settlement outcomes explain the exposure; they do not establish insurance-contract eligibility, a present value or the insurer's profit. Other benefits and costs remain outside this component example, and its held-pool assumption does not replace the standard's separate accounting for investment assets (Australian Accounting Standards Board, 2022g, paragraphs B66(a), B107–B108).

The accounting calculation therefore needs more than this payoff schedule. Paragraph B104 defines the variable fee as the entity's share of underlying fair value less fulfilment cash flows that do not vary based on underlying returns. Changes in the obligation to pass through the underlying fair value do not adjust CSM, whereas the entity-share changes relate to future service. Financial-guarantee changes have their own future-service treatment under B113(b). Paragraph 45 retains the loss-component and risk-mitigation exceptions, as well as new-contract, currency and service-allocation movements. Applying those rules requires the group's measured cash flows, discount and risk adjustments, remaining CSM and loss component, service information and any eligible mitigation designation. The original increase of 2 remains a valid share-component calculation; the complete CSM movement needs these additional inputs (Australian Accounting Standards Board, 2022g, paragraphs 32–38, 45, 48–50 and B104–B118).

38.9 Pricing the floor before releasing the margin

The settlement schedule identifies a liability that increases when the underlying assets fall. To measure that liability before settlement, we need a financial model. A small, explicitly specified market makes the distinction between a price and a forecast visible. The retained standard permits a replicating portfolio to measure cash flows that it matches in amount, timing and uncertainty; other techniques must remain consistent with observable market prices (Australian Accounting Standards Board, 2022g, paragraphs B44–B48).

Suppose a non-dividend-paying traded asset costs $S_0 = 100$. Its next observed value is either $S_u = 120$ or $S_d = 80$, after which it remains at that value until payment. Cash earns zero interest, so its accumulation factor is $R = 1$. There are exactly two financial states, unrestricted borrowing and shorting, no transaction costs and no counterparty failure. These are assumptions of the illustration. The financial observation precedes all supplied coverage units; the subsequent service interval contains the mortality risk. An actual period with service already provided would require its own dated allocation.

For payoffs X_u, X_d, let δ denote the units of the traded asset and b the amount deposited in cash. Matching the two payments requires

$$\delta S_u + bR = X_u, \qquad \delta S_d + bR = X_d.$$

Subtracting the second equation from the first gives $\delta = (X_u - X_d)/(S_u - S_d)$. Substitution then determines b and the portfolio's current cost:

$$b = \frac{S_u X_d - S_d X_u}{(S_u - S_d)R}, \qquad V_0 = \delta S_0 + b = \frac{qX_u + (1-q)X_d}{R}, \qquad q = \frac{RS_0 - S_d}{S_u - S_d}.$$

Thus the weight q follows from today's asset and cash prices. Positive state prices require $S_d < RS_0 < S_u$, with $R > 0$. Here $q = 1/2$. Additional financial states or restrictions that prevent the matching trades would invalidate this particular valuation method.

The policyholder's investment payment is $\max(0.9S, 90)$. Its top-up above the ordinary share is $(90 - 0.9S)^+$, where $x^+ = \max(x, 0)$. That top-up pays zero in the up state and 18 in the down state. The matching portfolio has $\delta = -0.45$ and $b = 54$: $-0.45(120) + 54 = 0$ and $-0.45(80) + 54 = 18$. Its cost is $-0.45(100) + 54 = 9$. Adding the ordinary share's value of 90 gives a financial-benefit value of 99. This derives the option value; no option price has been supplied as an unexplained input.

For a separate physical forecast, suppose the up state has probability 3/4. The expected top-up is then $18/4 = 4.5$, although its market-consistent price remains 9. Suppose also that a covered death triggers an additional payment of 20, with physical probability 1/10, independently of the market state. Its expected cost is 2. Nonfinancial risk adjustment of 1 is a separately supplied measurement assumption, with its entire release in the service result. The financial state prices, physical joint law and compensation for nonfinancial risk therefore occupy different records (Australian Accounting Standards Board, 2022g, paragraphs 33, 37, 81 and B42–B49).

38.9.1 The same measurement through service and payment

Use the retained historical Australian provisions for a conditional calendar-2024/25 example. Assume the contract has significant insurance risk, an enforceable link to the identified pool and the required direct-participation characteristics. The component rights terminate together; neither can be retained separately after the other matures. This supplies the specified interrelationship under B32(b), together with an assessment that no embedded derivative must be separated. These admission findings need actual terms before real use; neither q nor the 90% share proves them (Australian Accounting Standards Board, 2022g, paragraphs 10–13, B17–B23, B31–B32 and B101–B108).

The investment payment is due whether the insured lives or dies. There are no surrender, lapse, acquisition or tax cash flows in this example, and all figures are pretax. The insurer holds one unit of the underlying asset at fair value through profit or loss and makes no risk-mitigation election. The portfolio used to calculate the guarantee price is not its actual hedge. Insurance finance also enters profit or loss under paragraph 89(a). The investment component remains within the insurance measurement but is excluded from service revenue and expense under paragraph 85 (Australian Accounting Standards Board, 2022g, paragraphs 85, 89 and B115–B118).

Start with owner capital and cash of 50, and receive premium 110. Before receipt, fulfilment cash flows are $99 + 2 + 1 - 110 = -8$, giving CSM 8. After receipt, the remaining-coverage liability is 110: financial benefits 99, mortality estimate 2, risk adjustment 1 and CSM 8. Buying the underlying for 100 leaves cash 60 and investment 100 against liability 110 and equity 50.

At the financial observation, the future-service fee change is the change in the insurer's 10% share less the change in the top-up value (Australian Accounting Standards Board, 2022g, paragraph 45 and B111–B114). In the up state this is $2-(0-9)=11$, increasing CSM from 8 to 19. Fulfilment cash flows become $108+2+1=111$ and the liability becomes 130. Investment gain 20 and insurance finance expense 20 offset.

In the down state the fee change is $-2-(18-9)=-11$. The existing CSM absorbs 8; the excess 3 becomes a service loss. Fulfilment cash flows and the liability are $90+2+1=93$, with CSM zero and a loss component of 3 *within* that liability. Investment loss 20 and insurance finance income 20 offset, while the service loss reduces equity to 47. The loss component does not create a second liability, and the binding floor does not reopen the inception VFA classification (Australian Accounting Standards Board, 2022g, paragraphs 45, 48–52, 87(c) and B102).

Completing the service interval releases the expected mortality cost, risk adjustment and remaining CSM. Up-state revenue is $2+1+19=22$. In the down state the service-related release of 3 reverses the loss component, leaving revenue zero. Actual mortality expense is either zero or 20, first recognised through the incurred-claim liability and then paid. Selling the underlying and paying the investment component of 108 or 90 completes all balances.

Joint outcome	Physical weight	Profit	Final cash/equity	OCI
Up, survival	27/40	22	72	0
Up, death	3/40	2	52	0
Down, survival	9/40	0	50	0
Down, death	1/40	−20	30	0

Each row is a complete conditional history. Operating cash is premium 110 less the investment benefit and mortality payment; the investment purchase and sale give investing cash of +20 or −20. This classification assumes nontrading investments under the retained cash-flow provisions; fair-value measurement alone does not determine trading purpose (Australian Accounting Standards Board, 2023a, paragraphs 14–16). The physical expected profit is 14.5 under the supplied law, but the four rows preserve the separate downside exposures. An unknown mortality law, risk adjustment, dependence assumption or admission finding leaves a named unresolved measurement rather than a default probability or zero cost.

The exact state prices, journals, component changes and adverse alternatives are retained with the acquisition/treaty cases in `data/accounting-master/finite-method-decisions-20261015/insurance/`. The companion method, *Pricing the guarantee and retaining separate insurance groups*, binds valuation, probability and risk-adjustment records to the same group, date and evidence view, while keeping their meanings distinct. General market models and actuarial calibration require additional methods and validation.

38.10 The variable fee separates investment returns from service

The preceding guarantee valuation measures a contractual payoff. Direct participation adds a service question: which changes belong to the insurer's unearned fee? Admission requires all three B101 conditions, assessed using expectations at inception: an enforceable share of a clearly identified pool, a substantial share of fair-value returns payable to policyholders, and substantial variability of their payments with those returns. The assessment considers the contract's duration and probability-weighted present values, including scenarios in which a guarantee binds. It is reassessed only upon a relevant modification. Reinsurance issued or held cannot qualify as direct participation under B109 (European Commission, 2023a, IFRS 17.B101–109).

For an admitted group, the source separates the obligation equal to the underlying items' fair value from the variable fee deducted for future service. Changes in the former do not adjust CSM. Changes in the insurer's share and in eligible future-service costs do, subject to loss and risk mitigation conditions. The latter adjustments use current rates; a GMM locked-rate calculation cannot be copied into this model (European Commission, 2023a, IFRS 17.45,B104,B110–114).

Suppose a qualifying insurance group receives premium 100, invested in its identified pool. Policyholders are entitled to 80% of the pool's value regardless of whether an insured event occurs, together with separate insured benefits. Supply remaining nonfinancial claims of 10 and risk adjustment 2, two equal current/future service units, and zero discount, tax and FX effects. The 80% amount is an investment component under the stated terms. All eligibility and service-quantity findings are explicit case premises. With pool value 100, FCF is $80+10+2 = 92$, leaving CSM 8 and LRC 100. Opening cash and equity 200 become cash 200, pool asset 100 and liability 100 after receiving and investing the premium.

Before the first service allocation, the pool rises to 120. The policyholder's share rises by 16 to 96. The insurer's 20% share rises by 4, increasing CSM to 12. FCF becomes $96 + 10 + 2 = 108$, and total LRC becomes 120. The financial increase in the liability is therefore 20, not merely the policyholder's increase of 16. With all pool gains and insurance finance in profit, asset gain 20 offsets insurance finance expense 20; the extra fee 4 remains unearned within CSM.

Now supply expected and actual first-period insured claims 5 and risk adjustment release 1. Allocate the CSM 12 equally across the current and remaining service units, releasing 6. Revenue is $5 + 1 + 6 = 12$, claims expense 5 and service profit 7. Paying the claims leaves cash 195, pool asset 120 and LRC $96 + 5 + 1 + 6 = 108$, giving equity 207. In the second period, with unchanged pool value and the same claims and release, profit is another 7. Sell the pool for 120 and repay the investment component 96. Final cash and equity are 214. The repayment reduces the liability and cash; it is excluded from both insurance revenue and service expense (European Commission, 2023a, IFRS 17.81,83–85,B119).

38.10.1 Losses and risk mitigation change the CSM adjustment

Consider a separate adverse branch from the initial FCF 92 and CSM 8, before any service. Future nonfinancial costs rise by 15. Use the CSM 8 first, then recognise service loss 7 and a tracked loss component 7 within LRC 107. Equity falls from 200 to 193. A subsequent cost reduction of 6 reverses loss 6 and leaves loss component 1, CSM zero and LRC 101. A further reduction of 4 reverses the last loss 1 and restores CSM 3; FCF 97 plus CSM 3 brings LRC back to 100. Those loss reversals remain service results under the specified rules even when a qualifying source of the original loss was a decline in the insurer's underlying-item share (European Commission, 2023a, IFRS 17.45,48–50,87(c)).

A financial guarantee change needs a different cause record. From the same initial position, suppose its supplied future-service value increases by 6 while a derivative gains 6. Without applying the risk-mitigation choice, FCF rises to 98 and CSM falls to 2, leaving LRC 100. The derivative gain enters profit, increasing equity to 206. If B115–116's conditions are met and the insurer applies the choice to this change, CSM remains 8 and LRC rises to 106. Insurance finance expense 6 then offsets the derivative gain, leaving equity 200. The choice changes recognition timing, not the derivative's value or the obligation.

This treatment requires a previously documented risk-management objective and strategy, an economic offset and credit risk that does not dominate that offset. B115 specifies which derivatives, FVTPL non-derivative instruments or held reinsurance can mitigate each exposure. It cannot be elected retrospectively because a mismatch became inconvenient. B117A takes the relevant finance to profit for derivative/FVTPL mitigation and specifies the matched policy

for held reinsurance. When B116's conditions cease, B118 stops the treatment prospectively without undoing previous recognition (European Commission, 2023a, IFRS 17.B115–118).

38.11 Partial reinsurance and changes in insurance estimates

The following methods use IFRS 17 in the retained September 2023 EU compilation. They extend the preceding nonparticipating general-measurement examples. Contract rights, group assignments, recovery percentages, risk adjustments and discount rates are explicit conditional inputs. A change in one of these inputs must retain its cause: changed service expectations, financial risk, counterparty performance and a new legal agreement do not have interchangeable accounting consequences.

38.11.1 Only the covered contracts support loss recovery

IFRS 17.62–66B (European Commission, 2023a, IFRS 17.62–68 and B119C–F) keeps a held treaty separate from the issued contracts, with its own entry, coverage and recognition dates. Proportionate coverage can delay recognition until an underlying contract is recognised. B119C permits the initial onerous-loss recovery only if the treaty was entered into before or at the same time as the underlying onerous contracts were recognised. The record must distinguish these dates; the treaty's later cash payment does not identify when it was entered into.

When a treaty covers only some contracts within an onerous issued group, B119E requires a systematic and rational allocation of the group's loss to the covered contracts. B119D then multiplies that covered loss by the percentage of their claims expected to be recovered. A treaty's headline quota percentage must not be applied to uncovered losses. Preserve the contract-to-issued-group and contract-to-treaty joins, covered risks and periods, attachment points, limits and exclusions, the loss-allocation method, expected claims/recoveries and the separate transferred-risk adjustment. For excess-of-loss or capped cover, expected recoveries require the admitted claim distribution and treaty payoff; a constant quota share cannot be used without support.

Take an issued group with premium 100, expected claims 118 and risk adjustment 2, all measured with zero discounting at recognition. Initial loss is 20 and CSM is zero. A source-supported allocation in this synthetic contract set identifies a covered portion with premium 41, claims 48 and risk adjustment 1, giving loss 8; the uncovered portion has premium 59, claims 70 and risk adjustment 1, giving loss 12. These amounts sum to the group; they are not separate groups created to improve the result.

A treaty entered into on time covers half of the covered contracts' claims. Expected recovery is 24, transferred-risk adjustment is a separately supplied 0.5, and premium paid is 26.5. After payment, the ordinary fulfilment asset is 24.5 and cost CSM is 2. Loss recovery is

$$8 \times 50\% = 4,$$

so cost CSM rises to 6 and the measured held asset to 30.5. The loss-recovery component of 4 sits inside that asset; it is not another receivable. Starting from cash and equity 50, receive issued premiums 100 and pay treaty premium 26.5. Cash 123.5 plus held asset 30.5 equals issued liability 120 plus equity 34. Applying 50% to the entire loss would recognise 10 and overstate equity by 6. If this same treaty had been entered into only after the underlying recognition, the initial loss-recovery adjustment would not apply.

38.11.2 A changed loss estimate affects both groups without netting them

Before any service or time passes, reduce expected claims on the covered contracts from 48 to 44, leaving other assumptions unchanged. Issued fulfilment liability falls from 120 to 116. Paragraph 50(b) first reduces the loss component, from 20 to 16, and recognises a loss reversal of 4; there is no new issued CSM while that loss component remains.

Expected held recovery falls from 24 to 22. Under paragraph 66(c)(i) (European Commission, 2023a, IFRS 17.66(c),66(bb),B119F), this change follows the underlying change that did not adjust the issued CSM, so it reduces the held asset and creates expense 2 rather than being hidden in held CSM. Held fulfilment asset becomes 22.5, cost CSM stays 6 and total held asset becomes 28.5. The covered loss is now 4, so B119F's recoverable loss-component limit is 2 and the loss-recovery component falls from 4 to 2. That component change tracks the same recognised recovery loss; posting a second expense for it would double count the event. Equity becomes $123.5 + 28.5 - 116 = 36$, an increase of $4 - 2 = 2$.

If all remaining services are then provided and claims equal these revised estimates, issued liability 116 is released through insurance revenue 100 and loss reversal 16; claims expense and payment are 114. Issued service-period profit is 2. Held coverage asset 28.5 is released as premium allocation 26.5 and reversal of loss recovery 2, with incurred recovery 22 and its collection recorded separately. Held service-period loss is 6.5. The entity ends with cash and equity $123.5 - 114 + 22 = 31.5$ and no group balances. Initial combined loss 16, estimate-change gain 2 and service-period loss 4.5 sum to lifetime loss 18.5, equal to the change from opening equity 50. Reinsurance has not removed or netted away the issued obligation.

B119F requires later loss-recovery components to reflect changes in the underlying loss component and never exceed the portion expected to be recovered. Track separately adjustments already included in fulfilment-cash-flow changes and reversals under 66(bb) that are not. The cap does not authorise inventing recovery income. Under paragraph 67, a change in reinsurer non-performance risk does not adjust CSM as future service; preserve collateral and dispute assumptions required by paragraph 63. Transferred-risk adjustment must be measured under paragraph 64, not mechanically copied as a fraction of the issued adjustment.

38.11.3 Changed terms and changed estimates are different events

For a treaty change, retain the old and new legal terms, effective date, covered-contract links and consideration. An exercise of a right already contained in the contract is not a modification under paragraph 72. A new treaty requires its own recognition and grouping decision and cannot be backdated to create initial-loss recovery. For a modification, apply paragraph 72's tests (European Commission, 2023a, IFRS 17.72–77) to the actual changed contract: changed scope or separated components, a substantially different boundary, a different initial group, changed direct-participation status, or loss of eligibility for a previously used premium-allocation approach can require derecognition and a new contract. If none applies, paragraph 73 treats the resulting cash-flow changes as estimate changes, with the held-contract modifications in paragraphs 63–68 where applicable.

For required derecognition, paragraphs 76–77 remove the affected cash flows and risk adjustment, adjust the remaining group's CSM under the applicable model and revise remaining coverage units. A transfer or qualifying modification uses paragraph 77's actual transfer consideration or equivalent new-contract premium adjustment rather than simply deleting the old balance. The equivalent premium is a required valuation input, not the old carrying amount chosen for convenience. Missing terms or that valuation leaves the particular modification unresolved. The generic tests identify the required procedure; they do not price an unspecified changed treaty.

38.11.4 The replacement premium connects the old and new groups

A qualifying modification changes both the removed obligation and the new contract. Paragraph 77 prevents the insurer from choosing the new opening balance simply by copying the old one. For a contract derecognised under paragraph 72, determine the premium that would be charged for an equivalent new contract at the modification date, then subtract any additional premium actually charged for the modification. This difference adjusts the old group's CSM together with the removed fulfilment cash flows, and is treated as a premium received when measuring the new contract (European Commission, 2023a, IFRS 17.72–77).

Take a general-measurement group with FCF 100, comprising expected claims 90 and risk adjustment 10, and CSM 30. Cash is 200, so liability 130 leaves equity 70. A legally effective modification substantially changes one contract's boundary and requires a new contract under paragraph 72. Its old allocated FCF is 40, comprising claims 36 and risk adjustment 4. Assume zero discount rates and no intervening service, tax, FX or OCI. The independently supplied equivalent new premium is 60, of which 10 is actually charged and received as an additional premium. The new obligation has FCF 52, comprising claims 48 and risk adjustment 4. These values are case inputs; they are not inferred from the old carrying amount.

The deemed premium is $60 - 10 = 50$. Removing FCF 40 while transferring value 50 reduces the old group's CSM by 10. Its remaining FCF is 60 and CSM 20, giving liability 80. The new contract's total consideration of 60 less FCF 52 gives CSM 8. One accounting sequence debits the old liability 50 and credits the new liability 50, then debits cash 10 and credits the new liability 10 for the actual receipt. Cash 210 equals old liability 80, new liability 60 and unchanged equity 70. Recording a second cash receipt for the deemed 50 would invent liquidity.

If the old group previously had three equal remaining coverage units and the removed contract contributed one, it now has two; the new group has its own service assessment. A proportional removal of 40% of the old CSM would leave 18 instead of 20 and would disregard the equivalent-premium rule. Over the remaining service, assume actual claims equal the supplied estimates and all remaining risk adjustment and CSM are released. The old group produces profit $80 - 54 = 26$; the new group produces $60 - 48 = 12$. Final cash $210 - 54 - 48 = 108$ equals opening equity 70 plus profit 38.

The same removed FCF with an actual third-party transfer premium of 50 instead reduces the old liability by 50 and pays cash 50; there is no new issued contract in this separate case. Cash 150, liability 80 and equity 70 reconcile. If the original CSM were only 6, either adjustment requiring 10 would consume it and recognise service loss 4. For the modification, old liability 60 plus new liability 60 would then leave equity 90 from cash 210, compared with opening equity $200 - 106 = 94$. The resulting loss component 4 is tracked within the old liability, not added again. Paragraphs 44(c), 48–50 and 77 govern that limit (European Commission, 2023a, IFRS 17.44,48–50,77).

Where the removed contract carries allocated accumulated OCI, paragraph 91 adds a separate presentation decision. Amounts arising under the systematic allocation in 88(b) are reclassified to profit on the specified transfer or derecognition; amounts arising under the held-underlying-item policy in 89(b) are not. Preserve that origin and the allocated amount before closing the old contract. The no-OCI example above does not eliminate this step (European Commission, 2023a, IFRS 17.91).

38.11.5 Future service consumes CSM before creating a loss

For issued contracts without direct participation, paragraph 44 and B96–B97 (European Commission, 2023a, IFRS 17.44,48–52,B72,B96–97) distinguish future nonfinancial service

changes from financial changes and incurred-claim experience. Use initial-recognition discount rates for the specified CSM adjustments under B72(c), while fulfilment cash flows use current rates. Changes in current rates, financial risk and incurred-claim estimates do not enter CSM by simply labelling them future cash flows. Preserve the current and initial-rate valuations and the reason for each change; their difference is part of the finance analysis, not an unrecorded balancing amount.

To isolate the nonfinancial rule, assume no passage of time, zero rates, no new contracts and no service release. A group has fulfilment liability 80, CSM 20 and total liability 100. An adverse future-service estimate of 15 raises fulfilment liability to 95 and reduces CSM to 5, with no immediate profit effect. A further adverse change of 10 raises fulfilment liability to 105, exhausts CSM and creates loss 5 and a loss component of 5. A subsequent favourable change of 8 first reverses that loss 5; only the remaining 3 restores CSM. Closing fulfilment liability 97 plus CSM 3 is 100. Crediting all 8 to profit would overstate the reversal by 3. The loss component is an internal allocation of the liability, never an additional liability of 5.

38.11.6 Finance presentation preserves total income and expense

For the nonparticipating case in paragraphs 87–90 (European Commission, 2023a, IFRS 17.87–91 and B128–132), a portfolio-level policy can present insurance finance entirely in profit or disaggregate it using the required systematic allocation. Under B129–B131, the choice and the contract characteristics determine the method. If financial assumptions do not substantially affect the nominal amounts paid, the systematic allocation uses the initial-recognition rates specified by B72(e)(i). OCI is the difference between that allocation and total finance, not a discretionary smoothing reserve. Over the group's duration it totals zero under B130.

Consider a GMM group with a now-incurred, fixed claim of 121 payable in two years, no remaining coverage or CSM, and zero risk adjustment because no nonfinancial uncertainty remains in this illustrative obligation. Initial and current discount rates are 10%, so liability is $121/1.1^2 = 100$. The next year's current rate falls to zero; one year remains, and current liability becomes 121. Total first-year finance expense is 21. Under the admitted OCI policy, the initial-rate allocation is 10 in profit and the difference 11 in OCI. In the second year, with rate zero unchanged, current liability stays 121, so total finance is zero. The initial-rate allocation accretes the alternative carrying amount 110 to 121, giving profit expense 11 and OCI income 11. Paying 121 extinguishes the liability. Cumulative profit finance is 21 and OCI is zero.

With opening cash 150, liability 100 and equity 50, first-year equity falls to 29 and stays 29 in the second year because profit expense and OCI income offset. Final cash and equity are 29. Presenting all finance in profit gives expenses 21 and zero instead, with the same total equity and cash. The OCI choice changes presentation, not the contractual claim or current measurement. This is the B131 fixed-payment case; variable-payment B132, VFA/underlying-item B134–B136, PAA and actual curve/RA/process valuation retain their separate methods. A successful fixed-claim example does not establish those models.

38.12 Variable benefits require a different finance allocation

The fixed-claim example uses initial rates because financial assumptions do not substantially change the nominal payment. B132 covers a different case: financial assumptions substantially affect the amounts paid. For the specified OCI policy, the cash-flow finance allocation may use a rate that spreads revised expected finance over the remaining duration at a constant rate. Contracts using a crediting rate have the alternative based on amounts credited and expected

to be credited. Nonparticipating CSM finance retains its prescribed initial-rate calculation; any separately allocated risk-adjustment finance must follow the source's consistent allocation (European Commission, 2023a, IFRS 17.B130–132).

Take a group with no remaining service, CSM or nonfinancial uncertainty, but a payment that remains sensitive to financial assumptions. Its liability and the amount measured using the systematic allocation both start at 100. A new financial assumption raises the expected payment in two years from 121 to 144. Supply a market-consistent current valuation of $144/1.1^2 = 14400/121$, approximately 119.0083. Under the chosen constant-rate allocation, the remaining rate solves $100(1 + r)^2 = 144$, so $r = 20\%$. The remeasurement at that instant produces OCI expense approximately 19.0083; no time has yet elapsed for the revised allocation to accrue interest.

With the current discount rate staying at 10%, current liability rises to 144/1.1, approximately 130.9091, after one year. Total finance expense is approximately 11.9008. The separate allocation rises from 100 to 120, giving profit expense 20 and OCI income approximately 8.0992. In the final year, current liability rises to 144, giving total finance approximately 13.0909; the allocation rises from 120 to 144, giving profit expense 24 and OCI income approximately 10.9091. All amounts are checked as exact fractions before display rounding. Total profit finance is 44, and accumulated OCI is zero. Opening cash 200 and equity 100 close at cash and equity 56 after payment. Retaining the original 10% allocation would reach only 121 and leave an unexplained 23 at settlement.

For a general payment schedule, retain the revised dated cash flows, carrying amount under the allocation, remaining duration and solved rate, then reconcile the full expected allocation and OCI unwind. A numerical failure or non-unique admissible solution remains unresolved; it is not permission to use an arbitrary rate. Current measurement and systematic allocation are separate calculations even when their values happen to agree.

38.12.1 Held underlying items determine the matching presentation

When a direct-participation insurer holds the underlying items, paragraph 89(b) permits a different OCI policy. B134 places in profit insurance finance that exactly offsets the profit income or expense on those items, with both amounts presented separately. The residual insurance finance goes to OCI. Revisit the preceding pool increase of 20, but now supply a valid asset measurement that reports 8 in profit and 12 in OCI. The matching insurance finance is expense 8 in profit and expense 12 in OCI, eliminating both net financial movements. The insurer's fee increase still enters CSM under the service method; matching is not an additional fee adjustment (European Commission, 2023a, IFRS 17.87A–90,B134).

Actual holdings must be recorded separately from the pool named by the contract. If holdings change the available choice between 88(b) and 89(b), B135–136 preserves the previously accumulated OCI under a separate reclassification schedule based on the former approach and assumptions at the change date. Those assumptions are then frozen for that schedule. Prior comparatives are not restated and accumulated OCI is not recomputed as if the new policy had always applied. New-period finance follows the newly applicable policy. A method that simply replaces the old rate and reserve with a newly computed one would discard required history (European Commission, 2023a, IFRS 17.B135–136).

38.12.2 Loss releases and finance retain different origins

The single-service-period example releases the whole loss component at once. Across several periods, finance also changes the liability before services release it. Paragraphs 50–52 require a

systematic allocation of claim releases, risk-adjustment releases and insurance finance between the loss component and the remainder of the liability. The loss component must reach zero by the end of coverage (European Commission, 2023a, IFRS 17.49–52). Allocating only service releases can leave an unexplained balance when finance has changed the amount to release.

Consider a separate group with remaining-coverage liability 100, loss component 20 within that liability, and no CSM. Cash is 200 and equity is 100. Assume no new estimates, investment components, acquisition-cost presentation or tax, and present all finance in profit. For this group, supply a supported fixed proportional allocation of one fifth. Let L denote the liability and K its loss component, so $k = K/L = 1/5$. Positive finance expense h increases the liability; expected claims and risk-adjustment releases c and a reduce it. Under this declared allocation,

$$L' = L + h - c - a, \qquad K' = K + kh - kc - ka = kL'.$$

The last equality explains why exhausting the supported liability releases also exhausts the loss component. It does not prescribe proportional allocation for every group. The finance allocation kh remains a finance movement; the service reversal is $k(c + a)$. Revenue from these releases is $(1 - k)(c + a)$, excluding that loss reversal.

In the first period, finance expense 10 raises liability to 110 and its loss component to 22. Expected claim release 50 and risk-adjustment release 5 then release 11 from the loss component and produce revenue 44. Actual claims of 50 are incurred and paid separately. Liability closes at 55, loss component at 11, cash at 150 and equity at 95. Service profit is $44 + 11 - 50 = 5$; finance expense 10 makes period profit -5.

In the second period, finance income 5 reduces liability to 50 and the loss component to 10. Claim release 45 and risk-adjustment release 5 exhaust both: revenue is 40 and loss reversal is 10. Paying actual claims 45 leaves cash and equity of 105. Service profit is again 5; finance income makes this period's profit 10. Across both periods, service profit 10 less net finance expense 5 gives total profit 5. Treating all releases of 105 as revenue and then adding loss reversals of 21 would overstate profit by 21. The separate categories, rather than final cash alone, reveal the mistake.

The held treaty needs the same discipline. Recall the covered-loss example: issued liability 120 contains loss component 20, while a timely half-quota treaty covers contracts with loss 8. Held fulfilment asset 24.5 and cost CSM 6 give asset 30.5, containing a loss-recovery component of 4. With cash 123.5, equity is 34. Reducing covered expected claims by 4 reduces issued liability to 116 and loss component to 16. Held recovery falls by 2, giving held asset 28.5, unchanged cost CSM 6 and loss-recovery component 2. Equity rises to 36. Paragraph 66(c)(i) puts the held fulfilment change in profit because the related issued change did not adjust issued CSM. Paragraph 66(bb) addresses reversals *not already included* in held fulfilment changes; the same 2 cannot be charged again (European Commission, 2023a, IFRS 17.66(bb),66(c)(i),B119F). Subsequent issued and held service, claims of 114 and recoveries of 22 close at cash 31.5, as before. Tracking the recovery component does not create a second asset or a second expense.

38.12.3 A change in holdings leaves an old OCI schedule

Current insurance finance and the reclassification of old OCI answer different questions. Under paragraph 89(b), profit finance on a direct-participation group matches the profit income or expense of the underlying items actually held. Suppose total insurance finance expense is H, and those items produce profit income A. Insurance profit expense is then A, while insurance OCI income is $A - H$. Asset income and insurance expense remain separately presented, with net profit finance zero (European Commission, 2023a, IFRS 17.89–90,B134).

When a holding change moves the insurer between 88(b) and 89(b), B135–136 requires previously accumulated OCI to be reclassified using the former approach and the assumptions immediately before the change. Those assumptions are frozen, and comparatives are unchanged (European Commission, 2023a, IFRS 17.B135–136). The new policy governs new-period finance. Recalculating the old balance as if the new policy had always applied would erase the required history.

Use the terminal-payment arithmetic from the preceding variable-benefit example in a separate, supplied direct-participation case with no remaining service, CSM or nonfinancial risk adjustment. At the holding change, payment 144 is due in two years. Its frozen current value at 10% is 14400/121, while the old systematic carrying amount is 100 with allocation rate 20%. Old insurance OCI is therefore $100 - 14400/121 = -2300/121$. Qualification and actual holdings are supplied findings; these values cannot establish either on their own.

Under the old assumptions, after one year the current-value calculation would be 1440/11 and the systematic amount would be 120. Old OCI would be $-120/11$, reaching zero at payment. Thus the first year's increase in old OCI, and equal profit expense from reclassification, is

$$-\frac{120}{11} - \left(-\frac{2300}{121}\right) = \frac{980}{121}.$$

The second-year reclassification expense is 120/11. These amounts sum to the original expense 2300/121. They do not depend on the next period's actual current valuation.

To see both histories together, suppose the insurer now holds a matching underlying asset initially measured at 14400/121, with post-acquisition cash of 200. The acquisition cash is part of this supplied opening snapshot. In the first year, actual asset and liability both become 130. The asset reports profit income 1440/121 and OCI expense 10/11, which together reconcile its increase. New insurance profit finance exactly offsets that asset profit income; new insurance OCI is income 10/11, offsetting the asset OCI. The old 980/121 reclassification remains a separate profit expense and equal OCI income. Total equity is unchanged, although profit and OCI each contain that old history.

If the actual first-year asset and liability instead become 132, with the same asset profit income, asset OCI becomes income 12/11 and new insurance OCI becomes expense 12/11. The old reclassification is still 980/121. A method that changes this old amount in response to the new valuation has failed the frozen-assumption requirement. At terminal receipt and payment of 144, the asset, liability and both OCI schedules are exhausted; cash and equity remain 200.

The reverse direction also needs a former-policy calculation. Suppose a held underlying FVOCI asset has fair value 121, amortised cost 100 and original effective rate 10%, with a terminal payment 121 in two years. Under the supplied frozen zero-current-rate assumptions, its fair value remains 121 while amortised cost grows to 110 and then 121. A matching insurance history therefore has old OCI $-21, -11, 0$. After the holdings change to 88(b), the old balance reclassifies as expenses 10 and 11. The new systematic allocation begins at 121 with rate zero. If the actual liability becomes 130 and then returns to 121, new OCI is expense 9 and then income 9; it does not replace the old 10 and 11 schedule. Post-change cash 200 and liability 121 give equity 79, which returns to cash and equity 79 after settlement. The actual holdings change and asset accounting remain independently supplied.

38.12.4 Derecognition follows the origin of accumulated OCI

A transfer under paragraph 91 can have identical cash and total equity but different profit presentation. Take cash 200, liability 100 and equity 100, including accumulated insurance OCI expense 12. Assume a qualifying transfer at carrying, with no remaining service, CSM, risk

adjustment or pending policy-change runoff. Paying 100 and removing the liability leaves cash and equity 100.

If that OCI arose under 88(b), paragraph 91(a) reclassifies it to profit: expense 12 in profit and income 12 in OCI, leaving no accumulated balance. If it arose under 89(b), paragraph 91(b) prohibits that reclassification through profit (European Commission, 2023a, IFRS 17.91). In this finite presentation, the expense remains within equity in OCI; no separate within-equity transfer is assumed. A positive accumulated OCI amount reverses these signs but not the policy distinction. Selecting a rule from the current policy label alone would discard the fact that paragraph 91 needs.

The records consequently retain each balance's original policy, actual holding history and any frozen reclassification assumptions. They distinguish finance measurement, current presentation and old-balance reclassification before forming profit and total equity. Mixed-origin early termination during an unfinished policy-change schedule still requires a specific source-supported disposition; these homogeneous transfers do not settle that question.

38.13 A claim process needs joint probabilities

The preceding calculations take cash-flow and risk-adjustment values as inputs. Obtaining them requires a law for the dated obligations, including their dependence. Means alone can support a linear expected-claim total while giving the wrong answer for a treaty with a deductible. IFRS 17 requires probability-weighted discounted cash flows, with supportable non-market assumptions and consistency with market variables. It does not make independent claim draws a default when their dependence is unknown (European Commission, 2023a, IFRS 17.B37–41,B49–55).

Let claims X_1 and X_2 at two specified dates each be either 0 or 40, with probability one half for either amount. Assume zero discount and no counterparty loss solely to isolate dependence. Total expected claims are always 40. A treaty pays $\max(X_1 + X_2 - 60, 0)$. If both claims always occur together, total claims are either 0 or 80 and expected recovery is 10. If exactly one claim occurs, total claims are always 40 and recovery is zero. The two models have identical single-date distributions.

All joint laws on this declared binary support can be described by $a = \Pr(X_1 = 40, X_2 = 40)$. The two marginal constraints force the following probabilities:

	$X_2 = 0$	$X_2 = 40$	
$X_1 = 0$	a	$1/2 - a$	$0 \le a \le 1/2.$
$X_1 = 40$	$1/2 - a$	a	

Nonnegativity gives the displayed interval. Only the final cell produces a recovery, of 20, so its expectation is $20a$, ranging exactly from 0 to 10 within this finite family. Independent claims correspond to $a = 1/4$ and recovery 5; independence is one member, not a conclusion from the marginals. Substituting mean claims into the treaty would give $\max(40 - 60, 0) = 0$, which is wrong for every $a > 0$.

The same law supplies a sequential event procedure. Draw the first claim with probability one half. After observing 40, the conditional probability that the second claim is 40 is $2a$; after observing zero it is $1 - 2a$. These transitions reproduce the joint table and provide a conditional recovery forecast after the first event. This is an exact finite-law specification, not a simulated result. If the evidence leaves a unresolved, retain its admissible interval and separate conditional results; do not assign a probability distribution over possible values of a without additional evidence.

A general valuation method must similarly specify exposure and contract boundaries, event times

and transitions, observation conditioning, claim severity, lapse and expense dependence, payment delays, discount treatment and financial-variable dependence. Its version must distinguish changes in the law from new observations under the same law. Risk adjustment then requires an entity-supported compensation method and diversification assumptions; the mean, variance or stop-loss value alone does not identify that compensation (European Commission, 2023a, IFRS 17.B65,B74–78,B86–89). The exact family above supplies a small test of such a method's dependence handling. Selecting and calibrating a richer law still requires the target population, credible data and downstream validation specified in the empirical protocols.

38.14 Claims development estimates the remaining payments

The claims-development example above supplies an estimation input, not a recognition rule. Its point projection is 148, but the single observed final factor leaves its variance unresolved. In the general method the factors are

$$\widehat{f}_k = \frac{\sum_{i \in O_k} C_{i,k+1}}{\sum_{i \in O_k} C_{i,k}},$$

where O_k contains origin years with both required observed cells. Each incomplete row is projected through the remaining factors. The same triangle therefore has ultimate claims 180, 216 and 252, with unpaid amounts 0, 36 and 112. Tail development, calendar inflation, zero or negative cells and dependence need explicit extensions. The conditional means, variance scaling and independent-origin-year assumptions behind the inspected Mack error formulas (Mack, 1993, equations (1)–(3), Theorems 1–3) cannot turn this tiny triangle into a predictive law or a justified insurance risk adjustment.

38.15 A reserve estimate is not a future payment

The reserve of 148 leaves an important question unanswered: how much might the insurer ultimately pay? Repeating an estimation procedure can show how its fitted answer varies. Even if that answer were known exactly, however, the future claims would still vary. England's retained 2001 working addendum makes this distinction operational: first resample and refit to obtain future conditional means, then simulate payments conditional on those means. The inspected text is Actuarial Research Paper 138; its exact equivalence to the separately published 2002 journal addendum has not been established (England, 2001, Sections 2–3 and Appendix A).

Consider a small probability example, separate from the triangle. Suppose a fitted future mean, denoted by M, is either 2 or 8 with equal probability. These are authored possibilities, not estimates from an insurer. If $M = 2$, let a payment Y be 0 or 4, equally likely; if $M = 8$, let it be 4 or 12, equally likely. Within each fitted case, the payment has mean M and variance $2M$. Across cases, the fitted mean has expectation 5 and variance 9. The total variation in payments includes both the variation left within a case and the variation between case means:

$$\operatorname{Var}(Y) = \mathbb{E}[\operatorname{Var}(Y \mid M)] + \operatorname{Var}(\mathbb{E}[Y \mid M]) = \mathbb{E}[2M] + \operatorname{Var}(M) = 10 + 9 = 19.$$

One can check the result directly: Y is 0, 4 or 12 with probabilities 1/4, 1/2 and 1/4. Its mean is 5, and its variance is $(25/4) + (1/2) + (49/4) = 19$. Reporting the distribution of fitted means alone would instead give variance 9 and no payment above 8. The full authored payment law gives probability 1/4 to that event.

Even those two moments do not select a unique payment law. Keep the same two fitted means, but, conditional on $M = 2$, pay 1 or 6 with probabilities 4/5 and 1/5; conditional on $M = 8$, pay 6 or 16 with those probabilities. The conditional means remain 2 and 8. Their variances are $(4/5)1 + (1/5)16 = 4$ and $(4/5)4 + (1/5)64 = 16$, respectively. The overall mean and variance remain 5 and 19, but the chance of paying more than 8 is now 1/10. Matching a reserve and a prediction error therefore cannot determine a tail probability or the insurer's compensation for non-financial risk.

38.15.1 What the two stages require

The addendum starts from a particular model for incremental payments. For an observed incremental payment X_{ij} in origin period i and development period j, its fitted mean m_{ij} is strictly positive. The model assumes $\mathbb{E}[X_{ij}] = m_{ij}$, $\mathrm{Var}(X_{ij}) = \phi m_{ij}$ and $\log m_{ij} = c + \alpha_i + \beta_j$, with reference levels fixed to identify the parameters. This incremental model and Mack's conditional cumulative model have separate assumptions, even where their fitted reserve amounts agree. A common point estimate does not make their uncertainty models interchangeable (England, 2001, equations 2.1–2.3).

The first stage measures each observed increment's departure from its fitted mean using the Pearson residual $r_{ij} = (X_{ij} - m_{ij})/\sqrt{m_{ij}}$. With N observed cells and p fitted parameters, the paper estimates $\widehat{\phi} = \sum r_{ij}^2/(N - p)$ and resamples residuals after multiplying them by $\sqrt{N/(N - p)}$. The adjustment must be applied once. It cannot be followed by another correction of the same estimation variance (England, 2001, equations 2.4–2.6 and 3.1).

Start from the observed latest cumulative amounts, work backwards through fitted development factors, and difference the fitted cumulative amounts into increments. Calculate the residuals and dispersion from those increments. For each bootstrap draw, resample adjusted residuals with replacement and invert the residual relation to form pseudo-increments. Accumulate them, refit the development factors, project future cumulative claims and difference them into future means. Retain the same fitted draw across all cells. This produces a joint array of conditional means, not yet a joint array of payments (England, 2001, Section 2 and Appendix A, Triangles 1–9).

The second stage chooses a process law. For a positive future mean m and positive dispersion ϕ, the paper proposes ϕZ, where Z has a Poisson law with mean m/ϕ. Its mean is m and variance is ϕm. A gamma law with shape m/ϕ and scale ϕ has the same two moments but different support and tail. That is a modeling choice, not a change of random-number syntax. Each future payment is drawn conditionally, and aligned payments are summed by origin, payment period and total (England, 2001, Section 3 and Appendix A, Triangles 9–10).

Dependence must survive aggregation. In the first small example, suppose two cells share the same M but are independent conditional on it. Their conditional total variance is $4M$, with expectation 20, while their conditional total mean is $2M$, with variance 36. Their total variance is therefore 56. Independently redrawing each cell from its marginal payment law would give $19 + 19 = 38$, losing the dependence created by the shared fit. Marginal distributions are insufficient instructions for a joint simulation.

38.15.2 Sparse data, failed draws and implementation choices

The six observed increments in the earlier triangle fit the incremental model exactly. It has $p = 2(3) - 1 = 5$ parameters and only $N - p = 1$ residual degree of freedom, giving zero fitted dispersion. This arithmetic does not establish that future payments are deterministic.

It also does not identify Mack's final development variance. The proposed interface returns `DISPERSION_UNRESOLVED` for this illustrative input and retains the point reserve of 148 separately. More data or a justified additional assumption is required before a predictive distribution can be admitted.

Every draw records its model, source version, data vintage, residual pool, fitted parameters, process law and random stream. Nonpositive degrees of freedom, an unsupported negative mean, a zero projection denominator or non-finite arithmetic receives an explicit failure. Structurally absent triangle cells have a separate mask. A failed payment is not converted to zero or silently replaced by another draw. In the authored example, if the $M = 8$ case cannot be evaluated, its probability $1/2$ remains unresolved: the known contribution to the full mean is $\frac{1}{2}(2) = 1$, and a complete mean is withheld. In a finite bootstrap, attempted failure frequency is reported as such; it does not identify population probability.

The retained source itself needs a qualification. Its negative-mean suggestion draws X with mean $|m|$ and then subtracts $2m$. Read literally for $m < 0$, the resulting mean is $|m| - 2m = -3m$, not m; at $m = -4$ it is 12. That prescription is wrong relative to its stated mean target in the inspected working version. A translation or reflection can preserve a negative mean, but changes the support and requires a different variance model. AccDSL leaves that extension unsupported pending an explicit signed-claims model; this local finding does not reject the positive-mean two-stage procedure (England, 2001, Section 3, final paragraph).

Software names also conceal consequential choices. In the inspected ChainLadder 0.2.17 source snapshot, `od.pois` calls a negative-binomial generator for dispersion greater than one. The paper's scaled Poisson and that negative binomial can share a mean and variance while assigning different probabilities to possible payments. The same snapshot uses signed gamma draws and replaces missing process values by zero. An adapter must distinguish structural absence from failed computation, identify the exact law and its parameter domain, and preserve separate resampling and process streams before its output can satisfy this proposal. These are source-code observations, not completed runtime validation (Gesmann and contributors, 2023).

The paper compares 1,000 simulations with analytic and earlier bootstrap summaries. That provides a worked implementation comparison under its assumptions. It does not establish future tail coverage, select an insurer's risk adjustment or justify 1,000 draws as a universal budget. A later empirical protocol must address calendar effects, tail development, pooling, heldout diagonals, simulation uncertainty and consequences for measured insurance groups (England, 2001, Section 4, Tables 1–3 and Figure 1).

38.15.3 The measurement date changes the question

A lifetime payment distribution differs from a one-year change in estimated liabilities. The 2019 accepted manuscript by England, Verrall and Wüthrich defines the claims development result as opening reserves less payments during the year and less the revised closing reserve. The closing estimate uses the information that has emerged during that year (England et al., 2019, equations 5–7 and Section 3.2). If the opening reserve is 100, payments are 30 and the revised closing reserve is 80, the result is $100 - 30 - 80 = -10$. Subtracting cash alone would yield 70 and answer a different question. The scenario must carry both realized next-year payments and the prescribed re-estimation rule, with the correct information cutoff. A lifetime draw cannot be relabeled as a one-year result. Neither statistic by itself supplies the discounting, group allocation or risk compensation required for insurance measurement.

38.16 What a bootstrap must reproduce

Separating fitted means from future payments is necessary, but it still leaves the statistical target to be chosen. An insurer knows an observed claims triangle. A method assessment may instead hold only its latest diagonal fixed, allowing different earlier histories to lead to those same cumulative amounts. These are different conditioning questions. Let D denote the observed triangle, Q its latest diagonal, R the future reserve payment and $\widehat{R}(D)$ the estimate calculated from the triangle. A distribution of $R - \widehat{R}(D)$ conditional on Q allows the earlier entries of D, and therefore the estimate, to vary. Conditional on the whole D, that particular estimate is fixed.

A small authored example makes the distinction concrete. Suppose two earlier histories are equally likely given the same Q and produce estimates of 90 and 110. Let future payments be 80 or 120 with equal probability, independent of those histories conditional on Q. The prediction error given Q is then $-30, -10, 10, 30$, each with probability $1/4$: its mean is zero and variance 500. Fixing the history whose estimate is 90 instead gives errors $-10, 30$, mean 10 and variance 400. Both calculations are correct for their stated conditioning information. They cannot serve as interchangeable validation targets. Nor does holding an estimate fixed imply that all uncertainty about model parameters has disappeared; a Bayesian parameter distribution, for example, would be a further explicitly specified model.

Steinmetz and Jentsch analyze a Mack bootstrap against the prediction error conditional on the latest diagonal. Their asymptotic construction adds older accident years and additional development observations while retaining the current calendar boundary. It does not obtain more information merely by increasing the number of bootstrap draws. The framework assumes positive initial cumulative claims with finite moments, specified independence, development factors tending to one, sufficiently fast decay of development variances, and further moment and regularity conditions. It is distinct from the incremental overdispersed-Poisson procedure in the preceding section (Steinmetz and Jentsch, 2024, Sections 2–4, especially Assumptions 2.2–2.5).

Their analysis separates an unscaled process term from a smaller estimation term. Under the stated assumptions, the process term has a nondegenerate, generally non-Gaussian limit. Correctly reproducing its distribution requires the correct conditional family for future development factors; matching two moments within a wrongly chosen family is insufficient. The original bootstrap can reproduce that limit while missing the conditional mean and covariance of the estimation term at its own, square-root sample-size scale (Steinmetz and Jentsch, 2024, Assumptions 4.6–4.9, Theorems 4.8–4.11 and Remark 4.12).

There is no contradiction in these two findings. For illustration, let a process term be an equally likely -1 or 1, and an independent estimation term an equally likely $-1/n$ or $1/n$. Their sum has variance $1 + 1/n^2$. Dropping the estimation term produces the same limiting distribution for the unscaled sum, yet multiplying that omitted term by n would have left variance one. This example explains the distinction between scales; it is not a derivation of the paper's Mack limit or evidence that a finite insurer triangle is adequately approximated by it.

38.16.1 Reconstructing the past requires a backward law

The paper's proposed repair holds Q fixed, simulates earlier cumulative claims backwards, refits the resulting triangle, and centers simulated future reserve payments at that refitted estimate. Future development uses the originally estimated forward factors. The purpose of reconstructing the past is to retain the effect of conditioning on the observed diagonal when measuring estimation uncertainty. Theorem 6.3 requires Assumption 6.2, including conditional limit properties for the backward reconstruction. Appendix C invokes those properties; it does

not establish them for every positive distribution chosen by a programmer (Steinmetz and Jentsch, 2024, Section 5, Assumption 6.2 and Appendix C).

Forward conditional moments do not determine backward conditional moments. Let X be an earlier cumulative claim and Y the next cumulative claim. Suppose $X = 1$ or 2 with equal probability. Given $X = 1$, let $Y = 1$ or 3, equally likely. Given $X = 2$, let $Y = 3$ or 6 with probabilities 2/3 and 1/3. These positive claims satisfy $\mathbb{E}[Y \mid X] = 2X$ and $\mathrm{Var}(Y \mid X) = X$. Thus the forward factor Y/X has conditional mean 2 and variance $1/X$, the form of a one-step Mack moment relation.

Now observe $Y = 3$. The joint masses for $X = 1$ and $X = 2$ are 1/4 and 1/3, respectively. Their sum is 7/12, so Bayes' rule gives $\Pr(X = 1 \mid Y = 3) = 3/7$ and $\Pr(X = 2 \mid Y = 3) = 4/7$. The backward conditional mean is 11/7, and its variance is 12/49. Dividing the observed cumulative amount by the forward mean factor would instead give 3/2. Choosing a backward factor with mean 1/2 and variance 1/3 would imply a variance of 3 for X, also wrong for this joint law. The example establishes the missing implication; it does not assert that this finite distribution satisfies every assumption of an infinite-triangle theorem.

In the published paper, Section 5.1 proposes reciprocal-factor backward moments, while Remark 5.1 and Assumption 6.2 separately require adequate backward conditional behavior. The proposal therefore keeps such a generator as a hypothesis requiring its own justification. Given a joint model, the backward distribution must be derived from that model or approximated with a checked error criterion. Choosing a gamma family and matching reciprocal moments alone cannot establish the required conditional limit properties (Steinmetz and Jentsch, 2024, Section 5.1, Remark 5.1 and Assumption 6.2).

38.16.2 The center and the comparison must use the same quantity

Reserve payments exclude claims already paid. Write q_i for the latest cumulative payment in origin period i, U_i^+ for its simulated ultimate payment, and $\widehat{U}_i^+$ for the ultimate payment predicted from a reconstructed past. The reserve error must subtract like quantities:

$$R^+ - \widehat{R}^+ = \sum_i (U_i^+ - q_i) - \sum_i (\widehat{U}_i^+ - q_i) = \sum_i (U_i^+ - \widehat{U}_i^+).$$

For $q = 100$, simulated ultimate payment 130 and fitted ultimate payment 120, the reserve is 30, its estimate is 20, and their difference is 10. Subtracting the ultimate estimate 120 directly from the reserve 30 gives -90, shifting the error by the already-paid amount.

This distinction also resolves a literal inconsistency in the retained published text. Equation (5.3), printed on page 96, writes the centering quantity as a sum of projected ultimates without subtracting the latest diagonal, although the preceding step defines reserves net of that diagonal. Read literally, those two formulas produce the wrong reserve error. The adjacent product-difference formula and decomposition (6.1) instead cancel the diagonal and agree with the derivation above. The proposed implementation uses that explicitly derived reserve center; it does not copy (5.3) literally. No claim about the authors' unavailable execution code follows (Steinmetz and Jentsch, 2024, Section 5.1, equation 5.3 and equation 6.1).

The final journal article also changes the preprint's simulation comparison. Section 7.1.1 generates whole triangles to assess unconditional prediction interval coverage. Section 7.1.2 assesses conditional coverage using an approximate backward generator in equation (7.4), based on the same reciprocal-moment idea. The article acknowledges that the exact backward conditional law is generally difficult to obtain. Its conditional simulation therefore evaluates that constructed experiment; it does not by itself verify coverage under the exact backward law implied by every forward model. The published numerical differences remain reported

simulation observations, without a local replication or statistically supported ranking here (Steinmetz and Jentsch, 2024, Section 7.1, Table 3 and Figures 1–3).

For AccDSL, a reserving-method specification must identify the payment horizon, conditioning information, process family, parameter-estimation procedure, backward law when one is needed, and reserve-centering convention. Validation must distinguish whole-triangle repeated-sample coverage, coverage conditional on a diagonal, and prediction conditional on a full observed history. A simulator with a convenient reverse rule is a candidate model, not an independent reference for that same rule. Exact centering and support checks precede simulations; target-specific comparison, uncertainty estimation and downstream financial-statement checks precede any promotion. The small triangle remains an accounting and arithmetic illustration, with its unresolved dispersion preserved.

38.17 A payment distribution still leaves compensation to choose

A justified distribution answers how much the insurer might pay. It does not determine how much compensation the insurer requires for bearing that uncertainty. Two insurers can agree on every possible payment and its probability while choosing different compensation. The distinction also matters within one insurer: a payment quantile, an average of expensive outcomes and a charge for future capital answer different questions. England, Verrall and Wüthrich compare these choices in their retained accepted manuscript, keeping the risk measure and its tolerance level separate from the underlying reserve distribution (England et al., 2019, Sections 5–6).

Consider an authored finite example in which the discounted remaining payment is 0, 4 or 10, with probabilities 1/2, 1/4 and 1/4. All amounts use one currency and are present values at today's valuation date. These probabilities define the example, rather than being estimated from the small claims triangle. The mean payment is $0/2 + 4/4 + 10/4 = 7/2$. A method that returns a total risk measure of $31/4$ would therefore assign an additional amount of $31/4 - 7/2 = 17/4$. The choice of the total measure still needs explanation. For this finite specification, write X for the discounted payment and $m = \mathbb{E}[X]$ for its mean. If $\mathcal{R}(X)$ is the selected total risk measure, the candidate compensation is the relative amount

$$A_{\mathcal{R}} = \mathcal{R}(X) - m.$$

The subtraction separates expected payment from the additional allowance. It does not select $\mathcal{R}$, establish the probabilities, or admit this amount as an insurer's accounting risk adjustment. The entity's method and non-financial-risk perimeter must support that further use.

38.17.1 A tail average must retain the probability at its boundary

Suppose the insurer examines the upper forty percent of the example's payment distribution. The largest payment, 10, occupies only one quarter of all outcomes. Completing the upper forty percent requires another 3/20 of probability from the payment of 4. Discarding every outcome equal to the cutoff would average only the payments of 10; keeping all such outcomes would average half the distribution. Both would answer a different question.

To specify the cutoff precisely, let $F(x) = \mathbb{P}(X \le x)$ be the cumulative distribution function. We use its left quantile at level α,

$$Q_\alpha = \inf\{x : F(x) \ge \alpha\}, \qquad 0 < \alpha \le 1.$$

At $\alpha = 3/5$, the cumulative probability is $1/2$ at zero and $3/4$ at four, so $Q_{3/5} = 4$. This quantile is one possible total risk measure. Its level is an explicit preference or measurement input, not a number deduced from the three payments.

An expected-shortfall measure instead averages the quantiles in the upper $1-\alpha$ of the distribution. For distinct ordered payments x_j with positive probabilities p_j summing to one, put $q = Q_\alpha$. For $0 < \alpha < 1$, the finite formula is

$$\mathrm{ES}_\alpha(X) = \frac{1}{1-\alpha}\int_\alpha^1 Q_u\,du = \frac{\displaystyle\sum_{x_j>q} p_j x_j + q\big(F(q)-\alpha\big)}{1-\alpha}.$$

The first term keeps every payment strictly above the cutoff. The second keeps exactly the probability needed from the cutoff itself. In the example, the numerator is $10/4+4(3/4-3/5) = 31/10$; dividing by $2/5$ gives $31/4$. By comparison, $\mathbb{E}[X \mid X > 4] = 10$ and $\mathbb{E}[X \mid X \geq 4] = 7$. Neither equals the specified expected shortfall. At $\alpha = 0$, define the measure directly as the full mean, $\mathrm{ES}_0(X) = m$; the upper-tail average then includes the whole distribution. This quantile-integral convention makes explicit the treatment of probability atoms that the source's equal-upper-tail-weight description leaves implicit (England et al., 2019, Section 6).

The same atoms affect an equivalent confidence level. The amount 4 is the left quantile for every $\alpha \in (1/2, 3/4]$; it has no unique equivalent level. The amount $31/4$ is never a left quantile of this three-point law, because the only quantile values are 0, 4 and 10. Its cumulative coverage is nevertheless well defined: $F(31/4) = 3/4$. That coverage is not an exact quantile match, since $Q_{3/4} = 4$. A finite evaluator must therefore report the probabilities $F(v-) = \mathbb{P}(X < v)$ and $F(v) = \mathbb{P}(X \leq v)$ at the requested amount v, together with whether an exact quantile-equivalent level or interval exists. Interpolation would be a separately declared convention. This mathematical distinction does not decide the appropriate regulatory disclosure convention for an actual insurer.

38.17.2 Changing the weights changes the risk measure

A different approach gives additional weight to expensive outcomes throughout the distribution. For a nonnegative payment X, its survival probability $S(u) = \mathbb{P}(X > u)$ records how often the payment exceeds each amount u. The ordinary mean is the area under this survival function. The proportional-hazards transform replaces each survival probability by its $1/\rho$ power, with a chosen parameter $\rho \geq 1$:

$$H_\rho(X) = \int_0^\infty S(u)^{1/\rho}\,du.$$

At $\rho = 1$ this is the ordinary mean. Increasing ρ raises survival probabilities between zero and one, increasing the weight attached to large payments. The source introduces this integral for nonnegative losses (England et al., 2019, Appendix 2). Applying it to signed net cash flows requires a separately specified extension; silently clipping recoveries or shifting the distribution would change the stated calculation.

The finite weights follow from that same survival function. Order the payments increasingly, define $S_j = 1 - \sum_{k\leq j} p_k$, and set $S_0 = 1$. The transformed probability assigned to x_j is the fall in transformed survival probability across that payment:

$$w_j = S_{j-1}^{1/\rho} - S_j^{1/\rho}, \qquad H_\rho(X) = \sum_j w_j x_j.$$

The weights are nonnegative because survival decreases. Their sum telescopes from $S_0^{1/\rho} = 1$ to zero. At $\rho = 1$, each $w_j = p_j$ and the ordinary mean is recovered. Repeated payment values may

be combined by adding their probabilities; splitting one such mass into identical observations does not alter the result.

For an exact illustration of the transform, keep payments 0, 4 and 10 but change their probabilities explicitly to 5/9, 1/3 and 1/9. The mean is now 22/9. At $\rho = 2$, the transformed survival probabilities are $1, 2/3, 1/3, 0$, so all three transformed weights equal 1/3 and $H_2 = 14/3$. Computing the area directly gives the same answer: $4(2/3) + 6(1/3) = 14/3$. The example checks the transformation under its stated law; neither that law nor the parameter 2 is an insurer default.

The retained accepted manuscript prints the survival subtraction in the opposite order for observations after the first. On its page 34, that formula is wrong relative to the integral it defines. With two equally likely payments 0 and 4 and $\rho = 1$, the printed weights are 1/2 and $-1/2$. They sum to zero and give an unnormalised weighted sum of -2, whereas the integral is the mean, 2. The formula above uses the locally derived correction. The final journal edition has not been checked for this equation, and the printed error does not establish which formula produced the authors' numerical tables or was implemented in their software (England et al., 2019, Appendix 2, p. 34).

38.17.3 Tomorrow's information changes tomorrow's risk

A cost-of-capital calculation needs a sequence of future capital amounts. It is tempting to obtain each amount by taking today's simulated payments, removing the payments made before that future date, and measuring the spread of what remains. But the insurer will then know more than it knows today. Some of today's uncertainty will have become information, even when no cash has yet been paid.

Consider an authored two-period example with one payment at final settlement. The ultimate payment U is 0, 2, 8 or 14, each with probability 1/4. After the first period the insurer learns whether the payment belongs to the lower pair or the upper pair. Within either pair the two possibilities remain equally likely. Figure 38.1 separates the information arrival from final payment.

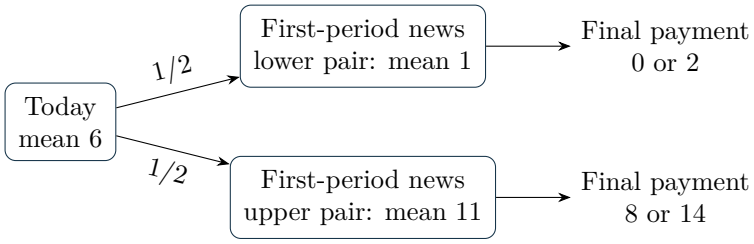


Figure 38.1: Information changes the remaining uncertainty before cash is paid. Within each revealed pair, the two final payments are equally likely.

Today's mean is 6 and variance is 30. After the news, the lower pair has mean 1 and standard deviation 1; the upper pair has mean 11 and standard deviation 3. The expected future standard deviation is therefore $(1 + 3)/2 = 2$. Taking the square root after averaging the future variances instead gives $\sqrt{(1 + 9)/2} = \sqrt{5}$. Keeping today's information throughout gives $\sqrt{30}$. These are three distinct quantities, not three estimates of one interchangeable target.

To generalise the distinction, let $\mathcal{F}_t$ denote all information available at time t, increasing through time, and let A_t be cumulative payments already made and known then. Remaining payments are $L_t = U - A_t$. Assume finite second moments and a final date T by which U is known. At a future time t, write $V_t = \mathrm{Var}(L_t \mid \mathcal{F}_t)$ for the conditional payment variance. Its square root is

the uncertainty remaining after that date's news. Viewed today, concavity of square root and conditional total variance give

$$\mathbb{E}[\sqrt{V_t} \mid \mathcal{F}_0] \le \sqrt{\mathbb{E}[V_t \mid \mathcal{F}_0]} \le \sqrt{\mathrm{Var}(L_t \mid \mathcal{F}_0)}.$$

The first inequality compares an average of standard deviations with their root mean square. For the second, total variance separates uncertainty remaining after the news from uncertainty about what the news will reveal:

$$\mathrm{Var}(L_t \mid \mathcal{F}_0) = \mathbb{E}[V_t \mid \mathcal{F}_0] + \mathrm{Var}\big(\mathbb{E}[L_t \mid \mathcal{F}_t] \mid \mathcal{F}_0\big).$$

The last term is nonnegative. In the example it is the variance of conditional means 1 and 11, namely 25, so $30 = 5 + 25$. These relations reconstruct the source's Lemma A3.1 in the present notation (England et al., 2019, Appendix 3). They concern expected future standard deviation. On the upper branch, the actual future standard deviation is $3 > \sqrt{5}$, so the middle quantity is not an upper bound in every future state. The argument also establishes no corresponding quantile ordering.

38.17.4 When future reserve revisions have additive variances

The previous claims development result subtracts both cash paid and the revised closing reserve. A useful ideal case assumes the reserve at every date is the true conditional expected remaining payment. Define $M_t = \mathbb{E}[U \mid \mathcal{F}_t]$ and $R_t = M_t - A_t$. Substituting these expressions into the result for period t gives

$$\begin{aligned} D_t &= R_{t-1} - (A_t - A_{t-1}) - R_t \\ &= (M_{t-1} - A_{t-1}) - (A_t - A_{t-1}) - (M_t - A_t) \\ &= M_{t-1} - M_t. \end{aligned}$$

Here D_t uses the source's profit sign: an increase in estimated ultimate payments is a loss and makes D_t negative. A capital calculation on losses must declare the opposite sign. The source's Table 7, for example, uses the negative of a lower-tail CDR quantile (England et al., 2019, equations 5–9 and Section 4).

Summing future results from $t+1$ to final settlement makes the intermediate conditional means cancel, leaving $M_t - U$. Its conditional mean at time t is zero. Since A_t is known then, $\mathrm{Var}(L_t \mid \mathcal{F}_t) = \mathrm{Var}(U \mid \mathcal{F}_t)$. Applying total variance once more therefore yields

$$\mathbb{E}[V_t \mid \mathcal{F}_0] = \mathrm{Var}\left(\sum_{k=t+1}^{T} D_k \,\middle|\, \mathcal{F}_0 \right).$$

Under the conditional-mean assumption, these future revisions have zero conditional covariance. To see why, take $k < l$: D_k is already known at $l-1$, while $\mathbb{E}[D_l \mid \mathcal{F}_{l-1}] = 0$ by iterated expectation. Thus $\mathbb{E}[D_k D_l \mid \mathcal{F}_0] = 0$, and each increment also has conditional mean zero. The variance above becomes

$$\mathbb{E}[V_t \mid \mathcal{F}_0] = \sum_{k=t+1}^{T} \mathrm{Var}(D_k \mid \mathcal{F}_0).$$

This is a local derivation under true conditional means. The source's Corollary A3.2 states a fixed-origin relation and then connects its variances to estimated MSEPs in a Bayesian chain-ladder representation (England et al., 2019, Appendix 3). That estimator claim is additional to the identity derived here. Using total ultimate payment U retains any dependence across origins; summing separate origin variances would require their covariance terms as well.

In the four-outcome example, the first result is 5 or -5. The second is 1 or -1 on the lower branch and 3 or -3 on the upper branch. The two variances are 25 and 5, and their covariance is zero. Nevertheless, they are dependent: the first result reveals the absolute size of the second. Zero covariance does not establish independence. Nor does an arbitrary refitted reserve rule necessarily equal the true conditional mean. With such a rule, preserve the joint sequence of results and its covariance; do not automatically replace its variance by a sum of MSEPs. The source itself warns that general re-reserving need not produce zero-mean or independent future results (England et al., 2019, Section 7).

38.17.5 Capital timing and entity choices remain separate inputs

Once a capital basis has been justified, a cost-of-capital calculation still needs the charge rate and its payment timing. Suppose K_t is capital held at the start of year t, c_t is the annual charge per unit of capital, and $d(0, t+1)$ discounts a charge paid at year end to time zero. For this stated timing convention, the margin is

$$\mathrm{RM}_0 = \sum_{t=0}^{T-1} d(0, t+1) c_t K_t.$$

The equation adds discounted capital costs. It does not supply the capital amounts or select their risk measure. A projected best-estimate runoff, a standard-deviation profile and a quantile profile can have the same opening capital and different future charges. Near equality in one source example does not make them equivalent. If capital or discount factors are random, the intended conditional expectation and their joint dependence must be specified; discounting expected capital generally differs from expecting discounted capital. Capital-charge timing also need not equal claim-payment timing.

The paper's chosen capital charges, discount rates and tolerance levels belong to its historical discussion and illustrative calculations (England et al., 2019, Sections 5–7). They are not AccDSL defaults. An actual insurance risk adjustment additionally requires the applicable accounting basis, the entity's compensation for non-financial risk, the relevant fulfilment cash flows and appropriate treatment of reinsurance and group allocation. The paper's payment-triangle example does not supply all those quantities, and its discussion leaves allocation and diversification beyond scope.

For implementation, the finite evaluation request must therefore identify the law and its provenance, information date, horizon, contract perimeter, currency and unit, payment or loss sign, discount convention, chosen functional and explicit parameter. It returns the mean, total risk measure, relative amount, boundary convention and cumulative coverage, together with any exact quantile-equivalence interval and unresolved condition. A negative relative quantile amount is retained and flagged if the declared compensation profile requires nonnegativity; silently flooring it would change the method. Unknown probabilities, mixed currencies, incomplete draws, missing preferences and signed inputs outside the nonnegative PH method block their dependent calculation. They cannot be repaired by borrowing a percentage from a paper. The next software boundary must preserve precisely the joint outcomes on which these calculations depend.

38.18 A zero payment and a failed draw need different records

The distinction between payments and fitted means must survive the software boundary. Suppose a reserving calculation requests two draws, each containing payments for three future cells. In

an authored example, the joint payment vectors are $(0, 4, 8)$ and $(2, 6, 10)$. Their totals are 12 and 18, and the sample mean is 15. These are possible nonnegative integer payments chosen to illustrate the exchange; they are neither fitted to the preceding triangle nor generated by a locally executed model.

Now let the last cell of the first draw fail. That draw's total is unresolved, while the second total remains 18. Reporting 18 as the requested sample mean would discard the failed draw. Replacing the failed value by zero would instead produce totals 4 and 18, and mean 11. Neither number is the mean of the requested complete sample. A diagnostic may state that the one completed draw has mean 18, provided it retains the failed draw and withholds the full sample mean. The fraction 1/2 describes completed computation, not a known probability mass assigned to unsuccessful economic histories.

This failure is relevant to the retained `BootChainLadder` source. Before forming its returned process triangles and totals, it replaces every process `NA` by zero. The returned array has no corresponding mask distinguishing failed future draws from structural cells. A valid payment vector $(0, 4, 0)$ and a vector $(0, 4, \text{failed})$ therefore become the same numeric vector and total. Inspecting the final array cannot recover the lost distinction (Gesmann and contributors, 2023, `BootstrapReserve.R`, process-triangle coercion and returned fields).

The unmodified output is consequently retained as source evidence but is unsupported as a complete predictive result under this proposal. A future instrumented implementation could preserve statuses before coercion, but would need its own source identity, dependency lock and numerical checks. Merely wrapping the returned array is insufficient. The documentary fixtures specify that boundary; they do not establish an implemented or tested R adapter.

38.18.1 The exchange retains the law and the whole draw

An optional `reserving.evaluate@1` signature uses the common operation envelope. Its declared profile names the claims triangle, method and process law, requested draw identifiers, information view and producer evidence. The present profile is a specification example that consumes authored evidence. It makes no financial-book mutation and grants no probability-law admission. Its request and result references use the same typed identity and dependency rules as the other operations.

The triangle carries the currency, base payment unit, scale, origin and development labels, payment-period mapping, valuation date and explicit cell mask. Observed cumulative amounts, future cells and unavailable observed amounts stay distinct. The inspected routine rejects unequal row and column counts, so the proposed initial adapter subset is square even though the native error message describes a broader shape. Rectangular support requires a separately inspected implementation.

Each replicate carries one fit identifier and its complete joint vector of future cells. A cell retains both its fitted mean and its process payment, with a status and reason if the payment is unavailable. A valid zero remains a value. Missing cells, repeated draw identifiers, a different fit inside one draw, or a currency mismatch prevent acceptance. Any failed cell leaves the draw total unresolved. Native evidence lacking the original status mask cannot be relabeled as complete by an adapter.

The process-law identity includes its formula. With positive mean m and dispersion $\phi > 1$, the inspected `od.pois` helper calls a negative binomial with mean m and size $m/(\phi - 1)$. At $\phi = 1$ it calls a Poisson distribution. For $0 < \phi < 1$, its negative-binomial size would be negative. A gamma method with positive shape and scale is a different possibility; the helper's limitation does not reject that separate law (Gesmann and contributors, 2023, `rpois.od` and the

process-law branches).

Nor can the negative-binomial path silently satisfy a request for the scaled-Poisson law from England's working addendum. For $m = 2$ and $\phi = 3/2$, both have mean 2 and variance 3. The scaled-Poisson support is $0, 3/2, 3, \ldots$, whereas the negative binomial can pay 1. The base payment unit thus matters to the model, not just to presentation. The positive-law profile excludes the package's separate signed handling; a different signed-claims model would need its own support and validation.

38.18.2 Stopping preserves what was actually computed

A result reports requested, attempted, completed, failed and pending draws separately. Its coverage domain is the set of computational replicate IDs. It is not the company's family of economic histories. Complete arrays do not establish calibrated forecasts, and an incomplete set of arrays does not justify a probability distribution conditional on computational success.

If cancellation follows the first complete draw in the example, retain its total 12 and mark the second draw unattempted. The diagnostic completed-draw mean is 12; the requested sample mean remains unresolved. A stop request alone does not establish that the worker has stopped. Unknown termination is reported as partial with unresolved worker state. Resume requires a durable checkpoint that binds the semantic manifest, original draw IDs and random generator state; the native function's seed argument supplies no such checkpoint.

The inspected source resets its supplied seed in both residual sampling and process generation. That behavior must retain its own method identity and cannot be described as verified independent streams (Gesmann and contributors, 2023, `sampleResiduals` and the process-law branches). A proposed replacement must specify generator version, stream derivation and restart behavior, and then check them in execution. Changing a method, law or request identifier creates a different request fingerprint. A resumed scheduling request must link to the old checkpoint under the persistence rules; it cannot silently alter the old request.

The retained snapshot's GPL terms, transitive dependencies and any future instrumented source remain part of the proposed runtime decision. No package installation, distribution clearance, crash-durability result or empirical performance follows from these exchanges. What they establish is narrower and necessary: a successful payment, a missing value and a failed computation remain distinguishable when reserving results enter the financial model.

38.19 Acquisition costs before the renewal contracts exist

An insurer may pay acquisition costs that help obtain both current contracts and their expected renewals. Some of the related future groups do not yet exist as recognised insurance balances. The retained AASB 17 provisions therefore require a separate acquisition-asset record for each related group, unless the permitted immediate-expense election is used. Qualifying costs are those directly attributable to the insurance portfolio; this is not a general permission to capitalise selling expenses. A systematic and rational allocation distinguishes the original group from groups expected to contain its renewals (Australian Accounting Standards Board, 2022g, paragraphs 28A–28B, B35A and Appendix A).

Consider a separate, authored example in currency units. Follow only 60 of qualifying costs already paid and allocated to expected renewals: 40 from one origination cohort to future group G1, and 20 from another cohort to future group G2. Other acquisition allocations are outside this example. Assume that both allocations are supported, neither future group has yet been recognised, the immediate-expense election is not used, and facts now indicate possible impairment. Those facts trigger the reporting-date recoverability assessment under paragraph 28E. An expected renewal remains a forecast, not a newly recognised premium receivable (Australian Accounting Standards Board, 2022g, paragraphs 28B, 28E and 35).

Each future group contains the relevant renewals together with other new contracts. For G1, suppose the whole group's expected net inflow is 35, of which the renewal cohort contributes 25. For G2, the corresponding amounts are 25 and 10. These are conditional measurements under paragraph 32(a), incorporating the required cash-flow, time-value and risk adjustments. They are neither gross premiums nor CSM. The acquisition cash already paid does not become a second future outflow. In this additive example, the other contracts contribute 10 to G1 and 15 to G2.

Paragraph B35D(a) first tests the asset against its related group's expected net inflow. G1's asset of 40 exceeds 35, so recognise impairment of 5 and retain 35. G2's asset of 20 is below its whole-group inflow of 25, so this test produces no loss for G2. Passing that first test does not settle whether the renewals themselves can recover the costs allocated to them.

The second test addresses that narrower population. Under B35D(b), the allocated renewal costs exceed the expected renewal inflows by $40 - 25 = 15$ for G1 and $20 - 10 = 10$ for G2. But 5 of G1's shortfall has already been recognised. B35D(b)(ii) requires that overlap to be deducted, leaving additional losses of 10 for G1 and 10 for G2. Total impairment is therefore $5 + 10 + 10 = 25$, and the remaining assets are 25 and 10. Adding the first loss to the two unadjusted renewal shortfalls would instead record 30, counting the same loss of 5 twice (Australian Accounting Standards Board, 2022g, paragraph B35D).

For these disjoint allocations, write A_g for the original asset, N_g for the whole-group expected net inflow and U_g for the relevant renewal inflow. With no prior impairment, the two losses and the resulting balance are

$$I_g = \max(A_g - N_g, 0), \qquad J_g = \max(A_g - U_g - I_g, 0), \qquad B_g = A_g - I_g - J_g.$$

The subtraction of I_g inside the second test preserves the already recognised loss. This local calculation relies on the identified, nonoverlapping renewal cohorts and positive inputs. A pool with several overlapping origins would need an explicit allocation of the relevant costs, inflows and prior losses before the same reasoning could be applied.

The entries also distinguish impairment from a cash payment. Debit impairment expense 5 and credit the G1 acquisition asset 5 for the first test. Then debit impairment expense 20 and credit each group's asset by 10 for the second. Cash does not move: the acquisition costs were already paid. For a small balance-sheet slice starting with cash and equity of 100 and no liabilities, paying the allocated costs leaves cash 40 and acquisition assets 60. After impairment, cash remains 40, acquisition assets are 35 and equity is 75. Paragraph 79 includes these acquisition assets within the related issued-contract portfolio carrying amounts; the separate records used here do not prescribe a separate line on the published balance sheet (Australian Accounting Standards Board, 2022g, paragraphs 28E and 79).

Recoverability can improve. At a later reporting date, suppose the full-group net inflows are 45 and 30, with renewal contributions of 35 and 15. Assume no intervening cash movement or recognition of either group. The allocation assumptions have been reviewed under B35B and remain unchanged; only the measurement inputs improve. Against the original allocations of 40 and 20, the remaining shortfalls are now 5 for each group. Paragraph 28F permits reversal to the extent the impairment conditions have improved. Reverse 10 for G1 and 5 for G2: debit the acquisition assets 15 and credit the impairment reversal in profit or loss 15. Assets become 35 and 15, totalling 50; cash stays 40 and equity becomes 90. The improvement restores 15 of the earlier loss of 25 without creating cash or recognising the expected renewals early (Australian Accounting Standards Board, 2022g, paragraphs 28F and B35B).

If the impairment conditions later disappear altogether, with these other facts unchanged, only the remaining loss of 10 can be reversed; the original allocated cost of 60 is the ceiling in this example. When related contracts are actually recognised and the acquisition cash flows enter their group measurement, the corresponding acquisition asset must be derecognised. B35C provides for partial incorporation when contracts are added over several periods. Keeping both the old asset and its incorporated amount would double count the same cost. The record must therefore preserve its origination cohort, future group, allocation history, impairment and reversal history, and the amount already incorporated into recognised contracts (Australian Accounting Standards Board, 2022g, paragraphs 28C–28F and B35B–B35C).

These are ordinary reporting-date impairment and reversal entries. They do not use the separate transition rules for acquisition assets. The historical edition, actual cost eligibility and supported renewal estimates must be checked before applying the example to an insurer.

38.20 When the acquisition asset enters the insurance group

Continue from the preceding reversal: cash is 40, the two acquisition assets are 35 and 15, and equity is 90. The next event is recognition of the related insurance groups. Assume that coverage begins and the first premiums become due at this date, with no earlier recognition trigger. All contemplated contracts are admitted together to their correctly identified groups. No service has yet elapsed and no claims have been incurred. Paragraphs 25–28 determine recognition; the later cash receipt is a separate event (Australian Accounting Standards Board, 2022g, paragraphs 25–28).

Keep the measured group net inflows of 45 and 30 from the previous example. To expose their components, suppose premiums of 100 and 60 are due immediately, the present values of future claims and expenses are 50 and 25, and each group has a nonfinancial risk adjustment of 5. Thus fulfilment cash flows before premium receipt are $50 + 5 - 100 = -45$ and $25 + 5 - 60 = -30$, using positive amounts for net outflows. These are supplied conditional measurements; present values do not imply a zero discount rate. Previously paid acquisition costs are not another future payment. Assume general measurement and no other assets, liabilities, taxes, reinsurance or intervening measurement changes.

Paragraph 38 requires initial CSM to prevent income or expense from the FCF, any initial cash flows and the derecognition of the acquisition asset, subject to its stated exceptions. The paid cost therefore affects initial CSM even though it is no longer a future cash flow. Let A denote the acquisition asset's remaining carrying amount, F the FCF and C the CSM. Removing the old asset without a gain or loss requires the new group balance to replace it. With positive L denoting a liability, the relation in this case is

$$L = F + C = -A, \qquad C = -F - A.$$

G1 consequently has CSM $45 - 35 = 10$ and a net group asset of 35; G2 has CSM $30 - 15 = 15$ and a net group asset of 15. Both margins are positive, so the onerous-group exception does not arise. There are no other previously recognised cash-flow assets to which paragraph B123A's revenue-and-expense exception would apply (Australian Accounting Standards Board, 2022g, paragraphs 28C, 32, 38, 47 and B123A).

An illustrative recognition entry debits the two group assets by 35 and 15 and credits the corresponding acquisition assets by those amounts. Cash is still 40 and equity is still 90. The separate acquisition balances are now zero. The amounts transferred are the carrying values after impairment and reversal, rather than the original allocations of 40 and 20. The earlier loss history remains available even though those assets have entered the group measurements (Australian Accounting Standards Board, 2022g, paragraphs 28C–28D and B35C).

Now receive the premiums in full, immediately after that measurement. Debit cash 160 and credit the group carrying balances by 100 and 60. Because the premiums are no longer future inflows, FCF becomes 55 for G1 and 30 for G2. CSM stays 10 and 15: no service or time has passed. The groups therefore carry liabilities of $55 + 10 = 65$ and $30 + 15 = 45$. Cash has increased to 200, and $200 - 65 - 45 = 90$ confirms unchanged equity. Paragraph 83 ties revenue to service, while B123(a)(i) expressly distinguishes premium cash inflows from the liability reductions that generate revenue. Later acquisition-recovery revenue and the equal service expense are allocated over time under B125; neither is created by this instantaneous receipt (Australian Accounting Standards Board, 2022g, paragraphs 40–41, 83–85 and B123–B125).

The three balance-sheet states make the transfer visible. Amounts are currency units; group assets and liabilities below are their net carrying amounts, not their separate FCF and CSM components.

	Before recognition	After recognition	After receipt
Cash	40	40	200
Acquisition assets	50	0	0
Group assets	0	50	0
Group liabilities	0	0	110
Equity	90	90	90

Receiving 160 has changed liquidity and the insurance balances without creating profit. Omitting the acquisition assets from the initial CSM calculation would instead give margins of 45 and 30, overstating total CSM by 50. Retaining the separate acquisition assets after the correct transfer would overstate assets and equity by 50.

Even a balanced statement can conceal both mistakes together. If the implementation leaves acquisition assets of 50 and uses the overstated margins, its post-receipt liability becomes 160. Cash 200 plus the retained asset 50 less liability 160 still equals equity 90. That identity passes, but the asset, liability and CSM are wrong relative to the stated measurement. Checking the components and the derecognition event is therefore necessary in addition to checking that the statement balances.

As a boundary case, if no acquisition asset exists, $A = 0$ gives $C = -F$ and a zero group carrying amount before premium receipt, under the same positive-margin assumptions. A general implementation must also handle onerous groups, other initial cash-flow balances and contracts added over time. The present calculation establishes the two-event bridge for the specified groups, rather than those additional cases or an insurer's actual valuation.

38.21 Acquisition costs and the timing of reinsurance

An acquisition asset can enter a group that is onerous at recognition. Reinsurance then raises a second question: how much of that initial loss may be recognised as recoverable, and when? Following the two groups separately prevents a recoverable loss from becoming either a second asset or a reduction of insurance revenue.

Consider a separate general-measurement group with opening capital and cash of 50. Before recognition, the insurer pays acquisition costs of 10, allocated wholly to that future group. At the prior assessment, expected premium 100, claims 80 and risk adjustment 5 give expected net inflow 15. The asset of 10 is therefore recoverable under the stated assumptions, leaving cash 40, acquisition asset 10 and equity 50. No renewal allocation is involved (Australian Accounting Standards Board, 2022g, paragraphs 28A–28F and B35A–B35D).

The expected claims increase to 100 when the contracts are recognised; premium remains 100 and risk adjustment remains 5. The example has no intervening unrecorded onerous period, discounting, tax, investment component or OCI. Before premium receipt, fulfilment cash flows are $100 + 5 - 100 = 5$. Removing the acquisition asset adds 10 to the initial loss, making 15, with CSM zero. The entry debits loss 15 and credits acquisition asset 10 and remaining-coverage liability 5. Receiving the premium then gives cash 140, liability 105 and equity 35 (Australian Accounting Standards Board, 2022g, paragraphs 28C, 38(c)(i) and 47).

The loss component of 15 is part of the liability of 105. Acquisition cash has already left the company: putting it back into future cash outflows would count the cost twice. Its original asset also disappears when it enters the group measurement.

A separate held treaty covers half of every claim of this group and costs 55. Assume no commissions, limits, disputes or counterparty shortfall. Expected recovery is 50; risk transferred is separately valued at 2.5. That supplied risk adjustment need not equal half the gross adjustment in a different contract. Ordinary cost CSM is $55 - 50 - 2.5 = 2.5$, so the held asset after premium payment would be 55 (Australian Accounting Standards Board, 2022g, paragraphs 62–65).

If the treaty was entered into at or before recognition of the onerous group, the specified initial loss recovery is $15(0.5) = 7.5$. It increases held cost CSM to 10 and the measured asset to $50+2.5+10 = 62.5$. The standard multiplies the recognised loss by the expected claims-recovery percentage; this does not mean the treaty separately reimburses half the acquisition invoice. The loss-recovery component of 7.5 is contained in the asset of 62.5. After paying the treaty premium, cash 85 plus that asset equals liability 105 plus equity 42.5 (Australian Accounting Standards Board, 2022g, paragraphs 66A–66B and B119C–B119F).

At the end of the single service interval, release the issued liability of 105 as revenue 90 and loss reversal 15. The ratio $15/105 = 1/7$ allocates the expected-claim and risk-adjustment releases to the loss component. Acquisition-cost presentation adds revenue 10 and an equal service expense 10 under B125. It creates neither cash nor a new acquisition asset. Recognising and paying actual claims of 100 therefore gives issued-group service-period profit $100 - 100 - 10 + 15 = 5$. Including the initial loss of 15, its lifetime loss is 10 (Australian Accounting Standards Board, 2022g, paragraphs 49–52 and B120–B125).

The held asset releases as premium allocation 55 and loss-recovery reversal 7.5, with incurred recovery and collection of 50 recorded separately. Its service-period result is $50-55-7.5 = -12.5$; the initial recovery of 7.5 makes its lifetime result -5. The company's final cash and equity are both 35:

$$50 + 100 - 10 - 55 - 100 + 50 = 35.$$

All issued and held balances are zero. Total operating cash is -15, profit is -15, and OCI is zero.

Premium allocation remains separate from issued revenue (Australian Accounting Standards Board, 2022g, paragraph 86).

If the otherwise identical treaty is entered into only after the issued group is recognised, B119C prevents recovery of that original loss. Its initial held asset is 55 and its cost CSM is 2.5. Equity is initially 35 and remains 35 after the combined subsequent service result of zero. The timely treaty instead gives initial equity 42.5 and a subsequent loss of 7.5. Equal final cash does not make those recognition histories interchangeable.

The transfer therefore requires the acquisition allocation, recoverability history and unconsumed asset version. The held measurement additionally requires treaty entry time, covered-contract mapping, claims recoverability and its own risk adjustment. Exact replay does not transfer the asset again; an unresolved treaty date does not default to timely recognition. Partial coverage or changing estimates require their stated allocations before the same procedure can be applied.

38.22 Reading acquisition costs in an actual report

Section 38.20 supplied the acquisition assets and the net inflows for two identified groups. A published insurer's accounts reveal a different level of detail. Munich Re's 2023 report shows contracts recognised for the first time during the year, separating concluded from acquired contracts and nononerous from onerous business. Note 27 covers segments predominantly using general measurement or the variable fee approach; it omits two predominantly PAA segments. These are segment disclosures, not a list of individual groups (Munich Re, 2024, printed pp. 252–254).

Consider the ERGO Life and Health Germany segment, identified in the paragraph preceding its table. Its newly recognised contracts combine GMM and VFA measurements. The following two columns reproduce its concluded-contract amounts in millions of euros. Preserve the report's signs: inflows are positive, outflows and the margin are negative. Its onerous CSM cell is blank; a dash below preserves that blank instead of assigning a reported zero (Munich Re, 2024, printed pp. 252–253).

	Nononerous	Onerous
Present value of future inflows	2,842	17
Future claims, expenses and investment components	−2,438	−12
Future acquisition costs	−243	−25
Risk adjustment	−8	0
Contractual service margin	−154	—
Reported total	0	−21

The nononerous amounts illustrate how expected future profit is deferred in CSM. The onerous column retains a loss-bearing population separately. Munich Re describes recognising an expected initial loss in profit or loss and keeping onerous contracts separate from profitable ones. Combining these columns into one net profit would erase that distinction (Munich Re, 2024, printed pp. 165–166).

Adding the displayed nononerous numbers gives −1, although the table reports zero. The onerous cash-flow and risk-adjustment rows give $17 - 12 - 25 - 0 = -20$, against a reported total of −21. The report states that amounts are rounded to millions and that totals may consequently differ. A possible set of unreported nononerous values is $2{,}842.3 - 2{,}437.8 - 242.8 - 7.8 - 153.9 = 0$; each rounds to its displayed amount. Similarly, $16.6 - 12.1 - 25.2 - 0.3 = -21$ is consistent with the onerous column if no CSM is assigned. Together these hypothetical values also reproduce

the table's outflow subtotals and aggregate rows after rounding. They demonstrate consistency under nearest-million rounding; they do not recover the issuer's unreported measurements or establish its exact rounding convention. Retain the original figures and blank cells (Munich Re, 2024, printed pp. 155, 253).

The acquisition-cost row appears within *future cash outflows*. Its reported total of $243+25 = 268$ therefore cannot simply be entered as the previous example's asset A, which arose from costs already paid before the related groups were recognised. Nor do the table's acquired-contract columns describe how sales acquisition costs were paid. Those columns concern a different distinction between contract populations. Their displayed zeros retain the same rounding qualification. The selected disclosure does not identify separate prior acquisition assets, their impairment history or their allocation to individual groups (Munich Re, 2024, printed pp. 252–253).

A third quantity appears in the segment's service revenue. For 2023, Munich Re reports €202 million as the portion of premium relating to recovery of acquisition costs. Its policy explains that acquisition costs enter insurance service expense in the same amount as the recovery included in revenue, allocated over the coverage period. This is a period service amount; the €268 million above is an estimate of future acquisition outflows for newly recognised contracts. Subtracting one from the other would not produce a remaining acquisition asset. The relevant populations, dates and movements must first be linked (Munich Re, 2024, printed pp. 177, 253, 264).

The segment's liability reconciliation makes that caution concrete: it shows €637 million in the service-expense row labelled "Recovery of acquisition costs", and €978 million of acquisition costs within cash flows. Thus the €202 million recovery in revenue does not describe the entire expense row. The difference of €435 million is an arithmetic residual; assigning its cause requires the relevant populations and expense treatments. Neither that residual nor cash less expense identifies a separate acquisition asset (Munich Re, 2024, printed pp. 230–231).

The same timing distinction applies to profit. The report gives €819 million of CSM for services provided during 2023 in this segment. That amount concerns service during the year, whereas €154 million is the margin attached to the newly recognised contracts in the selected table. It is not a release from that new-business amount alone. A reconciliation needs the opening CSM and every intervening movement. The segment's total insurance revenue also combines measurement approaches: €7,966 million under GMM/VFA and €1,976 million under PAA make the reported €9,942 million (Munich Re, 2024, printed pp. 253, 264).

Even within one insurer, an acquisition-cost policy needs its applicability conditions. Munich Re reports using the PAA option to expense acquisition costs directly for a substantial part of its property-casualty primary insurance business. The earlier general-measurement asset example therefore requires the relevant treatment and group facts before it can be applied to a real portfolio (Munich Re, 2024, printed p. 166). These are readings of the retained 2023 English translation, whose German original is binding. They identify reported aggregates and necessary further facts; the underlying group histories and valuations remain unresolved.

38.23 Following the margin through the year

Why can the segment release €819 million of CSM during 2023 when newly recognised contracts add only €154 million? The opening margin carries profit from earlier business. Munich Re's measurement-component reconciliation for ERGO Life and Health Germany begins with CSM of −10,449 million euros and ends with −9,569 million. Its signs present the margin as a negative balance: new business makes it more negative, while service release moves it towards zero. Keep that convention while reading the movements (Munich Re, 2024, printed p. 246).

The nonzero CSM movements are service provided during the year, contracts initially recognised, changes in estimates that adjust CSM, and derecognition or changes in the consolidated group. In that order, the displayed amounts give the complete bridge

$$-10{,}449 + 819 - 154 + 214 + 1 = -9{,}569.$$

In positive margin amounts, opening CSM of 10,449 receives 154 from new business, releases 819 for service, and falls by another 214 and 1 through the other two movements. The net reduction is 880. The table's CSM service subtotal of 879 already combines $819 - 154 + 214$; adding that subtotal to the individual movements would count them twice. The estimate movement here is the CSM column's 214, not a claim that every change in the underlying estimates is absorbed by CSM (Munich Re, 2024, printed p. 246).

The numbers connect three distinct disclosures. New-business CSM of 154 matches the initial-recognition table; the 819 release matches the segment's service revenue; and closing CSM of 9,569 is the balance used in the issuer's schedule of expected future releases. The latter match verifies which balance that schedule starts from, rather than the accuracy of the future releases. The comparative-year column also reconciles: $-9{,}412 + 948 - 221 - 1{,}764 = -10{,}449$, which becomes the 2023 opening amount (Munich Re, 2024, printed pp. 247, 253–254 and 264).

The larger liability totals require a different comparison. The measurement reconciliation includes the full GMM/VFA balances and the incurred-claims liabilities of PAA contracts. It excludes PAA remaining coverage. It would therefore be wrong to describe the whole table as GMM/VFA only. The footnote continued after its comparative table explicitly gives the omitted PAA remaining-coverage balance: −62 million euros in the comparative period and −1 million at the end of 2023 (Munich Re, 2024, printed pp. 241 and 247).

This scope difference accounts for the displayed opening and closing differences. The complete liability reconciliation starts at −109,556, against −109,494 in the measurement table: the difference is −62. Their closing amounts are −112,808 and −112,807, a difference of −1. Thus an apparently small missing balance is a disclosed excluded population, rather than permission to omit it or absorb it into rounding (Munich Re, 2024, printed pp. 230–231 and 246–247).

The cash rows preserve the same distinction. Premium receipts change the reported liability balance by −10,406 in the full table and −8,471 in the measurement table, leaving −1,935 attributable to the omitted PAA remaining-coverage portion at displayed precision. Acquisition cash movements of 978 and 502 leave 476 for that portion. These signs describe the liability reconciliation; they are not the debits and credits of a cash account. Nor are these differences all PAA cash flows, because PAA incurred claims are included in both tables (Munich Re, 2024, printed pp. 231, 241 and 246).

The 476 cash difference still does not explain the earlier acquisition expense difference of $637 - 202 = 435$. The two quantities differ by 41, but calling that amount a deferred acquisition asset would again invent a missing link. A cash payment, an expense and recovery included in service revenue can concern different dates and contract populations. The disclosed expensing election makes the treatment of PAA costs a relevant question; the selected tables do not establish that it alone explains 435. Preserve that unresolved attribution alongside the now-reconciled CSM and the identified PAA scope difference (Munich Re, 2024, printed pp. 166, 230–231, 246 and 264).

All these comparisons use the report's rounded euro millions. Exact addition of the CSM column does not establish unrounded measurement equality or exact conservation of every other column. The public reconciliation supplies movements and their reporting populations, while the group assignments, valuation inputs and detailed acquisition-expense allocation remain separate facts to obtain.

38.24 Adoption dates and the restatement of 2022

A standard's effective date and the economic date of a reported balance need not coincide. An insurer adopting new rules in 2023 may report a revised balance for December 2022 while retaining the earlier report as a record of what it originally disclosed. The system must preserve both versions and explain which one a calculation uses.

The Ministry of Finance's explanation dated 12 June 2025 describes the original ASBE 25 timetable. It specifies application from 1 January 2023 for enterprises listed both domestically and overseas, and for overseas-listed enterprises preparing financial statements under IFRS or ASBE. Other enterprises applying ASBE were scheduled to apply the revised standard from 1 January 2026. Its account of the 2025 notice says existing adopters, including enterprises that had adopted early under the 2020 notice, must continue applying it. Insurers facing genuine difficulties and seeking deferral must submit written reasons to the Ministry's Accounting Department and the relevant institutional supervision departments of the National Financial Regulatory Administration. That statement does not establish an automatic deferral merely because preparation is difficult (Ministry of Finance of the People's Republic of China, Accounting Department, 2025, answers on background and implementation arrangements).

The same explanation limits the new simplification choices to unlisted enterprises; matters outside those choices remain subject to ASBE 25. It names grouping, measurement, presentation and transition as affected areas, including comparative information and the transition date. These are source-supported descriptions of the Ministry's explanation. The operative 2020 and 2025 notices have not yet been directly verified in this collection, so the explanation does not supply the detailed conditions needed to implement those choices (Ministry of Finance of the People's Republic of China, Accounting Department, 2025, answer on the relationship between the notice and the standard).

Ping An provides a concrete adoption history. Its 2022 ASBE annual report states that the group would adopt the revised insurance standard for accounting years beginning on 1 January 2023. Its 2023 report confirms adoption on that date and restatement of comparative information (Ping An Insurance (Group) Company of China, Ltd., 2023, Note XIV.2, pp. 315–316) (Ping An Insurance (Group) Company of China, Ltd., 2024, Note IV.34, p. 192). The following balances all refer to the consolidated group at 31 December 2022, in RMB millions. The first column comes from the 2022 report; the second is the restated comparative in the 2023 report. Changes are calculated from those columns; the asset, liability and parent-equity changes also match the three rows of Note IV.34's reported bridge (Ping An Insurance (Group) Company of China, Ltd., 2023, pp. 146–148) (Ping An Insurance (Group) Company of China, Ltd., 2024, pp. 141–143 and 192).

Balance	Original	Restated	Change
Assets	11,137,168	11,009,940	$-127,228$
Liabilities	9,961,870	9,823,944	$-137,926$
Total equity	1,175,298	1,185,996	$10,698$
Parent shareholders' equity	858,675	869,191	$10,516$
Non-controlling interests	316,623	316,805	182

The distinction between total equity and equity attributable to the parent matters. Assets less liabilities rise by $-127,228 - (-137,926) = 10,698$. Of that increase, 10,516 belongs to parent shareholders and 182 to non-controlling interests. Treating the parent-equity row as total equity would leave an unexplained difference of 182. These are changes between reported accounting bases for the same date, not cash received in a later period.

The transition also changes how later growth is calculated. Ping An reports assets of 11,583,417 at the end of 2023. Subtracting its restated 2022 comparative gives a reported-balance increase of $11,583,417 - 11,009,940 = 573,477$. Subtracting the original 2022 amount instead gives 446,249, mixing the old and new bases and understating that comparable movement by 127,228 (Ping An Insurance (Group) Company of China, Ltd., 2024, p. 141). The first difference is a movement in reported asset balances; it is not a cash-flow measure or an explanation of every economic cause of growth.

An original-source query for assets at 31 December 2022, explicitly selecting the 2022 report, therefore returns 11,137,168. A query selecting the 2023 report's restated comparison for that date returns 11,009,940. Each assertion retains its report identity, consolidated scope, RMB-million scale, accounting basis and economic date. A link records that the later assertion restates the comparison; it does not delete the original. A request for a comparable annual movement must reject the mixed-basis pair unless an explicit reconciliation has first aligned it. Because the basis and disclosure version differ, these two amounts are not a same-context source conflict.

The information date remains a separate question. The 2023 report says its board authorised the statements on 21 March 2024 (Ping An Insurance (Group) Company of China, Ltd., 2024, Note I, p. 154). Neither that authorisation nor the 31 December reporting date establishes when the exact retained document became public. The corpus has archive-date leads but no verified publication date or historical byte vintage for either selected report. A query for information available to investors at 31 December 2023 must therefore retain an unresolved availability result for this source selection; it cannot silently import the later restated balance. The corpus's October 2026 revalidation timestamps describe local evidence handling, not investor knowledge in 2023.

Finally, Note IV.34 reports financial-asset reclassification, including use of the classification overlay for relevant assets derecognised during 2022. It also states that some insurance groups used modified-retrospective or fair-value transition because retrospective application was impracticable. The headline bridge consequently cannot be reconstructed by applying one percentage to every insurance liability. The example establishes how to retain and compare disclosed histories; reconstructing the transition itself still requires the affected groups, asset classifications and underlying measurement inputs (Ping An Insurance (Group) Company of China, Ltd., 2024, Note IV.34, p. 192).

38.25 Reconstructing balances at transition

A disclosed restatement tells us what balances an insurer reported under the new basis. It does not reveal the historical contract groups, assumptions or services needed to reproduce their opening measurement. The three transition methods address different limits on that reconstruction. The retained December 2022 AASB 17 compilation provides a useful, explicitly scoped account of those methods: its stated applicability covers annual periods beginning from 1 January 2023 to before 1 July 2026. These provisions explain a design requirement; they do not establish the rules governing Ping An's ASBE groups (Australian Accounting Standards Board, 2022g, Appendix C).

First distinguish the dates. Initial application begins the annual period in which the entity first applies the standard. Transition normally begins the immediately preceding annual period. Thus a calendar-year entity first applying the standard in 2023, with only 2022 presented as an adjusted comparative, starts its transition calculation on 1 January 2022. If it elects to present earlier adjusted comparatives, paragraph C25 instead takes transition back to the beginning of the earliest such period. Neither date establishes when the resulting report became public (Australian Accounting Standards Board, 2022g, paragraphs C2 and C25–C27).

Full retrospective application identifies and measures each group as if the standard had always applied. Acquisition-cash-flow assets are considered separately, balances that would not exist under the new rules are removed, and the resulting net difference is recognised in equity. This route needs the historical information required to reconstruct grouping, cash flows, measurement assumptions and service already provided. It is required unless application is impracticable or the specific paragraph C5A exception applies (Australian Accounting Standards Board, 2022g, paragraphs C3–C5).

Impracticability requires more than a missing historical workbook. Under AASB 108, the entity must have made every reasonable effort to apply the requirement. Relevant difficulties include effects that cannot be determined, assumptions that would have to be made about management's past intentions, and significant estimates for which the necessary contemporaneous evidence cannot be objectively distinguished from other information. Those conditions need to be established for the reconstruction at issue (Australian Accounting Standards Board, 2022a, paragraph 5).

An estimate prepared later can nevertheless describe an earlier period. The question is whether it reflects the circumstances that existed then, using evidence that would have been available when the earlier financial statements were authorised for issue. A newly assembled estimate can meet that test; replacing an earlier expectation with knowledge of what subsequently happened uses hindsight. The date of preparation alone therefore does not settle the question. Nor does authorisation establish when an outside investor could obtain the information: that requires the separate public-availability record. The insurance standard's specific transitional provisions govern their stated cases, including permitted information under the modified and fair-value routes described below (Australian Accounting Standards Board, 2022a, paragraphs 7, 19(a) and 50–53).

When full retrospection is impracticable for a group, the alternatives are modified retrospection and fair-value transition. Modified retrospection aims for the closest retrospective result possible using reasonable and supportable information available without undue cost or effort. Each specified modification is allowed only to the extent that the necessary retrospective information is missing. If the information needed even for this modified method cannot be obtained, paragraph C6(a) requires the fair-value approach. Missing history therefore does not license an unrestricted choice of present assumptions (Australian Accounting Standards Board, 2022g, paragraphs C5–C8).

For example, the modified route can reconstruct earlier expected cash flows from estimates at transition, or an earlier determinable date, adjusted for known intervening cash flows, including those of contracts that have expired. Its discount-rate approximations require the specified yield-curve evidence; its risk adjustment and allocation of past service require their own records. The source also prescribes nil amounts for particular acquisition-cash-flow items when the necessary information is unavailable. That narrow instruction cannot be generalized into a rule that every missing amount is zero (Australian Accounting Standards Board, 2022g, paragraphs C12–C16).

Fair-value transition takes a different starting point. Paragraph C20 determines the CSM or loss component from the difference between the fair value of the insurance group and its fulfilment cash flows, both measured at transition. The insurance group's fair value is distinct from the value of its underlying investment pool. Replacing one with the other would compute a different quantity. The calculation needs a supported group valuation as well as the cash-flow, discount and risk-adjustment inputs; C20 expressly excludes the demand-feature rule in AASB 13 paragraph 47. Its permitted information-date, grouping and accumulated-OCI choices also have specified conditions (Australian Accounting Standards Board, 2022g, paragraphs C20–C24B).

To interpret that group valuation, consider what would be paid to transfer the liability in an orderly transaction between market participants at the measurement date. The obligation remains outstanding, and the transferee must fulfil it; the thought experiment is not an immediate settlement with the policyholder. AASB 13 therefore calls for assumptions that market participants would use, in the accessible principal market or, in its absence, the most advantageous market. The unit being valued still comes from C20: the insurance group (Australian Accounting Standards Board, 2021c, paragraphs 2–3, 9, 13–24 and 34).

A quoted price is not required in every case. The valuation should maximise relevant observable inputs, while supported unobservable inputs can supply the participant assumptions for which market observations are unavailable. The insurer's own data can be a starting point, but require adjustment when reasonably available evidence indicates different participant assumptions or entity-specific features unavailable to other participants. A technique must also fit the circumstances and available data. These requirements make the source, date and interpretation of each valuation input consequential (Australian Accounting Standards Board, 2021c, paragraphs 61–67 and 87–89).

There is a further reason to retain separate fair-value and FCF calculations. AASB 13 includes non-performance risk, including the issuer's own credit risk, in liability fair value and assumes that risk is unchanged by the hypothetical transfer. AASB 17 expressly excludes the issuer's own non-performance risk from fulfilment cash flows for issued insurance contracts. Even when the two measurements draw on some of the same cash-flow records, their treatment of this risk differs. Neither a single imported FCF balance nor the value of the underlying investments supplies both C20 inputs (Australian Accounting Standards Board, 2021c, paragraphs 42–44) (Australian Accounting Standards Board, 2022g, paragraph 31).[1]

[1]The comparison uses AASB 108's 31 December 2022 compilation, applying to periods beginning from 1 January 2023, and AASB 13's 30 June 2021 compilation, applying from 1 July 2021. These are retained historical editions; later amendments and the actual entity's applicable requirements need their own check.

There is one important qualification to the ordinary method choice. Under C5A, a group with direct participation features may elect fair-value transition even when retrospective application is possible, if the entity chooses the B115 risk-mitigation option prospectively from transition and used qualifying mitigation before that date. The applicable instrument, qualification and designation conditions still matter. Treating every fair-value election as proof of impracticability would therefore be wrong (Australian Accounting Standards Board, 2022g, paragraphs C3(b), C5A and B115–B118).

The reconstruction must consequently retain the method and its basis for each group, each modification used, the information date and evidence supporting it, the separately measured FCF, CSM or loss component, relevant OCI and acquisition balances, and the opening-equity reconciliation. Some transition rules expressly permit transition-date information for specified assessments. That permission concerns the accounting restatement; it does not make a later report admissible in a forecast made before its publication. Detailed impracticability judgments under AASB 108 and valuation under AASB 13, together with actual group facts, remain necessary before this method description can produce a supported opening measurement.

38.25.1 Reconstructing the margin from cash and service history

The transition route must eventually produce a measured opening balance. Consider a separate nonparticipating group with three equal service periods, transition after the first, and supported zero discount rates. This example uses the retained EU 2023/1803 provisions; it does not infer EU applicability from the neighboring AASB or ASBE discussion. Assume no tax, foreign exchange, OCI, incurred unpaid claims or changes in remaining expectations. The zero rates and equal service units isolate reconstruction; they are supplied conditions, not defaults for an insurer's portfolio (European Commission, 2023a, IFRS 17.C3–C8,44,B119).

The insurer starts with capital and cash 100. Before recognition it pays acquisition costs of 10. A supported allocation assigns 6 to the recognized group and 4 to a future group, whose acquisition asset remains recoverable. Premium is 120, initial expected claims are 90 and initial risk adjustment is 6. The recognized group's initial margin is consequently

$$120 - 90 - 6 - 6 = 18.$$

That amount is CSM, not immediate profit. The acquisition payment appears once in cash; the allocated 6 reduces the margin, while the future group's 4 stays as a separate asset. These different destinations implement the acquisition and initial-measurement requirements (European Commission, 2023a, IFRS 17.28A–28F,38,C4).

Under full retrospective application, contemporaneous records show expected first-period claims of 30 and risk release of 2. Actual claims of 30 are paid. The one completed service unit releases $18/3 = 6$ of CSM, leaving 12. At transition, remaining claims 60 and risk adjustment 4 give FCF 64 and liability $64 + 12 = 76$. Cash is $100 - 10 + 120 - 30 = 180$; adding the future acquisition asset of 4 and subtracting liability gives equity 108. Equivalently, service releases and CSM revenue, less actual claims, give $30 + 2 + 6 - 30 = 8$, the increase from opening capital 100.

Suppose the prior accounting basis instead reports liability 80 and acquisition asset 10 at this same date, with the same cash 180. Prior equity is 110. Replacing those balances gives a transition equity adjustment of

$$(80 - 76) + (4 - 10) = -2.$$

There is no transition cash receipt or payment. C4(c) places the resulting net difference in equity; it does not supply a new cash flow to make the books balance (European Commission, 2023a, IFRS 17.C4).

Now change the available information, keeping the baseline cash history. Assume full retrospective application is impracticable, and that the evidence conditions permit the specified modified-retrospective calculations. C12 reconstructs initial expected claims as remaining claims 60 plus known past payments 30, giving 90. C14 reconstructs initial risk adjustment as the current 4 plus expected past release 2 supported by similar contracts issued at transition. C15(b) compares one past coverage unit with two remaining units. With the supported acquisition allocation, these inputs again yield initial CSM 18 and transition CSM 12 (European Commission, 2023a, IFRS 17.C6–C8,C12,C14–C15). The agreement follows from the supplied evidence and numbers; it is not a property of the route names.

A different actual claim payment makes the distinction visible. Suppose actual first-period claims were 35, while the full-history initial expectation remained 90 and the remaining expectation remained 60. Full retrospection retains CSM 12: the extra claim cost is past-service experience. Cash becomes 175 and equity 103. In the separately admitted modified information view, C12 instead reconstructs initial expected claims as $60 + 35 = 95$. Initial CSM is then $120 - 95 - 6 - 6 = 13$, of which $26/3$ remains. Liability is $64 + 26/3 = 218/3$, and equity is

$$175 + 4 - \frac{218}{3} = \frac{319}{3}.$$

The same cash supports different transition margins because the reconstruction uses different admissible information. Replacing the full-history expectation with the later payment would introduce hindsight; calling the modified result the recovered historical truth would overstate what its approximation supplies.

38.25.2 Prescribed missing-information treatments preserve actual cash

Missing acquisition allocation has a narrower consequence than missing cash. Return to the 30-payment case. Suppose the 10 payment is known, but reasonable and supportable information for the C14B allocation is unavailable. C14D prescribes nil for the specified group CSM adjustment and acquisition assets (European Commission, 2023a, IFRS 17.C14B–C14D). The modified initial CSM becomes $120 - 90 - 6 = 24$, leaving 16 at transition. Liability is 80 and the acquisition asset is zero, while actual cash remains 180. Equity is therefore 100. The payment was not erased; a specified transition measurement rule changed its asset and margin consequences. If the payment itself were unknown, this cash reconstruction would remain unresolved.

An onerous group brings reinsurance into the same calculation. Lower the premium to 80 and restore the supported acquisition allocation 6 and 4. Initial loss is $90 + 6 + 6 - 80 = 22$, with no CSM. For this unchanged, zero-rate group, the declared systematic allocation releases one third of the loss component with the first period's expected claim and risk releases. At transition the loss component is 44/3, contained within FCF 64. It is not added to the liability a second time. Cash is 140, so the acquisition asset of 4 gives equity $140 + 4 - 64 = 80$. C16 requires a systematic allocation under the modified route; the full route preserves the supported historical allocation (European Commission, 2023a, IFRS 17.49–52,C16).

Suppose a timely treaty covers half of all these claims. Its premium is 49, with separately supplied initial transferred risk adjustment 3 and transition risk adjustment 2. Ordinary initial cost CSM is $49-45-3 = 1$. The initial loss recovery is $22/2 = 11$, making total held cost CSM 12. After the first of three equal service periods, held cost CSM is 8. Remaining recovery 30 plus risk adjustment 2 gives held FCF 32 and total held asset 40. The remaining loss-recovery component 22/3 sits within that asset. This computation keeps ordinary treaty cost and recovery of the onerous loss as separate origins (European Commission, 2023a, IFRS 17.66–66B,B119C–B119F).

The treaty pays actual recovery 15 on the first-period claims. Combined cash is $140 - 49 + 15 = 106$, and transition equity is

$$106 + 4 + 40 - 64 = 86.$$

Under the admitted modified route, C16A gives the same remaining recovery component $(44/3)(1/2) = 22/3$. Adding it to ordinary remaining cost CSM 2/3 gives held cost CSM 8. Timely entry, covered-contract allocation and the expected claims-recovery percentage remain required findings; the numerical agreement does not establish them (European Commission, 2023a, IFRS 17.C16A–C16B).

If the specific information needed for C16A is unavailable, C16C instead requires that no loss-recovery component be identified. Suppose ordinary held valuation remains supported. Held FCF is still 32 and ordinary remaining cost CSM is 2/3, giving held asset 98/3 and equity $106 + 4 + 98/3 - 64 = 236/3$. This rule does not make the held asset zero or supply a missing ordinary valuation (European Commission, 2023a, IFRS 17.C16C).

Each calculation retains the information view, the permitted modification, known historical cash and the separate acquisition and treaty amounts. The reference exchange rejects hindsight in full retrospection and unsupported approximations in modified retrospection. General discount-curve reconstruction, direct-participation transition, changing service estimates and OCI still

need their specific procedures. These finite opening balances connect the route conditions to measurements without treating the unresolved methods as complete.

38.26 From transition measurements to opening equity

The fair-value approach produces a margin or a loss component, but those two outcomes enter the liability differently. Consider two distinct groups of issued contracts for which the grouping and eligibility for this transition method have already been established. Use the historical provisions just described and assume that both groups contain only remaining coverage. All amounts below are authored currency units at the same transition date. There are no incurred claims, reinsurance, acquisition balances, tax, accumulated OCI or other transition adjustments in this example. These simplifications isolate the effect of the two measurements; they do not supply actual insurer valuations or evidence of eligibility.

For group A, suppose the measured fulfilment cash flows are 90 and the separately measured fair value is 110. The FCF amount already includes the discounting and nonfinancial risk adjustment required by the measurement rules. Paragraph C20 makes the difference, $110 - 90 = 20$, the CSM. This is unearned profit associated with future services. Keeping it within the liability gives $90 + 20 = 110$; recording it as immediate profit would contradict that role (Australian Accounting Standards Board, 2022g, paragraphs 32–38, 40(a) and C20).

Now reverse the ordering of the inputs. For group B, let FCF be 120 and fair value 100. The difference identifies a loss component of 20, while CSM is zero. The liability is 120. Paragraphs 49–50 describe the loss component as part of the liability for remaining coverage, with its own subsequent allocation and reversal rules. Adding another 20 to the FCF balance would count that component twice. Replacing the liability by fair value, 100, would instead understate the required FCF balance (Australian Accounting Standards Board, 2022g, paragraphs 40, 47, 49–50 and C20).

Write F for measured future-service FCF, V for the group's fair value, C for CSM and K for the nonnegative loss component. Positive amounts denote liabilities. The two sign cases give the following local derivation:

$$C = \max(V - F, 0), \qquad K = \max(F - V, 0), \qquad L = F + C = \max(F, V).$$

Here L is the recognised remaining-coverage liability. The loss component K is tracked within L and is not added to it. Thus $C - K = V - F$, and at $V = F$ both components are zero. These relations apply to the stated transition case; later changes in FCF, service provision and finance effects follow the subsequent-measurement rules.

The numerical comparison makes the distinction visible:

Currency units at transition	Group A	Group B
Fulfilment cash flows	90	120
Fair value of the group	110	100
CSM	20	0
Loss component within the liability	0	20
Recognised liability	110	120

The recognised liability equals fair value for A and exceeds it for B. Calling the method fair-value transition therefore does not mean that every resulting liability is carried at fair value.

To see the opening-equity effect, suppose the old carrying liabilities are 105 for A and 115 for B. Assets of 500 remain unchanged, and these are the only liabilities. Before transition, equity is $500 - 105 - 115 = 280$. Afterwards, it is $500 - 110 - 120 = 270$. Each group's liability increases by 5, so total opening equity falls by 10. Paragraph C5 replaces the retrospective measurement requirement in C4(a); the removal of incompatible old balances and the net adjustment to equity in C4(b)–(c) still apply. The ordinary initial-recognition loss rule does not turn this transition adjustment into current-period operating expense (Australian Accounting Standards Board, 2022g, paragraphs C4–C5).

Illustrative component entries show the same reconciliation. For A, debit the old liability 105 and opening equity 5, then credit the FCF liability 90 and the CSM liability 20. For B, debit the old liability 115 and opening equity 5, then credit the FCF liability 120. Record B's loss component of 20 within that 120 for its subsequent allocation; it produces no additional credit. Neither entry changes cash. An actual transition would also need the acquisition, reinsurance, tax and OCI effects excluded here.

The order of calculation matters as much as the signs. If an implementation first combined the two already distinct groups, it would find total FCF of 210 and total fair value of 210. Applying the sign rule to those totals would produce neither CSM nor a loss component and would report a liability of 210. The correct group-by-group calculation gives 230. Pooling would therefore understate liability and overstate equity by 20. The programme must retain the admitted group boundary while calculating CSM and loss components, then aggregate the resulting balances. Whether different grouping is itself permitted requires the separate grouping rules; an aggregation shortcut cannot decide that accounting question.

The example supplies the arithmetic once the group's two measurements and old balances are known. A real transition disclosure may reveal only some of those quantities. The later Munich Re case shows both what can be read from an insurer's report and what still requires contract and valuation records.

38.27 The recoverable part of a transition loss

Group B's loss component of 20 describes the issued contracts. If the insurer also holds reinsurance, how much of that loss is expected to be recovered? Extend the preceding example by adding a separately recognised, qualifying group of reinsurance contracts held. Assume it covers all contracts in B and only B, with remaining coverage and no incurred claims, and uses general measurement. This changes the earlier no-reinsurance assumption. The issued obligation to policyholders remains 120, with its loss component of 20; purchasing reinsurance does not extinguish it (Australian Accounting Standards Board, 2022g, paragraphs 60–64 and 75).

The transition calculation has a specific timing exception. Ordinarily, paragraph B119C permits the recovery-income treatment in paragraph 66A only if the reinsurance contract was entered into before or when the onerous underlying contracts were recognised. Suppose the underlying contracts' recognition date, applying these rules, is 1 July 2020, the treaty was entered into on 1 October 2020 and both coverages remain in force at transition on 1 January 2022. The treaty does not meet that ordinary condition for the original underlying recognition. Paragraph C20A nevertheless expressly removes B119C's condition when determining the loss-recovery component at transition. The exception does not remove the treaty's other recognition, grouping, boundary or measurement requirements (Australian Accounting Standards Board, 2022g, paragraphs 61–66B, B119C and C20A).

Let q be the percentage of claims on the underlying contracts that the insurer expects to recover from this reinsurance group, expressed as a fraction. Assume $q = 0.60$ for this conditional illustration. A nominal treaty quota alone does not establish that expectation: coverage, limits, exclusions and collectability matter. The full reinsurance measurement also includes issuer non-performance risk and the nonfinancial risk transferred, under paragraphs 63–64. With $K = 20$ still denoting B's loss component, C20A gives the reinsurance loss-recovery component R:

$$R = Kq = 20 \times 0.60 = 12.$$

This 12 belongs to the reinsurance asset for remaining coverage. It is not the issued liability, the amount of expected claims, or a valuation of the whole reinsurance asset. If only some contracts in the underlying group were covered, the required allocation of the losses would also have to be established; the all-covered assumption avoids that extra calculation here (Australian Accounting Standards Board, 2022g, paragraphs 63–64, 66B, B119E and C20A).

The complete asset still needs the treaty cash flows, discounting, transferred risk and net cost or gain of purchasing reinsurance. The reinsurance CSM reflects that net cost or gain; held reinsurance is not classified as onerous under paragraph 68. Applying the issued-group CSM and loss formulas to the held contract would therefore be wrong. Presentation likewise keeps the issued and held-reinsurance portfolios separate (Australian Accounting Standards Board, 2022g, paragraphs 65–68 and 78–79).

To extend the opening balance sheet, supply the missing total explicitly. Assume a separate complete measurement gives a reinsurance asset of 50, already including the loss-recovery component of 12, against an old asset of 45. These are additional conditional inputs; the multiplication above does not derive them. Other assets remain 500 and the two issued liabilities still increase from 220 to 230. Total assets therefore rise from $500 + 45 = 545$ to $500 + 50 = 550$, while equity moves from $545 - 220 = 325$ to $550 - 230 = 320$. The asset increase adds 5 to equity against the issued-liability increase of 10, giving a net reduction of 5. An illustrative reinsurance entry debits the new asset 50, credits the old asset 45 and credits opening equity 5. Adding another asset of 12 would count a component of the supplied 50 twice and overstate equity by 12. The net transition change belongs in equity under C4(b)–(c), rather than being inferred as current recovery income from the ordinary recognition rule (Australian Accounting Standards Board, 2022g, paragraphs C4–C5 and C20A).

The component also imposes a useful later check. If B's loss component falls to 14 and the recoverable portion remains 60%, paragraph B119F limits the reinsurance loss-recovery component to $14 \times 0.60 = 8.4$. Leaving it at 12 would violate that bound. The bound alone neither selects a unique new component nor determines the full reinsurance asset or period's profit; the actual changes and allocation rules still matter. Similarly, $q = 0$ gives no loss-recovery component and $q = 1$ gives 20, without erasing the issued obligation or determining the entire asset (Australian Accounting Standards Board, 2022g, paragraphs 66–67 and B119F).

38.28 What an insurer's transition disclosure reveals

Munich Re's 2023 report shows how much a public disclosure can resolve. Its consolidated statements apply IFRS as adopted by the European Union, together with the stated German commercial-law requirements. The first application of IFRS 17 is 1 January 2023, with transition at 1 January 2022 and restated insurance comparatives. Financial-instrument comparatives generally remain under IAS 39, subject to the disclosed classification overlay. A single year label therefore does not imply a single measurement basis for every line in the comparative balance sheet (Munich Re, 2024, pp. 155 and 183–184).[2]

The report identifies full retrospection for certain property-casualty contracts and parts of life reinsurance. Where historical information is insufficient in life reinsurance, it mostly uses fair-value transition; it also uses that method for specified property-casualty reinsurance groups and German primary business. Modified retrospection and fair value together account for over 90% of net insurance liabilities at transition. That is a statement about a liability denominator at a particular date. It supplies neither a count of contracts nor the assignment and supporting history of every group (Munich Re, 2024, p. 183).

The opening-equity bridge makes the scale of the different effects visible. The reported margin component is much larger than the net equity reduction because the other measurement changes also enter the reconciliation. The following amounts reproduce the report's signs and billion-euro scale (Munich Re, 2024, p. 184).

Opening-equity reconciliation at 1 January 2022	EUR billions
Equity under IFRS 4	30.9
Modified measurement basis	25.3
Contractual service margin	−22.3
Risk adjustment for non-financial risk	−5.6
Loss component	−1.4
Deferred taxes	1.5
Reported change in equity	−2.5
Equity under IFRS 17	28.5

The five displayed change components sum to −2.5 billion, whereas the displayed endpoints differ by −2.4 billion. The report warns that rounding can cause such deviations; these figures are compatible under rounding to the displayed precision. The representation should retain both the reported endpoints and the bridge, with their precision, instead of changing one to force an exact identity (Munich Re, 2024, pp. 1, 155 and 184).

[2]The retained English report states that it is an informational translation and the German original is binding (p. 1). The discussion attributes the disclosed policies and amounts; the historical AASB texts above are comparison sources.

The separate margin tables also clarify which contracts are being followed. At 1 January 2022 the reported issued-contract CSM is −22,688 million and the held-reinsurance net cost/gain is 368 million. Adding those reported totals gives −22,320 million, consistent with the rounded −22.3 billion margin component above. The issued-contract transition table follows groups in force at transition, and its CSM amounts concern the general measurement model and VFA; the associated revenue also includes PAA contracts. Its later balances therefore cannot be substituted for the whole subsequent portfolio or used as if CSM and revenue had identical populations (Munich Re, 2024, pp. 210–211 and 254–255).

Finally, disclosure of a method does not supply all its inputs. Munich Re reports a 6% cost-of-capital rate for its non-financial risk adjustment, applied to projected capital requirements with diversification and discounting. Its equivalent one-year confidence level is about 90% at transition, compared with 88% for 2023 and 87% for the comparative period. These are dated properties of the insurer's calculation, not universal IFRS parameters. The rate also cannot be imported into the separate fair-value calculation merely because that discussion mentions cost of capital. Reconstructing an individual group still requires its cash flows, capital projections, curves, allocations and supported transition-method choice (Munich Re, 2024, pp. 165, 183 and 185).

38.29 Interfaces, exclusions and acceptance cases

Insurance events are group_recognize, premium_settle, service_release, claim_incur, claim_settle, assumption_change, onerous_change and treaty_change, each bound through the event-basis, authorization and commit contracts to the exact group/version, evidence and information view. Journals and component roll-forwards must reconcile independently; a component reallocation is not always a net liability change. Invalid grouping, missing rate/law, unsupported option valuation, ambiguous contract boundary or unknown service units produces the corresponding unresolved/outside-scope result. Failed numerical paths retain their coverage/mass, without successful-only normalization. Modified contracts need the applicable derecognition analysis; no new ID can erase unfavorable old-group obligations.

Initial transition requires an actual method choice and eligibility for full retrospective, modified-retrospective or fair-value transition under AASB 17 Appendix C, with acquisition, grouping, past service and historical-rate records. This collection does not claim an executed transition or a general VFA valuation. The nonzero-rate general case and onerous allocation through expiry above are finite specification examples. Required later acceptance still includes a priced guarantee, acquisition/reinsurance interaction, transition and actuarial backtesting. These retained obligations prevent the two-year illustration from being described as full insurance-domain readiness.

Chapter 39

Implementation contracts

39.1 Reference implementation and component boundaries

The proposed reference implementation uses Python for orchestration and a deterministic reference evaluator, exact decimal/rational arithmetic for booked amounts, SQLite for transactional metadata and versioned records, and columnar files for large scenario outputs. DuckDB is an optional analytical reader. These are convenience choices justified by inspectability and modest initial scale; they do not establish production performance. The IR and conformance tests allow a later compiled executor without changing accounting meaning. Committed journals, economic deltas, receipts and branch heads belong to the one transactional store, including for simulation branches. Columnar files hold derived analytical exports, never independently writable committed books.

```
accdsl/
  evidence/       documents, spans, filing contexts, extraction versions
  model/          typed IR, entities, books, symbols, contracts
  language/       grammar, parser, resolver, type checker, lowering
  constraints/    propagation, solver adapters, bounds and conflict cores
  accounting/     rule selection, posting validation, closing, consolidation
  economics/      contract and behavioural plugin interfaces
  execution/      scheduler, branch state, stochastic streams, run records
  reporting/      statements, notes, movement schedules, explanations
  storage/        versioned records, migrations, result readers
  interfaces/     Python API, CLI, structured import/export
  packages/       reviewed demonstration and accounting modules
```

The first service boundary is local and synchronous. A job abstraction can be added for extraction and simulation without requiring distributed execution. The model compiler returns an immutable execution plan; the executor receives that plan and run inputs. Neither should retrieve live market data implicitly. Input datasets and rule packages are selected and versioned before execution.

39.2 Persistent records

The table identifies logical storage relations and their obligations; exact versioned record schemas determine their payloads. Candidate calculations and diagnostics remain separate from the committed accounting relations.

Relation	Key information and constraints
`document`	Hash, original path/URI, filing ID, entity, publication and availability times, media type, revision links. Original bytes are retained.
`source_span`	Document hash, page/element locator, bounding box, text/table representation, extraction run ID.
`observation`	Context ID, value or missingness tag, precision domain, source-span IDs, semantic-reading ID, review status.
`context`	Entity/perimeter, concept, basis, currency/unit, instant or interval, dimensions and taxonomy version. Contexts are immutable.
`assertion_version`	Logical assertion ID, payload hash, economic validity, ingestion time, public availability, superseded version and reason.
`model_node`	IR node ID, type, serialised payload, dependency IDs, package/model version; no executable arbitrary text.
`constraint`	Expression, scope, source/assumption references, supported solver fragment, applicable alternative condition.
`alternative`	Parent model, added conditions, rationale, evidence, coverage declaration and rejection history.
`economic_event`	Stable entity/book/branch/business-event key; versioned event, contract/cohort, dates, quantities and dependencies.
`branch`	Branch key and immutable definition reference, including entity, book and required synthetic or real book mode.
`branch_head`	Branch owner, current state reference and generation; updated only by the expected-head transaction.
`posting_authorization`	Exact basis, posting profile, method, authority, findings, permission and historical view. A permission record alone does not establish a committed event.
`journal_batch`	Committed balanced postings with event key, book/basis, date, method and measurement references. Candidate journals are separate; membership here requires the complete accounting transaction.
`economic_delta`	Committed contractual and physical state change, bound to the same basis and authorization as its journal.
`committed_receipt`	Unique stable event key, full request fingerprint, original request, opening/closing state, journal and delta references. Published atomically with those records and the advanced head.
`review_view`	Immutable parent view, decisions and dependency selection revisions; preserves historical interpretation.
`dependency`	Typed source and derived-record edges, including candidate-set membership needed for transitive invalidation.
`search_checkpoint`	Semantic manifest, partition, explored/unresolved regions, frontier, used budget and result references, committed together.

Relation	Key information and constraints
`run`	Evidence and model hashes, cutoff, scenarios, branch selection, weight model, solver/numerical settings, seeds and environment.
`report_cell`	Run, branch/scenario, context, amount or result family, dependency references and completeness status.

Persistent identifiers are opaque; display labels can change without changing identity. References use foreign keys or validated content-addressed links. The full logical key of a reported fact includes its dimensions, not merely company, label and year. Duplicate assertions are stored as distinct source claims until a documented duplicate-resolution operation relates them.

Version records are append-only. A derived transaction-time interval can be closed by the next version event without deleting the original payload. Validity intervals can overlap when assertions conflict. Such overlap is diagnosed and resolved by the evidence view, not prohibited by a database constraint that would erase one assertion. Scenario states use structural sharing of immutable historical nodes and branch-local changes.

Large simulation output is indexed by run, explanation, parameter draw, scenario, innovation path, entity, book, date and account. Reducing this to means across all dimensions would destroy the product's central feature. Online summaries may compress draws within a declared distribution, while retaining conditional summaries and representative replay identifiers. A retention policy that removes individual traces must state which explanations can still be reconstructed.

39.3 Public API and result types

The following unversioned names describe the user workflow. They are conceptual shorthand, not the portable operation registry. The exact versioned requests are specified below; Python, command-line and any service binding must exchange the same typed values:

```
ingest(manifest, extractor_config) -> EvidenceVersion
compile(source_or_ir, evidence_view, package_lock) -> ModelPlan | Diagnostics
reconcile(plan, cutoff, solver_config) -> HistoryFamily | Diagnostics
condition(family, assumptions) -> ConditionalFamily | Infeasible
simulate(family, scenario_set, run_config) -> RunResult
render(run_id, report_request) -> ReportBundle
explain(node_id, evidence_view_or_run_id) -> ExplanationGraph
compare(run_a, run_b, alignment_spec) -> Comparison
```

`HistoryFamily` contains a model-scope declaration, shared variables, constraints, finite choices, feasibility status and solver-completeness status. It may contain witness histories without claiming that those witnesses enumerate the family. `RunResult` is a tagged union of concrete outputs, conditional samples, certified bounds, explored ranges and unresolved outputs. Probability-bearing results reference a probability-model record. Results without that reference cannot be presented as probabilities.

Comparison requires an alignment specification: entity/perimeter, dates, reporting basis, scenarios and shared innovations where applicable. Comparing IFRS and US GAAP books holds economic events fixed and changes accounting rules. Comparing management responses holds the specified scenario and initial information fixed and changes the behavioural model. These are different questions and must not be combined into an unexplained difference.

39.4 Versioned requests and independent results

The workflow names above become portable operations only when their arguments, input versions and effects are fixed. Each request names an exact operation, profile and signature; an immutable view; typed arguments; execution settings; and the expected head for a mutation. The body is closed:

```
OperationRequest {
  operation, request_id, workspace_key,
  profile_ref, signature_ref, view_ref, arguments,
  execution: {mode, budget, environment_ref},
  expected_head: null | {owner_key, state_ref, generation}
}
```

All fields are present. The mode distinguishes a specification example from a runtime request. Budget states operation-count and elapsed-time limits, or a reason either is inapplicable. The expected head appears only in this envelope. For accounting its owner is the entity/book/branch; for review it is the review namespace; for probability selection it is the family/evidence/time owner. Resolving a mutable "current" alias happens before constructing the request, and a mutation compares the resolved revision again.

The fingerprint covers the complete request, including its request identifier, expected generation and execution settings. In accounting the lookup namespace is still the business-event tuple in Section 34.5, not request ID. Changing the request ID for a committed event changes the fingerprint and conflicts. Source and review mutations use their own operation/workspace/request namespace. A transport correlation identifier and an economic event are not interchangeable identities.

The result preserves the submission even when decoding fails. Once exact-byte retention succeeds, it has the following fields:

```
OperationResult {
  submission_ref, request_ref, operation, input_view_ref,
  disposition, statuses, payload, diagnostics, coverage,
  unresolved, continuation_ref, commit_observation,
  receipt_ref, evidence_mode
}
```

The validated request reference remains null until all required input checks, including bindings and authorization, pass. Rejected input has null payload. If submission retention itself fails, the caller receives a transport failure with its original correlation, not a fabricated stored submission or a known commit outcome. A read can inspect an earlier committed receipt while its own commit observation says no new commit occurred.

Completed, partial, rejected, unsupported, cancelled, internal failure, unknown commit and replay are operation dispositions. They do not replace the separate structural, mathematical, computational, empirical, accounting, forecast, product and human statuses. Computing a zero normalizer can complete while conditioning on that event is undefined. A valid symbolic family can have only partly explored regions. A supported insolvency scenario can be fully evaluated while its company defaults. These distinctions prevent an infrastructure error from erasing an economically important path.

For example, let an admitted law assign mass 3/5 to a path with cash 120 and 2/5 to an unavailable cash calculation. The output retains both regions and their masses. If separate evidence bounds the unavailable cash by $[80, 110]$, the unconditional mean is bounded by

$$\tfrac{3}{5}(120) + \tfrac{2}{5}[80, 110] = [104, 116].$$

Without those bounds, even this mean interval is unsupported. Reporting a mean of 120 would silently condition on computational success.

Portable-only validation proceeds through decoding, field shape, reference integrity, cross-record binding and time context, capability and admission, calculation, transaction comparison and publication. Native interfaces preserve their own explicit stage order; the native accounting sequence checks recognizable dependencies before shape and source versions before bindings. At the earliest failed stage, return every independently decidable diagnostic in canonical field order. A malformed parent prevents its dependent checks, but does not hide unrelated sibling errors. Later stages are marked unevaluated.

Appendix A gives the exact operation names and argument meanings. Financial method names such as delivery or claim settlement are versioned members of a package; they are not invented new platform endpoints. In particular, continuous inference and causal methods need their own admitted extension signatures before the finite/exogenous initial profile can execute them.

39.5 Correction changes a view without rewriting its history

A correction can affect a result that never explicitly selected the corrected source. Suppose a query selected one available report because it was the only eligible candidate. Admitting a second report may create a conflict. The query therefore depends on candidate-set membership as well as on its selected record. Invalidation must follow both dependencies transitively.

Each correction creates an immutable new view with parent and changed decisions. Old frozen outputs retain their original views. A current query must check ancestor dependencies even for descendants created after an invalidation scan; otherwise delayed computation could revive a stale reading. Unrelated closures remain usable. Self-supersession, incompatible replacement scope and a missing required predecessor reject before mutation.

The mathematical correction and the audit of affected work are separate records. A `CorrectionBasis` depends on the unaffected base law, replaced reading, corrected source, target admission and replacement method. An `InvalidationReport` additionally depends on the inspected store snapshot and lists affected descendants. Computing another optional descendant changes the report, not the corrected law. This also keeps the content-reference graph acyclic.

Probability calculation, admission and selection are three effects. A selected law belongs to the owner consisting of family, target evidence view, target time and purpose. Correcting a source does not mutate the old owner's selected pointer or automatically select the new candidate posterior. In the new owner, an inherited stale selection returns selection superseded with an explicit readmission and reselection obligation. With no inherited selection it returns selection required. A historical query naming the old view remains valid in that frozen scope.

39.6 Durability and migration have observable obligations

One logical transactional store owns records, dependency edges, review views, branch heads, journals, economic deltas, receipts and search checkpoints. Original documents reside in immutable byte storage. Their exact bytes must be durable and verified before a metadata transaction refers to them. A crash between blob creation and metadata commit may leave an unreferenced blob; it must not leave an accepted reference to absent bytes. Analytical tables, caches and PDFs derive from a named snapshot and do not become separate writable versions of truth.

The proposed local store serializes commits even when several clients submit requests. A single

writer still needs expected-head comparisons. An event-log implementation is also possible if its commit marker, projection visibility and replay obey the same observable rules. Neither a document trace nor a choice of SQLite establishes physical-media durability or multiwriter throughput. Those require crash, reopen and fault-injection evidence in the selected environment.

A search checkpoint commits the semantic manifest, partition identity, explored and unresolved regions, frontier, used budget and result references together. A cell is not complete before its result and frontier transition are durable. Resume names that exact checkpoint and manifest. Increasing a scheduling budget can continue it; changing evidence, methods, scenario, family or traversal meaning requires a new manifest and an explicit reuse relation. Cancellation preserves the last committed frontier.

Export first stages a closed manifest, verifies every required record and blob, then publishes the manifest atomically. An incomplete export remains explicitly incomplete. Import verifies required bytes before committing metadata and grants no new financial admission. A failed report export cannot undo a journal or justify posting it again.

A representation-only migration retains old and new bytes and IDs plus a field/meaning mapping. Its input names the old and new schemas, migration method, source snapshot and declared semantic relation; the common request envelope supplies the expected head for its transaction. It must preserve typed quantities, context, uncertainty category, reference closure, admission, event identity and supported reference query results. Changing a rounding policy, calendar, recognition rule or probability law is a new model, not a format migration. A failed comparison leaves old heads and objects usable.

Exact reruns compare values on the same evaluated domain. An elapsed-time cap can visit different search cells on different machines, so equal partial coverage cannot be presumed. Reproducible prefix comparisons need a pinned traversal and operation-count budget; resume uses the saved frontier. Approximate methods use their declared financial error comparison, and random streams name economic owner, date, shock kind and draw index independently of branch order.

39.7 Publication effects and dependencies between profiles

An insurer can publish a valuation assessment without changing its accounts. A consolidation method can append a worksheet adjustment without posting to a subsidiary's legal book. A reporting request can publish a new report while leaving every financial balance unchanged. An implementation must distinguish these actions before it decides whether a later method has the state it needs. The retained component profiles make those distinctions differently, so a common interface cannot infer them from a field name alone.

39.7.1 What a successful invocation changes

`PublicationEffects/1` describes an invocation under one exact profile and schema lock. It retains the request, submission, result and original receipt, and compares the complete declared worlds before and after that invocation. Its four substantive effects are publication of immutable records and their decision index; changes to component heads; changes to legal-account heads; and appended journals with their native scope. The complete changed values and unchanged ownership domains remain available in the description.

For example, a successful bank assessment and a successful insurance assessment both publish a query result and receipt, yet their native Booleans differ. Neither changes an accounting head or appends an accounting journal. A group component's adjustment journal belongs to the

Native profile	Meaning of `additional_commit`
InsuranceComponents/1	A new component result or report was published.
GroupComponents/1	A new component result or report was published; the field is absent when a rejected result has a null payload.
BankAssessment/1	A new query result was published.
InsuranceAssessment/1	A financial commitment occurred; it remains false for the retained successful query publications.

Table 39.2: The same Boolean name does not denote the same effect across profiles.

consolidation worksheet. Its legal heads remain unchanged even when its successor component state refers to later immutable legal-book snapshots. Reading a later snapshot and changing the book are separate operations.

The description checks the newly added decision against its exact request, result and receipt. Previously published records and decisions cannot disappear or be overwritten. New records must equal the publication's dependency closure; an appended journal must equal the bound native result's journal. The closed world shape prevents an unfamiliar ownership field from being silently omitted. These checks establish the stated documentary effects. Recognition, measurement, applicable authority and actual source eligibility still require their own evidence.

Native submission encoding also matters. InsuranceComponents hashes the request body, whereas the other three profiles hash the complete immutable request record. Group and bank submissions bind a decoder and wire blob rather than containing a request-reference field. The comparison preserves these bindings instead of rewriting all four submissions into an invented common encoding.

39.7.2 Identity, replay and missing acknowledgement

Request identity is a pair within an exact profile namespace: a client or workspace identifier and a request identifier. The retained InsuranceComponents/1 index incorrectly joins its two allowed strings with a slash. The distinct pairs `("a/b","c")` and `("a","b/c")` both become `a/b/c`. Both pairs satisfy the native field schema, so that index is not faithful to its declared identity. The defect can falsely conflict distinct requests; it does not establish that a retained financial amount is wrong or that a duplicate posting occurred.

A separate metadata successor encodes the ordered pair as a canonical JSON array and binds the index to the exact profile. Decoding recovers both strings, so equality of the encoded arrays implies equality of the pair. Explicit conversion retains the old request, result and receipt references and refuses conflicting, missing or ambiguous decision history. It cannot reconstruct a decision already erased by an old collision without independent retained evidence. A missing lookup result therefore grants no permission to post. The execution-policy successor in Section 39.7.3 adopts this identity repair in complete financial requests while preserving their original bytes and the existing financial profile.

A replay describes the effects of the present invocation. It returns the original receipt and preserves the original diagnostics, coverage and result, while creating no additional publication or financial change. Treating the earlier receipt's journal as a new journal would double count the transaction. A refusal likewise changes no world and supplies no success receipt.

When acknowledgement is lost, the client cannot tell whether publication occurred. The description reports unknown publication knowledge and leaves the effects unknown. A conformance fixture may separately contain a committed world and an uncommitted world; neither hidden world becomes client knowledge. An exact retry publishes once in the uncommitted case and

replays the original receipt in the committed case. The retained cases check both possibilities without claiming that a physical store was exercised.

39.7.3 Changing the index while preserving an insurer's posting

An acquisition payment of 8 creates an acquisition asset of 8 and reduces cash by the same amount. A later recoverability assessment reduces that asset by 3 and recognises an expense of 3. Suppose the first request has identity `("a/b","c")` and the second has identity `("a","b/c")`. The old slash key falsely conflicts the second request, even when it supplies the correct opening owners and information view. Repairing the key must admit that second request while ensuring that a retry of the first never pays the acquisition cost again.

`InsuranceComponentsIdentity/1` supplies an explicit execution policy bound to the exact `InsuranceComponents/1` financial profile. Its identity record locks both the existing financial schemas and contract and the new identity schema and contract. The financial profile, operation names, original request bytes, immutable outputs and receipts retain their identities. Changing the financial profile reference inside an old request would create a different request; it would not constitute a retry of the old payment. Separating the execution policy makes the intended change precise.

Conversion adds the policy reference to the declared world and replaces each decision-map key with its canonical ordered pair. Every owner head, journal, published financial record, original decision reference, current view and unrelated ownership field remains unchanged. Each supplied historical decision must reproduce its original complete publication. Known missing, conflicting or ambiguous histories refuse conversion. The order of entries in a JSON map does not establish posting chronology: the converter checks the exact journal entries and their multiplicities against retained decisions, then preserves the original journal sequence. It never derives a new sequence from map order.

The new dispatcher checks the execution-policy binding and exact submission bytes before examining a prior decision. An identical request returns its original output, receipt, diagnostics and publication list with no additional commit, even after the current view and owner heads advance. Changed bytes under that same pair conflict. A fresh pair still has to pass the original full read set, evidence, authority, lineage and financial checks. The new identity format therefore removes a false conflict without weakening financial eligibility.

In the two-event example, the successor retains the original acquisition and impairment results and both journals. Each original request replays without another posting. Recovery also reproduces the original publication when the acknowledgement was lost before commitment, and returns the original receipt when commitment already occurred. The broader comparison preserves all 672 retained exchanges, including 84 publications, and converts 127 distinct before-and-after worlds. All 686 saved successor exchanges reconstruct. The checks also refuse a newly hashed but financially wrong proposal, duplicate journals and ambiguous complete publication histories.

These are finite documentary comparisons using the retained financial evaluator. They establish the checked protocol correspondence, not physical store durability or a new independent implementation of accounting. Conversion remains conditional on the supplied history being complete: an erased decision with no surviving independent evidence cannot be discovered from the index. The evidence and exact two-expression protocol change are retained under `data/accounting-master/insurance-identity-adoption-20261026/`. The policy does not translate these component owners into an insurance-assessment head; that receiving state still requires a separately justified adapter.

39.7.4 Declaring a dependency without inventing a state adapter

`RetainedProfileComposition/1` provides a finite text and typed description of dependencies between imported native exchanges. Each stage names a packet by its exact content hash and selects an exchange identifier. The resulting description preserves its profile, request, result, receipt, information view, native signature when present and complete publication effects. It contains no financial formula or arbitrary execution callback.

A sequence dependency requires the same exact profile and equality of the complete predecessor-after and successor-before worlds. A state dependency also names explicit paths to the produced and consumed native state references. Those references must agree, and the receiving request must actually declare the consumed head. Thus matching a closing cash amount cannot conceal a changed owner, information view, journal history or decision index. A receipt dependency requires the original receipt to occur in the receiving request's immutable dependency closure; that reference alone conveys no accounting authority.

The text form declares a composition name, ordered stages and dependencies. Its stage values are an identifier, packet hash and exchange identifier; its dependency values are source stage, destination stage, relation and two explicit head paths when the relation is a state handoff. All values are data. The typed form carries the same closed fields, and source-line locations are retained separately from semantic identity. The checked text and typed routes produce identical native findings. Existing annual-program authoring and declared-method requirements keep their own evaluation rules; this layer does not replace them.

Consider passing an insurance component's closing accounts to an insurance assessment. The former is an `OwnerState` containing a ten-account balance map and component history in a scenario-specific context. The latter expects an `AccountingHead` with a balance digest in a different entity, authority and information context. Both examples use USD, but that agreement supplies neither an ownership mapping nor a complete accounting-state conversion. The direct handoff is unsupported. An adapter would need explicit entity and book matching, balance and history mapping, temporal and source admission, applicable authority, and checked native request/result bindings.

Unknown acknowledgement makes dependent stages unresolved, including later stages whose immediately adjacent worlds otherwise match. A partial query can publish evidence without supplying a completed financial successor. A refused stage cannot supply a dependent input. The comparison preserves all these findings even when another defect determines the overall status. Duplicate publication decisions, missing imports, cycles, undeclared intervening changes and incompatible head references are refused. Stages with no declared edge are merely independent in the description; they are not certified for concurrent execution.

39.7.5 Reading an explicitly adopted profile

A corrected request index is useful only if the next interface understands which policy produced it. The original common reader expects the old insurer world and its slash key. Removing the new policy reference to satisfy that reader would hide a relevant execution condition. The exact-time bank profile presents a different issue: its request and result schemas change, while its submission schema remains the original one. A shared schema prefix cannot describe that deliberately mixed set of versions.

`PublicationEffects/2` retains the existing financial-effect meanings and adds an explicit execution binding. A provider names the exact financial profile, each record kind's schema, the identity policy when one applies, and the checked source packet and selected exchange. It invokes the corresponding native validity checks before describing the effects. The insurer provider preserves

the policy reference in both worlds and uses the ordered pair key. The bank provider preserves its exact-time rules and the original submission schema. Neither infers financial commitment from the Boolean alone.

Source identity also matters within a packet. Several insurer recovery examples have the same display label. Selecting them through a map keyed only by that label would silently discard earlier examples. The provider therefore uses the exact packet hash and a stable position-qualified identifier, while retaining the original label for explanation. This distinguishes every retained example without changing its request, receipt or financial state.

The composition language keeps its existing declarations and adds one explicit dependency check. Equal financial profiles with different execution policies require a policy adapter before their complete worlds can form a sequence. Financial identity still belongs to the financial profile and request pair: changing the policy does not create a second namespace in which the same payment may be posted again. Different financial profiles retain their separate state-adapter requirement. An unknown acknowledgement continues to make every dependent stage unresolved, including stages whose adjacent worlds happen to match.

The checked successor retains all 77 earlier complete text/typed composition results. Its 959 effect descriptions cover 672 retained insurer exchanges, 14 new exchanges, 168 unknown transport observations and 105 descriptions from 63 exact-bank packages. Ninety-one further compositions include the native insurer commit-to-report chains, bank publication/replay pairs and explicit incompatible, partial, refused and unknown cases. All saved descriptions and assessments recompute through fresh readers. A wrong policy, hidden head change, duplicate posting on replay or invented uniform bank schema is refused.

These comparisons preserve checked protocol meanings across the named providers. They remain conditional on the native financial evaluators and supplied facts; they do not supply a universal dispatcher or the missing insurer-to-assessment state conversion. The exact source changes, complete descriptions and dependency cases are retained under `data/accounting-master/adopted-profile-effects-20261026/`.

39.7.6 Checked examples and remaining scope

The effect comparison covers the retained 1,892 exchanges from the four profiles in Table 39.2, including 412 possible crash worlds and their retries. Semantic counterexamples change native flags, receipts, protected heads, journals, prior decisions and acknowledgement knowledge; changed records are rehashed so that integrity checking alone cannot hide a missing binding check. These are deterministic documentary comparisons, not observations of a deployed transaction service.

The composition examples include 66 retained native commit-to-report sequences and matched text/typed incompatible, partial, unknown and refused cases. No receiving request in the selected packets consumes a receipt, so the receipt relation has a specified rule and negative checks but no checked positive handoff. General method-input authoring, cross-profile adaptation, whole-company source admission and independent implementation acceptance remain unfinished.

The exact contracts, source locks, examples and verification manifests are retained under `data/accounting-master/`:

```
component-identity-repair-20261026
profile-publication-effects-20261026
profile-composition-boundary-20261026
```

Their separate evidence lets an implementation adopt a checked distinction without mistaking

a finite reference case for the completed simulator.

39.8 Plugin interfaces

Every plugin declares a version, supported types, input information dates, determinism, numerical measure and capabilities. A contract plugin implements `schedule`, `observe`, `transition` and `payoff`; it may return future conditional events. A behaviour plugin implements `decide` and declares which state and scenario information it uses. A measurement plugin implements `measure` and returns an amount or vector with a basis and numerical status. A reporting plugin maps classified journals and balances to output contexts.

The plugin response must include all inputs that materially affect results. An external calculator's global evaluation date, market-curve cache, calendar version or random seed cannot remain invisible. For example, QuantLib's floating coupon interface makes fixing-date conventions and pricing objects relevant inputs; the adapter records them rather than relying on process-global state. ACTUS risk-factor observations receive explicit dates and identifiers.

The constraint interface accepts a typed formula and a requested operation: feasibility, witness, enumeration of declared finite choices, or bound. It returns `sat`, `unsat` or `unknown`, plus evidence appropriate to the operation. Timeouts and unsupported theories return `unknown`. The history layer exposes that solver outcome as `SolverIncomplete`, retains the affected region and marks its coverage unresolved. These are layer-specific descriptions of incomplete computation, not infeasibility. A witness proves existence within the formula, not completeness of a recovered historical family.

39.9 Human review and model-assisted extraction

The review interface should put the source passage beside the proposed observation, context and reconciliation impact. An analyst can accept a mapping, add an alternative, reject an extraction, define a symbolic unknown or request more evidence. These actions create structured changes with reasons. The useful unit of review is a coherent disclosure and its links, not an isolated number detached from its note.

A language model can identify candidate tables, draft concept mappings, propose contract terms and suggest alternative readings. Its output must reference source spans and a schema. Deterministic checks reject fabricated source references, invalid units, inconsistent periods and unbalanced postings. A model-generated interpretation that passes those checks remains a proposal until its evidentiary status is established. Token probabilities and self-reported confidence are never automatically used as weights over economic histories.

For routine local research use, a lightweight interface and immutable source versions are sufficient. Multi-user approval hierarchies, cryptographic attestations and deployment security systems are outside this initial design. They can be added for a concrete operational requirement without changing the scientific representation.

39.10 Failure recovery and reproducibility

Extraction jobs and simulation jobs write explicit progress records and atomic results. Restarting a failed document conversion reuses its document hash and creates either the same idempotent extraction result or a new version under changed settings. Restarting a simulation resumes

only from a committed state whose model and input hashes match the run. Numerical failures retain the inputs needed for a focused diagnostic.

A run manifest records code revision when available, package lock, source hashes, cutoff rule, model hash, scenario data, priors or weights, seed/stream scheme, solver settings, decimal and rounding conventions, calendars, environment and result locations. Absence of a usable Git revision is recorded as unavailable with content hashes, rather than inventing a commit.

39.11 Review decisions have consequences for later results

A proposed reading starts in `Proposed`. Acceptance records a reviewer, date, reason, verified locator and context, together with the reconciliation result. Acceptance means that the stated layer has been reviewed: accepting an extraction of a company's sentence does not accept its accounting treatment. Competing defensible readings enter `AcceptedAlternative` under a choice group. Rejection preserves the source and reason. A correction creates a new reading; replacing an accepted reading requires an explicit `Supersede` action. A prior reading therefore remains reproducible without entering the latest model.

Suppose OCR reads Northstar's original 120 as 720. The source span supports 120; correcting extraction creates a new reading linked to the erroneous one. The later revenue correction from 120 to 110 is different: it changes the company's reported assertion. The original extraction remains faithful to its document. Finally, a reviewer may accept that two notes conflict without accepting either as a unique model constraint. The view then contains a conflict set requiring alternatives or further evidence.

A changed reading or mapping invalidates descendant constraints, families, measurements, reports and cached runs for a new view. Unaffected dependencies can be reused. Old frozen runs retain their original decisions and evidence IDs; they are marked stale for current use, not overwritten. A journal derived under the old view is never silently attached to a new source. The local workflow needs no multi-user approval hierarchy to enforce these rules.

39.12 Identity must survive harmless reordering

The portable serialisation in Section 33.3 preserves decoded Unicode code points without global NFC normalisation. It uses sorted ASCII keys and exact rational financial quantities, while integer metadata remains integer. Only schema-declared sets are sorted. Source identity hashes original bytes. The old Core 0 encoding survives only for its historical fixtures through an explicit legacy import; it does not define new portable digests. Neither encoding proves arbitrary economic equivalence.

SHA-256 with a domain/version prefix identifies three different closures. Evidence identity includes source bytes, locations, extraction and review versions. Model identity includes typed IR, assumptions, rules, calculator locks and resolution, while excluding formatting and source byte offsets. Run identity adds the evidence view, horizon, scenarios, outputs, calendars, numerical policy, random-stream scheme and environment. A corrected source can change evidence and run identity without changing the numerical model; a new rate assumption changes the model as well.

Each economic business event has a stable identity within its entity, book and branch. Its lookup key is the four-tuple in Section 34.5; event version, payload, expected head and request settings belong to the full fingerprint. Exact replay returns the original receipt before a new-event freshness check. Changed inputs under that key conflict. A new run label or version cannot

make the event post again. Independent events may reorder only when read/write sets are disjoint or a package proves commutation, including funding constraints. Alphabetical ordering cannot settle an economic timing ambiguity.

Stochastic keys contain seed, coupling group, entity, contract/cohort, economic date, shock kind and draw index. Branch enumeration order is absent. Shared coupling groups may use the same uniform draw with different marginal transforms. The reference fixture maps the first 64 bits of a SHA-256 key digest, interpreted as unsigned big-endian integer v, to $(v + 1/2)/2^{64}$. This fixes example replay; it does not validate a production Monte Carlo generator. Research runs select an assessed counter-based generator and record transforms and numerical environment. Inner measurement draws have different shock domains from outer realised defaults.

Schema migration preserves old and new payloads and a field/meaning mapping. A representation-only migration must demonstrate equivalent supported semantics; a changed recognition rule creates a new model. Restart resumes only a committed state whose locks and identities match. SQLite owns source, review, accounting and run metadata transactions; content-addressed files hold immutable bytes and JSON, while DuckDB reads versioned analytical exports. Each mutable transition has one owner.

39.13 Adapters exchange typed records

The portable interfaces described in this chapter govern new exchanges. The original `spec/core-contract.md` and `spec/core-ir.schema.json` retain the historical Core 0 examples. A named decoder and comparison link those examples to the portable records; their older fields do not silently extend a new closed signature. The source package includes those two historical files beside `spec/core-0.lark`. Their fixture and executable dependencies retain repository-relative paths; `spec/REFERENCE-ROOTS.md` gives the exact resolution and the independently relocated Core 0 component. The selected term-program component has its own versioned source and dependency manifest. Neither component supplies a migration between these distinct profiles or completes the remaining language interfaces. Every method returns a success record or a diagnostic with code, source IDs, entity/period, dependency path and retry status. The principal sequence is `ingest`, `review`, `compile`, `reconstruct`, `simulate`, `report`, with `bound`, `explain` and `replay` over frozen identities.

Arelle supplies candidate facts with QName, entity, context, period, dimensions, unit, accuracy, nil status and locator. Duplicates survive for review. Its archived `ModelInstanceObject` interface exposes these properties; validation success does not establish economic truth. Docling supplies page/table proposals, locations and conversion status. Missing headers and failed pages remain unresolved. Z3 receives named rational-linear assertions and finite choices; witnesses are checked against exact inputs and unknown results retain their reasons. The reference checks exercise Z3; the other selected adapters remain interface designs grounded in inspected official source snapshots.

ACTUS receives contract terms, calendars and dated risk factors and returns schedules, transitions and payoffs. AccDSL applies accounting separately. QuantLib receives an explicit evaluation date, instrument, curves, fixings, calendar/day-count and measure; it returns dated measurements with numerical status. No process-global valuation date may leak between branches. Beancount exports concrete journals for later differential checks; it does not own symbolic families. Import failures retain request ID and source hash so retries are idempotent. Licence/version evidence remains in the software ledger; integration acceptance requires pinned releases or source digests.

39.14 A successful extraction can still lose the fact

Consider an Inline XBRL amount whose literal is `1,234`, whose admitted numeric transform treats the comma as grouping, whose scale is -3, and whose context and unit identify a particular dollar observation. The resulting amount is $1234 \times 10^{-3} = 1.234$. Scaling that normalized value again gives 0.001234 and is wrong. An adapter must retain the literal, transform, scale, context, unit, nil status and original occurrence so that this error can be found.

The same principle applies to dates and dimensions. A lexical instant of 31 December remains distinct from the exclusive 1 January boundary used for an interval. Publication and ingestion are separate timestamps. Two equal numbers with different contexts remain separate occurrences. A nil fact is unknown, not zero. Dropping a typed dimension or context reference yields a preservation failure for that fact even if the resulting Markdown looks clear.

A PDF can preserve characters that the reader never sees. The retained HSBC 2023 annual report has a cover reading “Annual Report and Accounts 2023”, but both PyMuPDF and Poppler also return 2024 in the extracted cover text (HSBC Holdings plc, 2023, cover, physical PDF p. 1). Inspection of the drawing order explains the discrepancy: the earlier 2024 text is covered by an opaque dark rectangle, after which the 2023 text is painted. This is an observed difference between extracted characters and the visible title. It does not establish an accounting contradiction or explain why the file was constructed that way. Agreement between two text extractors would not resolve it.

The proposed exchange keeps the original PDF, the unchanged raw extraction, the page locator and a render from the same bytes. The annotation’s `literal` retains both years; separate `readings` record the raw extraction and the proposed visible title. A linked inspection record identifies the extractor versions and options, rendered image, relevant drawing order and reviewer. Thus a raw-text search still finds 2024, while the inspected cover supports a visible report-year reading of 2023. Rejecting 2024 as the visible report year must not delete its extracted occurrence. The current example has author inspection; independent adjudication remains pending.

If an earlier review had already accepted 2024 as the report year, the correction would require a new reading and an explicit `Supersede` decision. Results depending on that reading would need revalidation in the new evidence view; frozen earlier results would retain their original evidence. This follows the review rules already stated above. Even after the visible title is resolved, the cover year supplies no publication or availability timestamp for these particular bytes. An as-of query must therefore retain the unresolved availability condition. This single-page case motivates preservation and review behavior; it supplies no extraction-accuracy ranking or general visibility test for other pages.

The inspected Arelle interfaces support native XBRL contexts, dimensions, transformations and validation. Its internal object IDs do not replace stable source identity. The inspected Docling source offers useful layout and table proposals, but its examined XBRL graph path does not retain every required fact/context/unit/typed-dimension field. That path needs a preservation sidecar or must report unsupported conversion for affected records. These are source audit findings, not measured accuracy rankings of the tools.

A returned layout conversion can also conceal missing pages. At the pinned Docling implementation, the timeout branch assigns `PARTIAL_SUCCESS` when pages were expected, even if none finished. The public `convert_all` interface permits that status through `raises_on_error=True` if the remaining stages return normally. Thus absence of an exception cannot establish that a three-page request produced any extracted content. The assembly code can restore missing page numbers with empty page objects; counting those objects does not count successful processing (Docling contributors, 2026). These are source-derived failure cases, not observed failures on

Apple's filing.

The proposed exchange therefore keeps the original page inventory, requested range, successfully processed pages, empty placeholders and structured errors separate. If only page two was requested and processed, that is success on the declared subset; pages one and three remain outside the conversion claim. If the three-page request times out before any page finishes, all three remain unprocessed despite the partial-success status. Retain the source and the unfinished pages for another attempt. A deadline is also distinct from stopped execution: the inspected shutdown code waits for worker threads for a bounded time and can report that a thread is still alive. Later execution must confirm termination within its isolated process before treating the attempt as stopped (Docling contributors, 2026).

A requested processing stage may also remain unperformed. In the same Docling source, automatic OCR selects among engines according to platform, installed packages, languages and runtime availability. If it selects no engine and has recorded no language-specific rejection, it logs a warning; the subsequent call returns the pages without OCR. Language-specific rejection can instead produce an exception. Thus `do_ocr=True` records a request, not evidence that text recognition occurred. The proposed adapter records the requested mode, selected engine and model identity, and processing outcome for each page. A page that requires OCR but did not receive it remains unprocessed, even if other stages produced layout regions (Docling contributors, 2026).

Identifying the engine still leaves its model to be fixed. The inspected default layout preset names Heron at revision `main`, while the TableFormer downloader names the model repository's `v2.3.0` tag. Neither name supplies the hashes of the weights and preprocessing files actually used. The proposed adapter binds those files, configurations and dependency versions before use. Disabling remote inference services also does not disable model acquisition: the layout helper downloads when no artifact directory is supplied. With a supplied directory it requires the model's subfolder, but finding that folder does not check its contents against the requested revision. Local file verification and later isolated execution must therefore establish the intended offline behavior; a source setting alone cannot do so (Docling contributors, 2026).

The text supplied to table recognition also needs an identity. The retained `docling-ibm-models` 4.0.2 consumer uses OTSL cell and span tokens, converts them to HTML, and merges text from separately supplied PDF cells (Docling IBM Models contributors, 2026). This is a later representation than the original paper's HTML sequence. The Docling report explicitly describes a subsequent token-language refinement (Auer et al., 2024, Section 3); neither that statement nor a model-card score identifies the later files with the paper's experimental checkpoints.

The proposed adapter therefore retains the source page and crop, the supplied text cells and their locations, the predicted table cells, and the assignments and structural changes made during matching. If a correction combines columns or adds a cell for unmatched text, a reviewer must be able to recover the earlier structure and the text that prompted the change. Text obtained from an OCR engine also retains that engine's identity and outcome. A resulting table remains a proposal until its cell content, context and accounting interpretation have been checked; a well-formed HTML table alone cannot settle those questions.

Token syntax and cell geometry require separate checks too. In the inspected author version of OTSL, Lysak et al. (2023, Sections 4.1–4.3) use five tokens: C starts a cell, L joins the cell to its left, U joins the cell above, X joins both, and NL ends a row. Neighbor restrictions and equal row lengths allow invalid tokens to be detected during generation. The authors explicitly distinguish a syntactically valid sequence from a correct prediction.

Even those local rules need a further condition to ensure rectangular merged cells. Consider two rows, `C L NL` followed by `U C NL`. The L has C to its left, the U has C above, and the first-row,

first-column and equal-length rules are satisfied; no X occurs. Yet the first cell joins the top-left, top-right and bottom-left positions, leaving the bottom-right position to the second C. It is L-shaped. Representing it with both a two-row and a two-column HTML span would also occupy the second cell's position. This finite example shows why the six printed rules alone do not establish a nonoverlapping partition into rectangular cells. The proposed adapter checks that each merged cell occupies a complete rectangle, that cells do not overlap, and that every declared grid position belongs to a cell. Passing those checks still leaves correspondence with the source table to be established.

The later decoder needs its own verification. In the retained 4.0.2 source, a correction described as applying to the first row uses a row counter that is initialized to zero and never updated. It therefore also rewrites an affected token on later rows (Docling IBM Models contributors, 2026). This is a specific source-code defect relative to that stated scope; its frequency and effect on actual model predictions have not been measured. Neither the paper's syntax argument nor these partial corrections establish that a deployed converter enforces all the proposed checks.

The retained Apple annual filing makes the dependency problem concrete. Its issuer archive includes the instance, the issuer's schema and its calculation, definition, label and presentation linkbases. Those files also refer to US GAAP and SEC Reporting Taxonomy definitions, SEC schemas and XBRL definitions held elsewhere. The issuer archive alone therefore cannot supply every definition needed to interpret its tagged facts. For this particular archive, a static inspection now resolves the recorded imports, link references and typed-domain references to retained files, checks target namespaces and fragment identifiers, and checks that instance facts have declared concepts, contexts and units. This establishes a source dependency map. XML Schema and XBRL validation, the issuer's interpretation of each concept, and the completeness of the wider disclosure package remain separate questions.

An offline parser needs more than a map of intended filenames. At the inspected Arelle source version, the cache lookup can return a local pathname when the file is absent; subsequent loading records an `IOerror`. The proposed adapter must check each mapped file and its hash before loading, retain the missing dependency and affected records, and return an unresolved-dependency result. It must not infer successful retrieval from a nonempty pathname or silently substitute a newer taxonomy. Original schemas remain unchanged; an external resolver or isolated cache associates their declared addresses with the retained bytes. Any built-in cache fallback must belong to the same checked source set (Arelle contributors, 2026c).

Nor is a successful API return necessarily a validation verdict. In the same source version, the controller's return value takes accumulated validation errors into account only when `validationExitCode` is enabled. The proposed adapter sets it explicitly, requests validation, disables automatic formula execution, and examines the returned status, diagnostic messages and model errors together. It keeps the model open while copying the necessary facts, contexts, units and original occurrence references, then closes the session. One session runs at a time in its owning thread within a process; the inspected API uses shared package and plugin state. These are source-grounded interface requirements for a later implementation, not results from an executed Arelle adapter (Arelle contributors, 2026c).

39.14.1 An inline value is already a transformed reading

Return to the source spelling `1,234`. At the inspected Arelle version, an inline fact's `value` property first obtains assembled text and applies the selected transformation. In the ordinary non-nil numeric path, it then converts that result to a decimal number, applies the declared scale and applies the sign. Both `textValue` and `stringValue` delegate to this transformed value. Reading any of those three properties and applying scale again would therefore change the

reported amount (Arelle contributors, 2026b).

For a concrete constructed case, select `numdotdecimal` from the transformation registry dated 31 July 2011, scale −3 and a minus sign. The inspected function accepts `1,234` and removes the grouping comma, giving 1234. Scaling gives 1.234; applying the sign gives −1.234. If a consumer treats that returned value as the original literal and reapplies both scale and sign, it obtains +0.001234. The difference is a consumer error relative to the selected source path. The format's namespace and function name matter: the dispatcher selects a registry by namespace and can also invoke a custom transformation. The example establishes no equivalence among those choices (Arelle contributors, 2026b).

Nor does `rawValue` recover the original source spelling in every case. The inspected property assembles inner text, omits inline excluded content, can follow nonnumeric continuation links, and collapses XML whitespace when a format is present. An original spelling containing surrounding spaces and a line break can consequently yield the same `1,234` reading. Escaped content follows an additional markup-generation path with URI handling; one inspected branch can modify an object element's `codebase` attribute in the parsed tree. Original retained bytes and occurrence locators are therefore needed even when assembled text or a later tree serialization is available (Arelle contributors, 2026b).

Consider an unescaped nonnumeric fact containing text A, an excluded child containing X, and text B after that child, with a continuation containing C. Under the stated traversal conditions, its assembled text is ABC. The excluded X remains part of the source document, while B survives as the child's tail text. The helper follows continuation links already attached by the loader; it does not resolve the source's continuation identifier itself. If the link is absent, a nonempty AB can still be returned. The loader separately records missing-target, wrong-target and cycle diagnostics. Nonempty text alone cannot establish that the disclosure was completely assembled (Arelle contributors, 2026b).

The proposed adapter thus preserves the original bytes and nodes, the ordered text fragments and exclusions, the continuation identifiers and outcomes, and the assembled and transformed readings. It records transform identity, scale, sign and validation state so that each step can be reconstructed. Computing a value is distinct from validating it: the inspected property initializes an unvalidated state on first evaluation, and failures can set an invalid value and raise an exception. Nil facts take a separate branch and remain distinct from numeric zero. These source-derived requirements and constructed examples do not replace schema checks, an executed adapter or review of an actual issuer's facts (Arelle contributors, 2026b).

39.14.2 Equal amounts still have separate source occurrences

Suppose an issuer repeats one tagged amount in a statement and a note. A query may need a single amount, but an auditor still needs both locations and the evidence that they agree. These are different requirements. The proposed exchange preserves each occurrence before constructing a view that groups facts for a particular question. The inspected Arelle methods make the reason for that separation precise: displaying a value, comparing two values, grouping facts with matching aspects and removing repeated facts perform different operations (Arelle contributors, 2026a).

A displayed value is particularly unsuitable as the retained amount. The `effectiveValue` property produces a view: it rounds and formats numbers, represents nil with a string, and can return an error description. Even a convenient numeric property needs scrutiny. The inspected `vEqValue` converts numeric text to a binary floating-point number. The distinct integers $2^{53} = 9{,}007{,}199{,}254{,}740{,}992$ and $2^{53} + 1$ then have the same floating-point representation. This constructed example does not describe an issuer error; it shows why that conversion cannot

establish exact identity. The adapter therefore keeps the source literal, the typed value and the displayed reading as distinct fields, with the transformation between them recorded (Arelle contributors, 2026a).

The same caution applies to a method whose name contains *equal*. In the inspected `isVEqualTo`, the ordinary numeric comparison checks context and unit and follows the selected rounding mode, but the method does not itself compare concept names. Its nil branch returns whether the other fact is nil before comparing contexts. A caller needing facts about the same accounting quantity must establish that condition separately. Conversely, the item branch of `isDuplicateOf` checks concept, parent, context and numeric unit without comparing the item values. It can identify candidates for investigating a disagreement; its result does not resolve that disagreement. The tuple branch has additional child comparisons and must not be described as the same test (Arelle contributors, 2026a).

The dedicated duplicate-fact module supplies more explicit grouping. It checks concept, context and unit after forming hash buckets, and includes language for the relevant text-fact types. Its value tests nevertheless remain specific to the operation. Consider two hypothetical valid numeric occurrences with the same concept, context, unit and declared decimals of zero, whose value strings are `1` and `1.0`. In the inspected code, the ordinary value-equality key retains those strings, so the all-complete classification distinguishes them. The complete-subset reduction instead keys on declared decimals and the typed value; decimal one equals decimal one, so it retains the first occurrence. Thus even within this module, the classification and reduction do not share one universal equality key. The source strings and membership of a reduced view must remain recoverable (Arelle contributors, 2026a).

Reported accuracy creates a second distinction. For a finite declared decimal accuracy d, the inspected non-truncating interval helper assigns a numeric value v the inclusive bounds $v - \frac{1}{2}10^{-d}$ and $v + \frac{1}{2}10^{-d}$. These are the helper's comparison bounds, not a confidence interval for the true economic amount. Values zero and one, both declared to zero decimals, have bounds $[-0.5, 0.5]$ and $[0.5, 1.5]$. They touch. The interval option of `isVEqualTo` therefore admits overlap under its other required conditions. The duplicate-set consistency check rejects the pair because unequal typed values with the same declared decimal accuracy fail its additional condition. Substituting one test for the other would change the question being answered (Arelle contributors, 2026a).

Pairwise agreement also differs from agreement of the whole set. Take three constructed observations of the same quantity: 100 at zero decimals, and 99.6 and 100.4 at one decimal. The first has bounds $[99.5, 100.5]$; the other two have bounds $[99.55, 99.65]$ and $[100.35, 100.45]$. Each precise observation overlaps the coarse one, but the precise observations disagree and the three intervals have no common point. The inspected consistent-pairs reduction can remove the coarse occurrence while leaving both precise occurrences. The consistent-set path instead reports the inconsistent set and falls back to complete-subset reduction. Neither operation establishes which precise observation the accounting analysis should accept (Arelle contributors, 2026a).

These paths require their own admission checks. The initial grouping considers validation state and the availability of a concept, type and context; a singleton bucket is not itself a validity verdict. The numeric consistency path also expects an inferred decimal accuracy from its helper, which returns no value when the fact's decimals attribute is absent. An adapter must not silently reinterpret an unsupported or unvalidated case as an ordinary consistent numeric fact. It retains the declared decimals and precision, validation state, nil state, concept, full context and unit, together with the particular comparison operation and its options (Arelle contributors, 2026a).

Finally, the inspected deduplicated-instance export removes fact nodes from the model's underlying tree before saving. The controller performs it at the end and warns that subsequent

use of the modified model can be unreliable. The proposed exchange therefore copies every original occurrence and its diagnostics before constructing any optional reduced view. A view records its input members and selection rule; it does not replace the source record. This preserves disagreement for review while allowing a query to avoid counting an accepted repeated amount twice. These are requirements derived from pinned source inspection and independently checked constructed arithmetic; an executed adapter and actual-filing validation remain to be established (Arelle contributors, 2026a).

39.14.3 The same balance can survive different source identities

Apple's following-quarter filing provides an actual counterpart to the constructed examples. Its balance sheet repeats the 28 September 2024 comparative balances already examined in the company case. The retained annual HTML is a converted presentation without inline tags, so it cannot supply the proposed inline trace. Instead, the selected comparison follows the next-quarter inline comparative amounts to the retained annual XBRL instance. This compares reports of the same endpoint across two filings; it does not show that one file was extracted from the other (Apple Inc., 2025c, 2024c,b).

In the interim filing, the comparative total-assets occurrence has identifier `f-135`, value text `364,980`, scale 6 and declared decimals −6. Its format names `num-dot-decimal` in the 12 February 2020 transformation registry. The selected transform removes the grouping comma and canonicalizes the numeric string; scaling gives $364{,}980 \times 10^6 = 364{,}980{,}000{,}000$ US dollars. That reported amount agrees with annual-instance occurrence `f-169`. Scale converts the displayed millions to dollars. Declared decimals retain the source's million-dollar reporting accuracy; adding zeros to the string does not reveal an exact unrounded asset value (Apple Inc., 2025c, 2024c; Arelle contributors, 2026b).

The context identifiers differ: the interim uses `c-18`, while the annual instance uses `c-21`. Comparing the identifiers would reject a valid candidate match. Inspecting their contents instead shows the same CIK entity identifier and scheme, the same 28 September 2024 instant and no segment, scenario or dimensional additions in these selected contexts. The expanded concept name, USD unit and declared accuracy also agree. This is a checked comparison of these simple contexts, not a general context-equivalence algorithm or schema-validation result (Apple Inc., 2025c, 2024c).

The liabilities and equity comparisons use the same conditions and give 308,030 and 56,950 million dollars. Their reported totals reconcile: $308{,}030 + 56{,}950 = 364{,}980$. Equity, however, appears twice in the interim: `f-173` is the comparative balance-sheet amount and `f-176` is the following period's beginning equity. Both carry 56,950 million dollars under the same selected concept, context and unit. The annual instance also retains two matching occurrences, `f-207` and `f-256`. Amount and context matching therefore identifies a group, not a unique pairing of source locations. Adding the two interim occurrences would produce 113,900 million dollars and would double the reported equity balance. A balance query uses one accepted amount while preserving both locations and their statement roles (Apple Inc., 2025c, printed pp. 3–4); the annual counterparts are retained in the instance (Apple Inc., 2024c).

The accumulated other comprehensive loss exposes a different failure. Interim occurrence `f-171` contains numeric text `7,172`, with scale 6 and an explicit minus-sign attribute. Its reported value is consequently −7,172,000,000 dollars, matching annual occurrence `f-205`. Reading the digits alone loses the sign; reading the surrounding parentheses and then applying the sign attribute again would reverse it twice. The source text, markup attribute and displayed accounting reading must remain distinguishable (Apple Inc., 2025c, printed p. 3); the signed annual value is retained in the instance (Apple Inc., 2024c).

The bounded inventory retains all 24 interim and 32 annual-instance occurrences of the four selected concepts, then selects five occurrences from each source at the unsegmented September endpoint. Other dates and dimensional populations remain visible as outside this comparison. All selected reported amounts agree under the stated structural and arithmetic checks. The agreement supplies an actual preservation example; it establishes neither historical availability of the retrieved bytes nor schema validity, complete filing coverage or an executed adapter. Those distinct questions remain part of the wider acceptance work (Apple Inc., 2025c, 2024c).

Each adapter therefore pins code, dependency, model and taxonomy versions, licenses, input and output schemas, transformations and field-loss behavior. Preparation preserves bytes and receipts before parsing, inventories the package, attaches occurrences and contexts, records normalization, checks required fields independently and submits candidates for review. Partial conversion retains both successful units and the exact failures or unexplored frontier. Conversion counts cannot accept a package or an accounting treatment.

The proposed extraction comparison uses the same retained files, roots, budgets and taxonomy resources. Manual located source units and transparent native extraction provide the reference; applicable pdfplumber/Arelle configurations, plain Docling and explicit context/footnote repairs form the candidate ladder. Transcription, context, table structure, mapping, source support, retention, historical admission and cost are separate outcomes. Lost hashes or locators, look-ahead, silent omissions and untracked downloads veto the affected exchange. The current 31 development annotations cannot establish a heldout accuracy ranking, and cell counts within one report are not independent observations.

Reuse also has distribution conditions. Arelle's Apache and Docling's MIT code licenses do not by themselves clear every model, taxonomy or dataset. ACTUS technical-specification reuse retains its share-alike obligations; QuantLib, Z3 and ledger tools retain their own code and dependency terms. The Beancount and hledger comparison needs an explicit distribution boundary for copyleft components. License admission records actual versions, notices, models and data, rather than treating a repository's headline license as universal.

EasyOCR makes the dependency question concrete. Version 1.7.2 requires `python-bidi` without a version constraint and imports its `get_display` function when the OCR module loads. Although the call reorders Arabic recognition results, selecting English does not remove the import or package requirement. In the inspected `python-bidi` 0.6.11 source, this import selects a Rust-backed implementation; the older Python implementation has a different import path. Its own source states LGPLv3-or-later terms. The inspected release is a candidate for dependency analysis, and its source-level call signature alone does not establish compatibility with an installed OCR configuration (Jaided AI, 2024; Kriheli and contributors, 2026).

The same package shows why an accompanying notice still needs checking. Its third-party list names older PyO3 and `unicode-bidi` versions than its bundled Cargo lockfile. The exact locked sources retain MIT-or-Apache terms, so the version discrepancy does not demonstrate a licence conflict. Other locked packages add conditions: `target-lexicon` includes the LLVM exception to Apache 2.0, while generated `unicode-ident` tables require Unicode data-file terms alongside the chosen MIT or Apache licence (Kriheli and contributors, 2026; PyO3, Servo, Rust and dependency contributors, 2026). The adapter must therefore bind the selected build and retain each applicable notice. Verified source archives help prepare that decision; actual wheel contents, model rights and the intended distribution still need separate assessment.

The actual taxonomy notices show why that distinction changes the design. The inspected FASB notices permit specified financial reporting and unchanged incorporation in works that explain or assist implementation, subject to their notice and third-party conditions. They do not provide a general permission to edit taxonomy definitions. Keeping the original files and recording location mappings separately respects that distinction. The notices retrieved for this

inspection are unversioned; their relationship to the terms attached to the 2024 files still needs to be established before a distribution decision. XBRL's public-document notice separately directs software and schemas to a Software Notice, so permission for an explanatory document cannot simply be applied to a schema bundle. Exact dependency licences, component-specific terms and the intended distribution remain explicit preparation tasks (Financial Accounting Foundation, 2026b,a; XBRL International, 2026).

39.14.4 Choosing the first import path

The first optional importer uses the retained Arelle source (Arelle contributors, 2026c) for native XBRL, with the Apple 2024 instance as its concrete preparation case. Native facts already carry contexts, units and occurrence relationships that a flattened table cannot reconstruct. The selection therefore follows the preservation requirements established above. It selects a source and an interface for the first implementation slice; the source commit is not an installed release or a tested environment. The complete commit and body hashes are retained in the source-selection manifest.

Use the previously inspected session lifecycle with no optional plugins or additional package groups: request validation and its error-sensitive return, disable formula execution, and copy facts and their contexts before closing the session. Each owning process admits one session at a time. For this particular Apple archive, the recorded map has 27 resources and 3,551 reference edges. Resolving those static references does not establish XBRL validation or closure of every financial note.

The deterministic PDF comparator is the retained pdfplumber source, used for character and page geometry on machine-generated documents. Its output remains a proposal to be checked against the located page. A page that needs unadmitted OCR remains unresolved, with its original bytes and manual inspection route preserved. Docling, TableFormer and OCR are separate candidate paths. They do not become an automatic fallback that silently changes the selected method, model terms or resource budget.

There are three distinct forms of reproducibility here. A source selection identifies the code and technical evidence. A resolved environment identifies the actual platform, Python build, every direct and transitive package version and hash, enabled extras, model and taxonomy bytes, and applicable notices. An executed compatibility check establishes what those installed components actually do together. The present selection establishes the first. Arelle's 15 base requirement ranges and three build requirements remain an inventory; they are not an exact transitive lock. Before runtime, resolve that inventory for the selected platform and inspect the resulting compatibility and terms. Any changed dependency, transformation registry, taxonomy or configuration creates a new identity and invalidates dependent admission.

Offline access is exact URI-to-bytes resolution. An absent file, changed hash, wrong namespace or missing required fragment returns the affected dependency as unresolved before accepting its financial values. A cached pathname is insufficient. Original schemas remain unchanged, and the isolated cache or external resolver may provide only admitted bytes. Substituting a newer taxonomy would change the interpretation problem.

Terms also attach to different things: code, dependencies, issuer documents, taxonomies and model weights. The earlier source-specific dispositions remain binding evidence limits. In particular, an unversioned FASB page displaying 2027 does not establish the terms applicable to a retained 2024 package; the missing component-specific XBRL Software Notice remains a source dependency. Unresolved model-weight terms constrain the corresponding model path. They do not resolve or negate the separate native-XBRL source decision. No distribution or blanket legal clearance follows from this selection.

Eleven complete documentary exchanges test the proposed boundary without executing a converter. Every synthetic source body, context, unit and occurrence span is supplied. Raw "1,234" under an admitted grouping transform and scale −3 produces 1.234; precision metadata does not scale it again. A nil occurrence remains unknown. Equal values in different contexts stay separate. Unknown transforms, missing contexts or typed-dimension values, absent resources and changed hashes retain unresolved records instead of guessed facts.

A cancelled case retains its completed proposal and unprocessed frontier. Repeating an identical request preserves its proposal receipt; changing bytes under the same identity is a conflict. All results remain proposed or unresolved, and none claims package completeness, review acceptance or an accounting commit. These are authored expected exchanges with independently checked transformations. Their closed documentary envelope is a verification format for the proposal; it does not alter the portable product schemas or establish runtime translation into them. That translation and the exact environment remain prerequisites of the later implementation.

39.14.5 The source declarations constrain the first environment

The two selected source trees already impose a useful joint constraint. At retained pdfplumber commit `4c64b92d5caccd71c645e98e0fabb0c4dba7ff45`, its version file identifies 0.11.10. Its setup script reads requirements for pdfminer.six 20260107, Pillow at least 12.2.0 and pypdfium2 at least 5.9.0 (Singer-Vine and contributors, 2026). The selected Arelle tree requires Pillow at least 10 but below 13. The shared declared interval is therefore Pillow 12.2.0 or later, below 13. This is a necessary compatibility condition; an actual resolved and tested environment has not yet been established.

The apparent Python minimum also changes when the dependency is considered. pdfplumber's setup metadata says at least 3.8, while the pinned pdfminer release requires at least 3.10. Arelle likewise requires at least 3.10. The combined declarations therefore exclude Python 3.8 and 3.9 even before checking the remaining packages and platform (Arelle contributors, 2026c; pdfminer.six contributors, 2026). These statements come from the retained setup, requirement and release metadata; they do not recommend a latest release or assert that every later Python version works.

pdfminer adds charset-normalizer and cryptography. Its optional image dependency on Pillow is separate from pdfplumber's unconditional Pillow requirement. No extra is selected by this proposal. Likewise, a package with no declared Python dependency can still contain native libraries and third-party notices. The inspected pypdfium2 metadata expressly distinguishes its own licences from dependency licences (pypdfium2 contributors, 2026). An empty Python dependency field cannot certify the native runtime or its distribution terms.

Before admitting an environment, choose its actual platform and Python build, resolve all selected direct and transitive versions, retain their distribution hashes and notices, and test the exact source-preserving exchange. The source declarations make that task concrete. They leave the installation, native compatibility and component-specific terms as separate evidence obligations.

Chapter 40

Acceptance and implementation sequence

The reference cases now make several of the acceptance obligations executable as design checks. Northstar provides full statements and a known compatible history; Apple supplies real source contexts and a bounded marginal stress; bank, insurance, consolidation and inventory fixtures test the core's broader state requirements. They do not constitute a production implementation.

40.1 Three different questions require different evidence

Engineering correctness asks whether the implementation follows the specified types, transitions and arithmetic. Accounting validity asks whether the selected rules faithfully implement the applicable standards and disclosed policies. Scientific validity asks whether the economic assumptions and uncertainty representation support the intended forecasts or comparisons. A balanced journal answers only part of the first question.

The implementation must therefore maintain separate validation results. Compilation and deterministic examples cannot establish forecasting quality. Agreement with a public report cannot identify an undisclosed contract term. A lower extraction error on a document benchmark cannot establish that the engine preserved every materially different scenario response.

40.2 Core conformance cases

Test	Required case	Pass condition
T01	Exact debit/credit batches, including reversals and period closing	Every committed book conserves signed balances; imbalance rejected atomically.
T02	Money, quantities, currencies, instants and interval flows	Invalid additions and context combinations rejected; explicit conversions traced.
T03	Original filing, later correction and as-known cutoff	Earlier query reproduces the original evidence view; later view shows revision.

Test	Required case	Pass condition
T04	Annual and interim cumulative flows with revised comparatives	Quarter derivation only across compatible contexts and vintages.
T05	Fixed/floating borrowing with identical historical postings	Both explanations survive; next-year expense is 5 and 7 under the stated shock.
T06	Continuous fixed share $\alpha \in [0, 1]$	Exact model returns $I = 7 - 2\alpha$ and extrema 5, 7 without assigning probabilities.
T07	Duplicated encoding of one history	Canonical duplicates preserve results and declared weights; no extra probability mass from an additional encoding.
T08	Linked interpretation across debt classification and payment treatment	Impossible combinations rejected through shared constraints.
T09	Rounded totals, duplicate facts and contradictory notes	Correct interval compatibility; no automatic zero or balancing plug.
T10	Infeasible or unsupported constraint problem	Conflicting sources exposed; timeout/unsupported case reported as unknown.
T11	Zero-coupon bond and ordinary equity examples	Carrying amounts, payments, share quantities and report mappings match derivations.
T12	Foreign-currency asset, translation and cash movement	No mixed-unit balance test; translation effects separated from cash flows.
T13	Acquisition and intragroup balance elimination	Perimeter and adjustment book reproduce group totals without inventing capex.
T14	Conflicting recognition rules and unknown guards	Explicit conflict or symbolic partition; file order has no effect.
T15	Insufficient cash with constrained funding	Funding failure/default retained as an outcome; no hidden borrowing.
T16	Expected-loss measurement before future default is realised	Measurement uses only information allowed at that date.
T17	Repeated/reordered independent events and interrupted execution	Idempotent replay; branch isolation; identical committed outputs.

Test	Required case	Pass condition
T18	Conditional distributions and optional weighted mixture	Branch distributions retained; mixture agrees with declared weights and total probability; unweighted family never labelled probabilistic.

For synthetic cases, expected histories, excluded histories and scenario consequences are known by construction. Tests should include counterexamples that balance but have the wrong economic effect, such as treating a non-cash acquisition movement as operating cash. Such tests discriminate between arithmetic validity and the intended semantics.

Differential tests against an existing ledger check only the shared supported subset. ACTUS schedule comparisons check contractual events, and QuantLib comparisons check selected valuation calculations under matched conventions. Disagreements must be classified as different conventions, unsupported cases or implementation errors before one tool is treated as an oracle.

40.3 Extraction and reconstruction evaluation

A company corpus must include different standards, industries, languages, filing formats, extensions, interim calendars and restatements. Company and time splits are required so repeated report templates or revised versions do not leak into evaluation. Public benchmark tables are useful development material, while acceptance uses complete disclosure packages that include notes and their relationships.

Reviewers annotate source facts, contexts, mappings, required relationships, known ambiguities and unsupported assertions. Relevant measures include correct source-linked values and contexts, unresolved extraction volume, coverage of note obligations, reconciliation failures and invented facts. Report each separately. A single overall score can hide a serious error in a small but consequential financing note.

Historical reconstruction should test whether the known synthetic history remains feasible, whether known incompatible histories are rejected, and whether economically distinct alternatives are preserved. On real companies, public data generally cannot establish the true journal. Review instead assesses evidential compatibility, unsupported commitments and conditional economic consequences. Later disclosures may provide discriminating evidence for previously ambiguous terms.

No universal numerical threshold for semantic coverage or extraction accuracy is fixed in this document. Before a release evaluation, an accounting-reviewed test corpus and task-specific acceptance thresholds must be recorded in an evidence contract. Hard conformance failures, fabricated citations, silent discarding of valid alternatives and undocumented accounting adjustments veto acceptance regardless of an aggregate score.

40.4 Evaluate the DSL against the strongest practical baseline

The comparison uses the same IR, accounting examples and tasks. One group of models is authored through the typed Python API; another through the DSL. Both receive equivalent documentation and examples. Tasks include encoding a movement schedule, adding an alternative

interpretation, changing an effective date and tracing a simulated amount. Measure correctness, material omissions, time to a reviewed result, ease of explaining a change and maintenance effort.

This comparison tests whether the notation and compiler help users; it is not a comparison between careful DSL code and unstructured spreadsheets. If the DSL offers little advantage, retain the shared IR and library and narrow the textual language. Small pilots nominate improvements. A claim of superiority requires a predeclared comparison and uncertainty evidence, not a few favourable demonstrations.

40.5 Forecasting evaluation comes later

The proposed experiment asks whether a conditional company method adds out-of-time predictive information while preserving statement coherence and unresolved exposure. The comparator ladder uses the same available sources and targets: last-observation and seasonal naive forecasts; contractual runoff and accounting-identity forecasts; tuned classical statistical and conventional financial models; the structural model with one selected interpretation; the same structural model preserving alternatives; and declared extensions. This separates changes in economics from changes in the representation of ambiguity. Tuning stays within training windows. Test issuers or time blocks remain untouched, and unavailable predictions stay in the reported denominator.

For probabilities, predeclare proper scores, calibration, interval coverage, sharpness and dependence-sensitive joint checks. Unweighted feasible sets have coverage and width under a stated set interpretation, not probability intervals. Promotion needs a declared score difference with uncertainty evidence on matched eligible targets and no accounting or information-time veto. Issuer and time dependence require a justified paired resampling or equivalent inference design; overlapping reports are not independent replications. Runtime, a few losses and small-seed tail differences remain descriptive without that support. No such forecast experiment has been run for this proposal.

Look-ahead, broken conservation, a changed target or missing scope can invalidate an experiment. A valid comparison in which the proposed predictor performs poorly instead rejects that candidate and motivates its stated repair. The two outcomes must not be conflated when choosing the next experiment.

Forecast evaluation uses historical information cutoffs, fixed data vintages, declared economic scenarios and proper separation of training and testing periods. Every level of the comparator ladder above uses that same information boundary, so extra information cannot silently account for an apparent benefit.

Construct the inexpensive comparators before choosing a complex forecasting method. A no-growth operating forecast with scheduled contractual financing tests whether the model adds value beyond existing obligations. A constant margin and turnover model tests whether detailed operations improve on familiar financial ratios. A trailing-average growth model tests whether fitted dynamics improve on recent experience. A fixed-exposure contractual stress model tests whether estimated management responses help beyond repricing the existing book. Each comparator must generate coherent accounts through the same reporting rules and use the same historical information. In synthetic cases, the known hidden state also supplies an oracle conditional on the stipulated scenario.

Evaluate these comparisons separately under stable conditions, large rate resets, refinancing pressure, credit deterioration and changes in reporting perimeter. A pooled average can conceal failure in the situation the product is intended to explain. Supported underperformance against

a cheap comparator in a declared critical situation vetoes promotion of the complex forecasting method and must lead the result discussion. These checks are falsification tests, not training targets or evidence of superiority when merely passed.

Evaluate point forecasts where they are a stated target, distributional calibration when probabilities are justified, and decision-relevant robustness when only feasible sets are available. A wide feasible set is not automatically a good prediction, and an unweighted set cannot be assessed as if it were a 95% predictive interval. In multi-scenario stress tests, scenario severity and probability remain separate inputs.

Any serious numerical comparison must follow the repository's research plan, assumption audit, baseline, veto, uncertainty and result-record policies. The present document runs only deterministic checks of its own examples. It neither selects a scientific forecasting default nor reports comparative performance.

40.6 Fixing the target before evaluating a method

The following protocols specify the comparisons needed before selecting a forecasting method. The first question is what a successful prediction would mean. A correctly parsed filing, a retained feasible history and a useful forecast are different results. Each receives its own target and comparison.

Information available at the forecast date

The nine-issuer, 2018–2025 panel is a purposive development resource. Its 192 accepted reports do not form 192 independent observations. All existing issuer roles remain provisional: subsequent source work on JPMorgan and insurers has also exposed proposed test families. The 31 pilot annotations are development answers. The 136 unverified publication dates remain unresolved. Even a verified filing date is insufficient when it does not establish availability of the exact retained version at the intended cutoff. No date is imputed from a filename.

An eligible information record identifies original bytes and full hash, issuer and consolidated perimeter, report period, original/revised status, actual public availability evidence, clock precision, timezone and ingestion time. All language versions, mirrors and amended reports belong to the same issuer/version family. An as-of view excludes any document whose latest possible availability exceeds the cutoff. If only a day is known, the origin is after that day in the declared local clock; an intraday claim is unavailable. Later comparatives cannot repair training data retrospectively without defining a different retrospective task.

Forecast origin and target period must be distinguished. A December year-end report first available in March cannot support an entirely future prediction of the following January–December total: part of that total has already accrued. Such a task is a combined nowcast and forecast. The all-future annual task uses the first complete fiscal year whose start is strictly after the origin. A shorter future interval is admissible only if its realized quantity can be measured independently. Neither subtracting incompatible cumulative vintages nor silently calling an annual period a twelve-month forecast repairs this issue.

E1. Disclosure extraction and preservation

The unit is a complete declared disclosure package, with a source inventory covering statements, note spans, policies and incorporated references. Labels preserve raw tokens, entity/perimeter, concept, dimensions, unit, scale, sign, precision, period, availability and admissible readings.

Independent manual location and transcription supply the reference; a parser's output cannot label its own test. Disputed readings remain adjudication cases rather than convenient single-valued answers. Internal preparation precedes any separately authorized professional question.

Compare manual located transcription, transparent PDF/native-XBRL extraction, tuned pdfplumber or Arelle on their supported formats, plain Docling and the same converter with explicit context/footnote repair. Equal source bytes, referenced taxonomies, allowed network resources, annotations and tuning budgets are mandatory; PDF and native-XBRL strata are reported separately. A converter unable to accept a format is unsupported in that stratum, not assigned a fabricated accuracy. TableFormer/OTSL scores nominate layout methods only; they provide no accounting labels.

Prespecify separate denominators for required source units, resolved units, exact context/value tuples, unresolved units and materially wrong accepted assertions. Missing package content stays in coverage. Any fabricated locator, wrong admitted entity/unit/time, silent source loss or untracked download vetoes that candidate in the reviewed conformance set. It does not prove a zero population error rate when none occurs. For accuracy comparisons, aggregate paired differences first by report, then issuer; state whether the estimand weights issuers equally or weights deployment volume. Nine selected issuers cannot justify population-wide accuracy or a reliable rare-error bound.

E2. Reconstruction, query and conditional accounting

Use the same admitted evidence, finite template registry and unknown domains for an enumerated exact reference, a transparent typed-library model, the plain proposed reconstruction and any pruning/abstraction extension. Small rational polyhedra and finite history trees provide exact known feasible and infeasible cases. Test both soundness and completeness within that declared finite scope: each returned history satisfies every constraint, each reference history remains represented, and each query interval agrees with independently derived bounds. The continuous floating-share example has interest in [5,7]; duplicate encodings must leave that interval and any separately supplied probability measure unchanged.

Include cross-statement dependence, open endpoints, contradictory evidence, unknown solver results, interrupted enumeration, unsupported nonlinear terms, correction/replay and failed funding. The oracle for an interrupted case is an explicit incomplete result, not an invented full answer. Check journal and statement invariants plus economic timing and conditional response. A timeout is a resource outcome; representing it as infeasible is a correctness failure. On actual companies, evaluate evidential compatibility and retained alternatives; the hidden true journal is generally unavailable. Exact small cases do not prove completeness of unrestricted histories or the correctness of an entire solver.

E3. Conditional financial prediction

Corporate prediction begins with operating cash flow as the primary reported quantity and closing cash and financial debt as a prespecified joint secondary vector. Use the issuer's actual presentation, perimeter and accounting basis; cross-basis harmonization requires its own admitted mapping. The target record pins whether the outcome is the first released amount or a later corrected amount. Primary evaluation uses first release; restated values are a separately labelled sensitivity with their own release dates. Each unknown target remains in the eligibility/failure report, with a reason fixed independently of forecasts.

Every comparator receives the same as-of information and target. The ladder is last value/seasonal naive where frequency permits, scheduled contractual runoff with no-growth accounts,

tuned constant-margin/turnover and conventional statistical models, the structural model with one fixed interpretation, the same model preserving alternatives, then a named extension. Classical tuning uses training/validation periods and an equal stated resource cap. Model complexity, information and ambiguity representation are separate ablations. Neither a poorly tuned comparator nor selecting the best baseline after test results gives a fair primary comparison.

For an admitted probability law with finite first moment, use the negatively oriented continuous ranked probability score (smaller is better). If X and X' are independent draws from the predictive law and y is the realized amount,

$$L(F, y) = \mathbb{E}|X - y| - \tfrac{1}{2}\mathbb{E}|X - X'|.$$

Gneiting and Raftery (2007, sec. 4.2, eqs. 20–21) supply this representation and its conditions; the paper uses the opposite orientation before explicitly introducing this loss orientation. A point forecast reduces to absolute error. For probabilities one half on 5 and 7 and outcome 5, the first expectation is 1 and the second is 1, giving loss 0.5. A point forecast 6 has loss 1 on this observation. This one outcome is a calculation check, not evidence of superiority. For outcome 6 their losses reverse to 0.5 and 0.

For scale comparisons, divide all amounts for an issuer-origin by the same strictly positive, as-of known total-assets amount, fixed before the outcome. Missing or nonpositive scale prevents that scaled comparison; it is never replaced by a test-outcome-dependent scale. Report unscaled amounts as well. The joint secondary loss replaces absolute differences by Euclidean norms after fixed component scaling: the energy score with exponent 1 (section 4.3, equation 22). It requires a joint law and finite first moments; the vector of independently optimized marginal extrema is not a joint forecast. The source's kernel argument in section 5.1 supports propriety under its stated hypotheses; it supplies no assurance that this finite corporate panel can detect every dependence error. Prespecified cash shortfall and linked-statement event checks therefore accompany it.

For unweighted admissible sets, report inclusion of the realized quantity and width only under a declared observation/scenario interpretation. An outcome outside an assumed scenario does not refute conditional arithmetic; an always unbounded set is not useful just because it contains observations. Compare equal-scope sets and their decision implications without assigning 95% labels or manufacturing equal branch probabilities. Robustness across a prior class remains a range of scores or decisions, not one averaged forecast.

Fix the comparison population and weighting before scoring. A candidate missing eligible predictions cannot win on an easier surviving subset. Report paired scores on the prespecified common coverage and the full eligibility/failure denominator; a claim about a complete service requires a declared fallback with its own results. Never renormalize away unresolved probability mass. A probability forecast with unresolved mass is incomplete for the above proper-score target.

The primary effect is the paired mean loss difference, proposed minus the preselected strongest admissible comparator, with issuer and time dependence accounted for. Promotion requires a predeclared uncertainty interval to lie below minus a decision-justified improvement margin, while all validity vetoes pass. A zero margin can test statistical improvement but cannot establish material business value. The interval level, dependence model, time blocks, multiple-comparison rule and margin are locked using development evidence before test predictions. Nine purposively selected issuers with eight years and overlapping horizons do not automatically support a bootstrap approximation. Without a defensible uncertainty design, report differences descriptively and retain all screened viable candidates without ranking.

E4. Bank losses and insurance reserves

Bank prediction distinguishes contractual cash collections, realized defaults, recoveries and reported allowance measurements. An allowance is an accounting estimate, not an observed realization of latent credit loss. A bank protocol needs exposure definitions, vintage/cohort boundaries, default/cure policy, original rating history, censoring, collateral and recovery timing, and the information available at each measurement. Aggregate filings do not supply those fields merely because they disclose a total allowance. Until eligible cohort data exist, compare exact conditional transition examples and reported aggregate bridges; do not claim calibrated borrower-level PD/LGD/EAD.

Insurance validation separates ultimate reserve prediction conditional on an observed upper triangle from one-year paid claims plus re-estimated reserves. Use the retained Mack and bootstrap dossiers for the exact conditioning and net-reserve center. Deterministic chain ladder is a point baseline; moment-based Mack uncertainty, a fully specified process-law bootstrap, and any enhanced model are different comparators. Mack moments alone do not specify a simulation law. Preserve shared fitted-parameter dependence across accident years and the backward-law assumptions required by the inspected consistency argument. One-year validation refits after simulated/revealed development; it cannot substitute an ultimate prediction distribution. Missing increments, negative/zero bases, unverified development dates or absent process law veto the dependent comparison and trigger data/method repair. Financial-disclosure panels alone do not provide independent insurance triangles or actuarial calibration evidence.

E5. Numerical inference and scenario interpretation

Start with rational enumeration and the normal conditional model derived earlier: prior mean 0 and variance 4, observation 2 with variance 1 give posterior mean $8/5$ and variance $4/5$. Cash $100 + 10z$ then has mean 116 and variance 80. Check the measure, normalizer, transformations and downstream cash distribution before speed. Compare analytic/factored linear algebra, error-controlled quadrature where applicable, an ordinary target-specific sampler, and only then an enhanced sampler. Mixed branches require evidence weights, not chain counts. Exact-equality constraints require a declared constrained measure; adding noise changes the target and requires a separate comparison.

Numerical tolerances derive from the requested monetary enclosure or decision accuracy. Posterior agreement, unresolved mass, divergences, convergence and Monte Carlo standard error have declared veto roles before throughput is interpreted. Independent chain initials are required; seed repetition alone is not an independent target. Failure on an analytic target triggers implementation or target repair. Failure only on a complex target triggers the specified parameterization/tuning repair, not abandonment of all inference.

The earlier causal example supplies a separate semantic test. Under $U \sim \text{Bernoulli}(1/2)$, $X = U$ and $Y = 2X + U$, conditioning on $X = 1$ gives $Y = 3$; intervening to set $X = 1$ gives Y in $\{2,3\}$ with mean 2.5. The test checks the supplied structural model. A successful test does not identify a real corporate intervention from filings. Empirical causal promotion requires its own identification design, confounding assumptions, data and estimand; observational forecast performance cannot provide those premises.

E6. Language usability and reuse

Compare the DSL and typed-library authoring over exactly the same IR, accounting rules, error messages, documentation and tasks. Tasks include encoding a movement schedule,

retaining an alternative, changing an effective date and tracing an amount. Correctness and material omissions are primary; completion time applies to a correctly reviewed result, with censored/failed tasks retained. A later counterbalanced task order must separate practice effects and prior expertise. The same author trying both interfaces is a development pilot, not independent usability evidence. Participant number, target expertise and uncertainty design require a justified future study before a superiority claim. No participants are contacted by this proposal.

Reuse compatibility first uses documentary exchange fixtures and field-loss cases. Arelle, Docling and other selected adapters need exact dependency/model/ taxonomy locks and terms before runtime evaluation. A successful source review does not establish installation, compatibility, license compliance or speed.

Budget, seeds, repairs and future records

The future ladder is bounded by decisions, not by an invented universal sample size. B0 runs exact tractable/adverse cases only. B1 uses exposed development packets to estimate failure rates, resource costs and within-target variability; its results cannot promote a method. B2 permits a frozen, honestly labelled retrospective panel comparison after availability and exposure repair. B3 uses newly sequestered, task-appropriate data or participants and a justified sample/ uncertainty design for external claims. Failure of B0 invalidates the dependent run; a weak candidate in B1/B2 triggers its declared repair. A later rung begins only when its prerequisite artifacts exist and execution is authorized.

For stochastic B1 debugging, seeds 101, 202 and 303 are a reproducibility convenience, not a sufficient replication count. Paired arms share randomness only when the coupling preserves each arm's marginal law. Subsequent replication counts, independent data units, chains/draws, search trials, wall-time and memory caps are selected from target-specific development evidence and frozen before evaluation. Optional learned components need a separate objective/scaling, capacity, optimizer/search, budget, heldout and downstream protocol. No neural architecture or optimizer is selected by this general design. Insufficient budget produces under_budgeted, not a convenient inherited default.

Each future run saves eligibility and exclusions; data, source and exposure hashes; split and temporal embargo; target and outcome-vintage definitions; comparison/tuning budgets; random seeds; environment and CPU/GPU mode; exact command and commit; wall time; per-unit predictions and failures; independent validity checks; uncertainty calculations; decision and inference-status tables; and a repair/next-action record. Changes after outcomes are inspected receive a new exploratory identifier and cannot overwrite the frozen confirmatory result. Unknown critical source time, corrupted data, wrong target, missing essential diagnostics or exceeded resource limits stop that run. A valid poor candidate blocks its promotion and leaves independently ready repairs available.

40.6.1 The joined calculations retain separate validation targets

The sector group in Section 28.8 clarifies what an end-to-end check must preserve. The bank's expected loan cash, its contractual principal and coupon, the insurer's liability estimate, the parent's guarantee, and each taxpayer's usable loss have different meanings. A group statement can balance after one has been substituted for another. Its reference must therefore retain the separate legal books, gross external cash and origin of each adjustment, as well as the final totals.

Three kinds of comparison follow. For supplied economic and accounting facts, an independently

derived finite calculation tests the loan, tax, ownership, currency and consolidation rules. Constant FX and unchanged ownership remove one mechanism at a time: they test whether the calculation changes for the right reason. For estimated loan or insurance cash, E4 instead tests the stated conditional prediction against a credible tuned baseline. For numerical valuation, E5 checks the declared measure and error in the downstream quantity. A correct group bridge cannot establish the quality of a credit forecast or insurance risk adjustment used as its input.

The same separation matters elsewhere. A DTA may be arithmetically correct for supplied recoverable losses while its recovery forecast is unsupported. A POCI schedule may conserve cash while using a forecast made after the reporting date. A retrospective insurance method may follow its formula while using hindsight that the route does not permit. Each comparison must therefore bind the applicable law or policy, contractual and expected cash, historical information vintage, valuation assumptions and ownership interval before interpreting its result.

Some checks are exact within the stated domain: a hypothetical hedge option must not generate actual cash; a component's notional funding must not become new legal capital; and an NCI purchase after a close must not alter attribution before that close. Other quantities, including predictive losses, tail estimates and Monte Carlo accuracy, require their own uncertainty design. An error in the target or reference invalidates the dependent comparison. A valid but poorly performing forecast motivates the specified model repair. These outcomes should lead to different next steps.

The original E1–E6 protocols consequently remain distinct after composition. The newer examples add contract-loss order, dated credit receipts and write-offs, dependent EPS, actual versus aligned option values, insurance transition information and shared-sector attribution to their test populations. They supply tractable checks before a more expensive run. Eligible data, paired comparison budgets, dependence assumptions and decision-relevant margins must still be fixed for each actual empirical target.

40.7 Capability contracts for sector packages

Package	Required state and operations	Minimum acceptance evidence
Core corporate	Working-capital bridges, lots/assets, ordinary debt and equity, period closing and cash-flow mappings	Exact synthetic cases, reviewed public filing reconstruction, all statement reconciliations.
Financial instruments	Terms, calendars, resets, options, measurement categories, modifications and hedges	Contract schedule comparisons, valuation checks, paragraph-based posting cases and transition cases.
Banking	Credit cohorts, deposits, collateral, liquidity, regulatory capital, management actions	Allowance roll-forwards, repricing and runoff stresses, capital reconciliation and default/funding-failure cases.
Insurance	Policy groups, future cash flows, service and finance components, reinsurance and asset interaction	Cohort movement reconciliation, assumption change cases, service recognition and reviewed actuarial calculations.

Package	Required state and operations	Minimum acceptance evidence
Multi-basis reporting	Versioned IFRS/US GAAP/ASBE rules, entity elections and transition treatment	Matched economic-event cases with explained basis differences and independent accounting review.

Every package declares capabilities at operation level. “Supports bonds” is too vague; “supports fixed-rate, single-currency, non-callable amortised-cost issuance and redemption under this reviewed rule set” is implementable and testable. Unimplemented operations retain their source and contract representation while producing an explicit capability error when execution requires them.

40.8 Implementation milestones and dependency order

Stage	Concrete implementation deliverable	Exit evidence
P0	Portable records and contextual types; source/review/time selection; typed constructors and versioned IR; separate posting authorization and atomic journal/delta/state/receipt transactions for the running cases	T01–T04, T09 and T17; source inspection and exact receipt replay.
P1	Admitted history templates, finite choices, shared symbols, linear solver adapter and historical bridges; conditional simulation and report explanations	T05–T10, T14 and T18; synthetic family preserved.
P2	Matched debt text frontend with identical portable IR; connected corporate methods through the typed authoring API; document/XBRL import adapters	Parser/API equivalence; T11–T13; reviewed three-date public-company case.
P3	Contract and valuation adapters, dated standard packages, richer financing and tax/funding policies	Contract/valuation differential cases; accounting review; T15 and transition cases.
P4	Bank package, allowance models, regulatory reconciliation and management response alternatives	T16; banking-specific reviewed histories and stresses.
P5	Insurance cohort package, reinsurance and actuarial measurement adapters	Reviewed cohort roll-forwards, information-date and service/finance tests.
P6	Optional probabilistic inference, model-weight sensitivity and broader scenario research	Explicit targets, sampler/numerical validity, prior sensitivity and downstream evaluation.

These stages specify dependency order, not dates or staffing estimates. Exit checks are cumulative: a case can recur when a later stage changes the inputs or execution path that it

tests. Banking and insurance representation requirements influence P0 types, although their measurement engines arrive later. Probability weighting is optional; P1 already produces useful conditional cases and bounds. A failure of a chosen extractor or inference algorithm triggers a local replacement or repair, rather than abandoning the product's representation requirements. The parser/API comparison in P2 originally covered the matched debt fragment. The selected company source profiles in Section A.54.17 now author complete methods and ordered commands with retained typed-result comparisons. Their supplied resource and eligibility boundaries still limit which programs they admit. General corporate textual authoring and compatible profile migration remain further work; these selected extensions do not imply that every corporate operation has a text frontend.

40.9 Decisions and replaceable assumptions

The requirements fixed by this design are preservation of evidence and alternatives, explicit time and basis, balanced and admissible posting, economic state beyond journals, and distinct set/probability semantics. Replaceable choices include the storage engines, parser generator, solver, numerical libraries, aggregation resolution and user interface. Every scientifically material aggregation, behavioural parameter, prior or truncation rule must record provenance, purpose, failure mode and a diagnostic before it becomes a default.

The immediate work is to finish and review the governing proposal, including its real-source, authority, domain and research obligations. Product implementation follows completion of that work and explicit implementation authorization. In the table above, P0 establishes portable records, review/time selection, durable receipts and typed authoring; P1 builds admitted histories; P2 connects the corporate methods and matched textual frontend. These stages are a design sequence, not an already running product campaign.

40.10 Concrete conformance and corpus protocol

The fixture directory contains independently entered inputs and expected outputs for Northstar, rounded borrowing, evidence views and sector cases. `check_repair_conformance.py` recomputes exact journals and statements, constructs the joint linear bridges in Z3, checks strict bounds, and exercises replay, atomic rejection, finite Bayes updating and funding failure. `check_core_contract.py` checks grammar/schema examples, canonical identities, stable streams, information restrictions and source inventory. These scripts are reference oracles for portions of the design. They do not test an unimplemented full DSL compiler, all adapters or standards compliance.

Every conformance case must specify input files, the operation requested, expected values or typed errors, comparison convention and supported scope. Exact money, finite probabilities, linear witnesses and example journals compare as rationals. Approximate valuations require an error bound or a justified tolerance tied to the valuation scale. A missing expected result cannot become a passing test simply because the operation returned something. An implementation release must pass all applicable positive and negative fixtures and preserve unresolved operations; the arithmetic oracle alone does not release it.

The retained panel contains all 192 target annual/interim report slots for 2018–2025 across nine issuers and three evidenced accounting families, with eight consecutive reporting years per issuer. Here "complete" applies to the source-acquisition panel: every declared slot has a retained report version. It does not mean that each selected document contains every related filing section. For example, the selected JPMorgan shareholder annual reports for 2022–2025

explicitly omit Form 10-K pages (respectively 1–42 and 304, 1–44 and 322, 1–48 and 334, and 1–42 and 328). The complete issuer-hosted 10-K candidates and their bounded semantic annotations are preserved as a separate source layer; they do not retroactively change the selected panel or establish full package closure.

Apple, Caterpillar and JPMorgan Chase each contribute one annual and three quarterly reports per year; the other six issuers each contribute one annual and one half-year report. Thus $3 \times 8 \times 4 + 6 \times 8 \times 2 = 192$. This is source acquisition, not semantic closure or a representative random sample. The 244 retained original versions include historical alternatives to the active panel. Apple remains a detailed development package, with Caterpillar supplying a contrasting inventory policy; bank, insurer, IFRS and ASBE reports are also retained.

Publication dates remain unverified for 136 of the 192 accepted reports; 56 have filing-date metadata. A historical forecasting population needs the actual byte-availability evidence described in Section 29.13, not imputed publication dates. The 31 provisional annotations across four reports are exposed development material, not heldout answers. Policies, note links, incorporated documents, changed comparatives and adjudicated meanings still require task-specific preparation. Synthetic revisions provide controlled cases without substituting for a required real conflict. The corpus protocol and active panel preserve their distinct populations and versions.

Annotators first record literal/context readings without seeing the model's predictions, then mark source conflicts, mapping relationships, narrative qualifications and admissible alternatives. A second accounting-aware reviewer adjudicates material disagreements while preserving both initial annotations. Reports from the same issuer family and duplicated tables cannot be split between development and heldout evaluation. Forecast tests freeze the actual publication cutoff; later revisions are separately labelled outcomes and may not enter predictor inputs. Extractor upgrades rerun the same frozen corpus.

For the initial supported release, loss of a source span, wrong entity/unit/ period/basis, invented evidence, hidden financing, dropped feasible branch, incorrect cash reconciliation or unreported unsupported treatment blocks acceptance. All selected critical facts and operations must have exact reviewed answers or the expected unresolved response. These zero-tolerance conditions follow from the product's meaning-preservation requirements. Non-critical extraction quality is reported by stratum and error category; no arbitrary overall accuracy threshold certifies whole-report understanding. An extraction benchmark needs its own sample size and uncertainty protocol before making statistical accuracy claims.

The DSL and typed-library comparison uses identical packages, evidence, schemas and runtime. Users perform the same tasks: correct a context, add a shared uncertainty, represent a restatement, add a supported event and explain a forecast difference. Task order is counterbalanced and experience recorded. Both frontends receive the same validation and error messages. Measure successful completion, material semantic errors, time and maintenance effort with uncertainty appropriate to the participants; report no winner from the current syntax checks. A textual language earns its maintenance cost only if that comparison supports its use.

40.11 Workload limits and release evidence

The first integration workload is one local company package, up to 10,000 observations, 100 declared aggregate event slots per period, 64 materialised finite-choice cells and a 30-second budget per solver operation. These are provisional engineering limits for bounded failure behaviour, not measured capacity or scientific defaults. They must be configurable and recorded. Reaching a limit yields partial coverage with the unvisited formula/cells, never an exhaustive-

history claim. Peak memory and latency are measured only after an implementation exists; a slower but semantically correct result does not become a scientific failure.

Engineering release needs compiler/API equivalence, transactional replay and adapter subset checks. Accounting release additionally needs paragraph-level authority, dated applicability, exact fixtures and independent domain review for every promoted operation. Forecast promotion additionally needs preserved information sets, the stated baseline ladder, statistical uncertainty and critical-regime vetoes. Human comprehension review remains separate from all three. This version supplies a richer design candidate and reference checks while preserving these outstanding acceptance decisions.

Appendix A

Portable operation reference

These proposed versioned operations share the request and result envelope in Section 39.4. Required inputs are explicit even when a list is empty. An expected head mentioned below refers to the common envelope, never a second argument. The registry is a lookup reference after the financial and transaction meanings have been explained in the body. These are specified exchanges, not deployed endpoints.

A.1 The corporate exchange package

The Northstar example in Section 28.6 supplies a closed instance of `CorporateJointHistory/1`. Its package is retained at `data/accounting-master/corporate-portable-20261015/`. The files `record-schemas.json` and `portable-exchanges.json` contain the closed record bodies, signature-specific arguments, complete immutable records, required byte references, exact submitted request bytes, results and the before/after store worlds. The accompanying semantic definition is `docs/design/proposal/architecture/corporate-portable-20261015.md`. These files specify exchanges; generating or checking them invokes no product endpoint.

Each request uses the common envelope. Its arguments have exactly the two fields `tag` and `value`: the tag equals the versioned operation name, and the value obeys that signature's closed schema. The profile and signature references bind the interpretation of those fields. Every reference has a kind, schema and content digest. Its target must be present, with the correct type and every required dependency; the submitted raw bytes are preserved even when decoding or validation fails. A record's dependency list contains its body references in canonical order. Required source and schema blobs are checked separately by byte length and digest.

The eight source units retain the original inventory. Corrected interim and annual coverage points to the complete historical reconstruction; the superseded original statement remains a literal source with a separately typed revenue reading. The source-reference closure does not imply that every source line has been admitted as accounting truth. A separate package variant retains a missing incorporated debt agreement as an explicit obligation. The new qualitative and rounding examples have their own authored source bytes.

The principal record relationships are useful when locating an exchange:

Record	Required financial connection
`HistoryFamily`	Frozen module and view, shared symbol and constraints, opening state and complete historical reconstruction.
`Projection`	That family, the imposed scenario and corrected closing state used as its opening; conditional journals, final balances and obligations.
`Report`	One projection and statement map; the same period, view, family and scenario throughout all reported lines.
`StateTransitionReceipt`	Validated request, stable event identity, authorization, journal, independent quantity/money change and prior/new states.
`WorkReceipt`	Saved search owner, prior/new checkpoint state and unchanged semantic manifest and continuous partition.

The text frontend deliberately has a smaller authoring scope than typed construction. Its published grammar accepts the ordered `NorthstarDebt` declaration: principal, the rational interval for x, fixed and floating rates, and the stated interest formula. The frozen module supplies the context, year fraction, source and assumption dependencies. Both principal tokens, both interval endpoints and both rates must match those module records. The typed debt input identifies the same module; the full company input separately includes all operating templates and economic facts. An unsupported declaration has a located diagnostic. This is a specified new bounded frontend, not a claim that the earlier Core0 parser implements it.

Accounting commits change the accounting owner, event lookup, journals, independent deltas and receipts atomically. Review commits change their review owner and publish a new immutable view. Checkpoint commits preserve the completed and unresolved cells under their work owner. Import verifies all original record mappings and required bytes before changing its metadata owner; it admits no financial claim. Each exchange states every unaffected domain as well, permitting exact comparison of the two worlds.

Failure location matters. Malformed JSON, extra fields, missing required references, mismatched operation tags or corrupt supplied blobs leave the validated request and financial payload null. A transaction conflict or stale head after valid input retains the validated request and structural validity, while reporting no new commit. The result's coverage retains the continuous family for financial queries and the original partition on resume. Neither an unexplored cell nor an unsupported calculation becomes an empty family.

The independent checks reconstruct both half-years from preserved economic inputs, derive the annual bridges and 2025 affine coefficients, replay the proposed journals, and compare the permitted store changes. Deliberately altered rate sensitivity, inventory quantity, receipt history, publication date and unknown closing balance challenge those checks. Their scope is the finite documentary specification. Generic frontend and solver execution, admitted-law extensions, crash/reopen testing, accounting acceptance and human comprehension retain their separate evidence requirements.

A.2 Dated settlement annotations

Northstar's annual projection supplies amounts and an event sequence. The proposed `SettlementScheduleDraft/1` annotation adds dates and explicit funding knowledge without changing those amounts or the frozen company profile. It is a bounded documentary proposal; claim-level generation, behavioural-policy records and whole-company textual authoring remain further work. The existing `scenario.project@1` and accounting operations retain their distinct responsibilities.

The body has exactly five fields: `profile`, `projection_ref`, `horizon`, `funding_knowledge` and `events`. The projection reference identifies the existing immutable projection. The horizon has a start and exclusive end, at midnight UTC on the retained projection's fiscal boundary dates. Each event names one projection event, its instant and any required predecessors. An unknown instant carries a reason and named outstanding obligations. Each original event occurs once; neither an omitted event nor an invented payment is admissible.

Validate structure, the referenced projection and complete event coverage before considering execution. Dependencies must be acyclic and compatible with dates. Purchase precedes delivery and supplier settlement; delivery precedes collection; the income and tax events precede closing. A date outside the half-open horizon rejects. Two events at the same instant need an explicit order when their order matters. The bounded annotation conservatively retains an unresolved result for an unordered pair; it does not use event names or input-array position to decide who receives cash first.

The initial funding alternatives are an explicit scenario assumption that there is no eligible facility, or unknown facility access with its outstanding obligations. A committed facility needs a fuller contract describing limits, eligibility and draw timing. When cash suffices, payment can proceed without one. When cash is short and no facility is eligible, return an economic funding failure and leave the proposed payment unapplied. When access is unknown, retain unresolved funding instead. Subsequent bankruptcy or restructuring requires its own method; it is not inferred from this refusal.

The reference schema and fifteen complete schedules are retained in `data/accounting-master/simulator-timing-20261023/`. The cases distinguish collection-before-payment, payment-before-collection, unknown collection timing and unknown funding. They also exercise omitted or duplicated events, cycles, contradictory dates, invalid calendar dates, unresolved references and explicit same-instant ordering. The original accounting comparison retains both interest-rate scenarios at floating shares zero, one half and one. These checks establish the stated documentary examples, not a deployed scheduler.

A.3 Claims, funding and matched annual authoring

The early supplier payment also needs an identified obligation. Northstar's opening payables of 30 and new purchase of 170 make a payable control balance of 200. In the authored allocation, the payment of 160 reduces the new purchase claim to 10 while the opening claims remain 30. Opening receivables of 35 and sales of 300 are settled by collections of 10 and 270 respectively, leaving 25 and 30. These allocations agree with the retained annual totals; they are explicit scenario inputs, not invoice facts recovered from a filing.

A claim records its origin, recognised amount, adjustments where supported, settlements and outstanding amount. Recognition and settlement have separate events. Collecting an existing receivable does not recognise revenue again; settling a supplier does not recognise the purchase again. After each affected event, identified claims must sum to the receivable or payable control account. An excessive settlement rejects before changing either the claim or journal. The full-year example explicitly supports partial claim settlement; the earlier full-payment decision subprofile retains its narrower meaning.

Choosing and settling a draw. The proposed `SettlementDecisionDraft/1` separates cash, the identified claim, facility knowledge and the chosen action. A facility's usable capacity depends on its limit, prior drawings and reservations, with eligibility and the draw window kept explicit. A decision names a facility, amount and expected settlement instant, or explicitly

chooses cash alone. It observes only its permitted information. Unknown eligibility leaves a necessary draw unresolved; it does not supply a zero limit or establish default. Available cash may still support a cash-only decision.

Authorisation produces a proposal. Cash and debt change when a draw settles; the payable and cash change when the supplier is paid. These are separate events with separate atomic accounting transitions. A payment that fails after successful borrowing leaves the borrowing outstanding. The documentary checker can calculate a conditional successor assuming stipulated successful events, but that state is explicitly uncommitted. Durable storage and concurrent reservation semantics still require their own implementation evidence.

Authoring the complete conditional year. `FundedYearInputsDraft/1` imports an immutable annual projection and names its source hash, company, book, currency and floating-rate share. It then specifies the four claim allocations, facility capacity, new principal, annual rate, day-count convention, fees, tax method, assumptions and a dated event sequence. The example names zero fees and an eligible facility; it does not infer either from annual accounts. A nonzero fee is unsupported until a fee method is supplied. An unknown collection instant remains an unresolved input with a named obligation.

The restricted text form and typed JSON have identical fields and lower to the same closed body. Each text line declares a profile, reference, context, claim, facility, loan term, tax method, assumption or event. Amounts use reduced rationals; dates use UTC instants. The documentary lexical rule uses quoted tokens with an explicit semicolon terminator. The grammar and both complete inputs are retained in `data/accounting-master/funded-authoring-20261023/`. This grammar is a bounded proposed extension, not a change to the existing product compiler.

The pure reference executor consumes that body and the imported projection. It generates the added draw, interest and principal events, checks claim movements, derives tax from the accrued income accounts and closes those accounts. It does not obtain the result by copying expected closing balances. The 18 conditional comparisons combine the two retained stress rates, three floating shares and explicit facility rates of zero, five and seven percent. Every journal, cash prefix, closing account and statement agrees with the separate exact annual calculation. The loan's interest and tax effects are included; at zero interest the original closing accounts return exactly.

These comparisons establish the stated reference meaning under authored assumptions. They do not establish access to credit, customer payment behaviour, general economic-policy generation or complete company-language coverage. The annual operating and accounting model is still imported by reference. Broader facility terms, long-dated accruals, fees, refinancing and sector rules retain their separate specification obligations. The reference execution is conditional and uncommitted; production execution remains a later deliverable.

A.4 From operating inputs to the linked statements

Suppose Northstar's selling price changes from 10 to 11. A stored annual projection can display a new revenue figure, but that alone does not specify the customer claim, collection, tax or closing entry that should change with it. The operating-input successor makes those dependencies explicit. Its `CompanyYearInputsDraft/1` body imports the historical opening and authors the year's operating quantities and contract terms. Forecast journals are calculated from those inputs.

The purchase policy first combines demand of 30 units, target inventory of 8 units and opening inventory of 4 units. It purchases 34 units at the explicit homogeneous cost of 5, producing

inventory and a payable of 170. Delivery of 30 units at price 10 produces a customer claim and revenue of 300, with inventory cost of 150. The stated control-transfer assumption is required before that revenue entry can be proposed. An unknown control fact leaves the operation unresolved; a known failure of that condition does not silently use the ordinary sale method.

Claims are generated by those recognition events. The authored collection fraction of 90 percent settles 270 of the new claim, while a separately identified collection settles 10 of opening receivables. Changing price to 11 raises the new claim by 30 and collections by 27. At the demonstration's immediate 20 percent tax rate, tax rises by 6. Profit therefore rises by 24, cash by 21 and closing receivables by 3. The difference between profit and cash follows from the unpaid part of the increased sales, not a balancing adjustment. Changing demand instead changes purchases and cost of sales through their quantity rules.

Asset memory is also explicit. The opening plant cost of 122 comprises cohorts costing 100, 10 and 12, with useful lives of 10, 5 and 12 years. Their opening accumulated depreciation is 30, 1 and zero. All are in service at the start of the example year, so their annual charges total 13 under the stated straight-line, zero-residual method. The new expenditure of 20 enters service on 1 January 2026 and contributes no 2025 depreciation. If the unused 12-cost cohort also enters service in 2026, the 2025 charge falls by 1. Profit rises by 0.8 while cash tax rises by 0.2. The restricted method refuses a midyear service date until an accrual fraction is specified; it does not replace an unavailable convention with a full-year charge.

A shared input body. Text and typed records carry the same opening reference, source digest, context, fiscal period, inventory policy, claims, asset cohorts, debt mix, facility, tax method and event instants. The closed schema requires every field. Exact amounts use reduced rational strings, and supported unknowns retain named obligations. The evaluator verifies the opening's portable record identity as well as the supplied source bytes. Its permitted information mode is archival exogenous stress: the example does not pretend that its historical opening was available before publication.

Each conditional event records its method, input dependencies, proposed journal and economic-state changes. Claim balances, inventory units and asset cohorts reconcile with their control accounts after every event. The tax entry is derived from accrued income accounts and closing transfers their net balance to retained earnings. An explicit cash check precedes each payment. The draw and repayment retain their gross financing amounts even though their net principal effect is zero.

The 18 retained rate/share/borrowing-cost paths reproduce every journal, cash prefix, closing account and all 29 report lines of the independently retained comparison, including the additional interest and tax effects. Removing all forecast-projection records from the imported package leaves the new calculation unchanged. These checks establish the input-to-result meaning for this example; changing price, demand or asset service gives different results through the stated operation rules.

An interrupted year has a partial report. When payroll is moved before collections, the available cash is zero and the payment of 50 fails. The earlier draw and supplier payment remain in the conditional state; no payroll expense or cash payment is applied. A partial report identifies that last successful event and its dated balances. Opening cash and equity remain known, but requested annual cash, profit and the other uncomputed year-end lines remain unresolved. An unknown collection instant supplies no invented completed prefix. Complete reports bind every statement to the same input body and fiscal period, and reject an attempt to combine a changed-price input with an old result.

The exact schema, both authoring forms, reference semantics and checked comparisons are retained under `data/accounting-master/company-inputs-20261023/`. This successor uses one occurrence of each of fourteen event kinds in a full calendar year, synthetic treatment admission and the stated simple methods. The following extension defines repeated operations and revised multi-year assumptions without changing that earlier subset. Partial-period accruals, wider contracts and sector methods still require their own definitions. Historical reconstruction, general source-to-method admission and durable posting remain separate obligations. The result is conditional and uncommitted.

A.5 Repeated operations and consecutive company years

An unpaid customer claim does not disappear when the annual accounts close. Nor does a newly purchased asset lose its service date. A simulator that starts each year from account totals alone can balance the next set of statements while collecting the wrong invoice or charging the wrong depreciation. The company-path extension therefore carries the claims and asset cohorts alongside cash, inventory, debt and equity. It also replaces the single annual purchase and delivery with individually identified operations.

Each purchase supplies units and a new payable identity. Each delivery supplies units, price, the control-transfer assumption and a new receivable identity. Collections and supplier payments allocate explicit amounts to existing claims. Payroll, plant additions and dividends may repeat. The selected annual methods still require one depreciation calculation, legacy coupon, tax calculation and close, together with the one specified bridge loan. Repeating ordinary business operations does not repeat the tax or closing entry.

The distinction between recognition and settlement is preserved at each step. A customer payment reduces its receivable; it does not recognise revenue again. A supplier payment reduces its payable; it does not acquire inventory again. An operation checks every allocation, its journal and the resulting cash or stock before applying any change. If the second allocation overpays a claim, the first allocation in that same operation is also refused. Earlier completed operations remain in the conditional history. Reusing a settled claim's identity is invalid even when its outstanding balance is zero.

When splitting operations preserves the accounts. Under the example's homogeneous cost, a purchase of 34 units at 5 can be represented as two identified purchases of 17 units each. The payable total remains 170, but there are now two claims against which payments must be allocated. Corresponding splits of sales, receipts, supplier payments, payroll, capex and dividends reproduce all annual statements when their dates, dependency order, unit costs and asset lives remain unchanged. The checks compare every original journal for 18 rate/share/borrowing-cost paths and the aggregate journals of four proportional split examples. Balance alone is not the test.

Timing can nevertheless change feasibility. If half the baseline receipts, 140, arrives before bridge repayment and payroll, the company can pay principal 85.4, the small bridge interest charge and payroll 50. The remaining cash is positive. If only 40 percent, or 112, arrives, it can repay the bridge and pay the first payroll instalment of 20, but the remaining payroll of 30 fails by 23.4 plus bridge interest. The prior repayment, collection and payroll remain; the failed payment has no accounting effect. At a shared instant, the explicit operation order determines whether a collection precedes a payment. Future cash is not available merely because its timestamp matches the payment's.

The next year starts from economic memory. Northstar carries eight inventory units into 2026. The next year's authored demand of 32 and target stock of 10 require another 34 purchased units, costing 170. Selling 32 units at 11 produces revenue 352 and cost of sales 160. The company collects its 55 of opening receivables and 90 percent of the new sales, pays its 40 of opening payables and 160 of current purchases, and pays payroll 55. These are explicit synthetic choices, including collection before the supplier payment; they are not estimated operating behaviour.

The asset acquired for 20 in 2025 enters service on 1 January 2026 and contributes 2 of depreciation. The earlier cohorts still contribute 10, 2 and 1, so the annual total is 15. With legacy interest 6, pre-tax profit is $352 - 160 - 55 - 15 - 6 = 116$. The retained immediate 20 percent tax assumption gives profit 92.8. Operating cash is 87.6; after capex and dividends of 10 each, cash increases by 67.6. Every one of the 29 report lines is checked against independent arithmetic. A third authored year carries the same claim and cohort memory again. The joined case executes 63 operations across three years and retains 14 distinct claim identities, including those already settled.

A revision must say what changes. The annual authoring rule in Section A.4 collects 90 percent of new sales. Its price change from 10 to 11 therefore increases collections by 27, tax by 6 and cash by 21. A list of already authored settlement amounts has a different meaning. If only the delivery price changes, receipts stay fixed: profit rises by 24, receivables by 30 and cash falls by 6 through the additional tax. Neither result is a balancing convention. They answer different instructions about whether the collection policy is regenerated.

The same distinction persists across years. Increasing 2026's price from 11 to 12 while retaining its explicit collection amounts raises profit by 25.6, reduces cash by 6.4 and leaves 32 of additional receivables. Those claims remain in 2027 unless a later settlement is changed as well. The 2025 result is unchanged. Revising 2025 instead recomputes all affected later years from the original historical opening. Each conditional result identifies its input and predecessor; an old suffix cannot be joined to the revised opening. This is a new branch of assumptions, not a restatement of an observed historical book.

An unknown payroll amount at a known instant stops at that operation and retains the preceding completed operations. An unknown event instant prevents a reliable dated schedule. In either case an incomplete 2026 result cannot supply a 2027 opening. Reports keep the earlier complete years and any valid dated prefix, while uncomputed annual cash and profit remain unresolved.

A coherent path still needs the right information. The historical opening has its own source dependencies. The new evidence packet validates the complete reachable source and admission records, rather than just the opening balances. It also binds each of the fourteen operation methods to its payload definition, written rule, checked implementation and conformance evidence. Conditional closing records preserve full economic memory and identify their preceding result; they are not inserted into the historical Facts records. The packet contains 42 records and 35 exact byte payloads. These identities support reproducible local checking, not an external accounting approval or a signature establishing the truth of the inputs.

Source timing is consequential even in the synthetic example. Seven required source units carry a publication date of 2 March 2025, after the start of the first simulated year. They are ineligible under a 1 January cutoff. A cutoff during 2 March remains unresolved because the publication time is unknown; the next day passes that source-date test. System-as-of checks also distinguish ingestion from acceptance. At 12:30 UTC on 5 March, the sources have been ingested but two required acceptances at 13:00 UTC have not occurred.

Passing a later source cutoff does not establish when the future operating assumptions were authored or available to a decision-maker. The checked export therefore supports archival synthetic simulation. Investor-information or system-as-of use of the complete path remains unresolved until its assumptions and decision information are admitted as well. This preserves the distinction between a conditional accounting consequence and a forecast that could actually have been made at the stated date.

The typed definitions and reference calculations are retained in `data/accounting-master/company-operations-20261023/`, `data/accounting-master/company-operation-path-20261023/` and `data/accounting-master/company-path-admission-20261023/`. The first checks repeated operations, the second their multi-year composition, and the third complete-path evidence and method bindings. A corresponding bounded text frontend and portable partial-result and revision exchanges are developed in Section A.6. General policy expressions, time-admitted policies, production storage, wider accounting methods and sector interactions remain distinct requirements. The later group-company profile addresses a different part of that broader design.

A.6 Writing, revising and dating the company path

The interrupted second year creates two practical questions. What can another implementation recover from that result, and what must an author supply to continue it? The completed first year establishes opening cash and equity for the second. An unknown payroll amount leaves those openings known and preserves the operations already applied. It supplies neither a second-year closing nor a third-year opening. An exchange must retain that distinction even when all three years were requested together.

Each available annual outcome therefore identifies its exact input, predecessor, applied operations and report. A complete year also carries its conditional closing memory. An interrupted year has no such closing; the remaining periods stay explicitly unexecuted. In the payroll example, the first eleven second-year operations remain, while the 27 annual-result lines are unresolved. The two known opening lines are not erased. A supplier-payment failure has a different stopping reason and preserves its own shorter prefix.

Supplying the missing payroll amount creates a new assumption branch. Its revision record names the parent and child inputs, the affected year, the exact replacement and the reason. Comparing the full input bodies establishes that no other year or source opening changed. Earlier annual results remain identical; the affected suffix is recomputed from the original historical opening. Restoring the original payroll input reproduces the complete baseline path. This is an explicit repair of a conditional scenario, without changing what the company historically reported or claiming a durable database commit.

Content identities alone are insufficient for that conclusion. A deliberately changed cash report can be given internally consistent new hashes while still failing to follow its operations. The exchange distinguishes binding checks from trusted local replay. The first establishes which bytes refer to one another; the second recomputes their financial consequences. The extended example preserves the preceding complete evidence and exchanges five paths, five revisions and thirteen available annual outcomes, including both failures.

A collection rule changes a different instruction. Consider the second-year price revision from 11 to 12. Each of the two deliveries now recognises a receivable of 192. The fixed input still collects 79.2 per claim in each of two instalments, or 158.4 in total. Each claim therefore leaves 33.6 uncollected. This is the fixed-allocation result described in the preceding section, and it remains available unchanged.

An author may instead specify that 90 percent of each named invoice should be collected in two equal instalments. This optional rule generates 172.8 per claim, split into payments of 86.4, and leaves 19.2 uncollected. Relative to the fixed amounts, each of four allocations increases by 7.2. Cash consequently rises by 28.8 and receivables fall by 28.8. Revenue, cost, profit and the remaining annual report lines are unchanged. The rule alters settlement, not recognition. The 90 percent fraction is inherited from the illustrative example; it is not an estimate of how Northstar's customers would actually pay.

The policy names the parent path, year, deliveries, claims, collection events, fraction and instalment shares. Materialization creates a separate input path and records every old and generated allocation, invoice total and remainder. It changes no unspecified event date, older claim or following-year instruction. Unknown inputs remain unresolved. Another collection against a targeted claim outside these named events is unsupported until its priority and effect on the remaining amount have been specified. A generated path still needs accounting execution: a valid policy formula does not establish sufficient cash for all subsequent payments.

The text names the operation and its claim. The corresponding authoring subset preserves the typed operation path rather than hiding that settlement decision in a formula. Its outer declaration names the profile, company path and scenario assumption. An initial-year block gives the source opening, claims, assets and annual inputs. Continuation blocks give the following years' choices and events; they cannot replace the carried opening with newly authored balances. Every field required by the typed profile must be declared, including explicit markers for arrays.

For the price-12 scenario, one delivery and one allocation in its first fixed collection are written as follows:

```
operation delivery-1-a delivery 2026-01-03T09:00:00Z 16/1 12/1 true sales-ar-2026-a;
operation collection-1-a collection 2026-01-04T09:00:00Z;
allocate collection-1-a sales-ar-2026-a 396/5;
```

The listing wraps long lines for display; each declaration occupies one line in the executable text. The allocation names both the previously declared collection and the receivable it settles. Other claims and operations still require their own declarations. Materializing the illustrative 90 percent rule changes this allocation to 432/5, or 86.4. Editing the delivery price alone does not regenerate it.

Each declaration ends with a semicolon. Quoted text preserves spaces and punctuation. Amounts use canonical reduced fractions, truth values use `true`, `false` or `unknown`, and missing numeric values or instants remain `unknown`. The sequence of operation declarations preserves precedence at equal instants. Allocation lines attach to an already declared settlement; they neither create a claim nor reorder events. Duplicate operation or allocation identities and initial-opening declarations inside a continuation are rejected.

The complete baseline text contains all 63 operations over three years. Six examples, including both interrupted paths and the policy-generated path, lower to the exact saved typed inputs and reproduce their reports. Parsing still answers a narrower question than accounting execution. A well-typed negative purchase quantity is preserved so that execution can reject its economic meaning. The text parser evaluates no expressions, executes no shell commands and opens no source file named by the input. General expressions, endogenous branching and wider accounting profiles remain separate language requirements. The following extension supplies a bounded family wrapper and pure expressions for the declared collection policy.

The assumptions also have an information date. The source cutoff in the preceding section does not date the collection rule or other future inputs. An information declaration must identify the exact path header and each annual definition, then distinguish when those bytes were authored, publicly available, ingested and accepted. Changing one year changes that component's identity; its old timing evidence cannot admit the replacement. Unchanged years can retain their own evidence. This whole-year unit is coarse: it deliberately avoids claiming that an earlier subset admits a later complete definition.

The synthetic clock example authors the assumptions at 09:00 UTC on 6 March 2025, ingests them at 10:00 and accepts them at 11:00. Public availability is known only to the calendar day 7 March. A system-as-of query at 11:00 on 6 March can pass the synthetic ingestion and acceptance tests even though a public information query cannot yet do so. During 7 March, public availability remains unresolved; the next midnight is late enough for that day-precision declaration. Both views still require an eligible source opening.

Chronology matters independently of the cutoff. Exact equal timestamps can satisfy nondecreasing authoring, ingestion and acceptance order. Overlapping calendar-day bounds may leave the order unresolved. A publication day ends just before the following midnight, so authorship at that midnight cannot precede publication within the earlier day. Missing or untrusted observations cannot be repaired by assigning an early date in the declaration.

These times are authored test clocks, not observed historical records. Passing their checks establishes the selector's behavior on the example, without admitting a historical forecast or authenticating an author's identity. The separate settlement policy also requires its own information evidence; the path's dates cannot supply it automatically. The present design therefore retains archival synthetic simulation while making the missing temporal requirements explicit.

The executable references and complete examples are retained in four packages under `data/accounting-master/`: `company-path-exchanges-20261023`, `company-information-20261023`, `company-policy-20261023` and `company-text-20261023`. Together they check exchange consistency, assumption-time selection, declared settlement generation and text/typed correspondence. Broader methods and source evidence remain necessary for whole-company coverage. The next section keeps several authored choices together and makes their calculation recoverable.

A.7 Keeping company choices and their consequences together

The fixed collection amounts and the 90-percent rule answer different questions about the same company opening. An analyst comparing them needs each choice's cash, receivables, profit and later balances together. Selecting favorable amounts independently would describe a company that no stated choice produces. A named scenario family therefore retains the entire input path and result of each member, including its unresolved years.

The bounded family contains the baseline, two price revisions, the unknown payroll path, the supplier-payment failure and the policy-generated path. Four members complete their accounting; two remain partial. Every member has the same historical source, opening record, entity, book, currency and reporting periods. Opening claims, asset memory, drawn facility balance and debt principal must also agree. Future instructions may differ. These checks establish the common declared opening; they do not reconstruct an unobserved real history.

The text places an existing company path inside a named member block. Typed constructors embed the same path record. A family query returns each member's complete report vector for the requested year and retains its path and result identities. If a member never reached that year, its response says why the report is unavailable. The full family result also retains

the preceding and following years, so a later amount cannot be detached from the path that generated it.

These are authored exogenous scenarios. Six names imply neither equal probabilities nor exhaustive coverage of possible futures. Two declarations that produce identical results can remain separately named assumptions without acquiring extra probability mass. An empty authored selection means that the author supplied no members; it does not establish that no history is feasible. Historical reconstruction and justified probability laws retain their separate requirements from Chapter 32.

A policy expression has units and dependencies. The collection fraction can now be written as a calculation rather than supplied only as a final number. In the ordinary scenario it is one minus a declared discount of one tenth. A second branch supplies four fifths. Both branches use two equal instalments. The guard is an explicit Boolean assumption: it does not infer whether customers will behave normally.

The expression graph contains exact rational or Boolean inputs, literals, references and pure arithmetic, comparison and conditional operations. Text and typed construction preserve the same graph. Static analysis checks all references, operator arities, cycles and types, including both conditional arms. The resulting fraction and instalment shares pass through the existing settlement policy's domain and claim-allocation checks. The expressions do not create new collection dates or decide which contracts should exist.

Numeric types carry units and an explicit context. A quantity of goods times a price in dollars per good gives dollars; a quantity cannot be added to cash. Addition and comparison require matching dimensions and context. A context-free dimensionless fraction can scale a company amount. Multiplication does not silently convert currencies or combine different entities. A ratio of contextual amounts retains that context and cannot automatically become an unrestricted policy parameter.

For Northstar's two revised invoices, the total amount is 384. The ordinary fraction of nine tenths collects 345.6; four fifths collects 307.2. The latter therefore reduces cash by 38.4 and increases receivables by the same amount, with unchanged profit. The exact arithmetic agrees with the generated path's full accounting result. The example establishes the consequence of the two stated collection assumptions, without estimating which assumption is more likely.

A known guard evaluates only its selected arm, after both arms have passed static checks. An unknown guard leaves the policy output unresolved; it is not false. This bounded evaluator does not prove that equal-looking arms are symbolically equivalent. A valid but undefined calculation in an unselected arm is not evaluated, while division by known zero in a selected calculation is invalid. Missing declarations differ from declared unknown values. The former are invalid references; the latter remain visible dependencies.

Each output records all declared input dependencies and the subset actually evaluated. Thus an unselected branch's assumptions remain inspectable even when they do not determine the current value. Logical `and` and `or` conservatively propagate unknown operands in this subset; they do not introduce implicit short-circuit or SQL missing-value semantics. General functions, date arithmetic, source maps, richer types and endogenous event scheduling remain language work.

Reuse a calculation without changing its meaning. The collection example can express the ordinary fraction through a reusable function: given a dimensionless discount d, return $1 - d$. A second function selects the ordinary or stress fraction from the explicit Boolean assumption; a third returns the equal instalment share $1/2$. Calling these functions with the existing inputs

yields the same 9/10, 4/5 and 1/2 values as the earlier expression graph. Their reuse changes how an assumption is authored, not how it is estimated.

The optional `CompanyFunctions/1` declaration names each function's ordered, explicitly typed parameters, return type, local expression nodes and output. A call supplies argument-node identities in parameter order. Function bodies can read their parameters and literals; they cannot capture an undeclared company input. Expansion gives each call fresh local identities and lowers its operations to the existing expression evaluator. Parameter and return types must agree exactly, including units and company context. Every definition receives static checks, even if no selected output uses it.

Evaluation follows the existing graph's demand for values. An unused argument does not have to be evaluated, although its references and types must be valid. For example, a function that ignores one argument can return a known constant even when that argument contains a well-typed division by zero. If the result uses that division, it is invalid. Likewise, the stress collection branch can remain known when its unused discount is unknown. Each call retains all declared argument dependencies separately from the dependencies that reach its result; output evaluation records the inputs actually used.

The checked comparison preserves complete three-year results for the ordinary and stress policies. Moving from 9/10 to 4/5 reduces cash by 192/5 and raises receivables by 192/5, with unchanged profit. An unknown selected discount or guard prevents policy materialization. A negative generated fraction is rejected by the existing collection policy's domain checks. Compilation supplies neither a collection assumption nor a missing fact.

The bounded reference accepts at most 32 functions, 16 parameters per function and 256 source nodes per scope. Call depth is at most 16; both local and expanded dependency depth are at most 64. Validation and final expansion together permit at most 2,048 emitted nodes and 2,048 scope visits, including calls that merely pass an argument through. Serialized requests are limited to 262,144 UTF-8 bytes. These are explicit convenience limits, not measured scalability claims. Cycles, recursion and exceeded limits are refused; there is no implicit numerical solution or event scheduling.

The function request, compiler and generated legacy graph retain separate identities and definition/call origins. Their information evidence does not inherit the earlier policy's dates. This increment provides typed declarations; matched textual function syntax, parser spans, richer types and module dispatch remain further language work. Its 23 focused methods, 79 predecessor checks and 46 policy/accounting comparisons are retained in `data/accounting-master/company-functions-20261025/`.

Locate each declaration in the source that was supplied. The matched text extension now expresses these same bounded functions. For example, the ordinary collection fraction is written as follows:

```
function complement;
parameter discount '{"context":null,"dimensions":{},"kind":"number"}';
returns '{"context":null,"dimensions":{},"kind":"number"}';
param_ref d discount;
literal one '{"context":null,"dimensions":{},"kind":"number"}' 1/1;
node r sub one d;
result r;
end_function;
```

The quoted JSON type denotes a number with no dimensions and no company context. The parameter reference binds the local name `d` to the declared discount; `r` subtracts it from one. The result declaration identifies the returned node, and `end_function` closes the function. In the surrounding program, `call regular complement discount;` supplies the main program's discount

node and names the resulting call `regular`. Input declarations and the other functions remain necessary for the full collection policy. The displayed wrapping does not introduce extra source lines: each declaration occupies one physical line and ends in a semicolon.

Function parameters precede the return type and body; a result ends the body. Blocks cannot nest, but a call can name a function declared later. Main inputs, nodes and outputs use the earlier expression notation, within a `CompanyFunctions/1` program ending in `end_program`. The parser rejects duplicate names and JSON keys, incomplete blocks, unsupported declarations and semantic errors reported by the pinned typed compiler. Quoted command-looking characters are text: parsing performs no shell expansion, arithmetic evaluation or event generation. A validly typed division by zero can therefore compile and later fail when its value is requested.

Each accepted declaration records its path within the typed request, physical line and exact original UTF-8 byte interval. Byte offsets are zero-based and end-exclusive; line and Unicode-codepoint columns are one-based, with the ending column exclusive. The slice includes the terminal semicolon and excludes surrounding spaces, tabs and the line ending. Its hash and the complete source hash bind these locations to the supplied bytes. This distinction matters when non-ASCII names use several bytes per character, or a source uses CRLF rather than LF. Both line-ending forms are accepted without Unicode normalization.

Canonical rendering uses LF and deterministic quoting. It reproduces the same typed request but can have a different source hash and different locations. Consequently, original locations are never assigned to re-rendered text. The generated expression's function and node origins can be joined to these declarations; token-level debugging and runtime exception locations remain separate work. Source locations also supply no evidence of when a policy was written, published or used.

The text frontend bounds source size to 524,288 bytes, each physical line to 16,384 bytes and accepted declarations to 12,000 before the typed compiler's own limits apply. These are scoped engineering limits. Eighteen focused test methods, 23 typed-function regression methods, 79 expression checks and 285 source/financial comparisons cover six policy cases, preserving complete three-year results where available and unresolved or rejected paths elsewhere. The exact examples, declaration slices and financial comparisons are retained in `data/accounting-master/company-function-text-20261025/`. The checked extension still does not supply dates, polymorphism, module dispatch or endogenous economic decisions.

Let the next sale depend on the stock that remains. A reusable calculation becomes a company decision when it reads the current book and proposes an economic event. Consider a hypothetical company holding two units of a homogeneous, divisible inventory lot. The lot has a carrying amount and tax base of USD 20, and cash is USD 452. Management's authored rule keeps half a unit in reserve and sells the lesser of declared demand and stock above that reserve. Demand, the reserve and the price are scenario assumptions; the example constructor requires them to be nonnegative. The rule applies only when its separate Boolean switch is enabled.

With demand of one unit on each of three successive decision dates and a price of USD 15 per unit, the first decision sells one unit. The second reads the remaining unit and sells only half. The third reads the remaining half unit and proposes no sale. Total proceeds are USD 22.50, released inventory cost is USD 15 and profit before any subsequent tax assessment is USD 7.50. Cash becomes USD 474.50, while the retained half unit carries USD 5. Reusing the opening stock for all three decisions would give the wrong quantities. These exact fractions provide an independent check on the state read at each decision.

The bounded reference `StateDecisionEvent/1` separates this choice from accounting for its consequences. It receives the complete conditional annual book, its content hash, a decision

date, a typed function program and explicit input bindings. A binding can read the selected lot's current quantity or the book's cash balance, or supply a scenario value with provenance and an availability date. Quantity, currency and company context must agree with the program's declared types. Every program input starts as an unknown placeholder; the binding determines which value the rule may read.

A value dated after the decision, or with unknown availability, stays unknown for evaluation while its original record remains inspectable. The model itself also needs a declared availability date. These dates express the authored scenario; they do not establish when a real company knew a fact. The caller must further state that no intervening economic event is omitted between the snapshot and the decision. The reference neither verifies that premise against disclosures nor discovers event dates. Decision and sale share one calendar date within the current fiscal period; explicit order distinguishes decisions on the same day.

The program supplies three outputs: whether to act, the units sold and the proceeds. The compiler checks the entire program before evaluation. A false decision guard leaves the book exactly unchanged and need not read an unused future price. An unknown guard leaves the decision unresolved. Only a true guard requires the units and proceeds; unknown values remain unresolved, while invalid arithmetic refuses the proposal. This distinction prevents missing information from silently becoming a zero sale.

A known proposal still needs the existing sale method's six findings: proportional allocation selected, homogeneous units, divisible units, unchanged measurement basis, completed performance, and tax base equal to carrying amount. The decision rule supplies none of these findings by itself. Application recomputes the complete proposal against the same state and request, then dispatches it to the unchanged native sale transition. A stale snapshot or an altered proposal refuses even if the caller updates its hash. Unknown findings remain unresolved, false findings remain unsupported, and a refused proposal is retained separately from its accounting outcome.

In the checked single and repeated cases, the proposed operations, complete native transitions and financial statement fields equal those produced by manually authored sales. The report describes the recorded decision prefix and retains the annual interface's unresolved questions. A sequence stops at its first refusal, preserving earlier successful decisions, no-actions and remaining requests. Its decision cursor can advance through a no-action even though the book's last event date does not change; a later request cannot move behind that cursor. The reference accepts at most 64 decisions and 32 input bindings per decision. These are engineering limits, not a capacity claim.

Twenty-eight focused test methods, 14 native-accounting regression methods, 23 function regression methods and 64 exact comparisons across eight cases are retained in `data/accounting-master/state-decision-events-20261025/`. They check this composition of an authored rule and a conditional accounting method. The example leaves demand estimation, general financing decisions, simultaneous choices, event scheduling and actual historical execution open. No database commit, complete fiscal event population or forecast accuracy follows from the successful sequence.

Bind a scheduled decision when its date arrives. The preceding sequence fixes the state read by each decision, but its caller must supply the corresponding future book hashes. An authored simulation should instead supply its starting book and a calendar, then calculate those later states as it proceeds. The bounded `StateDecisionSchedule/1` extension does this by materializing each decision request only at its turn. Its expected state hash is the hash of the current book, after earlier successful events.

A schedule supplies a named rule, explicit priority and either a list of calendar dates or a start date, exclusive end date and positive integer number of days between occurrences. For example, daily decisions from 2 January 2027 up to, but excluding, 5 January produce the same three decisions as the explicit dates 2, 3 and 4 January. Calendar arithmetic includes leap days. It introduces no month-end adjustment, business calendar, timezone conversion or contractual day-count convention. All occurrences must fall within the current fiscal period and no earlier than the book's last event.

Different decisions on the same day need distinct explicit priorities. An unresolved tie stops scheduling before economic execution; rearranging the input array does not choose an order. Each decision and sale receives a stable identity derived from the declared root and schedule names and its occurrence number. The complete schedule also retains its own content identity, so an equivalent list of dates and recurrence can produce identical execution while remaining different authored documents. Every declared policy is statically checked before the first event, including a later policy that an earlier refusal might otherwise prevent the runner from reaching.

The checked calendar reproduces the manually assembled stock/reserve sequence: one unit sold, then half a unit, then no action. Cash is USD 474.50. The selected lot retains USD 5 of carrying amount, while the inventory control totals USD 65: the opening USD 80 also included USD 60 outside that lot. With a zero reserve, the same rule sells both units over two decisions and cash reaches USD 482; the next decision observes depletion and takes no action. Distinguishing a selected lot from the company's inventory control is necessary when checking the full balance sheet.

An unresolved or refused decision stops the sequence with the exact current book, earlier outcomes and pending calendar retained. The stop record includes the materialized request, its state hash and the proposed event where available. A no-action advances the decision cursor without changing the book's event date. The scheduler accepts up to 32 schedules and 64 total decisions; the existing request checker also bounds serialized size, nesting and structure. These limits keep the reference finite and do not establish production capacity.

Eighteen focused scheduler methods, 28 direct decision regression methods and 49 exact comparisons across five cases preserve manually assembled requests, complete native outcomes and all financial report fields. The evidence is in `data/accounting-master/state-decision-schedule-20261025/`. The calendar retains the explicit premise that no external economic event is omitted. Combining decisions with separately supplied financing, tax, purchase or period events requires those events' own timing and accounting rules. General calendar generation, historical completeness and durable restart remain separate work.

Declare the company book before applying its methods. The preceding native examples use a fixed company context and chart of accounts. A reusable calendar does not remove that restriction. A company may keep its cash under account `101`, another under `bank.alpha`; their contract records may also have different structures. The optional `CompanyBook/1` reference makes those choices explicit before a method reads balances or proposes an entry. It declares the entity, book, currency, conditional case, period, account classifications and selected cash account. Balances use the existing debit-positive convention and sum to zero. Each scalar in the method's memory has a declared type and an address within a closed schema.

The payment example separates the accounting relation from those local names. One synthetic company maps the cash and liability roles to `bank.alpha` and `note.alpha`, with the outstanding amount under `loan`. Another maps the roles to `101` and `230`, with the amount under `contracts/obligation`. Both start with cash of USD 100 and an outstanding liability of USD 30. A payment of USD 3.50 credits cash and debits the liability by the same amount, leaving cash of USD 96.50 and a liability of USD 26.50. The contract record falls to USD 26.50 and its

cumulative payments rise to USD 3.50. A second payment of USD 26.50 extinguishes the liability and leaves cash of USD 70. Other accounts and unrelated contract records are unchanged. The example assumes sufficient cash and a positive payment no larger than the outstanding amount; the generic book does not impose a universal funding policy.

`CompanyBookMethod/1` declares an injective mapping from account roles to the book's accounts, prerequisite predicates, monetary calculations, memory replacements and postconditions. Bindings select declared account balances, typed memory values or supplied event inputs. Their company, dimensions and scale must agree exactly with the method's input declarations. A dollar amount cannot become an inventory quantity merely because both are rational numbers. The existing monetary interpreter derives the journal and separate gross cash flows. The memory interpreter then replaces the specified scalar fields; this profile rejects dynamic paths, new collection members and container writes. Nested prerequisite and monetary definitions carry the enclosing method's identity and version. A literal memory replacement is interpreted in the destination's declared type.

`CompanyBookTransition/1` checks the entire request before assessing its prerequisites. It derives monetary effects, stages memory changes and constructs a new book from the same starting state. Only then does it evaluate the postconditions against that proposed book. In the payment case, a postcondition requires the credit balance of the liability account to equal the outstanding contract amount. A failed postcondition returns no applied book, although the staged calculation remains available for diagnosis. An unknown applicability finding remains unresolved; a false finding can make the chosen method unsupported. An inactive guarded prerequisite does not, by itself, cancel the economic event: a separate decision determines whether an event should occur.

The request identifies both the starting book and the complete method by digest. Application recomputes the proposal and compares its canonical JSON representation, including Boolean and numeric types. Changing a saved journal, memory value or authorization flag therefore cannot pass merely by replacing its digest. Event identities cannot repeat, dates cannot precede the current book, and each event must fall within the declared period. This is an atomic construction of a provisional value; the separate posting-authorization and durable-storage rules still govern any admitted or persisted transaction.

Five comparisons connect this reference to the retained native methods. Two current-tax payments and three complete-lot sales reproduce their native journals, gross flows and entire resulting states through explicit mappings. The mappings retain every native field outside the declared changes; both payment cases also reproduce the complete native financial reports. These checks concern the selected synthetic cases. General reporting and closing, source admission, dynamic contract collections, currency conversion and the broader corporate package still require their own methods. The declared book supplies a common place to compose those methods without weakening the older profiles' invariants.

Bind a named method to the event that actually occurred. The declared book separates company-specific accounts from a method's financial roles. Selecting that method also requires an unambiguous connection to the event. `CompanyMethodRegistry/1` names the available data-only method definitions, their schemas, expected account classifications and input types. Each entry identifies its complete method body. The optional `CompanyMethodCall/1` selects one entry under the book's company context and checks every entry's structure before evaluating the selected method's prerequisites. An invalid unused entry therefore cannot remain hidden until a later event happens to select it.

Some inputs are supplied facts; others describe the event itself. Keeping these sources distinct matters for a sale after a tax close. Suppose the event is dated 15 June, but a separately

supplied date says 1 September. A predicate comparing that supplied date with a 1 July close could accept the wrong timing. The named binder instead derives designated date and identity inputs from the event's own metadata. It rejects a supplied input with the same name. In the checked case, the 15 June sale consequently fails the closed-period prerequisite. A genuine change to the event date changes the predicate's input with it.

The binder materializes the ordinary company-book transition request, including the current state and selected method identities. The existing prerequisite, monetary and memory engines then determine the result. Two company payment layouts and a complete-lot sale reproduce their explicitly authored requests and entire provisional transitions; unknown and false applicability findings retain their different consequences. Repeated calls read the resulting book, so the next payment uses the reduced outstanding amount. Application recomputes the complete named call and uses the same type-sensitive JSON comparison as the underlying transition.

The current registry supports a directly dated event. Assessment intervals, measurement dates and period carry are different clock families and remain unsupported here. The registry also leaves native applicability to an explicit adapter. For example, a reusable inventory-sale adapter must validate all lot allocations, disposition and tax-base conditions, the selected sale method and the native close date, as well as preserve every unmapped field. Matching one native journal cannot establish those conditions. The selected comparisons establish the named binding mechanism; they do not turn the retained test projections into general native-method admission.

Carry the entire native book through the selected sale. A complete state check precedes this sale. The selected `NativeCompanySale/1` adapter extends the earlier test projection to the complete retained native whole-lot sale domain. It validates the native state's structure and accounting controls before importing every existing lot into a declared company book. The native account classifications and company context supply an explicit mapping. Fields outside the sale remain bound to the original state and are restored when the proposed book is exported. The adapter then validates the entire restored native state again.

Preserving every lot makes the comparison discriminating. In a two-lot example, the inventory account holds USD 20: USD 15 belongs to the selected lot and USD 5 to a second lot. Selling the selected lot for USD 40 raises cash from USD 454 to USD 494, reduces inventory to USD 5 and raises cost of sales from USD 56 to USD 71. The second lot is unchanged. Selling it later for USD 8 leaves cash of USD 502 and no inventory. The declared method reconciles the inventory account with the carrying amounts of all existing lots and other inventory before and after each sale; it replaces only the selected lot's units, carrying amount, tax base and disposition fields.

The generic monetary and memory interpreters compute these changes through the named method binder. The native transition remains the comparison calculation: a test disables it while the adapter still proposes and applies the sale. Across 23 accepted and refused cases, complete native states, journals, gross cash flows and refusal reasons agree. Unknown performance or proceeds remain unresolved; false performance and a fractional sale remain unsupported. Native cash-funding failure, duplicate identities, backdating, tax-close restrictions and inconsistent lot or control balances retain their native consequences.

This adapter preserves the fixed native profile's domain rather than inferring accounting applicability from the general book's type checks. It creates no new lot and supports only a complete-lot sale. Source admission, partial disposals, new collection members and a general company's full reporting package remain separate work. Its contribution is a checked route through the common book that preserves the native conditions and every field outside the selected transaction.

Disclosed cash constrains a history without identifying it. Importing a verified native book begins with supplied account and transaction facts. A public cash-flow statement supplies a different kind of evidence. The retained Apple annual statement reports opening cash of USD 30,737 million, a net decrease of USD 794 million and closing cash of USD 29,943 million. Those three figures constrain a possible movement history, but they do not identify its gross receipts, payments, transaction dates or legal accounts. The source record therefore remains distinct from a modeller's proposed explanation of the movement.

Consider two deliberately hypothetical borrowing and repayment histories. Let O and C denote opening and closing cash, and let B and P denote cash borrowing and principal repayment within the annual period, all in USD millions. Assume that these are the only cash movements in the example. The resulting cash relation is

$$C = O + B - P, \qquad 29{,}943 = 30{,}737 + 0 - 794 = 30{,}737 + 100 - 894.$$

The second history has an additional borrowing of 100 and an additional repayment of 100. It reaches the same closing cash while preserving a different gross event sequence. Both examples start with an assumed debt of 1,000 and balancing capital; the event dates and finite amount bounds are also supplied assumptions. These are counterexamples to identification from the selected cash figures, not disclosures of Apple's borrowing or histories shown to fit its other statements. The selected source model additionally assumes that unrounded values satisfy the cash bridge and uses the earlier proposed nearest-million rounding rule. That rule is not silently attributed to the issuer.

The optional `RetainedSourceHistoryAttachment/1` makes this distinction operational. Its request supplies the complete retained statement request, a typed history for each selected cash bridge, and explicit mappings from source quantities to history fields. It recomputes the source report, retains all 256 observations and 143 relation dispositions, and constructs each attachment from the original source identities, dates, monetary scale and basis context. The existing history and measurement methods then check both complete constraint systems and every supplied mapping. A history that fixes both annual cash movements to zero is feasible by itself but fails when attached to the annual decrease. Missing rounding or mapping premises remain unresolved; an exhausted computation budget yields an unknown result rather than an infeasibility conclusion.

The interim bridge illustrates a second distinction. Its opening 29,943, increase 356 and closing 30,299 permit hypothetical borrowing and repayment pairs $(356, 0)$ and $(456, 100)$. The annual closing and interim opening share a date and displayed amount, but their retained basis contexts differ. The two attachments consequently remain separate: equality of the displayed boundary cannot authorize a common cross-period history. Every candidate also retains the unresolved actual-transaction and unselected-disclosure obligations. A selected-cash witness therefore supplies no admitted legal book, cash-flow classification or compatibility conclusion for the whole filing.

Eleven complete exchanges and their semantic replay check this composition, including alternative exact witnesses, changed sources, missing mappings, incompatible contexts and unresolved premises. Fifteen focused tests and 65 existing source/history tests pass. The implementation and full exchanges are retained under the following path, split for legibility:

```
data/accounting-master/
retained-source-history-20261025/
```

The next step for a possible company history is to retain these source conditions when selecting its opening and continuation.

Another statement can reject a cash-compatible history. The preceding borrowing examples leave the other disclosed quantities unconstrained. That omission matters: mapping their assumed capital balance to the reported closing equity rejects the annual candidate, although the cash-history system and the balance-statement system are each feasible by themselves. Compatibility must be checked on their intersection, using one shared history, rather than inferred from separate successful calculations.

The optional `RetainedStatementHistoryAttachment/1` adds selected balance equations to the same history-and-cash constraints. It preserves the original source coefficients and contexts, moves the snapshot date into the explicit time type, and requires an explicit signed field mapping. All selected equations remain even when some operands have no history mapping; those unmapped quantities create unresolved obligations. The complete source report and both parent assessments remain attached. A different basis, monetary scale or quantity cannot be substituted because two amounts happen to agree.

In each annual and interim example, 27 reported closing-balance quantities join the three cash observations through ten selected relations. A constructed compatible history uses one signed account for each disclosed leaf caption and explicit sums for totals. It assumes unchanged noncash balances and specifies opening capital by reversing a hypothetical net capital contribution or return from the reported closing caption. These assumptions explain the construction; they are not recovered opening facts. Two gross contribution-and-return sequences can again fit the same selected closing values. Fourteen complete exchanges, 13 focused tests and 62 existing tests check the joint constraints, conflicting and unresolved cases, and exact replay. The complete exchanges are retained in

```
data/accounting-master/
retained-statement-history-20261025/
```

The reported opening balance sheet and equity movements provide further constraints that these examples have not yet used. They can reject the assumed opening or event sequence even after every selected closing caption agrees. The remaining task is therefore to attach those reported movements, while continuing to distinguish a conditional witness from actual history.

Reported openings test the assumed starting point. The retained statement network already expresses opening balances, equity roll-forwards, period totals and supported links between statements. The optional `RetainedNetworkHistoryAttachment/1` attaches that complete selected network to supplied histories, separating them by exact value context. Every selected observation and relation remains in one attachment. An unmapped quantity remains in the source equations and creates an explicit unresolved obligation; it is not silently removed from the calculation.

Adding reported opening balances rejects both of the constructed annual and interim histories. Adding the annual equity bridges also rejects the assumed annual history. The source measurements remain jointly feasible: the failure concerns the supplied history and its mappings. In particular, the earlier opening capital was constructed by reversing a hypothetical cash event, whereas the filing reports its own opening capital. Agreement with the closing balance sheet cannot replace this additional evidence.

Cash reveals a population mismatch as well. The retained annual opening balance-sheet caption is USD 29,965 million, while the cash-flow statement starts from USD 30,737 million. The latter label includes restricted cash and cash equivalents. Their difference, USD 772 million, is therefore not a cash-flow event and has not been identified as a legal account. Nor can all restricted cash simply be added to the balance-sheet caption: the same filing's financial-instruments note says that the 2024 cash-and-equivalents balance already includes some restricted escrow amounts. A

restriction is a classification that can overlap the balance-sheet population. The primary labels and note are visible in the retained annual PDF on PDF pages 36 and 40. Establishing the exact population relation precedes repairing the history.

The original PDF-only full network retains 236 observations and 101 relations, but remains unresolved because a reported dash appears in annual debt issuance. Section A.13.4 gives the separately selected numerical reading and full-network result without changing this original source record. An explicitly narrower selection has 229 observations and 100 relations and names the financing relation it excludes. Its conditional feasibility does not recover the omitted meaning or establish a whole-filing history. Eleven complete exchanges, ten focused tests and 44 existing tests check these distinctions, the opening conflicts and exact replay. The source network's unknowns remain present in the history result; they do not disappear when the selected mathematical constraints are feasible.

```
data/accounting-master/
retained-network-history-20261025/
```

Cash classifications can overlap without revealing their members. The source notes make the overlap concrete. In the retained 2024 annual filing, Note 7 on PDF page 42 (printed page 39) divides the USD 15.8 billion State Aid escrow into USD 2.6 billion classified as cash and cash equivalents and USD 13.2 billion classified as current marketable securities. Note 4 on PDF page 40 (printed page 37) also states that the restricted cash is included in cash and cash equivalents. Restricted status therefore cannot define a disjoint addition to that balance-sheet caption. The corresponding tagged cash disclosure retains its State Aid contingency dimension; it is not an unqualified total of every restriction.

To express what the two cash totals can tell us, let B denote the population under the balance-sheet caption and F the population under the cash-flow caption, at one date. Assume for this comparison that both use the same nonnegative additive monetary valuation. Write b for the value belonging only to B, f for the value belonging only to F, and i for the value in their intersection. These are aggregate values of membership classes, not identified bank accounts. If C_B and C_F are the two measured totals, the disjoint decomposition gives

$$C_B = b + i, \qquad C_F = f + i, \qquad C_F - C_B = f - b, \qquad b, f, i \geq 0.$$

The common value cancels on subtraction. Treating the 2023 displayed amounts as exact solely for this illustration, the difference of USD 772 million therefore fixes $f - b$. Both $(b, f, i) = (0, 772, 29965)$ and $(100, 872, 29865)$ satisfy the disclosed totals. An additional assumption $B \subseteq F$ forces $b = 0$ and hence $f = 772$ under that valuation, but the subtraction itself supplies no such membership evidence and does not identify the financial nature of f.

Equal totals do not settle the matter either. At the 2024 displayed amount of USD 29,943 million, both $(0, 0, 29943)$ and $(100, 100, 29843)$ satisfy the two equations. Moreover, even zero exclusive value does not prove that the exclusive set is empty when zero-value members are possible. The monetary constraints preserve this distinction between equal values and equal members. The 2023 annual filing independently displays the same earlier cash totals; that cross-edition agreement does not identify the missing members. No exact classification of the USD 772 million difference was located in the bounded inspection of the retained annual statements, cash notes and relevant tagged calculation arcs.

The optional `ConditionalPopulationOverlap/1` lowers two through four declared populations to the existing joint-measurement engine. It retains every nonempty Boolean membership class, the original source dimensions, explicit bounds and proposed projection premises. It requires a common additive valuation; signed or differently measured quantities need a separate

mapping. Bounds derived from displayed precision also remain proposed. For example, a lower-closed, upper-open nearest-rounding interpretation places the USD 2.6 billion cash disclosure in $[2550, 2650)$ million; it does not turn the displayed number into an exact underlying balance.

Under that rounded example, requiring the disclosed escrow cash both to belong to B and to be disjoint from B is infeasible. This rejects the proposed classification, not the filing. Eleven focused tests, 48 existing regression tests, ten complete exchanges and four separately checked rational witnesses cover the feasible alternatives, incompatible classifications, missing premises, context mismatches, resource limits and forged results. The implementation preserves unresolved membership and supplies no account postings or executable company opening. The evidence and exact source ledger are retained at

```
data/accounting-master/
cash-population-reconciliation-20261025/
population-overlap-20261025/
```

The overlap must also satisfy the original statement network. Separate feasible calculations are not enough when they share reported quantities. The retained-source composition attaches the region equations to the original observation identities at their exact instants. It preserves all 114 selected observations and 38 relations, including their unchanged units, contexts, quantities, intervals and rounding premises. Six new regions and four same-time equations yield a joint system of 120 observations and 42 relations. The complete source report, with 256 observations and 143 relations, remains attached together with the dispositions of unselected or unsupported content.

Two independently checked assignments change the exclusive and common region values while preserving every original measurement and equation. Forcing the 2023 opening populations to be equal is infeasible under their distinct retained rounding intervals. A cross-date or cross-context comparison is refused; a missing projection premise remains unresolved. Selecting the full supported network under the PDF-only interpretation still preserves the debt-issuance dash and its unresolved meaning. The separate source reading in Section A.13.4 changes that selected interpretation, not the cash-population identities or these preserved results. Six focused tests, 34 regression tests and eight complete exchanges check these outcomes and tampering. This is a composition of the existing temporal measurement engine, not a new cash-account reconstruction method. The evidence is retained in

```
data/accounting-master/
retained-population-overlap-20261025/
```

The resulting temporal constraints can be supplied to the existing history attachment, but they do not identify which real accounts or events should receive those constraints. That distinction remains essential when a feasible history is used to construct a conditional opening.

The selected numerical source reading also constrains the two supplied histories through the explicit composition in Section A.13.5. Their 180 unresolved obligations, separate contexts and monetary scale remain attached. The conditional-opening interface below still requires its own complete inputs and premises.

A conditional history gives a conditional opening. Importing a known book and selecting a possible history answer different questions. Suppose the disclosed constraints permit closing cash of USD 15. One compatible synthetic history starts with USD 10 and adds sales of USD 2 and USD 3; another starts with USD 8 and adds sales of USD 3 and USD 4. Both reach the same cash balance, yet their cumulative-sales memory is USD 5 and USD 7 respectively. A

continuation that reads that memory must retain which history supplied it. Equal closing statements do not erase the difference.

`ConditionalCompanyOpening/1` accepts an explicit witness for the typed history and measurement attachment. It replays the complete supplied constraints before constructing an opening. Every account maps once to a uniquely identified account with an explicitly declared classification; every numeric memory field maps once to a nonoverlapping path. Signed amounts, dimensions and scale are preserved. Missing or duplicate mappings refuse rather than creating an unallocated balancing amount. In the example, both openings contain cash of USD 15 and equity credit of USD 15, while their separate sales memories remain USD 5 and USD 7. A checked inventory example similarly retains one physical unit separately from inventory of USD 2 and cash of USD 21.

The boundary between the representations needs explicit premises. This first adapter selects the final declared end-of-day snapshot and starts the book on the next day, assuming no unrecorded event across that boundary. Entity and book must agree, and monetary amounts must already use the same currency at unit scale. The target context is explicitly synthetic. The source accounting-basis context and the stated mapping premise remain attached to it; changing the representation does not establish a basis conversion. Income and expense accounts must already have zero opening balances. Otherwise the adapter refuses: it does not invent fiscal closing entries.

The returned provenance retains the original history-and-measurement definition, selected witness, source snapshot, mapping premises, exclusions and unresolved obligations. Missing premises yield an unresolved result. Remaining target obligations or unmapped observations stay visible in a candidate opening. A valid opening balance report checks the mapped book; it does not admit the history or authorize a financial continuation. No investor information cutoff or forecast admissibility has been checked.

This distinction matters for receivables. An aggregate balance can be mapped exactly without supplying invoice identities, counterparties or due dates. Those facts must come from an explicit declared or source-supported contract population before an identified collection method can use them. The adapter therefore provides a checked conditional starting state while preserving the remaining method and contract obligations. Other compatible histories remain represented by the unchanged parent definition; this selected witness neither enumerates them nor assigns them probabilities.

Keep the selected history attached when the book changes. The conditional opening can support a further explicitly assumed transaction, provided that the method's premises are supplied and no target obligation remains unresolved in this bounded composition. The `ConditionalCompanyContinuation/1` wrapper recomputes the complete opening result and compares it with the supplied result before invoking the ordinary book transition. A stale state, altered witness record or different company context refuses. Missing opening premises or remaining target obligations return an unresolved result. The wrapper retains the complete opening evidence alongside the financial result; a copied provenance label alone would not establish that connection.

For the two preceding histories, assume a cash distribution of USD 1 on 2 January 2025. This synthetic method checks the explicitly selected applicability premise, a positive amount and sufficient cash. It credits cash and debits the declared equity account by USD 1, leaving cash of USD 14 and an equity credit of USD 14. The financing cash flow is minus USD 1 and profit is zero. In the statement reconciliation, the direct movement in reported equity is minus USD 1, although the ledger debit is positive. The common report rejects an explanation with the opposite sign. Neither history's sales memory changes: USD 5 and USD 7 still distinguish their

pasts.

The checked wrapper agrees exactly with the unchanged transition and report under both account charts. Unknown event inputs retain their reasons, false applicability remains unsupported, and insufficient cash refuses without changing the supplied book. This is a declared financial continuation of a selected conditional history. It does not establish lawful distributions, distributable reserves, accounting compliance or admitted execution. A future method that depends on unresolved invoice terms needs those terms explicitly; the successful distribution example cannot supply them.

Report the declared accounts and explain their movements. A common book also needs a reporting method that respects each company's chart. `CompanyBookReport/1` takes opening and closing books for the same company, period and account population, together with a declared statement map. Every account belongs to exactly one line of the same classification: asset, liability, equity, income or expense. A missing account, overlapping line or conflicting classification prevents the report. Different memory schemas can describe the opening and closing books, since a period may create a new lot; each book is validated independently, and reporting does not establish that memory migration.

The report follows the book's debit-positive convention. Assets and expenses use their stored signs; liabilities, direct equity and income reverse them. Profit equals income less expenses, and reported equity includes that profit. Contra balances remain signed. The first profile requires an opening book at the period start with closed income and expense accounts and no current-period cash flows. It therefore describes a recorded period, rather than an arbitrary comparison between two accumulated income statements.

Cash flows preserve each method's declared activity and origin. For operating, investing and financing activities separately, the report sums receipts and payments before computing the net movement. The aggregate must equal the change in the selected cash account. Each flow must refer to an event recorded after the opening history; a prior event cannot be reused to explain current-period cash. Direct equity movements require explicit account, event, date, amount and description. They must explain every change in the direct equity accounts. Profit enters the equity reconciliation separately, so a missing contribution or distribution cannot disappear into an unexplained residual.

An authored sale and contribution illustrate all four statements. Opening cash is USD 100, inventory USD 20, another asset USD 10, liabilities USD 30 and equity USD 100. Selling inventory with carrying amount USD 15 for USD 40 produces USD 25 profit. A subsequent USD 5 owner contribution brings cash to USD 145. Closing assets of USD 160 equal liabilities of USD 30 plus equity of USD 130; opening equity of USD 100, the USD 5 contribution and USD 25 profit explain that closing equity. Operating cash receipts are USD 40 and financing receipts USD 5. The same quantities reconcile under two distinct company charts. In a compatible retained annual case, every asset, liability, income and expense line and all direct cash-flow activities agree with the native report: profit is USD 49.50, cash USD 491.50 and equity USD 671.50.

These equalities establish the recorded amounts' consistency. They cannot reveal omitted offsetting events or establish the economic cause of a supplied equity explanation. The report therefore retains its recorded-period scope and never claims fiscal completeness. Note requirements remain explicitly missing, unsupported or unresolved; this profile cannot declare a note package complete. The indirect operating-cash bridge, source and accounting admission, comparative presentation and explicit closing and carry methods remain separate obligations.

Close the recorded balances under explicit carry policies. The period report leaves profit in the temporary income and expense accounts. `CompanyBookCarry/1` transfers those recorded balances to a named equity account and constructs the next provisional opening. The caller identifies that account explicitly; an account label cannot select it. Each temporary account receives the negative of its closing balance, while the selected equity account receives their sum. With debit-positive storage, profit therefore produces an additional equity credit and a loss an equity debit. The journal sums to zero, and cash and the other permanent accounts remain unchanged.

In the preceding company example, profit of USD 25 is transferred to equity, leaving USD 130 of opening equity for the next period and zero income and expense balances. An otherwise analogous USD 3 loss reduces equity by USD 3. If the company names a separate retained-income account, the contribution account remains unchanged. Zero profit produces no net retained-equity transfer, although offsetting income and expense balances still require closing entries. The next opening report has no current-period profit and reconciles to the preceding closing equity.

The method recomputes the period report and requires the closing snapshot to be dated the last included day before the period's exclusive endpoint. The next period starts at that endpoint and ends at an explicitly supplied later date; calendar years are not assumed. This date rule does not establish that the event record is complete. The method carries recorded balances provisionally and keeps fiscal-completeness and accounting-admission flags false.

Every typed memory leaf needs its own declared policy: preserve its value, reset it to an explicitly typed literal, or, for a date leaf, set it to the next period's start. Each policy records its rationale. Missing policies and type mismatches prevent carry. The engine does not silently reset payment counters, advance tax clocks or infer an opening tax basis. Such choices can be type-correct yet substantively wrong, so their provenance remains part of the proposed result. The existing scalar memory interpreter applies the declared replacements.

The next book clears prior-period cash flows but retains lifetime event identities and adds a new close identity. A returned archive record links the prior book, report, closing journal result, next opening and memory policies by digest. Application recomputes the full proposal. This is a returned provisional history record, not a durable storage commit. Comparisons reproduce the retained native carry's balances, period and imported lot memory; its other tax, interest and archive conventions remain separate native rules. Consecutive-period tests retain earlier event and close identities rather than making them available for reuse.

Run the declared company through successive periods. A company model needs to use these methods in one sequence, with each operation reading the book produced by its predecessors. The finite `DeclaredCompanyProgram/1` combines an opening book, named method registry, statement map and an authored list of events, reports and provisional carries. An event names its method and supplies dated, typed inputs. A report names its cutoff, direct equity explanations and unresolved note requirements. A carry also names the retained-equity account, next period endpoint and every memory policy. The program constructs each method request from the current book at that turn; future balance hashes and expected journals are not authored inputs.

An interim report can follow the last supplied transaction by several months. Its explicit cutoff advances only the observation date of a provisional copy; it creates no journal, cash flow or memory change. The trace records that advance, and a failed report leaves the preceding book unchanged. Reaching December's last day does not prove that all December events were supplied. Reports continue to use the actual opening of their current fiscal period. After a successful carry, its returned next opening becomes that basis; earlier income and cash flows do

not enter the new period's report. Supplied equity explanations and note requirements belong to each report rather than being copied silently across years.

Consider the preceding declared book before its illustrative owner contribution: cash is USD 100, inventory USD 20, another asset USD 10, liabilities USD 30 and equity USD 100. A supplied sale for USD 40 releases USD 15 of inventory and earns USD 25. After an interim report, paying USD 3.50 of principal leaves year-end cash USD 136.50, inventory USD 5, liabilities USD 26.50 and equity USD 125. Carry closes the temporary accounts. In the next year, selling the remaining inventory for USD 8 earns USD 3; paying the remaining USD 26.50 leaves cash USD 118, no inventory or liability, the other asset USD 10 and equity USD 128. A second carry retains the lifetime payment total of USD 30 and all event and close identities. These exact amounts, every constructed request and every engine result agree with separately written calls under two different account charts. The sale's aggregate carrying cost and applicability finding are supplied assumptions; this example does not infer a lot valuation or tax treatment.

The bounded program checks all command shapes and method definitions before execution, but evaluates economic prerequisites against the book at their actual turn. Same-day operations follow their explicit sequence positions. An unknown finding, false prerequisite, invalid date, repeated identity or failed carry returns the exact refused command alongside the entire successful prefix. Report and carry changes become visible together only after that command succeeds. Canonical replay recomputes the complete result, including report and archive references; changing an input or substituting a result prevents verification. The initial limits of 64 commands and an 8 MiB trace bound execution for this reference. Exceeding the trace limit retains the prefix and identifies the oversized result by digest; it makes no scalability or durable-recovery claim.

Author the sequence while retaining its declared definitions. The matched `DeclaredCompanyText/1` frontend expresses those same commands as text. Four separately checked content hashes identify the opening book, schemas, method registry and statement map. These definitions remain imported data; the following language authors their operation sequence. For example, after its program header and four import declarations, a sale can be written as:

```
event "sale.2027" method "sell" on "2027-02-01";
  input "proceeds" known "40/1" because "Synthetic assumption";
  input "cost" known "15/1" because "Synthetic assumption";
  input "approved" known true because "Synthetic assumption";
end_event;
```

Each declaration occupies one physical line. Each amount's type comes from the selected registry entry, including its company context, currency and units. A finding can instead be explicitly unknown, with its reason and provenance; compilation preserves it for the method to evaluate at the actual command.

A report command supplies its cutoff, direct-equity explanations and note requirements. A carry adds its retained-equity account, next period endpoint and one policy for each memory leaf. Preserve, typed reset and next-period-start policies retain the same meanings as their data-record counterparts. The compiler resolves reset types from the declared memory schema rather than guessing units. Missing policy coverage remains a carry refusal, so earlier successful commands remain visible. Repeated event identities and invalid chronology likewise retain the checked engine's exact refusal and preceding book.

Compilation verifies every imported hash and returns the resolved program, dependency identities, source hash and physical declaration locations. Unknown fields, repeated declarations,

malformed scalar values and unchecked imports are rejected. It neither evaluates Python nor treats a serialized object as a command. The two company charts reproduce their full data programs and execution results; an independently authored sale and interim report also recover cash USD 140, profit USD 25 and equity USD 125. Complete and stopped programs are compared, including unknown findings, missing carry policies and changed ordering. These checks establish the finite frontend's correspondence; authoring complete method algorithms, broader accounting policies and all company disclosures remains work for the wider language design.

Follow purchasing, sales and wages through one book. The next declared methods connect six ordinary operations: purchasing inventory on credit, delivering it on credit, collecting receivables, paying suppliers, accruing wages and paying those wages. Each method is a data definition interpreted by the same checked requirements, monetary and memory engines. An explicit role map identifies the accounts. Quantity and unit cost carry different dimensions, so their product is a currency amount; substituting a money input for units fails the type check. The selected synthetic premise is a positive, constant inventory unit cost stored in company memory. It does not establish an inventory valuation policy for heterogeneous lots.

Purchasing increases inventory and supplier payables without moving cash. Delivery increases receivables and revenue, while transferring its cost from inventory to cost of sales. Collection reduces receivables and increases cash. Supplier and wage payments reduce their respective liabilities and cash; wage accrual creates the expense and payable before payment. The methods check available inventory, receivables, obligations and cash at the event's actual turn. They also retain an explicit applicability finding: unknown remains unresolved, and false remains unsupported. Other accounts and all typed memory remain unchanged.

Start with cash USD 100, inventory USD 20 and equity USD 120, at a unit cost of USD 5. Purchasing six units creates inventory and payables of USD 30. Delivering five units for USD 50 creates that receivable and releases USD 25 of inventory. Collect USD 35, pay the supplier USD 20, accrue wages of USD 10 and pay USD 8. Cash ends at USD 107, receivables at USD 15 and inventory at USD 25. Supplier payables of USD 10 and wage payables of USD 2 leave equity of USD 135, including profit of USD 15. Assets of USD 147 equal liabilities of USD 12 plus equity. Operating cash of USD 7 also reconciles to profit less the USD 15 receivable increase and USD 5 inventory increase, plus the two payable increases of USD 10 and USD 2. This bridge explains the supplied cycle; it does not select a general indirect cash-flow policy.

Every journal, complete account balance and cash movement agrees with separate calls to the retained `DemoCorporate/1` methods under the same premises. A second chart produces identical financial quantities through its own explicit mapping, and the text frontend reproduces the data program. Additional checks retain nonzero plant, accumulated depreciation and debt while these operations run. This is monetary conformance for the selected methods, not a conversion of all native asset, history, tax or information-selection semantics.

A payment exceeding available cash preserves the preceding book. The current requirement interpreter returns a primitive invalid result for its failed cash condition. A separate diagnostic identifies an isolated liquidity shortage as an economic funding failure and retains that exact underlying result; another failed condition is never hidden by that explanation. A general typed default outcome remains a separate execution obligation. Maturity, ageing, impairment, tax and disclosure completeness also remain unresolved until their own facts and methods are supplied. The next step is to derive explicit working-capital roll-forwards from these verified events, rather than treating balanced totals as complete notes.

Explain working-capital movements from the verified history. The operating cycle now supports four reconciled movement schedules rather than only its closing totals. `DeclaredWorkingCapital/1` first recomputes the complete program result and selects one successful report, including the report inside a provisional carry. Its actual opening book defines the period. Earlier interim reports add no activity; a preceding carry establishes a new opening. If execution later stops, an earlier successful report can still be selected, while the failed command cannot stand in for a report.

For the illustrative cycle, receivables open at zero, increase by USD 50 of credit sales and fall by USD 35 of collections, closing at USD 15. Inventory opens at USD 20, receives USD 30 of purchases and releases USD 25 to cost of sales, closing at USD 25. Supplier payables increase from zero by USD 30 and fall by USD 20; wage payables increase by USD 10 and fall by USD 8. Their closing amounts are USD 10 and USD 2. Every movement retains its event identity, date, typed inputs, provenance, exact method definition and verified request/result references. Liabilities use positive presentation even though the book stores credits as negative balances. Purchases and deliveries also retain the supplied quantities and the declared unit-cost premise.

Classification depends on the complete checked method definition, not its registry nickname. Renaming a method key therefore preserves its meaning; changing the method cannot masquerade as the old operation. An unsupported movement in a scoped account prevents the schedule, even if another event offsets it. Gross posting checks also detect an unsupported debit and credit that cancel within one event. That last case exposed and repaired a first-draft error: inspecting only the net journal had hidden the movement. The preserved counterexample distinguishes the repair from a claim based only on successful ordinary examples.

After carry, collecting the remaining USD 15 and paying the remaining USD 10 of supplier balances produces the next period's movements, with no repetition of prior sales or purchases. Inventory remains USD 25 and wage payables USD 2. The four working-capital changes explain a USD 5 cash contribution. In general, this contribution is only part of an indirect cash reconciliation: a separate noncash expense can leave a residual between profit, these four changes and direct operating cash. The schedule displays that residual instead of forcing it to zero.

These schedules account for recorded movements. They retain unresolved receivable ageing and impairment, inventory valuation and quantity coverage, maturity and payment terms, related parties, tax and source-population completeness. An exact roll-forward supplies useful note evidence, but cannot make those absent disclosures complete. Both note-completeness and fiscal-completeness indicators remain false.

Accrue interest before changing the principal it prices. The same company book can now join its operating cycle to borrowing, repayment, interest, capital contributions and dividends. The illustrative borrowing has a fixed annual simple rate of 7.3 percent and an explicit actual-days/365 convention. These are declared synthetic terms. They provide an exact timing example; fees, effective-interest valuation, compounding, contractual maturity and source admission require their own methods.

Interest for each declared interval is principal times the annual rate times its calendar days divided by 365. A date stored in company memory marks the first day not yet accrued. Each interval starts at that date and ends exclusively at its declared boundary. The event occurs at the end of the preceding calendar day and advances the stored date to the boundary. Principal changes occur at the beginning of their event date and require that same date in memory. Thus a repayment cannot change the principal used to price an earlier, unaccrued interval. A second memory value must equal signed ledger debt; an unexplained principal posting prevents the next financing method.

Issue USD 100 on 1 January 2027. The 90 days to 1 April accrue $100 \times 0.073 \times 90/365 = 1.8$ dollars of interest. At that boundary, repay USD 40 and borrow USD 20: principal becomes USD 80. The next 91 days accrue USD 1.456, and the remaining 184 days accrue USD 2.944. Total interest is USD 6.2. The stored date reaches 1 January 2028, covering all 365 days. Repeating or skipping an interval, changing its event date, or changing principal before the preceding days have accrued is refused. The convention counts a leap day as another day while retaining the stated denominator of 365.

The method builder calculates each fraction from the exact dates and embeds it in the typed definition. The caller supplies neither an interest amount nor an unchecked year fraction. Rate and elapsed years have inverse dimensions, leaving a currency amount when multiplied by principal. Each interval remains one finite declared method; this is not yet a general instrument calendar. A report or provisional carry does not itself accrue omitted interest. Their balanced totals therefore remain separate from whether the stored date covers the reporting cutoff.

Keep financing, profit and direct equity distinct. Debt receipts and repayments change cash and principal. Interest accrual changes expense and interest payable; payment reduces that payable and cash. A capital contribution uses supplied shares times price. Dividend declaration reduces retained equity and creates a payable; payment settles it. The contribution and declaration need explicit explanations in the equity statement. A supplied applicability finding remains unresolved when unknown and unsupported when false. It does not establish a legal right to distribute retained earnings. Payments cannot exceed the recorded obligation or available cash.

Combine these financing events with the preceding six operating events. Add a USD 30 contribution, declare USD 12 of dividends, pay USD 5 of those dividends and USD 4 of interest. Cash closes at USD 208, receivables at USD 15 and inventory at USD 25. Debt is USD 80, unpaid interest USD 2.2 and unpaid dividends USD 7, alongside supplier and payroll payables of USD 12. Profit is USD 8.8 and equity USD 146.8: assets of USD 248 equal liabilities of USD 101.2 plus that equity. With interest payments explicitly classified as operating, operating cash is USD 3 and financing cash USD 105. Selecting the supported financing classification for interest changes the category, while preserving total cash and the expense.

Every monetary transition is compared with the retained corporate engine under the same premises. Separate calculations check the amounts and date coverage; the older engine's acceptance alone cannot establish timing because it accepts a supplied year fraction. Two charts and matched text produce the same financial results. Unpaid interest and dividends remain visible across provisional carry; maturity, valuation, outstanding-share reconciliation, lawful distributions and complete disclosures remain unresolved.

Compose bounded definitions without weakening each method's checks. The combined registry exceeded the original record's structural limit before any event ran. The successor `CompanyMethodRegistry/2` checks each method with the unchanged earlier materializer, including unused methods. Its enclosing record retains the one-megabyte input limit and the 32-entry limit. Each method retains its own depth, scalar and 4,096-node bounds. The complete registry identity remains bound to every call; selecting a valid method cannot conceal a malformed unused definition.

`DeclaredCompanyProgram/2` applies the same component-by-component validation to the opening, schemas, registry, statement map and commands. Its 64-command and 8 MiB trace bounds remain. `DeclaredCompanyText/2` uses the same command grammar and exact hashed imports with the successor registry and runner. Compatible smaller programs retain the complete monetary transition and report results; only the declared composition version and its provenance change.

This permits the demonstrated common book while leaving scalability and durable execution for separate validation.

Reconcile financing movements and show the dates still covered. A verified company trace can explain the financing balances at the same time as it explains the operating cycle. ‘DeclaredFinancingSchedules/1‘ first recomputes the v2 program and chooses a successful report. It then rebuilds the declared financing definitions from the exact policy, intervals and role map. Registry names do not classify an event; the method definition hash does.

The schedule records debt issues and repayments, interest accruals and payments, capital contributions, dividend declarations and dividend payments. Each row retains its event date, typed inputs, method, request and result hashes. Signed ledger credits are presented as positive liabilities. Opening plus increases less decreases equals the selected closing balance for debt, interest payable, capital, retained direct movements and dividends.

Interest coverage is a separate result. In the example, the cursor starts on 1 January 2027 and advances through intervals of 90, 91 and 184 days to 1 January 2028. The three rows therefore cover 365 days. A balanced report with a cursor that stops earlier remains a valid provisional balance report with an explicit accrual gap. A carry cannot silently complete the missing suffix. Unknown methods that change a scoped account, tracked debt or cursor refuse classification, even when their net journal is zero.

The four operating schedules remain compatible with the v2 program. Their partial cash bridge leaves the unpaid-interest residual visible under the selected operating-interest policy. This is an explanation for the synthetic example, not a general indirect cash-flow rule. Contract maturity, fees, valuation, outstanding shares, lawful distributions, source completeness and full notes remain unresolved.

Explain operating cash from profit under explicit classifications. The preceding schedules leave a broader reporting question: how does each recorded event reconcile profit with operating cash? Consider a separate synthetic sale of plant for USD 12 when its carrying amount is USD 10. The journal debits cash by 12, credits plant by 10 and credits gain income by 2. With the receipt explicitly classified as investing, operating cash is zero although profit is 2. An indirect explanation must remove that gain as well as account for changes in receivables, inventory and unpaid operating obligations. Balancing an unexplained difference would conceal which interpretation was used.

`DeclaredIndirectCash/2` supplies this explanation for a selected successful report in a verified company program. Its policy identifies each used method by its exact definition hash and classifies every noncash balance-sheet account and every income or expense account that the method actually posts to. A name such as *disposal* cannot substitute for those classifications. Working-capital movements, noncash profit adjustments, other nonoperating balance-sheet movements and direct owner movements are separate balance-sheet categories. Income and expense postings are separately designated operating or nonoperating. These are supplied premises for the explanation; the report preserves the financial statements and cash categories already produced by the company program.

To see why the explanation reconciles, first consider one event. All the following amounts are exact monetary scalars in the book’s currency, and signed journal movements use the existing debit-positive convention. Let P be profit, the negative sum of the event’s income and expense postings. Let W be the negative sum of postings classified as working-capital movements, and N the negative sum classified as noncash profit adjustments. Thus an increase in a receivable contributes negatively to W, while an increase in an operating payable contributes positively.

Let X be the signed sum of the remaining noncash balance-sheet postings, those classified nonoperating or owner-related. Finally, let C_o, C_i and C_f denote net operating, investing and financing cash receipts; payments are negative. The balanced journal implies

$$C_o + C_i + C_f - P - W - N + X = 0.$$

This identity alone leaves the nonoperating part of profit inside P. Define R as the signed sum of income and expense postings explicitly classified as nonoperating. A credited disposal gain therefore gives a negative R, whereas a debited disposal loss gives a positive R. The policy must satisfy a second, event-level check:

$$X + R + C_i + C_f = 0.$$

This checks that the excluded balance-sheet and profit movements account for the investing and financing cash. Substituting $X = -R - C_i - C_f$ in the balanced-journal identity gives $C_o - P - W - N - R = 0$, hence

$$C_o = P + W + N + R.$$

The implementation checks both relations for every included event, then sums the contributions over the selected period. Period profit and operating cash must agree with the retained income and direct cash-flow statements. No unexplained remainder becomes an adjustment.

In the plant-sale example, $P = 2$, $W = N = 0$, $X = -10$, $R = -2$ and $C_i = 12$. The exclusion check is $-10 - 2 + 12 = 0$, and operating cash is $2 - 2 = 0$. If the same plant instead sells for USD 8, profit is -2, the loss adjustment is $+2$, and $-10 + 2 + 8 = 0$ still reconciles the investing receipt. Depreciation of USD 5 gives $P = -5$ and $N = 5$, with no cash movement. An acquisition financed directly by debt can likewise have offsetting nonoperating balance-sheet postings and no cash receipt or payment; its gross postings remain visible.

Return to the combined operating and financing example. When interest payments are operating, profit is USD 8.8 and the working-capital adjustment is USD -5.8, including the increase in unpaid interest. The bridge gives USD 3 of operating cash. Under the separately supplied financing-interest classification, the payable movements are nonoperating and the expense is excluded from operating profit: $W = -8$ and $R = 6.2$. The same profit then gives $8.8 - 8 + 6.2 = 7$ dollars of operating cash. The USD 4 difference is the interest payment moved between cash categories. Total cash and profit are unchanged; neither reconciliation selects which policy is applicable.

The selected period begins at the original opening or the latest preceding successful carry. A stopped program can explain a report already present in its successful prefix, but cannot supply a later missing report. Opening, closing, requests and complete results must reproduce exactly before the explanation is accepted. The policy must cover exactly the methods used in that period and their gross posting accounts, including offsetting postings with zero net change. Missing or extra classifications refuse the explanation. One account has one category within a method; mixed treatment of that account's individual postings requires a further specification.

The finite checks cover both charts, period carry and a subsequent collection, noncash acquisition, depreciation, gross offsetting postings, and disposal gains and losses. They also preserve the earlier supported financial reports and operating-cash totals. Algebraically consistent decompositions can nevertheless disagree about whether a movement is working capital or another noncash adjustment. Classification admission therefore remains false. The result explains the selected recorded events under stated premises; it does not prove that those premises satisfy an accounting standard, that every required event was recorded, or that the full note package is complete.

Distinguish collecting a claim from assessing whether it is overdue. The same receivable balance can imply different collection problems at different dates. Consider two explicitly assumed claims: USD 60 due on 31 January 2027 and USD 40 due on 28 February. A receipt of USD 20 against the first claim on 15 January leaves USD 40 against that claim. The selected synthetic policy treats a remaining amount as overdue only after its due date. Thus neither claim is overdue at the end of January. A further receipt of USD 10 against the second claim in February leaves aggregate receivables of USD 70. At 28 February, USD 40 is overdue under this policy; the other USD 30 becomes overdue on the next day. These classifications describe the assumed payment terms. They do not measure an allowance or establish a right to write off either claim.

`DeclaredCollectionClock/1` keeps a fixed declared population of claims, with separate memory for each due date, remaining amount and last assessed amount. The opening claims must sum exactly to the selected receivable account. A collection identifies one claim, cannot exceed its recorded remainder, and posts debit cash and credit receivables. It changes that claim's remainder while preserving the previous assessment. An assessment instead reads the remainders at its declared date, stores the resulting overdue amounts and makes no journal entry or cash movement. Unknown required event inputs retain their reasons and stop that command; an explicitly false applicability finding refuses the method.

Assessment windows have an inclusive start and an exclusive end. The assessment occurs on the final included date, and the saved cursor must equal that window's start before it can advance to the end. Windows must be positive and contiguous. This makes a repeated, overlapping or skipped assessment detectable. Within one date, supplied event order is retained, but an assessment seals that date against a later collection: a payment intended to affect that observation must precede it. The journal engine also rejects events earlier than the current book date or outside the current fiscal period.

A report can advance its observation date without supplying the missing assessment. It then identifies the last assessment as historical, or records that none exists. Provisional period carry preserves unpaid claims, due dates and assessment memory; it cannot reset the cursor or turn year-end into a new assessment. In the example, a December assessment records overdue amounts of USD 40 and USD 30. After carrying those balances, collecting the first USD 40 in January 2028 leaves USD 30 outstanding. A declared January assessment then records overdue amounts of zero and USD 30. Cash has risen from USD 100 to USD 170, receivables have fallen from USD 100 to USD 30, and total assets and equity remain USD 200 throughout.

The existing command language can author these events, reports and carries through its hashed definitions. The guarded execution compares exact method definitions, so renaming a registry entry does not change its meaning. A foreign method that writes claim memory or posts to the receivable account refuses, including offsetting postings with no net balance change. Only the successful prefix remains available after refusal. The checked fragment supports eight declared opening claims and sixteen assessment windows within the existing work limits. New claim insertion, revisions to payment terms, complete invoice coverage, impairment, tax and source admission remain distinct requirements.

A credit sale must create the claim that can later be collected. The fixed population above can follow claims already present at opening. A new sale poses a different problem: the invoice identifier, customer and payment terms must enter the state when the receivable arises. Increasing only the aggregate receivable account leaves later collection without an identified claim. Reserving zero-valued invoices before they exist would instead confuse absence with an existing claim whose balance happens to be zero.

The additive `CompanyMemoryTypes/1` declaration distinguishes fixed records, exact scalar values

and finite maps of typed records. An empty map contains no claims. Once a sale inserts an invoice, each of its actual fields has an explicit type, including the company and currency context of its monetary amount. `CompanyBookTransition/2` checks that these declarations agree with the current book and its memory schema. It permits insertion of a complete map entry and replacement of a scalar field. It refuses duplicate identities, incomplete records, incompatible types and excess capacity. Map keys remain single identities even when they contain slashes or tildes. The frozen schema validator does not express the declared capacity, so the successor checks the typed population again after staging and before returning the proposed book. Monetary effects, memory changes and postconditions must all succeed; a failure leaves the supplied book unchanged.

Consider a separate synthetic company opening with cash of USD 100, inventory of USD 20 and equity of USD 120. It sells two units costing USD 5 each on 2 January 2027 for consideration of USD 30, due on 31 January. The declared sale creates the named customer's invoice for USD 30 and records its date and sale event. At the same step it debits receivables and credits revenue by USD 30, then debits cost of sales and credits inventory by USD 10. There is no cash movement. A receipt of USD 10 on 15 January debits cash, credits receivables and reduces only this invoice to USD 20. Receipts identify both the invoice and its declared customer; neither another customer's label nor a payment larger than the remainder is accepted.

`DeclaredReceivableLifecycle/1` requires the sum of invoice remainders to equal the receivable account before and after each successful command. It begins with an explicitly empty claim population; the preceding profile retains its separate opening-claim example. Assessment uses the same declared strictly-after-due-date convention and contiguous half-open windows. An assessment on 31 January records zero overdue; one on 28 February records USD 20. These operations update invoice memory without a journal or cash flow. Each invoice records whether it has been assessed and at which date. An invoice created after an earlier assessment therefore remains visibly unassessed until a subsequent assessment includes it. A receipt preserves the old snapshot rather than silently recomputing it. Assessment seals its date against later sales or receipts intended to change that observation.

With no further transactions in 2027, cash is USD 110, receivables USD 20 and inventory USD 10. Assets and equity are USD 140, profit is USD 20 and operating cash flow is USD 10. Provisional year-end carry transfers the profit to retained earnings and preserves every existing invoice field and the assessment cursor. Collecting the remaining USD 20 in January 2028 raises cash to USD 130, clears the receivable and produces operating cash flow of USD 20 in that year. It leaves the prior overdue snapshot historical until the next declared assessment records zero.

The versioned `DeclaredReceivableText/1` adapter uses the existing event, report and carry syntax with explicitly hashed lifecycle imports. Its checked executions reproduce the structured requests on two different charts of accounts. An unresolved or refused command retains earlier successful transactions and reports. These examples establish a bounded sale-to-collection composition, including identity creation and carry; they do not supply source-backed opening invoices, revised payment terms, impairment, tax, arbitrary method authoring or production durability. Those remain substantive requirements of the wider company simulator.

A claim needs evidence as well as a place in memory. The preceding lifecycle explains what an identified sale does to the book. It does not establish where the invoice identity, amount or due date came from. An implementer needs to preserve the source reading separately from the decision to recognize a sale. Otherwise a correction to a document can be mistaken for a second transaction, or a missing payment term can acquire an invented value merely because execution requires one.

The bounded `ReceivableSourceContract/1` supplies this separation for explicitly constructed

JSON documents. Each reading binds an exact UTF-8 byte span and its hash, the company/book/currency context, an assertion identity, and a stable claim and sale-event identity. Original and corrected assertions remain present. A named information cutoff controls which assertion was available; a separate ingestion cutoff controls which version the system had received. Unknown timing, competing corrections, altered source bytes or unreadable terms remain unresolved or refuse as appropriate. Recognition approval, receipts and assessment policy are separately supplied premises.

Extend the synthetic company by a second sale on 3 January 2027: consideration of USD 40, cost of USD 10, due on 28 February, to the same named customer but under a different invoice identity. After the earlier USD 10 receipt, the two claims are USD 20 and USD 40. Cash is USD 110, inventory is zero, profit is USD 50 and assets and equity are USD 170. Carry preserves both claims. The subsequent receipts of USD 20 and USD 40 raise cash to USD 170 and clear the claims, with no new profit and operating cash flow of USD 60. These balances follow from the declared transactions; they are not readings of a real company.

Now suppose a corrected assertion changes the first invoice's due date to 31 March. It becomes available at noon UTC on 15 March and is ingested at 09:00 UTC on 16 March. An earlier information or ingestion view retains the original assertion. A later view selects the correction and reconstructs a separate history from the same opening. Its February overdue assessment is zero instead of USD 20; its journals and financial statements are unchanged. Three assertions therefore still describe two claims. This operation corrects a reading of the original terms. A genuine renegotiation with its own effective date would require an amendment method; it cannot be represented by silently rewriting the original sale.

New payment terms leave earlier assessments intact. A prospective amendment answers a different question from a corrected source reading: what changes from the amendment date onward? Return to the earlier single-invoice example, before the illustrative second sale. The original invoice was due on 31 January 2027. Its remaining USD 20 was assessed as overdue on 28 February. Suppose the company and customer are now modeled as extending its due date on 10 March to 30 April. The February assessment remains USD 20 overdue under the original terms. A fresh assessment on 31 March records zero overdue under the extension. Replacing the old due date in place would lose the information needed to explain why both assessments are correct in their respective histories.

The bounded `ReceivableTermHistory/1` reference supplies this separate prospective operation. Its migration starts from a valid legacy claim book and an explicit premise that no earlier amendment occurred. On that premise, it imports the existing due date as the initial term and preserves the last stored assessment. It does not reconstruct earlier assessments. The imported assessment's event identity is recorded as unknown because the old summary does not identify that event. The source book and its hash remain available; the migration leaves its balances, cash flows, claim identities and original scalar observations unchanged. Consequently, even a successful migration leaves completeness of the earlier term history unresolved.

Each claim then has an append-only map of term versions and an append-only map of assessments. A term names its due date, effective date, originating event, predecessor and supplied premise. The current-term reference identifies the version used for a new assessment. Each assessment retains its own term reference, assessment date, remaining amount at that date and overdue amount. The validator checks the term applicable at that historical date; it does not recompute every old observation using the current due date or current balance. The familiar scalar assessment fields remain the summary of the latest assessment, while the current remaining balance can change independently.

For example, a second extension on 10 April moves the due date to 31 May. A receipt of USD 5

on 15 April reduces the claim to USD 15, while the March assessment still records the USD 20 that remained when it was made. The 30 April assessment records zero overdue; the 30 June assessment records USD 15 overdue. A 31 December assessment retains that amount. Year-end carry preserves the three term versions and all assessment records, and opens a new assessment window. Collection of the final USD 15 on 10 January 2028 clears the claim; the 31 January assessment records zero. Final cash is USD 130, receivables are zero and retained earnings are USD 20, as in the unamended single-invoice example: these date changes altered the assessment history, while the supplied sale and collection amounts determined the monetary result.

The immediate amendment takes effect on its event date. It requires the current state hash, existing claim and customer identities, the expected current term, a new term identity and explicit typed inputs. It refuses backdating, a due date before effectiveness, duplicate identities and two terms with the same effective date. An assessment seals its day against later amendments or collections. An amendment followed by assessment on that same day is allowed, so the order is explicit. Unknown required input remains unresolved. The bounded maps admit at most 32 terms and 64 assessments per claim; this reference does not silently discard old records when those limits are reached.

These constructed date-only amendments and assessments produce no journal entry. That is the declared scope of this method; an actual contract change may require a separately specified measurement and accounting treatment. Legal enforceability, modification gains or losses, interest and impairment remain outside this reference. The exact typed requests, historical states, receipts, carry and negative cases are retained for two account charts in `data/accounting-master/receivable-term-history-20261025/`. They connect the prospective operation to the preceding source-correction distinction without treating either operation as evidence for a real invoice population.

Term commands retain the meaning of the typed operations. An author can express this same lifecycle through `ReceivableTermText/1`. The text imports four explicitly hashed definitions: the current book, its schemas, the term-state configuration and the period-report context. The configuration preserves the migration premise and historical uncertainty already recorded in the state. The report context supplies the opening book and account mapping needed for carry. None of these facts is inferred from the spelling of a command.

Within those imports, an event selects `amend`, `assess` or `collect`, names its identity and date, and states each input with its provenance. For the March extension, these inputs identify the claim and customer, the expected current term, the new term identity and due date, the supplied premise and the explicit applicability choice. The compiler assigns their declared types and retains unknown input as unknown. A carry command names the closing identity, retained-earnings account and next period boundary. Its cutoff must equal the current book date; its preservation policy comes from the checked term operation. This profile does not let command text replace that policy or silently advance the closing book.

Malformed syntax, missing or altered imports, extra inputs and incompatible types prevent compilation. A later state-dependent refusal has a different meaning: the completed prefix and its state remain available, together with the command that stopped execution. An unknown applicability input, for example, stops at that event without turning the unknown into approval or undoing an earlier valid extension. Although the reused grammar can express standalone reports, this narrower term profile explicitly refuses that command.

The complete source examples and their imported definitions are retained in the following directory:

```
data/accounting-master/
```

```
receivable-term-authoring-20261025/
```

On both account charts, the nine events and year-end carry reproduce every request and complete result in the earlier typed evidence, including the historical assessment bases. This correspondence makes the authoring boundary testable; wider method composition and the usability of a full company language remain to be established.

A disclosed receivable may have an unidentified population. The retained Apple statements expose the next limit. Accounts receivable, net are displayed as USD 33,410 million at 28 September 2024 and USD 29,639 million at 28 December 2024. These are consolidated measurements with displayed precision, not gross invoices with customer identities and payment dates. The annual cash-flow entry of USD (3,788) million is a change in operating assets; it is not an identified customer collection. The receivable notes provide concentration information, but that information does not supply a complete invoice schedule.[1] (Apple Inc., 2024a, 2025c)

The distinction also affects method choice. Apple's vendor non-trade receivables arise from component sales to manufacturing vendors. The disclosed treatment excludes those sales from products net sales and recognizes gains as reductions in products cost of sales when the related final products are sold. The generic credit-sale lifecycle above recognizes revenue at its sale event, so applying it to these vendor transactions would compute a different treatment. Preserving the vendor caption is useful even while the required transaction facts and accounting method remain unspecified.

For such observations, `SourceClaimPopulationView/1` attaches the existing `MethodCollections/1` partial collection to the unchanged source-constrained company state. A partial collection can have no observed rows while the actual number of claims remains unknown. Its claim count, gross remainder and contractual due terms return explicit unknown results. This differs from an explicitly complete empty collection, whose count and total are zero. A complete constructed collection containing two uniquely identified remainders of USD 20 and USD 40 has count two and total USD 60, conditional on the supplied completeness premise. The retained-source adapter does not accept an override that could manufacture that premise.

The reported measurement remains available in all these cases, together with its source context, date, precision and unresolved obligations. An explicitly assumed rounding interval can constrain that same measurement; it supplies neither members nor a gross-to-net reconciliation. No balancing residual or legal-book allocation is inferred. Even a known contractual due date does not establish when a customer will pay: settlement behavior still needs its own evidence or scenario assumption. The attachment consequently exposes no executable opening and authorizes no posting.

The source-contract, retained-disclosure and partial-population references preserve complete typed requests, results and replay checks. Their reference directories are listed below under the common `data/accounting-master/` parent.

```
populated-source-contracts-20261025/
retained-receivable-boundary-20261025/
partial-claim-populations-20261025/
```

These references connect readings to useful conditional calculations without treating missing claim state as an empty book. Real-source admission, complete opening populations, impair-

[1] Retained annual Form 10-K, PDF pages 34, 36 and 41 (printed pages 31, 33 and 38); retained interim Form 10-Q, balance sheet and Note 4, Accounts Receivable. Original observations, reading-file entries and source hashes are retained in `data/accounting-master/retained-receivable-boundary-20261025/`.

ment, transaction-specific recognition and broader company composition remain substantive requirements.

A payment can change the decision that follows it. Separately supplied economic events can alter the state read by a scheduled rule. The bounded `MixedEventTimeline/1` joins directly dated native payments and inventory sales with the preceding decision calendar. It preserves the native accounting operation and materializes each decision against the book at its turn. Every action has an explicit date and priority; a same-day tie remains unresolved rather than inheriting the order of the input array.

An order-sensitive example starts with cash of USD 452 and tax payable of USD 10.25, split between carried vintages of USD 5.25 and USD 5. Management's authored rule permits a one-unit inventory sale at USD 15 only if cash is below USD 450. Paying USD 4 from the first tax vintage before the decision reduces cash to USD 448 and enables the sale. Cash then becomes USD 463, total inventory falls from USD 80 to USD 70, and the selected lot's carrying amount falls from USD 20 to USD 10. The sale contributes USD 5 of profit; the payment reduces cash and payable without creating a new expense.

Reverse the explicit same-day order and the decision sees cash of USD 452, so it takes no action. The later payment leaves cash of USD 448 and inventory of USD 80. In both cases tax payable is USD 6.25, with vintages of USD 1.25 and USD 5. These differences follow from the specified threshold and order, not an estimated company policy. The comparison checks the complete native journals, state histories and financial reports as well as these independent controls.

The initial adapters accept allocated payments of carried tax vintages and the existing proportional inventory sale. Their declared schedule date must match the date inside the native event. Structural checks cover every supplied entry before execution, but remaining cash, stock, payable and accounting findings are assessed against the book at the relevant action. A later unknown vintage or unavailable authorization therefore preserves the earlier successful prefix and the exact failed event. An unresolved decision after a successful payment retains that payment; it is not rolled back merely because the subsequent sale could not be evaluated.

Assessment and period-boundary operations need different treatment. A current tax assessment names an interval, a deferred measurement binds to a particular state, and a carry changes the fiscal period. Version 1 of the mixed reference leaves these operations, and nested return/lease envelopes, unsupported rather than assigning them an arbitrary event date or rewriting a stale measurement. This is an explicit implementation gap in the integrated timeline; the separate native methods retain their existing scope.

Seventeen focused methods, 18 scheduler regression methods and 61 exact comparisons across six cases are retained in `data/accounting-master/mixed-event-timeline-20261025/`. The runner bounds the combined population to 64 actions, including at most 32 supplied native entries, and retains the current book, successful outcomes, failed action and remaining timeline on refusal. Its closed-path premise covers both supplied events and decisions. General financing, richer calendars, historical completeness and production durability remain open.

An assessment closes the interval it measures. A tax assessment differs from a payment because it measures activity over an interval and constrains which events may follow. The optional `MixedEventTimeline/2` retains the same native annual book and decision rules while adding an explicit clock for current assessment. The interval has an inclusive start, an exclusive end and a final included activity date. That final date must be the day before the exclusive end, and it must equal the date supplied in the timeline. No extra `event.date` is introduced. The assessment reads the book at its actual turn, after earlier successful actions; static validation

does not calculate tax against the initial book.

Consider the same opening cash of USD 452, inventory of USD 80 and carried tax payable of USD 10.25, now with an explicitly enabled one-unit sale on 2 January 2027. The sale earns USD 15 against cost of USD 10. Under the constructed assumption of zero tax adjustments and a one-quarter tax rate, a subsequent assessment through that day measures profit of USD 5 and current tax of USD 1.25. Cash is USD 467, inventory is USD 70 and total tax payable is USD 11.50. Assessment creates no cash flow and preserves the carried vintages. These numbers illustrate the supplied tax model; they are not an assessment of a named company's liability.

Reversing the explicit same-day priorities gives a different admissibility result. Assessment first records zero current profit and closes the interval through 2 January. The later sale is refused because it would add activity inside an already assessed interval. The runner preserves the successful assessment, cash of USD 452 and the exact rejected sale. It neither silently moves the sale ahead of assessment nor rolls back the valid prefix.

A half-unit sale on the next day, after the successful sale-first sequence, adds proceeds of USD 7.50 and cost of USD 5. Cumulative profit becomes USD 7.50 and cumulative tax USD 1.875. Replacing the earlier assessment therefore records only USD 0.625 of additional expense and updates the same current vintage. Cash becomes USD 474.50 and inventory USD 65. This replacement rule prevents the earlier USD 1.25 charge from being counted again.

Twelve focused tests and 49 regression tests preserve eight complete exchanges in `data/accounting-master/assessment-timeline-20261025/`. The comparison checks the full materialized decision, native transition, journals and report, alongside independent arithmetic. Unknown rates retain their unresolved status after a successful sale; malformed clocks, stale initial state and ambiguous priorities refuse. The existing bounds of 32 native entries and 64 total actions are tested with actual boundary cases. In version 2, deferred measurement still requires its own state-binding rule, and carry still requires a fiscal boundary rule. Completing this declared timeline therefore establishes neither complete fiscal activity nor broader accounting admission.

A measurement belongs to a particular book. The optional `MixedEventTimeline/3` adds deferred assessment and an explicit terminal carry to the preceding timeline. Deferred assessment is dated by `measurement.at`; its supplied state digest must match the financial book when the operation is reached. The runner retains those measurement bytes. It does not replace the digest merely because an earlier action has changed the book.

This matters even for a payment that creates no expense. Start with the sale-first example, now assessed through 31 December 2027. Cash is USD 467, profit before tax is USD 5 and current tax is USD 1.25. A deferred measurement prepared for that state cannot be reused after paying USD 1 of a carried tax liability: cash and payable have both changed. The original request stops at the stale measurement and retains the payment. An author can instead supply a distinct measurement for the resulting book, including its own identity, date, tax bases and recovery assumptions. That is a new request, not an automatic renewal of the old evidence.

The constructed deferred calculation also tests a nonzero accounting effect. It begins with a deferred-tax asset and liability of USD 5 each. The supplied taxable temporary difference of USD 20 keeps the liability at USD 5 under the one-quarter rate. For the asset, the author now supplies only USD 12 of recognized deductions and matching recovery capacity, giving an asset of USD 3. Reducing the asset from USD 5 to USD 3 records USD 2 of deferred-tax expense without changing cash. After-tax profit is consequently USD $5 - 1.25 - 2 = 1.75$. The recovery assumption is hypothetical; the example does not estimate a company's future taxable income.

Carry occurs at the next period's start, which must equal the current period's exclusive end. It requires current assessment through the closing day and a fresh deferred assessment of the

same book. In the case without the intervening payment, carry on 1 January 2028 preserves cash of USD 467, inventory of USD 70 and current-tax payable of USD 11.50. It transfers profit of USD 1.75 to retained earnings, whose credit-signed balance changes from USD −11.75 to USD −13.50, and resets the new period's income and cash-flow movements. The old statements, tax vintages and assessment history remain in the period archive. The separately reported current-period profit is therefore zero immediately after carry; it does not replace the archived profit of USD 1.75.

This bounded calendar permits one carry, as its last action. Later decisions use another request bound to the new opening. Applying the retained inventory rule on 2 January leaves its half-unit reserve intact: one unit remains before the decision, so only half a unit is sold. Proceeds are USD 7.50 and cost is USD 5, leaving cash of USD 474.50 and total inventory of USD 65. The preceding year's report and tax vintages remain unchanged.

Thirteen focused tests and 61 regressions retain twelve complete exchanges:

```
data/accounting-master/fiscal-timeline-20261025/
```

They compare entire native transitions, journals and archives, including an expressly renewed measurement and a payment after deferred assessment that prevents carry. An unknown deferred finding remains unresolved; an absent required entry is invalid, and a known false prerequisite is unsupported. Provisional carry retains its own unknown or false finding statuses without granting accounting admission. Exact clocks and valid prefixes are preserved under the existing 32-native/64-total work limits. These examples specify timing and composition; complete fiscal activity, general multi-period calendars and actual-company tax findings remain separate obligations.

Available inputs do not establish a saved past calculation. The path's information date cannot also date the separate policy body or its materializer. The synthetic extension gives the policy public availability on 10 March 2025 and the method on 12 March, both at calendar-day precision. Even if the parent path is already available, the public dependency selection cannot pass until 13 March. The query binds observations to the exact policy and method bytes, then recomputes the generated path and its accounting result. Changing the rule or materializer prevents reuse of the old observations.

That result is a consequence calculated from the eligible synthetic inputs. It receives no invented authoring, publication, ingestion or acceptance date of its own. Showing that a calculation actually existed before a cutoff needs separate historical execution evidence. Eligible dependencies can also produce a partial accounting result: availability does not supply missing payroll, closing memory or a third-year opening.

The expression program and its assumptions introduce further dependencies. Their information evidence is not inherited from the directly supplied 90-percent policy. The current expression and family examples therefore retain archival scenario meaning, with historical, temporal and forecast admission unestablished. Their draft content hashes use the existing company engine's encoding, including its non-ASCII escaping. Import into the separately defined portable record encoding requires an explicit identity mapping; Section A.8 supplies that mapping for these legacy drafts and retains the separate verification levels.

Interrupted work differs from incomplete accounting. Suppose evaluation stops after two members. A checkpoint records those exact member results and the next member to evaluate, bound to the original family, source and method. Resume validates the stored prefix by replay and evaluates the requested number of new members. The ordering and identities

prevent a member from being skipped or counted twice. Different budget splits reproduce the same final family.

The unknown-payroll and supplier-failure members are finished evaluations with partial financial results. They are not unfinished computational work. Once stored, they remain part of the completed prefix, and recovery continues with the next member. This distinction is necessary to avoid repeatedly evaluating a known unresolved case while silently excluding it from the family. Recovery inside a member's year or event sequence remains outside this checkpoint.

The scheduling budget counts newly evaluated members. Validating a saved prefix also takes work, recorded separately; the budget is not a claim about total runtime. A changed source, model or method requires a new operation, and a rehashed false prefix fails trusted replay. Neither changed semantics nor a schema migration can be disguised as resume.

Publication of the checkpoint has its own work owner and generation. One reference transition publishes the new checkpoint and immutable request decision together. If acknowledgement is lost, either nothing was published or the complete transition occurred. Retrying the exact request publishes it in the first case and returns the saved response in the second. That response remains the same after a later request advances the work head. Reusing the identity with different request bytes conflicts; a new request with a stale head does not advance it.

The reference rejects a world with only part of the checkpoint, decision and generation update. It checks these publication meanings through explicit before-and-after states and complete replay, without establishing physical storage durability or changing the company's accounting state. A persistent implementation, efficient recovery and compatible migrations remain distinct engineering obligations.

The four extensions are retained under `data/accounting-master/`:

```
company-policy-information-20261023
company-family-20261023
company-expressions-20261023
company-recovery-20261023
```

They connect bounded authoring, information and recovery behavior to the same company example. Broader corporate and sector methods, actual source admission and independent implementation evidence remain necessary for the full financial-statement language.

Resolve a sector assessment without turning it into a common ledger. The common company world names a legal book and a component handle, but that name alone does not tell us what the handle contains. A bank assessment can carry account balances and an insurer assessment can carry only a head pointing to a separately recorded balance digest. Treating both as one generic state would lose the distinction that later accounting and reporting methods need.

The boundary therefore resolves one selected native case through its own content-addressed registry. The request names the component and the exact case; the resolver checks the native packet, schema, record identity, dependency closure, source blobs, owner, book, currency, state, source and information view. A changed record body, missing dependency, wrong schema, altered blob or different book refuses before any method is called. The resulting world retains the unrelated books and every unresolved obligation. Reference resolution is a copy-and-identity operation, not a calculation.

The retained bank case illustrates why the context travels with the state. It uses a synthetic US legal book, USD millions, a supplied US GAAP/CECL basis and a solo perimeter. Its selected accounting state contains balances, while its assessment profile declares `accounting_effect=none`.

The retained insurer case uses a synthetic legal book, USD illustrative millions and a historical EU IFRS 17 authority; its accounting head records a balances hash rather than materialized balances, and its assessment profile also has no accounting effect. These cases are not numerically compared. Their matching currency labels do not establish matching units, reporting bases, dates, legal facts or consolidation eligibility.

This distinction changes the implementation decision. A resolved assessment can be cited by a later method through its native profile and view, but it cannot post to the company book or authorize consolidation. The world remains conditional when source, law, entity admission or method obligations are unresolved. A later adapter must declare how a native state maps to the common book, which accounts and memory it changes, and which evidence proves that mapping. Until then, the exact native records are preserved beside the common world rather than flattened into a convenient balance sheet.

The checked resolver covers one named bank case and one named insurer case. It verifies 34 bank records and 11 insurer records in their selected closures, including 21 and 6 source blobs respectively, and retains their byte hashes. Those counts establish identity for the selected examples; they do not measure sector coverage, accounting compliance, valuation validity, or a complete bank or insurance engine. The native exchange validations remain the source of their own conditional method results. The common boundary contributes a safe route to those references and records the unresolved work required before posting.

Carry the premium-allocation groups into a dated interim book. The short-duration case in Chapter 38 provides another route from sector measurements to the shared book. It starts with cash and equity 20, receives premium 120, pays reinsurance premium 24, and observes half the supplied uniform service. Its purpose differs from the loss-component schedule that follows: both issued and held groups must remain identifiable through their coverage, claims, service and cash movements.

`InsurancePAADatedBook/1` proposes the eight-event sequence. It uses the unchanged interfaces `DeclaredCompanyProgram/2`, `CompanyBook/1` and `CompanyBookReport/1`. Its request contains a profile identifier, payload and payload digest. The payload supplies entity and currency, the explicit 2024 coverage calendar, distinct issued and held groups, their ownership and link, the selected measurements, eligibility findings, cash classification and exact source edition. Money and fractions use canonical rational strings. The full closed fields and serialized examples are included with the companion; unknown extra fields do not acquire meaning through an informal account label.

The bridge supports three retained measurement cases, a named or renamed account chart, and an optional unrelated equipment balance of 17 with matching equity. Group identifiers remain in typed memory and event inputs. Each method requires the correct date, preceding step and opening balances; the company engine checks the resulting journal, memory and state before the next event. The unrelated balances survive. The returned `proposed_conditional` result contains the complete compiled program, native execution, group records and unresolved conditions, including every intermediate journal and state.

The final report separately presents all mapped opening and closing accounts, assets, liabilities, income, expenses, equity movement and gross cash origins. The base case has cash 101, reinsurance coverage asset 12 and recovery asset 6 against issued coverage 60 and incurred claims 12. Profit is 27 and equity 47. The transferred-risk variant changes the recovery asset to 6.5; the separate non-performance variant changes it to 4. Cash remains 101. Both groups remain gross. The report uses the full coverage-year interval but observes only the 2 July state; it neither supplies the second half-year nor closes the year.

Missing eligibility or measurement findings return `unresolved` without a compiled execution. Contradicted findings or measurements outside the selected finite domain return `unsupported`. Malformed identities, links or digests return `invalid`. A refusal by the compiled native program preserves its stopped execution and preceding state; the bridge returns `native_refusal` without inventing a final report. The complete expected states and financial reports were authored separately before execution. They cover both charts, the unrelated balance and sequence/date/group failures.

These are proposed conditional entries. Source, group, chronology, accounting and posting admission remain false, as do durable commitment and fiscal or note completeness. The supplied uniform risk release, short claims timing, PAA eligibility and operating cash classification remain assumptions. The retained AASB 17 edition qualifies the same narrow example as the chapter; replay does not establish actual applicability. The exact contract, source qualification, expectations and internal reproduction archive are under `companions/insurance-paa-dated-book/` in this package. The original record remains under `data/accounting-master/insurance-paa-dated-book-20261027/`.

Map an exact insurer schedule into a proposed book. A different retained insurer calculation illustrates a loss-component posting sequence. It starts with cash of 200 and an insurance liability of 100, including a loss component of 20. Its six journals already specify exact amounts; the new adapter must show where those amounts go in the company book and what state must accompany them. This is a different synthetic case from both the resolved assessment and the premium-allocation example above. The original calculator continues to return a report without posting, while `InsuranceLossBook/1` proposes a separate sequence of declared company methods.

The request supplies the opening book, an injective mapping for six account roles, three typed memory locations and a statement mapping. The account roles are cash, insurance liability, finance expense, service revenue, loss-reversal income and claims expense. The memory records the loss component, the next step and the exact native request identity. Before constructing a method, the adapter recalculates the selected native request and checks the complete result, opening balances, account classes and memory. It accepts only the two retained requests; a plausible new schedule is outside this checked version.

The illustrative units require an explicit bridge as well. The native case uses `CU`, whereas the host book requires a three-letter currency field. The request binds both complete contexts and declares `XCU` to denote the same illustrative unit at scale one. No exchange-rate calculation occurs. The entity remains Synthetic Insurer, and the native group and historical rule stay attached to the original request. This syntactic alias does not admit a currency, accounting basis or real entity.

The following table gives all six signed journals for the nonzero-loss case. Amounts use the same illustrative units; debits are positive and credits negative. Each date has finance, liability release and claim-payment entries, in that order. A supplied premise classifies the claim payments as operating cash flows.

Date	Entry	Signed account changes
1 Jan 2026	Finance	Expense +10; liability −10
	Release	Liability +55; revenue −44; reversal −11
	Claims	Claims expense +50; cash −50
1 Jan 2027	Finance	Expense −5; liability +5
	Release	Liability +50; revenue −40; reversal −10
	Claims	Claims expense +45; cash −45

Every row balances without an opening-capital entry or a balancing amount. After the first three entries cash is 150, liability is 55 and profit is $44 + 11 - 10 - 50 = -5$. The next three leave cash at 105 and liability at zero; their profit is $40 + 10 - (-5) - 45 = 10$. The negative finance expense in the second date is retained explicitly, rather than replaced by a positive expense.

Let K denote the loss component measured in the same units as the liability. It is tracked within that liability, not added to it as a separate obligation. The retained calculation attributes its changes separately to finance, claim release and risk-adjustment release. Under its supplied allocation assumptions, the two date-end values are

$$K_1 = 20 + 2 - 10 - 1 = 11, \qquad K_2 = 11 - 1 - 9 - 1 = 0.$$

The release method combines the two release amounts in one memory update but keeps their original explanations separately. In the second retained case, K starts at zero and stays there. The liability releases are then revenue of 55 and 50, with zero loss reversals. Cash, liability and total profit match the first case, but the gross revenue and reversal accounts differ. Retaining the zero postings and separate origins makes that difference inspectable.

Every generated event checks its original date, scoped opening balances, loss component, progress and case identity before proposing its changes. Postconditions check the resulting balances and memory; skipped or repeated steps refuse. Unrelated equipment, equity and quoted or Unicode text in memory survive unchanged. Thus a successful calculation does not require replacing the whole company with the insurer example.

The host interval starts on 1 January 2025 and ends exclusively on 2 January 2027 so that both native event dates fit without alteration. Reports after the third and sixth entries are cumulative within that interval: profit is -5 and then 5, and operating payments are 50 and then 95. These reports do not close annual accounts or move 1 January events into the preceding year. Annual closing and reopening need their own explicit treatment.

Complete replay covers both native cases on two account charts, with and without an unrelated equipment balance. Unknown or false native findings, changed contexts, forged results, invalid mappings and inconsistent state prevent a valid proposal. The retained evidence is in `data/accounting-master/insurance-loss-book-20261025/`. It establishes this exact mapping and proposed execution. Source applicability, broader insurance measurements, durable posting and a complete statement-note package remain separate requirements; the supplied historical findings and allocation assumptions are preserved as such.

Carry the same insurer journals across explicit calendar years. The cumulative example can also support an explicitly supplied calendar-year interpretation. The first three native events still occur on 1 January 2026, and the next three on 1 January 2027. The host now observes and closes recorded balances on 31 December 2025, 2026 and 2027, reopening the following year each time. The first close has zero profit because the selected sequence contains no event in 2025. These are hypothetical observations of the supplied sequence; the event population remains unknown, so the empty first year is not evidence that an actual insurer had no business.

Inserting closing commands alone is insufficient. After the first three events, the cumulative finance-expense account is 10. Closing 2026 resets it to zero and carries the period loss of 5 into the selected retained-equity account. The unchanged fourth method would still require expense of 10 before posting its negative expense of 5. The checked naive-carry attempt stops at that condition, after preserving the valid earlier events and closing records.

The repair changes the balance conditions, leaving the journals intact. Write b^{cum} for a vector of temporary-account balances expected by a second-date method and c for those accounts'

balances closed at the end of 2026. Both vectors contain finance expense, service revenue, loss-reversal income and claims expense in that order, measured in the same illustrative units. The corresponding fiscal-period condition uses

$$b^{\text{fiscal}} = b^{\text{cum}} - c, \qquad c = (10, -44, -11, 50).$$

This subtraction applies to the opening and resulting balance conditions of each of the three 2027 methods. For example, finance expense must begin at $10 - 10 = 0$ and end at $5 - 10 = -5$. The method still posts exactly -5 to expense and $+5$ to liability. Reversing the 24 constant substitutions recovers all three original method definitions; their monetary and memory operations do not change. When the opening loss component is zero, the same procedure uses that case's own closed vector, including revenue of -55 and reversal of zero.

Every memory leaf has an explicit preservation policy. The loss component, next-step marker and case identity therefore survive closing, as does unrelated company memory. The retained account must be a supplied equity account outside the six insurer roles. Closing has no cash effect: it zeros the income and expense accounts, transfers their signed sum to retained equity and archives the report, closing balances and next opening together. A reset of required memory or an omitted carry cannot be treated as valid continuation.

For the case without unrelated equipment, the resulting year-end observations are as follows. Claim payments are shown as positive outflows; the closing liability is the positive statement amount.

Year	Profit	Claim payments	Cash	Liability	Equity
2025	0	0	200	100	100
2026	−5	50	150	55	95
2027	10	45	105	0	105

The two nonzero annual profits sum to the original cumulative profit of 5; payments sum to 95. Reopening 2028 preserves cash of 105, liability of zero and equity of 105, while period profit and period cash flows start at zero. An additional equipment balance of 17 with matching opening equity survives every step and raises each equity observation by 17 without changing profit or cash. The two account charts and both loss-component cases give the same correspondence under their explicit mappings.

`InsuranceFiscalBook/1` recomputes the complete cumulative proposal and checks its expected hash before compiling the fiscal sequence. It binds the calendar policy, selected retained account and complete memory policies, and retains the original native dates, gross postings and loss origins. Complete traces and closing archives are recorded in `data/accounting-master/insurance-fiscal-book-20261025/`. The result demonstrates recorded-balance carry under this supplied policy; actual fiscal completeness, source applicability and durable posting remain unestablished.

Explain the insurer's annual operating cash from the same journals. The earlier indirect bridge can now use each fiscal report directly. Under one supplied illustrative policy, liability movements from finance are noncash profit adjustments, liability releases are working-capital adjustments, and all six methods' income and expense postings remain in operating profit. These choices are premises of the explanation. They do not alter the native journals or the operating classification already supplied for claim payments.

Using the earlier meanings of P, W, N and R, the two nonempty years give

$$\begin{aligned} 2026: &\quad C_o = -5 - 55 + 10 + 0 = -50, \\ 2027: &\quad C_o = 10 - 50 - 5 + 0 = -45. \end{aligned}$$

The negative working-capital terms reflect the liability releases; the finance terms reverse the corresponding noncash expense or income. The selected 2025 report has no events and all terms are zero. The loss component K contributes no extra monetary adjustment because it is memory within the liability, whose postings are already included. Adding K again would count part of the same change twice.

A second supplied policy classifies all liability movements as noncash profit adjustments. It gives $W = 0$ and $N = -45$ in 2026, then $W = 0$ and $N = -55$ in 2027, with the same profit, cash and financial reports. Exact reconciliation therefore leaves this presentation choice unresolved. Both policies preserve every gross posting, including the zero loss reversal in the zero-loss case. Removing its classification is an error even though its amount is zero.

The comparison selects each period after its preceding carry and binds the exact fiscal method hashes. A pre-carry method hash or a classification from another year cannot stand in for the required policy. Complete exchanges under `data/accounting-master/insurance-statement-bridge-20261025/` cover the two charts, two retained native cases, two explicit policies and three years, with the unrelated equipment balance present. They use the existing indirect engine and reproduce the original annual financial reports. Actual classification validity and event completeness remain separate questions.

Credit accrual needs shared amounts before it can enter an exact book. The insurer examples use rational amounts. Dated credit accrual introduces a different problem: separate numerical bounds can enclose each entry correctly while losing the dependence that makes the journal balance. In the retained July credit example, let $F = (11/10)^{181/365}$ be the dimensionless accrual factor for the 181 days from 1 January to 1 July 2025. The signed entries are $100(F-1)$ to gross carrying amount, zero to allowance, and $-100(F-1)$ to interest income, all in the native currency units. They sum exactly to zero. Selecting every entry's upper bound from the numerical report instead leaves a residual of 10^{-16} currency units. A midpoint lies inside the factor's enclosure but fails its exact defining equation. Numerical accuracy alone therefore does not supply exact journal amounts.

One shared positive day factor preserves the dependence. For the supplied fixed annual rate $r > -1$, set $a = 1 + r$ and choose the unique positive z with $z^{365} = a$. A d-day accrual uses z^d; discounting uses z^{-d}. For any integer exponent e, write $e = 365q + s$ with $0 \le s < 365$. Then

$$z^e = (z^{365})^q z^s = a^q z^s, \qquad \sum_e c_e z^e = \sum_{s=0}^{364} b_s z^s.$$

Here the rational coefficients c_e are collected into rational coefficients b_s after the substitution. Negative q is valid because $a > 0$. Addition and multiplication followed by this reduction preserve evaluation at the same positive root. For monetary amounts the coefficients carry currency units; the powers of z are dimensionless. At zero rate, $z = 1$ and the expression reduces directly to a rational amount.

If every reduced coefficient is zero, equality is proved. A nonzero remainder does not prove inequality, because $z^{365} - a$ need not be irreducible. For example, the algebra-only choice $a = 32$ gives $(z^{73})^5 = 32 = 2^5$, so positivity implies $z^{73} = 2$ even though $z^{73} - 2$ has degree below 365. This example is outside the retained financial rate range; it tests the limit of the equality procedure. The reference returns an uncertified result whenever a required identity does not reduce to zero. It provides division by rational constants and inverse day powers, with no general polynomial inverse or ordering test.

The read-only `CreditLinkedSchedule/1` recomputes and binds each complete retained request and native result before reconstructing the expressions. Across nine schedules it preserves

57 stages and 63 journals, including partial and full write-off, prepayment, negative and zero rates, leap dates and irregular receipts. At each stage, net carrying amount equals gross less allowance, gross amounts reconcile to their dated origins, remaining expected cash discounts to net carrying amount, and cash plus net assets equals opening capital plus profit. Accumulated signed entries reproduce the same balances and income. Eighty independently authored targets and 652 exact identities pass; 986 certified numerical enclosures lie within the original bounds. The enclosures check consistency; the shared expressions establish the identities.

Complete exchanges and refusal evidence are retained under `data/accounting-master/` in the directory `credit-linked-measurements-20261025/`. Three retained cases also project exactly to rational amounts. The existing rational company book still requires a separate admission and account-mapping step. The other six cases need an explicitly specified exact amount extension or an authored quantization policy with its treatment of rounding differences. The checked expressions establish neither choice and make no ledger mutation.

A separate book retains exact credit amounts. The optional `CreditLinkedBook/1` now supplies that finite projection. It begins with a validated rational opening before acquisition, binds the complete exact credit result, and requires explicit account roles and an illustrative one-to-one mapping from native CU to host XCU. Cash begins at the supplied opening amount; gross carrying amount, allowance and the three income roles begin at zero. The original rational book remains unchanged. Unrelated rational balances become constant expressions under the same root, and unrelated memory remains byte-for-byte equal in its canonical encoding.

Allowance occupies a contra-asset account whose signed balance is $-L$. For example, after the first collection in the adverse-without-write-off case, gross carrying amount is 66 and allowance is 22, so the two signed asset balances contribute $66 - 22 = 44$. Classifying that credit balance as a liability would change the presentation. A write-off instead reduces gross and allowance together; its journal has zero cash movement. Interest, impairment and catch-up use signed income accounts, so the same roles accommodate both gains and losses.

The projection accumulates the original journals in their original order, including zero entries. Each native observation names the exact journal prefix that produces its balances. On 1 January 2026 in the partial-write-off case, accrual, collection, another accrual and the improvement observation correspond to prefixes 5, 6, 7 and 8. Sorting observations by date alone would lose that distinction. The complete native request and result retain expected cash, written-off portions and dated origins alongside the proposed monetary state.

Reports need only addition and the declared account classes. Let T_A, T_L, T_E, T_I, T_X denote the signed totals of asset, liability, equity, income and expense accounts, respectively, in host currency units. The zero journal sum preserves $T_A + T_L + T_E + T_I + T_X = 0$. Presentation therefore uses

$$A = T_A, \qquad L = -T_L, \qquad P = -T_I - T_X, \qquad E = -T_E + P, \qquad A = L + E.$$

Here P is cumulative profit since the opening and E includes that profit. These relations require no ordering test on an algebraic amount. In the constructed complement case, equipment of 17 and matching opening capital raise assets and equity by 17 throughout, while native cash and profit are unchanged. The period remains a cumulative interval containing the original dates; it does not establish annual completeness or a closing policy.

All nine schedules pass on two account charts with and without that complement: 36 complete exchanges preserve 252 journals and 228 observations. The checks certify 3,186 exact identities. Of the observations, 124 project to rational balances accepted by the unchanged host validator, including every observation in the three fully rational cases. This is a balance and type compar-

ison. The separate cash record preserves price, integral cost, fee and collection components, without assigning operating, investing or financing classes.

That distinction has a checked consequence. Asking the existing statement engine to explain the acquisition with no classified cash movements fails its cash reconciliation. An absent classification cannot be replaced by a zero movement. The optional reference consequently reports cumulative balances, profit and unclassified cash origins; a full cash-flow statement still needs an explicit classification policy. Full exchanges and refusals are retained under `data/accounting-master/` in `credit-linked-book-20261025/`. They establish a single-root proposed sequence with no durable posting, general multi-rate composition or accounting admission.

Cash classification is a supplied premise, not a balance-sheet inference. The separate `CreditCashClassification/1` layer makes that missing premise explicit. It imports an exact retained credit projection and asks the caller to identify the classification policy and its context. Each price, integral cost, integral fee and collection retains its native date, amount and journal origin. Effective-interest accrual and impairment changes are not cash flows merely because they affect income.

Consider the retained example with purchase price 98, integral cost 2, zero integral fee, and subsequent collections of 44 and 55. For a purely synthetic classification test, suppose the caller assigns price and collections to investing and cost and fee to operating. Investing then has receipts of 99, payments of 98 and net cash of 1; operating has payments of 2 and net cash of -2. Together they explain the cash reduction of 1. The example checks the consequences of those supplied labels. It does not select an IAS 7 policy for an actual lender or establish which part of a receipt is contractual interest.

A classification names a native origin by its position and content digest. The caller can declare separate components, but their exact signed amounts must sum to that origin. Positive and negative components cannot cancel within one supposedly gross receipt or payment. A principal/interest split therefore needs explicit supplied amounts and supporting policy; it cannot be copied from the effective-interest accrual. The original gross origin and each declared component remain available alongside the policy reference.

The classification follows journal prefixes, not dates alone. At each native observation it includes only cash origins already introduced by that journal prefix and reports gross receipts, gross payments and net amounts by activity. It preserves the exact financial balances and profits from the original credit projection. A changed activity label can change the cash-flow presentation but cannot change the native journal or the closing cash balance.

Missing classification remains unresolved even when its amount is zero. In the example, omitting the zero integral fee leaves an incomplete origin population; the unclassified net is zero, but that numerical equality does not prove complete classification. Likewise, an unknown later collection leaves earlier completed prefixes usable while making the affected later statement partial. Known activity totals remain labelled as the classified subset until every origin has a declaration. There is no balancing residual category.

For a rational financial state, the adapter supplies these explicit flows to the unchanged `CompanyBookReport/1`. Across the retained nine schedules, two account charts and presence or absence of the unrelated equipment balance, the three conditional classification examples produce 684 observations. The 372 rational observations agree with the host cash, balance, income and equity reports. The remaining 312 keep their single-root financial expressions; the adapter does not round them to obtain host-book acceptance. Counterexamples cover changed source and policy bindings, duplicate origins, wrong signs or totals, unsupported activities and concealed netting.

This supplies conditional classified reporting for the retained projections, while preserving the original unclassified report's refusal. Actual issuer facts and classification-policy eligibility, complete notes, fiscal completeness, the full indirect cash bridge and arbitrary multi-root contracts remain separate requirements. Exact cases, unknowns and refusals are retained under `data/accounting-master/credit-cash-classification-20261026/`.

A member's balances can depend on state held by the group. The corporate group introduces a different reason to preserve the native representation. Its parent and subsidiary have separate balances, but their state also includes shared inventory lots, returns, a compound instrument, tax calculations and ownership changes. Copying each balance dictionary into an independent company book would leave those dependencies behind. A later event could then use a valid-looking opening balance without the facts needed to determine its accounting consequences.

The reference adapter therefore selects a member within the complete group state. A selector names the exact state record and the path `/books/P` or `/books/S`; its identity differs from the identity of the group record itself. The common company's reference points to that selector. The result retains the selected balances, the whole native state and its information view together. Record schemas, dependency references and source bytes are checked before a selection is returned. The selector gives access to a member's balances without asserting that they form a separately executable book.

The two retained snapshots also show why time belongs in the relationship. On 1 January 2025, the supplied parent ownership fraction is four-fifths and the parent and subsidiary cash balances are 500 and 100. On 31 December 2025, ownership is nine-tenths and cash is 491.5 and 152.5. Binding either fraction to the entire supplied control interval would erase an ownership change. The adapter instead attaches it to the selected snapshot day. It preserves the supplied control premise and the archival, synthetic information view; it does not determine control, recalculate consolidation or admit current source evidence.

An elimination worksheet leaves the legal books intact. Separately valid company books permit a different operation. Consider a new constructed example in USD units at 31 December 2025. Parent P holds cash 80, an investment in subsidiary S of 100 and a receivable from S of 20; its equity is 200. S holds cash 70 and equipment 50, owes P 20 and has capital of 100. Both books balance. Assume that P controls and fully owns S, the books use compatible accounting policies and measurement bases, the reciprocal balances refer to the same obligation, and the investment equals the subsidiary's capital with no acquisition difference. These are supplied facts of the illustration. They do not describe the earlier four-fifths or nine-tenths ownership case.

Adding the legal books gives assets of 320, liabilities of 20 and equity of 300. The worksheet then records two explicit adjustments. First, it credits the investment by 100 and debits the subsidiary capital by 100. Second, it credits the receivable by 20 and debits the payable by 20. Under the declared signed-balance convention, each adjustment sums to zero. The resulting assets are cash 150 and equipment 50; liabilities are zero and equity is 200. Cash does not change. Neither legal book receives an entry: the adjustments belong to the separate worksheet, alongside the unadjusted contributions.

The implementation requires a complete map from each local account to the declared common chart. It checks the legal owners, case, currency, period, snapshot date and classifications, and preserves each book's memory and cash flow history. Every paired adjustment names its source accounts, expected full balances and economic premise. A source account can be used only once; the pair must come from different books and contain exactly opposite, nonzero balances. This

first operation supports the two declared pair roles, receivable–payable and investment–equity. It refuses cash elimination, partial allocations, unmatched balances, unresolved components and unsupported sector books rather than supplying a balancing amount. A worksheet with no declared adjustments remains an unadjusted combination.

Two versions of the illustration use different local account names and the same explicit common-chart meanings. Both reproduce the complete signed unadjusted balances, elimination deltas and adjusted balances; replay also checks that the legal books remain unchanged. These checks establish the finite worksheet calculation. The truth of the supplied economic premises still requires evidence, and classification equality does not prove policy equivalence. Noncontrolling interests, goodwill, tax, foreign exchange, intragroup profit and period cash-flow eliminations require additional methods. The result therefore remains a snapshot worksheet, not a complete consolidated statement package or an executable group book.

The reference selections and worksheet cases are retained in two directories under `data/accounting-master/`:

```
native-group-binding-20261025
compatible-book-composition-20261025
```

Correct net cash can conceal incorrect gross cash flows. The snapshot worksheet leaves a period question unanswered: which legal-book receipts and payments cross the group's boundary? An internal loan can produce one company's investing payment and another company's financing receipt. Adding the cash changes gives the right net amount, but retaining both flows would misstate the group's gross activity. Matching activity labels is therefore neither necessary nor sufficient to identify an internal transfer.

A separate constructed period illustrates the extension. At the start of 2025, P has cash 100, an investment in S of 100 and equity of 200; S has cash 100 and capital of 100. Continue to assume stable full ownership and control, compatible policies and measurement bases, and investment cost equal to S's capital. P then lends 30 to S, and S receives 20 from an external customer, recognising income of 20. At year end P has cash 70 and a receivable of 30; S has cash 150, a payable of 30 and income of 20. The investment and capital remain unchanged. This constructed period has no interest, impairment, acquisition difference or other transactions.

The opening worksheet removes the investment and capital of 100, leaving group cash and equity of 200. The closing worksheet also removes the reciprocal receivable and payable of 30, leaving cash 220 and equity including profit of 220. The period operation first obtains each legal book's own checked report. It preserves all three original cash flows and names the internal payment and receipt as a supplied pair. Removing that pair leaves an external operating receipt of 20, no external payment and a net cash increase of 20. Profit of 20 also reconciles opening and closing group equity. The result is the same whether both internal flows were labelled operating or the lender recorded investing and the borrower financing.

A pair identifies the owner, flow identity and exact expected record on each side, together with the supplied explanation connecting the transfers. The dates and currencies must match; the amounts must be opposite and nonzero; the owners must differ; and a flow cannot be removed twice. A changed record, partial amount, duplicate reference or unexplained cash or equity movement refuses. Opening and closing books must share the same period, chart, ownership perimeter and compatibility assumptions. Each legal report retains its own equity explanations and missing notes.

If the caller supplies no pair, the operation keeps the original gross flows. It does not infer that two equal amounts describe the same transaction. Likewise, an omitted closing reciprocal-balance adjustment leaves an extra receivable and payable in the worksheet, even though equity still reconciles. These cases remain visibly incomplete. By contrast, omitting the closing investment adjustment breaks the declared equity bridge and refuses. The distinction matters: reconciliation can expose contradictions without finding every missing internal transaction.

The checked extension covers two account charts and both activity classifications, with complete legal reports and group results preserved for replay under `data/accounting-master/` in `group-period-report-20261025`. It is a conditional period summary. Actual source evidence, the completeness of paired transactions, indirect operating cash flow, full notes and the wider group methods remain unresolved.

Carry the legal cash explanations through the same internal pairs. The period summary can also retain the event-level explanations of each legal book. `GroupIndirectCash/1` recomputes the group report and each supplied `DeclaredIndirectCash/2` result. Each legal bridge must select exactly the opening, closing, period and cutoff used by its group member. Its complete financial-report request and result must equal that member's retained report. A valid explanation for another company, chart or reporting date therefore cannot substitute for the required history. Both legal explanations remain in the result alongside the unchanged group report.

Simply adding the two operating-cash explanations would still retain any internal payment or receipt classified as operating. For the same period and currency, write P_j, W_j, N_j, R_j for the profit and adjustments already defined for legal book j. Let T be minus the signed sum of the operating cash flows in the group's explicitly removed pairs. Each receipt is positive and each payment negative. Summing the checked legal identities and applying that same removal gives group operating cash

$$C_o^{\text{group}} = \sum_j \bigl(P_j + W_j + N_j + R_j\bigr) + T.$$

Here the supported group method leaves profit unchanged: summed legal profit equals the group's reported profit. A method that eliminates intragroup profit would require a further derivation and contract. The adjustment T is calculated from the named gross cash records; it is never a balancing remainder.

The earlier two classifications both give $T = 0$. When both loan flows are operating, the payment of 30 and receipt of 30 cancel in this sum, although both still need removal from gross receipts and payments. When the payer records investing and the payee financing, neither flow enters operating cash. To test the correction itself, consider two additional supplied classifications. If P records the payment as operating while S records the receipt as financing, summed legal operating cash is $-30 + 20 = -10$. Removing the internal payment gives $T = 30$, so group operating cash is 20. If P instead records investing and S operating, summed operating cash is $30 + 20 = 50$ and $T = -30$, again leaving 20. These examples test category composition; they do not determine which classification is applicable to an actual loan.

The constructed event programs reproduce the previously supplied opening and closing books, including their memory and all balances. The adapter makes two explicit representational changes: it translates each old literal cash-flow identity into the execution engine's hash of event and local-flow identity, and adds the empty input schema required by a method with no memory writes. The corresponding pair references and report-request hashes change with those inputs. After these declared changes, the complete legal requests and results match, rather than only their cash totals. Both the original snapshot records and the translation are retained, so the correspondence does not rewrite the old provenance.

Eight cases cover two charts and the four category combinations. Changed dependencies, unrelated or forged legal traces, missing owners, stale pair records and repeated removal are refused. A zero correction still retains both gross sides of its pair. If the caller omits a pair, the report keeps those flows and explains the resulting conditional amount; it does not infer a match from equal values. Accounting, source, control, classification and completeness admission remain false. Wider consolidation methods and complete notes remain separate requirements.

A.8 Preserving a model before extending its accounts

The scenario family now keeps each company's instructions and consequences together. Exchanging those results with another implementation raises two further questions. Can it identify the same input despite a different text encoding? Can it add a new accounting method without creating another cash balance or losing the company's tax history? The first is a representation question. The second requires a richer account of company state.

Different bytes can express the same instruction. A company name containing an accented character may be written directly in UTF-8 or through a JSON escape. Both decode to the same sequence of characters, but their original files differ. Keeping the exact source file and identifying the decoded model therefore require separate identities.

The legacy import retains the original bytes and their hash, the declared local decoder and profile, and the existing company draft's decoded identity. A separate portable content record binds the canonical decoded value. Equivalent escaping or whitespace can share that content record while retaining distinct original imports. Different character sequences are not normalized into one name. The exchange itself changes when its original bytes change.

A field extracted from a saved result is not passed off as an original file. Its import identifies the parent and exact JSON pointer, and validation checks that the value is present there. Family-member records expose names, paths, results, contexts and annual outcomes; expression records connect the program, policy, selected parent, generated child and declared dependencies. Recovery records retain the same family and method with the completed member prefix. Missing dependencies and wrong reference kinds prevent import. This is an explicit reference to legacy drafts, rather than a conversion of every field into the full portable numeric and type language.

Identity checks alone cannot establish that cash was calculated correctly. One adverse example changes a cash amount and recomputes the affected hashes, including the saved recovery result. The resulting packet is internally consistent and passes content binding. Repeating the calculation with the trusted local engine rejects it. The two verdicts remain distinct: a complete dependency graph preserves what was supplied, while semantic replay checks what the selected method computes. Neither supplies an actual source's publication date, proves historical execution or establishes physical storage durability. Executable methods are selected by the caller; packet content cannot load code.

A module uses a company's existing book. Consider the retained return-right example. Its legal book already contains cash 100, a recovery asset 16 and a refund liability 40. A return module must refer to those balances. Adding its opening cash or capital to the company again would create resources that no event supplied.

The finite invocation therefore binds the original entity, legal book, branch, source context, information view, method and complete opening state. The trusted return method produces a proposed state transition. It uses the named book's cash, preserves other legal books and exposes the journal and gross cash effects. The checked example pays 48 and ends with cash 52. No commit is implied by the proposal. Changing the owner, method, source, state, event or account map prevents reuse; missing and false findings remain distinct from a successful calculation.

The retained owner is `Synthetic-return-expected`, with settlement on 30 June 2024. It is not Northstar. Northstar's current language has no return-right memory, refund/recovery accounts or return-settlement operation. Its homogeneous inventory and immediate-deductible tax method also cannot represent this case unchanged. The successful finite proposal identifies a usable method boundary; native company support needs the state design below. The separate sector-company method remains report-only and cannot be turned into a posting operation by relabelling its result.

Returned goods need their own carrying value. The original return contract sold ten units at cost 12, with a refund of 20 per returned unit. Its opening estimates recognize a refund liability 40 and recovery asset 16. These amounts depend on the supplied expected returns, handling and deterioration. The historical accounting source requires the refund liability and recovery asset to be distinguished, with the latter based on former carrying amount less expected recovery costs and loss of value (European Commission, 2023a, IFRS 15.B21–B27).

At final settlement, two units return. Their deterioration is 2 per unit, handling costs 4 per unit and the supplied net realizable value of the lot is 20. Handling is assumed to be selling

administration; costs needed to bring inventory to its location and condition would require a different treatment (European Commission, 2023a, IAS 2.15–16). The recovered lot is therefore carried at $\min\{2(12-2), 20\} = 20$, and cash payments are 40 for refunds and 8 for handling.

The settlement debits the refund liability 40 and inventory 20, credits the recovery asset 16 and cash 48, credits cost of sales 4 and debits selling expense 8. The journal balances and records a book loss of 4. A later sale of the whole recovered lot for 40 gives revenue 40 and cost of sales 20. The lot retains its original contract, quantity, carrying amount, tax base and date so this second event can consume the right stock once.

Simply adding two units to Northstar's homogeneous inventory would value them at its existing unit cost, which differs from the recovered lot's cost of 10. The extended book instead reconciles the original homogeneous lot and each recovered lot separately. Let q_0 and c_0 denote the homogeneous units and unit cost, and let V_ℓ denote each recovered lot's carrying amount, all for the same legal book and currency. The proposed inventory control is

$$I = q_0 c_0 + \sum_{\ell} V_\ell. \tag{A.1}$$

The sum preserves different carrying values without silently selecting a cost-flow method. With no recovered lots, equation (A.1) reduces to the existing inventory rule. Whole-lot sale is the initial subset; partial sales and changes of estimates require further specified methods.

Tax history changes the cash needed at the close. The same return appears in the joined company case of Section 28.7. Its legal parent P supplies a complete trace through settlement, later sale and two tax closes. P's opening, dates and tax premises are retained; they are not new facts about Northstar. The case assumes that the original sale and cost were taxed and deducted, refunds and handling become deductible on settlement, and recovered inventory enters tax stock at its carrying amount. The supplied tax rate is 25 percent. The required enactment and recoverability findings remain explicit. Tax bases, recognition conditions and recovery manner matter independently of book profit (European Commission, 2023a, IAS 12.7–10,24–25,46–51).

To see the difference, let F and R be the opening refund liability and recovery asset, n actual returned units, u refund per unit, h selling handling per unit and V recovered stock. Under these particular tax premises, the settlement's book profit B and taxable profit T are

$$B = F - nu - R + V - nh, \qquad T = -nu - nh + V, \tag{A.2}$$

$$T - B = R - F. \tag{A.3}$$

The tax deduction still includes the refund whose book expense was recognized before the opening; the book reversal of recovery value has its own earlier tax difference. Substituting $F = 40$, $R = 16$, $nu = 40$, $nh = 8$ and $V = 20$ gives $B = -4$, $T = -28$ and the one-time adjustment $T - B = -24$.

The opening liability's zero tax base gives a deferred tax asset of 10, while the recovery asset's zero tax base gives a deferred tax liability of 4. Their reversal adds deferred tax expense 6. P's first-half pretax profit 56 therefore produces taxable income 32, current tax 8 and after-tax profit 42. Annual pretax profit 66 gives taxable income 42 and total current tax 10.5; the second close recognizes only the remaining 2.5. Applying the -24 adjustment afresh to each interval would deduct it twice. The tax history binds the settlement event and subtracts tax already recognized from the cumulative calculation.

Multiplying first-half book profit by 25 percent would instead give current tax 14 and omit the separate deferred expense 6. A coincidentally equal total expense would not make the current cash payment correct. A complete native tax close consequently needs tax bases, timing,

rate, origin, recovery support and the full temporary-difference inventory. Without them the tax-dependent annual report remains unresolved, while earlier valid journals survive as a dated prefix. The example establishes no actual tax statute.

The statement bridge must retain opening obligations. P's annual profit is 49.5, but operating cash is 101.5. Depreciation 20 and deferred tax expense 6 are noncash charges. Inventory falls by 60, the recovery asset falls by 16 and the refund liability falls by 40. Another payment is easy to overlook: the matured opening debt includes 10 of interest expensed before this year. The selected historical interest policy classifies that payment as operating, although current finance expense is zero (European Commission, 2023a, IAS 7.31–33). The exact indirect bridge is

$$49.5 + 20 + 6 + 60 + 16 - 40 - 10 = 101.5. \tag{A.4}$$

Omitting the last term overstates operating cash by 10. Subtracting the whole debt payment instead would move 90 of financing principal into operating cash. Equation (A.4) agrees with the separate gross cash events. Opening cash 500, operating cash 101.5, investing cash -20 and financing cash -90 yield closing cash 491.5. Opening equity 622 plus profit 49.5 yields 671.5.

These are P's legal-book amounts. Its internal receipt from S remains a legal cash inflow and is eliminated only in the group report. The same distinction prevents an apparently balanced module from double-counting internal flows. Deferred tax is added back once in the indirect bridge; separately adjusting DTA and DTL again would repeat the same effect.

What the native extension must carry. The richer profile needs closed account and component schemas, explicit return/lot authoring, strict eligibility findings, one chronological shared journal executor, tax-origin history and extended report mappings. Every temporary account must enter profit and the closing transfer to retained earnings. A complete year carries remaining lots, rights, tax bases, event identities and source context forward. It resets period income and cash-flow totals without deleting the history needed to prevent a second settlement. A partial year supplies no complete next-year opening.

The existing company profile remains unchanged. Selecting the richer profile is explicit; it cannot redefine old record fields or silently replace a method. The design and exact legal-book trace specify the required repair. Section A.9 now supplies a closed component authoring profile and a finite reference for return, tax, legal reports and carry. Section A.10 adds a finite composed-company declaration and a separate partial-state description for actual disclosures. General composition and production integration remain unfinished. Broader corporate, group, bank and insurance methods retain their separate obligations.

The supporting reference records are under `data/accounting-master/`:

```
company-portable-authoring-20261023
company-module-boundary-20261023
company-native-return-design-20261023
```

They preserve original bytes, a finite shared-book proposal and the complete retained legal-book comparison. Independent implementation, domain and human acceptance remain pending.

A.9 From a return declaration to the next annual opening

P's recovered stock gives the language a concrete implementation problem. The return contract, the legal book and the tax calculation each remember something different about the same two

units. The contract records that the customer's rights have ended. The stock record keeps the recovered carrying amount until sale. The tax calculation remembers an adjustment arising from expense recognized before the year began. A company program must preserve all three without posting the opening balances again.

The finite reference now connects these records through the complete retained 2025 P trace. It implements return settlement, whole-lot sale, P's selected current and deferred tax method, legal statements and annual carry. Other corporate events continue to use the checked retained methods. This division makes the implementation comparison precise: the new return and tax calculations can be compared with the earlier company case, while the surrounding methods supply the same corporate transactions as before.

An author declares facts before choosing an event. The versioned `ReturnModule AuthoringDraft/1` describes a component of one legal book. It identifies the source packet and opening state, entity, currency, period and account roles. Its contract memory includes sold and expected-return units, unit refund and cost, estimated recovery costs, and outstanding refund and recovery balances. The recovered-stock declarations keep quantities, carrying amounts and tax bases separately. Tax declarations state the selected timing method, rate and required findings.

These declarations allocate existing accounts. For example, a refund balance of 40 must agree with the liability in P's opening book; it is not an instruction to create another liability. The deferred asset and liability must likewise agree with their declared opening allocations. The remaining company balances come from the shared opening, rather than from a second component wallet.

Typed records and text declarations contain the same fields. The text form uses named declarations for the component, lots, tax premises and dated settlement or sale. It contains no executable Python, external include or opaque JSON body. Both forms preserve canonical exact rational amounts, strict Boolean findings and source references. Rendering a typed declaration as text and reading it again must reproduce the same value. This is a matched authoring comparison with the typed-library baseline, not evidence that text is preferable for every user.

Unknown values are explicit. Suppose the tax rate has not been established but the author also supplies a refund liability of 41 while two expected returns at 20 require 40. The contradiction is already decidable and the record is invalid. The unknown tax rate cannot conceal it. If the amounts agree but a required recovery finding is false, the selected method is unsupported. If the required finding is unknown, the affected calculation remains unresolved. Arithmetic checks inspect their own dependencies rather than replacing unknowns with zero or postponing every known check.

Source correspondence is another question. A syntactically valid source pointer may name the wrong disclosure. The retained example therefore checks the entire declared projection against P's original packet, including owner, dates, amounts and opening identity. Changing P to Northstar fails that correspondence even though both names are valid text. A correctly bound local packet still needs separate evidence for external authenticity and any historical admission claim.

One event changes one shared book. The reference state combines P's account balances with return memory, recovered lots, dated tax adjustments, classified gross cash and lifetime event identities. Every account belongs to one financial-statement class; every temporary account enters profit. A proposed return first checks its component, quantities, findings and date. It then derives the settlement journal already shown in Section A.8, creates the separate

recovered lot and closes the return rights. The surrounding inventory allocation stays intact.

The proposal identifies the complete preceding state, the exact event and the selected method bundle. Applying it repeats the calculation with the trusted local method and compares the journal, gross cash and complete after-state. A changed cash amount cannot become valid merely by supplying a new hash. A stale proposal, repeated settlement or sale of an already disposed lot is refused. This is an in-process reference for checking a transition; durable publication and concurrent writers require the separate storage design.

Cash sufficiency is checked after deriving the complete proposed journal. If P cannot pay the refund and handling costs, the event returns a funding failure and leaves the preceding book intact. It does not create financing. Missing tax evidence has a different consequence: the valid non-tax events can remain as an exact dated prefix, but the tax-dependent annual statement and next-year opening remain unavailable. Thus a useful partial history survives without being presented as a complete year.

The retained adapter has an explicit, narrower role. It recomputes the other corporate events from their original state and checked methods, verifies that their P opening balances agree with the shared book, and supplies their journals and cash origins. Saved expected journals are comparison evidence, not instructions to the executor. S's book and the group ownership state remain in that retained method. Replaying them checks compatibility; it does not claim a second independent implementation of those same corporate rules.

Tax accumulates within a year and resets at its boundary. The settlement records its taxable-minus-book adjustment of -24 with its event identity and date. At either tax close, let $B_y(t)$ be P's cumulative pretax book income from the start of year y through cutoff t, let $A_y(t)$ be the sum of the selected method's dated tax adjustments in that interval, and let $C_y(t^-)$ be current tax already recognized in the year. Under the retained constant rate $r = 1/4$, the additional current tax is

$$\Delta C_y(t) = r\{B_y(t) + A_y(t)\} - C_y(t^-). \tag{A.5}$$

The final subtraction prevents an interim tax charge from being paid again. At the first close, $B_y = 56$, $A_y = -24$ and $C_y(t^-) = 0$, giving 8. At the annual close, $B_y = 66$ and the same cumulative adjustment remains -24; subtracting the existing 8 gives the remaining 2.5. Negative additional tax would require a refund or other tax method that this finite reference does not supply. The opening deferred balances reverse separately, producing deferred expense 6.

The annual legal statements reconcile profit 49.5, cash 491.5 and equity 671.5. Their operating cash of 101.5 agrees with both the gross payments and the indirect bridge in equation (A.4). In particular, the opening interest payment remains a cash adjustment even when current finance expense is zero. Legal receipts from S stay in P's cash flow; they are eliminated only by the separate group method.

At year end, the reference transfers profit to retained earnings and clears the income and period cash accumulators. It retains the source context, settled return component, disposed-lot identity and lifetime event history. The old tax adjustment remains available to explain 2025, but its date is outside $A_{2026}(t)$. In a deliberately inactive following year, equation (A.5) therefore gives zero additional tax and the deferred reversal is not repeated. This checks annual memory; it is not a prediction of an inactive business.

The complete reference compares every P account and gross cash origin after all thirteen dated transitions, then checks the annual statements and carry. It also preserves the existing six company-authoring paths. The next section exposes the selected methods, source state and schedule in a finite composed-company declaration, then distinguishes the partial inputs

available from actual disclosures. General changes in return estimates, partial lot sales, other tax treatments and production execution still need their own design and conformance cases.

The matched authoring and execution records are under `data/accounting-master/`:

```
company-return-schema-20261023
company-return-execution-20261024
```

They retain the original source comparisons and failure cases. The resulting methods are conditional finite references; independent implementation, domain review and human acceptance of the governing design remain pending.

A.10 A company program begins with the state it can justify

P's return example has an opening book, identified lots, dated events and explicit tax premises. That information allows the reference calculation to reach an annual result. An actual company's published statements usually offer a different starting point: amounts reported for a particular group and period, together with policies and notes that leave some underlying contracts and allocations unknown. The language must accommodate both starting points. An author should be able to select methods for a complete conditional example and also describe a source-constrained partial state whose missing facts still prevent those methods from running.

A.10.1 Selecting methods for a complete retained company

For P, the finite composed-company declaration names the source packet and opening, legal-book context, reporting period, return component and all thirteen events in their original order. Each event selects a declared capability. Return settlement and recovered-lot sale use the native return method; the two tax closes use the native tax method; the other transactions use the previously checked corporate methods. Annual reporting and carry have their own explicit selections. The caller provides the available implementations and their identities. A capability name written in a program cannot import arbitrary code, and an unavailable capability cannot silently select a different method.

Matched text and typed records express the same declaration. The relevant draft is `RetainedCompanyProgramDraft/1`; its example and evaluator are identified at the end of this section. Binding the program to the retained case means comparing its full source, opening, event schedule and method selections, rather than accepting a familiar company name as proof of identity. An author cannot rename P as Apple and retain the old source facts. Nor does a correctly hashed result establish its own correctness: the reference verifier recomputes the proposals, annual report and carry.

Review findings remain separate from immutable source facts. If a required tax finding is unknown, the first seven valid events remain available, the six subsequent events remain unexecuted, and no complete annual report or next opening is returned. A negative finding instead records that the selected treatment is unsupported. Editing a finding changes the authored program's identity; it does not edit the source packet. With the retained findings supplied, the composed calculation reproduces P's profit of 49.5, cash of 491.5, equity of 671.5 and operating cash flow of 101.5. This completes the finite declaration for the case while leaving broader company composition and method admission to their own specifications.

A.10.2 What the disclosed balance sheet leaves unresolved

Consider the retained Apple balance sheet at 28 September 2024. It reports cash and cash equivalents of 29,943, inventory of 7,286 and net property, plant and equipment of 45,680, in millions of US dollars (Apple Inc., 2024a, p. 31). These are useful observations. They do not, however, identify the unrestricted cash available to each legal entity, inventory quantities and lot costs, or the original costs, accumulated depreciation, service dates and remaining lives of asset cohorts. Supplying one invented unit cost or useful life would choose an economic history that the disclosure has not established.

The input-readiness reference therefore preserves the original statements and asks what each selected execution method still requires. It keeps all 139 package-inventory entries, the 102 annual and 100 interim readings, and the two collections of 27 opening or comparative balances. Fifteen dependencies connect those records and the original note obligations to company inputs, affected operations, reports and annual carry. Missing claims, asset cohorts, tax bases and dated events remain visible beside the known displayed amounts.

There are also restrictions in the selected methods themselves. The operations draft requires a full calendar year, a fixed account registry, homogeneous inventory cost, specified asset cohorts and a simple tax treatment. Apple's retained consolidated disclosures do not satisfy those requirements merely because their balance sheet balances. P's composed program has a different restriction: it binds exactly the retained synthetic case. The readiness result keeps missing company state and unsupported method scope as separate reasons for refusing execution. That refusal concerns the selected profiles; it does not establish that Apple cannot be represented by a broader model.

Known arithmetic can still be retained without pretending to solve those problems. The displayed current and noncurrent term-debt captions sum to $10{,}912 + 85{,}750 = 96{,}662$ at the annual date. This is a sum of two reported captions, not total financing obligations or an allocation to individual instruments. The source's dash for annual debt-issuance cash flow remains a dash. Unknown original-byte publication and ingestion clocks limit historical as-of admission, but they do not prevent reading the named retained source. The unresolved actual-company history and the earlier conditional subsystem calculations remain separate, as before.

A.10.3 Fiscal dates belong to the source

A partial-state declaration can carry the reported measurements without manufacturing an executable opening. The draft `SourceConstrainedCompanyStateDraft/1` binds the exact readiness request and result, declares its fiscal intervals and names each selected observation. It includes all 54 current annual and interim balance captions. Each binding keeps the original measurement date, consolidated perimeter, basis, units, precision and source-admission status. Prior comparisons and the remaining readings stay available in the referenced readiness result; they have not all become executable state constraints.

The retained annual source covers 1 October 2023 through 28 September 2024; the following interim source covers 29 September through 28 December 2024 (Apple Inc., 2024a, 2025c). Using the half-open interval convention already introduced in Chapter 29, these become

$$[\text{2023-10-01, 2024-09-29}), \qquad [\text{2024-09-29, 2024-12-29}). \tag{A.6}$$

The exclusive end is the day after the source's inclusive closing date. The first interval has 364 days and the second 91; they meet without a gap or overlap. A balance-sheet observation dated 28 September remains at that date, rather than moving to the exclusive boundary on 29 September.

Those date calculations specify reporting coverage. They do not select an interest day-count convention, a depreciation fraction or tax-year law. The generic structural validator admits explicit non-calendar periods; the retained-source binder separately rejects a period that disagrees with the original observation. Earlier fiscal-closing examples already permit short periods. The addition here makes fiscal intervals directly authorable alongside a partial source state, while preserving the stricter scope of the existing execution methods.

A.10.4 A rounding assumption constrains a measurement

The default interpretation of a displayed amount is deliberately modest: the source displays this value, while its unrounded underlying measurement is unresolved. Storing the displayed integer as an exact rational preserves the token; it does not establish exact equality with the company's underlying balance. Even an interval requires a rounding convention that the retained precision annotation does not by itself establish.

An author may supply a separate rounding hypothesis. Suppose the reporting quantum is $q > 0$, the displayed value d is an integer multiple of q, and the rule rounds to the nearest multiple with ties away from zero. Let x denote the underlying value of the same reported measurement: it has the same date, perimeter, unit and basis, not a newly inferred legal-account meaning. The midpoints adjacent to d are $d - q/2$ and $d + q/2$. At a midpoint the value with the larger absolute magnitude wins. It follows that the values producing d are

$$x \in \begin{cases} [d - q/2,\, d + q/2), & d > 0, \\ (d - q/2,\, d + q/2], & d < 0, \\ (-q/2,\, q/2), & d = 0. \end{cases} \tag{A.7}$$

For positive d, the lower tie rounds up to d, while the upper tie rounds away to $d + q$. For negative d, the endpoint inclusion reverses. Neither midpoint rounds to zero, which explains the two open endpoints in the last case.

Under that explicit hypothesis with $q = 1$ million dollars, the annual cash-and-equivalents caption of 29,943 permits $[29{,}942.5, 29{,}943.5)$. The displayed retained deficit of $-19{,}154$ permits $(-19{,}154.5, -19{,}153.5]$ (Apple Inc., 2024a, p. 31). These are conditional deductions, not evidence that Apple uses the proposed tie-breaking rule. The assumption has its own identity and provenance, and it remains distinct from accepted source facts. Without it, the declaration retains the displayed value and leaves the underlying measurement unresolved.

The intervals neither allocate cash to legal entities nor split inventory into lots. They are not assumed independent, and their Cartesian product is not asserted to be a feasible company history. Every missing-state dependency therefore survives the rounding calculation. The reference returns a partial description, with joint feasibility and actual history unknown, and supplies no executable opening. This lets the language retain what a source says while making the next evidence or modelling choice precise enough to review.

The closed typed records, named text examples and bounded reference checks are under `data/accounting-master/`:

```
company-composed-profile-20261024
real-company-input-readiness-20261024
source-constrained-state-20261024
```

Their checks cover source identity, complete dependency retention, matching text and typed interpretations, fiscal boundaries and signed rounding ties. They reject changed source dates, legal-book relabelling, invented exact balances and a forged executable result. The next section

extends the source description beyond current balance captions while preserving the difference between displayed relations and latent company state. General source binders, admissible state reconstruction, broader accounting methods and independent acceptance of the governing design remain unfinished.

A.11 Reading the statements together without inventing history

The balance sheet gives a dated description of the company. The income, cash-flow and equity statements explain different aspects of the movement between dates. Reading them together requires more than adding amounts with similar labels. An equity balance is measured at an instant, profit accrues over a period, a cash-flow adjustment can concern a different population of items from the balance-sheet caption, and a subtotal calculated by the analyst supplies no independent observation against which to check itself.

The retained Apple statement networks already record those relationships and unresolved differences. The complete context draft makes their roles explicit in the language. Its combined view preserves 256 source observations and all 143 original relation records, including their original explanations and diagnostic classifications. It gives 135 displayed differences, six derived subtotals and two calculations that remain unresolved because an operand is a reported dash. This accounts for the full retained network; it does not establish the missing underlying company history. Section A.13.4 separately checks a numerical reading from the retained XBRL instance; the original PDF-only result stays unchanged.

A.11.1 A reporting period can contain several balance dates

Apple's annual equity statement is a report for fiscal 2024, but its opening equity is measured at 30 September 2023 and its closing equity at 28 September 2024 (Apple Inc., 2024a, p. 32). The portable observation retains the annual report-period string. The original pointed reading separately records the measurement date. The adapter follows that pointer and retains both meanings: the date of the stock and the period of the statement in which it appears. It does not overwrite one with the other.

The same distinction matters for cash. An opening cash amount is a stock; operating cash flow is a movement during the period. A cash bridge joins the two endpoints and that intervening movement. A snapshot subtotal instead compares amounts measured at the same instant. The language therefore declares the relation's purpose as well as its operands. Same-period income comparisons, same-date equity comparisons, stock-flow bridges, net-asset changes and explanatory differences have different context rules.

For example, the displayed annual change in net assets is compared with the displayed change in equity, using both sets of reported endpoints. With A_t, L_t and E_t denoting assets, liabilities and equity in millions of dollars at the closing date, and subscript 0 denoting the opening date, the comparison is

$$\begin{aligned}(A_t - A_0) - (L_t - L_0) &= (364{,}980 - 352{,}583) - (308{,}030 - 290{,}437) \\ &= 12{,}397 - 17{,}593 = -5{,}196 \\ &= 56{,}950 - 62{,}146 = E_t - E_0. \end{aligned} \tag{A.8}$$

All operands come from the retained balance-sheet readings (Apple Inc., 2024a, p. 31). Recomputing the right side from the equity endpoints gives it a distinct observed basis. The corresponding interim comparison gives 9,808. These displayed reconciliations retain the origi-

nal precision and consolidated scope; they do not identify the transactions that produced the changes.

An annual-to-interim endpoint comparison also retains both source contexts. The annual closing equity and the next quarter's opening equity are reported at the same date, even though they appear in different reports. Their equality can be compared in the combined named-source view. An interim-only view cannot use the unselected annual observation. Its five cross-period comparisons therefore remain unresolved, with their original relation records still visible. No later source is silently supplied to complete them, and the annual and interim bases are not generally declared interchangeable.

A.11.2 A subtotal needs an independent target to be a check

The source audit exposed an evidentiary weakness in the earlier annual relationship ledger. Three analyst-calculated totals—other equity movements, operating reconciliation adjustments and operating asset/liability adjustments—were stored as fixed numeric targets and labelled displayed pass/fail checks. Reproducing those totals checks the calculation against its saved value. It does not supply an independent disclosed total or an additional accounting identity. The interim ledger already treated the corresponding calculations as explanatory subtotals.

The new interpretation classifies all six as derived quantities and returns no independent residual for them. Their original constants, coefficients and classifications remain in the retained records, so the reason for the change can be examined. The source amounts and earlier arithmetic are not rewritten. The correction concerns the strength of the evidence attributed to a calculated subtotal.

Other differences have a different purpose. The annual opening cash-flow population exceeds balance-sheet cash by 772. The annual payable comparison leaves 329, and interim tax expense less cash tax paid leaves −12,397. Those differences remain explanations to investigate, not amounts to post into a balancing account. Even the zero vendor-receivable difference does not identify a collection schedule, gross transactions or their dates. The full-context result retains each relation's populations and declared diagnostic purpose beside its arithmetic.

Two annual calculations cannot be completed under the PDF-only baseline because the debt-issuance operand is a dash. The earlier displayed-total calculation used an explicit no-contribution convention. That convention and its result remain in the historical evidence, while the new source interpretation preserves the dash and leaves the financing total and term-debt diagnostic unresolved. Using such a convention again would require its own explicit premise. Similarly, five interim relations retain the annual concepts absent from the interim table. Summing the rows that are actually listed does not establish that an omitted class of transaction was zero.

A.11.3 Local compatibility is not a company history

The optional rounding intervals in Section A.10 make it possible to ask a further, conditional question: could an individual measurement identity hold for values inside those intervals? This requires two separate premises. One specifies the rounding rule; the other asserts that the proposed identity concerns the same underlying measurements. A zero displayed residual alone supplies neither premise.

Let a proposed relation have residual $r = \sum_i a_i x_i$, including the target with coefficient -1. Before computing an interval for r, collect repeated occurrences of the same observation and add their coefficients. Thus $x - x = 0$ exactly, rather than the difference of two independently

varying copies of x. For a positive coefficient, the smallest contribution uses the lower endpoint and the largest uses the upper. For a negative coefficient those choices reverse. Summing the contributions gives the residual endpoints; an endpoint belongs to the result only if every nonzero contribution can attain its relevant endpoint.

For the annual current-assets relation, six positive displayed components sum to the displayed total of 152,987. Under the proposed nearest-rounding rule with quantum one million, write the component errors as $e_1, \ldots, e_6$ and the total's error as e_T. Every one of these positive displayed measurements has error in $[-1/2, 1/2)$. The residual of the underlying measurements is therefore

$$r = \sum_{i=1}^{6} e_i - e_T \in (-7/2,\ 7/2). \tag{A.9}$$

The lower extreme would require the total's excluded upper endpoint, and the upper extreme would require the components' excluded upper endpoints. Zero lies inside this range. Under the separately proposed identity, that establishes local compatibility with the intervals. It leaves the rest of the statement network untested.

A simple counterexample shows why this distinction is necessary. If $x \in [0, 1]$, the proposed equality $x = 0$ is locally compatible, and so is $x = 1$. No single value satisfies both. A collection of local interval tests is therefore not a witness for a joint feasible history. The displayed-statement reference retains joint feasibility as uncomputed and actual history as unknown. Section A.12 adds a separate conditional assessment of selected balance measurements. That assessment creates neither legal books nor the claims, lots, contracts and behavioral assumptions required to continue the company.

The complete observation and relation declarations have matched typed and named-text forms in `RetainedStatementContextDraft/1`. Each operand has its source identity, date or interval, unit and explicit alias. Relation roles, signs, targets, source omissions and retained dependencies are part of the declaration. Source binding rejects changed dates, units, populations, aliases and roles; result verification recomputes the complete description. The reference preserves per-share values and weighted-average share counts alongside monetary amounts without summing unlike units.

The checked records are under `data/accounting-master/`:

```
source-relations-20261024
statement-context-coverage-20261024
```

The former contains conditional interval diagnostics for the balance relations; the latter supplies complete displayed-statement contexts and explicit unresolved results. Together they make the source layer more precise. General source-to-state mappings, shared method requirements, admissible latent histories and independent acceptance remain subsequent design obligations.

A.12 One value for each retained measurement

The current-assets total appears in more than one balance-sheet relation. Its components must add to it, and it must combine with noncurrent assets to give total assets. If these relations are tested separately, each test can choose a different unrounded current-assets amount. A company has one amount at that date, so the question is whether the same choice can satisfy all the selected relations. This section supplies that narrower joint test before attempting to construct an accounting history.

Let d_i denote a retained displayed amount and x_i its proposed underlying amount, both expressed in millions of US dollars. The index identifies an observation, including its company,

perimeter, reporting basis and measurement date. Two uses of that observation share x_i; a different date or source identity does not acquire the same variable merely because its caption is similar. The retained balance example concerns Apple's consolidated amounts at 28 September and 28 December 2024. It preserves the two dates and does not equate their balances.

To constrain an underlying amount from its display, the example proposes rounding to the nearest million, with exact half ties rounded away from zero. This is the named hypothesis introduced for the partial-state example, not a rounding rule established from the issuer's disclosures. In displayed million-dollar units, it implies

$$I(d_i) = \begin{cases} [d_i - \frac{1}{2},\, d_i + \frac{1}{2}), & d_i > 0, \\ (d_i - \frac{1}{2},\, d_i + \frac{1}{2}], & d_i < 0, \\ (-\frac{1}{2},\, \frac{1}{2}), & d_i = 0. \end{cases} \tag{A.10}$$

For a positive display, its lower half tie rounds upward to that display, whereas its upper half tie rounds to the next larger integer. The endpoints reverse for a negative display. Neither half tie rounds to zero, which explains the open interval in the third line. If the hypothesis is absent, the source adapter supplies no interval and the joint question remains unresolved. A literal dash also remains nonnumeric; it is not converted into the third line.

A.12.1 A common assignment must satisfy every selected equation

Each selected relation supplies exact signed coefficients: components occur on the left and the reported target is moved to the left with a negative coefficient. A separate named hypothesis asserts that this displayed relation also describes the underlying unrounded quantities. With one row per selected relation and one column per distinct observation, these coefficients form a rational matrix A. The conditional question is

$$\mathcal{F}(A, I) = \{x \in \mathbb{R}^n : x_i \in I(d_i) \text{ for every } i, \quad Ax = 0\}. \tag{A.11}$$

An element of $\mathcal{F}(A, I)$ uses the same value everywhere a measurement recurs. This is the difference from checking whether each individual residual interval contains zero. For example, the two equalities $x = 0$ and $x = 1$ are separately compatible with $x \in [0, 1]$, but jointly require $0 = 1$. Their intersection is empty even though both local tests pass.

In the retained annual current-assets relation, the displayed components are cash and cash equivalents 29,943; current marketable securities 35,228; accounts receivable 33,410; vendor non-trade receivables 32,833; inventories 7,286; and other current assets 14,287. Their sum is

$$29{,}943 + 35{,}228 + 33{,}410 + 32{,}833 + 7{,}286 + 14{,}287 = 152{,}987. \tag{A.12}$$

Thus the displayed values themselves satisfy this selected equation and lie inside their proposed rounding intervals. The joint calculation also uses the same current-assets variable in the total-assets equation, and the same total-assets variable in the balance-sheet identity. It tests all these conditions together rather than supplying a new value to each equation.

The reference implementation `JointMeasurementConstraints/1` accepts explicit rational coefficients and intervals, including endpoint inclusion. It uses the existing exact solver to seek an assignment and checks each returned rational value independently with exact fraction arithmetic. For every variable, that check applies the original interval inequalities; for every equation, it substitutes all values into the original terms and requires exact equality. A valid witness establishes nonemptiness of this declared set. When the solver reports infeasibility, its conflicting constraints are checked again by the same solver. That recheck is useful consistency evidence, but it is not an independently verified formal proof.

The supplied computational budget is part of the request. An exhausted budget or an inconclusive solver response gives an unknown result, not infeasibility. Missing premises are reported before solving, after the structure and types have been checked. This prevents an unresolved assumption from concealing a malformed equation or a unit mismatch.

A.12.2 Preserve the observations when constructing the system

The finite source adapter `RetainedJointSourceRelations/1` receives the exact combined statement request, an explicit selection of supported relations and optional hypotheses. It recomputes the preserved statement description. Coefficients, observation identities, dates and units come from that description; the caller cannot replace them with a more convenient matrix or a fabricated source result. The currently supported selection is the nine current balance relations at each of the two retained dates.

The observation's retained digest identifies its shared variable. Its consolidated perimeter, reporting basis, original period description, measurement kind and exact instant remain attached to its type. The source unit *USD million* is represented as a US-dollar monetary dimension with scale 10^6. Coefficients and numerical interval endpoints remain in displayed million-dollar units, so the conversion does not multiply the amounts a second time. The complete original context is preserved alongside this typed representation.

All 256 observations and all 143 original relation dispositions remain in the returned description, including relations outside this first joint selection. The six derived subtotals still lack independent observed targets, and the two calculations that depend on unresolved dashes remain unresolved. Cross-period comparisons, omitted classes of transaction and explanatory differences retain their original meanings. None becomes an asserted underlying identity simply because another part of the balance sheet has a feasible assignment.

The earlier displayed-statement report therefore keeps its own annotation that joint underlying feasibility is uncomputed. A separate result identifies the selected conditional system, its hypotheses and its assessment. This separation lets a reader recover what the source description originally said while examining the additional mathematical question. An empty selection reports that no system has been asserted; an unasserted rounding or identity hypothesis leaves the selected question unresolved.

A.12.3 What the retained balance example establishes

With all 18 supported balance relations selected, the system contains 54 distinct monetary observations, 27 at each date. Under the explicit rounding and identity hypotheses, it has an exact rational witness. Independent checks substitute that witness directly into the retained source terms and separately derived rounding bounds. The displayed values themselves also provide a checked witness for this selected example. These two checks establish compatibility of the selected measurement equations under the proposed assumptions; they do not identify the issuer's unrounded amounts.

The local interval calculations are retained alongside the joint result using exactly the same premises. All 18 local residual intervals contain zero, and the joint witness resolves the further question of whether the same amounts can work throughout the selected network. The constructed $x = 0, x = 1$ example remains necessary: agreement between the two kinds of check in this company example does not make the local test sufficient in general.

Source changes, altered coefficients or dates, renamed hypotheses and changed hypothesis meanings are rejected. Result verification recomputes the deterministic source mapping and

checks the mathematical evidence. It permits another valid witness because a feasible system can have many solutions. It does not require the solver to choose the same assignment, conflict ordering or elapsed time on every replay. Recorded checks also exercise missing hypotheses, empty selections and zero computational budgets. The executable definitions and complete case results are retained under `data/accounting-master/`:

```
joint-measurement-constraints-20261024
joint-source-relation-binding-20261024
```

A.12.4 From compatible measurements to an executable company

A feasible balance-sheet assignment still leaves the simulator's main historical problem unresolved. Consolidated cash does not identify the available cash of a particular legal book; an inventory caption does not supply quantities, cohorts or cost layers; and aggregate liabilities do not specify contracts, maturities or payment conditions. Those distinctions affect which events can occur and how they change future statements.

The selected equations constrain measurements at two dates without supplying the events between them. They assign neither probabilities to possible histories nor behavioral rules to future transactions. Section A.13 first connects these measurements to dated stock-flow relations and a wider selected statement network. Reconstruction must then retain those constraints while representing the missing state and admissible event histories explicitly. Until that step has its own source, accounting and computational support, the actual history remains unknown and no executable company opening is produced.

A.13 Balances and movements retain their different times

A cash-flow statement connects a balance at the start of a period, movements during that period and a balance at its end. The three quantities have a common monetary unit but different time meanings. Requiring their complete measurement types to be identical would reject a valid stock-flow relation; deleting their dates would lose the reason that the relation is valid. The language therefore checks the roles of the quantities before solving their numerical equation.

The retained annual example reports cash-flow-statement opening cash of 30,737 at 30 September 2023, a change of -794 during the following fiscal year, and closing cash of 29,943 at 28 September 2024. The corresponding interim statement reports an opening amount of 29,943, change of 356 and closing amount of 30,299 at 28 December 2024. These are source-bound cash-flow captions. They do not identify the available cash of a particular legal entity or bank account. Treating their opening and closing captions as one population is an explicit proposed interpretation in the conditional calculation.

A.13.1 A relation can connect different measurement types

Each temporal measurement declares a value type, a time and a named quantity. The value type retains issuer, reporting perimeter, reporting basis, monetary dimension and scale. The time is either a dated snapshot or a half-open period. The named quantity distinguishes, for example, the cash population reported by a cash-flow statement from legal-book cash available to meet a payment. The source mapping preserves the complete original context, including its period wording and measurement-kind description, alongside this explicit type.

For date-only records, write the flow period as $[s, e)$, with its last included date immediately before e. The opening snapshot is dated the day before s, and the closing snapshot the day

before e. This is the convention used by the retained source records, not an intraday accounting rule. For a declared population q, a stock-flow relation has the form

$$B_q(e - 1\text{ day}) = B_q(s - 1\text{ day}) + \sum_{j=1}^{m} a_j F_j([s, e)). \tag{A.13}$$

The opening and closing measurements must name the same quantity q. Every movement must cover the declared period, and all operands must have compatible monetary value types. The coefficients a_j retain the source sign convention. A separately proposed identity specifies why those particular movements account for the change in the named population.

For the annual cash-flow example, s is 1 October 2023 and e is 29 September 2024. For the interim example, they are 29 September and 29 December 2024. Substituting the displayed amounts gives

$$30{,}737 + (-794) = 29{,}943, \qquad 29{,}943 + 356 = 30{,}299. \tag{A.14}$$

The negative annual movement reduces the closing balance; the positive interim movement increases it. These equalities are checked displayed arithmetic. Whether the same identities hold for the underlying unrounded measurements requires the explicit identity, population and rounding hypotheses.

The annual closing and interim opening have the same displayed value and date, but they remain distinct source observations. Their reporting bases differ: one belongs to the annual issuer report and the other to the condensed interim statements. The reference neither merges their identities nor imposes equality between their unrounded amounts merely because the displays agree.

The `TemporalMeasurementConstraints/1` reference accepts two relation forms. A same-time relation requires matching value types and time declarations for all operands. A stock-flow relation requires the opening, movement and closing roles just described. Once those checks pass, it lowers the numerical part of equation (A.13) to an opening coefficient of $+1$, the supplied movement coefficients and a closing coefficient of -1. One observation identity still means one unknown throughout the system. The original temporal definition and the generated scalar constraints are both retained, so the calculation can be traced back through that lowering.

This does not weaken the earlier same-context profile. It remains an error to add differently timed quantities through an ordinary same-time relation. Changing the opening date by one day, swapping opening and closing roles, using a movement over another period, or changing the endpoint population is rejected even when the numerical amounts happen to balance. Missing hypotheses leave a well-formed question unresolved; they do not conceal malformed types.

A.13.2 Construct the requested network before asking whether it fits

The larger retained description has 143 relation records. The adapter first classifies them by their existing meaning and context, before examining solver outcomes. Forty snapshot sums, 30 period sums, eight same-period comparisons, 16 same-date endpoint comparisons and ten stock-flow bridges supply 104 candidate equations. Their monetary units and invariant contexts match within each relation, and their dates fit the applicable temporal form.

Three candidates depend on source concepts omitted from the interim table. The capital and whole-equity bridges omit the annual concept of equity issued; the financing-cash relation omits debt issuance. The fact that a row is absent does not establish a zero movement. Without a separate interpretation for those omissions, the adapter leaves these three relations outside its supported selection and preserves the original omission records. It therefore supports 101 conditional equations in this version.

The other 39 source relations also retain their distinct roles: six derived subtotals, 24 explanatory differences, two endpoint differences, two cash population comparisons and five cross-period or cross-basis comparisons. A derived subtotal lacks an independently observed target; an explanatory difference is not automatically an identity. The population and cross-basis comparisons require their own interpretations. Classifying these records keeps their evidence available without silently asserting equations that their source roles do not justify.

Every selected relation requires named proposed hypotheses for its underlying identity and population interpretation. Rounding remains separately optional. The adapter recomputes the complete retained statement description and derives its coefficients and dates from that description. It accepts neither substitute source coefficients nor an arbitrary solver result. Original observation digests provide sharing across equations; equal captions, dates or displayed values do not merge different observations.

A.13.3 The PDF-only request remains unresolved on a reported dash

Selecting all 101 supported relations uses 236 distinct observations. One of them, the annual debt-issuance amount in the financing-cash relation, is a reported dash. Its source interpretation remains nonnumeric. Even with all proposed identity, population and rounding assumptions supplied, the requested system is therefore unresolved. The result identifies that observation and relation. It supplies no witness and does not replace the dash with zero.

There is a useful, different question: are the explicitly selected relations whose operands have numeric displayed amounts jointly compatible? That selection omits the annual financing-cash relation in advance, leaving 100 equations and 229 distinct observations. Under the proposed assumptions, this narrower system has an exact rational witness. Independent checks substitute the witness into every original source equation and separately check the signed rounding bounds. The result records the excluded relation and states that the supported network was not selected in full.

The distinction is substantive. A witness for 100 equations does not resolve the missing measurement in the 101-equation request. Nor are equations removed because the solver dislikes them: the narrower request follows the pre-existing nonnumeric source disposition. If a fully specified selected system is infeasible, that result concerns its declared assumptions and equations; it is not permission to drop inconvenient relations until a witness appears.

Both requests preserve all 256 observations and all 143 original dispositions. Seven observations used only by the omitted annual financing-cash relation fall outside the narrower numerical system, but remain in the preserved source description. Dashes, omitted concepts, derived subtotals and explanatory differences are therefore still available for subsequent interpretation. The result distinguishes selected relations, supported but unselected relations, and relations outside this version's capabilities.

A.13.4 A second source representation supplies a reported value

The unresolved dash concerns the selected PDF reading, not every available representation of the filing. A later inspection of the retained annual XBRL instance finds a numerical fact for the same caption. Occurrence `f-325`, under `ProceedsFromIssuanceOfLongTermDebt`, reports zero USD for 1 October 2023 through 28 September 2024. It identifies Apple's CIK, has no additional segment or scenario dimensions, and declares decimal accuracy `-6`. The filing-specific label is *Proceeds from issuance of term debt, net*, matching the annual cash-flow statement. Its presentation relationship places it in financing activities, and the calculation relationship gives it a positive contribution to the financing total (Apple Inc., 2024c, retained instance, occurrence f-325, context c-1; associated label, presentation and calculation relationships).

This is a separate source-linked reading. The PDF still contains a dash; its original observation and the earlier unresolved calculation remain intact. The selected interpretation retains both representations and their exact identities. A dash elsewhere, an absent interim row, a nil fact and a reported numerical zero do not become interchangeable. The XBRL value's declared accuracy also matters: a reported zero at million-dollar accuracy does not establish an exact underlying economic zero (XBRL International, 2021, Component Fact, value and decimals properties).

The neighboring years and repayment row provide useful checks on the match. The same issuance concept reports USD 5,228 million and USD 5,465 million in the two comparative years, agreeing with the visible columns. The FY2024 repayment fact reports USD 9,958 million as a payment; its negative calculation weight agrees with the parenthesized cash-flow amount. All thirty retained occurrences of the eight financing concepts, including duplicates, stay distinct. The seven signed FY2024 components then sum to $-5{,}441 - 15{,}234 - 94{,}949 + 0 - 9{,}958 + 3{,}960 - 361 = -121{,}983$ million dollars, matching the displayed financing total (Apple Inc., 2024a, p. 33). The earlier no-contribution convention is therefore no longer the only available basis for this displayed calculation: the selected numerical reading has its own inspected source support.

The term-debt comparison remains a different question. Closing current and noncurrent carrying amounts sum to USD 96,662 million, compared with USD 105,103 million at the opening date. Subtracting the selected reported issuance and the signed repayment gives $96{,}662 - 105{,}103 - 0 - (-9{,}958) = 1{,}517$ million dollars. This is an unattributed difference between carrying-value movement and the selected cash flows. It supplies no allocation to foreign exchange, hedges, fees or another account and creates no balancing journal.

The selected interpretation can also be carried into the whole conditional measurement system. It changes one unknown interval while preserving all 236 selected observations and all 101 equations. The complete parent still retains the PDF dash, all 256 source observations and all 143 relation dispositions. The rounding rule remains the earlier, explicitly proposed nearest-million rule with half ties away from zero. Under that assumption, the latent issuance amount for a reported zero lies in the open interval $(-1/2, 1/2)$ million dollars. This interval comes from the chosen rounding rule; the source accuracy alone is not used to impose its endpoints or an additional nonnegativity condition.

The full conjunction admits an exact rational witness. A separate checker substitutes every value into each original equation and interval. A second complete witness fixes the issuance measurement at USD 0.25 million and still satisfies the same original constraints. This is a useful discriminator: the numerical reading resolves the selected source uncertainty without pretending that the latent amount is known exactly. No financing equation is dropped to obtain either witness.

Missing interpretation or rounding premises remain unresolved. A changed issuer, period, source identity, original dash, equation set or rounding rule is refused; an exhausted computation budget remains unknown. The interface retains the original PDF-only result separately. These checks establish a conditional measurement composition, not an actual sequence of company events. Legal books, instrument populations, source availability and the remaining identity and population assumptions still require their own evidence.

The exact source selection and signed comparisons are retained under `data/accounting-master/debt-dash-source-20261027/`; the full network, independent witness checks and refusal cases are under `data/accounting-master/debt-reading-network-20261027/`.

A.13.5 The selected reading can constrain a supplied history

The numerical reading can now enter the two previously supplied historical candidates while retaining the complete statement network. The annual and interim candidates keep their separate source contexts. Each has nineteen declared accounts, two hypothetical capital transactions and thirty explicit links from reported quantities to historical amounts. The new composition preserves those histories and links, partitions every selected observation and equation by its original context, and attaches the selected issuance interval without replacing the PDF-only parent.

Both candidates satisfy their declared constraints. An independent rational calculation reconstructs the account movements and snapshots, checks the source-to-history links, and substitutes the resulting measurements into all 236 selected intervals and 101 equations. Reintroducing the earlier hypothesis that the opening cash populations are identical still makes that historical candidate incompatible. A numerical debt-issuance reading cannot repair a wrong cash-population mapping.

These candidates remain sparsely connected to the filing. Of the 236 selected measurements, 176 have no supplied history mapping. Their equations remain in the calculation, but they do not thereby identify an account or transaction. Four further obligations preserve the unknown actual history and wider source coverage in the two contexts. All 180 unresolved obligations remain attached to the result. Nor have the annual and interim candidates been joined into one company history. Their separate feasibility therefore supplies no executable legal-book opening or probability over possible histories.

`SelectedReadingHistoryAttachment/1` implements this explicit composition. It accepts the exact original source parent, selected reading, rounding premise, context-specific histories and declared computation budget. Missing premises remain unresolved; altered parents, equations or contexts are refused; an exhausted search remains unknown. The earlier interface still recomputes its own PDF-only source and retains its original results. Complete requests, independent checks and adverse cases are retained under `data/accounting-master/selected-reading-history-20261027/`. The companion `spec/source-opening-compatibility-v1.md` specifies the supported joins through conditional opening, financial transition, reporting and period carry, including the inputs that each join still requires.

A.13.6 What the temporal network contributes to reconstruction

The selected network connects balance-sheet, income, comprehensive-income, cash-flow and equity measurements using shared identities and explicit times. It extends the earlier isolated balance equations without inventing a sequence of company transactions. The recorded checks include altered dates, periods, source coefficients, quantity interpretations, hypotheses and completeness claims. They also compare the exact terms and intervals with the preceding balance and cash-bridge references. A zero computational budget returns unknown, and a valid alternative rational witness is allowed on replay.

The implementation and complete retained examples are under `data/accounting-master/`:

```
temporal-measurement-contexts-20261024
statement-network-contexts-20261024
```

They establish checked conditional measurement behavior. Their interpretation still depends on proposed identities and populations; feasibility does not promote those premises to issuer facts or accounting authority.

Section A.14 develops a finite conditional connection to authored states and event sequences. Actual historical reconstruction still requires the corresponding source and method admission. Several histories can produce the same balances and movements while leaving different contracts, tax attributes, inventory cohorts or liquidity constraints at the end. Those differences matter for the simulator's next event. A history-family representation must preserve them, state its search scope and retain uncomputed or missing cases. Until that connection is justified, the retained company has no executable legal-book opening and its actual history remains unknown.

A.14 Different histories can explain the same closing report

Knowing a closing balance does not determine the opening balance or the events that produced it. Consider a deliberately authored monetary example with closing cash of 15. An opening of 10 followed by receipts of 2 and 3 reaches that balance. So does an opening of 8 followed by receipts of 3 and 4. The closing report agrees, but the amounts received during the year are 5 and 7, and the intermediate balances differ. A simulator that keeps only closing cash loses information that a later question may require.

The example is a finite representation test. Its two supplied events use an authored linear template that increases cash, decreases a debit-positive equity account by the same amount, and increases a monetary cumulative-sales memory field. The template is not a claim about how an actual issuer should recognize sales. It makes the distinction between an observed balance and a complete history explicit while allowing every calculation to be checked by hand.

A.14.1 Keep the opening as a shared unknown

Let c_0 be opening cash at 31 December 2023, and let a and b be the amounts of the supplied events on 30 June and 31 December 2024. The authored scope gives $8 \le c_0 \le 12$ and $0 \le a, b \le 5$. Opening equity is the signed counterpart $-c_0$, and cumulative sales start at zero. Under the declared incidence, the dated states satisfy

$$c_{\text{June}} = c_0 + a, \qquad c_{\text{December}} = c_0 + a + b, \tag{A.15}$$

$$z_{\text{December}} = a + b, \qquad c_{\text{December}} = 15. \tag{A.16}$$

The same c_0 appears at both dates. It cannot be chosen independently for each observation. Substituting $(c_0, a, b) = (10, 2, 3)$ gives June cash 12 and cumulative sales 5; substituting $(8, 3, 4)$ gives June cash 11 and cumulative sales 7. Both have December cash 15. These are two members of a continuous conditional family, not two equally likely scenarios or an enumeration of every possible member.

More generally, let x_0 collect the declared opening account balances and monetary memory fields. For each supplied event j, let u_j be its nonnegative unknown amount, t_j its date and b_j its fixed incidence vector. A snapshot at date t is

$$x(t) = x_0 + \sum_{j:\, t_j \le t} b_j u_j. \tag{A.17}$$

The posting components of each b_j sum to zero, and the opening account balances also sum to zero. These conditions preserve the balance of the authored book. The memory components retain the specified monetary movements without treating them as additional accounts. A snapshot measurement applies its declared linear form to $x(t)$ and requires the result to lie within its supplied interval.

The reference `ConditionalMonetaryHistory/1` keeps x_0, event amounts, domains, templates, dates, measurements and proposed premises in the family definition. It accepts symbolic opening intervals rather than requiring one fixed opening assignment. Civil dates must be valid and chronological; events on the same date retain their supplied order for replay. The finite additive equations do not establish order-dependent liquidity or recognition preconditions. Supplied event slots and exclusions define the search scope; they are not evidence that other transactions did not occur.

Every reported witness is replayed independently with exact fraction arithmetic. The check starts from the original opening domains, applies each template to its event amount, verifies balanced postings and account totals, and recomputes every requested snapshot measurement. An altered after-state cannot substitute for that replay. The constant-opening fragment also agrees with the preceding finite reconstruction reference under identical supplied dates, templates, amounts and measurements.

This first profile uses a single explicit monetary dimension and scale for its accounts and memory. Inventory units, interest rates, nonlinear valuations and general contract or cohort structures require richer types and methods. An affine balanced history can also violate legal or operational conditions that its supplied constraints do not express. The reference therefore reports no native execution or accounting admission merely because its equations fit.

A.14.2 Attach a measurement family without choosing its witness

The measurement network from the preceding sections is a family of possible quantities, not a list of constants waiting to be copied into an opening book. Let $\mathcal{M}$ denote its complete set of assignments m, and let $\mathcal{H}$ denote the authored history family. Explicit mappings g_k connect selected measurements to the corresponding history quantities. The attachment asks for

$$\mathcal{J} = \{(h, m) \in \mathcal{H} \times \mathcal{M} : m_k = g_k(h) \text{ for every supplied mapping } k\}. \tag{A.18}$$

Both parent formulas remain in this set. The mapping equations add constraints; they do not replace either parent by a solver-selected assignment. Parent variables have separate namespaces until a declared equation relates them. Identical text identifiers in different parents do not create an implicit join.

Suppose the measurement family permits opening cash between 8 and 12, a period cash change between 3 and 7, closing cash 15, and the opening-plus-change identity. The assignment $(8, 7, 15)$ satisfies that family. If a candidate history instead fixes opening cash at 10, choosing $(8, 7, 15)$ first would make the attachment appear impossible. Keeping the full measurement formula allows the compatible assignment $(10, 5, 15)$. A feasible parent witness is therefore insufficient as the input to this join.

The opposite failure is also possible. Keep the history opening fixed at 10, but constrain the measurement opening to 8. Each parent can separately have a witness, yet the explicit opening mapping requires $10 = 8$. The attached system is conditionally infeasible. Separate feasibility cannot establish joint compatibility, just as separate residual intervals could not establish the earlier measurement network's compatibility.

A.14.3 Distinguish a snapshot mapping from a movement mapping

A snapshot mapping evaluates a declared account or monetary-memory linear form at the observation's exact instant. A movement mapping instead collects the signed contributions of events in the observation's half-open period. For an account component with incidence

coefficient b_{ij}, that movement is

$$\Delta_i([s,e)) = \sum_{j:\, s \le t_j < e} b_{ij} u_j. \tag{A.19}$$

It excludes the opening balance. In the first authored history, the annual cash movement is $2 + 3 = 5$, opening cash is 10 and closing cash is 15. Replacing the movement by the closing snapshot would answer a different question even though both are monetary quantities.

The mapping must match the observation's value context, dimension, scale and named quantity. Its snapshot date, or both period endpoints, must be present in the declared history timeline. A named proposed mapping premise states the intended population interpretation. Equal amounts or dates do not establish that interpretation. The checked reference rejects an attempted attachment of the retained real-source cash-flow context to this synthetic book; no legal-book cash mapping is inferred from consolidated captions.

The `HistoryMeasurementAttachment/1` reference reconstructs both parent formulas, adds the supplied mappings and solves the resulting conjunction. Independent verification checks the original temporal measurement equations, the original history replay and every mapping. Snapshot and period-flow checks use their respective definitions. Alternative valid joint witnesses are allowed; their different openings, events and memory remain visible.

A.14.4 A candidate witness need not be an admitted history

Three questions remain separate: does the declared numerical formulation have a candidate, has its requested search been completed, and are the target's economic and evidentiary requirements satisfied? Missing proposed premises leave the numerical question unresolved before solving. An explicit unresolved source or state obligation can instead permit a witness for the relaxed formulation while leaving target existence unknown. For example, satisfying the authored cash equations does not supply an admitted opening or the contract state needed by a native company method.

An observation without a supplied history mapping stays in the measurement parent and appears as an unresolved target obligation. It is not removed to make the attachment complete. Likewise, the history parent's unresolved obligations and the attachment's additional obligations survive the join. A complete solver search over the declared linear system cannot discharge those missing conditions. With all declared mappings and premises supplied, the result can establish a conditional witness for that limited model while source admission, accounting admission and native execution remain separate.

The recorded references exercise strict intervals, conflicting measurements, wrong chronology, unbalanced incidence, unknown premises, unmapped observations, namespace collisions, altered formulas and forged completeness. Zero computational budgets return unknown. Infeasibility uses the existing exact solver and its conflict recheck, which remains distinct from an independent formal proof. The definitions, complete cases and replay evidence are under `data/accounting-master/`:

```
conditional-history-family-20261024
history-measurement-attachment-20261024
```

These steps specify a concrete conditional connection between measurements and supplied linear histories. Section A.15 extends the authored history to typed economic quantities and explicit intermediate constraints. General candidate generation, broader dimensional state, nonlinear method conditions, legal-book mappings and source or treatment admission still

require their own evidence. The retained issuer's actual history remains unknown. Its missing contracts, inventory cohorts, tax attributes and usable liquidity cannot be replaced by either of the authored examples.

A.15 A balanced closing statement can hide a stock shortage

The preceding history records monetary balances and movements. A company also needs goods to deliver. A closing inventory value can agree with a closing quantity even when the proposed sequence sells goods before they are available. Testing that sequence requires quantities with their own units and conditions on the intermediate states.

Consider an authored example with opening cash of 20 dollars, no inventory and a credit equity balance of 20 dollars. Four units arrive at a fixed cost of 2 dollars each, and three units are sold for 3 dollars each. Both events are dated 30 June 2024; the supplied order says which occurs first. The dates and amounts are synthetic conformance inputs. The assumed postings illustrate a fixed-cost, cash-settled transaction, without establishing the recognition requirements of an actual inventory or revenue policy.

Let C denote cash in dollars, V inventory carrying amount in dollars, E the signed, debit-positive equity balance in dollars, and Q the number of units. Thus the opening state is $(C, V, E, Q) = (20, 0, -20, 0)$. For a restock quantity u and a sale quantity v, the assumed effects are

$$(\Delta C, \Delta V, \Delta E, \Delta Q)_{\text{restock}} = (-2u,\ 2u,\ 0,\ u), \tag{A.20}$$

$$(\Delta C, \Delta V, \Delta E, \Delta Q)_{\text{sale}} = (3v,\ -2v,\ -v,\ -v). \tag{A.21}$$

The coefficients 2 and 3 carry dollars per unit in the monetary components; the quantity coefficients are dimensionless. The three monetary changes sum to zero for each event. The quantity change is retained separately and is never added to dollars merely because both are represented by rational numbers.

Substituting $u = 4$ first gives $(12, 8, -20, 4)$. Substituting $v = 3$ next gives $(21, 2, -23, 1)$. The closing book balances because $21 + 2 - 23 = 0$, and the remaining unit has the assumed carrying amount 2. Now reverse the order. The sale first gives $(29, -6, -23, -3)$, and the later restock still produces $(21, 2, -23, 1)$. Every monetary posting balances in both orders. Only the second order requires an intermediate stock of minus three units.

A.15.1 Apply a stated economic condition at every event

Suppose the authored model permits between zero and 100 units at every state. This is an explicit stock condition, not something inferred from the balance sheet. Write x_j for the state just after event j, with x_0 the opening, and Q_j for its quantity component. The conditional family then requires

$$0 \le Q_j \le 100 \qquad \text{for } j = 0, 1, \ldots, n. \tag{A.22}$$

The restock-first path satisfies equation (A.22), with quantities 0, 4 and 1. The sale-first path fails at its first event, where $Q_1 = -3$. Testing only the final date would miss the violation. Same-day events therefore retain their declared sequence, even though a daily snapshot includes all events on that day.

The `TypedAffineHistory/1` reference permits a typed linear form of the accounts and memory to be bounded in this way. Each supplied condition applies to the opening and every post-event state. Its interval and proposed premise are retained with the family. If no stock condition

is supplied, the algebraic sale-first path remains a candidate and the result reports that the condition was not checked. A missing premise instead leaves the conditional question unresolved. Neither case establishes that selling before delivery is allowed for the company being studied.

The type of an event size need not be money. In equation (A.20) it is a quantity of goods, while each affected field has its own type. Before forming scalar equations, the checker requires

$$\operatorname{type}(b_{ij}) \times \operatorname{type}(u_j) = \operatorname{type}(x_i), \tag{A.23}$$

where b_{ij} is the fixed coefficient taking event size u_j into field i. Adding the coefficient's dimension exponents to those of the event type, and multiplying their scales, must yield the field's exact dimension and scale; their book contexts must agree. For a cash effect, dollars per unit times units gives dollars. A euro coefficient, a different book or an uncompensated thousand-unit scale fails this check before solving. A measurement coefficient is checked against its field and result type in the same way.

The reference retains the original types and definition after lowering to scalar rational equations. Independent replay starts with the original opening domains, applies the typed coefficients, checks every balanced posting and declared intermediate condition, then recomputes the snapshot measurements. Unmeasured economic memory remains a shared unknown. This extension is affine: prices and other coefficients are fixed authored values, while event sizes and openings may be unknown. Unknown prices multiplied by unknown quantities, integer lots, endogenous event discovery and general contract methods need additional methods. The declared stock condition checks one economic restriction; it does not supply all operational or accounting prerequisites.

A.15.2 Connect cash and quantity observations to the same path

The cash and stock observations can now restrict the same complete typed history. This matters when neither restock nor sale quantity is supplied as a point value. Retain the opening cash of 20 dollars and zero units, the fixed cost of 2 dollars per unit, the sale price of 3 dollars per unit and the restock-first sequence. Let both event quantities range from zero to ten units. Suppose the observations report closing cash of 21 dollars and one remaining unit. Opening cash, cash movement and closing cash form one temporal relation; opening units, unit movement and closing units form another.

Write u for restocked units and v for sold units. The original history's effects and the two closing observations jointly imply

$$u - v = 1, \tag{A.24}$$

$$-2u + 3v = 21 - 20 = 1. \tag{A.25}$$

Substituting $u = v + 1$ into the second equation gives $-2(v + 1) + 3v = 1$, hence $v = 3$ and $u = 4$. Cash alone would leave many possible pairs; the quantity observation supplies a distinct restriction. Both equations apply to the same unknown events. Solving each parent separately and passing an arbitrary witness from one into the other could discard their common solution.

The closed `TypedHistoryMeasurementAttachment/1` record retains both complete parent definitions and explicit mappings for the selected observations. For an instant observation $y_{k,t}$, the mapping has the form

$$y_{k,t} = \sum_i a_{ki} x_{i,t}, \qquad \operatorname{type}(a_{ki}) \times \operatorname{type}(x_i) = \operatorname{type}(y_k). \tag{A.26}$$

Here $x_{i,t}$ is a monetary account or economic-memory field at the observation date and a_{ki} is a fixed authored coefficient. A unit observation can use a dimensionless coefficient of one on

unit stock. A dollar valuation of that stock needs a stated dollars-per-unit coefficient. Every product must yield the observation's exact dimensions, scale and book context; the program supplies no implicit exchange rate, unit conversion or price. The mapping's named quantity and proposed interpretation premise remain part of its definition. Correct dimensions alone do not establish that a disclosure and a stock record refer to the same population of goods.

For a period movement, the mapping instead sums the event changes in the exact half-open period. It excludes the opening stock. If the example starts with two units and still restocks four and sells three, the unit movement remains one while closing stock is three. An instant snapshot includes every event on that date in its declared order. It cannot select the state after restocking but before a later same-day sale merely by using their shared calendar date.

The attachment includes the typed history's entire constraint system, not only its scalar monetary lowering. In particular, every opening and post-event stock bound remains. Reverse the example's event order: the same closing cash and stock observations still agree, but the intermediate quantity is minus three. The joined family rejects that path under the stated nonnegative-stock condition. Omitting the condition leaves the algebraic candidate and reports that the stock restriction was not checked.

Every observation and temporal relation remains present even when no mapping is supplied. Such an observation creates an unresolved target obligation; it does not disappear from its parent equations. Unknown economic memory also remains free within its declared domains. Missing mapping or parent premises prevent solving, while conflicting premise identities reject the definition. A feasible assignment replays the original typed history and the original temporal observation checker, recomputes the mappings with exact rational arithmetic and satisfies every compiled constraint. The source interpretations and complete actual company state still require their own evidence.

The retained examples include compatible alternative histories, individually feasible parents whose intersection is empty, explicit valuation and scale conversion, wrong-unit mappings and the intermediate shortage. Their complete definitions and replay evidence are in `data/accounting-master/typed-history-attachment-20261024`. This connection constrains authored paths with observations. It does not yet apply arbitrary accounting methods through changing legal and operational state.

A.15.3 Author the same family as text and as typed records

A conditional family must keep the same meaning when it is written as text. The closed `ConditionalFamilyExchange/1` authoring interface covers joint measurements, temporal measurements, monetary histories, typed affine histories and the existing history–measurement attachment. Declarations name observations, equations, accounts, memory, templates, events, measurements and intermediate conditions. Their payloads retain the complete typed records, intervals, premises and source references. The author supplies the computational budget and the exact digest of the selected complete definition.

For example, the two event declarations in the stock case begin as follows; the angle-bracketed domain markers here stand for the complete authored interval records and are explanatory placeholders, not accepted syntax:

```
event {"id":"restock","date":"2024-06-30",
  "template":"restock","amount":<four-unit domain>};
event {"id":"sale","date":"2024-06-30",
  "template":"sale","amount":<three-unit domain>};
```

Each complete declaration occupies one line in the actual source. The complete examples

contain the bounds, endpoint inclusion flags, named proposed premise and provenance that the excerpt abbreviates. Reversing the two declarations changes the event sequence and the selected definition's digest. It can change feasibility under the stock condition even though the closing statement agrees.

Attachment declarations refer to earlier complete family definitions. The parser embeds both parent formulas; it never substitutes their chosen witnesses. The existing attachment still accepts the monetary-history and temporal-measurement parents described in Section A.14. The separate typed attachment in Section A.15.2 is not yet a profile of this five-profile exchange. Its complete typed records therefore use the separate checked interface rather than that text grammar.

The text and byte decoders reject duplicate keys or declarations, unavailable profiles, malformed UTF-8, floating or non-finite numeric tokens, wrong selection digests, omitted budgets and trailing content. Rational quantities remain exact reduced strings. Explicit zero budgets retain the original unknown result; missing proposed premises remain unresolved. The parser treats declarations as data and does not execute their contents or retrieve source references.

Map keys are normalized for both typed and text exchange inputs, while every ordered list remains intact. This distinction matters because the preceding solver reports assign internal variable names in insertion order: semantically equal maps can otherwise produce different diagnostic text. Replay uses the normalized definition, and independent witness checks also agree with the original unsorted record. This normalization does not promise identical outcomes for arbitrary old requests stopped at tight resource limits.

The full grammar, strict error behavior and eight retained source-to-record-to-result examples accompany the implementation. The monetary embedding also preserves the earlier alternative histories, and adverse cases retain the stock shortage, missing premises and resource stops. The supporting specifications and checks are under `data/accounting-master/`:

```
typed-affine-history-20261024
conditional-family-authoring-20261024
```

These references make the declared state and authoring subset reproducible. Whole-company method coverage, richer contracts, source admission and an independent implementation of the broader language remain separate tasks.

A.16 A balanced return still needs a quantity limit

A return can be described by balanced monetary postings even when its quantity exceeds the number of units sold. The preceding histories preserve unknown amounts and explicit stock conditions. Applying an accounting method to those amounts also requires a connection between its monetary calculations and its selected economic prerequisites. This section develops that connection for a restricted, exact example of returned goods.

The example retains an authored settlement dated 1 February 2025. Let n be the returned quantity, p the refund price per unit, c the original unit cost, d the reduction in value per unit and h the handling cost per unit. Quantities are in units of goods; unit amounts are in dollars per unit. Let N be the total recoverable amount of the returned goods in dollars. The assumed calculations give refund cash R, handling cash H and inventory carrying amount V:

$$R = np, \qquad H = nh, \qquad V = \min\{n(c - d), N\}. \tag{A.27}$$

These are the retained synthetic method's calculations. Its applicability to an actual company's

policy and contract still requires the corresponding facts and authority. The purpose here is to preserve all conditional amounts within this specified method and test its selected prerequisites.

A.16.1 Keep both sides of the carrying-value minimum

Suppose $p = 20$, $c = 12$, $d = 2$ and $h = 4$ dollars per unit. The authored domains permit $0 \le n \le 12$ units and $5 \le N \le 30$ dollars, with endpoints included. Refund and handling cash are linear in n. The minimum is represented by two disjoint cases:

$$\begin{cases} V = 10n, & 10n \le N, \\ V = N, & N < 10n. \end{cases} \tag{A.28}$$

Equality belongs to the first case. Every possible pair (n, N) belongs to exactly one case, so the compiler neither drops equality nor counts it as a separate alternative. Each case retains the entire original input domain and all monetary calculations. Nested minima retain the combinations of their cases, subject to the declared finite branch limit.

The complete `PiecewiseMethodFamily/1` definition includes the selected `MethodMonetaryEffects/1` record and its digest, one explicit interval and premise for every input, output measurements, target obligations and the branch limit. Entity, book, date, dimensions and scale are checked before arithmetic is lowered to rational formulas. A closed singleton domain can supply a fixed coefficient while its original declaration remains in the definition. Thus an unknown n multiplied by the fixed $p = 20$ is supported. If both n and p vary, their product is outside this affine profile; it is reported as unsupported, rather than replaced by a guessed price.

The postings provide a separate algebraic check. Let L be the refund liability amount being released and A the recovery asset being removed, both expressed as positive dollar amounts. With debit-positive signs, the retained method posts L to the refund liability, $-A$ to the recovery asset, V to inventory, $R - L$ to revenue, $A - V$ to cost of goods sold, H to selling expense and $-R - H$ to cash. Their sum is

$$L - A + V + (R - L) + (A - V) + H - R - H = 0. \tag{A.29}$$

Each term cancels regardless of the proposed return quantity. The compiler requires this balance, and the cash-flow reconciliation, to simplify exactly; it does not add a balancing amount. Equation (A.29) also explains why balance cannot supply the missing quantity restriction.

A.16.2 Intersect the amounts with selected prerequisites

The selected `MethodRequirements/3` definition supplies that restriction through the sold quantity s, as well as conditions on dates, component identity, settlement state and rights. The joined `PrerequisiteMethodFamily/1` embeds both complete definitions. Every requirement input maps either to a shared monetary input or to an explicit external known or unknown value. A shared mapping requires exactly the same complete type and the original source reference on each side. Its proposed premise explains the correspondence; matching labels alone does not establish that two reported quantities refer to the same goods.

For the retained example, the relevant quantity and cash conditions include

$$0 < n \le s, \qquad np = R_{\text{cash}}, \qquad nh = H_{\text{cash}}, \tag{A.30}$$

where R_{cash} and H_{cash} are the supplied external cash facts. With $s = 10$, $R_{\text{cash}} = 40$, $H_{\text{cash}} = 8$, $p = 20$ and $h = 4$, the refund condition requires $20n = 40$, hence $n = 2$. The handling condition independently agrees: $4n = 8$. The quantity limit is also satisfied.

Both cases in equation (A.28) survive. For $20 \leq N \leq 30$, carrying amount is 20 dollars; for $5 \leq N < 20$, it equals N. The result therefore retains a family of possible carrying amounts and their dependence on recoverable value, even though refund cash has fixed the returned quantity. Neither the two cases nor the number of retained witnesses supplies a probability distribution.

Now fix $n = 11$, keep $s = 10$, and supply consistent refund and handling cash of 220 and 44 dollars. Also fix $N = 20$, $L = 40$ and $A = 16$. The method gives $V = \min\{110, 20\} = 20$. Its eight posting amounts are 40, −16, 20, 180, −4, 44, −220 and −44 dollars; they sum to zero. The monetary family consequently contains a balanced assignment. The selected quantity predicate requires $11 \leq 10$, so the joined conditional family is empty. All inputs are singleton in this counterexample, and constant evaluation leaves one minimum case to check. A hypothetical quantity inconsistent with an imposed constraint makes this intersection empty; it does not make the supplied record structurally malformed.

External findings retain a different role. In the selected settlement, a known false finding that all return rights have closed yields the original `unsupported` consequence. An unknown finding remains unresolved. The solver cannot choose either finding to rescue an otherwise attractive monetary assignment. Known dates and component identifiers likewise retain the exact original comparisons. Across fixed findings the original consequence priority is invalid, then unsupported, then unresolved.

The first joined profile evaluates fixed guards with the original evaluator. An inactive body adds no condition, while the separate requirement that an input be known still applies. A guard with unknown external support is unresolved; a guard depending on a varying monetary input is unsupported in this version. Numeric equalities and inequalities, conjunctions and products with at most one varying operand can restrict the shared quantities. General symbolic disjunctions and products of two varying quantities need further methods.

A value in a feasible assignment is known *within that hypothetical assignment*. This satisfies the selected requirement's known-value test during replay, but does not turn an unknown actual transaction into an observed fact. Missing external facts, missing mapping premises and residual source or native state obligations remain visible. A feasible numerical candidate with such target obligations cannot establish that the intended actual history exists.

A.16.3 Check the complete assignment and preserve unresolved search

Every feasible branch is replayed through the unchanged monetary evaluator and the unchanged prerequisite evaluator, using the same exact input assignment. The checker also substitutes that assignment into every generated constraint. It retains the parent branch identities, external values and reasons, mappings, original expressions and source references. Another valid witness is acceptable; reproducing the solver's first chosen value is not the membership criterion. An infeasibility result retains named conflicting constraints and rechecks their joint inconsistency.

A global finite cell budget and elapsed-time policy apply across the retained branches. A witness from one visited branch establishes candidate existence, while unvisited branches still prevent a complete-search claim. Conditional emptiness requires every retained branch to be checked infeasible. Zero or exhausted budgets leave the search unknown. Bounded Python validation and exact replay are not hard-preempted at the elapsed deadline, and repeated searches near that boundary can report different coverage.

The executable specifications, complete synthetic inputs and adverse cases are retained under `data/accounting-master/`:

```
piecewise-method-family-20261024
prerequisite-method-family-20261024
```

The intersection strengthens a conditional method family without selecting one history. The next section connects this method to one fixed complete opening and an ordered settlement and sale. General economic histories, actual legal and operational state, source interpretation admission, integer lots and broader nonlinear methods remain substantive simulator work.

A.17 Different intermediate histories can reach the same closing state

The preceding quantity restriction excludes a return of goods that were never sold. A feasible monetary assignment can still describe a payment that the company cannot fund, or a sale of a lot it does not hold. Applying the method therefore requires a complete opening state and an ordered change to that state. Keeping several possible carrying amounts also requires the later sale to use the amount created by its own preceding settlement.

We retain the same synthetic return, now within one complete opening book. Opening cash is 400 dollars and other inventory is 60 dollars. The refund liability and recovery asset are 40 and 16 dollars respectively. All other balances, component identities, existing lots, tax memory and prior event identities are supplied explicitly in the retained opening. They are not reconstructed from those four amounts. The two events are settlement on 1 February 2025 followed by sale of the newly recovered lot. The default case orders both events on that date; another checked case sells on 1 March.

A.17.1 Apply the conditional amount to a complete opening

The `ConditionalReturnStateFamily/1` definition contains the complete opening `state`, the selected `event`, numeric `event_domains`, a proposed `premise`, remaining `obligations` and a finite `branch_limit`. A settlement domain may vary an explicitly unknown returned quantity, unit handling cost, unit deterioration or recoverable value. For a sale, only explicitly unknown proceeds may receive such a domain. A domain cannot replace a known event fact. Each domain retains its exact interval, premise and reference to the original event record and field.

Dates, identities, fractions, Boolean findings and externally supplied cash facts keep their given known or unknown status. In particular, the solver cannot decide that return rights have closed, invent performance of a sale, or change refund cash to make its equations feasible. The original builders regenerate the full event-dated monetary types, prerequisite bindings and source references. An unknown required amount in the opening state remains unresolved in this fixed-opening profile.

The compiler preserves every selected prerequisite and every case of the carrying-value minimum. It also checks the native structural conditions: the event identifier must be new, a settlement cannot overwrite an existing lot, and a sale requires an existing undisposed lot. A valid opening does not by itself guarantee that the next event is funded. Write C_0 for opening cash and retain the earlier refund and handling amounts R and H. The cash after settlement, denoted C_1, must satisfy

$$C_1 = C_0 - R - H \geq 0. \tag{A.31}$$

The equality is the cash posting identity; the inequality is the selected native funding condition. In the example, $400 - 40 - 8 = 352$ dollars remains. If a valid opening instead contains only one dollar of cash, every proposed settlement consistent with those cash facts fails the condition. The resulting conditional family is empty. An opening that is already cash-negative has the original `funding_failure` status before this family is constructed. These are different questions, and neither creates an automatic loan.

A.17.2 Carry the same lot cost through the sale

The complete two-event definition, `SequentialReturnStateFamily/1`, contains the complete `first_family`, a fixed `sale_event`, an optional `proceeds_domain`, a continuity `premise`, `observations` and remaining `obligations`. Its first event must be settlement and its second event must sell the whole newly created lot. The sale date cannot precede settlement or fall outside the opening book's period; its identity must differ from the settlement and all prior events. Unknown performance or fraction remains unresolved. False performance and a fraction other than one retain their unsupported consequence in this restricted sequence.

No intermediate state is selected to construct the second event. Instead, every first-event branch retains its original variables and constraints. The sale's cost is the same symbolic carrying amount V created in that branch. The original sale expressions provide its cash, revenue and inventory postings. This lowering supports the inspected references, constants, addition and subtraction; a changed definition outside that syntax is unsupported. The explicit continuity record identifies the original family, the carrying calculation, the lot and both dates. Thus carrying amount at settlement becomes cost of the same lot at sale under a stated correspondence, without silently removing the date from either monetary type.

Let P be sale proceeds in dollars, let $I_0 = 60$ be the inventory already held, and use subscripts 1 and 2 for the states after settlement and sale. Selling the entire new lot removes exactly the inventory added at settlement:

$$I_1 = I_0 + V, \qquad I_2 = I_1 - V = I_0, \tag{A.32}$$

$$C_1 = C_0 - R - H, \qquad C_2 = C_1 + P. \tag{A.33}$$

Both cash balances must be nonnegative and proceeds must be nonnegative. A large later receipt cannot repair a negative C_1. Even when the events share a calendar date, their explicit order preserves that intermediate test.

The refund and handling facts still fix $n = 2$. With $N = 20$, the first minimum case gives $V = 20$; with $N = 5$, the second gives $V = 5$. These are two representative assignments within the retained family. Total inventory after settlement is therefore 80 or 65 dollars, not 20 or 5 dollars. With $P = 40$, both assignments end with cash 392 and inventory 60 dollars.

Their profit paths differ too. Let π_1 denote cumulative profit after settlement from the supplied zero opening income and expense balances, before any later tax-closing operation. The retained journal has revenue zero, cost of goods sold $16 - V$ and selling expense 8 dollars. The sale then adds revenue P and cost of goods sold V. Consequently,

$$\pi_1 = -(16 - V) - 8 = V - 24, \tag{A.34}$$

$$\pi_2 = (V - 24) + (P - V) = P - 24. \tag{A.35}$$

The carrying amount cancels from cumulative profit after the whole-lot sale. At the two representative assignments, π_1 is -4 or -19 dollars, while $\pi_2 = 16$ dollars in both. The retained replay also finds the same complete terminal native state, including the disposed lot and tax memory. The journals and intermediate states remain different. This cancellation follows from the selected two-event treatment; it does not remove the need to preserve the path or establish a general result about other methods.

A.17.3 Use observations without replacing the family

An observation contains its identifier, event prefix 1 or 2, named account, complete type and an exact interval with source reference and premise. Its date, entity, book, currency, dimensions

and scale must match that prefix account. Requiring total inventory to equal 65 dollars after settlement restricts $60 + V = 65$, hence $V = 5$. This excludes the first minimum case and retains the second. Requiring inventory 60 after sale retains both cases because equation (A.32) already gives that closing value.

Replacing the first family by its first witness, with $V = 20$, would wrongly discard the path consistent with the intermediate observation. The sequence instead intersects the observation with every complete first branch. Every new domain, observation and continuity premise remains in the result. Missing premises leave the formulation incomplete and unresolved; remaining source or state obligations keep actual history existence unknown.

For a feasible assignment, replay materializes only the declared unknown numeric event fields. The repaired one-event family and both original native transitions check the complete states, journals and flows. The original sale monetary evaluator, every compiled prefix balance, observation and constraint must agree exactly. An alternative valid assignment passes the same checks; matching the solver's first witness is unnecessary. Saved JSON results replay after serialization, including the original source bindings.

The global branch, cell and elapsed-time policies retain their earlier meaning. One valid path establishes conditional existence, while unvisited branches leave coverage partial. In the intermediate-inventory example, examining only the rejected first case yields unknown, not an empty family, because the valid second case remains unvisited. Emptiness requires checked infeasibility of every retained case. Preparation and replay are bounded Python procedures rather than hard interruption at the elapsed deadline.

The complete definitions, adverse cases and saved replays are under `data/accounting-master/`:

```
conditional-return-state-family-20261024-r2
sequential-return-state-family-20261024
```

The repaired first package preserves its failed serialization predecessor as historical evidence. Both successful profiles describe hypothetical, uncommitted states; source admission, accounting admission and method eligibility remain false. Arbitrary event sequences, variable lot collections, source-based legal and economic state, and wider accounting methods still require separate work.

A.18 Ask what follows from the complete family

The retained histories answer a financial question that a single simulated path cannot settle: which amounts could occur, and which amounts must occur under the supplied assumptions? In the preceding example, inventory of 65 dollars after settlement is possible. Inventory of 60 dollars after the whole-lot sale is necessary. The distinction depends on all admissible paths, including paths that a first witness does not select.

A.18.1 Separate a question from an observation

An observation of inventory of 65 dollars restricts the historical family. Asking whether inventory of 65 dollars is possible leaves that family unchanged. The query therefore carries no additional source assertion. It retains the original opening, event order, branch conditions, funding restrictions, observations and premises. It also retains unresolved obligations concerning the actual company.

Write $\mathcal{H}$ for this complete conditional family and h for one history within it. At a chosen prefix $t \in \{1, 2\}$, let $A_{a,t}(h)$ be the signed balance of account a after that event. The query asks about

the scalar amount

$$f_t(h) = \sum_{a \in \mathcal{A}} c_a A_{a,t}(h), \qquad h \in \mathcal{H}. \tag{A.36}$$

Here $\mathcal{A}$ is the set of distinct queried accounts, and each c_a is an exact rational, dimensionless coefficient. Every account must have the same full monetary type at that prefix: entity, legal book, date, currency, dimensions and scale. A threshold has that same monetary type. This is a same-date account sum, not permission to add differently dated or differently denominated amounts. The retained profit query applies the signed account convention to revenue, cost of goods sold and selling expense.

`SequentialOutcomeQuery/1` contains the full `family`, the chosen `prefix`, the complete `type`, the account `terms`, an `operator` and a rational `threshold`, alongside its `profile` identifier. The five comparisons are equality, strict less or greater, and weak less or greater. Arbitrary callbacks and compound expressions are outside this first interface.

A.18.2 Distinguish possible, necessary and empty

Let $Q(h)$ denote the selected comparison of $f_t(h)$ with its threshold. The implementation asks two existence questions on the unchanged family:

$$\mathcal{H}_Q = \{h \in \mathcal{H} : Q(h)\}, \qquad \mathcal{H}_{\neg Q} = \{h \in \mathcal{H} : \neg Q(h)\}. \tag{A.37}$$

A checked member of $\mathcal{H}_Q$ establishes possibility. A checked member of $\mathcal{H}_{\neg Q}$ is a counterexample to necessity. If both exist, the comparison is contingent. If the first exists and the second is proved empty, it is necessary within this family. If the first is proved empty and the second exists, it is impossible. When both sets are empty, the answer is `empty_family`; necessity is not applicable. In particular, an unfunded settlement does not make every financial assertion meaningfully true.

For inventory of 65 dollars after settlement, a path with $V = 5$ is a satisfying witness and a path with $V = 20$ is a counterexample. For terminal inventory of 60 dollars, equation (A.32) supplies the invariant, while the complete branch check proves that no counterexample meets the original conditions. Inventory of 64 dollars after settlement is impossible in this example. Adding the observation inventory of 65 dollars first would instead make that amount necessary in the smaller, observed family. These are different questions with different conditioning information.

Every witness replays both original native transitions. The verifier computes the account sum independently with exact rational arithmetic and checks which side of the comparison it satisfies. An empty region requires complete infeasibility evidence and a separately checked unsatisfiable constraint core. The checker accepts a different valid witness or a different valid core; reproducing the solver's first choice is not the criterion.

Search is divided into elementary regions. For equality, the regions are equal, less and greater; for each inequality, its complement uses the exact opposite strict or weak comparison. One shared cell-count and elapsed-time budget spans all regions of all parent branches. An unvisited region cannot prove emptiness. A found counterexample can already refute necessity, even if the search is otherwise incomplete; the result preserves that fact while leaving unproved conclusions unknown. The number of witnesses supplies no probability.

A.18.3 Bound the amount without selecting a history

The closed `SequentialOutcomeRange/1` request contains its `profile`, `family`, `prefix`, `type` and account `terms`. It asks for the tight lower and upper endpoints instead of a comparison. For a

nonempty family these are

$$L = \inf_{h \in \mathcal{H}} f_t(h), \qquad U = \sup_{h \in \mathcal{H}} f_t(h). \tag{A.38}$$

An endpoint is reported as attained only if an original admissible history reaches it. Thus an open interval has valid finite endpoints without supplying executable states at those endpoints.

The supported fragment has finitely many affine branches with explicitly finite numeric input domains. Each feasible branch is a convex region defined by rational linear equalities and strict or weak inequalities. Within a nonempty such branch, a closure point can be approached from an original feasible point: their convex combinations satisfy every strict inequality until the limiting closure point is reached. Optimizing over the closure can therefore locate its infimum or supremum. Certification additionally checks that the endpoint is feasible in that closure, no original value is better, and the original strict constraints either admit or exclude endpoint equality. An attained endpoint must also pass complete native replay.

The global lower endpoint is the smallest certified branch infimum; the upper is the largest certified branch supremum. This aggregation requires every original branch to be either checked infeasible or certified at both endpoints. A partial search returns local branch evidence with no certified global bounds. An empty family has no bounds. An unbounded answer is invalid in this explicitly finite fragment, rather than an authorized infinity result.

The amounts in the example follow directly from the shared carrying value. With returned quantity fixed at two, $V = \min(20, N)$. The supplied recoverable value domain is $5 \le N \le 30$, so it permits both minimum branches and the carrying envelope is $[5, 20]$. Substituting into equations (A.32) and (A.34) gives

$$I_1 = 60 + V \in [65, 80], \qquad I_2 = 60, \tag{A.39}$$

$$\pi_1 = V - 24 \in [-19, -4], \qquad \pi_2 = P - 24. \tag{A.40}$$

At proceeds $P = 40$, cumulative terminal profit is 16 dollars before tax closing. If proceeds instead range over $[0, 100]$, its envelope is $[-24, 76]$. Opening inventory of 60 dollars is included in both inventory queries. These equalities use the original signed postings; selecting two convenient witnesses would not establish the bounds.

If recoverable value is restricted to $5 < N < 20$, then $V = N$, and the intermediate inventory envelope becomes $(65, 80)$: neither endpoint is attained. In the original two-branch family, both branches support the upper bound of 80 dollars, but only the branch with $V = 20$ attains it. The other approaches 80 from below. Accordingly, `supporting_branches` and `attaining_branches` are different result fields. Global attainment requires at least one attaining branch at the global endpoint.

Even a tight outer envelope need not describe every feasible amount. A union of branches could permit only 65 and 80, leaving the intervening values unattainable while retaining the same endpoints. The range interface therefore reports an envelope and endpoint membership, not a general proof of interior feasibility. A predicate query can separately test a particular interior amount.

The reference definitions and saved complete requests/results are in `data/accounting-master/`:

```
sequential-outcome-query-20261024
sequential-outcome-range-20261024
```

Both interfaces preserve missing premises and unresolved target obligations. Where those obligations remain, an answer concerns the candidate family and the actual outcome stays unknown. Their states are hypothetical and uncommitted; source admission, accounting

admission and method eligibility remain false. General histories, nonlinear or integer-lot methods, actual-company state reconstruction and independent implementations remain outside these checked references.

A.19 Retain the evidence needed to reproduce an answer

An answer about a conditional family is useful only while its meaning can be recovered. The reported inventory range alone does not identify the opening book, permitted recoverable values, event order or unresolved source obligations. Retaining the complete definition and result allows a later reader to inspect those conditions and a compatible implementation to repeat the calculation. Storage must also distinguish a complete saved answer from an interrupted write.

AccDSL's earlier P07 reference already provides physical SQLite storage, immutable records, evidence views, event replay, retry handling and one schema migration for its original company engine. Its run construction and replay instantiate that older engine directly. A newer native state can be saved as a generic record without thereby becoming a state that the older engine can execute. The `SimulatorEvidenceArchive/1` reference uses the existing immutable nodes and transactions to retain evidence under a narrower, explicit meaning: the original bytes have been archived and their relationships checked.

A.19.1 Preserve exact records and their relationships

A complete range result contains both financial values and diagnostic metadata, including elapsed time. P07's original numeric contract refuses JSON floating numbers; its identity conventions also normalize Unicode text and treat certain named lists as sets. Those conventions serve its original interface but cannot promise exact retention of every newer JSON record. For example, two spellings of an accented name can have different UTF-8 bytes despite the same normalized text, and two differently ordered constraint lists can receive the same generic identity.

The archive retains the original JSON text as bytes inside a small closed envelope. Hex encoding preserves those bytes without applying text normalization or numeric conversion to their contents. A byte count and SHA256 digest detect changes relative to the stored record. The declared profile label is also kept as exact UTF-8 bytes. Whitespace, number spelling, list order, finite timing metadata and unresolved fields therefore survive retrieval. Different original bytes remain different records, even when an analyst considers their numerical content equivalent.

The closed `SimulatorEvidenceBundle/1` contains its `profile` and an ordered `records` list. Every record has a local `name`, a descriptive `role`, a declared `profile`, the original `json` text and `dependencies` naming earlier records in that bundle. Roles distinguish a definition, state, proposal, result or note; none confers authority to execute. Names must be distinct, and forward, missing, repeated or cyclic dependencies are refused. Declared dependency order is preserved even though the underlying physical dependency edges are unordered.

JSON must contain an object, have no duplicate keys, use valid UTF-8 scalar values and contain no nonfinite numbers. When the payload states a profile, it must equal its declared profile. An unfamiliar profile can still be retained as unverified evidence. Callers supply explicit maximum record count, individual byte size and total byte size. These limits control this reference's accepted input; they are not recommendations about model size or performance. Validation does not discard an inconvenient field or round an unsupported amount to make the record fit.

For the retained range example, one bundle contains the complete opening state, the query definition depending on that opening, and the complete answer depending on both. Another

preserves the candidate family's unresolved actual-source obligation. The receipt identifies every record in order. A compatible later query implementation can read the original request and repeat its semantic checks; the archive itself supplies no new financial interpretation.

A.19.2 Publish a complete bundle and recover it

Inserting several records independently can leave an incomplete group if the process stops between insertions. The archive therefore inserts the record nodes and their common receipt in one transaction. The receipt depends on every record, and each record depends on its named predecessors and the archive's version marker. An exact retry produces the same receipt. Changed bytes produce a new identity, rather than replacing an earlier record under the same name.

Before commit, the archive rereads the complete receipt and its records. This step matters because an existing content identity alone cannot prove that stored bytes remain intact. Retrieval checks closed envelopes, identities, byte counts, hashes, dependency edges, ordering and the original JSON conditions. A corrupted existing record therefore cannot pass an exact retry merely because its former identifier is still present.

A new archive is prepared and marked in a temporary file in the destination directory. After that file is closed, a non-overwriting hard link publishes it. An existing unrelated database is inspected read-only and refused before the older storage implementation can alter it. An unknown schema version or incompatible archive marker is also refused; this reference introduces no automatic migration. A hard process exit can leave an unreferenced staging file, which the archive does not sweep away automatically.

The retained interruption checks observe no destination before initial publication and a complete marked archive afterwards. For a bundle transaction, exit before commit leaves no new receipt; exit after commit leaves the complete bundle available on reopen. These are local process-recovery results. Power loss, concurrent writers, distributed storage and filesystems without the chosen publication operation require separate evidence. The unchanged P07 migration checks continue to concern its own schema and engine.

A.19.3 Check bytes separately from financial meaning

Consider a proposed settlement whose closing cash is increased by one dollar and whose signed original-equity balance is reduced by one dollar. The modified book still balances. Generic storage can retain it and pass its byte-integrity checks, but replaying the original settlement does not produce that state. The original native proposal checker rejects it. This example separates three questions: whether the bytes were preserved, whether the stated transition produces them, and whether the underlying company facts and accounting treatment have been admitted.

The archive answers the first question. Its result has `archived=true`; retrieval reports `byte_integrity_verified=true` after its checks. It keeps `semantic_verified`, `source_admission`, `accounting_admission`, `method_eligible`, `simulation_executed` and `native_committed` false. These are the archive's own conclusions, regardless of stronger assertions inside a retained payload. A stored digest is an integrity comparison, not proof of authentic evidence or a correct financial method.

The reuse investigation and reference cases are retained under `data/accounting-master/`:

```
simulator-storage-reuse-audit-20261024
simulator-evidence-archive-20261024
```

The reference preserves complete evidence across its tested restart conditions. Persisting an executable native event cursor, migrating arbitrary histories, joining sector methods and admitting real-company state remain separate design and conformance work.

A.20 Keep lease rights distinct within one company book

Preserving a proposal makes its calculations available for inspection. To reproduce them, the company model must also preserve the rights that generated the entries. A total lease liability and a total right-of-use asset do not tell us how much of either balance disappears when one right ends. Nor does a balanced calculation for one lease establish that the company can pay it after another lease has used the same cash.

Consider the retained synthetic example with two rights in one lease. Right A has a right-of-use asset of 90 and a liability of 100; right B has an asset of 30 and a liability of 50. Amounts are exact USD quantities in this example. The supplied findings support tracing the amounts to their individual rights. Terminating B therefore removes its asset of 30 and liability of 50, giving a gain of 20. Allocating the total asset of 120 in proportion to B's share of the total liability would instead remove

$$120\frac{50}{100+50} = 40, \qquad 50 - 40 = 10.$$

That gain of 10 is wrong for the stated traced-right inputs. The two rights have different asset-to-liability ratios. Their totals conceal that difference, so the simulator retains each right's amounts through termination and remeasurement.

This example uses the inspected `CorporateResiduals/1` methods and their historical synthetic source scope. The owner remains `Synthetic-heterogeneous-rights`, with the Legal book, base branch, USD currency and `historical_EU_IFRS_conditional` basis. The permitted execution window begins on 1 January 2024 and ends immediately before 1 January 2026. Complete opening balances, lease membership and method findings are supplied premises. They are not inferred from an actual issuer's reports or evidence of current accounting compliance.

A.20.1 Follow the rights through the complete lifecycle

Both opening rights have a supplied annual rate of 10 percent. On 1 January 2024, the example terminates B and replaces A's payment schedule with one payment of 121 at the annual period end. The original method discounts that payment for one year. For A, the liability rises from 100 to 110, and the remeasurement adds the same 10 to its asset:

$$L_A^{(1)} = \frac{121}{1.1} = 110, \qquad R_A^{(1)} = 90 + (110 - 100) = 100.$$

Here L_A is A's positive liability amount and R_A its right-of-use asset; the superscript identifies the completed step. B is now zero. Combining B's termination with A's remeasurement reduces the total asset by 20 and the positive liability by 40. With debit-positive signed accounts, the journal is therefore $\Delta\text{rou} = -20$, $\Delta\text{lease_liability} = +40$ and $\Delta\text{other_gains} = -20$. The entries sum to zero; the credit to gains records the 20 profit.

A second supplied modification occurs on the same date. It removes a further 10 of A's asset and 11 of its liability, producing a gain of 1. The remaining amounts before remeasurement are asset 90 and liability 99. Replacing the remaining annual payment with 99 gives:

$$L_A^{(2)} = \frac{99}{1.1} = 90, \qquad R_A^{(2)} = (100 - 10) + (90 - (110 - 11)) = 81.$$

The remeasurement reduces the remaining liability by 9 and makes the same reduction to the asset. Across this second step, the total asset falls by 19, the positive liability falls by 20, and the additional gain is 1. Tracing the removed amounts and then remeasuring the remainder explains all three changes.

On 1 January 2025, the supplied findings confirm one elapsed year with no intervening accrual or modification. The payment of 99 comprises interest of $90 \times 0.1 = 9$ and principal of $99 - 9 = 90$. Cash falls by 99, finance expense increases by 9, and the signed liability increases by 90 to zero. The retained cash-flow policy assigns 9 to operating outflow and 90 to financing outflow. Finally, a supplied systematic depreciation amount of 81 removes the remaining asset and records expense of 81. This depreciation amount is an input supported by the example's finding; the wrapper does not infer a useful life.

With opening cash 200, closing cash is $200 - 99 = 101$. Both lease balances are zero, and cumulative pretax profit from these four events is

$$\Pi = 20 + 1 - 9 - 81 = -69.$$

The gains, interest and depreciation explain the profit independently of the cash payment. This is a complete lifecycle for the supplied finite example; it does not supply an annual tax close or a complete financial report.

A.20.2 Apply each lease to the same cash balance

Now instantiate a second synthetic lease with the same supplied right-level amounts and its own identity and premise. A company opening containing both leases must reconcile to assets of 240 and positive liabilities of 300. Cash and equity belong to that one company opening. Adding an isolated lease calculator's opening cash or equity would capitalize the company again each time a method runs.

For example, opening company cash of 400 followed by one payment of 99 gives 301. Adding the component fixture's opening cash of 200 would give 501, an incorrect result. The shared-book proposal instead applies only the original journal deltas. It updates the selected lease's rights, preserves the other lease's memory, and leaves unrelated company accounts unchanged. Right A in the first lease and right A in the second lease are distinct: their identities are the pairs of lease identifier and local right identifier.

Funding must be checked in event order. Under the explicitly supplied `no_overdraft` policy, company cash of 300 funds both payments of 99 and leaves 102. Cash of 150 funds the first payment and leaves 51. The second payment then fails: its liability remains 90, while the first lease's liability remains settled. The successful earlier events survive; the failed event adds no journal, cash movement or after-state. This is a funding failure under the company premise. The original component calculator's ability to calculate a balanced entry ending in negative cash does not provide that funding policy.

A.20.3 Bind proposals to complete state and original methods

The `SharedLeaseBook/1` record contains context, original source-scope hash, execution period, current date, the full original account registry, all lease components, collection-completeness and instantiation premises, funding policy, prior event identities, gross cash movements and the conditional-state flag. Every right retains its asset, liability, rate and last measurement date. Canonical rational strings preserve exact amounts. Before and after each event, the signed balances sum to zero, the ROU account equals all named rights' assets, and the negative

lease-liability account equals all their liabilities. Unknown membership cannot establish those equalities merely because a supplied subset balances.

`LeaseBookEvent/1` binds a new event identity, lease identity, date, source scope and one of the original `lease.reduce`, `lease.pay` or `lease.depreciate` methods. Parameters must satisfy that method's closed schema and required findings. Dates are nondecreasing within the execution window; a selected right's measurement date cannot lie after the event. Payment retains the original exact annual-anniversary requirement. A missing required date does not create a dated prefix, and no leap-day convention is silently supplied.

The permitted account changes are explicit. Reduction affects ROU, lease liability and other gains. Payment affects cash, finance expense and lease liability. Depreciation affects ROU and depreciation expense. The wrapper checks these effects, reconciles the complete rights collection and enforces the company's cash policy. Unknown supported inputs remain unresolved; malformed records, false findings and unsupported schedules retain their distinct original refusal categories. No invented loan or balancing entry repairs a refusal.

A `LeaseBookProposal/1` binds the complete before-state, event and original method hashes to the hypothetical after-state, journal and gross cash effects. Applying it recomputes the original method and compares the entire proposal. A forged cash increase offset by equity can preserve balance while failing this comparison. Context, source, period, unrelated accounts and nonselected leases must remain unchanged. The result remains hypothetical: `committed`, `source_admission`, `accounting_admission` and `method_eligible` are false. Application here does not publish a database transaction; an evidence-archive receipt preserves bytes without replacing financial recomputation.

The original-scope audit and shared-book reference, including complete input and output examples, are retained under `data/accounting-master/`:

```
lease-module-composition-audit-20261024-r2
lease-shared-book-20261024
```

The checked composition preserves heterogeneous rights and one cash balance for these methods. General lease valuation, FX, tax and report integration, arbitrary module imports, and reconstruction of actual contract rights remain separate design dependencies.

A.21 One lease total can conceal different rights

The preceding lifecycle starts with a supplied asset of 30 for right B. Suppose that amount is unknown and only the combined right-of-use asset of 120 is given. Even if the two rights and their liabilities of 100 and 50 are known, the total asset does not identify its allocation. Assigning 20 to B leaves 100 for A; assigning 30 to B leaves 90 for A. Both allocations produce exactly the same opening signed account balances. Terminating B gives a gain of 30 in the first case and 20 in the second.

Let x be B's opening right-of-use asset, in the same exact USD units as the previous example. A's asset is then $120 - x$. Nonnegative asset amounts imply

$$0 \le x \le 120, \qquad R_A = 120 - x, \qquad R_B = x.$$

This is a family of supplied hypothetical openings. Exactly two identified rights, their liability allocation, the fixed 10 percent rate and the original method findings remain premises. Actual aggregate disclosures need not establish those premises. The family assigns no probabilities to its allocations.

A.21.1 Find where the supplied lifecycle can complete

Keep the original four events, including the annual payments and depreciation of 81. Only the unknown opening asset allocation and B's corresponding traced removal vary. Terminating B removes liability 50 and asset x, so its gain is $50 - x$. A's payment of 121 gives liability 110 and adds 10 to its asset. After the first event, total ROU is therefore $130 - x$.

The second reduction removes asset 10 and liability 11 from A. Remeasuring its remaining payment of 99 gives liability 90, which is 9 below the intermediate liability of 99. Its asset becomes

$$R_A^{(2)} = (130 - x) - 10 + (90 - 99) = 111 - x.$$

The original method requires a nonnegative remaining asset. For $x > 111$, this second reduction is outside its supported treatment. That refusal does not make the opening allocation impossible; it identifies a method limitation under the supplied event. The first completed event remains available.

For $x \leq 111$, the payment of 99 settles liability 90, records interest expense 9 and leaves company cash 101. The final event then proposes depreciation of 81. It can be applied only when that expense does not exceed the remaining asset:

$$81 \leq 111 - x \quad \Longleftrightarrow \quad x \leq 30.$$

Thus the complete supplied sequence works on $[0, 30]$. For $30 < x \leq 111$, it reaches payment but rejects the final depreciation. Replacing 81 with a smaller amount would change the supplied scenario; the reference keeps the refusal and its successful prefix instead.

The three regions partition the original allocation interval:

Allocation of B's asset	Outcome	Completed events
$[0, 30]$	Complete supplied lifecycle	Four
$(30, 111]$	Depreciation amount rejected	Three
$(111, 120]$	Second reduction unsupported	One

The endpoints matter. At $x = 30$, depreciation removes the remaining asset exactly. At $x = 111$, the second reduction leaves zero ROU, so it succeeds, but the later depreciation of 81 does not. Each failed event leaves the last successful state unchanged. A response containing only the first region would lose valid declared openings and conceal why their subsequent paths stopped.

A.21.2 Keep financial outcomes on the same allocation

Within the complete region, depreciation leaves asset $30 - x$. Pretax profit combines the first gain $50 - x$, the second gain 1, interest 9 and depreciation 81. Write R_{end} for final ROU and Π for cumulative pretax profit:

$$R_{\text{end}} = 30 - x, \qquad \Pi = (50 - x) + 1 - 9 - 81 = -39 - x, \qquad 0 \leq x \leq 30.$$

The possible asset amounts range from 0 to 30 and profits from -69 to -39. Both depend on the same opening allocation. Subtracting the two expressions eliminates x:

$$\Pi - R_{\text{end}} = (-39 - x) - (30 - x) = -69.$$

For example, $x = 20$ produces asset 10 and profit -59; $x = 30$ produces asset zero and profit -69. Asset 30 and profit -69 are individually feasible marginal endpoints, but they cannot occur together. The first requires $x = 0$; the second requires $x = 30$.

A hypothetical additional observation of final ROU equal to 10 would select $x = 20$ and profit -59, conditional on completion of the supplied lifecycle. The original unconditioned family still contains its rejected and unsupported regions. The query records the additional observation as a supplied premise; it does not present that number as a recovered issuer fact.

The interrupted paths also retain dependent financial amounts. In the rejected-depreciation region, the last successful state has ROU $111 - x$, profit $42 - x$, cash 101 and zero lease liability. In the unsupported-reduction region, the first event leaves ROU $130 - x$, profit $50 - x$, cash 200 and signed lease liability -110. These are prefix states, not completed-year statements. Keeping their dates and failure positions prevents a partially executed path from being reported as a complete result.

A.21.3 Represent the family without dropping interrupted paths

`LeaseAllocationFamily/1` binds the original source lock, a finite allocation interval within $[0, 120]$, explicit endpoint inclusion, an allocation premise, its source reference, a complete-membership finding and a `max_cells` budget. Amounts are canonical rational strings. The reference intersects the declared interval with all three regions. Empty intersections contribute no cell; every nonempty intersection remains visible. Missing membership or a missing allocation premise returns unresolved.

For each checked cell, the response retains its domain, original outcome category, completed-prefix length, failure index and reason, affine financial expressions and a complete native replay. The replay contains opening state, events, proposals and the last successful state. It checks a representative allocation against the unchanged original methods. The inspected affine derivation establishes the within-cell relationship; a few replayed allocations alone would not prove a bound over a continuous interval.

The cell budget limits those native replays. If it is exhausted, the remaining domains and categories stay present as unchecked cells. The response reports partial search, supplies no complete-family bounds and makes no assertion that all allocations complete. Once every retained cell is checked, any final-outcome bounds are explicitly conditional on complete execution. A family consisting entirely of interrupted paths has no such bounds.

`LeaseAllocationQuery/1` requires the explicit condition `complete_execution`. Its outcome constraints intersect one shared allocation interval, preserving dependence between asset, profit, cash and liability. The response retains the unconditioned family alongside the selected interval and its conditional ranges. An impossible combination has an empty selection and no bounds. A separate `max_replays` budget controls the query witness: with zero budget, a nonempty candidate interval remains visible but its selection and bounds are not reported as verified.

The source locks and full recomputation reject changed methods or forged saved results. Source admission, accounting admission, method eligibility, actual history identification and production commit remain false. The exact original dates, rates, schedules and depreciation are restrictions of this reference; unknown rates or liabilities, arbitrary membership and broader lease valuation require additional methods and cannot be handled by renaming this profile.

The derivation and complete saved requests and responses are retained under `data/accounting-master/`:

```
lease-allocation-family-audit-20261024
lease-allocation-family-20261024
```

They show how aggregate-compatible openings can remain distinct through a supported accounting calculation, and how unsuccessful paths remain part of the answer. Connecting that family

to actual disclosed rights, measurements and authority is still a separate source-reconstruction task.

A.22 What actual lease disclosures identify

The preceding allocation example starts with named rights, supplied contract findings and a hypothetical opening book. A company report usually supplies less. Before using it to select an opening, the simulator must distinguish what was reported from the contract state needed to calculate the next payment or modification. Henkel's retained 2025 annual report makes that distinction concrete. It gives several useful lease measurements, but those measurements neither enumerate the individual rights nor allocate their carrying amounts.

The following amounts are in millions of euros for the consolidated Henkel Group. The report describes its basis as IFRS Accounting Standards as adopted in the European Union. Its 2024 comparative and 2025 current figures belong to the same retained 2025 document; their appearance together supplies no evidence that these exact bytes were public in 2024. The basis note also says that individual figures are rounded and addition can differ from the indicated total. It supplies no rounding tie rule.[2]

At 31 December 2025, Henkel reports right-of-use assets of 594, lease liabilities of 666 and undiscounted contractual lease cash flows of 721. These amounts answer different questions. The first measures the remaining carrying value of the rights to use assets; the second measures the recognised liability; the third groups future undiscounted payments by maturity. The liability consists of 557 noncurrent and 109 current. The maturity bands are 114 due within one year, 351 due between one and five years, and 255 due after five years. Their displayed sum is 720, one below the reported total of 721.[3]

Different measurement bases explain why 594, 666 and 721 should not be forced to agree. Rounding creates a separate issue even within a single basis. The reported 2025 net right-of-use amounts for land and buildings, plant and machinery, and factory and office equipment are 530, 16 and 47. Those three numerical entries sum to 593; the total is 594. The construction category is printed as a dash. The inventory keeps that dash distinct from zero and records the difference between the total and the three numerical entries without inventing a fourth amount. Similarly, the 2024 gross right-of-use amount is 1,173 and accumulated depreciation and impairment is 543, while net carrying amount is 631. Treating every displayed integer as an exact underlying amount would demand both $1{,}173 - 543 = 630$ and a net amount of 631. That exact-value interpretation is wrong; it does not establish an error in the rounded report.

A.22.1 Read the liability bridge with its signs intact

The cash-flow reconciliation helps distinguish a payment from the expense recognised in the same year. Henkel presents liabilities with a negative sign in this reconciliation. A payment reducing the liability therefore enters positively; an addition or accrued interest increasing it enters negatively. For 2025, the displayed opening liability is -713. Cash changes are 158, comprising interest paid of 25 and redemption of 133. Interest expense is a separate movement of -25, additions are -139, acquisitions or disposals of subsidiaries contribute 1, foreign exchange contributes 50, and sundry changes contribute 2. The fair-value movement is printed as `-0`.

[2] Henkel, *Annual Report 2025*, p. 351. Page references in this section are printed numbers; physical PDF pages are two higher. The source audit cited at the end binds the retained PDF, its hash and exact source locations.

[3] Henkel, *Annual Report 2025*, pp. 370–374, 397 and 423–424. The maturity table is explicitly based on undiscounted cash flows.

Using the displayed numerical value of that last entry gives

$$-713 + 158 - 25 - 139 + 1 + 50 + 0 + 2 = -666. \tag{A.41}$$

This is exact arithmetic on the displayed entries, in EUR millions, for the calendar year 2025. The right side agrees with the negative of the reported closing liability. The sign of the printed zero remains in the source record even though its displayed rational value is zero.[4]

The subtotal 158 enters equation (A.41) once. Adding 25 and 133 again would count the same cash twice. The interest expense of -25 has a different role: it increases the liability that the interest payment reduces. The 2024 reconciliation follows the same sign convention, but the sum of its displayed opening and movements is -712, whereas its displayed closing balance is -713. The retained residual is -1. The rounding disclosure makes this a reason to examine precision, not authority to insert a missing transaction.

Nor does the bridge explain every lease-related cash flow. Henkel reports total cash paid for leases of 174 in 2025, exceeding the recognised-liability cash subtotal of 158 by 16. Short-term lease expense, low-value lease expense and excluded variable-payment expense are 11, 5 and 1, respectively. Their sum is 17, and the report does not identify them as the allocation of that cash difference. Expenses and cash flows can differ in both timing and scope. The source inventory retains those measurements separately.

The same care is necessary for the asset movement. Opening net right-of-use assets of 631, additions of 139, and depreciation and impairment of 132 give $631 + 139 - 132 = 638$, compared with a reported closing amount of 594. Those three movements are not a complete asset reconciliation. The difference of -44 does not identify a particular disposal, exchange effect or allocation to a right. It remains a diagnostic of the incomplete comparison.

A.22.2 Ask a conditional question about unrounded measurements

The useful next question is whether one set of unrounded measurements could satisfy selected relations while producing the displayed figures. Answering it requires an explicit rounding hypothesis and explicit premises about which underlying relations are exact. The report's general rounding statement does not choose these premises for us.

For illustration, suppose that each selected amount is rounded to the nearest million euros and exact half ties are rounded away from zero. Let d denote a displayed integer and x the corresponding unrounded amount, both measured in EUR millions. Under this proposed rule the inverse interval is

$$x \in I(d) = \begin{cases} [d - \frac{1}{2}, d + \frac{1}{2}), & d > 0, \\ (d - \frac{1}{2}, d + \frac{1}{2}], & d < 0, \\ (-\frac{1}{2}, \frac{1}{2}), & d = 0. \end{cases} \tag{A.42}$$

At the positive lower endpoint, a half tie rounds up to d; at the positive upper endpoint it rounds to $d + 1$. For a negative number the included endpoint is reversed. Neither half endpoint rounds to zero. Thus the zero case is open at both ends. Treating the printed `-0` numerically as zero imposes no additional sign restriction: the source has not established what sign rule, if any, its printed zero encodes.

This hypothesis can reconcile the 2024 gross, accumulated and net amounts without changing their displayed values. For example, underlying amounts 1,173.25, 542.50 and 630.75 satisfy $1{,}173.25 - 542.50 = 630.75$ and round to 1,173, 543 and 631 under the proposed rule. The

[4] Henkel, *Annual Report 2025*, pp. 450–451. The cash-flow statement on p. 347 reports redemption as an outflow of -133 in 2025 and -146 in 2024.

example is a conditional mathematical assignment, not a recovery of Henkel's actual amounts. A different rounding convention would require different endpoint rules.

Compatibility must also hold jointly. Let L_{25} be the positive closing liability, L_{25}^{nc} and L_{25}^{cur} its noncurrent and current portions, B_{25} its signed reconciliation closing, and O_{25} the opening of the 2025 reconciliation. Let B_{24} denote the signed closing at 31 December 2024. Selected underlying premises include

$$L_{25} = L_{25}^{\mathrm{nc}} + L_{25}^{\mathrm{cur}}, \qquad B_{25} = -L_{25}, \qquad O_{25} = B_{24}. \tag{A.43}$$

The same B_{25} appears as the closing amount in the 2025 stock-flow relation, and the same O_{25} appears there as its opening. Separate checks could choose incompatible values for these shared amounts. One joint system requires a common assignment. The opening is dated 31 December 2024, the movements cover [2025-01-01, 2026-01-01), and the closing is dated 31 December 2025; the opening symbol is indexed by the reconciliation year, not its measurement year.

The checked reference combines nineteen such relations over fifty-four selected measurements from the two years. With the proposed rounding and identity premises, it finds an assignment satisfying every interval and equation, checked with exact rational arithmetic. This establishes conditional consistency of the selected measurements. It neither chooses the actual unrounded amounts nor makes the rounded intervals independent. The 2024 displayed-as-exact counterexample fails because its three exact values cannot satisfy the gross-to-net relation; that failed hypothesis is distinct from a contradiction in the underlying accounts.

A.22.3 Preserve the disclosures before admitting a lease book

`HenkelLeaseMeasurements/1` binds this finite example to its retained source. Its closed request contains `profile`, `inventory_sha256`, `selected_relations`, `rounding_hypothesis` and `identity_hypotheses`. The hash selects the exact inventory. Relation IDs must be unique members of the supported registry; the identity map contains exactly those IDs. Each premise is absent or is the explicitly proposed premise for that relation. Callers cannot replace the amounts, units, dates, company or equations through this interface.

For each presentation year, the registry supports current plus noncurrent liability; gross less accumulated right-of-use amounts; depreciation categories; undiscounted maturity bands; cash subtotal components; the signed liability stock-flow; the closing-liability sign comparison; the redemption sign comparison; and the maturity table's separate carrying-value comparison. Opening-to-closing continuity between the two years supplies the nineteenth relation. Category sums involving unquantified dashes, the incomplete asset movement, attribution of the cash difference to expenses, and equality between discounted and undiscounted amounts are excluded.

Every response retains all 104 numerical-table and narrative entries, including eight dash cells, the twelve qualitative observations and fourteen unresolved state dependencies. The selected measurements carry their original source references, literals, dates, units and consolidation context. A stock-flow relation retains the roles of opening, period movements and closing; these roles are validated before lowering to the existing rational constraint engine. Shared observations keep one identity throughout the selected system.

By default, the adapter supplies no numeric inverse intervals. Missing rounding or identity premises leave the assessment `unresolved`. An empty relation selection reports `no_asserted_system` while retaining the source record. A verified assignment gives `conditionally_feasible`; a checked contradiction concerns only the declared conditional system. Resource exhaustion returns `unknown`, and invalid source bindings or declarations return `invalid`. The reference permits at most one conjunction cell, with an explicit bounded solver budget; zero cell budget supplies no verified witness. Saved results are rebuilt from the original request. Fixed fields

must agree, and saved witnesses must satisfy the exact intervals and equations; a different valid witness or elapsed runtime need not have identical bytes.

An actual lease book still needs individual lease and right identities, complete membership, legal lessees and owner books, unrounded allocations, contract and functional currencies, rates, commencement and measurement dates, payment schedules, modifications, depreciation and applicable findings. Henkel's description of country-, duration- and credit-specific discount rates does not supply those numeric inputs. Its description of optional lease periods does not supply contract-specific exercise findings, and its qualitative mid-triple-digit-million-euro payment description remains qualitative. Short-term and low-value recognition exemptions require separate treatment. Initial right-of-use measurement can also include payments, incentives and other adjustments beyond the liability; equality between the two is not a general allocation rule.[5]

The source adapter therefore leaves executable opening state absent and source, accounting and native-state admission false. It preserves useful measurements while keeping their missing contract explanation visible. The synthetic USD lease methods retain their original company, period and authority scope; they cannot become Henkel methods by changing a name or currency label. The full requests, relations and exact checked results are retained under `data/accounting-master/henkel-lease-measurements-20261024`, with the reference in `scripts/henkel_lease_measurements_v1.py`. The source audit, including the retained PDF path, SHA-256 and exact text and table locations, is in `data/accounting-master/henkel-lease-source-audit-20261024`. Connecting these measurements to actual contracts remains the next substantive reconstruction problem.

A.23 Return and lease methods in one company book

A customer refund and a lease payment draw on the same company's cash. Separate method examples can each balance and still fail to describe that competition. The lease example might spend money that the return example has already paid to customers. Combining their closing reports would discover the inconsistency too late: the second payment should have failed when it was proposed. This makes a common book, rather than a common calling convention, the next necessary part of the implementation.

The actual Henkel disclosures in the preceding section still lack the contracts needed for an executable opening. The present example therefore declares a new synthetic company. It uses retained return and lease calculations to test shared cash, journals and memory, while leaving actual-company reconstruction as a separate problem.

A.23.1 A common caller preserves different capabilities

The finite `ModuleInvocation/1` interface supplies one request form for four existing references. Typed returns and shared leases propose transitions; Henkel measurements assess conditional constraints; the sector reference produces a report. Their outputs have different meanings. A measurement result cannot become a posting merely because the caller uses the same envelope.

Each request names a registered module and operation, binds the complete payload, and identifies the inspected implementation and schema bundle. The caller selects code from its fixed registry. A supplied path or callback cannot select another function. Typed-return inputs

[5] Henkel, *Annual Report 2025*, pp. 373–374. The retained disclosure describes these policies; the present example does not independently certify their application to each contract.

also retain their original raw-byte hashes and duplicate-key checks. The complete native result survives, including journals, gross flows, memory changes and refusal reasons.

An outer status of `returned` means that the selected module answered. Its separate `native_status` may be proposed, unsupported, invalid or unresolved. A request to post the sector report, post a measurement assessment, or commit a lease proposal is refused before native invocation. Verification recomputes complete transition and report results. The measurement verifier instead permits another valid solver witness and another runtime while checking the underlying equations and all fixed information.

Identity binding does not authenticate source labels. In the retained return reference, changing a well-formed owner label or packet digest can still produce a proposal; source admission remains false. The lease reference fixes its original scope and refuses a changed scope digest. The common caller preserves both behaviors. It cannot manufacture the missing source-to-state judgment.

A.23.2 The original states are incompatible

The return state belongs to synthetic owner P and begins in 2025. The lease state belongs to a different synthetic owner and begins in 2024. Their schemas are closed, their legal-book labels differ, and their account registries contain 22 and 26 accounts respectively. The return state stores recovered lots, tax adjustments and fiscal history. The lease state stores named rights, their allocations and a premise that all leases are listed.

Even shared account names are insufficient. Cash needs one owner, currency, date and opening amount; inventory needs a complete allocation to stock; lease liabilities need their rights and payment history. A field union does not resolve those meanings and violates both original validators. Renaming the lease owner does not repair the original scope restriction. The read-only compatibility reference consequently rejects direct combination, preserves unknown membership as unresolved, and accepts only its explicit control: two valid identical snapshots of the same native profile. That control authorizes no joint transition.

A.23.3 One declared opening and explicit memory ownership

`SyntheticReturnLeaseBook/1` supplies a separate synthetic USD legal book. It retains the 22 return accounts and explicitly adds the right-of-use asset, lease liability and lease derecognition gains. Debit balances and expenses are positive; credit balances and income are negative. The six selected lease posting categories—cash, right-of-use asset, lease liability, finance expense, depreciation and other gains—have declared meanings in this 25-account registry. No mapping between unrelated return cost categories is inferred.

The opening supplies cash 500, inventory 60, a recovery asset of 16, a refund liability of 40, a right-of-use asset of 120 and a lease liability of 150, together with the surrounding accounts recorded in the complete fixture. Its signed original-equity balance is explicitly 20. These are synthetic premises chosen for the example; the program neither infers that opening from an issuer nor computes an equity plug. The two lease rights have right-of-use amounts 90 and 30 and liabilities 100 and 50. The declared collection contains exactly those rights.

The book owns one period, date, cash balance, event history and gross-flow history. Return methods own the return component, recovered lots and settlement tax adjustments. Lease methods own the lease-right collection. Other memory, including tax premises and prior fiscal history, must survive each selected event unchanged. Exact write sets enforce that separation.

For a dated event k, let B_k be the vector of signed account balances before the event and J_k its

journal in the same account order, both measured in USD. Let F_k contain its gross cash flows, with receipts positive and payments negative. Pure application checks the exact identities

$$B_{k+1} = B_k + J_k, \qquad \sum_a J_{k,a} = 0, \qquad J_{k,\text{cash}} = \sum_{f \in F_k} f.$$

The first identity applies the journal once. The second preserves the signed trial balance. The third prevents classified cash flows from disagreeing with cash. Component allocations are checked as well: recovered lots explain their inventory, and lease rights explain the right-of-use asset and lease liability. An aggregate balance alone cannot replace those allocations.

Return settlement and whole-lot sale reuse the checked typed prerequisite, monetary and memory programs. Their memory schema is extended explicitly to retain all 25 balances and the lease fields. The lease operations reuse the original component calculation kernel under the new declared context. The fixed-owner lease wrapper remains unchanged. This is a separately specified synthetic instantiation, not a lossless migration of the old owners.

A.23.4 Follow both methods through the same cash balance

At 1 January 2024 the lease reduction terminates right B and remeasures right A. The journal reduces the right-of-use asset by 20, reduces the credit liability by 40 and recognizes a gain of 20. No cash moves. The remaining right-of-use asset is 100 and its liability 110.

On 1 February the return method pays customers 40 and handling costs 8, releases the refund liability of 40 and recovery asset of 16, and recognizes recovered inventory of 20. The associated cost-of-sales credit is 4 and selling expense is 8. The two cash payments remain separate operating flows. A recovered lot records the units, carrying amount, tax base and acquisition date, while the settlement tax adjustment remains available for a later tax calculation.

On 1 March the entire recovered lot is sold for 30. Its carrying amount of 20 leaves inventory and enters cost of sales. On 1 January 2025 the lease payment of 121 comprises interest 11 and principal 110 under the supplied annual timing assumptions. Interest is an operating outflow and principal a financing outflow under the retained election. A separately supported depreciation amount of 10 then reduces the right-of-use asset. Neither that amount nor the discount rate is inferred from aggregate disclosures.

With no other cash events in the example, the closing amount follows directly:

$$\text{cash}_{\text{close}} = 500 - \underbrace{(40+8)}_{\text{refund and handling}} + \underbrace{30}_{\text{recovered-stock sale}} - \underbrace{121}_{\text{lease payment}} = 361.$$

The book closes this sequence with right-of-use assets of 90, no remaining lease liability and inventory of 60. Five event identities and five gross flows survive. Return events leave the lease memory unchanged; lease events preserve the return component, lots, tax adjustments and other return memory.

The low-cash case distinguishes genuine shared execution from two independent examples. Start with cash 130 and omit the recovered-stock sale. The signed original-equity amount is supplied as 390 in this alternative opening, keeping the declared trial balance balanced; the program makes no automatic adjustment. The return settlement leaves 82, so the later lease payment of 121 fails the shared funding check. The completed lease reduction and return settlement remain; the failed payment changes nothing. With the same opening cash but without the return settlement, the lease payment succeeds and leaves 9. Cash competition therefore changes the admissible sequence even though each method's arithmetic is unchanged.

A.23.5 What replay establishes

A proposal binds the whole before-state, event and implementation bundle. Application recomputes the proposal before returning a new in-memory state. An old proposal cannot apply to a changed cash balance, and a forged journal, flow, component change or admission flag fails full comparison. Event identities cannot repeat. An empty event list still requires a valid opening. Unsupported currency or tax operations are refused; missing return measurements or lease membership remain unresolved.

All five selected event kinds are also compared with the original native methods. The original return comparison stays in its 2025 period and original owner; the new book expressly uses different return dates and ownership. Journals and activity/amount cash flows match exactly for the same numerical component inputs. Component and lot memory match apart from the declared lot acquisition dates. The comparison supports reuse of those effects, not identity between the old and new companies.

This reference supplies a finite shared-book execution example. It preserves tax memory but does not close tax or produce complete financial statements for the new profile. General module substitution, arbitrary histories, group and foreign-currency methods, actual-source admission and physical database durability remain separate requirements. The existing database reference is tied to another engine; its transaction evidence does not transfer automatically. The full schemas, complete openings, requests, proposals and checked comparisons are retained under `data/accounting-master/` in `module-invocation-20261024`, `shared-state-compatibility-20261024` and `shared-book-successor-20261024`.

A.23.6 Sell part of a recovered lot without losing its remaining units

The whole-lot sale above leaves no recovered stock to track. A company that sells only part of that stock needs a quantity and cost record for what remains. The existing lot already records both; its carrying amount alone would not identify how many units were sold. A separately selected method therefore takes the sold quantity as an explicit input rather than deriving it from cash proceeds or inventory value.

This method assumes that the remaining units are homogeneous and divisible, that their measurement basis has not changed, and that proportional allocation is the selected treatment. The caller also supplies affirmative findings for completed performance and the retained equality of tax base and carrying amount. Those findings record explicit premises. Unknown findings remain unresolved, and false findings place the event outside this method's scope. A heterogeneous lot or a different tax treatment requires another method.

Immediately before a sale, let $N > 0$ be the remaining units, $C \geq 0$ their carrying amount in USD, and $T = C$ their tax base under the selected treatment. The event supplies sold units q, with $0 < q \leq N$, and cash proceeds $P \geq 0$ in USD. The fraction applies to the stock immediately before this event:

$$f = \frac{q}{N}, \qquad C_{\text{released}} = fC, \qquad T_{\text{released}} = fT.$$

The remaining lot keeps its identity and acquisition date, with units $N - q$, carrying amount $C - fC$ and tax base $T - fT$. It becomes disposed exactly when $q = N$. A positive-quantity lot whose carrying amount is zero remains live; zero cost does not imply that no goods remain.

The journal debits cash by P and cost of sales by fC, and credits revenue by P and inventory by fC. Its debit-positive sum is $P + fC - P - fC = 0$. Cash rises by P and inventory falls by fC, so net assets and profit both rise by $P - fC$. In the indirect operating cash calculation, adding back the inventory reduction to that profit effect gives $(P - fC) + fC = P$, the same receipt as

the direct cash record. The reduction in the lot's carrying amount equals the inventory posting; every other lot and the separate inventory balance remain unchanged. This sale assesses no tax expense and makes no new measurement of inventory.

After the February settlement in the shared-book example, the recovered lot has two units with carrying amount and tax base of 20. Cash is 452 and total inventory is 80. Selling one unit for 15 releases 10 of cost, leaves one unit with carrying amount and tax base of 10, and gives cash 467 and inventory 70. The prior lease gain of 20, cost-of-sales credit of 4 and handling expense of 8 give recorded profit of 16 before this sale; its margin of 5 raises profit to 21. Operating cash since the opening is $-48 + 15 = -33$. Assets of 827 less liabilities of 214 equal equity of 613, also opening equity of 592 plus profit of 21.

If the remaining unit is then sold for another 15 under the same conditions, the new fraction is one: one unit is sold from one remaining unit. The second sale releases the remaining 10, leaves cash 482 and total inventory 60, and marks the lot disposed. The combined account changes equal those of the original whole-lot sale for 30. The two event identities and cash receipts remain separate; equal aggregate effects do not make their histories identical.

A proposal retains the entire before-state, event, allocation and resulting book. Application recomputes the complete proposal. A changed remaining quantity, altered allocation or balanced but different journal fails that comparison. Selling more than the available units, selling a disposed lot, reusing an event identity or backdating a sale is refused without changing the successful preceding state.

This optional transition has checked quantity, journal and statement identities and reproduces the original full-sale state when all units are sold. The existing whole-lot method remains unchanged. Connecting the optional event to the mixed-book authoring and annual dispatch interfaces is separate work; these local controls do not establish that those interfaces already support it.

A.23.7 Carry unsold stock through the assessed annual book

The next composition supplies the annual execution step left open above. It keeps the full tax-adjusted company book, including income tax payable, assessment records, liability vintages, losses and earlier periods. The standalone inventory method accepts 25 accounts, whereas this financial book has 26. Removing tax payable or moving it into equity to reuse that smaller schema would change the company. The annual extension instead validates and updates the complete registered state.

The opening for this example comes from the checked retained-stock history, with its planned whole-lot sale omitted. At 1 January 2027 it has two unsold recovered units with carrying amount and tax base of 20, cash of 452 and tax payable of 41/4. New-period income starts at zero. Selling one unit for 15 on 2 January gives cash 467, inventory 70, a remaining unit with cost 10 and recorded profit 5. Operating cash is 15. The payable remains 41/4: an inventory sale does not pay or reassess a prior tax liability.

The sale retains the complete assessment and liability histories. If it occurs after a current assessment, that assessment's amounts remain recorded but its status records subsequent activity. Any prior deferred measurement becomes stale because it was bound to the earlier financial state. A sale inside the already assessed interval is refused under the existing timing rule. New activity is not silently inserted into a historical assessment.

Consider a hypothetical current assessment of 2 for the period containing the first sale. It adds a separate liability, so total payable becomes 49/4 and recorded profit becomes 3. With explicitly supplied fresh deferred measurement and the remaining carry conditions, a provisional

boundary transfers that profit to retained earnings and opens the next period with the unsold unit, its cost of 10, cash 467 and the same liability vintages. Selling that unit for 15 on 2 January 2028 then gives cash 482 and recorded profit 5. That period's operating cash is 15; the earlier receipt remains in the archived period. The new-period payable is still 49/4 until a payment or assessment changes it.

A separate sequence checks why fresh assessment matters. After a midyear assessment and deferred measurement, an inventory sale records further activity and invalidates the earlier measurement. Carry is unresolved at that point. A fresh cumulative current assessment followed by a measurement bound to the changed state permits the selected provisional carry; both earlier and replacement assessments remain available. Reusing the old measurement after that reassessment is invalid because its state digest no longer matches. Attempting measurement before the required current reassessment instead stops at that missing prerequisite. The two refusals answer different questions.

Ordinary operations continue through the protected annual evaluator. Their financial states, reports and traces are unchanged; the new program profile and request identities identify the extension. A full-quantity sale also matches the old annual whole-lot sale on the complete resulting book, journal, flows and report. Altering tax payable, an archived quantity or an allocation fails complete-result verification even if the alteration is balanced.

The optional operation now has a checked annual execution meaning. Its matched source declarations and annual-family dispatch still require separate extensions. The current authoring and family profiles do not acquire support merely because the financial transition exists. The examples remain conditional on the stated inventory and tax assumptions; actual source and accounting admission are unresolved.

Author inventory alternatives on the same annual book

The optional annual inventory method can now be described in source text and compared across named scenarios. These interfaces extend the checked execution above. They retain its opening balances, tax liabilities and recognition assumptions, so a change in how a sale is written cannot silently change the company being simulated.

`AnnualInventoryAuthoring/1` uses the annual declaration grammar with an explicit `partial_inventory_sale` operation. Its header names the sale; the body supplies the sale profile, date, allocation method, lot, units sold and cash proceeds. Each of the six inventory findings remains a separate known-or-unknown declaration. A known finding carries the text explaining its provenance; an unknown finding carries a reason. This inventory convention is preserved alongside the structured provenance used by the older tax operations. Neither form is silently translated into the other.

Compilation binds the declared opening request to its exact content digest and returns the same `AnnualInventoryProgram/1` request as the typed form. It does not execute a sale, assess tax or advance a period. Consequently, a false divisibility finding or an unresolved homogeneity finding survives as explicit data until the financial evaluator refuses the operation. Missing fields, duplicate declarations and malformed rational quantities fail earlier, when the source is read. The compiled record retains the original text hash and the lines containing each operation and field. Whitespace can change that text hash without changing the compiled financial request.

`AnnualInventoryFamily/1` then gives each alternative its own name, assumptions and ordered operations, all starting from the same declared annual book. Consider the earlier example with two recovered units costing 20 and cash of 452. The first alternative sells one unit for 15, assesses current tax of 2, refreshes deferred measurement and carries the remaining unit into

the following year. Its first-year profit is 3 and closing cash is 467. Selling the last unit for 15 in the next year gives that year's profit of 5 and closing cash of 482, while the carried tax payable remains 49/4. The two annual statement vectors belong to this one history and are returned together with its name.

A different alternative may fail before the carry because the homogeneous-lot finding is unknown, or may reach the second year and attempt to sell more units than remain. The family retains the precise valid prefix and the operations that were not executed. A report for a year that was never reached is unavailable; it is not a statement with zero balances. An explicit budget can also leave members unvisited, while preserving their names and assumptions. Even a zero member budget checks the common opening request. Equal financial results do not merge separately named alternatives.

Joint reports recompute the saved family before returning the complete archived or current-prefix statements. The reference checks cover inventory quantity, current-tax histories, residual-stock carry, source locations, rejected premises and retained alternatives. These are authored, unweighted scenarios under the selected proportional inventory model. They establish neither probabilities for those scenarios nor membership of an actual company's unknown history. General inventory policies and independent accounting acceptance remain open.

Resume the annual book from its last committed operation

A saved balance sheet is not enough to restart this simulation. After the first inventory sale, the remaining unit still carries its acquisition identity, while each tax payable belongs to an assessment and retains any settlements. The next operation must also be known: repeating the sale would change both stock and cash even if the repeated journal balanced. The optional `AnnualInventoryStore/1` therefore preserves the complete annual request, the full period book and the ordered operation history.

The local store records the evaluated opening before attempting annual operations. Each later transaction records one attempted operation together with its complete result and its position in the original request. A successful sale advances both the attempt position and the successful transaction history. A refused sale advances the attempt position but leaves the last valid financial state unchanged. The refusal and all remaining operations stay visible, and that immutable request stops. Repairing its assumptions creates a different request rather than rewriting the recorded history.

An explicit operation budget permits work to pause between transactions. While operations remain unattempted, the stored result says that it is paused; it does not turn a valid partial-year statement into a completed run. A caller can also supply the position it expected before a batch. If that batch has already committed, a retry returns the checked current result without applying another sale. If the caller expects a position that has not been reached, the store rejects the request.

In the running example, the first sale leaves cash of 467, one recovered unit costing 10, and the carried tax payable of 41/4. Closing and reopening the store retains all 26 accounts and their associated histories. Continuing through the declared tax assessment and period carry preserves the residual unit; its sale in the following year gives cash of 482 and a payable of 49/4, exactly as uninterrupted execution does. The complete statements and refused prefixes, not merely these selected balances, are compared in the reference checks.

Reopening checks the saved request, implementation identities, row order and content hashes, then recomputes each committed prefix through the same financial evaluator. This catches altered balances even when a stored hash has also been changed. Replayed calculation has no

external financial side effects. A new operation uses the loaded full state directly; reopening deliberately pays the cost of replay to validate that state.

The implementation uses one local SQLite writer and an atomic transaction for each operation result. In the process-exit checks, termination before commit leaves the preceding result, while termination after commit retains exactly one new result. A fresh process can then reopen and continue. These checks support local restart behavior for this finite annual profile. They do not establish hardware power-loss durability, concurrent-writer correctness, efficient scaling or authenticity against an attacker who controls the entire database. Family scheduling, migration and broader execution profiles remain separate work.

A.24 Recorded statements before annual completion

The mixed return and lease sequence already determines useful financial quantities. Its cash balance is 361, its lease liability has been paid, and its recovered stock has been sold. Waiting for a complete annual tax calculation should not make those recorded movements disappear. At the same time, a balanced trial balance cannot establish that all required tax, valuation or closing adjustments have been posted. The reporting question is therefore precise: what do the completed events explain, and what remains unassessed?

`RecordedSharedBookStatements/1` answers that question for the synthetic book. It replays the opening and requested event sequence, then constructs financial views of the completed prefix. The supplied closing balance is never an input. Each view identifies its last completed date, requested and completed event counts, any failed event, and the declared fiscal period. Even when every requested event succeeds, complete fiscal coverage and annual tax remain unresolved. A short successful list does not prove that nothing else happened.

A.24.1 A recorded balance sheet and equity movement

The opening must start at the declared period boundary, with zero income accounts and no prior event, cash-flow or settlement-tax-adjustment history in the current execution. All 25 accounts receive exactly one classification. The original return classes are extended by the right-of-use asset, lease liability and lease gains; those additions are not hidden in a residual. Owner-capital accounts must remain unchanged for these selected methods.

At the end of the five-event sequence, recorded assets total 701: cash 361, inventory 60, property, plant and equipment 100, investments 80, deferred-tax assets 10 and right-of-use assets 90. Other asset accounts are zero. Recorded liabilities total 104, comprising debt 100 and deferred-tax liabilities 4. The refund and lease liabilities are zero after their respective settlements. All amounts are USD in the supplied synthetic example.

Recorded profit is revenue and gains less the expenses actually posted. The sale contributes revenue 30 and the lease reduction a gain of 20. Cost of sales is 16 after the settlement credit and subsequent sale, selling expense is 8, interest is 11 and depreciation is 10. Current and deferred tax expense accounts contain no postings in this sequence. Thus recorded profit is

$$P_{\text{recorded}} = (30 + 20) - (16 + 8 + 11 + 10) = 5.$$

The zero tax postings describe the ledger; they are not a calculation of the tax charge required for the period. Tax assessment stays unresolved alongside the recorded result.

Let A and L be the totals of recorded assets and liabilities, and let E_0 be opening equity after reversing the debit-positive sign of the equity accounts. The unchanged owner accounts give

$E_0 = 592$. The two independent reconciliations then agree:

$$A - L = 701 - 104 = 597, \qquad E_0 + P_{\text{recorded}} = 592 + 5 = 597.$$

The first derives equity from the closing balance sheet; the second explains its change from the recorded income. Any disagreement refuses the report. No balancing reserve or owner contribution is inferred to make the equations hold.

A.24.2 Explain the same cash movement in two ways

Gross cash origins preserve the distinction between operating activity and financing. Operating receipts are 30 from the recovered-stock sale. Operating payments are 40 of refunds, 8 of handling and 11 of interest, totaling 59. Lease principal of 110 is a financing payment under the retained historical interest-classification assumption. There are no investing cash movements. The direct cash view therefore gives

$$\text{CFO} = 30 - 59 = -29, \qquad \Delta\text{cash} = -29 - 110 = -139 = 361 - 500.$$

The complete gross flow records remain attached to the result, so these totals can be traced back to the individual events rather than to a net cash plug.

The indirect view starts with recorded profit and removes its noncash effects. Depreciation of 10 is added back. The gain of 20 from reducing the lease is subtracted because the reduction produced no cash receipt. Releasing the recovery asset of 16 adds 16 in the working-capital bridge. Settling the refund liability reduces that credit balance from 40 to zero and uses 40 of cash. Inventory returns to its opening amount after the sale, and the other selected working-capital balances do not change. Consequently,

$$\text{CFO} = \underbrace{5}_{\text{recorded profit}} + \underbrace{10}_{\text{depreciation}} - \underbrace{20}_{\text{noncash gain}} + \underbrace{16}_{\text{recovery asset released}} - \underbrace{40}_{\text{refund liability settled}} = -29.$$

The sign of the refund term follows from the accounting convention. The signed liability moves from -40 to zero; subtracting its change gives -40 in the cash bridge. The credit-negative `other_gains` balance similarly removes the positive noncash gain from profit. The reporting code uses the account changes and gross flows to compute both reconciliations. The displayed values are the checked example, not constants used to construct a report.

The inherited bridge also retains recorded deferred-tax expense and any already-expensed opening interest paid as separate terms. The selected successor does not execute those additional methods. They remain zero in this example, and the fresh-opening requirement excludes prior paid interest from the opening. Full tax measurement requires its own supported method and facts.

A.24.3 A failed event leaves a smaller report, not a completed year

In the alternative opening with cash 130, the return settlement leaves 82 and the later payment of 121 fails. The recorded views describe only the lease reduction and return settlement that completed. They show profit 16 and operating cash outflow 48, with the failed payment and one unexecuted event still visible. Their last completed date is 1 February 2024. They do not show the lease as paid or describe the attempted payment date as a completed close.

If the first event is unsupported, a valid fresh opening can still be shown with zero completed events and its explicit refusal. An invalid opening supplies no financial views. An empty valid

event list supplies the opening view, but proves neither complete period coverage nor zero annual activity. These distinctions preserve information without manufacturing a closing state.

All results keep tax assessment, period completion and annual completion unresolved. The original native annual-report function retains its separate tax-close prerequisite and still refuses the incomplete-tax control. The new result is a recorded-prefix report with `annual_close_ready` and `complete_financial_statements` false. Its arithmetic can be exact even when required accounting adjustments remain unknown.

Complete replay checks statements, classifications, execution, gross origins, source identity and coverage together. A changed profit, dropped flow, omitted account, forged completion flag or altered component memory fails verification. The implementation and complete requests/results are retained in `scripts/shared_book_statements_v1.py` and `data/accounting-master/shared-book-statements-20261024`. This supplies the recorded-statement connection for the finite joint book; complete annual methods and actual-source admission remain further work.

A.25 Authoring the complete mixed book

The return and lease example can now be stated as a complete text program. An author supplies the opening accounts, the return obligation and inventory lots, the lease rights, the premises attached to each method, and the ordered events. The program ends with an explicit request for the recorded statements of Section A.24. These declarations lower to the same typed request used by that example. No opening balance, missing premise or additional event is supplied by the parser.

A.25.1 Declare the opening before the events

The opening has twenty-five signed account balances. All must be declared, including accounts whose balance is zero. The inventory and lease collections also have explicit opening and closing markers, even when empty. This makes an omitted collection an authoring error, rather than evidence that the company has no inventory lots or lease rights. The source description remains the stated synthetic premise; writing its name does not turn it into an issuer fact.

For example, the cash account and the two empty inventory delimiters in the checked opening appear as follows. This is an extract from the complete program: the other twenty-four accounts and all other opening fields are required.

```
account "cash" "500/1";
lots;
end_lots;
```

An amount such as `500/1` denotes an exact rational number. The same representation carries rates, costs and cash payments; JSON floating-point numbers are rejected. Structured values retain the return component, tax premises, previous event identifiers and cash-flow history without reducing them to account totals. The lease collection records each right's carrying amount, liability, rate and last measurement date.

After the complete opening, the event block declares lease reduction, return settlement, recovered-stock sale, lease payment and depreciation in that order. Each event names the finite return or lease module and supplies its full native record. The report declaration names the recorded-statement profile and the exact source-bundle, opening and event digests. A changed amount

therefore requires a correspondingly changed request identity. The compiler verifies that relationship; it does not authenticate the economic premise behind the amount.

Complete programs, including every opening field and event parameter, are saved under the shared-book-authoring evidence directory in the repository. The accompanying grammar and typed record define the same bounded language. Imports, include files, general expressions and user-supplied executable functions are absent from this subset.

A.25.2 Compilation and execution answer different questions

Compilation asks whether the author supplied a well-formed program for the selected methods. It checks closed field sets, exact scalar types, account names and completeness, required collection markers, supported methods and content digests. Duplicate JSON keys, accounts or lot keys are errors. A failed declaration identifies its line. Whole-source checks first reject invalid UTF-8, ambiguous line separators and excessive size or nesting.

Execution asks whether the declared opening and events support the proposed accounting movements. An unbalanced opening can be structurally well formed, yet fail the executor's balance check. Unknown lease membership remains unknown after compilation. An unsupported currency remains unsupported. Likewise, a duplicate event identifier or a payment without sufficient cash is judged by the original executor, with the same completed prefix and failure explanation as the typed request. Moving these judgements into the parser would change the meaning of a successful compilation.

This distinction also preserves partial knowledge. A missing recovery estimate that is explicitly recorded as unknown does not become zero. A depreciation finding recorded as null does not become an affirmative finding. A false tax premise remains false even if the recorded balance sheet balances. The text program states those premises; the selected accounting method determines what can follow from them.

A.25.3 Preserve the whole result across representations

The comparison uses all nineteen saved requests from the recorded-statement example. Sixteen belong to this finite grammar. For each, text and typed execution reproduce the complete original result: event trace, stopped prefix, component memory, gross cash origins, classifications, reconciliations and unresolved obligations. Equality of cash or profit alone would be insufficient. The other three requests are explicitly rejected at compilation: one names an unsupported tax module, one carries a stale opening digest, and one attempts to supply an extra closing-state field. They are retained in the comparison as refusals, rather than omitted from the evidence.

Text round trips preserve the complete typed program. Changing whitespace changes the text digest but leaves the program and report-request digests unchanged. Changing the program's name changes its program digest while leaving the financial request unchanged. Thus source spelling, program identity and the requested calculation remain distinguishable.

The cash competition remains visible through authoring. In the checked 130-cash opening, reduction and return settlement complete, leaving 82. The subsequent 121 lease payment fails. The report preserves the first two events and does not manufacture funds to finish the sequence. Separate checks retain a nonselected opening inventory lot through all five mixed events, and allow an explicitly empty lease collection with its separately supplied balanced opening.

The resulting language is an executable authoring route for the declared mixed book. Its complete-result comparisons support that bounded conclusion. They do not settle the remaining language design for other accounting methods or establish ease of use for human authors.

Recorded statements still carry the unresolved tax assessment and fiscal-period completion described in the preceding section.

A.26 Tax information before tax calculation

The mixed book has recorded profit, but it still lacks a tax assessment. The earlier return-only reference does contain a tax-close method. Whether that method can be reused is a substantive question: the new book contains lease rights and a modification gain that the old book did not represent. The unchanged method refuses the mixed state because its closed state shape does not admit those additional accounts and lease records. The two original tax examples still reproduce their complete results. Thus the refusal identifies an incompatible extension, rather than a regression in the original method.

A.26.1 A new income category changes the tax starting point

The original pretax function sums the original income accounts, excluding current and deferred tax. If that function is applied diagnostically to the mixed balances, it returns a loss of 15. The complete recorded report instead includes the lease-modification gain of 20. Under the signed-account convention, the additional `other_gains` balance is -20, so

$$P_{\text{old accounts}} = -15, \qquad P_{\text{mixed}} = -15 - (-20) = 5.$$

The difference is an omitted income category, not a rounding error. Replacing the state validator while leaving the original income list unchanged would therefore calculate the wrong pretax starting point for the stated mixed-book target. It would not preserve the meaning of the old calculation.

Even the corrected starting point does not determine tax. The return example has a signed reconciliation adjustment of -24. Mechanical reuse of that adjustment and the original illustrative rate would give

$$\frac{-15 - 24}{4} = -\frac{39}{4}, \qquad \frac{5 - 24}{4} = -\frac{19}{4}.$$

These expressions diagnose the effect of the omitted gain; neither is an assessed tax or a cash refund. The lease deduction timing, the tax treatment of the modification, the remaining right-of-use tax bases and the treatment of a loss have not been supplied. The original finite method refuses its negative-current-tax route. Copying the rate or reversing every deferred-tax balance would not supply the missing facts.

Dates create a separate incompatibility. The original close on 31 December stores 1 January as an exclusive tax endpoint. The mixed-book invariant currently requires that endpoint to be no later than the state date. Those two meanings cannot be combined by changing only the endpoint field. Moreover, the mixed example's execution window runs from 1 January 2024 to 1 January 2026: 731 days. That window contains the selected lease and return events, but does not establish a single annual tax interval or complete fiscal activity.

A.26.2 Keep missing declarations and rejected premises visible

A separate information assessor now accepts the unchanged recorded-report request, an explicitly stated tax interval and identified information declarations. The interval distinguishes its start, its last included calendar day and its exclusive endpoint. The first finite implementation requires

the start to equal the recorded opening and every requested event to lie inside the interval. The last recorded event may precede the asserted coverage date; that assertion remains a declaration, not evidence that all activity is known.

For this mixed example, the assessor identifies 61 required declarations. They cover jurisdiction and regime, the interval basis and coverage findings, rate, loss and cash-settlement treatment, deferred recognition and recovery, every pretax account's treatment and signed reconciliation adjustment, balance-sheet tax bases, and the tax bases and timing of each lease right. Zero balances remain in the account population. Both lease rights A and B remain relevant: the native state retains fully terminated B with zero asset and liability, and its earlier modification still requires a stated treatment.

Each supplied value carries an assumption or source-reference description. Each unknown carries its reason. A source reference is retained as supplied; the assessor does not authenticate it. With no information supplied, all 61 declarations remain unresolved. Removing any one from the hypothetical complete control again leaves an unresolved requirement. A false coverage or recognition finding remains false and is reported as unsupported. Extra fields, wrong scalar types, forged request identities and incompatible interval boundaries are invalid.

The complete native report accompanies the assessment. Its stopped event prefix, component memory, gross cash movements and unresolved annual coverage are preserved. When a requested payment fails, supplying every information declaration does not erase that failure: completion of the requested sequence remains unresolved. A first-year interval can be stated separately when its requested events fit that interval, without relabelling the two-year example.

A.26.3 Supplied information is not an executable tax rule

The positive result is deliberately called `information_supplied`. It means the finite declaration set is present and has the required types. The hypothetical controls exercise that representation; their supplied zero bases and adjustments are not inferred from the financial statements. A text field describing a deduction can still fail to specify a valid calculation. Likewise, individually typed amounts can be inconsistent when aggregated.

Accordingly, every result keeps calculation readiness, source and accounting admission, tax execution and annual-close readiness false. No tax amount is posted. The next calculation design must state how account adjustments, right-level tax bases, deferred recognition and settlements relate, and must check their joint consistency. This preserves useful progress: missing facts and competing declarations can be represented and examined now, while the remaining numerical and accounting questions remain explicit.

A.26.4 Reconcile a declared current-tax calculation

The next finite calculation makes one further assumption explicit: a single nonnegative rate applies to nonnegative taxable profit before credits or any other regime-specific adjustments. This is a hypothetical arithmetic model, named `uniform_rate_no_credits_v1`. It does not select a country's tax rules. The author must select that model affirmatively; a false selection is unsupported and an unknown selection remains unresolved.

Each declared account adjustment is now supported by identified allocation items. An item taken from the completed return history must retain its original event identity and signed amount, and must appear exactly once. Additional items have explicit assumption references. The items assigned to an income account must sum to that account's declared adjustment. Their category assignment is supplied by the author; the arithmetic check does not establish its

legal appropriateness. In particular, the native return adjustment is not added again after it has already been included in those account totals.

There are also four cross-field basis checks. At both opening and closing, the declared asset bases of all lease rights must sum to the aggregate right-of-use tax basis, and the declared liability bases must sum to the aggregate lease-liability basis. The zero-valued right B participates in those checks. A mismatch rejects the conditional calculation while preserving the complete recorded financial report and the inconsistent declarations for inspection. Neither carrying amount nor termination status supplies a tax basis.

Let P be recorded pretax income over all seven income accounts, a_i the identified signed adjustment items and r the explicitly supplied rate. The conditional taxable-profit expression and gross current tax are

$$T = P + \sum_i a_i, \qquad C = rT \quad \text{when } T \geq 0.$$

Here C is gross conditional current tax under the selected arithmetic rule. It is not an incremental expense, accrued payable or cash payment. The calculation does not subtract opening accruals, apply credits, recognize deferred tax or change the recorded closing state.

For the mixed example, $P = 5$ and the retained return adjustment is -24. Without any additional declared item, $T = 5 - 24 = -19$. The result is `loss_route_required`; no negative tax amount or refund is computed. A text declaration requesting a refund cannot change that result because it does not supply an executable loss rule. Two hypothetical controls illustrate the nonnegative branch. An expressly supplied additional adjustment of 19 gives $T = 0$ and $C = 0$. An additional adjustment of 40 gives $T = 21$ and, at the illustrative rate $1/4$, $C = 21/4$. Those extra adjustments are test assumptions, not inferred company facts or recommended tax treatments.

The comparison retains the entire financial report, including cash of 361 and recorded profit of 5, in every accepted arithmetic result. Inconsistent allocation or basis declarations retain that same report alongside their rejection. Tests also change native adjustment amounts, duplicate or omit identities, alter the model finding and tamper with computed results. These checks establish the specified arithmetic and preservation properties. A loss-utilization rule, tax settlement, deferred recognition and annual close remain separate work.

A.26.5 Carry a loss quantity without inventing a tax asset

The negative expression of -19 needs a rule before it can affect a later tax calculation. The finite extension supplies a conditional loss ledger. Each lot has an identity, an origin reference, an availability date, an explicit expiry declaration, original and remaining quantities, cumulative usage and expiry, and an eligibility finding. An unknown finding remains unresolved; a false finding is unsupported. A generated loss therefore becomes a quantity in this separate ledger only under an affirmative, traceable assumption. It is not a cash refund or a recognized deferred-tax asset.

The selected illustrative policy evaluates all declared usage on the last included calendar day of an interval. An exclusive expiry on that day makes the lot unavailable for usage. Expiry at the following endpoint permits usage on the preceding day, but the remainder expires before the next opening. No expiry is itself an explicit assumption. These conventions specify this finite model; another jurisdiction or an intraperiod method may require a different rule. Neither FIFO nor a legal utilization percentage is assumed. The author supplies the chosen lot allocations and an interval usage cap.

For each lot, original loss equals remaining, used and expired quantities. Across a step, with L_0 and L_1 denoting opening and closing remaining quantities, G generation, U usage and E expiry,

$$L_0 + G = U + E + L_1.$$

All quantities on this line are nonnegative exact rationals. A generated loss becomes available at the exclusive endpoint and cannot be used in its own generation interval. A usage cannot exceed its lot's remaining quantity, the declared interval cap or positive taxable profit. Zero and exhausted lots remain identified, preventing the same identity from being introduced again. Consecutive ledger entries must agree on both interval endpoints and their opening and closing quantities.

The checked negative example generates 19 under explicit eligibility and no-expiry premises. A separate hypothetical 2026 calculation supplies taxable profit of 21. Explicit use of 19 leaves taxable profit of 2 and gross conditional tax of 1/2 at rate 1/4. This connects the loss ledger across periods; it does not establish financial-book carry. The 2026 financial opening is independently declared for the test and is expressly not the carried closing state of the 2024–2026 example. Both complete conditional calculations remain attached.

Partial use makes the expiry rule consequential. Using 7 of 19 leaves 12 when the expiry is later than the next endpoint. If the exclusive expiry equals that endpoint, 7 may be used on the preceding day and the remaining 12 expires. If expiry instead equals the usage day, the usage is rejected; with no usage, all 19 expires. These outcomes follow the selected policy, rather than a silent decision to use the oldest lot or to preserve an expired balance.

The operation proposes a complete next loss state and per-lot movements. Applying it recomputes the proposal and compares the complete result, including the original financial calculation. Changed amounts, origins, eligibility, scope, cursors or even Boolean-to-integer substitutions are rejected. This is a pure in-memory application: it creates no durable commit, tax journal, deferred asset or annual authorization. Connecting the conditional tax amount and loss ledger to an actual financial tax balance still requires separately specified recognition, accrual and settlement methods.

A.26.6 Accrue current tax before choosing its payment

A conditional calculation supplies an amount, but recognition and payment change different financial balances. The next finite profile makes those changes explicit. It begins with the complete recorded return-and-lease calculation, recomputes the conditional loss result and requires an affirmative recognition premise. An unknown premise remains unresolved; a false premise does not produce a financial state. The profile applies only when the native period has no current-tax expense already recorded and its tax-close cursor still equals the assessment start. It therefore does not guess how an opening tax accrual or an earlier assessment should be allocated.

The successor adds a separate income-tax payable to the original 25 accounts. That liability is not trade payable. If the recomputed gross conditional tax after loss usage is C, the accrual debits current-tax expense and credits income-tax payable by C. Cash does not change. A zero amount produces a zero journal. Existing deferred-tax assets, deferred-tax liabilities and deferred-tax expense remain exactly as recorded; the loss balance does not create a new deferred asset.

Payment requires its own identity, date, positive amount and affirmative instruction. It debits income-tax payable and credits cash, subject to both the outstanding liability and available cash. The author supplies the amount; neither full payment nor a refund is inferred. A shortage

of cash blocks that payment while leaving a valid accrual available. This illustrative profile classifies its specified tax payments as operating cash movements.

Let S be cumulative payment against the newly accrued amount C, with $0 \le S \le C$. Relative to the original recorded statements, the changes are

$$\Delta\text{profit} = -C, \qquad \Delta\text{tax payable} = C - S, \qquad \Delta\text{cash} = -S.$$

Assets decrease by S, liabilities increase by $C - S$ and equity decreases by C, preserving the balance-sheet identity. In the indirect operating cash bridge, the lower profit and the increase in tax payable contribute $-C + (C - S) = -S$. This equals the separately recorded cash payment. Paying the liability does not charge the same expense again.

For the separate hypothetical 2026 calculation, loss usage leaves conditional tax of $1/2$. Accrual records expense and payable of $1/2$ while cash stays at 500. An explicit payment of $1/4$ leaves cash of $1999/4$ and payable of $1/4$; a second payment of $1/4$ clears the payable and leaves cash of $999/2$. The recorded deferred-tax asset of 10 and liability of 4 are unchanged. The original mixed example instead generates the loss quantity 19 and a zero tax journal. These are consequences of the declared synthetic inputs, not assessments of either a company or a jurisdiction.

The financial date and assessment coverage have separate meanings. Accrual is dated on the last included day; coverage ends at the following exclusive endpoint. The successor records that endpoint separately and leaves the native tax-close cursor unchanged. Payments cannot precede the last tax operation or fall outside the original financial period. An end-of-period accrual therefore cannot silently accept a payment in the next year: financial-period carry must first be specified.

The original native schema refuses the additional account and the exclusive close cursor, so this is a distinct financial profile. Its validation replays the retained calculation and every tax payment, then compares the complete state. Changed balances, dates, loss results, payment histories and proposal flags are rejected. Repeating the same accrual request produces the same pure proposal; it does not establish globally unique durable posting. Later ordinary return or lease events, financial-year carry, actual accounting admission and annual completeness remain separate requirements.

A.26.7 Continue ordinary events on the tax-adjusted book

The additional tax payable must survive when ordinary business resumes. A later return refund or lease payment has to use the cash left after tax payments, and its journal must leave the tax balances intact unless that method explicitly changes them. The continuation profile therefore retains all 26 accounts, the return and lease memories, the assessment result and one ordered event history. It extends the typed state and memory schemas directly; it does not remove the tax payable to fit the older book.

The selected return methods instantiate the existing monetary definitions over the complete account registry. Their memory operations still settle the return, create the recovered lot or dispose of that lot, while preserving the additional balances. The selected lease methods retain their original component equations and map the resulting journal into the same registry. Tax expense and deferred balances are unchanged by these ordinary methods. A separate, explicitly instructed payment reduces income-tax payable and cash.

Assessment coverage constrains the next event. An ordinary event must occur on or after the exclusive assessment endpoint and before the original financial period ends. Otherwise it would change activity already used in the retained tax calculation. A tax payment may occur on the

accrual day, but cannot precede an already completed operation. If assessment reaches the end of the declared financial period, no ordinary continuation is possible without a separately specified transition to the next opening.

An earlier assessment provides a concrete same-period test. The author declares the original two-year financial window, assesses the first interval through 31 December 2024 and places the later ordinary events on 1 January 2025. The taxable expression of 2 and tax of 1/2 are explicit hypothetical assumptions. Paying that tax, settling the return, selling the recovered lot and paying the lease use one cash balance:

$$500 - \tfrac{1}{2} - 48 + 30 - 121 = \tfrac{721}{2}.$$

After the accompanying depreciation, recorded profit is 9/2, assets are 1401/2, liabilities are 104 and equity is 1193/2. Operating cash is −59/2 and financing cash is −110. Direct cash, the indirect bridge, the balance sheet and the equity movement reconcile exactly. The return's later tax adjustment of −24 remains dated and identified for a subsequent assessment; this continuation does not silently fold it into the earlier tax.

Payment order becomes consequential when cash is scarce. With opening cash 485/4, the lease payment of 121 alone leaves 1/4. Paying tax of 1/2 first leaves 483/4, which cannot fund that lease payment. Reversing the order allows the lease payment but leaves too little for tax settlement. Each request stops at its first failed operation and retains the complete successful prefix. It neither restores spent cash nor supplies financing automatically.

The result preserves the complete original report, loss state and conditional tax calculation alongside the subsequent journals and gross cash movements. Applying it recomputes the ordered prefix and compares the whole result. Unknown findings remain unresolved, and repeated identities, changed balances and backdated activity are refused. These checks establish selected same-period execution under the declared assumptions. Further tax assessment, financial-year carry and actual accounting admission remain separate work.

A.26.8 Replace a provisional assessment without charging twice

Later activity changes the question that the tax calculation must answer. The continued book includes the return adjustment of −24, while its earlier assessment covers only the preceding interval. A transition to the next financial opening cannot make that unassessed activity disappear. The next finite method therefore supplies an explicitly selected cumulative replacement: a later calculation over the same original opening supersedes the earlier provisional assessment. This is a hypothetical tax model. Independent tax periods require a different method.

The replacement reads the actual continued 26-account book. It preserves the cash already paid, retains the separate payable and includes every pretax account and relevant lease right in the new declarations. Each native return adjustment must appear exactly once with its event identity and amount. Explicit additional assumptions must reconcile to the declared account totals, and right-level tax bases must agree with their aggregate balances. Missing declarations remain unresolved; incompatible scope, amounts or dates are refused.

The loss calculation returns to the original opening inventory. Using the provisional closing ledger as a new opening could count a loss twice or make it available inside the cumulative interval that generated it. The replacement instead applies the declared usage, cap, eligibility and expiry rule once to the original lots. The earlier result remains attached as superseded evidence. For example, a provisional generated loss of 19 replaced by a cumulative loss of 7 leaves 7, not 26. An original opening loss of 19, provisionally used by 1 and later used by 19 cumulatively, has total usage 19 rather than 20.

Let C_0 be current-tax expense already recorded, C_1 the replacement total after explicit loss usage, and S actual tax payments. The additional expense and resulting unpaid liability are

$$\Delta C = C_1 - C_0, \qquad L_1 = C_1 - S.$$

The method debits current-tax expense and credits tax payable by ΔC. A negative amount reverses those balances; it does not change cash. The selected profile requires $L_1 \geq 0$. If earlier payments exceed the new total, it preserves the calculation but refuses the financial proposal because a tax receivable or refund needs a separately specified method.

In the paid mixed example, expense and payments are both initially $1/2$. A declared replacement total of $21/4$ therefore adds only $19/4$ to expense and payable. Cash stays at $721/2$. Pretax profit of 5 less total current-tax expense of $21/4$ gives a recorded loss of $1/4$; assets of $1401/2$ less liabilities of $435/4$ equal equity of $2367/4$. The operating cash result remains $-59/2$. These amounts follow the explicit cumulative assumptions.

If no tax has been paid, replacing expense of $1/2$ with a total of $1/4$ reverses expense and payable by $1/4$ without moving cash. If $1/2$ was already paid, the same replacement would leave a receivable of $1/4$ and is refused by this profile. Both outcomes retain the complete continued financial history and the replacement calculation for inspection.

The new coverage endpoint is later than the provisional endpoint and no later than the original financial period boundary. The recorded events must fit within it. Validation recomputes the continued book, the cumulative calculation, the replacement losses and the incremental journal; applying a proposal compares the whole result. Deferred-tax balances remain unchanged. Further assessment chains, deferred-tax recognition, financial-period carry, source admission and annual completeness still require their own methods and evidence.

Begin another period with recorded balances and unresolved findings. A conditional simulation can need another period before all the preceding period's accounting questions have been resolved. For example, the scenario above has cash, an unpaid current-tax liability, lease rights and a recorded profit, but its deferred-tax assessment and the completeness of its historical inputs remain open. Those recorded quantities are sufficient to construct a provisional opening for selected subsequent transactions. They are insufficient to declare that the preceding annual financial statements are complete.

The explicit model `provisional_recorded_balance_carry_v1` makes that distinction part of execution. The request identifies the complete prior reassessment, a unique carry event and the next period's start and exclusive end. The prior current-tax assessment and loss history must reach the preceding financial endpoint, and the next period must start at that endpoint. The prior closing date must be its last included day. No calendar year or assessment of missing transactions is inferred from those dates. Separate findings describe event coverage, deferred-tax assessment, source admission and accounting admission. Unknown and negative findings remain visible in the resulting state; affirmative caller assumptions also leave annual readiness unestablished.

The transfer follows the book's debit-positive convention. Let P denote the recorded profit, R the signed retained-earnings balance, and b_k each signed account balance immediately before the boundary. Let $\mathcal{I}$ contain the temporary income and expense accounts. The new opening satisfies

$$P = -\sum_{k \in \mathcal{I}} b_k, \qquad R^{\text{open}} = R - P, \qquad b_k^{\text{open}} = 0 \quad (k \in \mathcal{I}).$$

Every other account keeps its recorded amount. A profit therefore increases the credit balance in retained earnings; a loss reduces it. The transfer journal sums to zero without a balancing plug. In the mixed scenario, recorded profit is $-1/4$ and retained earnings is -12, so the opening

signed balance is $-12-(-1/4)=-47/4$. Cash stays $721/2$ and the unpaid tax liability stays $19/4$. Positive-profit and zero-profit cases separately check the same sign convention.

The boundary changes period accumulators without discarding economic memory. Return components, recovered lots including disposed identities, lease rights, measurement dates, loss quantities and lifetime event identifiers survive exactly. The prior result retains its cash flows and tax adjustments together with hashes of the closing book and report. The next period begins with empty cash-flow and tax-adjustment lists, so an old receipt or deduction cannot be counted again. Opening deferred-tax allocations copy the recorded asset and liability; that copying does not assess their recognition or measurement. The new period's assessment cursor denotes an empty interval and its current tax is explicitly unassessed.

Subsequent statements use the actual carried opening. Paying the opening tax liability of $19/4$ reduces cash from $721/2$ to $1423/4$, eliminates that liability and leaves current-period tax expense at zero. In the indirect cash calculation, the reduction in the opening payable explains the operating cash outflow. It must not be charged to profit again. If depreciation of 10 follows, recorded profit is -10, assets are $2743/4$, liabilities are 104 and equity is $2327/4$. Depreciation is added back in the indirect bridge, leaving operating cash flow at $-19/4$. The balance sheet, equity movement and direct and indirect cash calculations reconcile to the same opening and closing state.

A second example carries an unsettled return component instead. Settlement pays 48, releases a refund liability of 40 and a recovery asset of 16, and recognizes recovered inventory of 20. Its cost-of-sales credit of 4 and selling expense of 8 reduce profit by 4. Selling that inventory for 30 then earns 10, so the two transactions together earn 6 and reduce cash by 18. Starting from cash of 500, the following-period cash is 482. The indirect calculation gives $6+16-40=-18$. The new return tax adjustment of -24 is retained for a later assessment; it is not absorbed into an asserted tax-complete closing balance.

Ordered execution retains the completed prefix if a later event fails. A payment still needs both an unpaid liability and actual shared cash; a reused identity or an event outside the next period is refused. Applying a result recomputes the entire predecessor and subsequent prefix and compares the complete result, including unresolved findings and archived history. The checked implementation covers this provisional boundary and selected return, lease and tax-payment operations. New-period tax assessment, repeated carry, complete event coverage and deferred-tax recognition require their own methods and evidence before the scenario can support a complete annual close.

Measure deferred balances from explicit recognition assumptions. Carrying a recorded deferred-tax asset and liability preserves the book, but a later assessment must explain whether their amounts should change. The mixed book records an asset of 10 and a liability of 4. Its current-tax declarations also contain tax bases. Neither fact establishes the amount of deductions that can be recognized, the applicable rate on recovery or settlement, or whether a movement belongs in profit or loss. The conditional deferred-assessment method therefore requires those decisions separately and binds them to the actual reassessed book.

The distinction follows the historical source treatment. IAS 12 paragraphs 7–9 distinguish asset and liability tax bases; paragraphs 24 and 27–29 condition asset recognition on available taxable profit and its relevant grouping and timing. Paragraphs 47 and 51 connect measurement to the applicable rate and manner of recovery or settlement. Paragraphs 58 and 61A distinguish profit-or-loss movements from tax related to items outside profit or loss (European Commission, 2023a, IAS 12, paras. 7–9, 24, 27–29, 47, 51, 58 and 61A). These retained historical provisions explain the method's boundaries. The example's rates, bases and recovery capacity remain explicit hypothetical premises, rather than findings about an actual company's tax treatment.

The selected model uses one declared entity, tax authority and recovery group, and keeps gross deferred assets and liabilities separate. Every nondeferred asset and liability account is represented. Inventory is divided between other inventory and retained lots; lease assets and liabilities are divided by right; each remaining tax-loss lot is included separately. A component and its aggregate cannot both contribute the same difference. Item bases must reconcile with the complete prior declarations, while recovered-lot bases must also agree with component memory. A finding about the absence of unrepresented tax-only items and credits makes this finite scope explicit.

Let C_i be the positive magnitude of an item's carrying amount, B_i its declared tax base and r_i its declared nonnegative rate. An asset has signed taxable difference $C_i - B_i$; a liability has $B_i - C_i$. A positive difference is taxable and a negative one is deductible. For a loss lot, the supplied model uses the negative of its remaining quantity, without consuming that quantity. Write d_i for the nonnegative deductible difference and t_i for the nonnegative taxable difference. The caller supplies a recognized deduction u_i and a common recovery capacity K, all in the same tax-base currency units:

$$0 \le u_i \le d_i, \qquad \sum_i u_i \le K, \qquad A_1 = \sum_i r_i u_i, \qquad L_1 = \sum_i r_i t_i.$$

The sums include deductible temporary differences and loss lots together, so one capacity cannot be reused for each item. The unrecognized amount $d_i - u_i$ remains visible. A deferred-tax liability does not automatically enlarge K: the timing, entity and income restrictions needed to support recovery must be supplied explicitly. The method does not produce that support by forecasting future taxable profit.

Before changing balances, the request allocates the recorded asset A_0 and liability L_0 completely across the same item population. Those allocations must sum exactly to the book's recorded gross amounts. They explain inherited balances rather than inventing a new assessment of the original opening. For the supported profit-or-loss route, the change in deferred expense is $(L_1 - L_0) - (A_1 - A_0)$. The journal debits the asset by $A_1 - A_0$, credits the liability by $L_1 - L_0$, and records the balancing expense or benefit. Negative changes reverse the corresponding balance. The request must support that allocation; an equity, other-comprehensive-income or business-combination route is refused by this particular implementation instead of being charged to profit.

In the checked example, property, plant and equipment has carrying amount 100 and tax base 60. At rate 1/4, its taxable difference of 40 gives a liability of 10. Debt has carrying amount 100 and tax base 60, so its difference is deductible. Recognizing 32 of the available 40 gives an asset of 8 and leaves 8 unrecognized. Relative to the recorded asset of 10 and liability of 4, the asset falls by 2 and the liability rises by 6; deferred expense is therefore 8. Cash stays 721/2, current tax and its unpaid liability are unchanged, and recorded profit becomes −33/4. Assets of 1397/2 equal liabilities of 459/4 plus equity of 2335/4. Adding the deferred expense back in the indirect cash calculation leaves operating cash at −59/2.

Loss recognition uses the same capacity discipline. Recognizing deductions of 32 for the debt and 10 for a loss lot needs capacity of at least 42; capacity of 32 is insufficient even if each item separately fits it. With capacity 42 and rate 1/4, their combined deferred asset is 21/2. The loss ledger still contains the original remaining quantity of 19: recognizing a tax benefit does not use the loss in the current-tax calculation. Matching bases, zero rates, partial recognition and opposite asset/liability differences have separate checked cases.

The proposal retains the entire prior result and changes only deferred balances, deferred expense and its new event identity. Cash flows, current tax, loss history, component state and assessment dates remain exact. Unknown necessary findings stay unresolved, and an inconsistent population,

allocation, base or capacity produces no changed financial state. Pure apply compares the complete recomputed proposal. Actual source and accounting admission, broader recovery and allocation methods and annual completeness remain open. The preceding frozen carry interface accepts a current-tax reassessment request; consuming this newer result requires an explicit successor interface that preserves the deferred journal and recognition schedule.

Carry the assessed book without renewing its recognition assumptions. The deferred journal changes the financial opening that a later period must inherit. Replaying the older current-tax-only request would restore the asset of 10 and liability of 4 and omit the deferred expense of 8. The resulting retained earnings would also be wrong for the assessed scenario. The successor carry therefore accepts the complete deferred-assessment request and recomputes that proposal before transferring its actual closing balances.

The transfer uses the same debit-positive convention as the preceding carry. Recorded profit is now $-33/4$, so retained earnings moves from -12 to $-12-(-33/4) = -15/4$. The deferred asset of 8, deferred liability of 10, cash of $721/2$ and current-tax payable of $19/4$ remain unchanged. Temporary income accounts, including the deferred expense, begin at zero. Thus the next period inherits the assessed economic amounts without charging the old expense again.

The complete predecessor remains attached. Its item-level differences, recognized and unrecognized deductions, recorded allocations, support findings and recovery capacity are retained with the original loss history. This archive explains the opening balances; it does not renew the capacity or establish new-period recognition. Opening deferred allocations copy the recorded 8 and 10, while both current and deferred tax in the next period are explicitly unassessed. Unknown or negative carry findings remain visible independently of the predecessor's conditional calculation.

If the next period pays the carried tax liability and records depreciation of 10, cash becomes $1423/4$ and recorded profit is -10. Assets of $2735/4$ equal liabilities of 110 plus equity of $2295/4$. Neither current nor deferred tax expense is repeated. Operating cash is $-19/4$: depreciation is added back, and the reduction in the opening tax payable explains the payment. The direct cash calculation and the equity movement use this same carried opening.

The request retains the prior assessment and loss endpoints, requires a contiguous next period and keeps lifetime event identities. Selected return, lease and payment operations then act on the full new state. A failed operation preserves the completed prefix. Applying a result compares the full recomputed predecessor, opening, recognition archive and subsequent operations. The checked scope remains provisional: a new current-tax assessment must distinguish payments of the opening liability from payments of the new period's tax, and new deferred recognition must reassess its own information. Repeated carry and complete annual source and event coverage remain separate requirements.

Keep carried tax payments separate from the new assessment. The first current-tax assessment after carry has a different opening from the same-period reassessment above. Current-period expense begins at zero, but the book can still owe tax from the preceding period. Paying that opening liability settles an old assessment; it does not reduce the new period's tax expense. Calculating paid tax as current expense less the outstanding payable would therefore be wrong for this opening.

The request binds the complete assessed-carry execution and begins its tax interval at the new financial opening and loss cursor. It requires a completed incoming execution and no previous assessment in the new period. Before that first assessment, the selected payment kernel can consume only the carried liability. Each payment can therefore be attributed to the

prior assessment from its exact event identity, date, amount and cash-flow record. A stopped incoming execution is retained for inspection but is not silently assessed.

Let P_0 denote the opening tax payable, S the completed payments against that carried liability and C the new conditional assessment. The two liability records reconcile to the financial book as

$$P_{\text{carried}} = P_0 - S, \qquad P_{\text{new}} = C, \qquad P_{\text{total}} = P_0 - S + C, \qquad 0 \le S \le P_0.$$

The first record retains the prior assessment's interval and identity with its new-period settlements. The second records the new interval, assessment and initially empty settlement history. The journal debits current-tax expense by C and credits the payable by the same amount. Cash, deferred balances and the old settlements remain unchanged. Even a fully paid old liability keeps its history rather than disappearing from the explanation.

For opening payable 19/4 and new tax of 5, paying all 19/4 leaves a total payable of 5. Paying only 1 leaves 35/4, and paying nothing leaves 39/4. The new expense is 5 in all three cases. In the fully paid example, after the new-period depreciation of 10, recorded profit is -15, cash is 1423/4, assets are 2735/4, liabilities are 115 and equity is 2275/4. The increase in the new payable cancels its expense in the indirect cash calculation; the old payment remains the operating cash outflow of 19/4.

Taxable profit is calculated from the new period's pretax accounts and explicit adjustments. Each native return adjustment must appear exactly once, and the complete tax-base declarations must reconcile at account and lease-right level. Those declarations do not themselves assess deferred tax. In a return example, new-period recorded profit of 6 and its native adjustment of -24 produce taxable loss of 18. The method retains a new loss quantity of 18, available at the interval's exclusive endpoint, without a refund or an automatic deferred asset.

Loss use starts from the actual carried inventory. If a loss of 19 remains available, explicit use against taxable profit of 20 leaves taxable profit of 1 and tax of 1/4 at the declared rate. The history continues from the prior cursor, with total use of 19. The selected close-day rule separately checks expiry: a loss expiring before the usage day cannot be used, while a use before expiry consumes only the declared quantity. Neither the previous deferred-tax recognition nor its recovery capacity consumes or replenishes the loss inventory.

Applying the proposal compares the full predecessor, tax calculations, loss movements, both liability records and financial statements. The checked scope is a first assessment without credits, refunds or prepayments. Once both tax vintages exist, subsequent payments need explicit allocation; the original aggregate payment interface no longer supplies enough information. New-period deferred assessment, repeated current assessments and later carry require their own compatible methods. Source admission and complete annual event coverage remain unresolved.

Name the tax liability settled by each payment

Once carried tax and newly assessed tax are both payable, the payment amount alone cannot identify which liability has been settled. A payment of 3 can reduce either the carried payable of 19/4 to 7/4, leaving the new payable of 5 unchanged, or reduce the new payable to 2, leaving the carried payable at 19/4. Both choices reduce cash and total payable by 3. Their remaining tax obligations have different histories, which the simulator must preserve.

The continuation therefore requires the caller to name each tax vintage and its allocated amount. Every amount must be positive, no greater than the named outstanding balance, and the allocations must sum to the payment. Actual shared cash must cover that sum. An unknown authorization, a duplicated vintage, an unknown liability or an excessive allocation prevents that operation. The method supplies no first-in-first-out convention or proportional split.

For example, take the hypothetical first assessment through 30 June 2026 inside the financial period ending on 31 December. Carried tax of 19/4 is still unpaid, new tax is 5, cash is 721/2, and recorded profit is −15. Allocate 1 of a payment to carried tax and 2 to new tax. Cash becomes 715/2; the two payables become 15/4 and 3, totaling 27/4. Tax expense remains 5 and profit remains −15. The payment is recorded in both affected settlement histories, with their distinct amounts and the same dated cash movement.

An ordinary event can follow only from 1 July, the assessment's exclusive endpoint. A further right-of-use depreciation charge of 2 reduces its asset from 80 to 78 and profit to −17, without changing either tax payable. Assets then total 1367/2, liabilities 467/4, and equity including profit 2267/4. Operating cash flow remains −3. This depreciation is explicitly listed as activity after the assessment; it has not received a new tax assessment.

The full book, prior loss inventory, deferred balances and unresolved findings remain attached to the result. If a later operation fails, earlier completed events and their settlement histories survive as a recorded prefix. Payments can occur on the recorded assessment day, but an assessment at the financial period's endpoint leaves no later date for ordinary events in that period. Cumulative reassessment, new-period deferred recognition and another period carry require their own compatible methods. The present calculation supplies conditional bookkeeping evidence rather than annual accounting acceptance.

Replace the assessment while retaining the payments

After further company activity, the new period's cumulative tax calculation may change. Its recorded payments do not change with it. A replacement must therefore start from the actual continued book and distinguish what was paid against the carried liability from what was paid against the current period's assessment. Using current expense less all tax payments would wrongly treat a payment of last year's tax as a reduction in this year's expense.

Let C_0 be recorded current-period tax expense, C_1 the replacement cumulative amount, S the actual payments allocated to that current vintage, and O the remaining carried liability. The replacement records only the expense difference:

$$\Delta C = C_1 - C_0, \qquad P_{\text{current}} = C_1 - S, \qquad P_{\text{total}} = O + C_1 - S.$$

The journal debits current-tax expense by ΔC and credits payable by the same signed amount. Cash and the carried liability remain unchanged. The current vintage retains the identity referenced by earlier payments; its old assessment and settlement record remain available alongside the replacement's provenance.

Continue the split-payment example above. Recorded current expense is 5, current-vintage payments are 2, and carried payable is 15/4. If the replacement cumulative tax is 7, the expense increment is 2, current payable becomes 5 and total payable becomes 35/4. Cash stays at 715/2 and recorded profit falls from −17 to −19. Assets remain 1367/2, while liabilities become 475/4 and equity including profit becomes 2259/4. If cumulative tax instead falls to 3, the expense reduction is 2 and total payable becomes 19/4.

A positive aggregate liability can conceal a different representation problem. If cumulative current tax falls to 1 after payments of 2, the current vintage would have a credit of 1 even though the carried liability leaves a positive aggregate of 11/4. The present positive-payable model refuses that replacement and preserves the preceding book. A credit or refund needs a specified account and treatment; it cannot be silently netted against the older obligation.

Losses are also recalculated cumulatively. Suppose the carried opening contains an available loss of 19 and the first assessment uses all 19 against taxable profit of 20. At a declared

rate of one quarter, current tax is 1/4. A replacement using the same cumulative profit and deduction again yields 1/4: it restarts from the original opening inventory, rather than using the provisional closing balance of zero or recording a second consumption of 19. Previous provisional calculations remain in the history. If that loss expires before the new assessment's usage date, the replacement must apply the stated expiry rule; it cannot retain the earlier deduction automatically.

The cumulative interval must cover every completed event and cannot shorten the previous assessment. A correction at the same endpoint is possible when no later event lies outside it. Native return adjustments enter the replacement exactly once; a worked return sequence has pretax income 6 and an adjustment of −24, producing a new loss of 18 without an automatic deferred asset. Fresh deferred recognition, another continuation or repeated replacement requires a compatible method. Source admission, complete fiscal coverage and annual acceptance remain unresolved.

Measure deferred tax from the newly assessed book

The current-tax journal changes the book on which deferred tax will be measured. In the cumulative example, payable was 27/4 before the replacement and 35/4 after it. The declarations used to prepare that current-tax calculation cannot silently serve as measurements of the later book. The new deferred request therefore names the exact recorded state and date, declares its jurisdiction and regime, and supplies fresh bases for every scoped account and current lease right. It also supplies the item-level bases and recognition decisions that must reconcile to those totals. An unchanged base may be declared again with its provenance; the method does not carry that premise forward automatically.

The selected hypothetical model retains the gross, single-recovery-pool, profit-or-loss calculation described earlier. The complete current population contains the ordinary asset and liability accounts, inventory subdivisions, lease rights and remaining loss lots. Fresh measurement and recognition support is required for every item. Recognized deductions across debt, losses and other deductible items share the one explicitly supplied recovery capacity. The older recognition schedule remains historical evidence. Its capacity cannot renew itself, and a deferred liability cannot silently supply additional capacity.

Recorded allocations must also reconcile to the balances actually carried into the period. In the worked book, those balances are a deferred tax asset of 8 and a deferred tax liability of 10. With property carrying amount 100 and a fresh base of 80, the taxable difference is 20. With debt carrying amount 100 and base 60, the deductible difference is 40; suppose fresh support permits recognition of only 20 against capacity 20. At a declared rate of one quarter, the desired asset and liability are both 5. The journal reduces the asset by 3, reduces the liability by 5, and records deferred-tax expense of −2.

After the preceding cumulative current-tax example, cash remains 715/2, current expense remains 7 and current-tax payable remains 35/4. Recorded profit rises from −19 to −17. Assets become 1361/2, liabilities 455/4 and equity including profit 2267/4; operating cash flow is still −3. All tax-vintage settlements, loss quantities, component states and earlier findings remain unchanged. The result retains both the fresh recognition premises and the previous period's schedule.

The capacity constraint matters when loss recognition is added. If the current loss inventory has 9 remaining and the caller recognizes all 9 as well as the 20 debt deduction, capacity must cover 29. At one quarter, the combined desired asset is 29/4. Capacity of only 20 refuses that combination; the same amount cannot support the debt deduction and then be used again for the loss. A loss quantity by itself does not establish a deferred asset.

The method compares the entire proposed result when applied, including the fresh measurements, item allocations, tax vintages and historical recognition record. Unknown bases or support remain unresolved, and unsupported equity, other-comprehensive-income and offset routes are refused. Another financial boundary must preserve these histories while distinguishing old settlements from the next period's cash movements. Conditional recognition here does not establish annual completeness or source and accounting admission.

Cross the next boundary without replaying old payments

The following-period book now contains current and deferred assessments, dated payments against distinct tax liabilities, and the history of a replaced assessment. Carrying this book into another period must preserve those histories as well as its closing balances. The earlier carry caller cannot stand in for that operation: it consumes a different predecessor and does not represent the separate liability vintages.

The new boundary requires the recorded current and deferred assessments to reach the financial endpoint, with the loss cursor at the same date and the closing book dated the preceding day. The next financial interval must be contiguous. This establishes the timing of a provisional carry, while unresolved source, event-coverage and accounting findings remain attached to it. A caller's assertion that those findings are true does not establish annual readiness.

In the fresh-deferred example, recorded profit is -17 and retained earnings has a debit-positive balance of $-15/4$. Closing the income accounts therefore sets retained earnings to $-15/4 - (-17) = 53/4$. All income accounts, including current and deferred tax expense, start the new period at zero. Cash stays at $715/2$, the deferred asset and liability stay at 5 each, and current-tax payable stays at $35/4$. Assets, liabilities and total equity are unchanged. The boundary journal balances, and the new opening has neither profit nor cash movement.

The two tax vintages retain their complete settlement and supersession records. Separate new-period records carry outstanding amounts of $15/4$ and 5 under the same liability identities. Each records the earlier vintage it came from, its new financial period and an initially empty list of current-period settlements. Earlier payments remain in the historical records; they do not enter the new period's cash flows. A fully paid vintage is retained with a zero opening balance rather than discarded.

The prior flows, tax adjustments, recognition calculation, measurement and recovery capacity are archived with the closing book. Components, inventory lots, lease-right dates, loss quantities and lifetime event identities remain intact. Both current and deferred tax in the new period are unassessed; carrying a recognition schedule does not renew its capacity or findings. This boundary profile accepts no subsequent events. A compatible continuation must first reconcile new payments against the new-period liability records while keeping all earlier settlements historical. General annual execution and actual fiscal completeness remain separate obligations.

Settle carried liabilities in the period that pays them

The new opening can now support ordinary events and tax payments. Its two liabilities still refer to earlier assessments, but the cash-flow statement must record only payments made in the new financial period. Reusing the old settlement list as current cash movement would count earlier payments again. The continuation therefore keeps each historical vintage unchanged and records new settlements against its separate current-period opening.

Suppose the company pays 3 on 1 January 2027 and explicitly allocates 1 to the liability opening at $15/4$ and 2 to the liability opening at 5. Their remaining amounts become $11/4$ and 3. Total

payable falls from 35/4 to 23/4, while cash falls from 715/2 to 709/2. The entry debits tax payable and credits cash by 3; it creates no current-tax expense. The new period's operating cash flow is −3. Historical payments remain available for inspection without entering that sum.

Each payment names the stable liability identities and supplies positive exact amounts whose sum equals the payment. Each allocation must fit that liability's remaining current-period amount, and the total must fit available cash. The caller also supplies affirmative payment authority. No oldest-first allocation, proration, borrowing or reassessment is inferred. A second payment can exhaust the remaining 11/4 and 3, leaving both liability records at zero and cash at 1395/4. Keeping the zero records preserves which assessments were settled.

Ordinary events act on the same full financial state. If depreciation of 2 follows the first payment, the right-of-use asset falls from 78 to 76 and recorded profit becomes −2. Assets are then 1351/2, liabilities 443/4 and equity 2259/4. The indirect operating cash calculation adds back depreciation and includes the reduction in opening tax payable, giving −3, the same amount as the dated cash payment. A retained inventory lot can also be sold through the selected return method; its history and tax adjustments remain attached to the resulting book.

Operations must have unused lifetime identities and chronological dates inside the new financial interval. A rejected allocation, missing authority or cash shortage stops that operation and preserves every preceding successful event, together with the exact unprocessed sequence. Applying a proposed result recomputes and compares the whole result, including its opening, financial history, liability records and recognition archive. A nonportable request, such as one containing a floating-point amount, is rejected before execution.

These events do not reassess current or deferred tax. The carried loss quantities, recognition findings and historical measurements remain intact, while both new-period assessments stay unassessed. A reusable annual design must next explain how a fresh assessment joins all outstanding liabilities and how another boundary preserves their complete histories.

Repeat the period with one explicit company book

The next assessment may have several earlier liabilities to preserve. A procedure that always separates one old liability from one current liability cannot represent that situation. The common period book instead retains an ordered collection of stable liability records, an immutable financial and loss opening, current financial state, assessment histories and archived periods. Each new period uses the same operations. The checked predecessor is imported once, and a finite sequence of operations supplies the later events and explicit assessment information.

A first current assessment adds one liability; a replacement changes that liability's cumulative assessment while keeping its identity and payments. Every earlier liability stays separate. Starting after the payment and depreciation example, cash is 709/2, the two liabilities are 11/4 and 3, and profit is −2. A hypothetical assessment of 4 adds a third liability, raises total payable to 39/4 and reduces profit to −6, without moving cash. Paying 4 with allocations of 1, 1 and 2 then leaves cash at 701/2 and the liabilities at 7/4, 2 and 2. If a fresh cumulative calculation replaces the current assessment of 4 with 3, only the difference of −1 enters current tax expense. The third liability becomes 1, total payable becomes 19/4 and profit becomes −5. Replacing the assessment with less than the 2 already paid requires a receivable or refund method; the selected no-credit model refuses that operation.

Loss replacement starts from the immutable loss opening for the period. For example, using 4 from an opening quantity of 9 leaves 5; replacing that calculation with a use of 3 leaves 6, not 2. The earlier use remains in the assessment history. Expiry is checked at the stated usage date, and a loss identity cannot be reused after its earlier generation has been superseded or carried

into another period. Lease-right populations likewise include identities observed in intermediate states, so a right that later disappears is not silently removed from the period's tax-information requirements.

Fresh deferred measurement follows the current assessment and binds the actual financial state. Its item bases, recorded balances, support and shared recovery capacity must reconcile under the selected model. Another payment, native event or current assessment makes that state-specific measurement stale. The earlier calculation remains available, but the next boundary requires a new measurement of the changed book. This dependency prevents an old recognition capacity from renewing itself merely because the program continues.

At the financial endpoint, the current assessment and loss cursor must reach the same exclusive date, the closing book must be dated the preceding day, and the deferred measurement must bind that exact closing state. The provisional carry closes actual profit into retained earnings and archives the period's balances, reports, liabilities, payments, assessments and findings. In the worked sequence, closing profit of -5 changes the debit-positive retained balance from $53/4$ to $73/4$. Cash remains $701/2$. The next period opens the three liabilities at $7/4$, 2 and 1, with no new settlements or cash movements. Both tax assessments begin unassessed.

A new assessment of 2 in that period adds a fourth liability and raises total payable to $27/4$. Profit is -2 and operating cash remains zero. The same deferred-measurement and carry operations can then close that period, changing retained earnings to $81/4$ while preserving all four liability identities. These exact recurrences check the selected reference book and its refusal behavior. Actual reporting completeness, admission of source evidence and broader company or sector models still require their own specified methods and evidence.

Author the same annual program in text and typed records

The common period book also needs an authoring form that preserves its explicit assumptions. A named annual program imports one checked root by its semantic digest and lists operations in order. The import is supplied as a record in the caller's local registry; it does not name a file to open or a service to query. The compiler verifies that the imported record has the stated digest and lowers the declarations to the same typed annual request.

After a first assessment creates the liability named `assessment:annual-first`, a payment block can state an intended allocation while preserving the fact that payment authority is unknown:

```
operation "allocated_tax_payment" "authored-payment";
date "2027-06-30";
amount "1/1";
authorization {"status":"unknown","reason":"Not supplied"};
payment_allocation {"vintage_id":"assessment:annual-first","amount":"1/1"};
end_operation;
```

The enclosing program supplies the financial book and preceding assessment. This declaration does not create the liability or authorize the payment. The finite grammar uses one declaration per source line. Other operation blocks expose native events, current-assessment information and adjustments, loss use, deferred-measurement items, recovery capacity and the next financial period as named fields or ordered entries. Empty collections remain explicit in the typed request. No field supplies an unstated accounting or behavioural assumption.

Structural compilation checks the closed operation shape, calendar dates, exact rational values, declaration types and duplicate identities. It preserves the distinction between known values with provenance and explicitly unknown information. Whether an authorization is true, a

payment fits available cash or an assessment has sufficient information remains a financial evaluation question. A program with an unknown tax rate therefore compiles and produces the same unresolved assessment as its typed counterpart. A cash shortage preserves the same successful prefix and unprocessed operations. Compilation itself executes no financial method.

The compiler retains both the exact source hash and the semantic request hash, with line locations for each operation and field. Adding blank lines changes the source hash and locations but leaves the lowered request unchanged. The canonical renderer reproduces the typed program, and both authoring paths produce equal complete financial results for the checked scenarios. These checks establish fidelity to the selected evaluator. They do not establish that its supported operations exhaust the financial-statement language or that a successfully parsed company program has achieved accounting admission.

Keep alternative annual histories and their statements together

The same opening book can support several explicitly assumed sequences. A company may have payment authority in one scenario and lack it in another; the tax rate or recovery capacity may remain unknown. Comparing those sequences requires keeping each company's balances, profit and cash movements together. A cash balance from one scenario cannot be combined with the profit or liabilities of another and presented as one company result.

An annual family names its members, states their assumptions and supplies one common opening request and a consecutive reporting horizon. Each member lists its own annual operations and planned period carries. The opening is bound by its digest, and the planned periods must agree. Structural validation checks these declarations without executing the financial methods. An unknown rate can therefore remain in an otherwise well-formed member; the financial evaluation determines where that member can proceed.

Each evaluated member uses the same annual evaluator as an individually submitted program. Its complete financial history is retained. In the worked sequence, the first period ends with cash 701/2, recorded profit −5 and operating cash flow −7. Its next period records profit −2, no cash movement and total tax payable 27/4. A fresh deferred measurement and another provisional carry retain cash 701/2 and change the debit-positive retained balance to 81/4. Both earlier periods remain available in the archive.

Other members need not reach those periods. If payment authority is false, the first assessment succeeds but the payment is refused. The assessment, its resulting balances and every remaining operation stay in that member's result. Its later period has no report because execution never reached it. An unknown rate stops before the first assessment; missing recovery capacity or a stale measurement stops later. An explicit payment exceeding cash is retained as a funding failure. None of these refusals removes the member or prevents an independent member from continuing.

A joint period query returns the whole report for each member, bound to its complete result. A period crossed by a successful provisional carry uses the archived report. The current period uses its completed-prefix report and states the actual date reached and the current and deferred assessment statuses. Completing all listed operations does not turn that prefix into an admitted annual statement. A period not reached is explicitly unavailable, rather than a report of zeros. The query recomputes the family before trusting saved results; changing cash and recomputing a content hash does not validate an altered financial history.

The caller supplies an explicit member-evaluation budget and traversal order. When that budget is exhausted, every unvisited member keeps its identity, assumptions and request digest, with no invented financial result. Even a zero member budget checks the common opening; that check

is outside the count of member evaluations. The count is not a guarantee about execution time or memory. Increasing it makes a new request for recomputation, not a durable continuation receipt.

These members are unweighted authored alternatives. Two names with different stated assumptions remain distinct even when their financial results agree. An empty authored selection does not establish that no feasible company history exists. Evaluating every supplied member establishes coverage of that list only. Reconstructing actual histories, proving an exhaustive feasible set or attaching a probability law requires additional evidence and methods.

The following group profile addresses composition and component migration under its own assumptions.

A.27 Joined company and explicit component migration

The company in Section 28.7 is specified by `CompanyComposition/1`. Its state contains both legal trial balances, the common currency and dates, ownership/control history, return and matured compound-instrument memory, the internal inventory lot, tax-close history and dated gross cash movements. The schema admits nine event kinds: redemption, return settlement, payroll, depreciation, internal inventory transfer, external inventory sale, reciprocal settlement, tax close and a purchase from NCI without loss of control. Every kind has a closed parameter record.

A basis binds the opening state, event, profile, information view and economic event identity. Authorisation checks these bindings and supplied findings. Proposal contains legal journals for both entities, the common closing state and dated cash origins. Commit publishes the proposal, closing state, receipt and immutable request decision together. An internal sale cannot publish only P's receivable while omitting S's payable. Unrelated owners remain unchanged.

An exact retry returns its stored result before current-head or current-view checks. Reusing the identity with changed bytes conflicts. A new request needs the exact current head and view. If an acknowledgement is lost, the observable outcome is unknown: either no commit exists, or the complete commit is durable. The retry commits once in the first world and replays in the second. The documentary examples specify both worlds; they do not measure runtime durability.

The report operation is read-only. It requires published receipts that connect the requested opening and closing in the complete date interval. It recomputes the consolidation worksheet, tax effects, gross external cash and ownership allocation. Reports carry the explicit joined-finite-company scope. A request for an uncomputed tax close, incompatible calendar or unsupported sector is not satisfied by adding independent component totals.

The source and schema locks bind exact bytes. Canonical JSON excludes duplicate keys, floats, non-finite values and unreduced rational quantities. Record identity includes schema and body; dependency lists are the exact transitive typed references. The retained independently checked packet is under `data/accounting-master/company-composition-20261016/attempt-03/`. It has 13 transactions, three reports and two migration demonstrations. Those counts describe the finite evidence, not a complete-company support measure.

Migration is deliberately narrower than method instantiation. The return projection maps the named `CorporateResiduals/1` opening state into a closed target containing its complete return memory, source context/view, owner/book/branch, dates and cash/equity accumulators. The compound projection does the same for the named matured `FinancingComponents/2` state. Nonzero balances outside the explicit map, unknown fields and foreign profiles are refused. Zero-only omitted accounts are reconstructed from the locked source schema in the round trip.

For these two operations, every quantity used by the subsequent settlement or redemption is preserved. Their exact sums, products and minimum therefore commute with the mapping: deriving the target result equals mapping the source result under the same findings. The checks exercise both the round trip and this following transition. They do not migrate old requests, annual-report labels or arbitrary methods. In particular, changing the original synthetic owner to P would be a new instantiation, not a lossless migration.

The separate `CorporateResiduals/1` namespace fixes the 17 finite methods in Section 35.11; its prior `CorporateComponents/5` requests remain bound to their original lock. The all-zero CGU allocation, full retrospective acquisition comparisons, general mixed-contract methods and current-law admission remain explicit residuals. No profile name itself provides interoperability or extends an earlier operation's accepted fields.

A.28 Real-source connections and partial answers

`RealDisclosureConnection/1` is the documentary extension for Section 28.24. Its closed record schemas, canonical request bytes, complete record registry and expected responses are retained in `data/accounting-master/real-package-connection-20261015/`. The semantic specification is `docs/design/proposal/architecture/real-package-connection-20261015.md`. The examples specify behavior; no product endpoint or live store was executed.

The five supported operations are source preparation, historical querying, conditional scenario projection, report construction and review proposal. Each uses the I1 request/result envelope with a profile-bound signature and tagged arguments. Record dependencies and separate byte references follow L1. The schema lock includes both schema bytes and the semantic specification. The 31 exchanges include the combined partial answer, named-source statement readings, original and later segment queries, unknown historical clocks, three conditional difference arms, six settlement paths and company-specific review proposals.

The family record represents an unresolved question about actual issuer histories. It admits neither the illustrative branches nor a probability law. Reported observations, reported dashes, unknown values and conditional derived amounts have distinct tags. A conditional amount references the assumptions under which it was calculated. The complete reported-token subquery does not upgrade the connected result's partial status.

Coverage partitions the requested claims. For a missing-date settlement request, eventual conditional amounts are completed and period timing is unresolved. These regions are neither probabilities nor proportions of all company histories. Future periods are the relative indices 1 and 2; an unknown period is null. An out-of-range period or allocation parameter is invalid. An unsupported depreciation or probability request retains its unresolved obligations instead of returning zero.

Decoding precedes operation-tag, signature, profile, issuer, view and parameter binding. Truncated bytes, a mismatched tag, invalid parameters and a cross-company binding return rejected input with no validated request, payload or coverage. Valid named-source results may only use observations eligible in their own view. Linking a later view in a connected result does not authorize an original-view query to read it.

Every exchange has null expected head, no commit receipt and unchanged before and after worlds for accounting, review, work and import owners. These empty worlds are explicit documentary examples of candidate/read-only effects; they do not establish persistence in a deployed service. Independent checks derive journal balances, all annual difference rows and all settlement-period effects, verify source spans and dependency closure, and preserve the actual source-conflict limits. They establish the stated finite specification and arithmetic, not

actual-company history feasibility or accounting admission.

A.29 source.prepare@1

package manifest, source snapshot, adapter/config lock, prior review, traversal budget and optional continuation → ExtractionPreparation

Acquires/preserves sources and constructs candidates under the source-preparation method; no semantic admission

A.30 review.propose@1

target Ref, layer, decision kind, evidence Refs, actor, rationale, replacement Ref or null → ReviewProposal

Candidate construction; accepts no layer

A.31 review.commit@1

ReviewProposal, expected review view/revision → immutable ReviewDecision and new view

Review transaction under the review and storage rules; admissible action/layer required

A.32 time.select@1

package/view, investor/system/exploratory mode, cutoffs, revision and unknown-time policy → selected assertions, conflicts and temporal obligations

Pure selection under the time-admission rules; no invented ingestion or publication time

A.33 model.compile@1

the versioned CompileRequest → candidate Module, source map, diagnostics and target admission/capability report

Candidate construction; no solver or posting

A.34 history.construct@1

admitted Module, opening, scope, template registry, query/horizon, resource budget → HistoryFamily, partition, witnesses and unresolved formulas

History-construction contract; witnesses are examples within the family, not its definition

A.35 history.refine@1

frozen family, additional admitted constraint or refinement, dependency view, budget → child family and inclusion relation

Candidate construction; record whether correction replaces constraints instead of intersecting them

A.36 history.query@1

family, typed joint observable, bound/feasibility request, capability manifest → symbolic/partial/exact answer, coverage and endpoint attainability

Pure; unsupported expressions remain explicit

A.37 uncertainty.evaluate@1

exact native uncertainty request, law/attachment/admission and operation-specific event or observation block → native UncertaintyResult and candidate derived bundle

Native admission/attachment/chaining rules; never selects its result automatically

A.38 uncertainty.admit@1

candidate bundle, target time/evidence/purpose, actor, rationale, requested assumption scope, expected selection revision → admitted attachment and decision

Admission transaction; author-stipulated measures keep that label

A.39 uncertainty.select@1

admitted attachment/receipt, rationale and expected selection revision → new selected-view revision

Selection transaction; no probability recomputation

A.40 scenario.project@1

family, exogenous scenario, policy/method lock, horizon/calendar, explicit information policy, resource budget → conditional event/journal/state proposals by history and scenario

Candidate construction; budgets preserve unresolved regions; no historical-book commit

A.41 accounting.assess@1

native assessment request, event, opening, context/time, authority/method, evidence/assumptions and admissions → TreatmentAssessment

Pure accounting assessment; inherited `assess_only` profile cannot post

A.42 accounting.authorize@1

AccountingEventBasis and posting-profile Ref → PostingAuthorization

Pure evaluation under a separately supported posting profile; an assess-only result cannot grant posting permission

A.43 accounting.propose@1

PostingAuthorization, AccountingEventBasis and posting/rounding method → JournalProposal and independent EconomicDelta

Candidate construction; revalidate bindings; unsupported method gives no journal

A.44 accounting.commit@1

AccountingCommitArguments/v1 and envelope expected branch head/generation → StateTransitionReceipt

Single atomic transaction under the event-commit contract; the separate commit schema, separate from native assess-only

A.45 report.build@1

frozen branch/family result, statement-map version, interval/view, typed requested statements and presentation policy → joint statement family, reconciliation and coverage

Pure; derive from retained state/journals, no balancing plug or hidden branch selection

A.46 record.explain@1

output Ref, its frozen view, requested lineage depth → source/method/assumption/decision paths and unresolved dependencies

Pure; depth truncation is explicit; cannot substitute today's view

A.47 work.resume@1

checkpoint Ref, original semantic manifest, scheduling budget → operation-specific resumed result

Resume the original operation; checkpoint ownership/atomicity under the durability contract; no model change hidden as resume

A.48 store.export@1

snapshot/root Refs, format/decoder, destination → closed manifest or explicit incomplete export

Publish a verified export manifest atomically under Section 39.6; no domain mutation

A.49 store.import@1

closed manifest, exact blobs/records, decoder and target namespace → import receipt and original-to-local mappings

Verify before metadata transaction; import alone grants no financial admission

A.50 store.migrate@1

migration request under Section 39.6 → preserved originals, new representation, comparison/mapping and receipt

Transaction only after required semantic comparisons; no changed-model-as-format claim

A.51 Closed accounting records

The following field lists are exact for the named versions. Every reference uses the portable kind/schema/digest type; plural references are schema-declared sets. Unknown fields reject. Source and admission sets can be empty only when the method's admitted assumption scope explicitly permits and explains that.

```
AccountingEventBasis/v1 {
  identity: {entity_id, book_id, branch_key, business_event_id},
  event_ref, opening_ref, context_ref, branch_ref, time_ref,
  chart_ref, authority_refs, method_ref, evidence_refs,
  admission_refs, assumptions_ref, view_ref
}
PostingAuthorization/v1 {
  basis_ref, posting_profile_ref, method_ref, authority_refs,
  status, findings, posting_permitted, evidence_level, view_ref
}
AccountingCommitArguments/v1 {
  basis_ref, authorization_ref, journal_ref, delta_ref
}
```

Identity fields are nonempty ASCII names. Each singular reference is nonnull. Authority references identify applicable support or a specifically typed synthetic authority for an explicitly synthetic example. Authorization status is eligible, unresolved, ineligible or outside scope; posting permission is a Boolean and can be true only for eligible status under a supporting profile. A branch's immutable book mode is synthetic or real, and a profile names its supported modes. Synthetic evidence cannot authorize a real-book posting. Journal and delta both identify the same basis and authorization. Their domain payloads obey the package's pinned journal and state schemas. No earlier record refers forward to the final receipt.

A.52 Method and extension definitions

A versioned financial method supplies the following nonempty definitions:

```
MethodDefinition {
  method_id, version, event_kind,
  state_schema, event_schema, journal_schema, delta_schema,
  output_schema, required_contexts, authority_requirements,
```

```
  preconditions, algorithm, account_roles, report_mappings,
  numerical_capability, examples, unsupported_cases
}
ExtensionPackage {
  package_key, version, record_schemas, operation_signatures,
  supported_contexts, state_requirements, authority_refs,
  method_refs, capability_requirements, migration_refs,
  example_refs
}
```

Each precondition has an identity, typed predicate, required input/admission references and separate false/unknown consequences. Domain state and event fields are specified with the financial methods in the body. Schema names and package members resolve through immutable locks. An empty authority list makes no accounting-authority claim. Registering a schema does not establish the method's entity, instrument, reporting-period or probabilistic applicability. The full machine-oriented contracts and finite exchanges accompany the proposal in the architecture and methods directories; their exact versions are pinned by the manuscript's dependency index.

A.53 What must be known before a method can run

A company model can name a tax method while still lacking the information that method needs. It may know the tax rate but not the relevant tax bases, or have a reported consolidated balance without an allocation to the legal book being simulated. Naming the implementation does not resolve either problem. The method definition therefore needs to say which quantities and findings it requires, how they are related, and what prevents the method from proceeding.

The preceding portable contract already calls for typed preconditions. The `MethodRequirements/1` reference gives a small part of that contract a common executable meaning. A requirement has an identity, a typed predicate, the exact input identities on which it depends, references to required admission findings, and different consequences for a false or unresolved predicate. The same interpreter can assess declarations for several methods. It does not replace their accounting algorithms, source checks or state transitions.

Consider the conditional annual-company example. Its floating-rate share is a fraction of the declared debt principal, so the selected method requires a value between zero and one. It also restricts the coupon interval to one year and the opening facility draw and loan fee to zero. A negative fraction or a share exceeding one contradicts the declared input domain. A nonzero fee instead requires a method that handles fees. Those are different findings: the first is invalid for the stated fraction; the second is outside the selected method's scope. Neither can be repaired by silently changing the supplied number.

A.53.1 A known contradiction survives an unrelated unknown

Let f denote the declared floating-rate share and r the tax-rate input in this finite annual profile. Its selected conditions include $0 \leq f \leq 1$ and $0 \leq r \leq 1$. These bounds belong to the declared profile; this example does not assert a universal model of tax measurement. Suppose the caller gives $f = 2$ and leaves r unknown. The supplied share already contradicts its meaning as a fraction. Uncertainty about tax does not remove that contradiction.

The common assessment evaluates every declared requirement and retains each finding. Its summary gives a known invalid input precedence over unsupported scope, and unsupported scope precedence over unresolved requirements. Successful selected requirements remain visible

as well. This ordering summarizes the results; it does not discard the reasons for an interrupted assessment.

The existing `CompanyOperationsDraft` validator follows a different order in this mixed case: it returns on a globally unknown input before reaching the fraction-domain test. It therefore reports unresolved for the same inputs. The new reference reports invalid while retaining the unresolved tax condition. The original validator is preserved, and this difference is recorded explicitly. The checked agreement concerns selected single-condition comparisons, rather than equivalence of every possible mixed input to the old validator.

Each input is separately marked known or unknown with a reason. An absent binding is unresolved; it is never zero or false. A declaration that refers to an undeclared input is structurally invalid. The literal text *unknown* can still be a known text value because the binding status is separate from its value. Caller findings also have their own role. A true finding records what the caller supplied; the evaluator does not establish that an accounting judgment is correct merely by reading a Boolean.

The Boolean operators use the following finite tables. Here U means an unresolved supplied input or predicate, not a probability and not a set of financial histories.

A	B	$\mathrm{all}(A, B)$	$\mathrm{any}(A, B)$
true	true	true	true
true	false	false	true
true	U	U	true
false	true	false	true
false	false	false	false
false	U	false	U
U	true	U	true
U	false	false	U
U	U	U	U

Negation exchanges true and false and leaves U unresolved. Every operand's dependency is retained even when another operand determines the result. Thus a false conjunction may still report the input missing from its other branch. Repeated references identify the same input; they introduce no independent draws or weights.

Arithmetic and comparisons with an unknown operand remain unknown in this fragment. In particular, an unresolved x leaves both $x = x$ and $x - x$ unresolved here. This conservative rule is deliberately different from symbolic deduction over an explicitly inhabited domain or the coefficient cancellation used in the source-relation calculation. Those procedures have additional premises. The requirement interpreter evaluates supplied inputs and does not decide entailment over admissible company histories.

A.53.2 The same requirement in text and typed form

Types carry explicit context maps. A rational amount can identify its currency, scale, entity, book, measurement date and accounting basis. Numeric operations and comparisons require exactly matching types and contexts. A reported consolidated balance of 100 and a legal-book balance of 100 are consequently not interchangeable merely because their numerals agree. An explicit allocation or conversion belongs to a separately specified method. Exact rational arithmetic also describes the calculation, not the precision or truth of a reported amount.

The first expression fragment contains references, typed constants, equality, ordered comparison, addition, subtraction and the Boolean operators. It has no arbitrary calls, imports, multi-

plication, quantifiers or collection traversal. These restrictions make the selected semantics inspectable while leaving the broader method definition open to later compatible extensions.

Text uses named, semicolon-terminated declarations and prefix expressions. For example, a requirement that a declared share does not exceed one can be written as follows. The context and input roles remain explicit.

```
input "share" rational assumption
  {"unit":"ratio"} "case:declared-share";
require "share.upper" invalid unresolved ["share"] []
  (le (ref "share")
      (const rational {"unit":"ratio"} "1/1"));
bind "share" known "1/2";
```

The display wraps declarations for reading; the reference text grammar places each complete declaration on one line. A complete document also contains the method identity and version, capability and residual-prerequisite declarations, an exact selected manifest digest and an end marker. The lower bound and other conditions must be declared separately; the upper-bound example supplies none implicitly. Rational values use reduced numerator/denominator strings. Boolean values are true or false, not integers or quoted words.

The frontend lowers these declarations to the same records used by the typed library and calls the same evaluator. Strict exchange decoding rejects duplicate JSON keys, including nested context keys. The text parser rejects duplicate declarations, unknown operators, malformed types, missing terminators and trailing content. Quoted delimiters and Unicode remain part of the exact supplied text values, without normalization. These checks establish a bounded authoring correspondence; they do not establish that users prefer this syntax.

A.53.3 Content identity leaves authority with the caller

The caller selects a manifest by its canonical SHA256 digest. Changing a predicate, context, source role or residual prerequisite changes that identity. A stale selection is rejected, and an unavailable definition or required capability is reported separately. The manifest loads no code. Transporting an availability list does not establish that an implementation is installed, and a content hash does not authenticate the sender. Registry ownership and capability availability remain responsibilities of the trusted caller.

A successful result is called `requirements_satisfied` within the scope `selected_requirements_only`. It retains the unassessed prerequisites and explicitly leaves method eligibility, execution, source admission and accounting admission false. For the checked company bridges, those remaining conditions include source binding, account and state controls, event order, unselected allocation equations and the actual method transition. A nonempty source reference or an empty admission-reference list cannot settle them.

The common interpreter has 35 checked selected-condition comparisons with the unchanged company-operation and return-module functions, complemented by independent truth tables, exact arithmetic and adversarial type/context cases. The separate frontend has exact typed/text and result comparisons across five examples, including constructed legal-book, reported-measurement and insurance-cohort contexts. The latter demonstrate language behavior, not new insurance or source-admission methods. The mixed unknown/contradiction case above remains a documented difference.

These references make prerequisites reusable and inspectable across the tested profiles. General method applicability, account-state conversion, source-to-latent reconstruction and independent acceptance still require their own evidence. The following section extends this reference to exact

record binding, dimensional arithmetic and guarded applicability. Complete method boundaries still require more than a partial predicate list before execution can proceed.

A.54 Binding a return obligation to its applicable equations

A return obligation connects several different quantities: the number of goods expected to be returned, the refund per unit, the carrying amount recoverable from each return, and the balances already recorded. Checking their relationship requires more than scalar bounds. The quantities must come from the intended record, their units must combine correctly, and the equation must apply to the state being examined. The successor requirement references make those three steps explicit while preserving the first reference's supplied-input semantics.

Consider the retained conditional return example for legal entity P and its 2025 annual period. In its unsettled opening, expected returns are two units, the refund is 20 dollars per unit, and the cost is 12 dollars per unit. Estimated handling and deterioration reduce recovery by three and one dollars per unit, respectively. These are supplied assumptions in a finite example. Its selected opening equations are

$$L_{\text{refund}} = qp = 2 \times 20 = 40, \tag{A.44}$$

$$A_{\text{recovery}} = q(c - h - d) = 2(12 - 3 - 1) = 16. \tag{A.45}$$

Here q is a quantity of goods; p, c, h, d are dollars per unit; and both balances are dollar amounts. The example's separate zero-tax-base conditions and supplied tax rate of one quarter give the selected deferred-tax amounts 10 and 4. Those premises belong to the retained conditional method. The equations alone establish neither the premises nor the broader recognition and measurement treatment.

A.54.1 Select the record before checking its amounts

An input such as the refund balance must identify the exact record and field from which it is read. Otherwise an equation could pass using a revised record, another legal entity, or a consolidated amount that happens to share the same number. The binder requires the caller to supply a plan, the selected method definition, a selected-field schema, and the raw record bytes. It resolves content identities from those supplied registries; it does not fetch files or follow executable references.

The plan names the definition's canonical digest. Each named record has a raw SHA256 digest and a schema digest, and each binding names an input, record and JSON pointer. The input's source reference must equal `sha256:RAW_DIGEST#POINTER`. The schema declares the field's complete type, role, encoding and missing-path policy. Type, context and role must agree exactly with the method input. All declared inputs must be mapped once, and every named record must be used. Missing registry entries are unsupported; stale hashes, duplicate mappings and undeclared field selections are invalid.

The raw decoder rejects duplicate keys, floating-point numerals, nonfinite values, malformed UTF-8 and non-ASCII object keys. Unicode string values remain available. Pointer escapes are checked before traversal, including when an earlier part of the path is absent. A malformed pointer is an invalid request, whereas an absent well-formed path follows the declared missing policy. Traversing a scalar as though it were a collection is invalid. These distinctions prevent a bad selector from being treated as merely unavailable financial information.

The three encoding choices also carry substantive meaning. A scalar is read as a known value of its declared type. The explicitly selected legacy convention treats the string *unknown* as

an unresolved value. A tagged binding supplies its own known value or unknown reason. The missing-path policy is either invalid or unresolved; unresolved paths receive the reason `missing_declared_path`. None of these conventions supplies zero. A text field using the scalar convention can contain the known word *unknown*.

A successful binding returns the values and their record, pointer, schema, type, role and source identity. The selected requirements are then assessed on those values. Binding can therefore succeed while a method condition is invalid or unresolved. Exact identity makes the calculation reproducible; the caller still has to justify the schema's economic meaning and the source's authority. The conformance examples that serialize supplied values into canonical records identify those records as derived projections, rather than original filing bytes.

A.54.2 Multiply quantities without erasing their units

The dimensional successor, `MethodRequirements/2`, adds multiplication and separates dimensions and scale from the context map. A rational type contains its kind, context, dimension exponents and a positive rational scale. The context identifies such matters as entity, legal book, component, period and basis. A dimension name denotes a caller-declared quantity such as dollars or return units. Zero exponents are omitted; exponents are bounded integers.

For two rational inputs in exactly the same context, let a, b be their rational coefficients, s_a, s_b their scales and $d_{a,k}, d_{b,k}$ the exponents of dimension k. Multiplication produces

$$(ab,\ s_a s_b,\ \{k \mapsto d_{a,k} + d_{b,k}\}).$$

Thus a return-unit quantity multiplied by dollars per return unit gives dollars: the unit exponent cancels and the dollar exponent remains one. This rule defines the local type calculation. It uses exact rational arithmetic and validates the resulting type; it performs no inferred currency or scale conversion.

Addition, subtraction and comparison require exact agreement of context, dimensions and scale. A dollar amount at scale one cannot be added directly to a dollar amount at scale one thousand. Two different currencies do not match, and a legal-book amount cannot be exchanged for a consolidated observation. A conversion or allocation needs a separately specified relation. An older v1 context string that happens to contain a currency name is never promoted automatically to this dimensional type.

The implementation lowers products to internal typed inputs, uses the preserved scalar evaluator for the remaining operations, and restores the original dependencies and unknown reasons in public findings. Internal product identities and the reserved context keys used to encode dimensions are unavailable to caller declarations in the dimensional versions. If a factor is unknown, the product remains unknown even when another factor is zero. This follows the declared conservative evaluation rule; it is not a symbolic proof about all admissible histories.

A.54.3 An inactive equation does not settle an unknown fact

The unsettled return equation is inappropriate once the example's obligation has been settled. `MethodRequirements/3` therefore adds a Boolean guard and its exact dependency list to each requirement. The body predicate retains its own dependency list. For the unsettled allocation, the guard is the negation of the supplied settlement indicator; a separate settled requirement checks the corresponding zero balance. The caller must author these conditions explicitly.

Guard evaluation has three distinct consequences. A true guard activates the body's ordinary true, false or unknown result. A false guard marks the requirement `not_applicable`, even when

the body is false or unknown. An unknown guard leaves the requirement unresolved, even when its body happens to be true. The result retains guard truth, body truth, activation and all original dependencies. The interpreter checks structure and types in every branch before assessment; an inactive branch cannot hide an unsupported operator or malformed value.

Some facts remain necessary independently of whether a particular equation is active. A separate `required_known` list declares those inputs. If the refund balance is required but unknown, making the unsettled equation inactive does not make the balance known. The result remains unresolved on that separate condition. When all requirements are inactive and no other condition fails, the summary is `no_active_requirements`. It does not assert that an accounting method has been satisfied or can run.

For the zero-tax-base example, an equation's false guard also must not hide a method limitation. Its zero-base scope checks are independently active: a known nonzero base remains unsupported by the selected method, even though the corresponding zero-base equation is inactive. This is why applicability, mandatory knowledge and supported scope have separate declarations. Their adequacy remains a question for the method's designer, supported by its source and state evidence.

A.54.4 Keep versions explicit through authoring and exchange

The new `MethodRequirementExchange/2` selects exactly one of the three requirement profiles. It carries the full definition, selected digest, values and available-capability names. The new text header names both the exchange and requirement versions; input and constant declarations carry complete type objects. The first two requirement versions use `require`; v3 uses `guarded`, including separate guard and body dependencies and expressions. Every v3 text document must contain exactly one `required_known` list, including an explicit empty list when intended. Omitting it is invalid.

The following fragment shows the guarded return relation after its quantity, price, balance and settlement inputs have been declared. As in the preceding reference, the display wraps one declaration for readability; the executable text places each complete declaration on one line.

```
guarded "refund_allocation" invalid unresolved
  ["settled"] ["expected","refund","refund_per_unit"]
  ["retained-unsettled-zero-tax-base-premises"]
  (not (ref "settled"))
  (eq (ref "refund")
      (mul (ref "expected") (ref "refund_per_unit")));
```

The complete document supplies the remaining declarations and its exact content digest. Changing the guard, a type, a required-known list or a source reference changes the selected definition. An unknown profile or incompatible declaration is rejected. Profile names select a closed set of preserved evaluators; they cannot name a caller-provided module or callback.

The corresponding `MethodInputBinding/2` requires a `RequirementInputSchema/2` that explicitly names the expected requirement profile. It retains the original full types and original plan, method, schema and raw-record identities. Internally, a separately identified v1 representation reuses the protected byte and pointer validation. This representation checks binding only and is never assessed as the method. The original selected evaluator receives the bound values, preserving dimensions, guards and unresolved reasons.

The recorded comparisons cover 32 cases, including 29 retained examples, and 59 combined unit-test methods. Full selected assessments agree across typed definitions, text, exchange and binding; independently authored syntax and adversarial changes supplement the roundtrips.

These are deterministic checks of the stated language fragment. Collection membership, identity order and selected shared-book conditions are made explicit in the following section. State effects, replay and reporting still require complete method semantics. Selected requirements can explain why an input is usable or why work must wait; they do not authorize those transitions.

A.54.5 A passing requirement still needs the intended native request

Consider a regulatory assessment for the synthetic bank's solo legal book. The supplied assessment date is 31 March 2026, the period starts on 1 January, and its exclusive endpoint is 1 April. The accounting state is also dated 31 March. Selected requirements can verify that the period is nonempty, that the assessment lies within it, and that the state and assessment dates agree. These statements remain true if the request names an obsolete accounting head or gives the operation a zero execution budget. The native request must still be refused in those cases.

An input binding therefore needs an explicit connection to the operation being requested. The proposed `NativeRequirementLink/1` supplies a finite example for `bank.regulatory.assess@1`. It names the exact request, submission, profile, operation signature, assessment basis and context. It also binds the requirement manifest and input-plan digests. The selected assessment and period dates come from the basis record; the state date comes from the accounting state referenced by that basis. Each selected scalar keeps its pointer, state role and full context, including bank, book, perimeter, currency, scale, accounting basis and jurisdiction. A matching currency alone cannot justify substituting another book or regulatory perimeter.

Two different content identities are involved. The native record digest includes its schema identifier, whereas the binder's raw digest identifies the supplied serialized bytes. The submission contains the complete canonical request record. The basis and state inputs are explicitly serialized from their retained native records; they are not original filing bytes. The link checks both kinds of identity and their correspondence. Replacing a signature, selecting a different record, weakening the selected definition or retaining a stale digest causes refusal rather than an implicit conversion.

The existing binder and original version-selected evaluator still perform the input and predicate work. The link adds no predicate language or dispatcher. Its success means that these particular selected requirements are tied to this particular native request. The native method must separately establish its last-day condition, permitted scenario window, evidence availability and ingestion times, source applicability, regulatory eligibility, current view and accounting head, environment and resource budget. Request identity and replay also remain part of the native operation. None of these duties disappears when the selected date requirements succeed.

The retained comparison makes the distinction observable. Across 271 regulatory exchanges, the native results include ten completed assessments, 75 replays, 50 unknown acknowledgements, 96 refusals and 40 partial assessments. Selected date requirements pass in 95 of the 96 refused exchanges and in every partial and unknown-acknowledgement case. An unknown acknowledgement leaves publication unresolved; it cannot be interpreted as a failed assessment with no effects. The original native response checks preserve those meanings. The three requirement versions produce 813 matching text and typed assessments, with independent date comparisons and malformed-input checks.

Missing evidence retains its own consequence. If a declared date is removed from an explicitly derived input record, the binder reports that date unknown. The derived record cannot then be represented as the unchanged native record. An unavailable selected capability remains unsupported, and a known date contradiction survives an unrelated unknown. Execution, source admission, accounting admission and full method eligibility remain false in these link results.

This example supplies a checked connection for one retained operation. The portable corporate purchase definition, whose preconditions are named strings, does not thereby become a definition of the bank assessment. General method authoring still requires explicit correspondences for each supported method, including its state and effect semantics. The checked source and result records are retained under `data/accounting-master/common-method-input-boundary-20261026/`.

A.54.6 Information cutoffs require time within the day

The assessment's civil date cannot determine whether a source was available at the decision time. Evidence arriving at 00:00:01 on 1 April is later than a cutoff at 00:00:00 on that date, although the two civil dates agree. The retained bank assessment checks the full timestamp. The common requirement versions, however, support ordered dates and rational numbers but have no timestamp type. Passing a date comparison therefore cannot discharge that native timing duty.

Precision matters even when a timestamp parser is available. The retained native schema accepts both `2026-04-01T00:00:00.0000001Z` and `2026-04-01T00:00:00.0000002Z`. These instants differ by one tenth of a microsecond, but its datetime-based conversion maps both to the same value. That conversion gives the wrong ordering for a claim of exact timestamp comparison. The retained bank examples themselves use whole-second timestamps; their checked comparisons are unaffected. The defect concerns extending the same conversion to all precision accepted by its schema.

The separate `ExactInstantProjection/1` addresses this boundary without changing the original requirement evaluators. First, the existing binder selects the timestamp strings from exact supplied record bytes. The projection then converts each known value to rational seconds from midnight UTC on 1 January of year 1, under an explicit proleptic Gregorian convention without leap seconds. It counts whole days and seconds with integers, adds the exact decimal fraction, and subtracts the supplied signed offset. No floating-point timestamp or microsecond truncation enters this calculation. The resulting coordinate serves ordering within the declared convention; physical elapsed time across leap seconds is a separate question.

For example, 01:00:00 with offset `+01:00` and 00:00:00 with suffix `Z` have equal coordinates. Their original strings and source identities remain distinct. The two fractional instants above retain their different rational coordinates, so the later one fails an available-by comparison against the earlier cutoff. The existing dimensional requirement evaluator performs that comparison. Each value has a seconds dimension and the same explicit time-axis identity; the complete original context is preserved separately from the time unit. A financial currency or scale in that context cannot become a unit of time.

The accepted syntax requires a full date, hours, minutes and seconds, followed by `Z` or an explicit signed hour-and-minute offset. An optional fraction has at most 18 decimal digits. That precision bound is a declared reference limit: more digits are refused, never rounded. Leap seconds, date-only values, missing offsets, the unknown-offset spelling `-00:00`, named timezones and UTC normalization outside years 1–9999 are unsupported. Invalid dates and out-of-range clock or offset fields are invalid. A date-only disclosure needs an explicit interval or availability policy before it could support an instant comparison.

Unknown bindings keep their original reasons. Conversion preserves the selected role, pointer, schema and raw-record identity together with the original string; it cannot upgrade a supplied timestamp into an authenticated fact. Source-view membership, the truth of availability and ingestion times, measurement dates and the current view still require separate evidence. A capability to compare times grants none of those admissions.

Independent integer and rational calculations cover 180 date/fraction points. Six explicit offset and precision examples and 16 retained native timing bindings agree with their stated comparators, and the associated text and typed requirements agree. Malformed encodings, incompatible contexts and changed bindings are refused. This optional conversion is adopted by the explicit bank successor in Section A.54.8; other profiles retain their own declared semantics. The counterexample and exact conversion evidence are retained under `data/accounting-master/exact-information-time-20261026/`.

A.54.7 Exact timing must retain the whole evidence dependency

Replacing timestamp arithmetic is useful only if the assessment still checks every source on which its result depends. A current value may depend on an earlier vintage whose evidence arrived too late. Checking only the current record would admit that dependency even when its own timestamp comparison is exact. The separately named `ExactBankInformation/1` therefore changes only the retained auditor's information check. It walks the same complete dependency closure and retains the original evidence-membership, ingestion and measurement-date obligations.

For each evidence record in that closure, the checker requires its exact reference to appear in the selected information view. Its available-at instant must be no later than the decision cutoff, and its ingested-at instant no later than the system cutoff. Its measurement date must be no later than the UTC civil day containing the decision cutoff. That day is obtained from the exact coordinate, including offset normalization. Unsupported or invalid timestamp encodings are refused explicitly; there is no fallback to the lossy parser.

All 271 retained regulatory responses agree under the original and exact auditors. Eleven additional information-check cases exercise later evidence, ingestion, membership, measurement dates, equivalent offsets and precision. Five outcomes intentionally change: later submicrosecond availability, later submicrosecond ingestion and later superseded evidence are refused, as are excess precision and the unknown-offset spelling `-00:00`. The superseded case uses a native vintage record, which actually declares its predecessor.

These eleven examples exercise the information submethod; they are not eleven new complete regulatory exchanges. A subsequent comparison checks ten complete derived regulatory exchanges and four exact replays, including five intended timing refusals. Its twelve adverse controls check the remaining request and publication obligations. The original profile, packet and auditor remain preserved. The separately identified profile below specifies conversion and replay without changing those originals. Correct timing also leaves source truth, signatures, state, budgets and accounting applicability to their existing checks. The complete comparisons and derived examples are retained under `data/accounting-master/exact-native-timing-20261026/`.

A.54.8 Changing the timing rule creates a new bank profile

A saved assessment records which rules were applied to which information. Changing the time comparator must therefore leave the old assessment interpretable. Consider evidence available one ten-millionth of a second after the decision cutoff. The original parser can erase that difference and accept the request; the exact comparator refuses it. Returning an old cached answer under the new interpretation would conceal the change, even if the arithmetic in the assessment were otherwise identical.

The documentary successor `BankAssessmentExactTime/1` gives that change an explicit owner. It supports the same three operations: regulatory assessment, conditional-loss query and policy description. Every evidence dependency must meet the exact availability and ingestion comparisons described above. Measurement, assessment and vintage/origin dates use the UTC

civil day of the exact decision instant. The parser retains the stated Gregorian calendar, known offset and eighteen-digit precision limits; unsupported precision is refused. The capital and loss methods retain their financial calculations. Only their two decision-day conversions change.

The new profile binds the exact schema and semantic-description bytes. Eight record kinds receive new schema identities because their typed references depend on the profile: the profile itself, operation signature, request, result, transport observation, query receipt, coverage domain and coverage region. Unchanged financial records retain their original schemas and, when their contents are unchanged, their original identities. A conversion creates a new request in a distinct workspace and new exact submission bytes. It publishes a new query receipt while preserving the old request, receipt and query-index entry. Replacing a supplied evidence timestamp additionally changes the identities of the records that depend on it.

Profile, schema, semantic-description and signature checks precede replay. An old request consequently cannot acquire the successor's meaning merely because it has a cached result. Within the successor, submitting the same request bytes returns the same output, receipt, statuses and coverage, even when the current view or accounting head has since advanced. Different bytes under an already used request identity are refused. A new request still needs the declared view, accounting head, execution environment and budget. A refusal has no result, receipt or coverage and changes no world state. A successful or partial query changes the immutable query records and index; it posts no journal and changes no accounting balance or owner head.

The finite conversion is checked against all 33 retained positive publications: 25 regulatory assessments, seven conditional-loss queries and one policy description. Expected outputs come from the saved publications with declared reference substitutions. The comparisons preserve entire outputs, including unknown amounts, undefined ratios, diagnostic codes, unresolved obligations and coverage. They do not compare only a selected ratio. Each converted publication coexists with its original query and receipt.

Thirty complete timing cases then exercise all three operations. Twenty-one outcomes intentionally change relative to a comparator that retains the old time methods: later fractional availability or ingestion, excessive precision in evidence or cutoff times, unknown offsets and eighteenth-digit ordering are refused. Equal times and equivalent offsets remain accepted. Forty-two exact replays and 28 additional controls check identity conflicts, stale locks, signatures, decoder descriptions, heads, views and publication effects. All 63 saved conversion packages reconstruct and pass their full response checks.

These examples establish a bounded conversion between two documentary profiles. They share the retained financial methods and synthetic inputs; they do not independently validate those methods against current banking rules or authenticate a real disclosure's publication time. Concurrent service execution, hardware-loss recovery, general profile migration and empirical forecast accuracy require separate evidence. The original profiles and examples remain preserved. The full-request comparisons are stored under `data/accounting-master/exact-bank-request-conformance-20261026/`; the successor contract, converted exchanges and checks are under `data/accounting-master/exact-bank-profile-adoption-20261026/`.

Keep accounting profiles and capital assessments distinct. The timing successor can be reproduced beside the other bank methods, but their common transport does not make them one accounting book. `BankComponents/1` represents a conditional historical Australian book with funded loans, credit memory, deposits, commitments, fees and settlement liquidity. Capacity and a funding request create no cash; confirmed receipts and withdrawals do. Its complete receipt chains support annual and lifetime reports. The separate `BankComponents/2` covers transfers, options and asset classification, including retained exposure, associated liabilities, interest,

impairment and OCI. It explicitly refuses the first profile as a foreign input. A combined book therefore needs a specified migration or composition preserving both sets of accounts, legal rights and credit memory.

`BankAssessment/1` instead contains the separate synthetic US book in Section 37.2.1. Its dated events reconstruct the opening and closing accounts before the prudential query uses them. The case's equity 23.6, goodwill deduction 2, admitted allowance 1.5 and risk-weighted assets 119.1 are different quantities. A deposit demand of 40 leaves 9 unpaid even though the supplied capital ratios are unchanged. A missing period-average asset input leaves leverage unresolved. These outcomes connect accounting, funding and prudential consequences within that profile; importing an Australian book by renaming its accounts would not establish the required US measurement or applicability findings.

The economic loss query likewise does not estimate the supplied CECL allowance. It preserves its declared joint cells and candidate laws; policy description selects no action and grants no distribution permission. The exact-time successor changes information comparisons and request identities while keeping those financial definitions. A successful or partial query publishes records and a receipt, without posting journals or changing the accounting head.

The internal companion at `companions/bank-interfaces/` reproduces the three retained packets and all saved exact-time cases from a separate declared input copy. It includes full reports, queries, refused requests, retries and lost-acknowledgement cases, with the original independent financial checks. The time successor shares those financial methods; its expected outputs are the retained publications with explicit reference substitution. Historical Git/index preservation checks remain earlier workspace evidence and are not recipient reproduction checks. The companion instead verifies every declared copied input before and after execution, with that difference stated explicitly.

The compatibility map and exact contracts identify which connections exist and which still require an adapter. Neither shared transport nor successful replay supplies broader bank composition, actual rule applicability, calibrated loss estimates or independent domain acceptance. The original consolidation record is under `data/accounting-master/bank-interface-consolidation-20261027/`.

A.54.9 Writing the complete method behind an event

A command that selects a registered payment method does not yet explain how an author writes that method. The payment needs a condition on available cash, a decrease in the cash account, a decrease in outstanding principal, a financing cash-flow explanation and a check that the resulting liability still agrees with the loan record. Leaving any of those parts behind a method name would leave a material part of the financial algorithm outside the authored text.

`CompanyMethodAuthoring/1` writes the complete retained `CompanyBookMethod/1` definition. Its header identifies the method and selects `CompanyBookTransition/1` or `/2`. The remaining declarations supply context and account mappings, guarded preconditions with their input bindings, monetary calculations and effects, ordered memory changes, guarded postconditions and residual prerequisites. The original validators still determine the meaning of each part. The text adds a way to author their inputs; it does not introduce another accounting evaluator.

The transition version is consequential. Version 1 supports replacement of existing scalar memory fields, such as outstanding principal. Version 2 also checks typed insertion, allowing a credit sale to create an identified invoice claim. Its compilation environment must therefore include the memory type definition. Both versions require an explicitly supplied book, event and schema set. Writing a method does not reconstruct those inputs from a company name or

from its disclosures.

Each declaration occupies one line and ends with a semicolon. Precondition and postcondition sections contain the existing requirement-exchange text, with empty event values and capabilities. Their selected content digests remain checked. The monetary section declares inputs, calculations, account roles, postings and gross flows. Expressions use references, constants, addition, subtraction, multiplication and minimum. Types and dependencies are explicit; the native monetary validator refuses an undeclared or forward reference, an incomplete dependency declaration or an incompatible context. The memory section names its schemas, binds its inputs and declares each ordered write. Binding records, types and memory templates use strict JSON data tokens.

For example, the payment's outstanding-principal calculation reads the existing principal and subtracts the supplied payment amount. A memory input then names that calculation as the value to write back. The liability posting uses the same payment amount with the book's signed-account convention, and the cash posting and financing flow both record its negative. A postcondition checks that liability plus outstanding principal is zero. The text therefore exposes the relationship between the accounting effect and the continuing contract state, rather than merely printing a balanced journal.

The checked full payment starts with cash 100 and liability -30, pays $7/2$, and finishes with cash $193/2$ and liability $-53/2$. A separately authored variant changes every monetary use of the payment to one half of that amount. Its cash becomes $393/4$, liability $-113/4$, outstanding principal $113/4$ and financing cash flow $-7/4$. The dimensionless multiplier retains the payment's context, as required by the existing product rule. This constructed variant tests whether text changes the algorithm itself; it does not propose a rule for how much an actual borrower should pay.

Four complete payment and claim-creating sale definitions, using two company charts, match their typed definitions, requests and full native results. The sales increase receivables by 30, reduce inventory by 10 and insert the identified claim with remaining amount 30; they create no cash flow. The comparisons include balances, memory, journals and gross flow records. Twenty syntax or environment refusal cases and twelve economic-outcome cases retain the expected distinctions: an unknown amount is unresolved, a negative amount is invalid, and false applicability is unsupported. A forged proposed after-state is refused by the existing application step.

Compilation itself executes nothing. Successful syntax lowering is distinct from compilation against an environment, and neither decision establishes source or accounting admission. Exact UTF-8 source slices retain their own hashes; whitespace and reordering of mapping declarations leave the semantic identity unchanged. Ordered calculations and writes keep their order. The bounded reference limits remain explicit, including one-megabyte text and 4096-node, 48-depth checks. The complete grammar, authored examples and results are retained under `data/accounting-master/complete-method-authoring-20261026/`.

This supplies complete authoring for the retained method definition. Broader method algorithms, registry loading, company-state reconstruction and independent implementation remain separate work. In particular, agreement with a shared native engine is evidence of consistent compilation, not independent proof of every accounting treatment implemented by that engine.

A.54.10 From an authored method to a sequence of company statements

A credit sale creates a claim; its collection later needs that claim to exist. The complete method text does not by itself determine when a method may be called or which method version a

command selects. The retained registry version 2 still routes its calls through transition version 1, which supports fixed scalar replacement. Its version number therefore cannot be used to infer support for claim insertion.

The separate `AuthoredMethodRegistry/1` records the transition version explicitly for each entry, along with the exact authored source, its hash and the complete method definition and hash. Lowering that source must reproduce the same method and version. Event-input types, account classes, metadata bindings and the direct-event-date clock remain explicit. A transition-v2 entry also supplies its memory type definition. The registry validates every entry against the current book, including entries not selected for the call; it then dispatches the selected event through the original transition engine.

This state dependence matters when composing commands. A collection method that reads an identified claim cannot be treated as usable before the sale creates that claim. `AuthoredCompanyProgram/1` therefore supplies a finite population of explicitly hashed registries and lets each event name the one it uses. The program first checks its complete command structure and source identities. At each event, the selected registry is validated against the book that actually exists at that point. Earlier commands never receive the benefit of a future state merely to make validation succeed.

Consider the constructed lifecycle used in the preceding claim examples. The sale increases receivables by 30 and creates the claim. A first receipt of 10 raises cash from 100 to 110 and leaves receivables and the claim at 20. Due-date assessments change the declared claim observations without creating cash. The annual report and period carry preserve that remaining claim while moving the year's profit into retained earnings under the supplied closing method. A receipt of 20 in the next year brings cash to 130 and clears both receivables and the claim. Next-period operating cash flow is 20. Two company charts reproduce this eleven-command sequence, including complete journals, memory, reports and the period-carry result.

Report and carry commands call the existing native methods with explicit cutoffs, mappings, equity movements, notes and per-leaf memory policies. The complete requests and results agree with the retained lifecycle, not only its cash totals. Method-input schemas remain distinct resources from the schemas supplied to those report requests. Every successful step also preserves the typed memory definition. Observing a book at a cutoff does not establish that all economically relevant events have been supplied.

The comparison has an important representational qualification. The original lifecycle builder specializes claim identity, customer and due date into its generated methods, while retaining redundant event fields. The registry requires exactly the declared inputs, so the derived comparison records every removed undeclared field and the changed request identity. Complete states, journals and flows still match the original lifecycle. Exact event-request comparison is against direct execution of the explicitly derived request. These methods do not yet supply arbitrary dynamic claim selection.

Failures preserve what happened earlier. If the first sale succeeds and the next collection amount is unknown, the result retains the sale, its claim and the unexecuted command suffix, with the collection unresolved. Negative amounts, false applicability, repeated identities, early reporting cutoffs, incomplete carry policies and invalid unselected entries retain their distinct refusals. Seven such cases preserve the exact successful prefix. Forged complete results fail recomputation. A separate authored edit halves the collection algorithm's payment: after the sale and that receipt, cash is 105 and receivables and the claim are 25. This intervention tests execution of the supplied algorithm.

The checked registry and program remain separate reference successors. They preserve the original engines and their provisional status. The program's 64-command and eight-

mebibyte trace limits are reference resource bounds, not production capacity claims. Full textual authoring of registry resources, general method specialization, broader financial methods and actual-company source admission remain open. The comparisons are retained under `data/accounting-master/authored-method-registry-20261026/` and `data/accounting-master/authored-program-composition-20261026/`.

A.54.11 A typed claim can still disagree with the book

The preceding successful histories do not establish that every authored method preserves the original lifecycle's economic controls. Suppose an otherwise unchanged sale posts receivables of 30 but creates a claim whose remaining amount is 31. Both values have the right monetary type, the journal balances, and the memory record satisfies its schema. The original lifecycle nevertheless rejects that state because the claims do not sum to receivables. Typed-book validation alone does not express this cross-record relation.

A focused audit changes one claim field at a time in the authored sale and then requests a report from the genuine period opening. Across two company charts, sixteen cases expose eight outer-control gaps: claim totals, the sale event's membership in the book, issue/due chronology, future issue dates, negative assessment amounts, nonzero unassessed placeholders, future assessment and an incorrect overdue snapshot. An empty customer string is already refused by the memory schema. That covered structural rule must not be counted as a missing economic invariant.

`BookStateInvariant/1` supplies the missing connection to the existing collection evaluators. Its explicit bindings select the book's date and receivable account, the identified rows of its claim map and its seen-event identities. Each scalar's type and context must agree with the declared input, including for an empty map. Values come from that exact supplied book; a caller cannot substitute a Boolean assertion that its controls passed.

The unchanged collection evaluator checks unique claim identities, references from each claim to its sale event and guarded conditions on every row. The unchanged collection-total evaluator sums remaining amounts and compares that total with receivables. An assessed claim must have an assessment date between issue and the current date. Its overdue snapshot equals the assessed amount when assessment occurs strictly after the due date, and zero otherwise. An unassessed claim retains zero assessment and overdue placeholders. Remaining, assessed and overdue amounts are nonnegative. These are declared controls of this constructed lifecycle, not a new impairment model.

The separate `GuardedAuthoredCompanyProgram/1` selects the invariant by its exact hash. It checks the opening, each current book and each proposed after-state. Only a proposed state that satisfies the selected controls enters the successful prefix. In the 31-versus-30 example, the proposed sale is refused, the opening book remains current, and the sale and report remain in the unexecuted suffix. A failed post-state check cannot quietly retain the invalid claim or its otherwise balanced journal.

Both eleven-command histories retain their complete original results, and all 24 opening and after-state books pass the original lifecycle control. All sixteen changed-method counterexamples are refused. Wrong pointers, contexts and field types also refuse. The original evaluators retain unresolved results for partial or unknown collection inputs, while an unrelated unknown does not erase a known contradiction. Unsupported capabilities and explicitly unsupported consequences remain distinct from invalid states. Neither a successful invariant nor a checked schema establishes source or accounting admission.

Completeness here means all members of the supplied internal map. It does not mean that

an actual issuer disclosed every claim. Event-specific duties also remain: selecting the right customer, preventing an inappropriate new event and respecting sealed assessment windows require their own prerequisites. The state-control repair therefore preserves the finite comparisons without claiming whole-lifecycle substitutability. Counterexamples and repair evidence are retained under `data/accounting-master/authored-program-invariant-audit-20261026/` and `data/accounting-master/authored-state-invariants-20261026/`.

A.54.12 Keep the event's intent when a method is specialized

A consistent closing book still cannot show whether a collection was permitted at the time it occurred. In the constructed lifecycle, an assessment on 31 January closes a window and advances its cursor to 1 February. A collection dated 31 January can satisfy ordinary nondecreasing book dates and all claim state invariants, yet violate that sealed-window policy. The retained lifecycle refuses it. A separately selected event contract is needed to preserve that meaning when the method is authored elsewhere.

Customer and interval assertions expose another boundary. The original full event states which claim and customer a receipt concerns, or which start and end dates an assessment covers. A specialized method may embed those values, so they no longer appear among its declared calculation inputs. Removing them from the projected event is acceptable only if the original intent remains available wherever eligibility and specialization must be checked. A method that happens to produce a consistent state cannot supply the missing assertion about which customer the caller meant.

`ContractedAuthoredCompanyProgram/1` therefore retains every complete original event beside the projected event. Its caller explicitly selects the retained receivable lifecycle contract, identified by the native implementation hash, roles, unit-cost pointer and policy. Before an event, that named owner checks the full input population, unknowns and applicability on the actual current book, then derives the intended method under its existing claim, customer and interval rules. The complete authored materialized request must equal that intended request after the declared input projection and permitted scalar-collection transition conversion. Both requests and the removed input values remain inspectable.

This contract is deliberately specific. The half-payment method previously authored is a valid alternative algorithm within its stated scope. It does not satisfy a caller's selection of the original full-payment lifecycle, so that selection refuses it. An explicit policy choice resolves the difference; a balanced journal or a valid type cannot choose between the two algorithms. The native specialization owner is named, not hidden behind an arbitrary callback or presented as a general declarative financial compiler.

Both eleven-command examples preserve complete states, statements, period carry and archives, and every full-intent native event request matches the retained lifecycle. Eighteen refusal comparisons across the two charts cover customer mismatch, missing or repeated claims, unknown inputs, false applicability, inconsistent assessment intervals, an early due date and the sealed-window event. Earlier successful commands remain unchanged when a later event refuses. Changed projected values and a method inconsistent with the selected intent are also refused; forged intent evidence fails recomputation.

These comparisons extend the checked refusal scope without proving every lifecycle rule or supplying actual company facts. General portable authoring of event contracts, wider financial operations and source/accounting admission remain unresolved.

The source boundary and counterexamples are retained under `data/accounting-master/authored-event-prerequisite-audit-20261026/`. The selected-contract evidence is retained

under `data/accounting-master/authored-event-contract-20261026/`.

A.54.13 Constructing a method from typed intent and ordered claim rows

A sale method must insert the claim the caller actually names. The same financial calculation can apply to a new customer, due date or invoice key, yet the inserted record and the later collection pointer must change. Writing one finished method for each saved event would merely move the native generator's results into a catalogue. Explicit parameters instead retain the complete method algorithm and identify exactly which values come from intent or current state.

`ParameterizedCompanyMethod/1` supplies that bounded construction step. Its template consists of literal, object, array and named-parameter nodes. Each parameter has a declared scalar type and source: a known full-intent input, event identifier or date, selected current-book metadata, or a pointer assembled from literal segments and earlier text parameters. A claim key containing a slash or tilde is escaped as a pointer segment; it is never reparsed as source. Unknown required input remains unresolved. Duplicate, unused, forward or unbound parameters are refused.

For a sale, parameters supply the claim key, customer, due date, event identifier and dates in the inserted row. For a collection, they supply the selected claim key and its remaining-balance pointer in both the precondition and the monetary input binding. The subtraction, postings and memory write remain visible in the full method. The input schema is compiled from its explicit typed definition, while the state schema comes from the actual book. Complete template, state and event identities bind the constructed request. The existing typed frontend and method-source renderer then check the whole method.

Four templates across the two retained charts reproduce six prior sale and collection requests and twelve new events with changed keys, customers, dates and amounts. Comparisons cover complete requests and transition results, and the emitted method source compiles back exactly. A deliberate edit halves the collection algorithm's cash, receivable, flow and remaining-balance effects. It produces exactly half the cash movement and preserves the claim balance relation, while differing from the explicitly selected full-payment contract. The construction mechanism therefore exposes a consequential algorithm rather than hiding a table of fixed financial outcomes.

Assessment adds a different need: the number of writes depends on the current claim population. `OrderedMethodExpansion/1` declares bounded row expressions and an ordered list of claim keys. In the retained state-only assessment, each row sets the assessment flag and date, copies remaining amount into the snapshot, and records overdue amount only when the due date is strictly earlier than the assessment date. A final write advances the cursor. Typed expressions can read a row field, key, event metadata or known event input, compare dates and choose between equally typed values. Both conditional branches and every write destination are checked even when the claim map is empty.

Order is explicit because it affects construction identity. Canonical JSON hashing ignores the order of object fields, but the native assessment's ordered write list follows its claim-map iteration. The new call therefore binds a separate key-order digest and requires the list to cover every current claim exactly once. Forward and reverse orders produce their corresponding native requests. Their disjoint writes yield the same financial state, but that agreement does not make the two ordered request records identical.

Eight retained and twelve new assessments match full requests and transition results, including empty maps, multiple claims, escaped keys and due-date boundaries. An authored inclusive comparison makes a claim overdue on its due date. The typed transition executes that different rule, while the selected strict lifecycle invariant rejects its snapshot. This distinction matters: types can establish that a date comparison is meaningful, but they cannot choose the caller's

required convention.

Both constructors work with the native specializer disabled in their checks. Their construction interface leaves full-intent eligibility separate; the next section supplies an explicit ordered alternative for the retained contracts. Fixed chart context, role mapping and unit-cost choices remain declared scope. The earlier complete program composition and invariant source continue to provide their own boundaries.

Complete constructions, controls and variants are retained under `data/accounting-master/parameterized-method-construction-20261026/` and `data/accounting-master/ordered-assessment-expansion-20261026/`.

A.54.14 Preserving the order of eligibility decisions

A customer can be unknown before a collection is authorised. Suppose the full intent supplies an unknown customer, false approval and a claim key absent from the book. The retained lifecycle first reports unresolved input. Once the customer is supplied, the same false approval produces an unsupported event; only after approval can the missing claim produce an invalid event. Combining all three predicates into one conjunction changes this behaviour: an invalid consequence can outrank the earlier unresolved or unsupported result. The individual predicates remain meaningful, but their conjunction computes a different decision.

`OrderedEventEligibility/1` makes this order explicit. A template declares its company context, operation, complete input types and ordered stages. Each stage supplies one existing guarded requirements manifest, exact bindings for its declared inputs, and the reason to report if it does not pass. The evaluator validates the typed book, event identity, chronology, full input population, types, values and provenance before interpreting the stages. Any unknown full input then returns unresolved, preserving the input names. Approval is the first semantic stage in each retained template. A stage with no active requirement is refused rather than treated as permission to continue.

For a sale or collection, the next stage checks that assessment has not sealed the event date. A sale then checks the due date and absence of the claim key. A collection checks presence of the key before comparing its recorded customer with the supplied customer. An assessment instead checks its starting cursor, nonempty interval and last included date, then its period boundary. Execution stops at the first stage that does not pass, retaining its status, reason and stage identifier. The existing guarded evaluator still determines each typed predicate; the ordered layer introduces no competing Boolean language.

The bindings supply facts rather than conclusions. They can read a full-intent input, event identifier or date, book date or period end, a declared memory leaf, map membership, or a selected row's scalar field. Membership is computed against the actual typed map using the supplied text key. A caller cannot replace that computation with an asserted Boolean. A row-field binding names an earlier stage whose unconditional predicate establishes membership in the same map under the same key. Compilation checks that relationship and the declared field type without requiring a row to exist. It also checks every later stage and guarded branch before execution, including when the map is empty or an earlier approval will fail.

Calendar adjacency is a narrow additional binding. After strict date validation, it compares the two date ordinals and returns true when the right date is one calendar day after the left. An assessment therefore relates its event date to its explicit end-exclusive date without constructing a successor that could overflow at the maximum representable date. Leap days and year changes follow the same relation. This is a civil-calendar condition, not a business-day adjustment or a fiscal-calendar policy. The unchanged chronology and interval checks remain separate

conditions.

The call binds the template, book, event, schema registry and memory definition. It also binds the order of full-intent inputs. Canonical JSON identity ignores map insertion order, whereas the retained unresolved result lists unknown inputs in that order. Reversing two unknown inputs therefore leaves the canonical event hash unchanged but changes the order digest and the returned list. Reusing the earlier call is refused. This is the same distinction between value identity and explicitly meaningful order that arose for assessment writes.

Six templates across the two retained charts admit all 38 previously checked events. Combined with the parameter and ordered-row constructors, they reproduce the complete saved requests and transitions while native lifecycle eligibility and construction functions are disabled. The original 18 refusals and 16 additional precedence cases preserve their statuses and reasons. Seventy controls include selected-row binding, skipped-stage compilation, changed rule truth, stale identities and calendar boundaries. A changed customer predicate can admit a wrong customer; that consequential edit demonstrates author control and deliberately differs from the selected contract.

These comparisons establish a finite declarative alternative for eligibility and method construction. Complete-program composition adds a different question: do reports, carry and invariant failures preserve the same current state and successful prefix? The following extension checks that execution boundary while retaining opening admission as a separate prerequisite. General company rules and independent implementation acceptance remain open.

The ordered rules, full comparisons, identity repair and control results are retained under `data/accounting-master/ordered-event-eligibility-20261026/`.

A.54.15 Executing declared event rules through a company program

A collection that works in isolation can still fail in a company program. An earlier sale changes the claim population and its state schema; a report advances the observation date; a year-end carry establishes the next opening book. The method for a later event must be constructed from that current state. Replaying a previously finished method cannot establish that these dependencies are handled correctly.

`DeclarativeEventCompanyProgram/1` composes the explicit mechanisms. The program supplies its opening book, typed memory definition, report mapping, authored state invariant and one eligibility/construction pair per operation. Event commands contain the full intent. An assessment also supplies an ordered list covering its current claim population. The program contains neither native event callbacks nor precomputed event results. Separate hashes bind the selected rules and invariant; program identity also retains the order of event inputs.

Before each command, the runner checks identity uniqueness, the typed book and the selected state invariant. For an event, ordered eligibility runs first. Only an eligible event reaches the parameter or ordered-row constructor, which uses the current book and full intent to produce the complete transition request. The unchanged typed transition then computes postings, flows and memory effects. A report uses the current period's opening and the book observed at its stated cutoff. A carry uses the existing closing and carry evaluator, with the explicitly selected policy preserving every memory leaf.

The book advances only after the resulting state also passes its type and invariant checks. Consider an authored sale whose postings increase receivables but whose inserted claim incorrectly records zero remaining amount. The typed transition can propose that state because each value and destination has a valid type. The separately selected claim-total invariant rejects it: claim amounts no longer equal the receivables balance. The runner retains the prior current

and opening books, records the refused command, and leaves the complete remaining suffix unexecuted. It adds no compensating journal or financing event. This is in-memory refusal behaviour, not a claim of durable database rollback.

The two retained eleven-command programs preserve all complete native command requests and results, including reports and carry archives, along with their before/after state identities and final books. Two further eleven-command programs start from the same declared openings but create three new claims each, using changed keys, customers and amounts. Their report, year-end carry and following-year assessment also agree. All native lifecycle functions are disabled during candidate execution. The saved result is checked again after serialization; no native after-state is supplied to the candidate.

Twenty-six adverse programs include the original eighteen refusal cases, combined unknown/approval/missing-claim conditions, and negative or excessive collections that fail financial preconditions after eligibility has passed. The latter retain the complete refused request and transition. Additional controls check stale selections, changed input order, duplicate identities, incomplete assessment order, a changed carry policy and the invariant-invalid sale. Each stopped run preserves its successful prefix, current opening, reports and unexecuted suffix.

The comparison concerns complete substantive command payloads. The candidate adds explicit eligibility, construction and invariant records; the native result carries different implementation and observation records. Those differences are retained and identified, rather than removed and described as whole-result equality. Native observations are separately recomputed from the candidate states. Shared transition, report and carry evaluators remain the owners of their respective financial calculations.

Opening admission remains a separate native prerequisite for this composer comparison. The original validation checks the retained opening/program before native functions are disabled. The next extension declares those opening conditions and binds them to the exact program selected for execution. The earlier company-source frontend retains its own native-contract target. General opening populations, a complete financial language, accounting validity and independent implementation acceptance remain separate questions.

The four complete programs, adverse cases and full comparisons are retained under `data/accounting-master/declarative-event-program-20261026/`.

A.54.16 Admitting the opening that the selected program will use

A balanced, typed opening book can still violate the selected company's operating conventions. Its claim map may be empty while receivables are nonzero; its unit-cost leaf may have the wrong dimensions; its account roles may overlap; or its statement mapping may omit an account. The opening check must establish these additional conditions before events run. Yet it must not silently promise that every future command is valid: an unknown future customer or an unsupported event is still decided at its execution boundary.

`DeclaredOpeningAdmission/1` records the retained empty-opening contract as ordered declarations. Its template specifies the input envelope, policy fields and values, exact claim-row and clock types, empty claim population, cursor equal to opening date, complete and distinct account roles, required account classes and cash identity. It then checks the unit-cost pointer, dimensions and strictly positive value, the opening receivable total, the opening statement report and the command-list limit. Each condition retains its native position and failure reason. All template stages compile before their evaluation; a malformed later declaration cannot disappear because an earlier condition would refuse the opening.

Structural conditions compare declared record fields, schema nodes, role maps and literal policy

values. Scalar date and monetary predicates use the existing guarded requirements evaluator, retaining their dimensional types until that evaluator lowers them. This separation preserves a subtle input boundary: a malformed carry-policy value fails the declared policy equality, rather than acquiring an earlier and different scalar-type error. Types remain essential where the selected contract requires typed numerical or date comparisons.

The empty-claim premise makes the opening claim-total condition particularly simple: the receivables balance must be zero. This is not a replacement for the row invariants of an operating book. Once events create claims, the complete authored state invariant again checks their totals, dates, references and assessment snapshots. The opening report retains its own evaluator, which checks the period start, opening flows and temporary balances, statement-map coverage and statement arithmetic. No statement rule is duplicated inside the opening template.

Two templates reproduce exact acceptance or refusal on 82 complete inputs, including the 58 preceding audit cases. Ten are admitted. Among them are inputs with a malformed future command, unknown operation or unknown customer that the original opening validator also accepts. The corresponding execution checks remain separate. A deliberate change from strictly positive to nonnegative unit cost admits a zero-cost opening; this proves that the declared predicate controls the decision and explicitly differs from the selected native contract. Dependency identities, input order and stale selections are checked alongside the substantive conditions.

Passing opening admission and passing execution independently would still leave a serious gap. One could admit a book under one account-role map but execute methods selected for another, or admit a different command list from the one that actually runs. `AdmittedDeclarativeProgram/1` therefore binds both evaluations to one explicit selection. That selection names the chart context, role map, unit-cost pointer, policy, opening template, complete rule set and state invariant. The call binds all supplied definitions and facts rather than accepting a caller's success flag.

After admission, the adapter compares the complete shared opening, schemas, statement mapping and memory definition. It checks the selected rules and invariant, then compares every projected event, report and carry command. Projection removes only the assessment-key order and memory carry-policy fields that the native-shaped admission input lacks; the composer checks those fields when they are used. Event-input order is retained explicitly because it affects the ordered list of unknown inputs. A mismatch refuses execution.

With all native lifecycle functions and the operating-method builder disabled, the combined path preserves the complete prior composer results for four successful programs and 26 adverse programs. The successful programs contain 44 commands whose complete substantive payloads already agree with the native baseline. Twenty-two binding controls reject changed facts, roles, cost selection, policy, rules, invariant, commands, order and stale identities. An opening refusal executes no command. Existing typed-memory, transition, requirements, report and carry evaluators remain the owners of their respective calculations.

This removes the native opening evaluator prerequisite for the selected combined path. The selection is still a finite reviewed chart contract, not a proof that arbitrary edited rules have equivalent meanings. The portable inputs use exact JSON values and rational strings. Earlier company-source frontends retain their older target. Section A.54.19 introduces the successor that expresses these algorithms directly. General nonempty opening populations, wider company rules, independent implementation and accounting acceptance remain open.

The opening comparisons and bound programs are retained under
`data/accounting-master/declared-opening-admission-20261026/` and
`data/accounting-master/admitted-program-binding-20261026/`.

A.54.17 Compose methods, state controls, and event intent in one authored program

The preceding examples established four separate boundaries: a method can be written in complete text, a registry can retain its source and identity, a state invariant can reject a bad book, and a selected event contract can retain the caller's full intent. A usable program must express those choices together. Otherwise a typed fixture may pass while the proposed language has no way to write the same program or to show which semantic owner supplied a missing field.

`AuthoredCompanyText/1` closes this composition boundary with a bounded source format. The program imports the opening book, report schemas, statement mapping and memory definition by exact SHA-256 identities. It selects one `BookStateInvariant/1` resource explicitly. These imports are facts and checked rule resources; they do not silently generate methods or claim that a small synthetic state is a complete disclosure history.

Each registry entry contains the complete `CompanyMethodAuthoring/1` source block. The entry records its lowered method digest, source digest, transition profile, input declarations, account classes, clock and separate entry-schema resources. A registry version therefore cannot hide a transition change. The method header supplies the transition profile, while the program author chooses the registry and method for each event. Entry schemas remain distinct from the shared report schemas because combining them would change the identity of a report request.

For the selected receivable contract, the same program also declares the native implementation identity, role-to-account map, unit-cost pointer and policy. Every event carries both its projected method inputs and its complete original intent. The latter retains the customer, claim and interval assertions that a specialized method may remove from its calculation inputs. The runtime then checks the selected native lifecycle on the current book and requires the authored projected request to equal the declared projection of that full request. An edited half-collection method therefore remains executable under the guarded target, where it changes the resulting claim and cash, but it is refused when the caller explicitly selected the full-payment native contract. This is a contract comparison, not an inference that either method is the correct accounting treatment.

Reports, notes, carry boundaries and memory policies reuse the existing command forms. A reset policy can obtain its type only from an explicitly present opening memory leaf. Unknown event inputs retain their runtime unresolved status. Duplicate declarations, unregistered aliases, changed resources and incomplete source blocks are errors with a stage and UTF-8 byte location. Method source is hashed from its original bytes; the verifier checks outer, method and declaration spans after Unicode and CRLF roundtrips.

Two complete eleven-command programs reconstruct their saved typed programs and runtime results exactly. The run includes nineteen structural refusals and six matched native refusals across the two charts, direct method editing, report-note preservation and failed-prefix checks. These finite comparisons show that complete methods, explicit registries, selected state rules and full event intent can share one source boundary. They do not provide general invariant authoring, whole-company source completeness, accounting admission, production readiness or human acceptance. The following successor supplies explicit source for the current finite invariant family.

The field inventory and grammar design are retained under `data/accounting-master/common-program-authoring-audit-20261026/`. The implementation and checks are under `data/accounting-master/common-program-authoring-20261026/`.

A.54.18 Write the invariant that a company program selects

Selecting a rule by its digest preserves its identity, but gives the author no syntax for changing the rule itself. The claim example makes the missing operation concrete. Receivables are 30 and the proposed claim is 31. An author must be able to state that all claim amounts sum to receivables, specify which state fields supply those amounts, and choose how an unknown total affects the program. Naming an imported algorithm alone does not express those choices.

`BookStateInvariantText/1` writes the existing finite invariant as explicit declarations. Global declarations identify the book date, balances or memory leaves. Each collection identifies a memory map or the book's seen-event sequence. Its fields declare their types, contexts, provenance and bindings to map keys, record fields or sequence values. Uniqueness, ordering and reference constraints name the collections and fields they compare. Per-row blocks contain guarded predicates and explicit row/global bindings. A final aggregate block names sums and global inputs, then states the predicates they must satisfy. Residual prerequisites remain attached to the row, collection, aggregate or whole invariant at which they apply.

The frontend lowers these declarations to the same `BookStateInvariant/1` definition. The retained collection, total and guarded-requirement evaluators still determine every predicate's meaning. No callback or second arithmetic engine is introduced. `AuthoredCompanyText/2` embeds that source between `begin_invariant_source` and `end_invariant_source`, replacing version 1's selection of an imported whole-rule resource. The compiler constructs the exact typed rule internally and passes it to the original program compiler. Opening books, schemas and other data resources remain explicit imports.

For the claim total, the aggregate predicate is equality between the sum of remaining claim amounts and the receivable balance. Changing its false outcome from `invalid` to `unsupported` changes the reason the program stops; the proposed sale still fails and its journal never enters the accepted prefix. A more consequential edit replaces equality with "receivables are no greater than the claim total." That weaker predicate permits the 31-versus-30 case. The source now controls the rule's truth, but the altered rule has lost the original accounting reconciliation. The comparison is an authoring test, not a recommendation to weaken that control. Selecting a well-formed rule never proves that it is the right rule for a company.

Compilation and state binding remain separate. A uniqueness rule over dates is a valid typed definition. If the author binds those dates to actual text map keys, the native binder refuses the mismatch, including when the map is empty. Partial populations and unknown amounts remain unresolved; an observed subtotal does not become the complete total. These distinctions are preserved by reusing the original typed evaluators rather than by treating a successful parse as evidence that the book satisfies its controls.

The two company sources reconstruct their original invariants and all 22 complete command results. All sixteen inconsistent-state examples retain their full refusal results, unchanged opening and unexecuted suffix. Fourteen source controls exercise duplicate, missing, unbound, incompatible and oversized declarations. Both direct rule edits agree with their explicitly edited typed counterparts, and the two partial/unknown cases retain their full unresolved results. Source and nested declaration spans are checked against UTF-8 bytes with both ordinary and CRLF line endings. Whitespace can change the source digest while leaving the typed program unchanged.

This is a bounded authoring language for the existing state rules. It does not add arbitrary algorithms, establish completeness of a real company's claim population or supply accounting authority. Wider rule families and independent implementation remain open. The grammar decision, source examples, complete comparisons and unchanged-source checks are retained under `data/accounting-master/invariant-source-authoring-20261026/`.

A.54.19 Write the selected company rules and commands in one source

A generated method can calculate the right journal while leaving the modeller unable to state the rule that generated it. In the claim example, the customer check, the source of a claim identifier and the order of assessment updates are substantive choices. Naming a template by its digest records which choice was made, but does not let an author write or change that choice. The selected company source therefore includes the opening conditions, state invariant, eligibility stages and construction recipes themselves. The earlier source frontends remain useful predecessor comparisons; this successor targets the declared opening and execution path together. The opening template in this profile still requires an empty claim map. Section A.54.20 extends that boundary for explicitly complete supplied carried claims.

Consider a collection whose claim identifier is absent. Reading that claim's customer immediately would fail for a missing row, obscuring the intended refusal that the claim does not exist. In `OrderedEligibilityText/1`, the author writes a membership stage before the customer stage. A binding such as `(map_contains "/claims" "claim_id")` supplies the membership predicate. A later `map_field` binding names the same map and event input, the customer field and the preceding presence stage. Compilation accepts that lookup only when the earlier stage unconditionally requires the matching membership fact. All stages compile, including ones that an earlier refusal would prevent from running. An empty map cannot hide a malformed field lookup.

The full event-input types, stage identifiers, failure reasons and guarded predicates are also declarations. The retained order first resolves the outer event structure and unknown inputs, then approval and the operation's specific conditions. Moving claim existence ahead of the sealed-day condition changes which reason is returned when both fail. Negating the approval predicate changes which events are eligible. These edits demonstrate that the source controls the selected rule; they are different rules, not equivalent ways to spell the retained contract. The six eligibility sources reconstruct their complete typed templates and preserve 38 eligible and 34 refused gate results.

Opening admission has a related but distinct job. `OpeningAdmissionText/1` states structural checks for the input fields, policy values, row and clock schemas, roles and report mapping, alongside typed predicates for the dates and amounts. The following excerpt is an actual declaration for the final command-count condition; its position follows the other opening checks.

```
stage "commands" commands "command_limit";
check 64;
end_stage;
```

Other structural stages declare their paths and expected values directly. Predicate stages use the same guarded-expression grammar as eligibility. Dependency declarations name the exact reused semantic implementations, while identities for locally authored predicates and templates are computed from the definitions. The two 21-stage sources preserve all 82 complete opening results. Ten inputs are admitted, including those whose future command will later fail. Neither successful parsing nor admitted opening conditions establish that the whole future program succeeds.

Construction recipes express the next step: how a permitted event supplies the values needed by a method. `MethodRecipeText/1` keeps the existing method declarations for monetary calculations, postings, cash flows, guarded conditions and memory writes, and adds explicitly typed parameters. For example, a customer parameter can read the event's customer input, and a memory write can refer to that parameter. No event value is interpreted as source code. A pointer parameter combines literal path segments and previously declared text parameters; the

existing pointer encoder escapes slashes and tildes in a claim identifier before using it as a path.

The distinction matters for the sale and collection methods. The sale writes the customer's name, due date, event identifier and issue date into a new claim. The collection reads the selected claim's remaining amount through a parameterized pointer. These values are supplied by the actual event and current book. The accounting expressions and account choices remain visible in the recipe. An author need not reproduce a serialized method tree or maintain its selection hash manually. Missing, unused, forward or mistyped parameters are refused by the original contextual constructor.

An assessment requires a sequence of writes over the current claims. `OrderedExpansionText/1` contains the complete base recipe, map path, row types, per-row replacements and final replacements. Each replacement declares its identifier, path and value expression. A row expression can read a field, the current key, an event input or event date, and choose between typed branches of a date condition. Both branches are checked even when only one will be evaluated. The explicit assessment-key order determines the write sequence; a duplicate or incomplete order refuses construction. A malformed row expression also refuses when the map happens to be empty.

Four scalar recipe sources and two ordered-expansion sources reconstruct their complete templates. Across 18 scalar and 20 expansion cases, the full construction results, transition requests and transitions equal the retained typed comparisons. These checks include identifiers containing slashes, tildes and Unicode. Source lowering reports syntax and structure only; the current book, event, schemas and memory definition remain necessary for contextual construction. Thus a reusable recipe is not prematurely described as a method valid for every possible state.

`SelectedCompanyText/1` joins these declarations. It imports the opening book, schemas, statement mapping and memory definition by exact content identity. Those four resources contain facts and types. The opening rule, invariant, eligibility and construction algorithms are inline source blocks. Roles, unit-cost pointer and policy define the selected company conventions. Events retain their full typed inputs, provenance and declaration order; reports and carry commands retain their dates, notes, equity movements and explicit memory policies. The native-shaped admission input is derived from these same facts and projected commands, rather than authored as an unrelated second copy.

Lowering creates the full program and its selection without running commands. Execution then reuses the opening-to-program binding and the existing eligibility, invariant, construction, transition, report and carry evaluators. For four successful programs, all 44 complete command results are preserved. For 26 adverse programs, the stopped prefix, current and opening states, refused request where present and unexecuted suffix are preserved as well. All native lifecycle functions and the operating-method builder are disabled during these comparisons. Changing the first sale's approval directly in the company source stops the program before any command succeeds.

One provenance distinction must remain visible. Canonical content identities ignore dictionary insertion order, whereas the outer execution binding also records the serialized order of the complete supplied program. The source parsers reconstruct some definition dictionaries in a different order. Their canonical content is identical, but that one outer order digest changes. The comparison recomputes and retains both order identities, and permits a difference only at `program_order_sha256`. Every other result field matches, including the separate digest of event-input name order, which must remain unchanged because it affects the order of reported unknown inputs. This is a precise provenance exception, not a relaxation of financial-result or failure comparison.

Source identity retains the original UTF-8 bytes. Whitespace and CRLF line endings can

change that identity while leaving the lowered program and its results equal. Errors inside a nested rule or recipe identify the corresponding bytes in the full company source. A lexical error can identify its token; a semantic error that is known only at stage or document scope is reported at that scope. The corrected nested-invariant reader described in Section A.54.20 also preserves quoted Unicode separator characters as data rather than treating them as source line endings. The checked sources reject duplicate or missing declarations, changed or unused imported resources and incompatible operation selections.

The comparison supports this finite synthetic company language. It does not make an edited rule accounting-equivalent to the retained rule, supply missing facts about a real company's claim population, or complete the broader fiscal, sector and measurement algorithms. Independent implementability and reader acceptance remain open. The complete source examples and result pairs are retained under `data/accounting-master/selected-company-source-20261026/`. The component evidence is under `data/accounting-master/ordered-eligibility-source-20261026/`, `data/accounting-master/opening-admission-source-20261026/` and `data/accounting-master/method-recipe-source-20261026/`.

A.54.20 Begin a reporting period with live claims

A new reporting period need not begin with an empty receivables register. Customers may still owe amounts from sales in the preceding year. The closing procedure carries those claims forward with their identifiers, issue dates, remaining amounts and assessment history. An opening rule that requires an empty register would reject this ordinary continuation even when the carried book and opening statements are internally consistent.

The retained examples make that conflict concrete. Two company charts carry one live claim with a remaining amount of 20 into 1 January 2028. Two further programs carry three claims whose remaining amounts are 13 each, giving a receivables balance of 39. The claims were issued during 2027 and assessed on 31 December 2027. The new reporting period ends exclusively on 1 January 2029; its period flows are empty, while the claim records and previously seen event identities remain. The original continuation contains three commands in each one-claim case and two in each three-claim case. Those ten commands supply an exact comparison for the opening extension.

The selected empty-opening template rejects all four books at `empty_claims`. Removing that stage alone exposes a second empty-specific condition: `claim_total` requires the receivables balance to be zero. Removing both conditions makes the remaining opening checks pass, but it also admits five deliberately corrupted books. The corruptions are a wrong receivables total, a negative remaining claim, a missing sale identity, an issue date after the opening and an inconsistent overdue amount. The full selected state invariant rejects each corruption. Thus the required change is not simply to permit a nonempty map. It is to retain the relationship between every supplied claim and the book as a whole.

`CompleteClaimsOpening/1` makes that change under an independently selected contract. The selection identifies the original empty-opening template and the full state invariant. The new contract carries both definitions, their canonical identities, the exact two excluded stage names and an explicit description of its scope. Before deriving the revised structural template, the adapter checks that `empty_claims` is the reviewed equality with the empty claim map and that `claim_total` is the reviewed zero-receivables predicate. Every other stage remains unchanged. A newly supplied predicate does not acquire reviewed status merely because the caller computes its digest.

The accompanying `SuppliedClaimPopulation/1` declaration states whether the supplied internal

claim map is complete, partial or unknown. It names `/claims`, identifies the exact opening book and supplies provenance. Only an explicit `complete` declaration proceeds under this profile; partial and unknown declarations return unresolved before any command runs. Completeness here concerns the supplied internal state. A populated map, a balanced book and a successful continuation cannot establish that a real issuer has disclosed every claim. That different question still belongs to source-constrained reconstruction.

Admission first runs the retained structural and opening-report checks on the same input. It then evaluates the full selected invariant over the unchanged book, schemas and memory definition. The invariant checks the aggregate receivables relationship and the claim conditions that the simple relaxation missed. A malformed schema or incomplete statement mapping therefore still fails its original check. A mid-period snapshot also remains unsuitable as a new-period opening even if its claims satisfy the invariant. This profile deliberately checks structure and the opening report before the full invariant; it does not promise an older profile's failure precedence when several conditions fail together.

The program wrapper, `AdmittedCompleteClaimsProgram/1`, retains the complete admission result. After admission succeeds, it delegates to the existing opening-to-program binding using the derived structural template and its corresponding internal selection. That binding again checks shared facts, schemas, mapping, memory definition, roles, policy, rules, invariant, complete projected commands and event-input order before the unchanged composer runs. A caller-supplied success flag cannot replace these checks. The retained delegated result exposes the requests, transitions, reports, carry state, trace and any stopped suffix.

The same distinction is explicit in company source. `CompleteClaimsCompanyText/1` surrounds the full selected company source with a reviewed-selection identity, population declaration and contract residuals. The declaration below supplies the population premise in a checked carried example; the opening-book identity is computed from the exact imported book rather than typed a second time.

```
population "SuppliedClaimPopulation/1" complete "/claims"
  "Exact retained synthetic carry; complete supplied internal map";
```

This display wraps one source declaration for readability. The complete program begins with the text and target profiles, then names the expected reviewed-selection digest. The caller supplies that reviewed selection separately. Between `begin_selected_company` and `end_selected_company`, the author writes the complete opening, invariant, eligibility, recipe and command source described in Section A.54.19. Its only imported resources remain the opening book, schemas, statement mapping and memory definition. An extra resource containing an opaque algorithm is refused.

Lowering checks that both the named identity and the nested source selection match the independently supplied selection. Editing an opening condition or invariant cannot silently replace that reviewed choice. Lowering also verifies the structural contract, but does not admit the opening or run commands. Execution invokes admission anew. UTF-8 bytes and line endings remain part of source identity; nested errors retain their original parent-source locations. The invariant parser supplies declaration locations, while the opening lexer can identify an offending token. The outer source preserves that difference instead of asserting precision the inner parser does not provide.

A source line ends at LF, with an optional preceding CR; a quoted string may contain other Unicode separator characters. The inherited invariant parser incorrectly split such strings at U+0085, U+2028 and U+2029. The corrected reader and nested rule renderer keep these characters inside the string. Nine comparisons place each character in an invariant residual, a

nested rule residual and an input source reference. Each reproduces the earlier parser failure and preserves the corrected typed program and complete result. All eight retained programs and 32 prior controls also remain exact. This repair changes source handling only; the financial evaluators are unchanged. The repaired invariant and complete company source implementations are under `data/accounting-master/invariant-unicode-repair-20261026/`.

Four carried and four empty-opening source programs reproduce the complete typed results. The carried programs retain all ten continuation commands; the empty-opening programs retain all 44 commands. The five corrupted states still refuse before execution. Native lifecycle callbacks and the operating method builder are disabled throughout these comparisons. The result comparison permits differences only in the outer and delegated `program_order_sha256` fields: both record the full program's serialization order, which source reconstruction can change while preserving canonical content. Both identities are independently recomputed and retained. Every other field, including complete execution and event-input-order identity, matches the typed comparison.

The checked successor therefore admits these complete supplied carried books without weakening the selected invariant. It leaves partial actual histories, arbitrary alternative financial rules and accounting applicability unresolved. The carried-state audit, typed contracts and full source comparisons are kept together with their explicit refusal cases under `data/accounting-master/nonempty-opening-audit-20261026/`, `data/accounting-master/complete-claims-opening-20261026/` and `data/accounting-master/complete-claims-source-20261026/`.

A.54.21 Keep each assessment attached to its historical terms

A claim can remain outstanding while its due date changes. Checking its current balance and current due date is therefore insufficient to explain an earlier overdue assessment. In the retained synthetic example, a claim issued on 2 January 2027 is initially due on 31 January. An imported assessment dated 28 February records 20 remaining and 20 overdue. Prospective extensions become effective on 10 March and 10 April, moving the due date first to 30 April and then to 31 May. Applying the current 31 May due date to the February assessment would incorrectly turn its overdue amount into zero. The historical assessment must remain attached to the initial term.

The term history is stored inside the claim. Each term has its own local identifier, effective date, due date, predecessor and event identity. Each assessment has a date, referenced term, remaining amount, overdue amount, origin and event identity. Local identifiers alone do not identify these records: another claim may use the same identifier. Nor does insertion order establish chronology. A declaration must name the parent claim and select a term using its effective date before comparing the financial amounts.

Expose the supplied rows without choosing a history. `NestedMapProjection/1` binds a declared parent map and nested child map to typed scalar rows. Bindings may read the parent key, child key, a parent field or a child field. The accompanying lineage retains both raw keys and escaped source paths. Each parent also has a group record with its child-map path, first row and row count. An empty child map still produces a group; an empty parent map produces no groups. This distinction lets a rule require terms for every supplied claim even when one claim has no terms.

Projection preserves supplied insertion order and makes no chronological selection. Declarations are checked against the memory definition even when there are no rows. An explicit population declaration identifies the exact book and both map paths; partial or unknown coverage remains

unresolved. Complete supplied internal maps are sufficient for these checks, but do not establish that an issuer's disclosures reveal all contracts or past events.

Select within one parent and one declared boundary. `GroupedRowQuery/1` provides count, typed match and greatest-row queries. The query names the parent key, escaped parent path and child-map path; all must agree. Count includes all rows in that group. Match compares one declared field with one value of exactly the same type. Greatest orders dates or exact rational values, optionally below an explicitly inclusive or exclusive upper bound. Units, scale and context must agree before a rational comparison.

The February assessment selects the greatest effective date no later than 28 February, which is 2 January. A predecessor query instead uses a strict upper bound: the predecessor of the term effective on 10 April is the term effective on 10 March. A query without an upper bound identifies the latest term. None of these operations sorts or replaces the stored source order. Candidate and winner indices retain that order, and a unique result includes the whole selected row and its source lineage.

Equal greatest dates produce an ambiguous result. Duplicate matches do the same; insertion order is never a tie breaker. A group absent from a complete supplied population is invalid. A present group with no qualifying row returns explicit `none` when the declaration permits it, or invalid when the row is required. An explicit unknown input remains unresolved; omitting a required input is a malformed declaration. Every evaluation recomputes its projection from the bound book, schemas, types and coverage. A caller cannot substitute previously computed rows after changing the source.

Use absence without treating uncertainty as absence. `QueryGuardedInvariant/1` binds query results into the preserved guarded requirement evaluator. Count becomes an exact dimensionless rational. A unique row supplies `has_row=true` and known selected fields. Allowed `none` supplies `has_row=false`, while its selected fields remain unknown. An ambiguous or unresolved query supplies unknown values, including an unknown presence indicator. Thus a tie cannot masquerade as an empty group. The declaration explicitly chooses whether ambiguity is invalid or unresolved.

A false guard makes a predicate inactive; it does not prove that predicate. The separate `required_known` declaration can require a selected field even when its predicate is inactive. Invalid queries fail structurally before the predicates run, and unresolved queries keep the overall result unresolved even under a false guard. Allowed absence is different: it can legitimately activate an empty-group rule while leaving a selected-row rule inactive. The result retains the original guard findings and distinguishes `no_active_requirements` from satisfied requirements.

Declare the rule once for every supplied claim and assessment. Finite queries with handwritten claim identities would miss newly supplied records. `NestedQueryInvariantTemplate/1` instead iterates over all groups or all child rows of a declared complete projection. Typed captures read the current parent or child identity, scalar fields and explicit book dates or memory values. Every query's full group identity is bound to that parent. Scalar substitutions can fill only a typed match value, typed bound or the value of a typed constant already present in a guarded predicate. They cannot replace operators, types, bindings or failure consequences. Duplicate or unsafe substitutions refuse before iteration, including on empty populations.

Each instance retains its captured values, source paths, resulting declaration and complete query and predicate results. The reviewed selection identifies the template as a whole. The instance selection is explicitly derived by the checked substitution procedure; it is not described

as an independent review of every resulting instance. The initial checked profile permits at most 64 instances and refuses overflow without a partially successful result. This bound is an implementation choice for the finite probe, not a restriction on the financial concepts that the product must eventually represent.

The term-row declaration uses a unique match at each effective date to reject ties. It selects the greatest strictly earlier term and checks the stored predecessor against that identity; when no predecessor exists, the explicit initial sentinel must be `none`. It also checks issue, effective, due and book dates. The assessment-row declaration checks uniqueness of its date, selects the greatest applicable term, preserves historical overdue, requires nonnegative assessed remaining amounts and distinguishes imported snapshots from recorded events. Parent-group declarations check current terms, latest assessments, assessed flags, summary amounts and empty-assessment sentinels.

Event membership uses `ProjectedStateCollections/1` to connect nested rows to the existing collection-reference evaluator. It binds complete term rows and the book's seen-event sequence. An explicitly typed equality filter selects assessments whose origin is `recorded_event`; their event identities must be seen. Imported snapshots have a separately declared unknown-event condition. A filtered collection is complete only relative to its complete supplied source and explicit predicate. The filter does not turn a partial source into a complete collection, and its types are checked even when it selects no rows.

Join the six semantic checks. `TermStateSemanticBundle/1` joins six reviewed components: common claim conditions and the receivables total, the collection-clock bound, term history, historical assessments, parent summaries and event membership. The clock reads the actual exclusive fiscal endpoint rather than assuming a calendar year. The common claim declaration retains nonnegative remaining amounts, sale-event membership, issue dates and the exact receivables sum. Historical assessment rules own overdue; the earlier scalar rule based on current due is not applied to these amended-term histories.

All 22 retained states satisfy this semantic bundle. Fifty-one complete comparisons preserve every subassessment, and 27 targeted semantic corruptions are rejected by both the independently run native validator and the declared checks. The candidate evaluations disable native financial callbacks. A separate discriminator changes the native migration premise: native validation rejects that premise while the book's declared semantic relations still hold. This difference identifies a separate structural condition rather than an alternative form of opening admission. The semantic bundle by itself grants no execution or accounting admission. Section A.54.22 supplies the selected structural contract, direct nested source and bounded event/carry composition while retaining actual-source and accounting limits.

Complete declarations, original-state comparisons, structural controls and the native-oracle outcomes are retained under `data/accounting-master/grouped-row-query-20261026/`, `data/accounting-master/query-guard-composition-20261026/`, `data/accounting-master/nested-query-template-20261026/`, `data/accounting-master/nested-obligation-completion-20261026/` and `data/accounting-master/term-semantic-composition-20261026/`.

A.54.22 Carry the historical terms through a complete command

The historical checks in Section A.54.21 answer whether the supplied terms and assessments fit together. They do not, by themselves, justify changing the claim. An amendment also needs a valid current state, an identified customer and prior term, an admissible effective date, and a complete description of the writes it will perform. A collection must change the remaining

receivable without rewriting its earlier assessments. At a period boundary, closing entries and the next assessment window must agree. These are connected requirements: accepting each snapshot in isolation would leave the correspondence between the event and the financial history unchecked.

The selected continuation described here joins those checks for the retained synthetic claim. It covers prospective due-date amendments, collections, assessments and period carry. The accounting premise is still deliberately narrow: an amendment changes the supplied terms without recognising a modification gain or loss, interest or impairment. Actual contractual applicability and the appropriate accounting treatment require their own evidence. Within this premise, the implementation question is precise: can explicit declarations produce the complete retained requests and results while the native financial builders are disabled?

Establish which supplied state the declaration accepts. The selected state contract combines structural checks with the six historical and financial checks already described. Structural checks require the exact state fields and profile, a typed book, the declared term-claim and clock schemas, the operating account roles, a positive cost leaf with the expected units, and the stated migration fields and premise. The account selection is explicit. A different cost pointer can be valid under the native validator and still fall outside this selected contract; refusal then means that the selection does not cover that configuration.

Existing admission stages evaluate most of these conditions on a normalized view in which the original `book` field is named `opening`. The original envelope is checked first, and normalization preserves every value and its insertion order. The reused stage engine's internal success label does not turn a mid-period snapshot into a new-period opening. The enclosing result reports satisfaction of the selected snapshot contract and retains false opening, source and accounting admission flags.

Two additional conditions are explicit. The migration source-book identifier must have the syntax of a SHA-256 digest, and every term premise must contain nonblank text. The distinction between text and nonblank text matters: a space-only premise can satisfy the generic text type while failing the native premise requirement. Checking a digest's syntax establishes neither the existence nor the truth of the referenced source. Similarly, complete supplied claim and nested-row populations remain a stated input scope. A partial population leaves the command unresolved before construction.

Resolve the event against the same history. For an amendment, eligibility first checks the event's identity and date, the complete typed inputs, and the supplied approval. Unknown inputs produce an unresolved result. An explicitly unselected approval produces an unsupported result. These outcomes differ from invalid state, stale selection, duplicate event or contradictory dates.

The event must name an existing claim belonging to the stated customer. Its prior-term input must equal the claim's current term. The new term identifier must be nonblank, unused and different from the sentinel `none`; the premise must also be nonblank. The new due date cannot precede the event's effective date. Finally, the effective date must be strictly later than that of the selected prior term. Equal dates would make the prospective sequence ambiguous and are refused, even when the supplied snapshot is otherwise valid.

A nested lookup is permitted only after the declaration establishes the presence of the parent claim and the selected child term. This is an ordering requirement on the checks, not an inference from a successful dictionary lookup. The following binding fragment names the prior term's effective date; the two referenced stages must appear earlier and establish the corresponding

memberships. The surrounding rule declares its inputs and comparison.

```
binding "previous";
  kind nested_map_field;
  pointer "/claims";
  key_input "claim_id";
  parent_after_stage "claim_exists";
  child_pointer "/terms";
  child_key_input "prior_term";
  field "effective";
  after_stage "prior_exists";
end;
```

The selected eligibility engine checks those dependencies when compiling the rule. It then resolves the bound values against the actual book and event. The preserved guarded-predicate evaluator owns the comparisons. Adding nested membership and field lookup does not introduce a second arithmetic or date language.

Construct the complete nested change. An eligible amendment constructs three ordered memory operations: insert the new term record, replace the claim's current-term identifier, and replace its current due date. The inserted record states due date, effective date, event identifier, predecessor and premise. Each value comes from an explicitly typed parameter. The parameterized constructor builds the full method and typed transition request; it does not import a stored native method as the answer for the current event.

An assessment has a different construction. For every supplied claim, it inserts a complete assessment record and replaces five summary fields: latest assessment, assessed flag, assessment date, assessed remaining amount and overdue amount. The record retains the applicable term identifier, so later amendments cannot change the basis of this assessment. A final operation advances the assessment cursor to the end of the checked window. In the retained one-claim case this produces seven writes. Two claims produce thirteen writes, with a single sequence of operation identifiers across both claims and the suffix.

The previous ordered expander admitted only scalar fields in each row and scalar replacements. It could not represent an insertion into the nested assessment map. Its successor therefore checks a declared nested row schema and recursively checks complete write templates. A record path must name a declared field; a dynamic map key must be typed text. Insertions target map members and supply the entire declared record. Replacements target scalar leaves with the correct type. Both conditional branches are checked, including the branch not taken by the current claim.

Validation precedes expansion even when the claim map is empty. An empty map then emits only the cursor suffix, while an invalid nested record template still fails. The existing 256-write limit is retained: a construction that would emit 257 writes refuses as a whole. This limit is an implementation boundary, not a conclusion that a larger company's history is economically impossible. Claim iteration order is explicit and is preserved in the writes; it does not determine which term was effective on an earlier date.

Check the proposed result before continuing. The command composition checks the selected snapshot before eligibility and again after the typed transition. A well-typed but incorrect predecessor or overdue write can pass construction and still fail the resulting historical requirements. In that case the result retains diagnostic transition evidence but does not return

an accepted top-level after-state. The caller cannot continue the sequence from that rejected proposal. The successful path remains a provisional calculation; it performs no durable commit and does not establish source or accounting admission.

Period carry follows the same discipline. Its report request must bind to the exact closing book. The selected policy preserves all typed memory leaves, including immutable term and assessment histories, and resets only the assessment cursor to the next period's start. The preserved report and carry engines produce the closing and opening books, which are checked against the selected snapshot contract. Completeness of the supplied population is carried forward only from the accepted pre-state through this declared transition; it is not manufactured for an unrelated supplied book.

Chronology is visible in the complete comparison. Each company's retained sequence contains nine events, with carry between the seventh and eighth events. The continuation uses the computed output of each command as the next input, including carry. Appending carry after all nine events, or resetting the input to a stored snapshot before each event, would not verify this chain. The final state agrees only after the commands are placed at their actual positions across the two fiscal periods.

Make the selected behavior directly authorable. The state-contract source spells out structural stages, nested projections, queries, captures, template substitutions, guarded requirements and collection constraints. Event-rule source separately spells out eligibility bindings, typed parameters, method conditions, monetary effects and nested writes. Carry policy has its own short declaration. These source forms contain the algorithm-bearing declarations rather than an opaque import of a completed financial algorithm. JSON tokens remain available for explicit types, contexts, schema expectations and scalar data.

The frontend preserves authored declaration order and exact Unicode strings. Only line-feed characters split source lines; carriage-return/line-feed input is also accepted without mistaking a Unicode line separator inside a string for a new declaration. Locations record physical line numbers and UTF-8 byte slices. Lexical errors identify tokens or declarations; failures reported by an opaque contextual validator identify the responsible source component, without pretending to identify a more precise field.

Parsing, eligibility compilation and contextual construction remain distinct states. A parsed construction is not reported as contextually compiled until its event and book have been supplied. Execution also requires the independently supplied selection of the combined contract and rules. Changing a source file does not authorize its changed declaration: the old selection refuses it. Source diagnostics are retained alongside the complete typed execution result, so adding location information does not obscure the comparison with that result.

The finite evidence establishes this selected continuation. Structural checks and direct state-contract source each reproduce 74 complete comparisons. The nested constructor reproduces ten retained assessment requests and transitions. Full command composition and its source counterpart each reproduce two continuous traces, comprising eighteen events and two carries. Malformed sources, incorrect bindings, partial populations, stale selections, nested write errors and the equal-effective-date boundary have separate controls. Native financial callbacks are disabled during candidate execution. Shared parser and typed-engine defects remain possible; this is bounded conformance evidence, not independent acceptance or whole-company coverage.

The complete sources, expected declarations, failures and results are retained under
`data/accounting-master/term-structural-admission-20261026/`,
`data/accounting-master/nested-contract-source-20261026/`,
`data/accounting-master/nested-assessment-construction-20261026/`,

`data/accounting-master/term-full-event-binding-20261026/` and
`data/accounting-master/term-event-source-20261026/`.

Run the selected commands as one program. An ordered program must preserve more than the last balance. Suppose a receivable's due date is extended twice, a partial receipt follows, and the unpaid balance crosses a year-end. A rejected later request must leave the earlier extensions, assessments and receipts intact. Reconstructing the input from a convenient stored snapshot, or recomputing a selection from changed source, would conceal a different history.

`SelectedTermProgram/1` therefore supplies the complete state-contract source, the four event/carry rule sources, an opening state, a declared supplied population and an ordered list of complete requests. An independently supplied selection binds the exact source bytes, program contents and declaration order. Every request retains its own expected-state identity. The program derives transport calls only after those identities agree; it neither authorizes a new selection nor rewrites a stale request to match the current state.

Compilation checks the whole source envelope before execution. Each successful command then passes its full state and returned population coverage to the next command. A first-command refusal returns no successful prefix; a later refusal returns the complete successful prefix, unchanged last accepted state and coverage, the failed request and result, and the untouched remaining requests. Unknown approval remains unresolved. A partial supplied population does not become complete because some commands can be evaluated. The program performs no durable commit.

The two retained ten-command programs reproduce every field of the earlier source executions, including period carry and its complete report result. Both end with cash 130 and receivables zero. More revealingly, the old February assessment retains overdue 20 under the initial January due date after both extensions. March's zero assessment remains tied to the first extension; the April and June assessments use the second. A receipt changes the amount outstanding but does not retrospectively change an assessment. A new assessment records zero after the next year's final receipt.

Fifteen adverse cases exercise changed source, opening and command order, unknown approval, partial population, malformed declarations and later failed requests. A relocated copy also recomputes both complete results with identical canonical and insertion-order identities while native financial builders are disabled. These checks make the selected program reproducible; they do not make it a general company grammar. Standalone arbitrary report commands, cross-profile migration, actual historical completeness and accounting applicability remain separate requirements. The profile caps a program at 64 commands, and inherited structural limits can refuse a smaller instance; neither limit is an economic capacity claim.

The complete envelope, failure semantics and expected results are retained in
`data/accounting-master/selected-term-program-20261027/`. The isolated component and its fresh-output command are in
`data/accounting-master/selected-term-bundle-20261027/attempt-02/bundle/`.

A.54.23 Preserve payment history when financing and remedies change

A company that pays late can end with the same cash, debt and equity as a company that pays on time. The ending balance sheet does not distinguish the two histories. A simulator must retain what was due, what remained unpaid when payment was missed, what was subsequently paid, and which contractual consequences were actually established. Otherwise a successful

funding decision can silently make a past failure disappear.

Consider the supplied zero-coupon promise used in the financing comparison. The opening book contains cash of 60, inventory of 100, debt with a carrying amount of 80 and equity of 80. Contractual face is 100, due on 30 June 2027. A single explicit measurement interval increases carrying debt and interest expense by 20 at maturity. Forty new shares are subscribed for 40 in cash. There are no fees, taxes, penalties or modification effects in this example; the price and funding date are supplied scenario assumptions. These choices make payment timing visible without requiring a financing equilibrium.

If the subscription settles on 29 June, the company pays all 100 at maturity. If it settles on 2 July, the company first pays its available 60 at maturity, records the unpaid 40 on 1 July, and pays the remaining 40 on 3 July. Both paths finish with cash and debt equal to zero, inventory of 100 and equity of 100 after the interest expense. Only the second path has a missed-payment history. The signed journal convention records debt as a credit balance; the following table displays its positive carrying amount.

Late-funding path	Cash	Debt	Unpaid claim	Past miss
Opening	60	80	100	0
Maturity, before payment	60	100	100	0
Partial payment, 30 June	0	40	40	0
Observation, 1 July	0	40	40	40
Share subscription, 2 July	40	40	40	40
Final payment, 3 July	0	0	0	40

The historical amount in the last column is the amount recorded at the missed-payment observation. It is not the current unpaid balance, and payment does not reduce it. Neither that amount nor the missed flag is, by itself, a finding that a particular legal event of default occurred.

Check the state and the change separately. The selected financing declaration uses the existing typed requirement evaluator for 26 snapshot conditions. Face, remaining claim, recorded arrears, historical missed amount and share counts must be nonnegative. The remaining claim cannot exceed face. Before maturity the unpaid claim remains face while the carrying amount may differ; after the selected maturity operation, carrying debt equals the remaining claim. A recorded miss requires a matured promise, an observation after its due date and a positive historical missed amount. The recorded arrears equal the remaining claim in this selected post-observation state. An unpaid state observed after the due date cannot pass without the explicit missed-payment observation.

These conditions are useful but insufficient. After full payment, a snapshot with a historical missed amount of 50 could satisfy the inequalities even though the recorded amount was 40. Likewise, a nonnegative share count can remain unchanged when a subscription should have increased it. The transition declaration therefore binds the old state, typed event inputs and new state. For a late payment of amount p, let U be the remaining claim, A the recorded arrears and H the historical missed amount. The selected change requires

$$U_{\text{after}} = U_{\text{before}} - p, \qquad A_{\text{after}} = A_{\text{before}} - p, \qquad H_{\text{after}} = H_{\text{before}}.$$

Cash falls by p, the signed debt balance increases by p, and the missed flag remains true. Original face, due date and old shares are preserved. A subscription has a different relation: new shares increase by the supplied share quantity, while cash and capital change by quantity

times issue price. Money and shares have different dimensions, so a money-typed value cannot be used as a share count merely because both are exact rationals.

The five declared transitions cover maturity, subscription, due-date payment, missed-payment observation and late payment. They also preserve unrelated memory, account balances and prior cash-flow records. Deliberately removing the share-memory or historical-miss write can still yield a balanced proposal from the generic engine. The selected transition relation rejects both. The failed proposal remains available for diagnosis, while the accepted current state and unexecuted command suffix retain the last valid prefix.

An internally consistent history still needs an opening premise. Opening assessment binds the selected method registry, schemas, reporting map, account roles and typed financial state. A separate supplied history record names the exact opening book and its debt-term fields, with nonblank provenance. Missing or partial history is unresolved. A paid opening that erases the missed flag and amount can be internally consistent, but it conflicts with the retained history record and is refused. A caller who supplies the same false account in both records can still satisfy this consistency test; source evidence must establish the historical facts.

A valid intermediate snapshot is also not automatically a fiscal opening. The existing statement engine requires the opening date to equal the fiscal period start, an empty current-period cash-flow sequence and zero temporary account balances. The first verification attempt incorrectly expected an intermediate funded state to qualify as an opening. Its refusal exposed the test error. The corrected comparison checks snapshot and supplied-history consistency separately while preserving the fiscal-opening restrictions. Closing and carrying the financing history require an explicit further step.

Keep current contractual findings apart from economic history. The conditional remedy examples introduce a separate record for an identified hypothetical series. It contains the selected clause finding, its observation date, the acceleration finding and the evidence identifier. The applicability of the selected clause is a supplied premise. A known unpaid principal amount must agree with the outstanding claim in the economic book; an unknown applicability finding remains unresolved rather than becoming false.

The retained conditional reference distinguishes seven kinds of question: clause satisfaction, authorized notice, new acceleration, remaining acceleration after rescission, protected-change consent, majority waiver and a finding about another series. Explicit typed predicates reproduce its 58 hypothetical classifications, including unknown facts and unresolved boundary-day order. Only three of these routes are composed into state updates here: a clause observation, a new acceleration assessment and a rescission assessment. The classification table is not a complete contract execution system.

In the composed example, the funded but still unpaid company first receives a supplied conditional principal-clause finding. That observation changes no cash and does not itself accelerate the obligation. A separate, supported declaration establishes acceleration. The existing typed settlement then pays the 40. A fresh paid-clause observation becomes unsatisfied, but a lack of a new acceleration trigger leaves the earlier acceleration finding intact. The selected rescission assessment changes that state only when all of its supplied conditions are resolved as required. The historical missed amount remains 40 throughout. These operations record conditional contractual status; priority, penalties, accelerated cash amounts and insolvency consequences need their own economic and accounting methods.

The initial contract-status record and its schema extension are explicit synthetic premises. They do not establish that the retained issuer description governs this promise. The source-dependent

conditions retain their original scope and unresolved operative-instrument dependencies. Known automatic and declaration routes also keep their different information requirements: the automatic branch does not require a notice finding that it does not use.

A matching date does not establish that a finding is current. The first composed candidate required a clause finding from the same date as the proposed acceleration. Its original trace passed, but a further comparison found a failure: record the unpaid clause, pay the full claim later in the same dated sequence, then request acceleration from the old finding. The date still matches although the economic state has changed. That candidate incorrectly accepts the stale finding relative to its own current-clause requirement.

The successor records a digest of a fixed economic-state projection with each resolved clause observation. The projection contains the book's identity and period, account definitions, balances, cash account, cash-flow history, non-contract memory and its types. Conditional-status memory and seen-event bookkeeping are excluded, so recording a remedy does not invalidate itself. The selected schema and types are checked separately. A new acceleration requires both the date and economic-state digest to match the stored observation.

The same-day payment now makes the stale request unresolved before any accepted write. A new observation of the paid claim supports the consistent assessment that there is no new trigger. Balanced account changes, changes to unrelated economic memory and revisions to cash-flow evidence also invalidate the old observation. This conservative rule avoids guessing which economic changes matter. Matching digests establish equality of the supplied record, not truth of the evidence or persistence of external legal facts. The relevant external findings still have to be supplied for the event.

The selected financing contract reproduces four complete financing paths on two account charts, including their statements. The repaired remedy composition reproduces eight complete transitions, with separate same-day failure and repair evidence. These checks support the stated finite declarations and preserve the rejected predecessor. Wider financing decisions, carry, changed rights, actual series admission and independent review remain necessary for the full design.

The independent settlement and conditional-source records are retained under `data/accounting-master/gap-program-20261026/`. The selected financial conditions and opening checks are under `data/accounting-master/financing-opening-invariants-20261026/`. The date-only failure and its preserved comparator are under `data/accounting-master/financing-remedy-binding-20261026/`; the checked successor is under `data/accounting-master/financing-remedy-state-binding-20261026/`.

A.55 A coherent book needs complete collections and ordered identities

The return equations can be arithmetically correct while the proposed sequence is impossible. A sale may precede the creation of its recovered inventory lot; the same lot may be sold twice; or a balanced book may contain negative cash. Each failure concerns information that a scalar equality does not express. The finite collection and state references extend the preceding requirements to those relationships, retaining a distinction between checking a supplied state and applying an accounting transition.

In the retained example, settlement closes the return component and creates a recovered lot. A later sale consumes that lot. The lot's carrying amount also contributes to inventory in the

shared legal book. Three different checks follow: the relevant records must be present and identifiable, their sequence must make the lot available when it is sold, and their amounts must reconcile with the book. Passing any one of these checks cannot establish the other two.

A.55.1 Which members are actually known

`MethodCollections/1` declares typed collections and selected constraints over their fields. Its first operations are uniqueness, exact-one reference resolution, nondecreasing order, and a guarded requirement evaluated for each row. Global inputs and row inputs have explicit mappings to the preserved guarded evaluator. Type, context and role must agree; a field name does not by itself establish economic equivalence.

A supplied collection is either complete with its rows, or partial with its observed rows and a reason for incomplete coverage. Missing collections are partial with no observed rows. Partial rows retain their relative supplied order, while additional members may be absent between them. An unknown field has a separate tagged reason. Thus an empty complete list says that there are no members in the declared scope; an empty partial list says that none has yet been supplied. These are different assertions, and neither proves that the scope describes an actual company's complete record.

Known duplicates contradict uniqueness even when another key is unknown. When no duplicate is observed, an unknown key or incomplete membership prevents a uniqueness conclusion. A reference must resolve to exactly one matching target of the same type and context. Two known matches already contradict that requirement. No match is a contradiction only when the target collection is complete and its keys are known. One known match remains unresolved if another unknown or unobserved target could create a second match. Target-wide uniqueness is a separate condition when needed.

Ordering checks preserve the caller's sequence. If two known dates are reversed, an unknown date between them cannot repair the reversal. If the known subsequence is ordered but another date or member is unknown, the full order remains unresolved. Equal dates retain their original order rather than being sorted by a hidden tie-breaker. Period predicates separately check the retained half-open interval: an event on the start date is included, while an event on the exclusive end date is outside the period.

Per-row checks retain all original guarded findings and indexed input mappings. A known invalid row remains visible despite another unknown row or incomplete membership. An inactive condition still undergoes type and structure checks, and its separately required facts remain required. A complete empty collection vacuously satisfies a universal row condition; a partial empty collection leaves it unresolved. These rules prevent missing records from being mistaken for a successful review of every record.

A.55.2 A sale needs an earlier live lot

Snapshot membership cannot show that a lot existed before a sale. The retained comparison makes this concrete by giving settlement and sale the same date and placing the sale first. Date ordering then passes, but the native method rejects the sale because the lot does not yet exist. A second sale and a second complete settlement expose similar limitations. These failures motivate an explicitly ordered identity check rather than a stronger interpretation of the snapshot.

`MethodIdentityLifecycle/1` declares named identity pools, their initial live and retired members, and a closed action mapping. Each action contains an ordered list of `create`, `require` or `consume`

operations. Create requires an identity absent from both live and retired memory. Require checks that an identity is live without changing it. Consume moves a live identity into retired memory. Retaining retired identities prevents an already used identifier from being silently created again.

For the conditional return example, settlement consumes the live return-component identity and creates a recovered-lot identity. Sale consumes that lot. Consuming the component makes a second complete settlement fail; consuming the lot makes a second whole-lot sale fail. The supplied order also makes a sale before creation fail, including when both events have the same date. The initial empty retired pools in this example are explicit assumptions of its synthetic opening. They are not evidence about an actual company's lifetime event history.

Each event's operations run against a private copy of the identity pools. Only when every operation succeeds does the diagnostic trace adopt that event's result. If a later operation fails or needs an unknown key, no partial identity change from that event survives. Earlier successful steps remain visible with their indices and after-states. This is local identity atomicity: no money is posted, no journal is authorized and no durable transaction is committed.

Unknown initial identities or incomplete initial pools stop the trace before its first event. An incomplete event sequence also stops before any observed event, because an omitted predecessor might create or consume a needed resource. With a complete sequence, an unknown action or operation key stops at that event. Later state is not inferred. All observed rows still undergo structural checks, and independent snapshot findings remain available. An otherwise complete identity trace can therefore coexist with an unresolved snapshot condition; the combined result does not report success.

The four retained failures—sale before creation on an equal date, a missing sale lot, repeated whole-lot sale and repeated complete settlement—are all rejected by this explicit identity sequence. The result establishes only those declared resource relationships. Prices, cash availability, tax consequences and accounting entries still require their own conditions.

A.55.3 Signed totals and the limits of balance

The shared legal book uses signed account balances. In the retained convention, assets normally carry positive coefficients and liabilities and equity negative coefficients. For the declared complete account registry $\mathcal{A}$, the selected balance condition is

$$\sum_{a \in \mathcal{A}} b_a = 0.$$

The registry matters as much as the sum. Omitting an account or adding an unclassified zero-balance account is a structural mismatch even if the displayed coefficients still add to zero. The retained declaration therefore checks unique account keys and reference membership in both directions against the pinned native account list. Account-classification coverage is checked separately against the implementation's classification lists.

Let $\mathcal{L}$ be the declared complete recovered-lot collection, c_ℓ its carrying amount for lot ℓ, and I_{other} the separately represented inventory. Selected allocation conditions include

$$b_{\text{inventory}} = I_{\text{other}} + \sum_{\ell \in \mathcal{L}} c_\ell, \tag{A.46}$$

$$b_{\text{refund liability}} = -L_{\text{refund}}, \tag{A.47}$$

$$b_{\text{recovery asset}} = A_{\text{recovery}}. \tag{A.48}$$

The minus sign in the refund equation follows the signed-book convention; it is not an absolute-value normalization. The retained lot tax method separately requires each lot's tax basis to equal its carrying amount. Other tax methods need different, explicitly selected conditions.

`MethodCollectionTotals/1` binds an assessment input either to a declared global value or to the sum of one declared rational collection field. It retains the original type, role and contributor indices. Complete, known inputs give an exact reduced rational sum; an empty complete collection gives zero. A missing or unknown field, or partial collection membership, leaves the total unknown. The observed subtotal remains available as a diagnostic but is never substituted for that unknown total. Even cancellation of all observed amounts cannot remove an unknown contribution.

Contexts, dimensions and scales must match exactly. There is no inferred currency conversion, group-to-legal allocation, rounding or balancing entry. Snapshot findings and the guarded scalar assessment remain visible together, so a correct sum cannot override a duplicate account or incomplete reference registry. Completeness remains a supplied declaration whose source support must be checked before using the result as a claim about a company.

Two counterexamples show the remaining limitation. One adverse fixture has a negative cash balance and an offsetting capital change, keeping the book balanced solely to isolate the funding check. Another marks a lot disposed while leaving its stock amounts unchanged. Both satisfy the selected totals. Neither is an admissible state under the retained method. The capital offset belongs only to the diagnostic fixture; the engine does not manufacture financing to repair a cash shortfall.

A.55.4 Distinguish funding failure from an invalid state

The additional state predicates check nonnegative cash and other inventory, settled return rights, lot identities and amount domains, disposition memory, lifetime event identities and tax-adjustment references. A disposed lot must have zero units, carrying amount and tax basis; a live lot must have positive units. Its stored identifier must match its collection key and its component must match the owner. Lifetime event IDs must be unique, and each uniquely named tax adjustment must refer to an event already in that history.

These conditions reuse guarded scalar predicates, collection constraints and exact totals. Known settlement and disposition facts are separately required; unknown is never interpreted as false. A narrow `MethodStateAssessment/1` layer preserves the economic category of an active, known-false cash condition. It reports `funding_failure` while retaining the original predicate result. It cannot relabel malformed values, inactive conditions, unknowns or unrelated failures as funding outcomes.

The complete findings summary orders invalid state, funding failure, unsupported method scope and unresolved information, retaining every finding. This differs from the native invariant's first-failure control flow. If cash is negative and a later lifetime event ID is duplicated, the native function first reports the cash shortfall; the common assessment reports invalid state and also retains the funding failure. Another explicit difference concerns disposition memory: the native function rejects an unknown disposition because it requires a concrete Boolean, whereas the tagged-input extension reports unresolved. Neither difference is hidden inside a claim of full native equivalence.

The saved comparisons repair both counterexamples: balanced negative cash is a funding failure and disposed stock is invalid. Selected adverse cases also cover identity mismatches, negative or empty live lots, outstanding settled rights and invalid tax references. The branch disposition record distinguishes these checked predicates from fixed implementation constants and the

still-separate validation of complete outer and nested record shape.

The resulting chain gives increasingly informative prerequisite assessments: typed facts, applicable equations, complete collections, ordered identity memory, signed reconciliations and state-domain findings. Each stage leaves source and accounting admission, full method eligibility and execution false. The next section connects these prerequisites to a selected return transition: complete raw-state structure, exact monetary effects, atomic memory updates and full proposal recomputation. Reporting and annual carry retain their own method conditions, and source correspondence remains distinct from authority.

A.56 From a valid record to a financial transition

A company can have a balanced opening book and enough information to assess the conditions of a method without yet having a journal for the next event. The method must determine the amounts, identify the accounts, and change the contract and inventory records consistently. The return example makes these steps concrete. At the settlement date, the company pays refunds and handling costs, extinguishes the remaining return rights, and receives a recovered lot. Cash, income, liabilities and inventory change together.

Begin with the complete preceding record. The concrete native-state profile specifies the outer record and every nested component, lot, balance, tax premise, cash flow and prior-period record. It rejects a missing field, an unexpected field, a noncanonical rational or a non-Boolean disposition flag before projecting selected values into a requirement. Fields that can retain an explicit unknown are named by the schema; unknown tax findings need not prevent an earlier non-tax event. Structural validity establishes that the record has the specified meaning and shape. Balance, allocation, financial domains and source authority remain separate questions.

A.56.1 Derive the amounts before changing the book

Suppose the company accepts two returned units. Each unit entitles the customer to a refund of USD 20. Its original unit cost is USD 12, deterioration is USD 2 per unit, and handling costs USD 4 per unit. The supplied net realisable value of the recovered lot is USD 20. The selected method treats handling as a selling expense and requires an explicit finding that all return rights are closed. These are the assumptions of this example, including the specified treatment of the recovered stock; a different accounting method needs its own definition.

Let n be the returned quantity, p the per-unit refund, c the original unit cost, d the per-unit deterioration, h the handling cost per unit and N the lot's net realisable value. All monetary inputs refer to the same currency and selected measurement context. The refund Q, handling payment H and recovered carrying amount V are

$$Q = np = 40, \qquad H = nh = 8, \qquad V = \min\{n(c-d), N\} = \min\{2(12-2), 20\} = 20. \tag{A.49}$$

Units multiplied by currency per unit produce currency. Subtracting $c - d$ requires matching units, scale and context; taking the minimum requires two amounts with the same full type. If N were USD 5 under a separately supplied scenario, the carrying amount would be USD 5. That changed scenario would no longer be the exact retained source event.

The preceding refund liability has a positive magnitude $F = 40$, and the preceding recovery asset is $R = 16$. These are balances carried from the preceding state, not estimates recalculated

at settlement. With debits positive, the selected journal is described by

$$\begin{aligned} \Delta\text{RefundLiability} &= F, & \Delta\text{RecoveryAsset} &= -R, & \Delta\text{Inventory} &= V, \\ \Delta\text{Revenue} &= Q - F, & \Delta\text{CostOfSales} &= R - V, & \Delta\text{SellingExpense} &= H, \\ \Delta\text{Cash} &= -Q - H. \end{aligned} \tag{A.50}$$

Thus the liability receives a debit of 40, the recovery asset a credit of 16, and inventory a debit of 20. Revenue has no net adjustment; cost of sales is credited by 4, selling expense debited by 8, and cash credited by 48. Adding the signed entries cancels F, R, V, Q and H exactly. The cash-flow record retains the refund outflow of 40 and handling outflow of 8 separately as operating flows. Netting them into one number would lose their distinct origins.

`MethodMonetaryEffects/1` expresses these calculations as a closed program of named inputs, ordered calculations, signed postings and gross flows. It reuses the dimensional product rules of the requirement language. Addition, subtraction, multiplication and binary minimum evaluate exact rational coefficients; no implicit rounding, scale conversion or currency conversion is performed. A calculation can reference an original input or an earlier named calculation. Its declared dependencies identify the original inputs even when intermediate names are used.

Unknown values remain unknown through this arithmetic. If N is unavailable, the program retains that reason and withholds a complete monetary proposal. Independent known calculations, such as the refund, remain available for diagnosis. Even multiplication by zero does not silently turn an unobserved input into an observed fact. The profile also checks finite expression, record and coefficient limits; exceeding a limit produces an explicit rejection.

Balance and cash agreement are necessary checks, but they do not establish the measurement. Replacing V by $V + 1$ in both affected postings still gives a balanced journal. The inventory and cost-of-sales amounts would nevertheless be wrong relative to equation (A.49). The implementation therefore compares the derived amounts and complete effects, rather than using journal balance as a substitute for the method.

A.56.2 Stage the whole event or retain the preceding state

Settlement also changes memory. The component's refund and recovery amounts become zero and its settled flag becomes true. A new lot records the two units, carrying amount 20, acquisition date and component identity. In this selected tax treatment its tax base equals its carrying amount, and the tax adjustment is $R - F = -24$. Those tax choices are retained conditional premises. They are not inferred from the mere existence of the accounting journal.

`MethodMemoryEffects/1` provides three operations over closed typed records: replace an existing object member, insert an absent map member, and append an array item. A value may be a literal, an exact read from the preceding state or event-derived inputs, or a constructed record or array. Reads always use the original records; they cannot accidentally observe an earlier staged write. The complete preceding and resulting states must satisfy the selected schema.

Paths use explicit tokens. A lot identifier supplied through an input is one map key, including any slash or tilde that the applicable schema permits. It cannot redirect an update into another branch of the record. Overlapping writes are rejected, apart from repeated appends to the same array, whose supplied order is retained. A duplicate lot insertion, missing parent, wrong value type or failed final schema check discards the staged result. The caller's original state remains unchanged.

For a subsequent whole-lot sale with proceeds P and carrying amount V, the selected method requires completed performance and a live, undisposed lot. It posts cash P, revenue $-P$, inventory $-V$ and cost of sales V. The lot remains in memory with zero units, carrying amount

and tax base, and a true disposition flag. Its identity cannot be reused for another sale. Events with equal dates retain supplied order: a sale before its lot is created fails, while a sale after settlement on that same date can satisfy the identity condition.

The complete transition checks the preceding book, evaluates the event's requirements, derives monetary effects, adds them to balances, stages memory and validates the resulting book. Cash below zero is a funding failure under this profile's absence of an additional financing policy. It is not repaired by inventing borrowing or by committing only the noncash entries. The preceding state and all its identity records remain available after a rejected event.

A.56.3 Recompute the proposal from its inputs

The selected `TypedReturnTransition/1` adapter binds exact raw state and event bytes to the two recorded programs. Its guarded requirements retain original input paths and raw identities. Dates must lie inside the reporting period and cannot precede either the state date or the tax close. Refund and handling cash facts must equal the quantities derived by the method. A false rights or performance finding is outside the selected method; an unavailable required finding remains unresolved.

The adapter carries preceding state amounts into the declared event measurement context. This is an explicit interpretation of the selected transition: the original state date and source path remain recorded. Equality of type contexts does not itself establish that information was available at the event date. The concrete profile is restricted to the retained USD legal book and the two selected operations. Other currencies and tax or surrounding operations require their own admitted method routes.

A proposal includes the raw and decoded input identities, selected program and schema identities, prerequisite findings, journal, gross flows and complete after-state. Applying it recomputes every field from the original inputs before returning the result. A modified journal, cash-flow origin, lot flag, input identity or extra proposal field fails that comparison. A matching hash supplied by a caller cannot replace the computation. Applying the same event to its resulting state also fails the lifetime-event uniqueness condition.

The reference adapter reuses the frozen native invariant before and after the transition. Agreement with the native implementation therefore provides an independent check of the typed effect derivation, but not an independent proof of that shared invariant. The separate state-requirement examples provide additional, explicitly scoped evidence. Where several requirements fail, the common assessment retains all findings and may summarize them differently from the native function's first-error order.

A.56.4 Correspondence with a source is a separate check

A source pointer with the right syntax does not show that an event was reported or that its preceding book follows from the source. The retained example has an additional finite binder. It accepts the exact original synthetic packet, checks the schema-prefixed identities of its records, and reconstructs the prescribed event projection. Legacy exact numbers such as `2` become canonical rationals such as `2/1`; both the original packet identity and the derived record identity remain available.

To establish the preceding book for a selected event, the binder initializes the opening and recomputes the prior events in the packet's supplied order. It uses the pinned native and surrounding financial methods, without adopting saved after-states or expected journals. It then requires exact agreement between that recomputed state, the prescribed event and the

supplied transition inputs. A balanced book with an extra unit of cash offset by capital is consequently rejected: balance alone does not establish correspondence with this history.

The projection includes authored interface identifiers and conditional tax premises. Its zero opening tax bases follow the retained taxable-history assumption; they are not observations extracted from a company's filing. The packet also contains nonfinancial build timing, handled by a separate bounded parser for these exact bytes. The strict financial-record decoder continues to reject floating monetary inputs and ambiguous encodings.

This chain supplies a reproducible transition for the selected synthetic case: complete state structure, explicit prerequisites, typed amounts, atomic memory, full proposal replay and exact source correspondence. It leaves accounting and source-authority admission false, and its in-memory application makes no durable storage or production-concurrency claim. Reporting, annual carry, alternative histories and real-company feasibility retain their distinct obligations. The checked implementation packages and their exact limitations are preserved in the accompanying transition and source-binding evidence records.

A.57 Ordered multi-series EPS report records

`OrderedEPS/1` specifies the report taught in Sections 36.2.4–36.2.5. Its closed Draft 2020-12 JSON schemas are retained as `ordered-eps/record-schemas.json` in the continuation evidence directory. Every field below is required; unknown fields reject. Braces denote objects, brackets arrays, and alternatives distinct shapes. No migration from `FinancingComponents/2` is inferred.

```
Request {
  wire: "OrderedEPS/1", operation: "report",
  context, context_ref, inventory, scope, basic, series
}
context {
  entity, book, head, view, population,
  period: {start, end_exclusive}, ordinary_class, share_basis,
  currency, currency_scale, rule_version
}
support {path, sha256, anchor}
inventory {status: complete | incomplete, ids: [id], support}
scope {
  nonparticipating_rights, independent_additive_effects,
  fixed_effects, support
}
basic (ready) {
  status: ready, context_ref, support, continuing_numerator,
  total_numerator, denominator,
  attribution_complete, chronology_complete
}
basic (unresolved) {
  status: unresolved, context_ref, support, reason
}
```

Names and reasons are nonempty strings, findings are Booleans, dates are valid ISO calendar dates, and intervals include their start but exclude their end. A reporting period has positive length. Digests are 64 lower-case hexadecimal characters. The context reference is SHA256 of UTF-8 JSON with keys sorted, compact separators, ASCII escaping and no trailing newline. Every basic or series record must carry that reference. A support reference identifies exact local bytes and a nonempty anchor. Source resolution checks the bytes; a caller must separately interpret the anchor and the finding.

Quantities are reduced rational strings: `0`, a signed nonzero integer, or a reduced numerator/positive-denominator fraction. Leading zeros, plus signs, negative zero, decimals, exponents and a redundant denominator of one reject. The basic denominator is positive. The context binds units and scale before arithmetic; the report performs no currency or share-basis conversion. Both completeness findings in a ready basic record must be true to return basic EPS. The inventory and series IDs must agree bijectively, with no duplicates. A complete empty inventory is permitted with support.

```
series (candidate) {
  id, resolution: candidate, context_ref, support,
  continuing_increment, total_increment, share_increment,
  earnings_complete, time_weighting_complete,
  potential_windows: [{start, end_exclusive}],
  actual_windows: [{start, end_exclusive}],
  end_kind: period_end | lapsed | converted, actual_in_basic
}
series (no effect) {
  id, resolution: no_effect, context_ref, support,
  continuing_increment, total_increment, share_increment,
  reason
}
series (unresolved or unsupported) {
  id, resolution: unresolved | unsupported,
  context_ref, support, reason
}
```

A candidate has a positive, already weighted share increment; earnings increments may be signed. Both completeness findings must be true for diluted calculation. It has at least one potential window. All windows have positive length, lie within the reporting period and do not overlap within that series. For a conversion, actual windows are nonempty, follow the potential windows, and `actual_in_basic` is true. For lapse or period-end treatment, actual windows are empty and that finding is false. A period-end series' final potential window ends at the period boundary. A no-effect record has all three increments zero. Reasons retain missing facts or the further method required; they never stand for a silently omitted series.

```
Result {
  wire: "OrderedEPS/1", request_sha256, context_ref,
  status, basic: EPSBundle | null, diluted: EPSBundle | null,
  trace: [Comparison], issues: [{code, path, series_id}],
  mutations: [], acceptance: "conditional_documentary_only"
}
EPSBundle {
  continuing_numerator, total_numerator,
  discontinued_numerator, denominator,
  continuing_eps, total_eps, discontinued_eps
}
Comparison {
  id, resolution: candidate | no_effect,
  continuing_increment, total_increment, share_increment,
  incremental_ratio,
  before: {continuing_numerator, denominator},
  after_if_included: {continuing_numerator, denominator},
  included, reason: strictly_dilutive | neutral |
                    antidilutive | no_effect
}
```

Result quantities remain reduced rational strings. The request digest uses the same serialization

over the complete request. The context reference is null only when no valid reference can be read. An issue identifies its code, JSON path and series ID or null. A no-effect comparison has a null ratio and unchanged amounts. Candidates appear in exact increasing-ratio order, IDs breaking ties; no-effect items follow in ID order. The schema permits null comparison states; the completed reference report supplies them.

`conditional_complete` returns both bundles, the complete trace and no issues. `unresolved` records missing findings or source bytes, incomplete inventory or unresolved items. `unsupported` records unsupported scope or series; `blocked` records both obstacles. These three statuses retain independently ready basic EPS, with null diluted EPS and an empty trace. `rejected` takes precedence for structural, binding, rational, sign, inventory, date or source-hash violations and returns neither bundle. Issues sort by path, code and series ID; mutations are always empty. Application methods remain responsible for live head freshness, source truth, accounting admission and future publication.

A.58 Company-to-EPS projection

`CompanyEPSJoin/1` is the finite read-only bridge illustrated in Section 36.2.7. Its request has exactly the fields `wire`, `operation`, `world`, `base_packet`, `premises` and `eps`. The operation is `report`; world is `conversion` or `redemption`. The base reference gives the path, SHA256 and anchor of the preserved CompanyComposition/1 attempt03 packet, pinned to that packet's recorded manifest hash. The premise record supplies the instrument, dates, rate, tax bases, tax consequences, opening share count and completeness findings taught in the example. The proposed EPS request obeys `OrderedEPS/1`. Exact closed Request/Result schemas and the complete premise record accompany the company-EPS evidence directory.

The bridge resolves protected source bytes and checks their hashes, reproduces the original company, merges the added journals in date order, recalculates tax and consolidates the resulting books. Its head is the hash of the complete closing legal books; its evidence view is the hash of the supplied premises. Both use sorted compact UTF-8 JSON with ASCII escaping and no trailing newline. The EPS record must exactly match the derived population, period, class, share basis, currency, earnings, weighted shares and potential/actual windows. Thus group profit, separate-parent profit and closing share counts cannot be substituted for the required ordinary numerator and weighted denominator.

The result has exactly `wire`, `request_sha256`, `status`, `company`, `eps`, `issues`, `mutations` and `acceptance`. Company contains both complete legal books, the financial statement reconciliation, every dated journal and cash origin, and the daily ordinary/potential-share history. EPS contains the complete ordered report. The accepted status is `conditional_complete`; a failed company input leaves both results null. A rejected proposed EPS projection can retain the independently calculated company result while returning null EPS. Missing bytes or incomplete findings are unresolved; different known methods are unsupported; invalid shapes, changed bytes or projection mismatches reject. Mutations are always empty and acceptance is `conditional_documentary_only`. The schemas also retain the selector's combined `blocked` status for a propagated unresolved/unsupported result.

This exchange checks two explicit documentary worlds. It supplies neither an arbitrary-company import nor a migration of the whole financing profile. The separate early-conversion policy in Section 36.2.6 is limited to the unchanged original equity right; this exchange still checks conversion at maturity. Other early-conversion terms, differing tax bases, broader rights, actual authority and live publication require their separate methods and evidence. Independent technical and human/domain acceptance remain pending.

Appendix B

Core grammar, records and execution details

B.1 Preserved Core 0 teaching grammar

The grammar governing the preserved Core 0 teaching examples is reproduced from `spec/core-0.lark`. It does not enlarge the first matched text frontend beyond the portable profile described in the body. Names are case-sensitive ASCII identifiers; strings are UTF-8 with JSON escaping. All literal words in the grammar are reserved. The companion contract specifies valid dates, exact rational literals, argument arity, type/context checks and package resolution. A syntax parse is not semantic validation.

```
// Normative Core 0 syntax, Lark EBNF. UTF-8 input; ASCII identifiers.
// Longest token wins; keywords cannot be declaration identifiers.
start: "module" NAME "version" STRING ";" declaration*
?declaration: use | source | entity | book | account | observation | symbol
            | choice | binding | constraint | alternative | contract | event
            | rule | scenario | report | opening
use: "use" NAME "version" STRING ";"
source: "source" NAME "=" expr ";"
entity: "entity" NAME "functional" NAME ";"
book: "book" NAME "entity" NAME "basis" NAME "version" STRING ";"
account: "account" NAME ":" NAME "book" NAME ";"
observation: "observation" NAME ":" type time_spec "=" expr "from" NAME ";"
time_spec: "instant" DATE | "period" DATE "to" DATE
symbol: "symbol" NAME ":" type "in" domain "because" STRING ";"
domain: "[" expr "," expr "]" | "{" expr ("," expr)* "}"
choice: "choice" NAME "in" "{" NAME ("," NAME)* "}" "because" STRING ";"
binding: "let" NAME [":" type] "=" expr ";"
constraint: "constraint" NAME ":" expr ";"
alternative: "alternative" NAME "{" assumption* "}"
assumption: "assume" expr ";"
contract: "contract" NAME ":" NAME "book" NAME "{" field* "}"
field: NAME "=" expr ";"
event: "event" NAME "(" [parameters] ")" "book" NAME ";"
parameters: NAME ":" type ("," NAME ":" type)*
rule: "rule" NAME "for" NAME "as" NAME ["overrides" NAME ("," NAME)*] "when" expr "{" rule_statement* "}"
?rule_statement: binding | require | post
require: "require" expr ";"
post: "post" "{" posting+ "}"
posting: SIDE NAME expr ";"
scenario: "scenario" NAME "{" setting* "}"
setting: "set" NAME "=" expr ";"
report: "report" NAME "for" NAME "{" field* "}"
opening: "opening" NAME "for" NAME "from" expr ";"
type: NAME ["<" type ("," type)* ">"]
?expr: disjunction
?disjunction: conjunction (OR conjunction)*
?conjunction: comparison (AND comparison)*
```

```
?comparison: sum [COMPARE sum]
?sum: product (ADD product)*
?product: unary (MUL unary)*
?unary: UNARY unary | postfix
?postfix: atom ("." NAME)*
?atom: number | STRING | DATE | BOOL | call | NAME | "(" expr ")" | array | record
call: NAME "(" [expr ("," expr)*] ")"
number: NUMBER [NAME | "per" "year"]
array: "[" [expr ("," expr)*] "]"
record: "{" [record_field ("," record_field)*] "}"
record_field: NAME ":" expr
SIDE: "debit" | "credit"
OR: "or"
AND: "and"
UNARY: "-" | "not"
COMPARE: "==" | "!=" | "<=" | ">=" | "<" | ">"
ADD: "+" | "-"
MUL: "*" | "/"
BOOL.2: "true" | "false"
DATE: /@[0-9]{4}-[0-9]{2}-[0-9]{2}/
NAME: /[A-Za-z_][A-Za-z0-9_]*/
NUMBER: /(?:0|[1-9][0-9]*)(?:\.[0-9]+)?/
STRING: /"(?:[^"\\\x00-\x1f]|\\(?:["\\/bfnrt]|u[0-9a-fA-F]{4}))*"/
%import common.WS
%ignore WS
%ignore /\/\/[^\n]*/
```

B.2 Required core record definitions

The historical closed machine-readable definitions remain in the original design directory at `spec/core-ir.schema.json`. They define contexts, source spans, observations, narratives, coverage units, mappings, partitions, entity/control records, books, accounts, symbols, choices, expressions, constraints, assumptions, alternatives, contracts, events, rules, packages, measurements, states, journals, scenarios, information views, review decisions, evidence views, model scopes, historical families, probability models, report mappings, result cells, branches, runs and diagnostics. For those historical records the schema fixes field spelling and structural shape, and its companion contract fixes reference resolution and operations. New portable records instead obey the exact versioned contracts introduced in Chapters 33, 34 and 39. No historical field list grants new posting authority.

For example, these selected records reproduce the schema's field names. Each field is required; a nullable link is represented by explicit null.

```
EvidenceView {
  id, cutoff, system_as_of, mode,
  documents[], observations[], decisions[],
  conflict_sets[][], transforms[]
}
Context {
  id, entity, perimeter, concept, basis, measurement,
  time, unit, currency, taxonomy_version, dimensions[]
}
Symbol {
  id, type, context, domain, uncertainty_kind,
  scope, dependencies[], admission, provenance[]
}
Journal {
  id, event, book, date, accrual_period,
  postings[], rule, measurements[], assumptions[],
  precondition_hash, explanation
}
ResultCell {
  id, context, history_condition, scenario, path, kind,
  value, expression, lower, upper,
  lower_attained, upper_attained, sample_ref,
  certificate, reason, dependencies[]
}
```

A value is a reduced rational, or a typed symbolic/domain reference. A posting is nonnegative with its side carrying the sign, in the book's functional currency. Original units and the valuation reference survive conversion. Monetary contexts require currency; a period flow requires a calendar and exclusive end boundary. Historical zero postings may preserve traceability but establish no economic event. New portable proposals omit zero lines and retain their explanation in the event and findings instead. The amount may have finer precision before the declared posting-rounding operation.

Schema validation rejects unknown fields but cannot prove that a treatment is permitted, a reference points to the correct kind of record, or two periods are comparable. Those semantic checks precede solver or runtime dispatch. The structural fixture and complete demonstration source/package are supplied separately so that a future implementation can check both layers explicitly. `fixtures/core-ir-demo.json` supplies an authored expected lowering; the reference checker evaluates its selected debt expressions. A general source-to-IR compiler and complete external-registry validation remain to be implemented. Domain records retain disclosed display text and rounding rules alongside their exact bounds.

B.3 Typing and partial-information rules

For expressions x and y in the same currency C, the money addition rule is

$$\frac{x : \mathrm{Money}\langle C\rangle \quad y : \mathrm{Money}\langle C\rangle}{x + y : \mathrm{Money}\langle C\rangle}.$$

Context compatibility is a separate premise for observations. There is no analogous implicit rule for different currencies. Explicit conversion uses a dated measurement and records its rate and purpose. A rate-times-principal calculation is typed through a declared accrual convention; the example's annual simple rate and year fraction produce money in the principal's currency.

Known expressions evaluate normally. Shared symbols evaluate to symbolic expressions subject to a joint constraint store. Missing source values cannot enter arithmetic until the model supplies a symbol or a conditional assumption. Boolean guards over symbols produce formulas, not the value false. This avoids the common error in which "unknown whether impaired" silently means "not impaired." Division requires a proven nonzero denominator or a partitioned domain that excludes zero; otherwise the expression remains unsupported for concrete execution.

Logical queries first require a nonempty feasible family. An empty family returns `Infeasible`; it must not produce vacuous answers to substantive questions. For a nonempty family, constraint evaluation has three results. `entailed` means every state in the declared feasible family satisfies the proposition. `refuted` means none does. `undetermined` means both remain feasible or the solver has not decided; the response distinguishes those causes. Probabilistic truth is a separate query requiring a measure. A language-level Boolean value must never be inferred from an arbitrary probability threshold.

For a decided answer, let $F(h)$ be the exact constraint representation of the whole family requested by the query, and let $P(h)$ be its proposition. Establish feasibility first. Then `entailed` requires that $F(h) \wedge \neg P(h)$ has no solution, while `refuted` requires that $F(h) \wedge P(h)$ has no solution. Feasible witnesses on both sides establish the mixed case. A timeout, unsupported operation, or unfinished search establishes neither universal conclusion. A proof for a selected cell applies to that cell; it applies to the parent family only with coverage of the remaining cells or a proof over the full family. The result records the scope, query kind and evidence for that status. Section 32.2 distinguishes this Boolean test from a certain-tuples view and a family of answer schedules.

B.4 Operational rule selection

Let $V(e, h)$ be the applicable rules for event e in concrete history h, after package/date selection and guard evaluation. Let $r \prec r'$ mean that r' validly overrides r through the declared acyclic graph. Define

$$\mathrm{Max}(V) = \{r \in V : \nexists r' \in V \text{ with } r \prec r'\}.$$

For one required recognition operation, a singleton selects that rule; the empty set yields `UnsupportedTreatment`; a larger set yields `RuleConflict`. Multiple distinct operations, such as measurement and recognition, are evaluated through separate declared interfaces. A rule that intentionally produces no postings must belong to a supported non-recognition event operation, whose economic state effect is still recorded.

Given a selected rule, the evaluator computes all local bindings and preconditions, emits proposed postings, verifies Equation (32.2), validates account/context restrictions and commits the event transaction. Failure returns the original state plus a diagnostic. This defines the atomicity requirement independently of the storage technology.

B.5 Example run request and response

The following request executes two named conditions under one scenario. It does not request an aggregate probability distribution.

```
{
  "model": "BorrowingExample@1",
  "evidence_view": "synthetic-2025-exact",
  "starting_state": "after-2025-close",
  "horizon": {"start": "2026-01-01", "end_exclusive": "2027-01-01"},
  "conditions": ["allFixed", "allFloating"],
  "scenarios": ["rateShock"],
  "mode": "conditional",
  "weight_model": null,
  "reports": ["balance_sheet", "income", "cash_flow", "equity", "journals"],
  "package_lock": "demonstration-packages-v1"
}
```

The response contains two separately labelled results with interest expense 5 and 7, cash 90 and 88, debt 100 in both, and equity -10 and -12. A request for the parent symbolic family returns the formula $7 - 2\alpha$ and, under exact linear bounds, $[5, 7]$ before monetary quantisation. The API rejects an unconditional mean request without a weight model. A user can instead request an arbitrary clearly labelled representative case, which does not stand in for the feasible family.

B.6 Requirement traceability

Requirement	Primary implementation area	Conformance cases
R1	Evidence registry and extraction adapters	T03, T09
R2	Context/version model and cutoff queries	T03, T04
R3	Types, mappings and package selection	T02, T12, T13
R4	Symbols, choices and constraint graph	T05–T10, T14
R5	Posting, closing and consolidation	T01, T11–T13
R6	Contract and behavioural interfaces	T05, T11, T15, T16
R7	Conditional output and probability types	T06, T07, T18
R8	Explanation graph and report dependencies	T09, T11–T13
R9	Capabilities and versioned modules	T02, T14–T16
R10	Run records, scheduler and replay	T03, T17, T18

Appendix C

Research basis and outstanding review

C.1 Source collection and reading scope

The initial research catalogue records 32 verified paper PDFs: 25 retrieved from external sources and seven preserved copies from the original collection. Later source inquiries retain additional papers and their technical reading in separate versioned records, including the reserve-risk and bootstrap studies used in Chapter 12. The initial count therefore describes that catalogue, not the full expanded library. Filenames use the title, first author's surname and year. Publication years are used where an archived working version underlies a later publication; the catalogue identifies the actual version read. The original collection has not been renamed or modified. Standards pages, contract specifications, software source snapshots and metadata are stored separately from papers.

The survey considered accounting algebra and DSLs, corporate and bank simulation, incomplete and temporal databases, provenance, inverse problems, partial identification, model averaging, causal scenarios, contract languages, probabilistic programming, document extraction and financial reasoning. OpenAlex and Crossref supplied dated metadata, while primary publisher, institutional and author copies supplied technical support. Public-index results were checked against the downloaded title pages; HTTP success alone was not accepted as a valid paper download.

Technical reading was targeted to the mechanisms used in this design. The full PDF is archived even where only specified sections support the argument. No claim is made that every theorem, appendix or empirical result in every archived paper was independently replicated. Representative support anchors are listed below; the source-support ledger contains the per-source scope and qualifications.

Source	Inspected technical support	Design use
Nehmer and Robinson	Sections 2–5: account states, transactions and aggregation	Conservation plus admissibility.
Doshi	Chapters 4, 7–9 of the 2024 thesis	Dates, logs, corrections and DSL scope.
Shahnazarian	Sections 2.3–2.8; relevant Chapter 3 equations; data and evaluation sections; Appendix B	Accounting/behaviour separation; limits of data transfer.

Source	Inspected technical support	Design use
Pierson	Sections 3–4 and reporting structures; supplementary Vensim equations	Accrual/payment bridges and reporting accumulators.
Caiani et al.	Selected Sections 2–5, Appendix A and author-linked consumption-firm code	Event order, counterpart stocks and explicit cash/debt-service definitions.
Gentry (1974 working paper)	Printed pp.1–9 and 12–13; Figure 1	Managerial objectives and financing decisions; return derivation unverified.
Libkin	Sections 2–5: incomplete states, conditions and certain answers	Shared unknowns and joint alternatives.
Green et al.	Definitions 3.1–3.2; Section 4	Composable relational provenance; limits outside positive operations.
Stuart	Section 2.2, observation equation and prior/regularisation discussion	Inverse-problem formulation, not imported convergence guarantees.
Giacomini and Kitagawa	Sections 2.1–2.6 and Theorem 1 in CWP12/20	Prior sensitivity and classes of posterior beliefs.
Hoeting et al.	Equations (1)–(3); prior and predictive-evaluation discussion	Optional model-weighted predictions.
Merigoux et al.	Sections 3–4, default calculus and translation figures	Scopes, exceptions, empty/conflict outcomes.
Peyton Jones et al.	Sections 3–5, combinators and valuation semantics	Contract composition separate from valuation assumptions.
FinQA / TAT-QA	Dataset operations, annotation, model and evaluation sections; FinQA evaluation source	Evidence-linked local arithmetic; benchmark boundaries.
DocFinQA / FinBen	Context/retrieval and financial-task evaluations	Long-document difficulty and task coverage.
PubTables / GTE	Canonicalisation, dataset construction, methods and evaluation	Table structure and financial-domain extraction.
Cosentino et al.	Sections 3.1–3.2, 4.6 and 5	Persistent RAG provenance; scope of auditability evidence.

C.2 Version and access qualifications

Several discrepancies mattered. Doshi's thesis is dated 2024 despite the original filename. The archived Mernik text is the 2003 CWI report underlying the 2005 article. Giacomini and Kitagawa's copy is the April 2020 working paper underlying the 2021 article. The Docling v5 copy lists Auer as first author. The probabilistic-programming tutorial is an arXiv version revised in 2021, first submitted in 2018; a merged index DOI was not used. The Libkin author PDF has a footer DOI that resolves to another paper; title/author metadata from the indexes resolves the cited work to DOI `10.1145/2594538.2594561`. These discrepancies remain

documented rather than being hidden by a cleaned bibliography.

The initial catalogue recorded McCarthy (1982), Imieliński and Lipski (1984), Tamer (2010), and Caiani and co-authors (2016) as blocked candidates. Access failures and rejected wrong-content downloads remain historical records. On 23 October, the correct Caiani journal paper was recovered from a university-hosted copy; selected technical sections and author-linked code now support the bounded comparison in Chapter 30. The Gentry working paper recovered the same day is dated 1974, and its equivalence to the 1979 journal version remains unestablished. The Francis–Rowell method remains uninspected after unsuccessful publisher and author routes. Uninspected technical claims are not used. More specialised work on accounting measurement, process mining, multi-company equilibrium, actuarial inference and learned forecasting is recorded as expansion work for the corresponding implementation stage. The survey supports the core architecture without pretending that those later modules have a complete literature or rule base.

The 25 October continuation recovered Gentry and Hoque's October 1990 working paper as a separate comparator. Its method, definitions, equations, sample, estimation discussion, conclusions, references and exhibits were inspected; the 46-page university-contributed scan matches the archive's recorded size and digest. The comparison in Chapter 30 uses that inspected paper without replacing the unresolved 1974/1979 or Francis–Rowell source questions. The retained ACTUS service still lacks a matched core: its README and Gradle dependency identify different versions, and its Dockerfile copies a local core absent from the inspected tree. The official developer route requests registration. No matched release is admitted on this evidence. Exact source findings and page anchors are retained in `data/accounting-master/simulator-survey-source-recovery-20261025/`.

OpenAlex retraction flags and available Crossref correction/version relations were inspected alongside local versions. No reviewed source was excluded for a discovered retraction. Missing correction metadata is marked unavailable; absence of a discovered notice does not certify that no notice exists. Citation counts are dated index metadata, and venue rankings were not available in this campaign. Neither is used as evidence of correctness.

C.3 Companion records and status

The six research ledgers are `source-support.json`, `citation-venue.json`, `backward-snowball.json`, `forward-snowball.json`, `claim-support.json`, and `omissions.json`. They record technical anchors, metadata sources, considered leads, omitted or blocked candidates and the distinction between source claims and this document's derivations. Citation leads with checked seed references are distinguished from cross-survey additions and topic expansion leads. Forward searches retain the most-cited and most-recent results returned by the recorded bounded queries; they do not establish exhaustive citation coverage. The separate `tools.json` records software evidence. Raw retrievals and text extracts permit a later reviewer to revisit the decisions.

The research library includes Mack's reserving paper, with its conditional-mean/variance assumptions, proofs, aggregate-error dependence and numerical comparisons inspected. It also archives official AASB texts incorporating IAS 2, IAS 32 and IFRS 17, distinguishing their Australian additions and applicable editions. The corporate filings are evidence of reported amounts and policies, rather than substitutes for every underlying standard. The ASBE 1 reproduction has not been verified against an accessible official current text, and official FASB/ASC access remains incomplete. These limitations are recorded at the specific operations they affect.

Each of the 179 unique candidates in the initial acquired forward-query files has an individual

title/metadata relevance decision. This initial population includes previously omitted discovery files associated with inaccessible seed papers. The initial generation script requires an explicit decision and stops on an unseen candidate. Later searches retain their own bounded populations and relevance decisions. Screening is still distinct from technical reading: deferred provenance, temporal-query, bank-optimisation, extraction and actuarial follow-ups remain identified expansion risks. An omitted-paper reason does not make that paper technically reviewed or make the survey exhaustive.

This document specifies the product; it does not report a completed engine. The companion review records cover deterministic example arithmetic, bibliography/build checks, rendered-page inspection and author review. Human comprehension review, independent accounting review and implementation conformance remain separate future evidence. They are not replaced by a clean LaTeX build or by the quantity of literature collected.

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